

EIGHTH EDITION

SHORTER OXFORD TEXTBOOK OF PSYCHIATRY

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Preface to the eighth edition

In the eight years since the previous edition of this book was published, the knowledge base, clinical practice, and nature of psychiatry have continued to evolve. The importance of understanding how to promote mental health and prevent and treat psychiatric disorders has become prominent not only for those working in healthcare but also across social, educational, and occupational sectors. Reflecting these developments, this edition has been significantly revised throughout. Several key changes are worth noting. We have incorporated the arrival of ICD-11 and its impact on diagnostic practice; we also include the mental health aspects of the Covid-19 pandemic. Coverage of global and public health has been strengthened and revised. In this regard, we welcome Kam Bhui to the author team. Kam brings a wealth of experience in these domains and in social psychiatry, including issues of race, ethnicity, and culture. This new edition has also been updated to reflect the increased

focus on topical issues regarding gender identity, digital media, the changing doctor–patient relationship, including changes to mental health legislation being implemented in the coming years.

The book continues to be aimed at psychiatrists in training but is also suitable for medical students and other professionals working across health and social care, and anyone seeking a concise, up-to-date, and evidence-based introduction to psychiatry. Further details of many subjects can be found in the third edition of the two-volume reference textbook, the *New Oxford Textbook of Psychiatry*, published in 2020.

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PJH, PJC, MF, and KB
Oxford, January 2026

Contents

1	Signs and symptoms of psychiatric disorders	1
2	Classification	21
3	Assessment	37
4	Ethics and civil law	71
5	Aetiology	89
6	Evidence-based approaches to psychiatry	119
7	Reactions to stressful experiences	137
8	Anxiety and obsessive–compulsive disorders	161
9	Depression	193
10	Bipolar disorder	231
11	Schizophrenia	251
12	Paranoid symptoms and syndromes	291
13	Eating, sleep–wake, and sexual disorders	305
14	Dementia, delirium, and other neuropsychiatric disorders	335
15	Personality and personality disorder	381
16	Child and adolescent psychiatry	403
17	Disorders of intellectual development	467
18	Forensic psychiatry	493
19	Psychiatry of the elderly	521
20	The misuse of alcohol and drugs	543
21	Suicide and self-harm	589
22	Psychiatry and medicine	609
23	Public mental health and global mental health	645
24	Psychological treatments	655
25	Drugs and other physical treatments	685
26	Psychiatric services	755

References 779

Index 841

CHAPTER 1

Signs and symptoms of psychiatric disorders

Introduction 1

General issues 2

Descriptions of symptoms and signs 4

Introduction

Psychiatrists require two distinct capacities. One is the capacity to collect clinical data as objectively and accurately as possible, then organize, apply, and communicate the information in a systematic and balanced way. The other is the capacity for the intuitive understanding of each patient as an individual, as a foundation for personalized medicine. These capacities reflect the science and the art of psychiatry, respectively. When a psychiatrist exercises the first capacity, they draw on their skills and knowledge of clinical phenomena; when they exercise the second capacity, they draw on both their knowledge of human nature as well as their experience with former patients to gain insights into the patient they are now seeing. Both capacities can be developed by listening to patients, and by learning from more experienced psychiatrists. A textbook can provide the information and describe the procedures necessary to develop the first capacity. The focus of this chapter on the first capacity does not imply that intuitive understanding is unimportant, but rather that the skills cannot be learned directly or solely from a textbook.

Skill in examining patients depends on a sound knowledge of how symptoms and signs are defined.

Without this grounding, the psychiatrist is liable to misinterpret or misclassify phenomena, compromising their ability to make accurate diagnoses. For this reason, this chapter is concerned with the definition of the key symptoms and signs of psychiatric disorders. Having elicited a patient's symptoms and signs, the psychiatrist needs to decide how far these phenomena fall into a pattern that has been observed in other psychiatric patients. In other words, they decide whether the clinical features conform to a recognized syndrome. This is done by combining observations about the patient's current state with information about the history of the condition, within the broader context of the person's background and circumstances.

The value of identifying a syndrome is that it helps to predict prognosis and to select an effective treatment, by directing the psychiatrist to the relevant body of accumulated knowledge about the causes, treatment, and outcome in similar patients. Diagnosis and classification are covered in Chapter 2, and also in each of the chapters dealing with the various psychiatric disorders. Chapter 3 discusses how to elicit and interpret the symptoms described in this chapter, and how to integrate the information to arrive at a diagnosis.

General issues

Before individual phenomena are described, some general issues are considered concerning the methods of studying symptoms and signs, and the terms used to describe them.

Psychopathology

The study of abnormal states of mind is known as *psychopathology*. The term embraces two distinct approaches to the subject—*descriptive* and *experimental*. This chapter is concerned almost exclusively with the former; the latter is introduced here but is discussed further in later chapters.

Descriptive psychopathology

Descriptive psychopathology is the objective description of abnormal states of mind avoiding, as far as possible, pre-conceived ideas or theories, and limited to the description of conscious experiences and observable behaviour. It is sometimes also called *phenomenology* or *phenomenological psychopathology*, although the terms are not synonymous, and phenomenology has additional meanings (Berrios, 1996). Likewise, descriptive psychopathology is more than just *symptomatology* (Stanghellini and Broome, 2014).

The aim of descriptive psychopathology is to elucidate the essential qualities of morbid mental experiences and to understand each patient's experience of illness. It therefore requires the ability to elicit, identify, and interpret the symptoms of psychiatric disorders, and as such is a key element of clinical practice; indeed, it has been described as 'the fundamental professional skill of the psychiatrist' (Oyeboode, 2022).

The most important exponent of descriptive psychopathology was the German psychiatrist and philosopher Karl Jaspers. His classic work, *Allgemeine Psychopathologie* (*General Psychopathology*), first published in 1913, still provides the most complete account of the subject, and the seventh edition is available in an English translation (Jaspers, 1963). A briefer introduction can be found in Jaspers (1968), and Oyeboode (2022) has provided a highly readable contemporary text on descriptive psychopathology.

Experimental psychopathology

This approach seeks to explain abnormal mental phenomena, as well as to describe them. One of the first attempts was *psychodynamic psychopathology*, originating in Freud's psychoanalytic investigations. It explains

the causes of abnormal mental events in terms of mental processes of which the patient is unaware (i.e. they are 'unconscious'). For example, Freud explained persecutory delusions as being evidence, in the conscious mind, of activities in the unconscious mind, including the mechanisms of repression and projection (see Chapter 5).

Subsequently, experimental psychopathology has focused on empirically measurable and verifiable conscious psychological processes, using methods such as cognitive and behavioural psychology and functional brain imaging. For example, there are cognitive theories of the origin of delusions, panic attacks, and depression. Although experimental psychopathology is concerned with the causes of symptoms, it is usually conducted in the context of the syndromes in which the symptoms occur. Thus its findings are discussed in the chapter covering the disorder in question.

Terms and concepts used in descriptive psychopathology

Symptoms and signs

In general medicine there is a clear definition of, and separation between, a symptom and a sign. In psychiatry the situation is different. There are few 'signs' in the medical sense (apart from, for example, the motor abnormalities of catatonia or the physical manifestations of anorexia nervosa), with most diagnostic information coming from the history and from observations of the patient's appearance and behaviour. Use of the word 'sign' in psychiatry is therefore less clear, and two different uses may be encountered. First, it may refer to a feature noted by the observer rather than something spoken by the patient (e.g. a patient who appears to be responding to a hallucination). Secondly, it may refer to a group of symptoms that the observer interprets in aggregation as a sign of a particular disorder. In practice, the phrase 'symptoms and signs' is often used interchangeably with 'symptoms' (as we have done in this chapter) to refer collectively to the phenomena of psychiatric disorders, without a clear distinction being drawn between the two words.

Subjective and objective

In general medicine, the terms *subjective* and *objective* are used as counterparts of symptoms and signs, respectively, with 'objective' being defined as something

observed directly by the doctor (e.g. meningism, jaundice)—even though, strictly speaking, it is a subjective judgement on his part as to what has been observed.

In psychiatry, the terms have broadly similar meanings as they do in medicine, although with a blurring between them, just as there is for symptoms and signs. ‘Objective’ refers to features observed during an interview (i.e. the patient’s appearance and behaviour). The term is usually used when the psychiatrist wants to compare this with the patient’s description of symptoms. For example, in evaluation of depression, complaints of low mood and tearfulness are subjective features, whereas observations of poor eye contact, psychomotor retardation, and crying are objective ones. If both are present, the psychiatrist might record ‘subjective *and* objective evidence of depression’, with the combination providing stronger evidence than either alone. However, if the patient’s behaviour and manner in the interview appear entirely normal, they record ‘not objectively depressed’, despite the subjective complaints. It is then incumbent on the psychiatrist to explore the reasons for the discrepancy and to decide what diagnostic conclusions to draw. As a rule, objective signs are accorded greater weight. Thus, a depressive disorder may be diagnosed if there is sufficient evidence of this kind, even if the patient denies the subjective experience of feeling depressed. Conversely, the psychiatrist may question the significance of complaints of low mood if there are none of the objective features associated with the diagnosis of depression.

Form and content

When psychiatric symptoms are described, it is useful to distinguish between form and content, a distinction that is best explained by an example. If a patient says that, when alone, they hear voices telling them they have AIDS, the *form* of the experience is an auditory hallucination (see ‘Types of hallucination’), whereas the *content* is the statement that they have AIDS. Another patient might hear voices saying that they are about to be killed. Again, the form is an auditory hallucination, but the content is different. A third patient might experience repeated intrusive thoughts that they have AIDS, but realizes that these are untrue. Here the content is the same as that of the first example, but the form is different.

Form is often critical when making a diagnosis. From the examples given above, the presence of a hallucination indicates (by definition) a psychosis of one kind or another, whereas the third example suggests obsessive-compulsive disorder. Content is less diagnostically useful, but can be very important in management; for

example, the content of a delusion may suggest that the patient could attack a supposed persecutor. It is also the content, not the form, that is of concern to the patient, whose priority will be to discuss the persecution and its implications, and who may be irritated by what seem to be irrelevant questions about the form of the belief. The psychiatrist must be sensitive to this difference in emphasis between the two parties.

Primary and secondary

With regard to symptoms, the terms *primary* and *secondary* are often used, but unfortunately with two different meanings. The first meaning is *temporal*, simply referring to which occurred first. The second meaning is *causal*, whereby primary means ‘arising directly from the pathological process’, and secondary means ‘arising as a reaction to a primary symptom’. The two meanings often coincide, as symptoms that arise directly from the pathological process usually appear first. However, although subsequent symptoms are often a reaction to the first symptoms, they are not always of this kind, for they too may arise directly from the pathological process. The terms primary and secondary are used more often in the temporal sense because this usage does not involve an inference about causality. However, many patients cannot say in what order their symptoms appeared. In such cases, when it seems likely that one symptom is a reaction to another—for example, that a delusion of being followed by persecutors is a reaction to hearing accusing voices—it is described as secondary (using the word in the causal sense). The terms primary and secondary are also used in descriptions of syndromes.

Understanding and explanation

Jaspers (1913) contrasted two forms of understanding when applied to symptoms. The first, called *Verstehen* (‘understanding’), is the attempt to appreciate the patient’s subjective experience: what does it feel like? This important skill requires intuition and empathy. The second approach, called *Erklären* (‘explanation’), accounts for events in terms of external factors; for example, the patient’s low mood can be ‘explained’ by childhood trauma or recent redundancy. The latter approach requires knowledge of psychiatric aetiology.

The significance of individual symptoms

Psychiatric disorders are diagnosed when a defined group of symptoms (i.e. a syndrome) is present. Almost any single symptom can be experienced by a healthy person; even hallucinations, often regarded as a hallmark of severe mental disorder, can be experienced by healthy people. An exception to this is that a delusion, even if isolated, is

usually considered to be evidence of psychiatric disorder—a psychosis of some kind—if it is unequivocal and persistent, and out of keeping with the person's cultural or religious beliefs and associated with impaired functioning. In general, however, the finding of a single symptom is not evidence of psychiatric disorder, but an indication for a thorough and, if necessary, repeated search for other symptoms and signs of psychiatric disorder.

The patient's experience

Symptoms and signs are only part of the subject matter of psychopathology. The latter is also concerned with the patient's experience of illness, and the way in which psychiatric disorder changes their self-view, their hopes for the future, and their view of the world (Stanghellini and Broome, 2014). This may be seen as one example of the understanding (*Verstehen*) mentioned above. A depressive disorder may have a very different effect on a person who has lived a satisfying and happy life and has fulfilled their major ambitions, compared with a person who has had many previous misfortunes but has lived on hopes of future success. To understand this aspect of the patient's experience, the psychiatrist has to understand a person in the way that a biographer understands their subject. This way of understanding is sometimes called the life-story approach. It is not something that can be readily assimilated from textbooks; it is best learned by taking time to listen to patients. The psychiatrist may be helped by reading biographies or works of literature that provide insights into the ways in which experiences throughout life shape the personality, and help to explain the diverse ways in which

different people respond to the same events. Other ways of understanding patient experience include narrative approaches to assessment, and creative arts methods, including creative writing.

Cultural variations in psychopathology

The core symptoms of most serious mental disorders are present in individuals from culturally diverse settings. However, there are cultural differences in how distress is appraised and expressed and, if recognized as symptoms, how these symptoms present in clinical settings. There are also cultural variations in the meanings that are attributed to them—also called explanatory models of illness or illness perceptions. For example, depression can present with prominent somatic symptoms, and these may be more attended to, less stigmatizing, and more prominent, in many East and South Asian populations. The content of symptoms can also differ between cultures. For example, for sub-Saharan African populations, delusions often centre upon being cursed, a rare delusional theme in Europeans. Cultural differences also affect the person's subjective experience of illness, and therefore influence that person's understanding of it (Fabrega, 2000). In many cultures, mental illness is greatly stigmatized, and can, for example, hinder prospects of marriage. In such a culture the effect of illness on the patient's view of self and their future will be different from the effect on a patient living in a society that is more tolerant of mental disorder. Indeed, in stigmatizing societies, discrimination may be common and undermine health and social function. Further discussion of cultural issues can be found in Chapter 23.

Descriptions of symptoms and signs

Disturbances of emotion and mood

Much of psychiatry is concerned with abnormal emotional states, particularly disturbances of mood and other emotions, especially anxiety. Before describing the main symptoms of this kind, it is worth clarifying two areas of terminology that may cause confusion, in part because their usage has changed over the years.

First, the term 'mood' can either be used as a broad term to encompass all emotions (e.g. 'anxious mood'), or in a more restricted sense to mean the emotion that runs from depression at one end to mania at the other. The former usage is now uncommon. The latter

is emphasized by the fact that, in current diagnostic systems, 'mood disorders' are those in which depression and mania are the defining characteristics, whereas disorders defined by anxiety or other emotional disturbances are categorized separately. In this section, features common to both 'mood' and 'other emotions' are described first, before the specific features of anxiety, depression, and mania are discussed separately.

The second point concerns the term 'affect'. This is now usually used interchangeably with the term 'mood', in the more limited meaning of the latter word (e.g. 'their affect was normal', 'they have an affective disorder'). However, in the past, these words had different nuances of meaning; mood referred to a prevailing and

prolonged state (like ‘climate’), whereas affect was linked to a particular aspect or object, and was more transitory (like ‘weather’).

Changes in the nature of emotions and mood

A person’s current emotional and mood state can be one of anxiety, depression, elation, or irritability and anger. In each case the distinction between normal and abnormal is a matter of degree, and the clinical significance of the emotional state is also impacted by its duration and the co-occurrence of other symptoms and signs. For example, mood that is significantly depressed is pervasive and often accompanied by other features of a depressive disorder; an anxiety state is accompanied by autonomic overactivity and increased muscle tension.

Changes in the way that emotions and mood vary

Emotions and mood vary in relation to the person’s circumstances and preoccupations. In abnormal states, this variation with circumstances may continue, but the variations may be greater or less than normal. Increased variation is called *lability* of mood; extreme variation is sometimes called *emotional incontinence*. Variability of mood is significant in various ways independent of the severity of depression (Broome et al. 2015).

Reduced variation is called *blunting* or *flattening*. These terms have been used with subtly different meanings, but are now usually used interchangeably. Blunting or flattening usually occurs in depression and schizophrenia. Severe flattening is sometimes called *apathy* (note the difference from the layperson’s meaning of the word).

Emotion can also vary in a way that is not in keeping with the person’s circumstances and thoughts, and this is described as *incongruous* or *inappropriate*. For example, a patient may appear to be in high spirits and laugh when talking about the death of their mother. Such incongruity must be distinguished from embarrassed laughter, which indicates that the person is ill at ease.

Clinical associations of emotional and mood disturbances

Disturbances of emotions and mood are the central feature of the mood disorders and anxiety disorders. They are also common in most other psychiatric disorders.

Anxiety

Anxiety is a normal response, but is abnormal when its severity is out of proportion to the threat of danger, or when it outlasts the threat. Anxiety includes somatic and autonomic components as well as the core subjective experience. All can be thought of as equivalent to the preparations for dealing with danger seen in other mammals, ready for flight from, avoidance of, or fighting with a predator. Mild to moderate anxiety enhances most kinds of performance, but very high levels interfere with it.

The anxiety response is considered further in Chapter 8. Its main components can be summarized as follows.

- *Psychological*. The essential feelings of dread and apprehension are accompanied by restlessness, narrowing of attention to focus on the source of danger, worrying thoughts, increased alertness (with insomnia), and irritability (i.e. a readiness to become angry).
- *Somatic*. Muscle tension and respiration increase. If these changes are not followed by physical activity, they may be experienced as muscle tension tremor, or the effects of hyperventilation (e.g. dizziness).
- *Autonomic*. Heart rate and sweating increase, the mouth becomes dry, and there may be an urge to urinate or defaecate.
- *Avoidance of danger*. A *phobia* is a persistent, irrational fear of a specific object or situation. Usually there is also a marked wish to avoid the object, although this is not always the case—for example, fear of illness (hypochondriasis). The fear is out of proportion to the objective threat, and is recognized as such by the person experiencing it. Phobic people feel anxious not only in the presence of the object or situation, but also when thinking about it (*anticipatory anxiety*).

Depression

Depression is a normal response to loss or misfortune, when it may be called grief or mourning. Depression is abnormal when it is out of proportion to the misfortune, or is unduly prolonged, or when there is no obvious reason for it. Depressed mood is closely coupled with other changes, notably a lowering of self-esteem, pessimistic or negative thinking, and a reduction in or loss of the experience of pleasure (*anhedonia*). A markedly depressed person often has a characteristic expression and appearance, with turned-down corners of the mouth, a furrowed brow, and a hunched, dejected posture. The level of arousal is reduced in some depressed patients (*psychomotor retardation*) but increased in others,

with a consequent feeling of restlessness or agitation. The psychopathology of depression is discussed further in Chapter 9.

Clinical associations

Depression as a symptom can occur in any psychiatric disorder. It is the defining feature of mood disorders, and commonly occurs in schizophrenia, anxiety, obsessive-compulsive disorder, eating disorders, and substance-induced disorders. It can also be a manifestation of an organic disorder.

Elation

Elation is an extreme degree of happy mood which, like depression, is coupled with other changes, including feelings of self-confidence and wellbeing, increased activity, and increased arousal. The latter is usually experienced as pleasant, but sometimes as an unpleasant feeling of restlessness. Elation occurs most often in mania and hypomania.

Irritability and anger

Irritability is a state of increased readiness for anger. Both irritability and anger may occur in many kinds of disorder, so they are of little value in diagnosis. However, they are of great importance in risk assessment and risk management, as they may result in harm to others and self. Irritability may occur in anxiety disorders, depression, mania, dementia, and drug intoxication.

Disturbances of perception

Specific kinds of perceptual disturbance are symptoms of severe psychiatric disorders. It is therefore important to be able to identify these symptoms and to distinguish them from the other, much less significant, alterations in sensory experience which occur. We therefore describe perceptual phenomena in some detail.

Perception and imagery

Perception is the process of becoming aware of what is presented through the sense organs. It is not a direct awareness of data from the sense organs, because these data are acted on by cognitive processes that reassemble them and extract patterns. Perception can be attended to or ignored, but it cannot be terminated by an effort of will.

Imagery is the awareness of a percept that has been generated within the mind. Imagery can be called up and terminated by an effort of will. Images are experienced as lacking the sense of reality that characterizes perception, so that a healthy person can distinguish

between images and percepts. A few people experience *eidetic imagery*, which is visual imagery so intense and detailed that it has a 'photographic' quality akin to a percept, although in other ways it differs from a percept. Imagery is generally terminated when perception starts. Occasionally, imagery persists despite the presence of percept (provided this is weak and unstructured). This sort of imagery is called *pareidolia*.

Percepts may alter in intensity and in quality. Anxious people may experience sensations as more intense than usual; for example, they may be unusually sensitive to noise. In mania, perceptions seem more vivid than usual. Depressed patients may experience perceptions as dull and lifeless.

Illusions

Illusions are misperceptions of external stimuli. They occur when the general level of sensory stimulation is reduced and when attention is not focused on the relevant sensory modality. For example, at dusk the outline of a bush may be perceived at first as that of a person, although not when attention is focused on the outline. Illusions are more likely to occur when the level of consciousness is reduced, as in delirium, or when a person is anxious. Illusions have no diagnostic significance, but need to be distinguished from hallucinations.

Hallucinations

A *hallucination* is a percept that is experienced in the absence of an external stimulus to the corresponding sense organ. It differs from an illusion in being experienced as originating in the outside world or from within the person's body (rather than as imagined). Hallucinations cannot be terminated at will.

Hallucinations are generally indications of significant psychiatric disorder, and specific types of hallucination are characteristic of different disorders, as outlined below. However, as noted above, hallucinations do occur in some otherwise healthy people. It is also common to experience them when falling asleep (*hypnagogic hallucinations*) or on waking (*hypnopompic hallucinations*). These two types of hallucination may be either visual or auditory, the latter sometimes as the experience of hearing one's name called. Such hallucinations are common in narcolepsy. Some recently bereaved people experience hallucinations of the dead person. Hallucinations can occur after sensory deprivation, in people with blindness or deafness of peripheral origin, occasionally in neurological disorders that affect the visual pathways, in epilepsy, and in Charles Bonnet syndrome.

Types of hallucination

Hallucinations can be described in terms of their complexity and their sensory modality (see Box 1.1). The term *elementary hallucination* refers to experiences such as bangs, whistles, and flashes of light, whereas *complex hallucination* refers to experiences such as hearing voices or music, or seeing faces. Most hallucinations of psychiatric significance are complex.

Auditory hallucinations may be experienced as noises, music, or voices. Voices may be heard clearly or indistinctly; they may seem to speak words, phrases, or sentences. They may seem to address the patient directly (*second-person hallucinations*), or talk to one another, referring to the patient as 'he' or 'she' (*third-person hallucinations*). Sometimes patients say that the voices anticipate what they are about to think a few moments later. Sometimes the voices seem to speak the patient's thoughts as they are thinking them (*Gedankenlautwerden*), or to repeat them immediately after they have thought them (*écho de la pensée*).

Visual hallucinations may also be elementary or complex. The content may appear normal or abnormal in size; hallucinations of dwarf figures are sometimes called *lilliputian*. Occasionally, patients describe the

experience of visual hallucinations located outside the field of vision, usually behind the head (*extracampine hallucinations*).

Olfactory hallucinations and *gustatory hallucinations* are frequently experienced together. The smells and tastes are often unpleasant.

Somatic hallucinations, sometimes called *tactile* or *haptic* hallucinations, may be experienced as sensations of being touched, pricked, or strangled. Sometimes they are felt as movements just below the skin, which the patient may attribute to creatures burrowing through the tissues. *Hallucinations of deep sensation* may be experienced as feelings of the viscera being pulled upon or distended, or of sexual stimulation or electric shocks.

An *autoscopical hallucination* is the experience of seeing one's own body projected into external space, usually in front of oneself, for short periods. The experience is reported occasionally by healthy people in situations of sensory deprivation, when it is called an out-of-body experience, or after a near-fatal accident or heart attack, when it has been called a near-death experience. Rarely, the experience is accompanied by the conviction that the person has a double (*Doppelgänger*).

Reflex hallucination is a rare phenomenon, in which a stimulus in one sensory modality results in a hallucination in another; for example, music may provoke visual hallucinations.

As noted above, *hypnagogic hallucinations* occur while falling asleep, and *hypnopompic hallucinations* occur on waking. They can occur in any modality and are of sudden onset such that the person believes that the perception woke them up. They are common in healthy people (despite meeting the criteria for true hallucinations) and also a feature of sleep disorders.

Clinical associations of hallucinations

Hallucinations occur in diverse disorders, notably schizophrenia, severe mood disorder, organic disorders, and dissociative states. Therefore the finding of hallucinations does not itself help much in diagnosis. However, as with delusions, there are certain kinds of hallucination that do have important implications for diagnosis of schizophrenia and other disorders.

- *Auditory hallucination*. Only clearly heard voices (not noises or music) have diagnostic significance. Third-person hallucinations (introduced above) are strongly associated with schizophrenia. Such voices may be experienced as describing the patient's intentions in a *running commentary* (e.g. 'He wants to make love to her') or actions (e.g. 'She is washing her face'), or

Box 1.1 Description of hallucinations

According to complexity

- Elementary
- Complex

According to sensory modality

- Auditory
- Visual
- Olfactory and gustatory
- Somatic

According to special features

- Auditory
- Second-person
- Third-person
- Gedankenlautwerden
- Écho de la pensée
- Visual
- Extracampine

Other

- Autoscopical
- Reflex
- Hypnagogic and hypnopompic

may make critical comments. Second-person auditory hallucinations (i.e. those that appear to address the patient) do not point to a particular diagnosis, but their content and the patient's reaction to them may do so. Thus voices with derogatory content (e.g. 'You are a failure, you are wicked') suggest severe depressive disorder, especially when the patient accepts them as justified. In schizophrenia, the patient more often resents such comments. Voices that anticipate, echo, or repeat the patient's thoughts also suggest schizophrenia.

- *Visual hallucinations* should always suggest the possibility of an organic disorder (i.e. a neurological or general medical disorder; see Chapter 2), although they can also occur in severe mood disorders, schizophrenia, and dissociative disorder. The content of visual hallucinations is of little diagnostic significance. Autoscopic hallucinations also raise suspicion of an organic disorder, such as temporal lobe epilepsy.
- *Hallucinations of taste and smell* are infrequent. They may occur in schizophrenia, severe depressive disorders, and temporal lobe epilepsy, and in tumours affecting the olfactory bulb or pathways.
- *Somatic hallucinations* are suggestive of schizophrenia, especially if they are bizarre in content or interpretation. The sensation of insects moving under the skin (*formication*) occurs in people who abuse cocaine.

Pseudohallucinations

This term refers to experiences that are similar to hallucinations, but which do not meet all of the requirements of the definition, nor have the same implications. The word has two distinct meanings. In the first meaning, pseudohallucination is a sensory experience that differs from a hallucination in not seeming to the patient to represent external reality, being located within the mind rather than in external space. In this way pseudohallucinations resemble imagery although, unlike imagery, they cannot be dismissed by an effort of will. In the second meaning, the sensory experience appears to originate in the external world, but it seems unreal. For a more detailed discussion, see Taylor (1981).

Both definitions of pseudohallucinations are difficult to apply clinically, because patients can seldom describe their experiences in adequate detail. In any event, it is usually sufficient to decide whether a perceptual experience is a 'true' hallucination or not, since it is only the former that carries diagnostic significance. If it is not a hallucination, the experience should be described, but

need not be labelled as one kind of pseudohallucination or the other.

Abnormalities in the meaning attached to percepts

A *delusional perception* is a delusion arising directly from a normal percept. This is sometimes erroneously considered to be a perceptual disturbance, but it is really a disorder of thought, and is therefore discussed in the next section.

Disturbances of thoughts

Disturbances of thoughts and thought processes are among the most diagnostically significant symptoms in psychiatry. As with disturbances of perception, therefore, this area of descriptive psychopathology merits relatively detailed description. It covers two kinds of phenomena:

1. *Disturbance of thoughts* themselves—that is, a change in the *nature* of individual thoughts. The category of delusion is particularly important. Disturbances of thought are covered in this section.
2. *Disturbance of the thinking process* and the linking together of different thoughts; this may affect the speed or the form of the relationship between thoughts. It can occur even if individual thoughts are unremarkable in nature. These phenomena are covered in the following section.

Delusions

A *delusion* is a belief that is firmly held on inadequate grounds, that is not affected by rational argument or evidence to the contrary, and that is not a conventional belief that the person might be expected to hold given their educational, cultural, and religious background. This definition is intended to distinguish delusions, which are cardinal symptoms of severe psychiatric disorder (and specifically of psychosis), from other kinds of abnormal thoughts and from strongly held beliefs found among healthy people. There are several problems with the definition, summarized in Box 1.2, but it suffices as a starting point for more detailed discussion.

Although not part of the definition, another characteristic feature of delusions is that they have a marked effect on the person's feeling and actions—in the same way that strongly held normal beliefs do. Since the behavioural response to the delusion may itself be out of keeping or even bizarre, it is often this that first brings the person to psychiatric attention, and leads to the delusion being elicited. For example, a person with the

Box 1.2 Problems with the definition of delusions

Delusions are firmly held despite evidence to the contrary

The hallmark of a delusion is that it is held with such conviction that it cannot be altered by presenting evidence to the contrary. For example, a patient who holds the delusion that there are persecutors in the adjoining house will not be convinced by evidence that the house is empty. Instead they may suggest that the persecutors left the house shortly before it was searched. The problem with this criterion for delusions is that some of the ideas of normal people are equally impervious to contrary evidence. For example, the beliefs of a convinced spiritualist are not undermined by the counterarguments of a non-believer. Strongly held but non-delusional beliefs are called *overvalued ideas*.

A further problem with this part of the definition of delusion relates to *partial delusions*. Although delusions are usually held strongly from the start, sometimes they are at first held with a degree of doubt. Also, during recovery it is not uncommon for patients to pass through a stage of increasing doubt about their delusions before finally rejecting them. The term 'partial delusion' refers to both these situations. It should be used during recovery only when it is known that the beliefs were preceded by a full delusion, and applied to the development of a delusion only when it is known in retrospect that a full delusion developed later. Partial delusions are not, in isolation, helpful in diagnosis—akin to the status of pseudohallucinations.

Delusions are held on inadequate grounds

Delusions are not arrived at by the ordinary processes of observation and logic. Some delusions appear suddenly without any previous thinking about the subject (*primary delusions*). Other delusions appear to be attempts

to explain another abnormal experiences—for example, the delusion that hallucinated voices are those of people who are spying on the patient.

Delusions are not beliefs shared by others in the same culture

This criterion is important when the patient is a member of a culture or subculture (including a religious faith), because healthy people in such a group may hold beliefs that are not accepted outside it. Like delusions, such cultural beliefs are generally impervious to contrary evidence and reasoned argument—for example, beliefs in evil spirits. Therefore, before deciding that an idea is delusional, it is important to determine whether other members of the same culture share the belief.

Delusions as false beliefs

Some definitions of delusions indicate that they are false beliefs, but this criterion was not included in the definition given above. This omission is because, in exceptional circumstances, a delusional belief can be true or can subsequently become true. A well-recognized example relates to pathological jealousy (see Chapter 12). It is not falsity that determines whether the belief is delusional, but the nature of the mental processes that led up to it. There is a further practical problem concerning the use of falsity as a criterion for delusion. It is that if the criterion is used, it may be assumed that, because a belief is highly improbable, it is false. This is certainly not a sound assumption, because improbable stories—for example, of persecution by neighbours—sometimes turn out to be true and arrived at through sound observations and logical thought. Therefore ideas should be investigated thoroughly before they are accepted as delusions.

These issues are discussed further in Butler and Braff (1991). See Garety and Freeman (2013) for a cognitive account of delusions.

delusion that they were being irradiated by sonic waves covered their windows with silver foil. Occasionally, however, a delusion has little influence on feelings and actions. For example, a patient may believe that they are

a member of the royal family while living contentedly in a bedsit. This separation is called *double orientation*, and usually occurs in chronic schizophrenia.

Types of delusions

Several types of delusions are recognized, categorized either by the characteristics or by the theme of the delusion (see Box 1.3). Many of the terms are simply useful descriptors, but a few of them carry particular diagnostic implications; for example, specific types of delusions are *first rank symptoms* of schizophrenia (Chapter 11). Most categories of delusions can be diagnosed reliably (Bell et al., 2006).

Primary and secondary delusions

A *primary* or *autochthonous* delusion is one that appears suddenly and with full conviction but without any mental events leading up to it. For example, a patient with schizophrenia may be suddenly and completely convinced, for no reason and with no prior thoughts of this kind, that they are changing sex. Not all primary delusional experiences start with an idea. Sometimes the first

experience is a delusional mood or a delusional perception (see below). Because patients do not find it easy to remember the exact sequence of such unusual and distressing mental events, it is often difficult to be certain which experience came first. Primary delusions are given considerable weight in the diagnosis of schizophrenia, and they should be recorded only when it is certain that they are present.

Secondary delusions are delusions apparently derived from a preceding morbid experience. The latter may be of several kinds, including hallucinations (e.g. someone who hears voices may believe they are being followed), low mood (e.g. a profoundly depressed person may believe people think they are worthless), or an existing delusion (e.g. a person who is convinced they are being ‘framed’ may come to believe that they will be imprisoned). Some secondary delusions seem to have an integrative function, making the original experiences more comprehensible to the patient, as in the first example above. Others seem to do the opposite, increasing the sense of persecution or failure, as in the third example. Secondary delusions may accumulate until there is a complicated and stable delusional system. When this happens the delusions are said to be *systematized*.

Delusional mood

When a patient first experiences a delusion, they often respond emotionally. For example, a person who believes that a group of people intends to kill them is likely to feel afraid. Occasionally, the change of mood precedes the delusion. This preceding mood is often a feeling of foreboding that some, as yet, unidentified sinister event is about to take place. When the delusion follows, it appears to explain this feeling. In German this antecedent mood is called *Wahnstimmung*. This term is usually translated as *delusional mood*, although it is really the mood from which a delusion arises.

Delusional perception

Sometimes the first abnormal experience is the attaching of a new significance to a familiar percept without any reason to do so. For example, the position of a letter that has been left on the patient’s desk may be interpreted as a signal that he is to die. This experience is called *delusional perception*, and it is an example of a first rank symptom of schizophrenia. Note that the perception is normal; it is the delusional interpretation that is abnormal. A delusional perception may follow delusional mood.

Delusional misidentification

This is the delusional misidentification of oneself or of specific other people. Several eponymous forms are

Box 1.3 Descriptions of delusions

According to fixity

- Complete
- Partial

According to onset

- Primary
- Secondary

Other delusional experiences

- Delusional memory
- Delusional mood
- Delusional perception

According to theme

- Bizarre
- Delusions concerning possession of thought:
 - Thought insertion
 - Thought withdrawal
 - Thought broadcasting

Delusions of control

Delusions of guilt

Delusions of reference

Grandiose (expansive)

Hypochondriacal

Nihilistic

Persecutory (paranoid)

Religious

Sexual or amorous

According to other features

- Shared delusions
- Mood congruency

described, and have been considered to be both symptoms and syndromes (see Chapter 12).

Delusional memory

In *delusional memory*, a delusional interpretation is attached to past events. In the commoner form, the past event was genuine, and the term ‘delusional’ refers to the significance that has now become attached to it. For example, a patient who believes that there is a current plot to poison them may remember (correctly) that they vomited after a meal, eaten long before their psychosis began, and now concludes (incorrectly) that they had been intentionally poisoned. Alternatively, a sudden (autochthonous) delusion arises, which is wrongly dated to a past event. This latter form might be viewed as a true delusional memory (i.e. the memory itself is the delusion), whereas in the first kind described, the memory is normal, but a delusional interpretation is placed upon it.

Shared delusions

As a rule, other people recognize delusions as false and argue with the patient in an attempt to correct them. Occasionally, a person who is close to a deluded patient comes to share their delusional beliefs. This condition is known as *shared delusions* or *folie à deux*. Although the second person’s delusional conviction is as strong as the partner’s while the couple remain together, it often recedes quickly when they are separated.

Delusional themes

For the purposes of clinical work, it is useful to group delusions according to their main themes, since the themes have some diagnostic significance. However, it is first worth considering the word ‘paranoid’, which is used widely but not always clearly in this context (see Box 1.4).

Persecutory delusions

These are most commonly concerned with persons or organizations that are thought to be trying to inflict harm on the patient in one way or another. Such delusions are common but of little help in diagnosis, because they can occur in delusional disorders, organic states, schizophrenia, and severe affective disorders. However, the patient’s attitude to the delusion may point to the diagnosis. In a severe depressive disorder, a patient with persecutory delusions characteristically accepts the supposed activities of the persecutors as justified by their own wickedness. In schizophrenia, however, they may resent these activities as unwarranted.

Delusions of reference

These are concerned with the idea that objects, events, or people that are unconnected with the patient have a personal significance for them. For example, the patient may believe that an article in a newspaper or a remark on television is directed specifically to them, either as a message or to inform others about them. Delusions of reference may also relate to actions or gestures made by other people which are thought to convey a message about the patient. Although most delusions of reference have persecutory associations, some relate to grandiose themes.

Delusions of control (passivity phenomena)

A patient who has a delusion of control believes that their actions, impulses, or thoughts are controlled by an outside agency. These are also called *passivity phenomena*. Delusions of control are strongly suggestive of schizophrenia, and have forensic implications, so particular care should be taken when eliciting and recording them. The symptom may be confused with voluntary obedience to commands from hallucinatory voices, with religious beliefs that God controls human actions, or with a metaphorical view of one’s free will. By contrast, a patient with a delusion of control firmly believes that their movements or actions are brought about by an outside agency (other than the divine), and are not self-generated. Other symptoms of schizophrenia are usually present as well.

Delusions concerning the possession of thought

Healthy people take it for granted that their thoughts are their own. They also know that thoughts are private experiences that become known to other people only if they are spoken aloud, or revealed in writing or through facial expression, gesture, or action. Patients with delusions concerning the possession of thoughts can lose these normal convictions in several ways, all of which are strongly associated with schizophrenia:

- *Thought insertion* is the delusion that certain thoughts are not the patient’s own but are implanted by an outside agency. Often there is an associated explanatory delusion—for example, that persecutors have used radio waves to insert the thoughts. This experience must not be confused with that of the obsessional patient, who may be distressed by thoughts that they feel are alien to their nature but who never doubts that these thoughts are their own.
- *Thought withdrawal* is the delusion that thoughts have been taken out of the mind. The delusion usually accompanies thought blocking, in which the patient

Box 1.4 The term 'paranoid'

The term 'paranoid' is often used as if it were equivalent to 'persecutory'. Strictly interpreted, however, the word 'paranoid' has a wider meaning (Lewis, 1970). It was used in ancient Greek writings to mean the equivalent of 'out of their mind'. For example, Hippocrates used it to describe patients with febrile delirium. Many later writers applied the term to grandiose, erotic, jealous, and religious delusions, as well as to persecutory delusions. Although for historical reasons it is preferable to retain

the broader meaning of the term, the narrower usage is now more common. Because the term 'paranoid' has two possible meanings, the term 'persecutory' is preferable when the narrow sense of paranoid is required to describe a symptom. The issue also affects the use of the word to describe syndromes in which such symptoms predominate; the older term 'paranoid psychoses' (or 'paranoid states') was replaced by 'delusional disorders', in part to avoid the ambiguities (see Chapter 12).

experiences a sudden break in the flow of thoughts and believes that the 'missing' thoughts have been taken away by some outside agency. Often there are associated explanatory delusions comparable to those that accompany delusions of thought insertion.

- *Thought broadcasting* is the delusion that unspoken thoughts are known to other people through radio, telepathy, or in some other way. In addition, some patients believe that their thoughts can be heard out loud by other people, a belief that also accompanies the experience of hearing one's own thoughts spoken (*Gedankenlautwerden*), described above in the section 'Types of hallucination'.

Grandiose delusions

These are beliefs of exaggerated self-importance. The patient may believe they are wealthy, endowed with unusual abilities, or a special person. Such expansive ideas occur particularly in mania, and in schizophrenia.

Bizarre delusions

Delusions with highly improbable content (e.g. of control by aliens who communicate via birds) are said to be *bizarre*. They are often given particular weight in the diagnosis of schizophrenia, but the category has problems of reliability and definition (Cermolacce et al., 2010), and it is not included in current diagnostic criteria.

Delusions of guilt

These beliefs are found most often in depressive illness, and for this reason are sometimes called *depressive delusions*. Typical themes are that a minor infringement of the law in the past will be discovered and bring shame upon the patient, or that his sinfulness will lead to retribution on his family.

Nihilistic delusions

These are beliefs that some person or thing has ceased, or is about to cease, to exist. Examples include a patient's delusion that they have no money, that their career is ruined, or that the world is about to end. Nihilistic delusions are seen in severe depression. Occasionally, nihilistic delusions concern failures of bodily function (e.g. that the bowels are blocked, or even that the patient is dead) and are sometimes referred to as *Cotard's syndrome*.

Hypochondriacal delusions

These are beliefs concerned with illness. The patient believes, wrongly and in the face of all medical evidence to the contrary, that they are suffering from a disease. Such delusions are more common in the elderly, reflecting the increasing concern with health among people in this age group. Other hypochondriacal delusions are concerned with the appearance of parts of the body. They must be distinguished from the health worries of hypochondriasis (Chapter 22), which are not delusional.

Mood-congruent and mood-incongruent delusions

If a delusion 'makes sense' in terms of the person's mood, it is said to be mood-congruent. Hypochondriacal and nihilistic delusions in psychotic depression, and grandiose delusions in mania, both fall into this category. In contrast, a delusion that is out of keeping with the prevailing mood is mood-incongruent, and is suggestive of schizophrenia. The concept of mood congruency can also be applied to hallucinations.

Delusions of jealousy

These are more common among men than women. Not all jealous ideas are delusions; less-intense jealous pre-occupations and obsessions are common. Jealous delusions are important because they may lead to aggressive behaviour towards the person(s) who is thought to be

unfaithful. A patient with delusional jealousy is not satisfied if they fail to find evidence supporting their beliefs; their search will continue.

Sexual or amorous delusions

These are rare, and more frequent in women than in men. Sexual delusions are occasionally secondary to somatic hallucinations felt in the genitalia. A person with amorous delusions believes that they are loved by a person who is usually inaccessible to them, and often of higher social status. In many cases they have never spoken to or met the person. Erotic delusions are the defining feature of *de Clérambault's syndrome* (Chapter 12).

Obsessional and compulsive symptoms

Obsessions

Obsessions are recurrent persistent thoughts, impulses, or images that enter the mind despite efforts to exclude them. One characteristic feature is the subjective sense of a struggle—the patient resists the obsession, which nevertheless intrudes into awareness. Another characteristic feature is the conviction that to think something is to make it more likely to happen. Obsessions are recognized by the person as their own and not implanted from elsewhere (in contrast to delusions of thought insertion). Another important distinction from delusions is that obsessions are regarded as untrue or senseless. They are generally about matters that the patient finds distressing or otherwise unpleasant. They are often, but not always, accompanied by *compulsions*.

The presence of resistance is important because, together with the lack of persistent or complete conviction about the truth of the idea, it distinguishes obsessions from delusions. However, in practice this distinction can be difficult, since the resistance tends to diminish when obsessions have been longstanding. Furthermore, when obsessions are very intense, patients may become less certain that they are false. However, a careful history, not only of the symptom but also of other relevant features (e.g. compulsions, other evidence of psychosis) helps avoid diagnostic difficulties. It is also necessary to distinguish clinically significant obsessions from similar thoughts that occur in healthy people, especially when they are tired or under stress. This requires evidence of dysfunction and persistence. Obsessional symptoms can also be a personality trait, as in anankastic personality disorder (Chapter 15).

Obsessions can take various forms (see Box 1.5).

- *Obsessional thoughts* are repeated and intrusive words or phrases that are upsetting to the patient—for

Box 1.5 Obsessional and compulsive symptoms

Obsessions
Thoughts
Ruminations
Doubts
Impulses
Obsessional phobias
Compulsions (rituals)
Obsessional slowness

example, repeated obscenities or blasphemous phrases coming into the awareness of a religious person.

- *Obsessional ruminations* are repeated worrying themes of a more complex kind—for example, about the ending of the world.
- *Obsessional doubts* are repeated themes expressing uncertainty about previous actions—for example, whether or not the person turned off an electrical appliance that might cause a fire. Whatever the nature of the doubt, the person realizes that the degree of uncertainty and consequent distress is unreasonable.
- *Obsessional impulses* are repeated urges to carry out actions, usually ones that are aggressive, dangerous, or socially embarrassing—for example, the urge to pick up a knife and stab another person, to jump in front of a train, or to shout obscenities in church. Whatever the urge, the person has no wish to carry it out, resists it strongly, and does not act on it.
- *Obsessional phobias*. This term denotes an obsessional symptom associated with avoidance as well as anxiety—for example, the obsessional impulse to injure another person with a knife may lead to consequent avoidance of knives.
- *Obsessional slowness*. Many obsessional patients perform actions slowly because their compulsive rituals or repeated doubts take time and distract them from their main purpose. Occasionally, however, the slowness does not seem to be secondary to these other problems but rather appears to be a primary feature of unknown origin.

Although the content (or themes) of obsessions are various, most of them can be grouped into one or other of six categories:

1. dirt and contamination
2. aggression

3. orderliness
4. illness
5. sex
6. religion.

Thoughts about *dirt and contamination* are usually associated with the idea of harm to others or self through the spread of disease. *Aggressive thoughts* may be about striking another person or shouting angry or obscene remarks in public. Thoughts about *orderliness* may be about the way objects are to be arranged, or work is to be organized. Thoughts about *illness* are usually of a fearful kind—for example, a dread of cancer. Obsessional ideas about *sex* usually concern practices that the individual would find shameful. Obsessions about *religion* often take the form of doubts about the fundamentals of belief (e.g. ‘does God exist?’) or repeated doubts about whether sins have been adequately confessed (‘scruples’).

Compulsions

Compulsions are repetitive and seemingly purposeful behaviours that are performed in a stereotyped way (hence the alternative name, ‘*compulsive rituals*’) in response to an obsession. They are accompanied by a subjective sense that the behaviour must be carried out and by an urge to resist it. The compulsion usually makes sense given the content of the obsession. For example, a compulsion to wash the hands repeatedly is usually driven by obsessional thoughts that the hands are contaminated. Sometimes obsessional ideas concern the consequences of failing to carry out the compulsion in the ‘correct’ way—for example, that another person will suffer an accident. Compulsions may cause problems for several reasons.

- They may cause direct harm (e.g. dermatitis from excessive washing).
- They may interfere with normal life because of the time they require.
- Although the compulsive act transiently reduces the anxiety associated with the obsession, in fact, the compulsions help to maintain the condition. Strategies to reduce them are central to behavioural treatments of obsessive–compulsive disorder.

There are four main types of compulsive acts.

1. Checking rituals are often concerned with safety—for example, checking repeatedly that the fire has been turned off, or that the doors have been locked.
2. Cleaning rituals often take the form of repeated hand washing but may involve household cleaning.

3. Counting rituals usually involve counting in some special way—for example, in threes—and are frequently associated with doubting thoughts such that the count must be repeated to make sure that it was carried out adequately in the first place. The counting is usually silent, so an onlooker may be unaware of the ritual.
4. In dressing rituals, the person lays out clothes, or puts them on, in a particular way or order. The ritual is often accompanied by doubting thoughts that lead to seemingly endless repetition.

Overvalued ideas

Overvalued ideas were first described by Wernicke in 1900, and reviewed by McKenna (1984). An overvalued idea is a comprehensible and understandable idea which is pursued beyond the bounds of reason. It may preoccupy and dominate a person’s life for many years, and affect their actions. It therefore shares some characteristics of delusions. However, it is essential to distinguish the two types of belief, as their diagnostic implications are very different. Overvalued ideas differ from delusions in two main ways.

1. The content of, and basis for, the overvalued idea is usually understandable when the person’s background is known, whereas delusions and the person’s explanation of them tend to be bizarre. For example, a person whose mother and sister suffered from cancer one after the other may understandably become convinced that cancer is contagious.
2. The theme also tends to be culturally common and acceptable, as in the overvalued ideas about body shape that characterize anorexia nervosa.

With an overvalued idea, there is a small degree of insight and willingness to at least entertain alternative views, even though this is not persistent and the patient always returns to and retains the belief.

Overvalued ideas must also be distinguished from obsessions. This is usually easier than the distinction from delusions, since there is no sense of intrusiveness or senselessness of the thought, nor is there resistance to it. Overvalued ideas differ from normal religious beliefs in that the latter are shared by a wider group, arise from religious instruction, and are subject to periodic doubts. Despite these differences, it can on occasion be difficult to recognize an overvalued idea and distinguish it unequivocally from a delusion, obsession, or normal belief. However, this should rarely lead to practical problems, because diagnosis depends on more than the presence or absence of a single symptom.

The beliefs concerning body shape and weight that are held in anorexia nervosa are perhaps the clearest example of overvalued ideas. According to McKenna (1984), the term also applies to abnormal beliefs in many other conditions, including dysmorphophobia, hypochondriasis, paranoid personality disorder, and morbid (pathological) jealousy. However, it is important to emphasize that overvalued ideas are defined by their form, not their content, and they have no inviolable relationship with, or implication for, any particular diagnostic category. Thus some cases of morbid jealousy are clearly delusional, whereas in hypochondriasis or dysmorphophobia the belief often has the character of an obsession or a worry, not of an overvalued idea.

Disturbances of thinking processes

Disturbances of the stream of thought

In disturbances of the stream (or flow) of thought, the amount and speed of thinking are changed. In *pressure of thought*, ideas arise in unusual variety and abundance and pass through the mind rapidly. In *poverty of thought*, the patient has few thoughts, and these lack variety and richness and seem to move slowly through the mind. Pressure of thought occurs in mania; poverty of thought occurs in depressive disorders. Either may be experienced in schizophrenia. Given that the phenomena are recognized through the person's use of language, they are also known as *pressure of speech* or *poverty of speech*.

Thought block

Sometimes the stream of thought is interrupted suddenly. The patient feels that their mind has gone blank, and an observer notices a sudden interruption in the patient's speech. In a minor degree this experience is common, particularly when a person is tired, anxious, or distracted. In thought blocking, the interruptions are sudden, striking, and repeated, and are experienced by the patient as an abrupt and complete emptying of their mind. Thought blocking is an important symptom, as it strongly suggests schizophrenia.

Disorders of the form of thought

Disorder of the form of thought (also known as *formal thought disorder*) is usually recognized from speech and writing, but is sometimes evident from the patient's behaviour—for example, they may be unable to file papers under appropriate category headings. Disorders of the form of thought can be divided into several kinds,

as described below. Each kind has associations with a particular mental disorder, but none of the associations is strong enough to be diagnostic.

Perseveration

Perseveration is the persistent and inappropriate repetition of the same thought(s). The disorder is detected by examining the person's words or actions. Thus, in response to a series of simple questions, the person may give the correct answer to the first question, but continue to give the same answer inappropriately to subsequent questions. Perseveration occurs in, but is not limited to, dementia and frontal lobe injury.

Flight of ideas

In flight of ideas, thoughts and speech move quickly from one topic to another so that one train of thought is not carried to completion before another takes its place. The normal logical sequence of ideas is generally preserved, although ideas may be linked by distracting cues in the surroundings and by distractions arising from the words that have been spoken. These verbal distractions are of three kinds, namely *clang associations* (a second word with a sound similar to the first), *puns* (a second meaning of the first word), and *rhymes*. In practice, it is difficult to distinguish between flight of ideas and loosening of associations (see below), especially when the patient speaks rapidly. When this happens it is often helpful to record a sample of speech. Flight of ideas is characteristic of mania.

Loosening of associations

This denotes a loss of the normal structure of thinking. To the interviewer the patient's discourse seems muddled, illogical, or tangential to the matter in hand. It does not become clearer when the patient is questioned further; indeed, the interviewer has the experience that the more they try to clarify the patient's thinking (or the longer he allows the patient to speak without interruption), the less they understand it. Several specific features of this muddled thinking have been described, but they are difficult to identify with certainty, and the most striking clinical impression is often a general lack of clarity, best described by recording an example of the speech and the impression made on the interviewer. This lack of clarity differs from that of people who are anxious or of low intelligence. Anxious people give a more coherent account when they have been put at ease, and people with low intelligence usually express their ideas more clearly when the interviewer simplifies the questions and allows more time for the reply.

Three characteristic kinds of loosening of associations have been described, all of which are seen most often in schizophrenia.

1. In *talking past the point* (*Vorbeireden*) the patient seems always about to get to the endpoint of the topic in question but then skirts round it and never in fact reaches it.
2. *Knight's move* or *derailment* refers to a transition from one topic to another, either between sentences or in mid-sentence, with no logical relationship between the two topics and no evidence of the associations described above under flight of ideas.
3. *Verbigeration* is said to be present when speech is reduced to the senseless repetition of sounds, words, or phrases. This abnormality can occur with severe expressive aphasia and occasionally in schizophrenia. When extreme, the term *word salad* is used.

Other disorders of thinking

Overinclusion refers to a widening of the boundaries of concepts, such that things are grouped together which are not normally regarded as closely connected.

Neologisms are words or phrases invented by the patient, often to describe a morbid experience. Neologisms must be distinguished from incorrect pronunciation, the wrong use of words by people with limited education, dialect words, obscure technical terms, and the 'private words' that some families invent for their own use. Before deciding that a word is a neologism, the interviewer should ask the patient what he means by it. Neologisms occur most often in chronic schizophrenia.

Depersonalization and derealization

Depersonalization is a change of self-awareness such that the person feels unreal, detached from his own experience, and unable to feel emotion (Sierra and David, 2011). *Derealization* is a similar change in relation to the environment, such that objects appear unreal and people appear as lifeless, two-dimensional 'cardboard' figures. Despite the complaint of inability to feel emotion, both depersonalization and derealization are described as highly unpleasant experiences. These central features are often accompanied by other morbid experiences, including changes in the experience of time, changes in the body image (e.g. a feeling that a limb has altered in size or shape), and occasionally a feeling of being outside

one's own body and observing one's own actions, often from above.

Because patients find it difficult to describe the feelings of depersonalization and derealization, they often resort to metaphor, and this can lead to confusion between depersonalization and delusional ideas. For example, a patient may say that they feel 'as if part of my brain had stopped working', or 'as if the people I meet are lifeless creatures'. Sometimes careful questioning is required to make the distinction; the 'as if' quality is a useful discriminator.

Depersonalization and derealization are experienced quite commonly by healthy people—especially when they are tired—as transient phenomena of abrupt onset. The symptoms have been reported after sleep deprivation and sensory deprivation, and as an effect of hallucinogenic drugs. They occur in anxiety disorders, post-traumatic stress disorder, depressive disorders, schizophrenia, and temporal lobe epilepsy. Depersonalization and derealization can also occur rarely as a primary disorder.

Motor symptoms and signs

Abnormalities of social behaviour, facial expression, and posture occur frequently in mental disorders of all kinds (e.g. motor slowing and agitation are important features of depressive disorder); motor symptoms and signs can also be side effects of medication. They are considered in Chapter 3, where the examination of the patient is described. Here we describe some specific motor signs. With the exception of tics, the signs can all be grouped together under the term *catatonia*. Catatonic signs are classically associated with schizophrenia, but they can also occur as a primary diagnosis of catatonia, or in organic disorders or secondary to drugs. Catatonia is covered in more detail in Chapter 11.

- *Tics* are irregular repeated movements involving a group of muscles—for example, sideways movement of the head or the raising of one shoulder.
- *Mannerisms* are repeated movements that appear to have some functional significance—for example, saluting.
- *Stereotypies* are repeated movements that are regular (unlike tics) and without obvious significance (unlike mannerisms)—for example, rocking to and fro.
- *Catalepsy* (*waxy flexibility*, *flexibilitas cerea*) is a term used to describe when a patient's limbs can be placed in a position in which they then remain for long periods while at the same time muscle tone is uniformly

increased. Patients with this abnormality sometimes maintain the head a little way above the pillow in a position that a healthy person could not maintain without extreme discomfort (*psychological pillow*). Catalepsy should not be confused with *cataplexy*.

- *Posturing* is the adoption of unusual bodily postures continuously for a long time. The posture may appear to have a symbolic meaning (e.g. standing with both arms outstretched as if being crucified), or may have no apparent significance (e.g. standing on one leg).
- *Grimacing* has the same meaning as in everyday speech. The term *Schnauzkrampf* (snout cramp or spasm) denotes pouting of the lips to bring them closer to the nose.
- *Negativism*. Patients are said to show negativism when they do the opposite of what is asked, and actively resist efforts to persuade them to comply.
- *Echopraxia* occurs when the patient imitates the interviewer's movement automatically, even when asked not to do so. *Echolalia* is imitating the interviewer's speech.
- *Mitgehen* ('going along with') describes another kind of excessive compliance in which the patient's limbs can be moved into any position with the slightest pressure.
- *Ambitendence*. Patients are said to exhibit ambitendence when they alternate between opposite movements—for example, putting out the arm to shake hands, then withdrawing it, extending it again, and so on repeatedly.

Disturbances of the body image

The body image or body schema is a person's subjective representation against which the integrity of their body is judged and the movement and positioning of its parts assessed. Specific abnormalities of the body image arise in neurological disorders. These abnormalities include the awareness of a *phantom limb* after amputation, *unilateral lack of awareness or neglect* (usually following stroke), *hemiasomatognosia* (in which the person feels, incorrectly, that a limb is missing), and *anosognosia* (lack of awareness of loss of function, often of hemiplegia). These abnormalities are described in textbooks of neurology and in David et al. (2009a).

Distorted awareness of size and shape of the body occurs occasionally in healthy people when they are tired or falling asleep. The experience, which includes feelings that a limb is enlarging, becoming smaller, or otherwise being distorted, also occurs in migraine, as

part of the aura of epilepsy, and after taking psychedelic agents. The person is aware that the experience is unreal. However, changes in the shape and size of body parts are described by some patients with schizophrenia; in this instance the symptoms are delusional or hallucinatory and there is no insight. *Coenestopathic states* are localized distortions of body awareness—for example, when the nose feels as if it is made of cotton wool.

A general distortion of the body image occurs in eating disorders, notably anorexia nervosa, where the patient is convinced that they are fat when in fact they are underweight.

The *reduplication phenomenon* is the experience that the body has doubled, or that part of the body has done so—for example, that there are two left arms. The experience is reported very occasionally in migraine, temporal lobe epilepsy, and schizophrenia. The related experience of autoscopic hallucinations was described earlier.

Disturbances of the self

The experience of self has several aspects. It is more than the awareness of the body; we have a feeling of unity between the various aspects of the self, we recognize our activities as our own, we recognize a boundary between the self and the outside world, and we have a feeling of continuity between our past and present selves. The concept of self is closely related to, but distinct from, that of the body image. Although the body image is usually experienced as part of the self, it can also be experienced in a more objective way, as when we say 'my leg hurts'. Some of these aspects of the self are changed in certain psychiatric disorders. The experiences are often associated with other abnormal phenomena, so that the account of abnormalities of the self overlaps with several other parts of this chapter.

Disturbed awareness of the unity of the self

Some patients lose the normal experience of existing as a unified being. Patients with dissociative identity disorder have the experience of existing as two or more selves, alternating at different times. In the experience of autoscopy and the related experience of the *Doppelgänger*, the person experiences two selves, present at the same time, but with the conviction that each is a version of the self.

Disturbances of the unity of the self

Although we recognize that we change over time, we retain a conviction of being the same person. Rarely, this feeling of continuity is lost in schizophrenia. For example, a patient may say that they are a different person

from the one who existed before the disorder began, or that a new self has taken over from the old one.

Disturbances of the boundaries of the self

This type of disturbance is experienced by some people after taking psychedelic or dissociative drugs, who may report that they felt as if they were dissolving. Hallucinations can be regarded as involving a loss of awareness of what is within the self and what is located outside. The same inability to determine what is part of the self, and what is not, is seen in passivity phenomena, in which actions willed by the patient are experienced as initiated from outside.

Disturbances of memory

The features of memory disturbance and its assessment are discussed in detail in Chapters 3 and 14. Here we briefly introduce some key terms and concepts.

Failure of memory is called *amnesia*. The related term *dysmnnesia* is occasionally used. *Paramnesia* is distortion of memory. Several kinds of disturbed memory occur in psychiatric disorders, and it is usual to describe them in terms of the temporal stages that approximate crudely to the scheme of memory derived from psychological research. A discussion of this subject can be found in textbooks of neuropsychology or in David et al. (2009b).

- *Immediate memory* concerns the retention of information over a short period, measured in minutes. It is tested clinically by asking the patient to remember a novel name and address and to recall it about five minutes later.
- *Recent memory* concerns events that have taken place in the past few days. It is tested clinically by asking about events in the patient's daily life which are known also to the interviewer directly or via an informant (e.g. what the patient has eaten) or in the wider environment (e.g. well-known news items).
- *Long-term (remote) memory* concerns events that have occurred over longer periods of time. It is tested by asking about events before the presumed onset of memory disorder.

When testing any state of memory, a distinction is made between spontaneous *recall* and *recognition* of information. In some conditions, patients who cannot recall information can recognize it correctly.

Memory loss most commonly occurs in dementia and delirium. It usually affects recall of recent events

more so than recall of distant ones. In dementia, it usually progresses with time and becomes severe, but is rarely total. Some organic conditions give rise to an interesting partial effect known as *amnesic disorder*, in which the person is unable to remember events that occurred a few minutes earlier, but can recall remote events. Total loss of all memory, or selective loss of personal identity, strongly suggests psychogenic causes or malingering. Some patients with memory disorder recall more when given cues. When this happens, it suggests that the disorder is concerned at least in part with retrieval.

After a period of unconsciousness, memory is impaired for the interval between the ending of complete unconsciousness and the restoration of full consciousness (*anterograde amnesia*). Some causes of unconsciousness (e.g. head injury, electroconvulsive therapy) also lead to inability to recall events before the onset of unconsciousness (*retrograde amnesia*).

Disturbances of recognition

Several *disorders of recognition* occur occasionally in neurological and psychiatric disorders.

- *Jamais vu* is the failure to recognize events that have been encountered before.
- *Déjà vu* is the conviction that an event is repeating one that has been experienced in the past, when in fact it is novel.
- *Confabulation* is the confident recounting of quite false 'memories' for recent events, and is characteristic of amnesic syndrome. For example, a patient with no memory of what they ate for breakfast may confabulate a completely incorrect menu. This is done in a plausible manner without any apparent difficulty, and without any awareness of the falsity of the information.

Recall of events can be biased by the mood at the time of recall. Importantly, in depressive disorders, memories of unhappy events are recalled more readily than other events—a process which adds to the patient's low mood.

Psychogenic amnesia

This is thought to result from an active process of repression which prevents the recall of memories that would otherwise evoke unpleasant emotions. The ideas arose from the study of *dissociative amnesia*, but the same factors may play a part in some cases of organic amnesia, helping to explain why the return of some memories is delayed longer than that of others.

Disturbances of consciousness

Consciousness is awareness of the self and the environment. The *level* of consciousness can vary between the extremes of alertness and coma. The *quality* of consciousness can also vary: sleep differs from unconsciousness, as does stupor.

- *Coma* is the most extreme form of impaired consciousness. The patient shows no external evidence of mental activity, and little motor activity other than breathing. They do not respond even to strong stimuli. Coma can be graded by the extent of the remaining reflex responses and by the type of electroencephalogram (EEG) activity.
- *Clouding of consciousness* refers to a state that ranges from barely perceptible impairment to definite drowsiness in which the person reacts incompletely to stimuli. Attention, concentration, and memory are impaired to varying degrees, and orientation is disturbed. Thinking appears to be muddled, and events may be interpreted inaccurately. Clouding of consciousness is a defining feature of delirium.
- *Stupor*, in the sense used in psychiatry, refers to a condition in which the patient is immobile, mute, and unresponsive, but appears to be fully conscious in that the eyes are usually open and follow external objects. If the eyes are closed, the patient resists attempts to open them. Reflexes are normal and resting posture is maintained. Stupor may occur in catatonia. Note that in neurology the term ‘stupor’ is used differently and generally implies an impairment of consciousness.
- *Confusion* means an inability to think clearly. It occurs characteristically in states of impaired consciousness, but it can occur when consciousness is normal. In delirium, confusion occurs together with partial impairment (clouding) of consciousness, along with other features. In the past, delirium was called a confusional state, with the result that the term ‘confusion’ was used to mean impairment of consciousness as well as muddled thinking.

Other terms that are used to describe states of impaired consciousness include the following:

- *oneiroid states*, in which there is dream-like imagery although the person is not asleep; if prolonged, this may be called a *twilight state*
- *torpor*, in which the patient appears drowsy, readily falls asleep, and shows evidence of slow thinking and narrowed range of perception.

In addition, there are sleep–wake disorders that can present with impaired consciousness, such as narcolepsy, in which there may also be recurrent, sudden, brief collapses owing to loss of muscle tone (*cataplexy*).

Disturbances of attention and concentration

Attention is the ability to focus on the matter in hand. *Concentration* is the ability to maintain that focus. The ability to focus on a selected part of the information reaching the brain is important in many everyday situations (e.g. when conversing in a noisy place). It is also important to be able to attend to more than one source of information at the same time (e.g. when conversing while driving a car).

Attention and concentration may be impaired in a wide variety of psychiatric disorders, including depressive disorders, mania, anxiety disorders, schizophrenia, and organic disorders. Therefore the finding of abnormalities of attention and concentration does not assist in diagnosis. Nevertheless, these abnormalities are important in management. For example, they affect a patient’s ability to give or receive information when interviewed, and can interfere with a patient’s ability to work, drive, or take part in leisure activities.

Insight

In psychopathology, the term *insight* refers to awareness of morbid change in oneself and a correct attitude to this change, including, in appropriate cases, a realization that it signifies a mental disorder. This awareness may be difficult for a patient to achieve, since it involves some knowledge of the limits of normal mental functioning and mental experience. Furthermore, insight has to be assessed against the background of knowledge and beliefs about psychiatric disorder—it is not the same as complete agreement with the views of the doctor. Insight is also influenced by cultural and cognitive factors (Nair et al., 2014).

Although, in the past, lack of insight was said to be a feature that distinguishes between psychosis (where it was said to be absent) and neurosis (where it is present), this distinction is neither reliable nor useful. The ‘lack of insight’ in psychosis is better conceptualized in terms of ‘impaired reality testing’ or ‘reality distortion’ (see Chapter 11).

Insight is not simply present or absent. It has several facets, each being a matter of degree.

- Is the patient aware of phenomena that others have observed (e.g. that they are unusually active and elated)?
- If so, do they recognize the phenomena as abnormal (rather than, e.g. maintaining that their unusual activity and cheerfulness are due to normal high spirits)?
- If so, do they consider that the phenomena are caused by mental illness (as opposed to, say, a physical illness or poison administered by enemies)?
- If so, do they think they need treatment?

The answers to these questions are more informative and more reliable than those of the single question as to whether insight is present or not. The value of

determining the degree of insight is that it helps to predict whether the patient is likely to accept the need for help and treatment.

In discussions of psychotherapy, insight has a different meaning from that considered so far, which is used in general psychiatry. In psychotherapy, insight is the capacity to understand one's own motives and to be aware of previously unconscious aspects of mental activity. The term *intellectual insight* is sometimes used to denote the capacity to formulate this understanding, whereas the term *emotional insight* refers to the capacity to feel and respond to the understanding.

For review of insight, see Landi et al. (2016).

Further reading

Berrios, G. E. (1996). *The History of Mental Symptoms: Descriptive Psychopathology since the Nineteenth Century*. Cambridge University Press.

Oyeboode, F. (2022). *Sims' Symptoms in the Mind. Textbook of Descriptive Psychopathology*, 7th edn. Elsevier.

Stanghellini, G., Broome, M., Raballo, A., et al. (2019). *The Oxford Handbook of Phenomenological Psychopathology*. Oxford University Press.

CHAPTER 2

Classification

Introduction 21

Concepts of mental illness 22

Criticisms of classification 24

The history of classification 26

Organizing principles of contemporary
classifications 26

Reliability and validity 29

Current psychiatric classifications 30

Other issues in psychiatric
classification 34

Introduction

Chapter 1 outlined the symptoms and signs of psychiatric disorder. In Chapter 3 we describe the psychiatric assessment, by which these symptoms and signs are elicited, interpreted, and used as the basis upon which diagnoses are made. Before doing so, in this chapter we discuss the principles of psychiatric diagnosis and classification, since this provides the framework within which this clinical process happens. The term *nosology* is sometimes used to refer to classification and its study.

Classification is needed in psychiatry for several purposes:

- to enable clinicians to communicate with one another about the diagnoses given to their patients;
- to aid patients and their families, by allowing clinicians to provide a framework for them to understand their symptoms and difficulties, and for proposed treatments;
- to understand the implications of these diagnoses in terms of their symptoms, prognosis, and treatment, and sometimes their aetiology;
- to relate the findings of clinical research to patients seen in everyday practice;

- to facilitate epidemiological studies and the collection of reliable statistics; and
- to ensure that research can be conducted with comparable groups of subjects.

Of these, the first four are the most relevant to clinical practice. Indeed, it is difficult to imagine, or justify, how psychiatry could be practised in any reasonable or evidence-based manner without the order that classification provides (Craddock and Mynors-Wallis, 2014). In this respect, the position of classification as one of the foundations of psychiatry is no different from that in the rest of medicine. However, in other respects psychiatric classification does raise particular challenges and controversies, largely because of the uncertain aetiology, or independent biological validation, of most disorders, which means that its diagnostic categories are almost entirely *syndromal*—a collection of symptoms and signs. The resulting difficulties are of two kinds. The first is conceptual, relating to the nature of mental illness and the question of what, if anything, should be classified. The second difficulty is a practical one, concerning how categories are defined and organized into a classificatory scheme. In this chapter, the conceptual issues and

criticisms are covered first, followed by a historical perspective to classification. We then describe and compare the two schemes in widespread use at present: Chapter 6 of the *International Classification of Diseases, 11th edition*

(ICD-11; World Health Organization, 2024), and the *Diagnostic and Statistical Manual of Mental Disorders, 5th edition* (DSM-5; American Psychiatric Association, 2013a).

Concepts of mental illness

In everyday speech the word ‘illness’ is used loosely. Similarly, in psychiatric practice the term ‘mental illness’ is used with little precision, and often synonymously with ‘mental disorder’. In this context, the terms ‘mental’ and ‘psychiatric’ are also used interchangeably.

A good definition of mental illness is difficult to achieve, for both practical and philosophical reasons, as outlined here. In routine clinical work, the difficulty is important mainly in relation to ethical and legal issues, such as compulsory admission to hospital. In forensic psychiatry the definition of mental illness (by the law) is particularly important in the assessment of issues such as criminal responsibility.

Diverse discussion of the concepts of mental illness can be found in Kendell (1975), Zachar and Kendler (2007), and Tyrer (2013).

Definitions of mental illness

Many attempts have been made to define mental illness, none of which is satisfactory or uniformly accepted. A common approach is to examine the concept of illness in general medicine and to identify any similarities or analogies with mental illness. In general medicine there are five types of definition:

- *Absence of health.* This approach changes the emphasis of the problem but does not solve it, because health is even more difficult to define. The World Health Organization (WHO), for example, defined health as ‘a state of complete physical, mental and social well-being, and not merely the absence of disease or infirmity.’ As Lewis (1953) rightly commented, ‘a definition could hardly be more comprehensive than that, or more meaningless.’ Many other definitions of health have been proposed, all unsatisfactory.
- *Disease is what doctors treat.* This definition has the attraction of simplicity but does not really address

the issue. The notion that disease is what doctors *can* treat has somewhat more merit, since there is evidence that, as a medical treatment for a condition becomes available, it becomes more likely that the condition will be regarded as a disease (Campbell et al., 1979).

- *Biological disadvantage.* The idea of defining disease in terms of biological disadvantage was proposed by Scadding (1967), and is the most extreme biomedical view of disease. Scadding never defined biological disadvantage, but the term has been used in psychiatry to include decreased fertility (reproductive fitness) and increased mortality. Viewing disease in terms of ‘evolutionary disadvantage’ is a similar concept (Wakefield, 1992).
- *Pathological process.* Some extreme theorists, most notably Szasz (1960), have taken the view that illness can be defined only in terms of physical pathology. Since most mental disorders do not have demonstrable physical pathology, according to this view they are not illnesses. Szasz took the further step of asserting that most mental disorders are therefore not the province of doctors. This kind of argument can be sustained only by taking an extremely narrow view of pathology. It is also arbitrary, based on current knowledge, and is increasingly incompatible with the evidence of a genetic and neurobiological basis to the major psychiatric disorders, and their associated morbidity and mortality.
- *Presence of suffering.* This approach has some practical value because it defines a group of people who are likely to consult doctors. A disadvantage is that the term cannot be applied to everyone who would usually be regarded as ill in everyday terms. For example, patients with mania may feel unusually well and may not experience suffering, although most people would regard them as mentally ill.

Biomedical versus social concepts

The above concepts may be divided into those that view mental illnesses in biomedical terms, and those that consider them to be social constructs or value judgements. This debate is still ongoing, and depends in part on one's opinion about their aetiology, but it is now generally accepted that value judgements play a part in all diagnoses, even if the disorders themselves are considered from a biomedical perspective (Fulford, 1989). For example, beliefs and emotions are central to most psychiatric disorders, yet it is a value judgement as to whether a given belief or emotion is 'reasonable' or 'unhelpful' for a given individual in their particular social context, and therefore what, if any, diagnostic significance it has.

Impairment, disability, and handicap

It is useful in medicine, and particularly in psychiatry, to describe and classify the consequences of a disorder. This approach is related to the concept of disease as involving dysfunction (Wakefield, 1992), as incorporated into the definitions of mental disorder used in ICD-11 and DSM-5. Three related terms, derived from medical sociology and social psychology, are used to describe the harmful consequences of a disorder.

1. *Impairment* refers to a pathological defect—for example, hemiparesis after a stroke.
2. *Disability* is the limitation of physical or psychological function that arises from an impairment—for example, difficulties with self-care that are caused by the hemiparesis.
3. *Handicap* refers to the resulting social dysfunction—for example, being unable to work because of the hemiparesis.

Incapacity may be seen as another harmful consequence of illness, although the term usually refers in a legal sense to the effect that illness has on one's competence to make treatment decisions, as enshrined in the United Kingdom by the Mental Capacity Act (see Chapter 4).

Diagnoses, diseases, and disorders

The term 'diagnosis' has two somewhat different meanings. It has the general meaning of 'telling one thing apart from another', but in medicine it has also acquired a more specific meaning of 'knowing the underlying cause' of the symptoms and signs about which the patient is complaining. Underlying causes are expressed in quite different terms from the symptoms. For example, the symptoms of acute appendicitis are distinct from the idea that will form in the doctor's mind that the appendix is inflamed and producing peritoneal irritation. To be able to make a diagnosis of this type is, of course, satisfying for the doctor and useful for the patient, since it immediately suggests what investigations and treatment are needed. Its clear utility also makes redundant most theoretical or philosophical concerns about classification. Unfortunately, for most psychiatric patients it is rarely possible to arrive at this type of diagnosis, the only exception to this being, by definition, 'organic' psychiatric disorders (see 'Organic and functional').

The lack of clear disease categories, in a medical sense, has led to the use of the more general term 'disorder'. Regarding ICD-11, the WHO defines psychiatric disorders as:

... syndromes characterized by clinically significant disturbance in an individual's cognition, emotional regulation, or behaviour that reflects a dysfunction in the psychological, biological, or developmental processes that underlie mental and behavioural functioning. These disturbances are usually associated with distress or impairment in personal, family, social, educational, occupational, or other important areas functioning. <https://icd.who.int/browse11/l-m/en#/http%3a%2f%2fid.who.int%2fcd%2fentity%2f334423054>

The DSM-5 definition of a mental disorder is similar (American Psychiatric Association, 2013a), and both definitions illustrate that most psychiatric disorders are based not upon theoretical concepts, or presumptions about aetiology, but upon recognizable clusters of symptoms and behaviours. This explains much of the debate about the reliability and validity of the categories being classified, as will be discussed later in this chapter.

Criticisms of classification

In contrast to the view that classification is an essential, albeit insufficient and imperfect, basis for clinical practice (Craddock and Mynors-Wallis, 2014), the use of psychiatric classification is sometimes criticized as being inappropriate or even harmful. In part, such criticisms arise from the various controversies outlined above: if the concept of mental disorder is itself disputed, then so will any classifications thereof. These criticisms were most prevalent and trenchant at the height of the ‘anti-psychiatry’ movement in the 1960s, although they continue to be expressed by a ‘critical psychiatry’ lobby (Bracken et al., 2012). Three main criticisms of this kind are made.

1. *Allocating patients to a diagnostic category distracts from the understanding of their unique personal difficulties.* However, these are complementary not competing approaches. The good clinician always considers a patient’s unique experiences, qualities, and circumstances when making diagnoses, not least because this contextual information often affects treatment and prognosis.
2. *Allocating a person to a diagnostic category is simply to label deviant behaviour as illness.* Some sociologists have argued that such labelling serves only to increase the person’s difficulties. There can be no doubt that terms such as epilepsy or schizophrenia attract stigma (see Box 2.1), but this does not lessen the reality of suffering and the need for care and treatment. However, it does emphasize that mental illness should not be defined solely in terms of socially deviant behaviour. The presence of the former must be separately established based on the psychiatric history and mental state examination. Moreover, if mental illness is inferred from socially deviant behaviour alone, political abuse may result. A serious example of the latter occurred in the former Soviet Union, where some psychiatrists colluded with the government in being willing to classify political dissent as evidence of mental illness. A further reason for excluding purely social criteria from the definition of mental illness, and from diagnostic criteria, is that many behaviours are appraised differently in different countries and at different times. For

example, homosexuality was considered to be a mental disorder in the United Kingdom until the 1970s.

3. *Individuals do not fit into the available categories.* Although it is not feasible to classify a minority of disorders (or patients), this is not a reason for abandoning classification for the majority.

It is certainly true that at times classification has been inappropriately used as part of a broader abuse of psychiatry, whether for political, financial, or other reasons. Although such abuses are fortunately rare, they are an extreme illustration of the fact that making diagnoses and classifying patients are not neutral acts but carry significant ethical and other implications. One of these implications concerns *stigma*, which remains a serious problem for patients with mental health problems, even if one does not accept the rest of the sociological thesis outlined above. It is incumbent on all those who use psychiatric diagnostic terms that they do so appropriately and judiciously, paying due attention to their correct use and purpose, and the context in which they are being applied. Doing so can help to reduce the problem of stigmatization, but cannot solve it, because stigma results from many other factors too (Thornicroft et al., 2022). The issue of stigma is discussed in Box 2.1.

Although these criticisms are important, they are arguments only against the improper use of, or over-reliance upon, classification. Disorders and their harmful consequences cannot be made to disappear by ceasing to give names to them. ICD-11 and DSM-5 emphasize that classification is a means of communication and a guide to decision making but acknowledge that they are provisional and imperfect schemes. Psychiatrists and other mental health professionals must use their clinical experience and common sense, as well as being guided by the descriptions of the disorders that make up the classifications.

Other criticisms of classification in psychiatry are mostly concerned with the specifics rather than the principles—for example, whether a particular diagnostic category is reliable and valid, and the severity threshold at which the diagnosis should be made. These issues are introduced later in this chapter, and at various points throughout the book.

Box 2.1 Stigma

People stigmatize others when they judge them not on their personal qualities but on the basis of a mark or label which assigns them to a feared or unfavoured group. The tendency to stigmatize seems to be deeply rooted in human nature as a way of responding to people who appear or behave differently. Stigmatization is based on fear that those who seem different may behave in threatening or unpredictable ways, and it is reduced when it becomes clear that the stigmatized person is unlikely to behave in these ways.

Stigma in psychiatry

People fear mental illness, and they stigmatize those who are affected by it. The reasons for this are complex. They include the perception that people with mental illness cannot control their own behaviour, and that they may act in odd, unpredictable, and possibly violent ways, and that they will never recover. Regrettably, much stigma is driven by portrayals of mental illness in films and novels, which are entertaining but often inaccurate. Stigma is reduced among those in contact with people with mental illness, but may be more common among particular cultural and ethnic groups where there is less knowledge about mental illness, and cultural idioms and beliefs may abound. Whatever the underlying psychological mechanisms, fear of mental illness makes people react to mentally ill individuals in the same cautious and unfavourable way—that is, to stigmatize them. This can have many adverse consequences, including discrimination, losing friends and social networks, as well as difficulties finding housing or employment. Stigmatization may also affect the allocation of resources for the care of people with mental illness, with a reluctance to give appropriate priority to mental health services. Stigma can also be internalized (self-stigma and negative attitudes to self), and manifested in professional stigma (psychiatrists, for example, may experience this from doctors in other specialties).

Diagnoses, as labels, have the potential to be stigmatizing (e.g. leprosy and AIDS). It has been suggested that the stigma of mental illness would be reduced if diagnoses such as schizophrenia were abandoned. This proposal misses the point that the basis of stigma is fear and ignorance, and that simply removing or changing the label does not reduce these factors. The mentally ill were stigmatized long before modern diagnostic terms were in use, and people who fear mental illness invent their own labels, which are far more stigmatizing than a diagnosis. To reduce stigma it is necessary to reduce fear and ignorance, and this requires accurate information about mental illnesses and better understanding of the people who suffer from them.

Reducing stigma

Campaigns to reduce stigma generally include:

- Information about the true nature of mental illness, and about the low frequency of dangerous behaviour;
- Encouragement to persuade public figures who have had a mental illness to speak out about their experiences; and
- A focus on young people, whose attitudes may be less fixed than those of their elders.

Although stigma can be reduced, this cannot be done easily or quickly. In the past, people with epilepsy were stigmatized, but as knowledge of the condition spread and as treatment improved, attitudes gradually changed. Changes are now beginning to be seen in the stigma attached to some psychiatric disorders. For example, autism is now generally a much less stigmatizing term than it was previously, whereas schizophrenia is not. Thus there is an ongoing need for public education campaigns to reduce the fear and misunderstanding that perpetuate stigma. However, anti-stigma programmes have had only modest benefits (Griffiths et al., 2014).

For review of stigma and its reduction, see Thornicroft et al. (2022).

The history of classification

Efforts to classify abnormal mental states have occurred since antiquity. One reason for including a chronological perspective here is that contemporary psychiatric classifications are, in part, a hybrid of various historical themes and opinions.

Early Greek medical writings contained descriptions of different manifestations of mental disorder—for example, excitement, depression, confusion, and memory loss. This simple classification was adopted by Roman medicine and developed by the Greek physician Galen, whose system of classification remained in use until the eighteenth century.

Interest in the classification of natural phenomena developed in the eighteenth century, partly stimulated by the publication of a classification of plants by Linnaeus, a medically qualified professor of botany who also devised a less well-known classification of diseases in which one major class was mental disorders. Many classifications were proposed, notably one published in 1772 by William Cullen, a Scottish physician. He grouped mental disorders together, apart from delirium, which he classified with febrile conditions. According to his scheme, mental disorders were part of a broad class of ‘neuroses’, a term that he used to denote diseases which affect the nervous system (Hunter and MacAlpine, 1963). Cullen’s classification contained an aetiological principle—that mental illnesses were disorders of the nervous system—as well as a descriptive principle for distinguishing individual clinical syndromes within the neuroses. In Cullen’s usage, the term neurosis covered the whole range of mental disorders, as well as many neurological conditions. The modern narrower usage developed later.

In the early nineteenth century, several French writers published influential classifications. Pinel’s *Treatise on Insanity*, which appeared in English in 1806, divided

mental disorders into mania with delirium, mania without delirium, melancholia, dementia, and idiocy. Pinel’s compatriot, Esquirol, wrote another widely read textbook, published in English in 1845, and added a new category, ‘monomania’, characterized by ‘partial insanity’, in which there were fixed false ideas that could not be changed by logical reasoning (i.e. delusions). Like other psychiatrists of the time, Pinel and Esquirol did not discuss neuroses (in the modern sense), because these conditions were generally treated by physicians.

Meanwhile, in Germany, Kahlbaum formulated two requirements for the definition of a mental illness and thus to classification: both the full clinical picture, and its entire course, were considered. These ideas were adopted at the end of the nineteenth century by Emil Kraepelin, who used these criteria to make the landmark distinction between manic-depressive psychosis (bipolar disorder) and schizophrenia. Successive editions of Kraepelin’s textbook made further refinements to the classification of mental illness, and his views remain the single greatest influence underlying today’s classificatory systems.

At the same time, separate developments in the emerging specialty of neurology led to decreasing medical interest in the ‘nervous patient’, a term used throughout the nineteenth century to refer to a large group of patients with varied complaints. These were gradually seen as a part of the new specialty of psychiatry alongside the major mental illnesses. The writings of Sigmund Freud and his contemporaries led to greater recognition of the psychological causes of nervous symptoms and ‘neurotic’ disorders, and to the modern concepts of hysteria and anxiety disorder.

For a review of nosological models in psychiatry, see Pichot (1994) and Zachar and Kendler (2007).

Organizing principles of contemporary classifications

As well as these historical roots, it is worth considering the major issues that contemporary classifications have faced regarding their organizing principles.

Organic and functional

The first issue concerns the distinction that is conventionally drawn between organic and functional

disorders. *Organic disorders* are those that arise from a demonstrable cerebral or systemic pathological process; the core disorders are dementia, delirium, and the various neuropsychiatric syndromes (David et al., 2009a). *Functional disorder* is consequently an umbrella or default term for all other psychiatric disorders. The organic–functional dichotomy has two main implications for classification.

1. It has a philosophical dimension, being inextricably linked to dualism and concepts of mind and body. At its extreme, the implication is that functional disorders have no biological basis, while psychological and social factors are irrelevant to organic disorders. This polarization can be reflected in the apparent divide between psychiatry and neurology. The same dualism may also unintentionally encourage psychiatrists to be either 'mindless' or 'brainless', rather than seeing that both aspects of aetiology always contribute. Equally, it has led to the suggestion that the two specialties should use a merged classificatory system (White et al., 2012).
2. It has practical implications, since the term 'organic' defines disorders aetiologically or pathologically, whereas all other psychiatric disorders are, by default, purely descriptive and based on clusters of symptoms and signs. This is not only unsatisfactory for psychiatry (Arango and Fraguas, 2016) but leads to inconsistencies and difficulties at the intersection; these are currently well illustrated regarding schizophrenia and organic schizophrenia-like disorders.

There is general agreement that, for these and other reasons, the organic–functional dichotomy is neither valid nor helpful (Spitzer et al., 1992). However, it has proved difficult to come up with an alternative. The ways in which ICD-11 and DSM-5 deal with the issue are discussed below.

Neurosis and psychosis

In the past, the concepts of neurosis and psychosis were important in most systems of classification. Although neither is used as an organizing principle in ICD-11 or DSM-5, in everyday clinical practice these terms are still useful as general descriptors, so it is of relevance to understand their history.

Psychosis

The term *psychosis* was suggested by Feuchtersleben in 1845. He proposed the use of the term for severe mental disorders, while he used the term *neurosis* for mental disorders as a whole. Thus, he wrote that 'every psychosis is at the same time a neurosis, but not every neurosis is a psychosis'. As the concept of neurosis narrowed, psychosis (also used in the plural, psychoses) came to be regarded as independent.

In modern usage, the term psychosis continues to refer broadly to severe psychiatric disorders, including

schizophrenia, and some organic and affective disorders. Numerous criteria have been proposed to achieve a more precise definition, but there are problems with all of them. Greater severity of illness is a common suggestion, but some cases are relatively mild (and some neuroses are severe and at least as disabling). Lack of insight is often suggested as a criterion, but insight itself is difficult to define. A somewhat more straightforward criterion is the inability to distinguish between subjective experience and external reality, as shown by the presence of delusions and hallucinations. Indeed, the presence of a delusion is sometimes regarded as sufficient to diagnose a psychosis. However, as well as the problems involved in fully defining these terms, the label 'psychosis' is unsatisfactory because the conditions embraced by the term have little in common, and it is usually more informative to classify the particular disorder concerned. Nevertheless, psychosis remains a convenient term for disorders that are usually severe, and which feature delusions, hallucinations, or unusual or bizarre behaviour (presumed to be secondary to these phenomena), especially when a more precise diagnosis cannot yet be made. The adjectival form is also useful and survives in ICD-11 categories of 'psychotic disorders' as well as in the context of antipsychotic drugs.

Neurosis

As noted, the term *neurosis* was introduced to denote diseases of the nervous system. Gradually the category of neurosis narrowed, first as neurological disorders with a distinct neuropathology (e.g. stroke) were removed, and later with the development of a separate category of psychosis.

The objections to the term neurosis are similar to the objections to the term psychosis and explain its removal as an organizing principle in current classification. First, the concept is difficult to define (Gelder, 1986). Second, the conditions that neurosis embraces have little in common. Thirdly, more information can be conveyed by using a more specific and descriptive diagnosis, such as 'anxiety disorder'. A further objection is that the term neurosis has been widely used with the unproven assumption of an aetiological meaning in the psychodynamic literature.

In the same way as for psychosis, the terms 'neurosis' and 'neurotic' remain useful as simple descriptors, especially if the specific disorder cannot yet be determined, to indicate disorders that are often comparatively mild, and usually associated with some form of anxiety.

Categories, dimensions, and axes

Categorical classification

Traditionally, psychiatric disorders have been classified by dividing them into categories that are supposed to represent discrete clinical entities. As already noted, in the absence of knowledge of underlying pathology, these categories can only be defined in terms of symptom patterns and course. Such categorization facilitates the decisions that have to be made in clinical work about treatment and management, but presents two problems.

1. Although definitions and descriptions can be agreed upon (to improve *reliability*), there is uncertainty about the extent to which these categories represent distinct entities or ‘carve Nature at her joints’ (*validity*).
2. A significant proportion of patients do not closely match the descriptions of any disorder, or meet criteria for two or more categories (*comorbidity*).

These are all significant points, and they are addressed further in the following sections. However, a more satisfactory and practical alternative system has not yet been devised.

Dimensional classification

Dimensional classification does not use separate categories, but characterizes the subject by means of scores on two or more dimensions. In the past, Kretschmer and several other psychiatrists advocated it, and subsequently it was promoted by the psychologist Hans Eysenck.

The concept of dimensionality has been revived and advanced by epidemiological surveys that have emphasized that there is a continuum between the healthy population and individuals with diagnosed psychiatric disorders. This applies, for example, to psychotic symptoms, and argues that even a severe disorder such as schizophrenia can be seen as occurring at one end of a dimension of psychotic-like experience (Linscott and van Os, 2013). The dimensional view of psychiatric disorder is comparable to that of hypertension and other medical diagnoses that are extremes of a normal distribution, and this view reflects the nature of the underlying genetic predisposition and presumed neurobiology much better than a categorical one. However, the problem with dimensions is that they are not of great value in clinical practice. For most patients, yes-no decisions need to be made, the most critical of these

being whether the person has a psychiatric disorder that merits treatment, and, if so, which one. These clinical imperatives strongly favour categorical approaches to classification.

The multiaxial approach

The term *multiaxial* is applied to schemes of classifications in which two or more separate sets of information (such as symptoms, aetiology, and personality type) are coded. Essen-Møller was probably the first to propose such a system for use in psychiatry, using one axis for the clinical syndrome and another for aetiology (Essen-Møller, 1971). Although attractive, there is a danger that multiaxial schemes are too complicated and time-consuming to be suitable for everyday use. However, a multiaxial scheme can play a helpful role in child and adolescent psychiatry, with the axes describing intellectual level, functional impairment, and psychosocial adversity (Rutter, 2011).

Hierarchies of diagnosis

Categorical systems often include an implicit hierarchy of categories. If two or more disorders are present, it has been conventional (although not always made explicit) to assume that one takes precedence and is regarded as the main disorder for the purposes of treatment and recording. For example, organic disorders ‘trump’ functional disorders, and schizophrenia takes precedence over affective disorders and anxiety. This type of assumption is justified because there is some clinical evidence for an inbuilt hierarchy of significance between the disorders. For instance, if a person has depression secondary to an endocrine disorder, the latter will need treatment as well as the depression. As another example, anxiety symptoms occur commonly with depressive disorders, and are sometimes the presenting feature. If the anxiety is treated, there is little response in the depressive symptoms, but if the depressive disorder is treated, there is often concurrent improvement in anxiety. Nevertheless, these issues must not obscure the importance of noting all disorders and symptoms that are present.

Comorbidity

In contemporary practice, less emphasis is placed on hierarchies of diagnosis, with greater weight being placed on comorbidity (also called *dual diagnosis*). This has occurred for three reasons. First, research has shown that comorbidity is very common (Kessler, 2004); if physical comorbidity is included, then it is

the rule not the exception. For example, over 50% of patients with bipolar disorder also meet the criteria for an anxiety disorder or substance use disorder. Secondly, it reminds the clinician to focus on all the various disorders that may be present, and not to assume that the disorder highest in the hierarchy is the only target for treatment. Thirdly, the diagnostic 'rules' used in current classificatory systems allow, if not encourage, multiple diagnoses to be made, and it has been argued that at least some psychiatric comorbidity is in fact an artefact of this (Maj, 2005), and that a simpler classificatory system that reduced it would be desirable (Goldberg, 2010).

The term *comorbidity* covers two different circumstances:

1. *Disorders that are currently considered to be distinct, but which are probably causally related.* In other words, there is one disease process, but there are two or more clinical manifestations, which are currently diagnosed separately owing to lack of knowledge or because of clinical convention.
2. *Disorders that are causally unrelated.* This refers to the chance co-occurrence of two disorders—for example, the onset of early-onset dementia in a person with longstanding panic disorder.

Note that comorbidity applies only when the criteria for two or more diagnoses are met. It should not be used for patients who fall between diagnostic categories but who do not meet the criteria for any one of them.

Reliability and validity

Reliability of psychiatric diagnoses

A prerequisite for any satisfactory classification scheme, whatever its organizing principle, is that the items (diagnoses) that are being classified can be recognized reliably (Kendell, 1975). However, although reliability is now known and is reasonable for most categories, this was not the case until relatively recently, for the reasons described below. Studies conducted in the 1950s and 1960s demonstrated substantial diagnostic disagreement between psychiatrists, which arose for two main reasons (Kreitman, 1961):

1. *The interviewing technique and characteristics of the psychiatrist.* This included the way in which symptoms and signs were elicited and interpreted and the weight attached to them. These elements in turn probably reflect many influences, including training, professional culture, etc.
2. *The differing use of diagnostic terms and criteria.* At the time, there were no widely accepted glossaries or definitions of key terms. Therefore it was impossible to ensure that psychiatrists were using the same criteria for symptoms and syndromes. A key study by Stengel (1959) illustrated 'the chaotic state of the classifications in current use' by collecting 28 classifications in a variety of languages. None of the 28 classifications was accompanied by any indication of the meaning of the constituent terms.

International studies of diagnostic criteria

The increasing concern about the extent of diagnostic disagreement between countries heralded international studies intended to identify the source of the variation, and then to improve the reliability. This work adopted the suggestion of the philosopher Carl Hempel that *operational definitions* should be developed—that is, the specification of a category (e.g. a symptom) by a series of precise inclusion and exclusion statements.

A key study was the US–UK Diagnostic Study (Cooper et al., 1972), which followed on from the demonstration that both diagnostic and admission rates for manic depression and schizophrenia differed considerably between the two countries. For example, the rate for manic–depressive illness in the United Kingdom was more than ten times that in equivalent mental hospitals in the United States, whereas the rate for schizophrenia was about twice as high in the United States as in the United Kingdom. Another seminal study was the International Pilot Study of Schizophrenia (IPSS), a large international collaborative study organized by the WHO. The IPSS demonstrated clearly that structured interviews could be translated and used in different cultures, enabling it to show that patients with typical symptoms of schizophrenia could be found in all countries studied (World Health Organization, 1973; see Chapter 11).

Standardized interview schedules

A major step towards improving diagnostic reliability came with the development of standardized interview

schedules that minimize the variations in interviewing technique and symptom rating between psychiatrists. This development was closely linked with the international studies mentioned above. For example, the US–UK Diagnostic Study used the Present State Examination (PSE), one of the first structured psychiatric interviews (Wing et al., 1974). The PSE was also notable since it used a computer program, CATEGO, to generate a diagnosis using the symptom ratings, eliminating any biases of the diagnosticians and reducing the chance for errors to be introduced. Subsequently, standardized schedules that specify the content and sequence of the interview, and provide scoring rules by which the presence and severity of symptoms are rated, have become widely used in different settings and diagnostic systems. Further examples are given in Chapter 3.

Validity of psychiatric diagnoses

The above discussion has focused upon the reliability of diagnoses, because without a reasonable level of inter-observer reliability it is not possible to test whether or not a concept is valid. Validity is a much more difficult topic. In a general sense, validity refers to the extent to

which a concept means what it is supposed to mean. For a discussion of reliability and validity in psychiatry, see Jablensky (2016).

Three forms of validity are usually recognized.

1. *Face validity* is the correspondence with the clinical concepts and descriptions currently accepted in clinical practice. This is fairly easy to achieve by the careful use of glossaries and lists of criteria (illustrating the fact that reliability and validity are not wholly separate).
2. *Predictive validity* is the extent to which disorders predict response to treatment and outcome. This has high utility.
3. *Construct validity* is the third and most fundamental form of validity, in which there is a demonstrable relationship between a disorder and its underlying aetiology and pathophysiology. Unfortunately, most psychiatric disorders have an unknown and probably low construct validity, reflecting the descriptive criteria upon which most are currently based.

To date, little progress has been made towards establishing the validity of the existing schemes of classification.

Current psychiatric classifications

The International Classification of Diseases (ICD)

The International Classification of Diseases (ICD) is produced by the WHO as an aid to the collection of international statistics about disease. The current version is the 11th edition (ICD-11), with chapter 6 devoted to psychiatry.

Mental disorders were included for the first time in 1948, in the sixth revision (ICD-6), but it was ICD-9, published in 1978, that provided the first satisfactory and widely used version. ICD-9 was the culmination of an extensive effort by the WHO to develop a usable and internationally acceptable system based on clinical descriptors of each disorder. ICD-10 followed in 1992. Chapter V of ICD-10, entitled ‘Mental and Behavioural Disorders’, divided psychiatric disorders into ten ‘F’ categories, each of which could be further divided using a numeric code (e.g. F3x = mood disorders; F31 = bipolar disorder; F31.7 = bipolar disorder, currently in remission).

ICD-11

ICD-11 came into force in January 2022, although it will likely be several more years before it replaces ICD-10 in all countries and settings (World Health Organization, 2024). Psychiatric disorders are covered in chapter 6, entitled ‘Mental, Behavioural and Neurodevelopmental Disorders’.

It retains many of the core concepts and categories of ICD-10 but has a different organization, new and revised diagnostic criteria, and certain other new features:

- Psychiatric disorders are organized into 21 diagnostic groupings. These are shown, and compared to those in ICD-10, in Table 2.1.
- The coding is complex. Each grouping begins ‘6’ (to reflect chapter 6), followed by a letter and further numbers and/or letter (together defining the diagnosis, subtype, and other qualifiers). For example, schizophrenia is coded 6A20; schizophrenia, first

Table 2.1 Diagnostic groupings for ICD-11 as related to ICD-10 categories

ICD-11 06 Mental, Behavioural and Neurodevelopmental Disorders chapter	ICD-10 F00–F99 Mental and Behavioural Disorders chapter
6D70–6E0Z Neurocognitive disorders (8A20–8A2Z Disorders with neurocognitive impairment as a major feature)	F00–F09 Organic, including symptomatic mental disorders
6C40–6C5Z Disorders due to substance use or addictive behaviours	F10–F19 Mental and behavioural disorders due to psychoactive substance use
6A20–6A2Z Schizophrenia or other primary psychotic disorders 6A40–6A4Z Catatonia	F20–F29 Schizophrenia, schizotypal and delusional disorders
6A60–6A8Z Mood disorders	F30–F39 Mood (affective) disorders
6B00–6B0Z Anxiety or fear-related disorders 6B20–6B2Z Obsessive-compulsive or related disorders 6B40–6B4Z Disorders specifically associated with stress 6B60–6B6Z Dissociative disorders 6C20–6C2Z Disorders of bodily stress or bodily experience	F40–F48 Neurotic, stress-related and somatoform disorders
6B80–6B8Z Feeding or eating disorders 6E20–6E2Z Disorders associated with pregnancy, childbirth, or the puerperium 6E40–6E40Z Physiological or behavioural factors affecting disorders or diseases classified elsewhere	F50–F59 Behavioural syndromes associated with physiological disturbances and physical factors
6C70–6C7Z Impulse control disorders 6D10–6D11.5 Personality disorders and related traits 6D30–6D3Z Paraphilic disorders 6D50–6D5Z Factitious disorders (7A codes: Sleep and sleep–wake disorders) (HA60–HA6Z Gender incongruence)	F60–F69 Disorders of adult personality and behaviour
6A00–6A00.Z Disorders of intellectual development	F70–F79 Mental retardation
6A00–6A06.Z Neurodevelopmental disorders	F80–F89 Disorders of psychological development
6C00–6C0Z Elimination disorders 6C90–6C9Z Disruptive behavioural or dissocial disorders	F90–F98 Behavioural and emotional disorders with onset usually occurring in childhood and adolescence
6E66–6E6Z Secondary mental or behavioural syndromes associated with disorders or diseases classified elsewhere	F99 Unspecified mental disorder

Diagnoses in brackets are contained in other ICD-11 chapters. Adapted from Gaebel, W., Stricker, J., & Kerst, A. (2020). Changes from ICD-10 to ICD-11 and future directions in psychiatric classification. *Dialogues in Clinical Neuroscience*, 22, 7–15.

episode, unspecified is 6A20.Z; recurrent depressive disorder, currently in remission is 6A71.7.

- Disorder groupings are based on shared aetiology, pathophysiology, and phenomenology.
- Each grouping includes a brief description of Essential (Required) Features, Boundary with Normality (Threshold), Boundaries with Other Disorders

(Differential Diagnosis), Course Features, Culture-related Features, Developmental Presentations, and Gender-related Features.

- There is no separate category of disorders with onset during childhood and adolescence.
- New diagnostic categories include catatonia, bipolar II disorder, complex post-traumatic disorder, and binge eating disorder.

- Major differences to diagnostic features include removal of the subtypes of schizophrenia, and removal of ‘mood episode’ as a diagnosis.
- Personality disorder subtypes are removed (except for borderline personality disorder), and replaced by a dimensional approach, with severity criteria and optional description of maladaptive traits.
- Sleep-wake disorders, and conditions related to gender and sexual health have been moved to separate chapters.

Field trials suggest ICD-11 has somewhat better reliability and greater ease of use compared to ICD-10 (Gaebel et al., 2020). For further details about ICD-11 and its major features, see Reed et al. (2019).

Diagnostic and Statistical Manual (DSM)

In 1952 the American Psychiatric Association published the first edition of the Diagnostic and Statistical Manual (DSM-I) as an alternative to the widely criticized ICD-6. DSM-I was strongly influenced by the views of Adolf Meyer and Karl Menninger, and reflected the prevailing acceptance of psychoanalytic ideas in the United States. DSM-II was published in 1968, and combined psychoanalytic ideas with those of Kraepelin.

DSM-III was published in 1980, and was an important step forwards, containing several innovations.

- Operational criteria were provided for each diagnosis, with explicit rules for inclusion and exclusion (Feighner et al., 1972). This was the first complete classification to do so, and the first to be based on criteria that had been field-tested.
- A multiaxial classification was adopted, with five axes (Axis I: Clinical syndromes; Axis II: Personality

disorders; Axis III: Physical disorders; Axis IV: Severity of psychosocial stressors; Axis V: Highest level of adaptive functioning in the last year).

- The nomenclature was revised, and some syndromes were regrouped.
- Its approach was empirical, and psychodynamic concepts were largely eliminated.
- For some conditions, duration of illness was introduced as a diagnostic criterion.

DSM-IV followed in 1994 and retained the basic structures and features from DSM-III.

DSM-5

When planning for DSM-V (renamed DSM-5) began, it was hoped that the classification could be based on aetiology (including the use of biomarkers) rather than description. It was also intended to make much greater use of dimensions rather than categories. However, it became apparent that for all major disorders both steps were premature and DSM-5, introduced in 2013, retains the same key elements as its predecessors, although without the multiaxial aspect. It includes some new and revised diagnostic criteria and other features that generated controversy for several reasons, including concerns about specificity and sensitivity, reliability, and conflicts of interest (Frances and Nardo, 2013).

For a history and critique of DSM, see Blashfield et al. (2014).

Comparing ICD-11 and DSM-5

The intention was to develop ICD-11 and DSM-5 in parallel and to avoid unnecessary differences. The efforts were largely successful, but there are some differences in their nature (Table 2.2) and in the diagnostic groupings (Table 2.3). There is also variation in the criteria

Table 2.2 Differences between ICD-11 and DSM-5

	ICD-11	DSM-5
Origin	World Health Organization	American Psychiatric Association
Use	Official global classification, for use by all health practitioners in all health settings	Mainly American psychiatrists, and psychiatric researchers
Presentation	Different versions for clinical work, research, and use in primary care	A single document
Languages	Available in all widely spoken languages	English version only
Structure	Part of overall ICD framework	Stand alone
Content	Clinical descriptors and guidance used	Operational criteria used

Table 2.3 Diagnostic groupings in ICD-11 and comparison with DSM-5

ICD-11	DSM-5
Neurodevelopmental disorders	Neurodevelopmental disorders
Schizophrenia spectrum and other primary psychotic disorders	Schizophrenia spectrum and other psychotic disorders
Mood disorders	No category
Bipolar and related disorders	Bipolar and related disorders
Depressive disorders	Depressive disorders
Anxiety and fear-related disorders	Anxiety disorders
Obsessive–compulsive and related disorders	Obsessive–compulsive and related disorders
Disorders specifically associated with stress	Trauma- and stressor-related disorders
Dissociative disorders	Dissociative disorders
Bodily distress disorders	Somatic symptoms and related disorders
Feeding and eating disorders	Feeding and eating disorders
Elimination disorders	Elimination disorders
Impulse-control disorders	Disruptive, impulse control, and conduct disorders
Disruptive behaviour and dissocial disorders	Disruptive, impulse control, and conduct disorders
Disorders due to substance use and addictive behaviours	Substance-related and addictive disorders
Paraphilic disorders	Paraphilic disorders
Factitious disorders	No category
Neurocognitive disorders	Neurocognitive disorders
Personality disorders	Personality disorders
Disorders associated with pregnancy, childbirth, and the puerperium	No category
Secondary mental and behavioural syndromes	No category
In ‘Sleep–wake disorders’ chapter	Sleep–wake disorders
In ‘Conditions related to sexual health’ chapter	Sexual dysfunctions
In ‘Conditions related to sexual health’ chapter	Gender dysphoria
In ‘Diseases of the nervous system’ and ‘External causes of morbidity or mortality’ chapters	Medication-induced movement disorders and other adverse effects of medication
In ‘Factors influencing health status and encounters with health services’ chapter	Other conditions that may be a focus of clinical attention

Adapted from Regier, D. A., et al. (2020). DSM-5 and ICD-11 classifications. In Geddes, J. R., Andreasen, N. C., & Goodwin, G. M. (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press.

for individual diagnoses, as described in the relevant chapter.

It is important to realize that the two classifications are complementary rather than in competition. ICD-11 results from an international effort and is designed for use in all healthcare settings (including primary

care) and in all countries with their varied cultures, professional needs, and traditions. DSM-5 is a national classification, and reflects the professional, educational, and financial priorities of its parent organization. For comparison of ICD-11 and DSM-5, see Regier et al. (2020).

Other issues in psychiatric classification

Many of the issues relating to classification discussed in this chapter continue to be topical and under active debate (Stein et al., 2022). This section raises some additional issues, especially those that may influence future developments (Box 2.2).

Cultural issues

Although ICD-11 and DSM-5 make national approaches to classification less important, local and cultural factors remain important in classification in several respects.

Psychiatrists and physicians in countries such as India, Pakistan, and China, which have their own long-standing and comprehensive systems of ideas about health and illness, sometimes complain that classifications developed in Europe and North America give too much emphasis to separation of mind and body, such as the concept of somatoform disorders. This approach causes problems in Western medicine and is not

understood at all elsewhere. Investigation of these issues is important but can be difficult if researchers do not fully appreciate significant cultural and local factors, or the varying ways in which emotions and behaviour are described in different languages.

Reflecting an increased focus on culture and mental health, ICD-11 includes ‘culture-related features’ for each diagnostic category. DSM-5 also pays greater attention to cultural issues than earlier versions, and distinguishes three concepts:

1. Cultural syndrome: syndromes characteristically found in a particular cultural group.
2. Cultural idiom of distress: terms, phrases, and ways of communicating suffering that are characteristic of a cultural group.
3. Cultural explanation or perceived cause: a label for, or attribution of, a cause of symptoms or distress that is accepted within a cultural group.

Box 2.2 Other national systems of classification

The widespread international acceptance of ICD and DSM schemes has diminished the importance of pre-existing national diagnostic traditions. However, the latter are of historical interest and, at times, still have some influence on educational programmes.

The descriptive concepts introduced by Kraepelin and Bleuler have been very influential in most European countries. In Scandinavia, emphasis has also been placed on the concept of psychogenic or reactive psychoses (Strömberg, 1985). In addition, Scandinavia was notable for its early concepts of multidimensional diagnoses. In France, Kraepelinian views of schizophrenia were less widely accepted, and two other diagnostic categories of psychosis not commonly used elsewhere

have persisted, namely *bouffée délirante* and *délires chroniques*. *Bouffée délirante* is the sudden onset of a delusional state with trance-like feelings, of short duration and good prognosis; *délires chroniques* are conditions that in ICD-11 would be classified as ‘persistent delusional disorders’.

Another example of international variation is the Chinese national classification (*Chinese Classification of Mental Disorders, 3rd edition, CCMD-3*), introduced in 2001. Although largely based upon ICD-10, it excludes almost all of the somatoform disorders, so that particular prominence can be given to the category of *neurasthenia*, which remains one of the most frequent diagnoses in Chinese psychiatry.

Research domain criteria

The scientific arguments for a dimensional rather than categorical approach to diagnostic classification have been outlined above. One manifestation of this was the move in 2010 by the United States' National Institute for Mental Health to advocate 'domains', and to require these, not diagnostic categories, to be used as the basis for research funding (Cuthbert and Insel, 2013). Such domains may include neuropsychological constructs (e.g. working memory, or reward sensitivity) or brain

systems (e.g. corticostriatal circuits), which underpin—and are thought to cut across—current diagnostic categories. The advent of these Research Domain Criteria (RDoC) had a major if controversial impact on psychiatric research, at least in the United States, but is not of clinical relevance without the discovery of domains that have the necessary utility and reliability.

For discussion of how dimensional, hierarchical, and other issues may be addressed in future psychiatric classifications, see Gaebel et al. (2020).

Further reading

American Psychiatric Association. (2013). *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)*. American Psychiatric Association.

Kendell, R. E. (1975). *The Role of Classification in Psychiatry*. Blackwell Scientific Publications.

World Health Organization. (2024). *Clinical Descriptions and Diagnostic Requirements for ICD-11 Mental, Behavioural and Neurodevelopmental Guidelines*. World Health Organization. The manual can be downloaded from <https://www.who.int/publications/i/item/9789240077263>.

CHAPTER 3

Assessment

Introduction 37

Psychiatric interviewing 38

The psychiatric history 41

The mental state examination 46

Other components of psychiatric assessment 51

Special kinds of psychiatric assessment 54

Patient characteristics that may affect the
interview 56

Integrating and evaluating the information 58

Recording and communicating information 61

Standardized assessment methods 67

Introduction

Psychiatric assessment has three main goals:

1. *To make a diagnosis.* Despite its limitations (see Chapter 2), diagnosis is central to the practice of psychiatry, since it provides the basis for rational, evidence-based approaches to treatment and prognosis (Craddock and Mynors-Wallis, 2014). A major goal of most assessments, especially those of new patients, is therefore to allow a diagnosis or differential diagnosis to be made. This in turn requires that the symptoms and signs of psychiatric disorder can be elicited, and their diagnostic significance appreciated.
2. *To understand the context of the diagnosis.* The psychiatrist needs to have sufficient information about the patient's life history, current circumstances, and personality. This is necessary in order to try and understand why the disorder has occurred in this person at this time; it also has a major bearing on decisions about management and prognosis.
3. *To establish a therapeutic relationship.* The psychiatrist must ensure that the patient feels able and willing to give an accurate and full history. Without this skill, the necessary diagnostic information is unlikely to be obtained. Thus, establishment of a therapeutic relationship is essential if the patient is to engage fully in

discussions about management, and to adhere to any treatment decisions which are agreed upon.

The process of psychiatric assessment, also known as the *psychiatric interview*, can be broken down into the following stages:

- *Preparation.* This includes having the interviewing skills necessary to achieve the above goals, acquiring as much background information as possible (e.g. the reason for referral, and whether informants are available) in order to help guide the assessment and identify areas for particular focus, and anticipating whether the interview may need to be adapted (e.g. due to language difficulties, or a shortage of time).
- *Collecting the information.* This is usually addressed by means of a series of headings covering the psychiatric history, mental state examination, and other components, as described below.
- *Evaluating the information* in order to arrive at a diagnosis or differential diagnosis. This is the hardest part of the process to describe to readers new to psychiatry, as it requires knowledge of the diagnostic significance of particular symptoms and symptom combinations. It is therefore suggested that this chapter should be returned to at regular intervals, in order to refine how

assessments are conducted to match one's developing knowledge and skills.

- *Using the information* to make treatment decisions and form prognostic opinions.
- *Recording and communicating the information collected, and the conclusions drawn.* The information must be shared with other health professionals involved with the care of the patient both now and in the future, and with the patient and their significant others. Various modes of communication are necessary (e.g. with regard to its nature and level of detail) depending on the circumstances of the assessment and the needs of the recipient.

This chapter covers these areas in turn, concentrating on the full initial assessment of a patient by a general adult psychiatrist. It also discusses how this process should be adapted in other circumstances. First,

however, the basic process of psychiatric interviewing is outlined. At the outset, it is worth emphasizing that the description of assessment in textbooks tends to make the process appear to be passive, even predetermined. In practice, however, assessment should be an active process in which diagnostic clues are pursued, hypotheses tested, and the focus of questioning adapted to the particular circumstances. This 'dynamic' aspect of assessment can only be learned from practical experience. It also requires a working knowledge of the main psychiatric syndromes, in order for the significance of specific symptoms or history items which emerge during the assessment to be appreciated. It was for this reason that psychiatric classification was covered in Chapter 2.

It is assumed that the reader is already competent in medical history taking and physical examination, and these topics are considered only briefly.

Psychiatric interviewing

Preparing for the assessment

Psychiatric assessments are conducted in many settings. The following recommendations should be followed as far as is practicable, but they cannot always be achieved in their entirety. In some locations, such as an Accident and Emergency department, the setting may be far from ideal. Nevertheless, it is important to do what is possible to ensure safety and confidentiality, and to put the patient at ease.

Before starting the interview, it is always worth finding out what is already known about the patient and the circumstances in which the assessment has come about. This will often be in the form of a referral letter from another doctor, or there may be notes available from previous assessments. This information will reveal areas of enquiry that may require particular emphasis, and may modify the diagnosis which is subsequently made. Equally, every assessment should of course begin with an open mind, not least since prior diagnoses may be incorrect, or the current presentation may be different.

Only a small minority of patients are potentially dangerous, but the need for precautions should be considered before every interview. The interviewer should always ensure that:

- Another person knows where and when the interview is taking place and how long it is expected to last. This is especially relevant to interviews in the community.

- Help can be called if it is needed. In hospital, check for an emergency call button and its position, and otherwise try to arrange for another person to be within earshot.
- Neither the patient nor any obstruction is between you and the exit.
- Any objects that could be used as weapons are removed.

If the risk is thought to be high, or if these requirements cannot be met, it may be necessary to defer the interview.

Starting the assessment

The psychiatrist should welcome the patient by name, give their own name and status, and explain in a few words the reason for and purpose of the assessment. If the patient is accompanied, the interviewer should greet the companion(s) and explain how long they should expect to wait and whether they will be interviewed. It is usually better to see the patient alone first, provided that they are able to provide an adequate history. The interviewer should explain that notes will be taken, and that these will be confidential. If the interview is for the purposes of a report to an outside agency (e.g. a medicolegal report), this should be made clear. The general structure of the interview should be explained, and the time

available stated; the latter is not only a matter of courtesy but is also valuable if the psychiatrist later needs to interrupt the patient, or ask them to give more concise answers, to ensure the assessment can be completed.

The patient should be comfortable, and the interviewer should not face the patient directly but arrange their chair at an angle. If possible, the interviewer should also avoid sitting at a higher level. Whenever possible the interviewer should make notes (whether on paper or electronically) during the interview, as writing up an assessment afterwards is time consuming and not always wholly accurate. However, it is usually better to delay note taking for some minutes until the patient feels that they have the interviewer's undivided attention. The patient should be placed at the left side of a right-handed interviewer. With this arrangement, the interviewer can attend to the patient and maintain an informal atmosphere while writing.

The following techniques have been shown to improve the results of the interview (Goldberg and Huxley, 1980). The interviewer should:

- Adopt a relaxed posture and appear unhurried—even when time is short;
- Maintain appropriate eye contact with the patient and not appear engrossed in note taking;
- Be alert to verbal and non-verbal cues of distress as well as to the factual content of the interview; and
- Maintain control of the interview if the patient is over-talkative or discursive.

Continuing and completing the assessment

The interview should begin with an open question (one that cannot be answered with a simple 'yes' or 'no'), such as 'Tell me about your problems or difficulties.' The patient should be allowed to talk freely for several minutes before further questions are asked. As the patient describes their problems, the interviewer should observe their responses and manner—for example, whether they are reticent, irritable, or appear confused.

The first step is to obtain a clear account of the patient's problems. It is important to separate symptoms from their consequences, and from other life problems which the patient may want to discuss. For example, a patient may have low mood, sexual difficulties, or financial worries as their presenting complaint. In each case the common denominator may prove to be depression, but it will require your assessment to discover this. Your priority at the start is to focus upon the symptoms and

signs of psychiatric disorder, leaving the other kinds of problem until later.

From the start, consider the possible diagnoses and, as the interview progresses, select questions to confirm or reject these diagnoses. For example, if the patient mentions hearing voices, this raises the possibility of schizophrenia and requires that, at some stage in the assessment, the other cardinal features of the disorder are sought, and their presence or absence recorded. The interviewer also considers what information is relevant to prognosis and treatment. Thus, as noted earlier, interviewing is not simply the asking of a routine set of questions; rather, it is an active and iterative process in which the focus of attention is directed by hypotheses formed from the information already elicited, and modified repeatedly as more information is collected. As the interviewer gains confidence and acquires more psychiatric knowledge, they will become better at thinking of possible diagnoses, and proceeding in a way that rules them in or out more rapidly and convincingly.

It is generally better to establish clearly the nature of the symptoms before asking how and when they developed. If there is any doubt about the nature of the symptoms, the patient should be asked to describe specific examples. When all of the presenting symptoms have been explored sufficiently, direct questions are asked about others that have not come to light but which may be relevant. In doing this, the interviewer uses their knowledge of psychiatric syndromes to decide what further questions to ask. For example, a person who complains of feeling depressed would be asked about suicidal thoughts and, if these are acknowledged, further questions should be asked regarding suicidal intent and previous attempts (see Chapter 21). Also ask about the impact that the symptoms have had on the patient's life, looking for evidence of functional impairment.

The onset and course of the symptoms are clarified next, together with their relationship to any stressful events or physical illness. Persistence may be needed to date the onset or an exacerbation of symptoms accurately. The patient's attempts to cope with the symptoms are noted—for example, increased drinking alcohol to relieve distress. If treatment has already been given, its nature, timing, and effects are noted, together with the patient's concordance with it.

The interviewer completes the relevant parts of the full assessment schedule described below. When time is adequate, the mental state is usually examined at the end of the interview, together with any relevant physical examination. If time is short, it may be better to examine the mental state after the presenting complaints

have been clarified. This can make it easier to select the key points to be asked about in the rest of the history.

Throughout the interview, allow the patient, as far as possible, to describe their problems spontaneously. In this way, unexpected material may be volunteered that might not be revealed by the answers to questions. However, questions may be needed to bring the patient back to the point after a digression, and to elicit specific information—for example, about the relationship between symptoms and stressful events. Whenever possible, the interviewer should use open rather than leading or closed questions (a leading question suggests the answer, whereas a closed question allows only the answers ‘yes’ or ‘no’). When there is no alternative to a closed question, the answer should be followed by a request for an example.

Before ending the interview, it is good practice to ask a general question such as ‘Is there anything that I have not asked you about, that you think I should know?’ It is also useful to summarize for the patient what you consider to be the key points, to check for any errors of fact, and to see whether they agree with your initial understanding of events.

Box 3.1 summarizes some techniques that promote effective interviewing.

Box 3.1 Some techniques for effective psychiatric assessments

- Help the patient to talk freely. This can be done using open questions, and by non-verbal cues such as nodding, or saying ‘Go on’ or ‘Tell me more about that.’
- Keep the patient to relevant topics. Again, non-verbal cues are useful, as well as specific interventions such as ‘At this point I’d like to ask you more about how you’ve been feeling. We can return to your money worries later.’
- Make systematic enquiries, but avoid asking so many questions that other, unanticipated issues are not volunteered.
- Check your understanding, and that you have enquired about all of the areas the patient thinks are important, by summarizing the key points of the history back to them. This step also helps you to begin to formulate your views on the diagnosis and causes.
- Be flexible in assessments, with regard to both their length and sequence. Select questions according to the emerging possibilities regarding diagnoses, causes, and plans of action.

Interviewing informants

Whenever possible, the patient’s history should be supplemented by information from a close relative or another person who knows them well. This is much more important in psychiatry than in the rest of medicine, for several reasons. Some psychiatric patients are unaware of the extent or significance of their symptoms. Others are aware of their problems but do not wish to reveal them. Patients and relatives may also give quite different accounts of personality characteristics, or have contrasting interpretations of recent events and symptoms. Interviews with a partner or relative are used not only to obtain additional information about the patient’s condition, but also to assess their attitudes to the patient and the illness, and often to involve them in the subsequent management plan. In addition, they provide an opportunity to learn what burdens the illness has placed on the partner or relative, and how they have tried to cope. A history from an informant is essential when the patient is unable to give an accurate account of his or her condition (e.g. because of impaired memory). Finally, when it is important to know about the patient’s early development or childhood, an interview with a parent is usually the best way to discover the relevant information.

Informants can either be seen separately from the patient, or invited to join the interview. The choice depends on both the assessor’s and the patient’s preference, but in both instances the patient must give their consent. If the patient refuses, explore the reasons for this and explain the difficulties that it will pose. There are a few situations in which the patient’s permission is not required before interviewing a relative or other informant—for example, if the patient is a child, or when adult patients are mute or confused. In other cases, the doctor should explain to the patient the reasons for interviewing the informant, while emphasizing that confidential information given by the patient will not be passed on. If any information needs to be given to a relative—for example, about treatment—the patient’s permission should be obtained. Questions from relatives should be dealt with in the same way.

The psychiatrist begins the interview with the informant by explaining its purpose. They may need to reassure the informant. For example, a relative may fear that they will be viewed as responsible in some way for the patient’s problems. The interviewer should be sensitive to such ideas and, when appropriate, discuss them in a reassuring way, but without colluding or becoming involved in ways that might conflict with their

primary duty to the patient. If the informant has been interviewed separately from the patient, the psychiatrist should not tell the patient what has been said unless the informant has given their permission. This is important even when the informant has revealed something that should be discussed with the patient—for example, an account of excessive drinking.

Sometimes it is necessary to speak to employers, educators, friends, police, or others in order to collect further information. This should be done only with the patient's permission, unless there are legal or safety issues which override this principle—for example, if the patient is in custody, or if there are grounds for concern that the patient may harm a third party.

The psychiatric history

The main parts of a psychiatric assessment are the *psychiatric history* and the *mental state examination*. The latter covers the symptoms and signs present during the interview, and the former deals with everything else. The assessment is then completed by the physical examination, and sometimes by further investigations. This section covers the psychiatric history, followed by the mental state examination, and then the other components of the assessment.

A commonly used scheme for history taking is given below. For ease of reference, the scheme is presented as a list of headings and items. More details, and some background information to the questions, are provided in the subsequent notes. As noted, much of the interview is designed to elicit diagnostic symptoms, but other questions are intended to obtain information about the patient's life and circumstances, while the interview as a whole must try to establish the rapport needed to achieve these goals and form the basis for a therapeutic relationship.

The following scheme is comprehensive and systematic, as an ability to conduct this form of assessment is essential before briefer interviews are attempted. Modification of the interview for use in different settings is described later. Moreover, although it is neither possible nor necessary to take a full history on every occasion, whatever information that has been elicited should always be recorded systematically and in a standard order. This practice helps the interviewer to remember all of the potentially important topics and to add further information later, and makes it easier for colleagues who need to refer to the notes. This order can be followed in the written or electronic record, even when it was not possible to elicit the information in the desired sequence. This and all other entries in the notes should be dated and, whenever possible, signed.

A scheme for history taking

Information is grouped under the headings shown in Box 3.2. For clarity, this section is written in the style of short notes and illustrative questions. The next section explains why these topics are relevant, and some of the problems that may occur when covering them.

Informant(s)

- Usually the principal informant is the patient. If not, state the reason.
- The name(s), relationship to the patient, and length of acquaintance of any other person(s) interviewed.
- The name of the referrer and the reasons for referral.

History of present condition

Also known as the *history of presenting complaint*, this section is in many ways the core of the interview, usually providing much or most of the key diagnostic information.

Box 3.2 Outline of the psychiatric history

Name, address, age, biological sex, and preferred gender pronoun
 Name(s) of informant(s) and their relationship to the patient
 History of present condition
 Family history
 Personal history (expanded in Box 3.3)
 Past illness
 Personality (expanded in Box 3.4)

- List the symptoms, with the onset, duration, severity, and fluctuation of each. Quantitative information is valuable. For example, a patient can be asked to rate their low mood on a 10-point scale.
- Ask about and record symptoms that might have been expected but that are *not* present (e.g. no suicidal ideation in a person with low mood; no hallucinations in a patient with delusions).
- The temporal relationship between symptoms and any physical disorder, psychological or social problems, or recent life events.
- The nature and duration of any functional impairment caused by the symptoms.
- Any treatment received, and its effects and side effects.

Family history

- Parents and siblings—age, current state of health or date and cause of death, occupation, personality, quality of relationship with patient. Psychiatric and medical family history.
- When the family history is complex and relevant, summarize it as a family tree.

Personal history

This is a particularly variable section of the history (Box 3.3). In the case of a young person with a disorder suspected to have origins early in life, it will be important and extensive; for older patients or some other disorders, only limited questioning may be necessary.

- Pregnancy and birth abnormalities (e.g. infections, prematurity, problems with labour).
- Early developmental milestones—walking, talking, etc.
- Childhood—any prolonged separation from the parents, and the patient's reaction to it. Any emotional problems (age of onset, course, and treatment). Any serious illness in childhood.
- Schooling and higher education—type, courses, qualifications, extracurricular achievements, relationships with teachers and other students. Note any experience of bullying (nature, duration, and impact) or school exclusions.
- Occupations—present job (dates, duties, performance, and satisfaction), earlier jobs (list them, with reasons for changes).
- Significant relationships —identity and gender of current partner, duration and nature of relationship. Partner's health and attitude to the patient's illness. Nature and number of previous relationships.

Box 3.3 Outline of the personal history

Mother's pregnancy and the birth
 Early development
 Childhood: separations, emotional problems, illnesses, education
 Gender identity
 Occupations
 Relationships and sexual partners
 Children
 Social circumstances
 Substance use
 Forensic history

- Sexual history—gender identity, sexual orientation, any sexual difficulties and their relationship to current symptoms.
- Children—identities, date of any abortions or stillbirths, temperament, emotional development, mental and physical health. Main child carers.
- Social circumstances—accommodation, household composition, financial situation.
- Use of alcohol, tobacco, illicit drugs—which ones, when, and how much. Problems arising from substance use.
- Forensic history—arrests, convictions, imprisonment. Nature of the offences, especially with regard to dangerousness. For a few patients, the forensic history is a key part of the assessment (see Chapter 18).

Past psychiatric and medical history

- Past psychiatric illnesses—nature and duration, and their similarity to current episode. Include any episodes of self-harm. Date, duration, nature, location, and outcome of any treatment.
- Past medical history—illnesses, operations, accidents, and drug treatments.
- Current medication, including over-the-counter medicines and alternative remedies. Any allergic or other adverse reactions.

Personality

By this stage in the interview, the patient's manner and description of their history will have provided some indication about their personality. However, a specific focus is also necessary, covering the domains noted in

Box 3.4 Assessment of personality

Relationships
Leisure activities
Prevailing mood and emotional tone
Character
Attitudes and values
'Ultimate concern'

Box 3.4. Personality can be a relatively difficult area of the history, as outlined below in the section 'Notes on history taking'.

- Relationships—friendships (few or many, superficial or close, with own or opposite sex), relationships with work colleagues and superiors.
- Use of leisure time—hobbies and interests.
- Predominant mood and emotional tone (e.g. anxious, optimistic, self-deprecating, overconfident, stable or fluctuating, controlled or demonstrative).
- Character traits (e.g. perfectionist, obsessional, isolated, impulsive, sensitive).
- Attitudes and values (e.g. moral or religious; attitude towards health).
- 'Ultimate concern'— who and what matters most in their life?

Notes on history taking

The scheme outlined above lists the items to be considered when a full history is taken, but has not indicated why these items are important or what sort of difficulties may arise when eliciting them. These issues are discussed in this section, which is written in the form of notes approximating to the headings used above.

The reason for referral

Only a brief statement need be given—for example, 'severe depression with somatic symptoms.' The reason for referral usually, but not always, proves to be the main focus of the interview. Check that the patient has the same understanding as to why they have been referred. If not, this in itself is useful information. For example, the patient may disagree that they are depressed, believing that their somatic symptoms are due to cancer. This may affect their willingness to engage fully in the assessment, or accept the treatment for depression which may be subsequently recommended.

History of present condition

Because it is central to the assessment, it is always worth spending sufficient time on this part of the history to identify the key elements. There will often be several such symptoms, and each should be characterized fully in order to appreciate its diagnostic significance. Record the severity and duration of each symptom, how and when it began, what course it has taken (increasing or decreasing, constant or intermittent), and what factors affect it. Indicate which symptoms co-vary. As noted above, record any symptoms or features that would have diagnostic significance but that are not present (e.g. lack of anhedonia in a person complaining of low mood).

Note any treatments that have been given during this episode, the response, and any adverse effects. If a drug was ineffective, ask whether the patient took it regularly and at the prescribed dosage.

Family history

A family history of psychiatric disorder may indicate genetic or environmental influences. A genetic contribution is more likely for some disorders than for others, and increases as more relatives are affected. Childhood adversity and maltreatment is an important factor in the development of mental health problems in adult life; therefore knowledge about the family's circumstances remains part of the basic information required for understanding the origin and presentation of the patient's problems.

Recent events in the family may have been stressful to the patient—for example, the serious illness or divorce of a family member. Events in the family may throw light on the patient's concerns. For example, the death of a sibling from a brain tumour may partly explain a patient's extreme concern about headaches.

Personal history

Pregnancy and birth

Events in pregnancy and delivery are most likely to be relevant when the patient has a disorder of intellectual development (learning disability), although they are also risk factors for several psychiatric disorders with onset in childhood, adolescence, or early adulthood. An unwanted pregnancy may be followed by a poor relationship between mother and child.

Child development

Few patients know whether they have passed through developmental stages normally. Failure to do so may be a sign of learning disability and also a risk factor for later disorders such as schizophrenia. However, this

information is usually more important if the patient is a child or adolescent, in which case the parents are likely to be available for interview. The effects of separation from the mother vary considerably, and depend in part on the age of the child, the duration, and the reason for separation. Questioning about the child's emotional development provides information about early temperament and emerging personality, and abnormalities or delays may serve as risk factors for, or early signs of, later problems. However, childhood behavioural characteristics as a rule are weak predictors of adult disorders, and only require detailed consideration when assessing children and adolescents. Assessment in child psychiatry is covered in Chapter 16.

Education

The school record gives an indication of intelligence, achievements, and social development. Ask whether the patient made friends and got on well with teachers, and about success at games and other activities. Bullying is a risk factor for later psychological difficulties, and other negative events such as exam failures may be important stressful memories. Similar questions are relevant to higher education.

Occupational history

Information about the present job helps the interviewer to understand the patient's current abilities, interests, and financial and social circumstances, and may be a potential source of stress. A list of previous jobs and reasons for leaving is relevant to the assessment of personality. If the status of jobs has declined, this may reflect chronic illness or substance misuse.

Relationship history

This heading includes all enduring intimate relationships. Ask about the current and any previous lasting relationships, preferably phrased in a way that does not assume the gender of the partner(s). Frequent broken relationships may reflect abnormalities of personality. The partner's occupation, personality, and state of health are relevant to the patient's circumstances and, like the nature of the relationship itself, will affect the partner's role in the care and management of the patient.

Sexual history

The interviewer should use common sense when deciding how much to ask the individual patient, depending on the response to initial questions, demographic factors (such as age and relationship status), and the nature of the presenting complaint. Usually the interviewer is

concerned to establish generally whether the patient's gender identity or sexual life is in any way involved in their current difficulties, whether as a cause, a correlate, or a consequence. If so, then more detailed enquiry is appropriate (see Chapter 13).

Judgement must also be used about the optimal timing and amount of detail of questioning about childhood abuse, especially sexual abuse. Unfortunately, such past experiences are sufficiently common to merit enquiry. However, often it may not be appropriate to raise the matter in a first interview, unless prompted by the patient or in the light of background information available to the interviewer. The decision to raise the topic also depends on the clinical suspicion, and the time and expertise available to the interviewer. Sensitivity is also necessary when deciding what information to record and with whom it should be shared. Safeguarding issues should also be considered.

Children

Pregnancy, childbirth, miscarriages, and terminations are events that are sometimes associated with adverse psychological reactions. Information about the patient's children is relevant to present worries and the pattern and characteristics of family life.

Consideration of the welfare and needs of any children is always integral to an assessment, as their health and care may be adversely affected by the parent's illness or its treatment. For example, if a seriously depressed woman has a young baby, appropriate steps to ensure the baby's wellbeing are essential, and the situation may influence decisions about the mother's care (e.g. about medication if she is breastfeeding). In turn, any concerns about the care or welfare of children should be clearly recorded, and followed by appropriate discussions and, if necessary, interventions. Such considerations are increasingly covered by safeguarding guidelines, protocols, and legislation (which differ from country to country), and the reader is advised to keep up to date in this respect.

Social circumstances

Questions about housing, finances, and the composition of the household help the interviewer to understand the patient's circumstances. Assets and resources (including potential carers) are assessed, as well as problems and sources of stress. There can be no general rule about the amount of detail to elicit, and this must be guided by common sense.

Substance use and misuse

This includes past as well as present consumption of alcohol and other substances, and the impact of their use on the patient's health and life. Misuse of prescribed drugs should also be considered. The patient's answers may be evasive or misleading, and may need to be checked with other informants and sources of information (e.g. urine screens, blood tests). See Chapter 20 for further information about interviewing in this area.

Past psychiatric and medical history

Many patients will have a past psychiatric history. A previous diagnosis increases the probability that the current diagnosis will prove to be similar, but it is important always to keep an open mind, as the diagnosis may have changed, or the previous diagnosis may have been incorrect. Patients or relatives may be able to recall the general nature of the illness and treatment, but it is nearly always appropriate to request information from others who have treated the patient.

The medical history is also important (Phelan and Blair, 2008). A medical disorder or its treatment may be directly related to the presentation (e.g. a recurrence of hypothyroidism presenting with lethargy, or mania induced by corticosteroids), but may also be indirectly relevant (e.g. via the psychosocial effects of chronic ill health, or as a sign of somatoform disorder).

Assessment of personality

This is important because:

- It helps the interviewer to understand the patient as a person, and to put their current difficulties into context;
- Personality traits can be a risk factor for psychiatric disorders (e.g. obsessiveness increases the risk of developing a depressive disorder);
- Personality traits can colour the presentation of psychiatric disorder; and
- Personality can be disordered, and personality disorder may be a differential or comorbid diagnosis.

Aspects of personality can be assessed by asking for a self-assessment, by asking others who know the patient well, and by observing behaviour. Good indications of personality can often be obtained by asking how the patient has behaved in particular circumstances, especially at times when social roles are changing, such as when starting work, or becoming a parent. However,

mistakes can arise from paying too much attention to the patient's own assessment of their personality, especially during a single interview. Some people give an unduly favourable or unfavourable account. For example, anti-social people may conceal the extent of their aggressive behaviour or dishonesty and, conversely, depressed patients often judge themselves negatively and critically. When assessing personality from behaviour at interview, consider the artificiality of the situation and the anxiety that it may provoke. These factors mean that it is essential to interview other informants whenever possible, and to avoid drawing premature or unjustified conclusions about personality. Personality tests are now rarely used in routine clinical practice, but interview schedules for diagnosis of personality disorders are sometimes used in forensic and other settings.

Relationships

Is the patient shy or do they make friends easily? Are their friendships close and are they lasting? Leisure activities can throw light on personality, by reflecting the patient's interests and preference for company or solitude, as well as their levels of energy and resourcefulness.

Mood and emotional tone

Ask whether the patient is generally cheerful or gloomy and whether they experience marked changes of mood, and if so, how quickly these moods appear, how long they last, and whether they follow life events. Information about prevailing mood and mood swings may also reveal evidence suggestive of mood disorder, which can be enquired about further.

Character

Common sense and experience will indicate the depth and focus of character assessment that are needed for each patient. If the patient (or informant) has difficulty describing their character with an open question, offer them options—for example, 'Would you call yourself an optimist or a pessimist? A loner or a socialite?' Do not focus entirely on negative attributes, but ask about positive ones, including resilience in the face of adversity. This is important not just to gain a balanced impression, but because strengths are usually better targets for intervention if personality proves to be therapeutically relevant.

The assessment of personality disorder has been described by Tyrer et al. (2015), and is discussed further in Chapter 15.

The mental state examination

The history records symptoms up to the time of the interview. The mental state examination is concerned with symptoms, signs, and behaviour during the interview, and is usually conducted after the history. Although the distinction is traditional, and conceptually useful, in practice the boundary between the history and the mental state examination is somewhat blurred. In particular, very recent symptoms and signs are often recorded in the mental state examination, even if the phenomena are not experienced or elicited during the interview. The mental state examination is also sometimes used to elicit and record symptoms and signs which, for whatever reason, have not been covered previously in the interview (e.g. whether the patient is suicidal). For a history of the mental state examination, see Huline-Dickens (2013).

The mental state examination uses a standard series of headings under which the relevant phenomena, or their absence, are recorded (Box 3.5). The symptoms and signs referred to in the following account are described in Chapter 1 and, with a few exceptions, are not repeated. Mental state examination is a skill that should be learned by watching experienced interviewers and by practising repeatedly under supervision. The mental state examination provides, in conjunction with the history of the present condition, most of the key diagnostic information. The ability to carry out and record an accurate and comprehensive mental state examination is therefore a core skill required by all psychiatrists and other mental health professionals. It is also specified by the United Kingdom General Medical Council as a requirement for all newly qualified doctors.

Box 3.5 Headings for the mental state examination

Appearance and behaviour
Speech
Mood
Thoughts—content, form
Perceptions
Cognitive function
Insight

Appearance and behaviour

General appearance

Much diagnostically useful information can be obtained from the patient's appearance and behaviour. Indeed, as discussed later, experienced clinicians often make provisional diagnoses within minutes of meeting a patient, relying heavily on this information. The process starts from the first moment you see the patient. For example, what is their manner and behaviour in the waiting room? Are they sitting quietly, pacing around, or laughing to themselves? When greeted, what is their response? As they walk towards the interview room, is there evidence of parkinsonism or ataxia? Note their general attire. A dirty, unkempt appearance may indicate self-neglect. Manic patients often dress brightly or incongruously. Occasionally an oddity of dress may provide the clue to diagnosis—for example, a rain hood worn on a dry day may be the first evidence of a patient's belief that rays are being shone on their head by persecutors. An appearance suggesting recent weight loss should alert the observer to the possibility of depressive disorder, anorexia nervosa, or physical illness.

Facial appearance and emotional expression

The facial appearance provides information about mood. In depression, the corners of the mouth are turned down, and eye contact is poor. Anxious patients have horizontal creases on the forehead, widened palpebral fissures, and dilated pupils. Facial expression may reflect elation, irritability, or anger, or the fixed 'wooden' expression due to drugs with parkinsonian side effects. The facial appearance may also suggest physical disorders (e.g. thyrotoxicosis).

Posture and movement

Posture and movement also reflect mood. A depressed patient characteristically sits with hunched shoulders, with the head and gaze inclined downwards. An anxious patient may sit on the edge of their chair, with their hands gripping its sides. Anxious people and patients with agitated depression may be tremulous and restless—for example, touching their jewellery or picking at their fingernails. Manic patients are overactive and restless. Other abnormalities of movement include tardive dyskinesia and the motor signs seen in catatonia.

Social behaviour

The patient's social behaviour and interactions with the interviewer are influenced by their personality and by their attitude to the assessment, as outlined above. However, their behaviour can also be influenced by psychiatric disorder, so it provides another potential source of diagnostic information. Manic patients tend to be unduly familiar or disinhibited, whereas those with dementia may behave as if they were somewhere other than in a medical interview, and a person with schizophrenia may be withdrawn and preoccupied. If the patient's social behaviour is highly unusual, describe what exactly is unusual, rather than simply using imprecise terms such as 'bizarre.'

Speech

Speech and thoughts are recorded in different parts of the mental state examination, even though it is only through speech that thoughts become known to the interviewer. By convention, this section covers the rate, quantity, volume, and flow of speech, and any delays in responding to questions or apparent difficulties producing or articulating speech. The content of speech, in the sense that it reveals the patient's thoughts is deferred until the 'Thoughts' section.

Rate and quantity

Speech may be unusually fast and increased in amount, as in mania (*'pressure of speech'*), or slow, sparse, and monotonous, as in depression. Patients with dementia or depression may pause for a long time before replying to questions, and then give short answers, producing little spontaneous speech. The same may be observed among shy people or those of low intelligence.

Difficulties with speaking

If the patient is having problems finding or articulating words, consider the possibility of dysphasia or dysarthria. For further details, see Box 3.2 and consult a neurology textbook.

Neologisms

Neologisms are private words invented by the patient, often to describe morbid experiences. If the interviewer is uncertain whether word is a neologism, ask the patient to repeat and spell the word, and explain its meaning.

Flow of speech

Abnormalities in the flow of speech may simply reflect an anxious, distracted patient, or one of low intelligence.

More significantly, such abnormalities may be evidence of disturbances in the stream or form of thoughts. For example, sudden interruptions may indicate thought blocking, and rapid shifts from one topic to another suggest flight of ideas. For description of these features, see Chapter 1. It can be difficult to be certain about these abnormalities; if possible, write down a representative example.

Mood

Conventionally, the mood section includes recording of other emotions such as anxiety, and also related phenomena, including suicidal thoughts.

Depression and mania

The assessment of mood begins with the observations of behaviour described already, and continues with direct questions such as 'What is your mood like?' or 'How are you in your spirits?' To assess depression, questions should be asked about a feeling of being about to cry (actual tearfulness is often denied), pessimistic thoughts about the present, hopelessness about the future, and guilt about the past. Suitable questions are 'What do you think will happen to you in the future?' or 'Have you been blaming yourself for anything?' Questions about elevated mood correspond to those about depression—for example, 'How are you in your spirits?' followed, if necessary, by direct questions such as 'Do you feel in unusually good spirits?' Note that the mood in mania can be irritable as well as cheerful. It can be useful to ask the patient to rate their current mood on a numerical or other scale.

As noted in Chapter 1, a distinction is sometimes drawn between 'objective' and 'subjective' mood. The latter is the patient's view of their own mood; the former refers to the interviewer's conclusion based upon their observation of the patient during the interview and the responses to their questions. On occasion the two may differ—for example, a severely depressed person may deny low mood. It is important therefore to record the presence of the various symptoms and signs of depression (or mania) during the mental state examination.

Both depressed and elevated mood, if clinically significant, are accompanied by other features—for example, anhedonia, tiredness, or poor concentration in depression. In practice, therefore, it is common to extend this part of the mental state examination to include questioning about other current features of mood disorder, if these have not been asked about already. The interviewer can choose to record them in the notes under this heading, or insert them into the relevant part of the history.

Fluctuating and incongruous mood

As well as assessing the prevailing mood, the interviewer should ascertain how mood varies. When mood varies excessively, it is said to be *labile*—for example, the patient appears dejected at one point in the interview but quickly changes to a normal or unduly cheerful mood. Any lack of emotional response, sometimes called blunting or flattening of affect, should also be noted.

Normally, mood varies during an interview in parallel with the topics that are being discussed. The patient appears sad while talking about unhappy events, angry while describing things that have irritated them, and so on. When the mood is not suited to the context, it is recorded as *incongruent*—for example, if the patient giggles when describing the death of their mother. Before concluding that such behaviour reflects an incongruous mood, consider whether it could be a sign of embarrassment or an effort to conceal distress.

Suicidal ideation

Some inexperienced interviewers are wary of asking about suicide due to fear that they may suggest it to the patient. There is no evidence to warrant this caution, and an explicit assessment of suicide risk should be part of every mental state examination. Nevertheless, questioning should be handled sensitively and in stages, starting with open questions such as ‘Have you ever thought that life is not worth living?’ and, if appropriate, going on to ask ‘Have you ever wished that you could die?’ or ‘Have you ever considered any way in which you might end your life?’, and then leading on to direct questioning about current suicidal intent or plans. The interviewer may also be concerned that they will not be able to cope if the patient does admit to being suicidal. A basic training in this topic, and knowledge of self-harm and suicide, should preclude these worries. Questions about suicide are considered further in Chapter 21.

Anxiety

Anxiety is assessed by asking about subjective feelings, by enquiring about the physical symptoms and cognitions associated with anxiety, and by observation. For example, the interviewer should start with a general question such as ‘Have you noticed any changes in your body when you feel upset?’ and then go on to enquire specifically about palpitations, dry mouth, sweating, trembling, and the various other symptoms of autonomic activity and muscle tension. Such features may also be apparent during the interview. To detect anxious thoughts, the interviewer can ask ‘What goes through your mind when you are feeling anxious?’ Possible

replies include thoughts of fainting or losing control. Many of these questions overlap with enquiries about the history of the disorder.

Depersonalization and derealization

These are usually symptoms of anxiety disorders, but occasionally they are diagnosed as a separate disorder. Their importance in the mental state examination is largely due to the fact that they are easily mistaken for psychotic symptoms and must be distinguished from them. Patients who have experienced depersonalization and derealization find them difficult to describe, and patients who have not experienced them may say that they have done so because they have misunderstood the questions. Try to obtain specific examples of the patient’s experiences. It is useful to begin by asking ‘Do you ever feel that things around you are unreal?’ and ‘Do you ever feel unreal or that part of your body is unreal?’ Patients with derealization often describe things in the environment as seeming artificial and lifeless. Patients with depersonalization may say that they feel detached, unable to feel emotion, or as if they are acting a part.

Thoughts

In this section, any predominant *content* of the person’s thoughts can first be noted. For example, there may be a preoccupation with persecutory themes, self-deprecating responses to questions, or a repeated return of the conversation to diet and body shape. This information may signify a delusional disorder, depression, and an eating disorder, respectively, and indicate areas for further questioning. However, the main purpose of this section is to determine the *nature* of the patient’s thoughts, and in particular to identify obsessions and delusions.

Thinking can also be abnormal in terms of the flow of thoughts, and the links from one topic to another, called *formal thought disorder*. This was introduced in Chapter 1.

Obsessions

Obsessions were defined in Chapter 1. An appropriate initial question to elicit them is ‘Do any thoughts keep coming into your mind, even though you try hard to stop them?’ If the patient says ‘Yes’, they should be asked for an example. Patients may be ashamed of obsessional thoughts, especially those about violence or sexual themes, so persistent but sympathetic questioning may be required. Before recording thoughts as obsessional, the interviewer should be certain that the patient accepts them as their own.

Compulsions

Many obsessional thoughts are accompanied by compulsive acts. Some of these can be observed directly (although rarely during the interview), but others are private events (e.g. repeating phrases silently), which are detected only because they interrupt the patient's conversation. Appropriate questions are 'Do you have to keep checking activities that you know you have completed?', 'Do you have to do things over and over again when most people would have done them only once?', and 'Do you have to repeat exactly the same action many times?' If the patient answers 'Yes' to any of these questions, the interviewer should ask for specific examples.

Delusions

A delusion cannot be asked about directly, because the patient does not recognize it as differing from other beliefs. Because of the difficulty that this poses for the interviewer, and the diagnostic significance of delusions, these were described in detail in Chapter 1.

The interviewer may be alerted to the possibility of delusions by information from other people or by events in the history. When searching for delusional ideas, it is useful to begin by asking what might be the reason for other symptoms or unpleasant experiences that the patient has described. For example, a patient who says that life is no longer worth living may be convinced that they are evil and deserve to die. Some patients hide delusions skilfully, and the interviewer needs to be alert to evasions, changes of topic, or other hints that information is being withheld. However, once the delusion has been uncovered, patients often elaborate on it without much prompting.

When ideas are revealed that could be delusional, the interviewer needs to determine whether they meet the criteria for a delusion using the approach described in Chapter 1. First, ascertain how strongly they are held. Achieving this without antagonizing the patient requires patience and tact. The patient should feel that they are having a fair hearing. If the interviewer expresses contrary opinions to test the strength of the patient's beliefs, their manner should be enquiring rather than argumentative. The next step is to decide whether the beliefs are culturally determined convictions rather than delusions. This judgement may be difficult if the patient comes from a culture or religious group whose attitudes and beliefs are not known to the interviewer. In such cases any doubt can usually be resolved by finding an informant from the same background, and asking this person whether others share the patient's beliefs.

Some types of delusion, which are characteristic of schizophrenia, present particular problems of recognition:

- *Delusions of thought broadcasting* must be distinguished from the belief that other people can infer a person's thoughts from their expression or behaviour. When eliciting such delusions an appropriate question is 'Do you ever feel that other people know what you are thinking, even though you have not spoken your thoughts aloud?' If the patient says 'Yes', the interviewer should ask how other people know this. (Some patients answer 'Yes' when they mean that others can infer their thoughts from their facial expression.)
- *Delusions of thought insertion.* A suitable question is 'Have you ever felt that some of the thoughts in your mind were not your own but were put there from outside?' A corresponding question about *delusions of thought withdrawal* is 'Do you ever feel that thoughts are being taken out of your head?' In each case, if the patient answers 'Yes', detailed examples should be sought.
- *Delusions of control* (passivity of thought) present similar difficulties. It is appropriate to ask, 'Do you ever feel that some outside force is trying to take control of you?' or 'Do you ever feel that your actions are controlled by some person or thing outside you?' Some patients misunderstand the question and answer 'Yes' when they mean that they have a religious or philosophical conviction that man is controlled by God or some other agency. Others think that the questions refer to the experience of being 'out of control' during extreme anxiety, while some say 'Yes' when in fact they have experienced auditory hallucinations commanding them to do things. Therefore positive answers should be followed by further questions to eliminate these possibilities.

Finally, the reader is reminded of the various categories of delusion described in Chapter 1, and be alert to the (rare) experiences of delusional perception and delusional mood. These issues only need to be addressed when there is already clear evidence of a psychosis, when they are useful in distinguishing schizophrenia from other psychotic disorders.

Perceptions

When asking about hallucinations, as with delusions, enquiries should be made tactfully in order to avoid distressing the patient, and to encourage them to elaborate on their experiences without being ridiculed. Questions can be introduced by saying 'Some people find that,

when their nerves are upset, they have unusual experiences.’ This can be followed by enquiries about hearing sounds or voices when no one else is within earshot. Whenever the history makes it relevant, corresponding questions should be asked about visual hallucinations and those in other modalities. Conversely, in assessments where there has been no previous evidence of psychosis at all, it may be appropriate to omit assessment of them altogether.

Auditory hallucinations

If the patient describes auditory hallucinations, certain further questions are required depending on the type of experience, because of their diagnostic significance. Has the patient heard a single voice or several? If there were several voices, did they appear to talk to the patient, or to each other about the patient in the third person? The latter experience must be distinguished from that of hearing actual people talking and believing that they are discussing the patient (an idea or delusion of reference). If the patient says that the voices are speaking to them, the interviewer should find out what the voices say and, if the words are experienced as commands, whether the patient feels that they must be obeyed. Note down examples of the words (and other sounds) spoken by hallucinatory voices.

Visual hallucinations

Visual hallucinations must be distinguished from visual illusions. Unless the hallucination is experienced at the time of the interview, this distinction may be difficult because it depends on the presence or absence of a visual stimulus which has been misinterpreted. Ascertaining whether there is an ‘as if’ quality to the image, or asking if it is seen ‘out there, or in your mind’s eye’ may aid the distinction. The interviewer should also distinguish hallucinations from dissociative experiences. The latter are described by the patient as the feeling of being in the presence of another person or a spirit with whom they can converse. Such experiences are reported by people with histrionic personality, although they are not confined to them, and are encouraged by some religious groups. They have little diagnostic significance.

Cognitive function

Early on in the interview, any significant cognitive difficulties will have already become apparent from the patient’s interactions with the interviewer and their responses to questions. If so, the assessment of cognitive function should be brought forwards, as the result may lead the interviewer to curtail the rest of the interview

or to postpone it until an informant is available. The Mini-Mental State Examination (MMSE; Folstein et al., 1975) and the Montreal Cognitive Assessment (MoCA; Nasreddine et al., 2005) are widely used cognitive screens in the elderly or where dementia or other organic disorder is suspected; see Chapter 14 for discussion of cognitive assessment in these contexts. Conversely, if the interview is nearing completion and no evidence or suspicion of any difficulties with memory or attention has arisen, cognition can be assessed very briefly.

Orientation

This is assessed by asking about the patient’s awareness of time, place, and person. Specific questions begin with the day, month, and year. When assessing the replies, remember that many healthy people do not know the exact date and that, understandably, patients in hospital may be uncertain about the day of the week or their precise location. If the patient cannot answer these basic questions correctly, they should be asked about their own identity; this is preserved except in severe dementia, dissociative disorders, or malingering.

Attention and concentration

While taking the history the interviewer should note the patient’s level of attention and concentration. In this way an opinion will already have been formed about these abilities before reaching the mental state examination. Formal tests add to this information, and can provide a semi-quantitative indication of changes between occasions. A useful first test is ‘*serial sevens*’: the patient is asked to subtract 7 from 100 and then to keep subtracting 7 from the remainder until the resulting number is less than 7. The time taken to do this is recorded, together with the number of errors. If poor performance seems to be due to lack of skill in arithmetic, the patient should be asked to state the months of the year in reverse order.

Memory

While taking the history the interviewer should compare the patient’s account of past events with those of any other informants, and be alert for gaps or inconsistencies. If memory is impaired, any evidence of *confabulation* should be noted. During the mental state examination, tests are given of short-term, recent, and remote memory. Since none of these is wholly satisfactory, the results should be assessed alongside other information about memory and, if there is any doubt, supplemented by standardized psychological tests. Objective evidence of memory impairment and its impact on normal activities (e.g. shopping, dressing) is also essential.

Short-term memory can be assessed by asking the patient to memorize a name and a simple address, to repeat it immediately (to make sure that it has been registered correctly), and to retain it. The interview continues on other topics for five minutes before recall is tested. A healthy person of average intelligence should make only minor errors. If recall is imperfect, memory can be prompted (e.g. by saying '35, Juniper ...' and the patient may then recall 'Street').

Memory for recent events can be assessed by asking about news items from the last few days.

Remote memory can be assessed by asking the patient to recall personal events or well-known items of news from former years. Personal items could be the birth dates of children or the names of grandchildren (provided these are known to the interviewer), and news items could be the names of well-known former political leaders. Awareness of the sequence of events is as important as the recall of individual items.

The reader is again referred to Chapter 14 for detailed assessment of cognitive functioning.

Insight

A note that merely records 'insight present' or 'no insight' is of little value. Instead the interviewer should enquire about the different aspects of insight discussed in Chapter 1. This includes the patient's appraisal of their difficulties and prospects, and whether they ascribe them to illness or to some other cause (e.g. persecution). If the patient recognizes that they are ill, do they think that the illness is physical or mental, and do they think that they need any treatment? If so, what are their views on medication, psychotherapy, or admission, as appropriate? The interviewer should also find out whether the patient thinks that stressful life experiences or their own actions have played a part in causing their illness. The patient's views on these matters are a guide to their likely collaboration with treatment.

Other components of psychiatric assessment

Although the psychiatric history and mental state examination are the main parts of the psychiatric assessment, several other elements may also be necessary. This section does not cover more specialized aspects of assessment (e.g. the use of rating scales; see 'Standardized assessment methods'), or those not directly linked to diagnosis or initial management (e.g. assessment for psychotherapy).

Physical examination

Physical examination provides three kinds of information:

1. It may reveal diagnostically useful signs (e.g. a goitre or absent tendon reflexes), and it is therefore particularly important in the diagnosis or exclusion of general medical conditions and organic disorders. Neurological and endocrine systems most commonly require detailed examination.
2. Psychotropic drugs may produce physical side effects, which need to be identified or measured (e.g. hypertension, parkinsonism, or a rash).
3. The patient's general health, nutritional status, and self-care may all be affected by psychiatric disorders.

For these reasons, a physical examination is an integral part of the psychiatric assessment. In practice,

however, its extent, and the medical responsibility for it, vary depending on the context. For example, in the United Kingdom, the general practitioner may have recently carried out an appropriate physical examination, and hence the latter is unnecessary for most patients seen in out-patient clinics or the community. In the case of inpatients, the psychiatrist is generally responsible for their physical as well as their mental health, and every newly admitted patient should have a full physical examination. Whatever the circumstances, the psychiatrist should decide what physical examination is relevant, and either carry it out themselves or ensure that this is or has been done by another doctor. Discussion with a neurologist or other physician is appropriate if the initial examination reveals equivocal or complex findings, or if a second opinion is sought.

The *neurological examination* merits specific mention, since there are many psychiatric presentations of neurological disorders, and some psychiatric disorders are accompanied by neurological signs (McLaughlin and Young, 2022). The examination may reveal localizing signs, signs of neurodevelopmental abnormality, or neurological side effects of medication (Benjamin and Lauterbach, 2018). As with physical examination in general, the neurological examination can range from a brief 'doorway' assessment to a highly detailed

Box 3.6 A basic neurological examination for psychiatrists

Domains

- Cranial nerves I–XII
- Motor system
 - Appearance
 - Tone
 - Power
 - Coordination
 - Reflexes
 - Gait
- Sensory system
 - Light touch
 - Pain
 - Temperature
 - Proprioception
 - Vibration
- Higher function
 - Consciousness and attention
 - Language
 - Perception
 - Memory
 - Executive function

Conducting the examination

- Observation
 - Attends with informant or alone
 - Head turning sign towards informant during history taking
 - Facies, posture
 - Abnormal involuntary movements (e.g. tremor, myoclonus)
 - Gait
 - Muscle wasting or fasciculation
- Listening
 - Dysphasia (fluent, non-fluent)
 - Dysarthria
- Formal examination
 - Applause sign
 - Imitation behaviour
 - Praxis (simple hand movements)
 - Limb tone (spasticity, rigidity)
 - Limb power (focal weakness)
 - Reflexes (tendon jerks, plantar responses)

See Lerner, A. J., Welch, K. A., & Carson, A. J. (2021). Neurological examination: What do psychiatrists need to know? *BJPsych Advances*, 27(5), 313–319 for explanation of these features and their significance.

procedure including neuropsychological testing. Box 3.6 summarizes the key domains and outlines a suggested minimum neurological examination for psychiatry (Lerner et al, 2021). For further information on conducting a neurological examination and a description of neurological symptoms and signs, see Kaufman et al. (2017) and Chapter 14.

Laboratory and other investigations

The need for laboratory and other investigations varies according to the nature of the differential diagnosis, the treatments that are being given, the patient's general health, and the resources available. At one extreme, no investigations may be necessary. At the other, extensive brain imaging, genetic testing, and biochemical screening may be needed—for example, if there is a strong suspicion of a treatable organic disorder or an early onset dementia. There is no one set of routine investigations applicable to every case although, by convention, routine blood tests (full blood count, renal, liver and thyroid function) are usually carried out on admission to hospital.

Investigations are discussed further in the chapters on individual syndromes and drug treatments.

Psychological assessment

Clinical psychologists and neuro psychological testing can contribute to psychiatric assessment in several ways. However, they are not required in most cases, and their availability is increasingly limited in many settings. We therefore only introduce the topic briefly here, illustrating the main forms and roles of psychological assessment.

Neuropsychological assessment

There are many psychometric tests available which measure different aspects of neuropsychological performance, ranging from overall intelligence to specific domains of memory, speed of processing, or tests that putatively assess functioning of a particular part of the brain (Lezak et al., 2012). Neuropsychological testing in psychiatry is primarily used in the following areas:

- In disorders of intellectual development, where IQ defines the severity of the condition;
- In dementia, where tests measure the severity and domains of cognitive impairment;

- If a decline in performance from premorbid abilities is suspected. In this instance, a discrepancy may be seen between verbal and performance IQ;
- To monitor the progress of neuropsychological deficits during the course of illness, by repeated administration of tests;
- To reveal deficits which may be subtle and neglected clinically, but which may be functionally important. For example, in schizophrenia there are persistent impairments in domains of memory and attention, and these predict poor outcome;
- If an organic disorder is suspected, the profile of test results may suggest the location of the lesion. However, this use of neuropsychological testing has largely been replaced by neuroimaging.

Cognitive and behavioural assessment

The term 'cognitive' is often used interchangeably with 'neuropsychological', or to refer to attention and memory. In the present context, however, the term refers to the assessment of a patient's cognitions (thoughts), assumptions, and patterns of thinking. It is used to determine the suitability for, and focus of, cognitive therapy.

Similarly, behavioural assessment is used to evaluate the behavioural components of a patient's disorder, especially as a prelude to behavioural therapy. For example, in a phobia, the elements of anticipatory anxiety, avoidance, and coping strategies, and their relationship to stimuli in the environment (e.g. heights), more general circumstances (e.g. crowded places), or internal cues (e.g. awareness of heart action).

Personality assessment

In the past, detailed personality testing, including the use of 'projective' tests such as the Rorschach test, was often part of psychiatric assessment. These tests are now obsolete, as they do not measure aspects of personality that are most relevant to psychiatric disorder, and they have not been shown to be valid predictors of diagnosis or outcome. Instead, personality is assessed descriptively as part of the history, supplemented in some settings with schedules for diagnosing personality disorders (see Chapter 15).

Risk assessment

Risk assessment is an essential part of psychiatric assessment. Risk in this context refers to risk of harm to others (through violence or neglect) and risk to self (through

suicide, deliberate self-harm, or neglect). A failure to carry out and document a risk assessment, and the resulting risk management plan, is a common criticism of enquiries that follow homicides and suicides involving psychiatric patients. All assessments, however brief, should include assessment of risk, and include a statement about the presence and magnitude of any acute risks in the record of the assessment.

Risk to self, and its assessment, is covered in Chapter 21. Here we consider assessment of the risk of violence to others. Three kinds of information are used to assess such risks—personal factors, factors related to illness, and factors in the mental state. These factors are summarized in Box 3.7, with the most important in each category indicated by means of an asterisk. Further information on risk assessment, in the context of forensic psychiatry, can be found in Chapter 18.

A history of violence is the best predictor of future violence. Therefore it is important to seek full information on this not only by questioning the patient but, in appropriate cases, from additional sources, including relatives and close acquaintances, previous medical and social services records, and in certain cases the police. Antisocial, impulsive, or irritable features in the personality are a further risk factor. Social circumstances at the time of any previous episodes of violence may reveal provoking factors, and should be compared with the patient's current situation. A parental history of violence, social isolation, and a recent life crisis also increase the risk. Among the illness factors, psychotic disorder and drug or alcohol misuse are important, and the combination of psychosis, substance misuse, and personality disorder is associated with the highest risk of violence.

The mental state factors in Box 3.7 require careful consideration. Thoughts of violence to others are very important, especially if they are concerned with a specific person to whom the patient has access. The entry concerning suicidal ideas refers to the occasional killing, usually of family members, by a patient with severe (usually psychotic) depression. Features of morbid jealousy and other delusional disorders may pose specific risks of harm against a perceived aggressor or rival (Chapter 12).

Situational factors are extremely important. Actual or perceived confrontational behaviour towards the patient by others may trigger violence, as may a return to situations in which violence has been expressed in the past. Enquiries should always be made about the availability of weapons.

Clinical experience and common sense have to be used to combine the risk factors into an overall

Box 3.7 Risk factors for harm to others

Personal factors

Previous violence*
 Antisocial, impulsive, or irritable personality traits
 Male and young
 Recent life crisis
 Poor social network
 Divorced or separated
 Unemployed
 Social instability
 Parent with history of violence

Illness-related factors

Psychotic symptoms*
 Comorbid substance abuse*
 Treatment resistant
 Poor compliance with treatment
 Stopped medication recently

Factors in the mental state

Planning of violence*
 Threats to people to whom patient has access*
 Irritability, hostility, anger
 Suspiciousness
 Thoughts of violence towards others
 Persecutory delusions
 Delusions of jealousy
 Delusions of influence
 Hallucinations commanding violence to others
 Suicidal ideas with severe depression
 Clouding of consciousness
 Lack of insight about illness

Situational factors

Ready availability of weapons*
 Confrontation and provocation by others
 Situations associated with previous violence

* Denotes the most important factors in each category.

assessment. Risk assessment schedules have been developed, but they complement and do not replace thorough and repeated clinical assessment. The assessment of risk should be shared among the members of the team treating the patient, or if the patient is in individual treatment it may need to be discussed with a colleague. In certain circumstances the assessment may need to be made known to an individual at risk. Risk assessments should be reviewed regularly, combining information from the members of the clinical team.

Although risk assessment is essential, and can be of value in a number of respects (Abderhalden et al., 2008), it is also important to put it in context. First, the vast majority of people with psychiatric disorders pose no risk to others, and are more likely to be the victims than the perpetrators of violence. The assumption that all patients with severe mental illness, especially schizophrenia, are potentially violent is unwarranted and contributes to stigma. Second, as the physicist Niels Bohr remarked, 'Prediction is very difficult, especially about the future.' Even a complete risk assessment provides only a weak guide to future harm to others. Many such acts are carried out by people with no past history of violence, and many of those who have multiple risk factors will never commit further acts of violence.

Assessment of needs

For patients with severe or enduring mental illness, particularly psychosis, it is important to focus on their needs in the broadest sense (e.g. physical health, hygiene, social isolation, domestic skills). The concern arises from findings that such needs are often substantial, and neither well recognized nor met. Although the conceptual status of needs is not as straightforward as is often implied, interest in its measurement is well established. The Camberwell Assessment of Need scale is widely used. This rates 22 domains as absent, present, or modified, and has versions for different age and patient groups (Slade and Thornicroft, 2020).

Special kinds of psychiatric assessment

So far, this chapter has been concerned with the complete assessment as carried out by a psychiatrist with a patient being seen in a psychiatric setting, and for whom sufficient time is available. However, most assessments do not meet all of these criteria. Many are

conducted by non-specialists in non-psychiatric settings, with limited time and imperfect surroundings. Nevertheless, a modified, well-focused interview can often yield a reliable diagnosis and plan of action in a surprisingly short time.

Interviews in an emergency

In an emergency, the interview has to be brief, focused on the key issues, and effective in leading to a provisional diagnosis and a plan of immediate action. These assessments generally involve acutely distressed or disturbed patients, and often take place in difficult settings, such as a police station. The diagnoses that are usually in question are the psychoses (schizophrenia, mania, drug-induced), and organic brain disorders such as delirium. Throughout, the interviewer should consider which questions are essential and which can be deferred. The most important issues are those that impact on immediate management decisions. The latter are likely to include whether the patient should be admitted or detained, whether laboratory investigations are indicated, whether medication should be given, and whether acute medical treatment is also required.

Because the assessment may be limited by the patient's ability or willingness to participate fully, seek out all available information first. Ask whoever is accompanying the person what they know about the patient and the recent events. Their belongings may reveal evidence of prior illness or medication. If the patient has a past psychiatric history, strenuous efforts should be made to access their records or to contact a professional involved in their care. The safety of the patient and those around them must also be actively considered when planning the nature and location of the interview.

An emergency assessment should, wherever possible, include the following core information:

- The presenting problem in terms of symptoms or behaviours, together with their onset, course, and present severity;
- Other relevant symptoms, together with their onset, course, and severity;
- History of psychiatric or medical disorder;
- Current medication;
- Use of alcohol and drugs;
- Recent stressful circumstances and possible precipitants;
- Social circumstances, and the possibilities of support; and
- Risk assessment, including the immediate risks of harm to self and others.

Interviews in general practice

In general practice, many presentations are with psychiatric disorders, notably depression and anxiety disorders,

as well as substance misuse and somatoform disorders. Such cases commonly present with physical complaints (e.g. chronic pain, fatigue), and the doctor needs to be aware of the possibility that the symptoms reflect an underlying psychiatric disorder (Chapter 22). The interplay of physical and psychological factors is emphasized by the finding that general practitioners who diagnose psychiatric disorder accurately have a good knowledge of general medicine (Goldberg and Huxley, 1980).

Whether the presentation is physical or psychological, it is a challenge to undertake an effective psychiatric assessment, given the very limited time that is usually available. There are two components to achieving this goal. First, the chances of detecting psychiatric disorder can be increased by attending to the way that the interview is conducted—for example, by being alert to cues in the patient's history, appearance, and behaviour. Secondly, screening questions can be used to detect the common disorders, and identify areas that may require more detailed assessment (see Table 3.1).

Goldberg and Huxley (1980) described how the assessment of a patient in general practice whose complaint may have a psychological cause should cover four areas:

1. *General psychological adjustment:* fatigue, irritability, poor concentration, and the feeling of being under stress.

Table 3.1 Brief screening questions for psychiatric disorders

During the past month	Screening for
1. Have you felt low in spirits?	Depression
2. Do you enjoy things less than you usually do?	Depression
3. Have you been feeling generally anxious?	Anxiety
4. Are you worried about your health or other specific things?	Health anxiety
5. Has your eating felt out of control?	Eating disorder
6. Do you drink alcohol? If so, ask the FAST or CAGE questions.*	Alcohol problem
7. Present three items and ask patient to recall them after 2 minutes.	Dementia/delirium

* FAST and CAGE each consist of four questions designed to detect alcohol misuse (see Chapter 20).

2. *Anxiety and worries*: physical symptoms of anxiety and tension, phobias, and persistent worrying thoughts.
3. *Symptoms of depression*: persistent low mood, tearfulness, crying, hopelessness, self-blame, thoughts that life is unbearable, ideas about suicide, early-morning waking, diurnal mood variation, weight loss, and loss of libido.
4. *The psychological context (i.e. the patient's personality and circumstances)*: although often known to the general practitioner from previous contacts, this should be reviewed and brought up to date.

Subsequently, more formalized sets of screening questions for psychiatric disorders were developed, such as those in Table 3.1, which are suitable for brief interviews in primary care, and which focus on the disorders that are commonly encountered in this setting. Further screening questionnaires are described later in this chapter. If the interview is conducted with psychological issues in mind, and screening questions are used appropriately, then it should be possible to identify many psychiatric disorders in the 10–15 minutes that are available in primary care. A preliminary plan can be made, and a further appointment arranged in order to evaluate the diagnosis and discuss management in more detail.

Interviews in a general hospital

Similar considerations apply to interviews that are conducted in a general medical setting, such as an Accident and Emergency department or an inpatient ward (Segal and Ranjith, 2016). In these situations, particular attention again needs to be given to physical symptoms and the possibility of somatoform disorder, and to the assessment of suicidal risk. Delirium and other organic disorders are also common reasons for a psychiatric consultation in a general hospital, and hence a focused assessment of cognitive function is often required.

Interviews in the patient's home

Many services assess a proportion of patients at home, particularly in the case of severely ill and disorganized patients who would otherwise fail to attend for assessment. Such a visit often throws new light on the patient's home life and gives a more realistic evaluation of the relationship between family members. Home visits are especially important in the assessment of elderly patients for whom psychiatric and physical difficulties frequently coexist. Before arranging a visit, the psychiatrist should, if possible, talk to the general practitioner, who often has first-hand knowledge of the patient and their family. The safety of the interviewer should always be considered before embarking on a domiciliary visit. The interviewer should ensure that another member of the team is aware of the location and time of the visit. If there is any concern about potential risk, a joint assessment should be made.

Interviews with the family

In child and adolescent psychiatry an interview is usually conducted with several family members together to find out their attitudes to the patient and the illness, and the nature of any conflicts within the family (Chapter 16). The interviewer should remember that the family has usually tried to help the patient, and may be feeling demoralized, frustrated, or guilty.

The interviewer should ask the following questions:

- How has the illness affected family life, and how have the family tried to cope?
- How do different members of the family perceive the illness and its origins?
- Are the family members willing to try new ways of helping the patient?

Family interviewing in the context of family therapy is covered in Chapter 24.

Patient characteristics that may affect the interview

Interviews can prove difficult for a variety of reasons. Many of these reflect the situation (e.g. noisy surroundings, lack of time) or interviewer characteristics (e.g. inexperience, tiredness). However, other problems arise from the characteristics of the

patient, and these are outlined here. Remember that such problems can be diagnostically useful—for example, a patient may be monosyllabic because of depression, and a disturbed patient may have delirium.

Anxious patients

Although anxiety may be part of the patient's disorder, it may also relate to the interview. If the person seems unduly anxious, the interviewer can say that many people feel anxious when they first take part in a psychiatric interview, and then go on to explore the patient's concerns.

Taciturn patients

Taciturn patients can be encouraged to speak more freely if the interviewer shows non-verbal expressions of concern (e.g. leaning forwards a little in the chair with an expression of interest), in addition to verbal expressions that are part of any good interview.

Garrulous patients

It is not easy to curb an over-talkative patient. If efforts to focus the interview are unsuccessful, the interviewer should wait for a natural break in the flow of speech and then explain that, because time is limited, they will need to interrupt from time to time to help the patient to keep focused on the issues that are most important.

Confused patients

When the patient gives a history in a muddled way, or appears perplexed, the interviewer should test cognitive functions early in the interview. If there is evidence of impaired cognition or consciousness, the interviewer should try to orientate and reassure the patient, before starting the interview again in a simplified form. In such cases every effort should be made to interview an informant.

Uncooperative or 'difficult' patients

Some patients are reluctant to be interviewed and have come at the insistence of a third party (e.g. their partner). If the patient seems unwilling to collaborate, the interviewer should talk over the circumstances of the referral, and try to persuade them that the interview will be in their own interests. Remember that lack of cooperation may occur because the patient does not realize that they are ill (e.g. some patients with delirium, or with psychosis). In such cases it may be necessary to interview an informant before returning to the patient.

Some patients try to dominate the interview, while others adopt an unduly friendly approach that threatens to convert the interview into a social conversation. In either case, the interviewer should explain why they need to guide the patient towards relevant issues. In the longer term, developing a manner which is confident and assertive, but not domineering, helps to avoid such problems.

Unresponsive patients

If the patient is mute or stuporous, it is essential to interview an informant who can describe the onset and course of the condition. With regard to the mental state, the interviewer can only observe behaviour, but this can be informative. Because some stuporous patients change rapidly from inactivity to violent overactivity, it is wise to have help available when seeing such a patient.

Before deciding that a patient is mute, the interviewer should allow adequate time for reply, and try several topics. If the patient still fails to respond, attempt to persuade them to communicate in writing. Apart from observations of behaviour, the interviewer should note whether the patient's eyes are open or closed. If they are open, note whether they follow objects, move apparently without purpose, or are fixed. If the eyes are closed, find out whether the patient opens them on request, and if not, whether attempts at opening them are resisted. A physical examination, including neurological assessment, is essential, including for signs of catatonia.

Patients with disorders of intellectual development

When interviewing people with disorders of intellectual development (learning disability), certain points should receive particular attention. Questions should be brief and worded in a simple way. It may be difficult to avoid closed questions, but if they are used the answers should be checked. For example, if the question 'Are you sad?' is answered 'Yes', the question 'Are you happy?' should not be answered in the same way. People with learning disability may have difficulty in timing the onset and course of symptoms and to obtain this and other information it is important to interview an informant.

For further discussion of the assessment of patients with disorders of intellectual development, see Chapter 17.

Patients from another culture

The interviewer may be unfamiliar with the patient's culture and heritage, their degree of acculturation, their cultural identity, or their health beliefs. The clinician may also need to accommodate various levels of English language literacy and appropriately skilled interpreters, as well as the patient's health literacy more generally, and familiarity with local health systems. Without understanding these sources of cultural context (including relevant influence of religious, race, tribe, caste, and migration history), the clinician may find it difficult to ascertain the significance of the patient's expressions, behaviours, and appearance, and the extent to which these indicate 'psychopathology' as signs of mental illness rather than culturally sanctioned expressions of distress.

Certain factors may have different implications for a patient from another culture to those that they would have in the interviewer's culture. The stigma of mental illness may be greater, leading to the patient being guarded and less willing to share experiences of distress for fear of judgement or that others might discover their need for mental health care. Priorities within the family may be different, with the wellbeing of the family outweighing that of its individual members, and decision making and consent being more sociocentric and shared rather than being based on individual consent and responsibility. Emotional disorder may be experienced and expressed more readily in the form of physical complaints and symptoms, or indeed in interpersonal, social, or religious terms. Expectations and fears about treatment may be based on knowledge and experience of health services in the country of origin. Ideas about causation may differ, an extreme example being that distress may be ascribed to the actions of evil spirits. It may be particularly difficult to decide whether strongly held ideas are delusional or normal within the culture or subculture.

Overall, the clinician needs time to explore these possibilities, with reference to people in the patient's social and cultural and religious networks, as well as using knowledge hubs such as those offered via charities or specialist departments available in some clinical services. ICD-11 and DSM-5 offer some guidance in the form of a cultural formulation and ensuring diagnoses that are globally relevant and useful (Gureje et al., 2019). Finally, there is also more awareness that cultural concepts are contested, and that culture exists for everyone from all ethnic and racial backgrounds, so exploring cultural issues in non-migrants, including the White British population, can also be fruitful.

Patient–interviewer language differences

If the patient and interviewer do not speak the same language, it will need to be decided who the best choice of interpreter is. There are advantages to a family member interpreting, as they are often trusted by the patient and can potentially help the interviewer understand some of the complexities of a different cultural context. However, there are disadvantages, including incorrect translations and possible reluctance of the interviewee to share sensitive information with their relative. If an official interpreter is used, it is valuable if they are also a healthcare professional as they can then assist in the assessment itself.

The presence of a third person, and the process of translation, affect the interview and lengthen it considerably. It may therefore be necessary to focus on the essential topics in a first assessment, and complete the process in a second interview.

Children and the elderly

Psychiatric assessment of children and adolescents is described in Chapter 16. Assessment in elderly patients is described in Chapter 19.

Integrating and evaluating the information

So far this account of assessment has been largely about data gathering. The following section explains how the facts that emerge are evaluated and integrated with knowledge of psychiatry, in order to arrive at a diagnosis, provide prognostic information, and determine

treatment decisions. We emphasize again that, in practice, information is evaluated as it is collected, with hypotheses being tested as they arise during the assessment. Clearly, this skill takes time to learn and also requires a solid foundation of psychiatric knowledge.

Drawing conclusions and making decisions

The areas in which an opinion must be formed, or a decision made, are listed in Box 3.8. It may also be useful to think in terms of a set of rhetorical questions that need to be answered:

- What is the diagnosis?
- What are the effects on the patient's life?
- Are there immediate risks that need to be managed?
- What are the patient's current circumstances?
- Does the patient have any dependents, or carers?
- Why has the disorder occurred?
- What treatment is indicated?
- What is the prognosis?
- What other information is needed to answer these questions?
- What does the patient need and want to know?

What is the diagnosis?

The first step is to make a diagnosis using information about the symptoms and signs that has been obtained from the history and mental state examination and, in

relevant cases, from the physical examination and any investigations that have been performed. This information is then used to make a diagnosis based upon knowledge of psychiatric classification and the criteria for each diagnostic category. Sometimes a diagnosis is provisional, or several diagnostic possibilities entertained, until further information becomes available. The diagnosis of specific psychiatric disorders is discussed in subsequent chapters; here we are concerned with the general approach to assessment. Diagnosis is usually accompanied by an assessment of the severity of the disorder in order to determine whether it is mild, moderate, or severe. Remember that the outcome of some assessments is that the patient does not have evidence of any psychiatric disorder.

What are the effects on the patient's life?

When patients describe their problems, they will include both symptoms and other matters. A key purpose of the assessment is to identify and characterize the psychopathology, since this underlies the diagnosis. However, it is also necessary to enquire about the effects that the patient's symptoms are having on their life, in part since evidence of impaired functioning is relevant to, or an essential part of, most diagnoses. It is therefore helpful to ascertain the patient's usual level of functioning and how far the current state differs from it.

Are there immediate risks that need to be managed?

As we have discussed, risk assessment is a core part of the psychiatric assessment. Once any risks have been identified, they need to be managed. For example, if the patient is at risk of self-neglect or self-harm, this will influence whether compulsory admission is necessary. If a risk of harm to another party has been identified, that individual may need to be warned.

What are the patient's current circumstances?

Knowledge of the patient's accommodation, finances, interests, values, and relationships may influence both management and prognosis. For example, a homeless person will require a different care package from a patient with the same disorder who has a stable home and a carer. The current circumstances may also act as maintaining factors for the disorder, and therefore be a target for intervention.

If the patient has children or other dependents, their welfare must be considered. If necessary, further information should be obtained, and ensure

Box 3.8 Topics to be evaluated in a psychiatric assessment

The patient's problem and its consequences

- Diagnosis
- Functional and social impacts
- Risk to self and others
- Effects on others

The patient and their circumstances

- Personal history
- Current circumstances
- Personality

Aetiology

The response to the patient's problem

- Treatment
- Prognosis

The patient's understanding of the above

that the dependents are included in the management plan being developed. This is both clinically appropriate and, in the United Kingdom, is a safeguarding requirement.

Why has the disorder occurred?

Aetiology is discussed in Chapter 5. Here we are concerned with how aetiological factors are applied to the individual patient. A useful approach is a chronological one, with causes being divided into those that are *predisposing*, *precipitating*, and *perpetuating*. Predisposing factors may be genetic or related to temperament, or to adverse childhood experiences. Precipitating factors are often stressful life events. Perpetuating (maintaining) factors may be continuing stressors, or related to the way that the patient attempts to cope with stressors (e.g. avoiding anxiety-provoking situations, misusing alcohol). Perpetuating factors are highly relevant to treatment. Intersecting this chronological approach to causation, one can divide the causal factors into *biological*, *psychological*, and *social*.

A comment about personality is appropriate here. Clear appreciation of a patient's personality is part of the full understanding of their life history and their psychiatric disorder. For this reason we emphasize the collection of sufficient and reliable information about personality during the assessment. As well as aiding the aetiological formulation, this knowledge is useful when planning management and predicting outcome. For example, comorbid personality disorder worsens the prognosis of many conditions, and may also influence decisions about psychological treatment.

What treatment is indicated?

A key decision to be made is what, if any, treatment the patient requires. If treatment is indicated, the options should be discussed with the patient, and the evidence for each possible treatment presented in terms of efficacy, side effects, etc. These issues are discussed in later chapters, especially Chapters 24 and 25.

What is the prognosis?

The prognosis depends primarily on the disorder concerned. It also depends on the individual characteristics of the patient (e.g. their age, severity of symptoms, comorbid conditions). Not all of this information may be available after an initial assessment. However, it is usually possible to make a cautious prediction about the short-term outcome at this stage, and most patients will expect and appreciate this. The patient can be told that more accurate and longer-term prognostic judgements will be possible as further information is obtained, and

the course of the disorder is followed over subsequent weeks and months.

What other information is needed to answer these questions?

For many patients, assessment is not complete after one interview. Assessment should be viewed as an iterative process in which opinions and conclusions undergo continuing review (both within the initial interview and thereafter), as further information about the patient and their illness is obtained. For descriptive purposes, this chapter focuses on the diagnostic purpose of assessment, but in practice there is no firm distinction between this and the other goals of assessment (e.g. to establish a therapeutic relationship, assess risk, or monitor response to treatment). Each interaction between the psychiatrist and the patient contains a mixture of assessment, evaluation, review, and decision making. Assessment, and its revision, is a process that continues throughout treatment and follow-up.

What does the patient need and want to know?

The guiding principle is that, as a rule, a patient (and their carer) should be given as much information as possible about the diagnosis, and the likely prognosis and presumed causation, and should be fully involved in, and aware of, all decisions that affect them. The practical implications are introduced below, and are discussed in later chapters with regard to specific disorders and treatments.

Other issues that affect how the information is evaluated

Disease and illness

These concepts were introduced in Chapter 2, but a brief mention is required here. *Illness* refers to a patient's experience, and *disease* refers to the pathological cause of this experience. Patients can have a disease without feeling ill, or feel ill without having an identifiable disease. In general medicine, patients tell doctors about their experience of illness, and doctors seek to discover the disease that is causing the illness, as this will guide treatment. Patients and their relatives also want to find a cause and a treatment for their illness, but they do not always understand how a medical diagnosis helps in finding these. Psychiatrists also search for disease as a cause of illness, but should always look for other causes, too. The patient's experience can be understood sometimes as an extreme variation from the norm, sometimes

as a reaction to circumstances, and sometimes as a combination of both. Other mental health professionals may emphasize one or other of these factors to a greater extent, depending on their background. For example, psychologists focus on variation from the norm, whereas social workers focus on the role of reaction to circumstances and the place of the patient in their family and social context. These differences in emphasis can lead to apparent disagreements during multidisciplinary assessment. It is important that psychiatrists are aware of this possibility, so that they can help to integrate the social, psychological, and medical approaches and thereby use the various skills of the multidisciplinary team to best help the patient.

Evaluations by experienced psychiatrists

Throughout this chapter, a systematic and logical approach to assessment has been advocated, in which information is collected carefully, and eliciting

of symptoms and signs forms the basis of diagnosis. Anyone new to psychiatry should follow this approach closely until they have mastered the process and have a good grounding of knowledge. However, studies show that experienced psychiatrists actually carry out assessments rather differently. They make rapid diagnoses, often within the first few minutes of the interview, that presumably reflect the predictive power of the patient's initial appearance, behaviour, and utterances (Kendell, 1975). The psychiatrist may not realize the cues and clues that they are using to form these diagnostic judgements. Schwartz and Wiggins (1987) called this process *typification*. The rest of the interview functions primarily to confirm and refine this diagnostic opinion. Regardless of their seniority and experience, the interviewer should always ensure that they have not come to premature conclusions about a case, causing them to fail to gather relevant information, to interpret findings in a biased way, or to disregard contrary evidence.

Recording and communicating information

Having completed the assessment, it is necessary to record and communicate your understanding of the patient, their disorder, and its management. These records and communications take a variety of forms (e.g. written, verbal, electronic), and differ according to their source, purpose, and intended recipients. As with assessment itself, it is helpful to learn the 'traditional' and relatively detailed formats for communicating information (e.g. case summaries and formulations), while also appreciating that in practice many other—and briefer—modes of communication are the norm. Regardless of format, all communications should follow certain basic principles.

- Information should be presented clearly and concisely. Include important negatives (e.g. 'not suicidal'). Avoid repetition of information that is already known to the recipient. Use subheadings to highlight key points (e.g. 'Diagnosis: ...', 'Current medication ...', 'Acute risks: ...').
- In many countries, patients are entitled to read what is written about them. This information may also be used for legal purposes. In some countries, including the United Kingdom, it is now expected that patients are copied into all correspondence between health professionals; alternatively, the patient is the primary recipient and the letter copied to the other health

professional. This principle applies equally to correspondence from psychiatrists, unless there is a compelling reason not to do so or the patient declines. Ensure that all information is accurate, and that any opinions or inferences you make are reasonable, and avoid unwarranted or unnecessarily personal comments. It is sometimes better to communicate verbally with the other health professional(s) in order to expand on some details of the case.

- All communications must be kept confidential.

Here we consider a range of ways in which information is recorded and shared—within the psychiatric team, and between the psychiatrist and other doctors, but first, between the psychiatrist and the patient.

Discussing the diagnosis and management plan with the patient

When discussing the conclusions from your assessment with the patient, and with their relatives, it may be useful to begin by briefly summarizing the key points for them. This helps to ensure that you have understood the history correctly and have not omitted anything which in their opinion is significant. This process is also helpful in that it demonstrates your engagement and empathy

with the patient, and sets the scene for discussion about the diagnosis and how you propose to proceed.

When introducing the diagnosis, do not use medical terms without explaining them. The patient may misunderstand their meaning or make incorrect assumptions about their implications. Explain the significance of the diagnosis in terms of cause, prognosis, and treatment options—and any diagnostic uncertainty that remains.

When discussing the proposed plan of management, it is useful to begin by asking what treatment the patient has been expecting, and whether they have strong views—for example, with regard to the use of medication. Whether drug treatment or psychological treatment is planned, there are issues to raise and discuss (see Chapters 24 and 25). It is important to set aside enough time for this explanation during the interview, as it is likely to improve concordance with the treatment plan. If, after they have been given a full explanation, the patient refuses to accept part of the plan, the interviewer should document this, and try to negotiate an acceptable alternative.

Box 3.9 summarizes the points that should be considered when communicating with patients and their relatives. In the National Health Service (NHS) in the UK, this information will usually be recorded on a pro forma as part of the Care Programme Approach (CPA) for any patients who require either prolonged or complex secondary care (see Chapter 26).

The importance of case notes and clinical records

Good case records, whatever their format, are important in every branch of medicine. In psychiatry they are vital because a large amount of information is collected from a variety of sources. It is important to summarize the information in a way that allows essential points to be grasped readily by someone new to the case. Case notes are not just an *aide mémoire* for the writer—they are important for others concerned with the patient. They are also of medico-legal importance. If a psychiatrist is called upon to justify their actions after a serious untoward incident, after a complaint has been made, or in court, they will be greatly assisted by notes that are comprehensive but concise and well organized. Indeed, imagining being in a situation of this kind may help the psychiatrist to judge what constitutes appropriate record keeping, not just for case notes but for other forms of communication.

Box 3.9 Communicating with patients and relatives

The diagnosis

- What is the diagnosis? If uncertain, what are the main possibilities?
- What further information or investigations are required?
- What are the implications of the diagnosis?
- What may have caused the condition?

The care plan

- What is the plan, and how will it help?
- Are any legal powers to be used? If so, why, and what is their effect?
- What needs to be communicated about medication or psychological treatments?

Who does what?

- What is the role of the psychiatrist, and how often will they see the patient?
- Who is the key worker and what will they do?
- What is the role of other members of the psychiatric team?
- What is the role of the general practitioner?
- What can the family (or other carers) do to help?

Emergencies

- What are the likely acute risks, and how might they be avoided?
- Are there possible early warning signs?
- Who should be approached in an emergency, and how can they be contacted?

Inpatient notes

Admission notes

When a patient is admitted to hospital urgently, the doctor often has only limited time available. The admission note should contain at least the following:

- The reasons for admission;
- Any information that is required for a decision about provisional diagnosis and immediate treatment, which may include a physical examination and laboratory investigations; and

- Any other relevant information that will not be available later, such as details of the mental state on admission (and prior to medication), and information from any informant who may not be available again.

If there is time, a systematic history can be added. However, inexperienced interviewers sometimes spend too much time on details that are not essential to the immediate decisions, while failing to record details of the mental state that may be transitory and yet of great importance in diagnosis. The rest of the assessment can always be completed over the next few days. The admission note should end with a brief plan of immediate management, which has been agreed with the senior nurses caring for the patient at the time.

Continuation (progress) notes

Continuation notes for both inpatients and patients in the community need to be succinct, clear, and regularly updated, and to contain specific information if they are to be of value. Instead of recording merely that the patient feels better or is behaving more normally, record in what ways they feel better (e.g. less preoccupied with thoughts of suicide). The notes should also record treatments, including both medication and other interventions, accompanied by a brief explanation of their rationale, and the temporal relationship between a change in treatment (e.g. dose of drug) and change in clinical state. A note should be made of any information or advice given by the doctor to the patient or their relatives.

Observations of progress are made not only by psychiatrists but sometimes by nurses, occupational therapists, clinical psychologists, and social workers. These professionals may also keep separate notes, but it is desirable that important items of information are written in the medical record. A note should also be kept of formal discussions between members of the team.

Discharge notes

It is important to set out clearly the plans made for the patient's further care on discharge from hospital, including the identity of the key worker, and how the patient or their carer can contact the team in an emergency. These arrangements are important, in part because it is recognized that the days after discharge from inpatient care are a particularly high-risk period for suicide. A phone call to the patient's general practitioner is desirable if there are any acute concerns about the patient's welfare, or if there is any delay anticipated in production

of the discharge letter or case summary. A statement of the diagnosis at the time of discharge is also important, since it may have changed since admission.

Case summaries

Case summaries are generally used for inpatients, but they are valuable for all patients with complex, long-standing, or recurrent disorders. Summaries should be brief but comprehensive, written in telegraphic style using a standard format, to help other people to find particular items of information. A completed summary should seldom need to be any longer than two typewritten pages. An example is given in Box 3.10. The completed summary will provide a valuable record should the patient become ill again, especially if they are under the care of a different team.

Some of the items in the summary call for comment.

- The reason for referral should state the problem rather than anticipate the diagnosis—for example, 'found wandering in an agitated state, shouting about the devil', rather than 'schizophrenia.'
- The description of personality is important. The writer should strive to find words and phrases that characterize the person in a non-judgemental way.
- Unless an abnormality has been found, the results of the physical examination and investigations can be summarized briefly.
- When the mental state is recorded, a comment should be made under each heading as to whether or not any abnormality has been found.
- Diagnoses should be made using ICD-11 or DSM-5 criteria. If the diagnosis is uncertain, alternatives can be listed, with an indication of which is judged to be more likely.
- The summary of treatment should include the dosage and duration of any medication. The roles of the mental health team and of the general practitioner and any other agencies in the treatment plan should be made clear.
- The prognosis should be stated briefly but as definitely as possible. Statements such as 'prognosis guarded' are of little help to anyone.

Formulations

A formulation is an exercise in clinical reasoning that helps the writer to think clearly about the diagnosis,

Box 3.10 Example of a case summary

Patient: Ms AB (she/her). *Date of birth:* 7 February 1991

Consultant: Dr C. Summary compiled by Dr D. (CT2 core trainee).

Admitted: 27 June 2025 *Discharged:* 12 August 2025

Reason for referral: Attempted suicide by hanging. Continuing suicidal intent and low mood.

Family history: Father, 66, retired gardener, well, mood swings, not close to patient. Mother died from cancer when patient aged 4. Sister, 37, divorced, well. Mental illness: father's brother—'manic depression'; father's sister—depression.

Personal history and current circumstances: Birth and early development normal. Childhood health good. School 6–16, uneventful, 5 GCSEs. Few friends. Worked 16–22, shop assistant. Several boyfriends; married at 22, husband 2 years older, lorry driver. Unhappy in last year after discovering his infidelity. Children: Jane, 7, well; Paul, 4, epileptic. Current financial worries.

Previous illness: Aged 31, (postnatal) depressive illness lasting a few weeks, saw counsellor. Hypothyroidism, on thyroxine.

Previous personality: Cyclothymic, but usually anxious and with low self-esteem. Interested in supernatural phenomena. Drinks occasionally, non-smoker, no illicit drugs.

History of present illness: For 3 months, increasingly low and tearful, waking early, worse in morning. Anhedonic, loss of appetite and weight. Worrying about marriage, and has become preoccupied that she is responsible for her son's epilepsy. Strong suicidal ideation, culminating in abortive attempt to hang herself on night of admission. Getting worse despite venlafaxine 75–150mg/d for 2 months, taken regularly. On waiting list for CBT.

On examination: Physical: n.a.d., no evidence of myxoedema. Mental state: dishevelled, agitated, psychomotor retardation. Speech: slow, quiet, normal form. Thoughts: the preoccupations noted above are held intensely, and the belief that she caused her son's epilepsy (because she did not take enough care in pregnancy) is delusional in nature. Prominent suicidal ideation, and intent, wishing she were dead. Mood: subjectively and

objectively very depressed, with self-blame, hopelessness. Perceptions: imagines she sees and hears her dead grandmother talking to her 'as if she was still alive'. Cognition: attention and concentration poor. Memory not formally tested. Insight: accepts she might be ill, consents to admission; her views on treatment are unclear.

Special investigations: Haemoglobin, electrolytes, TSH, n.a.d.

Treatment and progress: Initially on close observations. Venlafaxine increased to 225 mg per day, augmented by olanzapine, 10mg nocte. However, condition deteriorated, and given course of 6 right unilateral ECTs (with consent). Mood improved after several treatments, suicidal ideation reduced, and no longer delusional.

Condition on discharge: Moderate residual depressive symptoms. No suicidal thoughts and realizes her son's epilepsy was not her fault. Hopeful but worried about future of marriage. Childcare—no concerns, husband and patient's sister (lives nearby) are supportive.

Diagnosis: (ICD-11) 6A71.4 Recurrent depressive disorder, current episode severe, with psychotic symptoms. Now resolving.

Prognosis: Short-term prognosis is good, but at risk of further depressive episodes. Prognosis will be improved by compliance with treatment, and by resolving marital problems

Further management:

- Under CPA. Key worker is Jo Smith, community psychiatric nurse, who will visit on 17 August 2025.
- Continue venlafaxine 225 mg daily and olanzapine 10mg nocte. GP to prescribe, and to monitor for metabolic syndrome (due in 1 month). Plan to remain on antidepressant for at least 9 months; olanzapine to be reviewed after 3 months.
- On waiting list for CBT—expected to start in a few weeks.
- Patient and husband will arrange marriage counselling.
- Patient has a copy of her care plan, and she and her husband know who to contact in an emergency.
- First review with psychiatrist in 2 weeks.

aetiology, treatment, and prognosis (Box 3.11). Although now uncommon in clinical practice, formulations are extremely useful for developing skills in diagnostic reasoning, for understanding how psychiatric knowledge can

be applied to the individual patient, and for learning how to prioritize and focus on the problems that require attention. We therefore encourage psychiatrists-in-training to use formulations as an integral part of their work. Even

Box 3.11 The formulation

Statement of the problem
Differential diagnosis
Aetiology
Plan of treatment
Prognosis

for an experienced psychiatrist, a formulation remains a valuable exercise.

A formulation is divided into several sections, as in the example shown in Box 3.12. A short opening

statement is followed by the differential diagnosis. This consists of a list of reasonable possibilities in decreasing order of probability, together with a note of the evidence for and against each alternative. To produce a good differential diagnosis, it is necessary to have elicited and interpreted the key symptoms and signs, and to know the cardinal features of the major psychiatric disorders and how they are classified. Under aetiology, predisposing, precipitating, and perpetuating causes should be identified; this requires a grounding in the causation of psychiatric disorders. There follows a list of outstanding problems and any further information that is needed. Next a concise plan of management is outlined, and finally there is a statement about prognosis.

Box 3.12 Example of a formulation

(Note: This formulation refers to the case summarized in Box 3.10, at the time of admission. By comparing the two ways of condensing information, the reader can appreciate the difference between the two approaches.)

Opening statement

Ms AB is a 34-year-old married woman with 3-month history of worsening depression, now with psychotic features. Admitted following serious suicide attempt.

Differential diagnosis

Recurrent depressive disorder (current episode: psychotic). For: subjective and objective low mood, anhedonia, anergia, poor concentration, early-morning waking, loss of appetite and libido, self-blame. Suicidal. Delusional belief that she caused her son's epilepsy. Past history of depressive episode. *Against:* none, unless current depressive episode is better explained by one of the diagnoses listed below.

Bipolar disorder (current episode: depressive). For: depressive symptoms as listed above; long-standing mood variability, family history of bipolar disorder. *Against:* no evidence of past clinically significant mood elevation, nor current hypomanic or mixed affective symptoms.

Organic mood disorder. For: history of hypothyroidism. *Against:* no current symptoms or signs of hypothyroidism (or other abnormalities on physical exam); TSH normal.

Schizophrenia. For: delusional belief regarding her son's epilepsy; hears and sees her dead grandmother. *Against:* the delusion is mood congruent given her severe depression. The perceptual abnormalities are not true

hallucinations, and are consistent with her supernatural beliefs. She has no first-rank symptom of schizophrenia.

Conclusions: Psychotic depression. 6A71.4: Recurrent depressive disorder, current episode severe with psychotic symptoms.

Aetiology

Predisposing: Genetic—family history of mood disorder. Death of mother when aged 4. Unhappy childhood and not close to father. Personality traits. Previous depressive episode.

Precipitating: Discovery of husband's infidelity.

Perpetuating: Ongoing marital arguments, worry about debts.

Treatment

Given her diagnosis, and non-response to an adequate trial of venlafaxine, plan to increase dose and augment with an antipsychotic (olanzapine). ECT may be required if no response. Is on waiting list for CBT. Joint interviews with the patient and her husband may help to reduce the marital problems. Suggest sources of financial advice to reduce this stressor.

Prognosis

Ms AB has a good chance of a full therapeutic response to the planned treatment plan, especially if the marital problems can be improved. However, given her history and the predisposing factors noted above, she is at increased risk of further depressive episodes, particularly at times of stress.

Problem lists

A problem list is useful in cases where there are multiple issues to be addressed, especially complex social problems. The list identifies the range of problems (or needs), summarizes the solution(s) proposed, states who is responsible for each action, and gives a review date. It is best to draw up the list with the patient, to ensure that they are fully involved in the identification of problems and in how these will be addressed.

Table 3.2 shows a problem list that might be drawn up following the initial assessment of a man who had taken a serious overdose and was found to have anxiety symptoms, distress from his psoriasis, and social problems including concerns about how he and his partner were caring for their son. As progress is made in dealing with problems in this list, new problems may be added or existing problems modified or removed.

Letters to the general practitioner

When a letter is written to a general practitioner after an initial assessment, consider what they already know about the patient, and what questions were asked when they referred them. If the referral letter outlined the salient features of the case, there is no need to repeat these in the reply. If the patient is less well known to the general practitioner, or if the referral has come from another source, more detailed information should be given. The letter should state clearly the diagnosis (if one has been made), or the range of differential diagnoses that are still being considered.

Treatment and prognosis are dealt with next. When discussing treatment, the dosage, timing, and duration of any drug treatment should be stated. Name any key worker, therapists, and other agencies involved, and

the nature of their involvement. State the date of the patient's next appointment, and whether you have advised the patient to see the general practitioner in the interim (e.g. for blood tests). Responsibility for physical health care, and for prescribing, should be made clear. Ideally, all these details should be discussed and agreed with the general practitioner by phone before the letter is written to confirm them. The letter should emphasize a collaborative approach, and encourage the general practitioner to get in touch if they are unclear about the arrangements, or are concerned about the patient.

The same principles apply to subsequent correspondence during psychiatric care. That is, keep letters concise, focusing only on those issues which have changed since the last letter (e.g. symptoms, circumstances, treatments, and what is expected of the general practitioner). It is also helpful to highlight at the top of the letter the diagnosis, current state, current treatments (especially drugs and doses), and date of the next appointment.

As noted earlier, it is now standard practice for patients (and their carers, if requested) to be sent copies of all correspondence about them. All letters and other documents should be prepared with this in mind.

Electronic records and other new technologies

A switch from paper to electronic health records is occurring in many settings and poses significant challenges. For example, it may raise extra concerns about confidentiality, responsibility for backdating and maintaining the electronic records, and a patient's access to their record (Mehta et al., 2019). Equally, developments in electronic records and other aspects of 'digital psychiatry' provide many opportunities. Potential advantages include having access to patient records from any location and at all times, and the ability to keep

Table 3.2 A problem list

Problem	Action	Agent	Review
1. Concern over son's care	Initially, discuss with GP	Key worker and GP	2 weeks
2. Relationship problems	Seek counselling	Patient and their partner to arrange	3 months
3. Housing unsatisfactory	Visit housing department	Patient	2 weeks
4. Panic attacks	Assess for CBT	Psychological services	2 months
5. Psoriasis	GP to review	Patient	1 month

in contact with patients remotely via mobile phone or internet (e.g. for monitoring of symptoms using apps, or to send text prompts for clinic appointments). Psychiatrists need to be aware of the policies,

procedures, and regulations in their own clinical setting, and to keep up to date with the rapid and significant developments that will undoubtedly occur in the coming years (Torous et al., 2021).

Standardized assessment methods

In research, and sometimes in clinical practice, it is helpful to use standardized methods to assess symptoms and syndromes, as well as disabilities and other consequences of psychiatric disorders. Such methods improve reliability and facilitate comparison of findings across time and between psychiatrists. For a review of rating scales in psychiatry, see Tyrer and Methuen (2007).

Standardized methods of assessment are of four main types. Schedules for assessment of need were considered earlier; the other three are considered in turn below:

1. Those that rate symptoms in order to make a diagnosis.
2. Those that rate the severity of a symptom or group of symptoms.
3. Those that assess the overall evidence for and effects of psychiatric disorder; these are called *global rating scales*, and they include quality-of-life measures.

Standardized diagnostic assessments

A range of diagnostic assessment schedules have been developed. Commonly used ones are mentioned here. An important distinction is between those schedules designed for use by interviewers with training in psychiatry, and those for use by interviewers without such training. Schedules in the latter category (generally used in large-scale epidemiological research) require more precise rules for detecting symptoms, and for diagnosing syndromes.

Present State Examination (PSE) and Schedules for Clinical Assessment in Neuropsychiatry (SCAN)

Though now of limited use, the Present State Examination (PSE) is the archetypal interview schedule for use by trained psychiatrists (Wing et al., 1974). It became widely used, notably in seminal international studies on the diagnosis of schizophrenia (see Chapter 11). The interviewer identifies abnormal phenomena that have been

present during a specified period of time and rates their severity. Each of the 140 items is defined in detail in a glossary. Computer programs generate a symptom score, a diagnosis (CATEGO), and a measure of the severity of non-psychotic symptoms (the Index of Definition). The PSE was subsequently incorporated into the Schedules for Clinical Assessment in Neuropsychiatry (SCAN), used to diagnose a broader range of disorders, including eating, somatoform, substance abuse, and cognitive disorders.

Structured Clinical Interview for Diagnosis (SCID)

The Structured Clinical Interview for Diagnosis (SCID) is a diagnostic assessment procedure designed to make DSM diagnoses (Spitzer et al., 1987). It can be used by the clinician as part of a normal assessment procedure, or in research or screening as a systematic evaluation of a whole range of medical states. SCID-5 is the version released for use with DSM-5.

Composite International Diagnostic Interview (CIDI)

The Composite International Diagnostic Interview (CIDI) combines aspects of an earlier interview schedule, the Diagnostic Interview Schedule (DIS), with the PSE (Robins et al., 1988). It was designed for lay interviewers to make diagnoses according to ICD and DSM criteria. It is available in many languages and intended for use in different cultures. The interview includes questions about symptoms and problems experienced at any time in life, as well as questions about current state.

Mini International Neuropsychiatric Interview (MINI)

The Mini International Neuropsychiatric Interview (MINI) is a short structured interview (15–20 minutes, compared to 90 minutes for CIDI) for lay interviewers to make ICD and DSM diagnoses (Lecrubier et al., 1997). It focuses on current disorders only, and is widely used in surveys, and screening for clinical trials.

Instruments for measuring symptoms

In addition to instruments designed to make diagnoses, other scales measure the severity of symptoms or their fluctuation with time. Some such instruments rate one or a few symptoms, while others rate a broad group of symptoms as an overall measure of the severity of a disorder.

Ratings of affective symptoms

Hamilton Rating Scale for Depression (HAM-D)

The Hamilton Rating Scale for Depression (HAM-D) (Hamilton, 1967) (available in 17- and 21-item versions) is filled in by a trained interviewer who uses an unstructured interview. It measures the severity of the depressive syndrome rather than the symptom of depression.

Beck Depression Inventory (BDI)

The Beck Depression Inventory (BDI) (Beck et al., 1961) is a 21-item inventory completed by the patient. Each item has four statements, and the patient chooses the one that applies best to their feelings during the previous week.

Montgomery-Åsberg Depression Rating Scale (MADRS)

The Montgomery-Åsberg Depression Rating Scale (MADRS) inventory (Montgomery and Åsberg, 1979) has ten items rated on a four-point scale by an interviewer using definitions for each point. Along with the HAM-D, it is widely used in antidepressant drug trials.

Patient Health Questionnaire 9 (PHQ-9)

The Patient Health Questionnaire 9 (PHQ-9) (Kroenke et al., 2001) self-report questionnaire rates each of the nine DSM-IV criteria for depression on a scale of 0 to 3. A score of ≥ 10 has a sensitivity and specificity of 88% for major depression. The PHQ-9 is widely used as a screening tool for depression in primary care, and in monitoring of outcome. Versions with eight, four, and two questions are available; the PHQ-2 has two questions, covering depressed mood and anhedonia (Kroenke, 2021).

Quick Inventory of Depressive Symptomatology (QIDS)

The Quick Inventory of Depressive Symptomatology (QIDS) (Rush et al., 2003) is an increasingly popular 16-item scale for rating of depressive symptoms. It is usually used in a self-report format (QIDS-SR). It has good psychometric properties, is sensitive to change, and useful for clinical and research purposes.

Ratings of mania

Severity of manic symptoms can be rated by patient and clinician using the 11-item Young Mania Rating Scale (YMRS; Young et al., 1978), or by patients using the briefer Altman Self-Rating Mania Scale (Altman et al., 1997). A past history of manic-like episodes is captured by the Mood Disorder Questionnaire (MDQ) and used to screen for bipolar disorder (Hirschfeld et al., 2000).

Ratings of anxiety symptoms

Hamilton Anxiety Scale (HAS)

In the HAS (Hamilton, 1959), 13 items are rated by an interviewer on five-point scales, each on the basis of a brief description. The interviewing method is for the rater to decide. Some depressive symptoms are included so that the scale is in fact a measure of the severity of the anxiety syndrome and not of the symptoms of anxiety.

The State–Trait Anxiety Inventory (STAI)

The State–Trait Anxiety Inventory (STAI) (Spielberger et al., 1983) is a self-rating scale with 20 statements, which is completed in two ways—as the person feels when they are completing the scale (state), and how they feel generally (trait).

Hospital Anxiety and Depression Scale (HADS)

The Hospital Anxiety and Depression Scale (HADS) (Zigmond and Snaith, 1983) has seven questions about anxiety and seven about depression, and is used to assess symptom severity and caseness in psychiatric and medical patients, and in the general population.

Generalized Anxiety Disorder Assessment (GAD-7)

The Generalized Anxiety Disorder Assessment (GAD-7) (Spitzer et al., 2006) is a rapid self-administered questionnaire used as a screening tool and severity measure for anxiety symptoms and Generalized Anxiety Disorder.

Ratings of other symptoms

Ratings used in the assessment of cognitive impairment and dementia

The MMSE (Folstein et al., 1975) and MoCA (Nasreddine et al., 2005) have already been introduced. Other scales that are used to assess cognitive impairment, and the behavioural symptoms of dementia, are discussed in Chapter 14.

Ratings of symptoms of schizophrenia

These are discussed in Chapter 11. The Positive and Negative Syndrome Scale (PANSS) (Kay et al., 1987) and Brief Psychiatric Rating Scale (BPRS; Overall and Gorman 1962) are widely used.

Ratings of motor symptoms

A range of scales are available, especially related to side-effects of antipsychotic medication. They include the Extrapyramidal Symptoms Rating Scale (ESRS; Chouinard et al., 1980) and the Barnes Akathisia Rating Scale (BARS; Barnes, 1989).

Ratings of personality and its disorders

These are discussed in Chapter 15.

Ratings of broad groups of symptoms

General Health Questionnaire (GHQ)

The General Health Questionnaire (GHQ; Goldberg and Hillier, 1979) is designed for use as a screening instrument in primary care, general medical practice, or community surveys. The original instrument contained 60 items, but shorter versions (e.g. GHQ-12 and GHQ-30) have been developed. Even the full version can be completed within ten minutes. The symptom ratings are added to a score that indicates overall severity, which is expressed by whether a psychiatrist would judge the patient to be a 'case' or a 'non-case'.

Global rating scales

Clinical Global Impression (CGI)

The Clinical Global Impression (CGI) rating scale has two items. The main one, *global severity*, requires the clinician to rate the overall severity of the patient's illness at the time of interview, relative to other patients with the same diagnosis. The *global change* item rates change relative to a baseline assessment. The CGI is often used as a simple measure of efficacy in clinical trials.

Global Assessment of Functioning (GAF)

The Global Assessment of Functioning (GAF) is a 100-point scale, on which the clinician rates the overall functioning of the patient (Phelan et al., 1994). Each decile has a brief description of psychological, social, and occupational performance.

WHO Disability Assessment Schedule 2.0 (WHODAS 2.0)

The WHO Disability Assessment Schedule 2.0 (WHODAS 2.0) is a generic assessment instrument for assessing level of disability. It covers six domains of functioning (e.g. self-care, life activities, cognition) and can be administered in less than 20 minutes.

Health of the Nation Outcome Scales (HoNOS)

The Health of the Nation Outcome Scales (HoNOS) (Wing et al., 1998) is a 12-item scale that rates clinical problems and social functioning. It was developed as an instrument to measure progress towards a UK NHS target to improve the health and social functioning of mentally ill people, and used routinely for this purpose. It was designed to be applicable to all psychiatric disorders, so what it gains in applicability it loses in specificity. HoNOS has attracted criticism, but this has not limited its use. It has been revised and adapted for use in different countries and for specific clinical groups (James et al., 2018).

Quality-of-life scales

Quality of life is a key parameter of outcomes throughout medicine, and a range of scales are used to measure it. Quality-of-life scales that are commonly used in psychiatry include the relatively simple EuroQol EQ-5D, the SF-36, SF-6D, and the Lancashire Quality of Life Scale. Such scales can be helpful in research (e.g. in health economic evaluations) but are of limited value in clinical practice.

Patient-Reported Outcome Measures (PROMS)

A patient-reported outcome is a measurement of a patient's health that comes directly from the patient and is not interpreted by anyone else. Reflecting a rapid increase in the importance of these measures in clinical practice and research, a range of PROMS scales have been developed, but none is yet well established (Gelkopf et al., 2021).

Further reading

Poole, R., and Higgs, R. (2017). *Psychiatric Interviewing and Assessment*, 2nd edn. Cambridge University Press.

CHAPTER 4

Ethics and civil law

Introduction 71

General issues 71

Ethical problems in psychiatric practice 73

The Mental Health Act 81

Some aspects of civil law 85

Introduction

This chapter is concerned with the ways in which general ethical principles, relating to matters such as confidentiality, consent, and autonomy, are applied in the care of people with mental disorders. We assume that the reader has already studied ethical aspects of general medicine, otherwise they should consult Hope et al. (2016) or a comparable textbook of medical ethics (Beauchamp and Childress, 2019).

Ethical considerations are important in all branches of medicine. However, in psychiatry they have additional importance because some patients lack the capacity to make judgements about their own need for care, and because of the powers of the Mental Health Act in the UK (and legislation in other countries) that permit involuntary admission and treatment. Questions about capacity to consent to treatment of psychiatric illness commonly arise, and psychiatrists are sometimes asked by colleagues in other specialties, and by the courts and the criminal justice system,

for advice about capacity to consent to treatment for physical illness.

In this chapter we also consider some aspects of the Mental Health Act and the law relevant to psychiatry. These aspects are necessarily general because the specific provisions of the law differ from one country to another. It is important that the reader makes themselves familiar with legal provisions in their own country and, if in any doubt, always seek expert advice.

The chapter is divided into four parts:

1. General issues
2. Ethical problems in psychiatry
3. The Mental Health Act
4. Some aspects of civil law that are relevant to the practice of psychiatry.

Ethical problems that relate specifically to issues discussed elsewhere in this book are considered in those chapters (see Box 4.1).

General issues

The conclusion that it is ethically right to act in one way rather than another should be:

- based on agreed ethical approaches and principles;
- logically sound;
- consistent across decisions.

Ethical approaches

In psychiatry, as in ethics generally, two broad approaches to ethical problems are employed—a duty-based approach and a utilitarian approach.

The *duty-based approach* sets out clinicians' obligations in a series of rules—for example, doctors must not

Box 4.1 Ethical problems discussed in other chapters

Classification (Chapter 2)

Diagnosis on the basis of moral or political judgements (pages 43, 400, 545, 94)

Evidence-based approaches to psychiatry (Chapter 6)

Placebo-controlled trials and randomization (pages 80, 124)

Eating, sleep-wake and sexual disorders (Chapter 13)

Compulsory treatment in anorexia nervosa (page 312)

Personality and personality disorder (Chapter 15)

Stigma and patient involvement (pages 383, 385)

Child and adolescent psychiatry (Chapter 16)

Conflict of interest, confidentiality, and consent (page 464)

Disorders of intellectual development (Chapter 17)

Normalization, autonomy, and conflict of interests (page 490)

Consent to treatment, and consent to research (pages 490, 491)

Drugs and other physical treatments (Chapter 25)

Drug treatment (page 691)

ECT and psychosurgery (pages 745, 751)

Suicide and self-harm (Chapter 21)

Patients who harm themselves and refuse treatment (Box 21.10)

Psychiatry and medicine (Chapter 22)

Patients who refuse treatment (Box 22.14)

Genetic counselling (page 111)

HIV (page 635)

exploit their patients, nor have sexual relationships with their patients. These rules may be brought together in a code of practice, leaving clinicians in no doubt about their duties. However, this approach is difficult to apply to more complex situations.

The *utilitarian approach* balances judgements about benefit and harm. Instead of applying specific rules, each case is assessed individually on the assumption that the correct action is the one with the best foreseeable *overall* consequences. This approach considers clinical complexities, but can result in a contested conclusion, as people may give different weight to the benefits and harms in a particular case. Thus, this approach may produce ethical dilemmas for which there are no easy solutions, so careful discussion and dissection of the benefits and harms must inform decision making. Doing so as a team, working across disciplines and under legal and

ethical advice, can help clinicians to weigh up different perspectives to reach the best decision for the best outcome for the patient.

In practice the two approaches overlap. A duty-based approach may include a duty to do that which will result in the best outcome, and in a utilitarian approach the best outcome may result from deliberation about specific benefits and harms, and testing of or modification of rules.

Ethical principles

Three ethical principles are widely adopted in medical ethics (Beauchamp and Childress, 2019).

1. *Beneficence (and non-malevolence)*: doing what is best for patients (and not doing harm). In practice this usually means doing what the body of professional opinion judges to be best.
2. *Respect for autonomy*: involving patients in healthcare decisions, informing them so that they can make the decisions, and respecting their views.
3. *Justice*: acting fairly and balancing the interests of different people.

Ethical principles can, and do, regularly conflict with each other. In psychiatry this most often occurs when beneficence is in conflict with respect for autonomy—for example, when a patient refuses treatment that professional opinion judges to be essential, and the Mental Health Act has to be used (see The Mental Health Act). The essay ‘Two Concepts of Liberty’ by the philosopher Isaiah Berlin outlines this dilemma in distinguishing between ‘freedom from’ and ‘freedom to’ (Berlin, 1958). Autonomy is not always served simply by ensuring the absence of coercion if the individual is not in a position to act on their liberty. The Nobel Laureate Amartya Sen argues that our sensitivity to injustice informs our balancing of ethical principles, and not vice versa (Sen, 2009). This argues for clinicians being sensitive to injustices and to the process of deliberating on dilemmas.

Codes of practice

Codes of practice and guidelines are prepared and overseen by professional organizations such as the American Psychiatric Association (American Psychiatric Association, 2013). Much advice and codes of practice are now available on websites and are updated regularly. In the UK, the General Medical Council sets out 32 pieces of ethical guidance for conduct of doctors, students, and physician associates, as well as more specific

guidance on consent issues, decision making, consent, end-of-life care, remote consultations, coronavirus provisions, safeguarding, and prescribing. The Royal College of Psychiatrists update their codes of practice on a regular basis, and the latest guidance in 2017 addresses the following areas of practice:

- Direct patient care
- Healthcare purposes outside direct patient care (secondary uses)
- Uses and disclosure for non-healthcare purposes
- Requests for information
- Requests for reports
- Vulnerable patients
- Children and young people
- Media, audio and video recording
- Protection and security of patient information.

In some countries ethical codes are enforced not by the professions but by government. Loss of confidence in the professions has led to some demands for this arrangement in the UK.

The other professions involved in psychiatric care, such as nursing and psychology, have their own codes

of practice, and these are not identical in every respect to those of the medical profession. Such differences may sometimes complicate decisions in multidisciplinary teams.

Values-based practice has recently been proposed as a complement to *evidence-based practice* (Fulford, 2011). Ethical values can be in conflict, and routinely need to be balanced. The relative weight assigned to different values by the members of multidisciplinary teams (as well as the patient) may vary considerably. Values-based practice encourages the exploration of this in arriving at decisions, aiming to ensure that the traditional hierarchy in such teams does not inhibit their expression. For example, the psychiatrist may privilege treatment effect, whereas a nurse may privilege the patient's self-respect and the social worker may privilege the patient's rights. All of the team members acknowledge the relevance of all of these values, but they may weight them differently. Best practice and effective teamworking requires that all these values are considered. Indeed, this description of professional values is somewhat oversimplistic, as many psychiatrists argue for patients' rights while social workers may seek better clinical outcomes to optimize social function in the family and community.

Ethical problems in psychiatric practice

In the following sections of this chapter we shall discuss ethical problems relating to:

- the doctor–patient relationship
- confidentiality
- consent
- compulsory treatment
- research.

As mentioned previously, ethical problems that arise in particular aspects of practice are discussed in other chapters (see Box 4.1).

The doctor–patient relationship

A relationship of trust between doctor and patient is the basis of ethical medical practice. This relationship should be in the patient's interests and based on the principles outlined above of respect for autonomy, beneficence, non-maleficence, and justice.

Abuse of the relationship

The more intense the doctor–patient relationship, and the more vulnerable the patient, the more readily that relationship can become abusive or exploitative. This can happen consciously or inadvertently, when unconscious processes can lead to 'enactments.' For these reasons, particular care has to be taken to manage professional practice and feelings in all therapeutic relationships but especially in more intensive therapeutic work, for example, in psychotherapy settings or inpatient environments. Doctors and other mental health professionals abuse the doctor–patient relationship when they:

- *Impose their own values and beliefs* on their patients—for example, when counselling about termination of pregnancy. This influence may be overt—for example, when a doctor refuses termination of pregnancy, stating that it is morally wrong. Alternatively, it may be covert—for example, when the doctor expresses no opinion but nevertheless gives more attention to the

arguments against termination than to those for it. Similar problems may arise, for example, in marital therapy when therapists' values may intrude on their approach to the question of whether a couple should separate.

- *Put the interests of third parties before those of patients*—for example, the interests of family or employers. This does not mean the interests of patients should be considered in isolation. When allocating scarce resources or with potentially dangerous patients, the interests of the patient have to be balanced against those of others. These difficult decisions are discussed further on pages 76, 768.
- *Exploit patients sexually.* Medical codes of practice contain an absolute prohibition against sexual relationships with a patient. Particular care is needed if sexual problems and intimate relationships are prominent in psychotherapy.
- *Exploit patients for financial gain*—for example, by prolonging treatment in private practice beyond that which is truly necessary.

Confidentiality

Confidentiality is central to the trust between patients and doctors. It is particularly important in psychiatry because information often concerns private and highly sensitive matters. As a general rule, information should not be disclosed without the patient's explicit consent. As will be discussed later, there are specific exceptions to this rule. The rule is an ancient one, stated in the Hippocratic Oath:

Whatever, in connection with my professional practice or not in connection with it, I see or hear in the life of men, which ought not to be spoken abroad, I will not divulge, reckoning that all such should be kept secret.

The rule was restated in 1948 in the Declaration of Geneva, which added the important point that the obligation of confidentiality continues after a patient's death:

I will respect the secrets which are confided in me, even after the patient has died.

In many countries, national professional organizations publish guidelines that deal with common clinical situations such as protecting records, sharing information with relatives, and disclosing information to third parties (see, for example, General Medical Council guidance on confidentiality: <https://www.gmc-uk.org/ethical-guidance/ethical-guidance-for-doctors/confidentiality>). Confidentiality is also enforced by law and by employment contract. Although professional

guidelines do not have the force of law, they are taken seriously by the courts as evidence of generally accepted standards.

Many countries have laws of privacy, and laws that govern the ways in which written and electronic records can be held and that set out patients' rights to see their personal information. Psychiatrists should be aware of the ethical and legal requirements in the place in which they are working. In the UK, the relevant legislation includes the 1998 Data Protection Act and the 2018 General Data Protection Regulation.

In English law, it is in the *public* interest, as well as that of the individual, that patients should be able to trust their doctors to maintain confidentiality. Whether it is unlawful for a doctor to breach confidentiality is generally answered by balancing two conflicting public interests rather than a private against a public interest.

Ethical principles relating to confidentiality

Safeguarding information

Personal information must be safeguarded and records kept securely, and consideration must be given to the security of mobile phones or email. Unintentional disclosure should be avoided by carrying out consultations where they cannot be overheard, sometimes difficult on home visits or on a medical ward. Discussions between professional staff should be avoided in places where they might be overheard. Personal data should be stored in secure servers with appropriate protections, and access permitted for legitimate reasons. People also have a right to access, correct, or erase their records if they wish. Breaches of protocols should be recorded and handled with candour, following professional and organizational guidance.

Consent to disclosure of information

Confidentiality is not breached when a patient has given informed consent to disclosure. Informed consent implies consent based on a full understanding of the reasons for disclosure, what will be disclosed, and the likely consequences of disclosure. In principle, confidentiality is not breached when the patient cannot be identified—for example, in a disguised case report. Nevertheless, it is usual for journals and books to require that patients give informed, signed consent to publication, even when their identities have been disguised. This obligation, unlike libel laws, continues after death. Poorly anonymized case reports of deceased patients,

from which these individuals have been identified, have led to professional complaints from families.

Confidentiality in the care of children

These and other ethical problems in the care of children are discussed in Chapter 16. Here it is sufficient to note that children over a stated age (usually 16 years) have the same rights of confidentiality as adults. For younger children, clinical information is usually shared with parents, who have a legal duty to act in their children's best interests and therefore need to be properly informed. In older children, their ability to understand consent and capacity to give consent are considered alongside parental or guardian consent.

Situations in which there may be problems regarding confidentiality

Seeking information from others

As a rule, patients' permission should be obtained before information is sought from other individuals. If the patient is too unwell to give an account of himself, the psychiatrist must decide whether to seek it without the patient's consent. The guiding principles are to act in the patient's best interests, and as far as possible to obtain the information from close relatives.

Disclosing information to others

Information should be disclosed only with the patient's consent, except in the special circumstances discussed below (see Exceptions to the rule of confidentiality). Disclosure should be limited to the minimum necessary information. Relatives often ask for information, but this should usually be given only with the patient's consent. This is a very contentious issue in the era of community care when the family members may be the main carers and run the greatest risk (see Care in the community). When the patient is unable to consent to disclosure (e.g. as a result of dementia), information may be shared provided that disclosure is in the patient's best interests.

Assessment on behalf of a third party

Often a patient is assessed on behalf of a third party; for example, an assessment of fitness to work carried out for an employer. It is essential to ensure at the outset that the patient understands the purpose of the assessment and the obligations of the doctor towards the third party. Written consent should be obtained. Otherwise

disclosure can be made only in the public interest (see Exceptions in the public interest), usually to prevent death or serious harm.

Care in the community

Patients should know from the start that information will be shared as necessary with other members of the care team, and these team members must follow the principles of confidentiality. Some members of the team (e.g. social workers) may be required to discuss information with their supervisors, or to pass on information to other agencies (e.g. when helping patients with housing). It is important to keep such matters in mind as treatment plans develop, and to discuss them with the patient when this becomes necessary. When visits are made to patients, neighbours may become aware of the visits. Moreover, outreach programmes may need to ask neighbours about high-risk patients who have not kept appointments. These and similar potential problems should be anticipated as far as possible and discussed with the patient so that the necessary permissions are obtained in advance. Indeed, a first and subsequent assessment should review documented consents. In an emergency, the right to privacy has to be balanced against the risks to the patient or to others, should the necessary enquiries not be made.

If a patient refuses the disclosure of information despite clear information about the likely consequences of non-disclosure, their wishes must be respected unless this would put others at serious risk.

Group therapy

Group therapy presents special problems of confidentiality because patients reveal personal information not only to the therapist but also to other members of the group. Before treatment begins, the therapist should explain the requirement to treat as confidential the revelations of the other group members from everyone in the group.

Therapy with couples

Often couples' therapy is preceded by an interview with each partner separately. Information obtained in this way should not be revealed without an agreement. It is much better if all relevant information is revealed by the partner concerned during the joint sessions. If, during therapy, an individual interview is needed (e.g. to assess symptoms of depression), it may be better to ask a colleague. Similar problems may arise in family therapy.

Exceptions to the rule of confidentiality

Exceptions in the patient's interest

In exceptional circumstances, doctors may disclose information to a third party without the patient's consent, when this disclosure is in the patient's best interest. An example is when information is requested urgently by another doctor who is dealing with the patient in an emergency, or when a patient is incapable of giving informed consent because of severe mental or physical illness, and the disclosure is essential for their care.

Exceptions in the public interest

Although there is a general legal obligation for doctors to keep confidential what patients tell them, there are special circumstances in which doctors are obliged to disclose information. There are *statutory obligations* to do so in relation, for example, to communicable disease, the use of certain controlled drugs, unfitness to drive, and suspected child abuse. There is also an obligation to disclose relevant information in response to a Court Order, and when there is evidence of serious crime, usually a crime that will put some person at risk of death or serious harm (e.g. the abuse of a child). *Child protection* has become a particularly sensitive area for professional confidentiality. Society's tolerance of the potential abuse or neglect of dependent children is almost zero now in most countries. The current expectation is that professionals' threshold for breaking confidentiality should be very low in cases where there are any concerns, even without clear evidence. Every effort should be made to persuade the patient to agree to disclosure but, if they refuse, the reasons for the disclosure should be explained and written down. If there are doubts, ethical and legal advice may be sought. Each healthcare provider will also have a lead person who carries responsibility for supporting decision making, and advising when breaches of confidentiality are being considered for the purposes of safeguarding.

Exceptions for legal representatives

A patient's legal representative, like the patient, may read their clinical notes and letters, although there may be restrictions in relation to the possibility of harm to others and for information given in confidence by informants.

For further information about problems related to confidentiality see Hope et al. (2016).

Consent to treatment

Consent is relevant to the whole of medical practice, and in the account that follows we assume that the reader is familiar with the relevant concepts and procedures from their general medical training. Any readers who are not should consult the latest General Medical Council guidelines if they are working in the UK (see <http://www.gmc-uk.org>) or the equivalent document if they are working elsewhere, and read a textbook of ethics such as that by Hope et al. (2016). In the UK, the General Medical Council, British Medical Association, and Royal College of Psychiatrists offer regular updates on their respective websites.

Obtaining consent

The patient should:

1. Have a *clear and full understanding* of the nature of the condition to be treated, the procedures available, and their probable side effects.
2. *Agree freely* to receive the treatment.
3. Be *competent* to take decisions (i.e. to *have legal capacity*; see below). (Note that, in the United States, the word 'capacity' is generally used in the clinical sense rather than in the legal sense adopted in the United Kingdom).

Maintaining consent

Some patients consent to treatment of chronic illness but later fail to collaborate with it. When this happens, the clinician should seek to re-establish consent and collaboration. To achieve these, offers of additional help are justified, but threats that help will be reduced are not. This is particularly complex in the long-term community care of individuals with severely disabling disorders such as psychoses. The difference between an 'offer of additional help' and a 'threat that help will be reduced' is not always that clear in practice, and certainly not to the patient. Is a case manager's comment 'I can only really recommend you for this tenancy if I can say you are taking treatment' an offer, or is it a threat? There are gradations of treatment pressures or coercion in mental health care and all should be vigilant (Szmukler and Appelbaum, 2008).

When explicit consent is not required

There are special situations in which explicit consent is not needed (their precise nature depends on local law and precedent). These include:

- *Implied consent*—for example, when a patient holds out his arm to have his blood pressure measured.
- *Necessity*—that is, a circumstance in which grave harm or death is likely to occur without intervention, and there is doubt about the patient's competence (see below).
- *Emergency*—to prevent immediate serious harm to the patient or to others, or to prevent a crime.

When consent to medical treatment is refused

In the United Kingdom, the law distinguishes between consent to medical treatment and consent to psychiatric treatment. This section is concerned with refusal of *medical* treatment; refusal of psychiatric treatment is discussed on pages 77, 80, 464, 512, 606, 691, 739. Competent adults have a right to refuse medical treatment, even if this refusal results in death or permanent disablement.

If a patient refuses medical treatment, the doctor needs to make two judgements before accepting that the patient has the right to refuse.

1. Does the patient lack competence, i.e. the legal capacity (see Refusal by mentally disordered patients) to consent to or refuse treatment, by reason of mental illness?
2. Has the patient been influenced by others to the extent that a refusal has been coerced or is not voluntary?

Refusal by competent patients

Refusal may, of course, be a rational decision, having weighed up the pros and cons (pain, discomfort, separation from family versus improvement in symptoms or increased life expectancy). However, it is often the result of misunderstanding or fear about the illness and its treatment. Clinicians should set aside the time needed to understand the patient's concerns and understanding of their condition before explaining the medical issues once more. With such an approach, agreement can usually be reached on a treatment plan that is both medically appropriate and acceptable to the patient. Nevertheless, some competent adult patients continue to refuse treatment even after a full discussion, and it is their right to do so. If the clinician is concerned about a potentially serious irreversible consequence, they may seek legal advice, and a judgement by a court hearing.

Refusal by incompetent patients

In the United Kingdom and most other countries the doctor has the right to act in the patient's best interests and give immediate treatment in *life-threatening emergencies* when the patient lacks capacity to consent, as when they are unconscious or delirious. Whenever possible, other health workers should be consulted, and a careful record should be kept of the reasons for the decision. Doctors should know the relevant law in the country in which they are working. For further discussion of this issue (Hope et al., 2016)

Refusal by mentally disordered patients

If a patient who refuses medical treatment has an accompanying mental disorder that appears to impair their ability to give informed consent, the mental disorder should be treated, if necessary and appropriate, under compulsory legal powers. In English law, such powers do not allow treatment of the physical illness against the patient's wishes. However, they do allow the treatment of the mental disorder. After successful treatment of the mental disorder, the patient may then give informed consent for the treatment of the physical illness that was previously refused but, if not, their decision must be respected no matter how perverse it may seem.

Mental capacity

The legal concept of *capacity* to consent relates to a patient's ability to

- comprehend and retain information about the treatment
- to believe this information
- and to use it to make an informed choice.

Patients may lack the capacity to give informed consent by reason of the following:

1. Young age—parents give consent for children and adolescents below the age at which, in law, they are able to consent (consent to treatment for children and adolescents is considered in Chapter 16).
2. Learning disability.
3. Mental disorder. Psychiatrists may be asked to assess patients who are thought to be in this third group; they should do so by making the enquiries listed in Box 4.2.

Judgements of legal capacity to consent are *specific to the particular decision*. Thus a patient with a severe

Box 4.2 Assessment of competence of adult patients

Step 1 Identify the decision required and the information relevant to it:

- the decision to be made
- the alternative reasonable decisions
- the pros and cons of each reasonable decision.

Step 2 Assess cognitive ability.

Assess whether the patient has the cognitive ability to:

- understand the information
- retain the information
- evaluate the information and reach a decision.

Step 3 Consider the possible causes of impaired cognitive ability:

- delirium
- dementia
- other neurological disorders that may impair cognition
- learning disability.

Step 4 Assess other factors that may interfere with capacity.

Mental illness:

- delusions
- hallucinations
- mood disorder.

Lack of maturity: assess emotional and cognitive maturity.

mental disorder may be incompetent in other respects, but nonetheless competent to decide whether to consent to a particular treatment. For example, a patient with schizophrenia and paranoid delusions may be capable of deciding about medical treatment of a heart attack.

Mental Capacity Act

The 2005 Mental Capacity Act for England and Wales addresses the previously neglected area of the treatment of incompetent patients who are not actively refusing treatment but who cannot be considered to have consented. It also addresses those who may have previously indicated (when competent) what they would have wished. In 2014, additional guidance was published (<https://www.gov.uk/government/collections/mental-capacity-act-making-decisions>).

Advanced statements

Advance statements, sometimes called 'living wills', are accepted in many countries. They are designed to ensure that those who previously had the capacity to take decisions but then lose it (usually because of dementia) are treated as they would have wished when competent. To make an advance statement, the person must be competent and well informed at the time. Advance statements respect the principle of autonomy and are generally considered by doctors and patients to be helpful.

Since the 2005 Mental Capacity Act (MCA) regularized the situation, advance statements now have some legal authority in England (the Act does not apply to Scotland).

The MCA distinguishes between an advance statement and an 'Advance Decision to Refuse Treatment' (ADRT). Advance statements are relatively informal and may be recorded in notes or care plans (e.g. 'Only use second-generation antipsychotics if I am compulsorily treated', 'Do not inform my sister if I am admitted'). They can even be verbal. They are not legally binding, but it is good practice to follow them if possible and also to encourage patients to update them. If they are not followed, the reasons should be recorded. Because of the relapsing nature of many severe mental illnesses, advance directives are often based on the patient's previous experiences. In general medicine they are more often based on how the individual assumes that they would feel about treatment if and when they lost capacity, usually through dementia. The incorporation of advance statements as part of 'joint care plans' has been tested in some services. Although patients seem reluctant to make them, they do appear to improve relationships and may improve treatment even if they have not been shown to reduce relapses (Thornicroft et al., 2013). The importance of advanced statements is further reinforced by a recent review in the United Kingdom of mental health legislation, the proposal being to place advanced statements on a statutory footing to ensure all patients have an opportunity to use them and professionals must comply (Jankovic et al., 2020).

However, the ADRT is binding, even if the decision may result in death. To have this enforced it must be a written and witnessed statement with confirmation that the individual had capacity at that time. The provision also allows for the naming of a 'surrogate decision-maker', who can give consent on behalf of the patient. The ADRT can be overridden if the Mental Health

Act is used, apart from the advance decision to refuse electroconvulsive therapy (ECT), which is absolute. Compulsory treatment overriding an advance decision may be used in acutely unwell patients most often (but not only) when there are risks to others. Psychiatry is the only area of medicine where this override is possible, and some view this as discriminatory.

Deprivation of liberty and the 'Bournewood Gap'

An adult man with autism was admitted to Bournewood Hospital against the wishes of his informal carers, but did not himself actively object to being in hospital. An individual was then only considered to have their liberty restricted if they were actively objecting to their treatment. The Bournewood case indicated a 'gap' in the legislation, namely the failure to protect an individual who was deprived of his liberty although not resisting it. This was ruled to conflict with European human rights legislation. The distinction between *deprivation of liberty* and *restriction of liberty* can be a fine judgement. To comply with European legislation it must be clearly in the patient's best interests (i.e. not for the protection of others). A consequence of this legislation is the Deprivation of Liberty Safeguards, which were introduced in the 2007 amendments to the Mental Health Act. These are complex and usually dealt with by a 'Best Interests Assessor', who may call on a psychiatrist to advise on the capacity or diagnosis of such patients. The implication for psychiatrists is to pay attention to any restrictions on incapacitous patients, even if they are not objecting.

Liberty Protection Safeguards (LPS) were introduced in 2022 to replace Deprivation of Liberty Standards. LPS seek to improve outcomes for people deprived of their liberty, if they are aged 16 or above, to permit treatment or care when they lack capacity. These are intended to protect those with dementia, autism, and learning disability, seeking to avoid illegal detention when people are not obviously suffering with a mental disorder and thus care cannot be provided under the powers of the Mental Health Act. Each country will have its own legal protections, so practitioners need to avail themselves of the most recent ethical and legal guidance in their particular jurisdictions. In the UK (England and Wales for the purposes of the legislation), LPS requires:

- A capacity assessment
- A 'medical assessment' to assess the presence of mental disorder
- Necessary and proportionate assessment to evaluate harms and benefits and if the arrangements are proportionate.

Families should be involved, and for those 16–17 years of age, an application must be made to the court of protection. These arrangements are new and subject to review.

Consent by a proxy

In English law, if an adult patient does not have the capacity to consent to treatment, proxy consent on behalf of that patient can only be given if there is a written advance statement. In some other jurisdictions there is provision for a form of proxy consent such as, in Scottish law, the appointment of a guardian. When there is no legal provision for proxy consent and the patient has not made an advance directive, and a decision as to whether to give treatment must be made, this is done on the basis of the patient's best interests. Best interest is judged by the responsible clinician in accordance with general medical opinion. It is wise to consult relatives and to discuss the case with other professional staff. Detailed notes should be kept of the reasons for the decision and of the consultations that took place.

Ethics of research

Psychiatric research is bound by the ethical principles that apply to all medical research. These derive from the internationally agreed guidelines on research involving people, referred to as the 'Helsinki Declaration'. The main issues that should always be considered are summarized in Box 4.3. These principles followed the revelations of the appalling abuse of medical research in Nazi Germany, and were first published by the World Medical Association in 1964 (see <https://www.wma.net/what-we-do/medical-ethics/declaration-of-geneva/> for World Medical Association updates). Since then there have been several national and international guidelines, generally enforced by national and local research ethics committees. The ethical problems of psychiatric research are the same as those of other medical research as set out, for example, in the UK General Medical Council guidelines (see <http://www.gmc-uk.org>) or the website of the UK Health Regulatory Authority (<http://www.hra.nhs.uk>). They pay particular attention to issues related

Box 4.3 Assessment of some ethical issues relating to research

Note: Ethical problems related to recruitment for and conduct of clinical trials are considered on page 123.

Scientific merits

- Will the findings be of value?
- Are the methods and the size of the groups likely to achieve the aims?
- What are the sources of financial and other support, and are there any potential conflicts of interest?
- Are there any potential conflicts of interest for any of the investigators?
- Could the aims be achieved in an ethically better way?

Safety

- Are the procedures safe? If there is a risk, are all of the necessary precautions being taken?
- Is the assessed level of risk acceptable to investigators, subjects, and relatives?

Consent

- Will the subjects be competent to give consent?
- Will the subjects receive clear and sufficient information?
- Will they have adequate time to consider and, should they wish to do so, to withdraw consent?
- Will it be clear that refusal will not affect the quality or quantity of care provided?
- Is the relationship between subject and investigator potentially coercive (e.g. a supervisor and a student)?
- Is any payment to subjects excessive and acts as an overwhelming incentive to consent?
- Is the researcher under any pressure to recruit subjects (e.g. receiving payments from a sponsoring company)?

Confidentiality

- Have subjects consented to the use of confidential information in the research?
- Will the data from the research be kept securely?

Adapted from Hope, T., Savelescu, J., & Hendrick, J. (2008). *Medical Ethics and Law: The Core Curriculum*, 2nd edn. Copyright with permission from Elsevier.

to consent by competent but vulnerable adults who may feel under pressure to consent, and by those who lack capacity.

Informed consent to research

Informed consent is crucial to the ethical conduct of research.

- Patients must be made specifically aware that the research is not being conducted for their individual benefit.
- Patients must be free from any coercion or inducement.
- Patients have the right to withdraw from the study at any time without any kind of penalty or impact on their broader treatment.
- In addition to the investigator, a family member or other suitable person should be encouraged to monitor the patient's condition and report to the investigator if there are concerns.
- In placebo-controlled trials, patients must understand clearly the probability of receiving placebo, the lack of improvement that might result, and the possibility of symptomatic worsening.

Research involving people who cannot give informed consent

Most psychiatric patients can give informed consent, but a few have a disorder that impairs judgement and decision making. To exclude these patients from all research could deprive future similar patients of the benefits of research. Ethical protocols for involving such patients have received intense scrutiny in the design of treatment trials for unconscious trauma victims. The decision as to whether to include such patients is made after considering the following:

- Any potential benefit of the research to the person who is being asked to consent;
- Any possible discomfort or risk to this person;
- The potential benefit to others who have a similar problem and incapacity; and
- Any signs or other indications that suggest unspoken objection.

It is advisable to consult with relatives or others who may be able to take an informed view of the patient's situation. Jurisdictions vary in the extent to which this is a legal requirement, and investigators should take great care to follow local ethical and legal procedures.

The Mental Health Act

Compulsory treatment for mental disorders

Compulsory treatment has always been a feature of psychiatry and a legitimate and important responsibility for psychiatrists. In most countries outside the United Kingdom its imposition of compulsory admission or treatment is by a legal body (judge, magistrate, etc.). However, the reality in most legislations is that the medical recommendation carries considerable weight. Several states distinguish between compulsory detention and compulsory treatment. In the Netherlands, Austria, and parts of Switzerland a patient can be detained but still must agree to treatment.

There appears to be marked variation in rates of detention by country, which are largely unexplained (de Stefano and Ducci, 2008). In some instances measures to reduce compulsion lead to more compulsory treatment, and evidence suggests that greater spend on healthcare in high-income countries is associated with more compulsion. There appear to be complex interactions between the legislation, the way it is implemented, clinical practice, and local contexts (Sheridan Rains et al., 2019).

When using the Mental Health Act it is essential that psychiatrists familiarize themselves with the local provisions and requirements. Not only do these vary from one country to another, but they can vary within countries. For example, the Scottish and Northern Ireland Mental Health Acts differ in important details from that for England and Wales, each state in the United States has its own legislation, and different cantons of Switzerland impose different conditions. Mental health legislation is often long and obscure—the 2007 amendment runs to several hundred dense pages, and the accompanying Code of Practice likewise. Practising clinicians generally only need to know the main provisions and procedures for the practices outlined in Box 4.4. The major provisions of the 1983/2007 Mental Health Act for England and Wales will be used as the reference point for outlining the principles and practices in what follows. Local training with regular updates is essential and is now obligatory in the United Kingdom. New legislation proposed in England and Wales will also require careful scrutiny and a period of implementation pilots.

Criteria for detention

Until recently all compulsory care required admission to a psychiatric hospital, so the expression ‘detained under the Mental Health Act’ is still widely used to indicate involuntary care, whether in or outside hospital. The UK acts are ‘permissive’—that is, they outline what a doctor *may* do but not what they *must* do. The decision to admit or treat a patient against their will is still a clinical decision. The wording of the Act allows considerable scope for clinical discretion. The conditions for detention are worded differently for treatment and assessment, but there are essentially four of them.

1. The responsible clinician considers that the patient is likely to be suffering from a mental disorder (detention may sometimes be necessary to establish this).
2. Detention is necessary for the health or safety of the patient or of others.
3. The patient refuses voluntary admission.
4. Appropriate treatment is available (for treatment orders).

Until the 2007 revision there were four classes of mental disorder subject to detention (mental illness, mental impairment, severe mental impairment, and psychopathic disorder), but these were replaced by a single category of mental disorder. This was to remove individual requirements for specific disorders. The UK criteria are quite paternalistic by international standards—they prioritize the potential for patient benefit. Many countries (especially those with strong libertarian civil rights traditions, such as the United States) emphasize risk rather

Box 4.4 Key clinical procedures in the Mental Health Act

- Criteria for detention
- Assessment orders
- Treatment orders
- Transfer from prisons and courts
- Police powers and powers of entry
- Community Treatment Orders.

than patient benefit. In many US and some European states, patients must pose an *immediate risk* to physical safety (their own or that of others). Finally, patients must always be given the opportunity to accept voluntary treatment ('the least-restrictive option') before compulsion can be used.

Historically families (or their formal representatives) made applications for detention, but the 1959 Mental Health Act permitted social workers to do this on their behalf. Over time this has become the accepted practice, although families can still appeal to discharge their members. The social worker has a specific role in ensuring that the patient's civil rights are properly protected and that less restrictive alternatives have not been overlooked. Apart from emergencies, two medical opinions are needed, preferably one who knows the patient (often their GP) and one who is a qualified psychiatrist. Note that this is *preferable*. Although it is worth aiming for, it is not essential. Patients or their families should not be put at risk by delays in trying to comply rigidly with it. Reviews of the legality of detention are conducted by Mental Health Tribunals, which follow automatically at set periods and also if requested by patients. The main sections of the Act used by clinicians are summarized in Table 4.1.

Assessment and treatment sections

The Assessment Order, Section 2, is for four weeks (in emergencies, Section 4 is for 72 hours). It allows patients who are not already known and diagnosed to be admitted and assessed if their condition is serious and warrants this. It cannot be repeated. Necessary medication to manage the assessment safely is permitted, but if the diagnosis is confirmed and involuntary treatment is to be instituted, the patient should be transferred to Section 3. There is considerable variation in how much intervention is permitted while the patient is on Section 2. Some social services have advocated for the use of Section 2 as a matter of course, even when patients and their disorders are known to the team. This is to 'assess' their response to this admission. There seems to be little logic to this, and generally a Section 3 should be used when the patient is known to those admitting them, or if the clinical picture is clear and a confident diagnosis can be made. Section 3 is for up to six months in the first instance, and can be repeated. Patients on Section 3 can be allowed out of hospital 'on leave' while still under compulsion using Section 17. There is no specific time limit for Section 17 beyond that imposed by the Section 3, which stays in force. The Code of Practice proposes that it should be reviewed every seven days, and the

alternative of a Community Treatment Order considered if extended beyond seven days. Voluntary patients can be detained for assessment if their condition deteriorates and they attempt to discharge themselves under the provision of Section 5.

Restriction orders and transfers

Patients can be transferred from courts on Section 37 (equivalent to a Section 3) for treatment. Their management is the same as for a Section 3, and they can be discharged when well. Similarly, a prisoner who becomes seriously unwell can be transferred on Section 47 for treatment (again following the same principles as Section 3), but has to be returned to prison when they have recovered. The equivalent assessment order (up to 28 days) is Section 35. The Home Office can impose a restriction order (37/41) on a patient. This means that the psychiatrist cannot discharge them from their section without permission. However, they can be managed

Table 4.1 Sections of the England and Wales Mental Health Act commonly used by clinicians

Section number	Purpose	Duration
Section 2	Assessment	28 days
Section 3	Treatment	6 months, repeatable
Section 4	Emergency assessment	72 hours
Section 5	Detention of informal inpatient	72 hours
Section 17	Leave while on Section 3	Section 3 still in force
Section 17a	Treatment in community (Community Treatment Order)	6 months, renewable
Section 35	Transfer for assessment	28 days
Section 37	Transfer for treatment	6 months
Section 37/41	Restriction order	Imposed by Home Office
Section 135	Forced entry	72 hours
Section 136	Police powers to convey	72 hours

on leave in the community with the Home Office's permission.

Police powers and power of entry

The police have the power to detain anyone whom they find in a public place who is manifestly disturbed and whom they consider to need urgent psychiatric assessment. In the UK this is Section 136 of the Act, and it allows the police to bring the patient to a place of safety for assessment by a psychiatrist and an approved mental health professional (AMHP). The person can be held for up to 72 hours, but the Code of Practice recommends assessment within six hours. In the past, police stations were the usual place of safety (and they sometimes still are), but it is considered good practice for assessment to take place in an agreed psychiatric assessment suite. Section 136 is predominantly used in metropolitan settings, and there is little reason to believe that it is over-used (the police have quite high thresholds for using it).

The right of individuals to be undisturbed in their own homes is a very ancient one, and one that is vigorously protected. Unless there is overwhelming evidence of immediate risk, entry cannot be forced. Where it is believed that a known patient is deteriorating and requires assessment, but they are refusing entry to their home, a warrant from a magistrate is needed. This is Section 135, usually obtained within a day by the social worker. The warrant allows admission by the police officer, accompanied by a psychiatrist and an AMHP, who generally assess for Section 2 or 3.

Community Treatment Orders

The most significant change introduced by the 2007 amendment of the 1983 Mental Health Act was the introduction of Community Treatment Orders (CTOs), often referred to as Section 17a, in official documents. These introduced an effective mechanism for ensuring treatment in the community for patients with established severe disorders and poor compliance with treatment ('revolving-door patients'). Forms of compulsory treatment in the community had been in widespread use in the United States, Canada, Australia, and New Zealand for over three decades (Dawson, 2005). They have attracted strong clinician and family support, although the scientific evidence is that they do not improve outcomes. The three randomized controlled trials found no impact on relapse and readmission. This has not, however, slowed their use (Burns et al., 2013). The UK provision can only be applied when a patient is

already detained on Section 3. They are subject to a legal requirement to attend for monitoring in the community.

The CTO mirrors the inpatient treatment order (i.e. six months repeatable in the first instance, with repeat renewals of 12 months subsequently, and regular tribunal assessments). It is strengthened by a requirement for an independent second opinion in all cases. Beyond this statutory condition for a second opinion, other clinically determined conditions can be added. The commonest are to reside at a stated address and to allow regular access by the clinical team. CTOs do not permit any force outside hospital, but allow for patients to be returned to a clinical setting for up to 72 hours for assessment and treatment. About 4,000 CTOs per year are initiated in England and Wales.

CTOs in the UK are generally imposed for substantial periods, and during the first years of their introduction very few of them have been revoked. This reflects the United Kingdom's 'paternalistic' approach, with CTOs being used to stabilize functioning in patients with frequently relapsing psychoses. In the United States, most are imposed for a short period while patients pose an active risk. However, those who remain on them beyond this early phase stay on them for months or even years.

A New Mental Health Act (England & Wales)

In 2018, an independent review of the mental health act recommended changes to modernize the act. Following consultation with professionals and patient groups, a series of recommendations were being scrutinized by a joint committee of Parliament (2023). In summary, the recommendations improve the language to be recovery orientated and less stigmatizing, and in addition propose the following (relevant for chapters on psychiatric services, Chapter 18; and on forensic psychiatry, Chapter 26):

- Greater rights of appeal and better choice over care experience
 - Advance choice statements and care and treatment plans are encouraged by local providers, rather than be mandated in law.
 - Patients can challenge decisions more easily, and earlier.
 - Patients can choose a nominated person they want to represent their interests.
 - More frequent access to the independent tribunal.
 - Privacy of individual rooms if detained in hospital.

- The initial period of detention for treatment under section 3 is to be reduced to three months from six months set out under the current MHA. Section 3 can be renewed for a further 6 months rather than one year.
- Race equality
 - The right to challenge the detention decision, which is proposed to address ethnic disparities in use of the Act.
 - A patient and carer race equality framework was recommended, so each organization must assess, manage, and reduce disparities.
 - The role of independent mental health advocates will be expanded with an opt-out system.
 - All detained patients will have a statutory personalized care plan.
 - Criteria restricted to risk of serious harm and reasonable prospect of therapeutic benefit from detention.
- Autism and learning disability
 - The Act will limit the period of detention of people with learning disability or autism to 28 days (section 2), in the absence of a defined mental illness. Hospital sentence options for prisoners were retained as were the provisions for hospital transfers.
 - Cannot be detained for treatment under section 3, unless there is clear evidence of a co-occurring mental illness.
 - Introduce commissioners to better manage crises in those with learning disability and autism.
- Criminal justice system
 - Remove police stations and prisons as places of safety.
 - A new community restriction for offenders who are ready to leave hospital but need restrictions in the community.
 - Prison mental health care and rapid diversion was supported, by introducing a 28-day time limit for transfers from prison to hospital for prisoners with severe mental illness.

Informal coercion

Traditionally research into coercion in psychiatry has been based on rates of legal detention. Investigations into how patients perceived their care, however, revealed that it is not so clear-cut. Coercion is endemic in psychiatric practice, perhaps in relation to a constant tension to manage risks through restrictions and entice and encourage engagement and voluntary participation

in care (Sashidharan et al., 2019). Coercion in mental health care is defined as a broad range of practices from pressure to accept certain treatments (implicit or explicit) to forced practices including involuntary admission and treatment, seclusion and restraint (Barbui et al., 2021). Coercion raises numerous ethical dilemmas. It is considered legitimate in exceptional circumstances, when restrictions (e.g., freedom of movement, short-term autonomy) are the only means to achieve important goals such as patient's safety, and building up to greater autonomy (Chieze et al., 2021). In many countries, coercive practices are no longer confined to involuntary hospital admissions; they are becoming increasingly prevalent within community-based care and during 'voluntary' admission to hospitals.

Staff training, shared decision making around involuntary admission, and integrated care all reduce the experience of coercion (Barbui et al., 2021). Coercion may be experienced during any contact and when making any therapeutic or clinical offer. Burns et al. (2015) suggest a five-step model to reduce coercion and questions if clinical offers are ethical. The five steps are:

1. Decide whether the proposal is a threat or an offer.
2. Decide whether the offer is likely to be unwise.
3. Decide whether the offer exploits the patient.
4. Decide whether the offer raises broader questions of fairness.
5. Decide whether the offer compromises the voluntariness of the patient's decision making.

More research is needed, and better tools are required to measure coercion. Much of the research has used a rating scale developed by a US research group—the MacArthur Perceived Coercion Scale (Gardner et al., 1993). This simple 15-question scale has three components—perceived coercion, negative pressures, and procedural justice. Initially developed to measure the experience of psychiatric admission, the scale has been adapted for community care. A striking finding is that how a compulsory admission is handled (procedural justice—whether the patient feels listened to and understood) makes a substantial difference in how coercive they rate the admission (Bonsack and Borgeat, 2005).

Hierarchy of pressures and leverage

With the growing awareness that many community patients experienced their care as coercive, attempts have been made to classify the different types of coercion and to measure some of them. Szmulker and Applebaum

(Szmukler and Appelbaum, 2008) have proposed a hierarchy of treatment pressures arranged on five levels (Box 4.5). Not all levels of persuasion are considered coercive but certainly the last two are.

What Szmukler and Appelbaum refer to as inducement has been called 'leverage' in later literature. Leverages are when access to benefits (or avoidance of unwanted consequences) are *explicitly* linked to adherence to treatment. Leverage is when the clinician says to the patient 'if you take your treatment regularly, I can help you get a flat, or better benefits, or probation rather than prison'. A survey of leverage with 1,011 patients in public mental health services in five states in the United States found that nearly half of them reported feeling pressured in this way into cooperating with treatment (Monahan et al., 2005). The leverages were classified as financial, housing, and criminal justice (usually avoiding or reducing a sentence). A similar pattern, albeit at lower levels (around 30–35%) has been reported in the United Kingdom (Burns et al., 2011) and Switzerland (Jäger & Rössler, 2009). Access to separated children has also emerged as an important leverage, particularly in individuals with substance abuse (Burns et al., 2011)

Contingency management is a special form of leverage where a direct reward is provided for adherence to treatment. The most controversial schemes involve paying patients to cooperate and have been used mainly with socially marginalized individuals such as homeless sufferers from tuberculosis and even to encourage antenatal care for pregnant teenagers. The approach generates very strong feelings, and many clinicians find it quite unacceptable as it seems to devalue the therapeutic relationship. One trial of paying psychosis patients to

Box 4.5 Levels of persuasion

- Persuasion (includes providing information, appreciation of patient's preferences, and discussion and collaboration between clinician and patient).
- Interpersonal leverage (based on a patient's emotional dependency on clinician; patient agrees to adhere to please clinician).
- Inducement (clinician offers reward for adherence).
- Threats (clinician proposes negative or prevents positive consequences of a patient's non-adherence).
- Compulsion (involuntary).

Reproduced from Szmukler G and Appelbaum P. (2008). Treatment pressures, leverage, coercion and compulsion in mental health care. *J Ment Health*, 17(3), 233–244. Copyright with permission from Taylor & Francis.

comply with their depot injections did find an improvement (Priebe et al., 2013), but there is little evidence of any widespread acceptance of the approach and the findings were not sustained over time albeit the intervention seemed to be cost-effective (Priebe et al., 2016).

The recent focus on informal coercion has underlined just how common it is and also that it is a legitimate part of the therapeutic relationship. All human relationships involve some degree of trying to influence each other, either explicitly or implicitly. The recent attention has made clear there is a need for more structured discussion to permit the emergence of some consensus on what is good practice and what may be overstepping the mark.

Some aspects of civil law

Civil law deals with the rights and obligations of individuals to one another, including family law, which is a particular concern of child psychiatrists. In this respect civil law differs from criminal law, which is concerned with offences against the state (although some are directed against an individual, e.g. homicide). As well as the issues discussed earlier in this chapter, civil law is concerned with property, inheritance, and contracts. Proceedings in civil law are undertaken by individuals or groups who believe that they have suffered a breach of the civil law, in contrast to criminal law proceedings, which are undertaken on behalf of the state.

Psychiatrists are sometimes asked to submit written reports on a patient's mental state in relation to a civil case. Such reports should be prepared only after full discussion with the patient, and only with the patient's informed consent, and they should be concise and factual. The structure should follow the plan set out on pages 418, 518 (Box 18.10). It is advisable to seek legal advice about the relevant aspects of law relating to the case. The following sections are based on the law in England and Wales, although the principles apply more widely.

Testamentary capacity

This term refers to the capacity to make a valid will. If someone is suffering from mental disorder at the time of making a will, its validity may be in doubt and other people may challenge it. The will may still be legally valid if the testator was of ‘sound disposing mind’ (see below) at the time of making it. Psychiatrists may be asked to report in relation to two issues:

1. Testamentary capacity.
2. The possibility that the testator was subjected to undue influence.

To decide whether or not a testator is of sound disposing mind, the doctor should decide whether the person making the will:

- Understands what a will is and its consequences;
- Knows the nature and extent of his property (not necessarily in detail);
- Knows the names of close relatives and can assess their claims to his property; and
- *Is free from an abnormal state of mind* that might distort feelings or judgements relevant to making the will. A deluded person may legitimately make a will, provided that the delusions are unlikely to influence it.

To decide these matters, the doctor should first interview the testator alone and then interview relatives or friends to check the accuracy of factual statements.

Assessment of undue influence is more complex, and requires an assessment of the relationship between testator and beneficiary, the mental state of the testator, and what is known of the person's earlier intentions.

Power of attorney and receivership

If a patient is incapable of managing their possessions by reason of mental disorder, alternative arrangements need to be made, particularly if the incapacity is likely to last for a long time. Such arrangements may be required for patients who are living in the community as well as for those who are in hospital. In English law two methods are available—power of attorney and receivership.

Power of attorney is the simpler method, and requires only that the patient gives written authorization for someone else to act for them during their illness. In signing such authorization, the patient must be able

to understand what they are doing. They may revoke it at any time.

Receivership is the more formal procedure, and is more likely to be in the patient's interests. In England and Wales an application is made to the Court of Protection, which may decide to appoint a receiver. The procedure is most commonly required for the elderly. The question of receivership places special responsibility on the psychiatrist. If a patient is capable of managing their affairs on admission to hospital, but later becomes incapable by reason of intellectual deterioration, then it is the doctor's duty to advise the patient's relatives about the risks to property. If the relatives are unwilling to act, then it is the doctor's duty to make an application to the Court of Protection. The doctor may feel reluctant to act in this way, but any actions taken subsequently are the Court's responsibility and not that of the doctor.

Aspects of family law

A *marriage contract* is not valid if, at the time of the marriage, either party was so mentally disordered as not to understand its nature. If mental disorder of this degree can be proved, a marriage may be decreed null and void by a divorce court. If a marriage partner becomes of ‘incurably unsound mind’ later in a marriage, this may be grounds for divorce, and a psychiatrist may be asked to give a prognosis. A doctor may also be asked for an opinion about the *capacity of parents or a guardian* to care adequately for a child.

Torts and contracts

Torts are wrongs for which a person is liable in civil law as opposed to criminal law. They include negligence, libel, slander, trespass, and nuisance. If such a wrong is committed by a person of unsound mind, then any damages awarded in a court of law are usually only nominal. In this context the legal definition of unsound mind is restrictive, and it is advisable for a psychiatrist to take advice about it from a lawyer.

If the person makes a contract and subsequently develops a mental disorder, the contract remains binding. If a person is of unsound mind when the contract is made, in English law a distinction is made between the ‘necessaries’ and ‘non-necessaries’ of life. A contract made for necessaries is always binding and that for *non-necessaries* is binding unless it can be shown both that the person did not understand what they were doing and that the other person was aware of the incapacity.

Personal injury

Psychiatrists may be asked to write reports in relation to claims for compensation by patients with post-traumatic stress disorder or one of the other psychological sequelae of accidents (see pages 143, 146). Such reports should be set out to accord with the relevant local legal procedures and should state clearly the sources of information, the history of the trauma, the psychiatric and social history, and the post-accident course. They should include a detailed assessment of function and of the relationship between the trauma and any subsequent symptoms and disability.

Fitness to drive

Questions of fitness to drive arise in relation to many psychiatric disorders. Dangerous driving may result from suicidal inclinations or manic disinhibition, panicky or aggressive driving may result from persecutory delusions, and indecisive or inaccurate driving may be due to dementia. Concentration on driving may be impaired in severe anxiety or depressive disorders. The question

of fitness to drive also arises in relation to the sedative and other side effects of some psychiatric drugs, such as some anxiolytic or antipsychotic drugs in high dosage.

Doctors should be aware of and follow the legal criteria for fitness to drive in the places in which they are working. These criteria may differ somewhat for drivers of cars and drivers of heavy goods vehicles (the latter are stricter). UK holders of driving licences have a duty to inform the Driver and Vehicle Licensing Agency (DVLA) if they have a condition that may affect their safety as a driver. Doctors should inform patients if they have such a condition, and make sure that they understand their duty to report it. When deciding on a patient's fitness to drive, doctors should consider whether the patient's condition or its treatment is liable to cause loss of control, impair perception or comprehension, impair judgement, reduce concentration, or affect motor functions involved in handling the vehicle. If, after full discussion, a patient with such a condition continues to drive, a doctor working in the UK should disclose the relevant medical information to the medical adviser of the DVLA.

Further reading

In many countries, national medical and psychiatric organizations publish guidelines about the ethical and legal aspects of practice.

Bloch, S., & Green, S. (Eds.). (2021). *Psychiatric Ethics*, 5th edn. Oxford University Press. (A series of reviews by leading writers covering the principal theoretical issues.)

Wilkinson, D., Herring, J., & Savulescu, J. (2019) *Medical Ethics and Law. A Curriculum for the 21st Century*. Elsevier.

Montgomery, J. (2002) *Health Care Law*, 2nd edn. Oxford University Press. (A useful work of reference.)

CHAPTER 5

Aetiology

Approaches to aetiology in psychiatry 89

General issues relating to aetiology 89

Aetiological models 92

The historical development of ideas of aetiology 96

The contribution of scientific disciplines
to psychiatric aetiology 98

Relationship of this chapter to those on
psychiatric syndromes 118

Approaches to aetiology in psychiatry

Psychiatrists are concerned with aetiology in two ways. First, in everyday clinical work they try to discover the causes of the mental disorders presented by individual patients. Secondly, in seeking a wider understanding of psychiatry they are interested in aetiological evidence obtained from different kinds of research, such as

clinical studies, community surveys, or laboratory investigations. Correspondingly, the first part of this chapter deals with some general issues relating to aetiology in the assessment of the individual patient, while the second part deals with the various scientific disciplines that have been applied to the study of aetiology.

General issues relating to aetiology

Aetiology and intuitive understanding

When the clinician assesses an individual patient, they draw on a common fund of aetiological knowledge that has been derived from the study of groups of similar patients, but they cannot understand the patient in these terms alone. They must also use everyday insights into human nature. For example, when assessing a depressed patient, the psychiatrist should certainly know what has been discovered about the psychological and neurochemical changes that accompany depressive disorders,

and what evidence there is about the aetiological role of stressful events, and about genetic predisposition to depressive disorder. At the same time they will need intuitive understanding to recognize, for example, that this particular patient apparently feels depressed because they have failed to gain a promotion at work.

Common-sense ideas of this kind are an important part of aetiological formulation in psychiatry, but they must be used carefully if superficial explanation is to be avoided. Aetiological formulation can be done properly only if certain conceptual problems are clearly understood. These problems can be illustrated by a case history.

For four weeks a 38-year-old woman became increasingly depressed. Her symptoms started soon after her partner left her, saying that he wanted to live by himself. In the past the patient's mother had received psychiatric treatment on two occasions, once for a severe depressive disorder and once for mania; on neither occasion was there any apparent environmental cause for the illness. When the patient was 14 years old, her father went away to live with another woman, leaving his children with their mother. For several years afterwards the patient felt rejected and unhappy, but she eventually settled down. She later married, and she had two children, aged 13 and 10 years, at the time of her illness. Two weeks after leaving home the patient's partner returned, saying that he had made a mistake and really loved his wife. Following his return the patient's symptoms persisted and worsened. She began to wake early, gave up her usual activities, and spoke at times of suicide.

When thinking about the causes of this person's experience and life story, the clinician attends to symptoms and draws on knowledge of aetiology derived from scientific enquiries. Genetic investigations have shown that, if a parent suffers from mania as well as depressive disorder, a predisposition to mood disorder is particularly likely to be transmitted to the children. Therefore it is possible that this patient received this predisposition from her mother.

Clinical investigation has also provided some information about the effects of separating children from their parents. In the present case, the information is not helpful because it refers to people who were separated from their parents at a younger age than the patient. On scientific grounds, therefore there is no particular reason to focus on the departure of the patient's father, but *intuitively* it seems very likely that this was an important event.

From everyday experience it is understandable that a person should feel very upset if their partner leaves them. It is also understandable that they are likely to feel even more distressed if this event recapitulates a related distressing experience in their own childhood. Therefore the clinician would recognize intuitively that the patient's depression is likely to be a reaction to the partner's departure and having to manage not only the complexity of her emotions but also the everyday adjustments now needing to take place in managing work, a household, and children alone. The same sort of intuition might suggest that the patient would feel somewhat better when her husband came back, but in the event she did not recover.

This simple case history illustrates some important aetiological issues in psychiatry:

- the complexity of causes
- the classification of causes
- the concept of stress
- the concept of psychological reaction
- the relative roles of intuition and scientific knowledge in aetiological formulations.

The complexity of causes in psychiatry

In psychiatry, the study of causation is complicated by three problems. These problems can be encountered in other branches of medicine, but usually to a lesser degree.

Lack of temporal association

The first problem is that causes are often *remote in time* from the effects that they produce. For example, it is widely believed that childhood experiences partly determine the occurrence of emotional difficulties in adult life. It has been difficult to test this idea, because the necessary information can only be gathered either by studying children and tracing them many years later, which is difficult, or by asking adults about their childhood experiences, which is unreliable.

Cause and effect

The second problem is that a single cause may lead to *several effects*. For example, being deprived of parental affection in childhood has been reported to predispose to antisocial behaviour, suicide, depressive disorder, and several other disorders. Conversely, a single effect may arise from several causes. The latter can be illustrated either by different causes in different individuals or by multiple causes in a single individual. For example, learning disability (a single effect) may occur in several children, but the cause may be a different genetic abnormality or environmental stressor in each child. On the other hand, depressive disorder (a single effect) may occur in one individual through a combination of causes, such as genetic factors, adverse childhood experiences, and stressful events in adult life.

Indirect mechanisms

The third problem is that aetiological factors in psychiatry rarely exert their effects directly. For example, the genetic predisposition to depression may be mediated in part through temperamental factors that make it more likely that the individual concerned will experience adverse life events. Thus aetiological effects are usually

mediated through complex intervening mechanisms, which also need to be investigated and understood.

The classification of causes

A single psychiatric disorder, as just explained, may result from several causes. For this reason a scheme for *classifying* causes is required. A useful approach is to divide causes chronologically into those that are *predisposing*, *precipitating*, and *maintaining*.

Predisposing factors

These are factors, many of them operating from early in life, that determine a person's vulnerability to causal factors acting close to the time of the illness. They include *genetic endowment* and the *environment in utero*, as well as *physical, psychological, and social factors in infancy and early childhood*. The term 'constitution' is often used to describe the mental and physical make-up of a person at any point in their life. This make-up changes as life goes on under the influence of further physical, psychological, and social influences. Some writers restrict the term constitution to the make-up at the beginning of life, while others also include characteristics that are acquired later (this second usage is adopted in this book). The concept of constitution includes the idea that a person may have a predisposition to develop a disorder (e.g. schizophrenia) even though the latter never manifests itself. From the standpoint of psychiatric aetiology, one of the important parts of the constitution is the *personality*.

When the aetiology of an individual case is formulated, the personality is always an essential element. For this reason, the clinician should be prepared to spend sufficient time talking to the patient and to people who know them to build up a clear picture of their personality. This assessment often helps to explain why the patient responded to certain stressful events, and why they reacted in a particular way. The obvious importance of personality in the individual patient contrasts with the small amount of relevant scientific information so far available. Therefore, when evaluating personality it is particularly important to acquire sound clinical skills through supervised practice.

Precipitating factors

These are events that occur shortly before the onset of a disorder, and which *appear to have induced it*. They may be physical, psychological, or social. Whether they produce a disorder at all, and what kind of disorder, depends partly on constitutional factors in the patient (as mentioned above). Examples of physical precipitants

include cerebral tumours and drugs. Psychological and social precipitants include personal misfortunes such as the loss of a job or an important relationship, and changes in the routine of life, such as moving home. Sometimes the same factor can act in more than one way. For example, a head injury can induce psychological disorder either through physical changes in the brain or through its stressful implications for the patient.

Maintaining factors

These factors *prolong the course of a disorder* after it has been provoked. When planning treatment, it is particularly important to pay attention to these factors. The original predisposing and precipitating factors may have ceased to act by the time the patient is seen, but the maintaining factors may well be treatable. For example, in their early stages many psychiatric disorders lead to secondary demoralization and withdrawal from social activities, which in turn help to prolong the original disorder. It is often appropriate to treat these secondary factors, whether or not any other specific measures are carried out. Maintaining factors are also called *perpetuating factors*.

The concept of stress

Discussions about stress are often confusing because the term is used in two ways. First, it is applied to events or situations, such as working for an examination, which may have an adverse effect on someone. Secondly, it is applied to the adverse effects that are induced, which may involve psychological or physiological change. When considering aetiology it is advisable to separate these components.

The first set of factors can usefully be called *stressors*. They include a large number of physical, psychological, and social factors that can produce adverse effects. The term is sometimes extended to include events that are not experienced as adverse at the time, but which may still have adverse long-term effects. For example, intense competition may produce an immediate feeling of pleasant tension, although it may sometimes lead to unfavourable long-term effects.

The effect on the person can usually be called the *stress reaction*, to distinguish it from the provoking events. This reaction includes *autonomic responses* (e.g. a rise in blood pressure), *endocrine changes* (e.g. the secretion of adrenaline and noradrenaline), and *psychological responses* (e.g. a feeling of being keyed up). Much current research is involved in studying the effects of stressors on the brain, and in particular how stressors affect the

mechanisms involved in the regulation of mood and processing of emotional information (see Chapter 7).

The concept of a psychological reaction

As already mentioned, it is widely recognized that psychological distress can occur as a reaction to unpleasant events. Sometimes the association between event and distress is evident—for example, when a person becomes depressed after the death of their partner. In other cases it is far from clear whether the psychological disorder is really a reaction to an event, or whether the two have coincided fortuitously—for example, when a person becomes depressed after the death of a distant relative.

Jaspers (1963, p. 392) suggested three criteria for deciding whether a psychological state is a reaction to a particular set of events:

1. The events must be adequate in severity and closely related in time to the onset of the psychological state.
2. There must be a clear connection between the nature of the events and the content of the psychological disorder (in the example just given, the person should be preoccupied with ideas concerning their distant relative).
3. The psychological state should begin to disappear when the events have ceased (unless, of course, it can be shown that perpetuating factors are acting to maintain it).

These three criteria are useful in clinical practice, although they can be difficult to apply (particularly the second criterion).

Understanding and explanation

As already mentioned, aetiological statements about individual patients must combine knowledge derived

from research on groups of patients with intuitive understanding derived from everyday experience. Jaspers (1963, page 302) has called these two ways of making sense of psychiatric disorders ‘Erklären’ and ‘Verstehen’, respectively.

In German, these terms mean ‘explanation’ and ‘understanding’, respectively, and they are usually translated as such in English translations of Jaspers’ writing. However, Jaspers used them in a special sense. He used ‘Erklären’ to refer to the sort of causative statement that is sought in the natural sciences. It is exemplified by the statement that a patient’s aggressive behaviour has occurred because he has a brain tumour. He used ‘Verstehen’ to refer to psychological understanding, or the intuitive grasp of a natural connection between events in a person’s life and his psychological state. In colloquial English, this could be called ‘putting oneself in another person’s shoes’. It is exemplified by the statement ‘I can understand why that person became angry when their children were shouted at by a neighbour.’

These distinctions are reasonably clear when we consider an individual patient, but confusion sometimes arises when attempts are made to generalize from insights obtained in a single case to widely applicable principles. Understanding may then be mistaken for explanation. Jaspers suggested that some psychoanalytical ideas are special kinds of intuitive understanding that are derived from the detailed study of individuals and then applied generally. They are not explanations that can be tested scientifically. They are more akin to insights into human nature that can be gained from reading great works of literature. Such insights are of great value in conducting human affairs. It would be wrong to neglect them in psychiatry, but equally wrong to see them as statements of a scientific kind.

Aetiological models

Before considering the contribution that different scientific disciplines can make to psychiatric aetiology, attention needs to be given to the kinds of aetiological model that have been employed in psychiatry. A model is a device for ordering information. Like a theory, it seeks to explain certain phenomena, but it does so in a broad and comprehensive way that cannot readily be proved false.

Reductionist and non-reductionist models

Two broad categories of explanatory model can be recognized. *Reductionist models* seek to understand causation by tracing back to increasingly simple early stages. Examples include the ‘narrow’ medical model, which is described below, and the psychoanalytic model. This

type of model can be exemplified by the statement that the cause of schizophrenia lies in disordered dopamine neurotransmission in a specific area of the brain.

Non-reductionist models try to relate problems to wider rather than narrower issues. The explanatory models that are used in sociology are generally of this kind. In psychiatry, this type of model can be exemplified by the statement that the cause of a patient's depression lies in their family; the patient is the most conspicuous element in a disordered group of people. In the same way it can be asserted that certain depressive states are associated with indices of social deprivation and isolation, and can be best understood as being caused by these factors.

The neuroscience approach

The technical and conceptual advances in brain sciences have led to what is often called the *neuroscience approach*. Kandel (1998) outlined the key assumptions underlying this approach to aetiology, which can be summarized as follows.

- All mental processes derive from operations of the brain. Thus all behavioural disorders are ultimately disturbances of brain function, even where the original 'cause' is clearly environmental.
- Genes have important effects on brain function and therefore exert significant control over behaviour.
- Social and behavioural factors exert their effects on the brain in part through changes in gene expression (epigenetics). Changes in gene expression leading to altered patterns of synaptic connectivity underlie the ability of experiences to produce learning and psychotherapy to change behaviour.

The latter concept derives from the ability of a wide range of environmental stimuli to modulate gene expression by various mechanisms, including epigenetic modifications (see Epigenetics). Thus while genes coding for particular proteins are inherited, environmental and developmental influences are involved in determining whether and to what extent a particular gene is expressed. This provides a plausible mechanism by which nature and nurture interact in the production of a behavioural phenotype.

The neuroscience approach therefore seeks to comprehend the role of social, family, and personal factors in behaviour by relating them to changes in brain function. For example, in understanding the effect of childhood neglect on liability to adult depression, it may be important to find out how adverse childhood experiences might alter relevant brain mechanisms (e.g. the

endocrine response to stress), and how this abnormality might predispose to depression when the individual is exposed to difficulties in adulthood. Thus, although a neuroscience approach encompasses the importance of social and personal factors, it seeks to understand their consequences in a *reductionist* way.

Medical models

Several models are used in psychiatric aetiology, but the so-called *medical model* is the most prominent one. It represents a general strategy of research that has proved useful in medicine, particularly in studying infectious diseases. A disease entity is identified in terms of a consistent pattern of symptoms, a characteristic clinical course, and specific biochemical and pathological findings (see Chapter 2 regarding models of disease). When an entity has been identified in this way, a set of necessary and sufficient causes is sought. In the case of tuberculosis, for example, the tubercle bacillus is the necessary cause, but it is not by itself sufficient. However, the tubercle bacillus in conjunction with either poor nutrition or low resistance is sufficient cause.

The importance of personal and social factors in the presentation and course of illness is well recognized in general medicine, and modern medical models are therefore considerably broader than those based on the elucidation of the mechanism of infectious disease. Modern medical models also recognize that much illness is characterized by quantitative rather than qualitative deviations from normal (e.g. high blood pressure). This also applies to certain disorders in psychiatry, particularly anxiety and milder depressive disorders, which can therefore be accommodated in a broad medical model.

Difficulties with the medical model arise, particularly in relation to disorders that are characterized principally by *abnormalities of conduct and social behaviour*, such as antisocial behaviour and substance misuse. As mentioned above, current neuroscience approaches would seek to understand these disorders through changes in relevant brain systems. This is because causal factors in abnormal social behaviour, such as childhood maltreatment and deprivation, must ultimately express their effects on behaviour through changes in brain mechanisms.

Although the latter view appears theoretically valid, the key decision for both clinician and policymaker is at what level the disorder is best understood and managed. For example, it is possible to understand problems in substance misuse as arising from a defect in brain reward systems which, in a vulnerable individual, results in sporadic experimentation with illicit substances leading to

substance misuse, with adverse personal and social consequences. Equally, one can see excessive drug misuse in society as a ‘symptom’ of social deprivation and family disruption (see Chapter 20). Both aetiological models may be valid, but each would suggest different forms of research activity and therapeutic intervention.

These ideas are brought together on the *biopsychosocial* model, first developed by George Engel in 1977. This proposes that for a proper understanding of psychiatric disorders it is necessary to consider biological, psychological, and sociological factors as well as the interactions between them (Bolton, 2022). This is particularly applicable to the formulation of individual cases and their management.

The behavioural model

As explained above, certain disorders that psychiatrists treat, particularly those defined in terms of *abnormal behaviour*, do not fit readily into the medical model. The latter include deliberate self-harm, the misuse of drugs and alcohol, and repeated acts of delinquency. The behavioural model is an alternative way of comprehending these disorders. In this model the disorders are explained in terms of factors that determine normal behaviour. These include drives, reinforcements, social and cultural influences, and internal psychological processes such as attitudes, beliefs, and expectations. The behavioural model predicts that there will not be a sharp distinction between the normal and the abnormal, but a continuous gradation. This model can therefore be a useful way of considering many of the conditions that are seen by psychiatrists.

Although the behavioural model is mainly concerned with psychological and social causes, it does not negate the importance of genetic, physiological, or biochemical causes. This is because normal patterns of behaviour are partly determined by genetic factors, and because psychological factors such as reinforcement have a basis in physiological and biochemical mechanisms. Also, the behavioural model employs both *reductionist* and *non-reductionist* explanations. For example, abnormalities of behaviour can be explained in terms of abnormal conditioning (a reductionist model) or in terms of a network of social influences.

Developmental models

Medical and behavioural models incorporate the idea of predisposing as well as precipitating causes (i.e. the idea that past events may determine whether or not a current cause gives rise to a disorder). Some models place

even more emphasis on past events in the form of a sequence of experiences leading to the present disorder. This approach has been called the ‘life story’ approach to aetiology. One example is Freud’s psychoanalysis (see Box 5.1), and another is Meyer’s psychobiology. These ideas are considered further below.

Political models (‘antipsychiatry’, ‘critical psychiatry’)

The models outlined above rely on a scientific approach to psychiatric aetiology. This implies that psychiatric disorders, like other medical conditions, can be studied and understood in an objective and empirical way using the methods of natural sciences. In the history of psychiatry this view has often been regarded as far from self-evident, and other conceptual frameworks have been advocated. For example, it was customary in the Middle Ages to explain mental illness in terms of demonic possession and witchcraft (see below). Over the past 50 years, however, criticisms of scientific approaches to aetiology have most often taken the view that psychiatric illness is defined by *social and political imperatives* and represents at best a cultural value judgement and at worst an abusive means of controlling those whom society finds inconvenient or troubling.

Arguments of this nature were put forwards strongly by the French philosopher, Michel Foucault (1926–1984), and were further developed by psychiatrists such as R. D. Laing (1927–1989), who employed a phenomenological approach to argue that schizophrenia is an understandable response of an individual to a culture of exploitation and alienation. Lack of faith in a scientific approach to mental experience was exemplified by the psychiatrist Thomas Szasz (1920–2012), who commented that ‘There is no psychology; there is only biography and autobiography.’ These ideas are sometimes called ‘antipsychiatry’ to emphasize their fundamental contrast with the natural science approaches employed by conventional psychiatry.

Most psychiatrists do not believe that these formulations have advanced the understanding of mental illness, and in fact provide rather poor explanations of the range of clinical psychopathology. For example, they seem unable to account for the existence of schizophrenia in all human societies (see Chapter 11). However, *political and social perceptions* of mental illness are important because they have powerful effects on the way that mentally ill people are treated, and the services provided (see Chapter 23).

Box 5.1 Psychoanalysis

Psychoanalysis is derived from clinical experience rather than from basic sciences, and consists of elaborate theories of both normal and abnormal mental development. Compared with experimental psychology it is more concerned with *unconscious* aspects of mental activity.

Psychoanalytic theories are derived from data obtained in the course of psychoanalytic treatment, and relate to the patient's thoughts, dreams, and fantasies, as well as their memories of childhood experiences and relationships with carers and siblings. Therefore psychoanalytic theories are forms of intuitive understanding ('*Verstehen*') rather than scientific explanation ('*Erklären*'). Subsequently some theories derived from psychoanalytic thinking have been supported by scientific enquiry, for example, the role of sexual trauma in psychosis, while others seem outdated and difficult to locate in contemporary psychiatric practice.

Freud developed a number of ideas about the *unconscious mind*, from which he believed all mental processes originated. The unconscious mind has three key features that are important in the genesis of psychological problems.

1. It is inaccessible to conscious awareness and may seem unrelated to real events or conscious beliefs.
2. It is dynamic and contains powerful instinctual forces that are repressed but might break through into conscious awareness in therapy or stressful conditions.
3. Unconscious processes may lead to behaviours (enactments) that seem to not be helpful and do not align with the persons' conscious wishes or self-image

Freud believed that the unconscious mind could be probed by *dream analysis* and *free association*. The manifest content of a dream can be traced back to a latent content, which is an infantile wish. This wish is disguised by *dreamwork*, which, through various psychological mechanisms, transforms the latent content into a manifest form that is more acceptable to consciousness.

The unconscious mind, according to Freud, is the seat of powerful instinctual drives (e.g. sexual and aggressive impulses). The forces in the unconscious mind struggle

against the conscious mind, which can give rise to anxiety. This anxiety can be reduced by a variety of *defence mechanisms*, which can be discerned in the behaviour of healthy people as well as those with psychological disorders (see Chapter 24).

Freud described sexual impulses as being present from the earliest stages of child development, with progression through different stages of organization (oral, anal, and genital) as the child grows. Failure to pass through these stages effectively can lead to sexual energy (*libido*) becoming fixated, which results in the individual exhibiting infantile patterns of behaviour or regressing to them when under stress. The object of libido also changes with development, first focusing on the self and then focusing on the mother. In the course of this maternal attachment, boys experience angry feelings towards their father (the *Oedipus complex*), while girls develop reverse attachments. Difficulties in resolving these conflicted attachments can give rise to problems in interpersonal relationships later in life.

Freud's theories were further developed by many others. Melanie Klein's work, which addresses pre-Oedipal aggressive and sexual drives in a theory of *object relations*, has been influential in psychoanalytic practice in the United Kingdom.

While formal psychoanalysis has little direct influence on the practice of general psychiatry, psychoanalytic concepts promote and facilitate reflection on unconscious processes, and how to relate to contradictions and paradoxes in the ways people present and talk about themselves. These can be especially helpful in understanding people presenting with personality difficulties.

Psychoanalytic models of treatment are mostly provided in private practice or in some specialist referral units. Nonetheless, the basic concepts of psychoanalysis can help psychiatrists and care teams work with complex and challenging patient and their families. Indeed, anyone in the care system might be impacted by vicarious trauma and unconscious processes that can lead to stigmatizing or negative reactions towards patients.

There is no doubt that horrifying abuse of psychiatric patients has occurred, as demonstrated by the cooperation of many leading German psychiatrists with the euthanasia programmes of the Nazi regime (Torrey and Yolken, 2010). While this is an extreme and abhorrent

example, political analyses of psychiatric practice highlight the need for the *rights of patients to be respected*, and their experiences understood in a personal and social context as outlined above. Particular sensitivity is required when treating patients under Mental Health

Act legislation where careful balancing of complex ethical issues is mandatory (Chieze et al., 2021). For a

discussion of the ‘critical psychiatry’ approach to mental illness, see Middleton and Moncrieff (2019).

The historical development of ideas of aetiology

From the earliest times, theories of the causation of mental disorder have recognized both *somatic* and *psychological* influences. Greek medical literature referred to the causes of mental disorders, mainly in the Hippocratic writings (fourth century BCE). Serious mental illness was ascribed mainly to physical causes, which were represented in the theory that health depended on a correct balance of the four body ‘*humours*’ (blood, phlegm, yellow bile, and black bile). Melancholia was ascribed to an excess of black bile. Most of the less severe psychiatric disorders were thought to have supernatural causes and to require religious healing. An exception was hysteria, which was thought to be physically caused by the displacement of the uterus from its normal position.

Roman physicians generally accepted the causal theories of Greek medicine, and developed them in some respects. Galen accepted that melancholia was caused by an excess of black bile, but suggested that this excess could result either from cooling of the blood or from overheating of yellow bile. Phrenitis, the name given to an acute febrile condition with delirium, was thought to result from an excess of yellow bile.

Throughout the Middle Ages these early ideas about the causes of mental illness were largely neglected, although they were maintained by some scholars, such as Bartholomeus Anglicus. The causes of mental illness were now formulated in theological terms of sin and evil, with the consequence that many mentally ill people were persecuted as witches. It was not until the middle of the sixteenth century that beliefs in the supernatural and witchcraft were strongly rejected as causes of mental disorder, notably by the Flemish writer Johan Weyer (1515–1588) in his book *De Praestigiis Daemonum*, published in 1563. Earlier, the renowned physician Paracelsus (1493–1541) had emphasized the natural causes of mental illness.

In the seventeenth and eighteenth centuries, a more scientific approach to the causation of mental illness developed as physicians became interested in mental disorders, mainly hysteria and melancholia. The English physician Thomas Willis attributed melancholia to ‘passions of the heart’, but considered that madness (illnesses with thought disorder, delusions, and hallucinations) was due to a ‘fault of the brain’. Willis realized

that this fault was not a recognizable gross structural lesion, but a functional abnormality. In the terminology of the time, he referred to a disorder of the ‘vital spirits’ that were thought to account for nervous action. He also pointed out that hysteria could not be caused by a displacement of the womb, because the organ is firmly secured in the pelvis.

Despite this renewed medical interest in the causes of mental disorder, the most influential seventeenth-century treatise was written by a clergyman, Robert Burton (1577–1640). This work, *The Anatomy of Melancholy*, published in 1621, described in detail the psychological and social causes (such as poverty, fear, and solitude) that were associated with melancholia and seemed to cause it.

Aetiology depends on *nosology*. Unless it is clear how the various types of mental disorder relate to one another, little progress can be made in understanding causation. From his observations of patients with psychiatric disorders, the Italian physician Giovanni Battista Morgagni (1682–1771) became convinced that there was not one single kind of madness, but many different ones; further attempts at classification followed. One of the best known was proposed by William Cullen, who included a category of neurosis for disorders not caused by localized disease of the nervous system.

The idea that individual mental disorders are caused by lesions of particular brain areas can be traced back to the theory of phrenology proposed by Franz Gall (1758–1828) and his pupil Johann Spurzheim (1776–1832). Gall proposed that the brain was the organ of the mind, that the mind was made up of specific faculties, and that these faculties originated in specific brain areas. He also proposed that the size of a brain area determined the strength of the faculty that resided in it, and that the size of brain areas was reflected in the contours of the overlying skull. Thus the shape of the head reflected a person’s psychological make-up. Although the last steps in Gall’s argument were erroneous, the ideas of cerebral localization were to develop further. An increased interest in brain pathology led to theories that different forms of mental disorder were associated with lesions in different parts of the brain.

It had long been observed that serious mental illness ran in families, but in the nineteenth century this idea took a new form. Benedict Augustin Morel (1809–1873), a French psychiatrist, put forward ideas that became known as the ‘theory of degeneration’. He proposed not only that some mental illnesses were inherited, but also that environmental influences (such as poor living conditions and the misuse of alcohol) could lead to physical changes that could be transmitted to the next generation. Morel also proposed that, as a result of successive effect of environmental agents in each generation, illnesses appeared in increasingly severe forms in successive generations. It was inherent in these ideas that mental disorders did not differ in kind but only in *severity*, with neuroses, psychoses, and mental handicap being increasingly severe manifestations of the same inherited process.

These ideas were consistent with the accepted theories of the inheritance of acquired characteristics, and they were widely accepted. They had the unfortunate effect of encouraging a pessimistic approach to treatment. They also supported the Eugenics Movement, which held that the mentally ill should be removed from society to prevent them from reproducing. These developments are an important reminder that *aetiological theories may give rise to undesirable attitudes to the care of patients*.

Mid-nineteenth-century views of the causation of mental illness can be judged from the widely acclaimed textbooks of Jean-Étienne Esquirol (1772–1840), a French psychiatrist, and of Wilhelm Griesinger (1817–1868), a German psychiatrist. Esquirol focused on the causes of illness in the individual patient, and was less concerned with general theories of aetiology. He recorded psychological and physical factors, which he believed to be significant in individual cases, and he distinguished between predisposing and precipitating causes. He regarded heredity as the most important of the predisposing causes, but he also stressed that predisposition was acted on by psychological causes and by social (at that time called ‘moral’) causes such as domestic problems, ‘disappointed love’, and reverses of fortune. Important physical causes of mental disorder included epilepsy, alcohol misuse, excessive masturbation, childbirth and lactation, and suppression of menstruation. Esquirol also observed that age influenced the type of illness. Thus dementia was not observed among the young, but mania was uncommon in old age. He recognized that personality was often a predisposing factor.

In *Pathology and Therapy of Mental Disorders*, Griesinger maintained that mental illness was a *physical disorder of the brain*, and he considered at length

the neuropathology of mental illness. He paid equal attention to other causes, including heredity, habitual drunkenness, ‘domestic unquiet’, disappointed love, and childbirth. He emphasized the multiplicity of causes when he wrote:

A closer examination of the aetiology of insanity soon shows that in the great majority of cases it was not a single specific cause under the influence of which the disease was finally established, but a complication of several, sometimes numerous causes, both predisposing and exciting. Very often the germs of the disease are laid in those early periods of life from which the commencement of the formation of character dates. It grows by education and external influences (Griesinger, 1867).

British views on aetiology in the late nineteenth century can be judged from *A Manual of Psychological Medicine* by Bucknill and Tuke, published in 1858, and from *The Pathology of Mind* by Henry Maudsley, published in 1879. Maudsley described the causes of mental disorder in terms similar to those of Griesinger. Thus causes were multiple, while *predisposing causes* (including heredity and early upbringing) were as important as the more obvious *proximal causes*. Maudsley held that mistakes in determining causes were often due to ‘some single prominent event, which was perhaps one in a chain of events, being selected as fitted by itself to explain the catastrophe. The truth is that in the great majority of cases there has been a concurrence of steadily operating conditions within and without, not a single effective cause’ (Maudsley, 1879, page 83).

Although these nineteenth-century writers and teachers of psychiatry emphasized the *multiplicity of causes*, many practitioners focused narrowly on the findings of genetic and pathological investigations, and adopted a pessimistic approach to treatment. However, Adolf Meyer (1866–1950), a Swiss psychiatrist who worked mainly in the United States, emphasized the role of psychological and social factors in the aetiology of psychiatric disorder. Meyer applied the term *psychobiology* to this approach, in which a wide range of previous experiences was considered and then common-sense judgements used to decide which experiences might have led to the present disorder. Meyer acknowledged the importance of heredity and brain disorder, but emphasized that these factors were modified by life experiences, which often determined whether or not a particular disorder would be clinically expressed. Meyer’s approach remains the basis of the evaluation of aetiology for the individual patient.

The aetiological theories considered so far have been mainly concerned with the major mental illnesses.

Less severe disorders, particularly those that came to be called neurosis, hysteria, and hypochondriasis, and milder states of depression, were treated mainly by physicians. Pierre Charcot, a French neurologist, carried out extensive studies of patients with hysteria and of their response to hypnosis. He believed that hysteria resulted from a functional disorder of the brain and could be treated by hypnosis. In the United States, Weir Mitchell proposed that conditions akin to mild chronic depression were due to exhaustion of the nervous system—a condition he called *neurasthenia*.

In Austria, another neurologist, Sigmund Freud, tried to develop a more comprehensive explanation of nervous diseases, first of hysteria and then of other conditions. After an initial interest in physiological causes, Freud proposed that the causes were psychological, but hidden from the patient because they were in the unconscious part of the mind. Freud took a developmental approach to aetiology, believing that the seeds of adult disorder lay in the process of child development (see Box 5.1).

Psychoanalysis became increasingly influential, particularly in American psychiatry, where it predominated

until the 1970s. Since that time there has been renewed interest in genetic, biochemical, and neuropathological causes of mental disorder—an approach that has become known as *biological psychiatry* (Guze, 1989).

Perhaps the most important lesson to learn from this brief overview of the history of ideas on the causation of mental disorder is that each generation bases its theories of aetiology on the scientific approaches that are most active and plausible at the time. Sometimes psychological ideas prevail, sometimes neuropathological ones, and sometimes genetic ones. Throughout the centuries, however, observant clinicians have been aware of the *complexity of the causes of psychiatric disorders*, and have recognized that neither aetiology nor treatment should focus narrowly on the scientific ideas of the day. Instead, the approach should be broader, encompassing whatever psychological, social, and biological factors seem to be most important in the individual person. Modern psychiatrists are working in an era of rapid development of the neurosciences, but they need to keep the same broad clinical perspective of aetiology while assimilating any real scientific advances.

The contribution of scientific disciplines to psychiatric aetiology

The main groups of disciplines that have contributed to the knowledge of psychiatric aetiology are shown in Box 5.2. In this section each group is discussed in turn, and the following questions are asked:

- What sort of problem in psychiatric aetiology can be answered by each discipline?

Box 5.2 Scientific disciplines that contribute to psychiatric aetiology

Clinical descriptive studies
Epidemiology
Social sciences
Clinical psychology
Neuropsychology (cognitive neuroscience)
Genetics
Biochemical studies
Pharmacology
Endocrinology
Physiology
Neuropathology

- How, in general, does each discipline attempt to answer the questions?
- Are any particular difficulties encountered when applying its methods to psychiatric disorders?

Clinical descriptive studies

Before reviewing more elaborate scientific approaches to aetiology, attention is drawn to the continuing value of simple *clinical investigations*. Psychiatry was built on case studies, such as the seminal series by Aubrey Lewis on ‘melancholia’ (Lewis, 1934).

Another good example of this approach is the study conducted by Judd and colleagues (2002) who assessed the natural history of bipolar disorder by prospectively following up 146 patients with bipolar I disorder over a mean of 12 years. Their work showed that bipolar disorder is not, as has often been thought, an illness of discrete episodes of mood disturbance followed by clinical remission. Instead, despite treatment, the patients were symptomatic almost half the time. The majority of this morbidity was attributable to depressive symptomatology. This work stimulated the search for better treatments of depression in bipolar disorder (see Chapter 10).

Clinical cohort studies are playing an important role in elucidating the psychiatric and neuropsychiatric sequelae of the Covid-19 pandemic, where follow up studies have revealed elevated rates of fatigue and cognitive dysfunction, as well as anxiety and depression (Premraj et al., 2022). Well-conducted clinical enquiries are likely to retain an important place in psychiatric research for many years to come and, via qualitative methodologies, will be used increasingly to generate a richer understanding of the experiences and values of patients (Bhui et al., 2021).

Epidemiology

Epidemiology is the study of the distribution of a disease in space and time within a population, and of the factors that influence this distribution. Its concern is with disease in groups of people, not in the individual person.

Concepts and methods of epidemiology

The basic concept of epidemiology is that of *rate*, or the ratio of the number of instances to the numbers of people in a defined population. Instances can be episodes of illness, or people who are or have been ill. Rates may be computed on a particular occasion (*point prevalence*) or over a defined interval (*period prevalence*).

Other concepts include *inception rate*, which is based on the number of people who were healthy at the beginning of a defined period but became ill during it, and *lifetime expectation or risk*, which is based on an estimate of the number of people who could be expected to develop a particular illness in the course of their whole life. In *cohort studies*, a group of people are followed for a defined period of time to determine the onset of, or change in, some characteristic with or without previous exposure to a potentially important agent (e.g. lung cancer and smoking).

Three aspects of method are particularly important in epidemiology:

1. defining the population at risk
2. defining a case
3. finding cases.

It is essential to *define the population at risk* accurately. Such a population can be all the people living in a limited area (e.g. a country, an island, or a catchment area), or a subgroup chosen by age, gender, or some other potentially important defining characteristic.

Defining a case is the central problem of psychiatric epidemiology. It is relatively easy to define a condition such as Down syndrome, but until recently the

reliability of psychiatric diagnosis has not been satisfactory. The development of standardized diagnostic criteria (see Chapter 2) has greatly improved the reliability and validity of epidemiological studies.

Two methods are used for *case finding*. The first is to enumerate all cases known to medical or other agencies (*declared cases*). Hospital admission rates may give a fair indication of rates of major mental illnesses, but not, for example, of most mood or anxiety disorders. Moreover, hospital admission rates are influenced by many variables, such as the geographical accessibility of hospitals, attitudes of doctors, admission policies, and the law relating to compulsory admissions.

The second method is to search for both declared and undeclared cases in the community. In community surveys, the best technique is often to use two stages—preliminary screening to detect potential cases with a self-rated questionnaire such as the General Health Questionnaire (Goldberg, 1972), followed by detailed clinical examination of potential cases with a standardized psychiatric interview.

Aims of epidemiological enquiries

In psychiatry, *epidemiology* attempts to answer three main kinds of question:

1. What is the prevalence of psychiatric disorder in a given population at risk?
2. What are the clinical and social correlates of psychiatric disorder?
3. What factors may be important in aetiology?

Prevalence can be estimated in community samples or among people attending general practitioners or hospital cases. Studies of prevalence in different locations, social groups, or social classes can contribute to aetiology. Studies of associations between a disorder and *clinical and social variables* can do the same, and may be useful for clinical practice. For example, epidemiological studies have shown that the risk of suicide is increased in elderly men with certain characteristics, such as living alone, misusing drugs or alcohol, suffering from physical or mental illness, and having a family history of suicide.

Causes in the environment

Epidemiological studies of aetiology have been concerned with predisposing and precipitating factors, and with the analysis of the personal and social correlates of mental illness. Among *predisposing factors*, the influence of heredity has been examined in studies of families, twins, and adopted people, as described below in the section on genetics. Other examples are

the influence of maternal age on the risk of Down syndrome, and the psychological effects of parental loss during childhood. Studies of *precipitating factors* include life-events research, which is described below in the section ‘Social sciences’.

Epidemiological approaches to aetiology can be illustrated by the results of studies of environmental correlates of mental disorders. For example, it has been apparent for many years that schizophrenia is more common in urban environments, particularly in disadvantaged inner-city areas. This finding could be of aetiological importance, or it could be a consequence of the experience of schizophrenia, with, for example, people in the early stages of illness isolating themselves. In a study of this question, van Os et al. (2003) confirmed that the prevalence of psychosis increased linearly with the degree of urbanicity (overall odds ratio, 1.57; 95% confidence interval [CI], 1.30–1.89). This significant effect remained after adjustment for factors such as age, gender, education level, parental psychiatric history, and country of birth.

As expected, there was, in addition, an independent and highly significant influence of a family history of psychosis on the risk of an individual developing psychosis (odds ratio, 4.59; 95% CI, 2.41–8.74). Further analysis showed that the effect of urbanicity on increasing the risk of psychosis was much greater in individuals with a family history of psychosis than in those without such a history. These findings suggest an important interaction between genes and environment, such that the adverse environmental effects of urbanicity are expressed particularly in individuals with a genetic predisposition to psychosis.

Social sciences

Many of the concepts used by sociologists are relevant to psychiatry (see Table 5.1). Unfortunately, some of these potentially fruitful ideas have been used uncritically—for example, in the suggestion that mental illness is no more than a label for socially deviant people, the ‘myth of mental illness’. This development points to the need for sociological theories to be tested in the same way as other theories by collecting appropriate data.

Some of the concepts of sociology overlap with those of social psychology—for example, attribution theory, which deals with the way in which people interpret the causes of events in their lives, and ideas about self-esteem. An important part of research in sociology, namely the study of life events, uses epidemiological methods.

Transcultural studies

Studies conducted in different societies may help to make useful causal distinctions. Biologically determined features of mental disorder are likely to be similar in different cultures, whereas psychologically and socially determined features are likely to be dissimilar. Differing prevalences of a disorder might also provide clues to causation.

For example, depressive disorders have a wide range of prevalence. The World Health Survey studied the prevalence of depression in 53 countries using the Composite International Diagnostic Interview (CIDI) to generate ICD diagnoses of depressive episode. Prevalence ranged from 0.4% in Vietnam to 15.7% in Morocco. Interestingly, economic variation between countries accounted for relatively little of the contrasting prevalence rates, and individual factors seemed more important. The latter included female gender, being separated, divorced, or widowed, as well as having fewer years of education or personal material assets (Rai et al., 2013).

The study of life events

Epidemiological methods have been used in social studies to examine associations between illnesses of various kinds and events in a person’s life. Holmes and Rahe (1967) developed a list of 41 kinds of life event (e.g. in the areas of work, residence, finance, and family relationships), and weighted each according to its apparent severity (e.g. 100 for the death of a spouse, and 13 for a spell of leave for a serviceman). They found that the higher the score, the more likely it was that the person would become ill.

Table 5.1 Some applications of social theory to psychiatry

Concept	Application
Social class and subculture	Epidemiology of substance misuse
Stigma and labelling	Analysis of social exclusion of the seriously mentally ill in the community
Institutionalization	Negative behavioural and emotional effects of institutions
Social deviance	Delinquent behaviour
Abnormal illness behaviour	Psychological consequences of physical illness

In later developments the study of the psychological impact of life events has been further improved in a number of ways.

- To reduce memory distortion, limits are set to the period over which events are to be recalled.
- Efforts are made to date the onset of the illness accurately.
- Attempts are made to exclude events that are not clearly independent of the illness (e.g. losing a job because of poor performance).
- Events are characterized in terms of their nature (e.g. losses or threats) as well as their severity.
- Data are collected with a semi-structured interview and rated reliably.

Although they are significant, life events taken in isolation may be less important than first appears to be the case. For example, in one study, events involving the loss or departure of a person from the immediate social field of the respondent ('exit events') were reported in 25% of patients with depressive disorders, but in only 5% of controls. This difference was significant at the 1% level and appears impressive, but Paykel (1978) questioned its real significance on the basis of the following calculation.

The incidence of depressive disorder is not accurately known, but if it is taken to be 2% for new cases over a six-month period, a hypothetical population of 10,000 people would yield 200 new cases. If exit events occurred for 5% of people who did not become cases of depressive disorder, in the hypothetical population, exit events would occur for 490 of the 9,800 people who were not new cases. Among the 200 new cases, exit events would occur for 25% (i.e. 50 people). Thus the total number of people experiencing exit events would be 490 plus 50, or 540, of whom only 50 (less than 1 in 10) would develop depressive disorders. Thus the greater part of the variance in determining depressive disorder must be attributed to something else; that is, life events trigger depression largely in *predisposed individuals*.

This idea leads us on to the consideration of *vulnerability* and *protective factors*. However, at this point it is also worth noting that studies of genetic epidemiology have taken life events research a stage further by showing that the tendency to experience adverse life events is itself partly genetically determined. For example, individuals differ genetically in their liability to 'select' those environments that put them at relatively higher risk of experiencing adverse life events. Presumably this is one way in which the genetic vulnerability to depression may be expressed (see Kendler et al., 2004).

Vulnerability and protective factors

People may differ in their response to life events for three reasons. First, the same event may have different meanings for different people, according to their *previous experience*. For example, a family separation may be more stressful to an adult who has suffered separation in childhood. Thus adverse experiences that are remote in time from the adverse life event itself may predispose to the later development of psychiatric disorder.

Causes in the family

It has been suggested that some mental disorders are an expression of emotional disorder within a whole family, not just a disorder in the person seeking treatment (the 'identified patient'). This approach is important in *childhood psychiatric disorders*. Although family problems are also common among adults with psychiatric disorder, their general importance in aetiology is almost certainly overstated in this formulation, as emotional difficulties in other family members may be the result of the patient's problems, rather than its cause. In addition, emotional difficulties in close relatives may result from shared genetic inheritance. For example, the parents of children with schizophrenia are more likely to have similar but milder traits (see Chapter 11). It seems more likely that family difficulties may modify the course of an established disorder. For example, high levels of 'expressed emotion' from family members increase the risk of relapse in patients with schizophrenia (see Chapter 11). However, in terms of aetiology, twin studies show that shared (family) environment is less important than shared genes in explaining familial clustering in most psychiatric phenotypes.

Migration

Moving to another country, or even to an unfamiliar part of the same country, is a life change that has been suggested as a cause of various kinds of mental disorder. A number of possible mechanisms have been identified:

- *Selective migration*. People in the early stages of an illness such as schizophrenia may migrate because of failing relationships in their country of origin.
- *Process of migration*. Events relating to the process of migration itself (e.g. physical and emotional trauma, prolonged waiting periods, exhaustion, and social deprivation and isolation) may cause several different kinds of stress-related disorder.
- *Post-migration factors*. Many factors come into play after migration that could influence the risk of developing mental illness. These include *social adversity*

caused, for example, by *racial discrimination* and *acculturation*, in which the breakdown of traditional cultural structures results in loss of self-esteem and social support. *Social exclusion* and *poverty* are also common problems for migrants. Disparities between aspiration and achievement may also cause stress and depression. Finally, immigrants may be exposed to unfamiliar viruses, which could conceivably affect intrauterine development and predispose to psychiatric disorder in the next generation.

It is fairly well established that immigration is associated with higher rates of psychosis in several ethnic groups, but the mechanisms involved are unclear (see Chapter 7). The effects of immigration on other psychiatric disorders are less consistent, and some groups experience a relative improvement in mental health compared with their native populations. Clearly, refugees who have fled persecution are likely to have elevated rates of stress-related symptomatology, and many of them will meet the formal diagnostic criteria for post-traumatic stress disorder. It is important that such symptoms are interpreted sensitively in the context of the relevant cultural ways of dealing with trauma, but this should not prevent appropriate evidence-based treatment being offered.

Experimental and clinical psychology

The psychological approach to psychiatric aetiology has a number of characteristic features:

- The idea of *continuity between the normal and abnormal*. This idea leads to investigations that attempt to explain psychiatric abnormalities in terms of processes that determine normal behaviour.
- Concern with the *interaction between the person and their environment*. The psychological approach differs from the social approach in being concerned less with environmental variables and more with the person's ways of processing information that is coming from the external environment and from their own body.
- An emphasis on *factors that maintain abnormal behaviour*. Psychologists are less likely to regard behavioural disorders as resulting from internal disease processes, and more likely to assume that persisting behavioural problems are maintained by unhelpful coping mechanisms (e.g. by anxiety-reducing avoidance strategies).

Neuropsychology (cognitive neuroscience)

Neuropsychological approaches share common ground with biological psychiatry in attempting to identify the

neurobiological substrates for psychological phenomena. Various methodologies are employed, but the aim is to understand psychopathology in the context of brain science. Investigations may therefore involve animal experimental work or a range of human studies, including neurological patients with defined brain lesions and patients with psychiatric disorders.

For example, animal experimental models have shown that there is a crucial role for the *amygdala* in fear conditioning. Furthermore, because of its connections to the thalamus, the amygdala is activated by threatening stimuli and can produce autonomic fear responses before there is any conscious awareness of threat. LeDoux (2014) has related this circuitry to traumatic anxiety by proposing an imbalance between the *implicit (unconscious) emotional memory system* involving the thalamus and amygdala and the *explicit (conscious) declarative memory system* in the temporal lobe and hippocampus.

In addition to animal experimental studies, neuropsychological investigations also involve different groups of human subjects. Valuable information may be gained from subjects who have *well-defined brain lesions*. For example, patients with bilateral amygdala lesions can recognize the personal identity of faces, but not the facial expression of fear. This supports the notion that the amygdala is important in the processing of fear-related stimuli.

Current neuropsychological approaches also make extensive use of *functional brain imaging techniques*. This allows localization of the brain regions and neural circuitry involved in specific psychological processes, and facilitates comparisons between healthy subjects and patients who experience abnormalities in the processes concerned. For example, in a magnetic resonance imaging (MRI) investigation it was found that when patients with depression were shown pictures of sad facial expressions, they exhibited greater activation than controls in brain circuitry related to the processing of emotion, including the amygdala. This increased activation was attenuated by treatment with antidepressant medication. This suggests that increased activity of the amygdala may play a role in the emotional preoccupations characteristic of depression, and that antidepressants may act by decreasing activity in amygdala and related circuitry (Ma, 2015).

Information processing

The *information theory approach* to psychology proposes that the brain can be regarded as an information channel, which receives, filters, processes, and stores information from sense organs, and retrieves information

from memory stores. This approach, which compares the brain to a computer, suggests useful ways of thinking about some of the abnormalities in psychiatric disorders. There are various mechanisms involved at different stages of information processing and therefore different points at which dysfunctional processing could give rise to psychiatric disorder. Two of these mechanisms are *attention* and *memory*, changes in which have been linked to psychiatric symptomatology.

Attention

Attention is viewed as an active process of selecting, from the mass of sensory input, the elements that are relevant to the processing that is being carried out at the time. There is evidence that attentional processes are disturbed in some psychiatric disorders. For example, anxious patients attend more than non-anxious controls to stimuli that contain elements of threat, and depressed mothers attend less to the cues of their infants. This can be shown experimentally as a disruption of psychological performance where the task involves ignoring threat-related words. One example is the use of a modified Stroop test, where subjects have to name the colour of a background on which a word is written. When the word is a threatening one (e.g. 'kill'), the latency taken to name the background colour is increased, and this increase is exaggerated in anxious subjects.

Subsequent studies have made two additional observations that are clinically important. First, the attentional bias in anxiety disorders is probably due to a failure to disengage attention from threat-related stimuli, rather than to excessive initial orientation towards them. Secondly, anxious subjects still produce greater responses to threat-related stimuli than controls, even when the stimuli are 'masked' so that they are received outside conscious awareness. Masking is achieved by presenting the stimulus for a very short time (less than 40ms), immediately followed by the longer presentation of another stimulus (the mask). The fact that masked stimuli elicit greater behavioural responses in anxious subjects suggests that the abnormal attentional mechanisms in anxiety involve the non-conscious threat-processing pathways associated with the amygdala (LeDoux, 2014). Although these findings are of interest, it is important to remember that they may in fact be a *consequence* of the anxiety disorder rather than a *causal* mechanism. However, even in the former case they could still play a role in maintaining symptomatology.

Memory

The *information-processing model* has been applied fruitfully to the study of memory. It suggests that there are

different kinds of memory store, namely *sensory stores* in which sensory information is held for short periods while awaiting further processing, a *short-term store* in which information is held for only 20s unless it is continually rehearsed, and a *long-term store* in which information is retained for long periods. There is a mechanism for retrieving information from this long-term store when required, and this mechanism could break down while memory traces are intact. This model has led to useful experiments. For example, patients with the amnesic syndrome (see Chapter 14) score better on memory tests that require recognition of previously encountered material than on tasks that require unprompted recall. This finding suggests a breakdown of information retrieval rather than of information storage.

It is well established that low mood facilitates recall of unhappy events. This can be demonstrated in healthy subjects undergoing a negative mood induction as well as in depressed patients. Once again, it is not clear whether in depressed patients this phenomenon is a manifestation of depressed mood, or one of its causes. However, it is possible that it could play a role in maintaining the depressive state. More recent research has focused on the way that patients with mood disorders recall personal memories. For example, when asked to think of a specific event associated with the word 'happy', a depressed patient may give the response 'when I used to go for long walks by myself', which is a rather general reply. In contrast, a non-depressed person is more likely to respond quite specifically—for example, 'when I went for a walk in Leighton Forest last Sunday with my family'. This over-generalized style of memory recall is associated with a history of negative life events, and might also be linked to impaired problem-solving ability (Hitchcock et al., 2019).

Beliefs and expectations

The information processing model also predicts that responses to information, including emotional responses, are determined by *beliefs* and *expectations*. This idea proposes that behaviour of all individuals is guided by their beliefs, and that psychopathology is associated with altered *content of beliefs* about the self and the world. Cognitive psychology assumes that such beliefs are organized into *schemas*. Schemas have important properties in relation to different kinds of psychopathology.

- They influence information processing, conscious thinking, emotion, and behaviour.

- Although not necessarily accessible to direct introspection, their content can usually be reconstructed in verbal terms (known as *assumptions* or *beliefs*).
- In patients with psychiatric disorders, these beliefs are dysfunctional, resistant to refutation, and play a part in the aetiology and maintenance of the disorder.

These ideas have been used in the development of *cognitive therapy*, where researchers aim to identify the dysfunctional beliefs associated with particular disorders and apply techniques that help the patient to re-evaluate and change them. For example, experimental work has shown that patients with panic disorder (see Chapter 8) have inaccurate expectations that sensory information about rapid heart action predicts an imminent heart attack. This expectation results in anxiety when the information is received, with the result that the heart rate accelerates further and a vicious circle of mounting anxiety is set up. Changing these expectations can help alleviate panic attacks.

Ethology and evolutionary psychology

Many psychological studies involve quantitative observations of behaviour. In some of these investigations, use is made of methods that were originally developed in the related discipline of ethology. Complex behaviour is divided into simpler components and counted systematically. Regular sequences are noted as well as interactions between individuals (e.g. between a mother and her infant). Such methods have been used, for example, to study the effects of separating infant primates from their mothers, and to compare this primate behaviour with that of human infants separated in the same way.

More recent applications of ethology have used insights from the field of *evolutionary psychology* to understand both normal and abnormal behaviour in an evolutionary context. This approach attempts to explain why various behaviours might have arisen in terms of evolutionary *adaptation* (see Nesse, 2023).

For example, because depressive states are ubiquitous in human societies, it is reasonable to ask what their adaptive value may be. One suggestion is that depression may reflect a form of subordination in people who have lost rank in a social hierarchy. Rather than fighting a losing battle, the depressed individual withdraws and conserves their emotional resources for another day.

Such ideas are not readily testable experimentally, but can give rise to hypotheses concerning possible brain mechanisms. One theoretical difficulty is that psychiatric disorders often appear to represent *maladaptive* rather

than adaptive behaviours. For example, Wolpert (1999) drew an analogy with cancer, in which the consequences of abnormal cell growth are clearly maladaptive and injurious to the individual. As cancer can be regarded as normal cell division 'gone wrong', so depression might be normal emotion (e.g. sadness) 'gone wrong'. From this viewpoint the question is not what the adaptive value of the abnormal behaviour is, but rather what is the adaptive value of the normal behaviour (in this case sadness) to which the abnormal state is related.

Genetics

Most psychiatric disorders have a genetic contribution, and a significant amount of aetiological research is currently devoted to identifying the genes concerned, and the mechanisms by which they influence the risk of illness or other *phenotypes* (an observable characteristic, such as a personality trait, or a cognitive ability). The concepts, methods, and terminology of psychiatric genetics are complex, and will only be outlined briefly here. An article by Mizen (2023) provides a useful introduction to psychiatric genetics.

The genetic contribution to psychiatric disorders

The first clue that a disorder has a genetic component usually comes from studying aggregation in families. In psychiatry, this is often complemented by adoption studies. However, it is twin studies that provide the most compelling evidence. Positive findings then provide the impetus to use techniques of molecular genetics to locate and identify the genes concerned.

Family studies

In family studies, the investigator determines the risk of a psychiatric condition among the relatives of affected individuals and compares it with the expected risk in the general population. The affected individuals are usually referred to as *index cases* or *probands*. Such studies require a sample that has been selected in a strictly defined way. Moreover, it is not sufficient to ascertain the current prevalence of a psychiatric condition among the relatives, because some of the population may go on to develop the condition later in life. For this reason, investigators use corrected figures known as *expectancy rates* (or *morbidity risks*).

Family risk studies have been used extensively in psychiatry. Since families share environments as well as genes, these studies by themselves cannot clearly reveal the importance of genetic factors. However, by demonstrating that the disorder of interest shows familial

clustering, they are a valuable first step, pointing to the need for other kinds of investigation.

Adoption studies

Adoption studies provide another useful way of separating genetic from environmental influences. The basic method is to compare rates of a disorder in biological relatives with those in adoptive relatives. Three main designs are used:

1. *Adoptee study*: the rate of disorder in the adopted-away children of an affected parent is compared with that in adopted-away children of healthy parents.
2. *Study of the adoptee's family*: the rate of disorder in the biological relatives of affected adoptees is compared with the rate in adopted relatives.
3. *Cross-fostering study*: the rate of disorder is measured in adoptees who have affected biological parents but unaffected adoptive parents, and compared with the rate in adoptees who have healthy biological parents but affected adoptive parents.

Adoption studies are affected by a number of biases, such as the reasons why the child was adopted, the consequences of adoption itself, the non-random nature of the placement (i.e. efforts are made to match the characteristics of the child to those of the adoptive parents), and the effects on adoptive parents of raising a challenging child. A more fundamental limitation is that adoption studies do not control for the prenatal environment, which may be important for disorders associated with intrauterine factors or birth complications. The value and limitations of adoption studies are perhaps best illustrated in schizophrenia research (see Chapter 11).

Twin studies

In twin studies the investigator seeks to separate genetic and environmental influences by comparing rates of *concordance* (i.e. where both co-twins have the same disorder) in uniovular (monozygotic, MZ) and binovular (dizygotic, DZ) twins (Kendler, 2001). If concordance for a psychiatric disorder is higher in MZ twins than in DZ twins, a genetic component is presumed; the greater the difference in concordance, the greater the *heritability* (see below). As well as showing the size of the genetic contribution, modern twin studies allow the environmental contribution to be divided into that which is unique to the individual (*'non-shared'*) and that which reflects the common (*'shared'*) environment experienced by the twins. This is usually done using a statistical approach called structural equation modelling.

Despite their key role in genetic epidemiology, the results of twin studies should not be accepted

Box 5.3 Considerations in the interpretation of twin studies in genetic epidemiology

- Has zygosity been accurately determined? Although MZ co-twins are virtually genetically identical, there can be minor differences.
- It is assumed that MZ and DZ twin pairs both experience the same degree of environmental sharing, and this '*equal environments assumption*' appears to hold for most disorders. However, this may not necessarily be true, especially with regard to the prenatal environment; counterintuitively, this is more dissimilar for MZ than for DZ co-twins. It should also be noted that gene and environment effects interact with each other, and so their effects are not simply additive. For example, parents with antisocial personality traits may pass on genes that increase the risk that their children will inherit a liability to conduct disorder, but may also produce a family environment that itself increases the risk of behavioural disturbance.
- Most twin studies are now population based, rather than being derived from psychiatric case registers. This reduces the biases of the latter, but does mean that relatively few cases are detected even in large samples, resulting in estimates that can have wide confidence intervals. In addition, the reliability of diagnoses may be less certain.
- Being a twin might in itself affect the risk of developing a psychiatric disorder. However, there is little evidence of this.

uncritically, as they make several assumptions, which are outlined in Box 5.3.

Heritability

Heritability is a measure of the extent to which a phenotype is 'genetic'. More precisely, it refers to the proportion of the liability to the phenotype that is accounted for by additive genetic effects (Visscher et al., 2008). Recent estimates for common psychiatric disorders, based on population-based twin studies, are shown in Table 5.2 (see also Box 5.4). The data show that most psychiatric disorders—like most biological and behavioural traits—are heritable to a degree, and many show a substantial heritability.

Table 5.2 Heritability estimates for selected psychiatric disorders

Disorder	Heritability estimate (%)
Major depressive disorder	35
Anxiety disorder	40
Conduct disorder	55
Alcohol use disorder	60
Autistic spectrum disorder	60
Bipolar disorder	60
Attention deficit hyperactivity disorder	85
Schizophrenia	80

Source: Geschwind, D. H., & Flint, J. (2015). Genetics and genomics of psychiatric disease. *Science*, 349(6255), 1489–1494. Copyright (2015), American Association for the Advancement of Science. Rounded to nearest 5%; see caveats in Box 5.3.

Given the importance of the concept, some further comments on heritability estimates and their interpretation are relevant here.

- Estimates of heritability may vary in different populations under different environmental conditions.
- Estimates such as those in Table 5.2 are not precise. For example, the 95% confidence intervals for schizophrenia range from 73–95%.
- Heritability cannot be applied to an individual, only to a population. For example, some cases may not have a genetic predisposition, and are called *phenocopies*.
- Heritability does not give any indication as to the number, nature, or mechanisms of the genes involved.
- Heritability should not be confused with concordance, or with penetrance. A phenotype can show high concordance in MZ twins without being genetic (e.g. religious faith or football team supported). Equally, modest concordance rates between MZ twins may still denote high heritability—it is the *difference* in concordance rates between MZ and DZ twins that denotes heritability. *Penetrance* refers to the likelihood that a specific phenotype occurs in people who carry a particular genotype. This does not always happen, probably reflecting protective genetic or environmental factors, or ‘stochastic factors’ (i.e. chance). Indeed, only a few conditions are fully penetrant.
- Heritability estimates sometimes refer to that which is attributable to single nucleotide polymorphisms (see Mutations and polymorphisms). These figures are

Box 5.4 Some reasons why finding genes for psychiatric disorders is difficult

- Starting with the ‘wrong’ clinical phenotype. Genes are highly unlikely to map on to current diagnostic categories, yet samples are usually collected based upon the latter. Stronger genotype–phenotype relationships may be seen if categories are broadened (e.g. ‘psychosis’ rather than schizophrenia and bipolar disorder) or decomposed (e.g. schizophrenia into cognitive deficits and psychotic symptoms, etc.).
- No ‘major genes’ exist. Each gene on its own contributes only a small fraction of the heritable risk.
- Different genes may affect risk in different people (*genetic heterogeneity*).
- Within a given gene, different variants may affect risk in different people (*allelic heterogeneity*).
- The presence of phenocopies and *de novo* mutations.
- Gene–gene interactions (epistasis).
- Gene–environment interactions.

lower than those given from twin studies, since the latter captures other forms of genetic variation.

The mode of inheritance

A useful intermediate step between finding that a disorder or other phenotype clusters in families and is heritable, and applying molecular methods to find the gene(s) responsible, is to determine the mode of inheritance. In essence, the question is whether the disorder has the characteristics of a Mendelian trait (i.e. whether the family history shows a classic Mendelian pattern of dominant, recessive, or X-linked inheritance). If it does, the disorder can be assumed to be caused by a single major gene. Indeed, in the rare instances where this pattern is observed in psychiatry, causative genes have in many cases already been discovered (e.g. for familial Alzheimer’s disease and Huntington’s disease). However, such examples are very rare. Most psychiatric disorders, like most common medical disorders, do not show classic Mendelian patterns of inheritance. They are called *non-Mendelian* or *complex genetic disorders*. No gene is either necessary or sufficient to cause the disorder; these *susceptibility genes* are best considered as risk factors that set the genetic threshold of vulnerability.

Within the large category of non-Mendelian disorders, the ‘genetic architecture’ of most psychiatric disorders (i.e. the number of genes and genetic variants, and how they operate to increase risk) is unclear. Most of the heritability is thought to come from genetic variants (*polymorphisms*; see Mutations and polymorphisms) that are common in the population and which, although they are important epidemiologically, confer only a very small increase in risk to the individual (the *common disease–common variant* model). There may also be genetic variants that are rare in the population, but which, when present, put the individual at a more substantial risk (*rare variants* and *copy number variants*). The relative importance of these two forms of genetic variation is unclear, and likely differs between psychiatric disorders (Smoller et al., 2019).

Linkage and association

The distinction between Mendelian and non-Mendelian disorders has implications for how best to find the genes involved. The former are best studied using *genetic linkage*, in which affected and unaffected individuals within large families are compared with genetic markers to identify which region (*locus*) of which chromosome *segregates* with (shows linkage to) the illness. The result is usually expressed as the *logarithm of the odds of detection of linkage* (LOD) score. A LOD score of 3 or more is conventionally regarded as reasonable evidence for linkage. Having identified the locus, the gene itself is then sought using other approaches.

In contrast, linkage does not work well in non-Mendelian disorders (largely because there is no one gene, nor therefore one locus, to be found). Hence, linkage strategies were unsuccessful in psychiatry with the exception of some familial dementias. In this situation, *genetic association* is a more feasible and powerful approach. The basic design involves comparing cases with unaffected, unrelated control subjects to find out whether the groups differ in the frequency of specific genetic polymorphisms (see Mutations and polymorphisms), measured using a chi-squared test or another similar test of proportions. If a significant difference is found, that variant is said to be genetically and statistically associated with the disorder concerned; whether the association reflects a true aetiological role of the variant in the disorder requires further study. Genome-wide association studies (GWAS) have been the work-horse of psychiatric genetics and are described further below.

Types of genetic variation and their psychiatric relevance

Genetic differences between individuals are of several different types. Each is relevant in the causation of psychiatric disorders.

Chromosomal (cytogenetic) abnormalities

Cytogenetics is concerned with identifying structural abnormalities in chromosomes. The abnormality can be in the number of chromosomes (*aneuploidy*), a *deletion* or *duplication* of part of a chromosome, or *translocation* of part of one chromosome to another. Cytogenetic abnormalities are usually suspected on the basis of a characteristic physical appearance. They are often associated with learning disability, and can be diagnosed relatively easily by clinical geneticists using *karyotyping*, in which the chromosomes are visualized.

A good example in psychiatry is Down syndrome, which is usually caused by an additional copy of chromosome 21 (trisomy 21, an example of aneuploidy), although some result from translocation of part of chromosome 21 with a portion of another chromosome. Other examples involve the X and Y chromosomes, such as Turner syndrome (XO) and Klinefelter syndrome (XXY). Prominent examples of a more subtle cytogenetic disorder are fragile X syndrome, in which part of the X chromosome is abnormal, and velocardiofacial syndrome (VCFS), in which part of one copy of the long arm of chromosome 22 is deleted.

Although cytogenetic abnormalities are extremely rare causes of psychiatric disorders other than those involving learning disability, their occurrence provides important clues as to where susceptibility genes may be located. For example, it was the observation that Alzheimer’s disease occurs earlier and more commonly in Down syndrome, which encouraged investigators to search chromosome 21 for genes that might cause the disease in non-trisomic subjects. In this way the putative role of the amyloid precursor protein (APP) gene was identified (see Chapter 14). Similarly, the markedly increased frequency of psychosis in people with VCFS has focused attention on genes located in that region of chromosome 22 as predisposing to psychosis in general (see Chapter 11).

Mutations and polymorphisms

Most genetic variation between individuals, and their vulnerability to disease, is attributable to changes in one or a few nucleotides (bases) of DNA sequence. These are called mutations or polymorphisms. These and related

terms are discussed in Box 5.5, which gives examples of polymorphisms that are relevant to psychiatry.

Copy number variation

Between the extremes of a cytogenetic abnormality and a single nucleotide polymorphism (SNP), an intermediate type of genetic variation has become apparent, called *structural variation* or *copy number variation* (CNV). CNVs are duplications or deletions of stretches of DNA ranging in size from thousands to millions of nucleotides. VCFS is an example of a large CNV.

CNVs are a feature of the normal genome, but they are also genetic risk factors for some psychiatric disorders, notably schizophrenia, autism, and learning disability. The pathogenicity of CNVs depends on their size, their co-occurrence with other variants, and their location in the genome. Large CNVs, and those that disrupt key genes, are more likely to be harmful. Such CNVs are rare (e.g. less than 1 in 5,000 people), yet, when present, typically have a significant effect on disease risk (e.g. an odds ratio of 10); these features contrast with SNPs, which (by definition) are common yet have only a very small effect on risk of a disorder (typically odds ratio of less than 1.1). CNVs can be inherited but many occur *de novo* (i.e. are not seen in either parent). For a review, see Kirov et al. (2015).

Genetic association studies

We have already noted that multiple SNPs appear to underlie much of the genetic predisposition to psychiatric disorders, and that their role is investigated using genetic association. Genetic association studies are of two main types.

Candidate gene studies

As noted earlier, an association study measures the frequency of a genetic polymorphism in a group of individuals who have the phenotype of interest, and compares it with the frequency in a group of matched healthy controls. Such studies have been widely used in psychiatry, because it is relatively easy to collect samples (DNA can be extracted from blood or a cheek swab), and then to *genotype* using methods based on the polymerase chain reaction (PCR). Initially, for technical and financial reasons, only a very limited number of genes and SNPs were studied at a time. A gene would be selected for study because researchers considered it to be a *candidate gene*—that is, one that they considered to be a plausible candidate for contributing to the phenotype in question (e.g. a dopamine receptor for schizophrenia and the 5-HT transporter for depression). Within the gene, the specific polymorphism was chosen because it was

Box 5.5 The terminology of molecular genetic variation: polymorphisms, alleles, and mutations

- No two people share precisely the same *genome* (the total genetic information contained on our 23 pairs of chromosomes, about 3 billion base pairs of DNA). Each person has a unique DNA sequence (with the exception of MZ twins). DNA sequence variants are called *polymorphisms* or *allelic variants*. Most polymorphisms involve a change in a single nucleotide, hence the term *single nucleotide polymorphism* (SNP). SNPs occur on average about every 1,000 base pairs (around 2 million SNPs in total), both within genes and in the stretches of DNA between genes. A given polymorphism can be rare, or both variants (*alleles*) can occur at equal frequency in the population.
- The term *mutation* can be used to refer to any very rare polymorphism (i.e. one present in less than 1% of people). However, it is often used in a more restricted way to denote a change in DNA sequence which is by definition harmful—that is, it causes a disease. This differs from polymorphisms in general, which, as noted above, often have no consequences at all; if they do, they can be beneficial or harmful, and any effects are not deterministic but merely change the probability of a particular phenotype. The term mutation is therefore used largely with reference to Mendelian disorders.
- Because we have two copies of every autosomal gene (*autosomes* are chromosomes other than X or Y), one from each parent, at any point in the genome the two alleles may be identical (*homozygosity*), or they may differ (*heterozygosity*).
- SNPs that are close together tend to be inherited together. They are said to be in *linkage disequilibrium*, and the SNPs together comprise a *haplotype*. These properties and concepts are important for the conduct and interpretation of genetic association studies. For example, linkage disequilibrium means that if one SNP is measured and shown to be genetically associated with a disorder, it cannot be concluded that it is the biologically important variant; it might simply be a marker for another SNP within the haplotype that is in fact the important one.

common (providing more statistical power), easy to measure, or had been the subject of a previous positive report.

There were a huge number of candidate gene association studies in psychiatry. However, positive findings (other than apoE4) have not been reliably replicated, which reflects several major limitations of the candidate gene approach.

- The fundamental problem is that there are about 25,000 protein-coding genes in the human genome, and over two million SNPs. The prior probability that any one gene, let alone any one polymorphism, is truly associated with the phenotype that one is measuring is therefore very small, unless there is already compelling evidence implicating the gene (which is rarely the case in psychiatry). There is therefore a high probability of obtaining a false-positive result, especially as these are more likely to be published than is a negative result.
- The groups that are being compared must have closely similar ethnic backgrounds, as the frequency of polymorphisms can vary markedly. This can lead to artefactual group differences owing to *ethnic stratification*. One method of avoiding this problem is not to use a control group, but to genotype the parents of the cases to see whether affected children inherit alleles more often than would be expected by chance.
- Other problems include genotyping errors and inadequate sample sizes, bearing in mind that in polygenic disorders the impact of an individual gene on the behavioural phenotype will be very small.

Genome-wide association studies

Candidate gene association studies have now been supplanted by genome-wide association studies (GWAS), in which hundreds of thousands of SNPs, selected to cover the whole of the genome, can be tested at the same time. This is done using a silicon ‘chip’, which contains probes for the SNPs, and on to which the person’s DNA is added. A scanner then ‘reads’ the genotype at each SNP. The fundamental advantage of GWAS is that the whole genome is surveyed without the need to have prior hypotheses (or biases). The major problem is that of multiple testing. Because so many SNPs are tested, the criterion for statistical significance between groups must be correspondingly lowered, to approximately $P \leq 10^{-8}$. This in turn requires very large samples to provide the necessary power (tens of thousands of cases, and a similar number of controls).

Studies of this size, and meta-analyses of these, have now been reported for most psychiatric disorders, with

significant progress being made in psychosis, bipolar disorder and autism, as well as in some neurodegenerative disorders. It now appears that genetic vulnerability to psychiatric illness, in most cases, results from the impact of at least several hundred genes, each of which individually is of small effect. Moreover, the risk genes thus far identified often cross traditional diagnostic boundaries so that high genetic correlations are seen between disorders that have been regarded as clinically distinct for example, bipolar disorder and schizophrenia (Smoller et al., 2019).

The increasing number of predisposing genes detected by GWAS has led to the construction of *polygenic risk scores* (PRS) where the individual genes relevant to a specific disorder are aggregated and weighted by their effect size to produce a single estimate of genetic risk for an individual. Such risk scores have potential to become useful clinical tools for the prediction of illness risk and treatment response. However, at present PRS predict only a small amount variance in liability to illness, for example, in the case of schizophrenia, around 6%. Larger GWAS studies combined with whole genome sequencing to assess all DNA variations in the genome (see below) should increase the clinical value of PRS prediction. (Plomin and Von Stumm, 2022). For a review of GWAS concepts and developments, see Horwitz et al. (2019) and Tam et al. (2019).

Exome and genome sequencing

The latest technologies move on from sampling or ‘tagging’ the genome (using SNPs) to sequencing the DNA. *Exome sequencing* is the term used to describe sequencing of the stretches of DNA that encode expressed genes (i.e. present in messenger RNA). *Whole genome sequencing* is where the genome is sequenced in its entirety. These methods are rapidly being adopted in genetics research and clinical diagnostics, and data are beginning to emerge in autism and some other psychiatric disorders. Exome sequencing is revealing many more genetic differences between individuals than anticipated, but it is not always clear which of these are of any functional or pathological significance. For review, see Wang et al. (2019).

Other aspects of psychiatric genetics

Relationships between genotype and phenotype

Genes do not code for psychiatric disorders—they code for RNA and protein, and it is these gene products which in turn influence the functioning of cells, tissues, organs, and, ultimately, individuals. This simple point has several implications. First, it is not surprising that genes do not map closely on to our current diagnostic

categories. For example, one gene can contribute to various phenotypes (*pleiotropy*). Secondly, it may contribute to the difficulty in finding genes for disorders because there are so many intervening steps that may mask the relationship. In response to this problem, researchers often study *endophenotypes* (also called *intermediate phenotypes*), which are features that are thought to be more closely related to a disorder's underlying genetic basis than is the clinical syndrome itself. For example, eye-tracking dysfunction, impaired working memory, and neuroimaging abnormalities are all putative endophenotypes in schizophrenia.

In terms of molecular mechanisms, there are two main ways in which genetic variation can affect phenotype and the risk of a disorder. If the variant alters the sequence of the encoded protein, the function of that protein may be impaired. Even a single amino acid change can be significant—for example, the APP

mutations that cause familial Alzheimer's disease, or the apoE4 SNP mentioned in Box 5.6. Larger deletions or insertions within proteins caused by CNVs are also likely to impair, inactivate, or change the function of the protein.

However, most SNPs that have been associated with psychiatric disorders to date do not alter the protein sequence, and may not even be located within a protein-coding gene at all (see Box 5.6). Their effects on disease risk probably occur because they alter the way in which the gene is regulated (i.e. the amount, timing, and location of gene expression and the synthesis of the protein), or they impact on the function of the large number of *non-protein-coding genes* now recognized.

Epistasis

Epistasis refers to the non-additive interactions between two or more genes. For example, an SNP in gene A and

Box 5.6 Examples of polymorphisms in psychiatry

Apolipoprotein E4 in Alzheimer's disease

The apoE gene on chromosome 19 exists in three common forms (*alleles*): apoE2, apoE3, and apoE4. ApoE3 is the most common variant in the population. Since 1993, dozens of studies involving thousands of people have shown an unequivocal association between apoE4 and Alzheimer's disease—a higher proportion of patients have the apoE4 variant of the gene than do age-matched subjects without the disease. In some populations, apoE2 is protective. ApoE4 is thus said to be genetically associated with Alzheimer's disease and, as such, is a genetic risk factor for it. Individuals with one copy of apoE4 (*heterozygotes*, with their other chromosome carrying apoE3 or apoE2) are two to three times more likely to develop Alzheimer's disease, and the risk is over fivefold greater in apoE4 *homozygotes* (in whom both copies of the gene are apoE4). In other words, apoE4 accounts for about a third of all cases of Alzheimer's disease. However, about half of all Alzheimer's disease occurs in people without an apoE4 allele, and some apoE4 homozygotes never develop it.

ApoE4 represents the best established and most important known genetic risk factor for a common neuropsychiatric disorder. However, it also emphasizes that apoE4, like most genes involved in psychiatric disorder, acts as a risk factor, not a determinant; it is neither necessary nor sufficient. The apoE genotype also affects the

risk of developing certain other neurological conditions, illustrating the fact that genes can have effects across different disorders. For a further discussion of apoE4, see Chapter 14.

5-HT transporter gene, stress, and depression

The serotonin (5-HT) transporter (5-HTT) regulates synaptic 5-HT availability and is the target of selective serotonin reuptake inhibitors (SSRIs). Its gene contains a polymorphism in its 'upstream' promoter region (which regulates the expression of the gene). The polymorphism is unlike the above examples in two ways. First, it is non-coding (i.e. it does not change the amino acid sequence of the protein). Secondly, it is not an SNP, but is a polymorphism in the length of the DNA, the two alleles being called short (S) and long (L) where the short form gives rise to less expression of the serotonin transporter.

In a highly cited paper, Caspi et al. (2003) reported that people with an S allele of the transporter were more liable than L allele carriers to experience major depression in the presence of life stress. This is an example of a gene–environment interaction (see below). This study is also a prominent example of the difficulties in replication in the field, since several subsequent investigations have failed to confirm the original findings (see Border et al., 2019).

an SNP in gene B each confer an odds ratio of 1.2 for a disorder. However, individuals who happen to have both of these risk SNPs have an odds ratio of 6. Epistasis has a major role in the genetics of cancer and many other complex traits. It may be similarly important in the heritability of psychiatric disorders, but the data are limited for various technical and statistical reasons. For a review of the role of epistasis in psychiatric disorders see Webber (2017).

Gene–environment interactions and correlations

Genes and the environment are sometimes considered separately, but they are inextricably linked in aetiology. That is, genes affect our susceptibility to environmental factors (*gene–environment interaction*). They also affect our exposure to particular environments (*gene–environment correlation*). Although this importance is clear from epidemiological studies, few interactions between a specific gene variant and a specific environmental factor have yet been demonstrated in psychiatry (see Box 5.6 for one controversial example), probably because they require very large samples and also careful measurements of the environment. Note that heritability estimates such as those in Table 5.2 can be affected by gene–environment interactions and correlations (Plomin et al, 2016)

Epigenetics

Epigenetics describes chemical modifications of DNA and of its binding proteins (called *histones*), which regulate gene activity without changing the DNA sequence. Important examples are methylation of cytosine nucleotides in the promoter region of genes, and the acetylation of specific histone amino acids. A range of environmental factors, including drugs, childhood abuse, and stress, have been shown to affect these modifications. As such, epigenetic regulation provides one mechanism, perhaps the most important one, by which genes and environment interact. There is also interest in the possibility that some epigenetic ‘marks’ may be heritable. For a review of this subject, see Alameda et al. (2022).

Clinical and ethical implications of psychiatric genetics

For some inherited neuropsychiatric conditions (e.g. Huntington’s disease and Down syndrome) genetic counselling and testing are regarded as helpful. This is also now increasingly relevant in paediatrics and learning disability psychiatry, since a genetic cause can be identified in over a quarter of cases of developmental disorder (Wright et al., 2015). For the major psychiatric disorders, there is still no established clinical role for genetic testing though this may change if and when

the predictive value of polygenic risk scores increases, or when the causal role of specific CNVs or rare variants become clearer. The Royal College of Psychiatrists recommends testing for CNVs in patients with schizophrenia who have co-occurring neurodevelopmental disorders (Royal College of Psychiatrists, 2023).

The growing availability of knowledge of this nature will require much ethical discussion about issues such as parental genetic testing, the rights of family members to genetic information about each other, as well as preimplantation screening and selection of embryos. Fundamental to such debate are questions about the impact (both positive and negative) of psychiatric disorder for affected individuals, their families, and society. Here the role of stigma must be closely examined. It is sobering that ‘genetic’ explanations of psychiatric disorder do not guarantee that the stigma of illness will be lessened—if anything, the reverse may be the case. For a discussion of these issues, see Morris et al. (2022).

Biochemical studies

Biochemical studies can be directed either to the causes of diseases or to the mechanisms by which disease produces its effects. The methods of biochemical investigation are too numerous to consider here, and it is assumed that the reader has some knowledge of them. The main aim here is to consider some of the problems of using biochemical methods to investigate psychiatric disorder.

It will be clear from the above account that the scope for molecular genetic studies is greatly enhanced by the presence of a *biochemical abnormality* that reliably distinguishes patients with a particular psychiatric disorder. The value of such an abnormality would be greater still if the biochemical abnormality concerned played a significant role in the cause of the illness or its pathophysiology. However, the nature of the biochemical changes associated with most psychiatric disorders remains unknown. This is due both to our lack of knowledge about the biochemical complexities of the normal brain and to the difficulty of investigating the neurochemistry of the living human brain directly. Moreover, because most psychiatric disorders do not lead to death (other than by suicide), post-mortem material is not widely available except among the elderly.

Because of these problems, workers have often adopted a variety of *indirect methods* involving sampling of peripheral tissues and fluids, such as cerebrospinal fluid (CSF), blood cells, and urine. These studies, although more feasible to carry out, are not always easy to interpret. For example, concentrations of neurotransmitters and their metabolites in lumbar CSF have an

uncertain relationship to the corresponding functionally active neurotransmitter in the brain. Equally, biochemical (and epigenetic) measures in blood platelets and lymphocytes often appear to be regulated in a different way to their brain counterparts. Finally, measures in plasma and urine are very susceptible to confounding dietary and behavioural changes (see Peripheral measures).

The reader will find accounts of the results of biochemical research in subsequent chapters, especially those on mood disorders and schizophrenia. At this point a few examples will be given of the different kinds of investigation that are used.

Post-mortem studies

Post-mortem studies of the brain can provide direct evidence of chemical and molecular changes within it and have been widely used to measure levels of transmitters, metabolites, mRNAs, proteins, and other parameters. Unfortunately, interpretation of findings can be difficult, because it must be established that findings were not affected by perimortem events, or related to the immediate cause of death. Moreover, differences between cases and controls may be due to medication or other factors (e.g. smoking or alcohol use) rather than to the disease of interest. For these reasons, case-control studies using post-mortem brains for research of this kind are now rarely performed; such tissue is now used primarily to study the mechanisms by which risk genes for psychiatric disorders exert their effects.

Brain biochemistry and brain imaging

Over the past few years, effective methods of studying biochemical events in the living brain have become available and have been used in some studies of psychiatric disorders. These methods include the following:

- magnetic resonance imaging (MRI)
- single-photon emission tomography (SPET)
- positron emission tomography (PET).

The use of these techniques to measure cerebral structure and blood flow is discussed below under the relevant headings. However, brain imaging can also be employed to measure aspects of brain biochemistry. For example, it is possible to carry out *in vivo* receptor binding in different groups of psychiatric patients using positron-labelled ligands and PET or SPET imaging.

Receptor binding with PET and SPET

The 5-HT_{1A} receptor plays an important role in the regulation of 5-HT neurotransmission and is an important target for antidepressant medications. Using PET imaging in conjunction with a positron-labelled 5-HT_{1A}-receptor

antagonist, a number of groups have found that the binding of 5-HT_{1A} receptors in the brain is decreased in patients with major depression. Moreover, this abnormality appears to persist in patients who have recovered from depression and are no longer taking medication. This suggests that low 5-HT_{1A}-receptor binding might represent a trait marker for vulnerability to depression. Alternatively, the diminished receptor availability could be a consequence of having been depressed (See Wang et al., 2016).

For reasons of cost, studies employing PET are likely to remain restricted to a small number of specialist research centres. While SPET neuroreceptor imaging is potentially more widely available, it is currently less use in psychiatric research though SPET imaging of the dopamine transporter is clinically useful in the diagnosis of Parkinson's disease.

Neurotransmitter release *in vivo*

Studies using PET and SPET in conjunction with specific dopamine-receptor ligands have enabled estimation of dopamine release *in vivo*. The principle is to scan subjects on two occasions—after administering a drug that modulates endogenous dopamine release (e.g. amphetamine) and after administering placebo. Amphetamine increases dopamine release presynaptically, and the increased levels of endogenous dopamine compete with the tracer ligand for access to postsynaptic receptors. Therefore the specific binding of the tracer is reduced and the difference in tracer signal between the amphetamine and placebo scans provides a measure of how much dopamine was released by the amphetamine.

Use of this approach has led to the conclusion that dopamine release is increased in patients with acute schizophrenia (see Chapter 11). Current studies are investigating how these techniques can be applied to the release of other neurotransmitters such as serotonin.

Magnetic resonance imaging

MRI has the advantage over SPET and PET that subjects are not exposed to radiation. Although MRI has proved to be an excellent tool for structural brain imaging and, more recently, for the examination of cerebral blood flow (see below), its application to the study of brain biochemistry (magnetic resonance spectroscopy; MRS) has been somewhat limited by lack of sensitivity. However, there are growing numbers of applications of MRS to the study of psychiatric disorders and their treatment:

- Proton (¹H) MRS can be used to detect a number of compounds of neurobiological interest, including the important amino acid neurotransmitters,

Table 5.3 Neuronal metabolites and transmitters measured by MRS

1H-NMR	31P-NMR
N-Acetyl-aspartate (NAA)	ATP
Creatine	Phosphocreatine
Myoinositol	Inorganic phosphate
GABA	Phosphodiesteres
Glutamate	Phosphomonoesters

gamma-aminobutyric acid (GABA) and glutamate (see Table 5.3).

- MRS can also be used to identify the spectrum of phosphorus-containing compounds, and thus can provide information about energy metabolism and intracellular pH.
- MRS has also been used to image lithium in the human brain, where it appears that brain levels of lithium are about half those seen in plasma.

One reasonably consistent finding from proton MRS is that patients with schizophrenia have decreased levels of glutamate in anterior cingulate cortex (Godlewska et al., 2021). This has shed some new light on a condition where aetiological hypotheses have been dominated by the dopamine theory (see Chapter 11).

Peripheral measures

CSF and blood

There have been longstanding doubts as to whether changes in the composition of neurotransmitters in the CSF reflect functionally significant changes in the brain. However, there are reasonably reproducible links between lowered CSF levels of the serotonin metabolite, 5-hydroxyindoleacetic acid (5-HIAA) and impulsive aggressive behaviour across various psychiatric disorders (Moberg et al., 2011). This suggests that CSF 5-HIAA does correlate with certain defined aspects of behaviour. More recently, CSF sampling has been studied as an early way of diagnosing Alzheimer's disease where CSF levels of amyloid peptides and tau may be useful predictors of transition from mild cognitive impairment to Alzheimer's disease (Stiffel et al., 2022). The major limitation of CSF studies is that it is often ethically and practically difficult to obtain CSF samples from psychiatric patients. In addition, it is not feasible to monitor time-dependent changes in neurotransmitter metabolism by repeated sampling.

Ingenious attempts have been made to infer biochemical changes in the brain from measurements of substances in the blood. For example, it is known that the rate of synthesis of serotonin depends on the concentration of the 5-HT precursor tryptophan in the brain. Several studies have shown that plasma tryptophan levels are decreased in patients with major depression, a finding that supports the hypothesis that brain serotonin function may be impaired in depressive disorders. However, a similar reduction in plasma tryptophan levels is found when healthy people lose weight by dieting. Therefore it is quite possible that the decrease in plasma tryptophan levels found in depressed patients is, in part, a consequence of concomitant weight loss.

A more recent example of this approach is provided by assay of serum levels of brain-derived neurotrophic factor (BDNF) in the hope that such measures might correlate with the elaboration of BDNF in the brain, and therefore permit examination of the neurotrophic hypothesis of depression (see Chapter 9). Indeed, the majority of published studies suggest that serum BDNF levels are lowered in depressed patients and increased after antidepressant treatment. However, before the serum measures can be taken as a valid index of BDNF changes in relevant brain regions, it will be necessary to clarify the origin of serum BDNF (Shi et al., 2020). There is also interest in the use of blood markers of inflammation, which have the potential, for example, to identify depressed patients where specific anti-inflammatory treatments may be of value (see Drevets et al., 2022).

Peripheral biochemical measures are of established value in the field of learning disability, where measurement of metabolites in blood and urine can provide a valid picture of the abnormalities present in the brain, as well as valuable diagnostic tests. A good example is phenylketonuria (Chapter 17).

Accessible peripheral cells

Peripheral blood cells such as platelets and lymphocytes possess receptors for neurotransmitters that often resemble the analogous receptor binding sites in the brain. There have been many studies of monoamine receptors in the platelets of depressed patients, but the findings tend to be inconsistent and easily confounded by factors such as drug treatment. In addition, it is far from clear whether abnormalities found in these peripheral binding sites will necessarily also be present in the brain. Indeed, those studies that have looked simultaneously at peripheral receptor binding and *in vivo* receptor imaging have not found correlations. Similar comments apply to the use of blood cells to investigate neurotransmitter-linked second messengers and gene expression.

Peripheral tissue studies of psychiatric disorder are entering a renaissance because of the ability to create *induced pluripotent stem cells* (iPSCs) from accessible tissues such as skin fibroblasts or keratinocytes. iPSCs can in turn be ‘reprogrammed’ into neurons of various subtypes and also used to create three-dimensional ‘brain’ organelles (‘cerebral organoids’). For example, it is possible to create serotonin neurons from patients with mood disorders and compare their morphology and functional properties with non-affected controls. iPSCs also permit high-throughput screening for identification of genetic and pharmacological targets. There are considerable technical challenges in this work and the reliability of findings between different laboratories has yet to be established. For a review, see Unterholzner et al. (2021).

‘Multi-omics’ in aetiological research

Advances in molecular and cellular biology coupled with the capacity to manage and analyse large data sets (‘big data’) have given rise to numerous high throughput technologies able to produce a global examination of genes and their molecular products (Hasin et al., 2017). Such methodologies are used increasingly in medical research. The GWAS method, as described above (*genomics*), uses very large numbers of genetic markers in thousands of individuals to detect the multiple genetic variants associated with complex diseases. *Transcriptomics* assesses genome-wide RNA levels measuring both the nature of the DNA transcripts and their abundance. *Proteomics* is used to assesses the quantity and distribution of thousands of proteins from cells and body fluids, while *metabolomics* quantifies multiple small molecules including amino acids, carbohydrates and fats. For a review of how these approaches may be applied to psychiatric aetiology and biomarker development see García-Gutiérrez et al. (2020)

There has been great recent interest in the related field of microbiomics which assesses the genes (the microbiome) and products of microorganisms in the gut, skin, and mucosa of individuals. Variations in gut bacteria, in particular, have been linked to various psychiatric conditions including autism and depression (see Chen et al., 2021).

‘Big data’ studies by their nature are ‘data’ rather than ‘hypothesis’-driven. Therefore, when trying to distinguish profiles of patients from control groups, there is inevitably a high risk of false positive observations because of the large number of comparisons carried out. This makes rigorous statistical correction and replication of findings particularly important.

Pharmacology

The study of effective treatment of disease can sometimes throw light on aetiology. In psychiatry, because of the great problems of studying the brain directly, research workers have examined the actions of effective psychotropic drugs in the hope that the latter might indicate the biochemical abnormalities in disease. Of course, such an approach must be used cautiously. If an effective drug blocks a particular transmitter system, it cannot be concluded that the disease is caused by an excess of that transmitter. The example of parkinsonism makes this clear—anticholinergic drugs modify the symptoms, but the disease is caused by a deficiency in dopaminergic transmission and not an excess of cholinergic transmission.

Two striking examples of this aetiological approach are the ‘dopamine hypothesis’ of schizophrenia and the ‘monoamine hypothesis’ of depression. The dopamine hypothesis of schizophrenia arose in part because drugs found helpful in the treatment of psychosis were demonstrated subsequently to block post-synaptic dopamine receptors. More recent investigations using PET imaging have demonstrated that patients with acute psychosis indeed have excessive pre-synaptic release of dopamine (Howes et al., 2015). Thus in the case of psychosis, the pharmacological action of effective medications did provide a helpful clue to aetiology (Howes et al., 2015). However, it should be noted that other aspects of psychosis, for example, negative symptoms and cognitive impairment, are not explained by increased dopamine activity (see Chapter 9).

The monoamine hypothesis of depression arose from the ability of effective antidepressant drugs such as tricyclic antidepressants to potentiate the action of noradrenaline and serotonin. The hypothesis therefore posited that depression was associated with decreased availability of these monoamines. However, direct evidence that serotonin or noradrenaline metabolism in the brain is abnormal in depression is lacking (see Chapter 9).

In general, as can be seen from the foregoing, aetiological theories based on the actions of current pharmacological treatments invoke ‘neurotransmitter’ theories of psychiatric disorders. Such a view is inconsistent with the growing influence of the ‘systems level’ approach to aetiology in psychiatry; the latter, however, is more compatible with the complex brain circuitry and numerous neurochemical modulators likely to be involved in the aetiology of psychiatric conditions.

Endocrinology

Changes in circulating concentrations of hormones can have profound effects on mood and behaviour, while abnormalities in endocrine function are responsible for a number of well-defined clinical syndromes, some of which have characteristic neuropsychiatric presentations (e.g. depression in Cushing's disease). The onset of puberty is associated with a sharp increase in rates of anxiety and depression, particularly in women (Lewis et al., 2018).

Despite these intriguing associations, measurement of basal plasma hormone levels in psychiatric disorders has not, in general, shown consistent abnormalities in psychiatric patients or thrown much light on aetiology. The exception is major depression, in which a significant proportion of patients hypersecrete cortisol. Elevated cortisol levels in depression have been postulated to play a role in the pathophysiology of depression, perhaps leading to cellular and synaptic neuropathology. It is also possible that persistent elevation in cortisol may be a factor in the medical comorbidities associated with depression, such as diabetes, cardiovascular disease, and cognitive impairment. This would be consistent with the observation that the secretion of cortisol on waking remains persistently increased in depressed patients following clinical recovery (Cowen, 2015).

Physiology

Physiological methods can be used to investigate the cerebral and peripheral disorders associated with disease states. The following methods have been used:

- Psychophysiological methods, including measurements of pulse rate, blood pressure, blood flow, skin conductance, and muscle activity.
- Studies of cerebral blood flow.
- Electroencephalographic (EEG) studies.

Psychophysiological measures

Psychophysiological measures can be interpreted in at least two ways. The first interpretation is straightforward. The data are used to provide information about the activity of peripheral organs in disease—for example, to determine whether electromyographic (EMG) activity is increased in the scalp muscles of patients who complain of tension headaches. The second interpretation depends on the assumption that peripheral measurements can be used to infer changes in the state of arousal of the central nervous system. Thus increases

in skin conductance, pulse rate, and blood pressure are taken to indicate greater arousal.

Measurement of cerebral blood flow, metabolism, and neuronal function

Advances in brain imaging methods have led to increasing sophistication in the measurement of cerebral blood flow in psychiatric disorders. The use of tomographic techniques allows a three-dimensional measurement of regional cerebral blood flow to be achieved both 'at rest' and following various kinds of challenges, usually either neuropsychological or pharmacological. The aim is to detect abnormal cerebral blood flow, which can then be linked to particular psychiatric disorders or symptom clusters.

Functional MRI

An important development is the demonstration that MRI techniques that use the water proton signal are sufficiently sensitive to define regional increases in cerebral blood flow following neuronal activation. This technique is usually referred to as *functional MRI (fMRI)*. The principal method of fMRI is *blood oxygenation-level-dependent (BOLD)* imaging. The use of BOLD depends on the fact that deoxyhaemoglobin is paramagnetic, and therefore aligns with an applied magnetic field, making the local magnetic field stronger. By contrast, oxygenated haemoglobin is only slightly diamagnetic, and creates weak local field disturbances.

Increases in neuronal activity are associated with increases in local cerebral blood flow, which cause *decreases in deoxyhaemoglobin*. This is because under normal conditions of activation there is a relatively greater increase in blood flow than neuronal oxygen consumption. The change in *local* deoxyhaemoglobin levels can be imaged and measured to provide an indirect measure of cerebral blood flow and thereby of local neuronal activity. The advantages of fMRI are that it has greater spatial and temporal resolution than PET and SPET, and does not require the use of radioactivity.

It is also possible to use fMRI to measure cerebral blood flow more directly than the BOLD signal allows. One method, *dynamic susceptibility contrast*, uses intravenous injection of a gadolinium-based contrast agent. Another less invasive approach is provided by *arterial spin labelling (ASL)*, which uses a radiofrequency inversion pulse to label or magnetically 'tag' the blood water in arterial vessels. A second scan in which the blood is not labelled enables cortical blood flow to be estimated by subtraction. This technique allows the rapid generation of a whole brain map of cortical blood flow.

PET imaging

PET imaging can be used to measure either *cerebral metabolism* or *cerebral blood flow*. Usually the two measures are closely correlated. In the adult brain, functional activity is almost entirely dependent on oxidative metabolism, which requires glucose and oxygen as substrates. Therefore rates of cerebral metabolism can be determined by measuring the utilization of oxygen or the accumulation of *deoxyglucose*. Measurements of regional cerebral blood flow can be made by assessing the accumulation of radioactivity in the brain during inhalation of suitably labelled CO_2 or H_2O .

SPET

Measurement of blood flow with SPET employs lipophilic radiotracers such as technetium-labelled hexamethylpropyleneamine oxime ($^{99\text{m}}\text{Tc-HMPAO}$). Following intravenous administration, these compounds are retained in the brain in a stable form for several hours. This enables high-resolution images to be obtained with the use of a conventional detector such as a rotating gamma camera. The uptake of $^{99\text{m}}\text{Tc-HMPAO}$ is linearly related to cerebral blood flow. However, unlike PET, SPET cannot provide an absolute measure of regional cerebral blood flow. Therefore, the results of SPET studies are often expressed by comparing the radioactive counts in each brain region of interest with a reference area, usually either whole brain or cerebellum.

Cerebral blood flow in psychiatric disorders

There have been many '*resting studies*', that is, studies of 'resting' blood flow, in various psychiatric disorders, but the results of different investigations have often been contradictory. To a large extent the conflicting data may result from the considerable methodological difficulties in standardizing the imaging conditions and the patient population. Despite these difficulties, more carefully controlled investigations in rigorously assessed drug-free patients have reached a greater level of consensus. For example, PET studies of patients with depression have revealed decreased metabolic activity and blood flow in brain regions associated with the *regulation* of emotion (the dorsomedial and dorsolateral prefrontal cortex) and increased blood flow in areas concerned with the *perception and experience* of emotion (the ventrolateral prefrontal cortex, anterior cingulate cortex, and amygdala) (see Price and Drevets, 2012).

More recent work using fMRI has investigated the *functional connectivity* of different brain regions when subjects rest quietly in the camera. Appropriate mathematical modelling can then be used to delineate various *resting state networks* which show synchronized

fluctuations in the BOLD signal and presumptively, therefore, in neural activity. These networks often subsume well-characterized functions of brain activity (e.g. the motor and visual networks). Of particular interest to psychiatry is the *default mode network* (DMN), consisting of the precuneus, medial frontal, inferior parietal, and temporal regions. This network is more active when subjects are at rest, and has been linked to cognitive activities such as mind wandering and the integration of cognitive and emotional processing. Studies in depressed patients suggest hyperconnectivity both within the DMN and between the DMN and dorsolateral prefrontal cortex (see Scheepens et al., 2020).

Psychological activation paradigms have been widely used in PET and fMRI studies of healthy volunteers to map the brain regions and distributed neuronal circuits involved in fundamental processes such as memory, language and emotional response.

For example, when normal control subjects undertake reward-based learning tasks, there is an increase in activity in ventral striatum as measured by changes in the fMRI BOLD signal. In patients with depression this pattern of striatal activation is lessened both to the anticipation of reward and its experience (Keren et al., 2018). This neural deficit is likely to underpin the subjective feeling of anhedonia, a key symptom of major depression.

Investigations that integrate candidate genetic polymorphisms with variance in cognitive performance and changes in regional cerebral blood flow have been a popular research area. However, as noted above, as with most candidate gene studies, the problem of replication remains substantial (see Murphy et al., 2013).

Electroencephalography

Methods

The electroencephalograph (EEG) provides a measure of *cortical neuronal activity* through detection of potential differences across the scalp. The following techniques are relevant to studies of aetiology in psychiatry:

- standard (analogue) EEG
- quantified (digital) EEG
- sleep EEG (polysomnogram)
- magnetoencephalography (MEG)
- evoked potentials.

Standard EEG

The standard clinical EEG is a *qualitative* assessment of a paper trace by a trained observer using visual inspection. These kinds of recordings have been most helpful when

studying the relationships between epilepsy and psychiatric disorders, but otherwise have not been particularly informative about aetiology. About 30% of psychiatric patients who are referred for an EEG are reported to have an abnormal recording, but the relevance of this has proved elusive. Artefacts from drug treatment are probably common. The standard EEG has good temporal but relatively poor spatial resolution.

Quantified EEG

The EEG signal can also be examined *quantitatively* using a number of different mathematical approaches. The most commonly used method employs power spectral analysis with Fourier transformation. Characteristic spectral patterns have been reported for certain disorders, although relating these to underlying brain mechanisms is not straightforward. Statistical removal of EEG artefact is also problematic. So far, the main clinical research application has been in the analysis and detection of the effects of different drugs, with the hope of developing an objective method of screening for novel psychotropic compounds. More recently quantitative EEG has been used as a predictor of response to antidepressant treatment (see Grzenda and Widge, 2020).

Sleep EEG (polysomnogram)

During sleep the EEG shows a characteristic recurrent pattern of waves, which can be divided into stages. The fundamental distinction is between *rapid eye movement* (REM or *dream sleep*) and *non-REM* (*quiet sleep*). The sleep EEG or polysomnogram shows fairly consistent abnormalities in depressed patients, notably a decrease in the *latency to the onset of REM sleep*. Some of these abnormalities persist into clinical remission and may indicate vulnerability to mood disorder.

The main disadvantage of polysomnography has been the need for a specialized facility (a 'sleep laboratory'). However, the development of home-based monitoring with ambulatory equipment has been helpful in this respect. The polysomnogram has also been useful for measuring the effects of drugs on sleep quality and architecture, and can be helpful in the diagnosis and management of sleep disorders.

Magnetoencephalography (MEG)

MEG is able to measure changes in extracranial magnetic fields to detect ion fluxes in cortical neurons. Like EEG, MEG has the ability to detect changes in physiological signals over time intervals of the order of milliseconds. It can provide better localization of signals than EEG, but the most useful information may come from using the techniques in combination, or by combining MEG

with other imaging modalities. In this way superior temporal and spatial resolution of cortical processing can be obtained. Neither MEG nor EEG is generally helpful in identifying changes in subcortical neuronal activity. MEG can be used to measure cortical oscillations, and particular frequencies have been linked to the activity of the large-scale neuronal networks identified in resting state fMRI studies (see Hirano and Uhlaas, 2021).

Evoked potentials

EEG techniques can also be used to detect changes in brain electrical activity in response to environmental stimuli. These evoked (or event-related) potentials can be detected by computerized averaging methods, and can be identified as waveforms occurring at particular times after the stimulus. For example, the P300 response is a positive deflection that occurs 300ms after a subject has identified a target stimulus embedded in a series of irrelevant stimuli.

The P300 wave probably corresponds to the cognitive processes required for the recognition, retrieval from memory, and evaluation of a specific stimulus. In patients with schizophrenia, the amplitude of the P300 wave is reduced. It is notable that the same abnormality can be found in first-degree relatives of schizophrenic patients and those with schizotypal personalities (see Parker et al., 2021).

In these subjects, the change in the P300 response is likely to stem from an abnormality in information processing, and may represent a vulnerability trait marker factor for the development of schizophrenia. However, these changes are not specific in that they can also be found in patients with other disorders, such as bipolar disorder and alcohol misuse. In addition, interpretation of evoked potentials in terms of brain mechanism is not easy, because the potential recorded from the scalp is far from its generational source and is likely to reflect the activity of many different neural systems operating in parallel.

Neuropathology

Neuropathological studies attempt to answer the question of whether a *structural change* in the brain accompanies a particular kind of mental disorder. Brain structure can now be studied in life, usually with MRI scans, as well as by the traditional direct post-mortem examination of the brain. MRI structural imaging continues to develop, and *diffusion tensor imaging* (DTI) can be used to obtain detailed images of white matter tracts *in vivo* by measuring the diffusion of water in neural tissue. A review of studies in mood disorders indicated consistently reduced

anisotropy of white matter in the frontal and temporal lobes, which suggests a loss of the integrity of white matter, although the precise pathological cause has yet to be established (see Jiang et al., 2017).

Neuropathology has been central to the understanding of dementia and a few other psychiatric disorders in which lesions can readily and reliably be found and, if necessary, quantified. It has not shown equivalent diagnostic kinds of lesion in other psychiatric disorders, a factor that contributed to the conventional view that most psychiatric conditions were functional as opposed to organic disorders. However, the advent of MRI and improved neuropathological methods has shown that there are structural correlates of many psychiatric disorders. For example, the brain is smaller and lighter in schizophrenia, associated with changes in its cellular and synaptic composition (see Chapter 11). Similarly, alterations in volume of parts of the limbic system and its cytoarchitecture have been reported in depression (see Chapter 9). Although none of these changes can yet be used for diagnostic purposes (because of overlap in each parameter with comparison subjects, and across

diagnostic boundaries), they do argue strongly against the functional versus organic dichotomy.

These research advances are also a useful reminder that the methods of investigation that are available at a particular time may fail to detect relevant biological abnormalities even when the latter are present. For example, Alois Alzheimer (1864–1915) spent a decade searching for the neuropathology of schizophrenia before he came across the case of presenile dementia and identified the lesions which now define the disease that is named after him. In addition, as neuropathological investigations embrace the molecular level, drawing distinctions between ‘functional’ and ‘structural’ disorders becomes somewhat arbitrary. Finally, it is worth noting that progress in determining aetiology is most likely to be made through the combination of genetic, pathological, biochemical, and neuroimaging investigations, and combining these with epidemiological ascertainment and careful clinical, psychological, and social characterization of subjects. In this way the various approaches can be used to inform and guide each other, and a more integrated view of psychiatric aetiology can ultimately emerge.

Relationship of this chapter to those on psychiatric syndromes

This chapter has reviewed diverse approaches to aetiology. It may be easier for the reader to put these approaches into perspective when reading the sections

on aetiology in the chapters on the different psychiatric syndromes, especially those on depression (see Chapters 9 and 10) and schizophrenia (see Chapter 11).

Further reading

Charney, D. S., Buxbaum, J. D., Binder, E. B., Gordon, J. A., Picciotto, M. R., & Nestler, E. J. (2025). *Charney and Nestler's Neurobiology of Mental Illness*, 6th edn. Oxford University Press.

Jaspers, K. (1963). *General Psychopathology* (trans. J. Hoenig and M. W. Hamilton). Manchester University Press. pp. 301–11, 355–64, 383–99. (The classical text: these selected pages

explain the concepts of meaningful connections and psychological reactions.)

Schulze, T., & McMahon, F. (2018) *Psychiatric Genetics: A Primer for Clinical and Basic Scientists*. Oxford University Press.

CHAPTER 6

Evidence-based approaches to psychiatry

What is evidence-based medicine? 119

Individual treatment studies 121

Systematic reviews 124

Implementation of evidence-based medicine 129

Evaluation of evidence-based medicine 131

Other applications of evidence-based medicine 131

Big data 133

Qualitative research methods 133

Evidence-based medicine in psychiatry 135

What is evidence-based medicine?

Evidence-based medicine (EBM) is a systematic way of obtaining clinically important information about aetiology, diagnosis, prognosis, and treatment. The evidence-based approach is a *process* in which the following steps are applied:

- formulation of an answerable clinical question;
- identification of the best evidence;
- critical appraisal of the evidence for validity and utility;
- implementation of the findings; and
- evaluation of performance.

The principles of EBM can be applied to a variety of medical procedures. For psychiatry, the main use of EBM at present is to assess the value of *therapeutic interventions*. For this reason, in the following sections the application of EBM will be linked to studies of treatment. Applications to other areas, such as diagnosis and prognosis, are discussed later.

History of evidence-based approaches

Examples of what we now might call ‘evidence-based approaches’ to the investigation of treatments have a

long, if sporadic, history in medicine. For example, in 1747 a naval surgeon, James Lind, studied six pairs of sailors ‘as similar as I could have them’ who were suffering from scurvy. The sailors who received oranges and lemons recovered within a few weeks, in contrast to those who simply received the same housing and general diet. Lind’s study was not carried out ‘blind’, but in 1784 Benjamin Franklin applied blindfolds to the participants in a mesmerism study, who were therefore unaware whether or not the treatment was being applied. The ‘blinding’ abolished the treatment effect of mesmerism, providing strong evidence that its effects were mediated by suggestion (Devereaux et al., 2002).

The application of modern randomized trial methodology to medicine is attributed to Sir Austin Bradford Hill (1897–1991), who designed the Medical Research Council (MRC) trial of streptomycin treatment of tuberculosis in 1948. Subsequently, Bradford Hill lent his influence to the application of randomized trials in the evaluation of psychiatric treatments, often in the face of vociferous opposition from the profession. The first psychiatric trial to use this methodology was carried out at the Maudsley Hospital in 1955 by David Davies and Michael Shepherd, who demonstrated that, relative to placebo, reserpine had beneficial effects in anxiety and depression. A few

years later, Ackner and Oldham (1962) used double-blind randomized methods to debunk insulin coma therapy (see Chapter 25). Subsequently, in 1965, an MRC group reported the first large-scale, multicentre, randomized controlled trial (RCT) in psychiatry, in which imipramine and electroconvulsive therapy (ECT) were shown to be therapeutically superior to placebo in the treatment of hospitalized depressed patients (see Tansella, 2002).

More recent developments in evidence-based approaches owe much to Archibald Cochrane (1909–1988), an epidemiologist and author of an influential book, *Effectiveness and Efficiency: Random Reflections on Health Services*, which was published in 1972. Cochrane emphasized the need, when planning treatment provision, to use evidence from RCTs because it is more reliable than any other kind. In a frequently cited quotation (Cochrane, 1979), he wrote: ‘It is surely a great criticism of our profession that we have not organized a critical summary, by specialty or subspecialty, adapted periodically, of all relevant randomized controlled trials.’

Cochrane’s views were widely accepted, and two further developments enabled his vision to be realized. First, the availability of electronic databases and computerized searching made it feasible to find all (or nearly all) of the relevant randomized trials when gathering evidence on particular therapeutic questions. Secondly, the statistical techniques of meta-analysis enabled randomized trials to be combined, providing greater power and allowing a reliable quantification of treatment effects. Results from studies using these methodologies are called ‘systematic reviews’ to distinguish them from the more traditional, less reliable, ‘narrative reviews’ in which the judgement of the authors plays a major role in deciding what evidence to include and what weight to give it. The *Cochrane Collaboration*, which was formed in 1993, is now the largest organization in the world engaged in the production and maintenance of systematic reviews (<http://www.cochrane.org>). In the United Kingdom, the Centre for Reviews and Dissemination, based at the University of York, maintains an up-to-date database of systematic reviews of healthcare interventions (<http://www.york.ac.uk/inst/crd/index.htm>). Similarly, the *Campbell Collaboration* provides systematic reviews of evidence-based social interventions in the field of education, the criminal justice system, social welfare, and international development (<http://www.campbellcollaboration.org/>).

Why do we need evidence-based medicine?

There are two main related problems in clinical practice that can be helped by the application of EBM:

1. The difficulty in keeping up to date with clinical and scientific advances.
2. The tendency of practitioners to work in idiosyncratic ways that are not justified by the available evidence.

With the burgeoning number of clinical and scientific journals, the most assiduous clinician is unable to keep up to date with all the relevant articles even in their own field. In fact, it has been estimated that to accomplish this task would require scrutiny of at least 20 publications a day! Clinicians therefore have to rely on information gathered from other sources, which might include, for example, unsystematic expert reviews, opinions of colleagues, information from pharmaceutical companies, and their own clinical experiences and beliefs. This can lead to wide variations in practice—for example, those described for the treatment of depression in secondary care (Paton et al., 2020).

Kinds of evidence

The fundamental assumption of EBM is that some kinds of evidence are *better* (i.e. more valid and of greater clinical applicability) than others. This view is most easily elaborated for questions about therapy. A commonly used ‘hierarchy’ is shown in Box 6.1.

In this hierarchy, evidence from randomized trials is regarded as more valid than evidence from non-randomized

Box 6.1 Hierarchy of the quality of research about treatment

- Ia Evidence from a systematic review of randomized controlled trials
- Ib Evidence from at least one randomized controlled trial
- IIa Evidence from at least one controlled study without randomization
- IIb Evidence from at least one other type of quasi-experimental study
- III Evidence from non-experimental descriptive studies, such as comparative studies, correlation studies, and case–control studies
- IV Evidence from expert committee reports or opinions and/or clinical experience of respected authorities

trials, while *systematic reviews of randomized trials* are seen as the gold standard for answering clinical questions. This assumption has itself yet to be tested systematically, and some argue that large trials with simple clinically relevant endpoints may be more valid than meta-analyses (see Furukawa, 2004). It is certainly important that clinicians are trained in critical evaluation of systematic reviews before they apply their results to clinical practice.

More recent formulations of this conventional evidence hierarchy suggest that clinical experience should not be regarded as the 'lowest' form of evidence but rather a complementary kind of knowledge to be used in conjunction with EBM. This allows EBM to be applied to the individual circumstances of each patient and to take account of the patient's experience and values (Paez, 2018).

Individual treatment studies

Validity

The key criterion for validity in treatment trials is *randomization*. In addition, clinicians entering patients into a therapeutic trial should be unaware of the treatment group to which their patients are being allocated. This is usually referred to as *concealment of the randomization list*. This means that clinicians must not be aware of the likely treatment allocation of the next patient. Such awareness could influence their decision as to whether or not to enter a patient into the trial. Without concealed randomization, the validity of a study is questionable, and its results may be misleading.

Other important points to consider when assessing the validity of a study include the following:

- Were all the patients who entered the trial accounted for at its conclusion?
- Were patients analysed in the groups to which they were allocated ('intention-to-treat' analysis)?
- Were patients and clinicians blind to the treatment received (a different question to that of *concealed randomization*)?
- Apart from the experimental treatment, were the groups treated equally?
- Did the randomization process result in the groups being similar at baseline?

Presentation of results

Odds ratios and relative risk

When the outcome of a clinical trial is an event (e.g. admission to hospital), a commonly used measure of effectiveness is the *odds ratio* (OR). The OR is the odds of an event occurring in the experimental group divided by the odds of it occurring in the control

group. The OR is given with 95% confidence intervals (which indicate the range of values within which we have a 95% certainty that the *true* value falls). The narrower the confidence intervals are, the greater is the precision of the study.

If the OR of an event such as admission to hospital is 1.0, this means the rates of admission do not differ between the control and experimental groups. Therefore if the confidence interval of the OR of an individual study includes the value of 1.0, the study has failed to show that the experimental and control treatments differ from each other.

Relative risk also measures the relative likelihood of an event occurring in two distinct groups. It is regarded as a more intuitive measure of effectiveness than the OR. For example, if action A carries a risk of 99.9% and action B carries a risk of 99.0%, the relative risk is just over 1, which seems intuitively correct for two such similar outcomes. However, the calculated OR is almost 10! With relatively infrequent events, the OR and relative risk become more similar. Measures of relative risk cannot be used in case-control designs and are hard to adjust by covariance for confounding variables.

Effect sizes

In many studies the outcome measure of interest is a continuous variable, such as a mean score on the Hamilton Rating Scale for Depression. It is possible to use the original measure in the meta-analysis, although more often an estimate of *effect size* is made because it is more statistically robust.

Effect sizes are obtained by dividing the difference in effect between the experimental group and the control group by the standard deviation of their difference. The clinical interpretation of the effect size is discussed below.

Clinical utility of interventions

Risk reduction and number needed to treat

An important part of EBM involves using the results of randomized trials of groups of patients to derive the impact of an intervention at the level of the individual patient. A useful concept when assessing the value of a treatment is that of *absolute risk reduction* (ARR). This compares the proportion of patients receiving the experimental treatment who experienced a clinically significant adverse outcome (e.g. clinical relapse) with the rate in patients receiving the comparison treatment. These rates are known as the *experimental event rate* (EER) and *control event rate* (CER), respectively, and are calculated as percentages. The difference between these two outcome rates is the ARR.

The ARR can be converted into a more clinically useful number, known as the *number needed to treat* (NNT). The NNT is the reciprocal of the ARR, and it tells us how many patients would need to be treated to experience one more positive outcome event compared with a comparator treatment (or no treatment) (see Box 6.2). Like ORs, NNTs are usually given with 95% confidence intervals.

Box 6.2 Indices for translating research results into clinical practice

	Experimental treatment, X	Control treatment, Y
Positive outcome	a	b
Negative outcome	c	d

Control event rate (CER): $b/(b + d)$

Experimental event rate (EER): $a/(a + c)$

Absolute risk reduction (ARR): The difference in the proportions with a positive outcome on treatments X and Y = $(CER - EER)$

Relative risk: EER/CER

Odds ratio (OR): The ratio of the odds of a positive outcome on treatments X and Y = $(a/c)/(b/d) = ad/bc$

Number needed to treat (NNT): The number of patients that need to be treated with treatment X to obtain one more positive outcome than would be expected on treatment Y = $1/ARR$

Example

Paykel et al. (1999) randomized 158 patients with residual depressive symptoms following an episode of major depression to either standard clinical management or clinical management with 18 sessions of cognitive behaviour therapy (CBT). Over the following 68 weeks the relapse rate in the CBT-treated group (29%) was significantly less than that in the clinical management group (47%; $P = 0.02$).

The ARR in relapse with CBT is $47 - 29 = 18\%$. The NNT is the reciprocal of this number, which is approximately six (usually the NNT is rounded up to the next highest integer). This means that six patients with residual depressive symptoms have to be treated with CBT to avoid one relapse. In general, an NNT of less than ten denotes a useful treatment effect. However, interpretation of the NNT will also depend on the nature of the treatment, together with the extent of its therapeutic and adverse effects. The NNTs for some common psychiatric treatments are shown in Table 6.1. Overall NNTs of psychiatric interventions are comparable to those employed in general medicine (Leucht et al., 2012).

If the outcome measure of an intervention is a beneficial event (e.g. recovery) rather than avoidance of an adverse event, the effect of the intervention is calculated as the *absolute benefit increase* (ABI) in the same way as

Table 6.1 Examples of number needed to treat for interventions in psychiatry

Intervention	Outcome	NNT
Maintenance antidepressant treatment	Prevention of depression relapse	4
Ketamine infusion in resistant depression	Clinical response (50% decrease in symptom score at one week)	4
Antipsychotic drugs in schizophrenia	Relapse prevention	3
Family therapy in schizophrenia	Relapse at 1 year	7
Atypical antipsychotic augmentation of SSRI-resistant depression	Clinical response (50% decrease in symptom score at six weeks)	9
Fluoxetine to treat depression in adolescence	Clinical response (50% decrease in symptom score at six weeks)	10

SSRI, Selective serotonin reuptake inhibitor.

the ARR (see above), with the NNT being similarly computed. A concept related to NNT is the *number needed to harm* (NNH), which describes the adverse risks of particular therapies (e.g. extrapyramidal symptoms with antipsychotic drugs).

Computing the NNT from ORs

If a study or meta-analysis provides an OR, it is possible to compute an NNT that may be more relevant to the clinical circumstances of the practitioner and their patient. For example, in the example given above (Paykel et al., 1999), relapses occurred in 35 of 78 subjects in the clinical management group, compared with 23 of 80 subjects in the CBT group. This gives an OR of the risk of relapse between the two treatments of 0.49. To obtain an NNT from the OR it is necessary to know, or to estimate, the expected relapse rate in the control group. This is known as the *patient expected event rate* (PEER). The PEER is combined with the OR in the following formula:

$$\text{NNT} = \frac{1 - \text{PEER} (1 - \text{OR})}{(1 - \text{PEER}) \text{PEER} (1 - \text{OR})}$$

If we take the relapse rate in the patients who were in the clinical management group in the above study (45%), we have:

$$\text{NNT} = \frac{1 - 0.45 \times (1 - 0.49)}{(1 - 0.45) \times 0.45 \times (1 - 0.49)}$$

This gives an NNT of about 6, which we also derived from the other method of calculation involving the ARR. However, if from a local audit we know that in our own service the relapse rate of patients with residual depressive symptoms is about 20% (rather than the figure of 45% reported by Paykel et al., 1999), using the above formula the NNT becomes about 11. This means in our own service we would need to treat 11 patients with CBT to obtain one less relapse. Thus ORs can be used to adjust NNTs to local clinical conditions, thereby aiding decisions about the applicability of interventions.

Clinical relevance of effect size

Like the OR, the effect size is not easy to interpret clinically. A useful approach is to use the effect size to estimate the degree of overlap between the control and experimental populations. In this way we obtain the proportion of control group scores that are lower than those in the experimental group. (A negative effect size

simply means that scores in the control group are *higher* than those in the experimental group.)

Effect sizes have been classified in the following way:

- 0.2 = small
- 0.5 = moderate
- ≥ 0.8 = large.

For example, the relief of the symptoms of Parkinson's disease by L-DOPA has large effect size (about 0.9) while the ability of statin treatment to prevent major cardiovascular events is much less (around 0.15). The effect size of antidepressant medication relative to placebo to produce a clinical response is about 0.4 (Leucht et al., 2015). At the kind of response levels seen in antidepressant-treated patients (response rate of around 30% in the placebo group and 60% in the experimental group), an effect size of 0.2 is equivalent to an NNT of about 10. With an effect size of 0.5, the NNT falls to 5.

Ethical aspects of therapeutic trials

Randomization

As we have seen, randomization is a key process in the conduct of an evidence-based clinical trial, because it is the best way of avoiding bias owing to chance and random error. However, a clinician may feel uncomfortable about randomization if, for example, they have a strong belief in the efficacy of one of the treatments that is being assessed. Randomization is ethical where there is genuine *uncertainty* about the best treatment for the individual concerned. In fact, EBM suggests that this situation is more common than clinicians may realize, in that many strongly held beliefs about the efficacy of therapeutic interventions are based on anecdotal experience rather than systematic evidence.

Use of placebo

The use of drug placebo in trials of psychotropic agents is controversial. However, such studies are required by many drug-licensing authorities before, for example, a new antidepressant drug is licensed. The arguments for the use of placebo in antidepressant drug trials have been summarized as follows (see Kader and Pantelis, 2009):

- The placebo response in major depression is variable and unpredictable, and is not infrequently equivalent in therapeutic effect to active treatment.
- Placebo is required to establish the efficacy of new antidepressants. Comparison against an active treatment is not methodologically sufficient because, while a finding of 'no difference' in antidepressant activity

might mean that the new and established treatments have equivalent efficacy, it might also mean that neither treatment was actually effective under the particular trial conditions employed.

- The lack of placebo-controlled design in antidepressant drug development might lead to the marketing of a drug that is ineffective, thereby harming public health.

These arguments have to be weighed against the knowledge that antidepressants are generally somewhat more effective than placebo in the treatment of depression. Therefore a patient who is treated with placebo in a randomized trial is not receiving the best available therapy. One way of trying to deal with this is to ensure that patients in such trials receive particularly close clinical monitoring, which will result in their being withdrawn from the study if they are not doing well.

An unintended consequence of this problem is that patients in placebo-controlled trials may not be representative of the patients who will receive the treatment concerned in real-world conditions. For example, in placebo-controlled studies of depression, patients are recruited by advertisement, and there can be inflation of depression scores by raters prior to treatment so that individuals will meet the cut-off score for recruitment into the trial. Such patients tend to do well, whether they are receiving placebo or active treatment, making it difficult to show convincing clinical differences between drug and placebo. In addition, the clinical characteristics

of such patients differ significantly from those treated routinely (Wisniewski et al., 2009).

Informed consent

The role of informed consent is crucial to the ethical conduct of randomized and placebo-controlled trials. This raises difficulties with some psychiatric disorders in which the judgement and decision-making abilities of patients may be impaired. Kader and Pantelis (2009) outlined a number of important factors.

- Patients must be made specifically aware that the trial is not being conducted for their individual benefit.
- With placebo treatment there must be clear specification of the probability of receiving placebo, the lack of improvement that might result, and the possibility of symptomatic worsening.
- Patients must be free from any coercion or inducement.
- Patients have the right to withdraw from the study at any time without any kind of penalty.
- In addition to the investigator, a family member or other suitable person should be encouraged to monitor the patient's condition and report to the investigator if there are any concerns.

The key issues therefore are *open and explicit information sharing* with the patient and their family, and all necessary measures to *avoid placebo treatment leading to harm to the subject*. However, the issue remains controversial.

Systematic reviews

Validity

The most common aim of a systematic review is to obtain all of the available *valid* evidence about a specific procedure or intervention, and from this to provide a more precise quantitative assessment of its efficacy. Other uses of systematic reviews (e.g. to examine prognosis) are discussed later in this chapter in the section 'Other applications of evidence-based medicine'. Two advances have greatly increased the feasibility of systematic reviews—first, the availability of electronic databases such as Medline and Scopus and, secondly, new statistical techniques through which results from different studies can be combined in a quantitative manner. Because a meta-analysis uses all of the available valid data, its *statistical power* is greater than that of an individual study. It may therefore demonstrate moderate but

clinically important effects of treatment that were not apparent in individual randomized studies.

A problem with systematic review is that, particularly in areas such as psychiatry, where clinical trials are difficult, it can lead to much randomized evidence being excluded from consideration because of perceived lack of quality. For example, a systematic review of the effects of methylphenidate treatment on attention deficit/hyperactivity disorder (ADHD), considered 185 trials of over 12,000 participants, but only six trials, with 183 participants, were regarded as adequate. Such reduction of evidence can lead to much uncertainty about the benefits and harms of widely used interventions. In addition, systematic reviews are rarely able to provide endpoints that are meaningful to clinicians, patients, and their families. This indicates the need, when assessing the value of interventions, to integrate perspectives from

multiple research designs, including observational and qualitative studies. This approach may be more helpful in capturing the treatment effects that are most relevant to patients and those caring for them (Fazel, 2015a).

Systematic reviews of treatment, like single therapeutic studies, have to be tested for *validity* and *quality*. The following questions should be posed.

- *Is it a systematic review of relevant and randomized studies?* We have already seen that the first task in the EBM process is to ask a clearly formulated question. It is therefore necessary to determine whether the subject of the systematic review is truly relevant to the therapeutic question that needs to be answered. The next step is to make sure that only randomized studies have been included. Systematic reviews that contain a mixture of randomized and non-randomized studies may give misleading results.
- *Do the authors describe the methods by which relevant trials were located?* Although electronic searching greatly facilitates the identification of clinical trials, up to half of the relevant studies may be missed because of mis-coding. It is therefore important for authors to make clear whether they supplemented electronic searching with hand-searching of appropriate journals. They may also, for example, have contacted authors of trials, as well as relevant groups in the pharmaceutical industry. In general, negative studies are less likely to be published than positive ones, which can lead to falsely optimistic conclusions about the efficacy and tolerability of particular treatments.
- *How did the authors decide which studies should be included in the systematic review?* In a systematic review, authors have to decide which of the various studies that they identify should be included in the overall analysis. This means defining explicit measures of quality, which will be based on the factors outlined above. Because these judgements are in part subjective, it is desirable for them to be made independently by at least two of the investigators.
- *Were the results of the therapeutic intervention consistent from one study to another?* It is common to see differences in the size of the effect of a therapeutic intervention from one study to another. However, if the effects are mixed, with some studies showing a large clinical effect while others find none at all, the trials are said to show heterogeneity. Sometimes heterogeneity can be accounted for by factors such as lower doses of a drug treatment or differences in patient characteristics. As noted above, systematic reviews should carry a measure of *quality*, and sometimes effects are found only in the high- or low-quality studies. If there is no

likely explanation for heterogeneity, the results of the review must be considered tentative.

- *Did the review follow Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines?* The PRISMA guidelines are aimed at standardizing the information provided in publications about how a particular systematic review and meta-analysis was conducted, analysed, and presented (see Moher et al., 2009).
- *Did the review employ the Grades of Recommendation, Assessment, Development, and Evaluation (GRADE) system?* The GRADE system provides a more fine-grained assessment of the quality of studies in a systematic review including, for example, study design, risk of bias, magnitude of effect, and dose response gradients (Djulgovic and Guyatt, 2017). Where randomized trials would be inappropriate, GRADE also permits the rating of certain observational evidence as providing high-quality and clinically relevant information, for example, the use of insulin to treat diabetic ketoacidosis.

Presentation of results

Combining ORs

The results of meta-analyses are often presented as a 'forest plot', in which the findings of the various studies are shown in diagrammatic form (see Figure 6.1). As noted above, studies in which the outcome is an event are presented as ORs with 95% confidence intervals.

The aim of meta-analysis is to obtain a *pooled estimate* of the treatment effect by combining the ORs or effect sizes of all the studies. This is not simply an average of all the ORs, but is *weighted* so that studies with more statistical information and greater precision (with narrower confidence intervals) contribute relatively more to the final result. The pooled OR also has a 95% confidence interval. Once again, if this interval includes the value of 1.0, the experimental intervention does not differ from the control.

In Figure 6.1 some of the individual studies tend to show an antidepressant benefit of increasing the daily dose of selective serotonin reuptake inhibitor (SSRI) treatment over the lowest licensed dose while others show that a low fixed dose is preferable. Overall, however, the meta-analytic statistic shows no difference between the two treatment regimes.

We have already seen that the studies in a meta-analysis may indicate *heterogeneity*. This can be tested statistically with a modification of the chi-squared test. If significant heterogeneity is present as in Figure

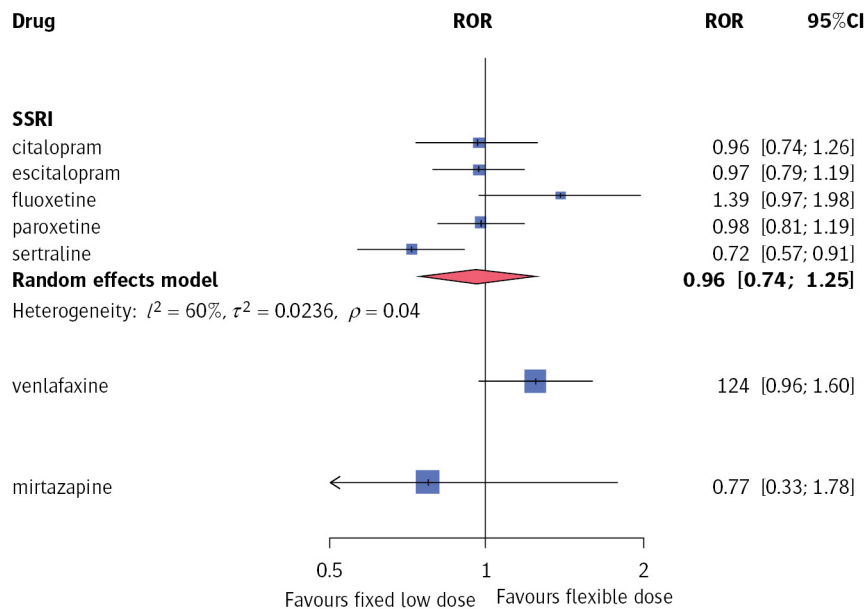


Figure 6.1 Odds ratio of clinical response of SSRIs and other antidepressants in studies using the fixed a minimum license dose and those titrating the dose upwards. Reproduced from Furukawa, T. A., Salanti, G., Cowen, P. J., et al. (2020). No benefit from flexible titration above minimum licensed dose in prescribing antidepressants for major depression: Systematic review. *Acta Psychiatr Scand*, 141(5), 402–409. Copyright (2020), with permission from John Wiley and Sons.

6.1, the most appropriate meta-analytic technique is a *pooled random effects model*. This model assumes that different treatment effects will occur in different studies, and takes this into account in the pooled estimate. This usually results in wider confidence intervals. If the studies suggest a single underlying population treatment effect (i.e. lack of heterogeneity), the pooled treatment analysis should use a *fixed effects model*. This estimate has narrower confidence intervals that may, however, be misleading in the presence of significant heterogeneity.

Effect sizes

As noted above, where the outcome measure is a continuous variable, the usual method of calculating results is to use effect sizes. As with ORs, the effect sizes can be combined to give a pooled estimate of greater precision.

Clinical utility

The clinical utility of meta-analyses is assessed as described for individual studies above. Meta-analyses will often provide figures for the NNT. However, as shown above, it is also possible to calculate NNT values from meta-analysis data using ARR or ORs.

Network meta-analysis

Network meta-analysis (NMA), or multiple treatments meta-analysis, is a development in meta-analysis that allows indirect comparisons between multiple treatments even when particular treatments have not been compared directly themselves. This approach provides a useful means of assessing competing interventions and ranking them for efficacy and acceptability. NMA therefore extends the usual pair-wise comparison of two interventions by producing a connected network of interventions in which information flows both directly and indirectly. This provides the important possibility of ‘indirect comparison’. As an example we can consider two treatments, A and B, that have both been compared with a common comparator, C. In this case the effectiveness of A versus B can be estimated indirectly through the common comparator C (Mavridis et al., 2015).

An example of a network plot of different treatments for acute mania is given in Figure 6.2, and the efficacy and acceptability results of this particular analysis are given in the section on the pharmacological treatment of acute mania (see Chapter 10, Figure 10.1). A problem for NMA is the fact that indirect comparisons cannot control for possible differences between included studies in trial methodology and patient population. The assumption made is that differing trial characteristics

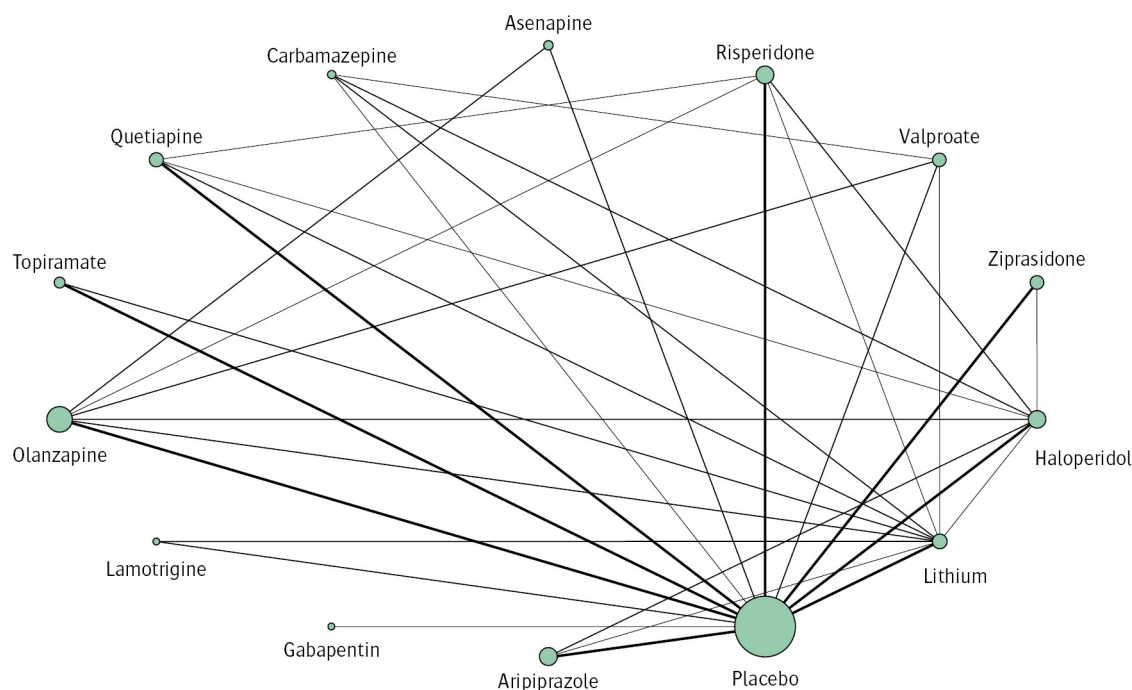


Figure 6.2 Network of eligible comparisons for a multiple-treatments meta-analysis of efficacy and acceptability of different treatments for acute mania. The width of the lines is proportional to the number of trials comparing every pair of treatments, and the size of every node is proportional to the number of randomized participants (sample size). Reproduced from Cipriani, A., Barbui, C., Salanti, G., et al. (2011). Comparative efficacy and acceptability of antimanic drugs in acute mania: A multiple-treatments meta-analysis. *Lancet*, 378(9799), 1306–1315. Copyright (2011), with permission from Elsevier.

are evenly distributed across studies or do not affect trial outcome. This assumption is known as *transitivity* (Mavridis et al., 2015).

Patient level meta-analysis

Traditional meta-analysis combines data from different investigations by aggregating overall findings from each study, as in Figure 6.2. An increasingly popular approach is to use *individual patient data*, where the original data for each participant is obtained and used in the meta-analysis. In ideal situations, results from patient level and aggregate meta-analyses should be very similar. However, patient level meta-analyses have the ability to apply consistent inclusion and exclusion criteria across different studies and to harmonize and allow for important patient characteristics, including baseline prognostic factors. This form of analysis also facilitates the inclusion of unpublished data, and the identification of duplicate reporting. Finally, consistent statistical analysis can be applied to the individual studies. The main disadvantage of the approach is the time and practical difficulty involved in obtaining individual data from

each investigator, as well as ethical concerns regarding the maintenance of confidentiality.

Problems with meta-analysis

Biased conclusions

Apart from a systematic location of evidence, the aim of meta-analysis is to combine data from multiple studies into a single estimate of treatment effect. There are a number of ways in which the results of such an exercise can be misleading.

- **Publication bias.** Evidence indicates that studies that show positive treatment effects are more likely to be published than negative studies. If negative studies are not included in the meta-analysis, the effect of treatment will be inflated.
- **Duplication of publication.** Just as negative treatment studies may go unpublished, positive studies may be published several times in different forms, sometimes with different authors! This again will falsely elevate treatment effects if the same study is included more than once.

- *Heterogeneity of studies.* As noted above, individual studies may vary widely in the results obtained, because of quite subtle differences in study design, quality, and patient population. If such heterogeneity is not recognized and accounted for in the meta-analysis, misleading conclusions will be drawn.

How accurate is meta-analysis?

There are some well-known examples where the results of meta-analyses have been contradicted subsequently by single, large, randomized trials. For example, a meta-analysis that showed that intravenous magnesium improved outcome in patients with myocardial infarction was later decisively refuted by a single, large, randomized trial of 58,000 patients. The misleading result of the meta-analysis was later explained on the basis of publication bias, poor methodological quality in the smaller trials, and clinical heterogeneity (Baigent et al., 2010).

Reviews of this area have generally found that about 80% of meta-analyses agree with single large trials in terms of direction of effect of treatment, but the size and statistical significance of the effect often differ between the two methods. Furthermore, separate meta-analyses of the same therapeutic intervention may reach rather different conclusions. For example, using the network meta-analysis approach, Cipriani et al. (2009) found clinically significant differences in the efficacy and acceptability of a range of newer antidepressants, whereas Gartlehner et al. (2008), using essentially the same data and approach, found no important differences between the various medications.

Funnel plots

One way of improving the reliability of meta-analyses is through the use of *funnel plots*. The funnel plot is based on the assumption that the precision (confidence interval) of the estimated treatment effect will be greater in studies with a larger sample size. Therefore the effect sizes of larger studies should cluster around the overall mean difference between experimental and control groups. By contrast, the results from smaller studies should be more dispersed around the mean. Therefore, when the precision of individual studies is plotted against their ORs or effect sizes, the resulting graphical plot should resemble a symmetrical inverted funnel (the funnel plot). Statistically significant deviations from this plot suggest the presence of *publication bias* (e.g. that negative studies of treatment are not fully represented). In this case, the findings of the meta-analysis should be treated with caution.

Umbrella reviews

Umbrella reviews are overarching systematic reviews of previously published systematic reviews and meta-analyses. They are used to synthesize results in fields where several previous systematic meta-analyses have been conducted, particularly where the results of these meta-analyses have been contradictory. By defining the variables of interest and estimating a common effect size, umbrella reviews allow an assessment of the impact all the relevant factors identified in the previous individual meta-analyses.

For example, in an umbrella review of 23 previous systematic review and meta-analyses, Barbui et al. (2021) found that staff training and shared decision making significantly lowered the use of coercive treatment in mental health services. In contrast, community treatment orders and adherence therapy did not lower compulsory admission rates. For a review of the methodology of umbrella reviews, see Fusar-Poli and Radua (2018).

Large-scale randomized trials

As noted above, the advantage of meta-analysis is that by combining individual studies it can assemble sufficient patient numbers to allow detection of moderate-sized but clinically important therapeutic effects. Another way of detecting moderate-sized treatment effects is to randomize very large numbers of patients to a single study. These large-scale randomized (simple) trials (or *mega-trials*) have advantages over meta-analysis in that all of the patients can be allocated to a single study design. Such studies need numerous collaborators and therefore require a *simple study design* and a *clear endpoint*.

This methodology has been most successfully applied to areas of medicine, such as cardiovascular disorders, where interventions can sometimes be simple (e.g. one dose of aspirin daily) and endpoints (e.g. cardiac infarction, or death) clearly identified (Baigent et al., 2010). An outstanding, recent example of such a study was the RECOVERY Trial which showed a significant reduction in mortality following dexamethasone treatment in Covid-19 patients requiring respiratory support (Horby et al., 2021). The challenge for psychiatric trials is to adapt such methodology to conditions for which interventions are more complex and endpoints more subtle. Furthermore, in populations under 18, the additional complexity of navigating hurdles of consent/assent as well as a requirement for parental consent, can hinder needed studies taking place.

Applicability

A general problem when applying evidence from randomized trials and meta-analysis to routine clinical work is that clinical trials are often conducted in rather 'ideal' conditions, which in a number of respects may not match routine clinical work.

- *Patient population.* Patients in controlled trials may differ systematically from those in routine clinical care in being less severely ill and having fewer comorbid difficulties. Thus trials may be conducted on patients who are in fact rather unrepresentative of a usual patient population.
- *Level of supervision.* In drug trials, concordance is regularly monitored by frequent review and supervision. Thus patients are less likely to drop out of treatment even when drugs are not particularly well tolerated.
- *Therapist variables.* Particularly in psychotherapy trials, treatment may be administered by skilled and experienced therapists. In routine practice, treatments may be given by people with less experience. Furthermore, in trials the performance of therapists is often monitored closely to ensure that it conforms to the treatment protocol. Everyday practice may match the protocol less well.

Pragmatic trials

To overcome these limitations, it has been suggested that *pragmatic trials* might be a more appropriate way to study the effect of certain psychiatric interventions.

Such studies aim to conduct randomized trials in 'real-life' situations. Methodologically they have much in common with the mega-trials described earlier, in that they are designed to answer simple and important clinical questions. As far as possible, pragmatic trials are conducted in a *routine clinical setting*. Other important features include the following:

- Randomization of very large numbers of subjects is necessary to take account of the fact that most advances in treatment will yield only moderate effect sizes. Blinding is seen as less important than randomization, particularly where active treatments are compared.
- The process of recruitment is simplified by avoiding restrictive entry criteria. The principal criterion is that both doctor and patient should feel substantial uncertainty as to which of the trial treatments is best.
- Assessments are streamlined so that they fit in with routine clinical practice: 'Many trials would be of much greater scientific value if they collected 10 times less data ... on 10 times more patients' (Baigent et al., 2010).
- Clinically relevant outcome measures are used. For example, in a trial of a therapeutic intervention in schizophrenia, a rating by a patient and family member on a simple scale of wellbeing may carry more clinical relevance than a score on a standardized rating scale.

Implementation of evidence-based medicine

Implementing evidence-based medicine for the individual patient

Having obtained the best evidence on a therapeutic intervention and decided that it is valid and therapeutically useful, it is necessary to decide how applicable it is to the individual patient you are considering. In large measure this depends on the answers to the questions on 'applicability' listed above. The key issues are as follows.

- How similar is the patient to those in the randomized trials? (This might be particularly relevant to psychiatry, where patients with more severe illness may be underrepresented in randomized trials.) A survey in Finland and Sweden of 25,000 patients with

schizophrenia spectrum disorder found that almost 80% would not meet standard eligibility criteria for entry into a clinical trial of antipsychotic drug treatment (Taipale et al., 2022)

- Can the local service deliver the intervention successfully? (For example, it is no use recommending applied relaxation for generalized anxiety disorder if there are no trained therapists available to carry it out.)

When making the decision about implementation, it may be useful to adjust the NNT for local clinical conditions if the relevant information is available (see above). A further way of taking more information into account in clinical decision making is provided by

the concept of ‘likelihood of being helped or harmed (LHH)’. For example, a meta-analysis of the addition of atypical antipsychotic medication to ineffective SSRI treatment found that the NNT to produce remission was 9, while the NNH to discontinue medication owing to side-effects was 17. The LHH is calculated as the ratio of (1/NNT) to (1/NNH) or (1/9):(1/17), which is about 2 to 1 in favour of the addition of atypicals in these circumstances. It is also possible to weight the NNT and NNH with factors that incorporate the patient’s attitude to the value of improving depression relative to that of experiencing adverse effects. Whether such efforts at quantification add significantly to a careful clinical assessment and discussion with the patient is questionable. The important point is that results from randomized trials need to be adapted to the differing needs of individuals.

Implementing evidence-based medicine at a service level

The following stages are important in the implementation of a new treatment:

- *Efficacy*. Does the intervention work under carefully controlled (‘ideal’) conditions?
- *Effectiveness*. Does the intervention work when provided under the usual circumstances of health care practice?
- *Efficiency*. What is the effectiveness of the intervention in relation to the resources that it consumes (i.e. its cost-effectiveness or cost–benefit)?

Ideally, the full implementation of EBM would involve successful negotiation of all these stages, and only interventions that have satisfied the three criteria of efficacy, effectiveness, and efficiency would be used. In practice, many therapeutic interventions in psychiatry (particularly drug treatment and cognitive behaviour therapy) are of proven *efficacy*, but there is often uncertainty about their *effectiveness* and *efficiency*. For example, lithium treatment is *efficacious* in the prophylaxis of bipolar disorder, but appears to have disappointing *effectiveness*, perhaps because under standard clinical conditions many patients find it difficult to take lithium reliably.

The discipline of implementation science has developed to better understand the barriers and facilitators to implement research findings into practice; placing

emphasis on incorporating these insights at the outset of a research trial and actively addressing implementation barriers throughout the research process and dissemination of findings (see Beidas et al., 2021).

Clinical practice guidelines

In some medical fields there is a substantial amount of evidence of different kinds, but still considerable clinical uncertainty about the best therapeutic management. In this situation it may be worth developing *clinical practice guidelines*, which are explicitly evidence-based.

Such guidelines are best developed in the following ways:

- A guideline development group, composed of a multidisciplinary group and patient representatives, decides on the precise clinical questions to be answered.
- The available evidence is systematically reviewed and classified according to the hierarchy shown in Box 6.1.
- The guideline development group makes recommendations, explicitly demonstrating how their recommendations are linked to the available evidence.

Clinical guidelines are best developed at a national level by appropriate professional organizations in collaboration with patients, but usually require modification to take local clinical conditions into account. Guidelines will only be effective if they are actively disseminated and implemented.

In the United Kingdom, the National Institute for Health and Care Excellence (NICE) has taken a prominent role in analysing and promulgating evidence about therapeutic interventions in the form of national guidelines (<http://www.nice.org.uk/>). We cite many of these guidelines in this book. The success of this process in changing clinical practice is not certain, and may depend on other factors such as the strength and stability of evidence, and cost issues. A further problem is that sometimes the evidence used in guideline development is based on a few trials whose relevance to the real world may be questionable. Nevertheless, workable guidelines need relatively definitive advice; this can lead to the issuing of rather arbitrary guidance together with a diminished probability that more informative studies will be carried out subsequently. Clinicians

Box 6.3 Evaluation of utility of therapeutic guidelines

- Did preparation and publication avoid significant conflict of interest?
- Is the guideline concerned with an appropriate topic and patient group?
- Did the guideline panel include multidisciplinary, relevant expertise as well as patient 'lived experience'?
- Have subjective judgements been made explicit and justified?
- Have all relevant data been obtained and evaluated?
- Has the evidence been properly synthesized and are results obtained in keeping with the guideline conclusion?
- Are variations in medical practice addressed?
- Is the guideline clinically relevant, comprehensive, and flexible?
- Does the guideline consider the acceptability, affordability, and practicality of its recommendations?
- Does the guideline describe how it should be disseminated, implemented, and reviewed?

Source: data from Greenhalgh, T. (2019). *How to Read a Paper: The Basics of Evidence-Based Medicine and Healthcare*, 6th edn. Copyright John Wiley and Sons.

need to be aware of the key guidelines relating to the conditions that they manage; they should be able to

justify pursuing different approaches where they judge this to be necessary in individual cases (Box 6.3).

Evaluation of evidence-based medicine

EBM itself needs to be evaluated through randomized trials of effectiveness. Individual practitioners can also evaluate their EBM performance by:

- auditing what proportion of their clinical decisions are evidence-based;
- recognizing gaps in practice that require a search for and appraisal of relevant evidence; and
- auditing the effectiveness of evidence-based practice changes.

In this way the process of EBM can become an integral part of continuing professional development and the audit cycle.

Other applications of evidence-based medicine

The foregoing account has focused on the use of EBM in the assessment of therapeutic interventions in psychiatry. Other applications of EBM include the assessment of evidence relating to diagnosis, prognosis, and aetiology. These applications require rather different methodologies from the randomized trials considered above, and diagnosis and prognosis will be discussed in the remainder of the chapter. Approaches to aetiology have been discussed in Chapter 5. All of these applications start with a focused question, which, as with treatment-related questions, must:

- be directly relevant to the identified problem; and
- be constructed in a way that facilitates searching for a precise answer.

Diagnosis

If a practitioner is trying to assess the value of a particular study assessing a diagnostic test, they need to consider a number of questions, including:

- Was there an independent, blind comparison of the test with a diagnostic gold standard?
- Did the sample include the range of patients to whom the test is likely to be applied in clinical practice?

		Alcohol abuse or dependency		
		Yes	No	
Number of positive answers to the four CAGE questions	3 or 4	60 (true +)	1 (false +)	61
	2, 1, or none	57 (false +)	400 (true +)	457
		a+c	b+d	a+b+c+d
		117	401	518

$$\text{Sensitivity} = \frac{a}{(a+c)} = \frac{60}{117} = 0.51 \text{ or } 51\%.$$

$$\text{Specificity} = \frac{d}{(b+d)} = \frac{400}{401} = 0.998 \text{ or } 99.8\%.$$

Figure 6.3 The CAGE questions for screening for alcohol abuse/dependency. Source: Sackett, D. L. (1996). Evaluation of clinical method. In D. J. Weatherall, J. G. G. Ledingham, & D. A. Warrell (Eds.), *Oxford Textbook of Medicine*, 3rd edn. (pp. 15–21). Copyright (1996), Oxford University Press.

- What is the sensitivity and specificity of the test?
- Will it aid the management of my patients?

Example

Question: How useful is the CAGE questionnaire in detecting problem drinking in medical and surgical inpatients?

The CAGE questionnaire is a simple four-item questionnaire designed to detect patients with alcohol misuse (see Chapter 20). Sackett (1996) describes a study in a community-based teaching hospital in Boston where the CAGE questionnaire was administered to 518 patients. The gold standard to which the CAGE questionnaire was compared was an extensive social history and clinical examination supplemented by liver biopsy where indicated. We can therefore be reasonably confident that cases of alcohol misuse were reliably identified.

On clinical (gold standard) grounds, 117 patients met the criteria for alcohol misuse or dependence. Of the 61 patients who scored positively on the CAGE questionnaire (scores of 3 or 4), 60 were found to have gold standard evidence of alcohol misuse. In this sample of patients the CAGE is therefore highly specific. However, the remaining 57 patients with alcohol misuse were not identified by the CAGE. The CAGE therefore has only a modest sensitivity (see Figure 6.3).

These results show that the CAGE is a useful screening instrument for problem drinking in a general hospital setting, in that a positive response is highly predictive of alcohol problems. However, the test would have to be applied in the knowledge that a negative CAGE response does not rule out alcohol misuse, particularly if there is other evidence of problem drinking.

Prognosis

Studies relating to prognosis should be assessed by considering the following questions.

- Was a defined, representative sample of patients identified at a common point, early in the course of the disorder?
- Was the follow-up sufficiently long and was it complete?
- Were objective outcome criteria applied in a blind fashion?
- Are these follow-up data likely to apply to my own patients?

A common problem with prognostic studies is lack of complete follow-up. As a rule of thumb, less than 5% dropout is ideal, and more than 20% makes the study of questionable validity. As with treatment trials, the applicability of the study will depend critically on the extent to which the patients in it resemble those whom the practitioner is considering.

Example

Question: How much of the time during long-term follow-up do patients with bipolar disorder experience affective symptomatology, and what is the pattern of symptoms?

Judd et al. (2002) recruited 146 patients with a diagnosis of bipolar disorder and a current episode of major mood disturbance from five tertiary care centres in the United States. They were followed up with interviews every six months for the first five years, and annually thereafter. At interview, affective symptoms were elicited using LIFE Psychiatric Status Rating scores, which were linked to the Research Diagnostic Criteria (RDC). Affective symptoms that did not meet the criteria for RDC diagnosis were assigned to sub-syndromal categories of depression or mania.

The mean follow-up period was 14.2 years, and 93% of the subjects were followed up for more than two years. Patients were symptomatically ill for about half of the time. Overall, about 90% of the patients spent one or more weeks during follow-up with depressive symptoms, and almost the same number (86%) experienced at least one week of manic or hypomanic symptoms. However, depressive symptoms during follow-up (32%)

of follow-up weeks) were about three times as common as manic or hypomanic symptoms (9.3%). Most of these depressive states were classified as subsyndromal depression or minor mood disorder rather than major depression. At least one week of mood cycling or mixed affective states was noted in about 48% of patients.

This study suggests that patients with bipolar disorder who are referred to a tertiary centre with an episode of major mood disturbance will have some symptoms of mood disturbance for about half of the time over the next few years. Overall, depressive symptoms

predominate, particularly minor and subsyndromal depressive states. However, over time patients can experience considerable fluctuation in symptoms (both manic and depressive, or mixed). This study also has some aetiological implications, because it suggests that bipolar disorder and other milder depressive and hypomanic states are different expressions of the same underlying disorder. In terms of applicability, we would note that the patients in the study are tertiary referrals, so the findings might not apply, for example, to patients in primary care.

Big data

Big data takes advantage of the increasing amount of digital information available about health-related experiences, behaviours, interventions, and outcomes. Linkage of these diverse sources of information can provide valuable insights into the kind of questions addressed by EBM but without the need for complex and resource-intensive clinical trials and investigations.

The notion is that because of the large numbers involved, the fact that patients are not randomized to treatment becomes a less pressing issue, provided appropriate control is made for potential confounding factors. It must be acknowledged that the latter often present difficulties. Also, the quality and reliability of the data collected is less certain than that obtained for a specific research protocol.

An example of a relevant study is provided by Joas et al. (2017), who combined several Swedish national registries to assess the effect of different pharmacological treatments on the risk of psychiatric hospital

admission in 35,022 patients with bipolar disorder. After adjusting for potential confounds (such as previous time spent in psychiatric inpatient care), the data showed that treatment with lithium was associated with the largest decrease in risk of admission, assessed by hazard ratio.

Another growing use of big data is the application of machine learning to build predictive models. For example, Chekroud et al. (2016) used 164 patient reported variables from a clinical trial of an SSRI in 4,000 depressed patients. Treatment response was predicted largely from 25 variables which were replicated with data obtained from an independent trial.

The problem with this latter approach is that is largely atheoretical and therefore exclusively data driven. It cannot provide clues to mechanism and replication of predictors in future studies has proved challenging.

Qualitative research methods

Qualitative research methods are used to collect and analyse data that cannot be easily represented by numbers (see Brown and Lloyd, 2001). Although current evidence-based approaches in psychiatry have focused on the use of RCTs with quantitative endpoints, the use of qualitative approaches has a long history and encompasses, for example, the classificatory work of Kraepelin and the case studies of Freud. More modern case series, such as that of Russell (1979) describing bulimia nervosa, also rely on a qualitative approach.

The key differences between qualitative and quantitative research are summarized in Table 6.2.

When should qualitative methods be used?

There are a number of circumstances in which the application of qualitative methods is appropriate in psychiatric research and service development:

- In the initial stages of research, to conceptualize and clarify the relevant questions and to generate hypotheses;
- To gather and collate the attitudes, beliefs, and experiences of service users, carers, and professionals;

Table 6.2 Key differences between quantitative and qualitative research

	Quantitative research	Qualitative research
Fundamental aim	Objectivity, reliability, scientific truth	Understanding through personal encounter and observation; 'permeability'
Sampling	Random selection; designed to represent general population of subjects and avoid bias	'Purposive;' subjects chosen deliberately to possess characteristics relevant to study question
Data collection	Standardized, 'objective'; validated rating scales, biochemical endpoints, etc.	Observation and interview, often interactive and open-ended; systematic study of written material
Analysis of data	Quantitative, statistical, hypothesis testing	Narrative, generation of categories and themes; analysis occurs iteratively with data collection

- In the development of assessment tools and rating scales; and
- To examine the use of evidence-based interventions in practice and to understand problems with their implementation.

From this it can be seen that qualitative and quantitative approaches are not conflicting, but have particular uses in defined situations. For example, Rogers et al. (2003) combined a qualitative methodology with a quantitative randomized trial, which aimed to improve the management of antipsychotic medication in patients with schizophrenia. The qualitative study indicated that the trial participants did not readily recall the details of the interventions to which they had been exposed. On the other hand, they valued the opportunity provided by the trial for greater communication and contact with professionals. This was associated with greater feelings of self-efficacy and clinical improvement.

Evaluation of qualitative research

Like quantitative research, qualitative research needs to be relevant and valid. Similar general principles apply to the processes of participant selection, which should be clear and justified. It is important to be explicit about the reasons for choosing a qualitative approach, and to give a clear description of the methods of data collection and analysis. The *concept of permeability*, or the extent to which observations have influenced understanding, is important in qualitative research. The following factors influence study permeability.

- *The degree of engagement with the material.* How far were theories generated by direct contact (e.g. through interviews, naturalistic observation, and familiarity with textual sources)?
- *Iteration.* Did the investigators continually reformulate and re-examine their interpretations in the light of continuing observations?
- *Grounding.* Were the procedures for linking interpretations with observations clearly presented, and were illustrative examples given?
- *Context.* Were the values and expectations of the investigator disclosed? Was the cultural context of the research and its meaning to the participants made explicit?

Validity in qualitative research

The concept of validity in qualitative research refers to the soundness and reasoning of interpretation, rather than comparison with an objective external criterion. Validity also differs according to role. For example, readers of a study will look for coherence and internal consistency of material, whereas participants must feel that their experiences are accurately described by the interpretations. Finally, the authors of the research need to consider the effect that the research process itself might have on the data that they have collected ('*reflexivity*').

Qualitative research has a valuable role to play in answering important questions about the experience of clinical care and service delivery. An increasing number of high-quality studies are adopting a mixed methodology, including both qualitative and quantitative components, to better elucidate the topic under investigation and enhance implementation of findings. The drive to involve patients in the planning and undertaking of research studies means that researchers who might employ largely quantitative methodologies

also need an understanding of how to assess and utilize qualitative data. For a further discussion of the

role of qualitative research in the biomedical field, see Greenhalgh (2019).

Evidence-based medicine in psychiatry

Practising according to evidence-based principles is an important aspect of modern psychiatry, and is followed as far as possible in this textbook. EBM is probably best viewed as an 'approach' whereby the key tasks for the practitioner are to formulate a relevant clinical question and then identify the best evidence with which to answer it. In addition, EBM provides

an important means by which psychiatrists can work in a way that improves both consistency and quality of treatment. However, clinicians work with individual patients with particular needs in specific settings. Therefore EBM needs to be tempered with the clinical skills that an expert practitioner brings to each clinical encounter.

Further reading

Greenhalgh, T. (2019). *How to Read a Paper*. Wiley Blackwell.

CHAPTER 7

Reactions to stressful experiences

Introduction 137

The response to stressful events 137

Classification of reactions to stressful events 140

Acute stress reaction and acute stress disorder 141

Post-traumatic stress disorder 143

Response to specific kinds of severe stress 148

Adjustment disorders 150

Special kinds of adjustment 152

Introduction

Stressful events frequently precede the onset of psychiatric disorders. Such events can also provoke emotional reactions that are distressing, but not of the nature or severity required for the diagnosis of an anxiety disorder or a mood disorder. These less severe reactions are discussed in this chapter, together with post-traumatic stress disorder (PTSD), which is an intense and prolonged reaction to a particularly severe stressor. With the exception of normal grief reactions, the conditions described in this chapter are listed as disorders in ICD-11 and DSM-5.

The chapter begins with a description of the various components of the response to stressful events, including coping strategies and mechanisms of defence. The classification of reactions to stressful experience is discussed next. The various syndromes are then described, including acute stress reactions, PTSD, other forms of response to severe stress, and adjustment disorders. The chapter ends with an account of special kinds of adjustment reaction, including adjustment to bereavement (grief) and to terminal illness, and some of the longer-term consequences for adults who have experienced sexual abuse in childhood.

The response to stressful events

The response to stressful events has three components:

1. An emotional response, with somatic accompaniments.
2. A coping strategy.
3. A defence mechanism.

Coping strategies and *defence mechanisms* are overlapping concepts, but they originated from different schools of thought, and for this reason they are described separately in the following account. Individuals have

to adapt to stresses, whether they are trying to make sense of specific one-off events, or facing ongoing difficulties that require a change of expectations. The process of coming to terms with stresses is often referred to as ‘working through’ them. This term originated in psychoanalysis to designate the subconscious processing or ‘work’ that was necessary for change. It is now used to mean any coming to terms with emotional difficulties that involves reflection and re-evaluation.

Emotional and somatic responses

These responses are of two kinds:

1. *Anxiety* responses, with autonomic arousal leading to apprehension, irritability, tachycardia, increased muscle tension, and dry mouth.
2. *Depressive* responses, with pessimistic thinking and reduced physical activity.

Anxiety responses are generally associated with events that pose a *threat*, whereas depressive responses are usually associated with events that involve separation or *loss*. The features of these responses are similar to, but less intense than, the symptoms of anxiety and depressive disorders (described in Chapters 8 and 9, respectively).

Coping strategies

Coping strategies serve to reduce the impact of stressful events, thus attenuating the emotional and somatic responses and making it more possible to maintain normal performance at the time (although not always in the longer term; see below). The term *coping strategy* is derived from research in social psychology; it is applied to activities of which the person is aware—for example, deliberately avoiding further stressors. (Responses of which the person is unaware are called *defence mechanisms*; see below.)

Coping strategies are of two kinds: *problem-solving strategies*, which can be used to make adverse circumstances less stressful; and *emotion-reducing strategies*, which alleviate the emotional response to the stressors.

Problem-solving strategies include:

- *Obtaining information or advice* that would help to solve the problem.
- *Solving problems*—making and implementing plans to help manage the problem.
- *Confrontation*—defending one's rights, and persuading other people to change their behaviour.

Emotion-reducing strategies include:

- *Ventilation of emotion*—talking to another person and expressing emotion.
- *Evaluation of the problem*—to assess what can be changed and try to change it (by problem-solving), and what cannot be changed and try to accept it.
- *Positive reappraisal of the problem*—recognizing that it has led to some good (e.g. that the loss of a job is an opportunity to find a more satisfying occupation).

- *Avoidance of the problem*—by refusing to think about it, avoiding people who are causing it, or avoiding reminders of it.

Coping strategies are generally useful for reducing the problem or lessening the emotional reaction to it. However, they are not always adaptive. For example, avoidance may not be adaptive in the early stages of physical illness, because it can lead to delay in seeking appropriate treatment. Therefore, a person needs not only the ability to use coping strategies, but also the ability to judge which strategy should be used in particular circumstances.

Maladaptive coping strategies

These strategies reduce the emotional response to stressful circumstances in the short term, but lead to greater difficulties in the long term. Maladaptive coping strategies include the following:

- *Use of alcohol or unprescribed drugs* to reduce the emotional response or to reduce awareness of stressful circumstances.
- *Self-harm*, either by drug overdose or by self-injury. Some people gain relief from tension by cutting their skin with a sharp instrument to induce pain and draw blood. Others take overdoses to withdraw from the situation or to show their need for help.
- *Aggressive behaviour* for some can provide immediate release of feelings of anger and distress. In the longer term, it may increase the person's difficulties by damaging relationships.

Coping styles

When particular coping mechanisms are used repeatedly by the same person in different situations, they are said to constitute a coping style. Some people change their coping strategies according to the circumstances—for example, they may use problem-solving strategies at work but employ avoidance when unwell. Some people habitually use maladaptive strategies—for example, they repeatedly misuse alcohol or other substances or take overdoses of drugs when under stress. More recent research has distinguished between *coping style*, which is seen as a relatively enduring behavioural trait, and *coping response*, which is much more specific to particular stressful environments.

Defence mechanisms

Defence mechanisms (the main mechanisms are described in Box 7.1) are conceptualized as unconscious

Box 7.1 Defence mechanisms

Repression

This is the exclusion from consciousness of impulses, emotions, or memories that would otherwise cause distress. For example, especially painful aspects of the memory of distressing events, such as sexual abuse in childhood, may be kept out of full awareness for many years.

Denial

This is a related concept, which is inferred when a person behaves as if they are unaware of something that they may reasonably be expected to know. For example, on learning that they are dying of cancer, a patient may continue to live normally as if they are unaware of the diagnosis. In this example, denial might be adaptive for that individual (although less so for close family members), as it can help to reduce depression. However, in the early stage of illness, denial may delay help-seeking or lead to refusal of necessary investigations and treatment. In this second example, denial is maladaptive.

Displacement

This is the transfer of emotion from a person, object, or situation with which it is properly associated, to another source. For example, after the recent death of a partner, a person may blame the doctor for failure to provide adequate care, and may thus avoid blaming themselves for putting their work before their partner's needs during their last months of life.

Projection

This is the attribution to another person of thoughts or feelings similar to one's own, thereby rendering one's own thoughts or feelings more acceptable. For example, a person who dislikes a colleague may attribute reciprocal feelings of dislike to them; it is then easier to justify their own feelings of dislike for the colleague.

Regression

This is the adoption of behaviour appropriate to an earlier stage of development—for example, dependence on others. Regression often occurs among physically ill people. In the acute stages of illness it can be adaptive, enabling the person to acquiesce passively to intensive medical and nursing care. If regression persists into the stage of recovery and rehabilitation, it can be maladaptive because it reduces the patient's ability to make efforts to help themselves.

Reaction formation

This is the unconscious adoption of behaviour that is the opposite to that which would reflect the person's true feelings and intentions. For example, negative attitudes to sexuality can sometimes (but not always) be a reaction to sexual urges that that person, themselves, find uncomfortable.

Rationalization

This is the unconscious provision of a false but acceptable explanation for behaviour that has a less acceptable origin. For example, a person may leave their partner at home because they do not enjoy their company, but reassure themselves falsely that their partner is shy and would not enjoy going out.

Sublimation

This is the unconscious diversion of unacceptable impulses into more acceptable outlets—for example, turning the need to dominate others into the organization of good works for charity. (There are, of course, many other motives for charitable work.)

Identification

This is the unconscious adoption of the characteristics or activities of another person, often to reduce the pain of separation or loss. For example, a person may undertake the same voluntary work that their partner used to do.

responses to external stressors as well as to anxiety arising from internal conflict. They were originally described by Sigmund Freud and later elaborated by his daughter, Anna Freud. Defence mechanisms are *unconscious processes* (i.e. people do not use them deliberately and are unaware of their own real motives, although they may become aware of these later through introspection or through another person's comments). Freud

identified defence mechanisms in his study of the 'psychopathology of everyday life', a term that he applied to slips of the tongue and lapses of memory. The concept of defence mechanisms remains useful in understanding many aspects of the day-to-day behaviour of people under stress, notably those with physical or psychiatric illness. Freud also used the concept of mechanisms of defence to explain the aetiology of mental disorders,

but this extension of his original observations has not proved useful.

Present circumstances, previous experience, and response to stressful events

The response to a stressful life event is modified by present circumstances and by past experience. Some current

circumstances make a person more vulnerable to stressful life events—for example, the lack of a confidant with whom to share problems. Such circumstances are called *vulnerability factors*. Previous experience can also increase vulnerability. For example, the experience of losing a parent in childhood may make a person more vulnerable in adult life to stressful events involving loss. It is difficult to examine these more remote associations scientifically.

Classification of reactions to stressful events

Although they are included within the classifications of diseases, not all reactions to stressful events are abnormal. Grief is a normal reaction to the stressful experience of bereavement, and only a minority of people have a very severe or abnormally prolonged reaction. There is also a normal pattern of reaction to a dangerous or traumatic event such as a car accident. Most people have an immediate feeling of great anxiety, are dazed and restless for a few hours afterwards, and then recover; a few people have more severe and prolonged symptoms—an abnormal reaction. It is difficult to decide where to draw the separation between normal and abnormal reactions to stressful events in terms of severity or duration, and in practice the division is arbitrary. Similarly, among patients who are in hospital for medical or surgical treatment, most are anxious, but a few are severely anxious and show extreme denial or other defence mechanisms that impair cooperation with treatment.

ICD-11 and DSM-5 reactions to stressful experiences are classified into three main groups (see Table 7.1).

Acute reactions to stress

This category is for immediate and brief responses to sudden intense stressors in a person who does not have another psychiatric disorder at the time. The DSM-5 definition of *acute stress disorder* states that the onset should occur while or after experiencing the distressing event, and requires that the condition lasts for at least two days and for no more than four weeks. The ICD-11 diagnostic criteria for *acute stress reaction* are similar.

Both diagnostic systems require that the stressor must be of an ‘exceptional’ nature and that actual or threatened serious injury to self or others or sexual violation has occurred. DSM-5 also requires the presence of clinically significant distress, impairment in social or occupational spheres. ICD-11 states that acute stress

reaction is, ‘... usually, but not necessarily, accompanied by substantial subjective distress and/or interference with personal functioning’.

Post-traumatic stress disorder (PTSD)

This is a prolonged and abnormal response to exceptionally intense stressful circumstances such as a natural disaster or a sexual or other physical assault. It is described further below.

Table 7.1 Classification of reactions to stressful experience

ICD-11	DSM-5
Acute stress reaction	Acute stress disorder
Post-traumatic stress disorder	Post-traumatic stress disorder
Complex post-traumatic stress disorder	
Adjustment disorder	Adjustment disorder
	● With depressed mood
	● With mixed anxiety and depressed mood
	● With anxiety
	● With disturbance of conduct
	● With mixed disturbance of emotions and conduct
	● Unspecified

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11), Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (DSM-5), Copyright (2013), American Psychiatric Association.

Adjustment disorder

This is a more gradual and prolonged response to stressful changes in a person's life than acute stress reaction or acute stress disorder. In DSM-5, adjustment disorders are subdivided, according to the predominant symptoms, into depressive, anxiety, mixed anxiety and depressive, with disturbance of conduct, and with mixed disturbance of emotions and conduct. ICD-11 does not have separate descriptive categories of adjustment disorder but advises that, in addition to worry and preoccupation with the stressor, patients may manifest 'externalizing' behaviours such as impulsivity and substance misuse.

For all of these stress-related diagnostic categories, the presumptions are:

- They arise as a *direct consequence* of either acute stress or continued unpleasant circumstances.
- It is judged that the disorder would not have arisen without these factors.
- The disorder will remit once the stressor and its consequences are removed.

Coding grief reactions

In ICD-11, *prolonged grief disorder* is classified under the general heading of 'disorders specifically associated with stress'. DSM-5 codes abnormal grief reactions under adjustment disorder. A more specific set of bereavement-related symptoms is called *persistent complex bereavement disorder*, which is coded under 'other specified trauma and stressor-related disorder'. Persistent complex bereavement disorder is also listed under 'conditions for further study'.

Acute stress reaction and acute stress disorder

Clinical picture

The *core symptoms* of an acute psychological response to stress are anxiety, depression, or despair. Anxiety is the response to threatening experiences, and depression and despair are responses to loss. Anxiety and depression often occur together, because stressful events often combine danger and loss—an extreme example is a road accident in which a companion is killed. Other symptoms include feelings of being numb or dazed, feeling unreal, difficulty in remembering the whole sequence of the traumatic event, insomnia, restlessness, poor concentration, and physical symptoms of autonomic arousal, especially sweating, palpitations, and tremor. Anger or histrionic behaviour may be part of the response. Occasionally there is a flight reaction—for example, when a driver runs away from the scene of a road accident.

Coping strategies and defence mechanisms are also part of the acute response to stressful events. Avoidance is the most frequent coping strategy, where the person avoids talking or thinking about the stressful events, and avoids reminders of them. The most frequent defence mechanism is denial.

Usually avoidance and denial recede as anxiety diminishes; memories of the events can be more readily accessed, and the person is able to think or talk about them with less distress. This sequence allows the person to work through and come to terms with the stressful experience, although there may be continuing difficulty in recalling the details of highly stressful events.

Variations in the clinical picture

Not all responses to acute stress follow this orderly sequence, in which coping strategies and defences are maintained for long enough to allow the person to function until anxiety and depression subside, and are then abandoned so that working through can occur. Not all coping strategies are adaptive—an example is the excessive use of alcohol or drugs to reduce distress. Defence mechanisms may also be of a less adaptive type, such as regression or displacement. Sometimes defence mechanisms persist for longer than is adaptive—for example, denial may persist for so long that 'working through' is delayed. Sometimes vivid memories of the stressful events intrude into awareness as *images* and *flashbacks* or *disturbing dreams*. *Dissociation* with depersonalization and derealization can also occur.

Diagnostic conventions

As noted above, 'acute stress reaction' in ICD-11 and 'acute stress disorder' in DSM-5 capture psychological responses to stress of an extreme nature. The DSM-5 definition of acute stress disorder emphasizes *dissociation and re-experiencing*. People who develop acute stress disorder are more likely to experience subsequent PTSD (indeed, the symptomatology of PTSD in DSM-5 is similar to that of acute stress disorder, the main difference being in the timing and duration of symptoms). However, around 50% of those who eventually develop

PTSD after a trauma do not meet the criteria for acute stress disorder soon after it.

Both systems of classification describe typical symptoms of the disorder. In DSM-5 the diagnosis of acute stress disorder requires marked symptoms of anxiety or increased arousal, re-experiencing of the event, and three of the following five 'dissociative' symptoms:

1. A sense of numbing or detachment
2. Reduced awareness of the surroundings ('being in a daze')
3. Derealization
4. Depersonalization
5. Dissociative amnesia.

IN DSM-5, there must also be avoidance of stimuli that arouse recollections of the trauma, and significant distress or impaired social functioning.

In ICD-11, dissociative and other symptoms are not required to diagnose the disorder. The ICD classification emphasizes symptoms such as confusion, sadness, anxiety, despair, and social withdrawal, though amnesia and depersonalization experiences are mentioned. Autonomic systems of anxiety are common.

The terms *acute stress reaction* and *acute stress disorder* are used only when the person was free from these symptoms immediately before the impact of the stressful event. Otherwise the response is classified as an exacerbation of pre-existing psychiatric disorder. In both classifications, symptoms should have largely resolved within about a month of exposure to the stressor (unless the stressor or its consequences are ongoing).

Epidemiology

Rates in the general population are unclear. The rate of acute stress disorder has ranged from around 15% in motor accident survivors to over 50% in women victims of sexual assault. After the Wenchuan earthquake in China, about 30% of the survivors met criteria for acute stress disorder (Zhao et al., 2008).

Aetiology

Many kinds of event can provoke an acute response to stress—for example, involvement in a significant but brief event (e.g. a motor accident or a fire), an event that involves actual or threatened injury (e.g. a physical assault or rape), or the sudden discovery of serious illness. Some of these stressful events involve life changes to which further adjustment is required (e.g. the serious injury of a close friend involved in the same accident).

Not all people who are exposed to the same stressful situation develop the same degree of response. This variation suggests that differences in constitution, previous experience, and coping styles may play a part in aetiology. A history of psychiatric disorder and particularly of depression and disassociation prior to the trauma are predisposing factors.

Treatment

Planning for disaster

Planning is needed to ensure an immediate and appropriate response to the psychological effects of a major disaster. Such a response can be achieved by enrolling and training helpers who can support victims and are willing to be called on at short notice, and by agreeing procedures for contacting these helpers promptly. At the time of the disaster, priorities have to be decided between the needs of the victims of the disaster, those of relatives (including children), and those of members of the emergency services, who may be severely affected by their experiences. The essential elements of psychological assistance for victims of disaster have been described by Alexander (2005) (see Box 7.2).

Debriefing

After a major incident, counselling has often taken the form known as debriefing, or critical incident stress debriefing (CISD), provided either individually or in a group. In debriefing the victim goes through the following stages, after the counsellor has first explained the procedure:

- facts—the victim relates what happened
- thoughts—they describe their thoughts immediately after the incident

Box 7.2 The principal components of psychological first aid

- Comfort and consolation
- Protection from further threat and distress
- Immediate physical care
- Helping reunion with loved ones
- Sharing the experience (but not forced)
- Linking survivors with sources of support
- Facilitating a sense of being in control
- Identifying those who need further help (triage)

- feelings—they recall the emotions associated with the incident
- assessment—they take stock of their responses
- education—the counsellor offers information about stress responses and how to manage them.

Debriefing has been widely used, but current evidence suggests that single-session ‘stand-alone’ debriefing is not helpful in lowering subsequent psychological distress, and might even be harmful for some (Forneris et al., 2013). In healthcare settings, team debriefing is widely used to assist with learning and support after critical incidents. This process is primarily designed to improve practice and not for the prevention of trauma-related psychological disturbance. Those showing such symptoms during debriefing should be offered specific psychological therapies (Kolbe et al, 2021).

Management

After a traumatic event, many people talk informally to a sympathetic relative or friend, or to a member of the professional staff dealing with any physical injuries that originated during the incident. If anxiety is severe, an anxiolytic drug may be prescribed for a day or two, and if sleep is severely disrupted a hypnotic drug may be given for one or two nights. Since

in most cases stress reactions will resolve with time, a policy of watchful waiting is appropriate, although it is good practice to offer a follow-up appointment around two weeks after the trauma to identify people whose symptoms are not settling and who are therefore at increased risk of developing the more long-term and disabling PTSD.

If patients show persistent and troublesome symptoms of acute stress disorder at this point, consideration should be given to treatments that may help prevent the onset of PTSD. Approaches of demonstrable efficacy involve verbal exposure to the original event and include *prolonged exposure therapy* (which aims to minimize continuing avoidance) and *trauma-focused cognitive behaviour therapy* (Howlett and Stein, 2016). These treatments are superior to less-specific psychological approaches such as supportive counselling. Effective pharmacological treatments would have the advantage of easy implementation, but none are currently established and the use of selective serotonin reuptake inhibitors (SSRIs) in acute stress disorder to prevent the development of PTSD has not generally proved helpful. On the other hand, research studies indicate that there may in future be a role for treatments that increase corticosteroid levels (Florida et al., 2023).

Post-traumatic stress disorder

This term denotes an intense, prolonged, and sometimes delayed reaction to an intensely stressful event. The three essential features of a post-traumatic stress reaction are as follows:

1. Re-experiencing of aspects of the stressful event.
2. Hyperarousal.
3. Avoidance of reminders.

Examples of extreme stressors that may cause this disorder are natural disasters such as floods and earthquakes, man-made calamities such as major fires and serious transport accidents, or the circumstances of war, and rape or serious physical assault on the person. The original concept of PTSD was of a reaction to such an extreme stressor that any person would be affected by it. Epidemiological studies have shown that not everyone who is exposed to the same extreme stressor develops PTSD; thus personal predisposition plays a part. In many disasters the victims suffer not only psychological distress but also physical injury,

which may increase the likelihood of PTSD. Other predisposing factors are reviewed below in the section on Aetiology. People may experience trauma-related symptoms after serious stressors of a somewhat less intense nature, for example, a relationship break-up. Conventionally, in ICD-11 and DSM-5, such symptomatology would be diagnosed as an adjustment disorder rather than PTSD.

The condition now known as PTSD has been recognized for many years, although under other names. The term PTSD originated during the study of American servicemen returning from the Vietnam War. The diagnosis meant that affected servicemen could be given medical and social help without being diagnosed as suffering from another psychiatric disorder. Similar psychological effects have been reported (under other names) among servicemen in both world wars, and among survivors of peacetime disasters such as the Tohoku earthquake and tsunami in 2011. For a historical review of the concept of PTSD, see Jones and Wessely (2014).

Other reactions to severe stress

PTSD occurs only after exceptionally stressful events, but not every response to such events is PTSD. Six months after a serious accident, major depression may actually be more frequent than PTSD. Also, PTSD is commonly comorbid with other psychiatric disorders, most notably depression, alcohol dependence, substance misuse, and anxiety disorders.

Clinical picture of post-traumatic stress disorder

The clinical features of PTSD can be divided into three groups (see Box 7.3). The most characteristic and diagnostically important symptoms are related to *re-experiencing* (also called *intrusion*) and include flashbacks, recurrent nightmares, and intrusive images or other sensory impressions from the event. The second group of symptoms is concerned with *avoidance*, and includes difficulty in recalling stressful events at will, avoidance of reminders of the events, a feeling of detachment, and inability to feel emotion ('numbing'). The third group of symptoms is related to *hyperarousal*, and includes persistent anxiety, irritability, insomnia, and poor concentration.

Maladaptive coping responses may occur, including persistent aggressive behaviour, the excessive use of alcohol or drugs, and deliberate self-harm and suicide.

Other features

Depressive symptoms are common, and guilt and shame are often experienced by the survivors of a disaster. There can be diminished interest in activities and an inability to experience positive emotions. After some traumatic events, survivors feel forced into a painful reconsideration of their beliefs about the meaning and purpose of life. Some develop exaggeratedly negative views of the world as well as of themselves and others. Dissociative symptoms such as depersonalization are also prominent in some patients.

Onset and course

Symptoms of PTSD may begin very soon after the stressful event, or after an interval, usually of days, but usually within three months. It is now accepted, however, that a minority of cases of PTSD can have a delayed onset or, more accurately, that subthreshold symptoms may not develop to fulfil diagnostic criteria for PTSD for many months or even years. In DSM-5, PTSD cannot be diagnosed until at least one month of symptomatology has elapsed; until then, the condition is regarded as an acute stress disorder. However, in these circumstances it

is doubtful whether the diagnosis of stress-related disorder and PTSD represents two separate conditions. About one third of cases of PTSD remit within three months, but about 40% of patients have a chronic course (Santiago et al., 2013).

Diagnosis

The diagnostic criteria in ICD-11 and DSM-5 are similar, although the latter are broader, with ICD-11 requiring only symptoms of *re-experiencing*, avoidance, and hyperarousal. In DSM-5, in addition to these symptoms, patients must also have some symptoms of *altered cognitions and mood*. By convention, PTSD can be diagnosed in people who have a history of psychiatric disorder before the stressful events.

Differential diagnoses include the following:

- Stress-induced exacerbations of previous anxiety or mood disorders.
- Acute stress disorder (distinguished by the time course).
- Adjustment disorders (distinguished by the different pattern of symptoms and/or nature of the stressor, which is not required to be extreme).

PTSD may present as deliberate self-harm or substance misuse which have developed as maladaptive coping strategies (see above).

Box 7.3 The principal symptoms of post-traumatic stress disorder

Hyperarousal

Persistent anxiety
Irritability
Insomnia
Poor concentration

Re-experiencing

Intense intrusive imagery
'Flashbacks'
Recurrent distressing dreams

Avoidance

Difficulty in recalling stressful events at will
Avoidance of reminders of the events
Detachment
Inability to feel emotion ('numbness')
Diminished interest in activities

Prevalence

Estimates of PTSD in the general population have mainly been obtained from the United States, where lifetime rates (using older diagnostic criteria) have been between 6% and 9%. Rates in high-risk groups (for example, soldiers exposed to combat) can be much higher, up to 40% (Sareen, 2014).

Aetiology

The stressor and predisposing factors

The necessary cause of PTSD is an exceptionally stressful event. It is not necessary that the person should have been harmed physically or threatened personally; those involved in other ways may develop the disorder—for example, the driver of a train in whose path someone has thrown himself for suicide, and the bystanders at a major accident. DSM-5 describes such events as involving actual or threatened death or serious injury or a threat to the physical integrity of the person or others.

Rates of PTSD are higher in those who have been exposed to interpersonal violence, particularly rape. Even so, not all of those affected by the stressor will develop PTSD, a finding that indicates that some form of personal vulnerability plays a part. Such vulnerability might be genetic or acquired. Epidemiological research has revealed that important risk factors for PTSD include female gender, lower educational level, and previous interpersonal trauma. A previous history of an anxiety or mood disorder also increases the risk of developing PTSD (Shalev et al., 2019).

Genetic factors

Studies of twins suggest that differences in susceptibility to PTSD are in part genetic. True et al. (1993) studied 2,224 monozygotic and 1,818 dizygotic male twin pairs who had served in the US armed forces during the Vietnam War. After allowance had been made for the amount of exposure to combat, genetic variation accounted for about one third of the variance in susceptibility to self-reported PTSD. Self-reported childhood and adolescent environment did not contribute substantially to this variance. The genetic liability to PTSD is partly explained by a genetic effect on *personality*, which modifies the propensity of individuals to engage in risky behaviours. However, even when allowing for genetic effects on personality, there is an additional genetic influence on the liability to experience PTSD after a given trauma. Several genome-wide association studies (GWAS) of PTSD have been carried out with the aim of

identifying causative genes, but reliable findings have not yet emerged (Gottschalk, 2020).

Neurobiological correlates

Research to date on the neurobiology of PTSD has focused on monoamine neurotransmitters and the hypothalamic–pituitary–adrenal (HPA) axis, both of which are involved in mediating defensive responses to stressful events. In addition, brain imaging studies have implicated changes in the *hippocampus*, a brain region that is important in memory formation, and the *amygdala*, which plays a role in non-conscious emotional processing (see Box 7.4). These findings suggest that hippocampal dysfunction prevents adequate memory processing, while increased activity in noradrenergic innervation of the amygdala increases arousal and facilitates the automatic encoding and partial recall of traumatic memories. Functional imaging studies in PTSD suggest overactivity of the amygdala

Box 7.4 Neurobiological abnormalities in PTSD

Hypothalamic–pituitary–adrenal axis

Evidence, albeit somewhat contradictory, for low plasma cortisol levels and increased sensitivity to dexamethasone suppression. Increased levels of corticotropin-releasing hormone in cerebrospinal fluid (CSF).

Noradrenaline

Increased sympathetic tone. Increased startle response. Increased levels of 3-methoxy-4-hydroxy-phenylglycol (MHPG) in CSF. Increased anxiety response to noradrenaline challenge.

Brain imaging

Smaller volume of the hippocampus (which may be a vulnerability factor), overactivity of the amygdala in response to traumatic psychological stimuli, as well as decreased activity in anterior cingulate cortex and prefrontal cortex. More recent ‘network’ models have implicated diminished activity in the central executive and default mode networks coupled with increased activity in the salience network. These neural changes could correlate with psychological symptoms of avoidance and dissociation, and the ready triggering of flashbacks and intrusive memories (Abdallah et al., 2019).

in the setting of decreased regulatory control of the amygdala and other limbic regions by the ventromedial prefrontal cortex.

Psychological factors

Fear conditioning

Some patients with PTSD experience vivid memories of the traumatic events in response to sensory cues, such as smells and sounds related to the stressful situation. This finding suggests that classical conditioning may be involved, as well as failure to *extinguish* conditioned responses.

Cognitive theories

These suggest that PTSD arises when the normal processing of emotionally charged information is overwhelmed, such that memories persist in an unprocessed form in which they can intrude into conscious awareness. In support of this idea, patients with PTSD tend to have incomplete and disorganized recall of the traumatic events. Individual differences in response to the same traumatic events are explained as being because of differences in the appraisal of the trauma and of its effects. Similarly, differences in the appraisal of the early symptoms may explain why these symptoms persist for longer in some individuals. Negative interpretations of intrusive thoughts (e.g. 'I am going mad') after road accidents predict the continuing presence of PTSD after one year. The cognitive model of PTSD has been reviewed by Beierl et al. (2020).

Maintaining factors

As noted above, symptoms of PTSD may be maintained in part by negative appraisals of the early symptoms. Other suggested maintaining factors include avoidance of reminders of the traumatic situation (which prevents deconditioning and cognitive reappraisal), suppression of intrusive memories (which is known to make them more likely to recur), and rumination

Assessment

This should include enquiries about the nature and duration of symptoms, previous personality, and psychiatric history. When the traumatic events have included head injury (e.g. in an assault or transport accident), a neurological examination should be performed. Feelings of anger and thoughts of self-harm are common in PTSD, and an appropriate risk assessment needs to be carried out. Secondary complications such as substance

misuse may require treatment in their own right (see Chapter 20).

Treatment

By the time patients with PTSD are assessed for treatment, the disorder may have been present for many months or even years, and comorbidity with depression and substance misuse often complicate management. Psychological treatments are generally preferred in the treatment of PTSD, although pharmacotherapy has a role in patients presenting with significant comorbid depression or where psychological approaches are not beneficial or unavailable. Psychological treatments such as *trauma-focused cognitive behaviour therapy* (CBT), *eye movement desensitization and reprocessing* (EMDR), and *narrative exposure therapy* (NET) are more effective than less-specific treatments such as stress management, supportive therapy, and hypnotherapy. It should be noted that the majority of studies for treating PTSD have been conducted following 'single-episode' events and that those exposed to multiple traumatic events, such as humanitarian disasters, conflict, and prolonged sexual abuse might need different treatments. Patients who have undergone repeated traumas can be classified in ICD-11 as having complex PTSD (see below).

Cognitive behaviour therapy

CBT is the most appropriate treatment. It has several components when applied to PTSD:

- Information about the normal response to severe stress, and the importance of confronting situations and memories related to the traumatic events.
- Self-monitoring of symptoms.
- Exposure in imagination and then *in vivo* to situations that are being avoided.
- Recall of images of the traumatic events, to integrate these with the rest of the patient's experience. When first recalled these images are often fragmentary and are not clearly related in time to the other contents of memory.
- Cognitive restructuring through the discussion of evidence for and against the appraisals and assumptions.
- Anger management for people who still feel angry about the traumatic events and their causes.

A meta-analysis of psychotherapy studies of PTSD suggests that CBT treatments with a strong exposure element have a therapeutic effect size of around 1.5 when

compared with waitlist control, and around 1.2 when compared with treatment as usual (McLean et al., 2022). At the end of psychological treatment, around 50% of patients no longer meet the criteria for PTSD, although many are still symptomatic to some degree. Appropriately modified trauma-focused CBT can be delivered via the internet with good therapeutic effects though dedicated therapist time is still required to derive full benefit (Ehlers et al., 2023).

EMDR was designed specifically for the treatment of PTSD. Treatment trials using this technique in subjects with PTSD have shown similar effect sizes to those obtained with CBT, but the evidence base is not as large. Some have questioned whether the eye movements associated with this treatment add any specific therapeutic value to the element of exposure.

NET is a more recently developed treatment that aims to enhance autobiographical memory processing of the trauma by embedding it in a chronological life narrative developed collaboratively by patient and therapist. *NET* was developed originally for those suffering from PTSD following multiple traumatic events and was first trialled in refugee camps. It is a treatment that has been successfully delivered by lay therapists and it has been adapted for use in children. It is being increasingly used in high-income settings as well as low- and middle-income countries, and has a growing evidence base. For a review, see Lely et al. (2019).

Medication

Anxiolytic drugs such as benzodiazepines should be avoided in patients with established PTSD, because prolonged use may lead to dependence. A number of antidepressant drugs have shown efficacy in clinical trials, including SSRIs, serotonin and noradrenaline reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), and monoamine oxidase inhibitors (MAOIs). There are also more preliminary data supporting the efficacy of mirtazapine, and augmentation with atypical antipsychotic drugs such as olanzapine may have a place in treatment-resistant patients, particularly those with marked sleep disturbance. The anticonvulsant drug topiramate also appears effective but based on fewer studies (Cipriani et al., 2018).

Meta-analyses reviewed by the National Institute for Health and Care Excellence (2018b) indicate that, in patients with PTSD, structured psychotherapies generally have higher effect sizes than drug treatment, and medication should therefore not be a first-line approach unless the patient expresses a preference for

Box 7.5 National Institute for Health and Care Excellence (NICE) guidance for the management of PTSD and related traumatic syndromes

1. The routine use of a brief screening instrument for PTSD at one month post incident should be considered for all people who have been involved in a major disaster.
2. Where symptoms are mild and have been present for less than four weeks after the trauma, watchful waiting—as a way of managing the difficulties presented by individual sufferers—should be routinely considered by healthcare professionals. A follow-up contact should be arranged within one month.
3. Trauma-focused CBT should be considered for those with severe post-traumatic symptoms during the first month after the traumatic event. All PTSD sufferers should be offered a course of trauma-focused psychological treatment.
4. Drug treatments for PTSD should not be used as a routine first-line treatment in preference to a trauma-focused psychological therapy.
5. Drug treatments such as SSRIs and venlafaxine should be considered for the treatment of PTSD where a sufferer expresses a preference not to engage in a trauma-focused psychological treatment, or such treatment has proved ineffective. Antipsychotic drugs either alone or in combination with antidepressants may be considered by specialists.

Source: National Institute for Health and Care Excellence (NICE). (2018b). Post-traumatic stress disorder. NICE guideline NG116. <https://www.nice.org.uk/guidance/ng116>. Copyright (2018), National Institute for Health and Care Excellence.

it or psychotherapy is not available or is ineffective (Box 7.5).

Complex PTSD

ICD-11 has an additional category of PTSD called complex post-traumatic stress disorder (CPTSD). In this condition, patients suffer typical symptoms of PTSD (re-experiencing, avoidance and hyperarousal) but also manifest significant problems in three other psychological domains: affect regulation, self-image, and intimate relationships (Box 7.6).

Box 7.6 ICD-11 criteria for complex post-traumatic stress disorder

All criteria for PTSD are met, plus:

1. *Severe and pervasive problems in affect regulation.* For example, heightened emotional reaction to minor stresses, violent outbursts, self-destructive behaviour, emotional numbing with inability to experience positive emotions.
2. *Beliefs about the self as defeated or worthless.* For example, pervasive feelings, of shame, guilt, and failure, particularly related to the traumas experienced.
3. *Difficulties in sustaining relationships and feeling close to others.* For example, avoidance of relationships, diminished interest in pursuing and sustaining relationships, generally restricted social engagement.

The disorder results in significant impairment in personal, occupational or social functioning. If function is maintained, it is only through significant additional effort.

Additional clinical features can include suicidality, substance abuse, psychotic symptoms and somatic complaints.

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Traditional concepts of PTSD viewed the disorder as a reaction to a single horrifying event. CPTSD is typically a response to a series of prolonged or repetitive traumas from which escape is not possible. Such experiences are encountered, for example, in repeated childhood maltreatment, slavery, torture, and prolonged domestic abuse. Nevertheless, it is possible for people exposed to a single trauma to develop CPTSD and, conversely, some who have undergone repeated traumatic experiences develop PTSD rather than CPTSD. This presumably reflects differences in individual susceptibility. Not surprisingly, CPTSD is associated with a greater burden of symptomatology and problems in daily living than PTSD (Cloitre, 2020).

Some of the features of CPTSD, for example, affective instability, are also characteristic of borderline personality disorder. However, currently the two conditions are regarded as distinct. People with CPTSD have a negative but stable self-image as opposed to the fluctuating self-concept of those with borderline personality. Also, the relationship difficulties experienced in CPTSD are characterized by avoidance rather than the vacillating engagement and disengagement seen in borderline personality.

Currently, there are no specific treatments for CPTSD. It is recommended that CBT, as for PTSD, should be provided, although the number of treatment sessions required will probably be greater because of the higher burden of symptomatology. Additional specific psychological treatment is likely to be needed to improve affect regulation and increase self-compassion. For a review, see Maercker et al. (2022)

Response to specific kinds of severe stress

Rape and physical assault

Many victims of rape or other sexual or physical assault experience acute reactions to stress, PTSD, anxiety and depressive disorders, and psychosexual dysfunction. In one survey of 4,000 women in Australia, just over a quarter reported exposure to one or more events of 'gender-based violence' (rape and sexual assault, intimate partner violence, or stalking). Following the trauma, around 25% experienced PTSD, but rates for mood and anxiety disorders were even higher (34% and 38%, respectively). Substance use disorder was also common (29%) (Rees et al., 2014).

As well as experiencing symptoms of PTSD, victims of rape and assault often feel humiliated, ashamed, and remain vulnerable to further attack. They lose confidence and self-esteem, question why they were chosen as victims, and blame themselves for putting themselves in unnecessary danger. To these problems are added issues of betrayal and secrecy when the assault is carried out by a family member or a friend. The victim may experience problems in trusting, persistent anger and irritability, and excessive dependence. These problems were first described among female victims of rape, but similar difficulties have been described among male victims of sexual assault.

Problems are more likely to persist when there has been an actual or perceived threat to life, previous psychological and social problems, past victimization, particularly abuse in childhood, past psychiatric illness, or substance abuse, or a lack of social support.

Treatment

Social support is important in providing opportunities for the victim to talk over the problem and to regain self-esteem. Specific treatment for trauma-related symptoms is similar to that for other kinds of PTSD, including prolonged exposure by reliving the events in imagination, with additional emphasis on overcoming feelings of vulnerability and self-blame. There is some support for the use of early interventions based on trauma-focused CBT to decrease the risk of subsequent PTSD (Lomax and Meyrick, 2022).

Combatants

'Shell shock', 'battle fatigue', and 'war neurosis' were terms used during the First World War to describe psychological reactions to battle among British and American servicemen. Most of the reactions appear to resemble cases now diagnosed as PTSD; others seem to have resembled panic disorder or depressive disorders. Cases with panic attacks and concerns about the heart, now diagnosed as panic disorder, were known then as Da Costa's syndrome or disorderly action of the heart. Army psychiatrists were few in number, and patients with shell shock were treated mainly by neurologists or psychologists.

Treatment

At first, shell shock was treated with the methods in use at the time for neurasthenia, namely, evacuation from the combat zone, rest, isolation, massage, and diet, but these methods had a low success rate. Hypnosis achieved some dramatic cures, but was not generally effective. Medical psychologists tried psychotherapeutic methods advocated by Freud, including the recall of stressful events to remove repression and the expression of associated emotion. Later, there was an increasing emphasis on early treatment, keeping the soldier with his colleagues at the front with the explicit aim of rapid return to action ('the soldier's interest and the army's interest are the same'). This approach has been termed '*forward psychiatry*' (Jones and Wessely, 2003), in which psychotherapy is combined with military drill to maintain general fitness and morale. This combined treatment led to

improved results. It became apparent that chronicity increased dramatically if soldiers were evacuated from the theatre of war.

More recent approaches have included pre-deployment training in the recognition and management of acute stress reactions in combat. In the operational setting a behavioural stepped-care approach is used with specialist providers linked to combat brigades providing psychological first aid. This is designed to address stress-related symptoms with the aim of rapidly restoring military performance. For a review, see Matson et al. (2022),

Problems of refugees and victims of torture

Refugees and other forced migrants may have experienced a wide range of traumatic events, either in their countries of origin, in their journey to a place of potential refuge, and then in settling in a new location. The potential exposure to traumatic events can include the following:

- The impact of armed conflict directly on refugees and on their families and communities, usually through violence.
- Loss by death or separation from relatives and friends.
- Loss of home and possessions.
- Physical injury either from the actions of war or from assault, rape, or torture, and the witnessing of violence to others.
- Experience of refugee camps or immigration detention facilities and threats by immigration authorities of being returned to their countries of origin.

Those involved may develop any of the reactions to stressful events, especially PTSD, depressive and anxiety disorders, and substance misuse. These conditions have been identified in refugees from many cultures, although there are important cultural influences on the experience, meaning making, and therapeutic processes of these conditions (see Patel and Hall, 2021).

It is important to remember that stressors related to a refugee's current situation can be just as difficult to deal with as those that led to flight in the first place. In a systematic review, Blackmore et al. (2020) found that around 30% of refugees had symptoms consistent with PTSD. A similar proportion met criteria for major depression. In many individuals these disorders coexisted.

Treatment

Treatment should combine physical and psychiatric care, and requires a number of skills and sources of expertise (Bhui, 2022). The latter should be introduced carefully, since it may be resisted as stigmatizing, not only by the refugee but also by humanitarian workers. Special care is needed to establish a trusting relationship. Practical help is often an essential preliminary to engagement. Many refugees have problems related to separation, bereavement, loss of material possessions, and immigration status, so it is important not to focus narrowly on PTSD.

Generally psychological treatments based on CBT and NET approaches seem most beneficial for the treatment of PTSD and associated anxiety and depression in refugees. In victims of torture it is helpful, if the patient agrees, to document the episodes of torture that they have experienced. As well as providing valuable information for purposes of advocacy and legal proceedings, the documentation can be developed into *testimonial therapy*, in which patients form a narrative of their experience which can be delivered to others, including their household (Duffy and Kelly, 2015).

Health beliefs and understanding of the ‘normal’ psychological response to stress may well differ between a health worker from one culture and a forced migrant from another. Prior to psychological treatment, or where its use does not seem suitable, provision of language classes, informal support, and acquisition of computer skills can be very beneficial (Duffy and Kelly, 2015). In refugee children and adolescents, psychological interventions have been delivered in community, school,

and camp settings with variable effects. The strongest positive effects are for group or individual exposure-based therapies, including trauma-focused CBT and NET (Tyrer and Fazel, 2014).

Care is needed when working through interpreters, especially if these interpreters come from the same cultural group and occasionally from the group representing the ‘other side’ of the conflict. There are advantages and disadvantages in using family members as interpreters, but it is important to ensure that some therapeutic time is made available with a non-family, non-acquaintance performing the interpretation as the refugee is unlikely to speak of what they regard as shameful experiences in front of family members, community elders, or others they might perceive as likely to speak to people they know.

This point is especially important in situations in which the person may have experienced sexual assaults (both men and women), for these may be construed in some cultures as bringing shame to the whole family. Ideally, such problems should be dealt with by a same-gender mental health professional who understands the patient’s language and culture, but this may be difficult to arrange. If conducting exposure-based therapy, it is best to try and arrange for all the sessions to be conducted with the same interpreter, who might themselves need emotional support after any sessions (especially if they have also come from an area of armed conflict).

For further information about the psychiatric problems of refugees and victims of torture, see Duffy and Kelly (2015).

Adjustment disorders

This term refers to the psychological reactions that arise in relation to adapting to new and challenging circumstances. Such circumstances include divorce and separation, a major change of work and abode (e.g. transition from school to university, or migration), and the birth of a child with disabilities. Bereavement, the onset of a terminal illness, and childhood sexual abuse are associated with specific adjustment difficulties, which are discussed below.

Clinical features

The symptoms of an adjustment disorder include anxiety, worry, poor concentration, depression, and irritability, together with physical symptoms caused by autonomic arousal, such as palpitations and tremor. There may be

outbursts of dramatic or aggressive behaviour, single or repeated episodes of deliberate self-harm, or the misuse of alcohol or drugs. The onset is more gradual than that of an acute reaction to stress, and the course is more prolonged. Social or occupational function is impaired. The *impairment in social or occupational function*, as well as the *intensity of distress*, is what distinguishes adjustment disorder from normal adaptive reactions.

A further requirement for diagnosis is that the disorder starts soon after the change of circumstances. DSM-5 requires that the disorder starts *within three months*, and ICD-11 indicates that it usually begins within *one month*. An essential point is that the reaction is understandably related to, and in proportion to, the stressful experience when account is taken of the patient’s previous

experiences and personality. Once the stressor or its consequences are removed, the symptoms should resolve within *six months*.

Diagnostic conventions

Stressful life events may precipitate depression, anxiety, schizophrenia, and other psychiatric disorders. For this reason, the diagnosis of adjustment disorder is not made when diagnostic criteria for another psychiatric disorder are met. Therefore, if a stressful experience leads, for example, to symptomatology that meets criteria for a depressive disorder, then the diagnosis is depression rather than adjustment disorder. While patients with adjustment disorder may have significant depressive symptoms, these symptoms should not meet the threshold for a diagnosis of major depression.

As explained previously under 'Classification of reactions to stressful events', DSM-5 lists six types of adjustment disorder (see Table 7.1). ICD-11 does not describe specific types but comments that in adjustment disorder, the emotional reactions to the stressor include excessive worry, upsetting thoughts about the stressor, and rumination about its implications. Depression can also occur, as may externalizing symptoms such as aggressive behaviour and drug misuse.

Epidemiology

The prevalence of adjustment disorder in the community has been little studied. A community study in older people found that the prevalence of adjustment disorders was about 2%. It is presumed that in certain settings, such as the general hospital and primary care, prevalence rates are increased. For example, in studies of general hospital outpatients the prevalence ranges from 5% to 20%, while in hospital psychiatric consultations, the rates may be as high as 50% (Kilpatrick et al., 2020). High rates are also seen in people who make suicide attempts (see page 593).

Aetiology

Stressful circumstances are the necessary cause of an adjustment disorder, but individual vulnerability is also important, because not all people who are exposed to the same stressful circumstances develop an adjustment disorder. The nature of this vulnerability is unknown; it seems to vary from one person to another, and may relate in part to previous life experiences.

Prognosis

Clinical experience suggests that most adjustment disorders last for several months, and a few persist for years if the stressor or its consequences cannot be removed. In a systematic review of the literature, Morgan et al. (2022) concluded that, while the prognosis for adjustment disorder in adults is generally better than that of other mental health conditions, about 30% of patients subsequently developed a more serious psychiatric disorder. The risk of developing general health conditions, such as cardiovascular disease, dementia, and Parkinson's disease, was also increased. A diagnosis of adjustment disorder was associated with an elevated risk in overall mortality, as well as a separate increased risk of suicide.

The latter findings support the traditional clinical advice that persistence of adjustment disorder should lead to a review of diagnosis in case symptomatology for a formal anxiety or depressive disorder has developed. As noted above, adjustment disorders can be associated with suicidal ideation and behaviour. Therefore, clinical assessments should include careful questioning about risk.

Treatment

Treatment is designed to help the patient to resolve the stressful problems if this is possible, and to aid the natural processes of adjustment. The latter is done by reducing denial and avoidance of the stressful events, encouraging problem solving, and discouraging maladaptive coping responses. Anxiety can usually be reduced by encouraging the patient to talk about the problems and to express their feelings. Occasionally, an anxiolytic or hypnotic drug is needed for a few days.

Problem-solving counselling (see Chapter 24) encourages the patient to seek solutions to stressful problems, and to consider the advantages and disadvantages of various kinds of action. The patient is then helped to select and implement a course of action to solve the problem. If this action succeeds, another problem is considered. If the first attempt fails, another approach to the original problem is tried. If problems cannot be resolved, the patient is encouraged to come to terms with them.

More formal CBT and pharmacological approaches have also been employed in adjustment disorder. However, the quality of the relevant clinical trials has been assessed as low, and currently there are no evidence-based guidelines to inform treatment approaches (Constantin et al., 2020).

Special kinds of adjustment

Adjustment to physical illness and disability

Appraisal of illness

Adjustment to illness cannot be understood simply in terms of the facts about the disease and its objective consequences. Adjustment depends on the patient's beliefs about their disorder and its effects on their life—in other words, on their appraisal of their illness. This appraisal may be similar to that of the professionals who are treating them, or it may be very different because it is based on false information or on emotions rather than on facts, or influenced by cultural beliefs. The appraisal may be reinforced by members of the family who share the patient's views, or it may be contradicted by them, thus adding to the patient's distress. Two terms are much used in the discussions of adjustment to illness and handicap—*illness behaviour* and the *sick role*. These terms will be considered next.

Illness behaviour

Mechanic (1978) suggested the term *illness behaviour* to describe behaviour associated with adjustment to physical or mental disorder, whether adaptive or not. Illness behaviour includes consulting doctors, taking medicines, seeking help from relatives and friends, and giving up inappropriate activities. These behaviours are adaptive in the early stages of illness, but may become maladaptive if they persist into the stage of convalescence when the patient should be becoming independent. Illness behaviour results from the person's conviction that they are ill, rather than from the objective presence of disease, and it may develop when no disease is present. Illness behaviour without disease is an important problem in general practice and, once firmly established, can be difficult to treat. The concept of illness behaviour overlaps with that of the sick role, but the two are described separately because they have different origins.

The sick role

Society bestows a special role on people who are ill. The sociologist Talcott Parsons (1951) called this the *sick role*, which is made up of two privileges and two duties:

1. Exemption from certain social responsibilities, and the right to expect help and care from others; and

2. The obligation to seek and cooperate with treatment, and the expectation of a desire to recover.

While the person is ill, the sick role is adaptive. If they continue in the sick role after the illness is over, recovery is delayed as they continue to avoid responsibilities and depend on others instead of becoming independent.

Adjustment to the onset of physical illness

When a person becomes physically ill, they may feel anxious, depressed, or angry. Usually this emotional reaction is transient, subsiding as the patient comes to terms with the new situation. As in other adjustment reactions, denial or minimization can protect the patient against overwhelming anxiety when the diagnosis is first known. Although helpful in this way, denial can be maladaptive—in the early stage of illness it may lead to delay in seeking help, and at a later stage it may lead to poor compliance with treatment. Other coping strategies can be divided into emotion-reducing and problem-solving groups. Coping strategies that reduce emotion are often appropriate in the early stages of illness, but should give way to problem-solving coping. Coping may fail when demands are very great or when coping resources are limited either in the long term, or as a temporary result of disease or, trauma to, the brain.

Physical illness as a direct cause of psychiatric symptoms

As well as acting as a stressor, physical illness may induce psychiatric symptoms directly. Anxiety, depression, fatigue, weakness, weight loss, or abnormal behaviour may all be caused directly by physical disorders; common examples are listed in Chapter 22, Box 22.5). Similarly, sexual function may be impaired by physical illness or its treatment (Chapter 13). Any of these symptoms may be the reason for referral, and psychiatrists should always be alert for the possibility of undetected physical illness in their patients.

Psychiatric symptoms due to treatments for physical illness

Some drugs that are used in the treatment of physical illness may affect mood, behaviour, and consciousness. The drugs most likely to have these effects are listed in Chapter 25, Table 22.4.

Help for people who are adjusting to physical illness

Most people adjust well to physical illness, but when adjustment is slow and incomplete, psychological treatment may be needed. This treatment need not be complicated, and can usually be provided effectively by the general practitioner or the hospital doctors or nurses who are managing the physical illness. Generally, the psychiatrist has a role in treating only the more severe problems or in supporting the medical and nursing staff.

The first step is to identify patients who are adjusting badly (i.e. failing to cope). This is generally done by the professional staff who are managing the physical illness. They can do this most easily by looking out for patients who are progressing less well than would be expected on the basis of the severity of the disease. Mood disorders are a common cause of slow progress, but may be dismissed as normal responses to the problems of the illness. Screening questionnaires can be used to detect mood disorders among these patients, but the results should be checked at least by a brief interview, which should be conducted, if possible, in surroundings in which their replies will not be overheard. Generally, one or more members of the family should be interviewed to obtain information about the patient's previous adjustment to problems and illness, and to discover how the family views the illness.

Some patients require medication, but for many counselling (see Chapter 24) is more appropriate. Counselling requires a trusting relationship with the patient, and this in turn requires adequate time for the interviews. Counselling begins with an explanation of the nature of the illness and its treatment. The patient is then helped to accept the implication of the diagnosis, to adjust to the illness, and to give up any maladaptive behaviours, such as excessive dependence on others or denial of the need for treatment. Graded activities, motivational interviewing, and anger management may be useful in some cases. More structured psychological treatment in the form of CBT may also be useful.

If the reaction to physical illness is an anxiety or depressive condition, treatment appropriate to that disorder should be given (see Chapters 8 and 9). Randomized trials of cancer patients with depression have shown that, compared to treatment as usual, a collaborative intervention by cancer nurses, psychiatrists, and primary care physicians substantially improved depressive symptomatology as well as diminishing pain and fatigue over the following six months (Sharpe et al., 2014).

Adjustment to terminal illness

Among patients referred to palliative care services symptoms of anxiety and depression are common. Particularly in the final days of life, cognitive impairment and delirium can complicate management. Determinants of emotional reactions include the patient's personality, and the amount and quality of support from family, friends, and carers. Understandably, emotional reactions are more common among young dying patients than among the elderly. Surveys indicate that loss of independence and dignity and the management of pain are major concerns.

Anxiety

Anxiety may be provoked by the prospect of severe pain, disfigurement, or incontinence, by fear of death, and by concerns about the future of the patient's family. Families and carers sometimes try to spare the patient anxiety by concealing the truth about the condition. Since most patients become aware of the diagnosis, attempts at concealment only serve to increase their fear of possible consequences of the disease, such as pain or incontinence.

Depression

Depression may be provoked by the prospect of separation from family and friends and the loss of valued activities. Changes in physical appearance caused by the illness, the effects of surgery, and the debilitating effects of radiotherapy are other causes of low mood.

Guilt and anger

Some patients experience guilt because they believe that they are making excessive demands on relatives or friends. Patients with religious beliefs may believe that illness is a punishment for previous wrongdoing; conversely, anger may be felt about the unjustness of impending death; this anger may be displaced on to doctors, nurses, and relatives, making care more difficult (see below).

Symptoms induced directly by illness or its treatment

Psychological symptoms may also be induced by the disease or by its treatment, and these may add to the patient's distress. There is a particularly strong association between dyspnoea and anxiety, and drugs may cause or add to delirium. The more frequent associations between disease and psychological symptoms are summarized in Box 22.5. The associations between drug

treatment and psychological symptoms are summarized in Table 22.3.

Treatment

In the UK about 60% of deaths occur in hospitals and care homes, and less than 10% in specialist hospice care (Galappathie and Khan, 2016). Dying patients are usually helped to adjust by the staff who are managing the physical illness. Psychiatrists are called upon only when there are special problems (see below) or to assist with staff support and training.

The aims of treatment

Surveys have shown that patients believe the following factors are important in achieving a 'good death':

- managing symptoms
- avoiding prolongation of dying
- achieving a sense of control
- relieving burdens placed on the family
- the strengthening of relationships.

Advance directives can help achieve these aims by preparing for the end of life and communicating future wishes; this might be particularly helpful if there is subsequent loss of capacity.

Kubler-Ross (1969) formulated an influential model which described five phases of psychological adjustment to death. The phases do not necessarily occur in the same sequence, and some of them may not occur at all, but they are a useful guide for professionals who are helping dying patients. They can be summarized as follows:

1. Denial and isolation
2. Anger
3. Partial acceptance ('bargaining for time')
4. Depression
5. Acceptance.

This model has been criticized as being rather mechanistic and failing to capture the complex and unique life experience that each patient and their family bring to the imminent prospect of death. A qualitative study of Danish hospice patients found that patients faced imminent death with sorrow for the separation to come from family and close friends, and concern for how loved ones would fare without them. However, they did not suffer undue anxiety and continued to take significant interest in the daily life of the hospice (Moestrup and Hansen, 2015).

Reducing symptoms

Adequate control of pain and breathlessness and the reduction of confusion due to delirium are particularly important. Anxiety and depression may diminish as pain and breathlessness are controlled. The causes of delirium are listed in Box 14.2. Among dying patients, important remediable causes are dehydration, the side effects of drugs, secondary infection, cardiac or respiratory failure, and hypercalcaemia.

Helping the patient to adjust

It is essential to establish a good relationship with the patient so that they can talk about their problems and ask questions. The nature of the illness should be explained honestly and in simple language. Sometimes doctors are apprehensive that such an explanation will increase the patient's distress. Although excessive detail given unsympathetically can have this effect, it is seldom difficult to decide how much to say about diagnosis and prognosis, provided that patients are allowed to lead the discussion, express their worries, and say what they want to know. If patients ask about the prognosis, they should be told the truth; evasive answers undermine trust in the carers. If a patient does not seem to wish to know the full extent of their problems, it is usually better to save this information until later. At an appropriate stage the patient should be told what can be done to make their remaining time as comfortable as possible.

While the whole account should be truthful, the amount that is disclosed on a single occasion should be judged by the patient's reactions and by their questions. If necessary, the doctor should be prepared to return for further discussion when the patient is ready to continue. It is important to bear in mind that most dying patients become aware of their prognosis whether or not they are told directly, because they infer the truth from the behaviour of those who are caring for them. They notice when answers to questions are evasive and when people avoid talking to them. Patients who are anxious, angry, or despairing need to be able to express these feelings and to discuss the ideas that induce them.

Informing the staff

All of the staff should be made aware of the information that has been given to the patient, otherwise conflicting advice and opinions may be offered. If all those involved know what has been said, they will feel more at ease when talking to the patient. Otherwise they will draw back from the patient, isolating them, and thereby increasing their difficulty in adjusting.

Informing and supporting the relatives

Relatives need to know what has been said to the patient so that they will feel less ill at ease when talking to them. They may need as much help as the patient. They may become anxious and depressed, and they may respond with guilt, anger, or denial. Such reactions make it difficult for them to communicate helpfully with the patient or the staff. Relatives need information, and opportunities to talk about their feelings and to prepare for the impending bereavement, otherwise the patient and their family may become increasingly distant and alienated. Patients might also need specific advice as to how to explain their illness to their children, as sensitively informing children of parental illness is important (Dalton et al., 2019).

Specialist services

In many hospitals, *specialist nurses* work with the family doctor and with the hospital staff who are caring for dying patients. These nurses are trained in the psychological as well as physical care of the dying. Such care is increasingly provided in *hospices*, where it is possible to provide close attention to the details of care that improve quality of life for the dying person. These hospices care for patients when home care is impractical, and provide periods of respite care to relieve those who are caring for the patient at home. Local hospice teams help with care of people who are dying at home.

Referral to a psychiatrist

Referral to a psychological medicine team and/or psychiatrist is appropriate when psychiatric symptoms or behaviour disturbance start to emerge. The referrals are concerned with the assessment and management of:

- *Depressed patients*, to determine the cause and whether the patient requires medication or more structured psychotherapy.
- *Uncommunicative patients* who will not talk about the illness.
- *Uncooperative patients* who do not accept the social restrictions imposed by the illness, will not make appropriate plans, or will not take the necessary decisions.
- *Longstanding problems that are made worse by the illness* and that are related to personality or family conflicts.
- *Other symptoms*: although anxiety and delirium are common, these problems are more often dealt with appropriately by medical staff than referred to a psychiatrist; the exception is delirium with paranoid symptoms.

Management of depressive disorders

Depressive disorders may be caused by symptoms such as pain or breathlessness, and all such symptoms should be treated appropriately. Any drugs that can cause depression (see Table 22.4) should be reviewed and, if possible, given at a lower dose or replaced. Some symptoms of depressive disorder are difficult to evaluate in patients with advanced cancer, as weight loss, anorexia, insomnia, loss of interest, and fatigue may be caused by the physical illness. Early-morning waking, extreme hopelessness, and self-blame are more reliable guides to diagnosis. Suicidal ideation should be assessed carefully. If counselling and improved medical management do not improve the low mood, antidepressant drugs should be prescribed with careful supervision. The starting dose should be small, and medication should be changed if necessary to find a compound that is well tolerated.

Liaison with medical and nursing staff

Liaison with medical and nursing staff is important. Often these staff can provide treatment when the psychiatrist has formulated a plan. For further information about end-of-life care, see Galappathie and Khan (2016) and Box 7.7.

Box 7.7 Priorities for care of the dying person

1. The possibility that a person may die within the next few days or hours is recognized and communicated clearly, decisions made, and actions taken in accordance with the person's needs and wishes, and these are regularly reviewed and decisions revised accordingly.
2. Sensitive communication takes place between staff and the dying person, and those identified as important to them.
3. The dying person, and those identified as important to them, are involved in decisions about treatment and care to the extent that the dying person wants.
4. The needs of families and others identified as important to dying person are actively explored, respected, and met as far as possible.
5. An individual plan of care, which includes food and drink, symptom control, and psychological, social, and spiritual support, is agreed, coordinated and delivered with compassion.

Source: Leadership Alliance for the Care of Dying People, Copyright (2014).

Grief and adjustment to bereavement

Terminology

Although the words ‘bereavement’, ‘mourning’, and ‘grief’ are sometimes used interchangeably, they have separate meanings, which incorporate distinctions that are useful in psychiatry.

- *Bereavement* is the loss through death of a loved person.
- *Grief* is the involuntary emotional and behavioural response to bereavement.
- *Mourning* is the voluntary expression of behaviours and rituals that are socially sanctioned responses to bereavement. These behaviours and rituals differ between societies and between religious groups both in their form and in their duration.

The systems of classification do not make these distinctions in consistent ways. In ICD-11, uncomplicated bereavement is coded appropriately as QE.62—that is, as one of the ‘factors influencing health status and contact with health services’. In DSM-5, however, while bereavement itself is not coded, ‘*persistent complex bereavement*’ (a severe grief reaction that has persisted for at least 12 months) is listed under ‘conditions for further study’—thus the term is used to denote the response to bereavement, rather than the event itself. ICD-11 codes grief under adjustment disorders, but uses the term ‘prolonged grief disorder’. Mourning, which is a form of social behaviour, is not a disorder and, appropriately, is not listed in the index to either classification. In this chapter, the term *bereavement reaction* is used to denote all responses to bereavement, both normal and abnormal. Normal reactions are referred to as grief, while abnormal reactions include abnormal grief (also referred to as pathological or complex grief), depressive disorders, and adjustment disorders.

Grief

Grief is a continuous process, but for clarity can be described as having three stages (see Box 7.8).

The *first stage* lasts from a few hours to several days. There is denial, which is manifested as a lack of emotional response (‘numbness’), often with a feeling of unreality, and incomplete acceptance that the death has taken place. The bereaved person may be restless, as if searching for the dead person.

The *second stage* usually lasts from a few weeks to about six months, but may be much longer. There may be extreme sadness, weeping, loneliness, and often overwhelming waves of yearning for the dead person.

Anxiety is common; the bereaved person is anxious and restless, sleeps poorly, lacks appetite, and may experience panic attacks. Many bereaved people feel guilt that they failed to do enough for the deceased. Some feel anger and project their feelings of guilt, blaming doctors or others for failing to provide optimal care for the person who has died. Many bereaved individuals have a vivid experience of being in the presence of the dead person, and about one in ten experience brief hallucinations. The bereaved individual is preoccupied with memories of the dead person, sometimes in the form of intrusive images. Withdrawal from social relationships is frequent, and complaints of physical symptoms are common.

In the *third stage*, these symptoms subside, and everyday activities are resumed. The bereaved person gradually comes to terms with the loss and recalls the good times shared with the deceased in the past. Often there is a temporary return of symptoms on the anniversary of the death.

Although these stages are a useful guideline, individual responses vary, and no one feature is universal. Furthermore, the experiences may be more complex

Box 7.8 Normal grief reaction

Stage 1: Hours to days

Denial, shock, disbelief
‘Numbness’

Stage 2: Weeks to six months

Sadness, weeping, waves of grief
Somatic symptoms of anxiety
Restlessness
Poor sleep
Diminished appetite
Guilt, blame of others
Experience of a presence
Illusions, vivid imagery
Hallucinations of the dead person’s voice
Preoccupation with memories of the deceased
Social withdrawal

Stage 3: Weeks to months

Experiences of grief diminish
Social activities resumed
Memories of good times
Symptoms may recur at anniversaries.

and a strict expectation of progression through stages while theoretically attractive, may be distressing when the experiences do not follow a linear process (Avis et al., 2021).

Abnormal grief

Grief is considered to be abnormal if it is delayed or inhibited, or unusually intense and prolonged, when it is referred to as *complicated* or *pathological*. The usual criterion for delay is that the first stage of grief has not occurred by two weeks after the death. The criterion for abnormal duration is that the response lasts for more than six months. However, many people grieve for longer than this and judgement needs to be used as to whether feelings of grief are continuing to resolve, albeit at a slow rate. In all forms of abnormal grief, persistent avoidance of situations and of other reminders of death are common.

Depression in the course of a grief reaction. Depressive symptoms are a frequent component of normal grief, and around 30% of bereaved people meet the criteria for a depressive disorder at some time during their grieving. Most of these depressive disorders resolve within six months, but about 20% persist for longer. It might be argued that, if about one third of bereaved people meet the criteria for depressive disorder at some time, the threshold has been set too low. However, people who meet the criteria for a depressive disorder are more likely to have poor social adjustment, to visit doctors frequently, and to use alcohol. Also, they can respond to medications and psychotherapies employed in depression. Therefore, it is of practical value to use the criterion and to record the additional diagnosis of a depressive disorder in these cases.

If there is doubt whether depressive disorder should be recorded, particular attention should be paid to symptoms of retardation, global loss of self-esteem, and guilt (the latter clearly of greater intensity than the common regrets about omissions of care during the terminal illness), because these features are seldom present in uncomplicated grief. It is also important to assess whether *suicidal feelings* are present. In complicated grief, suicidal thinking usually concerns a wish to be united with the person who has died, whereas in depression the wish to be dead is associated with a pervasive sense of hopelessness.

Complicated grief. Complicated grief occurs in about 10% of those bereaved (Szuhany et al., 2021). People with complicated grief have an intense and prolonged reaction to bereavement. This can include strong yearnings for the person who has died, together with the thought that life now lacks all purpose and meaning. There is

disbelief and anger about the death, and reminders of the loss are avoided or, conversely, persistently sought out to the detriment of other activities and relationships. Complicated grief is also associated with sleep disturbance, increased substance use, and impairment in work and social functioning. Many of these features are captured in the DSM-5 category of *persistent complex bereavement*, although here the necessary duration of grief since the bereavement is 12 months, rather than the six months stipulated in the ICD-11 diagnosis of prolonged grief disorder. In DSM-5-TR, this condition is categorized as Prolonged Grief Disorder, as in ICD-11.

Causes of abnormal grief

Abnormal grief is generally thought to be more likely to occur when:

- The death was sudden and unexpected.
- The bereaved person had a very close, or dependent, or ambivalent relationship with the deceased.
- The survivor is insecure, or has difficulty in expressing their feelings, or has suffered a previous psychiatric disorder.
- The survivor has to care for dependent children and so cannot show their grief easily.

Morbidity after bereavement

Several studies (systematically reviewed by Ennis and Majid, 2021) have shown an increased rate of mortality among bereaved spouses and other close relatives, with the greatest increase being in the first six months after bereavement. Most studies report increased rates of death from heart disease, and some have reported increased rates of death from cancer, liver cirrhosis, suicide, and accidents. The reasons for these associations are uncertain, and are likely to be different for different conditions.

Management of grief

Grief is a normal response, and most people pass through it with the help of family, friends, spiritual advisers, and the rituals of mourning. In some western societies, many people may not have links with a religion, the rituals of mourning may be attenuated, and family may not be close at hand. For these and other reasons, family doctors have an important part to play in helping the bereaved. Psychiatrists may be asked to help people with abnormal grief.

Although bereaved people have some problems in common, they also have problems that are individual. For example, a young widow with small children will

have many difficulties that are not shared by an elderly widow whose adult children can support her. A mother who is grieving for a stillborn child will have special problems (see below). When planning management it is important to consider the individual circumstances of the patient, as well as the general guidelines outlined below.

Coping strategies, rituals, and expectations vary between religious and cultural groups, and require sensitive exploration and understanding when making judgments about what might be helpful (for more on cultural and global perspectives on grief, see Smid et al., 2018).

Counselling

When counselling is appropriate, it is similar to counselling for other kinds of adjustment reaction. The bereaved person needs to talk about the loss, to express feelings of sadness, guilt, or anger, and to understand the normal course of grieving. It is helpful to forewarn a bereaved person about unusual experiences such as feeling as if the dead person is present, illusions, and hallucinations, otherwise these experiences may be alarming. Help may be needed to:

- accept that the loss is real
- work through the stages of grief
- adjust to life without the deceased.

The bereaved person may need help to progress from the first stage of denial of loss to the acceptance of reality. Viewing the dead body and putting away the dead person's belongings help this transition, and a bereaved person should be encouraged to perform these actions. Practical problems may need to be discussed, including funeral arrangements and financial difficulties. A young widow may need help with caring for young children, and in supporting them without inhibiting her own grief excessively. As time passes, the bereaved person should be encouraged to resume social contacts, to talk to other people about the loss, to remember happy and fulfilling experiences that were shared with the deceased, and to consider positive activities that the latter would have wanted survivors to undertake. For further information about psychological interventions for grief, see Johannsen et al. (2019).

Parents who are grieving for a stillborn child need special help. Stillbirth is associated with increased rates of depression and trauma-related conditions in parents (Westby et al., 2021). Previous practice has been to advise physical contact with the stillborn baby, and this is something that many parents understandably desire. However, there is growing evidence that holding the body of the baby is associated with greater risk of adverse

psychological outcomes, so current advice is that parents should not view or hold the body unless they express the wish to do so (Sun et al., 2023). Stillbirth increases the risk of subsequent relationship breakdown; lack of support by the partner at the time of the stillbirth and having held the stillborn infant increase this risk.

Medication

Drug treatment cannot remove the distress of normal grief, but it may be needed in specific circumstances. In the first stage of grief, a hypnotic or anxiolytic drug may be needed for a few days to restore sleep or to relieve any severe anxiety. In the second stage, antidepressant drugs may be beneficial if the criteria for depressive disorder are met, although such usage has not been widely evaluated in this special group. Antidepressant medication may also sometimes be helpful in the treatment of complicated grief but is best used in conjunction with a structured psychotherapy.

Support groups

Support groups have been developed to help recently bereaved people, particularly young widows and widowers. One such organization in the United Kingdom is known as Cruse. By sharing their experience with others who have dealt successfully with bereavement, recently bereaved people can share their grief, obtain practical advice, and discuss ways of coping.

Psychotherapy

It is not practicable, nor is there evidence that it is helpful, to provide structured psychotherapies for most bereaved individuals. For abnormal grief, dynamic psychotherapy has a clear rationale and approach, but its effectiveness has not been formally evaluated. However, *complicated grief treatment* (CGT), which combines aspects of CBT and interpersonal therapy, has been reported to be helpful in the management of abnormal grief and more effective than interpersonal therapy or treatment with the SSRI citalopram. (Szuhany et al., 2021).

CGT emphasizes the development of a narrative about the loss, which includes the bereavement itself as well as the positive and negative aspects of life with the deceased. Psychoeducation about grief and its complications is supplemented with exposure to aspects of the bereavement that are being avoided. Cognitive restructuring to re-evaluate maladaptive thinking, for example, that grieving is the only way of honouring the deceased is also employed. Some of the techniques of CGT can be applied by non-specialists; for example, discussion of the bereavement and the previous relationship with the deceased as well as advice for the bereaved person not to avoid events such as family

reunions, which will bring about reminders of the person and the loss (Szuhany et al., 2021).

Long-term adjustment to sexual abuse in childhood

When sexually abused, children may experience disruption in multiple areas of their lives, including within their families and in their relationships with peers and adults as well as in their education. They are also at risk of anxiety, depression, and PTSD. These effects can persist into adult life with strongest psychiatric associations with conversion disorder, borderline personality disorder, anxiety, and depression. Furthermore, sexual abuse in childhood may be followed by persistent low self-esteem and psychosexual difficulties, whether or not a psychiatric disorder develops (Hailes et al., 2019).

Very occasionally, adults who had been sexually abused in childhood but do not actively remember details, can suddenly recall or be reminded of the abuse in a vivid and disturbing way. Sometimes this recall occurs spontaneously, often when the person has encountered a reminder of the events. It may also occur during counselling or psychotherapy, at a time when childhood experiences are being discussed. Some of these recollections may be confirmed by other evidence, but often, especially when time has passed, it can be harder to obtain additional evidence about the abuse. This can add to the distress of the experiences, especially if any perpetrators deny the allegations. Furthermore, it has been suggested that some of these unconfirmed reports of abuse are not accurate memories, and that some have been induced by questions, suggestions, or interpretations from the therapists. This phenomenon has been termed false memory syndrome (Brewin, 2009).

Recovered memory and false memory

Victims of sexual abuse, and of other severe stressful events, can have partial amnesia for the most stressful parts of the experience, even though they have suffered no head injury that could lead to post-traumatic amnesia. Indeed, partial amnesia is part of the clinical picture of PTSD. However, complete amnesia is less frequent and, in the view of many psychiatrists, complete amnesia for repeated stressful events followed by their recall is improbable, especially when there is no supporting evidence for the events from another source. This doubt is increased by evidence that ‘memories’ of single non-abusive childhood events can be implanted by suggestion in about 30% of subjects (Scoboria et al., 2017).

Evidence for the proposition that true memories can be inaccessible for many years and then be recovered

comes mainly from clinical reports. These reports suggest that around 25–50% of people who report childhood sexual abuse describe long periods during which they did not remember the abuse. Furthermore, clinical studies have shown that around 20–60% of people who report childhood sexual abuse state that there were periods during their life when they could not remember that the abuse had taken place. In addition, some people have recovered memories of abuse prior to engaging in therapy, so the effect of suggestion could not have been an influence in these cases.

In the absence of conclusive evidence about the status of memories that are recovered during counselling or psychotherapy, the clinician carrying out these procedures should:

- take special care not to suggest memories of sexual abuse
- consider most carefully apparent recovered memories that arise for the first time in therapy before concluding that they are true memories of actual events.

The state of the current scientific evidence indicates that practitioners should keep an open mind about the possibility and likely frequency of recovered memories of sexual abuse and, of course, of other kinds of traumatic experience. For a review see Otgaar et al. (2022).

Epidemiology

Adults who report having experienced sexual abuse in childhood have higher rates of psychiatric disorder in adult life. It is not clear what proportion of children who were sexually abused in childhood go on to develop these disorders in adult life, but some appear to make a good adjustment. An important factor in this kind of resilience is the experience of at least some good-quality relationships through childhood and adolescence (Fuller-Thompson et al., 2020).

In a prospective study in Australia of over 2,750 sexually abused children (80% female) followed for up to 40 years, Cutajar et al. (2010) found that 24% of the abused group subsequently had contact with mental health services, compared with 8% of controls. Relative to controls, victims of abuse had higher rates of several adult psychiatric disorders, notably PTSD, substance use disorders, and personality disorders; however, they were also at increased risk of mood and anxiety disorders, and psychosis. More severe sexual abuse and an older age of occurrence were significant predictors of adult psychiatric disorder. An umbrella review of the effects of childhood sexual abuse on adult psychiatric outcomes came to a similar conclusion (Hailes et al., 2019).

Chen et al. (2014) studied over 6,000 Han Chinese women with recurrent major depression and a similar number of unaffected controls. They found that any form of reported childhood sexual abuse produced a fourfold risk in the likelihood of experiencing recurrent depression, and the risk increased with the severity of the abuse. Abuse was also associated with suicidality, as well as feelings of guilt and worthlessness.

Aetiology

Childhood sexual abuse may be a direct cause of vulnerability to adult psychiatric disorder, or it may be a marker of some other factor, such as disturbed relationships within the family, which could be the primary cause of the excess psychiatric disorder in adult life. It has become apparent, for example, that sexual abuse frequently coexists and is therefore highly correlated with other kinds of *childhood adversity*, such as other forms of *maltreatment* (physical abuse, emotional abuse, and neglect), *parental loss* (death and other kinds of separation such as divorce), and *parental maladjustment* (violence, criminality, substance abuse, and parental mental illness). Thus studies assessing a single form of childhood adversity on the risk of adult psychiatric disorder may overestimate the impact of that particular kind of adversity. More recent studies have employed measures of a range of childhood adversities followed by multivariate modelling to estimate the relative importance of different kinds of adversity and how far their effects on the risk of developing subsequent psychiatric disorders are additive rather than shared.

In a systematic review of 96 studies on the long-term effects of childhood maltreatment on subsequent mental health, Gardner et al. (2019) found that several forms of childhood maltreatment (sexual abuse, physical abuse, emotional abuse, neglect, and exposure to intimate partner violence) increased the risk of subsequent anxiety and depressive disorders and PTSD. Of the various forms of maltreatment, sexual abuse was the strongest predictor of anxiety disorders and PTSD. The consequences for adverse mental health outcomes

increased as the number of forms of maltreatment experienced by the child increased.

Treatment

The late effects of childhood sexual abuse have been treated with counselling, dynamic psychotherapy, cognitive therapy, and group treatments. The various methods have several common features.

- The general aim is to help the patient to understand the earlier experiences and the effects of these on their life to improve present adjustment.
- The therapeutic relationship is used to help the patient to feel trusted, understood, and respected, and to increase their self-esteem.
- The patient is allowed to set the pace at which they talk about the experience of being abused, otherwise they may be overwhelmed by an extreme emotional response to the memories of abuse, and withdraw from treatment.
- Present problems of adjustment are identified, especially any avoidance of problems and difficulties in expressing anger. Help is given to overcome these difficulties.
- Some patients need help with psychosexual problems.

The main difference between the dynamic and cognitive behavioural approaches is the greater emphasis given in the former to understanding the effects of the trauma on self-esteem and emotional expression, and the greater emphasis given in the latter to more precise specification of the ways in which current patterns of thinking affect present behaviour. In a meta-analysis of 44 studies, Taylor and Harvey (2010) found that many different psychological treatments, ranging from group analytic therapy to individual CBT, produced moderate benefits on a range of symptomatology in adults who had suffered sexual abuse in childhood. Generally the largest effect size was seen with CBT treatment, particularly for symptoms related to PTSD. For many patients, the treatments being developed for complex PTSD will be relevant (see above).

Further reading

Beck, J. G., & Sloan, D. M. (2023) *The Oxford Handbook of Traumatic Stress Disorders*, 2nd edn. Oxford University Press.
National Institute for Health and Care Excellence (NICE). (2018b). Post-traumatic stress disorder. NICE guideline NG116. <https://www.nice.org.uk/guidance/ng116>

Parkes, C. M., & Prigerson, H. G. (2010). *Bereavement: Studies of Grief in Adult Life*, 4th edn. Penguin.

CHAPTER 8

Anxiety and obsessive–compulsive disorders

Terminology and classification 161

Anxiety disorders 161

Generalized anxiety disorder 163

Phobic anxiety disorders 169

Panic disorder 178

Transcultural variations in anxiety disorder 182

Obsessive–compulsive disorder 183

Obsessive–compulsive-related disorders 188

Terminology and classification

The symptom of anxiety is found in many disorders. In the anxiety disorders, it is the most severe and prominent symptom, and it is also prominent in the obsessional disorders, although these are characterized by their striking obsessional and compulsive symptoms. In both ICD-11 and DSM-5 obsessive-compulsive disorder (OCD) is classified separately from anxiety disorder. However, for convenience, both types of disorder are covered in this chapter.

Clinical anxiety disorders are principally either anxiety or fear-related. In conceptual terms the emotion

of *fear* describes an emotional reaction to a currently encountered object or situation, while *anxiety* concerns worry about potential future threats. A *phobia* is an irrational, persistent and uncontrollable fear about a specific object or situation. In practice many patients can experience both anxiety and fear-related symptomatology. For example, a person with a fear of flying (a specific phobia) may have understandable *anticipatory anxiety* about the prospect of a planned aeroplane journey.

Anxiety disorders

Anxiety disorders are abnormal states in which the most striking features are mental and physical symptoms of anxiety, occurring in the absence of organic brain disease or another psychiatric disorder. The symptoms of anxiety are described in Chapter 1, and are listed for convenience in Box 8.1. Although all of the symptoms can occur in any of the anxiety disorders, there is a characteristic pattern in each disorder,

which will be described later. The disorders share many features of their clinical picture and aetiology, but there are also differences:

- In generalized anxiety disorders, anxiety is continuous, although it may fluctuate in intensity.
- In phobic anxiety disorders, anxiety is intermittent, arising in particular circumstances.

Box 8.1 Symptoms of anxiety**Psychological arousal**

Fearful anticipation
Irritability
Sensitivity to noise
Restlessness
Poor concentration
Worrying thoughts

Autonomic arousal**Gastrointestinal**

- Dry mouth
- Difficulty in swallowing
- Epigastric discomfort
- Excessive wind
- Frequent or loose motions

Respiratory

- Constriction in the chest
- Difficulty inhaling

Cardiovascular

- Palpitations
- Discomfort in the chest
- Awareness of missed beats

Genitourinary

- Frequent or urgent micturition
- Failure of erection
- Menstrual discomfort

Muscle tension

Tremor
Headache
Aching muscles

Hyperventilation

Dizziness
Tingling in the extremities
Feeling of breathlessness

Sleep disturbance

Insomnia
Night terror

- In panic disorder, anxiety is intermittent, but its occurrence is unrelated to any particular circumstances.

These differences (and some exceptions to these initial generalizations) will be explained further when the various types of anxiety disorders are described.

The development of ideas about anxiety disorders

Anxiety has long been recognized as a prominent symptom of many psychiatric disorders. Anxiety and depression often occur together and, until the last part of the nineteenth century, anxiety disorders were not classified separately from other mood disorders. It was Freud, in 1895, who first suggested that cases with mainly anxiety symptoms should be recognized as a separate entity under the name ‘anxiety neurosis’.

Phobic disorders have been recognized since antiquity, but the first systematic medical study of these conditions was probably that of Le Camus in the eighteenth century. The early nineteenth-century classifications assigned phobias to the group of monomanias, which were disorders of thinking rather than of emotion. However, when Westphal first described agoraphobia in 1872, he emphasized the importance of anxiety in the condition.

In the 1960s, the different responses of certain phobias to behavioural methods suggested a grouping into simple phobias, social phobia, and agoraphobia, and these groups were also found to differ in their age of onset. (Simple phobias generally begin in childhood, social phobia in adolescence, and agoraphobia in early adult life.) At about the same time, it was observed that, when phobias were accompanied by marked panic attacks, they responded poorly to behaviour therapy and better to imipramine (Klein, 1964). These cases were subsequently classified separately as panic disorder. This advance led to the present scheme of classification into generalized anxiety disorder (GAD), phobic anxiety disorder (simple, social, and agoraphobic), and panic disorder.

The relationship between OCD and anxiety disorders has been and remains uncertain. Freud initially thought that phobias and obsessions were closely related. He later proposed that anxiety is the central problem in both conditions and that their characteristic symptoms (phobias and obsessions) resulted from different kinds of defence mechanisms against anxiety. Others considered that obsessional disorders were a separate group of neuroses of uncertain aetiology.

As explained above, the two major diagnostic systems no longer classify OCD with anxiety disorders—in DSM-5, OCD is classified together with other ‘obsessive–compulsive-related disorders’, which include ‘body dysmorphic disorder’, ‘hoarding disorder’, ‘trichotillomania’, and ‘excoriation (skin picking) disorder’. ICD-11 follows a similar plan.

The classification of anxiety disorders

The classification of anxiety disorders in ICD-11 and DSM-5 is shown in Table 8.1. The schemes are broadly similar, but ICD-11 includes two childhood anxiety disorders, separation anxiety and selective mutism

Table 8.1 Classification of anxiety disorders

ICD-11	DSM-5
Anxiety or fear-related disorders	Anxiety disorders*
Agoraphobia	Agoraphobia
Social anxiety disorder	Social anxiety disorder
Specific phobia	Specific phobia
Panic disorder	Panic disorder
Generalized anxiety disorder	Generalized anxiety disorder
Separation anxiety disorder	
Selective mutism	

* The order of presentation has been altered to facilitate comparison of the schemes.

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022). World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (DSM-5). Copyright (2013), American Psychiatric Association.

The epidemiology of the various anxiety disorders is considered under each condition. The 12-month prevalence rates from a meta-analysis of 85 European studies are shown in Table 8.2 to illustrate the relative frequency of the different disorders in population studies.

Table 8.2 Twelve-month prevalence rates of anxiety disorders and obsessive–compulsive disorder in population studies in the European Union

Diagnosis (DSM-IV)	12-month prevalence range (%)	Median (%)
Specific phobia	3.1–11.1	4.9
Social phobia	0.6–7.9	2.0
Panic disorder*	0.6–3.1	1.2
Generalized anxiety disorder	0.2–4.3	2.0
Agoraphobia†	0.1–3.1	1.2
Obsessive–compulsive disorder	0.1–2.3	0.7

* With and without agoraphobia.

† Without panic disorder.

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Generalized anxiety disorder

Clinical picture

The symptoms of GAD (see Box 8.2) are persistent and are not restricted to, or markedly increased in, any particular set of circumstances (in contrast to phobic anxiety disorders). All of the symptoms of anxiety (see Box 8.1) can occur in GAD, but there is a characteristic pattern, which consists of the following features:

- *Worry and apprehension* that are more prolonged than in healthy people. The worries are widespread and are not focused on a specific issue as they are in panic disorder (i.e. on having a panic attack), social phobia (i.e. on being embarrassed), or OCD (i.e. on

contamination). The person feels that these widespread worries are difficult to control.

- *Psychological arousal*, which may be manifested as irritability, poor concentration, and/or sensitivity to noise. Some patients complain of poor memory, but this is because of poor concentration. If true memory impairment is found, a careful search should be made for a cause other than anxiety.
- *Autonomic overactivity*, which is most often experienced as sweating, palpitations, dry mouth, epigastric discomfort, and dizziness. However, patients may complain of any of the symptoms listed in Box 8.1. Some patients ask for help with any of these symptoms

Box 8.2 Symptoms of generalized anxiety disorder

Worry and apprehension
Muscle tension*
Autonomic overactivity*
Psychological arousal*
Sleep disturbance*
Other features:

- Depression
- Obsessions
- Depersonalization

* See Box 8.1.

without mentioning spontaneously the psychological symptoms of anxiety.

- *Muscle tension*, which may be experienced as restlessness, trembling, inability to relax, headache (usually bilateral and frontal or occipital), and aching of the shoulders and back.
- *Hyperventilation*, which may lead to dizziness, tingling in the extremities and, paradoxically, a feeling of shortness of breath.
- *Sleep disturbances*, which include difficulty in falling asleep and persistent worrying thoughts. Sleep is often intermittent, unrefreshing, and accompanied by unpleasant dreams. Some patients have night terrors and wake suddenly feeling extremely anxious. Early-morning waking is not a feature of GAD, and its presence strongly suggests a depressive disorder.
- *Other features*, which include tiredness, depressive symptoms, obsessional symptoms, and depersonalization. These symptoms are never the most prominent feature of GAD. If they are prominent, another diagnosis should be considered (see the section on Differential diagnosis).

Clinical signs in the patient may include:

- The face appears strained, the brow is furrowed, and the posture is tense.
- The person is restless and may tremble.
- The skin is pale, and sweating is common, especially from the hands, feet, and axillae.
- Being close to tears, which may at first suggest depression, reflects the generally apprehensive state.

Diagnostic conventions

There is no clear dividing line between GAD and normal anxiety. They differ both in the extent of the symptoms and in their duration. The diagnostic criteria for both extent and duration are arbitrary. Both DSM-5 and ICD-11 require the presence of *excessive worry about several different matters* for a period of several months (specifically more than six months in DSM-5). In ICD-11, pervasive and prolonged general apprehensiveness can also be a diagnostic criterion. DSM-5 requires the presence of at least three from six other anxiety symptom. ICD-11 is less prescriptive but states that to meet the diagnosis of GAD, worry or apprehensiveness should be accompanied by additional characteristic symptoms of anxiety, such as muscle tension, increased sympathetic autonomic activity, feeling on edge, irritability, difficulty concentrating, and sleep disturbance. Both DSM-5 and ICD-11 require that the symptoms cause clinically significant distress or problems in functioning in daily life.

Comorbidity of anxiety symptoms

Anxiety and depression

The two classifications differ in their approach to cases that present with significant symptoms of both depressive disorder and GAD without meeting the full criteria for either condition considered separately. ICD-11 has a separate category for these cases, namely *mixed depressive and anxiety disorder*, which is listed under 'depressive disorders'. These conditions are discussed further in Chapter 9. It is also important to note that depressive symptoms are commonly also present in diagnosed anxiety disorders, and that patients with GAD may suffer from comorbid major depression (Baldwin et al., 2014). In ICD-11, when a patient presents with both GAD and a depressive disorder, GAD should be diagnosed only if the requirements for GAD were met *before* the onset of the depressive episode.

Generalized anxiety disorder and other anxiety disorders

In both DSM-5 and ICD-11, the emphasis that is placed on the worrying ideas in GAD makes it possible to diagnose GAD when those ideas are present even in the presence of symptoms of one of the other three anxiety diagnoses. When this convention is followed, comorbidity between GAD and other anxiety disorders is frequent (social phobia in 23% of cases of GAD, simple

phobia in 21%, and panic disorder in 11%) (Tyrrer and Baldwin, 2006).

Differential diagnosis

GAD has to be distinguished not only from other psychiatric disorders but also from certain physical conditions. Anxiety symptoms can occur in nearly all psychiatric disorders, but there are some in which particular diagnostic difficulties arise.

Depressive disorder

Anxiety is a common symptom in depressive disorder, and GAD often includes some depressive symptoms. The usual convention is that the diagnosis is decided on the basis of the severity of two kinds of symptom and the order in which they appeared. Information on these two points should be obtained, if possible, from a relative or other informant as well as from the patient. Whichever type of symptoms appeared first and is more severe is considered primary. An important diagnostic error is to misdiagnose the agitated type of severe depressive disorder as GAD. This mistake will seldom be made if anxious patients are asked routinely about symptoms of a depressive disorder, including depressive thinking and, when appropriate, suicidal ideas. Depressive disorders are often worst in the morning, and anxiety that is worst at this time suggests a depressive disorder. As noted above, in some patients a depressive disorder and GAD coexist and both diagnoses can be made.

Schizophrenia

People with schizophrenia sometimes complain of anxiety before the other symptoms are recognized. The chance of misdiagnosis can be reduced by asking anxious patients routinely what they think caused their symptoms. Patients with schizophrenia may give an unusual reply, which leads to the discovery of previously unexpressed delusional ideas.

Dementia

Anxiety may be the first abnormality to present in a person developing dementia. When this happens, the clinician may not detect an associated impairment of memory, or may dismiss it as the result of poor concentration. Therefore memory should be assessed in middle-aged or older patients who present with anxiety.

Substance misuse

Some people take drugs or alcohol to relieve anxiety. Patients who are dependent on drugs or alcohol sometimes believe that the symptoms of drug withdrawal are those of anxiety, and take anxiolytic or other drugs to

control them. The clinician should be alert to this possibility, particularly when anxiety is severe on waking in the morning, which is the time when alcohol and drug withdrawal symptoms tend to occur. (Anxiety that is worst in the morning also suggests depressive disorder; see above.)

Physical illness

Some physical illnesses have symptoms that can be mistaken for those of an anxiety disorder. This possibility should be considered in all cases, but especially when there is no obvious psychological cause of anxiety, or there is no history of past anxiety. The following conditions are particularly important:

- In *thyrotoxicosis*, the patient may be irritable and restless, with tremor and tachycardia. Physical examination may reveal characteristic signs of thyrotoxicosis, such as an enlarged thyroid, atrial fibrillation, and exophthalmos. If there is any doubt, thyroid function tests should be arranged.
- *Phaeochromocytoma* and *hypoglycaemia* usually cause episodic symptoms, and are therefore more likely to mimic a phobic disorder or panic disorder. However, they should also be considered as a differential diagnosis of GAD. If there is any doubt, appropriate physical examination and laboratory tests should be carried out.

Generalized anxiety disorder that is mistaken for physical illness

When this happens, extensive investigations may be carried out, which increase the patient's anxiety. Although physical illness should be considered in every case, it is also important to remember the diversity of the anxiety symptoms. Palpitations, headache, frequency of micturition, and abdominal discomfort can all be the primary complaint of an anxious patient. Correct diagnosis requires systematic enquiries about other symptoms of GAD, and about the order in which the various symptoms appeared.

Epidemiology

Estimates of incidence and prevalence vary according to the diagnostic criteria used in the survey, and whether a clinical significance criterion is used. The Adult Psychiatric Morbidity Survey found a one-week prevalence of 5.9% for GAD symptoms in England (McManus et al., 2016) with rather lower prevalence 12-month prevalence figures in European countries (around 2%; Wittchen et al., 2010). Rates in women are about twice

as high as those in men. GAD is associated with several indices of social disadvantage, including lower household income and unemployment, as well as divorce and separation (McManus et al., 2016). Rates of anxiety, as well as expressions of anxiety, vary across development, and the anxiety disorders *are the most common child mental disorders*, as discussed in Chapter 16.

Aetiology

In general terms, GAD appears to be caused by stressors acting on a personality that is predisposed to anxiety by a combination of genetic factors and environmental influences in childhood. However, evidence for the nature and importance of these causes is incomplete.

Stressful events

Clinical observations indicate that GADs often begin in relation to *stressful events*, and some become chronic when stressful problems persist. A study by Kendler et al. (2003) showed that stressful life events characterized by loss increased the risk of both depression and GAD. However, life events characterized by ‘danger’ (where the full import of the event was yet to be realized) were more common in those who subsequently developed GAD.

Genetic causes

Early twin studies, such as that by Slater and Shields (1969), showed a higher concordance for anxiety disorder between monozygotic than dizygotic pairs, which suggests that the familial association has a genetic cause. However, the genes involved in the transmission of GAD appear to increase susceptibility to other anxiety conditions, such as panic disorder and agoraphobia, as well as to major depression. Overall, the findings suggest that genes play a significant although moderate role in the aetiology of GAD, but that the genes involved predispose to a range of anxiety and depressive disorders, rather than GAD specifically. Genome-wide association studies (GWAS) studies have not thus far identified reliable genetic makers (Meier and Deckert, 2019).

Early experiences

Accounts given by anxious patients of their experience in childhood suggest that *early adverse experience* is a cause of GAD.

Brown and Harris (1993) studied the relationship between adverse experience in childhood and anxiety disorder in adult life in 404 working-class women living in an inner city. Adverse early experience was assessed from patients’ accounts of parental indifference and of physical or sexual abuse. Women who reported early adversity had increased rates of GAD (and also of

agoraphobia and depressive disorder, but not of simple phobia). Parenting styles characterized by *overprotection and lack of emotional warmth* may also be a risk factor for GAD, as well as for other anxiety and depressive disorders in offspring.

Other important early family factors in the development of anxiety disorders include domestic violence, parental alcohol and drug use, and parental separation. School bullying is associated with the development of anxiety disorders in youth.

Cognitive behavioural theories

Conditioning theories propose that GAD arises when there is an inherited predisposition to excessive responsiveness of the autonomic nervous system, together with generalization of the responses through conditioning of anxiety to previously neutral stimuli. This theory has not been supported by a body of objective data.

Cognitive theory. Particular coping and cognitive styles may also predispose individuals to the development of GAD, although it is not always easy to distinguish predisposition from the abnormal cognitions that are seen in the illness itself. As noted above, it seems likely that people who lack a sense of control of events and of personal effectiveness, perhaps because of early life experiences, are more prone to anxiety disorders. Such individuals may also demonstrate trait-like cognitive biases in the form of increased attention to potentially threatening stimuli, overestimation of environmental threat, and enhanced memory of threatening material. This has been referred to as the *looming cognitive style*, which appears to be a general psychological vulnerability factor for a number of anxiety disorders (Reardon and Williams, 2007).

More recent cognitive formulations have focused on the process of worry itself. It has been proposed that people who are predisposed to GAD use worry as a positive coping strategy for dealing with potential threats, whereby the individual cannot relax until they have examined all of the possible dangers and identified ways of dealing with them. However, this can lead to ‘worry about worry’, when a person comes to believe, for example, that worrying in this way, although necessary for them, is also uncontrollable and harmful. This ‘metacognitive belief’ may form a transition between excessive but normal worrying, and GAD (Wells, 2013).

Personality

Personality traits. Anxiety symptoms are associated with *neuroticism*, and twin studies have shown an overlap between the genetic factors related to neuroticism and those related to GAD.

Personality disorder. GAD occurs in people with anxious–avoidant personality disorders, but also in individuals with other personality disorders.

Neurobiological mechanisms

The neurobiological mechanisms involved in GAD are presumably those that also mediate normal anxiety. The mechanisms are complex, involving several brain systems and several neurotransmitters. Studies in animals have indicated a key role for the *amygdala*, which receives sensory information both directly from the *thalamus* and from a longer pathway involving the *medial prefrontal cortex* and *anterior cingulate cortex*. *Cortical involvement* in anxiety is important because it indicates a role for cognitive processes in its expression. The *hippocampus* is also believed to have an important role in the regulation of anxiety, because it relates fearful memories to relevant present contexts.

Animal experimental studies have led to an understanding of the regulation of anxiety in the brain by neurotransmitters and neuromodulators. *Noradrenergic neurons* that originate in the locus coeruleus increase arousal and anxiety, whereas *5-HT neurons* that arise in the raphe nuclei appear to have complex effects, and serve both to signal the presence of anxiety-producing stimuli in the environment and also to restrain the associated behavioural responses. *Gamma-aminobutyric acid (GABA) receptors*, which are widely distributed in the brain, are inhibitory and reduce anxiety, as do the associated benzodiazepine-binding sites. There is probably also a role for *corticotropin-releasing hormone*, which increases anxiety-related behaviours and is found in high concentration in the amygdala. However, although pharmacological manipulation of 5-HT and GABA mechanisms can be helpful in the treatment of generalized anxiety, there is no firm evidence that changes in these neurotransmitters are fundamentally involved in the pathophysiology of the disorder.

Functional imaging of the brain during the presentation of aversive stimuli (e.g. fearful faces) has shown inconsistent changes in amygdala reactivity in patients with GAD. There is more reliable evidence of altered activity in cortical regulatory regions such as the *ventrolateral prefrontal cortex* and altered connectivity between this region and the amygdala. Also, patients with GAD appear to have difficulties engaging prefrontal cortical areas and anterior cingulate cortex during tasks of emotional regulation. This picture is consistent with deficient ‘top-down’ control of emotional responses, which might explain why patients with GAD use *worry* as an emotional coping strategy (Goossen et al., 2019).

Prognosis

One of the DSM-5 criteria for GAD is that the symptoms should have been present for six months (at least several months in ICD-11). One of the reasons for this cut-off is that anxiety disorders that last for longer than six months have a poor prognosis. Thus most clinical studies suggest that GAD is typically a *chronic condition* with low rates of remission over the short and medium term. Evaluation of the prognosis is complicated by the frequent comorbidity with other anxiety disorders and depression, which worsen the long-term outcome and accompanying burden of disability. In the Harvard–Brown Anxiety Research Program, which recruited patients from Boston hospitals, the mean age of onset of GAD was 21 years, although many patients had been unwell since their teenage years. The average duration of illness in this group was about 20 years and, despite treatment, the outcome over the next three years was relatively poor, with only one in four patients showing symptomatic remission from GAD (Yonkers et al., 1996).

However, the participants in the above study were recruited from hospital services, and may not be representative of GAD in community settings. In a naturalistic study in the United Kingdom, Tyrer and colleagues (2004) followed up patients with anxiety and depression identified in primary care and found that, 12 years later, 40% of those initially diagnosed with GAD had recovered, in the sense that they no longer met the criteria for any DSM-III psychiatric disorder. The remaining participants continued to be symptomatic, but in only 3% was GAD still the principal diagnosis. In the majority of patients, conditions such as dysthymia, major depression, and agoraphobia were now more prominent. This study confirms the chronic and fluctuating symptomatic course of GAD in many clinically identified patients.

Treatment

Self-help and psychoeducation

A variety of forms of *self-help* have been studied in patients with anxiety disorders, including GAD. Such approaches typically include written and electronic materials with information about the disorder, and practical exercises to carry out based on the principles of cognitive behaviour therapy (CBT). Typically self-help has minimal therapist input, but it is also possible for self-help for anxiety disorders to be guided by a trained practitioner (*guided self-help*). Another form of educational treatment takes place in group form, so-called *group psychoeducation*, where one therapist works with up to

a dozen clients in about six weekly sessions of interactive learning and shared experience. The evidence for the benefit of these forms of treatment is limited and the effects, although superior to no treatment, appear to be modest. There is better evidence for CBT-based digital approaches both guided and unguided by a therapist (Pauley et al., 2023).

If practised regularly, *relaxation* appears to be able to reduce anxiety in less severe cases. However, many patients with such disorders do not practise the relaxation exercises regularly. Motivation may be improved if the training takes place in a group, and some people engage more with treatment when relaxation is taught as part of a programme of yoga exercises. A structured therapy, known as *applied relaxation*, does appear to be effective in lowering anxiety over 12–15 sessions guided by a trained therapist. A critical element of this treatment is the application of learned relaxation skills to anxiety-provoking situations.

Cognitive behaviour therapy in GAD

This treatment combines relaxation with cognitive procedures designed to help patients to control worrying thoughts. The method is described in Chapter 24. Compared with treatment as usual, CBT produces quite substantial benefits in terms of symptom resolution, with relatively few dropouts. However, the outcome obtained with CBT does not appear to differ from that obtained with other kinds of psychological interventions, such as applied relaxation and non-directive counselling, and there are few data on longer-term outcomes (Cuijpers et al., 2014a).

Medication

Anxiolytic drugs are described in Chapter 25. Here we are concerned with some specific points about their use in GADs. Medication can be used to bring symptoms under control quickly while the effects of psychological treatment are awaited. It can also be used when psychological treatment has failed. However, medication is often prescribed too readily and for too long. Before prescribing, it is appropriate to recall that, even though GAD is said to have a poor prognosis, in short-term studies of medication, pill placebo treatment in the context of the clinical care provided by a controlled trial is beneficial for a significant proportion of patients. For example, in a 12-week, placebo-controlled trial of escitalopram and paroxetine, just over 40% of patients responded to placebo, and around 30% reached remission (Baldwin et al., 2014).

Short-term treatment. One of the *longer-acting benzodiazepines*, such as diazepam, is appropriate for the

short-term treatment of GAD—for example, diazepam in a dose ranging from 5 mg twice daily in mild cases to 10 mg three times daily in the most severe cases. Anxiolytic drugs should seldom be prescribed for more than three weeks, because of the risk of dependence when they are given for longer. *Buspirone* is similarly effective for short-term management of GAD and is less likely to cause dependency, but has a slower onset of action. *Beta-adrenergic antagonists* are sometimes used to control anxiety associated with sympathetic stimulation. However, they are more often used for performance anxiety than for GAD. If one of these drugs is used, care should be taken to observe the contraindications to treatment, and the advice given in Chapter 25 and in the manufacturer's literature.

Long-term treatment. Because GAD often requires lengthy treatment, for which benzodiazepines are unsuitable (see above), and is often *comorbid* with depression and other anxiety disorders, treatment guidelines usually recommend *selective serotonin reuptake inhibitors* (SSRIs) as the initial choice of medication (National Institute for Health and Clinical Excellence, 2011b). *Serotonin and noradrenaline reuptake inhibitors* (SNRIs) such as duloxetine and venlafaxine are also effective, but are somewhat less well tolerated than SSRIs. The anticonvulsant *pregabalin* is also licensed for the treatment of GAD in the United Kingdom. It has a different side effect profile to SSRIs and SNRIs, and might therefore be useful in patients who cannot tolerate the latter agents. However, because of the risk of dependence and misuse, pregabalin is now classified as a controlled drug. Where patients with GAD respond to medication, the risk of relapse is substantially reduced if treatment is maintained for at least six months, and probably longer. A network meta-analysis of pharmacological treatments suggested that duloxetine, pregabalin, venlafaxine, and escitalopram were the most effective agents for the treatment of GAD, with good overall acceptability (Slee et al., 2019).

Management

In primary care, many patients are seen in the early stage of an anxiety disorder before a formal diagnosis of GAD can be made. In these circumstances, simple steps such as education and self-help can be tried first. If anxiety is severe, a short course of a benzodiazepine can bring rapid relief. Psychiatrists are more likely to encounter established cases. The steps in the management of such patients can be summarized as follows:

- *What patients need to know.* It can be explained to patients that GAD is a common mental health

problem where worries become excessive and out of control. This can make it hard to carry out usual occupational and social activities. People with generalized anxiety can also experience symptoms of tension and exhaustion and can feel physically very unwell. There are many effective psychological and drug treatments available and discussion with a health care practitioner will assist in finding the best approach for them. It might be helpful for a partner or member of the family to be given an explanation of the symptoms the patient is experiencing and how they can best help.

- *Check the diagnosis and comorbidity*, especially *depressive disorder*, *substance abuse*, or a *physical cause* such as *thyrotoxicosis*. If any of these are present, treat them appropriately.
- *Be aware of the high comorbidity between GAD and autistic spectrum disorders (ASD)*. The presence of ASD may influence the description and presentation of GAD but the response to treatment is similar (Nimmo-Smith et al., 2020)
- *Evaluate psychosocial maintaining factors* such as persistent social problems, relationship conflict, and concerns that physical symptoms of anxiety are evidence of serious physical disease.
- *Explain the evaluation and proposed treatment*, especially the origins of any physical symptoms of anxiety. Discuss the way in which the patient might describe the condition to employers, friends, and family. Self-help books reinforce the explanation and describe simple cognitive behavioural techniques that people can use on their own or as part of a wider plan of treatment.
- *Offer structured psychological treatments, such as CBT or applied relaxation*. For patients who do not respond to these initial approaches or who have significant functional disability, benefit can be obtained by using a structured treatment such as CBT administered by a trained therapist.

Box 8.3 Stepped-care approach* for generalized anxiety disorder

1. Identification and assessment: education about GAD and treatment options; active monitoring.
2. Low-intensity psychological interventions: pure self-help and guided self-help, group psychoeducation.
3. Choice of a high-intensity psychological intervention (CBT or applied relaxation) or a drug treatment.
4. Specialist treatment (complex drug and psychological regimens): input from multiagency teams, crisis services, or day hospitals.

* Initiate treatment at Step 3 if the patient presents with more severe symptomatology and/or significant functional impairment.

Source: National Institute for Health and Care Excellence. (2011b). Generalised anxiety disorder and panic disorder in adults: Management. Clinical guideline CG113. <https://www.nice.org.uk/guidance/cg113>. Updated June 2020. Copyright (2011), National Institute for Health and Care Excellence.

- *Consider the use of medication*. A short course of benzodiazepines may be prescribed to reduce high levels of anxiety initially, but should seldom be given for more than about three weeks. Where psychological treatment is not available or has failed, medication—usually with an SSRI initially—is appropriate. The main uses and side effects of medication should be discussed, as when using the same drugs in the treatment of depression (see Chapter 9).
- *Discuss the plan with the patient, the general practitioner, and the community team* and allocate tasks and responsibility appropriately. Plans should recognize that GAD is often a long-term problem.

A guideline from the National Institute for Health and Clinical Excellence (2011b) describes a stepped-care approach to the treatment of GAD, shown in Box 8.3.

Phobic anxiety disorders

Phobic anxiety disorders have the same core symptoms as GAD, but these symptoms occur only in specific circumstances. In some phobic disorders these circumstances are few and the patient is free from anxiety for most of the time. In other phobic disorders many

circumstances provoke anxiety, and consequently anxiety is more frequent, but even so there are some situations in which no anxiety is experienced. Two other features characterize phobic disorders. First, the person *avoids* circumstances that provoke anxiety and, secondly,

they experience *anticipatory anxiety* when there is the prospect of encountering these circumstances. The circumstances that provoke anxiety can be grouped into *situations* (e.g. crowded places), *objects* (a term that includes living things such as spiders), and *natural phenomena* (e.g. thunder). For clinical purposes, three principal phobic syndromes are recognized—specific phobia, social phobia, and agoraphobia. These syndromes are described next.

Classification of phobic disorders

In ICD-11 and DSM-5, phobic disorders are divided into specific phobia, social phobia, and agoraphobia. In both ICD-11 and DSM-5, panic disorder and agoraphobia are coded as two separate diagnoses. A patient who meets criteria for both panic disorder and agoraphobia has both diagnoses assigned.

Specific phobia

Clinical picture

A person with a specific phobia is inappropriately anxious in the presence of a particular object or situation. In the presence of that object or situation, the person experiences the symptoms of anxiety listed in Box 8.1. Anticipatory anxiety is common, and the person usually seeks to escape from and avoid the feared situation. Specific phobias can be characterized further by adding the name of the stimulus (e.g. spider phobia). In the past it was common to use terms such as arachnophobia (instead of spider phobia) or acrophobia (instead of phobia of heights), but this practice adds nothing of value to the use of the simpler names.

In DSM-5, five general types of specific phobia are recognized, which are concerned with:

1. animals
2. aspects of the natural environment
3. blood, injection, medical care, and injury
4. situations (e.g. aeroplanes, lifts, enclosed spaces)
5. other provoking agents (e.g. fears of choking or vomiting).

The following specific phobias merit brief separate consideration.

Phobia of dental treatment

Around 5% of adults have a fear of dental treatment. These fears can become so severe that all dental treatment is avoided, and serious caries develops. A meta-analysis

of 38 studies of behavioural treatment found a significant reduction in fear, with, on average, 77% of treated individuals attending for dental treatment four years after the treatment (Kvale et al., 2004).

Phobia of flying

Anxiety during aeroplane travel is common. A few people experience such intense anxiety that they are unable to travel in an aeroplane, and some seek treatment. This fear also occurs occasionally among pilots who have had an accident while flying. Desensitization treatment (see Chapter 24) is provided by some airlines, and self-help books are available. Virtual reality programmes have been used to replace actual and imagined exposure. Good results have been reported, but it is unclear whether the treatment needs to be supplemented with other psychological measures, such as psychoeducation (Da Costa et al., 2008).

Phobia of blood and injury

In this phobia, the sight of blood or of an injury results in anxiety. However, the accompanying autonomic response differs from that in other phobic disorders. The initial tachycardia is followed by a vasovagal response, with bradycardia, pallor, dizziness, nausea, and sometimes fainting. It has been reported that individuals who have this kind of phobia are prone to develop neurally mediated syncope even without the specific blood injury stimulus. Treatment consists of exposure *in vivo* together with the use of muscular tension to help to prevent syncope (Ayala et al., 2009).

Phobia of choking

People with this kind of phobia are intensely concerned that they will choke when attempting to swallow. They have an exaggerated gag reflex and feel intensely anxious when they attempt to swallow. The onset is either in childhood, or after choking on food in adulthood. Some of these individuals also fear dental treatment, while others avoid eating in public. Treatment consists of desensitization.

Epidemiology

While specific fears are very common in the general population, specific phobias that require the presence of avoidance and impairment for diagnosis have a lifetime prevalence of around 7% in men and 17% in women (Kessler et al., 2005a). The age of onset of most specific phobias is in childhood; however, a second peak in incidence of specific phobias is seen in women in early adult life during the child-bearing years (Eaton et al., 2018).

Aetiology

Persistence of childhood fears

Most specific phobias in adulthood are a continuation of childhood phobias. Specific phobias are common in childhood (see Chapter 16). By the early teenage years most of these childhood fears will have been lost, but a few persist into adult life. Why they persist is not certain, except that the most severe phobias are likely to last the longest.

Genetic factors

It has been estimated that specific phobias have a heritability of about 30%. Genetic influences appear to be more relevant to the development of specific phobias in adulthood than in those with a childhood onset. The genes that predispose to specific phobias appear different to those involved in the development of other phobic disorders and GAD (Eaton et al., 2018).

Conditioning and cognitive theories

Conditioning theory suggests that specific phobias arise through association learning. A minority of specific phobias appear to begin in this way in relation to a highly stressful experience. For example, a phobia of horses may begin after a dangerous encounter with a bolting horse. Some specific phobias may be acquired by observational learning, as the child observes another person's fear responses and learns to fear the same stimuli. Cognitive factors are also involved in the maintenance of the fear, especially fearful anticipation of and selective attention to the phobic stimuli.

Prepared learning

This term refers to an innate predisposition to develop persistent fear responses to certain stimuli. Some young primates seem to be prepared to develop fears of snakes, but it is not certain whether the same process accounts for some of the specific phobias of human children.

Neural mechanisms

Functional imaging studies have revealed hyperactivity of the amygdala upon *presentation* of the feared stimulus, which appears to diminish with successful treatment. *Anticipation* of a phobic stimulus activates the anterior cingulate cortex and the insular cortex. Generally, imaging studies indicate that specific phobias are characterized by increased activation in the regions linked to emotional appraisal and fear (amygdala, insula, anterior cingulate), with a concomitant failure to recruit prefrontal regions such as ventromedial prefrontal cortex, that regulate limbic brain regions (Goodkind et al., 2013).

Differential diagnosis

Diagnosis is seldom difficult. The possibility of an underlying depressive disorder should always be kept in mind, since some patients seek help for longstanding specific phobias when a depressive disorder makes them less able to tolerate their phobic symptoms. Obsessional disorders sometimes present with fear and avoidance of certain objects (e.g. knives). In such cases a systematic history and mental state examination will reveal the associated obsessional thoughts (e.g. thoughts of harming a person with a knife).

Prognosis

The prognosis of specific phobia in adulthood has not been studied systematically. Clinical experience suggests that specific phobias that originate in childhood continue for many years, whereas those that start after stressful events in adulthood have a better prognosis.

Treatment

The main treatment is the *exposure form of behaviour therapy* (see Chapter 24). With this treatment, the phobia is usually reduced considerably in intensity and so is the social disability. However, it is unusual for the phobia to be lost altogether. The outcome depends importantly on repeated and prolonged exposure, and up to 25% of phobic patients decline exposure-based therapies. Some patients seek help soon before an important engagement that will be made difficult by the phobia. When this happens, a few doses of a benzodiazepine may be prescribed to relieve phobic anxiety until a treatment can be arranged.

Exposure usually takes place over several one-hour sessions, but it can be carried out in a single very long and intensive session lasting for several hours. Virtual reality exposure may also be of benefit (Eaton et al., 2018). In a meta-analysis of 33 studies, Wolitzky-Taylor et al. (2008) found that exposure-based treatments were superior to other kinds of therapy, for example, relaxation therapy. Exposure treatment *in vivo* outperformed other kinds of exposure therapy in the short term but not at follow-up. Exposure therapy seemed equally effective across the range of different specific phobias. A significant weakness of most studies was a failure to include dropouts from therapy in the analysis of results.

Although pharmacotherapy has not been regarded as useful in the treatment of specific phobias, there is some evidence that *D-cycloserine*, a partial agonist at the glutamate N-methyl-D-aspartate (NMDA) receptor, may be helpful in augmenting the effectiveness of exposure treatment of phobias (Rodrigues et al., 2014). In animals,

D-cycloserine facilitates fear extinction, and it is possible that a similar mechanism may be involved when D-cycloserine is combined with behaviour therapy in humans.

Social phobia (social anxiety disorder)

Clinical picture

In this disorder, inappropriate anxiety is experienced in social situations, in which the person feels observed by others and could be criticized by them. Socially phobic people attempt to avoid such situations. If they cannot avoid them, they try not to engage in them fully—for example, they avoid making conversation, or they sit in the place where they are least conspicuous. Even the prospect of encountering the situation may cause considerable anxiety, which is often misconstrued as shyness. Social phobia can be distinguished from shyness by the levels of personal distress and associated social and occupational impairment (Leichsenring and Leweke, 2017).

The *situations* in which social phobia occurs include restaurants, canteens, dinner parties, seminars, board meetings, and other places where the person feels observed by other people. Some patients become anxious in a wide range of social situations (*generalized social phobia*), whereas others are anxious only in specific situations, such as public speaking, writing in front of others, or playing a musical instrument in public. In DSM-5 ‘performance only’ social phobia, anxiety is restricted to speaking or performing in public. (In DSM-5 and ICD-11, the term ‘social anxiety disorder’ is preferred to ‘social phobia’.)

People with social phobia may experience any of the anxiety *symptoms* listed in Box 8.2, but complaints of blushing and trembling are particularly frequent. Socially phobic people are often preoccupied with the idea of being observed critically, although they are aware that this idea is groundless.

The *cognitions* centre around a fear of being evaluated critically by others. These cognitions are described in more detail in the section on aetiology below.

Other problems. Some patients take alcohol to relieve the symptoms of anxiety, and alcohol misuse is more common in social phobia than in other phobias. Cannabis may also be employed to ameliorate anxiety symptoms. Comorbid depressive disorders as well as other anxiety disorders are also common (Kessler et al., 2005b).

Onset and development. The condition usually begins in the early teenage years. The first episode occurs in

a public place, usually without an apparent reason. Subsequently, anxiety is felt in similar places, and the episodes become progressively more severe with increasing avoidance.

Diagnostic conventions

Box 8.4 shows, in summary form, the criteria for the diagnosis of social anxiety disorder in ICD-11. The requirements are similar for both ICD-11 and DSM-5. DSM-5 has the criterion that symptoms must have been persistent, lasting for at least six months while in ICD-11, the symptoms should have lasted for ‘several months’.

Differential diagnosis

Agoraphobia and panic disorder

The symptom of social phobia can occur in either of these disorders, in which case both diagnoses can be made, although it is generally more useful for the clinician to decide which syndrome should be given priority.

Generalized anxiety disorder and depressive disorder

Social phobia has to be distinguished from the former by establishing the situations in which anxiety occurs, and from the latter from the history and mental state examination.

Schizophrenia

Some patients with schizophrenia avoid social situations because they have persecutory delusions. However, when they are in the feared situation, people with social phobia may feel convinced by their ideas that they are

Box 8.4 Abbreviated diagnostic criteria for social phobia in ICD-11

- Marked and excessive fear or anxiety that occurs in one or more social situations. Concern of acting in a way or showing anxiety that will lead to negative evaluation by others (be humiliating, embarrassing, or will lead to rejection).
- Relevant social situations are consistently avoided or endured with intense fear or anxiety.
- Interferes with functioning or causes marked distress.
- Not secondary to, or better accounted for, another disorder.

Source: *International Classification of Diseases, Eleventh Revision (ICD-11)*. Copyright (2022), World Health Organization. <https://icd.who.int>

being observed, when they are away from the situation, they know that these ideas are false.

Body dysmorphic disorder

People with this disorder may avoid social situations, but the diagnosis is usually clear from the patient's account of the problem.

Avoidant personality disorder

Social phobia has to be distinguished from a personality characterized by lifelong shyness and lack of self-confidence. In principle, social phobia has a recognizable onset and a shorter history, but in practice the distinction may be difficult to make, as social phobia usually begins in adolescence, and the exact onset may be difficult to recall. Many people have disorders that meet the criteria for both diagnoses.

Inadequate social skills

This is a primary lack of social skills, with secondary anxiety. It is not a phobic disorder, but a type of behaviour that occurs in personality disorders, in schizophrenia, and in some people with learning difficulties. People with ASD lack social communication skills and ASD and social phobia not infrequently coexist.

Normal shyness

Some people who have none of the above disorders are shy and feel ill at ease in company. As noted above, the diagnostic criteria for social phobia set a level of severity that is intended to exclude these individuals.

Epidemiology

The National Comorbidity Survey Replication reported a lifetime prevalence rate of social phobia in the community of around 12%. The rate is therefore not much lower than that of specific phobia. Social phobias are about equally frequent among the men and women who seek treatment, but in community surveys they are reported rather more frequently by women (Kessler et al., 2005a,b). As noted above, social phobia is associated with depression and alcoholism.

Aetiology

Genetic factors

Genetic factors are suggested by the finding that social phobia is more common among the relatives of people with social phobia than in the general population, and the risk is greatest in first-degree relatives (about a four-fold increase in incidence), decreasing as the degree of relatedness diminishes. The concordance rate of social phobia in monozygotic twins (around 24%) is higher

than that seen in dizygotic twins (around 15%) and *heritability* has been estimated to be around 55%, with shared (familial) environment making relatively little contribution. Avoidant personality disorder also appears more common in families of people affected by social phobia, suggesting that the two conditions may have a shared genetic aetiology.

Conditioning

Most social phobias begin with a sudden episode of anxiety in circumstances similar to those which become the stimulus for the phobia, and it is possible that the subsequent development of phobic symptoms occurs partly through conditioning.

Cognitive factors

The principal *cognitive factor* in the aetiology of social phobia is an undue concern that other people will be critical of the person in social situations (often referred to as a *fear of negative evaluation*). This concern is accompanied by several other ways of thinking, including:

- excessively high standards for social performance
- negative beliefs about the self (e.g. 'I'm boring')
- excessive monitoring of one's own performance in social situations
- intrusive negative images of the self as supposedly seen by others.

People with social phobia often adopt *safety behaviours*, such as avoiding eye contact, which make it harder for them to interact normally. For a review of cognitive and meta-cognitive models of social phobia, see Nordahl and Wells (2017).

Neural mechanisms

Functional neuroimaging studies have found that patients with social phobia have increased *amygdala* responses to presentation of faces with expressions of negative affect. The *anticipation* of public speaking in individuals with social phobia produced activation in limbic and associated regions, including the *amygdala*, *hippocampus*, and *insula*, while activation of cortical regulatory areas such as the *prefrontal cortex* was diminished. The *insula* is thought to represent interoceptive cues, and increased *insula* activity in patients with social phobia may underpin the preoccupation of patients with bodily autonomic changes; for example, flushing and sweating (Goodkind et al., 2013). Apart from the increased activation in the *insula*, the pattern of changes in social phobia is similar to that seen in simple phobias.

Course and prognosis

Social phobia has an early onset, usually in childhood or adolescence, and can persist over many years, sometimes even into old age. Only about 50% of people with the disorder seek treatment, usually after many years of symptoms (Pilling et al., 2013).

Treatment

Psychological treatment

CBT is the psychological treatment of choice for social phobia (for a description of this treatment, see Chapter 24). The original cognitive procedures were based on those used successfully to treat agoraphobia and panic disorder and, when added to exposure, did not greatly increase the benefit. A modified form of CBT appears to be more effective. This modified treatment is based on the particular cognitive abnormalities that are found in social phobia (see the section on Aetiology above), coupled with measures to reduce safety behaviours, and using video or audio feedback. CBT can be given in a group format, but this may not be as effective as individual treatment (Pilling et al., 2013).

Dynamic psychotherapy. There is limited evidence that dynamic psychotherapy given weekly over six months can improve symptoms of social phobia relative to a waitlist control (Pilling et al., 2013). Clinical experience suggests that this treatment may help patients whose social phobia is associated with pre-existing problems in personal relationships. However, there have been no controlled trials to test this possibility.

Drug treatment

SSRIs. Treatment guidelines generally recommend SSRIs as the first choice of pharmacological treatment in the management of social phobia. All of the SSRIs, as well as venlafaxine (an SNRI), have been shown to be effective, although the data for fluoxetine are slightly less consistent. With any of these drugs, the onset of action may take up to six weeks. Medication is usually continued for at least six months, and often for longer, because the risk of relapse is high if medication is discontinued after an acute response (Baldwin et al., 2014). When medication is reduced, this should be done slowly.

Other drugs. The monoamine oxidase inhibitor *phenelzine* is more effective in the treatment of social phobia than placebo. *Moclobemide*, the reversible inhibitor of monoamine oxidase, can also be prescribed, but reported response rates vary, and in some studies the drug was not more beneficial than placebo. Benzodiazepines are

effective and can be used for short-term relief of symptoms, but should not be prescribed for long because of the risk of dependency (see Chapter 25). The main use of benzodiazepines is to help patients to cope with essential social commitments while waiting for another treatment to take effect. *Beta-adrenergic blockers* such as propranolol may achieve short-term control of tremor and palpitations, which can be the most handicapping symptoms of specific social phobias, such as performance anxiety among musicians. However, it is doubtful whether beta-adrenergic blockers are generally effective in social phobia (Baldwin et al., 2014). There are also positive controlled trials of the antidepressant *mirtazapine* and the anticonvulsants *pregabalin* and *gabapentin*. For a review of the pharmacological treatment of social phobia, see Baldwin et al. (2014).

Management

What patients need to know

Patients need to understand that, although constitutional factors may play a part, the extent and severity of their social anxiety are a result of adopting *maladaptive ways of thinking and behaving* when they are socially anxious. These patterns of thinking and behaviour can be reversed either with psychological treatment or with medication. Self-help books can inform patients and help them to use simple cognitive behavioural approaches while awaiting further help. It is important not to misuse alcohol or other substances to deal with the anxiety produced by social situations.

The choice of treatment

Guidelines from the National Institute for Health and Care Excellence recommend *individual CBT* specifically developed for the treatment of social phobia as first-line treatment. If CBT is declined, a further psychological treatment should be offered; for example, an approach based on CBT principles applied using supported self-help. Psychodynamic psychotherapy treatment may be used as a third-line psychological option.

Medication

Medication should be reserved for patients who decline psychological treatment or where psychological treatment is ineffective or only partly effective. If medication is to be used, SSRI treatment with either escitalopram or sertraline is recommended, with venlafaxine being offered if SSRIs are not effective. Phenelzine or moclobemide can be employed as third-line drug choices (see Pilling et al., 2013).

Agoraphobia

Clinical features

People with agoraphobia are anxious when they are away from home, in crowds, or in situations that they cannot leave easily. They avoid these situations, feel anxious when anticipating them, and experience other symptoms. Each of these features will now be considered in turn.

Anxiety

The anxiety symptoms that are experienced by agoraphobic patients in the phobic situations are similar to those of other anxiety disorders (see Box 8.1), although two features are particularly important:

1. panic attacks, whether in response to environmental stimuli or arising spontaneously
2. anxious cognitions about fainting and loss of control.

Situations

Many situations provoke anxiety and avoidance. They seem at first to have little in common, but there are three common themes, namely *distance from home*, *crowding*, and *confinement*. The situations include buses and trains, shops and supermarkets, and places that cannot be left suddenly without attracting attention, such as the hairdresser's chair or a seat in the middle row of a theatre or cinema. As the condition progresses, the individual increasingly avoids these situations until in severe cases they may be more or less confined to their home. Apparent variations in this pattern are usually due to factors that reduce symptoms for a while. For example, most patients are less anxious when accompanied by a trusted companion, and some are helped by the presence of a child or pet dog. Such variability in anxiety may suggest erroneously that when symptoms are severe, they are being exaggerated.

Anticipatory anxiety

This is common. In severe cases anticipatory anxiety appears hours before the person enters the feared situation, adding to their distress and sometimes suggesting that the anxiety is generalized rather than phobic.

Other symptoms

Depressive symptoms are common. Sometimes these are a consequence of the limitations to normal life caused by anxiety and avoidance, while in other cases they seem to be part of the disorder, as in other anxiety disorders. *Depersonalization* can also be severe.

Onset and course

The onset and course of agoraphobia differ in several ways from those of other phobic disorders.

Age of onset. In most cases the onset occurs in the early or mid-twenties, with a further period of high onset in the mid-thirties. In both cases this is later than the average ages of onset of simple phobias (childhood) and social phobias (mostly the teenage years).

Circumstances of onset. Typically, the first episode occurs while the person is waiting for public transport or shopping in a crowded store. Suddenly they become extremely anxious without knowing why, feel faint, and experience palpitations. They rush away from the place and go home or to hospital, where they recover rapidly. When they enter the same or similar surroundings, they become anxious again and make another hurried escape. However, not all patients describe such an onset starting from an unexplained panic attack. It is unusual to discover any serious immediate stress that could account for the first panic attack, although some patients describe a background of serious problems (e.g. worry about a sick child), and in a few cases the symptoms begin soon after a physical illness or childbirth.

Subsequent course. The sequence of anxiety and avoidance recurs during the subsequent weeks and months, with panic attacks experienced in a growing number of places, and an increasing habit of avoidance develops. However, sometimes avoidance can occur without the development of panic attacks.

Effect on the family. As the condition progresses, agoraphobic patients become increasingly dependent on their partner and relatives for help with activities that provoke anxiety, for example, shopping. The consequent demands on the partner often lead to relationship difficulties. Alternatively, the partner may become over-involved in supporting the patient, and difficulties in relinquishing this role may complicate efforts at treatment.

Diagnostic conventions

Most, but not all, patients with agoraphobia have panic attacks, which may be situational or spontaneous, and many of the latter individuals meet the criteria for panic disorder as well as for agoraphobia. Thus, in both ICD-11 and DSM-5, patients with agoraphobia who also experience spontaneous panic attacks (i.e. attacks not confined to specific feared situations), can be diagnosed with both agoraphobia and panic disorder. The ICD-11 criteria for the diagnosis of agoraphobia are shown in Box 8.5. The DSM-5 criteria are similar though specify

Box 8.5 Abbreviated diagnostic criteria for agoraphobia in ICD-11

- Marked and excessive fear or anxiety that occurs in, or in anticipation of, multiple situations where escape might be difficult or help not available, such as public transport, being outside the home alone, being in shops, or in cinemas.
- Consistent fear or anxiety about these situations due to fears of a negative outcome such as a panic attack, or other incapacitating or embarrassing physical symptoms. The situations are actively avoided or endured only with intense anxiety.
- The symptoms persist for an extended period of time, for at least several months.
- The symptoms cause significant distress or significant impairment in functioning.
- Not the result of another disorder, or of cultural beliefs.

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>

that the marked fear and anxiety should arise in at least two of five specific situations. The latter include public transport, open spaces, enclosed places, crowds, being away from home and being by oneself. The fear, anxiety, and avoidance should have persisted for at least six months.

Differential diagnosis

Social phobia

Some patients with agoraphobia feel anxious in social situations, and some people with social phobia avoid crowded buses and shops, where they feel under scrutiny. Detailed enquiry into the current pattern of avoidance and also the order in which the two sets of symptoms developed will usually decide the diagnosis.

Generalized anxiety disorder

When the agoraphobia is severe, anxiety may develop in so many situations that the condition resembles GAD. In these cases, the history of development of the disorder will usually point to the correct diagnosis.

Panic disorder

Agoraphobia often includes panic attacks and patients may meet criteria for both disorders, as discussed above.

Depressive disorder

Agoraphobic symptoms can occur in a depressive disorder, and many agoraphobic patients have depressive symptoms. Enquiry about the order in which the symptoms developed will usually point to the correct diagnosis. Sometimes a depressive disorder develops in a person with longstanding agoraphobia, and it is important to identify such cases and treat the depressive disorder (see below).

Paranoid disorders

Occasionally a patient with paranoid delusions (arising in the early stages of schizophrenia or in a delusional disorder) avoids going out and meeting people in shops and other places. The true diagnosis is usually revealed by a thorough mental state examination, which generally uncovers delusions of persecution or of reference.

Epidemiology

A community investigation in Europe, using DSM-IV criteria, estimated a lifetime prevalence of agoraphobia without panic of 0.6%. However, minor variations in the diagnostic criteria increased the incidence to 3.4% (Wittchen et al., 2010). In the United States a population study of over 9,000 participants using DSM-IV criteria estimated the lifetime risk of agoraphobia to be 2.6% while the risk for panic disorder (with and without agoraphobia) was 5.2% (Kessler et al., 2012). The risk in women for both disorders was two to three times higher than that in men. In clinical samples, agoraphobia without panic appears to be rare, but in the community, it may well be more frequent.

Aetiology

Theories of the aetiology of agoraphobia have to explain both the initial anxiety attack and its spread and recurrence. These two problems will now be considered in turn.

Theories of onset

Agoraphobia begins with anxiety in a public place—generally, but not always—as a panic attack. There are two explanations for the initial anxiety.

1. The *cognitive hypothesis* proposes that the anxiety attack develops because the person is unreasonably afraid of some aspect of the situation or of certain physical symptoms that are experienced in the situation (see the section on Panic disorder, below). Although such fears are expressed by people with established agoraphobia, it is not known whether they were present before the onset.

2. *The biological theory* proposes that the initial anxiety attack results from chance environmental stimuli acting on an individual who is constitutionally predisposed to over-respond with anxiety. There is some evidence for a genetic component to this predisposition, in that relatives of probands with agoraphobia are at increased risk of experiencing an anxiety disorder themselves (Meier and Deckert, 2019).

Theories of spread and maintenance

Learning theories. Conditioning could account for the association of anxiety with increasing numbers of situations, and avoidance learning could account for the subsequent avoidance of these situations. Although this explanation is plausible and is consistent with observations of learning in animals, there is no direct evidence to support it.

Personality. Agoraphobic patients are often described as dependent, and prone to avoiding rather than confronting problems. This dependency could have arisen from overprotection in childhood, which is reported more often by agoraphobic individuals than by controls. However, despite such retrospective reports, it is not certain that the dependency was present before the onset of the agoraphobia.

Family influences. Agoraphobia could be maintained by family problems, and clinical observation suggests that symptoms are sometimes prolonged by overprotective attitudes of other family members, but this feature is not found in all cases.

Prognosis

Although short-lived cases may be seen in general practice, agoraphobia that has lasted for one year generally remains for the next five years, and usually the illness runs a chronic course. Episodes of depression are common in the course of chronic agoraphobia, and clinical experience suggests that people are more likely to seek help during these episodes (Wittchen et al., 2010).

Treatment

Much of the available treatment has been developed for panic disorder and for panic disorder with agoraphobia probably because patients with agoraphobia without panic are not common in clinical samples. There has been little systematic investigation of treatment for agoraphobia without panic (Baldwin et al., 2014).

Psychological treatment

Exposure treatment was the first of the behavioural treatments for agoraphobia. It was shown to be effective, but

more so when combined with anxiety management (see Chapter 8).

CBT for panic and agoraphobia is described in Chapter 24. Clinical trials (reviewed under panic disorder) indicate that, in the short term, cognitive therapy is about as effective as medication, and that in the long term it is probably more effective.

Medication

The drug treatment of agoraphobia resembles that for panic, except that medication is usually combined with repeated practice in re-entering situations that are feared and avoided. This exposure may account for some of the observed change. Most studies of drug treatment include both agoraphobic and panic disorder patients, and it is difficult to separate the treatment response of the two disorders. The following account should be read in conjunction with the subsequent discussion of medication for panic disorder.

Anxiolytic drugs. *Benzodiazepines* may be used for a specific, short-term purpose such as helping a patient to undertake an important engagement before other treatment has taken effect. Anxiolytic drugs should not be prescribed for more than a few weeks because of the risk of dependence. Indeed, guidelines issued by the National Institute for Health and Clinical Excellence (2011b) for panic disorder and agoraphobia suggest that benzodiazepines should not be used at all, because they may worsen the long-term outcome of the condition. However, in some countries, although seldom in the United Kingdom, *alprazolam*, the high-potency benzodiazepine, is used to treat agoraphobia with frequent panic attacks. Some authorities believe that short-term benzodiazepine treatment does have a role in panic disorder—for example, while the patient is being established on more suitable drug treatment. However, it is possible that benzodiazepines could impair the response to psychological treatments (National Institute for Health and Clinical Excellence, 2011b).

Antidepressant drugs. As well as the obvious use to treat a concurrent depressive disorder, antidepressant drugs have a therapeutic effect in agoraphobic patients who are not depressed but who have frequent panic attacks. *Imipramine* was one of the first agents to be used in this way, but similar effects have been reported with *clomipramine*. The treatment regime is the same as that described for panic disorder. In addition, several *SSRIs* and *venlafaxine* have been shown to be effective in panic disorder with and without agoraphobia (Baldwin et al., 2014). SSRIs are generally recommended as the most suitable first-line treatment because of their safety and

tolerability relative to tricyclic antidepressants (National Institute for Health and Clinical Excellence, 2011b). As with other drug treatments for anxiety, maintaining the medication for several months after a clinical response has been obtained significantly lowers relapse rates.

Management

What patients need to know

Patients with agoraphobia and those around them usually have difficulty in understanding the nature of agoraphobia, and may think of it as the result of lack of determination to overcome normal anxiety. A two-stage explanation, starting with the panic attacks, is generally helpful. *Panic attacks* can be likened to *false alarms* occurring in an oversensitive intruder-alarm system. They can occur in most people under stressful circumstances. The excessive sensitivity can be explained in terms of constitution or chronic stress, whichever fits the patient's history. *Avoidance* can be explained in terms of conditioning, with examples such as anxiety after falling from a bicycle or following a car accident. Partners, friends, and relatives can usually understand the principles of behaviour therapy, but may be unsympathetic to drug treatment and puzzled when antidepressants are prescribed for anxiety. In response the patient can say that the medication is to reduce the sensitivity of the 'alarm system', and explain that some antidepressant drugs can do this.

Patients also need to know that medication is only likely to be effective if accompanied by determined and persistent efforts to overcome avoidance. The therapist should explain how to do this, but must emphasize that

the result will depend on the patient's own efforts. Self-help books are a useful source of information about the disorder and about the ways in which people with agoraphobia can help themselves.

Behavioural management

In early cases, the patient should be strongly encouraged to return to the situations that they are avoiding. The treatment of choice for established cases is probably a combination of exposure to phobic situations with cognitive therapy for panic attacks. If there is a waiting list for cognitive therapy, the referring clinicians should supervise exposure treatment. Several self-help manuals have been published that reduce the time that therapists need to spend in doing this.

Medication

Medication can be offered as a first treatment, especially when panic attacks are frequent and/or severe. However, it needs to be accompanied by repeated exposure to previously feared and avoided situations. In the United Kingdom, the medication is usually an antidepressant, generally an SSRI. Any medication that has proved beneficial should be discontinued only gradually.

Patients who have relapsed after drug treatment can be offered CBT, although no controlled trials have been carried out specifically with such individuals. Most patients improve, but few of them lose the symptoms completely following treatment. Relapse is common, and patients should be encouraged to seek further help at an early stage should relapse occur.

Panic disorder

Although the diagnosis of panic disorder did not appear in the nomenclature until 1980, similar cases have been described under a variety of names for more than a century. The central feature is the occurrence of *panic attacks*. These are sudden attacks of anxiety in which physical symptoms predominate, and they are accompanied by fear of a serious medical consequence such as a heart attack.

In the past, these symptoms have been variously referred to as irritable heart, Da Costa's syndrome, neurocirculatory asthenia, disorderly action of the heart, and effort syndrome. These early terms assumed that patients were correct in fearing a disorder of cardiac function. Some later authors suggested psychological causes, but it was not until the Second World War (when

interest in the condition revived) that the cardiologist Paul Wood (1907–1962) showed convincingly that the condition was a form of anxiety disorder. From then until 1980, patients with panic attacks were classified as having either generalized or phobic anxiety disorders.

In 1980, the authors of DSM-III introduced the new diagnostic category of *panic disorder*, which included patients whose panic attacks occurred with or without generalized anxiety, but excluded those whose panic attacks appeared in the course of agoraphobia. In DSM-IV, all patients with frequent panic attacks were classified as having panic disorder, whether or not they had agoraphobia. In DSM-5 and ICD-11 agoraphobia and panic disorder are diagnosed separately and patients who have both conditions receive both diagnoses.

Clinical features

The symptoms of a panic attack are listed in Box 8.6. Not every patient has all of these symptoms during the panic attack, and for a diagnosis of panic disorder, DSM-5 requires the presence of only four or more symptoms. The important features of panic attacks are that:

- anxiety builds up quickly
- the symptoms are severe, and
- the person fears a catastrophic outcome.

Some people with panic disorder hyperventilate, and this adds to their symptoms.

Hyperventilation is breathing in a rapid and shallow way that leads to a fall in the concentration of carbon dioxide in the blood. The resulting hypocapnia may cause the symptoms listed in Box 8.7. The last symptom in the list, the feeling of breathlessness, is paradoxical, as the person is breathing excessively. It is important because it leads to a further increase in breathing, which worsens the condition. Hyperventilation should always be borne in mind as a cause of unexplained bodily symptoms. The diagnosis can usually be made by watching the pattern of breathing when the symptoms are present.

Diagnostic criteria

In DSM-5 the diagnosis of panic disorder is made when: 1) panic attacks occur recurrently (at least twice) and unexpectedly (i.e. not in response to an identified phobic stimulus); and 2) at least one attack has been followed by four weeks or more of persistent fear of another attack and worry about its implications (e.g. having a heart attack), *and/or* a significant maladaptive change in behaviour (e.g. avoiding exercise or public transport). The criteria in ICD-11 are similar, where for a diagnosis of panic disorder, panic attacks must be followed by persistent concern or worry for 'several weeks' about their recurrence or their health implications

Differential diagnosis

Panic attacks occur in GAD, phobic anxiety disorders (most often agoraphobia), depressive disorders, and acute organic disorder. Two of the DSM-5 diagnostic criteria help to distinguish these secondary attacks from panic disorder—first, the presence in panic disorder of a persistent marked concern about having further attacks, and secondly, worry about the potentially catastrophic consequences of the attacks.

Box 8.6 Symptoms of a panic attack

Sudden onset of:
 Palpitations
 Choking sensations
 Chest pain
 Dizziness and faintness
 Shortness of breath
 Chills or hot flushes
 Nausea or abdominal distress
 Tingling or lack of feeling in extremities
 Depersonalization
 Derealization
 Fear of dying, losing control, or going mad

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>

Box 8.7 Symptoms caused by hyperventilation

Dizziness
 Tinnitus
 Headache
 Feeling of weakness
 Faintness
 Numbness
 Tingling in the hands, feet, and face
 Carpopedal spasms
 Precordial discomfort
 Feeling of breathlessness

Epidemiology

The National Comorbidity Survey Replication found a 12-month prevalence rate of DSM-IV panic disorder of 2.7% and a lifetime risk of 4.7% (Kessler et al., 2005a,b). These figures include panic disorder with agoraphobia, which accounts for about 50% of the cases in the general population. In Europe the 12-month prevalence of panic disorder was somewhat less, at around 1.2%. In most studies, the prevalence in women is about twice that in men. Patients with panic disorder have increased rates of other anxiety disorders, major depression, and alcohol misuse.

Aetiology

Genetics

Panic disorder is familial, with about a fivefold increase in risk in first-degree relatives. Rates in monozygotic twins are higher than those in dizygotic twins, indicating that the family aggregation is likely to be at least partly owing to genetic factors, with a heritability of about 40% (Bienvenu et al., 2011). GWAS studies have not yet identified significant risk loci but there appears to be some genetic correlation between panic disorder, major depression, and trait neuroticism (Forstner et al., 2021).

Biochemistry

Chemical agents, notably sodium lactate and the noradrenaline α_2 -adrenoceptor antagonist yohimbine, but also flumazenil (a benzodiazepine-receptor antagonist), and a 5-HT receptor agonist, m-chlorophenylpiperazine, can induce panic attacks more readily in patients with panic disorder than in healthy individuals (Ballenger, 2009). The multitude of chemical agents that provoke panic attacks in panic disorder patients make it difficult to identify a single causal mechanism, and the abnormal responses could reflect altered *psychological attribution*, as described below. However, changes in brain 5-HT, noradrenaline, and GABA-ergic mechanisms have been suggested. There is most evidence for changes in GABA, with lowered cortical GABA levels measured by magnetic resonance spectroscopy, as well as diminished benzodiazepine-receptor binding in the parietotemporal regions in unmedicated patients with panic disorder (Hasler et al., 2008). Positron emission tomography (PET) studies also suggest a lowering of cortical 5-HT_{1A} receptor binding in patients with panic disorder, similar to that found in depression. However, this could be attributable to the high comorbidity between panic disorder and depression.

Neural mechanisms

Animal experimental studies suggest that the neural circuitry of fear and particularly escape behaviour (which panic attacks resemble to some extent) involves the amygdala, periaqueductal grey, hippocampus, hypothalamus, and brainstem nuclei, including the locus coeruleus, the origin of noradrenaline cell bodies. The ventromedial prefrontal cortex plays a regulatory inhibitory role. Structural imaging studies in panic disorder have reported altered volumes of cortical areas including dorsolateral prefrontal cortex and insula (Wu et al., 2018).

Functional imaging studies have described abnormalities both in baseline perfusion and during panic

provocation in various elements of this fear-related neural circuitry, but the findings have been variable. The best evidence for changes is in the insula, underlining the important role for interoceptive processes in the pathophysiology of panic disorder (Chavanne and Robinson, 2021).

Cognitive theories

The cognitive hypothesis is based on the observation that fears about serious physical or mental illness are more frequent among patients who experience panic attacks than among anxious patients who do not have panic attacks. It is proposed that there is a spiral of anxiety in panic disorder, as the physical symptoms of anxiety activate fears of illness and thereby generate more anxiety. ‘Safety behaviours’ prevent disconfirmation of these fears. These observations have led to an effective cognitive treatment for panic disorder.

Hyperventilation as a cause

A subsidiary hypothesis proposes that hyperventilation is a cause of panic disorder. Although there is no doubt that voluntary overbreathing can produce a panic attack, it has not been shown that panic disorder is caused by involuntary hyperventilation.

Course and prognosis

Follow-up studies have generally included patients with panic attacks and agoraphobia as well as patients with panic disorder alone. Earlier studies that used categories such as ‘effort syndrome’ found that most patients still had symptoms 20 years later, although most had a good social outcome. More recent studies of patients diagnosed with panic disorder also reveal a lengthy course, with fluctuating anxiety and depression. About 30% of patients remit without subsequent relapse, and a similar proportion show useful improvement, although with persistent symptomatology (Ballenger, 2009). The prognosis of panic disorder without agoraphobia is somewhat better, and results from controlled trials suggests effective treatment can improve the prognosis of both panic disorder and panic disorder with agoraphobia. Mortality rates from unnatural causes and, among men, from cardiovascular disorders are higher than average, and may be linked to changes in the regulation of the sympathetic nervous system (Davies et al., 2010).

Treatment

Apart from supportive measures and help with any causative life problems, treatment is with medication or

cognitive therapy. A number of different kinds of medication can be used.

Benzodiazepines

When given in high doses, benzodiazepines control panic attacks. In these doses, most benzodiazepines cause sedation, but alprazolam, a high-potency compound, is an exception. For this reason it has been used to treat panic disorder, and its effectiveness over placebo has been demonstrated in many controlled trials (see Ballenger, 2009). At the end of treatment, alprazolam should be reduced very gradually to avoid withdrawal symptoms, because even when it is reduced over 30 days, about one third of patients report significant withdrawal symptoms. This treatment was developed in the United States and has not been widely adopted in the United Kingdom, where benzodiazepines are not recommended for the treatment of panic disorder (National Institute for Health and Clinical Excellence, 2011b).

Antidepressants

Imipramine was the first antidepressant to be shown to be effective for the treatment of panic disorder, and other tricyclic antidepressants, including clomipramine and lofepramine, are superior to placebo in controlled trials. SSRIs are also beneficial in the treatment of panic disorder, as is the SNRI venlafaxine, and the selective noradrenaline re-uptake inhibitor reboxetine (Baldwin et al., 2014). The initial effect of antidepressants in patients with panic disorder is often to produce an unpleasant feeling of apprehension, sleeplessness, and palpitations. For this reason the initial dose should be small (e.g. 5mg of citalopram), increasing gradually each week until an effective dose is reached. Where patients respond to medication, maintaining treatment for at least six months helps to prevent relapse (Baldwin et al., 2014).

In a network meta-analysis, Chawla et al. (2022) found that the most effective pharmacological treatments for panic disorder, with or without agoraphobia, were benzodiazepines, tricyclic antidepressants, and SSRIs. However, when adverse effects were considered, SSRIs emerged as the most acceptable pharmacotherapy, particularly escitalopram and sertraline.

Cognitive therapy

Cognitive therapy reduces the fears of the physical effects of anxiety, which are thought to provoke and maintain the panic attacks. Common fears of this kind are that palpitations indicate an impending heart attack, or that dizziness indicates impending loss of consciousness. In treatment, the physical symptoms

that the patient fears are induced by hyperventilation or exercise. The therapist points out the sequence of physical symptoms that leads to fear, and explains that a similar sequence occurs in the early stages of a panic attack. The therapist goes on to question the patient's belief in the feared outcome. The procedure is described further in Chapter 24.

Controlled studies have shown that cognitive therapy is as effective as antidepressant medication in the treatment of panic disorder (Cuijpers et al., 2013). Combined treatment with medication and psychotherapy may result in a better response in the acute phase than either treatment modality given alone, but whether this persists in the longer term is uncertain (Cuijpers et al., 2014b). The most effective components of CBT therapy appear to be face-to-face therapy together with interoceptive exposure and cognitive restructuring. Muscular relaxation and virtual reality exposure have limited efficacy (Pompoli et al., 2018)

Management

What patients need to know

Patients need to be able to explain the disorder to relatives and friends, and the explanation outlined in relation to agoraphobia (see above) is usually appropriate. Patients also need to understand how particular ways of thinking and behaviours increase and prolong the disorder (see the section on cognitive therapy, above). Self-help books based on cognitive behaviour principles are available and can be helpful.

Choice of treatment

As cognitive therapy and medication have similar effects on symptoms, the choice of treatment depends on the patient's preference, the availability of cognitive therapy, and considerations of cost and long-term benefit (cognitive therapy is more costly, but probably has more lasting effects). If medication is chosen, one of the SSRIs are preferred to tricyclics because they have fewer side effects. Alprazolam has been used in some countries, but is not recommended in the United Kingdom (see above). Information about the main effects and side effects should be the same as that given when the same medication is given for other reasons, including the delayed onset of action, and the need to withdraw medication gradually (see Chapter 25). If there is no improvement after about 12 weeks, the treatment can be changed to an antidepressant of a different class, or cognitive therapy can be started. If cognitive therapy has not been effective, medication can be tried.

Transcultural variations in anxiety disorder

Anxiety disorders are universal but their prevalence and presentation show important cultural differences (Agorastos et al., 2012). In several cultures the presenting symptoms of anxiety disorder are more often somatic than psychological, although this may be an artefact of observations that are not sensitive to local indigenous concepts of distress and mistake somatic metaphors, or physiological–emotional interactions, as indicating somatization.

Recent research shows that cultural idioms are often recruited to express distress and so the presentation of symptoms may not conform to usual expectations; these may include somatic metaphors, or alternative explanatory models of distress; experiences of social and psychological and biological distress can trigger the same cognitions, which might then attract a range of diagnostic labels or be called cultural bound syndromes. There is a risk of reifying diagnosis once proposed, as if these are real entities, rather than a function of othered labelling and superficial phenomenology (Chowdhury and Brahma, 2020). When faced with unusual beliefs clinicians may wish to take more time exploring health beliefs and the mapping of conventional diagnostic categories to lay and indigenous idioms of distress before reaching a diagnosis (Bhui et al., 2021).

Sometimes this difference parallels the different words used to describe anxiety in the corresponding languages. Thus there is no word for anxiety in some African, Oriental, and Native American languages, and instead a phrase denoting bodily function is used. For example, in Yoruba, an African language, the phrase is ‘the heart is not at rest’. In addition, several conditions have been described that may be transcultural variants of anxiety disorders, although their exact relationship to these disorders is uncertain. It does appear that in non-western cultures anxiety is experienced more somatically.

Koro can occur among men from India, South China, and Japan, and has similarities to panic disorder. Cantonese people call it *suk-yeong*, which means ‘shrinking of the penis’. There are episodes of acute anxiety, lasting from 30 minutes to one or two days, during which the person complains of palpitations, sweating, pericardial discomfort, and trembling. At the same time he is convinced that the penis will retract into the abdomen,

and that when this process is complete, he will die. Most episodes occur at night, sometimes after sexual activity. To prevent the feared outcome, the patient may tie the penis to an object, or ask another person to hold the organ. This belief parallels the conviction held by patients during a panic attack that the heart is damaged and that they will die. Epidemics of *koro* have been described among people who are experiencing anxiety because of social stressors (Tseng, 2006).

Variants of social phobia have been described in the east, originally among people in Japan, where it is known as *taijin-kyofu-sho* or ‘phobia of interpersonal relations’. There is intense anxiety in social situations and an intense conviction, bordering on the delusional, that the person is being thought of unfavourably by others. Other symptoms include fears of producing body odours, dysmorphophobia, and aversion to eye contact (Agorastos et al., 2012).

It is also important not to ascribe to a psychiatric disorder expressions of anxiety that are culturally appropriate. For example, in collectivist cultures, public speaking or voicing dissent may be associated with a high but understandable level of anxiety. Also, in some cultures, there may be appropriate fears of leaving the home by oneself, for example, where discrimination or public violence is a risk or where women are expected to be accompanied in public spaces. In such circumstances a diagnosis of agoraphobia should be considered only when anxiety about being in public exceeds cultural norms.

Tseng (2006) pointed out that, although earlier research in this area tended to focus on ‘exotic’ presentations of disorder in unfamiliar cultures, all psychiatric syndromes are *culturally influenced*. Therefore a knowledge of how social context alters the presentation of psychological distress is relevant to *all cultures*, particularly in a historical period when cultures in many countries are becoming increasingly diverse. Indeed, depression and anorexia have been proposed to be a western cultural bound syndrome that is spreading across the globe. Social context and current stresses are particularly important when assessing anxiety disorders in ethnic and religious minorities but especially in recent migrants, where fear and anxiety may well be appropriate to personal circumstance.

Obsessive–compulsive disorder

The concise description of OCD, contained in ICD-9, is still valuable:

The outstanding symptom is a feeling of subjective compulsion—which must be resisted—to carry out some action, to dwell on an idea, to recall an experience, or ruminate on an abstract topic. Unwanted thoughts, which include the insistency of words or ideas, ruminations or trains of thought, are perceived by the patient to be inappropriate or nonsensical. The obsessional urge or idea is recognized as alien to the personality but as coming from within the self. Obsessional actions may be quasi ritual performances designed to relieve anxiety, e.g. washing the hands to deal with contamination. Attempts to dispel the unwelcome thoughts or urges may lead to a severe inner struggle, with intense anxiety.

Clinical picture

OCD is characterized by obsessional thinking, compulsive behaviour, and varying degrees of anxiety, depression, and depersonalization. Obsessional and compulsive symptoms are listed in Box 8.8. They were described in Chapter 1, but the reader may find it helpful to be reminded of the main features.

Obsessional thoughts are words, ideas, and beliefs that are recognized by patients as their own, and that intrude forcibly into the mind. They are usually unpleasant, and attempts are made to exclude them. It is the combination of an *inner sense of compulsion* and of *efforts at resistance* that characterizes obsessional symptoms, but the effort at resistance is the more variable of the two. Obsessional thoughts may take the form of single words, phrases, or rhymes, are usually unpleasant or shocking to the person, and may be obscene or blasphemous. Obsessional images are vividly imagined scenes, often of a violent or disgusting kind (e.g. involving sexual practices that the person finds abhorrent).

Obsessional ruminations are internal debates in which arguments for and against even the simplest everyday actions are reviewed endlessly. Some obsessional doubts concern actions that may not have been completed adequately (e.g. turning off a gas tap or securing a door), while other doubts concern actions that might have harmed other people (e.g. that driving a car past a cyclist might have caused him to fall off his bicycle). Sometimes doubts are related to religious convictions or observances ('scruples')—a phenomenon well known to those who hear confession.

Obsessional impulses are urges to perform acts, usually of a violent or embarrassing kind (e.g. leaping in front of a car, injuring a child, or shouting blasphemies at a religious ceremony).

Obsessional rituals include both mental activities (e.g. counting repeatedly in a special way, or repeating a certain form of words) and repeated but senseless behaviours (e.g. washing the hands 20 or more times a day). Some rituals have an understandable connection with the obsessional thoughts that precede them (e.g. repeated hand washing following thoughts about contamination). Other rituals have no such connection (e.g. arranging objects in a particular way). The person may feel compelled to repeat such actions a certain number of times, and if this sequence is interrupted it has to be repeated from the beginning. People who use rituals are usually aware that these are illogical, and usually try to hide them. Some people fear that their symptoms are a sign of incipient madness, and are greatly helped by reassurance that this is not so.

Obsessional slowness. Although obsessional thoughts and rituals lead to slow performance, a few obsessional patients are afflicted by extreme slowness that is out of proportion to other symptoms.

Obsessional phobias. Obsessional thoughts and compulsive rituals may worsen in certain situations—for example, obsessional thoughts about harming other people may increase in a kitchen or other place where knives are kept. The person may avoid such situations

Box 8.8 Principal features of obsessive–compulsive disorder

Obsessional symptoms
Thoughts
Ruminations
Impulses
'Phobias'
Compulsive rituals
Abnormal slowness
Anxiety
Depression
Depersonalization

because they cause distress, just as people with phobic disorders avoid specific situations. Because of this resemblance, the condition is called an obsessional phobia.

Anxiety. This is a prominent component of OCDs. Some rituals are followed by a lessening of anxiety, while others are followed by increased anxiety.

Depression. Obsessional patients are often depressed. In some patients, depression is an understandable reaction to the obsessional symptoms; in others, depression appears to vary independently.

Depersonalization. Some obsessional patients complain of depersonalization. The relationship between this distressing symptom and the other features of the disorder is unclear.

Relationship to obsessional personality. Obsessional personality (obsessive–compulsive personality disorder in DSM-5) is described in Chapter 15. There is no simple, one-to-one relationship between OCD and this kind of personality. Although obsessional personality is over-represented among people who develop OCD, about one third of obsessional patients have other types of personality (as noted in a classic paper by Lewis, 1934). Furthermore, people with obsessional personality are more likely to develop depressive disorders than OCDs.

Diagnostic criteria

In DSM-5 the diagnosis of OCD requires the presence of *either* obsessions or compulsions *or* both. In addition, the obsessions and/or compulsions are required to be *time-consuming* (e.g. taking more than one hour daily) and/or to cause *clinically significant distress* and/or *impairment in social or occupational function*. It is also necessary that the obsessive–compulsive symptoms are not attributable to the effects of a substance or another medical condition, and that the disturbance is not better explained by the symptoms of another mental disorder. In DSM-5, it is also possible to specify the degree of insight that a patient possesses, ranging from ‘good or fair’ to ‘absent’ or ‘delusional’. Under these diagnostic criteria, therefore, it is not necessary for the patient with OCD to believe that their obsessions and compulsions are unfounded or nonsensical.

The diagnostic criteria in ICD-11 are similar and stress repetitive behaviours together with cognitive phenomena such as obsessions, intrusive thoughts and pre-occupations. The level of insight can vary from ‘good’ to ‘absent’. Neither ICD-11 nor DSM-5 emphasize the subjective sense of *resistance* to obsessional thoughts and compulsions, mentioned in the ICD-9 description given above. However, clinically this remains a useful phenomenon, helpful in the identification of OCD. It is

also worth noting that compulsive acts, while they may relieve tension, are not in themselves *intrinsically pleasurable*. This distinguishes compulsions from acts that are associated with immediate gratification, such as those associated with addictions.

Differential diagnosis

OCD must be distinguished from other disorders in which obsessional symptoms occur.

Anxiety disorders

The distinction from GAD, panic disorder, or phobic disorder is seldom difficult provided that a careful history is taken and the mental state is examined thoroughly.

Depressive disorder

The course of OCD is often punctuated by periods of depression in which the obsessional symptoms increase. When this happens the depressive disorder may be overlooked. Furthermore, obsessional symptoms sometimes occur in the course of a primary depressive disorder and may disappear when the depression is successfully treated.

Schizophrenia

Occasionally, the symptoms of OCD may resemble delusions because the content of the obsessional thoughts is peculiar, or the rituals are exceptionally odd, and resistance is weak. The disorder can then be mistaken for a delusional disorder or schizophrenia. In such cases, the correct diagnosis can be made following thorough history-taking and a careful search for other symptoms of psychosis. Obsessive–compulsive symptoms are not uncommon in schizophrenia, and may require separate treatment (Schirmbeck and Zink, 2022).

Organic disorders

Obsessional symptoms are occasionally found in organic cerebral disorders. They were observed most strikingly in chronic cases of encephalitis lethargica following the epidemic in the 1920s (see page 455).

Childhood-onset disorders

In children, obsessive–compulsive symptoms are frequently comorbid with developmental disorders such as *Tourette syndrome* and *autism* (see page 455).

Epidemiology

Previous community surveys in the United States estimated the *lifetime prevalence* of OCD to be around 2%

(Ruscio et al, 2010). However, more recent studies, using strict criteria to eliminate subthreshold cases, yield a lower prevalence of around 1.5% (Carmi et al, 2020). OCD is more common in women (about 1.5:1.0) but the course of OCD in men is associated with a greater functional disability. The age of onset is typically in in adolescence.

Aetiology

Healthy individuals commonly experience occasional intrusive thoughts, some of which are concerned with sexual, aggressive, and other themes similar to those of obsessional patients. It is the frequency, intensity, and, above all, the persistence of obsessional phenomena that have to be explained, as well as the subjective distress that they cause.

Genetics

In the small number of twin studies that have been conducted, the concordance rate was greater in monozygotic than in dizygotic pairs, indicating that at least part of the familial loading is genetic. Heritability has been estimated at around 50%. Family studies have reached a similar conclusion, and indicate that the risk of OCD in first-degree relatives is increased approximately four-fold compared with control rates. Such studies have also found increased rates of OCD-related disorders in relatives of people with OCD such as body dysmorphic disorder, trichotillomania, pathological skin picking, tic disorders, and obsessive compulsive personality disorder (Nestadt and Samuels, 2020).

Candidate molecular genetic studies reported a number of associations between OCD and various genetic polymorphisms, including those coding for the serotonin transporter, the 5-HT_{2A} receptor and the neuronal glutamate transporter; however, these associations have not been confirmed in GWAS studies. GWAS studies themselves have not produced reliable genetic associations, probably because of sample size limitations (Nestadt and Samuels, 2020).

Evidence of a brain disorder

Two kinds of evidence suggest a disorder of brain function in OCD—first, associations with conditions that have known effects on brain function, and secondly, evidence from brain imaging.

Associations with other brain disorders. As noted above, obsessional symptoms were recorded frequently among patients affected by encephalitis lethargica after the Spanish ‘flu pandemic of 1918–1919. Also, Gilles de la Tourette included obsessional symptoms in his original

description of the disorder that now bears his name, and more recent studies have confirmed this observation (Carmi et al., 2020). In childhood, 70% of cases of Sydenham’s chorea, which affects the caudate nucleus, are reported to have obsessive–compulsive symptoms. The condition, which is referred to as PANDAS (paediatric autoimmune neuropsychiatric disorder associated with streptococcal infection), is considered further in Chapter 16. In addition, the development of OCD in some children has been linked to group A streptococcal infections, although the findings are inconsistent and other throat infections in children can produce neuropsychiatric symptoms, including obsessions and compulsions (Orlovskaya et al., 2017).

There is also an increased prevalence of ASD traits in children and adolescents with OCD. Increased ASD traits in this population is associated with a greater impairment in global functioning (Tiley and Kyriakopoulos, 2021).

Brain imaging studies. Structural imaging in patients with OCD has revealed rather variable changes, but the most consistent are greater volume of the pallidum, decreased volume of hippocampus, and parietal cortical thinning (Veltman, 2021). Resting state functional imaging studies in OCD reveal abnormal connectivity between caudate, striatum, and thalamus (Liu et al., 2022). Task-based fMRI studies have shown increased activation in orbitofrontal and anterior cingulate cortex as well as the caudate nucleus. These abnormalities are reversed by successful treatment (Goodman et al., 2021).

The neuroimaging findings have supported proposals that the symptoms of OCD are related to dysfunction in the well-described *cortico–striatal–thalamic loops*, which support key neuropsychological domains such as affective and reward processing, working memory and executive function, and response inhibition. In OCD, current theories propose dysfunction in cortico–striatal–thalamic loops involving orbitofrontal cortex, ventromedial prefrontal cortex, anterior cingulate, and ventral striatum (see Goodman et al., 2021)

Abnormal serotonergic function

The finding that obsessive–compulsive symptoms respond to drugs that increase 5-HT function suggests that *5-HT mechanisms* might be abnormal in OCD. The full effect of 5-HT uptake inhibitors on obsessive–compulsive symptoms takes several weeks, so the late effects are likely to be most relevant. However, these late effects are complex, and it is not known which of them are important. In any case, the response of obsessive–compulsive symptoms to drugs that affect 5-HT function does not prove that 5-HT function is abnormal in OCD.

Attempts to assess 5-HT function in OCD have used both neuroendocrine tests and measures of serotonin transporters and serotonin receptors using PET and single-photon emission tomography (SPET). The results of these studies have been inconsistent (Goodman et al., 2021).

Neuropsychological function

In psychological terms ‘*compulsivity*’ describes a tendency to perform repetitive acts in a habitual or stereotyped manner to prevent perceived negative consequences. Compulsivity has been contrasted with *impulsivity*, which is seen as a failure to inhibit inappropriate behaviours that are driven by reward and gratification. A tendency to compulsive responding can be detected by neuropsychological tasks that require an altered response once a ‘habit’ has been established by previous learning. Consistent with this, patients with OCD perform less well on tasks of ‘set-shifting’ and ‘reversal learning’ than controls, and a similar abnormality has been detected in first-degree relatives of OCD patients. This suggests that, in people with OCD and those at risk of OCD, actions may be mediated by a relative preponderance of *habit learning* over *goal-directed learning*. This might underpin the development of compulsive behaviours.

It is also worth noting that although *compulsivity* and *impulsivity* are often regarded as opposite behaviours, they can coexist. An anatomical basis for this is that, like compulsivity, impulsivity is thought to arise from dysfunction in *cortico–striatal–thalamic loops*. There is therefore significant scope for interaction between the circuits regulating impulsive and compulsive behaviour. Indeed, neuropsychological tasks suggest that patients with OCD and their first-degree relatives manifest significant difficulties in impulse control; for example, by failing to inhibit incorrect responses in tasks assessing motor performance. Thus part of the symptomatology of OCD may arise from failure to inhibit inappropriate responses as well as a compulsion to carry them out. For a review, see Goodman et al. (2021).

Cognitive theory

This theory starts with the premise that it is not the occurrence of intrusive thoughts that has to be explained (as these are experienced at times even by healthy people; see above), but rather the obsessional patient’s *inability to control them* and see them in perspective. For example, people with obsessional disorder often respond to such thoughts as if they were personally responsible for their possible consequences (e.g. for harm to another person). It is suggested that this feeling of responsibility leads to excessive attempts to ward off the thoughts and their supposed consequences by adopting compulsive

Box 8.9 Some key cognitive processes in obsessive–compulsive disorder

1. *Thought–action fusion*. Magical thinking—for example, the belief that if one *thinks* of harming others, one is likely to *act* on the thought or might have done so in the past.
2. *Responsibility*. An inflated sense of responsibility for preventing harm to others; the belief that one has power that is pivotal to bringing about or preventing crucial negative outcomes.
3. *Compulsions and safety-seeking behaviours*. Compulsions—whether behavioural or mental—are reinforcing because they reduce anxiety temporarily. They strengthen the belief that, had the compulsion not been carried out, discomfort would have increased, and harm might have occurred.
4. *Overestimation of the probability of a harmful outcome*.
5. *Difficulty managing uncertainty and ambiguity*.
6. *The need to feel in control*.

Source: Veale, D. (2007). Cognitive–behavioural therapy for obsessive–compulsive disorder. *Adv Psychiatr Treat*, 13(6), 438–446. Copyright (2007), Cambridge University Press.

behaviours and avoidance, and seeking repeated reassurance. Although the theory is unproven, it directs attention to aspects of the disorder other than the obvious obsessions and compulsions (see Box 8.9). For a review of the cognitive approach to OCD, see Veale (2007).

Prognosis

About two thirds of cases improve to some extent by the end of one year. Of the cases that last for more than a year, some run a fluctuating course, but others are chronic. The prognosis is better when there has been a precipitating event, social and occupational adjustment is good, and the symptoms are episodic. The prognosis is worse when there is a personality disorder, and onset is in childhood. Male gender, tic-related forms of OCD, and overvalued ideas about the obsessions also predict a poor prognosis (Carmi et al., 2020).

Marcks et al. (2011) studied over 100 patients with OCD, who were recruited to a research programme and followed up for 15 years. Most received treatment with medication and psychotherapy at some point in their course and all had at least one comorbid anxiety disorder.

Remission was defined as being essentially free of symptoms for a period of at least eight weeks. Remission rates in the first year of illness were low (16%), but gradually increased during follow-up to just over 40% at 15 years. The presence of comorbid major depression diminished the chance of recovery.

Treatment

Medication

Clomipramine. The use of clomipramine is described first because of its historical importance in the development of pharmacological treatment for OCD. However, it is not now used as a first-line agent. Clomipramine is a tricyclic antidepressant with potent 5-HT-uptake-blocking effects. Most patients tolerate the treatment, but typical tricyclic side effects are common, and at high doses a few patients develop seizures. A clinically useful effect may not be reached until after about six weeks of treatment, and further improvement may take another 6–12 weeks. Many patients relapse during the first few weeks after the drug is stopped, but the relapse rate is reduced if clomipramine is combined with exposure, and if drug treatment is maintained. Other tricyclic antidepressants that are less potent 5-HT-uptake blockers, for example, amitriptyline, do not possess the therapeutic effect of clomipramine in OCD (National Institute for Health and Clinical Excellence, 2005).

SSRIs. These are effective in reducing obsessional symptoms, and it seems unlikely that there are clinically important differences between them, although individual patients may respond better to one SSRI than another. There is some evidence that higher doses are somewhat more efficacious (Fineberg et al., 2021). Overall, SSRIs appear to have similar efficacy to clomipramine. However, clomipramine treatment is associated with more dropouts from treatment (National Institute for Health and Clinical Excellence, 2005). As with clomipramine, relapse is common during the first few weeks after an SSRI has been stopped, and longer-term maintenance treatment is advisable.

Since only about 50% of the treated patients improve substantially, attempts have been made to improve the response rate by adding a second drug to the SSRI. The only approach with a consistent, though rather limited, therapeutic effect involves the addition of an antipsychotic agent, usually at a low dose. Beneficial effects have been seen with both typical and atypical antipsychotic drugs, but the latter are usually better tolerated. A meta-analysis suggested that significant, albeit

modest, benefit was most likely with the addition of low doses of risperidone and aripiprazole (Veale et al., 2014).

Anxiolytic drugs. These drugs give some short-term symptomatic relief but should not be prescribed continuously for more than two to four weeks at a time.

Cognitive behaviour therapy

Exposure and response prevention. Obsessional rituals usually improve with a combination of response prevention and exposure to any environmental cues that increase the symptoms. About two thirds of patients with moderately severe rituals can be expected to improve substantially, although not completely. When rituals respond to this treatment, the accompanying obsessional thoughts usually improve as well. The results seem to be at least comparable with those of treatment with clomipramine and SSRIs (National Institute for Health and Clinical Excellence, 2005).

Behavioural treatment is less effective for obsessional thoughts that occur without rituals. The technique of thought-stopping has been used for many years, but there is no good evidence that it has a specific effect. Cognitive approaches to obsessions may well be more successful (see below).

Cognitive therapy. This therapy seeks to reduce the patient's attempts to suppress and avoid obsessional thoughts, as such attempts have been shown to increase, rather than decrease, the frequency of these thoughts. The patient is helped to record the frequency of obsessional thoughts to compare the effects of suppression and distraction. As suppression and avoidance appear to be driven by the conviction that to think something is to make it happen, attempts are made to weaken this conviction by reviewing the evidence for and against it. These techniques may be combined with exposure to audio-recorded repetition of the thoughts, and discussion of any other cognitive distortions along the general lines of cognitive therapy (see Chapter 24). Cognitive therapy by itself may not be as effective as behaviour therapy, but incorporating elements of cognitive approaches into behaviour therapy may enhance the effects of the latter treatment). Current treatments generally incorporate elements of both approaches (National Institute for Health and Clinical Excellence, 2005).

Dynamic psychotherapy

Exploratory and interpretative psychotherapy seldom helps obsessional patients. Indeed, some are made worse because these procedures encourage painful and unproductive rumination about the subjects discussed during treatment.

Neurosurgery and deep brain stimulation

The immediate result of neurosurgery for severe OCD is often a striking reduction in tension and distress. However, the long-term effects are uncertain, as no prospective controlled trial has been carried out. In a systematic review and meta-analysis, Lai et al. (2020) studied the effect of ablative surgeries in 459 patients with severe, treatment-resistant OCD. The overall clinical response rate was around 50% with best results being obtained with capsulotomy. The most common side effects were headache, cognitive deficits, and behavioural disturbances.

Deep brain stimulation (DBS) has also been employed to treat intractable OCD. A variety of stimulation sites have been identified, including ventral striatum, nucleus accumbens, and internal capsule. A meta-analysis of eight randomized trials with 85 treatment-refractory OCD patients found that, relative to sham stimulation, active stimulation produced significantly greater decreases in OCD symptomatology. There was no difference in efficacy between the various stimulation sites. Adverse events were relatively frequent and included hypomania, sleep disturbances, irritability, and apathy. There were also medical complications, including infection, headache, and weight change (Martinho et al., 2020).

For an account of the use of DBS, see Chapter 25.

Management

In treatment, it is important to remember that some cases of OCD run a fluctuating course, with long periods of remission. Also, depressive disorder frequently accompanies OCD, and in such cases effective treatment of the depressive disorder often leads to improvement in the obsessional symptoms. For this reason, a thorough search for depressive disorder should be made in every patient who presents with OCD.

What patients need to know

Treatment should begin with an explanation of the symptoms, and, if necessary, with reassurance that these

symptoms are not an early sign of insanity (a common concern of obsessional patients). If the patient agrees, their partner or another close relative can be involved in educational sessions about the nature and treatment of the disorder. Patients and family members are often helped by reading a book on OCD.

Obsessional patients may attempt to involve other family members in their rituals. If this happens, the relatives should be helped to find ways of resisting such requests in a way that is firm and appropriately sympathetic. In particular, they need to understand that repeated reassurances about obsessional ruminations do not help.

Choice of treatment

As noted above, medication controls the symptoms of OCD in many cases, but when it is stopped many patients relapse. Exposure with response prevention seems to produce better long-term results, but it is difficult to achieve response prevention when symptoms are severe, and a significant number of patients are unable to tolerate the technique. Guidance from the National Institute for Health and Clinical Excellence (2005) advises that patients with mildly impairing OCD symptoms should receive first-line treatment with CBT (including exposure and response prevention). For patients with greater degrees of impairment it may be more helpful to combine medication and response prevention (Skapinakis et al., 2016).

First-line medication should generally be an SSRI, and, if this is not effective, a switch to a second SSRI may be considered. If this is unsuccessful, a trial of clomipramine or augmentation with an antipsychotic drug may be considered. CBT should be tried if not used previously. Because patients with OCD may fare better on higher doses of antidepressant medication, it is worth exploring the full recommended dose range if tolerance permits this. If patients respond to medication, long-term maintenance treatment may be needed.

For an account of the management of OCD, see Fineberg et al. (2021).

Obsessive–compulsive-related disorders

We have already seen that in both DSM-5 and ICD-11 a number of disorders are grouped with OCD, including *body dysmorphic disorder*, *hoarding disorder*,

trichotillomania, and *excoriation (skin picking)*. These conditions are listed with OCD principally because all display prominent compulsive features. ICD-11 also

includes hypochondriasis (health anxiety disorder), and olfactory reference disorder. Here we briefly outline the clinical features, diagnosis, epidemiology, and management of hoarding disorder, trichotillomania, and skin picking. Body dysmorphic disorder and hypochondriasis are discussed in Chapter 22.

Hoarding disorder

Clinical features

Historically, *pathological hoarding* has been recognized for well over a century, when it became associated with the psychoanalytical concept of ‘anal personality’. Subsequently, hoarding behaviour was seen as characteristic of *obsessive–compulsive personality disorder* and, more recently, in DSM-IV, as a symptom of *OCD*. However, in DSM-5, hoarding is now recognized as a disorder in its own right.

Hoarding is a persistent phenomenon typically arising in a middle-aged or elderly person and is driven by emotional difficulties in parting with or discarding possessions, no matter what their actual value. While normal individuals may collect items in an assiduous way, this collection is systematic and purposeful and does not interfere with the conduct of the individual’s life. In hoarding disorder the clutter of material fills up and congests living areas to the extent that their intended use, for example, as a kitchen or bedroom, becomes impossible. The most commonly hoarded items are newspapers, magazines, clothing, and paperwork. Some patients also hoard animals in insanitary and unsuitable ways. A form of hoarding in older people that is associated with extreme self-neglect and domestic squalor has been called *Diogenes syndrome* or *senile squalor syndrome* (see page 540). It is important to note that the latter may be associated with dementia.

Despite the interference of the hoarding behaviour with everyday life and the attendant distress this can cause, affected individuals do not resist the urge to hoard, but feel anxious, upset, or angry if an attempt is made to reduce the accumulated clutter. Unlike *OCD*, thoughts about hoarding are not experienced as intrusive or repetitive or distressing, and the hoarding behaviour is not seen as senseless or inappropriate. Hoarding remains one of the criteria for *obsessive–compulsive personality disorder* in DSM-5; however, hoarding has a low correlation with the other symptoms of obsessive–compulsive personality, and many patients who hoard do not have other features of personality disorder.

Patients with hoarding behaviour have high rates of psychiatric comorbidity, particularly mood and anxiety

disorders. There is also significant co-morbidity with *OCD* (Pertusa et al., 2019).

Diagnosis

In both ICD-11 and DSM-5 the diagnosis of ‘hoarding disorder’ requires that patients show a longstanding problem with getting rid of worthless items and marked distress at the prospect of parting with them. The result is an accumulation of clutter that takes over living areas and makes them impossible to use for their intended purposes, and which eventually can render the home unhygienic and unsafe. When a person meets the diagnosis for both hoarding disorder and obsessive–compulsive personality, both diagnoses should be recorded. Other conditions, for example, organic brain disorders and psychosis, can also lead to cluttered living spaces and should be ruled out before a diagnosis of hoarding disorder is made.

Epidemiology

A moderate degree of hoarding appears common in the general population, so estimates of the population frequency of pathological hoarding vary according to the diagnostic criteria used. A point prevalence between 2% and 6% is probable, with an increasing incidence with age. Rates of hoarding disorder appear equal in men and women (de la Cruz and Mataix-Cols, 2020).

Aetiology

There is a familial disposition to hoarding behaviour and a small twin study estimated a heritability of around 50%. Stressful life events may trigger the onset of hoarding. Neuropsychological studies show evidence of impaired memory, attention, and executive functioning, though the presence of frequent psychiatric comorbidity makes interpretation of these findings difficult (Pertusa et al., 2019). Studies of patients with organic brain disease who show hoarding has implicated damage in cortical regulatory regions including ventromedial prefrontal cortex and anterior cingulate cortex. Similar brain regions have been implicated in neuroimaging studies in patients with a primary hoarding condition (de la Cruz and Mataix-Cols, 2020).

Management

Management of patients with hoarding disorder is usually difficult because, in the view of most patients, their hoarding is a not a problem. However, improvements in hoarding behaviour compared to waitlist control have been shown with cognitive behaviour therapies modified for the treatment of hoarding disorder, given in either

individual or group format. There have also been investigations of pharmacotherapy in pathological hoarding, with SSRIs and venlafaxine showing benefit. However, the published studies were open label, and randomized controlled trials are needed (Pertusa et al., 2019).

Trichotillomania

Clinical features

The key feature of trichotillomania is recurrent hair pulling, which results in hair loss. The hair loss is usually visible but in some patients is less apparent because single hairs are pulled from widely distributed sites, or because make-up or a wig is employed. Trichotillomania can involve any site on the body where hair grows, but most often affects the scalp, eyebrows, or eyelashes. Hair pulling can occur in brief episodes throughout the day or in more sustained periods of several hours. Prior to hair pulling, often during a time when the impulse to pull is being resisted, there are increasing feelings of tension, while the act of hair pulling itself is associated with a sense of relief. Extracted hair can be chewed and swallowed, thereby causing hair balls in the stomach (trichobezoars) with a risk of bowel obstruction.

Trichotillomania can have an onset in childhood, where there is often less evidence of tension prior to hair pulling. However, it can also begin in adolescence and adulthood. The disorder is more common in women, and the course may be episodic or chronic. In clinical samples, trichotillomania is often comorbid with mood and anxiety disorders, as well as impulse control disorders such as substance misuse. Individuals with trichotillomania frequently exhibit other repetitive behaviours involving the body, such as lip chewing or skin picking.

Diagnosis

The DSM-5 diagnosis for trichotillomania requires that the patient repeatedly pulls their hair, despite repeated attempts to stop doing so, and that this hair loss causes distress and/or significant impairment in occupational or social functioning. It is also necessary that the hair loss not caused by another medical condition or mental disorder. Diagnosis in ICD-11 is similar, and requires that the hair pulling be ‘recurrent and habitual’.

Epidemiology

The prevalence of trichotillomania defined by current diagnostic criteria remains to be clarified, but is around 1–2%. Co-morbidity with OCD has been reported to be around 12%. Family studies indicate that trichotillomania is associated with increased rates

of obsessive–compulsive and grooming disorders (trichotillomania, nail biting, and skin picking) among first-degree relatives. Conversely, first-degree relatives of patients with OCD have higher rates of grooming disorders, OCD, and trichotillomania (Carmi et al., 2020).

Management

Psychological therapies are currently the best validated management for trichotillomania. There are positive studies with behavioural treatment employing habit reversal, and in one study CBT proved more effective than clomipramine and placebo. However, when behaviour therapy has been compared to a psychological control procedure, such as progressive muscular relaxation, it has not been clearly superior.

In a meta-analysis of 24 randomized trials of 857 participants with trichotillomania, Farhat et al. (2020) found that the most effective treatment was behaviour therapy with habit reversal training. SSRIs were not efficacious but there is some evidence of benefit from clomipramine, N-acetylcysteine, and olanzapine.

Excoriation (skin picking) disorder

Clinical features

The key feature is recurrent, compulsive picking of normal or abnormal skin, the latter often involving previously damaged areas. The most common sites affected are face, arms, and hands but some patients may pick from multiple body sites. In most cases skin picking is carried out with fingernails, although instruments such as tweezers and pins may also be used. Skin lesions may be readily apparent or concealed by make-up. Skin picking is often triggered by an unpleasant emotional state, such as anxiety or boredom, which is relieved to some extent by the skin picking behaviour. On other occasions, however, skin picking occurs at a more automatic level with limited conscious awareness.

Skin picking often occupies a significant amount of time causing missed appointments and limiting opportunities for occupational and social activities. Damage to the skin can be associated with serious medical complications, including localized and systemic infections and the need for skin grafting.

Skin picking may present at various ages but most often has an onset during adolescence. It is sometimes triggered by the development of acne or some other skin condition. The disorder is more common in women, and the course is usually chronic with some waxing and waning. Psychiatric comorbidity, particularly with

mood and anxiety disorders, is common (Jafferany and Patel, 2019).

Diagnosis

DSM-5 requires that there is recurrent skin picking that results in lesions and that there have been repeated attempts to decrease or stop the behaviour. The skin picking should be associated with significant distress or impairment in social or occupational function. There are the usual stipulations that the condition should not be the result of a substance or medical illness and that it is not explained by another psychiatric condition. ICD-11 describes skin picking as one of a number of OCD-related disorders ‘which are primarily characterized by recurrent and habitual actions directed at the integument’. ICD-11 notes that such conditions often lack a prominent cognitive component, such as the obsessions that characterize OCD. The diagnostic criteria for excoriation disorder in ICD-11 are similar to those of DSM-5.

Epidemiology

Estimates for the prevalence of skin picking vary but have been estimated between 1.4–5.4%. There is significant comorbidity with OCD (about 16%) (Carmi et al., 2020). There are also significant associations between skin picking, trichotillomania, and body dysmorphic disorder. Some patients have co-existing skin picking and body dysmorphic disorder. In these circumstances distinguishing which is the primary condition can be challenging.

Management

The best-established treatment currently is CBT with habit reversal training. The components have been described in Jafferany and Patel (2019). There is currently little high-quality evidence for the efficacy of pharmacotherapy, though in a randomized trial, N-acetylcysteine produced significant therapeutic benefit compared to placebo (Grant et al., 2016).

Further reading

Baldwin, D. S., Anderson, I. M., Nutt, D. J., et al. (2014). Evidence-based pharmacological treatment of anxiety disorders, post-traumatic stress disorder and obsessive-compulsive disorder: A revision of the 2005 guidelines from the British Association for Psychopharmacology. *J Psychopharmacol*, 28(5), 403–439.

Fineberg, N., & Robbins, T. W. (2021). *The Neurobiology and Treatment of OCD: Accelerating Progress*. Springer.

National Institute for Health and Care Excellence (NICE). (2011b). Generalised anxiety disorder and panic disorder in adults: Management. Clinical guideline CG113. <https://www.nice.org.uk/guidance/cg113>

National Institute for Health and Clinical Excellence. (2005). Obsessive-compulsive disorder and body dysmorphic disorder: Treatment. Clinical guideline CG31. <https://www.nice.org.uk/guidance/cg31>

CHAPTER 9

Depression

Introduction 193

Clinical features 193

Transcultural factors 199

Classification of depression 199

Differential diagnosis of depressive disorders 203

The epidemiology of depressive disorders 203

The aetiology of depression 204

Course and prognosis of depression 215

The acute treatment of depression 216

The longer-term treatment of depression 221

The assessment of depressive disorders 222

The management of depressive disorders 223

Introduction

The next two chapters describe depressive disorders and bipolar disorder. These conditions are also called *mood disorders* because one of their main features is abnormality of mood. Nowadays the term is usually restricted to disorders in which this mood is depression or elation, but in the past some authors have included states of anxiety as well. In this book, anxiety disorders are described in Chapter 8. Mood disorders have in the past been referred to as ‘affective disorders’, a term that is still used fairly widely.

Depressive disorders

It is part of normal experience to feel unhappy during times of adversity. The symptom of depressed mood is a component of many psychiatric syndromes, and is also

commonly found in certain physical diseases (e.g. in infections such as viral hepatitis, and some neurological disorders). In this chapter we are concerned neither with normal feelings of unhappiness nor with depressed mood as a symptom of other disorders, but with the syndromes known as *depressive disorders*.

The central features of these syndromes are:

- depressed mood
- negative thinking
- lack of enjoyment
- reduced energy
- slowness.

Of these, depressed mood is usually, but not invariably, the most prominent symptom.

Clinical features

Depressive syndromes

The clinical presentations of depressive states are varied, and they can be subdivided in a number of different

ways. In the following account, disorders are grouped by their *severity*. The account begins with a description of the clinical features of an episode of severe depression, together with certain important clinical variants that

can influence the presentation of depressive disorders. Finally, the special features of the less severe depressive disorders are outlined.

What constitutes an ‘episode’ of clinical depression is inevitably a somewhat arbitrary concept. The symptoms listed for the diagnosis of ‘depressive episode’ in the ICD-11 classification are shown in Box 9.1. Both ICD-11 and DSM-5 require the presence of either persistently depressed mood or loss of interest or pleasure in usual activities. Many patients will have both symptoms. ICD-11 and DSM-5 have a similar list of associated depressive symptoms shown in Box 9.1. In both DSM-5 and ICD-11 it is stipulated that a total five symptoms or more are required for the diagnosis of a ‘depressive episode’. ICD-11 and DSM-5 also require that the symptoms cause

clinically significant distress or impairment in social, occupational, or other important areas of functioning. ICD-11 adds that if functioning is maintained, ‘it is only through significant additional effort’. ICD-11 and DSM-5 classify depressive episodes as ‘mild’, ‘moderate’, or ‘severe’ depending on the number of symptoms experienced, their severity, and their impact on functioning.

Severe depressive episode

In a severe episode of depression, the central features are *low mood*, *lack of enjoyment (anhedonia)*, *negative thinking*, and *reduced energy*, all of which lead to decreased social and occupational functioning.

Appearance

The patient’s appearance is characteristic. Dress and grooming may be neglected. The facial features are characterized by a turning downwards of the corners of the mouth, and by vertical furrowing of the centre of the brow. The rate of blinking may be reduced. The shoulders are bent, and the head is inclined forwards so that the direction of gaze is downwards. Gestures and movements are reduced. It is important to note that some patients maintain a smiling exterior despite deep feelings of depression.

Mood

The mood of the patient is one of *misery*. This mood does not improve substantially in circumstances where ordinary feelings of sadness would be alleviated—for example, in pleasant company or after hearing good news. The low mood is, in this sense, *pervasive*. Moreover, the mood is often experienced as different from ordinary sadness. Patients sometimes speak of a black cloud pervading all mental activities. Some patients can conceal this mood change from other people, at least for short periods. Some try to hide their low mood during clinical interviews, which makes it more difficult for the clinician to detect. The mood is often worse first thing in the morning when the patient wakes, improving as the day wears on. This is called *diurnal variation of mood*.

Loss of interest and pleasure

Lack of interest and enjoyment, or *anhedonia*, is frequent, although spontaneous complaints may not occur. Patients show no enthusiasm for activities and hobbies that they would normally enjoy. They feel no zest for living and no pleasure in everyday things. They often withdraw from social encounters. *Anhedonia* is an important symptom because it is an important way of distinguishing depression of at least moderate severity from milder disorders. It is also a key symptom of *melancholic* depression (see ‘Classification by symptomatic picture’ below).

Box 9.1 Symptoms needed to meet the criteria for ‘depressive episode’ in ICD-11

Concurrent presence of at least five of the following characteristic symptoms occurring most of the day, nearly every day, for a period lasting at least two weeks. At least one symptom must be from the affective cluster.

Affective cluster

- Depressed mood reported or observed (in children or adolescents depressed mood may manifest as irritability).
- Markedly diminished interest or pleasure in usual activities.

Cognitive-behavioural cluster

- Difficulty concentrating
- Feelings of worthlessness or inappropriate guilt
- Hopelessness about the future
- Recurrent thoughts of death or suicide.

Neurovegetative cluster

- Reduced energy or fatigue
- Psychomotor agitation or retardation
- Changes in sleep
- Changes in appetite.

Note: The symptoms are not better accounted for by bereavement, and are not due to a general medical condition, medication, or a substance.

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>

Depressive cognitions (cognitive-behavioural symptoms)

Negative thoughts ('depressive cognitions') are important symptoms that can be divided into four groups:

1. worthlessness
2. pessimism
3. guilt
4. suicidal thinking.

In feeling *worthless*, patients think that they are failing in what they do and that other people see them as a failure; they no longer feel confident, and discount any success as a chance happening for which they can take no credit. *Pessimistic thoughts* concern future prospects. Patients expect the worst. They foresee failure in work, the ruin of finances, misfortune for family, and an inevitable deterioration in health. These ideas of *hopelessness* are often accompanied by the thought that life is no longer worth living and that death would come as a welcome release. These gloomy preoccupations may progress to thoughts of, and plans for, *suicide*. It is important to ask about these ideas in every case (the assessment of suicidal risk is considered further in Chapter 21).

Feelings of *guilt* often take the form of unreasonable self-blame about minor matters—for example, patients may feel guilty about past trivial acts of dishonesty or letting someone down. Usually, these events have not been in patients' thoughts for years, but when depressed, these thoughts flood back into the patient's memory, accompanied by intense feelings. Preoccupations of this kind strongly suggest depressive disorder. Some patients have similar feelings of guilt but do not attach them to any particular event. Other memories are focused on unhappy events; patients remember occasions when they were sad, when they failed, or when their fortunes were at a low ebb. These gloomy memories become increasingly frequent as the depression deepens. Patients blame themselves for their misery and incapacity, and attribute it to personal failing and moral weakness.

Complaints of *poor memory and concentration* are also common; depressed patients commonly show deficits on a wide range of neuropsychological tasks, but impairments in the retrieval and recognition of recently learned material are particularly prominent. Sometimes the impairment of memory in a depressed patient is so severe that the clinical presentation resembles that of dementia. This presentation, which is particularly common in the elderly, is sometimes called depressive pseudodementia (Chapter 19).

Neurovegetative symptoms

Psychomotor retardation is frequent in severe depression (although, as described later, some patients are agitated rather than slowed). The retarded patient walks and acts slowly. Slowing of thought is reflected in their speech; there is a significant delay before questions are answered, and pauses in conversation may be unusually prolonged. *Agitation* is a state of restlessness that is experienced by the patient as inability to relax, and is seen by an observer as restless activity. When it is mild, patients are seen to be plucking at their fingers and making restless movements of their legs; when it is severe, they cannot sit for long, and instead pace up and down.

There is an important group of symptoms that is often described as 'biological', 'somatic', or 'vegetative'. These symptoms include loss of energy, sleep disturbance, diurnal variation in mood, loss of appetite, loss of weight, constipation, loss of libido, and, among women, amenorrhoea. They are very common in depressive disorders of more severe degree (but less usual in mild depressive disorders). Some of these symptoms require further comment

Reduced energy is characteristic (although depression is sometimes associated with a degree of physical restlessness that can mislead the observer). Patients feel lethargic, find everything an effort, and leave tasks unfinished. Understandably, many patients attribute this lack of energy to physical illness.

Sleep disturbance in depressive disorders is of several kinds. The most characteristic type is early morning waking, but delay in falling asleep and waking during the night also occur. Early morning waking occurs two or three hours before the patient's usual time of waking. They may not fall asleep again, but lie awake feeling unrefreshed and often restless and agitated. They think about the coming day with pessimism, brood about past failures, and ponder gloomily about the future. It is this combination of *early waking and depressive thinking* that is important in diagnosis. It should be noted that some depressed patients sleep excessively rather than waking early, but they still report waking unrefreshed.

Weight loss in depressive disorders often seems to be greater than can be accounted for merely by the patient's reported lack of appetite. However, in some patients the disturbances of eating and weight are towards excess—that is, they eat more than usual and gain weight. Usually eating brings temporary relief of their distressing feelings.

Complaints about *physical symptoms* are common in depressive disorders. They take many forms, but complaints of constipation, fatigue, and aching

discomfort anywhere in the body are particularly common. Complaints about any pre-existing physical disorder usually increase, and *hypochondriacal preoccupations* are common.

Other features

Several other psychiatric symptoms may occur as part of a depressive disorder, and occasionally one of them dominates the clinical picture. They include *depersonalization*, *obsessional symptoms*, *panic attacks*, and, rarely, *dissociative symptoms* such as fugue or loss of function of a limb.

Anxiety is frequent, although not invariably present, in severe depression. (As described later, it is common in less severe depressive disorders.) Another common symptom is *irritability*, which is the tendency to respond with undue annoyance to minor demands and frustrations, and can be a core presenting feature in adolescents in particular.

Psychotic depression

As depressive disorders become increasingly severe, all of the features described above occur with greater intensity. There is complete loss of function in social and occupational spheres. Inattention to basic hygiene and nutrition may give rise to concern about the patient's wellbeing. Psychomotor retardation may make interviewing difficult or impossible. In addition, there may be delusions and hallucinations, in which case the disorder is referred to as *psychotic depression* (depressive psychosis is a synonym). As with other psychotic states, insight is impaired and patients (usually of blameless character) regard themselves as wicked and being justly punished for their misdeeds.

The delusions of severe depressive disorders are concerned with the same themes as the non-delusional thinking of moderate depressive disorders. Therefore they are termed *mood-congruent*. These themes are worthlessness, guilt, ill health, and, more rarely, poverty. Such delusions have been described in Chapter 1, but a few examples may be helpful at this point. Patients with a delusion of guilt may believe that some dishonest act, such as a minor concealment when filling in a tax return, will be discovered and that they will be punished severely and humiliated. They are likely to believe that such punishment is deserved. Patients with hypochondriacal delusions may be convinced that they have cancer or venereal disease, while patients with a delusion of impoverishment may wrongly believe that they have lost all their money in a business venture.

Persecutory delusions also occur. Patients may believe that other people are discussing them in a

derogatory way or are about to take revenge on them. When persecutory delusions are part of a depressive syndrome, typically patients accept the supposed persecution as something that they have brought upon themselves. In their view, they are ultimately to blame. This can be a useful diagnostic feature for distinguishing severe depression from non-affective psychosis (see Chapter 11). Some depressed patients experience delusions and hallucinations that are not clearly related to themes of depression (i.e. they are 'mood-incongruent'). Their presence appears to worsen the prognosis of the illness.

Particularly severe depressive delusions are found in *Cotard's syndrome*, which was described by the French psychiatrist Jules Cotard in 1882. The characteristic feature is an extreme kind of nihilistic delusion. For example, some patients may complain that their bowels have been destroyed, so they will never be able to pass faeces again. Others may assert that they are penniless and have no prospect of ever having any money again. Still others may be convinced that their whole family has ceased to exist and that they themselves are dead. Although the extreme nature of these symptoms is striking, such cases do not appear to differ in important ways from other severe depressive disorders.

Other clinical variants of depressive disorders

Agitated depression

This term is applied to depressive disorders in which *agitation* is prominent. As already noted, agitation occurs in many severe depressive disorders, but in agitated depression it is particularly severe. Agitated depression is seen more commonly among middle-aged and elderly patients than among younger individuals.

Retarded depression

This name is sometimes applied to depressive disorders in which *psychomotor retardation* is especially prominent. There is no evidence that they represent a separate syndrome, although the presence of prominent retardation is said to predict a good response to electroconvulsive therapy (ECT). If the term is used, it should be in a purely descriptive sense. In its most severe form, retarded depression merges with *depressive stupor*.

Depressive stupor

In severe depressive disorder, slowing of movement and poverty of speech may become so extreme that the patient is motionless and mute. Such depressive stupor is rarely seen now that active treatment is available. Therefore the description by Kraepelin (1921, p. 80) is of particular interest: 'The patients lie mute in bed, give no

answer of any sort, at most withdraw themselves timidly from approaches, but often do not defend themselves from pinprick. [...] They sit helpless before their food; perhaps, however, they let themselves be spoon-fed without making any difficulty.'

Kraepelin commented that recall of the events that took place during stupor was sometimes impaired when the patient recovered. Nowadays, the general view is that, on recovery, patients are able to recall nearly all of the events that took place during the period of stupor. It is possible that in some of Kraepelin's cases there was clouding of consciousness (possibly related to inadequate fluid intake, which is common in these patients). Patients in states of depressive stupor may exhibit catatonic motor disturbances (see Chapter 1).

Atypical depression

The term *atypical depression* is generally applied to disorders of moderate clinical severity. The precise meaning of the term has varied over the years, but currently it is applied to disorders characterized by:

- variably depressed mood with mood reactivity to positive events
- overeating
- oversleeping
- extreme fatigue *and heaviness in the limbs* (lead paralysis)
- rejection sensitivity.

Rejection sensitivity refers to a lifelong tendency to react in an exaggerated way to perceived or real rejection, although this character trait can be exacerbated by the presence of a depressive disorder. Patients with atypical depression also have an earlier onset of illness and a more chronic course. The importance of recognizing atypical depression is that, because of their interpersonal sensitivity, patients with this disorder can be hard to manage and may be regarded as having 'difficult' personalities rather than depressive disorder.

In DSM-5 depression with 'atypical features' can be used as an illness specifier. For criteria for this specifier to be met, DSM-5 requires the presence of mood reactivity, as well as at least two of the other 'atypical' symptoms listed above. Traditionally, atypical depression was associated with a poor response to tricyclic antidepressant (TCA) treatment but had a better outcome with monoamine oxidase inhibitors (MAOIs). However, there is little evidence that the diagnosis of atypical depression predicts response to modern antidepressant drug treatment such as selective serotonin reuptake inhibitors (SSRIs). There is, however, evidence that atypical

depression may be particularly associated with immunometabolic dysregulation (Lamers et al., 2018).

Mixed depression

Mixed affective states have long been recognized in bipolar patients (see Chapter 10), where depressive symptoms are not infrequently detected in patients in whom the main presentation is mania. However, it is possible for patients with major depression to exhibit symptoms that might be seen in mania but do not reach the threshold for diagnosis of bipolar disorder. The most common symptoms in mixed depression are irritable mood, mood lability, racing thoughts, distractibility, agitation, and impulsivity. These symptoms qualify for a diagnosis of mixed episode in ICD-11.

DSM-5 has a specifier for major depressive episode 'with mixed features' with rather different symptomatic criteria than those above. The DSM-5 specifier includes more typical manic symptoms, such as elevated and expansive mood, inflated self-esteem, pressure of speech, and decreased need for sleep. The aim of this categorization was to avoid symptoms such as irritability that can occur in both depression and mania; however, this scheme may thereby exclude patients who previously would have been regarded as having mixed affective features and its clinical utility has been questioned (Tundo et al., 2020).

Not surprisingly, mixed symptoms, however defined, are more common in depressed patients with a family history of bipolar disorder, and depressed patients with mixed features are more likely themselves to develop bipolar disorder in the future.

Milder depressive states

It might be expected that *mild depressive disorders* would present with symptoms similar to those of the depressive disorders described above, but with less intensity. To some extent this is the case, but in mild depressive disorders there are frequently additional symptoms that are less prominent in severe disorders. These symptoms have been characterized in the past as 'neurotic', and they include *anxiety, phobias, obsessional symptoms*, and, less often, *dissociative symptoms*. As noted above, in terms of classification, both DSM-5 and ICD-11 have categories of *mild depression* where criteria for a depressive episode are met but the depressive symptoms are fewer and less severe.

Apart from the 'neurotic' symptoms that are found in some cases, mild depressive disorders are characterized by the expected symptoms of *low mood, lack of energy and interest*, and *irritability*. There is *sleep disturbance*, but not the early morning waking that is so characteristic

of more severe depressive disorders. Instead, there is more often difficulty in falling asleep, and periods of waking during the night, usually followed by a period of sleep at the end of the night. 'Biological' features (poor appetite, weight loss, and low libido) are not usually found. Although mood may vary during the day, it is usually worse in the evening than in the morning. The patient may not appear obviously dejected, or slowed in their movement. Delusions and hallucinations are not present.

In their mildest forms, these cases merge into the *minor mood disorders* considered below. As described later, they pose considerable problems of classification. Many of these mild depressive disorders are brief, starting at a time of personal difficulty and subsiding when fortunes have changed or a new level of adjustment has been achieved. These may be diagnosed as adjustment disorders (see Chapter 7). However, some cases persist for months or years, causing considerable suffering, even though the symptoms do not increase. These chronic depressive states have been termed *dysthymia*, which is characterized in ICD-11 as the persistence over at least two years of depressive symptoms that are not severe enough to meet criteria for a depressive episode. However, it is not uncommon in such patients for periods of more severe depression to supervene, in which case the diagnosis of depressive episode is made.

DSM-5 has a similar category, called 'persistent depressive disorder (dysthymia)', which requires fewer symptoms than those needed to diagnose 'major depressive disorder' but also stipulates that the symptoms must have persisted for at least two years. In DSM-5, this category also includes patients who meet full criteria for 'major depression' that has persisted for more than two years.

Minor affective disorders (mixed depressive–anxiety disorder).

We have already seen that anxiety and depressive symptoms often occur together. Indeed, earlier writers considered that anxiety and depressive disorders could not be separated clearly even in patients who had been admitted to hospital with severe disorders. Although most psychiatrists now accept that the distinction can usually be made among the more severe forms that present in psychiatric practice, the distinction is less easy to make in the milder forms that present in primary care.

Classification

As psychiatrists have worked increasingly with general practitioners (GPs), the importance of minor

anxiety–depressive disorders has been recognized, but without any agreement about classification.

ICD-11 includes a category of 'mixed depressive and anxiety disorder', which can be applied when neither anxiety symptoms nor depressive symptoms are severe enough to meet the criteria for an anxiety disorder or a depressive disorder, and when the symptoms do not have the close association with stressful events or significant life changes that is required for a diagnosis of acute stress reaction or adjustment disorder.

According to ICD-11, patients with this presentation are seen frequently in primary care, and there are many others in the general population who are not seen by doctors. In ICD-11 this diagnosis appears among the mood disorders, a view that is reflected in the alternative term *minor affective disorder*.

In DSM-5, no comparable diagnosis appears in the classification, although the category 'unspecified depressive disorder' addresses depressive symptoms that cause distress or impairment in social and occupational functioning without meeting criteria for any specific depressive disorder. Although little is known about these conditions or about their relationship to other disorders, patients commonly present to primary care doctors with this group of symptoms. A suitable category is needed even if it is not possible to write strict criteria for diagnosis.

Clinical picture

One of the best descriptions of minor affective disorder is that given by Goldberg et al. (1976), who studied 88 patients from a general practice in Philadelphia. The most frequent symptoms were:

- fatigue
- anxiety
- depression
- irritability
- poor concentration
- insomnia
- somatic symptoms and bodily preoccupation.

A similar range of symptoms was found in the Adult Psychiatric Morbidity in England survey (McManus et al., 2016). Mixed depression and anxiety was the most common mental health problem in the community with a one-week prevalence of around 8%.

Patients with minor anxiety–depressive disorders commonly present to medical practitioners with *prominent somatic symptoms*. The reason for this is uncertain; some symptoms are autonomic features of anxiety, and it is possible that patients expect somatic complaints to be

viewed more sympathetically than emotional problems. Another point of clinical relevance is that minor affective disorders can be prolonged and, in some cases, may cause quite disabling difficulties in personal and occupational

function. Thus the term ‘minor’ may not capture the serious consequences of the disorder for an individual. In some patients minor affective disorders may represent residual symptoms of a major mood disturbance

Transcultural factors

There are cultural variations in the clinical presentation of depressive states, but in most countries, depression appears to be underdiagnosed, particularly in primary care. In fact, sadness, joylessness, anxiety, and lack of energy are common symptoms of depression in most cultures. While somatic presentations of depression are undoubtedly found in all societies, they are apparently more frequent and prominent in non-western cultures, including in some immigrant groups in western societies. An important factor leading to somatization of depression appears to be the extent to which different cultures stigmatize mental illness (Bagayogo et al., 2013). In some cultures, for example, the expression of

suicidal thinking and behaviour may be strongly prohibited so such phenomena may not be readily expressed.

It is necessary, however, to distinguish between somatization (see Chapter 22) and somatic metaphors for a painful emotional state. For example, Punjabi women living in London have been found to use expressions such as ‘weight on my heart’ and ‘feelings of heat’ to express emotional suffering, the presence of which they were well aware. Particular cultures have their own ways of dealing with painful emotions produced, for example, by loss, and the diagnosis of depression should take this into account by placing suitable emphasis on issues such as degree of distress and level of disability.

Classification of depression

There is no general agreement about the best method of classifying depressive disorders. A number of approaches have been tried, based on the following:

- presumed aetiology
- symptomatic picture
- course.

Classification by presumed aetiology

Historically, depressive disorders were sometimes classified into two kinds—those in which the symptoms were caused by factors within the individual, and were independent of outside factors (*endogenous depression*), and those in which the symptoms were a response to external stressors (*reactive depression*). However, it has been recognized for many years that this distinction is unsatisfactory. For example, Lewis (1934) wrote that ‘... every illness is a product of two factors—of the environment working on the organism—whether the constitutional factor is the determining influence or the environmental one ... is never a question to be dealt with as either/or’.

As noted in Chapter 5, when considering the aetiology of individual cases of depression, the relative contributions of a variety of aetiological factors must be considered. Neither ICD-11 nor DSM-5 contains categories of reactive or endogenous depression.

Classification by symptomatic picture

Melancholic depression

It is well recognized that episodes of depression vary in symptomatic profile both within and between subjects. In the section on clinical description it was noted that some depressive conditions are characterized by prominent ‘biological’ symptoms, such as loss of appetite, psychomotor changes, weight loss, constipation, reduced libido, amenorrhoea, and early morning waking. These symptoms have sometimes been termed *melancholic*, and they have been used to delineate a specific subgroup of depressive disorders, namely major depression with *melancholia* in DSM-5, or depressive episode with *melancholia* in ICD-11 (see Box 9.2). The difficulty with this classification is that most patients have melancholic

Box 9.2 Clinical features of depression with 'melancholic' features

- Loss of interest or pleasure in usual activities
- Lack of emotional reactivity to normally pleasurable surroundings and events
- Early morning waking (two hours or more before usual time)
- Depression worse in the morning
- Marked psychomotor agitation or retardation
- Significant anorexia or weight loss
- Distinct quality of depressed mood (DSM-5 only)
- Excessive guilt (DSM-5 only).

Note: In DSM-5, at least four of these symptoms are required to make a diagnosis of depression 'with melancholic features'. DSM-5 also specifically requires either 'loss of interest' or 'lack of emotional reactivity' to be present. ICD-11 requires that 'several' of the above symptoms have been present during the worst period of the present episode.

symptoms of some kind, although a careful search may be required to reveal them. Therefore, the number of symptoms that are needed to fulfil the criterion for depression with melancholia is somewhat arbitrary.

Melancholia does seem to identify a more severe form of depression as well as patients with relatively high rates of psychomotor retardation and lifetime suicidal ideation. Otherwise, the clinical differences between melancholic and non-melancholic depressed patients seem slight, though patients with melancholia may respond better to TCAs than to SSRIs (see Tondo et al., 2020).

Psychotic depression

As noted above, severe depression can also manifest with *psychotic features* (although in depressive psychosis the features of melancholia are almost invariably present as well). The presence of psychotic features indicates that treatment with antidepressant medication alone is unlikely to be successful, and that combination with antipsychotic drugs is usually needed (Kruizinga et al., 2021).

Non-melancholic depression

In this classification by symptom profile, the remaining forms of major depression ('*non-melancholic*' depression) include several different kinds of clinical disorder—for example, milder depressive episodes and atypical depression. These depressions are more likely to have a relative prominence of features, such as anxiety, hostility,

phobias, and obsessional symptoms. In the past, because of these symptoms, non-melancholic depressions were sometimes called 'neurotic depression', but this term does not appear in current diagnostic classifications. As noted above, *atypical depression* has particular clinical characteristics and in DSM-5 atypical depression, like melancholic depression, can be a specifier for a major depressive episode.

Classification by course

Unipolar and bipolar disorders

Mood disorders are characteristically recurrent, and Kraepelin was guided by the course of illness when he brought mania and depression together as a single entity. He found that the course was essentially the same whether the original disorder was manic or depressive, and so he put the two together in a single category of *manic-depressive psychosis*.

This view was widely accepted until 1962, when Leonhard and colleagues suggested a division into three groups:

1. Patients who had had a depressive disorder only (*unipolar depression*).
2. Those who had had mania only (*unipolar mania*).
3. Those who had had both depressive disorder and mania (*bipolar*).

Nowadays, it is the usual practice not to use the term 'unipolar mania', but to include all cases of mania in the bipolar group on the grounds that nearly all patients who have mania eventually experience a depressive disorder (see Chapter 10).

Seasonal affective disorder

Some patients repeatedly develop a depressive disorder at the same time of year, usually the autumn or winter. In some cases the timing reflects extra demands placed on the person at a particular season of the year, either in work or in other aspects of their life. In other cases there is no such cause, and it has been suggested that seasonal affective disorder is related in some way to the changes in the seasons (e.g. to the number of hours of daylight). Although these seasonal affective disorders are characterized mainly by the time at which they occur, some symptoms are said to occur more often than in other mood disorders. These symptoms include:

- hypersomnia
- increased appetite, with craving for carbohydrate
- an afternoon slump in energy levels.

The most common pattern is onset in autumn or winter, and recovery in spring or summer. This condition is also called ‘winter depression’. Some patients show evidence of hypomania or mania in the summer, which suggests that they have a seasonal bipolar illness. This pattern has led to the suggestion that shortening of daylight hours is important in the pathophysiology of winter depression, and treatment methods include exposure to bright artificial light during hours of darkness. Both DSM-5 and ICD-11 have specifiers for ‘seasonal pattern’, which can be applied to recurrent major depression with an established seasonal onset. The use of bright light treatment is reviewed in Chapter 25.

Table 9.1 Classification of depressive disorders

ICD-11	DSM-5
Single episode depressive disorder	Major depressive episode
Mild	Mild
Moderate	Moderate
Moderate with psychotic symptoms	
Severe	Severe
Severe with psychotic symptoms	with psychotic features
Recurrent depressive disorder	Major depressive disorder
Currently mild	Recurrent
Currently moderate	
Currently severe	
Currently severe with psychosis	
In partial or full remission	
Dysthymic disorder	Persistent depressive disorder
	Other specified depressive disorder
Mixed depressive and anxiety disorder	Recurrent brief depression

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (DSM-5). Copyright (2013), American Psychiatric Association.

Recurrent brief depression

Some individuals experience recurrent depressive episodes of short duration, typically two to seven days, that are not of sufficient duration to meet the criteria for major depression or depressive episode. These episodes recur with some frequency, about once a month on average. There is no apparent link with the menstrual cycle in female sufferers. Although the depressive episodes are short, they are as severe as the more enduring depressive disorders, and can be associated with suicidal behaviour. Thus recurrent brief depression is associated with much personal distress and social and occupational impairment. Individuals with recurrent brief depression often receive treatment with antidepressant medication, but its value is questionable (see Baldwin and Sinclair, 2015).

Classification in DSM and ICD

The main categories in the sections on depressive disorders in DSM-5 and ICD-11 are shown in Table 9.1 and Box 9.3. Broad similarities are evident. The first

Box 9.3 Additional specifiers for depressive disorders in DSM-5 and ICD-11

DSM-5	ICD-11
With anxious distress	With prominent anxiety symptoms
With mixed features	With panic attacks
With melancholic features	With melancholia
With seasonal pattern	With seasonal pattern of episode onset
With psychotic features	With persistent current episode (at least two years)
With atypical features	
With catatonia	
With peripartum onset	

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (DSM-5). Copyright (2013), American Psychiatric Association.

Table 9.2 A systematic scheme for the clinical description of depressive episodes

The episode	
Severity	Mild, moderate, or severe
Special features	With melancholic symptoms
	With atypical symptoms
	With prominent anxiety
	With psychotic symptoms
	With agitation
	With retardation or stupor
The course	Single or recurrent episodes
Aetiological factors	Biological
	Family history of mood disorder
	Personal history of mood disorder
	Childhood adverse experiences
	Personality
	Social support
	Life events

similarity is that both systems contain categories for single episodes of mood disorder as well as categories for recurrent episodes. The second is that both recognize milder but persistent depressive states (dysthymia) although, as noted above, in DSM-5 dysthymia has been subsumed into the category of 'persistent depressive disorder', which also includes 'chronic major depression'. In DSM-5, mood disorders that are judged to be secondary to a medical condition are included as a subcategory of mood disorders, and ICD-11 they are subsumed under 'secondary mood syndrome'.

Both ICD-11 and DSM-5 classify depressive episodes on the basis of *severity* and whether or not *psychotic features* are present. DSM-5 also allows the diagnosis of recurrent brief depression (see Table 9.2). As noted above, DSM-5 has some additional clinical specifiers which may have implications for treatment and

Table 9.3 The aetiology of mood disorders

Area of investigation	Relevant studies
Genetic	Genetic epidemiology
	Molecular genetics
Personality	Temperament
	Cognitive style
Early environment	Parental deprivation
	Childhood adversity and abuse
Social environment	Life events
	Chronic difficulties
Psychological	Psychodynamic
	Cognitive
Biological	Monoamines
	Hypothalamic–pituitary–adrenal axis
	Neuropsychology and brain imaging
	Neuropathology
	Neurogenesis, neurotrophins, synaptic plasticity

prognosis. ICD-11 follows a similar pattern, though the specifiers show some differences (Table 9.3).

Classification and description in everyday practice

Although neither DSM-5 nor ICD-11 is entirely satisfactory, it seems unlikely that further rearrangement of descriptive categories would be an improvement. A solution will only be achieved when we have a better understanding of aetiology. Meanwhile, either ICD-11 or DSM-5 should be used for statistical recording. For most clinical purposes, it is better to describe disorders systematically than to classify them.

This can be done for every case by referring to the severity, the distinguishing symptomatic features, and the course of the disorder (Table 9.2), together with an evaluation of the relative importance of known aetiological factors (see Table 9.3).

Differential diagnosis of depressive disorders

Depressive disorders have to be distinguished from the following:

- normal sadness
- adjustment disorder
- anxiety disorders
- schizophrenia
- organic brain syndromes.

As already explained, the distinction from normal sadness is made on the basis of the presence of other symptoms of the syndrome of depressive disorder and their duration and severity. The presence of functional impairment is also important in diagnosing a clinical disorder. Depressive disorders have high rates of comorbidity with a wide range of other disorders—for example, anxiety disorders, eating disorders, substance misuse, and personality disorder. In all of these cases it is important to recognize and treat the depressive disorder. *Adjustment disorders* following a stressor (see page 150) often present with depressed mood, together with tearfulness and feelings of hopelessness. However, the full diagnostic criteria for major depression are not met.

Anxiety disorders

Mild depressive disorders are sometimes difficult to distinguish from anxiety disorders. Accurate diagnosis depends on assessment of the relative severity of anxiety and depressive symptoms, and on the order in which they appeared. Similar problems arise when there are prominent phobic or obsessional symptoms, or when there are *dissociative* symptoms. In all of these cases, the clinician may fail to identify the depressive symptoms and so offer the wrong treatment.

Schizophrenia

The distinction from schizophrenia depends on a careful search for the characteristic features of this condition (see Chapter 11). Difficult diagnostic problems may arise when the patient has *depressive psychosis*, but here again the distinction can usually be made on the basis of a careful examination of the mental state, and of the order in which the symptoms appeared. Information about the past psychiatric history may also be useful. Particular difficulties also arise when symptoms characteristic of depressive disorder and of schizophrenia are found in equal measure in the same patient; these so-called *schizoaffective disorders* are discussed in Chapter 11.

Dementia and other organic conditions

In middle and late life, depressive disorders are sometimes difficult to distinguish from *dementia*, because some patients with depressive symptoms complain of considerable difficulty in remembering. In fact, patients with severe depression can perform very badly on tests of cognitive function, and distinction between the two conditions purely in terms of the nature of the cognitive impairment may not be possible. Here the presence of depressive symptoms is the key to diagnosis, which should be confirmed with improvement of the memory disorder as normal mood is restored. Numerous other general medical conditions can present with depressive features (see Chapter 22). The key to diagnosis is a careful history and physical examination, supplemented by special investigations where appropriate.

The epidemiology of depressive disorders

It is difficult to determine the prevalence of depressive disorder, partly because different investigators have used different diagnostic definitions. More modern investigations have used structured diagnostic interviews linked to standardized diagnostic criteria, including the Diagnostic Interview Schedule (DIS) and the Composite International Diagnostic Interview (CIDI) (see Chapter 3).

Major depression

Defining the boundaries of depressive episodes in community surveys presents a number of difficulties. However, if the DSM-5 criteria for major depression are applied, recent surveys in industrialized countries suggest that:

- The 12-month prevalence of major depression in the community is around 2–5%.
- The lifetime rates in different studies vary considerably (from 4–30%) and the true figure probably lies in the range 10–20%.
- The mean age of onset is about 27 years (though a substantial proportion of cases have an onset before the age of 21).
- Rates of major depression are about twice as high in women as in men, across different cultures.
- There may be increased rates of depression in people born since the 1970s.
- Rates of depression are higher in the unemployed and divorced.
- Major depression has a high comorbidity with other disorders, particularly anxiety disorders and substance misuse.

The reasons for *higher rates of major depression among women* are uncertain. They could be due in part to a greater social acceptability of women admitting to having depressive symptoms. However, such selective reporting is unlikely to be the whole explanation. It is possible that some depressed men misuse alcohol and are diagnosed as suffering from alcohol-related disorders rather than depression, with the consequence that the true number of major depressive disorders is underestimated. Again, misdiagnosis of this kind is unlikely to account for the whole of the difference. Furthermore, in most societies, women are subject to various kinds of social disadvantage and, for example, are much more likely than men to experience sexual abuse and domestic violence. Factors of this sort are also likely to be implicated in an increased risk of depression. It is also important to note that

the preponderance of depression in women emerges at puberty when it seems linked to changes in sex hormone levels rather than chronological age (Lewis et al., 2018).

Although it used to be thought that the risk of depressive disorders increased with age, recent surveys suggest that major depression is most prevalent in the 18–44-year age group. For a review of the epidemiology of mood disorders, see Kessing (2020).

Dysthymia and persistent depressive disorder

The lifetime risk for *dysthymia* ranges from 1–6% and the risk for persistent depressive disorder is similar (about 4%). Rates of dysthymia are higher in *women* and in the *divorced*. In comparison to episodic depression, people with persistent depressive disorders have greater childhood adversity, higher rates of chronic depression in relatives, and an earlier age onset of depressive symptoms. Chronic depression is also associated with greater functional impairment and higher suicide rates (Schramm et al., 2020).

Minor depressive–anxiety disorders

Estimates of the frequency of *minor depressive–anxiety disorders* show wide variations because the different studies have not defined cases in the same way. However, these are probably the most prevalent psychiatric disorders in the community. We have already seen that in the Adult Psychiatric Morbidity Survey in England, McManus et al. (2016) found that mixed depression and anxiety was the most common mental disorder in the community, with a one-week prevalence of around 8%.

The aetiology of depression

There have been many different approaches to the aetiology of depression. There is substantial knowledge about the *genetic epidemiology* of depression and how certain *childhood experiences* can lay down a predisposition to depressive disorders in adulthood. There is also a good understanding of the role of *current life difficulties and stresses* in provoking depression in predisposed individuals. There is much less knowledge about the neurobiological mechanisms involved in the translation of these predisposing and provoking factors into clinical symptomatology.

When trying to elucidate these mechanisms (which, of course, have important implications for treatment), investigators have employed two main conceptual approaches, which can be broadly termed *psychological and biological*. It is likely, of course, that these approaches represent different levels of enquiry that will eventually inform each other. The aetiology section in the current chapter is structured so as to illustrate to the reader the many ways in which research into the causation of psychiatric disorder can be approached (see Box

9.4). It may therefore be helpful for this section to be read in conjunction with Chapter 5.

Genetic causes

Family and twin studies

It is well recognized that depression tends to run in families, and the risk of depression in a first-degree relative of a proband is increased about threefold. Environmental influences are also important in the aetiology of depression, but the effect of environment is mainly mediated through effects specific to the individual rather than through shared family experiences (Merikangas and Merikangas, 2020). Relatives of patients with *unipolar depression* do not have increased rates of bipolar disorder or schizoaffective disorder.

Twin studies confirm that the aggregation of depressive disorders in families is in part due to genetic factors, with the *concordance rate* being higher in monozygotic twins (about 45%) than in dizygotic twins (about 20%). The heritability of major depression has been estimated at around 35%, which is considerably less than that of bipolar disorder or schizophrenia (Geschwind and Flint, 2015). Twin studies have also suggested that susceptibility to major depression and generalized anxiety disorder involves similar genes but different environmental risk factors (Hettema et al., 2005).

Mode of inheritance

The familial segregation of depression does not fit a simple Mendelian pattern. The female preponderance of depression is well established, and it is possible that genetic factors play a somewhat greater aetiological role in women since the heritability of depression in women is higher than that in men (Kendall et al., 2021). Overall, it seems likely that the genetic liability to depression results from the combined action of multiple genes of modest, or small, effect—so-called *polygenic inheritance*—in a context of gene–environment interaction. Although many important environmental contributions to depression have been identified (see below), the level at which they interact with genetic predisposition remains unclear.

Molecular genetics

The monoamine theory of depression suggests that allelic variation in genes coding for monoamine synthesis or metabolism or specific receptors may contribute to the risk of mood disorders. There have been numerous

Box 9.4 Multifactorial origin of mood disorders

- An important genetic contribution to mood disorder is made by multiple genes of small individual effect. This genetic contribution may be expressed directly through modification of relevant cortical circuitry, or indirectly through effects on personality and psychological coping mechanisms.
- Adverse early life experiences also shape personality and limit subsequent attachment behaviour and ability to access social support. In addition, adverse early experience may affect development of the hypothalamic–pituitary–adrenal (HPA) axis and neurobiological responses to stress in adulthood.
- Depressive disorders are often triggered by current life events, particularly in people who lack social support. The impact of life events is modified by early life experience, personality, and genetic inheritance. The interaction of these factors determines the resilience or vulnerability of an individual to a life event, and the subsequent risk of clinical mood disturbance.
- Depression is associated with changes in the activity of the neural circuitry involved in the expression and regulation of emotion. Some structural and functional brain abnormalities in mood disorder suggest persistent biological vulnerability, probably produced by genetic inheritance or early developmental factors.

association studies of such *candidate genes*, although the results have not been compelling because of the difficulty of replication. In an analysis of population-based and case-control samples, Border et al. (2019) found no evidence for an association between commonly studied candidate genes and depression with the possible exception of the gene for the dopamine D₂ receptor.

Reliable genome-wide association study (GWAS) findings from groups of patients with depression have been slower to emerge than similar studies in schizophrenia and bipolar disorder. However, more recently larger samples have identified around 180 replicated loci. The genes concerned are involved in neuronal growth, synaptic function and inflammation. New identified loci also implicate GABA and glutamate neurotransmission (Als et al., 2023; Kendall et al., 2021).

The genes involved in the heritability of depression show substantial overlap with those predisposing to anxiety and neuroticism. There are also significant, somewhat lesser genetic correlations between depression, bipolar disorder, schizophrenia, and low educational attainment (Als et al., 2023).

Molecular genetic data confirm that depression is a highly polygenic disorder where genes interact with important environmental determinants such as childhood trauma, socio-economic adversity and social support. Current polygenic risk scores (PRS) for depression explain only a small amount of genetic liability (about 4%). At present this makes PRS scores for depression of limited clinical utility, though they have the potential in the future to play a role in predicting treatment response as well as helping unpick the clinical and biological heterogeneity of the depressive syndrome. For example, higher PRS scores for depression may distinguish patients with a predisposition to recurrent rather than single episodes of depression and a greater risk of experiencing comorbid psychiatric disorders (Als et al., 2023).

Personality

Certain kinds of personality could be associated with predisposition to mood disorder. For example, it is a common clinical observation that patients with depression often seem to have high levels of premorbid anxiety. Aspects of personality could be associated with depression in a number of ways.

1. What is recognized as a personality characteristic may in fact represent a milder form of the illness. For example, while dysthymia is now regarded as a mood disorder, previously people with such symptoms might have been classified as having a 'depressive personality'.
2. Some personality features might influence the way in which people respond to adverse circumstances, and thus make depressive disorders more likely. For example, a cognitive style that is characterized by *sociotropy* (a strong need for approval) is associated with increased risk of depression after adverse life events. Similarly, high trait perfectionism resulting in self-criticism and disproportionate concern about making mistakes has been linked to the development of depression (Ferrari et al., 2018).
3. Certain kinds of personality development and depressive disorder may share common genes. For example,

neuroticism, as measured by the Eysenck Personality Questionnaire, predisposes to major depression, but twin studies suggest that neuroticism and major depression have genes in common (Fanous and Kendler, 2004).

Overall, current findings suggest that part of the genetic risk of depression can take the form of inheritance of particular character traits and cognitive styles, which predispose to psychiatric illness in the presence of specific kinds of life stresses.

Early environment

Parental deprivation

Psychoanalysts have suggested that childhood deprivation of maternal affection through separation or loss predisposes to depressive disorders in adult life. Overall, however, epidemiological studies do not suggest that loss of a parent by death in childhood increases the risk of depressive disorder in adulthood. By contrast, there is more support for the proposal that depressive disorder in later life is associated with *parental separation*, particularly divorce. The key factor here appears to be not so much the loss itself as the discord and diminished care that result from it. Indeed, family discord and lack of adequate care predispose to depression even in families where separation does not occur (Brown, 2009).

Relationships with parents

It is clear that gross disruption of parent-child relationships, as occurs, for example, in *physical and sexual abuse*, is a risk factor for several kinds of adult psychiatric disorder, including major depression. It is less certain whether more subtle differences in parental style may also predispose to depression. One problem is the difficulty of determining retrospectively what kind of relationship a patient may have had with their parents in childhood. The patient's recollection of the relationship may be distorted by many factors, including the depressive disorder itself. However, it appears that lack of care, particularly by fathers, predisposes to depression in adulthood (Kullberg et al., 2020). Mothers with postnatal depression may manifest a rearing style that is characterized by neglect and emotional indifference. This could lead to longer-term deleterious effects on self-esteem and attachment style in the child, thus increasing the risk of depression in the subsequent generation (Ramchandani et al., 2009).

Precipitating factors

Recent life events

It is an everyday clinical observation that depressive disorders often follow *stressful events*. However, several other possibilities must be discounted before it can be concluded that stressful events cause the depressive disorders that succeed them. First, the association might be coincidental. Secondly, the association might be non-specific—there might be as many stressful events in the weeks preceding other kinds of illness. Thirdly, the association might be spurious—the patient might have regarded the events as stressful only in retrospect when seeking an explanation for their illness, or he might have experienced them as stressful only because they were already depressed at the time. Finally, the depression itself might have caused the life event.

Research workers have tried to overcome each of these methodological difficulties. The first two problems—whether the events are coincidental, and whether any association is non-specific—require the use of control groups suitably chosen from the general population and from people with other illnesses. The third problem—whether the association is spurious—requires two other approaches. The first approach is to separate events that are undoubtedly independent of illness (e.g. losing a job because a whole factory closes) from events that may have been secondary to the illness (e.g. losing a job when no one else is dismissed). The second approach is to assign a rating to each event according to the consensus view of healthy people about its stressful qualities.

Methodologically reliable research has shown that:

- There is a sixfold excess of adverse life events in the months before the onset of depressive disorder.
- An excess of similar events also precedes suicide attempts, and the onset of anxiety disorders and schizophrenia.
- In general, ‘loss’ events are associated with depression and ‘threat’ events are associated with anxiety.
- Life events are important antecedents of all forms of depression, but appear to be relatively less important in established melancholic-type disorders and where there is a strong family history of depression.

Events that lead to feelings of *entrapment and humiliation* may be particularly relevant to the onset of depression, including peer victimization through bullying. In contrast, remission from depression is often associated with ‘fresh-start’ life events (e.g. establishing a new relationship or starting an educational course; Brown, 2009). It is also important to note that genetic factors may be

involved in individual liability to experience life events. Thus certain individuals seem to be more prone to select risky environments, and genetic factors also play a role in how life events are perceived by a particular individual, perhaps through the personality mechanisms discussed above.

In general, the importance of life events in the onset of a depressive episode decreases as the number of episodes increases. This suggests that, once a depressive disorder is clearly established, depressive episodes can occur in the absence of major environmental precipitants (Monroe et al., 2019).

Vulnerability factors and life difficulties

It is a common clinical impression that the events immediately preceding a depressive disorder act as the ‘last straw’ for a person who has been subjected to a long period of adverse circumstances, such as an unhappy marriage, problems at work, or unsatisfactory housing.

In general, there is good evidence that *poor social support*, measured as lack of intimacy or social integration, is associated with an increased risk of depression (Brown, 2009). The mechanism of this association is unclear, and is open to different interpretations. First, it may be that a lack of opportunities to confide makes people more vulnerable. Secondly, it may indicate that depressed people have a distorted perception of the degree of intimacy that they achieved before becoming depressed. Thirdly, some other factor, presumably an abnormality in personality, may result in difficulty in confiding in others, and thus lead to vulnerability to depression.

The effects of physical illness

All medical illnesses and their treatment can act as non-specific stressors, which may lead to mood disorders in predisposed individuals. However, sometimes certain medical conditions are believed to play a more direct role in causing the mood disorder (e.g. brain disease, certain infections, including HIV, and endocrine disorders). The resulting mood disorders are known as organic mood disorders (for a further discussion of this subject, see Chapter 22).

Inevitably, the above distinction is arbitrary. For example, major depression occurs in about 50% of patients with *Cushing’s disease*. Since not all patients with Cushing’s disease suffer from depressive disorder, it follows that variables other than raised plasma cortisol levels are involved. However, organic mood disorders can give clues to aetiology. For example, depressive disorders in patients with Cushing’s disease remit after cortisol levels are restored to normal, a finding that led to the proposal that increased cortisol levels may play

a pathophysiological role in depression more generally (see hypothalamo–pituitary–adrenal (HPA) axis below). It is also worth noting here that the puerperium, although not an illness, is associated with an increased risk of mood disorders (see Chapter 22).

Psychological approaches to aetiology

These theories are concerned with the psychological mechanisms by which recent and remote life experiences can lead to depressive disorders. Much of the literature on this subject fails to distinguish adequately between the symptom of depression and the syndrome of depressive disorder. The main approaches to the problem are derived from the ideas of psychoanalysis (see Box 9.5) and cognitive behavioural theories.

Cognitive theories

Depressed patients characteristically have recurrent and intrusive *negative thoughts* ('automatic thoughts'). Beck (1976) proposed that these depressive cognitions reveal negative views of the *self*, the *world*, and the *future* (the depressed patient usually reviews the past in a similar vein). These automatic thoughts appear to persist because of illogical ways of thinking, which Beck called *cognitive distortions*. These include:

- *Arbitrary inference* (drawing a conclusion when there is no evidence for it and even some evidence against it).
- *Selective abstraction* (focusing on a detail and ignoring more important features of a situation).
- *Overgeneralization* (drawing a general conclusion on the basis of a single incident).
- *Personalization* (relating external events to oneself in an unwarranted way).

Although most psychiatrists regarded these cognitions as secondary to a primary disturbance of mood, Beck suggested that another set of cognitions precede depression and predispose to it. These cognitions are *dysfunctional beliefs* or *schemas* such as 'No one really likes me.' Schemas of this sort are established early in life, usually through childhood adversity, and affect the way in which a person responds to stress and adversity. Such schemas can become activated by 'matching' life experiences. For example, a failure in a relationship is more likely to provoke depression in a person who holds the dysfunctional belief described above (Clark and Beck, 2010).

Experimental studies have confirmed the presence of negative cognitive biases in people with depression.

Box 9.5 Psychoanalytical theory

- In his seminal paper 'Mourning and melancholia', Freud drew attention to the resemblance between the phenomena of mourning and symptoms of depressive disorders, and suggested that their causes might be similar (Freud, 1917).
- Freud suggested that, just as mourning results from loss by death, so melancholia results from loss of other kinds. Since it was apparent that not every depressed patient had suffered an actual loss, it was necessary to postulate a loss of 'some abstraction' or internal representation, or, in Freud's terms, the loss of an 'object'.
- Freud pointed out that depressed patients often appear to be critical of themselves, and he proposed that this self-accusation was really a disguised accusation of someone else for whom the patient 'felt affection'. In other words, depression was thought to occur when feelings of love and hostility were present at the same time (*ambivalence*). When a loved 'object' is lost, the patient feels despair; at the same time, any hostile feelings attached to this 'object' are redirected against the patient as self-reproach.
- John Bowlby (1907–1990) showed that the rearing abilities of the main caregiver play an important role in giving the infant a secure emotional 'attachment', which is of critical importance in the development of satisfactory interpersonal relationships. Insecure attachments can increase the risk of various kinds of adult psychopathology, including depression.

For a review of the psychoanalytical model of depression and its application to treatment, see Taylor (2008).

These involve a range of psychological processes sensitive to emotional state including self-referential thinking, attention and memory. There is also evidence for deficient 'top-down' cognitive control of mood-congruent material which results in increased maladaptive emotional regulation strategies such as rumination. These mechanisms can be seen as playing an important role in maintaining the depressive state (see Figure 9.1). How far they might induce it remains to be determined. For a review, see LeMoult and Gotlib, 2019)

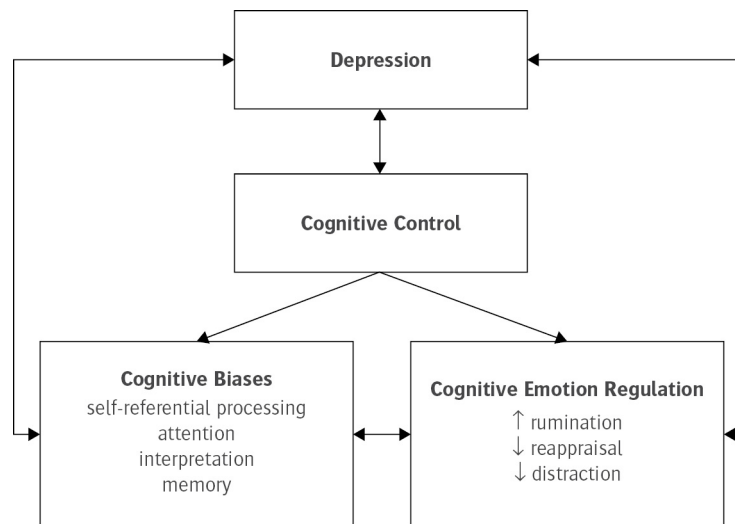


Figure 9.1 Depression is associated with 1) cognitive biases in self-referential processing, attention, interpretation, and memory; 2) the use of maladaptive versus adaptive, cognitive emotion regulation strategies; and 3) deficits in cognitive control over mood-congruent material, which in turn, contributes to cognitive biases and the use of maladaptive, emotion regulation strategies, all of which exacerbate and sustain symptoms of depression. Reproduced from LeMoult J, Gotlib IH. (2019). Depression: A cognitive perspective. *Clin Psychol Rev*, 69, 51–66. Copyright (2019), with permission from Elsevier.

Neurobiological approaches to aetiology

Whatever their aetiology, the clinical manifestations of depressive disorders must presumably ultimately be mediated through changes in brain neurochemistry and the circuitry involved in emotional regulation. Biochemical investigations in depressed patients have focused on the *monoamine neurotransmitters* because monoamine pathways appear to play an important role in the actions of effective antidepressant drugs (see below). Of course, the actions of antidepressant drugs may not reverse the cause of depression, but may merely change the expression of symptoms. However, another reason for studying monoamines is that they play an important role in mediating adaptive responses to stressful events, and depressive disorders can be viewed as a failure of these adaptive responses.

Monoamine pathways, particularly those involving noradrenaline and 5-hydroxytryptamine (5-HT), innervate cortical and subcortical brain regions thought to be involved in mood regulation. Recent studies have used structural and functional imaging techniques to elicit changes in the neural circuitry that underpins the expression of clinical affective symptomatology. In hand with this has come the realization that mood disorders, despite their fluctuating and remitting clinical course, are associated with distinct and persistent

neuropathological changes in relevant brain regions. Some of these changes are likely to be developmental in origin; this could encompass either genetic inheritance or the consequences of adverse childhood experiences.

There is currently much ongoing research effort in both animal and human studies to assess how traumatic experiences in development might affect the subsequent maturation of the HPA axis and thereby influence responses to environmental stress. A final point worth considering is that the prognosis of mood disorders worsens as the number of episodes increases. Although this phenomenon may simply represent an association between severity and poor prognosis, it is also possible that acute episodes of mood disturbance themselves produce neurobiological effects that worsen the subsequent prognosis (a phenomenon that is sometimes known as ‘scarring’).

The monoamine hypothesis

This hypothesis suggests that depressive disorder is due to an abnormality in a monoamine neurotransmitter system at one or more sites in the brain. In its early form, the hypothesis suggested a changed provision of the monoamine; more recent elaborations postulate alterations in receptors as well as in the concentrations or the turnover of the amines. Three monoamine transmitters have been implicated—*serotonin (5-HT)*, *noradrenaline*, and *dopamine*. The latter two neurotransmitters are called catecholamines. This chapter considers the

evidence for abnormalities in monoamine neurotransmitters in untreated depressed patients (see Box 9.6).

5-HT function

Plasma tryptophan. The synthesis of 5-HT in the brain depends on the availability of its precursor amino acid, L-tryptophan. Plasma tryptophan levels are decreased in untreated depressed patients, particularly in those with melancholic depression. Studies in healthy subjects have shown that weight loss through dieting can lower plasma tryptophan levels, and this factor appears to explain some, but not all, of the reduction in plasma tryptophan concentration that is seen in depression (Pu et al., 2021).

Studies of cerebrospinal fluid (CSF). Indirect evidence about 5-HT function in the brains of depressed patients has been sought by examining CSF. Numerous studies have been carried out, but overall the data do not suggest that drug-free patients with major depression have a consistent reduction in CSF concentrations of 5-hydroxyindoleacetic acid (5-HIAA), the main metabolite of 5-HT formed in the brain.

Neurochemical brain imaging studies. Recent developments in brain imaging with selective labelled ligands have enabled the assessment of certain brain 5-HT receptor subtypes *in vivo*. The most reliable finding is for a reduction in the number of midbrain 5-HT reuptake (serotonin transporter) sites in depressed individuals (Kambeitz and Howes, 2015).

Box 9.6 Evidence for abnormalities in monoamine activity in depression

5-HT

- Decreased plasma tryptophan
- Blunted 5-HT neuroendocrine responses
- Decreased brain 5-HT reuptake (serotonin transporter) sites (SPET and PET)
- Clinical relapse after tryptophan depletion.

Noradrenaline

- Blunted noradrenaline-mediated growth hormone release
- Clinical relapse after α -methyl-para-tyrosine (AMPT).

Dopamine

- Decreased homovanillic acid (HVA) levels in cerebrospinal fluid
- Clinical relapse after AMPT.

Neuroendocrine tests. The functional activity of 5-HT systems in the brain has been assessed by giving a substance that stimulates 5-HT function, and by measuring an endocrine response that is controlled by 5-HT pathways, usually the release of prolactin. Neuroendocrine challenge tests have the advantage that they measure an aspect of brain 5-HT function. However, the 5-HT synapses that are involved presumably reside in the hypothalamus, which means that important changes in 5-HT pathways in other brain regions could be missed.

Studies in unmedicated depressed patients have shown consistent evidence that 5-HT-mediated endocrine responses to challenge with 5-HT re-uptake inhibitors such as clomipramine and citalopram are blunted in depressed patients. It is possible that this abnormality persists into clinical recovery, which suggests a persistent dysfunction of some aspects of 5-HT neurotransmission in those at risk of depression (Cowen, 2015).

Tryptophan depletion. Although the findings outlined above provide strong evidence that aspects of brain 5-HT neurotransmission are abnormal in depression, they do not reveal whether these changes are central to pathophysiology or might instead represent some form of epiphenomenon. To assess this, it is necessary to study the psychological consequences of lowering brain 5-HT function in healthy subjects and those at risk of mood disorder.

As mentioned above, the synthesis of brain 5-HT is dependent on the availability in the brain of its amino acid precursor, L-tryptophan. It is possible to produce a transient lowering of plasma tryptophan and brain 5-HT function over a few hours by administering a mixture of amino acids that lacks tryptophan. This procedure is called *tryptophan depletion*. Tryptophan depletion in subjects with no personal or family history of mood disorder has little measurable effect on mood, and does not produce significant clinical depressive symptomatology. By contrast, unmedicated euthymic patients with a personal history of mood disorder undergo a rapid but temporary depressive relapse when exposed to tryptophan depletion. People with a strong family history of depression but no previous illness show some lowering of mood after tryptophan depletion, but these changes are relatively mild and not of clinical severity. The following conclusions can be drawn:

- Low brain 5-HT function is not sufficient to cause depression, because tryptophan depletion fails to alter mood in those who are not vulnerable to mood disorder.
- Low brain 5-HT function appears to interact with *other vulnerability factors* to cause depressive symptoms in some people.

The nature of these other vulnerability factors remains conjectural. It is possible that the 5-HT pathways of vulnerable individuals react abnormally to precursor deficit. Alternatively, there may be pre-existing deficits in the central mood-regulating circuitry that are 'revealed' in the presence of low brain 5-HT states. For a review of tryptophan depletion studies in depression, see Ruhé et al. (2007).

Noradrenaline function

Metabolism. There is no consistent evidence that brain or CSF concentrations of noradrenaline or its major metabolite 3-methoxy-4-hydroxy-phenylethylene glycol (MHPG) are altered in depressed patients.

Neuroendocrine tests. Increasing brain noradrenaline function elevates plasma concentrations of adrenocorticotrophic hormone (ACTH), cortisol, and growth hormone. There is fairly consistent evidence that the growth hormone *response* to both the noradrenaline reuptake inhibitor desipramine and the noradrenaline-receptor agonist clonidine is blunted in patients with melancholic depression. Clonidine acts directly on postsynaptic α_2 -adrenoceptors in the hypothalamus to increase plasma growth hormone, and therefore the blunted response in depressed patients suggests a decreased responsiveness of hypothalamic postsynaptic α_2 -adrenoceptors.

Catecholamine depletion. It is possible to lower the synthesis of catecholamines by inhibiting the enzyme tyrosine hydroxylase, which catalyses the conversion of the amino acid tyrosine to L-DOPA, a precursor of both noradrenaline and dopamine. The drug that is used to achieve this effect is α -methyl-para-tyrosine (AMPT). In healthy subjects, AMPT produces sedation but not significant depressive symptoms. However, as with tryptophan depletion, when administered to recovered depressed patients off drug treatment it causes a striking clinical relapse in depressive symptomatology (see Ruhé et al., 2007). This could be mediated by diminished function of either dopamine or noradrenaline, or by combined inhibition of both of these neurotransmitters.

These findings suggest that some subjects at risk of mood disorder are vulnerable to decreases in both 5-HT and catecholamine neurotransmission. This is consistent with the clinical evidence that drugs that act selectively on noradrenaline or 5-HT pathways are effective antidepressant treatments.

Dopamine function

The function of dopamine in depression has been less well studied than that of 5-HT or noradrenaline, but there are a number of reasons for believing that

dopamine neurons may be involved in the pathophysiology of the depressed state.

- Dopamine neurons in the mesolimbic system play a key role in incentive behaviour and reward. These processes are disrupted in depression, particularly melancholic states.
- Antidepressant treatments in animals increase the expression of dopamine receptors in the part of the mesolimbic system called the nucleus accumbens.

There is some evidence that suggests that dopamine function may be abnormal in depression. For example, CSF levels of the dopamine metabolite homovanillic acid (HVA) are consistently low in depressed patients, and some brain imaging studies in depressed patients have found increased binding of dopamine D_2/D_3 receptors in striatal regions. However, this may be restricted to patients with psychomotor retardation. There is also limited evidence for regional reductions in the density of dopamine D_1 receptors (Meyer, 2013). These findings, taken together with the effect of AMPT in causing relapse in recovered depressed patients, suggest that impaired dopamine function may play a role in the manifestation of the depressive syndrome and in the effects of antidepressant drug treatment.

Role of monoamines in depression

There is now good evidence that unmedicated depressed patients have abnormalities in certain aspects of monoamine function. However, these abnormalities vary in extent from one case to another, and the changes are not large and are not sufficiently sensitive to be diagnostic. Some abnormalities may also persist into clinical recovery, suggesting that they are related to vulnerability to illness rather than the acute depressive state.

The most convincing studies that show a key role for monoamines in the pathophysiology of depression are the 5-HT and catecholamine depletion paradigms. This is because lowering of 5-HT and noradrenaline and dopamine function is sufficient to cause clinical depression in those at risk by virtue of a previous illness. It is unclear what neurobiological changes produce this psychological vulnerability to monoamine depletion, but it seems likely that they are at least in part consequences of previous episodes of illness and perhaps their treatment.

Amino acid neurotransmitters

Developments in magnetic resonance spectroscopy (MRS) have facilitated measurement of the amino acid neurotransmitters *gamma-aminobutyric acid* (GABA) and *glutamate* in the brain in patients with mood disorders. Overall, there is evidence for decreased levels of

both neurotransmitters in the anterior brain regions in depressed patients (Godfrey et al., 2018). However, current MRS protocols are unable to distinguish the GABA and glutamate involved in neurotransmission from that in intermediary metabolism.

Endocrine abnormalities

Abnormalities in endocrine function may be important in aetiology for two reasons:

1. Some disorders of endocrine function are followed by mood disorders more often than would be expected by chance, suggesting a causative relationship.
2. Endocrine abnormalities found in depressive disorder indicate that there may be a disorder of the hypothalamic centres that control the endocrine system.

Endocrine pathology and depression

About 50% of patients with *Cushing's syndrome* suffer from major depression, which usually remits when the cortisol hypersecretion is corrected. Depression also occurs in Addison's disease, hypothyroidism, and hyperparathyroidism. Endocrine changes may account for depressive disorders that occur premenstrually, during the *menopause*, and after *childbirth*. These clinical associations are discussed further in Chapter 22.

HPA axis

Much research effort has been concerned with abnormalities in the control of cortisol in depressive disorders. In about 50% of patients whose depressive disorder is at least moderately severe, plasma cortisol secretion is increased throughout the 24-hour cycle. This increase in cortisol secretion is associated with enlargement of the adrenal gland and increased cortisol response to ACTH challenge.

Depressed patients may also show impaired suppression of cortisol in response to the synthetic corticosteroid dexamethasone, and this test was once proposed as a laboratory marker of melancholic depression. While this hope has not been borne out, the abnormality does suggest that depressed patients have a reduction in the sensitivity of glucocorticoid receptors involved in HPA feedback (Menke, 2019).

Current approaches have made use of salivary cortisol measures and in particular the increase in salivary cortisol that occurs on waking in response to endogenous ACTH release. The waking increase in cortisol is increased in patients with acute depression but also in patients who have recovered. This suggests that some vulnerable individuals may have fairly enduring abnormalities in HPA axis regulation (Cowen, 2015).

In experimental animal studies, early adverse experiences produce longstanding changes in HPA axis regulation, indicating a possible neurobiological mechanism whereby childhood trauma could be translated into increased vulnerability to mood disorder.

Sustained increases in cortisol could be linked to some of the medical comorbidities associated with persistent depression such as cardiovascular disease, diabetes, obesity, and cognitive decline. On the other hand, there is little current evidence that lowering cortisol activity with, for example, metyrapone or antagonists of corticotropin releasing hormone, produces useful therapeutic effects in acutely depressed patients (Menke, 2019).

Depression and the immune system

There is growing evidence that patients with depression manifest a variety of disturbances of immune function. Earlier studies found decreases in the cellular immune responses of lymphocytes in depressed patients, but more recent research has produced evidence of immune activation, with increases in particular in the release of certain proinflammatory *cytokines* (see Box 9.7). Cytokines are known to provoke HPA axis activity, and therefore changes in immune regulation may play a role in HPA axis dysfunction in depression.

It is also possible that the changes seen in immune activity are secondary to other depressive features (e.g. lowered food intake and diminished self-care). However, the fact that medical administration of some cytokines such as interferon can cause significant depressive symptoms suggests that, in some situations, changes in immune function may have a more direct role in provoking mood disorders. Current formulations propose that cytokines can induce expression of the tryptophan-metabolizing enzyme *indoleamine 2, 3-dioxygenase*. This

Box 9.7 Immune changes in depression

- Lowered proliferative responses of lymphocytes to mitogens
- Lowered natural killer cell activity
- Increases in positive acute phase proteins such C-reactive protein
- Increases in inflammatory cytokine levels (e.g. IL-6 and TNF α)
- Induction of indoleamine 2,3-dioxygenase
- Activation of brain microglia.

lowers tryptophan levels and increases kynurenine metabolites some of which, like quinolinic acid, have neurotoxic effects.

Depressed patients also have indications of central inflammation shown by PET imaging of activated microglia, though peripheral inflammatory markers do not correlate with central changes. It is also worth noting that raised levels of C-reactive protein in the blood have been linked to unsatisfactory therapeutic responses to antidepressant medication, particularly SSRIs (Jha et al., 2017). For a review of inflammation and depression, see Drevets et al. (2022).

Sleep and depression

We have already seen that disturbed sleep is an important symptom of depression. Recordings of the sleep EEG (polysomnogram) in depressed patients have shown a number of abnormalities in *sleep architecture*:

- impaired sleep continuity and duration
- decreased deep sleep (stages 3 and 4)
- decreased latency to the onset of rapid eye movement (REM) sleep
- an increase in the proportion of REM sleep in the early part of the night.

A further link between REM sleep and depression is that many (but not all) effective antidepressant drugs decrease REM sleep time and the latency to its onset. Intriguingly, in view of the sleep difficulties experienced in depression, *sleep deprivation* can produce an alleviation of mood in some depressed patients (Ioannou et al., 2021.) This effect, while occasionally striking, is transient and dissipates when the patient next sleeps. The biological mechanisms involved are obscure

It is also important to note that insomnia is a risk factor for the development of depression, increasing the incidence approximately twofold. Indeed, genetic studies indicate a significant overlap between the heritability of insomnia and depression. Clinical trials have shown that cognitive behaviour therapy (CBT) for insomnia decreases current depressive symptoms as well as the future risk of developing major depression. For a review, see Plante (2021).

Brain imaging in mood disorder

Structural brain imaging

Changes in brain volume. Magnetic resonance imaging (MRI) studies have found a number of abnormalities in patients with depression, particularly in those with more severe and chronic disorder (Goodwin and Browning, 2020). The most consistent findings are:

- decreased hippocampal volume, and
- decreased grey matter volume in the anterior brain areas, including frontal cortex, orbitofrontal cortex, and cingulate cortex.

The origin of these structural abnormalities is unclear, but might be related to the cellular neuropathological abnormalities that have been described in depression, such as loss of interneurons and glial cells in anterior brain regions (see Stockmeier and Rajkowska, 2004) The *neurotrophic hypothesis* of depression suggests that stress (perhaps aided by cortisol hypersecretion) can lead to atrophy and death of neurons and downregulation of neurogenesis, particularly in the hippocampus (Jaggar et al., 2019). Conceivably processes of this kind could lead to the kind of structural deficits noted above.

White matter hyperintensities. Hyperintense MRI signals can be detected in a number of regions in both normal ageing subjects and older patients with major depression. The usual sites are in the deep white matter and periventricular white matter. In major depression, increased deep white matter hyperintensities are associated with the following:

- late onset of depressive disorder
- greater illness severity and poorer treatment response
- apathy, psychomotor slowness, and retardation
- the presence of vascular risk factors.

It has been proposed that major depression with these clinical and radiological features is likely to be caused by vascular disease, which presumably impairs functioning in the pathways involved in mood regulation.

Cerebral blood flow and metabolism

Cerebral blood flow can be measured in a number of ways—for example, with single-photon emission tomography (SPET), positron emission tomography (PET), or functional magnetic resonance imaging (fMRI). PET can also be used to measure cerebral metabolism. Cerebral blood flow and metabolism are normally highly correlated.

Numerous studies have examined both cerebral metabolism and blood flow in groups of depressed patients. The abnormalities found support a circuitry model in which depression is associated with abnormal interactions between several brain regions, rather than a major abnormality in a single structure. The circuitry that is implicated involves regions of the medial prefrontal cortex, anterior cingulate cortex, amygdala, ventral striatum, thalamus, and hypothalamus (Price and Drevets, 2012). Some tentative correlations between

Box 9.8 Some neuropsychological correlates of altered cerebral perfusion and metabolism in depressed patients

Dorsolateral and dorsomedial prefrontal cortex

Cognitive dysfunction (particularly executive dysfunction). Impaired voluntary regulation of emotion.

Medial prefrontal cortex

Abnormal emotional processing.

Anterior cingulate

- Impaired attentional processes
- Altered emotional salience.

Amygdala

Abnormal emotional processing.

Ventral striatum

- Impaired incentive behaviour
- Psychomotor disturbances.

these brain regions and clinical depressive features are listed in Box 9.8.

Functional MRI

Chapter 5 described how fMRI can be used to assess both 'resting-state' functional connectivity in the brain and the neural response to psychological tasks relevant to the psychiatric condition being studied. In resting state studies depressed patients show increased connectivity within the *default mode network* (DMN), which comprises the medial prefrontal cortex, posterior cingulate cortex, and parietal regions. The DMN becomes active at rest, for example, during 'mind-wandering', and shows decreased activity during specific cognitive tasks. In depressed patients, the default mode network remains active even during task-based activity, and this has been proposed to correlate with depressive ruminative thinking and impaired performance on cognitive tasks (Kaiser et al., 2015).

Task-based fMRI in depression has employed presentation of salient emotional stimuli (such as the image of a sad face) and compared the neural response of patients with depression to non-depressed controls. The usual pattern in depression is an increased response in

limbic areas such as the amygdala, which are involved in emotional appraisal while cortical regulatory areas such as dorsolateral prefrontal cortex have reduced activity. While this seems a plausible account, similar models have been proposed for anxiety disorders, so their specificity for depression is unclear (Goodwin and Browning, 2020).

Neuropsychological changes in mood disorder

Patients with acute depression show poor performance on several measures of *neuropsychological function*. Impairment is typically seen over a wide range of neuropsychological domains, including *attention*, *learning*, *memory*, and *executive function*. There is disagreement as to whether these defects are best regarded as global and diffuse, or whether there may be some selectivity with regard to the changes that are seen. However, some authors have suggested that deficits in executive function may be particularly prominent, which would be consistent with the abnormalities seen in prefrontal perfusion in imaging studies.

Some of the cognitive impairments resolve as the mood disorder remits, but a growing literature attests to the persistence of cognitive defects, particularly in processing speed, selective attention, and working memory in otherwise recovered depressed patients (Semkovska et al., 2019). Such deficits may worsen with repeated episodes of illness.

Conclusions

The *predisposition* to develop depressive disorders has an important genetic contribution but the identity of the genes and how they operate to affect risk are not clear. They could act via effects on the regulation of neural responses to emotion, or through more remote factors such as temperament and the psychological response to stressful life events. Adverse early experiences such as parental conflict, or abuse of various kinds, may play a part in shaping features of personality which, in turn, determine whether, in adulthood, individuals are able to access emotional support to help to buffer the stressful effects of social adversity. In addition, early life experiences could programme the HPA axis to respond to stress in a way that might predispose to the development of mood disorder.

The *precipitating causes* of mood disorders are stressful life events and certain kinds of physical illness. Some progress has been made in discovering the types of event that provoke depression, and in quantifying

their stressful qualities. Such studies show that *loss* can be an important precipitant, but not the only one. The effects of particular events may be modified by a number of background factors that may make a person more vulnerable—for example, being bullied or socially isolated. As noted in the preceding paragraph, the impact of potentially stressful events also depends on early life experience, personality factors, and probably genetic inheritance.

Two kinds of *pathophysiological mechanism* have been proposed to explain how precipitating events lead to the phenomena observed in depressive disorders. The first mechanism is psychological and the second is neurobiological. The two sets of mechanism are not mutually exclusive and, as the neural basis of emotional processing becomes better understood, the two approaches are converging in a way that promises a richer understanding of pathophysiology. In particular it seems that abnormalities in the neural processing of emotion can account for the emotional biases that are characteristic of depression, and can probably account for much of the subjective symptomatology. The relevant brain regions receive a strong innervation from monoamine neurons, and manipulation of 5-HT and noradrenaline neurotransmission is capable of altering emotional processing at both a behavioural and neural level (Harmer et al., 2017). This might account for the ability of monoamine manipulation to produce symptomatic relief in some depressed patients.

Using sophisticated statistical techniques it is possible to derive quantitative estimates of the roles of different *risk factors* in the development of depressive disorders. For example, from a prospective study of 1,942 female

twin pairs, Kendler et al. (2002) calculated that about 50% of the susceptibility to an episode of major depression is attributable to the following factors (in order of relative importance):

- stressful life events
- neuroticism
- genetic risk
- past history of major depression.

The same study found that adverse childhood experiences also played a significant role in the risk of subsequent depression, but this was expressed indirectly through an increased risk of life events and difficulties and diminished social support. Interestingly, some of the impact of genetic factors was mediated through an increased risk of early adverse experiences. This suggests that part of the genetic risk associated with depression is expressed through greater likelihood of exposure to an adverse family environment. This in turn could be linked to temperamental factors in parents and children.

Taken together, the findings of recent studies show that major depression is a disorder with important genetic, environmental, and interpersonal determinants. These factors do not interact in a simple additive manner, but modify each other in both direct and indirect ways. Although this formulation precludes the use of simple models to explain the aetiology of major depression, it does correspond more closely with clinical experience. In addition, it suggests that a number of different kinds of intervention could be useful for decreasing the liability of individuals to develop depression.

Course and prognosis of depression

Major depression

When considering course and prognosis of major depression it should be remembered that any sample of patients with unipolar depression will contain a proportion of patients who in fact have a bipolar disorder, which has not yet declared itself (since mania or hypomania have not yet occurred). It has been estimated that about 10% of patients who present with a depressive disorder will eventually have a manic episode. Such patients are more likely to have a family history of mania or to show brief, mild manic mood swings during antidepressant treatment.

- The age of onset of major depression varies widely and can occur at any point in the lifespan. About half of all cases occur before the age of 21. The relative contribution of different aetiological factors probably varies between early and late-onset cases.
- The average length of a depressive episode is about six months, but around 25% of patients have episodes that last for more than one year, and around 10–20% develop a chronic unremitting course.
- About 80% of patients with major depression will experience further episodes (i.e. have recurrent major depression).

- Over a 25-year follow-up, patients with recurrent major depression experience on average about four further episodes.
- The interval between episodes becomes progressively shorter.
- About 50% of depressed patients do not achieve complete symptom remission between episodes, and experience continuing subsyndromal depressive symptomatology of fluctuating severity.
- The longer-term prognosis of recurrent major depression is modest. For example, only about 25% of patients with recurrent depression achieve a period of five years of clinical stability with good social and occupational performance.

Dysthymia

Dysthymia is by definition a chronic disorder that lasts for many years. Despite this, about 50% of outpatients may be expected to show a clinical recovery over a five-year follow-up. Over the lifespan, some patients with dysthymia develop major depression (so-called *double depression*), while some patients who originally present with major depression subside into dysthymia. The development of mania is rare.

Minor depressive disorders

Minor depressive conditions are depressive disorders that do not meet threshold criteria for major depression, even of mild severity ('unspecified depressive disorder' in DSM-5). These conditions show a recurrence rate similar to that of major depression. Patients who meet the criteria for minor depression at one point in follow-up may then subsequently be diagnosed as suffering from major depression, and vice versa. Thus, in a similar manner to dysthymia, minor depression is likely to be a risk factor for major depression, and may also be a residual state following remission of major depression. Overall, the longitudinal data suggest that major and minor depression and dysthymia are not distinct conditions, but represent part of a *spectrum of depressive disorders* (Angst, 2009).

Mortality of depressive disorders

Mortality is significantly increased in patients with depression, largely, although not exclusively, due to *suicide*. The standardized mortality ratio in mood disorders is about twice that found in the general population. Apart from suicide, excess deaths are due to accidents, cardiovascular disease, and comorbid substance misuse. There is now compelling evidence that depressive disorders increase the risk of general medical conditions such as diabetes and cardiovascular disease. Epidemiological studies suggest that treatment lowers the mortality in patients with depression (Angst et al., 2013).

Rates of suicide in patients with depression are at least 15 times higher than those in the general population, and tend to be higher in unipolar than in bipolar disorder (Angst, 2009). A 30-year follow-up of depressed patients treated in hospital found that the risk of dying by suicide was 7% in men and 4% in women (Kessing, 2020).

Prognostic factors

The best predictor of the future course is the history of *previous episodes*. Not surprisingly, the risk of recurrence is much higher in individuals with a history of several previous episodes. Other factors that predict a higher risk of future episodes include the following:

- incomplete symptomatic remission
- early age of onset
- poor social support
- poor physical health
- comorbid substance misuse
- comorbid personality disorder.

The various risk factors, particularly previous pattern of recurrence and the extent of current remission, have important implications for the use of longer-term maintenance treatments (see below). In many patients, depressive disorders are best conceptualized as chronic relapsing conditions that require an integrated long-term approach to management.

The acute treatment of depression

This section is concerned with the *efficacy* of various forms of treatment in the acute management of depression. Details of treatment with drugs and ECT are provided in Chapter 25, which should be consulted before

reading this section. Advice on the selection of treatments and the day-to-day care of patients is given in the section on management (see 'The management of depressive disorders' below).

Antidepressant drugs

Antidepressant drugs are effective in the acute treatment of major depression. The largest effects relative to placebo are seen in patients with major depression whose symptoms are of at least *moderate severity*. Short-term response rates in controlled trials are about 50% for patients on active treatment, and about 30% for those on placebo; the number needed to treat (NNT) is between 5 and 7 (Cleare et al., 2015). In terms of efficacy there is little to choose between the various antidepressants, although their side effect profiles differ significantly. This has important implications when discussing choice of treatment with patients.

Similar clinical response rates are seen in *dysthymia*, where again several classes of antidepressant drugs, including TCAs and SSRIs, have shown therapeutic efficacy. A meta-analysis by Levkovitz et al. (2011) showed a response rate in patients receiving active treatment of 52%, compared with 29% in those on placebo (NNT = 5). Antidepressants do not appear to be more effective than placebo in the treatment of minor depression (Barbui et al., 2011).

SSRIs and serotonin and noradrenaline reuptake inhibitors (SNRIs)

SSRIs have undergone extensive trials both against placebo and against comparator antidepressants. There is good evidence that they are beneficial in patients with major depression, with little significant differences

between individual medications; however, in a comprehensive network-meta-analysis, escitalopram and sertraline seemed the most effective and acceptable (Cipriani et al., 2018). The same meta-analysis indicated that the SNRIs venlafaxine and duloxetine were among the most efficacious antidepressant drugs but were somewhat less well tolerated than SSRIs.

Other newer antidepressants

A variety of other antidepressant drugs are available (see Chapter 25 and Table 9.4), all of which have established efficacy relative to placebo, though in head-to-head comparisons of active treatments, reboxetine and trazodone were the least effective (Cipriani et al., 2018). The main differences between the various preparations are in side effect profile. Apart from reboxetine, anticholinergic-type side effects are uncommon. Trazodone and mirtazapine are sedating. Agomelatine may have a different mechanism of action, involving the activation of melatonin receptors (Whiting and Cowen, 2013). Agomelatine appears the best tolerated of the antidepressants but has the disadvantage of requiring repeated liver function tests in the early phases of treatment.

Tricyclic antidepressants (TCAs)

TCAs have long been recognized to be effective antidepressant agents, particularly in patients with more severe depressive symptoms including those with melancholic depression. However, due their anticholinergic side

Table 9.4 Clinical characteristics of some antidepressant drugs

	Anticholinergic	Sedation	Weight gain	Sexual dysfunction	Toxicity in overdose
Amitriptyline	+++	+++	+++	+	+++
Lofepamine	+	0	0	+	0
SSRIs	0	0	+	+++	0*
Vortioxetine	0	0	+	+	0
Venlafaxine	0	0	+	+++	++
Duloxetine	0	0	+	+++	+
Trazodone	0	+++	+	0	+
Reboxetine	+	0	0	+	0
Mirtazapine	0	+++	+++	0	0
Agomelatine	0	+	0	0	0

* Citalopram and escitalopram may be somewhat more toxic than other SSRIs because of an effect to prolong the QTc interval. 0, none; +, mild; ++, moderate; +++, marked.

effects they are difficult to tolerate at a therapeutic dose. This, together with their cardiac toxicity and danger in overdose, means that TCAs are rarely used now in the pharmacological treatment of depression. However, they retain a role in patients with resistant depression and, at low doses, are widely used for the treatment of neuro-pathic pain. Of the available compounds, lofepramine is relatively safe in overdose.

Monoamine oxidase inhibitors (MAOIs)

There is extensive evidence that MAOIs are effective antidepressants and of equal therapeutic activity to TCAs for moderate to severe depressive disorders. Like TCAs, MAOIs are now rarely used because of their liability to cause dangerous reactions with other drugs and some foods. However, MAOIs can be effective in depressed patients who have not been helped by a wide variety of other treatments and for this reason they retain some occasional use (see Chamberlain and Baldwin, 2021).

The reversible type-A MAOI *moclobemide* has the advantage of not requiring adherence to a low-tyramine diet. However, it can still produce hazardous interactions with other drugs (see Chapter 25). Controlled trials have shown that moclobemide is more effective than placebo in the treatment of uncomplicated major depression. However, it is doubtful whether moclobemide at standard doses is as effective as conventional MAOIs for patients with resistant depression.

Lithium

Lithium as a sole treatment

This section is concerned only with lithium as a treatment for depressive disorders. Placebo-controlled trials suggest that lithium may have some antidepressant efficacy in bipolar depression, but its effects in unipolar depression as a sole treatment are not established (Goodwin et al., 2016).

Lithium in combination with antidepressants

Despite its limited utility as a sole drug treatment for depression, lithium can produce useful therapeutic effects when added to antidepressant medication in treatment-resistant patients (*lithium augmentation*). In a meta-analysis, Nelson et al. (2014) found that about 40% of depressed patients responded to lithium augmentation of their antidepressant regimen, compared with about 15% of patients who were given placebo (NNT = 5).

Although some studies have reported a rapid amelioration of the depressed state within as little as 48 hours after the addition of lithium, the more usual pattern of response is a gradual resolution of symptoms over two to three weeks. The effects of lithium augmentation in depression do not appear to be restricted to any specific class of antidepressants.

Anticonvulsants

Anticonvulsants such as *carbamazepine*, *valproate*, and *lamotrigine* are useful in the management of bipolar disorder, and in these circumstances can prevent episodes of major depression. Whether these agents also have acute antidepressant efficacy in unipolar depression is less clear. Lamotrigine has been shown to have antidepressant effects in placebo-controlled trials in bipolar depressed patients, particularly those with higher levels of symptomatology, and this therapeutic effect may extend to unipolar depression (Goh et al., 2019).

Atypical antipsychotic drugs

Antipsychotic drugs are often combined with antidepressant drugs in the treatment of patients with depressive psychosis, and there is also evidence that atypical antipsychotic agents, used at relatively low dose, can be of benefit when combined with antidepressants in non-psychotically depressed patients who have failed to respond to antidepressant treatment alone. In a meta-analysis of trials involving 3,500 patients, Spielmann and colleagues (2016) found that, relative to placebo, the addition of drugs such as aripiprazole, quetiapine, and risperidone to ineffective SSRI treatment was significantly more likely to result in clinical remission (NNT = 9). Olanzapine addition was also of benefit, but the effect was less (NNT = 19).

Electroconvulsive therapy

This treatment is described in Chapter 25, where its unwanted effects are also considered. The present section is concerned with evidence about the therapeutic effects of ECT for patients with depressive disorders.

Comparison with simulated ECT

Six double-blind controlled trials have compared the efficacy of ECT and simulated ECT (anaesthesia with electrode application but no passage of current) in patients with major depression. Five of these studies found ECT to be more effective than the simulation. In

the study that did not find the full procedure to be more effective, unilateral low-dose ECT was used, a procedure that is considered on other grounds to be relatively ineffective. The overall response rate is about 70% for ECT and 40% for simulated treatment (NNT = 3–4) (UK ECT Review Group, 2003; see also Chapter 25).

Comparison with other treatments

Several studies have compared depressed inpatients receiving ECT with those receiving antidepressant drugs. In a total of nine comparisons with TCAs, ECT was therapeutically more effective in six studies and equally effective in the remaining three. In five comparisons with MAOIs, ECT was superior in each trial and worked more quickly. These data suggest that, in severely depressed inpatients, ECT is probably superior to antidepressant drug treatment, at least in the short term (UK ECT Review Group, 2003; see also Box 25.11).

Indications for ECT

Clinicians generally agree that the therapeutic effects of ECT are greatest in severe depressive disorders, especially those in which there is *marked weight loss, early morning waking, retardation, and delusions*. From the trials comparing full ECT with simulated ECT, it appears that delusions and (to a lesser extent) retardation are the features that distinguish patients who respond to full ECT from those who respond to placebo (UK ECT Review Group, 2003).

Other studies have established that patients with *depressive psychosis* respond better to ECT than to TCAs or antipsychotic drugs given alone. However, combined treatment with antidepressants and antipsychotic drugs may be about as effective as ECT, although no direct comparisons have been made. Another point of practical importance is that ECT may often prove effective in depressed patients who have not responded to full trials of medication, whether or not psychotic features are present. However, in such patients relapse rates are high (Heijnen et al., 2010).

Psychological treatment

All depressed patients, whatever other treatment they may be receiving, require psychotherapy in a general sense, which provides education, reassurance, and encouragement. These measures ('clinical management') can provide some symptomatic relief, and can also increase the likelihood that pessimistic patients will adhere to specific treatments. Education and reassurance

can also be given helpfully to the patient's partner, close family members, and other people involved in their care.

The psychological treatments used for depressive disorders can be divided into the following categories:

- supportive psychotherapy (problem solving)
- CBT
- interpersonal psychotherapy
- behavioural activation
- couples therapy
- counselling
- dynamic psychotherapy
- mindfulness meditation.

These psychotherapies can be employed as alternatives to antidepressant medication, or as adjuncts. Psychotherapies have been less well evaluated than antidepressant medication in major depression; the use of 'waitlist' controls or 'treatment as usual' may inflate apparent efficacy of treatment. In general, specific psychotherapies are more effective than treatment as usual in the management of depression, particularly in primary care. There appear to be few differences in response rates between the various therapies, though non-directive counselling and psychodynamic therapy fare somewhat less well in meta-analyses (Cuijpers et al., 2021). In randomized trials, structured psychotherapies perform as well as drug treatment in depressed patients (National Institute for Health and Clinical Excellence, 2022b).

Supportive psychotherapy and problem-solving

Supportive psychotherapy goes beyond clinical management in focusing on the identification and resolution of current life difficulties, and in using the patient's strengths and available coping resources. A development of this approach is *problem solving*, in which the therapist and the patient identify the main problems of concern and devise feasible step-by-step ways of tackling them. Randomized trials suggest that problem-solving treatment is better than treatment as usual in depressed patients in primary care. (National Institute for Health and Clinical Excellence, 2022b).

Cognitive behaviour therapy

For depressive disorder, the essential aim of CBT is to help the patient to modify their ways of thinking and acting in relation to life situations and depressive symptoms (for further information, see Pages 658, 659). There have

been numerous studies of CBT in acute major depression, which have been reviewed (National Institute for Health and Clinical Excellence, 2022b). Current conclusions can be summarized as follows:

- There is strong evidence that CBT is superior to a waiting list control in relieving depressive symptomatology.
- CBT is not generally superior to other structured psychological treatments, such as behavioural activation and interpersonal therapy.
- CBT is as effective as pharmacological treatment in moderately depressed outpatients.
- Combined CBT and pharmacological treatment is better than pharmacological treatment alone.

There are also several studies of CBT delivered by the internet (iCBT) for the treatment of depression. The evidence suggests that this can be an effective approach, particularly when the programme involves behavioural activation (see below) and therapist guidance (Furukawa et al., 2021).

Behavioural activation

Behavioural activation uses the principles of operant conditioning by tracking the links between actions and emotional outcomes. The goal of therapy is to assist patients, through scheduled activity, to engage in behaviours that will lead to a positive effect on mood (see Chapter 24). Unlike CBT, it is claimed that therapist training in behavioural activation techniques is simple and quickly accomplished. A meta-analysis of 28 randomized trials of behavioural activation in depression showed that the technique was superior to control procedures (the majority of which were waitlist control). However, study quality was generally rated as low (Stein et al., 2021).

Interpersonal psychotherapy

Interpersonal therapy is a systematic and standardized treatment approach to personal relationships and life problems (see page 665). It has been less studied than CBT in depression, but seems to be as effective. In a meta-analysis of interpersonal therapy in patients with depression, Cuijpers et al. (2016) concluded that interpersonal therapy:

- is more effective than placebo with clinical management and GP treatment as usual
- is as effective as antidepressant medication
- is more effective when combined with antidepressants than when given alone.

Couples therapy

Couples therapy can be offered to depressed patients for whom *interactions with a partner* appear to have contributed to causing or maintaining the depressive disorder. The aim of the intervention is to understand the nature of these interactions and modify them so that the relationship becomes more mutually supportive. There are few randomized trials available, but the limited evidence suggests that couples therapy is significantly more effective than a waiting list control and as effective as CBT. In practice, antidepressant treatment and couples therapy are often used together, but the combination has not been well evaluated.

Dynamic psychotherapy

Dynamic psychotherapy has a different aim from the treatments described so far, in that it aims to resolve underlying developmental conflicts and attendant life difficulties that are believed to be causing or maintaining the depressive disorder. In the management of depression, '*brief dynamic therapy*' (see Chapter 24) is usually employed. There are different views as to whether brief dynamic therapy may be less effective than CBT in the treatment of depression (Cuijpers et al., 2021; Driessen et al., 2015). Therapist training and patient selection are likely to play a particularly important role in the efficacy of psychodynamic therapy.

Other treatments

Bright light treatment

Over 50% of patients with recurrent winter depression respond to bright light treatment (about 10,000 lux). Treatment is usually given for an hour or two in the morning, but the timing of light treatment is not always critical, and evening light or even midday exposure can be effective. The duration of exposure usually needs to be one to two hours.

Designing placebo-controlled trials of bright light for winter depression presents problems, because most patients are aware before treatment that bright light is believed to be the important therapeutic ingredient. Within this limitation, most studies have found that dim light is less effective than bright light. The usual onset of the antidepressant effect of bright light is within two to five days, but longer periods of treatment appear to be needed in some patients. Patients with 'atypical' depressive features such as *overeating* and *oversleeping* appear to respond best. To avoid relapse, light treatment usually needs to be maintained until the usual time of natural remission, in the early spring.

Some studies have shown that bright light treatment may be effective in the initial treatment of non-seasonal depression when combined with antidepressant medication (Geoffroy et al., 2019). However, the durability of such an effect has not been established.

Exercise

Exercise programmes are often recommended for people with depressive symptoms; however, it has been less clear how effective they might be for people with diagnosed depressive disorders. Morris et al. (2018) carried out a meta-analysis of various forms of exercise treatment

that involved 455 depressed patients recruited through mental health services. They found that aerobic exercise delivered for about 45 minutes three times weekly produced a substantial beneficial benefit in relieving depressive symptoms. Similar effects were seen whether exercise was delivered in an individual or group format.

Assessment of the effects of exercise in depression is hampered by the difficulty of devising suitable control conditions and taking expectancy effects into account. Also, patients with more severe depression, who are likely to experience problems of low motivation and fatigue, tend not to be included in such studies.

The longer-term treatment of depression

Follow-up studies have shown that mood disorders often recur and that, if left untreated, they have a rather poor long-term prognosis. For this reason there is now increasing emphasis on *long-term management*.

Prevention of relapse and recurrence

Strictly speaking, the term *relapse* refers to the worsening of symptoms after an initial improvement during the treatment of a single episode of mood disorder, whereas *recurrence* refers to a new episode after a period of complete recovery. Treatment to prevent relapse should be called *continuation treatment*, and treatment to prevent recurrence should be called *prophylactic* or *maintenance treatment*. In practice, however, it is not always easy to maintain the distinction between these two kinds of treatment, because a therapy may be given initially to prevent relapse, and may then be used to prevent recurrence.

Drug treatment of unipolar depression

Continuation treatment

It is now well established that stopping antidepressants soon after a treatment response has been obtained is associated with a high risk of *relapse*. About one third of patients who are withdrawn from medication will relapse during the next year, with the majority of the relapses occurring in the first six months. Placebo-controlled studies of the role of continuation therapy have reached the following conclusions (Cleare et al., 2015):

- Continuing antidepressant treatment for six months past the point of remission halves the relapse rate.
- Treatment should be at the originally effective dose of medication if possible.

- In patients who are at low risk of further episodes, continuation of antidepressant treatment for longer than six months confers little extra benefit except in the elderly, where continuation therapy for 12 months is more appropriate.

Maintenance treatment

Controlled studies involving patients with recurrent depression (usually defined as at least three episodes over the past five years) have shown that *maintenance antidepressant treatment* can substantially reduce relapse rates. For example, Geddes et al. (2003) found in a systematic review that one to two years of continued antidepressant treatment lowered the relapse rate from 41% on placebo to 18% on active medication, with an NNT of 4 (see Figure 9.2). Maintaining the dose of medication at the level that was required to achieve remission appears most effective in prophylaxis if tolerability permits.

Lewis et al. (2021) studied antidepressant discontinuation using double-blind, placebo-controlled conditions in 478 depressed patients in primary care who had been treated with medication for over two years. Patients were well and felt ready to discontinue treatment. During a one-year follow-up, 56% of patients taking placebo relapsed compared to 39% of those maintained on medication, yielding an NNT of 6 to prevent relapse.

Lithium carbonate has also been used in the prevention of recurrent unipolar depression but, while some patients show a clear response, the overall evidence for its efficacy is less robust than for the prevention of bipolar disorder. However, where a therapeutic response to lithium augmentation has been achieved, maintaining lithium together with the antidepressant appears worthwhile (Cleare et al., 2015).

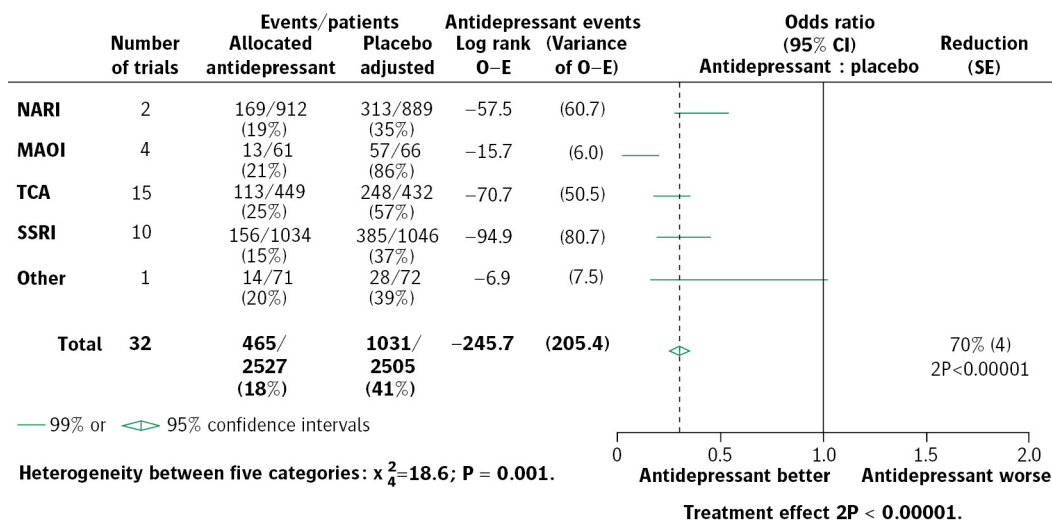


Figure 9.2 Meta-analysis of rates of depressive relapse in patients randomly allocated to receive continuation treatment for on average about one year with antidepressant medication or placebo. Reproduced from Geddes, J. R., Carney, S. M., Davies, C., et al. (2003). Relapse prevention with antidepressant drug treatment in depressive disorders: A systematic review. *Lancet*, 361(9358), 653–661. Copyright (2003), with permission from Elsevier

Psychotherapy

Cognitive therapy

As noted above, there is some evidence that cognitive therapy given during an acute phase of depression leads to a more sustained improvement in depressive symptomatology and lessens the risk of subsequent relapse compared to antidepressant drug treatment (National Institute for Health and Clinical Excellence, 2022b). There is also growing interest in the use of continuation and maintenance treatment with cognitive therapy, particularly in patients who have *residual depressive symptomatology* and are therefore at increased risk of relapse. There is good evidence that CBT continued (or started) after remission prevents relapse, perhaps to a greater extent than maintenance medication. Cognitive therapy also reduces the risk of relapse on medication withdrawal (Breedvelt et al., 2021).

Mindfulness-based cognitive therapy (MBCT) integrates CBT with meditation techniques designed to lower

stress by facilitating acceptance and self-compassion. There is evidence that MBCT lowers the risk of relapse in patients with recurrent depression and is as effective as maintenance antidepressant treatment in this respect (McCartney et al., 2021).

Interpersonal therapy

Combining interpersonal therapy with medication in the treatment of the acute episode appears to decrease relapse rates over the following 12 months. A benefit of the combined treatment over medication alone was seen with nortriptyline but not with paroxetine (Reynolds et al., 2006). These findings suggest that, in terms of recurrence prevention, the benefit of combined interpersonal therapy treatment over medication alone is equivocal. However, in patients who achieve remission from depression with interpersonal therapy as a sole treatment, continuation psychotherapy does seem to be helpful in preventing recurrence (Cuijpers et al., 2016).

The assessment of depressive disorders

The aims of the various steps of assessment are as follows:

- To decide whether the diagnosis is depressive disorder.
- To judge the severity of the disorder, including the risk of suicide.
- To form an opinion about the causes.
- To assess the patient's social resources.
- To gauge the effect of the disorder on other people.

Diagnosis depends on taking a thorough history and examination of the physical and mental state, and has been discussed earlier in this chapter. Particular care should be taken not to overlook a depressive disorder in patients who do not complain spontaneously of being depressed ('masked depression'). It is equally important not to diagnose a depressive disorder simply on the grounds of prominent depressive symptoms, as the latter could be part of another disorder, such as a general medical condition. It should also be remembered that certain drugs—both legal and illegal—can induce depression (Chapter 25).

The *history of previous mood disturbance* is important in assessment. Some patients will have had recurrent episodes of mood disorder. A history of these episodes often provides clues to the probable course of the current disorder and its response to treatment. It is particularly important to ask about possible previous episodes of *mania* and *hypomania*, even if these were mild and short-lived. If there is a history of mania, the mood disorder is bipolar. Interviews with relatives and close friends may help to establish whether such episodes have occurred.

The *severity* of the disorder is judged from the symptoms. Considerable severity is indicated by 'biological' symptoms, hallucinations, and delusions, particularly the latter two. It is also important to assess how the depressive disorder has reduced the patient's capacity to work or to engage in family life and social activities. In this assessment, the duration and course of the condition should be considered, as well as the severity of the present symptoms. Not only does the length of the history affect the prognosis, but it also gives an indication of the patient's capacity to tolerate further distress. A long-continued disorder, even if not severe, can bring the patient to the point of desperation. The *risk of suicide* must be judged in every case (the methods of assessment are described in Chapter 21).

Aetiology is assessed next, with reference to precipitating, predisposing, and maintaining factors. No attempt

need be made to allocate the syndrome to an exclusively 'endogenous' or 'reactive' category. Instead, the importance of all relevant risk factors should be evaluated in every case.

Precipitating factors may be psychological and social (the 'life events' discussed earlier in this chapter), or they may be physical illness and its treatment. In assessing such cases, it is good practice to enquire routinely about the patient's work, finances, family life, social activities, general living conditions, and physical health. Problems in these areas may be recent and acute, or may take the form of chronic background difficulties such as prolonged marital conflict, problems with children, and financial hardship.

The patient's social resources are considered next. Enquiries should cover family, friends, and work. A loving family can help to support patients through a period of depressive disorder by providing company, encouraging them when they have lost confidence, and guiding them into suitable activities. For some patients, work is a valuable social resource, providing distraction and comradeship. For others it is a source of stress. A careful assessment is needed in each case.

The effects of the disorder on other people must be considered carefully. The most obvious problems arise when a severely depressed patient is the mother of young children who depend on her. This clinical observation has been confirmed in objective studies, which have demonstrated that depressive disorder in either parent is associated with emotional disorder in the children (Ramchandani et al., 2009). It is important to consider whether the patient could endanger other people by remaining at work (e.g. as a bus driver). If there are *depressive delusions*, it is necessary to consider what would happen if the patient were to act on them. For example, severely depressed parents may harm their children because they believe them to be doomed to suffer if they remain alive.

The management of depressive disorders

This section starts by considering the management of patients with a depressive disorder of moderate or greater severity (referred to in NICE, 2022b as 'more severe' depression). The first question concerns the level of care and supervision that may be required. The answer depends on the *severity of the disorder* and the quality of the patient's *social resources*. When judging severity, particular attention should be paid to the *risk of*

suicide (or any risk to the life or welfare of family members, particularly dependent children) and to any failure to eat or drink that might endanger the patient's life. Provided that these risks are absent, most patients with a supportive family can be treated at home, even if they are severely depressed. Patients who live alone, or whose families cannot care for them during the day, may need intensive community treatment or day-patient care.

Inpatient treatment can be considered for patients who are not improving with these measures, or where safety issues raise particular concern. *Involvement of the family* wherever possible is also likely to improve the outcome (see Box 9.9).

Box 9.9 What patients and families want to know

Particularly for patients with a first episode of more severe depression, questions such as the following are important:

1. What is wrong with me?
2. Can I recover?
3. What treatment do I need?
4. What can I do to improve the situation?
5. Can my family be helped?

The answers to these questions require a good knowledge of patients and their circumstances. Patients will understand and appreciate that a careful practitioner will need to gather information before giving definitive answers to important questions. However, some general advice is given below.

- This form of depression is a medical condition and not any fault of the individual concerned. Feelings of guilt are common and are seen as part of the condition.
- The prognosis for individual episodes of depression is good, although people with depression are naturally pessimistic about the future.
- Treatment consists of a comprehensive package with psychological support and medication if indicated. The treatment plan will be developed as a collaborative exercise between the patient and the clinical team, and the patient's rights and choices will be respected.
- Part of the treatment plan will involve a full discussion about the most appropriate level of occupational and social activity for the current circumstances. This will be reviewed frequently to allow adjustment to the patient's changing clinical state.
- It is very useful for the family to be fully involved in discussions about the nature of depression and the treatment plan, as their support can be invaluable. Conversely, critical comments can be demoralizing for depressed people. One of the aims of family involvement is to help to improve mutual communication and support.

The next question is whether the patient should continue with their studies or go to work. If the disorder is of moderate severity, study and work can provide a valuable distraction from depressive thoughts, and be a source of companionship. If the disorder is more severe, retardation, poor concentration, and lack of drive are likely to impair performance at work, and such failure may add to the patient's feelings of hopelessness. In severe disorders, there may be dangers to other people if patients remain at their jobs.

In considering specific treatment it is worth noting that the most effective intervention for depression of greater severity is a *combination* of individual CBT and antidepressant medication (Cuijpers et al., 2020). However, depending on patient choice and the availability of psychotherapy, either treatment modality alone can be provided. The presence of melancholic features suggests the advisability of antidepressant medication, as do particular depressive symptoms such as agitation and cognitive impairment, which may make it hard for patients to engage beneficially in psychotherapy as a sole treatment.

For less severe depression, antidepressants are regarded as having a diminished benefit–risk ratio and a variety of group and individual psychotherapies are recommended. Group exercise may also be helpful (National Institute for Health and Clinical Excellence, 2022b) (Box 9.10). However, where a patient with a clear previous history of major depression develops symptoms of mild depression, antidepressant medication can be considered. Guidelines for a 'matched care' approach to the management of depression have been developed by the National Institute for Health and Clinical Excellence (2022b). Here the choice of treatment is determined by the severity of the problem, the patient's choice, and their previous experience of treatment (see Box 9.11).

Choice and use of antidepressant drug

Several kinds of antidepressant drug treatment are available, and the choice should be made in collaboration with the patient, with particular consideration of the likely side effects. These are fully described in Chapter 25.

- *SSRIs* are the usual first choice, unless the patient considers that they have done well previously with a different agent. It is also important to be aware of medical contraindications to SSRI drugs, including, for example, concomitant treatment with non-steroidal anti-inflammatory drugs, which can increase the risk of bleeding. *Lofepramine*, a TCA that is relatively safe

Box 9.10 Treatment options for less severe depression

1. Guided self-help
2. Group CBT
3. Group behavioural activation
4. Individual CBT
5. Individual behavioural activation
6. Group exercise
7. Group mindfulness and meditation
8. Interpersonal psychotherapy
9. SSRI antidepressants
10. Counselling
11. Brief psychodynamic psychotherapy

Note: Treatment options are listed in descending order of recommended use, based on Committee interpretation of clinical and cost effectiveness together with ease of implementation.

Source: National Institute for Health and Clinical Excellence (2022b).

in overdose, is another non-sedating compound with a different side effect profile. *Reboxetine* is also a possibility (see Table 9.4).

- *Mirtazapine* may be considered if the patient needs concomitant sedation or if there are relative contraindications to the use of SSRIs (e.g. concomitant treatment with non-steroidal anti-inflammatory drugs).
- TCAs such as amitriptyline are rarely used now as first-choice agents, even for patients with severe depression.

The dosage of these drugs, the precautions to be observed when using them, and the instructions to be given to patients are described in Chapter 25. Here it is necessary only to emphasize again the importance of explaining to the patient that, although side effects will appear quickly, the therapeutic effect is likely to be delayed for two weeks, with maximum improvement occurring over 6–12 weeks. During this time the patient should be seen regularly to provide suitable clinical management; those with more severe disorders may need to be seen every few days, and other patients once a week. It is important to make sure that the drugs are taken in the prescribed dose. The patient should be warned about the effects of taking alcohol. They should be advised about *driving*, particularly that they should not drive while experiencing sedative side effects or any other effects that might impair their performance in an emergency.

Box 9.11 Guidelines for the management of depression

1. All patients presenting with depression should receive assessment, psychoeducation, and active monitoring.
2. Antidepressants are not recommended as first-line treatment for less severe depression.
3. Patients with less severe depression should initially be offered low-intensity psychological interventions, such as guided self-help or group-based psychotherapies.
4. For patients with persistent, less severe, depressive symptoms that do not respond to these measures, consider a higher-intensity psychological treatment, such as individual CBT or behavioural activation, or drug treatment with an SSRI.
5. Patients who present with more severe depression should be treated with a combination of antidepressant medication and a high-intensity psychological intervention, or either of these treatments alone.
6. Patients who respond to antidepressant medication should continue treatment for at least six months. Patients at high risk of relapse should be advised to continue antidepressant treatment for two years.
7. Consider CBT or mindfulness-based cognitive therapy for patients who relapse despite antidepressant treatment, or for patients who are well but who wish to discontinue medication.

Source: National Institute for Health and Care Excellence (NICE). (2022b). Depression in adults: Treatment and management. NICE guideline NG222. <https://www.nice.org.uk/guidance/ng222>. Copyright (2022), National Institute for Health and Care Excellence.

The use of ECT

ECT will very rarely be part of the first-line treatment of depression, and in such circumstances will usually be considered only for patients who have already been admitted to hospital. The only indication for ECT as a first measure is the need to bring about improvement as rapidly as possible. In practice this applies to two main groups of depressed patients:

1. Those who are unable to drink enough fluid to maintain an adequate output of urine (including the rare cases of depressive stupor).
2. Those who present a highly dangerous suicidal risk.

Occasionally, ECT is indicated for a patient who is suffering such extreme distress that the most rapid form of treatment is deemed justifiable. Such cases are rare. It should be remembered that, with the exception of patients who are unresponsive to antidepressant drugs, the effects of ECT differ from those of antidepressant drugs in terms of greater rapidity of action, rather than the final therapeutic result. In patients with *depressive psychosis*, ECT is considerably more effective than an antidepressant given alone, but probably about the same therapeutic effect can be achieved, albeit more slowly, if a combination of an antidepressant drug and an antipsychotic drug is used.

Activity

The need for *suitable activity* should be considered for every patient. Depressed patients give up activities and withdraw from other people. As a result they become deprived of social stimulation and rewarding experiences, and their original feelings of depression are increased. It is important to make sure that the patient is occupied adequately, although they should not be pushed into activities in which they are likely to fail because of slowness or poor concentration. Thus there is a fairly narrow range of activity that is appropriate for the individual depressed patient, and the range changes as the illness runs its course. If the patient remains at home, it is important to discuss with the relatives how much they should be encouraged to do each day. For inpatients, this question should be decided by the clinical team.

Psychotherapy

The appropriate kind of psychological treatment should also be decided in each case. As noted earlier, all depressed patients require support, encouragement, and a thorough explanation that they are suffering from illness and not moral failure. Similar counselling of partners and other family members is often useful.

The use of one of the more specific psychological treatments discussed earlier should also be considered. These treatments can be used as the sole therapy for patients with less severe depression (National Institute for Health and Clinical Excellence, 2022b) (Box 9.10). The kind of psychological treatment that is used depends largely on the availability of a suitably trained therapist and the preference of the patient, although the more structured therapies (such as CBT and behavioural activation) have a greater evidence base with regard to the treatment of more severe depression. The therapeutic

response to antidepressant drugs is usually faster than that to psychotherapy.

Current recommendations are that in patients with more severe depression it is helpful to add CBT to antidepressant medication (National Institute for Health and Clinical Excellence, 2022b). In practical terms, some time is usually needed to arrange CBT, and in any event some initial improvement produced by antidepressant medication may enable the patient to make more use of psychological treatment. It is also important to tackle any psychosocial problems that are making an important contribution to the depression. In particular, *couples therapy* can be a helpful addition in depressed patients where problems with a partner are playing a role in maintaining the disorder.

If the depressive disorder is severe, too much discussion of problems at an early stage is likely to increase the patient's feelings of hopelessness. Therapy directed towards self-examination is particularly likely to make the disorder worse. During intervals between acute episodes, such therapy may be offered to patients who have recurrent depressive disorders that are largely caused by their ways of reacting to life events.

Failure to respond to initial treatment

If a depressive disorder does not respond within a reasonable time to a chosen combination of antidepressant drugs, graded activity, and psychological treatment, the plan should be reviewed. The first step is to check again that the patient has been taking medication as prescribed. If not, the reasons for this should be sought. The patient may be convinced that no treatment can help, or may find the side effects unpleasant. The diagnosis should also be reviewed carefully, and a check made that important stressful life events or continuing difficulties have not been overlooked.

If this enquiry is unrevealing, antidepressants should be continued at an increased dose if possible. In general, SSRIs do not have clear dose-response relationships, although some patients respond to higher doses, particularly if a partial response has been observed at a standard dose.

If it becomes clear that the patient is not improving, a number of further steps can be taken (see Box 9.12). There is not a fixed order in which treatments should be offered. Prescribing decisions should be made in collaboration with the individual patient, considering factors such as specific symptomatology, how far the condition may have shown a partial response, and the side effect profile of the various treatment options.

Box 9.12 Some pharmacological treatments for treatment-resistant depression

- Increase antidepressant to the maximum dose, if tolerance permits; if the patient has depressive psychosis, add an antipsychotic drug; try a different class of antidepressant drug, including venlafaxine and TCAs.
- Try an antidepressant combination (e.g. an SSRI or venlafaxine with mirtazapine)
- Add an atypical antipsychotic drug (usually quetiapine or aripiprazole) to an SSRI or venlafaxine.
- Add lithium to antidepressant drug treatment.
- Add lamotrigine to antidepressant drug treatment.
- Add triiodothyronine to antidepressant drug treatment.
- MAOIs (can be usefully combined with lithium).
- Consider ketamine or esketamine.

Change in antidepressant drug treatment

If a patient does not respond to one antidepressant, the first step is usually to stop the first medication and try another. Most published studies of this approach have studied patients in an open sequential way; clearly this cannot control for the placebo effect or the possibility of spontaneous remission. Overall, however, there is reasonable evidence that switching to a second antidepressant can produce benefit in about 50% of patients who are unresponsive to an initial medication trial (Cleare et al., 2015).

If a patient has not responded to one kind of antidepressant, it would seem sensible to switch to an antidepressant that has a *different pharmacological profile*. However, it must be acknowledged that studies have shown good response rates when patients who failed to respond to one SSRI were switched to another. A meta-analysis of randomized studies suggested that switching from an SSRI to a different class of drug (mirtazapine, venlafaxine, or bupropion) was marginally better in terms of remission rate (SSRI, 23.5% versus non-SSRI, 28%; NNT = 22) than switching to a second SSRI, but this advantage is of uncertain clinical significance (Papakostas et al., 2008). Switching from a drug with serotonergic properties to another serotonergic compound should be carried out cautiously because of the risk of serotonin toxicity. In practice this means that the first compound should be fully withdrawn if at all possible, and the second one started at a half-dose or

less. In the case of fluoxetine, because of its long duration of action, at least one week should elapse before starting a second serotonergic agent. However, when switching between agents with different pharmacological properties (e.g. from citalopram to mirtazapine or reboxetine), cross-tapering can be employed. Detailed instructions are provided by the Maudsley Prescribing Guidelines (Taylor et al., 2021).

Duloxetine, venlafaxine, and amitriptyline appear to be slightly more effective than SSRIs in patients with severe depression; these drugs are therefore worth trying at some point in the management of patients who are unresponsive to initial medication trials (Cowen and Anderson, 2015). Similar comments apply to the use of conventional MAOIs, which clearly have a beneficial effect in some patients with resistant depression (Chamberlain and Baldwin, 2021). However, the drug and food restrictions associated with MAOIs mean that it is unusual to use them early in drug-resistant depression unless a patient has responded well to them in the past.

Combination treatment with antidepressants

Combination strategies aim to supplement the antidepressant effect of an ineffective or partially effective medication with another antidepressant agent. This approach can therefore be considered an augmentation strategy, although if the patient's condition remits it may be unclear whether the response is due to the combined effect of the two antidepressants or to the second agent acting alone.

The pharmacological rationale of combination treatment is the use of two agents to produce a broader spectrum of activity on monoamine pathways than either agent could produce alone. In practice this means that SSRIs or venlafaxine are usually combined with noradrenergic-promoting agents such as bupropion or mirtazapine (which increases noradrenaline function through blockade of auto-inhibitory α_2 -adrenoceptors).

The evidence for any of these strategies is limited, although they are endorsed in case series and expert reviews. A recent meta-analysis suggested therapeutic benefit when SSRIs or SNRIs were combined with either mirtazapine or trazodone (Henssler et al., 2022). Combination of a TCA with SSRIs must be approached with caution because of the risk of elevation of TCA levels (see Chapter 25).

Augmentation of antidepressant drug treatment

When switching antidepressant preparations, one problem is that withdrawal of the first compound may not

be straightforward. Patients may have gained some small benefit from the treatment (e.g. improved sleep or reduced tension), and this improvement may be lost. Also, if the first medication is stopped quickly, withdrawal symptoms may result (see Chapter 25). However, if the dose is reduced gradually, the changeover in medication may be protracted and may not be easily tolerated by a depressed and despairing patient.

For this reason, in patients who are unresponsive or partly responsive to first-line medication it may be more appropriate to add a second compound to the antidepressant—so-called *augmentation therapy*. A disadvantage of augmentation therapy is the increased risk of adverse effects due to drug–drug interaction.

Antipsychotic drug augmentation

As noted above, in patients with *psychotic depression* it is usually best to prescribe a combination treatment of antidepressant and antipsychotic medication (Kruizinga et al., 2021). In non-psychotic resistant depression, conventional antipsychotic drugs are of little value except, at low doses, for ameliorating agitation and distress. However, there is good evidence that some *atypical antipsychotic drugs* may have antidepressant effects when used in combination with SSRIs or SNRIs in patients with non-psychotic depression (Aronson et al., 1996).

The use of atypical antipsychotics to augment SSRIs employs lower doses than would be used to treat schizophrenia, perhaps because the key pharmacological mechanism probably involves 5-HT₂ receptor antagonism rather than dopamine D₂ receptor blockade. Despite this, olanzapine and quetiapine, even at low doses, can cause troublesome sedation and weight gain, while concomitant use of risperidone also produces some degree of weight gain together with hyperprolactinaemia. Aripiprazole is less likely to cause metabolic side effects, but is associated with agitation and restlessness. In randomized studies of SSRI augmentation, the *discontinuation rate* due to adverse effects was significantly greater with atypical antipsychotics than with placebo (Spielmans et al., 2016). This suggests that, although atypical antipsychotic drug addition can be useful in SSRI-resistant depression, the clinical utility of this approach may be limited by its tolerability.

Lithium augmentation

We have already seen that data from randomized trials indicate that lithium can be effective in the treatment of resistant depression. Provided that there are no contraindications, the addition of lithium to antidepressant drug treatment is usually safe and well tolerated. About 50% of depressed patients will show a useful response

over one to three weeks. Lithium can be added to any primary antidepressant medication with good effect, although combination with SSRIs and venlafaxine should be undertaken with caution because of the risk of *5-HT neurotoxicity* (see Chapter 25). In the latter case it is appropriate to start lithium at a low dose (200 mg daily) and increase by a maximum of 200 mg weekly. In augmentation treatment, the aim should be to obtain a lithium concentration within the range used for prophylactic purposes (0.5–0.8 mmol/l); the lower end of the range is appropriate for combination with serotonergic antidepressants. It has been suggested that the combination of lithium with MAOIs or with clomipramine (sometimes supplemented with tryptophan) can be effective in depressed patients who are otherwise refractory to treatment (Cowen and Anderson, 2015).

Tri-iodothyronine augmentation

Some open and controlled studies have suggested that the addition of tri-iodothyronine (T3) in doses of 20–40 µg daily to ineffective TCA treatment can bring about a useful clinical response, even where underlying thyroid activity is normal. However, a meta-analysis of four published randomized trials that assessed the efficacy of T3 addition to ineffective TCA treatment was of borderline significance (Aronson et al., 1996). The evidence for the efficacy of T3 augmentation of newer antidepressant drugs, such as SSRIs, is inconsistent. Despite this, some depressed patients do appear to be helped by the addition of T3, which can start at a dose of 10 µg daily and be increased to 20 µg after one week, if tolerance is good. At this dose the side effects are mild, but tremor and sweating can occur. It is prudent not to use T3 addition in patients with evidence of cardiovascular disease.

Ketamine in depression

Ketamine is a dissociative anaesthetic that blocks the N-methyl-D-aspartate (NMDA) subtype of glutamate receptor. Intravenous administration of a sub-anaesthetic dose of ketamine can produce a rapid (within hours) antidepressant effect in depressed patients who have not responded to conventional medications (Bahji et al., 2021). This effect of ketamine can last for several days but ultimately dissipates. Patients can then respond to further intravenous treatments and thus the response maintained.

While the need for repeated intravenous therapy is cumbersome, a nasally administered form of ketamine, called esketamine, has been licensed for patients who have failed to respond to conventional antidepressant medication. However, esketamine, like intravenous ketamine, causes significant dissociative symptoms, which

require it to be administered under close clinical supervision. Also, the efficacy of esketamine in depressed patients with more significant degrees of treatment resistance has not been demonstrated.

The use of ECT in resistant depression

If severe depression persists, ECT should be considered. ECT is often beneficial to patients who have not responded to antidepressant drugs, and it is probably more effective than drugs in the most severe depressions, particularly when psychotic features are present. Whether medication resistance lowers the response to ECT has been debated. However, a meta-analysis of observational studies suggested that the overall response rate to ECT in patients who failed pharmacotherapy (48%) was significantly lower than the rate in those who did not (65%) (Heijnen et al., 2010).

It is important to note that, among patients who have not responded to full-dose antidepressant medication, the relapse rate in the year after ECT may be as high as 50%. This may be because patients are often continued on the same antidepressant medication to which they previously failed to respond. In these circumstances it seems reasonable to use an antidepressant of a different class, or lithium, for continuation treatment after ECT, but there is currently no unequivocal evidence that this procedure lowers the relapse rate. However, a randomized study of 200 patients found post-ECT prophylaxis with a combination of lithium and nortriptyline to be as effective as maintenance ECT in sustaining remission in the six months following a successful course of treatment. Despite this, just over 50% of the patients in each group relapsed (Kellner et al., 2006).

Transcranial magnetic stimulation (TMS) in resistant depression

TMS is non-convulsive form of neuromodulation in which a powerful magnetic field is used to produce current flow in the brain. The technique is described further in Chapter 25. For the treatment of depression, an electromagnetic TMS coil is placed in a cortical area (usually left prefrontal cortex), and repetitive stimuli are given usually for about 30 minutes per session. TMS sessions are usually given five days a week for four to six weeks, though various accelerated forms of treatment are under development.

Most studies of TMS in depression have been carried out in patients who have not responded to pharmacological treatment. Evidence suggests that TMS is superior to sham treatment. Overall remission rates in TMS-treated

patients are about 16% compared to about 6% in the sham treated group, with an NNT of 10 (Schatzadeh et al., 2019). The durability of the antidepressant effect of TMS is not yet clear. Also, methods of application and refining coil placement continue to evolve.

Other important measures

Continuing support is essential for patients who do not respond to treatment. Lack of improvement increases the pessimism experienced by the depressed patient. Therefore it is important to give reassurance that depressive disorders have a good chance of recovering eventually, whether or not treatment speeds recovery. Meanwhile, if the patient is not too depressed, any problems that have contributed to their depressed state should be discussed further. Particularly in patients who have not received high intensity psychological treatments, the place of structured psychotherapies should be considered. A study of primary care patients who had failed to respond adequately to antidepressant medication found that CBT was markedly better than treatment as usual in bringing about a therapeutic response (Wiles et al., 2013).

In patients with prolonged depression, it is particularly important to watch carefully for suicidal intentions. For a very small number of patients with severe symptoms and particularly longstanding refractory illness, other treatment approaches such as *deep brain stimulation* and *neurosurgery* can be considered. These are discussed in Chapter 25.

Prevention of relapse and recurrence in depression

After recovery, the patient should be followed up for several months by the psychiatric team or GP. If the recovery appears to have been brought about by an antidepressant drug, that drug should usually be continued for about six months and then gradually withdrawn over a period of several weeks. If residual symptoms are still present, it is safer not to withdraw medication. As noted above, psychotherapies such CBT and mindfulness-based therapies can be helpful in lowering the risk of relapse.

At follow-up interviews, a careful watch should be kept for *signs of discontinuation reactions* or *relapse*. It is helpful to discuss with the patient the possible early signs of relapse, and to develop a plan of action should any of these signs appear. If the patient agrees, it is helpful to involve the relatives in this plan.

Maintenance treatment

Major depression is often recurrent, and long-term maintenance treatment may need to be considered. The risk factors for recurrence have been noted above. In addition, the clinician will need to consider the following individual factors:

- the likely impact of a recurrence on the patient's life
- the previous response to drug treatment
- the patient's view of long-term drug treatment.

It is estimated that, among patients who have had three episodes of major depression, the likelihood of another episode is 90%. The usual recommendation is that maintenance drug treatment should be considered if a patient has had two previous episodes of depression within a five-year period, particularly if there is a family history of recurrent major depression, or personal and social factors predictive of recurrence (Cleare et al., 2015). *Persistent residual symptoms* are also an indication for maintenance treatment.

Choice of medication

For most patients, the choice of antidepressant will be derived from their response in the acute or continuation phase of treatment. In this case, the same medication can be continued, if possible at the same dose. As noted earlier, newer antidepressant drugs are better tolerated in the longer term, and may be preferred for maintenance treatment. If a change needs to be made because of adverse effects (e.g. sexual dysfunction with SSRIs), an alternative choice should be determined on the basis of side effect profile.

Lithium can also be effective for long-term maintenance treatment of recurrent depression in some patients. Its adverse-effect profile and the need for plasma monitoring mean that it will not be a first choice for most patients, but its use should be considered if there is any history of manic episodes, even if these have been mild and transient. In addition, lithium will sometimes be effective in patients with recurrent depression who do not respond to maintenance treatment with antidepressants. Importantly, lithium does seem to lower risk of suicidal behaviour in patients with mood disorders, an action that is not explained completely by its ability to prevent depressive relapse (Cleare et al., 2015).

Other measures

If a depressive disorder was related to self-imposed stressors, such as overwork or complicated social relationships, the patient should be encouraged to change to a lifestyle that is less likely to lead to further illness. These readjustments may be helped by psychotherapy, which may be individual, couples, or group. Community nurses and nurse therapists can play a valuable role in delivering treatment of this kind.

GPs play a key and increasing role in the long-term monitoring of patients, and should always be involved in treatment planning. The evidence for important comorbidities between depression and general medical conditions such as cardiovascular disease and diabetes (Nemeroff, 2022) gives added importance to the collaboration between mental health services and GPs. Annual monitoring of body weight, blood pressure, glucose, and lipids, as recommended for bipolar patients (see Box 10.4), is advised.

Further reading

Lewis, A. J. (1934). Melancholia: A clinical survey of depressive states. *J Ment Sci*, 80, 277–378. [Reprinted in Lewis, A. J. (1967). *Inquiries in Psychiatry*. Routledge and Kegan Paul. pp. 30–117].
Nemeroff, C. B., Schatzberg, A. F., Rasgon, N., & Strakowski, S. M. (Eds.). (2022). *The American Psychiatric Association Publishing*

Textbook of Mood Disorders, 2nd edn. American Psychiatric Association.

Styron, W. (2001). *Darkness Visible: A Memoir of Madness*. Vintage.

CHAPTER 10

Bipolar disorder

Introduction 231

Clinical features 231

Transcultural factors 234

Classification 234

Differential diagnosis of bipolar disorder 235

Epidemiology 236

Aetiology 237

Course and prognosis 239

Treatment of mania 240

Treatment of bipolar depression 241

Longer-term treatment of bipolar disorder 243

Practical management of bipolar disorder 245

Introduction

Chapter 9 considered depressive disorders, and we now turn to bipolar disorders, another group of conditions in which depressive episodes are prominent. In bipolar disorder, however, the course is marked by at least one episode of *mania* or *hypomania*. Kraepelin (1921) brought mania and depression together as *manic depressive psychosis*, because he believed that the longer-term clinical course, with its tendency to recurrence of mood disturbance, was similar, whether patients presented with mania or depression. However, Leonhard et al. (1962) pointed out that bipolar disorder tends to show a distinct familial clustering. In addition,

there are other epidemiological differences between unipolar depression and bipolar disorder (Table 10.1). However, the differences are not great and there must be overlap between the two groups, because a patient who is classified as having unipolar depression at one time may have a manic disorder subsequently. In other words, the unipolar group inevitably contains some bipolar cases that have not yet declared themselves. Despite this limitation, the division into unipolar and bipolar cases is a useful classification because it has implications for treatment, particularly that of bipolar depression.

Clinical features

Mania

The central features of the syndrome of mania are *elevation of mood*, *increased activity*, and *self-important ideas*.

Mood

When the mood is elevated, the patient appears cheerful and optimistic, and may have a quality described

by earlier writers as ‘infectious gaiety’. However, other patients are irritable rather than euphoric, and this irritability can easily turn to anger. The mood often varies during the day, although not with the regular ‘diurnal’ rhythm that is characteristic of many severe depressive disorders. In patients who are elated, it is not uncommon for high spirits to be interrupted by brief episodes of *depression*.

Table 10.1 Epidemiology of bipolar I and unipolar (depressive) disorder

	Bipolar I disorder	Unipolar disorder
Lifetime risk	About 1%	About 15%
Sex ratio (M:F)	1:1	1:2
First-degree relatives		
Lifetime risk for bipolar disorder	About 10%	About 2%
Lifetime risk for unipolar disorder	20–30%	20–30%
Average age of onset	18 years	27 years
Patients with recurrence	90%	80%
Average number of episodes	10	4

Appearance and behaviour

The appearance of patients often reflects their prevailing mood. Their clothing may be brightly coloured and ill assorted. When the condition is more severe, the patient's appearance is often *untidy and dishevelled*. Manic patients are overactive. Sometimes the persistent overactivity leads to *physical exhaustion*. Manic patients start many activities but leave them unfinished as new ones attract their attention. Appetite is increased, and food may be eaten greedily with little attention to conventional manners. Sexual desires are increased, and sexual behaviour may be uninhibited and quite out of character. Precautions against pregnancy may be neglected. Sleep is often reduced. Patients wake early, feeling lively and energetic, and often get up and busy themselves noisily, to the surprise (and sometimes, the annoyance) of other people.

Speech and thought

The speech of manic patients is often *rapid and copious* as thoughts crowd into their minds in quick succession. When the disorder is more severe, there is *flight of ideas* (see Chapter 1), with such rapid changes that it is difficult to follow the train of thought. However, the links are usually understandable if the speech can be recorded and reviewed. This is in contrast to thought disorder in schizophrenia, where changes in the flow of thought may not be comprehensible even on reflection.

Expansive ideas are common. Patients believe that their ideas are original, their opinions important, and their work of outstanding quality. Many patients become *extravagant*, spending more than they can afford

(e.g. on expensive cars or jewellery). Others make reckless decisions to give up good jobs, or embark on plans for ill-considered and risky business ventures.

Sometimes these expansive themes are accompanied by *grandiose delusions*. Some patients may believe that they are religious prophets or destined to advise statesmen about major issues. At times there are *delusions of persecution*, when patients believe that people are conspiring against them because of their special importance. Delusions of reference and passivity feelings also occur. Schneiderian first-rank symptoms (see Box 11.2) have been reported in around 10–20% of manic patients. Neither the delusions nor the first-rank symptoms last for long, most of them disappearing or changing in content within a period of days.

Perceptual disturbances

Hallucinations occur. These are usually consistent with the mood, taking the form of voices speaking to the patient about their special powers or, occasionally, of visions with a religious content.

Other features

Insight is invariably impaired in more severe manic states. Patients see no reason why their grandiose plans should be restrained or their extravagant expenditure curtailed. They seldom think of themselves as ill or in need of treatment.

Most patients can exert some control over their symptoms for a short time, and many do so when the question of treatment is being assessed. For this reason it is important to obtain a history from an informant whenever possible. Henry Maudsley (1879, p. 398) expressed the problem well:

Just as it is with a person who is not too far gone in intoxication, so it is with a person who is not too far gone in acute mania; he may on occasion pull his scattered ideas together by an effort of will, stop his irrational doings and for a short time talk with an appearance of calmness and reasonableness that may well raise false hopes in inexperienced people.

Manic stupor

In this unusual disorder, patients are *mute* and *immobile*. Their facial expression suggests elation, and on recovery they describe having experienced a rapid succession of thoughts typical of mania. The condition is rarely seen now that active treatment is available for mania. Therefore, an earlier description by Kraepelin (1921, p. 106) is of interest:

The patients are usually quite inaccessible, do not trouble themselves about their surroundings, give no answer, or at

most speak in a low voice [...] smile without recognizable cause, lie perfectly quiet in bed or tidy about at their clothes and bedclothes, decorate themselves in an extraordinary way, all this without any sign of outwards excitement.

On recovery, patients can remember the events that occurred during their period of stupor. The condition may begin from a state of manic excitement, but at times it is a stage in the transition between depressive stupor and mania.

Criteria for manic episodes in ICD-11 and DSM-5

The symptoms that are required to make a diagnosis of 'manic episode' in ICD-11 are listed in Box 10.1. The criteria for manic episode in DSM-5 are very similar, although the number of manic symptoms required for diagnosis is specified more precisely. In both diagnostic systems, patients are required to have *both* altered mood (euphoric, expansive, or irritable) *and* increased activity, or the subjective experience of increased energy. Also, in both DSM-5 and ICD-11 manic symptoms that occur during treatment with antidepressant medications and persist at a syndromal level despite the antidepressant being stopped are regarded as meeting criteria for a manic *episode*, rather than being coded as a drug-induced manic *illness*.

DSM-5 and ICD-11 note that it is common for patients with mania present with *psychotic symptoms*, in which case the clinical picture described in Box 10.1 is typically more severe, with inflated self-esteem and grandiose ideas developing into *grandiose delusions*. At the same time irritability and suspiciousness may result in *delusions of persecution*. Sustained physical activity and excitement may result in aggression or violence, and neglect of eating and drinking and personal hygiene can lead to a dangerous state of dehydration and self-neglect.

Criteria for hypomanic episodes in ICD-11 and DSM-5

Hypomania refers to a state of elevated mood that is of lesser extent than mania. The criteria in ICD-11 and DSM-5 are similar:

- There is persistent mild elevation of mood for at least several days (in DSM-5, at least four days) with a subjective sense of increased energy and activity. In some patients, irritability of mood may predominate.
- There is increased sociability, talkativeness, and over-familiarity, increased sexual energy, and decreased need for sleep. Distractibility may be present.
- There may impulsive or reckless behaviour and overspending.

Box 10.1 Clinical features of a manic episode in ICD-11

1. Euphoria, expansiveness or irritability, *and* increased activity or subjective experience of increased energy
2. Plus 'several' of the following symptoms:
 - a. increased talkativeness or pressured speech
 - b. flight of ideas
 - c. increased self-esteem or grandiosity
 - d. decreased need for sleep
 - e. distractibility
 - f. Impulsive reckless behaviour
 - g. Increase in sexual drive, sociability, or goal-directed activity.

The episode should last for at least one week (unless shortened by treatment) and should cause significant impairment in functioning, require treatment to prevent harm to self or others, or is accompanied by delusions or hallucinations.

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>

- The mood disturbance, although associated with an unequivocal change in function, which is observable to others, is not sufficiently severe to cause marked impairment in social or occupational activities, or to necessitate hospital admission.
- Psychotic features are absent.

Other clinical presentations of bipolar disorder

Mixed mood (affective) states

The diagnostic features of mixed affective states were described in Chapter 9, where was noted that depressive and manic symptoms sometimes occur at the same time. Patients who are overactive and over-talkative may be having profoundly depressive thoughts. In other patients, mania and depression follow each other in a sequence of rapid changes—for example, a manic patient may become intensely depressed for a few hours and then return quickly to his manic state. These changes were mentioned in early descriptions of mania by psychiatrists such as Wilhelm Griesinger (1817–1868), and have been re-emphasized in recent years because they appear to predict a better response to certain mood stabilizers, such as valproate.

Rapid cycling disorders

Some bipolar disorders recur regularly, with intervals of only days or weeks between episodes. In the nineteenth century these regularly recurring disorders were designated *folie circulaire* (circular insanity) by French psychiatrist Jean-Pierre Falret (1794–1870). At present, the frequent recurrence of mood disturbance in bipolar patients is usually termed *rapid cycling disorder*. These recurrent episodes may be depressive, manic, or mixed. The main features are that recurrence is frequent (by convention at least four distinct episodes a year), and that episodes are separated by a period of remission or a switch to an episode of opposite polarity. A number of clinical features of rapid cycling disorder are important in management and prevention:

- They occur more frequently in women.
- Concomitant hypothyroidism is common.
- They can be triggered by antidepressant drug treatment.

The lifetime risk of rapid cycling in bipolar populations varies between studies, but is probably in the range 25–35%. Rapid cycling may be a temporary phenomenon, and in most patients, it remits within about two years. For a review, see Roosen and Sienaert (2022).

Cyclothymia

The term *cyclothymia* disorder refers to a persistent instability of mood in which there are numerous periods of

mild elation or mild depression that do not meet severity criteria for either major depression or hypomania. It is seen as a milder variant of bipolar disorder. It is not unusual, however, for episodes of more severe mood disorder to supervene in the course of the disorder.

Depression

Depressive episodes are common in the course of bipolar disorder and most patients with bipolar disorder present initially with an episode of major depression. The ability to predict which patients first presenting with depression will eventually develop bipolar illness is currently limited, although family history of bipolar disorder can provide a useful clue. The presence of any *hypomanic or mixed symptomatology* at initial presentation has some predictive value, but the majority of depressed patients who convert to bipolar illness do not have hypomanic symptoms during the initial episodes of depression. Other clinical features associated with subsequent development of bipolar illness include early age of onset and clinical severity, particularly the presence of *psychosis*.

There is a high degree of overlap between the clinical symptomatology of unipolar and bipolar depression; however, psychomotor changes, cognitive impairment, and psychotic features are reportedly more common in patients with bipolar disorder (Vöhringer and Perlis, 2016).

Transcultural factors

The rates of bipolar disorder are relatively consistent across countries, although there is some evidence for an increased prevalence of bipolar disorder in certain minority groups within a country, such as in Black and ethnic minority groups and in immigrants. Moreover, patients from these minority groups are more likely to present with a first episode of mania and prominent psychotic features than a comparable white population with bipolar disorder. The reason for this

difference is not fully understood, but social adversity and exclusion are likely to be relevant factors, particularly in relation to late access to services (Watson and Bhugra, 2020).

The content of grandiose beliefs is strongly influenced by cultural context. However, it is important to determine how far such beliefs are part of normative cultural experience before regarding them as evidence of delusional thinking.

Classification

Both DSM-5 and ICD-11 delineate hypomania from mania on the basis of duration of symptoms, absence of

psychotic features, and lesser degree of social and occupational impairment (see above). A manic episode can

be further subdivided according to severity and whether or not psychotic symptoms are present. In both DSM-5, and ICD-11 the presence of a single episode of mania is sufficient to meet the criteria for bipolar disorder. DSM-5 and ICD-11 also categorize bipolar disorder as follows (Table 10.2):

- Bipolar I, in which *mania* has occurred on at least one occasion.
- Bipolar II, in which *hypomania* has occurred, but mania has not. However, to make the diagnosis of bipolar II disorder an episode of major depression must also have occurred.

The diagnosis of bipolar II disorder is intended to indicate the importance of detecting mild hypomanic episodes in patients who might otherwise be diagnosed as having recurrent major depression. The presence of such episodes may have implications for treatment response. There is debate about where the line should be drawn between unipolar depression and bipolar disorder, because some patients with definite depressive episodes appear to have features of bipolarity but do not meet the full DSM-5 criteria for hypomania. Such patients may, for example, have elevations of mood that last for less than the required four days, or that have little discernible effect on functioning.

In DSM-5, the diagnosis of 'bipolar and related disorder' can be made for such disorders and the reasons why the full criteria fail to be met can be either 'specified' or 'unspecified'. In clinical practice, the term *bipolar spectrum* is also used for these conditions. The pharmacological treatment of bipolar depression differs to some extent from that given to patients with recurrent unipolar depression (see 'Treatment of bipolar depression' below), so making the diagnostic distinction between

Table 10.2 Classification of bipolar disorder

ICD-11	DSM-5
Manic episode	Hypomanic episode
Hypomanic episode	Manic episode
Mixed episode	Mild
	Moderate
	Severe
	Severe with psychosis
Bipolar Type 1 and type II disorders	Bipolar I and bipolar II disorders
Currently hypomanic	Current (or most recent episode)
Currently manic	Hypomanic
Currently depressed	Manic*
Currently mixed	Depressed
In remission	Mixed*
Cyclothymic disorder	Cyclothymic disorder

* Excludes bipolar II. In ICD-11, manic, mixed and depressive states are classified 'with or without' psychotic symptoms.

Source: *International Classification of Diseases, Eleventh Revision (ICD-11)*. Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)*. Copyright (2013), American Psychiatric Association.

unipolar and bipolar depression can have important implications for management. However, it is not clear whether depressed patients in the 'bipolar spectrum' have a better outcome when treated as having bipolar depression rather than unipolar depression (for a discussion, see Goodwin et al., 2016).

Differential diagnosis of bipolar disorder

Manic disorders have to be distinguished from *schizophrenia*, *organic brain disease* involving the frontal lobes (including brain tumour and HIV infection), and *states of brief excitement* induced by amphetamines and other illegal drugs.

Schizophrenia

The distinction from schizophrenia can be most difficult. Auditory hallucinations and delusions, including

some that are characteristic of schizophrenia, such as delusions of reference, can occur in manic disorders. However, these symptoms usually change quickly in terms of content, and seldom outlast the phase of overactivity. When there is a more or less equal mixture of features of the two syndromes, the term *schizo-affective* is often used. This term is discussed further in Chapter 11. Further clues to diagnosis can often be elicited by a careful personal and family psychiatric history.

Organic brain disorder and drug misuse

An organic brain lesion should always be considered, especially in middle-aged or older patients with expansive behaviour and no past history of affective disorder. In the absence of gross mood disorder, extreme social disinhibition (e.g. urinating in public) strongly suggests *frontal lobe pathology*. In such cases, appropriate neurological investigation is essential. In younger adults, infection with HIV or head injury may lead to the manifestation of mania.

The distinction between mania and excited behaviour caused by *drug misuse* depends on the history and on an examination of the urine for drugs, which is needed before treatment with psychotropic drugs is started. Drug-induced states usually subside quickly once the patient is in hospital. However, it should be remembered that a significant proportion of patients who have bipolar disorder misuse alcohol and other drugs.

Recurrent depression

The distinction between bipolar disorder and recurrent unipolar depression depends on the eliciting of symptoms of mania or hypomania in the past history. However, distinguishing recurrent depression from bipolar II disorder can be particularly challenging because brief periods of hypomania can be difficult to identify or

may not be remembered clearly by the patient. Moreover, at the time of diagnosis, such symptoms may not yet have declared themselves and it is only subsequently, with the development of mania or hypomania, that the bipolar disorder can be identified. Another point is that patients with bipolar II disorder, while commonly seeking treatment for depressive episodes, tend not to regard spells of hypomania as being illness.

Borderline personality disorder

One of the diagnostic features of borderline personality disorder is marked affective instability, which may at times be difficult to distinguish from rapid cycling bipolar disorder. A family history of bipolar disorder may be a useful pointer. Also, shifts in mood in borderline personality disorder can be very rapid over hours and days, which is less common in bipolar disorder. Classic symptoms of mania, such as increased energy and grandiosity, are not usually present. Finally, mood disturbances in borderline personality are usually triggered by interpersonal issues such as the fear of loss or rejection, while in rapid cycling bipolar disorder such psychosocial triggers are not prominent.

It has been estimated that between 10–20% of patients have both bipolar disorder and borderline personality disorder. Unsurprisingly such patients have increased an increased burden of psychopathology and disability (Zimmerman et al., 2021).

Epidemiology

The epidemiology of bipolar disorder depends on the diagnostic criteria used to define it. Including patients in the bipolar spectrum increases rates of bipolar disorder at the expense of recurrent major depression. In addition, as noted above, distinguishing bipolar disorder from recurrent unipolar depression can be challenging.

Community surveys (using ICD-10 or DSM-IV criteria) in industrialized countries have suggested that:

- The lifetime risk for bipolar I disorder is around 1.0%.
- The lifetime risk for bipolar II disorder is around 1.1%.
- In bipolar I disorder the prevalence in men and women is approximately equal. In bipolar II disorder

the excess of women is similar to that seen in unipolar depression.

- A somewhat larger group of people (2.4%) have 'sub-threshold' symptoms of bipolar disorder.
- The mean age of onset of bipolar disorder is about 17 years in community studies.
- Bipolar disorder is highly comorbid with other disorders, particularly anxiety disorders and substance misuse, as well as general medical conditions such as cardiovascular disease.

For a review of the epidemiology of bipolar disorder, see Kessing (2020).

Aetiology

Overview

In earlier studies of aetiology, particularly of depression, the distinction between recurrent unipolar depression and depression in the context of bipolar disorder was not always made. The fact that first-degree relatives of patients with bipolar disorder have an increased risk of recurrent unipolar depression suggests that there is some aetiological overlap between the two conditions. Alternatively, it may be that such relatives make up one particular subtype of the heterogenous disorder which is currently called major depression.

One of the most striking features of bipolar disorder is its high heritability and, accordingly, much recent work has employed molecular genetic techniques in an attempt to implicate genetic loci relevant to pathophysiology. In addition, advances in structural and functional imaging have allowed a clearer description of the brain basis of bipolar disorder and its possible distinction from recurrent unipolar depression. However, none of the brain changes yet identified is sufficiently reliable and distinct to be used as a diagnostic biomarker. In addition, increasing knowledge of possible pathophysiology has not yet led to new treatments, which continue to be developed on an empirical basis.

Genetic causes

Family and twin studies

The risk of both bipolar and unipolar mood disorders is increased in first-degree relatives of bipolar probands. Relatives of bipolar probands also have increased risks of *schizoaffective disorder*. Twin studies have shown that the concordance rate for mood disorder in the monozygotic co-twin of a proband with bipolar disorder is around 60%, but for dizygotic twins the rate is only about 20%. This indicates an important degree of heritability, which has been estimated at around 60% (Geschwind and Flint, 2015).

Mode of inheritance

The familial segregation of bipolar disorder does not fit a simple Mendelian pattern. Overall, the genetic liability to the condition results largely from the combined action of multiple genes of modest, or small, effect—so-called *polygenic inheritance*. Rare variants with larger effects may also contribute. A concordance rate of monozygotic twins for bipolar disorder of around 60%

indicates that environmental factors are also important in pathophysiology.

Molecular genetics

Genome-wide association studies (GWAS) have identified risk loci for bipolar disorder. A large analysis including over 40,000 patients identified 64 loci (Mullins et al., 2021), with the genes implicated involved in synaptic signalling pathways, including the voltage-gated calcium channel *CACNA1C* (Harrison et al., 2020b). There is a substantial overlap of these risk loci with those affecting schizophrenia risk and, to a lesser extent, major depression and some other disorders. The loci together explain some of the heritability of bipolar disorder but many more remain to be identified. In addition to the common genetic variants identified by GWAS, there are a few very rare gene mutations which, if present, confer a much greater risk of bipolar disorder. Finally, in contrast to schizophrenia, in bipolar disorder there is not a clear excess of copy number variants, providing one of the aetiological differences between the two conditions. For a review, see Gordovez and McMahon (2020).

Environment

The concordance rates for bipolar disorder in monozygotic twins noted above show that environmental factors are likely to be involved in the pathophysiology of bipolar illness. Most of the environmental factors identified thus far do not seem specific for bipolar disorder but rather confer risk of psychiatric disorder more generally.

For example, compared to healthy controls, patients with bipolar disorder report higher rates of childhood adversity, including sexual abuse. Childhood sexual abuse is known to be linked to the development of psychosis generally in adulthood and in one study in bipolar patients was associated with an increased risk of auditory hallucinations, but not delusions (Uptegrove et al., 2015). Generally, patients with bipolar disorder who report childhood abuse have a worse course of illness with greater symptom severity and number of episodes as well as increased risk of suicide attempts and rapid cycling (Alloy et al., 2020).

Life events have been shown to precipitate episodes of both depression and mania in bipolar patients, although the effect appears more pronounced in early stages of the illness, suggesting a possible ‘kindling’ effect. Negative life events can trigger episodes of both

depression and mania, although mania can also apparently be precipitated by life events associated with goal attainment. One way that life events might lead to illness is through effects on circadian rhythms, for example, decreasing sleep and disrupting regular cycles of sleep and activity (Alloy et al., 2020).

Neurobiological approaches to aetiology

Neurochemistry

An influential hypothesis links *excessive dopamine activity* to the pathophysiology of schizophrenia (see Chapter 11). It has been proposed that manic states may also be attributable to dopamine overactivity because drugs of abuse, particularly those acting to increase central dopamine activity, are associated with many of the symptoms of acute mania, such as elevated mood, decreased need for sleep, talkativeness, and the unguarded pursuit of rewarding behaviours. Also, drugs that block dopamine receptors are commonly employed in the acute treatment of mania.

Studies of dopamine metabolism and function have provided little direct evidence for overactivity of dopamine pathways in non-psychotic patients with manic illness. However, manic patients with *psychotic symptoms* show increased dopamine synthesis in the striatum similar to patients with schizophrenia (Ashok et al., 2017).

Many of the drugs used in the longer-term management of bipolar disorder are anticonvulsants, which alter *brain glutamate* levels. Several magnetic resonance spectroscopy (MRS) investigations have been carried out in bipolar patients at various stages of illness. There are suggestions that glutamate levels might be increased in bipolar patients, which makes an interesting point of contrast with findings in major depression (see Chapter 9). The reliability of this finding and its possible utility in diagnosis require further study (Chabert et al., 2022).

Endocrine abnormalities

Cortisol

As in major depression, disturbances of the hypothalamic–pituitary–adrenal (HPA) axis with hypersecretion of cortisol have been reported in patients with bipolar disorder. However, the patients studied have usually been taking medication, making interpretation difficult. There have been suggested links between excessive cortisol secretion in bipolar disorder and the presence of cognitive impairment, with beneficial effects on cognition being reported following treatment with glucocorticoid antagonists. It is also well recognized that

administration of exogenous corticosteroids in the treatment of general medical conditions can give rise to an acute manic illness. For a review see Young and Juruena (2021).

Oestrogen and progesterone

The immediate postpartum period is a time of heightened risk for the development of acute psychotic illness in mothers—usually presenting as a form of bipolar disorder (see Chapter 22). There has therefore been great interest in whether this might be attributable to the abrupt fall in circulating levels of sex steroids that occurs at this time; however, no correlation has been identified. It is possible that women vulnerable to postpartum psychosis are particularly sensitive to such hormonal fluctuations (Jones, 2020).

Sleep and circadian rhythm

Sleep disturbances are a hallmark of bipolar disorder. Decreased sleep is an important symptom of mania and may be an early sign of an impending manic episode. In some patients sleep deprivation may trigger mania. Conversely, bipolar depression is associated with hypersomnia. Decreased sleep efficiency can persist into remission in bipolar patients and may be a risk factor for relapse (Meyer et al., 2020).

Disruptions of circadian rhythm occur in all phases of bipolar disorder, including remission, and a delayed sleep phase is common. Jet travel across time zones, particularly west to east, has been associated with the development of mania, though here the effect of sleep deprivation should be considered. Treatments aimed to improve sleep and stabilize circadian rhythms may be useful in the management of bipolar disorder. For a review, see Ahmad et al. (2021).

Brain imaging

Structural brain imaging

Changes in brain volume measured by magnetic resonance imaging (MRI) in bipolar disorder have been inconsistent, and stage of illness and drug treatment, particularly the use of lithium, appear to be important confounding factors. There are reliable reductions in grey matter and white matter volume in prefrontal cortex in bipolar disorder. These reductions are associated with illness duration and are greater in bipolar I than bipolar II disorder. Studies using diffusion tensor imaging (see Chapter 5) in bipolar patients have revealed abnormalities in white matter tracts, perhaps representing abnormal myelination or orientation of axons in predominantly frontal and temporal white matter regions (see Phillips and Drevets, 2020).

Similarly to patients with major depression (see Chapter 9), there is evidence that neuropathological changes occur in patients with bipolar disorder. For example, there is a reduction in glial cell density in subgenual anterior cingulate cortex, and some loss of gamma-aminobutyric acid (GABA) interneurons in prefrontal cortex (Harrison et al., 2020a). The mechanisms responsible for these changes are unclear.

Functional imaging

As in patients with unipolar depression, bipolar depressed patients show elevated neural responses in the amygdala and abnormally reduced activity in regulatory lateral and medial prefrontal cortical regions in response to negative emotional cues. Unlike unipolar depressed patients, euthymic bipolar patients show elevated amygdala responses and decreased activity in cortical regions to positive emotional stimuli such as happy faces. Euthymic bipolar patients also demonstrate enhanced reactions in ventral striatal and orbitofrontal ‘reward’ circuitry during tasks that engage reward processing (Phillips and Drevets, 2020). Generally, functional imaging studies show a greater range of neural dysregulation in bipolar disorder than in unipolar depression, with abnormalities encompassing not only neural responses to negative emotional stimuli but also positive stimuli, including the experience of reward.

Neuropsychological changes

Cognitive deficits are present in bipolar disorder during both acute illness and periods of euthymia. Deficits are apparent in first-episode patients, suggesting that they may form an integral part of the illness. In comparison to patients with schizophrenia, the deficits are relatively modest but cover a wide range of neuropsychological domains, including *executive function*, *verbal*

memory, *attention*, and *visual spatial working memory*. The deficits appear similar in bipolar I and bipolar II patients (King et al., 2019). At the moment there is no clear evidence that cognitive deficits in bipolar disorder increase through the course of the illness (Ehrlich et al., 2022), though patients are at greater risk of dementia (see below).

Conclusions

The *predisposition* to develop bipolar disorder has a major genetic contribution and the relevant genes, which may number many thousands, are starting to be identified. If such genes converge on a number of discrete biological pathways, it will greatly aid elucidation of aetiological mechanisms and may provide clues to new treatments.

In terms of *environmental factors*, adverse early experiences, particularly sexual abuse, may also predispose to bipolar disorder as it does to many other psychiatric conditions. Early episodes of illness are often precipitated by life events, but these may become less important as the illness progresses. The psychosocial environment can modify the course of bipolar illness and there may well be an important role for factors that disrupt sleep–wake activity.

Brain imaging studies show functional abnormalities consistent with disordered emotional regulation and reward processes; unlike the abnormalities seen in major depression, these changes also involve exaggerated functional responses to positive emotional stimuli. Modest but widespread neuropsychological deficits can be seen in bipolar disorder, and these appear to extend into periods of euthymia.

Course and prognosis

From recent hospital and population-based studies, the following general conclusions can be drawn (Kessing, 2020):

- Bipolar disorder usually begins as depression, with the first manic episode manifesting about five years later.
- The length of most manic episodes is between three and six months.
- At least 90% of patients with mania experience further episodes of major mood disturbance.
- Over a 25-year follow-up, on average bipolar patients experience about ten further episodes of major mood disturbance.
- The interval between episodes becomes progressively shorter with both age and the number of episodes.
- Over long-term follow-up, patients with bipolar disorder experience mood-related symptomatology of varying severity for about half of the time. In both bipolar I and bipolar II disorder, this most commonly takes the form of depressive symptoms.

- Occupational and social disability is common in bipolar disorder. This is related as much to cognitive dysfunction as affective symptomatology.
- People with bipolar disorder have a two- to threefold increased risk of developing dementia. This risk may be decreased by long-term lithium treatment.

Mortality of bipolar disorder

Mortality is significantly increased in patients with bipolar disorders. The high mortality in bipolar disorder is caused not only by suicide and accidents but also general medical conditions such as cardiovascular disease and the secondary consequences of comorbid substance misuse, including

smoking. A study of 2,200 Danish patients with bipolar disorder found a reduction in life expectancy of about *13 years in men* and *9 years in women*. About two thirds of this reduction was accounted for by natural causes, with the remainder being due to suicide and accidents. The absolute risk of committing suicide within three to four decades after the first psychiatric contact has been estimated to 8% in men and 5% in women (Kessing, 2020).

Cardiovascular disease is an important cause of increased mortality in bipolar patients, which highlights the importance of attending to the medical needs of this population. The metabolic effects of extended treatment with psychotropic medication, particularly antipsychotic drugs, needs careful assessment.

Treatment of mania

General measures

The treatment of mania presents a formidable clinical challenge, which often tests the management skills of the psychiatric team. Drug treatment plays a pivotal role in the management of mania, and has the aim of reducing physical and mental overactivity, improving features of psychosis, and preventing deterioration in health caused by exhaustion, sleep deprivation, and poor fluid intake. It is worth noting that before the advent of modern drug treatment the mortality of mania in the hospital setting was over 20%; nearly 50% of these patients died from exhaustion (Derby, 1933).

Medication

Typical antipsychotic drugs

Several randomized controlled trials have shown the efficacy of *chlorpromazine* and *haloperidol* in treating mania, whether or not patients have clear psychotic features. However, the use of typical antipsychotic drugs has limitations. Manic patients often receive high doses, and may be particularly susceptible to extrapyramidal side effects. In addition, conventional antipsychotic drugs do not protect against the depressive downswings that can follow resolution of a manic illness.

Atypical antipsychotic drugs

Because of their improved tolerability profile, atypical antipsychotic agents are now used predominantly in the treatment of mania. Placebo-controlled trials

support the efficacy and acceptability of aripiprazole, olanzapine, quetiapine, and risperidone in patients with mania (Kishi et al., 2021). However, it should be noted that more severely ill patients may not be included in placebo-controlled trials, which could limit the generalizability of these findings.

Lithium

Five placebo-controlled trials have shown that lithium is effective in the acute treatment of mania (Cipriani et al., 2011). In the only study that compared lithium with both placebo and an active comparator, Bowden et al. (1994) found a response rate of 49% for lithium, 48% for valproate, and 25% for placebo. In general, lithium is as effective as an antipsychotic medication, but its onset of action is slower. Moreover, in highly active states, antipsychotic medication may be preferable (Goodwin et al., 2016). Prominent depressive symptoms and psychotic features predict a poorer response to lithium alone, as does a rapid cycling disorder.

Carbamazepine

Assessment of the efficacy of carbamazepine in acute mania has been limited by problems with regard to study design. These include small numbers, the use of adjunctive medications, and mixed diagnostic groupings. However, more recent placebo-controlled trials of an extended-release form of carbamazepine found a clinically significantly antimanic effect (Goodwin et al., 2016). Carbamazepine induces drug-metabolizing enzymes in the liver, which can lead

to lower plasma levels of concomitantly administered drugs.

Valproate

Several randomized controlled trials have shown that valproate possesses antimanic activity that is greater than that of placebo and equivalent to that of lithium (Kishi et al., 2021). Valproate may be more effective than lithium in the subgroup of patients with prominent dysphoric symptoms and rapid cycling.

One advantage of valproate is that it has an earlier onset of activity than other mood stabilizers. For example, with 'valproate loading' (20 mg/kg/day) the onset of the antimanic effect can occur as early as within one to four days. This is probably because the tolerability of valproate allows rapid dose escalation, whereas lithium and carbamazepine generally need to be introduced more gradually.

Benzodiazepines

Benzodiazepines are useful *adjuncts* in the treatment of mania, because they can rapidly diminish overactivity and restore sleep. They have been used as sole therapy in the treatment of mania, but this carries a risk of disinhibition. Their most useful role is as an adjunct to mood stabilizers, because the latter drugs can take several days to become effective. Benzodiazepines can also be given in combination with antipsychotic drugs, because this lowers the dose of antipsychotic drug required to calm agitated and overactive patients. The benzodiazepines that are usually employed are the high-potency agents lorazepam and clonazepam. Use should be on an 'as needed' basis and for as short a time as possible to minimize the risk of tolerance and dependence (Goodwin et al., 2016).

Electroconvulsive therapy

In the past, ECT has been widely used to treat mania, although there have been only two prospective trials

against pharmacological treatments in the modern era. In one of these trials, bilateral ECT was found to be superior to lithium (Small et al., 1988). In the other study, unilateral and bilateral ECT were compared with a combination of lithium and haloperidol in patients who had not responded to antipsychotic drugs alone. The response rate of these patients to ECT (13 of 22) was greater than that to combined drug treatment (none of five). The efficacy of unilateral and bilateral ECT did not differ (Mukherjee et al., 1994). Five other studies have compared different modalities of ECT administration where bitemporal, bifrontal, and right unilateral electrode placements have been found equally effective.

Overall, the response rate to ECT in patients with mania is about 80%, even in those unresponsive to medication. ECT is also effective in *mixed affective states*. Previously, ECT was often given at shorter intervals in the treatment of mania than in the treatment of depression, but there is no evidence that this regimen is necessary or that it speeds up the treatment response. For a review of the use ECT in mania, see Elias et al. (2021).

Continuation treatment of mania

It is a common clinical experience that too rapid a reduction in doses of drug treatment can lead to the sudden recrudescence of an apparently treated manic disorder. Since a manic episode can last for several months, it seems prudent to continue some form of medication for at least this period. There are numerous studies involving atypical antipsychotic drugs which show that, compared with placebo, continuation treatment decreases the risk of manic relapse (Goodwin et al., 2016). Patients who have been severely ill may well be taking both a mood stabilizer and an antipsychotic agent. Because of the adverse effects of antipsychotic drugs, it will often be appropriate to withdraw this form of treatment slowly first.

Treatment of bipolar depression

For many years bipolar depression was treated similarly to unipolar depression, and, for a number of major issues (e.g. clinical assessment and general psychological support), this still holds. However, in recent years it

has become apparent that best pharmacological treatment of depression occurring in the context of bipolar disorder can differ significantly from that employed in unipolar depression.

Medication

Antidepressant drugs

Treating depression in the context of a bipolar disorder can be problematic because standard antidepressant treatments have a number of disadvantages, which include:

- a possibly lower efficacy than in unipolar depression
- a risk of inducing mania
- a risk of inducing rapid cycling.

Whether conventional antidepressant medication is efficacious in bipolar depression has been controversial. However, more recent meta-analyses indicate therapeutic benefit for some antidepressant agents. For example, in a network meta-analysis, Bahji et al. (2020) found that fluoxetine and venlafaxine, as well as the older MAOI tranylcypromine, appeared effective in the treatment of bipolar depression. However, compared to studies in unipolar depression, trials are few in number and involve patients with bipolar I and II disorders as well as old and newer antidepressant drugs. It may be that antidepressants as monotherapy are more clinically useful in bipolar II depression than in bipolar I (McIntyre et al., 2020).

It is generally agreed that treatment with tricyclic antidepressants increases the risk of manic switching in bipolar depressed patients. However, the risk with selective serotonin reuptake inhibitors (SSRIs) seems less and only slightly greater than that seen with placebo. Venlafaxine is thought more likely than SSRIs to destabilize mood. Observational studies tend to find higher switching rates with antidepressant treatment than controlled trials (Allain et al., 2017).

Lithium

Lithium may have a role in the *prevention* of depression in bipolar disorder (see page 243), but evidence for its effectiveness in acute bipolar depression is limited and is based on older trials with unsatisfactory designs. More recent placebo-controlled trials in bipolar depression have found lithium less efficacious than other agents, such as certain atypical antipsychotic drugs (Bahji et al., 2020) (Table 10.3).

Atypical antipsychotic drugs

There is growing evidence for the short- and longer-term efficacy of certain *atypical antipsychotic drugs* such as *quetiapine* in the treatment of bipolar depression, and similar effects have been reported for *lurasidone* and *cariprazine*. The effects of olanzapine as a monotherapy

Table 10.3 Meta-analysis of placebo-controlled trials of selected pharmacological treatments in bipolar depression

Drug	Drug:placebo remission odds ratio	95% CI
Olanzapine + fluoxetine	2.3	1.48–3.57
Quetiapine	1.93	1.50–2.49
Lurasidone	1.78	1.04–3.03
Olanzapine	1.66	1.07–2.58
Cariprazine	1.63	0.96–2.76
Lamotrigine	1.54	1.07–2.21
Lithium	1.27	0.65–2.47
Aripiprazole	0.83	0.49–1.41

Source: Bahji, A., Ermacora, D., Stephenson, C., Hawken, E. R., & Vazquez, G. (2020). Comparative efficacy and tolerability of pharmacological treatments for the treatment of acute bipolar depression: A systematic review and network meta-analysis. *J Affect Disord*, 269, 154–184. Copyright (2020), Elsevier.

in bipolar depression seem slightly less than those of quetiapine, although in combination with fluoxetine it appears at least as effective (Table 10.3). In contrast, aripiprazole does not appear useful in bipolar depression (Bahji et al., 2020).

Anticonvulsants

Some *anticonvulsant mood stabilizers* are helpful in the treatment of bipolar depression, with positive evidence from randomized trials for *lamotrigine* and *valproate*. Both these drugs would be expected to be free from the risk of inducing mania or rapid cycling. A recent large pragmatic trial found benefit at both 12 and 52 weeks for the addition of lamotrigine to ineffective quetiapine therapy. At 12 weeks remission in the lamotrigine group was 31% versus 16% in the placebo group where the number needed to treat (NNT) was 6.6 (Geddes et al., 2016). An interesting observation, made possible by the factorial design of the trial, was that coadministration of *folic acid* appeared to antagonize the therapeutic effect of lamotrigine.

Electroconvulsive therapy

For resistant depression in bipolar patients, *ECT* can be of benefit. There are few randomized studies of patients specifically diagnosed with bipolar illness. However, in a recent systematic review of both randomized and

observational studies, Bahji et al. (2019) found similar responses rates to ECT (77% versus 74%) in patients with bipolar depression as in those with unipolar illness. Mania is a possible adverse effect of ECT treatment in bipolar depression though rates of switching from the various

studies are variable. How far the rate of manic switching during ECT in bipolar depressed patients is increased over that occurring spontaneously is not clear (Kessler, 2019).

For a review of the treatment of bipolar depression, see McIntyre et al. (2020).

Longer-term treatment of bipolar disorder

Medication

Lithium

There is substantial evidence for the efficacy of lithium in the maintenance treatment of recurrent mood disturbances in patients with bipolar disorders. In a systematic review of controlled trials, Geddes et al. (2003) found that lithium was significantly more effective than placebo in preventing relapses of all mood disorders (mean relative risk, 0.65; 95% confidence intervals [CI], 0.5–0.84). However, although lithium was clearly effective in preventing manic relapse, its effect in preventing depressive episodes was more equivocal (mean relative risk, 0.72; 95% [CI], 0.49–1.07).

In general, about 50% of bipolar patients who are treated with lithium respond well. The following are predictors of a *relatively poorer* response to lithium maintenance treatment (Hui et al., 2019):

- rapid-cycling disorders or chronic depression
- mixed affective states
- alcohol and drug misuse
- psychotic features
- absence of family history of bipolar illness.

There is also evidence that use of lithium in patients with recurrent mood disorders is associated with a *significant reduction in mortality from suicide*. This effect is most often seen in dedicated lithium clinics, and the mechanism and specificity of the association require further study. Nevertheless, it is clinically important that the carefully supervised use of lithium is associated with a lowered risk of suicidal behaviour. This effect is not necessarily shared by other mood stabilizers (Del Matto et al., 2020).

Valproate

While valproate has been commonly used in the maintenance treatment of bipolar disorder, the evidence for its effectiveness is less complete than that

of lithium. Bowden et al. (2000) randomized 372 patients to one year of maintenance treatment with valproate, lithium, and placebo. There were no significant differences between the three treatment groups on the primary outcome measure of time to recurrence of any mood disorder. However, the number of patients who left the study because of the occurrence of a severe mood episode was significantly less among those taking valproate compared with those on placebo. Patients who were taking valproate experienced more tremor, alopecia, and weight gain than those on placebo.

In a pragmatic study in bipolar patients, the relapse rate on lithium (59%) was significantly less than that with valproate (69%) over 24 months. The combination of lithium and valproate was associated with the lowest relapse rate (54%) (BALANCE Study Group, 2010). A network meta-analysis suggested that while valproate prevented the recurrence of manic and hypomanic episodes, its effectiveness at preventing depression was no greater than placebo (Kishi et al., 2021). Increasing concern about the teratogenic effects of valproate (page 665) will probably lead to a diminution in its use.

Lamotrigine

Prospective placebo-controlled trials have shown that lamotrigine has prophylactic effects in patients with bipolar illness. Although it may have some modest benefit in the prevention of mania, lamotrigine has a clearer prophylactic effect against depression. Its profile of activity therefore contrasts with those of lithium and valproate, which are more effective in preventing mania (Kishi et al., 2021).

Carbamazepine

Although the number of patients studied in randomized controlled studies is relatively few, carbamazepine appears to have efficacy in the prophylaxis of bipolar disorder. However, it seems to be less effective than lithium in 'classical' bipolar illness (Goodwin et al., 2016; Kishi et al., 2021). Therefore, carbamazepine can be considered

in patients who respond poorly to lithium, particularly those with rapid-cycling disorders. It can be given alone or in combination with lithium. Carbamazepine can induce hepatic metabolizing enzymes, leading to lower levels of co-administered drugs. Therefore, it should be used with caution in combination treatments.

Antipsychotic drugs

Patients with bipolar disorder have sometimes been maintained on *typical antipsychotic drugs*, usually given in addition to mood-stabilizing agents. It is usually wise to minimize this form of treatment where possible, because conventional antipsychotic drugs do not protect against depression, and may be more liable to cause tardive dyskinesia in bipolar patients.

Atypical antipsychotic drugs are also being employed in the longer-term maintenance treatment of bipolar disorder, either as a sole treatment or as an adjunct to mood stabilizers. There is evidence from randomized studies for the efficacy of olanzapine and quetiapine in this respect in the prevention of both depression and mania (Goodwin et al., 2016; Kishi et al., 2021). However, a common design of these studies is to continue antipsychotic treatment in patients who have responded to short-term, open-label treatment, and to compare this with placebo substitution. This sample, enriched for antipsychotic treatment responders, may favour active treatment unduly, and therefore the findings may not be generally applicable to other clinical situations.

Psychotherapy

Studies of structured psychotherapies in bipolar disorder have usually been directed to patients outside acute episodes of mood disturbance. In this situation, therefore, psychotherapy has the aim of improving and sustaining recovery. Psychotherapies are usually administered as ‘adjunctive’ treatment to maintenance pharmacotherapy.

Cognitive behaviour therapy

Psychotherapy has been less studied in bipolar patients, but, theoretically, cognitive behavioural techniques may be valuable in helping patients to accept their illness and the need for medical treatment, as well as improving the management of emotionally stressful situations. Some randomized studies have shown that cognitive techniques combined with education about early signs

of relapse reduced the rates of relapse in bipolar patients (Chatterton et al., 2017).

Family-focused approaches

The mood swings experienced by bipolar patients, particularly mood elevation, can be very challenging for their families; this may lead to high levels of ‘expressed emotion’, which in turn worsen the clinical course. Family-based therapy aims to improve relationships and support for the patient within the family by using psychoeducation as well as training in communication skills and problem solving. In two studies substantial effects of this approach were apparent, with reductions in both hospitalizations and symptomatology by about a third. Provision of psychoeducation appears to be an important component of successful family treatment (Miklowitz et al., 2021).

Interpersonal and social rhythm therapy

We have already seen that mood disturbance in bipolar disorder is almost invariably associated with sleep disruption; however, the converse is also the case, and it is well recognized that sleep deprivation in bipolar patients can trigger episodes of mania. Interpersonal and social rhythm therapy was derived from interpersonal therapy (Chapter 24) and focuses on helping bipolar patients to maintain regular sleep-activity schedules with careful attention to sleep hygiene. This approach appears to be beneficial effect in improving recovery from depressive and manic symptoms (Miklowitz et al., 2021).

Psychoeducation

Psychoeducation focuses on increasing knowledge about bipolar disorder and its treatment. Along with this, psychoeducation fosters active skills training, for example, in monitoring for prodromal mood symptoms, increasing medication adherence, stabilizing the sleep-wake cycle, and avoiding recreational drug use.

Group psychoeducation attempts to deliver such psychotherapy in a cost-effective way by educating groups of bipolar patients in a medical understanding of their illness and the best ways of achieving mood stability. Psychoeducation can also be provided in conjoint and family settings where it has demonstrated beneficial effects on reducing illness recurrence as well as lowering treatment drop-out rates (Miklowitz et al., 2021).

Practical management of bipolar disorder

Acute mania

Assessment

In the assessment of mania, the important steps can be summarized as follows:

- Decide the diagnosis.
- Assess the severity of the disorder.
- Form an opinion about the causes.
- Assess the patient's social resources.
- Judge the effects on other people.

Diagnosis depends on a careful history and examination. Whenever possible, the history should be taken from relatives as well as from the patient, because the latter may not recognize the extent of their abnormal behaviour. Differential diagnosis has been discussed earlier in this chapter. It is always important to remember that mildly disinhibited behaviour can result from frontal lobe lesions (e.g. tumours) as well as from mania. There should be a urine screen for illegal substances.

Severity is judged next. For this purpose it is essential to interview an informant as well as the patient. Manic patients may exert some self-control during an interview with a clinician, and may then behave in a disinhibited and grandiose way immediately afterwards. At an early stage of mania, professionals can easily be misled and may lose the opportunity to persuade the patient to enter hospital before causing themselves long-term difficulties (e.g. owing to ill-judged decisions or unjustified extravagance). Where possible, it is important to identify any life events that may have provoked the onset of manic illness. Some manic episodes follow physical illness, treatment with drugs (especially steroids or antidepressants), or operations. *Sleep deprivation* may trigger mania in some susceptible individuals.

The patient's resources and the effect of the illness on other people are assessed in the ways described in Chapter 9 for depressive disorders. Even the most supportive family will find it extremely difficult to care for a manic patient at home for more than a few days unless the disorder is exceptionally mild. The patient's responsibilities with regard to the care of dependent children, or at work, should always be considered carefully.

Management of mania

Hospital admission

The first decision is whether to *admit the patient to hospital*. In all but the mildest cases, admission is nearly always advisable to protect the patient from the consequences of their own behaviour, though intensive treatment in the community may also be possible. If the disorder is more severe, compulsory admission is likely to be needed.

General clinical management

Development of a *therapeutic alliance* with the manic patient is an important goal of treatment, although lack of insight into the condition and anger at involuntary detention can make this a testing and time-consuming process. It is best to use an understanding, yet firm, approach that minimizes confrontation. For example, it is often possible to avoid an argument by taking advantage of the manic patient's easy distractibility; instead of refusing demands, it is better to delay until the patient's attention switches to another topic that they can be encouraged to pursue. In hospital, nursing staff play a key role in this kind of management, where they attempt to provide manic patients with a low-stimulus and safe environment that limits confrontations and impulsive behaviour.

Supportive, reality-orientated psychotherapy is an important element of treatment, and may need to be extended to the patient's partner and family, who often will have experienced great strain during the manic episode. *Psychoeducational sessions* are helpful when the patient is more settled. Patients may need much practical help to limit the financial, legal, and occupational repercussions of the illness.

Medication

Medication plays an important role in the acute management of mania. However, all current drug treatments have limitations in terms of their efficacy and tolerability. In the United Kingdom, *antipsychotic drugs* are generally used as primary agents in the management of acute mania, with *mood stabilizers* being reserved for longer-term prophylaxis or where the initial response to antipsychotic drugs is unsatisfactory. A network meta-analysis suggested that the most useful acute antimanic drugs in terms of efficacy and tolerability are olanzapine, quetiapine, risperidone, and haloperidol (Cipriani et al., 2011; Figure 10.1).

For patients who are not already taking mood stabilizers, the drug management can be summarized as follows:

- Begin treatment with either olanzapine, quetiapine, or risperidone (a modest dose of haloperidol is an alternative).
- Where necessary, use adjunctive high-potency benzodiazepines (e.g. lorazepam or clonazepam) to avoid the need for high-dose antipsychotic therapy, and to ensure satisfactory sleep.
- Add a mood stabilizer if the response to this treatment is unsatisfactory (use lithium for 'classical' mania, and valproate for mania with prominent dysphoric or mixed states, or where lithium is contraindicated).

If it is decided to use a mood stabilizer as the initial treatment, the choice previously lay between lithium and valproate. However, the use of valproate is now highly restricted because of its teratogenicity. Current advice in the United Kingdom is that people under the age of 55

(including males) should not be treated with valproate unless two independent specialists have documented that valproate treatment is appropriate because there is no reasonable pharmacological alternative.

As before, benzodiazepines can be used as an adjunct to reduce overactivity and permit sleep, while antipsychotic agents can be reserved for patients who are unresponsive to these measures. If a patient presents with a manic episode and is already taking a mood stabilizer, the first step is to optimize the mood stabilizer treatment in terms of dosing and adherence. Concomitant treatment with antipsychotic drugs will then probably be needed; addition of a second mood-stabilizing drug is an alternative (Goodwin et al., 2016).

Progress of a manic episode can be judged not only by the *mental state* and *general behaviour*, but also by the *pattern of sleep* and the regaining of any weight lost during the illness. As progress continues, antipsychotic drug treatment is reduced gradually. It is important not to discontinue the drug too soon; otherwise, there may

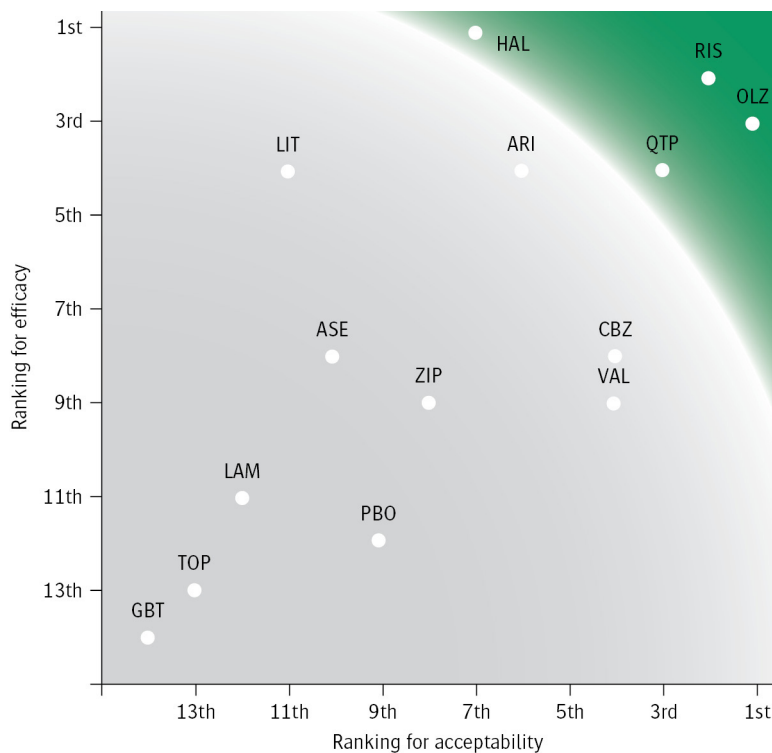


Figure 10.1 Antimanic drugs have been ranked for efficacy (vertical axis) and acceptability (horizontal axis). Treatments in the green section combine the best profile. ARI = aripiprazole; ASE = asenapine; CBZ = carbamazepine; GBT = gabapentin; HAL = haloperidol; LAM = lamotrigine; LIT = lithium; OLZ = olanzapine; PBO = placebo; QTP = quetiapine; RIS = risperidone; TOP = topiramate; VAL = valproate; ZIP = ziprasidone.

Adapted from Cipriani, A., Barbui, C., Salanti, G., et al. (2011). Comparative efficacy and acceptability of antimanic drugs in acute mania: A multiple-treatments meta-analysis. *Lancet*, 378(9799), 1306–1315. Copyright (2011), with permission from Elsevier.

be a relapse and a return of all the original problems of management.

ECT was used to treat mania before antipsychotic drugs were introduced, but reliable evidence about its effectiveness is still limited (see above). It is appropriate to consider the treatment for the unusual patients who do not respond to drug treatment even when this is given at maximum doses. In such cases, clinical experience suggests that a relatively short course of ECT may often be followed by a significant reduction in symptoms sufficient to allow treatment to be continued with drugs. ECT may also be helpful for *mixed affective states* in which depressive symptoms are prominent.

Whatever treatment is adopted, a careful watch should be kept for symptoms of *depressive disorder*. It should be remembered that transient but profound depressive mood change, accompanied by depressive ideas, is common among manic patients. The clinical picture may also change rapidly to a sustained depressive disorder. If either change happens, the patient may develop suicidal ideas. A sustained change to a depressive syndrome is likely to require additional treatment unless the disorder is mild. For National Institute of Health and Care Excellence (NICE) recommendations on the pharmacological treatment of mania and hypomania, see Box 10.2

Continuation treatment

Since the duration of an untreated manic episode can be several months (see above), some form of *continuation treatment*, as in the management of depression, is advisable for at least six months. As a guide, treatment should not be withdrawn finally until the patient has been asymptomatic for at least eight weeks. Withdrawal of lithium should be particularly cautious (e.g. by not lowering the daily dose by more than about 100 mg each week), because sudden discontinuation can lead to 'rebound' mania (Goodwin et al., 2016).

Bipolar depression

Assessment

Depressive episodes are common in bipolar illness and are associated with the majority of the illness burden in both bipolar I and bipolar II disorder. Depression may occur as the presenting feature of a new episode of bipolar illness or immediately following an episode of mania. The assessment of depression should be carried out as described in Chapter 9, with particular attention to *suicidal thinking*, which may be especially prominent after a sudden switch from mania to depression or during the course of a *mixed affective state*.

Box 10.2 NICE guidance on the pharmacological treatment of mania and hypomania

1. If a person develops mania or hypomania and is taking an antidepressant:
 - a) consider stopping the antidepressant, and
 - b) offer an antipsychotic regardless of whether the antidepressant is stopped.
2. If a person develops mania or hypomania and is not taking an antipsychotic or mood stabilizer, offer haloperidol, olanzapine, quetiapine, or risperidone, considering any advance statements, the person's preference, and clinical context (including physical comorbidity, previous response to treatment, and side effects).
3. If the first antipsychotic is poorly tolerated at any dose (including rapid weight gain) or ineffective at the maximum licensed dose, offer an alternative antipsychotic from the drugs listed above, considering any advance statements, the person's preference and clinical context (including physical comorbidity, previous response to treatment, and side effects).
4. If an alternative antipsychotic is not sufficiently effective at the maximum licensed dose, consider adding lithium. If adding lithium is ineffective, or if lithium is not suitable (e.g. because the person does not agree to routine blood monitoring), consider adding valproate adhering to current restrictions and cautions.
5. If the person is already taking a mood stabilizer such as lithium or valproate, consider increasing the dose, up to the maximum therapeutic level if necessary, depending on clinical response. If there is no improvement, consider adding haloperidol, olanzapine, quetiapine, or risperidone, depending on the person's preference and previous response to treatment.

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Medication

Current guidelines suggest that antidepressant drugs should not be a first line choice for the treatment of bipolar depression. In patients taking mood stabilizers

Box 10.3 NICE guidance on the pharmacological treatment of bipolar depression

1. If a person develops moderate or severe bipolar depression and is not taking a drug to treat their bipolar disorder, offer fluoxetine combined with olanzapine, or quetiapine on its own, depending on the person's preference and previous response to treatment.
2. If the person prefers, consider either olanzapine (without fluoxetine) or lamotrigine on its own.
3. If there is no response to fluoxetine combined with olanzapine or quetiapine, consider lamotrigine on its own.
4. If a person develops moderate or severe bipolar depression and is taking either lithium or valproate, first (if tolerance permits) increase the mood stabilizer to within the maximum permitted by the therapeutic range. If this is unsuccessful add one of the options above.¹

¹ Carefully follow the guidance of the *British National Formulary* when adding lamotrigine to valproate because of the likelihood of elevated lamotrigine levels.

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such as lithium and valproate, it is important to optimize blood levels as far as tolerance permits. For patients not receiving mood stabilizer treatment, the initial choice lies between treatment with quetiapine or the combination of olanzapine and fluoxetine. Both may cause problems with sedation and weight gain. Lurasidone is an alternative if these side effects prove too problematic (Box 10.3).

Lamotrigine is an alternative as a monotherapy (and can be added to ineffective quetiapine), but the slow dose escalation required to minimize the risk of rash means that the usual therapeutic dose of around 200 mg daily will not be reached for several weeks. Lamotrigine probably does not induce mania but may not protect against it.

Where bipolar patients with depression are already taking mood stabilizers, the treatments outlined above can be added. However, where lamotrigine is initiated in a patient taking valproate, lamotrigine dosing must be particularly cautious, and the *British National Formulary* consulted.

Some patients with bipolar disorder do seem to require conventional antidepressants to achieve a therapeutic response, and short-term treatment can be justified although it is advisable for patients with bipolar I disorder to be on concomitant treatment with a mood stabilizer or an antimanic dose of an atypical antipsychotic. If benefit is obtained consideration should be given to stopping the antidepressant treatment after about 12 weeks to lessen the risk of mood destabilization. Whether the same cautions around antidepressant use should apply to depressed patients with bipolar II disorder is not established (McIntyre et al., 2020). ECT can be considered in patients whose depressive symptoms are severe and require urgent treatment or where sustained attempts at pharmacotherapy have been ineffective. Whether ketamine may have a role in the treatment of bipolar patients with refractory depression is not clear (Dean et al., 2021)

Psychotherapy

There are very few studies examining the role of structured psychotherapies in acute bipolar depression. In the circumstances, depending on the severity of the depressive disorder and the preferences of the patient, it seems reasonable to offer a structured psychotherapy; for example, cognitive behaviour therapy or interpersonal therapy, that has been shown to be effective in unipolar depression (National Institute for Health and Care Excellence, 2014c).

Prevention of relapse and recurrence

Medication

As mentioned, the majority of patients with bipolar disorder will experience recurrence, suffering both depressive and manic episodes, although the pattern in individual patients is variable. For patients who have had two or more episodes of illness in less than five years, particularly where the illnesses have proved personally disruptive or hazardous, longer-term maintenance treatment should be considered. This will usually involve long-term treatment with a mood stabilizer or an atypical antipsychotic drug such as quetiapine or perhaps a combination of both. It is worth noting that *valproate* taken during pregnancy is established to be a significant cause of *neurodevelopmental defect*. The restrictions attached to valproate use were outlined above. Where valproate is prescribed a pregnancy prevention programme must be in place.

People with bipolar disorder are often reluctant to consider long-term treatment with medication. There are usually a number of reasons for this, and it is necessary to spend time in discussion with patients so that their point of view can be fully appreciated and understood. An excellent personal account has been provided by Jamison (1997). The reasons for reluctance to consider such treatment may include:

- Difficulties in accepting a diagnosis of a lifelong condition and the need for maintenance treatment with medicines; issues of stigma are usually important.
- A belief that it should be possible to control mood without medication, and that the use of medication for this purpose is a sign of personal weakness.
- A fear that medication will blunt the patient's emotional life, and sometimes specifically that mood-stabilizing drugs will decrease feelings of joy and creativity. If the latter feelings are associated with hypomania, it is indeed possible that they will be lessened by medication. For some individuals this can represent a significant loss, which needs to be balanced against the prevention of disabling episodes of depression.

In the United Kingdom, lithium is still regarded as the first choice of mood stabilizer. In addition, there is evidence that the supervised use of lithium is associated with a decreased risk of suicidal behaviour in bipolar illness. Real-world observational studies in large patient cohorts continue to show that lithium is the most effective agent in reducing important adverse outcomes, for example, hospital admission, in patients with bipolar disorder (Joas et al., 2017).

The practical management of using lithium to prevent further episodes is discussed in Chapter 25. There are two aspects of maintenance that require emphasis. First, the patient should be seen and their plasma lithium levels and renal and thyroid function assessed at regular intervals. Secondly, some patients stop lithium of their own accord because they fear that the drug may have harmful long-term effects, or because it makes them feel 'flat'. The risk associated with stopping lithium suddenly is that it may lead to acute relapse. Therefore, patients should be advised to discontinue lithium slowly under supervision.

With carefully supervised follow-up, the likelihood of relapse can be substantially reduced by maintenance treatment, although lesser degrees of mood change often continue. These mood changes may require adjunctive antidepressant or antipsychotic

Box 10.4 Monitoring physical health in bipolar disorder

Ensure that the physical health check for people with bipolar disorder, performed at least annually, includes:

- weight or BMI, diet, nutritional status, and level of physical activity
- cardiovascular status, including pulse and blood pressure
- metabolic status, including fasting blood glucose, glycosylated haemoglobin (HbA_{1c}), and blood lipid profile
- liver function, and
- renal and thyroid function, and calcium levels, for people taking long-term lithium.

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drug treatment. For patients who do not show a good response to lithium, alternative mood-stabilizing drugs can be used. It is possible to use combinations of mood stabilizers, although the risk of adverse reactions is increased. When using combinations of mood stabilizers it is helpful to bear in mind that lamotrigine appears to be more effective in preventing depression than in preventing mania, while the reverse is true for lithium and valproate. Quetiapine appears to be effective in preventing both depression and mania, although its longer-term use is associated with weight gain and metabolic abnormalities.

Current guidelines stress the importance of *regular clinical and biochemical monitoring of the physical health of bipolar patients*, as well as monitoring of their psychological status (Box 10.4).

Psychosocial approaches

Bipolar disorder is often a lifelong, disruptive illness which carries a substantial morbidity. There are several areas of psychosocial function where problems may arise:

- adjustment to the diagnosis, and the need for lifestyle limitations
- interpersonal and relationship difficulties, and occupational problems
- misuse of both illegal and legal substances
- problems relating to concordance with medication.

Box 10.5 Psychological approaches to bipolar disorder

- Advice about lifestyle (regular social and sleep routines, avoidance of illegal drugs).
- Identification and avoidance of triggers for relapse (e.g. sleep deprivation, substance misuse).
- Identification of early subjective signs of relapse (e.g. feeling driven, sleeping badly), with contingency plans for action.
- Education about the importance of medication, and discussion of sensitivity to side effects and active measures to reduce these.

Psychological treatment approaches designed to help with these important problems employ a number of theoretical approaches (see above). However, they have a number of features in common which centre around education about the illness and enhancing self-management (Goodwin et al., 2016) (Box 10.5). Such approaches have useful effects to improve the course of bipolar disorder and should be considered an integral part of management (Miklowitz et al., 2021).

Further reading

Jamison, K. R. (1997). *An Unquiet Mind*. Picador.
Kraepelin, E. (1921). *Manic-Depressive Insanity and Paranoia* (trans. R. M. Barclay). Churchill Livingstone [reprinted in 1976 by Arno Press].

Young, A. H., & Juruena, M. F. (2021). *Bipolar Disorder: From Neuroscience to Treatment*. Springer Nature.

CHAPTER 11

Schizophrenia

Introduction 251

Clinical features 251

Diagnosis and classification 257

Epidemiology 262

Aetiology 264

Neurobiology 273

Neurodevelopmental model 277

Course and prognosis 278

Treatment 280

Management 285

Discussing schizophrenia with
patients and carers 290

Introduction

Schizophrenia has often been considered to be the disorder at the heart of psychiatry, yet of all the major psychiatric syndromes, it is perhaps the most difficult to define and describe. This partly reflects the fact that, in the last century, widely divergent concepts were held in different countries and by different people. Although there is now a greater consensus,

schizophrenia remains the best example of the fundamental issues with which psychiatry continues to grapple—concepts of disease, classification, and aetiology. Here, we start with an introduction to the major symptoms and other important clinical features. For reviews, see Marder and Cannon (2019) and Jauhar et al. (2022).

Clinical features

In this section, it is assumed that the reader has read the descriptions of symptoms and signs in Chapters 1 and 3, which include definitions of many of the cardinal features of schizophrenia. These have classically been divided into two groupings:

1. Positive symptoms include delusions and hallucinations, the most florid and well-known types of symptoms of schizophrenia. Particular types of delusion and hallucination (called *first-rank symptoms*, discussed below) have been used to help distinguish schizophrenia from other psychotic disorders. *Disorganized thinking (formal thought disorder)* and

inappropriate affect are also characteristic and usually considered to be positive symptoms but sometimes considered to be a separate grouping.

2. Negative symptoms reflect a loss of normal functioning. They are often listed as ‘four As’: *alogia* (decreased spontaneous speech), *avolition* (decreased motivation), *affective flattening* (lack of emotional expressivity, but not depression), and *anhedonia*. Although less dramatic than positive symptoms, negative symptoms are often considered to be more fundamental to the syndrome, as discussed below (Marder and Galderisi, 2017).

In recent years, a third cluster, *cognitive symptoms*, has also been recognized, as the magnitude and significance of impairments in attention and memory in schizophrenia have become apparent.

The predominant symptoms differ between *acute schizophrenia* and *chronic schizophrenia*, and the further description of the clinical syndrome is divided on this basis. Briefly, the acute syndrome is dominated by positive symptoms. Many patients recover from the acute illness, but progression to the chronic syndrome is also common. Chronic schizophrenia is characterized by negative symptoms; once the chronic syndrome is established, few patients recover completely. Note, however, that the acute versus chronic distinction is an oversimplification; all features of schizophrenia can occur, and co-occur, at any phase of the illness.

Acute schizophrenia

In acute schizophrenia, positive symptoms predominate, and are usually florid. Behavioural disorganization, especially formal thought disorder, is also prominent. Negative symptoms are less common and cognitive deficits less apparent, although this is partly because they are masked by the positive symptoms.

The vignette in Box 11.1 illustrates several common features of acute schizophrenia, including persecutory delusions, with accompanying hallucinations, gradual social withdrawal, and impaired performance.

In appearance and behaviour some patients with acute schizophrenia seem entirely normal. Others seem changed, although not always in a way that would immediately point to psychosis. They may be preoccupied

with their health, their appearance, religion, or other intense interests. Social withdrawal often occurs—for example, spending a long time in their room, perhaps lying immobile on the bed—and relationships are not maintained. Some patients smile or laugh without obvious reason. Some appear to be constantly perplexed, while others are restless and noisy, or show sudden and unexpected variability of behaviour.

The speech often reflects an underlying *thought disorder*. In the early stages, there is vagueness in the patient's talk that makes it difficult to grasp their meaning. Some patients have difficulty in dealing with abstract ideas. Other patients become preoccupied with vague pseudoscientific or mystical notions. Thought disorder is reflected in the *loosening of association* between expressed ideas, and may be detected in illogical thinking (e.g. '*knight's move*' thinking) or talking past the point (*vorbeireden*). In its severest form, the structure and coherence of thinking are lost, so that utterances are jumbled (*word salad* or *verbigeration*). Some patients use ordinary words or phrases in unusual ways (metonyms or paraphrases), and a few coin new words (*neologisms*). Disorders of the form (or stream) of thought include pressure of thought, poverty of thought, thought blocking, and thought withdrawal.

Auditory hallucinations are among the most frequent symptoms. They may take the form of noises, music, single words, brief phrases, or whole conversations. They may be unobtrusive, or so severe as to cause great distress. Some voices seem to give commands to the patient. Some patients hear their own thoughts apparently spoken out loud either as they think them (*Gedankenlautwerden*) or immediately afterwards (*echo de la pensée*), and some voices discuss the patient in the third person or comment on his actions; these are first-rank symptoms (see Boxes 11.2 and 11.3). Visual hallucinations are less frequent, and usually occur together with other kinds of hallucination. Olfactory, gustatory, and somatic hallucinations are reported by some patients. They are often interpreted in a delusional way—for example, hallucinatory sensations in the lower abdomen are attributed to unwanted sexual interference by a persecutor.

Delusions are almost invariable in acute schizophrenia, although primary delusions are infrequent and are difficult to identify with certainty. Delusions may originate against a background of so-called primary delusional mood (*Wahnstimmung*). Persecutory delusions are very common, but are not specific to schizophrenia, as they occur in all psychoses and characterize delusional disorder (see Chapter 12). Less common, but of greater diagnostic value, are delusions of reference and

Box 11.1 A vignette of acute schizophrenia

A previously healthy 20-year-old student had been behaving in an odd way. At times she appeared angry and told her friends that she was being persecuted; at other times she was seen laughing for no apparent reason. For several months she had seemed increasingly preoccupied with her own thoughts and her academic work had deteriorated. When interviewed, she was restless, suspicious, and exhibited odd mannerisms. She described hearing voices commenting on what she was doing. She believed that the police had conspired with her tutors to read her thoughts and interfere with them.

Box 11.2 Schneider's symptoms of the first rank

First-rank symptoms (shown in Box 11.3) were described by Kurt Schneider (1887–1967), not to be confused with Carl Schneider, who categorized types of thought disorder. First-rank symptoms were an attempt to make the diagnosis of schizophrenia more reliable by identifying a group of symptoms characteristic of schizophrenia and rarely found in other disorders. Unlike Bleuler's fundamental symptoms (Box 11.4), Schneider's first-rank symptoms were not supposed to have any central psychopathological role, merely that they were valuable in making the diagnosis.

First-rank symptoms became a widely adopted concept, and formed part of diagnostic criteria in earlier versions of ICD and DSM. However, though they remain in common clinical usage, they are not included in ICD-11 or DSM-5, due to concerns about their diagnostic specificity and lack of predictive value.

of control (*passivity*), and delusions about the possession of thought. The latter are delusions that thoughts are being inserted into or withdrawn from one's mind, or 'broadcast' to other people. Some of these symptoms are first-rank symptoms (Box 11.3).

Insight is almost always impaired. Most patients do not accept that their experiences result from illness, but usually ascribe them to the malevolent actions of other people. Orientation is usually normal, although this may be difficult to determine if there is florid thought disorder or if the patient is too preoccupied with their psychotic experience to attend to the interviewer's questions.

Alterations in mood are common and are of three main kinds. First, there may be symptoms of anxiety, depression, irritability, or euphoria. These can be clinically significant, but if such features are sufficiently prominent and sustained, the possibility of schizoaffective disorder or affective psychosis should be considered. Secondly, there may be blunting (or flattening) of affect—that is, sustained emotional indifference or diminution of emotional response. Thirdly, there may be incongruity of affect, in which the expressed mood is not in keeping with the situation or with the patient's own feelings.

Finally, we emphasize the variability of the clinical picture. Few patients experience all of the symptoms

Box 11.3 First-rank symptoms of schizophrenia

- Hearing thoughts spoken aloud
- Third-person hallucinations
- Auditory hallucinations in the form of a 'running commentary'
- Somatic (bodily, tactile) hallucinations
- Thought withdrawal or insertion
- Thought broadcasting
- Delusional perception
- Feelings or actions experienced as made or influenced by external agents (passivity)

introduced above, while others already have features of the 'chronic' syndrome at first presentation. Moreover, the overall pattern and duration of features are also considered before making a diagnosis. These issues, together with some additional clinical features, are discussed later in the chapter.

Chronic schizophrenia

Although the positive symptoms of the acute syndrome may persist, the chronic syndrome is characterized by the *negative symptoms* of underactivity, lack of drive, social withdrawal, and emotional apathy. The vignette in Box 11.4 illustrates several of the negative features of what is sometimes called a 'defect state' or *deficit syndrome*. The most striking feature is diminished volition—that is, a lack of drive and initiative. The patient may remain inactive for long periods or engage in aimless and repeated activity. They may withdraw from social encounters, and their social behaviour deteriorates. Self-care may be poor, and the style of dress and presentation somewhat inappropriate. Some patients collect and hoard objects, so that their surroundings become cluttered and dirty. Others break social conventions by talking intimately to strangers or shouting obscenities in public.

Speech is often abnormal, showing evidence of thought disorder of the kinds found in the acute syndrome described above. Affect is generally blunted, and when emotion is shown, it is incongruous or shallow. Hallucinations and delusions can occur but tend to be held with little emotional response. For example, the patient may be convinced that they are being persecuted but show neither fear nor anger.

Box 11.4 A vignette of chronic schizophrenia

A middle-aged man lives in a supported hostel and attends a day centre. He spends most of his time alone. He is dishevelled and unshaven, and cares for himself only when encouraged to do so by others. His social behaviour seems odd and stilted, his speech slow, and its content is vague and incoherent. He shows few signs of emotion. For several years this clinical picture has changed little except for brief periods of acute symptoms, which are usually related to upsets in the ordered life of the hostel.

Various disorders of movement can occur, including stereotypies, mannerisms, and other catatonic symptoms, and dyskinesias (see below). The latter are primarily but not entirely due to antipsychotic medication. Cognitive impairment is common, if not universal, in chronic schizophrenia and, together with the negative symptoms, contributes to a low level of functioning and poor outcome. However, the cognitive deficits are rarely of sufficient magnitude, nor of concern to the patient, to be apparent unless detailed cognitive testing is undertaken.

As with acute schizophrenia, the symptoms and signs of the chronic illness are variable. At any stage, positive symptoms may recur or become exacerbated; this may be in response to life events, or to non-compliance with, or discontinuation of, medication.

Subtypes of schizophrenia

Schizophrenia has conventionally been divided into several descriptive subtypes. Although these terms are not used in ICD-11 or DSM-5 they are mentioned here since, like first-rank symptoms, they remain in clinical parlance.

- *Paranoid schizophrenia* is the most common form. It is characterized by persecutory delusions, often systematized, and by persecutory auditory hallucinations. Thought disorder and affective, catatonic, and negative symptoms are not marked. Personality is relatively well preserved.
- In *hebephrenic schizophrenia*, also called *disorganized schizophrenia*, thought disorder and affective symptoms are prominent. The mood is variable, with behaviour often appearing silly and unpredictable. Delusions and hallucinations are fleeting and not

systematized. Mannerisms are common. Speech is rambling and incoherent, reflecting the thought disorder. Negative symptoms occur early and contribute to a poor prognosis.

- In *catatonic schizophrenia*, the most striking features are motor symptoms, as noted in Chapter 1, and changes in activity that vary between excitement and stupor. In ICD-11, catatonia is now a separate syndrome (see below).
- *Undifferentiated schizophrenia* describes cases that do not fit readily into any of the above subtypes, or where there are prominent features of more than one of them.
- *Residual schizophrenia* refers to a stage of chronic schizophrenia when, for at least a year, there have been persistent negative symptoms but no recurrence of positive symptoms.

Three clinical sub-syndromes

In another approach to identifying sub-syndromes of schizophrenia, Liddle (1987) studied patients with chronic schizophrenia and, based on clustering of symptoms, described three overlapping clinical syndromes, which he called reality disturbance, disorganization, and psychomotor poverty (see Table 11.1). In many respects, these are comparable to the groupings of positive symptoms, behavioural disorganization, and negative symptoms introduced earlier. They have been well replicated in later research. These symptom clusters were subsequently linked to distinct patterns of neuropsychological deficit and to regional cerebral blood flow (Liddle et al., 1992), with the most reproducible finding being a link between psychomotor poverty, impaired performance on frontal-lobe tasks, and decreased frontal blood flow.

Other aspects of the clinical syndrome

Cognitive features

Despite Kraepelin's original term *dementia praecox*, cognitive impairment was for many years a neglected component of schizophrenia. However, contemporary studies have emphasized its extent and importance. Deficits are seen across all domains of learning and memory, with disproportionate involvement of semantic memory, working memory, and attention. The deficits average almost one standard deviation below expected performance, and are present at the first episode; IQ is also reduced (Mesholam-Gately et al., 2009). Executive

Table 11.1 Cerebral and psychological correlates of three sub-syndromes of chronic schizophrenia

Syndrome	Symptoms	Regional cerebral blood flow correlates	Impaired psychological performance
Reality disturbance	Delusions, hallucinations	Left medial temporal lobe, cingulate cortex	Disorders of self-monitoring
Disorganization	Formal thought disorder, inappropriate affect, bizarre behaviour	Anterior cingulate, right ventral frontal cortex, bilateral parietal regions	Tests of selective attention
Psychomotor poverty	Flat affect, poverty of speech, decreased spontaneous movement	Underactivity of frontal cortex ('hypofrontality')	Word generation tasks, planning tests

function and attention may be the core deficits, in that they are seen even in patients with otherwise intact cognition. Patients also have deficits in *social cognition*—the perception, storage, and use of information about other people and ourselves—leading to difficulties in social interactions and daily life (Green et al., 2019).

The time course of neuropsychological involvement in schizophrenia is complex and variable. On average, people who later develop schizophrenia already have reduced IQ scores during childhood; the onset of illness is associated with further cognitive impairments (Jonas et al., 2022). Subsequently, additional decline may be seen, and later in the illness the risk of overt dementia is markedly increased (Stroup et al., 2021). The dementia of schizophrenia is not attributable to Alzheimer's disease or other recognized neurodegenerative disorder. Like many other features of schizophrenia, impaired cognitive performance can be observed in attenuated form in unaffected first-degree relatives.

Cognitive aspects of schizophrenia are currently being emphasized for several reasons (Green et al., 2019). First, they are a major determinant of poor functional outcome. Second, they are potential targets for both drugs and psychological therapies (*cognitive remediation*). Third, cognitive features are increasingly conceptualized as being central to the disorder and as underlying the psychotic symptoms (Kahn, 2020).

Methods have been developed to promote and standardize cognitive assessments in schizophrenia (e.g. the MATRICS Consensus Cognitive Battery; Nuechterlein et al., 2008); however, this has not become widespread in clinical practice. Failure to achieve this goal, together with uncertainty about their diagnostic specificity, contributes to the fact that cognitive deficits continue to be omitted from diagnostic criteria for schizophrenia (though they are used in ICD-11 as a descriptive sub-category), despite acceptance of their importance.

Depressive symptoms

Depressive symptoms commonly occur in schizophrenia, in any phase of the illness. They are also features of the prodrome (see below). About 25% of patients exhibit persistent and significant depression, sometimes called *post-schizophrenia depression*. Comorbid depression in schizophrenia worsens the functional outcome.

There are several reasons why depressive symptoms may be associated with schizophrenia.

- Depression may be an integral part of schizophrenia. This view is supported by the observation that about 50% of patients with acute schizophrenia experience significant depressive symptomatology, which improves as the psychosis remits.
- In the post-psychotic phase, depressive symptoms may be a response to recovery of insight into the nature of the illness and the problems to be faced. Again, this may happen at times, but it does not provide a convincing general explanation.
- Depression may be a side effect of medication. This is not the only explanation, as depressive symptoms can occur in the absence of antipsychotic drug therapy.

Depressive symptoms in schizophrenia should not be confused with negative symptoms nor for the parkinsonian side effects of antipsychotic medication. These distinctions are important, but can be difficult to make (Krynicky et al., 2018). The Calgary Depression Scale for Schizophrenia can be useful in this regard.

Neurological signs

Neurological signs are a neglected clinical feature of schizophrenia (Chan et al., 2010). They are called 'soft signs' because they do not localize pathology to a particular tract or nucleus. They include abnormalities in sensory integration, coordination, and sequencing of complex

motor acts; catatonic features and dyskinesias may also be considered under this category (Koning et al., 2010). Neurological signs are seen in unmedicated, first-episode patients (and are therefore at least partially separate from the side effects of antipsychotic drugs), but are more common in chronic schizophrenia. The presence of neurological signs correlates with cognitive dysfunction, evidence for developmental anomaly, and diffuse brain pathology, and are thought to be a manifestation of the neurodevelopmental origins of schizophrenia (Liddle, 2013). A neurological examination that focuses upon the extrapyramidal system should be part of the routine assessment of a person who presents with schizophrenia (Chapter 3).

Olfactory dysfunction

Patients with schizophrenia have deficits in olfactory function, which affect the identification and discrimination of odours (Carnemolla et al., 2020). The deficits are not attributable to medication or smoking, and are associated with cognitive and negative symptoms.

Pain insensitivity

A diminished sensitivity to pain in patients with schizophrenia has long been noted by clinicians, and can on occasion be extreme (Engels et al., 2014). Its cause is unknown. Proposed explanations include occult thalamic lesions, and the possibility that psychotic symptoms render the patient less concerned by, or distracted from, the pain.

Factors that modify the clinical features

The social and cultural background of the patient affects the content of symptoms. For example, in many countries religious delusions are less common than they were a century ago, and have been replaced by delusions about contemporary concerns such as terrorism and social media. Age also seems to modify the picture. In adolescents and young adults, the clinical features often include thought disorder, mood disturbance, passivity phenomena, thought insertion, and withdrawal. With increasing age, paranoid symptomatologies with systematized delusions are more common.

Intelligence also affects the clinical features, and the psychiatrist's ability to elicit them. Patients with disorders of intellectual development usually present with a simple clinical picture, sometimes referred to as *profposchizophrenie*. In contrast, highly intelligent people develop complex delusional systems, and are also better able to articulate, or conceal, their experiences.

The amount of social stimulation has a considerable effect. Under-stimulation is thought to increase negative

symptoms, whereas over-stimulation precipitates positive symptoms. Psychosocial approaches to treatment, discussed below, are designed to avoid both extremes.

The prodrome of schizophrenia

Increasing attention is being paid to the prodrome of schizophrenia—that is, the period of time during which psychosis is ‘brewing’, when there are identifiable symptoms but before diagnosable criteria are met, and before the patient typically presents for help. The terms ‘at-risk state’ or ‘high-risk state’ are also used to describe the prodrome. The focus on the prodrome reflects the fact that in many patients this period lasts for several months or years, and the longer the duration of untreated psychosis, the worse the outcome (Howes et al., 2021). There are several possible explanations for this association:

- A prolonged, insidious onset is a feature of a severe and treatment-resistant form of schizophrenia.
- Untreated psychosis is 'neurotoxic' and makes the illness less responsive to treatment.
- Illness is more severe by the time treatment is initiated and therefore is harder to treat.
- A longer prodrome leads to greater psychosocial impairments and loss of functioning, hampering full recovery.

These considerations led to the development of ‘early intervention’, with services designed to detect and then more effectively treat, if not prevent, emerging cases of schizophrenia (Fusar-Poli et al., 2020). The therapeutic and service implications are discussed later in this chapter and in Chapter 26.

The prodrome is characterized by insidious and shifting profiles of largely non-specific symptoms, including mild psychotic-like positive and negative symptoms, and also commonly depressive and anxiety symptoms. Cognitive and social functioning also deteriorates. Some people exhibit brief, limited intermittent psychotic symptoms (BLIPS) lasting a few days.

Various criteria are available to define, assess, and rate the prodrome, including the Comprehensive Assessment of At-Risk Mental State (CAARMS). Importantly, many people who are considered to be prodromal do not progress to overt psychosis; typically, less than 25% do so during a two- to three-year follow-up. This has important implications for treatment interventions. Studies are attempting to identify the predictors of conversion to psychosis (Fusar-Poli et al., 2017).

Diagnosis and classification

This section is concerned mainly with the diagnostic criteria for and classification of schizophrenia as specified in ICD-11 and DSM-5. However, the current approach—and its problems—can be understood better with some knowledge of the historical perspective.

Historical development of ideas about schizophrenia

The development of ideas about schizophrenia is discussed in Box 11.5. To a large extent, they mirror the development of ideas about psychiatric illness in general, reflecting the central position in psychiatry held by schizophrenia during the last century.

In the 1960s it was noticed that there were wide divergences in the use of the term schizophrenia, and marked differences in diagnostic practice. This was unsurprising, given the multiple views and traditions summarized in Box 11.5. For example, in the United Kingdom and continental Europe, psychiatrists generally employed

Schneider's first-rank symptoms to identify a narrowly delineated group of cases. In the United States, however, interest in psychodynamic processes led to diagnosis on the basis of mental mechanisms, and to the inclusion of a much wider group of cases. First-admission rates for schizophrenia were also much higher in the United States than in the United Kingdom.

These discrepancies prompted two major cross-national studies of diagnostic practice, discussed in Chapter 2, which proved highly influential not just for schizophrenia but also for the development of diagnostic criteria and classificatory systems, namely the US–UK Diagnostic Project (Cooper et al., 1972), and the International Pilot Study of Schizophrenia (IPSS) (World Health Organization, 1973). These findings led to a consensus that agreed diagnostic criteria were required, and the development of standardized methods by which these could be defined and identified, and, with refinement, are reflected in the current ICD-11 and DSM-5 criteria.

Box 11.5 Development of schizophrenia concepts and terminologies

In the nineteenth century, one view was that all serious mental disorders were expressions of a single entity, which Griesinger called *Einheitspsychose* (unitary psychosis). The alternative view, put forwards by Morel in France, was that mental disorders could be separated and classified. Morel searched for specific entities, and argued for a classification based on cause, symptoms, and outcome. In 1852 he gave the name *démence précoce* to a disorder he described as starting in adolescence and leading first to withdrawal, odd mannerisms, and self-neglect, and eventually to intellectual deterioration. A few years later, Kahlbaum (1863) described the syndrome of *catatonia*, and Hecker (1871) wrote an account of a condition he called *hebephrenia*.

Emil Kraepelin (1855–1926) derived his ideas from studying the course and outcome of the disorder. His observations led him to argue against the idea of a single psychosis, and to propose a division into *dementia praecox* and *manic–depressive psychosis*. This grouping

brought together as subclasses of *dementia praecox* the previously separate entities of hebephrenia and catatonia. (His adoption of the word *dementia* emphasizes the prominence he attributed to the cognitive impairments of the disorder.) Kraepelin's description of *dementia praecox* appeared for the first time in 1893, in the fourth edition of his textbook, and was expanded in subsequent editions. He described the illness as occurring in clear consciousness, and consisting of 'a series of states, the common characteristic of which is a peculiar destruction of the internal connections of the psychic personality' (Kraepelin, 1919). Kraepelin divided *dementia praecox* into three subtypes (catatonic, hebephrenic, and paranoid). He separated the condition that he named *paraphrenia* (Box 12.2) on the grounds that it started in middle age and was free from the changes in emotion and volition found in *dementia praecox*. It is commonly held that Kraepelin regarded *dementia praecox* as invariably progressing to chronic deterioration. However, he reported

that, in his series of cases, 13% recovered completely (although some relapsed later) and 17% were ultimately able to live and work without difficulty.

Eugen Bleuler (1857–1959) based his work on that of Kraepelin, and in his own book wrote ‘the whole idea of dementia praecox originates with Kraepelin’ (Bleuler, 1911). Compared with Kraepelin, Bleuler was concerned less with prognosis and more with the mechanisms of symptom formation. Bleuler proposed the name *schizophrenia* to denote a ‘splitting’ of psychic functions, which he considered to be of central importance. He believed in a distinction between *fundamental* and *accessory* symptoms. Fundamental symptoms included four features sometimes known as Bleuler’s ‘four A’s: *Associations* (thought disorder), *Affect* (blunting, flattening, and incongruity of affect), *Ambivalence* (lack of motivation), and *Autism* (social withdrawal and failure of theory of mind). It is interesting that, in Bleuler’s view, some of the most frequent and striking symptoms were accessory (secondary)—for example, hallucinations, delusions, catatonia, and abnormal behaviours. Since Bleuler was preoccupied more with psychopathological mechanisms than with symptoms themselves, his approach to diagnosis was less precise than that of Kraepelin. In addition, Bleuler

took a more optimistic view of the outcome, but still held that ‘as yet I have never released a schizophrenic in whom I could not still see distinct signs of the disease’ (Bleuler, 1911).

In Scandinavian countries, cases of psychosis arising after stressful events received much attention from psychiatrists such as Langfeldt. The terms *reactive psychosis* or *psychogenic psychosis* are still sometimes applied to conditions which appear to be precipitated by stress, are to some extent understandable in their symptoms, and have a good prognosis. In current diagnostic schemes such disorders are classified as *brief psychotic disorder* or *schizophreniform disorder*.

Current debates about schizophrenia nomenclature and classification centre on two issues. Firstly, whether the word schizophrenia should be abolished, as has already occurred in some countries. This is advocated to reduce stigma, and to replace it with a term which better fits current views about its nature and boundaries, such as *psychosis spectrum syndrome* or *altered perception syndrome* (Mesholam-Gately et al., 2021). Secondly, whether schizophrenia should be a dimensional rather than categorical diagnosis (see Chapter 2). For review see Lawrie et al. (2016a).

In summary, a wide range of views about schizophrenia have been held over the past century since Kraepelin and Bleuler established the concept. With the advent of current classificatory systems, the term now refers to a syndrome which can be diagnosed reliably, a prerequisite for evidence-based clinical practice. However, as noted in Chapter 3, reliability is not in itself sufficient. Because the causes of schizophrenia are still largely unknown, the syndrome remains of uncertain validity. Until this fundamental question is answered, there will continue to be dispute as to its most important features, diagnostic boundaries, and internal subdivisions.

Classification of schizophrenia in ICD-11 and DSM-5

The classification of schizophrenia and schizophrenia-like disorders in ICD-11 and DSM-5 is outlined and compared in Table 11.2.

ICD-11

ICD-11 places schizophrenia within a broader category of ‘schizophrenia or other primary psychotic disorders’. The ‘primary’ emphasizes that the psychotic symptoms

are their defining feature, and that they are not secondary to another identified condition (e.g. a medical disorder or substance use). The ICD-11 requirements for schizophrenia are shown in Box 11.6. The other disorders listed in Table 11.2 are covered later in the chapter, except for delusional disorder, which is in Chapter 12. As noted, ICD-11 does not use subtypes of schizophrenia, but does allow for a more detailed descriptive characterization of schizophrenia (and the other disorders listed in Table 11.2) in several ways:

- Course specifiers—whether first episode, multiple episodes, or continuous illness
- Current severity—symptomatic, partial remission, or full remission
- Current symptoms—positive, negative, depressive, manic, cognitive, psychomotor; each of these is rated as none, mild, moderate, or severe.

DSM-5

In DSM-5, schizophrenia is defined in terms of the symptoms, course, and impaired functioning. The key symptoms and criteria are broadly similar to ICD-11 but with some additional requirements. Notably, at least two of the symptoms have been present for at least

Table 11.2 Classification of schizophrenia and schizophrenia-like disorders in ICD-11 and DSM-5^a

ICD-11 ('Schizophrenia and other primary psychotic disorders' section)	DSM-5 (Schizophrenia chapter)
Schizophrenia	Schizophrenia
Schizoaffective disorder	Schizoaffective disorder
Delusional disorder	Delusional disorder
Acute and transient psychotic disorder	Brief psychotic disorder
	Schizophreniform disorder
Other specified schizophrenia or primary psychotic disorders	Other specified schizophrenia spectrum and other psychotic disorder
Schizotypal disorder	Unspecified schizophrenia spectrum and other psychotic disorder
	Substance/medication-induced psychotic disorder
	Psychotic disorder due to another medical condition

^a The order of categories in the two systems has been adjusted to compare their features more clearly. In ICD-11, schizophrenia-like disorders secondary to substance use or medical conditions are coded separately. In DSM-5, schizotypal disorder is categorized as a personality disorder.

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (DSM-5). Copyright (2013), American Psychiatric Association.

Box 11.6 Essential diagnostic features for schizophrenia in ICD-11

At least two of the following, present for at least one month, including at least one of items a–d:

- a. Persistent delusions
- b. Persistent hallucinations
- c. Disorganized thinking and speech
- d. Experiences of passivity or control
- e. Negative symptoms
- f. Grossly disorganized behaviour
- g. Psychomotor disturbances (catatonia)

Symptoms are not secondary to a medical condition. (If so, the term is *secondary psychotic syndrome*; see Chapter 14).

Symptoms are not due to the effects of a substance or medication, including withdrawal. (If so, the term is *substance-induced psychotic disorder*).

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>

one month (unless treated), and there must have been continuous signs of disturbance for at least six months. Patients who have schizophrenia-like symptoms of less than the one- and six-month duration criteria are classified as suffering from *schizophreniform disorder* or *brief psychotic disorder*. Other related disorders include

delusional disorder (see Chapter 12) and psychotic disorders not otherwise classified (see below). After at least one year has elapsed since the onset of psychotic symptoms, DSM-5 allows the disorder to be classified by the number of acute episodes which have occurred, and whether the patient is current in an episode, or in partial or full remission.

As in ICD-11, DSM-5 includes catatonia as a syndrome that can be diagnosed as occurring as part of schizophrenia or as a separate disorder.

Schizophrenia-like disorders

Whatever definition of schizophrenia is adopted, there will be cases that resemble schizophrenia in some respects and yet do not meet the criteria for diagnosis. In ICD-11 and DSM-5 these disorders can be considered within five groupings:

1. Delusional disorders (also called paranoid psychoses).
2. Brief psychotic disorders.
3. Psychotic disorders accompanied by prominent affective symptoms (schizoaffective disorder).
4. Schizophrenia-like symptoms in enduring and attenuated form (schizotypal disorder).
5. Catatonia.

Delusional disorders are discussed in Chapter 12. The latter four groups are discussed here.

Brief psychotic disorders

Classification of disorders in which psychotic symptoms occur with an acute onset and subside within days or weeks has proven difficult, and a confusing range of terms and groupings have been used, such as *bouffée délirant*, *cycloid psychosis*, and *psychogenic psychosis*, reflecting differing views about the nature and origin of the condition. For review of brief psychotic episodes, see Fusar-Poli et al. (2022).

ICD-11 takes a simpler approach and uses a single category, *acute and transient psychotic disorder*. These disorders are of rapid onset and reach their maximum severity within two weeks. Symptoms fluctuate markedly from day to day. The episode does not exceed three months, and typically lasts from a few days to a month. There is no prodrome, no negative symptoms, and usually a full return to premorbid function. An acute stressor is often reported prior to onset, but this is not a diagnostic requirement. As with schizophrenia, if the episode is attributable to substance use or a medical condition, it should be diagnosed accordingly. If symptoms persist for more than three months, the diagnosis should be changed to schizophrenia or other disorder depending on the profile of symptoms.

DSM-5 uses the term *brief psychotic disorder* to refer to a syndrome characterized by at least one of the criteria a, b, and f shown in Box 11.6. The disorder lasts for at least one day but not more than one month, by which time recovery has occurred, including full return to premorbid level of functioning. The disorder may or may not follow a stressor but, as in ICD-11, psychoses induced by the direct physiological effects of drugs or medical illness are excluded.

In DSM-5, *schizophreniform disorder* is a syndrome similar to schizophrenia that has lasted for more than one month (and so cannot be classified as brief psychotic disorder), but for less than the six months required for a diagnosis of schizophrenia to be made. Unlike for schizophrenia, social and occupational dysfunction are not required to make the diagnosis.

Schizoaffective disorder

Some patients have a significant mixture of both schizophrenic and affective symptoms. Such patients are classified under *schizoaffective disorder* in both ICD-11 and DSM-5.

In ICD-11, the diagnosis of schizoaffective disorder requires that the patient meets criteria for schizophrenia and for a mood episode (depressive, manic, or mixed) within the same episode of illness, either simultaneously

or within a few days of each other. DSM-5 requires that there should have been an uninterrupted period of illness during which there is a major depressive or manic episode concurrent with symptoms that meet criteria for schizophrenia. During this time, acute psychotic symptoms must have been present for at least two weeks in the absence of prominent mood symptoms. However, the episode of mood disturbance must have been present for a substantial part of the illness.

Importantly, the label ‘schizoaffective’ should not be applied just because a patient has a few symptoms consistent with both diagnoses, nor because the assessment has not been sufficiently detailed to identify the primary diagnosis. Neither should it be used for patients with schizophrenia who develop depressive symptoms, patients with depressive disorder who develop psychotic symptoms, or patients who meet criteria for schizophrenia in one episode but criteria for depression or mania in a different episode of illness. In practice, these various distinctions can be hard to make, and the diagnosis has only moderate reliability (Peterson et al., 2019).

For a review, see Miller and Black (2019).

Schizotypal disorder

There are some people who from an early age have had an enduring pattern of eccentric behaviour, schizophrenia-like symptoms (e.g. ideas of reference, persecutory beliefs, and unusual types of thinking) and a lack of interest or capacity for social relationships. These features cause distress or impairment, and ICD-11 uses the term *schizotypal disorder*; the distinction from schizophrenia is based on the relative mildness of symptoms and their long-term stability, and DSM-5 classifies the condition as a personality disorder, *schizotypal personality disorder*, reflecting the view of *schizotypy* as a personality trait.

Catatonia

Catatonic symptoms are manifested as different types and co-occurrences of increased, decreased, or abnormal psychomotor activity. They were introduced in Chapter 1 and summarized in Box 11.7. Autonomic instability can also occur. *Catatonia* is the term given to a syndrome when these psychomotor features are prominent. In ICD-11, catatonia comprises three conditions:

1. *Catatonia associated with another mental disorder*. Schizophrenia is the best-known example (as reflected in the catatonic schizophrenia subtype mentioned earlier) but catatonia is also recognized

Box 11.7 Features of catatonia

Decreased psychomotor activity

Stupor
Ambitendency
Negativism
Mutism
Staring

Increased psychomotor activity

Hyperactivity or agitation
Non-purposeful movements
Extreme emotional reactions

Abnormal psychomotor activity

Mannerisms
Grimacing
Posturing
Stereotypies
Rigidity
Catalepsy ('waxy flexibility')
Echolalia and echopraxia
Verbigeration

as occurring in autism spectrum disorder and severe mood disorder.

2. *Catatonia induced by substances or medication.* Several illicit drugs (e.g. phencyclidine, LSD, cocaine) as well as licensed drugs (e.g. antipsychotics, steroids, disulfiram) can lead to catatonia.
3. *Secondary catatonia syndrome.* A wide range of acute medical conditions can produce catatonia, including diabetic ketoacidosis, hepatic encephalopathy, and encephalitis.

For review of catatonia, see Walther et al. (2019).

Comorbidity

Many patients with schizophrenia also meet the criteria for another psychiatric diagnosis, complicating diagnosis and treatment, and worsening the outcome. The example of depression (which occurs in up to 50% of cases) has already been mentioned, and substance use disorder is present in over 40% (Hunt et al., 2018). Other psychiatric disorders often comorbid with schizophrenia

include post-traumatic stress disorder, obsessive-compulsive disorder, and panic disorder. For a review of psychiatric comorbidity in schizophrenia, see Buckley et al. (2009). Physical comorbidities are discussed later in this chapter.

Differential diagnosis

We have already described how current classifications include schizophrenia-like disorders as well as schizophrenia, but the boundary between them is blurred and to some extent arbitrary. Similarly, there is no clear distinction between disorders that are considered to be variants of schizophrenia, and some which are viewed as being part of its differential diagnosis. For example, delusional disorder and schizotypal disorder are often included in both categories, and catatonia can be seen as a form of schizophrenia or as a separate diagnosis. Such difficulties reflect the unknown validity of the syndrome(s), and are unlikely to be resolved until the classification is based upon aetiology or other empirically validated markers (Lawrie et al., 2016b). In this regard, the distinction between schizophrenia and organic schizophrenia-like disorders is also not only unhelpful (Chapter 2), but also becomes increasingly blurred as potential causes of some cases of 'schizophrenia' are discovered, such as copy number variants (CNVs) and anti-NMDA receptor antibodies (see below). The important point is to ensure appropriate assessment and investigation of all patients presenting with schizophrenia-like disorders.

With these caveats in mind, current diagnostic practice requires schizophrenia to be distinguished from a number of other disorders, summarized in Box 11.8.

Organic syndromes. Acute schizophrenia can be mistaken for *delirium*, especially if there is pronounced thought disorder and a rapidly fluctuating affect and mental state. Careful observation is needed for clouding of consciousness, disorientation, and other features of delirium (see Chapter 14).

Schizophrenia-like disorders can also occur, in clear consciousness, in a range of neurological and medical disorders. These conditions are referred to as a *secondary psychotic syndromes* in ICD-11, *psychotic disorder due to another medical condition* (DSM-5), or simply as *secondary schizophrenia*. Classic examples include temporal lobe epilepsy (complex partial seizures), general paralysis of the insane, and metachromatic leukodystrophy (see Chapter 14, and Hyde and Ron, 2011). Visual, olfactory, and gustatory hallucinations are suggestive

Box 11.8 Differential diagnosis of schizophrenia

Cases meeting some but not all criteria for schizophrenia

Brief psychotic disorder
Delusional disorder
Schizotypal disorder

Cases with affective and psychotic symptoms

Schizoaffective disorder
Psychotic depression
Bipolar disorder

Schizophrenia-like disorders attributable to a specific cause

Schizophrenia-like disorder due to psychoactive substance use
Organic schizophrenia-like disorder

Other

Autism and related disorders
Severe obsessive compulsive disorder
Body dysmorphic disorder
Schizoid personality disorder
Paranoid personality disorder

of an organic aetiology. The occurrence of organic schizophrenia-like disorders emphasizes the importance of a careful medical history, physical examination, and investigations. One study in London found that 10 out of 268 cases (3.7%) of 'first-episode

schizophrenia' had an organic cause other than substance misuse (Johnstone et al., 1987).

Drug-induced states (including substance misuse). Certain prescribed drugs, especially steroids and dopamine agonists, can cause florid psychotic states. Psychoactive substance misuse, particularly with psychostimulants or phencyclidine, and also alcohol, should always be considered in the presentation of schizophrenia-like psychoses. Urine or hair testing can be helpful in diagnosis, as is the temporal association between drug use and symptoms. However, the high prevalence of recreational drug use in young adults means that a clear distinction between a drug-induced psychosis and schizophrenia is not always possible. See also Chapter 20.

Mood disorders with psychotic features. The distinction of schizophrenia (and schizoaffective disorder) from affective psychosis depends on the degree and persistence of the mood disorder, the relationship of any hallucinations or delusions to the prevailing mood, and the nature of the symptoms in any previous episodes. The distinction from mania in young people can be particularly difficult, and sometimes the diagnosis can only be clarified by longer-term follow-up.

Delusional disorder. These disorders are characterized by chronic, systematized paranoid delusions, but lack other symptoms of schizophrenia, and many areas of the mental state are unremarkable.

Personality disorder. Differential diagnosis from people with paranoid and schizoid personality traits can be difficult. Prolonged observation may be required to detect genuine symptoms of psychosis, and the additional features indicative of schizophrenia. Some patients with borderline personality disorder also exhibit transient psychotic symptoms, although the presence of affective instability and other features means that there should rarely be diagnostic confusion with schizophrenia.

Epidemiology

Schizophrenia is a disorder with a low incidence and a high prevalence. Estimates of both have varied, reflecting the methodology and diagnostic criteria used, as well as probable differences between populations. Recent systematic reviews and meta-analyses have helped to clarify the estimates and their precision.

Incidence

The annual incidence is estimated at 0.16–1.00 per 1,000 population using a broad definition; for more restrictive diagnoses, the incidence is two to three times lower (Jablensky et al., 2011). A meta-analysis reported a mean

incidence of 0.24 per 1,000 population but a median incidence of 0.15 per 1,000 (McGrath et al., 2008). The difference reflects a skew, with many estimates in the upper tail of the distribution, which in turn results from studies that sampled from within populations with a high incidence (e.g. migrant groups, as discussed below). Another meta-analysis found an incidence of schizophrenia in England of 15.2 per 100,000 person years (Kirkbride et al., 2012a).

These and other recent analyses show that schizophrenia does not have a similar incidence (and prevalence) in all populations, a view which had arisen from the WHO Ten Country Study (Jablensky et al., 1992). Indeed, there is a fivefold global variation in incidence, with rates higher in developed than developing countries (McGrath et al., 2008). There are also variations in incidence related to a number of other environmental factors such as migration (see below).

Suggestions made in the latter part of the last century that the incidence of schizophrenia was falling have not been borne out; indeed, the incidence of diagnosed early-onset schizophrenia has, if anything, increased over recent decades (Okkels et al., 2013).

Prevalence

The lifetime prevalence is estimated at about 5 per 1,000 (Simeone et al., 2015), and a lifetime morbid risk of 7.2 per 1,000 (McGrath et al., 2008). These are median values because the data are skewed; the means are higher, at 5.5 and 11.9 per 1,000, respectively. The latter figure is more consistent with the frequent statement that about 1 in 100 people develop schizophrenia, but the value of 7 in 1,000 is statistically more appropriate.

Gender

Schizophrenia is more common in men than in women, with a male to female incidence ratio of 1.7 (Ochoa et al., 2012). The difference is more marked for severe cases, and is not due to differences in referral, identification, or age of onset. It has been attributed to the neuroprotective effects of oestrogens, but the evidence is inconclusive (Riecher-Rossler et al., 2018). Gender differences in genetic and epigenetic risk factors may also be relevant.

Age at onset

Schizophrenia is extremely rare in children under 10 years of age. The usual onset is between 15 and 54 years of age, most commonly in the mid-twenties; more detailed analyses suggest that there are two peaks, one at 20 years of age, and a second peak at 33 years of age. Gender differences in age of onset may explain these observations, as men have a single peak in their early twenties, whereas women have a broader range of age of onset, and a second peak in their fifth decade. However, the gender difference in mean age of onset is smaller and less robust than often stated (Jauhar et al., 2022).

For review of schizophrenia in children see Chapter 16; for review of late-onset schizophrenia, see Chapter 19.

Fertility

Whether patients with schizophrenia show decreased fertility has been a controversial topic, not least because of its implications for debate about whether schizophrenia is a 'disease', and also with regard to its genetic basis. A meta-analysis showed a substantial reduction in fertility in people with schizophrenia (about 40% of that expected), more so in men than in women (Bundy et al., 2011). These figures are difficult to interpret as being an intrinsic feature of schizophrenia, since opportunities for patients to have children have historically been limited by institutionalization, as well as by the amenorrhoea and sexual dysfunction caused by antipsychotic drugs. On the other hand, fertility is also decreased before the onset of psychosis (Zimbron et al., 2014), and is slightly reduced in the unaffected siblings of patients (Bundy et al., 2011). There are various theories but no explanation why schizophrenia persists in the population despite the apparent reduction in fertility.

Other aspects of schizophrenia epidemiology

Epidemiological aspects relevant to aetiology are considered in the next section. The course and outcome of schizophrenia are discussed later in the chapter.

Aetiology

Overview

Views about the aetiology of schizophrenia have been inextricably linked with the controversies regarding its nature and classification. Schizophrenia therefore exemplifies the whole range of biological, psychological, and social factors considered to be important in psychiatric causation, and the methods that have been applied to try to identify them. This section summarizes the current knowledge and theories in each domain, and also mentions some outdated but influential views. Table 11.3 lists the main evidence-based contemporary aetiological factors and theories.

Key aspects of the present consensus regarding the aetiology of schizophrenia can be summarized as follows. The most important influence is genetic, with 60–80% of the risk being inherited. The mode of inheritance is complex, and the genes—some of which have recently been identified—act as risk factors, not determinants of illness. A number of environmental factors contribute too, many of which act early in

life, and which interact with the genetic predisposition. Together, these and subsequent risk factors lead to a neurodevelopmental disturbance that renders the individual vulnerable to the later emergence of symptoms, and that manifests itself premorbidly in a range of behavioural, cognitive, and biological features. In schizophrenia, there are minor but clear differences in brain structure and function (though none of which are diagnostic), and their characteristics support the view that the syndrome is a disorder of brain connectivity. Acute psychosis is associated with excessive dopamine neurotransmission in the basal ganglia, which may be secondary to abnormalities of the glutamate system. Various psychosocial factors significantly influence the onset, and course, of illness. Finally, it is emphasized that there are few certainties about the aetiology of schizophrenia, and even where the facts are robust, their interpretation often remains unclear.

For review, see Birnbaum and Weinberger (2017) and McCutcheon et al. (2020).

Table 11.3 Schizophrenia: aetiological factors and theories

Category	Examples
Genetic	Single nucleotide polymorphisms
	Copy number variants
	Rare variants
Early environment	Maternal malnutrition
	Maternal infection
	Birth complications
	Urban birth and upbringing
Social	Migration
	Ethnic minority status
Other	Early cannabis use
Hypotheses	Neurodevelopmental
	Dopamine
	Glutamate
	Dysconnectivity
	Immune/inflammatory

Genetics

Family, twin, and adoption studies have all been applied extensively in schizophrenia, and provide irrefutable evidence for a major genetic contribution to the syndrome. Such studies set the context for the ongoing genomic and molecular studies designed to identify the genes and genetic mechanisms by which the risk for schizophrenia is mediated. For a review, see Owen et al. (2025). The reader is referred to Chapter 5 for an introduction to genetics and genetic terminology.

Family studies

The first systematic family study was conducted in Kraepelin's department by Rudin, who showed that the rate of dementia praecox was higher among the siblings of probands than in the general population. Gottesman (1991) reviewed the combined results of many studies and provided estimates for the lifetime risk of schizophrenia in various classes of relatives (Table 11.4). While some of these estimates are questionable, they do provide clear evidence of a familial aetiology, and the greater concordance rates in monozygotic compared to dizygotic twins indicates that this arises primarily from shared genes. Equally, the higher incidence in dizygotic twins than in siblings supports a role for shared environmental factors.

Table 11.4 Lifetime risk of schizophrenia in relatives of a proband with schizophrenia

Relative type	Lifetime risk of schizophrenia
Monozygotic twin	48%
Dizygotic twin	17%
Sibling	9%
Half-sibling	6%
Child with one affected parent	17%
Child with two affected parents	46%
First cousin	2%
Baseline risk	1%

Source: Gottesman I, (1991). *Schizophrenia Genesis: The Origins of Madness*. Copyright (1991), W. H. Freeman.

Family studies can also be used to determine whether the liability to schizophrenia and other disorders (e.g. bipolar disorder) is transmitted independently, which should be the observed pattern if the two disorders are separate syndromes with differing aetiology. Large-scale studies indicate clearly that there is no such independent transmission. For example, Lichtenstein et al. (2009) showed in the Swedish population that schizophrenia and bipolar disorder co-exist in families. It is also known that the risk of schizophrenia, schizoaffective disorder, and schizotypal and paranoid personality disorders is increased in first-degree relatives of patients with schizophrenia. These findings support the concept of a *schizophrenia spectrum*, in which the familial predisposition is to a range of disorders, not simply schizophrenia itself. This is also consistent with recent findings from genome-wide studies, discussed below.

Twin studies

Twin studies were introduced in Chapter 5. They have been of considerable importance in research on schizophrenia, providing unequivocal evidence of its heritability.

The first substantial twin study was conducted in Munich by Luxenberger in the 1920s. He found concordance in 11 of his 19 monozygotic (MZ) pairs and in none of his 13 dizygotic (DZ) pairs. Subsequent investigations show that concordance is several-fold higher in MZ than in DZ twins, with representative estimates of 40–50% concordance for MZ twins and about 10–15% for DZ twins (Cardno and Gottesman, 2000).

Modern twin studies also produce estimates of heritability (the proportion of liability to schizophrenia in the population which can be attributed to genes; see Chapter 5), and can separate environmental factors into those that are unique to the individual and those that are shared with others. A meta-analysis of twin studies (Sullivan et al., 2003) confirmed the substantial heritability of schizophrenia (81%; 95% CI, 73–90%). These data show that inheritance (i.e. genes) contribute the majority of the risk for schizophrenia. However, there are caveats about the interpretation of heritability figures. For example, the estimates make some assumptions about the ‘genetic architecture’ (i.e. how the genetic factors operate) and include gene–environment interactions. The meta-analysis also showed that most of the environmental contribution comes from shared rather than individual-specific influences (11%; 95% CI, 3–19%). The identity of these shared environmental factors is unknown. It is also worth noting that estimates for schizophrenia heritability from population studies are somewhat lower (e.g. 64% in the study by Lichtenstein et al., 2009) than those of twin studies, for reasons that are not entirely clear.

Among discordant MZ twins, the risk of schizophrenia is increased equally in children of the unaffected and the affected co-twin. This indicates that the unaffected co-twin had a similar genetic susceptibility to developing schizophrenia as the affected twin, but for some reason did not express the phenotype. This is most probably due to environmental protective factors, or chance (‘stochastic processes’), affecting the penetrance or expression of the genetic predisposition. Moreover, as a group, unaffected identical co-twins do exhibit some mild features of schizophrenia (in terms of symptoms and biological findings) that fall short of being diagnostically significant, but which are on average greater than those seen where neither member of a twin pair has schizophrenia. This probably reflects a partial expression of the risk genotype.

Adoption studies

Adoption studies provide another way to try and disambiguate genetic from family environmental factors. Heston (1966) studied 47 adults who had been born to mothers with schizophrenia and separated from them within three days of birth. As children they had been brought up in a variety of circumstances, although not by the mother’s family. At the time of the study their mean age was 36 years. Heston compared them with controls matched for circumstances of upbringing, but whose mothers had not suffered from schizophrenia. Among the offspring of the affected mothers, five

were diagnosed as having schizophrenia, compared with none of the controls. The rate for schizophrenia among the adopted-away children was comparable with that among children with a schizophrenic parent who remained with their biological family.

Further evidence came from a series of Danish studies that started in the 1960s (Petersen and Sorensen, 2011). In one project, two groups of adoptees were identified—a group of 33 adoptees who had schizophrenia, and a matched group who were free from schizophrenia. Rates of disorder were compared in the biological and adoptive families of the two groups of adoptees. The rate for schizophrenia was higher among the biological relatives of the adoptees with schizophrenia than among the relatives of the controls, a finding which supports the genetic hypothesis. Furthermore, the rate for schizophrenia was not increased among couples who adopted the affected children, which suggests that environmental factors were not of substantial importance. Follow-up studies confirmed that biological first-degree relatives of patients with schizophrenia have an approximately tenfold increased risk of suffering from schizophrenia or a related ('spectrum') disorder (Kety et al., 1994).

The data from adoption studies thus support the view that genetic factors explain the familial clustering of schizophrenia. However, it should be noted that they cannot control for the prenatal environment, which other studies, discussed below, suggest is important. Nor can they rule out environmental influences within the adoptive family.

The mode of inheritance

As pointed out in a seminal paper by Gottesman and Shields (1967), the frequency profile of schizophrenia among people with different degrees of genetic proximity to the proband does not fit any simple Mendelian pattern, as would be expected if the disorder was caused by a single major gene. Instead, the pattern indicates that schizophrenia likely arises from the cumulative effect of many genes each of small effect, as a so-called *polygenic disorder*, also known as a *complex genetic* or *non-Mendelian* disorder. The liability to schizophrenia lies along a continuum in the population, and is expressed when a certain threshold of genetic susceptibility is exceeded. No genes are either necessary or sufficient, and they act as risk factors, not determinants. Within a polygenic model, there can be a range of different kinds of genetic variation and genetic mechanisms, and many of which apply to schizophrenia (Smeland et al., 2020).

Schizophrenia susceptibility genes

Using the molecular approaches introduced in Chapter 5, findings show that three types of genetic variation contribute to schizophrenia risk:

1. Single nucleotide polymorphisms (SNPs)
2. Copy number variants (CNVs)
3. Rare variants (also called ultra-rare variants).

These vary in terms of their rarity and their impact (Figure 11.1).

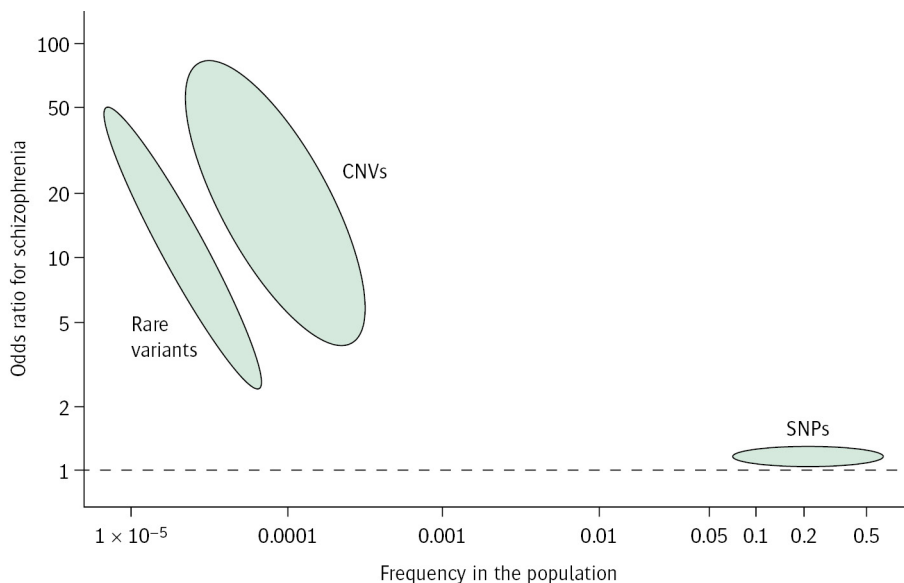


Figure 11.1 The allelic spectrum of schizophrenia. Both axes are logarithmic scales.

Single nucleotide polymorphisms (SNPs)

Much of the genetic risk for schizophrenia comes from SNPs, as revealed by genome-wide association studies (GWAS; see Chapter 5). The largest study to date, in over 75,000 cases and 240,000 controls, found SNPs in 287 distinct genomic loci to be associated with schizophrenia (Trubetskoy et al., 2022). Each SNP has a very small effect on risk, with odds ratios mostly less than 1.10. The implicated genes include a number which support prior hypotheses about schizophrenia (e.g. affecting synapses, glutamate, and neurodevelopment). The results are a major advance in schizophrenia genetics, but they leave many questions unanswered, as discussed below.

CNVs

CNVs were introduced in Chapter 5 (Kirov et al., 2015). There is now strong evidence that at least 20 specific CNVs are associated with schizophrenia (CNV and Schizophrenia Working Groups of the Psychiatric Genomics Consortium, 2017). The best known CNV is hemideletion (i.e. loss of one copy) of chromosome 22q11 (*velocardiofacial syndrome*; Box 11.9). Each CNV is very rare, yet has a substantial effect on risk. For example, hemideletion of 2p16.3 has an odds ratio of 14 for schizophrenia, being present in 0.17% of schizophrenia cases and 0.015% of controls.

About half of the CNVs associated with schizophrenia are inherited, the others are *de novo*—that is, they are not seen in either parent of the affected person; in effect, they are sporadic mutations, and the person's illness has not been inherited. However, if the affected person has children, each child has a 50% chance of inheriting the CNV.

Most schizophrenia-associated CNVs are also associated with a risk of other neuropsychiatric and neurodevelopmental phenotypes. However, there is not a clear excess of CNVs in bipolar disorder (Green et al., 2016), and this may be one of the points of separation in the aetiology of these disorders

Rare variants

A further small proportion of cases of schizophrenia are due to very rare coding variants in individual genes which, if present, confer a high risk of the disorder. The largest study to date used exome sequencing in over 24,000 cases and 97,000 controls and found rare variants in ten genes associated with schizophrenia with odds ratios between 3 and 50 (Singh et al., 2022). The variants, which are a single amino-acid change, are predicted to cause a loss-of-function in the encoded protein, mostly by truncating it. As with CNVs, some rare variants are inherited, some occur *de novo*, and they are

Box 11.9 Velocardiofacial syndrome: a schizophrenia-associated CNV

Unlike most CNVs associated with schizophrenia which were discovered recently, *velocardiofacial syndrome* (VCFS, also called *DiGeorge syndrome*) was already well known, being detectable using standard clinical genetic techniques due to its large size. VCFS is caused by deletion of one copy of chromosome 22q11 (hence its alternative name, *22q11 hemideletion syndrome*). It is a relatively common cytogenetic anomaly, occurring in 1 in 4,000 live births, and causes a range of physical abnormalities. Of relevance here, psychosis (either schizophrenia-like or affective) occurs in about 30% of individuals, with an odds ratio of 68 (CNV and Schizophrenia Working Groups of the Psychiatric Genomics Consortium, 2017). 22q11 hemideletion is also associated with other neurodevelopmental and neuropsychiatric syndromes (e.g. intellectual disability, ADHD), illustrating how one CNV can produce multiple phenotypes (*pleiotropy*). For review, see Zinkstok et al. (2019).

commoner in cases of schizophrenia with coexisting epilepsy or intellectual disability. The overall importance of rare variants (and CNVs) in schizophrenia will become apparent as exome and genome sequencing are carried out in larger samples.

Disrupted in schizophrenia-1 (*DISC-1*)

DISC-1 deserves mention because it has a unique place in schizophrenia genetics. It was identified from extensive studies of a large Scottish family in which a translocation between chromosomes 1 and 11 is linked with a high incidence of schizophrenia and several other disorders. *DISC-1* is the gene affected ('disrupted') by the translocation, and is thus the likely cause of the observed phenotypes, and experimental studies show that *DISC-1* is involved in brain development and function. However, no good evidence has emerged (from GWAS, or CNV or rare variant studies) that *DISC-1* has a role in schizophrenia outside this one family.

The genetic architecture of schizophrenia

We have seen that the different forms of genetic variation involved in schizophrenia differ in their frequency and their influence on risk. This is illustrated in Figure 11.1,

and is sometimes termed the *genetic architecture* or *allelic spectrum*. It highlights that there are some ultra-rare genetic variants of large effect, and a large number of common variants of very small effect.

The ‘missing heritability’

Much of the estimated 60–80% heritability of schizophrenia is still unexplained (‘missing’), despite the large numbers of SNPs, CNVs, and rare variants already identified (Owen et al., 2025). It has been estimated that SNPs collectively will explain about 30% of the genetic risk, CNVs another 5–10%, and rare variants a smaller amount. The remainder may arise from several sources:

- Many additional genetic variants, especially rare variants and small CNVs, remain to be identified.
- *Gene-gene interactions (epistasis)*. Epistasis is common for many complex traits and disorders, but studies have not been powered to investigate it in schizophrenia, and so its importance remains unclear (Webber, 2017).
- *Gene-environment interactions*. Many genes may affect schizophrenia risk indirectly, by affecting an individual’s response to environmental factors, as discussed in the following section (Wahbeh and Avramopoulos, 2021). Since environmental data are rarely collected in genomic studies, most of these interactions will have been missed.
- Epigenetic factors, some of which are heritable. See Smigielski et al. (2020).
- The heritability may have been overestimated.

The relative importance of these factors in explaining the missing heritability for schizophrenia (and other psychiatric disorders) is unknown.

The biology of schizophrenia genes

The discovery of multiple loci and genes associated with schizophrenia has been followed by attempts to understand the biological implications, and the molecular mechanisms (Harrison, 2015). Although these are at an early stage, several broad conclusions can be drawn:

- The genes converge on several functional networks and biochemical pathways. This applies not only to genes within the many loci identified by GWAS, but there is also an overlap between GWAS, CNV, and rare variant findings. In particular, there is a genetic influence on NMDA glutamate receptor-mediated signalling and synaptic plasticity. There is also a convergence on genes involved in immune function,

Box 11.10 Convergence of schizophrenia genes: the examples of NMDA receptor signalling and immune function

NMDA receptor signalling. A role for glutamate and NMDA receptor hypofunction has long been postulated in schizophrenia (Coyle et al., 2020). It is now supported by clear evidence that genes involved in glutamatergic signalling (Harrison and Bannerman, 2023), and synaptic function more generally (Hall and Bray, 2022), are over-represented among the genes affected by schizophrenia risk SNPs, CNVs, and rare variants.

Immune function. The major histocompatibility complex (MHC) locus on chromosome 6 shows strong statistical association to schizophrenia. There are two main interpretations of this finding. Firstly, that it supports a role for immune, autoimmune, or inflammatory factors in schizophrenia, consistent with a range of epidemiological and biological evidence discussed in this chapter. However, it is also known that many ‘immune genes’ also play a role in brain development, influencing synaptogenesis and other neural processes. Sekar et al. (2016) showed that the genetic signal at the MHC locus arises mainly from the complement component 4 (C4) gene, which is expressed by developing neurons and synapses. They also show that individuals with the risk forms of C4 express the gene at higher levels, and that (at least in mice) this is associated with more pruning of synapses during brain development.

voltage-gated ion channels, and neurodevelopmental processes. See Box 11.10, also Hall and Bray (2022).

- CNVs likely exert their effects mainly through gene dosage. That is, a deletion leads to insufficient levels of expression of the gene(s) affected (*haploinsufficiency*), and duplications to excess. Deletions may also be pathogenic by revealing the effects of a harmful recessive allele on the undeleted chromosome.
- The mechanism by which SNPs affect the risk of schizophrenia is more complicated to determine, for several reasons. Firstly, since a GWAS does not identify the causal gene or genetic variant, merely one that is ‘tagging’ the genetic risk. Secondly, even where the

causal variant has been inferred, very few are coding substitutions (in contrast to *ApoE4* and Alzheimer's disease; see Box 5.6), and so their functional significance is unclear. The mechanism is likely to be that the SNPs affect how the gene is regulated—for example, when and where it is transcribed or spliced—particularly during early brain development (Birnbaum and Weinberger, 2017).

Clinical and therapeutic implications of schizophrenia genetics

At present there is no accepted role of genetic testing for schizophrenia in clinical practice, since none of the genetic findings are diagnostic, and they do not have clearly established prognostic or therapeutic implications. However, this is under active debate, and the Royal College of Psychiatrists has recommended consideration of routine testing for CNVs in patients presenting with schizophrenia who also have coexisting conditions (such as congenital abnormalities, neurodevelopmental disorders, or marked cognitive impairment), and in childhood-onset schizophrenia (Royal College of Psychiatrists, 2023). The report concludes there is no current clinical indication to screen for rare variants, or for polygenic risk scores. The importance of attention to ethical issues (Morris et al., 2022), and access to genetic counselling, is emphasized.

Genetic discoveries are also providing new potential drug targets and encouraging investment in the field, but seem unlikely to lead directly to new pharmacotherapies in the near future because of the many complexities involved (Harrison et al., 2022).

Environmental risk factors

Despite the prominent genetic component, the twin studies discussed earlier, along with many epidemiological and other studies, show that environmental factors are also important in the aetiology of schizophrenia. A range of factors, especially pre- and perinatal ones, have been identified (Davies et al., 2020), and the major ones are highlighted in Box 11.11. For an overview, see Radua et al. (2018).

Several issues are worth bearing in mind when considering the individual risk factors to be discussed:

- The odds ratios, when known, are mostly between 1.3 and 3.
- The causality of most environmental risk factors for schizophrenia is often unknown, and differing interpretations of the reported associations are possible.

Box 11.11 Environmental risk factors for schizophrenia

Maternal infections
Maternal malnutrition
Birth (obstetric) complications
Winter birth
Advanced paternal age
Urban birth and upbringing
Childhood trauma and adversity
Being an immigrant
Being in an ethnic minority
Cannabis use
Tobacco smoking
Life events

- Many of the risk factors relate to early and prenatal life, and have contributed to the neurodevelopmental hypothesis of schizophrenia.
- Many of the risk factors are overlapping rather than independent, for example, cannabis use and tobacco smoking.
- Many of the environmental risk factors are not specific to schizophrenia but, like the genetic factors, are shared by other disorders.
- The distinction between genetic and environmental factors is an over-simplification, as noted earlier, there are gene–environment interactions. Some specific examples are given in the following sections.

Obstetric complications

Rates of schizophrenia are increased in individuals who experienced obstetric complications (also called *birth complications*), compared with their unaffected siblings or normal controls. Meta-analyses suggest an odds ratio of about 2 (Cannon et al., 2002). A range of exposures are associated with increased risk, including antepartum haemorrhage, low birth weight, needing an incubator, and Rhesus incompatibility, and it is unclear which are the most important and the most robust.

The association of these diverse obstetric complications with schizophrenia has several possible explanations. They might be directly causal (e.g. via intrapartum foetal hypoxia), or a reflection of pre-existing foetal brain abnormality, or even a reflection of maternal characteristics. A common factor and key mechanism may be dysfunction of the placenta in response to stress, which

amplifies the effect of schizophrenia genetic risk, mediated through placental expression of a subset of these genes (Ursini et al., 2018). The effect is greater in males and might contribute to the sex differences in psychosis.

Infective and inflammatory factors

Several studies suggested that fetuses exposed during the second trimester to influenza, especially the 1957 influenza A2 pandemic, have an increased risk of schizophrenia. Some biological support for this association was provided by animal studies which showed that prenatal influenza affects foetal brain development, and by the demonstration that serological evidence of influenza infection during early pregnancy was associated with an increased risk of schizophrenia in the offspring. However, other studies yielded negative results and a relationship between influenza and schizophrenia has been hard to establish. Several other maternal (and childhood) infections have also been associated with schizophrenia, notably toxoplasmosis (Cheslack-Postava and Brown, 2022). A number of studies have measured inflammatory markers (such as interleukins, cytokines, and C-reactive protein) in serum collected during pregnancy, and related this to schizophrenia risk in the offspring (Kappelmann et al., 2022). Although these markers are often taken as supportive of an infective origin, inflammation may occur for various reasons (Cannon et al., 2014). There may also be an interaction of genetic factors with inflammatory processes contributing to schizophrenia, notably via interleukin IL-6 (Williams et al., 2022).

Covid-19 infection has been associated with an increased risk of psychosis, including schizophrenia, in the following two years (Taquet et al., 2022). The clinical features and outcome of these cases, and the mechanism, remains unclear.

Maternal malnutrition

Children born to mothers who experienced famine early in their pregnancy have an increased risk of schizophrenia, with an odds ratio of about 2. This was shown initially in children born after the Dutch 'Hunger Winter' of 1944, and subsequently in two Chinese famines in the 1960s. The mechanism is proposed to be epigenetic—malnutrition alters DNA methylation and other processes regulating gene expression in the affected offspring (Kirkbride et al., 2012). It is not known whether the cause is related to general malnutrition or lack of specific micronutrients (e.g. vitamin D), nor is it clear whether less-extreme fluctuations in nutritional status during pregnancy affect schizophrenia risk. For review, see Brown and Susser (2008).

Winter birth

Schizophrenia is slightly more frequent among people born in the late winter than among those born in the summer (Davies et al., 2003). This season of birth effect has been shown in both hemispheres, and becomes more prominent at higher latitudes. The explanation is unknown. It has been linked to the prevalence of influenza earlier in the winter, and to sunshine and vitamin D levels around the time of birth, but it is independent of genetic risk for schizophrenia (Escott-Price et al., 2019).

Paternal age

A replicated finding is that schizophrenia is associated with paternal age, especially in those without a family history of psychosis. A meta-analysis found a pooled risk estimate of 1.31 (Olderoid et al., 2018). The favoured explanation is that the frequency of *de novo* mutations or epigenetic changes in sperm increase with age. However, it may also be related to how personality or psychosocial relationship factors modify the age at which a man first fathers a child (Khachadourian et al., 2021).

Child development

Parnas et al. (1982) reported a study of 207 children of mothers with schizophrenia who were first assessed when they were 8–12 years of age and then again, as adults, 18 years later, by which time 13 individuals had developed schizophrenia and 29 had 'borderline schizophrenia.' Of the measures that were used on the first occasion, those that predicted schizophrenia were poor rapport at interview, being socially isolated from peers, disciplinary problems mentioned in school reports, and parental reports that the person had been passive as a baby.

In a national birth cohort of more than 16,000 children prospectively studied over a 16-year period, those who developed schizophrenia could be distinguished at the age of 11 years, if not earlier, by greater hostility towards adults, and by speech and reading difficulties (Done et al., 1994). This difference was apparent compared to those who grew up to develop a neurotic illness (anxiety or depression), as well as those who remained well. In another large prospective cohort study, Jones et al. (1994) found that children who eventually developed schizophrenia showed delayed milestones and speech problems, together with lower education test scores and less social play. A graded relationship between delayed milestones (e.g. age at walking) and schizophrenia has been confirmed in several cohorts. The behaviour of children (including motor acts in infants), as

recorded in home movies and videotapes, is also associated with later schizophrenia (Schiffman et al., 2004).

These and other studies provide good evidence that individuals who will develop schizophrenia show increased rates of intellectual and motor dysfunction and poor social competence in childhood. However, it is unclear how specific these changes are, nor how they are related to the subsequent development of the illness. Most of the children who were destined to develop schizophrenia were not considered clinically abnormal at the time, and many children who perform poorly on these indices do not develop schizophrenia. Nevertheless, the findings support an early neurodevelopmental contribution to schizophrenia, as outlined below.

Substance use

The high prevalence of drug and alcohol use in patients with schizophrenia is well established, but whether substance misuse plays a causal role in schizophrenia is more controversial. If substance use is considered to have directly caused or precipitated the psychosis, it is diagnosed as such, and not as schizophrenia. However, there is evidence that prior use of some drugs, particularly cannabis, is associated with an increased risk of later developing schizophrenia. Andreasson et al. (1987) followed up over 45,000 Swedish conscripts for 15 years, and found that the relative risk of developing schizophrenia was 2.5 times higher in subjects who used cannabis, with a sixfold increase in risk for heavy users. Subsequent longitudinal studies show a consistent association between cannabis use and psychosis (with an odds ratio of about 2), which may be modified by interaction with genetic factors, and with the strength of the cannabis (the tetrahydrocannabinol content). Cannabis use is also associated with an earlier age of onset of psychosis, and with violence in patients with psychosis. However, the causality and specificity of the relationships between cannabis use and psychosis, and hence the implications for public health policy, remain controversial (Ganesh and D'Souza, 2022; see also Chapter 20). For review, see Hasan et al. (2020).

Tobacco smoking is also a risk factor for development of psychosis. As with cannabis use, it is likely, but not certain, that this is a causal relationship (Hunter et al., 2020).

Social and psychosocial factors

Although neglected in recent years compared to biological risk factors, social and psychosocial factors are also important in schizophrenia and are now being integrated into biopsychosocial models. Some studies are

seeking to identify whether particular symptom clusters are linked to different biological and social pathways (Betz et al., 2023).

Occupation, social class, and income

Schizophrenia is over-represented among people of lower social class (Gallagher, 2018). In Chicago, Hollingshead and Redlich (1958) found the incidence and prevalence of schizophrenia to be highest in the lowest socioeconomic groups. At first, these findings were thought to be of aetiological significance, but they could be a consequence of schizophrenia. For instance, Goldberg and Morrison (1963) found that people with schizophrenia were of lower social status than their fathers, and that this was usually because they had changed status after the onset of the illness. However, Castle et al. (1993) found that, compared with controls, patients with schizophrenia were more likely to have been born into socially deprived households. Those authors proposed that some environmental factor of aetiological importance was more likely to affect individuals of lower socioeconomic status. Schizophrenia is also associated with poverty and income inequality (Burns, 2014).

Place of birth and residence

Faris and Dunham (1939) studied the place of residence of mentally ill people in Chicago, and found that individuals with schizophrenia were over-represented in the disadvantaged inner-city areas. This distribution has been confirmed in other cities, and it has been suggested that unsatisfactory living conditions can cause schizophrenia. These findings have also been ascribed to the occupational and social decline ('*social drift*') described above, or to a search for social isolation by people who are about to develop schizophrenia.

However, schizophrenia is also associated with place of birth and early upbringing, findings which cannot be explained as a consequence of illness. Specifically, population-based studies in several countries show that urban birth is associated with an increased risk of schizophrenia (Vassos et al., 2012). Larger cities confer a higher odds ratio than small towns. The cause and interpretation of the association remains unclear, and may relate to social deprivation, migration, infections, stress, or interactions between genetic vulnerability and urban environments (Abrahamyan Empson et al., 2020).

Migration and ethnicity

In a study of Norwegians who had migrated to Minnesota, Ødegaard (1932) found that the inception rate for schizophrenia was twice that of Norwegians living in Norway. Many subsequent studies have confirmed high rates of

schizophrenia among migrants (Selten et al., 2020). The relative risk ranges from about 2–4.5, and tends to be higher in migrants from lower- and middle-income countries. This literature is complemented by findings of ethnic inequalities in diagnosis of psychosis (Halvorsrud et al., 2018).

A particularly controversial aspect of migration, ethnicity, and schizophrenia concerns the African-Caribbean population in the United Kingdom. This group has an increased incidence of schizophrenia, particularly in the ‘second generation’, who were born in the United Kingdom. The relative risk for all psychosis ascertained in the multi-centre first-onset AESOP study was 6.7 for the African-Caribbean population, and 1.5 for the Asian population, compared with the white population (Fearon et al., 2006). Overall, rates in Caribbean-born migrants and their descendants in England are increased over fivefold (Halvorsrud et al., 2019).

The increased rate of schizophrenia in migrants has sometimes been attributed to a disproportionate migration of people who are unsettled because they are becoming mentally ill. Alternatively, the effect of migration could be due to exposure to environmental factors in the host country, especially in those who are genetically or otherwise predisposed. Thus both ‘social selection’ and ‘social causation’ may contribute. Among the latter category, evidence indicates that experiential factors likely play a significant part. Studies show that the high incidence in ethnic minority patients varies according to where they grew up in the host country. Controlling for known risk factors (e.g. cannabis use and deprivation), the rates are still more than doubled when these individuals grow up in areas where they are members of a small minority as opposed to areas where they are a relatively large minority. The mechanism by which this differential ‘ethnic density’ impacts upon rates of schizophrenia is unclear (Shaw et al., 2012). The explanation for the increased incidence of schizophrenia in migrants likely includes a range of additional factors too, including institutional racism and perceived discrimination (Morgan et al., 2019; Schwartz and Blankenship, 2014).

Life events and difficulties

Life events and difficulties have often been proposed as precipitants of schizophrenia. Brown and Birley (1968) used a standardized procedure to collect information from 50 patients newly admitted with a precisely datable first onset or relapse of schizophrenia. By comparison with a control group, the rate of ‘independent’ events in those with schizophrenia was increased during the three weeks before the onset of the acute symptoms. Paykel

(1978) calculated that experiencing a life event doubles the risk of developing schizophrenia during the subsequent six months. A meta-analysis yielded an odds ratio of 3.2 for life events and psychosis (Beards et al., 2013).

Childhood adversity

There is an increased prevalence of childhood trauma and abuse, including bullying, in individuals who develop schizophrenia, with an odds ratio of about 3. The risk is greater for multiple and severe adversities. The relationship is not explained fully by genetic factors and is likely to be at least partly causal; however, most evidence comes from retrospective studies which are subject to recall bias (Jauhar et al., 2022). For review of childhood adversity and schizophrenia, see Sideli et al. (2020).

Psychological factors

Several early writers, including Bleuler, commented on the frequency of abnormalities of personality preceding the onset of schizophrenia. In the 1930s, following Bleuler, Kretschmer viewed ‘schizoid personality’ as a partial form of schizophrenia. The evidence for both views is weak, although Kretschmer’s ideas are similar to the current concept of the schizophrenia spectrum, in which schizophrenia represents the most severe end of a continuum. However, many people with schizophrenia have an unremarkable premorbid personality, and only a minority of those with schizotypal or schizoid personalities develop schizophrenia. For review of personality and psychosis, see Franquillo et al. (2021).

Psychological theories

In the past, psychological theories of schizophrenia were based on psychodynamic concepts and often centred on a pathogenic role of the mother and family (see Box 11.12). Such theories are now only of historical interest. However, it is important to know about them because they help focus attention on interpersonal and developmental aspects, and because they remain influential with some families and health professionals. They also contributed to the family therapy approaches discussed later. Current psychological models of schizophrenia are more grounded within neuroscientific and cognitive theories, as discussed elsewhere in this chapter.

An early neuropsychological model proposed that positive symptoms arise from a failure to integrate stored memories with current stimuli (Gray et al., 1991). The *aberrant salience* model of Kapur (2003), mentioned later, shares some features with Gray’s model and

Box 11.12 Family dynamics as a cause of schizophrenia

In the middle of the last century, the family was proposed to be a cause of schizophrenia, either through *deviant role relationships* (associated with Fromm-Reichmann and Lidz) or *disordered communication* (associated with Bateson and Wynne). The role of the family in influencing the *course* of schizophrenia is discussed later.

The '*schizophrenogenic*' mother, a cold and distant figure unable to foster emotional growth, was suggested by the analyst Fromm-Reichmann, while Lidz and his colleagues used intensive psychoanalytical methods to study the families of 17 wealthy patients with schizophrenia, and suggesting two abnormal family patterns:

1. *marital skew*, in which one parent yielded to the other's (usually the mother's) eccentricities, which dominated the family; and
2. *marital schism*, in which the parents maintained contrary views, and consequently the child had divided loyalties.

A *double bind* (associated with Bateson) was said to occur when an instruction is given overtly, but is contradicted by a second more covert instruction. For example, a mother may tell her child to come to her, while conveying by her manner and tone of voice that she rejects them. Double binds are only pathogenic within families that implicitly prohibit the acknowledgement of this contradiction. Double binds trap the child in ambiguous responses and inhibit the development of logical thinking.

An association between abnormal social communication in parents and schizophrenic illness in children may in fact be a consequence of a shared genetic inheritance or a response to the stresses of relating to a child who is exhibiting symptoms or is in the prodrome of psychosis. These and other speculations about the causative role of family relationships have had the unfortunate consequence of inducing unjustified guilt in parents.

links neuropsychology to the dopamine hypothesis of schizophrenia.

Frith (1996) argued that in schizophrenia there is a breakdown in the internal representation of mental events. For example, a failure to monitor and identify

one's own willed intentions might give rise to the idea that thoughts and actions arise from external sources, resulting in delusions of control. This theory has a counterpart in models of aberrant connectivity and brain networks to be discussed below.

Neurobiology

Schizophrenia is at the forefront of attempts to understand psychiatric disorders in terms of brain structure and function, utilizing the whole range of contemporary neuroscientific concepts and techniques introduced in Chapter 5. For review, see McCutcheon et al. (2020) and Zoghbi et al. (2023).

Structural brain changes

Whether there is a neuropathology associated with schizophrenia has been a matter of debate for over a century. The search began with Alzheimer, who spent a decade studying the brains of patients with dementia praecox (Box 11.4) before he reported the case of presenile dementia with which his name is associated. The failure of Alzheimer and others to identify a neuropathology was central to the view of schizophrenia

as a functional disorder rather than an organic one. However, evidence has accrued over the past 40 years which disproves the null hypothesis that there are no structural differences in the brain in schizophrenia, albeit their details and interpretation remain poorly understood, and the findings are not clinically useful in the diagnosis of individual patients. For review, see Lawrie (2025).

Structural brain imaging

In a landmark study, Johnstone et al. (1976) used the novel technique of computerized tomography, and found significantly larger ventricles and cerebral atrophy in 17 elderly hospitalized patients with schizophrenia than in eight healthy controls. A large number of subsequent imaging studies, mostly using MRI, and meta-analyses thereof, have confirmed the findings and

Box 11.13 Summary of structural brain changes in schizophrenia

Brain imaging

- Decreased brain volume
- Decreased intracranial volume
- Enlarged lateral and third ventricles
- Smaller hippocampus and thalamus
- Less grey matter
- Reduced cortical thickness
- Altered white matter pathways

Neuropathology

- Decreased brain weight
- Absence of neurodegenerative changes or gliosis
- Reductions in synaptic and dendritic markers
- Decreased markers of some interneurons
- Fewer thalamic neurons

extended them to a range of other structural parameters (Box 11.13). A few points are worthy of note.

- Ventricular volumes are increased by about 30%, grey matter volume decreased by about 2% (7% in fronto-temporal regions), and intracranial volume by 2%. The latter reduction implies an early developmental origin to the brain volume reduction. This interpretation is consistent with the fact that similar but lesser reductions in brain volume are seen in those at genetic risk (de Zwarte et al., 2019) and linked to subsequent transition to psychosis.
- There is no evidence for an ‘organic’ subtype of schizophrenia. Rather, there is a unimodal shift in ventricular and brain volumes. There are few clear clinico-pathological associations, although particular structural alterations have been associated with specific symptoms, for example, thought disorder with smaller superior temporal gyri.
- The imaging findings in Box 11.13 are seen in first-episode patients as well as in chronic schizophrenia (Kuo and Pogue-Geile, 2019). The progression of structural brain changes after the first episode is controversial. The view that there is no progression is an oversimplification. There is further loss of brain tissue exceeding that seen in healthy controls, but its timing, magnitude, and explanation remain unclear (Weinberger and Radulescu, 2016). Use of antipsychotic medication contributes to the changes during the illness, but cannot explain it entirely.

- The effect sizes for structural brain differences in schizophrenia are small to moderate, with significant overlap between patients and controls. Furthermore, they are not diagnostically specific—for example, some of the findings are also seen in bipolar disorder. For these reasons, structural brain imaging is not usually considered a routine investigation in people presenting with schizophrenia (although this remains under debate; see below and Blackman et al., 2023).

Neuropathology

Post-mortem neuropathological studies have sought to explain the cellular and molecular basis for the neuro-imaging findings, especially the reductions in grey matter. For review, see Harrison (1999). Though the positive findings are less robust than those from structural imaging, a few findings deserve mention (Box 11.13):

- Brain weight is decreased by 2–3%.
- There is no good evidence for any neurodegenerative processes, such as gliosis, neurofibrillary tangles, or β -amyloid plaques.
- The main positive findings are reductions in some markers of synapses and dendrites, and in specific neuronal populations (e.g. the parvalbumin subclass of interneurons, and layer III pyramidal neurons) (Berdens von Berlekom et al., 2020).
- The nature of the neuropathological changes strongly implies an origin early in life, and has contributed to the neurodevelopmental hypothesis of schizophrenia and involvement of aberrant synaptic pruning in that process (Johnson and Hyman, 2022). However, virtually all of the patients whose brains are studied post-mortem had been ill, and medicated, for many years. It is therefore difficult to determine the extent to which the findings in schizophrenia are confounded by the consequences of illness, antipsychotic medication, and other factors. In this respect, the recent development of a PET tracer enabling measurement of a synaptic protein, SV2A, has been valuable, and results shows that synaptic pathology is present early in the illness (Howes et al., 2023).

Functional brain imaging

As described in Chapter 5, a number of imaging techniques, particularly PET and fMRI, have been used to assess patterns of brain activity in schizophrenia. Study designs usually compare patients with controls, either at rest or during performance of different kinds of cognitive task. For review, see Howes et al. (2023).

Cerebral blood flow and metabolism

The first PET study was conducted by Ingvar and Franzen (1974), who used injections of radioactive xenon, and found decreased perfusion of the frontal cortex compared with the posterior regions in chronic, medicated patients. This ‘hypofrontality’ has sometimes been considered to be a feature of schizophrenia and, although many studies have given negative results, was confirmed in a meta-analysis (Hill et al., 2004). However, stronger relationships with hypofrontality are seen if the phase of illness and symptom profile are considered (e.g. the association with psychomotor poverty) (see Table 11.1). Blood flow is now usually assessed using MRI methods, and the results also suggest reductions in frontal regions in those with negative symptoms, whereas positive symptoms are associated with subcortical hyperperfusion (Percie du Sert et al., 2023).

Glucose use is a marker of brain metabolism and can be measured using PET with 18-fluorodeoxyglucose (FDG). Meta-analysis shows moderate to large reductions at rest in chronic schizophrenia, especially in frontal lobe. However, findings were related to antipsychotic use and were not significant in first episode cases (Townsend et al., 2023).

N-acetylaspartate (NAA) is a metabolite that indexes neuronal activity, and can be measured using magnetic resonance spectroscopy. Similar to the FDG findings, NAA is lower in chronic schizophrenia, but it is also reduced in several regions in first-episode patients and during the prodrome (Whitehurst et al., 2020).

In summary, there is evidence from different modalities for reductions in cortical metabolism and activity in schizophrenia. However, their relationships to the structural imaging and neuropathological findings remains unclear.

BOLD signal on fMRI

Altered frontal activity as measured by the BOLD response (Chapter 5) is apparent in fMRI studies of schizophrenia, particularly in the prefrontal cortex during performance of working memory tasks, such as the Wisconsin Card Sorting Test and the N-back task. The profile of changes is complex (Minzenberg et al., 2009). When patients are matched with controls for performance, patients require more frontal cortex activation to achieve the same level of performance as controls, which suggests that there is a reduced ‘efficiency’ of cortical processing (Callicott et al., 2003).

Other fMRI studies have looked at the cerebral correlates of specific symptoms—for example, the cortical regions that are activated during auditory hallucinations.

There are also alterations in ‘resting state’ activity in particular brain networks (Dong et al., 2018).

Aberrant connectivity

Bleuler conceptualized the fundamental symptoms of schizophrenia as ‘psychic splitting’ or a failure of integration of mental functions, and Wernicke and other neuroanatomists of the late nineteenth century took a ‘connectionist’ view (Collin et al., 2016). A similar view is currently framed in terms of ‘dysconnectivity’, whereby the activity of different brain circuits or networks is aberrant in schizophrenia (Stephan et al., 2009). Some variants of this model also involve a structural component (changes in the ‘wiring’ of the brain), consistent with the structural imaging and neuropathology findings mentioned earlier. Recent studies focus on brain oscillations and network activity, the coordinated firing of ensembles of neurons between brain areas underlying cognition and other brain operations (Gonzalez-Burgos et al., 2015).

Neurophysiological findings

Neurophysiological function in schizophrenia has been examined using a range of methods. For review, see Kim et al. (2020).

Electroencephalography and evoked potentials

The resting electroencephalogram (EEG) shows an increased preponderance of slow rhythms in schizophrenia (Boutros et al., 2008), but more interesting findings have emerged from sensory evoked potentials.

The P300 response is an auditory evoked potential which occurs 300 milliseconds after a subject has identified a target stimulus embedded in a series of irrelevant stimuli. The ‘oddball’ response provides a measure of auditory information processing. In patients with schizophrenia, and a proportion of their first-degree relatives, the amplitude of the P300 wave is reduced. Abnormalities in another evoked potential, the P50 wave, are also observed in patients with schizophrenia and their relatives. Moreover, molecular genetic studies in families with schizophrenia have shown that P50 deficits are linked to the gene coding for a subunit of the nicotinic cholinergic receptor. This suggests that cholinergic neurotransmission could underlie abnormalities in information processing in schizophrenia.

The P50 and P300, as well as other evoked potentials, are also altered in people at risk of schizophrenia (Perrotelli et al., 2021), and in psychotic bipolar disorder, and have potential use as biomarkers (Parker et al., 2021).

Biochemical findings

Dopamine

Two early lines of research converged and implicated dopamine as having a key role in schizophrenia. The first concerned the effects of amphetamine which, among other actions, releases dopamine at central synapses. Repeated use of amphetamine at high doses can induce a disorder similar to acute schizophrenia and acute amphetamine administration worsens psychotic symptoms in people with schizophrenia. The second approach starts from the finding that all antipsychotic drugs are dopamine-receptor antagonists, and that their affinity at dopamine D₂ receptors is the property that correlates best with their clinical potency. These observations led to the well-known ‘dopamine hypothesis’ of schizophrenia, which emerged during the 1970s and has persisted.

Despite the prominence and longevity of the theory, evidence that dopamine neurotransmission is excessive or otherwise abnormal in schizophrenia was initially hard to obtain, in part because of the difficulty in distinguishing drug effects from disease effects in medicated patients. However, the availability of PET and SPET to image dopamine receptors and dopamine synthesis in various ways revealed good evidence to support the presumed ‘hyperdopaminergia’ in acute schizophrenia. There is a slight increase of dopamine D₂ receptors, and an increased occupancy of the receptors, but the main finding is that the synthesis and release of dopamine in the striatum are markedly increased (Howes et al., 2012). Moreover, the extent of the increase predicts the response to antipsychotic medication. The abnormality develops during the prodrome and is not seen in patients during remission. Contrary to expectations, the excess dopamine function occurs primarily in the associative and not the ventral striatum (i.e. in the mesocortical rather than the mesolimbic dopamine pathway).

An influential phenomenological account about the role of dopamine in psychosis was proposed by Kapur (2003). He argues that the known roles of dopamine in cognition and behaviour suggest that it mediates the ‘salience’ of external events and internal representations, and that in schizophrenia the dopamine abnormalities lead the patient to misattribute stimuli and their meaning. Delusions and hallucinations are viewed as the consequence of these experiences and the patient’s attempt to make sense of them. Later research has shown that dopamine does more than simply signal the salience of events, but has a complex role in detecting differences between expectation and outcomes, called *prediction error*. In schizophrenia there is evidence for altered prediction error signals, likely another reflection of the

dopaminergic abnormalities (Millard et al., 2022). For review, see Feldman and Schmack (2025).

Other aspects of schizophrenia may be associated with different changes in dopamine function. In particular, various lines of evidence suggest that there is deficient dopaminergic activation of the prefrontal cortex, relevant to the cognitive deficits. There are plausible pathways and mechanisms to link these opposing dopaminergic abnormalities in striatum and frontal cortex (Weinberger, 1987; 2022).

The cause of dopaminergic involvement in schizophrenia are unclear. There is a genetic association to the D₂ receptor, but current theories emphasize the dopamine dysregulation as being secondary to glutamatergic and developmental abnormalities (McCutcheon et al., 2020).

Glutamate

The excitatory neurotransmitter amino-acid glutamate is second only to dopamine in neurochemical theories of schizophrenia (Coyle et al., 2020). The key finding which triggered this interest was that antagonists of the N-methyl-D-aspartate (NMDA) type of glutamate receptor (e.g. phencyclidine) can induce a schizophrenia-like psychosis. There are now various lines of evidence supportive of an ‘NMDA receptor hypofunction’ model, including the involvement of glutamate receptor genes noted earlier (Box 11.14). Some cases of schizophrenia are also associated with anti-NMDA receptor antibodies; the findings are intriguing and their causal and therapeutic significance are being investigated (Lennox, 2022).

Box 11.14 Evidence for glutamatergic involvement in schizophrenia

NMDA-receptor antagonists induce or exacerbate schizophrenia symptoms

NMDA-receptor co-agonists (e.g. D-serine) reduce some symptoms

Alterations in glutamate markers and receptors in schizophrenia brain

Altered levels of glutamate in schizophrenia brain shown using spectroscopy

Schizophrenia risk genes affect glutamate receptors and glutamate signalling

Anti-NMDA receptor antibodies in some cases

Glutamate and dopamine are intricately linked, and their dysfunction in schizophrenia is also thought to reflect shared factors. As noted above, glutamate is generally considered to be the primary or 'trait' abnormality, with the dopamine abnormality downstream and 'state' related. Alternatively, there may be differential involvement of glutamate and dopamine in different patients. For review, see McCutcheon et al. (2020).

Gamma-aminobutyric acid (GABA)

GABA is the major inhibitory transmitter in the brain, and it is implicated in schizophrenia for several reasons (Gonzalez-Burgos et al., 2015). There is a decrease in its synthetic enzyme, glutamic acid decarboxylase (GAD), in the cerebral cortex, and reductions in the connections made by a type of GABA neuron (parvalbumin-containing interneurons). There are also alterations in the expression of GABA_A receptors, and reduced GABA levels as measured by spectroscopy. The relationship between GABA and glutamate alterations in schizophrenia is unclear; one point of convergence is that NMDA receptors on GABA neurons may be especially vulnerable. A related notion is that there is an imbalance between excitation and inhibition ('E/I imbalance') in schizophrenia (Howes and Shatalina, 2022).

Other neurotransmitters

A role for *serotonin* (5-HT) in schizophrenia has long been considered, because the hallucinogen, lysergic acid diethylamide (LSD), is an agonist at 5-HT₂ receptors. Other evidence comes from alterations in 5-HT receptor densities in schizophrenia, notably a reduction in the number of frontal cortex 5-HT_{2A} receptors and an increase in 5-HT_{1A} receptors.

Cannabinoid signalling and CB1 receptors are of interest, because of the associations of cannabis use with psychosis, and the potential therapeutic value of cannabidiol. However, direct evidence that the cannabinoid system is affected in schizophrenia is limited (Borgan et al., 2021).

Inflammatory markers

The possible involvement of inflammation and immunity in schizophrenia has been mentioned with regard to risk genes and prenatal factors. There are also many studies showing alterations in various inflammatory markers in blood, and some in brain, suggestive of a pro-inflammatory state. Such activation can in turn influence the dopamine, glutamate, and 5-HT systems (de Bartolomeis et al., 2022). These findings have contributed to an emerging view of schizophrenia as a multisystem disorder (Pillinger et al., 2019).

Neurodevelopmental model

The notion that schizophrenia is a disorder of brain development was proposed by Clouston in 1892 and by others, including Kraepelin, at the turn of the last century. Contemporary interest can be traced primarily to Weinberger (1987). Specific forms of the

neurodevelopmental model focus on the second trimester *in utero*, adolescence, or an interaction between the two. Evidence in favour of the model has accrued from several sources (Box 11.15). Many of the factors have

Box 11.15 Findings that support the neurodevelopmental model of schizophrenia

Schizophrenia risk genes affect brain development
Structural brain changes present at or before illness onset
Limited progression of brain changes after onset
Motor, cognitive, and social impairments in children who later develop schizophrenia
Neuropathological changes without gliosis implies a prenatal timing

Most environmental risk factors relate to pre- and perinatal period
'Soft' neurological signs
Increased frequency of minor physical anomalies
Increased frequency of large cavum septum pellucidum
Findings from animal models

already been discussed in this chapter. For review, see Birnbaum and Weinberger (2017).

There are no serious challenges to the neurodevelopmental model as a hypothesis of schizophrenia pathogenesis. This partly reflects its vagueness, which

allows most findings to be interpreted as being consistent with it. Its weaknesses include a failure to readily explain the variable course of the disorder, or provide a convincing explanation for late-onset schizophrenia.

Course and prognosis

The course and long-term outcome of schizophrenia is a fiercely debated issue, in part due to differing ways in which diagnosis was made, patients followed up, and recovery defined.

Kraepelin initially believed that dementia praecox had an invariably poor outcome, although later reported that 17% of his patients eventually were socially well adjusted. Manfred Bleuler (1974), son of Eugen Bleuler, personally followed up 208 patients who had been admitted to his hospital in Switzerland between 1942 and 1943. He concluded that the outcome was not so gloomy: 20 years after admission, 20% exhibited complete remission, 35% had good social adjustment, but 24% remained severely disturbed. When full recovery had occurred, this was usually within the first two years. Subsequent studies have broadly supported these figures. A meta-analysis of outcomes in the five to seven years after first-episode psychosis found remission in 58% and recovery in 38% (Lally et al., 2017). At the other extreme, a ten-year follow up of a UK cohort found 23% had experienced an unrelenting illness (Morgan et al., 2014). Comparison of outcome figures across time is fraught with difficulty, and compounded by possible geographical differences mentioned below, but the lack of an unequivocal or marked improvement in outcomes over the past century is disappointing and highlights the need for more effective, and preventative, interventions (Davies et al., 2018).

For a review of the course and clinical outcome of schizophrenia, see an der Heiden and Häfner (2011). Kooyman and Walsh (2011) reviewed the societal outcomes.

Physical health, mortality, and suicide in schizophrenia

In an early meta-analysis, Harris and Barraclough (1998) found that the risk of death from all causes was increased 1.6-fold in people with schizophrenia. Subsequent studies have found a higher figure, with a risk ratio of about 2.5, and with a higher risk in first-episode cases

and those with comorbid substance use (Correll et al., 2022). Expressed another way, men with schizophrenia die approximately 20 years prematurely, and women 15 years prematurely.

About 60% of the excess early mortality in schizophrenia is accounted for by unnatural causes (suicide and accidents), and the rest from natural causes. The latter are especially cardiovascular disease, partly reflecting the high rates of smoking, as well as poor diet, sedentary lifestyle, and higher rates of obesity, type 2 diabetes, and other physical illnesses. Lower rates of diagnosis and less active treatment of these general medical disorders may also contribute. Medication, especially atypical antipsychotics, has been postulated to contribute significantly to this excess mortality via their metabolic effects, and hence the importance of metabolic monitoring and management in people taking these drugs (Vancampfort et al., 2019; see Chapter 25). However, these risks should be set against the fact that antipsychotic use is protective against all-cause mortality, even at high doses, compared to no treatment (Correll et al., 2022).

The lifetime risk of suicide in schizophrenia is often quoted as 10% or more, but meta-analyses suggest a figure of about 5%. The risk is greatest early in the illness, and in those with affective symptoms, a history of suicide attempts, recent discharge from hospital, and number of psychiatric admissions (Popovic et al., 2014).

For review of mortality in schizophrenia, see Laursen (2019).

Predictors of outcome

The outcome of schizophrenia is, as noted earlier, heterogeneous and still largely unpredictable (an der Heiden and Häfner, 2011). However, epidemiological data, and some characteristics of the first presentation, have identified a number of poor prognostic factors, including younger age of onset, male sex, poor premorbid functioning, a prolonged duration of untreated psychosis, and early negative symptoms and cognitive impairment. See Box 11.16 for a list of factors influencing outcome in schizophrenia. However, caution should always be

Box 11.16 Factors predicting a poor outcome in schizophrenia

Demographic factors

Male
Single
Younger age at onset
Family history of schizophrenia
Comorbid substance misuse

Clinical features

Poor premorbid adjustment
Insidious onset
Long duration of untreated psychosis
Hebephrenic (disorganized) features
Negative symptoms
Cognitive impairment
Absence of affective symptoms
Poor insight

Other factors

High expressed emotion in family
Poor adherence with treatment

exercised when asked to give an opinion on the outcome of individual cases. Although a range of predictive biomarkers are under investigation, none is yet validated for use in clinical practice; this includes prediction of outcome after a first episode of psychosis (Lee et al., 2022).

International studies suggest that the course and outcome of schizophrenia may differ between countries. The IPSS (World Health Organization, 1973) found the two-year outcome was better in India, Colombia, and Nigeria than in Western countries, particularly for the proportion achieving complete remission. A 15- and 25-year follow-up study confirmed the stability of geographical differences in course, and suggested that a difference in the severity and type of illness at first presentation is an important contributory factor (Harrison et al., 2001). Nevertheless, the better outcome of schizophrenia in lower- and middle-income countries, and its explanation, remains controversial.

Social stimulation

In the 1940s and 1950s, clinicians recognized that among people with schizophrenia living in institutions, many clinical features were associated with an unstimulating environment. Wing and Brown (1970) investigated

patients at three mental hospitals, and devised a measure of 'poverty of the social milieu', which considered little contact with the outside world, few personal possessions, lack of constructive occupation, and pessimistic expectations on the part of ward staff. Poverty of social milieu was found to be closely related to three aspects of the patients' clinical condition: social withdrawal, blunting of affect, and poverty of speech. The causal significance of these social conditions was supported by a further survey four years later; improvements had taken place in the environment of the hospitals, and these changes were accompanied by corresponding improvements in the three aspects of the patients' clinical state. Although dated, the study is important since similar factors may also apply to patients living in the community, where environments can also be impoverished or understimulating in various ways.

Family life and expressed emotion

Over-stimulation also appears to be associated with poorer outcomes. Brown and colleagues found that patients with schizophrenia returning to their families generally did worse than those entering hostels. Relapse rates were greater in families with *high expressed emotion* ('*high EE*') (Brown et al., 1962). This consisted of making critical comments, expressing hostility, and showing signs of emotional over-involvement. In such families the risk of relapse was greater if there were high levels of contact (measured as more or less than 35 hours a week). Arising from this work, a behavioural programme involving family sessions was developed, which reduced high EE and led to decreased relapse rates (Leff et al., 1985). Subsequent work supported its effectiveness, and family interventions are included in clinical guidelines (see below). However, it has not been widely adopted in routine practice, partly due to lack of resources. For review, see Hahlweg and Baucom (2023).

Effects of schizophrenia on the family

Relatives caring for patients with schizophrenia describe two main kinds of problems: social withdrawal and episodes of disturbed behaviour. Patients tend not to interact with other family members; they seem slow, lack conversation, have few interests, and neglect themselves and stay in bed. Qualitative studies with families indicate that this self-neglect is experienced as a greater burden than the problems of more obviously disturbed or socially inappropriate behaviour, and occasional threats of violence.

Relatives of patients with schizophrenia may feel anxious, depressed, guilty, or bewildered. They may be understandably uncertain about how to deal with

difficult and odd behaviour. Further difficulties can arise from differences in opinion between family members, and from a lack of understanding and sympathy among neighbours and friends. While much less common than in the past, parents do still sometimes feel blamed or

blame themselves, in part due to the discredited theories of family causation (Box 11.12). Support and help for the relatives and carers of patients with schizophrenia is often poor, and more services are called for. For a review, see Shiraishi and Reilly (2019).

Treatment

Antipsychotic drugs are the mainstay of treatment for schizophrenia, but are rightly used with increasing caution, and complemented by specific psychological and psychosocial interventions. This section is concerned with the evidence regarding individual treatments. The following section on management deals with how these treatments are used together and delivered in clinical practice.

Clinical guidelines for schizophrenia

Several recent consensus guidelines are available for the treatment and management of schizophrenia. There is broad agreement on most key issues, but also some differences:

- NICE have produced UK guidelines for the management of schizophrenia in adults (National Institute for Health and Clinical Excellence, 2014a) and in children and adolescents (National Institute for Health and Clinical Excellence, 2013a). Compared to the other guidelines mentioned below, the NICE guidelines are relatively short on details, and the adult guideline has been criticized for over-emphasizing psychological and psychosocial interventions (Taylor and Perera, 2015).
- The British Association for Psychopharmacology guidelines (Barnes et al., 2020) are clear and comprehensive, and consider a wide range of clinical situations (e.g. treatment of aggression, at-risk mental states, pregnancy, and comorbid substance misuse).
- The American Psychiatric Association guidelines are concisely summarized in Keepers et al., (2020).

Pharmacological treatment

There is a strong evidence base supporting the use of antipsychotic drugs in the treatment of schizophrenia and for prevention of relapse. However, there are

important limits to their effectiveness, as well as significant side effects and other potential harms. Some of the key issues are listed in Box 11.17 and the evidence summarized below, together with data regarding other physical treatments used in schizophrenia. As noted, discussion of how medication is used in practice and integrated with other aspects of treatment is deferred to the following section.

The pharmacology of antipsychotic drugs is covered in Chapter 25.

Effectiveness and side effects of antipsychotic drugs

The effectiveness of antipsychotic medication in the treatment of acute schizophrenia is well established by multiple placebo-controlled and active comparator studies. The median effect size compared to placebo is 0.44 (Leucht et al., 2013). About two thirds of patients show a significant therapeutic response, but at present it is not possible to predict how an individual patient will respond. Importantly, antipsychotic drugs only treat the positive symptoms of schizophrenia. They have little or no effect on negative (Fusar-Poli et al., 2015) or cognitive (Nielsen et al., 2015) symptoms. For review, see Haddad and Correll (2018).

There are no substantial differences in efficacy between most antipsychotics (with the exception of clozapine), nor between typical and atypical antipsychotics. However, this does not mean that all antipsychotics are the same. There are some differences in efficacy, and more substantial differences in tolerability. The latter reflects the propensity of a drug to cause a range of side effects (see Chapter 25), including weight gain and metabolic dysregulation (Pillinger et al., 2020), sedation, QTc prolongation, orthostatic hypotension, prolactin elevation, and extrapyramidal side effects such as akathisia and tardive dyskinesia (Stroup and Gray, 2018).

In a network meta-analysis, Huhn et al. (2019) ranked 32 antipsychotics on a range of efficacy measures

Box 11.17 Key points in the pharmacological treatment of schizophrenia

Acute episode

- Initiate antipsychotic medication at lower end of the licensed dose range.
- Do not use loading doses ('*rapid tranquilization*'). If sedation is needed, use adjunctive benzodiazepines.
- The choice of drug should be based on patient preference, effects of previous treatments, and relative liability of the drug to cause serious side effects.
- Titrate dose within licensed range, monitoring for effects and side effects.
- Aim to achieve optimum dose with good adherence for two weeks. If no response at that time, consider changing drug.
- Record the indications for medication, the anticipated benefits and time course, and discussions with patient and carers.
- Consider psychological interventions whenever medication is being introduced or changed.

Maintenance and relapse prevention

- Continue medication for at least a year, or longer, using the same principles as for the acute episode.
- Ensure that dose, duration, and adherence are adequate before switching drug.
- Drug withdrawal should be gradual, and the mental state monitored during this time.
- Continuous treatment is more effective than intermittent treatment.
- Monitor adherence regularly.
- Monitor for side effects, including metabolic syndrome, regularly
- Consider long-acting injectables, especially if adherence is a problem.
- Always consider psychological interventions together with pharmacological options

and side effect liabilities, and the reader is referred to the article to help guide prescribing decisions. For example, in some patients, avoidance of QTc prolongation might be a priority, but for others it may be weight gain or sedation.

Onset of action

Antipsychotics do not have a delayed onset of action, as is sometimes believed, but a gradual onset of effect. If there has not been a 20% improvement in symptom score after two weeks, the chances of a later response to that drug and dose are small (Samara et al., 2015).

Dosage

Studies using PET and SPET to measure dopamine D2 receptor occupancy, discussed in Chapter 25, provide a rationale for the dosing of antipsychotic drugs. This evidence complements clinical guidelines which emphasize that antipsychotic drugs should be used at the lowest effective dose. The latter is lower in the first episode than subsequently (Kahn and Sommer, 2015), and applies particularly to adolescents requiring antipsychotic medication (National Institute for Health and Clinical Excellence, 2013b). The dose–response of antipsychotics varies between drugs, but in general dosing at or above the licensed maximum is best avoided, since it produces no more efficacy but more side effects (Leucht et al., 2020).

Clozapine is the only antipsychotic for which it is often worth measuring plasma levels (see below).

Maintenance treatment and prevention of relapse

Since the original demonstration by Pasmanick et al. (1964), controlled trials and meta-analyses have confirmed the effectiveness of continued antipsychotic medication in preventing relapse. One meta-analysis reported a one-year relapse rate of 65% in those who discontinued medication, compared to 27% in those who had continued it, with an NNT of 3 (Leucht et al., 2012). Antipsychotic continuation was also associated with a better quality of life. For review, see Goff et al. (2017).

The effectiveness of maintenance antipsychotic treatment in schizophrenia is noted in all clinical guidelines, with the risk–benefit ratio in favour of their long-term use (Correll et al., 2018). However, there is less evidence as to how long maintenance treatment should last after an acute episode, reflected in differences between guidelines: National Institute for Health and Clinical Excellence (2014a) suggests one to two years, whereas Barnes et al. (2020) recommend at least two years after remission. For patients with multiple episodes, prolonged or indefinite treatment is common.

Treatment adherence

Partial or non-adherence to treatment with antipsychotic drugs is frequent, and is associated with worse outcomes. A meta-analysis reported non-adherence rates of up to 52%, with non-adherence being related to poor insight and negative attitudes to medication; other variables, such as severity of illness and side effects, showed inconsistent effects (Sendt et al., 2015). Frequency of administration is also an important factor, with adherence greatest for once-daily dosing.

Long-acting injectables (depot antipsychotics)

Long-acting injections of antipsychotics (previously called *depots*) were introduced to deal with the problem of non- (or uncertain) adherence to treatment. They appear to be more efficacious than oral medication in preventing relapse, presumably due to improved compliance (Kishimoto et al., 2021; Tiihonen et al., 2017), though a network meta-analysis has not confirmed this benefit (Schneider-Thoma et al., 2022). Long-acting injectables are prescribed to about 30% of patients with schizophrenia in the United Kingdom (Barnes et al., 2009).

Clozapine

At least 30% of patients do not respond to antipsychotics, or are intolerant of them. The only proven drug intervention for this group is clozapine, which is effective in about one third of such patients, as shown by the key trial by Kane et al. (1988). Meta-analyses of clinical trials confirm that clozapine is at least as effective as other antipsychotics, and more effective than most, in the acute treatment of schizophrenia (Huhn et al., 2019). Real-world data also show a greater efficacy of clozapine compared to other oral antipsychotics (Tiihonen et al., 2017). Clozapine may also have benefits with regard to suicide risk, aggression, and substance misuse, although earlier suggestions that it might improve persistent negative symptoms have not been borne out. Nevertheless, its beneficial properties give clozapine a unique place in the treatment of schizophrenia, despite its risks and side effects (see Chapter 25, and below).

There is little evidence to guide the treatment of patients who are unresponsive to, or unable to take, clozapine (Siskind et al., 2018). There are two common augmentation strategies. The first is to add an antipsychotic that has a high affinity for the dopamine D_2 receptor such as amisulpride; there is also a vogue for

using aripiprazole. This is one of the few situations which justifies prescribing two antipsychotics at once. The other strategy is to add a mood stabilizer, especially lamotrigine. Occasionally, ECT is used to augment clozapine, and a systematic review suggests that it merits greater consideration (Lally et al., 2017).

Other drug classes used in schizophrenia

Antidepressants. Despite the prevalence of depression in schizophrenia, and the frequent use of antidepressant drugs, there is limited evidence regarding their efficacy in this situation, and recommendations for their use vary between clinical guidelines. A meta-analysis shows clear but modest beneficial effects against depressive and negative symptoms, especially for SSRIs (Galling et al., 2018).

Mood stabilizers. The value of mood stabilizers in treating schizophrenia is unproven. There is no evidence that they have an antipsychotic effect, and the beneficial effects reported in some trials is likely due to treatment of affective symptoms. The use of mood stabilizers in enhancing clozapine response was mentioned earlier.

Benzodiazepines. There is no role for benzodiazepines in augmenting antipsychotic drugs (Dold et al., 2013), but they are useful acutely as part of rapid tranquilization (see below).

Muscarinic agonists. A drug combining xanomeline, a muscarinic M1/M4 receptor agonist, with trospium, which blocks peripheral side-effects, was recently launched in the United States, though not in the United Kingdom at the time of writing. Named Cobenfy, it is notable for being the first licensed non-dopaminergic antipsychotic drug, but its place in clinical practice remains to be established.

Future drug treatments. A diverse group of medications are under investigation, especially for negative and cognitive symptoms. These include glutamatergic drugs, cannabidiol, non-steroidal anti-inflammatories, and polyunsaturated fatty acids. Some have clinical trial data suggesting efficacy, but none is yet licensed for clinical use.

Electroconvulsive therapy (ECT)

The traditional indications for ECT in schizophrenia are catatonic stupor, severe comorbid depressive symptoms, and severe behavioural disturbance. In each situation the clinical impression is that the effects of ECT can be dramatic, but data are limited and it is not recommended in guidelines (Pompili et al., 2013). ECT is used very infrequently in the United Kingdom in the

treatment of schizophrenia. However, it remains more widely used in some countries and, as noted above, it may have value in clozapine augmentation.

Psychosocial approaches

Concerns about the limitations of medication have led to a renewed emphasis on psychosocial interventions in the treatment of schizophrenia. They are of several different kinds (Box 11.18), but share the following aims:

- enhancement of interpersonal and social functioning, including promotion of independent living in the community, and
- attenuation of symptom severity and associated comorbidity.

Note that many of the psychosocial interventions to be described here are not widely available. For review of psychosocial interventions in schizophrenia, see Solmi et al. (2023). *Assertive community treatment* is sometimes also considered a psychosocial intervention (Marder and Cannon, 2019) but is discussed in Chapter 26.

Family therapy

Family therapy, both formal and informal, is often employed at various stages of treatment. The most systematically evaluated intervention is that designed to decrease high EE in family members, introduced earlier. The procedure is usually combined with education about the illness and its consequences, together with practical advice on management (Box 11.19). A recent meta-analysis suggested that family psychoeducation alone is as effective as more complex (and resource-intensive) family interventions (Rodolico et al., 2022).

Cognitive behavioural therapy

The use of CBT in schizophrenia is based on the rationale that positive psychotic symptoms are amenable to structured reasoning and behavioural modification. Several trials, and meta-analyses, have found CBT to be effective in schizophrenia, following Tarrier et al. (1998), who reported that it was more effective than supportive counselling in decreasing positive symptoms (see Turner et al., 2014). In patients with delusional beliefs, for example, individual ideas are traced back to their origin and alternative explanations are explored. Similarly, it may be possible to modify a patient's beliefs about the omnipotence and origin of auditory hallucinations,

with a resulting decrease in the distress that accompanies the experience and, perhaps, in the intensity of the hallucinations. Although normally used as an adjunct to medication, CBT may also be effective in patients who choose not to take an antipsychotic drug (Morrison et al., 2014).

All current guidelines recommend CBT as part of an integrated treatment package for schizophrenia. However, there have also been studies with negative findings, and other reviews have come to more cautious conclusions (Taylor and Perera, 2015). Indeed, recent systematic reviews concluded that there is no advantage for CBT over simpler psychosocial therapies for schizophrenia (Jones et al., 2018) and guidelines have been criticized for overemphasizing the evidence for, and role of, CBT in psychosis (Jauhar et al., 2019). CBT is not of clinical value in clozapine-resistant cases (Morrison et al., 2018).

Box 11.18 Psychosocial interventions for schizophrenia

Family therapy (psychoeducation)
Cognitive–behaviour therapy
Cognitive remediation
Art therapy
Social skills training
Illness management skills
Supported employment
Exercise
Integrated treatment for comorbid substance misuse

Box 11.19 Elements of family intervention in schizophrenia

Education about schizophrenia
Improving communication
Lowering expressed emotion
Expanding social networks
Adjusting expectations
Reducing number of hours of daily contact

Cognitive remediation

As noted above, cognitive impairments are important determinants of poor outcome in schizophrenia, and current pharmacological strategies have little (if any) effect. At present, the evidence is stronger for psychological approaches. Most of these rely on mental exercises and training to improve performance through practice. The most extensively tested is *cognitive remediation therapy*, which has beneficial effects on cognitive performance and on functional outcomes (Vita et al., 2021). It is yet to be widely available in most settings.

Other psychosocial interventions

Social skills training uses a range of approaches to teach the patient complex interpersonal skills, including behavioural reversal, feedback, and training. It is not recommended in current UK guidelines (National Institute for Health and Clinical Excellence, 2014a) but is in US guidelines (Keepers et al., 2020), and meta-analyses show some efficacy for improving social skills and negative symptoms (Granholm and Harvey, 2018).

Arts therapies are included in NICE guidelines among psychosocial interventions, which should be considered for patients with negative symptoms. However, there is minimal evidence for their effectiveness (Laws and Conway, 2019).

Dynamic psychotherapy is not recommended in schizophrenia. In addition to the lack of evidence for the underlying psychodynamic theories of the disorder (Box 11.12), there is no convincing evidence for efficacy and there are concerns that dynamic psychotherapy may cause overstimulation and relapse.

Exercise is effective in improving a range of symptoms including cognition, quality of life, and global functioning in schizophrenia (Dauwan et al., 2016). It is also valuable more generally for physical health, and for counteracting the weight gain propensity of many antipsychotic drugs.

Supported employment, delivered as part of a broader psychosocial package, is effective in helping patients with schizophrenia find employment or remain in education (Rosenheck et al., 2017) and is recommended in treatment guidelines.

Adherence therapy (previously referred to as *compliance therapy*) is a brief intervention using motivational interviewing, which aims to improve outcome by enhancing adherence to treatment using. It is not recommended in treatment guidelines. However, a meta-analysis found that adherence therapy had a significant effect

on reducing symptoms, even though it did not improve adherence or alter attitudes to medication adherence (Gray et al., 2016).

Virtual reality forms of psychological intervention for psychosis are showing promise, and may well become part of the therapeutic options, for example, for reducing paranoia (Freeman et al., 2023).

Treatment of other primary psychotic disorders

The preceding sections have focused on schizophrenia. Here we summarize treatment of the other disorders listed in Table 11.2, except for delusional disorders, which are covered in Chapter 12. Note that clinical trials of the following disorders are very limited, and evidence-based guidelines are not available. Most treatment decisions are based on extrapolation from schizophrenia, and on clinical experience.

Acute and transient psychotic episodes. Antipsychotic drug treatment of a brief psychotic episode follows the principles described for first-episode schizophrenia. However, there is uncertainty as to how long it should be continued given that about half of patients will not have a later recurrence, and whether medication should be introduced if symptoms are abating by the time of presentation (Fusar-Poli et al., 2022). Although recommended, psychotherapy is rarely provided to patients with a brief psychotic episode (Minichino et al., 2019).

Schizoaffective disorder. Many trials of schizophrenia include patients with schizoaffective disorder, allowing for generalizability of findings to the latter condition and similar use of antipsychotic medication. The affective component should be treated following guidelines for mania, depression and mood stabilization discussed in other chapters. For review, see Miller and Black (2019).

Catatonia. Benzodiazepines are effective in 70–80% of cases, and a marked reduction in symptoms to a 1–2 mg lorazepam challenge is considered a diagnostic test. ECT is also effective, and used either in benzodiazepine resistant cases, or as first-line treatment, or in combination. If catatonia occurs in the context of schizophrenia or other underlying disorder, the latter should also be treated. For a review, see Rogers et al. (2023).

Schizotypal disorder. People with schizotypal disorder rarely seek treatment, and there is little evidence regarding effective interventions. Low-dose antipsychotic medication is sometimes used, and social skills training and cognitive remediation may be of value (McClure et al., 2019).

Management

Successful management of schizophrenia depends on first establishing a good relationship with the patient. This can be difficult because of the nature of the illness, but with skill and patience, progress can usually be made. All plans made should be realistic, and acceptable to the patient and their carers. The importance of medication needs to be carefully explained, as well as its limitations and its side effects. The same principle applies to all other interventions. Such discussions should be informed by an up-to-date knowledge of the evidence. Novel modes of service delivery are likely to impact on the management of schizophrenia over the next few years, and it is important to keep up to date with these developments, too.

Box 11.20 summarizes key elements of the management of schizophrenia. When reading the following sections, the reader is referred to the various current clinical guidelines available to inform and guide management decisions (Barnes et al., 2020; Keepers et al., 2020; National Institute for Health and Clinical Excellence, 2014a).

Box 11.20 Principles and components of the management of schizophrenia

Therapeutic partnership with patient and carers
 Comprehensive management plan, regularly reviewed
 Early intervention for the first episode, and prompt intervention for later episodes
 Integrated, multidisciplinary working, involving primary and secondary care
 Antipsychotic drugs for treatment and maintenance, with attention to side effects and adherence
 Trial of clozapine for all patients who meet the criteria
 Psychosocial interventions including psychoeducation and CBT
 Family interventions
 Cognitive remediation
 Assertive outreach for vulnerable patients
 Regular assessment of needs
 Crisis resolution and home treatment teams as alternatives to admission
 Maintain realistic therapeutic optimism

The acute illness

Admission to hospital for assessment and treatment is often needed for the first episode of schizophrenia and for severe relapses, and not infrequently requires use of the Mental Health Act. Hospital admission allows a thorough assessment, provides a safe environment, and gives a period of relief to the family, who will often have experienced considerable distress during the emergence of the illness. For milder episodes, crisis resolution teams (Chapter 26) can enable patients to receive assessment and treatment at home, although their effectiveness remains unclear (Werbelloff et al., 2017).

A drug-free observation period is a counsel of perfection, and is rarely attempted now, in part because of the pressure on inpatient beds. The main difficulty in establishing the diagnosis is often to elicit all of the symptoms from a withdrawn or suspicious patient. This procedure may require several interviews, as well as information from relatives or close friends and careful observations by nursing staff.

A physical examination and routine investigations (see Chapter 3) should be carried out as soon as the mental state permits, for two reasons. First, to exclude the possibility of an 'organic' or drug-induced psychosis. Second, to ensure that medical comorbidities are detected and to provide baseline data for metabolic syndrome monitoring. The role of brain imaging is controversial; MRI scanning is not routinely carried out in many countries, including the United Kingdom. However, a recent meta-analysis found a 5.9% prevalence of clinically actionable findings in patients with first-episode psychosis, arguing that MRI should be part of the initial clinical assessment (Blackman et al., 2023).

While the diagnosis is being established, a social assessment should be carried out. This includes a history of the patient's personality, level of functioning, work record, accommodation, and leisure pursuits, and the attitudes to the patient of relatives and any close friends. The current social functioning and capabilities can be assessed by other members of the multidisciplinary team, notably occupational therapists and social workers. Discussions about the use and choice of psychosocial interventions can also be held, although the patient is unlikely to be able to engage fruitfully in specific psychological therapies at this stage.

Drug treatment of the acute episode

Antipsychotic medication is the mainstay of the treatment of schizophrenia and other psychoses. It will usually be initiated soon after the diagnosis is made—or even while this remains uncertain—and will often continue for several years. Therefore the use of medication, and appropriate involvement of the patient and their carer in decision making, are key aspects of management.

If the patient is acutely disturbed, *rapid tranquilization* may be required (Chapter 25). In other situations, treatment with low to moderate doses of an oral antipsychotic drug should be started. In general, in patients presenting with a first episode of psychosis, antipsychotic drug doses should be towards the lower end, certainly at the initiation of treatment. As discussed earlier, the choice of drug should be based on an understanding of their differences in likely side effect profile, drug interactions, presence of any contraindications, and other clinical considerations.

Metabolic syndrome is prevalent even in unmedicated first-episode psychosis, being present in about 10% of patients (Mitchell et al., 2013a). The rate rises significantly later in the illness, presumably as a result of antipsychotic medication and lifestyle factors (Vancampfort et al., 2015). Hence baseline examination and investigations for metabolic syndrome (Chapter 25) should be carried out before initiating antipsychotic medication, and at regular intervals thereafter (Cooper et al., 2016).

As soon as the patient's mental state permits it is essential to explain the need for drug treatment and to obtain their full consent, especially in view of the risk of serious and long-term side-effects. If the patient is judged to be unable to make decisions of this nature, the appropriate legal safeguards under the Mental Health Act and its code of practice should be employed. The patient's family should be involved as far as possible in all decisions about treatment as a matter of principle, and because they will have a role in encouraging adherence. The use of patient advocates can be helpful.

In the early stages of treatment, the choice and timing of doses should be adjusted if necessary, from one day to the next, until the most acute symptoms have been brought under control. It is important to be alert for acute dystonic reactions and neuroleptic malignant syndrome in the early days of treatment or after a dose increase. For the elderly or physically ill, observations of temperature and blood pressure should be made in order to detect hypothermia or postural hypotension.

Although patients usually become rapidly more settled after starting antipsychotic drugs, improvement in

psychotic symptoms is usually gradual, with full resolution taking a number of weeks. Current trends are to maintain the dose of antipsychotic drugs at a modest steady level and not to escalate the dose in the hope of speeding up the rate of improvement. For patients in whom agitation and distress continue to cause concern, (re)introduction of short-term intermittent treatment with a benzodiazepine is a better option than increasing the dose of antipsychotic.

If there are doubts about whether the patient is swallowing tablets, the drug can be given as syrup or as an orally dispersible tablet. Short-acting injectable preparations of some antipsychotics are available, but are not recommended in patients whose tolerance of antipsychotics has not been determined. Anticholinergic drugs should be prescribed if extrapyramidal side effects are troublesome, but should not be given routinely or for sustained periods. As noted earlier, the acute symptoms of excitement, restlessness, irritability, and insomnia can be expected to improve rapidly after instituting treatment. Delusions, hallucinations, and other psychotic symptoms also begin to respond during this time, with improvement developing further over a period of several weeks.

By the time that meaningful symptomatic improvement has occurred, and prior to discharge, the team should have formulated a provisional plan for continuing care, based upon the social assessment that has been carried out.

Care plans and the Care Programme Approach

Care plans are important because of the need for coordinated multidisciplinary working, risk management, and clear documentation of responsibilities for their delivery. In the United Kingdom, this responsibility is formalized in the Care Programme Approach (CPA; see Chapter 26). CPA is particularly appropriate for those with schizophrenia because of the frequent need for complex and long-term support. The essential elements of a care plan are to identify and record:

- a systematic assessment of health and social needs
- a treatment plan agreed by the relevant staff, the patient, and relatives
- the allocation of a key worker whose task is to maintain contact with the patient, monitor their progress, and ensure that the treatment programme is being delivered
- a regular review of the patient's progress and needs.

Management after the acute episode

Once the acute episode has settled, the priorities for management are to maximize and maintain recovery. This will usually require maintenance medication which, as noted earlier, should normally continue for at least one to two years, and for longer after a second or subsequent episode. For some patients, long-acting injections, mentioned earlier, are a helpful way to ensure adherence with medication (Box 11.21).

The emphasis on medication in no way diminishes the importance of multidisciplinary care in the long-term management of schizophrenia. Indeed, it is important to use medication as part of an integrated package of care. This can involve supporting realistic structured activity, whether it be attending a day centre, pursuing education, or seeking a job. Advice should be given about avoiding obviously stressful events. Psychosocial interventions (Box 11.20) should continue, or be reconsidered if they have not previously been implemented. The patient should be seen regularly as an outpatient or at home by a member of the community mental health team (often a psychiatric nurse).

The extent and duration of ongoing care after an acute episode of schizophrenia depend on the severity and nature of the residual symptoms and impairments. At one extreme, following a single episode with excellent

recovery, medication and other interventions can gradually be tailed off, and the patient returned to the care of the GP with a cautiously good prognosis. At the other end of the spectrum, the patient whose schizophrenia proves refractory to treatment requires long-term and sometimes intensive support, as described below.

Between these extremes lie the majority of patients, whose illness is relapsing and remitting, who do not return to their premorbid level of functioning, and who have persistent difficulties or problems that need to be identified and managed. This group includes patients with prominent negative symptoms, poor social adjustment, and cognitive and behavioural deficits characteristic of chronic schizophrenia. In patients who fail to return to their premorbid level of functioning and activities, despite resolution of their positive symptoms, it is important to consider the cause. For example, it may be due to negative symptoms, depressive symptoms, cognitive impairments, or the side effects of medication. If a cause is identified, appropriate action should be taken; this may include interventions for depression, a change in medication, or referral for psychological treatment.

Despite the above steps, the reality is that many patients progress to an illness state for which few evidence-based treatment options exist. Once a clinical picture of chronic schizophrenia has become established, the main emphasis is on a programme of support tailored to the needs of the patient. An occupational therapist, social worker, and clinical psychologist can all make valuable contributions. The following specific issues require consideration.

Box 11.21 Use of long-acting injectable antipsychotics

For a number of reasons, some patients do not take their drugs reliably, and in these circumstances, intramuscular long-acting injectable (depot) preparations may be useful. At the start of treatment a small test dose is given to find out whether serious side effects are likely to occur with the full dose, although this is not considered necessary with atypical antipsychotics preparations (Taylor et al., 2021). The appropriate maintenance dose is then established by observation and careful follow-up. Since depot preparations have long half-lives, it may take several weeks for maximum plasma concentrations to be reached. This has implications for the rate at which dose increases and decreases should be made, and also for the tapering of doses of oral antipsychotic medication once depot treatment has started.

- Living accommodation and circumstances. Will the patient be able to live alone or will they return to live with their parents or other relatives? Will accommodation need to be arranged?
- Occupational activities. Will the patient be able to undertake some form of paid employment or voluntary work?
- Activities of everyday living. Does the patient need help to re-learn and implement basic living skills such as cooking, budgeting, shopping, housework, and personal hygiene?
- If available, consider cognitive remediation, social skills training, or related therapies aimed at improving cognition and social functioning.
- Monitoring and management of any comorbid condition or substance misuse.
- Monitoring and management of physical health problems.

- Optimization of the efficacy and tolerability of medication, while recognizing that any further response beyond that achieved earlier in the illness is unlikely.
- The family, and any other carers, must be fully supported, to help them to understand the patient's illness, to have realistic expectations of what the patient can accomplish, and to decrease their level of emotional involvement where appropriate. Support groups and psychoeducation for families and carers are valuable and can reduce distress.

Some patients with chronic schizophrenia understandably tend to withdraw from treatment and follow-up. For those who are judged to be at continuing risk of relapse, more assertive outreach (see Chapter 26) has a useful role to play.

Treatment-resistant schizophrenia

About a quarter of patients with schizophrenia come into this category, also known as *refractory schizophrenia* (Siskind et al., 2022). Management is particularly challenging because of the nature of the illness and the lack of evidence-based treatment options other than clozapine. The use of clozapine is hampered by the need for regular blood monitoring due to the risks of agranulocytosis, as well as by other serious side effects, notably weight gain, sedation, and hypersalivation (Chapter 25). Nevertheless, because of its efficacy, clozapine should be recommended to all patients who do not respond to or cannot tolerate at least two other antipsychotics. In practice, only a minority of suitable patients have a trial of the drug, and often this does not occur until many years into the illness.

Assessment involves a detailed review of the history and establishment of the current symptom profile (Box 11.22). It is also worth trying to establish that the patient is refractory to treatment and not simply non-adherent; one study found that half of patients considered treatment resistant had low or undetectable antipsychotic plasma levels (McCutcheon et al., 2015).

The use of clozapine requires skill and experience to optimize adherence and efficacy, and to minimize and manage the common side effects and the rare but serious adverse events. Plasma levels are worth measuring if there are concerns about non-response or non-compliance. A minimum trough level of 200µg/l should be attained; higher levels, up to 500µg/l or more, may increase response, but at the cost of more side effects. For practical advice on clozapine use and monitoring, see Chapter 25 and Taylor et al. (2021).

Box 11.22 Assessment of treatment-resistant schizophrenia

- Review diagnosis (e.g. possibility of organic psychosis)
- Review comorbidities (e.g. substance misuse, physical health)
- Review treatment history (e.g. drugs, doses, psychological therapies)
- Rate current symptoms and severity using established scales (e.g. BPRS, PANSS, CGI)
- Physical examination (e.g. for metabolic syndrome)
- Discuss patient's views regarding treatment
- Check for treatment adherence (e.g. drug levels)
- Consider routine blood tests and further investigations (e.g. imaging)

Despite the focus on clozapine, management of treatment-resistant schizophrenia should also pay due attention to optimizing all other aspects of care, including physical health and psychosocial functioning, as described above for other patients with chronic schizophrenia.

For discussion of treatment-resistant schizophrenia, see Howes et al. (2017).

Other aspects of management

Early intervention services

As noted earlier, there is a consensus that early treatment of schizophrenia may improve the long-term prognosis, based on a number of lines of evidence.

- Patients have often been psychotic for many months before the diagnosis is made and treatment is initiated. Patients with a longer duration of untreated psychosis have a poorer longer-term prognosis.
- The treatment response can become slower and less marked during subsequent psychotic episodes.
- Much of the deterioration in social function takes place during the first two years after diagnosis.

These observations imply that it is worth attempting to identify patients with psychosis as early as possible. Early Intervention in Psychosis Service (EIS) teams are mandated in the United Kingdom and endorsed in NICE guidelines (National Institute for Health and Clinical Excellence, 2014a). These teams are described

in Chapter 26. EIS services improve short-term outcomes (Puntis et al., 2020) and may modestly reduce the duration of untreated psychosis (Albert and Weibell, 2019) but their longer-term effectiveness remains unclear (Hansen et al., 2023).

In some places, the early intervention approach has been extended to individuals thought to be in the prodrome of illness but who do not meet diagnostic criteria, and even to people who have no symptoms but are considered to be at clinical high risk of developing psychosis. Evidence regarding intervention in the prodrome is limited (Bosnjak Kuharic et al., 2019), and there is no evidence to favour preventive interventions (Davies et al., 2018). The fact that only about 25% of people at clinical high-risk transition to psychosis within three years should also be considered (Salazar de Pablo et al., 2021).

Substance misuse

Many people with schizophrenia misuse alcohol or other drugs. They tend to have a worse symptomatic course and to be more difficult to engage in treatment. In addition, substance misuse is associated with increased risks of homelessness, violent behaviour, poor general health, and a substantial excess early mortality (Correll et al., 2022). Antipsychotic drugs, especially clozapine, show some efficacy in reducing comorbid cannabis misuse (Wilson and Bhattacharyya, 2016). Comprehensive outpatient and assertive community programmes that address both substance misuse and psychotic illness in an integrated way have been proposed, although evidence is limited (Lees et al., 2021).

The violent patient

Overactivity and disturbances of behaviour are common in schizophrenia, and although major violence towards others is not common, it does occur. Men and women with schizophrenia are 4.5 and 10 times, respectively, more likely to perpetrate violence (Whiting et al., 2022). Over a 35-year period, the absolute risk of violence by men with schizophrenia is 1 in 4, and 1 in 20 in women. The risk of violence is markedly higher in those with comorbid substance misuse.

General management for the potentially violent patient is the same as that for any other patient with schizophrenia, although forensic assessment and compulsory treatment are more likely to be required. Much can be done by providing a calm, reassuring, and consistent environment in which provocation is avoided. Threats of violence should be taken seriously, especially if there is a history of such behaviour. The risk of violence usually resolves as acute symptoms are brought under control, but a few patients pose a continuing threat. In patients

with persistent hostility and violence, a trial of clozapine is recommended (Barnes et al., 2020). The management of violence is considered further in Chapter 18.

Suicide risk

We have already seen that the rate of suicide is markedly increased in schizophrenia. As in other psychiatric disorders, demographic risk factors apply, such as male gender, social isolation, unemployment, and a previous episode of suicidal behaviour (Popovic et al., 2014). Other risk factors include:

- comorbid depression and hopelessness
- frequent exacerbations of psychosis
- being in the early stage of the disorder
- during hospitalization or shortly after discharge.

Patients with pronounced negative symptoms are less likely to make suicide attempts. Although suicide may be associated with active positive symptoms such as delusions of control or auditory hallucinations, more often it occurs in those with previously high levels of educational attainment who are relatively well at the time. Such patients often have depressive symptoms in the context of high premorbid expectations of themselves and dismay at how their illness has damaged their future prospects.

Lowering the risk of suicide in schizophrenia involves being aware of and managing the relevant risk factors, and treating depressive symptoms vigorously. Clozapine is the antipsychotic most clearly associated with reduced risk of suicide, although rates are also lower in patients taking antipsychotic medication compared to those not on treatment (Taipale et al., 2021).

Management of schizophrenia in children and adolescents

There is relatively little evidence to guide management in this population. Guidelines agree that the same principles apply as to adults with schizophrenia, but with greater caution concerning the use and dosage of antipsychotic medication (Yee et al., 2022). Psychological interventions may be considered preferable, but evidence for efficacy is limited (Gergov et al., 2022). See also Chapter 16.

Management of schizophrenia in older patients

See Chapter 19.

Discussing schizophrenia with patients and carers

There is probably more stigma and misunderstanding associated with schizophrenia than with any other psychiatric diagnosis (Box 2.1), reflecting the range of views about its nature and its portrayal in the media. It is therefore particularly important that the term is used carefully and sensitively, and all patients and their carers should have regular opportunities to ask questions, express their views, and raise any concerns that they may have. The involvement of the family in its management has already been mentioned.

There should be discussion about the nature of the illness, identification of any misapprehensions, and an explanation of the current meaning of the term schizophrenia. It is worth discussing what is known about the aetiology, in part to ensure that misconceptions do not lead to unnecessary guilt. For example, some parents worry, incorrectly, that their behaviour may have caused the illness. Mention of its heritability may lead to questions about the risks of schizophrenia in other relatives or future generations. These can be answered using the figures in Table 11.4, acknowledging the increased risk while emphasizing the probability that offspring and unaffected relatives will remain well. If, after discussion, the family continue to have very different concepts of schizophrenia from those of the clinical team, such views should be respected.

The prospects for recovery should be portrayed realistically but optimistically, given that some people wrongly believe schizophrenia is untreatable and inevitably follows an unremitting or deteriorating course. The key role of medication, together with psychosocial factors (e.g. family environment, substance misuse), in influencing outcome should be emphasized. Note that the family may have very different views about the

meaning of recovery. The 'Recovery Approach' (sometimes referred to as the 'Recovery Movement') has achieved prominence and influence in UK policy (Slade et al., 2014). Originating in the United States and well established in New Zealand, it is a relatively loose collection of ideas and principles that emphasize the priority of the patient's aspirations and the goal of living life as fully as possible with the illness. Recovery advocates often criticize psychiatrists for having too pessimistic a view of the prognosis in schizophrenia, and insist on the need for optimism.

A further area in which there may be a difference in emphasis between psychiatrist and carer or patient is the focus upon the content, rather than the nature, of the patient's beliefs. That is, the psychiatrist is usually more concerned as to whether a thought is or is not delusional, or whether a perception is or is not a hallucination, whereas the patient wishes to discuss the thought or percept itself and its meaning for them. The psychiatrist should give the patient or carer adequate time to voice these concerns while also explaining why it is necessary to ask detailed questions about the nature of the experiences.

Other than noting these broad themes, the psychiatrist must be prepared to adapt to the circumstances of each case. As noted, the full involvement of the patient and their carers should be an integral part of the management of schizophrenia. It is reasonable to assume that establishing and maintaining a genuine therapeutic alliance will increase the likelihood that the patient maintains engagement with psychiatric services, receives appropriate treatments, and thereby has an improved prognosis.

Further reading

Lieberman, J. A., Stroup, T. S., Perkins, D. O., & Dixon, D. B. (Eds.). (2020) *The American Psychiatric Association Publishing Textbook of Schizophrenia*, 2nd edn. American Psychiatric Association.

Weinberger, D. R., & Harrison, P. J. (Eds.). (2011). *Schizophrenia*, 3rd edn. Wiley-Blackwell.

CHAPTER 12

Paranoid symptoms and syndromes

Introduction 291

Paranoid symptoms 292

Paranoid personality disorder 294

Paranoid symptoms in psychiatric disorders 295

Delusional disorders (paranoid psychoses) 296

Specific delusional disorders 297

Assessment of paranoid symptoms 301

Treatment of paranoid symptoms and delusional disorder 302

Prognosis of delusional disorder 303

Introduction

The term *paranoid* can be applied to symptoms, syndromes, or personality types. *Paranoid symptoms* are overvalued ideas or delusions that are most commonly persecutory, but not always so (see Chapter 1). *Paranoid syndromes* are those in which paranoid delusions form a prominent part of a characteristic constellation of symptoms; hence, the term *delusional disorders* used in ICD-11 and DSM-5. In *paranoid personality disorder*, there is excessive self-reference and undue sensitiveness to real or imaginary humiliations and rebuffs, often combined with self-importance and combativeness. Thus the term paranoid is descriptive; if we recognize a symptom or syndrome as paranoid, this does not constitute making a diagnosis, but it is a preliminary to doing so. In this respect it is like stupor or depersonalization.

Paranoid syndromes present problems of classification and diagnosis. The difficulties can be reduced by dividing them into two distinct groups:

1. Paranoid symptoms occurring as part of another psychiatric disorder, such as schizophrenia, mood disorder, or an organic mental disorder.
2. Paranoid symptoms occurring without evidence for an underlying disorder. It is this group, which includes disorders such as *morbid jealousy* and *erotomania*, that has caused persistent difficulties—regarding their terminology, their relationship to schizophrenia, and their forensic implications.

This chapter begins with definitions of the common paranoid symptoms, expanding upon their descriptions given in Chapter 1, and then reviews the causes of such symptoms. Next there is a short account of paranoid personality disorder. This is followed by a discussion of primary psychiatric disorders with which paranoid symptoms are frequently associated, and the differentiation of these disorders from delusional disorders. The general features of delusional disorder and its major subtypes are then reviewed. A historical perspective is also given, with particular reference to two related terms, *paranoia* and *paraphrenia*. The chapter ends with a summary of the assessment, treatment, and prognosis of patients with paranoid symptoms.

Paranoid symptoms

Although the vast majority of paranoid delusions are persecutory, the term is also applied to the less-common delusions of grandeur and jealousy, and sometimes to delusions concerning love, litigation, or religion. It may seem puzzling that such varied delusions should be grouped together. The reason is that the central abnormality implied by the term paranoid is a distortion of beliefs or attitudes concerning relationships between oneself and other people. If someone believes falsely or on inadequate grounds that they are being victimized, deceived, or loved by a famous person, then in each case they are construing the relationship in a morbidly distorted way.

The varieties of paranoid symptoms were discussed in Chapter 1, but important ones are also outlined in Box 12.1 for convenience.

Causes of paranoid symptoms

When paranoid symptoms occur as part of another psychiatric disorder, the main aetiological factors are those that determine the primary illness. However, the question still arises as to why some people develop paranoid symptoms, while others do not. This has usually been answered in terms of premorbid personality and social isolation.

Premorbid personality

Many writers, including Kraepelin and Kretschmer, have held that paranoid symptoms are most likely to occur in patients with premorbid personalities of a paranoid type (see Box 12.2 and next section). Supporting this view, Kay and Roth (1961) found paranoid or hypersensitive personalities in over half of a group of 99 subjects with late-onset paraphrenia.

Box 12.1 Some paranoid symptoms

Ideas of reference

Ideas of reference are held by people who are unduly self-conscious. The subject cannot help feeling that people take notice of them in buses, restaurants, or other public places, and that they observe things about them that they would prefer not to be seen. The person realizes that this feeling originates within themselves, and that they are no more noticed than other people, but all the same they cannot help the feeling, which is quite out of proportion to any possible cause. Ideas of reference are a form of overvalued idea.

Delusions of reference

Delusions of reference consist of an elaboration of ideas of reference, to the point that the beliefs become delusional. The whole neighbourhood may seem to be gossiping about the subject, far beyond the bounds of possibility, or they may see self-references in the media. The person may hear someone on the radio say something connected with a topic they have just been thinking about, or they may seem to be followed, their movements observed, and what they say recorded. The importance of distinguishing

a delusion of reference from an idea of reference is that the former is a symptom of psychosis.

Delusions of persecution

When a person has delusions of persecution, they believe that a person, organization, or power is trying to kill or harm them in some way, or damage their reputation. The symptom may take many forms, ranging from the direct belief that they are being hunted down by specific people to complex, bizarre, and impossible plots.

Delusions of grandeur

These may be divided into delusions of grandiose ability and delusions of grandiose identity. The subject with delusions of grandiose ability thinks that they have been chosen by some power, or by destiny, for a special purpose or because they have unusual talents. The person may think they are much cleverer than anyone else, or have invented machines or solved mathematical problems beyond other's comprehension. The subject with delusions of grandiose identity believes that they are famous, rich, titled, or related to prominent people.

Box 12.2 Historical background: paranoia and paraphrenia

The terms *paranoia* and *paraphrenia* have played a prominent part in psychiatric thought. Although the classic studies are now several decades old, much can still be learned from reviewing the issues and the conceptual difficulties involved.

The term *paranoia*, from which the adjective *paranoid* is derived, has a long and chequered history. It has probably given rise to more controversy and confusion of thought than any other term used in psychiatry. A comprehensive review of the large body of literature, which is mostly German and from the period before the 1970s, was provided by Lewis (1970) (see also Box 1.4). The word is derived from the Greek *para* (beside) and *nous* (mind). It was used in ancient Greek literature to mean 'out of mind' (i.e. of unsound mind or insane), but when the term came into prominence in the nineteenth century, in German psychiatry, it became particularly associated with conditions characterized by delusions of persecution and grandeur. There were many different conceptions of these disorders. The main issues, most of which remain today, can be summarized as follows.

- Did these conditions constitute a primary disorder, or were they secondary to another disorder?
- Did they persist unchanged for many years, or were they a stage in an illness that later manifested deterioration of intellect and personality?
- Did they sometimes occur in the absence of hallucinations?
- Were there forms with a good prognosis?

Kahlbaum raised these issues as early as 1863, when he classified paranoia as an independent primary condition; the classification would remain unchanged over the years. Kraepelin had a strong influence on the conceptual history of paranoia, although he was never comfortable with the term, and his views changed strikingly over the years. In 1896 he used the term only for incurable, chronic, and systematized delusions without severe personality disorder. In the sixth edition of his textbook, he wrote:

The delusions in dementia praecox [schizophrenia] are extremely fantastic, changing beyond all reason, with an absence of system and a failure to harmonize them with events of their past life; while in paranoia the delusions are largely confined to morbid interpretations of real events, are woven together into a coherent whole, gradually becoming extended to include even events of recent date, and contradictions and objects are apprehended and explained.

(Kraepelin, 1904, p. 199)

In later descriptions, Kraepelin (1919) used the distinction made by Jaspers (1913) between personality development and disease process. He proposed paranoia as an example of the former, in contrast to the disease process of dementia praecox. In his final account, Kraepelin (1919) developed these ideas by distinguishing between dementia praecox, paranoia, and a third paranoid psychosis, namely paraphrenia. He made the following suggestions:

- Dementia praecox had an early onset and a poor outcome, ending in mental deterioration.
- Paranoia was restricted to patients with late onset of completely systematized delusions and a prolonged course, usually without recovery, but not inevitably deteriorating. An important point was that the patient did not have hallucinations.
- Paraphrenia was somewhat intermediary, in that the patient had unremitting systematized delusions but did not progress to dementia. The main difference from paranoia was that the patient with paraphrenia had hallucinations.

Bleuler's (1911) concept of the paranoid form of dementia praecox (which he later called paranoid schizophrenia) was broader than that of Kraepelin. Bleuler did not regard paraphrenia as a separate condition, but as part of dementia praecox. However, he accepted Kraepelin's view of paranoia as a separate entity, although he differed from Kraepelin in maintaining that hallucinations could occur in many cases. Bleuler was particularly interested in the psychological development of paranoia; at the same time he left open the question of whether paranoia had a somatic pathology.

From this time, two views of paranoia predominated. The first was that paranoia was distinct from schizophrenia and was psychogenic in origin. The second view was that paranoia was part of schizophrenia. Some celebrated studies of individual cases appeared to support the first theme. For example, Gaupp (1974) made an intensive study of the diaries and mental state of Ernst Wagner, who murdered his wife, children, and eight other people as part of a careful plan to revenge himself on his supposed enemies. Gaupp concluded that Wagner suffered from paranoia in the sense described by Kraepelin. At the same time, he believed that Wagner's first recognizable delusions developed as a psychogenic reaction. The most detailed argument for psychogenesis was presented by Kretschmer (1927), who believed that paranoia should

not be regarded as a disease, but as a psychogenic reaction occurring in people with particularly sensitive personalities. However, Kolle (1931) analysed a series of 66 patients with so-called paranoia and, based on symptomatic and genetic grounds, concluded that paranoia was a mild form of schizophrenia.

Considerably less has been written about paraphrenia. However, it is interesting that 50 of Kraepelin's series of 78 paraphrenic patients later developed schizophrenia. Indeed, paraphrenia has increasingly been regarded as late-onset schizophrenia or schizophrenia-like disorder of

good prognosis. Kay and Roth (1961) used the term 'late paraphrenia' to denote paranoid conditions in the elderly that were not due to primary organic or affective illnesses. These authors found that a large majority of their 99 patients had the characteristic features of schizophrenia.

In current classifications, the term paranoia has, in effect, been replaced by delusional disorder. Paraphrenia does not feature, but it continues to be used clinically to describe chronic, atypical, paranoid psychoses of middle and later life. In ICD-11, the term paranoid is not used either.

Social isolation and deafness

Social isolation may also predispose to the emergence of paranoid symptoms. Prisoners (especially those in solitary confinement) and migrants have been considered to be prone to paranoid symptoms and syndromes, with social isolation being the common factor. However, there are no data that unambiguously support this interpretation, and there may be other explanations.

There is better evidence that the social isolation produced by deafness increases the risk of paranoid

symptoms, as originally noted by Kraepelin. For example, Kay and Roth (1961) found hearing impairment in 40% of patients with late-onset paraphrenia. Subsequent studies have confirmed that hearing impairment is a risk factor for disorders in which paranoid symptoms occur, especially in the elderly (Blazer, 2020). However, it should be remembered that the great majority of deaf people do not become paranoid, and that many deaf people are not socially isolated.

Paranoid personality disorder

As discussed in Chapter 15, the category of paranoid personality disorder (like other personality disorder subtypes) has been removed from ICD-11, but it is present in DSM-5 and remains a clinically useful concept. It is characterized by the following:

- extreme sensitivity to setbacks and rebuffs;
- suspiciousness;
- a tendency to misconstrue the actions of others as hostile or contemptuous; and
- a combative and inappropriate sense of personal rights.

This definition embraces a wide range of types. At one extreme is the excessively sensitive young person who shrinks from social encounters and thinks that everyone disapproves of them. At the other is the assertive and challenging individual who flares up at the least provocation.

Because of the implications for treatment, it is important to distinguish paranoid personality disorder from the delusional disorders to be described later. The distinction can be very difficult to make, and is based on the fact that in paranoid personality disorder there are no delusions (only overvalued ideas). Considerable skill is needed to separate paranoid ideas from delusions. The criteria for doing so were given in Chapter 1, and exemplified by the comparison between ideas of reference and delusions of reference (Box 12.1). Evidence of a time when the person did not have similar beliefs may also be helpful in the distinction. In reality, the conditions probably lie along a continuum. Family studies indicate a genetic relationship between paranoid personality disorder and delusional disorder, and individuals with paranoid personality traits are at increased risk of developing delusional disorder.

For review of paranoid personality disorder and its management, see Carroll (2009).

Paranoid symptoms in psychiatric disorders

Paranoid symptoms are often secondary to a primary psychiatric disorder. Thus when paranoid symptoms are elicited it is important to assess for the other features of these disorders. The diagnosis of delusional disorder, to be considered below, is in many respects a ‘residual’ category, used for patients whose delusions cannot be attributed to one of these other conditions. As the primary disorders are described at length in other chapters, they will be mentioned only briefly here.

Paranoid symptoms in organic disorders

In ICD-11, these disorders come under the heading of ‘Secondary psychotic syndrome, with delusions.’ An organic aetiology for paranoid symptoms should always be considered, especially in the elderly or in cases where there is other evidence for an acute medical illness. In such situations, paranoid symptoms are often a feature of *delirium*. Impaired grasp of what is going on around the patient may give rise to apprehension and misinterpretation, and so to suspicion. Delusions may then emerge, which are usually transient and disorganized; these may lead to disturbed behaviour. Similarly, persecutory delusions commonly occur at some stage in *dementia*, and are occasionally the presenting feature. Paranoid symptoms and delusional disorders may occur with *focal brain lesions* of various causes, including tumour, stroke, and trauma. Some examples are given later in this chapter when the specific delusional disorders are considered. Finally, paranoid symptoms can occur in learning disability and other neurodevelopmental disorders.

Paranoid symptoms in substance misuse disorders

Paranoid symptoms occur in many substance misuse disorders, especially those associated with amphetamines, cocaine, and alcohol. An important example is the association between alcohol misuse and morbid jealousy, described below. Some therapeutic drugs, such as L-DOPA, can also precipitate paranoid symptoms.

Paranoid symptoms in mood disorders

Paranoid symptoms are common in patients with severe depressive disorders, and paranoid delusions

are a feature of psychotic depression. Conversely, depressive symptoms often occur in delusional disorders, although the diagnostic criteria for the latter require that their total duration is relatively brief (see Box 12.3). In practice, it is sometimes difficult to determine whether the paranoid symptoms are secondary to depressive disorder, or vice versa, as both scenarios are common. The distinction is of some importance, as the two disorders differ with regard to treatment and prognosis. A depressive disorder is likely if the mood changes have occurred earlier and are of greater intensity than the paranoid features. Previous psychiatric history and family history may also be useful pointers. Finally, in depressive disorder the patient typically accepts the persecution as justified by their own guilt or failings. This is a useful point clinically, as it contrasts with non-affective psychoses, in which such persecutions are bitterly resented.

Paranoid symptoms also occur in mania, and are typically mood-congruent and thus grandiose rather than persecutory.

Box 12.3 Types of delusional disorder

Type	Synonymous with, or includes
Jealous	Morbid jealousy, pathological jealousy, erotic jealousy, sexual jealousy, Othello syndrome.
Erotic	Erotomania, de Clérambault’s syndrome.
Somatic	Monosymptomatic hypochondriacal psychosis, delusional body dysmorphic disorder.
Querulous	Persecutory
Shared	Induced delusional disorder, folie à deux, communicated insanity.
Other	Delusional misidentification syndrome, Capgras syndrome, Fregoli delusion, intermetamorphosis, syndrome of subjective doubles.

Paranoid symptoms and paranoid schizophrenia

Paranoid schizophrenia is described in Chapter 11. Its distinction from delusional disorder has been problematic, both conceptually and practically (see Box 12.2; Gonzalez-Rodriguez and Seeman, 2022). Recent studies emphasize a number of ways in which delusional disorder differs from paranoid schizophrenia, including a later age of onset and better premorbid adjustment and overall functioning, but more severe delusions, and a poorer response to antipsychotic medication (Muñoz-Negro and Cervilla, 2016; Peralta and Cuesta, 2016). When diagnosing an individual case, it is useful also to focus on the differences in the core features of delusional disorder compared to paranoid schizophrenia:

- In schizophrenia, delusions tend to be fragmented and multiple, whereas in delusional disorder they are *systematized* and based around a single, internally consistent theme. In delusional disorder, the delusional system is also characteristically *encapsulated*, such that the rest of the mental state can appear remarkably normal, in contrast to schizophrenia.
- Patients with paranoid schizophrenia often have auditory hallucinations, and the content of these is often unrelated to their delusions. In delusional disorder, hallucinations are very rare, and when they do occur, they are fleeting and clearly related to the delusions.
- A common clinical view is that the diagnosis of paranoid schizophrenia is more likely if the delusions are particularly odd in content (*bizarre delusions*), whereas delusional themes are more ‘plausible’ in delusional disorder. However, this distinction has not been shown to be reliable, and it is not used in ICD-11 or DSM-5.

Paranoid symptoms in schizophrenia-like syndromes

Paranoid symptoms are features of several schizophrenia-like syndromes, which were discussed in Chapter 11. These include the ICD-11 grouping of ‘Acute and transient psychotic disorders, and the DSM-5 categories of brief psychotic disorders and schizophreniform disorder.

Delusional disorders (paranoid psychoses)

As mentioned in the introduction and explored in Box 12.2, the terminology and classification of psychoses which are neither affective, organic, nor schizophrenia have been disputed for many years. This section describes the core features of delusional disorders—the current terminology for these conditions. The specific types of delusional disorder are covered in the following section.

Classification of delusional disorder

ICD-11 and DSM-5 use the term *delusional disorder* to describe a condition with one or more persistent delusions that cannot be attributed to any other disorder (e.g. schizophrenia or mood disorder). It is synonymous with the widely used term *paranoid psychosis*. The essence of the modern concept of delusional disorder is that of a stable delusional system that develops insidiously in a person in middle or later life. The delusional system is encapsulated, and there is no impairment of other mental functions. The patient can often continue working,

and the person’s social life may be maintained fairly well. ICD-11 requires symptoms to have been present for three months, whereas DSM-5 requires one month.

Delusional disorder can be further specified based on the theme of the delusion(s): persecutory, grandiose, erotomanic, somatic, and mixed. Specific delusional disorders can also be diagnosed as various eponymous syndromes, as outlined below.

For review of delusional disorder, see Marneros (2020).

Epidemiology of delusional disorder

Delusional disorder is regarded as being uncommon, although there are relatively few data. Kendler (1982) reviewed the literature and reported an incidence of 1–3 per 100,000 per year, with delusional disorder constituting 1–4% of all psychiatric admissions. In a large population survey, delusional disorder had a lifetime prevalence of 0.18% (Perälä et al., 2007). In a large case series, the mean age of onset of delusional disorder was

46 years, and the diagnosis remained reasonably stable over a ten-year period, with 21% being re-diagnosed with schizophrenia (Marneros et al., 2012).

Aetiology of delusional disorder

What little is known about the aetiology of delusional disorder is based upon its relationship to, and comparison with, schizophrenia, paranoid personality disorder, and depressive disorder (Kendler, 1982). This question has been addressed by family and neurobiological studies. However, the relatively small sample sizes and varying diagnostic definitions mean that few conclusions can be drawn. Psychological explanations for delusional disorder centre upon the delusions themselves (Garety and Freeman, 2013).

Family studies of delusional disorder

First-degree relatives of patients with delusional disorder have an increased incidence of paranoid personality

disorder (Kendler et al., 1985). The familial relationship of delusional disorder to schizophrenia is less clear. Although the risk of delusional disorder is increased in first-degree relatives of patients with schizophrenia, relatives of patients with delusional disorder do not have an increased risk of schizophrenia or schizotypal personality (Kendler and Walsh, 1995; Tienari et al., 2003). No individual loci or genes associated with delusional disorder have been identified. There is also a familial clustering of delusional disorder and alcoholism (Kendler and Walsh, 1995), which could contribute to the association between morbid jealousy and alcohol misuse.

Neurobiological studies

Very little is known regarding biological correlates of delusional disorder, nor the extent to which they overlap with or differ from schizophrenia. A study of younger patients with delusional disorder found alterations in brain activation and cortical thickness, especially in the insula and cingulate cortex (Vicens et al., 2016).

Specific delusional disorders

Subtypes of delusional disorder are recognized, on the basis of the content of the predominant delusion(s) (see Box 12.3). Historically, these symptoms have been of particular interest to French psychiatrists. Classification in this area is confusing for two reasons.

1. Some of the disorders are often referred to by older, eponymous terms or by categories that are not included in ICD-11 or DSM-5, but which remain in common usage.
2. Some of the syndromes can be viewed as symptoms (e.g. delusional misidentification).

In this section we also consider stalking and persistent litigants, as both behaviours may be secondary to delusional disorder. For review of specific delusional disorders, see Marneros (2020).

Pathological jealousy

Pathological or *morbid jealousy* (other synonyms are listed in Box 12.3) is described first and in most detail as it is the archetypal delusional disorder; it is also the most common and, importantly, appears to carry the greatest risk of dangerousness.

The essential feature is an abnormal belief that the patient's partner is being unfaithful. The condition is

termed pathological because the belief, which may be a delusion or an overvalued idea, is held on inadequate grounds and is unaffected by rational argument. The belief is often accompanied by strong emotions and characteristic behaviour, but these do not in themselves constitute pathological jealousy. A person who finds their partner in bed with a lover may experience extreme jealousy and behave in an uncontrolled way, but this should not be called pathological jealousy. The term should be used only when the jealousy is based on unsound evidence and reasoning.

Pathological jealousy is more common in men, with surveys finding a ratio of about 2:1. The frequency of the condition is unknown, but it is not uncommon in psychiatric practice. Each case merits careful attention, not only because of the great distress that the condition causes within relationships, but also because these individuals may be dangerous.

The classic paper on morbid jealousy is by Shepherd (1961).

Clinical features of pathological jealousy

As indicated, the main feature is an abnormal belief in the partner's infidelity. This may be accompanied by other abnormal beliefs—for example, that the partner is plotting against the patient, trying to poison them,

or infecting them with a sexually transmitted disease. The mood of the pathologically jealous patient may vary with the underlying disorder, but often it is a mixture of misery, apprehension, irritability, and anger.

Typically, the behaviour involves an intensive search for evidence of the partner's infidelity—for example, looking through diaries and examining bed linen and underwear. The patient may follow the partner about, or engage a private detective. The jealous person often cross-questions the partner incessantly. This may lead to violent quarrelling and paroxysms of rage in the patient. Sometimes the partner becomes exasperated and worn out, and is finally goaded into making a false confession. If this happens, the jealousy is inflamed rather than assuaged. An interesting feature is that the jealous person often has no idea who the supposed lover may be. Moreover, they may avoid taking steps that could produce unequivocal proof one way or the other.

Behaviour may be strikingly abnormal. A successful city executive carried a briefcase that contained not only financial documents but also a machete for use against any lover who might be detected. A carpenter installed an elaborate system of mirrors in their house so that their partner could be watched from another room. A third patient avoided waiting alongside another car at traffic lights in case their partner, sitting in the passenger seat, might surreptitiously make an assignation with the other driver.

Aetiology of pathological jealousy

Pathological jealousy, like other paranoid symptoms and syndromes, can occur in a range of primary disorders (see Box 12.4). Different surveys have reported a range of figures, probably reflecting the population studied and the diagnostic scheme used. For example, paranoid schizophrenia has been reported in 17–44% of patients, depressive disorder in 3–16%, neurosis and personality disorder in 38–57%, alcoholism in 5–7%, and organic disorders in 6–20%.

Box 12.4 Disorders associated with pathological jealousy

- Schizophrenia
- Mood disorder
- Organic disorder (including Parkinson's disease)
- Substance misuse (especially alcohol)
- Paranoid personality disorder

The role of personality in the genesis of pathological jealousy should be emphasized. It is often found that the patient has a pervasive sense of inadequacy, together with low self-esteem, and a discrepancy between their ambitions and attainments. Such a personality is particularly vulnerable to anything that may threaten this sense of inadequacy, such as loss of status or development of sexual dysfunction. In the face of such threats the person may project the blame on to others, and this may take the form of jealous accusations of infidelity. Freud believed that unconscious homosexual urges played a part in all jealousy, but clinical studies do not support an association between homosexuality and pathological jealousy. Similarly, although pathological jealousy has sometimes been attributed to the onset of sexual difficulties, there is no good evidence of such an association.

Prognosis of pathological jealousy

Little is known about the prognosis of pathological jealousy. It probably depends on a number of factors, including the nature of any underlying psychiatric disorder and the patient's premorbid personality. The clinical impression is that the prognosis is often poor.

Risk of violence

Although there are no reliable estimates of the risks of violence, there is no doubt that people with pathological jealousy can be dangerous. In addition to homicide, the risk of physical injury inflicted by jealous patients is considerable. In one series, around 25% had threatened to kill or injure their partner, and 56% of men and 43% of women had been violent towards or threatened the supposed rival (Mullen and Maack, 1985). Schanda et al. (2004), studying convicted murderers in Austria, found that delusional disorder (subtype not specified) is associated with homicide, with an odds ratio of 6. There is also a risk of suicide, particularly when an accused partner finally decides to end the relationship.

Assessment of pathological jealousy

The assessment of a patient with pathological jealousy should be particularly thorough, and should always include the partner, who should be interviewed separately whenever possible. Risk assessment is of particular importance. If there appears to be a risk of violence, the doctor should warn the partner, even if this involves a breach of confidentiality.

The partner may give a much more detailed account of the patient's morbid beliefs and actions than can be elicited from the patient. The doctor should try to find out tactfully how firmly the patient believes in the

partner's infidelity, how much resentment they feel, and whether they have contemplated any vengeful action. What factors provoke outbursts of accusations and questioning? How does the partner respond to such outbursts? How does the patient respond in turn to the partner's behaviour? Has there been any violence so far?

In addition to these enquiries, the doctor should take a detailed relationship and sexual history from both partners, and assess for underlying psychiatric disorder, as this will have implications for treatment.

Treatment of pathological jealousy

The treatment of pathological jealousy, as with other delusional disorders, is in principle fairly straightforward, the mainstay being antipsychotic drugs. However, in practice, treatment can be very difficult because of the patient's lack of insight and their reluctance to collaborate with the treatment plan. Furthermore, evidence from clinical trials is lacking.

Adequate treatment of any associated disorder, such as schizophrenia or a mood disorder, is a first requisite. If alcohol or other substance misuse is present, specific treatment will be needed. If the jealousy appears to be delusional in nature, a careful trial of an antipsychotic drug is worthwhile, although the results are often disappointing. As noted above, even when depressive disorder is not the primary diagnosis, it frequently complicates pathological jealousy and may worsen it. Treatment with an antidepressant may help in these circumstances, and also when the jealousy appears to be an overvalued idea rather than a delusion.

Psychotherapy may be attempted in cases where the jealousy appears to arise from personality problems, though again there is a lack of good-quality evidence. One aim is to reduce tensions by allowing the patient (and their partner) to vent their feelings. Behavioural methods include encouraging the partner to produce behaviour that reduces jealousy, for example, by refusing to argue, depending on the individual case. Cognitive and cognitive analytical approaches have also been attempted.

In some cases the safest procedure is to advise separation. It is not uncommon for feelings of pathological jealousy to wane once a relationship has ended. Sometimes, however, the problem re-emerges if the patient enters a new relationship.

Erotomania and erotic delusions

Erotic delusions can occur in any psychotic disorder, especially paranoid schizophrenia, but they are the predominant and persistent symptom in a form of delusional

disorder called *erotomania*. It was a French psychiatrist, G. G. de Clérambault, who in 1921 proposed that a distinction should be made between paranoid delusions and delusions of passion (erotic delusions). The latter differed in their pathogenesis and in being accompanied by excitement. This distinction is of historical interest only, but the syndrome is still sometimes known as *de Clérambault's syndrome*.

Erotomania is rare and appears to occur predominantly in women. However, it can manifest in men, with Taylor et al. (1983) reporting four cases in a series of 112 men charged with violent offences. The woman, who is usually single, believes that an exalted person is in love with her. The supposed lover is usually inaccessible, as he is already married, or is a famous person. According to de Clérambault, the infatuated woman believes that it is the supposed lover who first fell in love with her, and that he is more in love with her than she is with him. She derives satisfaction and pride from this belief. She is convinced that the supposed lover cannot be a happy or complete person without her. The patient often believes that the supposed lover is unable to reveal his love for various unexplained reasons, and that he has difficulties in approaching her, has indirect conversations with her, and has to behave in a paradoxical and contradictory way. The woman may cause considerable nuisance to the supposed lover. She may be tenacious and imperious to reality. Other patients turn from a delusion of love to a delusion of persecution, become abusive, and make public complaints about the supposed lover. This was described by de Clérambault as two phases—hope, followed by resentment.

There are few data regarding the treatment and outcome of erotomania. For review, see Sampogna et al. (2020). For a review of the concept of erotomania, see Berrios and Kennedy (2002).

Stalking

A proportion of 'stalkers' appear to suffer from delusional disorders, including erotomania, which is why the topic is mentioned here.

There is no clear consensus about the definition of stalking. Most formulations contain the following elements:

- a pattern of intrusive behaviour;
- the intrusive behaviour is associated with implicit or explicit threats; and
- the person being stalked experiences fear and distress.

Stalkers typically follow their victims around and loiter outside their house or place of work. Unwanted

communications by telephone, letter, or graffiti, and, in more recent times, by email and social media, are very common. Behaviour can then become more threatening, with hoax advertisements or orders for services, scandalous rumour mongering, damage to the victim's property, threats of violence, and actual assault.

Stalkers are a heterogeneous group with differing underlying psychopathologies (Dressing et al., 2006). One study found that 71% of stalking offenders met criteria for a psychiatric diagnosis, with personality disorder and substance use disorder being most common; psychotic disorder was present in 10% (Nijdam-Jones et al., 2018). Stalkers have often had a relationship with their victim that may have been quite superficial; in other cases, however, a serious relationship has cooled. A previous history of domestic violence in the relationship puts the victim at particularly high risk of assault and injury. Whether or not the victim suffers actual assault, they invariably experience severe psychological stress, which can lead to anxiety and mood disorders and post-traumatic stress disorder. Risk assessment is important.

For a review of stalking, see Mullen et al. (2006).

Somatic delusional disorder

People with somatic delusional disorder believe that they suffer from a physical illness, deformity, or infestation (e.g. delusional parasitosis, also called Morgellons disease). The term encompasses *monosymptomatic hypochondriacal psychosis*, as there is often a single, intense delusional belief of this kind. Somatic delusional disorder needs to be distinguished from the hypochondriacal delusions (and somatic hallucinations) that can occur in other disorders (e.g. schizophrenia, psychotic depression, cocaine abuse), and from genuine somatic symptoms that occur secondary to organic disorders (e.g. the pruritus of hepatic failure). It must also be distinguished from the common occurrence of obsessional thoughts or overvalued ideas about bodily issues. A specific example of the latter is body dysmorphic disorder (also called dysmorphophobia; see Chapter 22). In fact there is much overlap clinically, and perhaps therapeutically, between delusional and non-delusional forms of body dysmorphic disorder (Phillips et al., 2006).

Because of the content of the belief, somatic delusional disorders often present to the relevant medical specialism—for example, body dysmorphic disorder to plastic surgeons, or delusional parasitosis to dermatologists. To ensure correct identification and appropriate treatment, it is necessary that the physician recognizes

the nature of the disorder, and is able either to treat it or to involve a psychiatrist in its management. However, this does not always occur, and in any event, patients are often reluctant to accept the diagnosis.

Querulant delusions and reformist delusions

Querulant delusions were the subject of a special study by Krafft-Ebing in 1888. Patients with this kind of delusion indulge in a series of complaints and claims lodged against the authorities. Closely related to querulant patients are paranoid litigants, who undertake a succession of lawsuits, and become involved in numerous court hearings, in which they may become passionately angry and may make threats against the magistrates. The characteristics of persistent litigants have been reviewed by Lester et al. (2004).

Baruk (1959) described 'reformist delusions', which are based on religious, philosophical, or political themes. People with these delusions constantly criticize society and sometimes embark on elaborate courses of action. Their behaviour may be violent, particularly when the delusions are political. It is extremely important that this diagnosis is made on clear psychiatric grounds rather than on political grounds, as occurred in the former Soviet Union.

Delusional misidentification syndrome

Another group of delusions involves different aspects of misidentification, either of the self or of others. Like all delusions, they often occur in other psychotic disorders, especially schizophrenia and organic disorders, but they can also occur in isolation, and have been given the collective label of delusional misidentification syndrome (Christodoulou, 1991). The category is not named in ICD-11 or DSM-5, but constitutes an example of 'other persistent delusional disorders'. One argument for bringing them together is that they all appear to be 'face-processing disorders', and are associated with abnormalities in the posterior part of the right hemisphere, where the systems responsible for face recognition are located. Note also the seemingly close relationship of these disorders to the neurological category of *prosopagnosia* (the inability to recognize familiar faces). Interestingly, the delusions are specific to a few, usually familiar, people, and recognition of other faces (and objects) is not impaired. Although the

beliefs are delusional, the patient is aware that something is wrong with the ‘replacement’ person. The patient may be extremely distressed, and may occasionally act against individuals whom they believe to be impostors.

Four main variants of delusional misidentification are recognized. In each case there is debate as to whether they constitute a symptom or a syndrome.

Capgras syndrome

In this rare condition (Oyeboode, 2021)—which is really a symptom rather than a syndrome, hence its alternative name, *Capgras delusion*—the patient, usually a woman, believes that a person closely related to her (often her partner) has been replaced by a double. She accepts that the misidentified person has a strong resemblance to the familiar person, but still believes that they are different people. Some patients with Capgras syndrome may behave dangerously by attacking the presumed double. A history of depersonalization, derealization, or déjà vu is not unusual. The syndrome is an example of a reduplicative paramnesia. In a systematic review (Pandis et al., 2019), 43% of cases were attributed to an organic cause (such as dementia). Of the remainder, schizophrenia was much the commonest diagnosis. Organic cases had less additional psychopathology and were significantly older.

The name derives from the classic description by Capgras and Reboul-Lachaux in 1923, who called it *l’illusion des sosies* (illusion of doubles). However, as noted, it is a delusion, not an illusion.

Fregoli syndrome

Also known as the *Fregoli delusion*, the patient believes that one or more individuals have changed their appearance to resemble familiar people, usually in order to persecute the patient in some way. The symptom is usually associated with schizophrenia or with organic brain disease. Its name derives from an actor called Fregoli who had remarkable skill in changing his facial appearance. For review, see Langdon et al. (2014).

Intermetamorphosis

In this syndrome, the patient believes that one or more individuals have been transformed, both physically and psychologically, into another person or people, or that people have exchanged identities with each other. As with the other forms of delusional misidentification, note that intermetamorphosis is not a hallucination—the abnormality is one of interpretation, not misperception.

The syndrome of subjective doubles

In the syndrome of subjective doubles, the patient has the delusion that another person has been physically transformed into his own self, like a doppelgänger.

Shared or induced delusional disorder

Sometimes a person who is in a close relationship with someone who already has an established delusional system develops similar ideas. A commonly used term is *folie à deux*, and the condition has also been called *communicated insanity*. The frequency of induced psychosis is not known, but it is low. Sometimes more than two people are involved (*folie à plusieurs*), but this is exceedingly rare. It has also been speculated that some apocalyptic cults involve phenomena of this kind.

Over 90% of reported cases are members of the same family. Usually there is a dominant partner with fixed delusions who appears to induce similar beliefs in a dependent or suggestible partner, sometimes after initial resistance. The beliefs in the recipient may or may not be truly delusional. Generally the two people have lived together for a long time, often in isolation from the outside world. Once established, the condition runs a chronic course.

It is usually necessary to advise separation of the affected individuals. This may lead to resolution of the quasi-delusional state in the recipient; the original patient should be treated in the usual fashion for delusional disorder. For a review of shared delusional disorder, see Silveira and Seeman (1995).

Assessment of paranoid symptoms

The assessment of paranoid symptoms involves two stages—first, the recognition of the symptoms themselves, and secondly, the diagnosis of the underlying condition.

Sometimes it is obvious that the patient has persecutory ideas or delusions. At other times recognition of paranoid symptoms may be exceedingly difficult, and considerable skill is required by the interviewer. The

patient may be suspicious or angry. They may be very defensive, say little, or speak fluently about other topics while steering away from persecutory beliefs or denying them completely. The psychiatrist should be tolerant and impartial, acting as a detached but interested listener who wants to understand the patient's point of view. The interviewer should show compassion, but not collude in the delusions or give promises that cannot be fulfilled. When an apparently false belief is disclosed, considerable time and effort may then be needed to determine whether or not it meets the criteria for a delusion rather than an overvalued idea or other form of belief. This is of crucial diagnostic significance, as the presence of a delusion is likely to be the symptom upon which a diagnosis of psychotic disorder is based, whereas non-delusional thoughts which may be similar in content are consistent with a range of other diagnostic categories.

If delusions are detected, the next step is to diagnose the type of psychosis, based upon the diagnostic features of the disorders noted earlier in this chapter. It is also important to determine whether the patient is likely to try to harm the alleged persecutor. A full risk assessment is needed. This calls for close study of the patient's personality, history of violence, and the characteristics of their delusions and any associated hallucinations. Hints or threats of homicide should be taken seriously. The doctor should be prepared to ask about possible homicidal plans and preparations to enact them. In many ways the method of enquiry resembles the assessment of suicide risk: 'Have you ever thought of doing anything about it?', 'Have you made any plans?', and 'What might prompt you to do it?'

The assessment of dangerousness is discussed further in Chapter 18.

Treatment of paranoid symptoms and delusional disorder

General principles

Management of paranoid symptoms and delusional disorder is frequently difficult. Patients typically regard their delusional beliefs as valid and justified, and therefore see no need for treatment. Or, they may be suspicious and distrustful, believing that psychiatric treatment is intended to harm them. Particular tact and skill are needed when dealing with such patients, not only to encourage them to describe their symptoms fully, as discussed above, but also to persuade them to accept treatment and then to adhere to it. Sometimes treatment can be made acceptable by offering to help non-specific symptoms such as anxiety or insomnia, or by pointing out the harmful consequences of the beliefs. Thus a patient who believes that they are surrounded by persecutors may agree that their nerves are being strained as a result, and that this needs treatment.

A decision must be made as to whether to admit the patient for inpatient care. This may be indicated if there is a significant or immediate risk of violence to others, or of suicide. When assessing such factors, it is important to consult other informants and obtain a history of the patient's behaviour. Compulsory admission may rarely be justified to protect the patient or other people, although this is likely to add to the patient's resentment.

Drug treatment

Paranoid symptoms in delusional disorder are treated with antipsychotic drugs just as in other psychoses, although there are few randomized trial data to guide decision making (Munoz-Negro and Cervilla, 2016). Pimozide has a specific indication for delusional disorders and has sometimes been advocated as the drug of choice, but the evidence is weak, and any high-potency, non-sedating antipsychotic is suitable (e.g. risperidone), always starting with a low dose. Signs of improvement, notably a decrease in preoccupation with the delusion(s) and a reduction in agitation, may be seen within a few days. The importance of establishing a good therapeutic relationship to improve collaboration with treatment has already been emphasized.

With regard to the delusional form of body dysmorphic disorder, some data suggest that selective serotonin reuptake inhibitors (SSRIs), rather than antipsychotics, should be used as first-line treatment, with antipsychotic augmentation for those patients who do not respond (Phillips et al., 2006). The role of antidepressants in other delusional disorders remains unclear, although they are often used at some stage in treatment, reflecting the frequency of comorbid depressive symptoms, and their emergence during treatment. The risk of suicide should be monitored regularly.

Psychological treatment

Patients with paranoid symptoms require support, encouragement, and reassurance. This form of non-specific psychological treatment is an integral part of management, and essential if the patient is to be persuaded of the benefits of more targeted interventions. Of the latter, drugs are the mainstay of treatment, but

specific psychological therapies may have a role, too. In particular, cognitive therapy, as used for the treatment of delusions in schizophrenia, may be worth trying if a sufficiently good therapeutic rapport exists. Interpretative psychotherapy and group psychotherapy are unsuitable, because suspiciousness and hypersensitivity may easily lead the patient to misinterpret what is being said.

Prognosis of delusional disorder

There are no reliable data on long-term outcomes. Clinical impression suggests that the prognosis in delusional disorder is poor, although, compared to schizophrenia, long-term outcome and overall functioning are relatively good (Marneros et al., 2012). In some patients, medication can be reduced or stopped without ill effects, while in others

(probably the majority) delusions recur rapidly on discontinuation, and treatment must be maintained for prolonged periods. This issue can be judged only by a careful clinical trial with regular monitoring of the patient's mental state, and it requires discussion with the patient about the risks and benefits of long-term medication.

Further reading

Enoch, M. D., Puri, B. & Ball, H. N. (2020). *Uncommon Psychiatric Syndromes*, 5th edn. Edward Arnold.

Lewis, A. (1970). Paranoia and paranoid: A historical perspective. *Psychologic Med*, 1, 2–12.

Oyeboode, F. (2021). *Psychopathology of Rare and Unusual Syndromes*. Cambridge University Press.

CHAPTER 13

Eating, sleep–wake, and sexual disorders

Introduction 305

Eating disorders 305

Sleep–wake disorders 316

Disorders of sexual function and paraphilic disorders 321

Introduction

As discussed in other chapters, the drives to eat, sleep, and have sex can all become impaired or dysfunctional in many psychiatric disorders or can occur as a side-effect

of their treatment. They can also all be primary disorders, and it is the latter which are the focus here. The chapter also considers issues related to gender identity.

Eating disorders

Eating disorders are characterized by abnormalities in the pattern of eating and the amount and nature of food eaten that are not developmentally appropriate nor culturally sanctioned. The disorders share a distinctive core psychopathology, which is best described as an overevaluation of weight and shape, such that patients judge their self-worth in these terms, and in their ability to control them. The main disorders are anorexia nervosa and bulimia nervosa, together with a number of related conditions.

Until the late 1970s, eating disorders were believed to be uncommon. Following the description of bulimia nervosa, they have become conspicuous and their significant morbidity apparent. It remains uncertain whether the marked rise in presentation and diagnosis reflects a true increase in incidence or an increase in detection and diagnosis. Many eating disorders go clinically unrecognized, and it is estimated that only about 50% of the cases of anorexia nervosa in the general population are detected in primary care; for bulimia

nervosa the figure is substantially less, and the majority of individuals are untreated. Within secondary care, eating disorders are managed by general psychiatrists as well as by specialist eating disorder services; gastroenterology and paediatric teams also contribute.

For a review of eating disorders, see Treasure et al. (2020), and for eating disorders in children and adolescents see Eddy et al. (2021). For an overview of eating disorder epidemiology, see Halmi (2020), for their treatment, see McElroy et al. (2020), and for their outcome, see Solmi et al. (2024).

Classification

ICD-11 and DSM-5 take a very similar approach to classification of what are termed ‘Feeding and eating disorders’ (Table 13.1). Although anorexia nervosa and bulimia nervosa are the archetypal eating disorders, binge eating disorder and other specified feeding and eating disorders are considerably more common (Santomauro et al.,

Table 13.1 Diagnostic classification of eating disorders

ICD-11 Feeding and eating disorders	DSM-5 Feeding and eating disorders
Anorexia nervosa	Anorexia nervosa
Bulimia nervosa	Bulimia nervosa
Binge eating disorder	Binge eating disorder
Avoidant-restrictive food intake disorder	Avoidant/restrictive food intake disorder
Pica	Pica
Rumination–regurgitation disorder	Unspecified feeding or eating disorder
Other specified feeding or eating disorders	Other specified feeding or eating disorder

Source: *International Classification of Diseases, Eleventh Revision (ICD-11)*. Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)*. Copyright (2013), American Psychiatric Association.

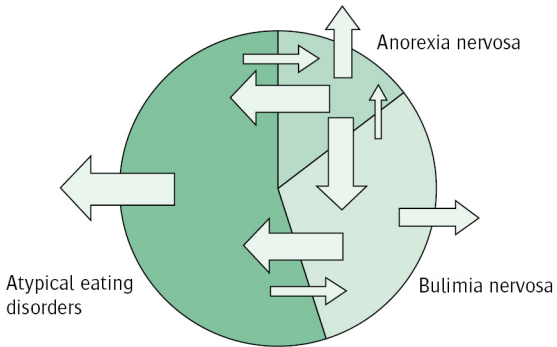


Figure 13.1 Schematic representation of temporal movement between the eating disorders. The size of the arrow indicates the likelihood of movement in the direction shown. ‘Atypical eating disorders’ include binge eating disorder and cases with mixed features. Arrows that point outside of the circle indicate recovery. Reproduced from Fairburn, C. G., & Harrison, P. J. (2003). Eating disorders. *Lancet*, 361(9355), 407–416. Copyright (2003) with permission from Elsevier.

2021). Moreover, many patients ‘migrate’ between eating disorder categories over time (Figure 13.1), suggesting that they share a common pathophysiology, and that the boundaries between them are largely arbitrary (Fairburn and Harrison, 2003), leading to a transdiagnostic theory of eating disorders (Cooper and Dalle Grave, 2017).

Anorexia nervosa

Although there were many prior descriptions, anorexia nervosa was first named in 1868 by the English physician William Gull, who emphasized the psychological

causes of the condition, the need to restore weight, and the role of the family.

The main features of anorexia nervosa are:

- Very low body weight, which is maintained by restriction of energy intake and not explained by another condition nor the unavailability of food. A commonly used threshold is a body mass index (BMI) of 18.5kg/m², or a BMI below the 5th centile for a child.
- A persistent pattern of restrictive eating or other behaviours aimed at achieving or maintaining low body weight, typically associated with extreme concern about weight and shape, an intense fear of gaining weight, and a strong desire to be thin.
- A preoccupation with body weight or shape and an undue influence of body weight or shape on self-evaluation.
- Lack of recognition of the seriousness of low body weight.

Most patients are young women. The condition usually begins in adolescence, although childhood-onset and older-onset cases are encountered. It generally begins with ordinary efforts at dieting, which then get out of control. The central psychological features are the characteristic overvalued ideas about body shape and weight.

The *pursuit of thinness* may take several forms. Patients generally eat little and set themselves very low daily calorie limits (often between 600 and 1000 kcal). Some try to achieve weight loss by inducing vomiting, exercising excessively, and misusing laxatives (*purging*). Patients are often preoccupied with thoughts of food, and sometimes enjoy cooking elaborate meals for other people. Others may have episodes of binge eating.

Amenorrhoea is one of several characteristic physical abnormalities, though it is no longer incorporated in diagnostic criteria. It occurs early in the development of the condition, and in about 20% of cases it precedes obvious weight loss, although careful history-taking generally reveals that these patients had already started food restricting. Some cases first come to medical attention with amenorrhoea rather than disordered eating.

Other symptoms. Depressive, anxiety, and obsessional symptoms, lability of mood, and social withdrawal are all common. Three quarters of patients report a lifetime history of major depressive disorder. Lack of sexual interest is usual.

Anorexia nervosa can be subdivided into those with a restricting pattern, or a binge–purge pattern, depending on whether episodes of the latter occur. It can also be divided based on body weight, because of the medical significance of the latter. A BMI of 18–14 is considered ‘significantly low body weight’ and a BMI below 14 constitutes ‘dangerously low body weight’. At the other extreme, ‘anorexia nervosa with normal body weight’ can be used

for people in the year following attaining a normal BMI and cessation of the anorectic behaviours, but with persisting characteristic ideas about weight and shape.

For a review of anorexia nervosa, see Mitchell and Peterson (2020). See Downs and Kelly (2025) for a review of anorexia nervosa in males.

Physical consequences

A number of important symptoms and signs of anorexia nervosa are secondary to starvation. Several body systems can be affected, with increasing involvement as BMI falls (see Box 13.1). Some cases first come to medical attention because of one of these complications rather than because of disordered eating itself.

Epidemiology

The incidence of anorexia nervosa based upon primary care and mental health surveys is about 5 per 100,000.

Rates in the community are considerably higher. Incidence is greatest among young women, with 40% of all incident cases occurring in 15- to 19-year-old females. Reported incidence rates increased from the beginning of the twentieth century up to the 1970s, but remained fairly stable subsequently until a Covid-19 pandemic-associated increase (Taquet et al., 2022). Anorexia nervosa is uncommon in children under 13 years of age, and in this age group the sex ratio is closer to one.

Prevalence estimates vary depending on the method of ascertainment, country, and age range studied, and whether atypical anorexia nervosa is included. If so, population studies find a lifetime prevalence of between 0.9% and 4% in women; rates in men are always considerably lower, but the sex ratio is probably less than the 10:1 figure often stated. The condition is thought to be more common in those from high socioeconomic status

Box 13.1 Main physical features of anorexia nervosa

Physical symptoms

Heightened sensitivity to cold
Gastrointestinal symptoms—constipation, fullness after eating, bloating
Dizziness and syncope
Amenorrhoea
Poor sleep, with early-morning wakening

Physical signs

Emaciation
Stunted growth and failure of breast development (if onset is prepubertal)
Dry skin, with orange discoloration of the palms and soles
Fine downy hair (*lanugo*) on the back, forearms, and sides of face
Salivary gland swelling
Erosion of the inner surface of the front teeth (perimyolysis) in those who vomit frequently
Cold hands and feet; hypothermia
Peripheral oedema
Proximal myopathy

Physical investigations

Endocrine abnormalities

Low luteinizing hormone, follicle-stimulating hormone, and oestradiol levels

Low T_3 , and raised reverse T_3 , with T_4 in low normal range and normal concentrations of thyroid-stimulating hormone (low T_3 syndrome)
Increase in cortisol and dexamethasone non-suppression
Raised growth hormone concentration

Cardiovascular abnormalities

Bradycardia; hypotension; cardiac arrhythmias (especially in those with electrolyte abnormalities)
QT interval prolongation

Gastrointestinal abnormalities

Delayed gastric emptying
Decreased colonic motility (if chronic laxative misuse)

Haematological abnormalities

Mild normocytic normochromic anaemia
Mild leukopenia with relative lymphocytosis
Thrombocytopenia

Other metabolic abnormalities

Hypercholesterolaemia
Raised serum carotene
Dehydration
Electrolyte disturbances, notably hypokalaemia (in those who vomit frequently or misuse laxatives or diuretics), also low sodium, phosphorus, magnesium, fasting glucose, albumin

Other abnormalities

Osteopenia and osteoporosis

groups, and less common but increasing in low- and middle-income countries, and in minority ethnic groups in western countries. Sexual and gender minority groups appear to be at increased risk (Nagata et al., 2020).

For review of anorexia nervosa epidemiology, see van Eeden et al. (2021).

Onset, course, and prognosis

In its early stages, anorexia nervosa often runs a fluctuating course, with exacerbations and periods of partial remission. The long-term prognosis is difficult to judge owing to incomplete follow-up, or because there may be normalization in weight or menstrual function but persistent abnormalities of eating habits and attitudes to weight and shape. Long-term follow-up of clinically ascertained cases shows that about a third have a persistent eating disorder, with a median duration of ten years (Dobrescu et al., 2019). Poor prognostic factors include onset before puberty or in adulthood, delayed presentation (>3 years), premorbid personality problems, comorbid substance misuse, and childhood obesity. In addition to their chronic psychopathology and physical health problems, patients may have significant impairments in social functioning and employment.

Anorexia nervosa has the highest mortality rate of any major psychiatric disorder, with an almost sixfold increase in mortality compared to the general population (Semchishen et al., 2026). One in five deaths is from suicide; the others reflect the many adverse health consequences of the disorder, notably cardiac events and sepsis.

Aetiology

Genetics

Anorexia nervosa is familial, with a reported heritability of 50–60%. Eight genetic loci have been identified, and implicate an underlying metabolic dysfunction to the disorder, independent of BMI, as well as correlations with other psychiatric disorders and traits (Watson et al., 2019). The genetic risk may vary between genders and ethnicities, and with other factors such as age of onset (Huckins et al., 2022).

Neurobiology

There have been many brain imaging and other neurobiological studies of anorexia nervosa, and a range of structural, functional, and biochemical abnormalities reported (Su et al., 2021). These include reductions in brain volume, and alterations in the 5-HT (serotonin) system. However, it is often difficult to determine whether abnormalities

are causal, or are the result of starvation and weight loss. Nevertheless, some findings are of interest.

- Cognitively, there are impairments in several neuropsychological domains, including visuospatial abilities and planning (Stedal et al., 2021).
- Grey matter volume is increased in the orbitofrontal cortex and insula, regions known to be involved in assessing reward and in introspection, respectively.
- Functional neuroimaging also indicates involvement of brain regions involved in responses to food rewards.

Linking neuroimaging with neurochemistry, one theory is that the restrictive eating of anorexia nervosa is a maladaptive attempt to reduce the negative affect caused by an imbalance between the aversive (serotonergic) and reward (dopaminergic) systems of the brain (Kaye et al., 2013).

Sociocultural factors

The fact that anorexia nervosa is more common in certain societies suggests that environmental factors play a part in its development. Important among such factors is likely to be the notion that thinness is desirable and attractive. Surveys in affluent societies show that most schoolgirls and female college students diet at one time or another. However, when other risk factors are considered, people who develop anorexia nervosa have no greater exposure to factors that increase the risk of dieting. This suggests that the problem is more due to how an individual reacts to dieting than to dieting itself. Evidence supports the importance of media and peer groups in influencing weight and shape concerns in adolescents, especially through social media (Dane and Bhatia, 2023). A recent example of a probable sociocultural influence is the Covid-19 pandemic which, as noted earlier, was associated with an increased incidence and severity of eating disorders (Taquet et al., 2022a).

Individual psychological causes

Bruch (1974) was one of the first writers to discuss the psychological antecedents of anorexia nervosa. She suggested that these patients are engaged in ‘a struggle for control, for a sense of identity and effectiveness, with the relentless pursuit of thinness as a final step in this effort’. These clinical observations are supported by epidemiological studies, which implicate low self-esteem and perfectionism in the development of the disorder (Fairburn, 1999). It has been suggested that these premorbid personality traits can make it particularly difficult for an individual to negotiate the demands of adolescence.

Causes within the family

Disturbed relationships are often found in the families of patients with anorexia nervosa, and some authors have suggested that they have an important causal role. Minuchin et al. (1978) held that a specific pattern of relationships could be identified, consisting of ‘enmeshment, overprotectiveness, rigidity and lack of conflict resolution’. Although views of this kind lack strong evidence and are now less prominent, a role for family dynamics in the origin of anorexia nervosa continues to be considered a potential contributory factor.

Epidemiological studies suggest that people who develop anorexia nervosa are more likely than others to have been exposed to a range of childhood adversities, including poor relationships with parents and parental psychiatric disorder, particularly depression. However, these risk factors are not specific to anorexia nervosa, but are found with equal frequency among people who subsequently develop other psychiatric disorders (Fairburn, 1999). It seems likely that these general risk factors interact with specific factors within the individual, such as perfectionism and low self-esteem, to increase the risk of developing anorexia nervosa. For review, see Treasure et al. (2020).

Assessment

Most patients with anorexia nervosa are reluctant to change their behaviour, let alone see a mental health professional, so it is important to establish a good relationship. This means listening to the patient's views, explaining the treatment alternatives, and being willing to consider compromises—though it is also important to be clear that there is little alternative to the calorific intake needed to achieve a healthier weight. A thorough history should be taken of the development of the disorder, the present pattern of eating and weight control, and the patient's ideas about body weight (see Boxes 13.2 and 13.3). In the mental state examination, particular attention should be given to depressive symptoms, as well as to the characteristic psychopathology of anorexia nervosa itself. More than one interview may be needed to obtain this information and gain the patient's confidence. In children and adolescents, parents should be interviewed, and information gathered regarding early feeding and the parents' own weight and shape concerns.

Medical complications are an integral part of anorexia nervosa, and it is essential to perform a full physical examination (Gibson et al., 2019). Particular attention should be paid to the degree of emaciation (including measurement of weight and BMI), cardiovascular status

(blood pressure, heart rate), and temperature. Routine investigations include full blood count, urea and electrolytes, blood glucose, liver function tests, and ECG. The results may reveal that the patient is at high risk of medical complications and requires further urgent investigations or even inpatient treatment (Box 13.4). The MEED (Medical Emergencies in Eating Disorders) guidelines provide up-to-date advice about the assessment, and the treatment, of severe anorexia nervosa (Royal College of Psychiatrists, 2022).

A range of medical and psychiatric disorders may present with weight loss or loss of appetite, and lead to the possibility of anorexia nervosa being raised (Box 13.5). However, there should rarely be significant diagnostic uncertainty once the presence or absence of the cardinal psychopathology of anorexia nervosa has been determined.

Box 13.2 Assessment of eating and dieting behaviours

Mealtimes and diet

What is a typical day's eating? What are the mealtime arrangements? Are there rituals?

To what degree is the patient attempting restraint?

Do they avoid particular foods? If so, why?

What is their experience of hunger or of any urge to eat?

Bingeing and purging

Do they binge? How large and how often? Do they feel out of control?

Do they make themselves vomit? If so, how?

Do they take laxatives, diuretics, emetics, or appetite suppressants? If so, with what effects?

Exercise and other weight-loss behaviours

Do they exercise? What type, duration, frequency?

What are the goals of exercise? To 'burn off calories', affect body shape?

Is exercise pleasurable? Or is the person compelled to do it?

Source: Palmer, B. (2014). *Helping People with Eating Disorders: A Clinical Guide to Assessment and Treatment*, 2nd edn. Copyright (2014), John Wiley and Sons.

Box 13.3 Assessment of psychological issues related to eating disorders

What does the patient feel about their body, weight, and shape?

What do they think would happen if weight or eating were not controlled?

What are the rewards of their current weight and behaviours?

What are the thoughts and feelings before, during, and after bingeing or purging?

How do they view their eating and weight, and what should be done about it?

Source: Palmer, B. (2014). *Helping People with Eating Disorders: A Clinical Guide to Assessment and Treatment*, 2nd edn. Copyright (2014), John Wiley and Sons.

Treatment of anorexia nervosa

Treatment of anorexia nervosa is almost exclusively based on psychological interventions. Some key principles, based on the current National Institute for Health Care Excellence (2017) guideline, are summarized in Box 13.6; see also Hay et al. (2014). However, anorexia nervosa remains a difficult condition to treat, and existing therapies are markedly less effective than for bulimia nervosa and binge eating disorder. There is no good evidence that any one form of psychotherapy shown in Box 13.6 is superior to any other in the treatment of anorexia nervosa, nor do they clearly outperform treatment as usual (Solmi et al., 2021). The choice of therapy is also dependent on its availability and acceptability.

Psychotherapy

Family therapy has been widely advocated, reflecting the belief that family factors are important in the origins of anorexia nervosa. Various kinds of family therapy have been used, especially for children and adolescents. Family-based treatments that focus on the eating disorder and related issues are more effective than those addressing broader family processes, and more so than individual psychotherapy (Lock, 2015).

In adults, individual treatments predominate. Generic CBT has only modest benefits, but a specifically tailored form ('CBT-ED', also called CBT-E) has some efficacy in weight restoration and in weight maintenance (Fairburn et al., 2013). As discussed below, CBT-ED is also highly effective for bulimia nervosa, and is

Box 13.4 Abnormalities requiring urgent follow-up or intervention in anorexia nervosa

General

Weight loss >1 kg per week for two consecutive weeks

BMI <13 kg/m² or <70% of expected BMI in under-18-year-olds

Temperature <35.5°C

Muscle weakness

Acute food refusal for >24 hours

Cardiovascular

Bradycardia (<40 beats/minute)

Blood pressure <80/50 mmHg

Postural hypotension >20 mmHg, or with syncope

Any significant ECG abnormality (e.g. prolonged QT interval)

Blood tests

Hypokalaemia (<2.5 mmol/l)

Hypophosphataemia (<0.5 mmol/l)

Hypoglycaemia

Neutropenia

Source: Hay, P., Chinn, D., Forbes, D., Madden, S., Newton, R., Sugener, L., Touyz, S., & Ward, W. (2014). Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for the treatment of eating disorders, *Aus N Z J Psychiatry*, 48(11), 977–1008. Copyright (2014), SAGE Publications; Zipfel, S., Giel, K. E., Bulik, C. M., Hay, P., & Schmidt, U. (2015). Anorexia nervosa: Aetiology, assessment, and treatment. *Lancet Psychiatry*, 2(12), 1099–1111. Copyright (2015), Elsevier.

the first evidence-based 'trans-diagnostic' eating disorder therapy. Other psychological interventions include MANTRA (see Box 13.6) and focal psychodynamic psychotherapy. Nevertheless, as noted above, all therapies are of limited effectiveness—a problem exacerbated by the fact that it can be difficult to engage patients in treatment of any kind.

Medication

Both antidepressants and antipsychotics can be used in anorexia nervosa, with antidepressants sometimes prescribed in high dosage. However, systematic reviews show no clear effect of antidepressants on weight gain, maintenance, or psychological symptoms, and they

Box 13.5 Differential diagnosis of anorexia nervosa

Medical disorders

Neoplasia (e.g. gastrointestinal, hypothalamic, lymphoma)
 Inflammatory bowel disease
 Malabsorption syndromes (e.g. coeliac disease, chronic pancreatitis)
 Hyperthyroidism
 Chronic infection
 Diabetes mellitus
 Pituitary failure
 Cystic fibrosis

Psychiatric disorders

Other eating disorders
 Depression
 Somatoform disorders
 Obsessive–compulsive disorder

are not recommended for routine use. Small trials have suggested possible benefit from olanzapine in terms of weight gain but without important impacts on core psychopathological symptoms (Attia et al., 2019).

Antidepressants are also used to treat depression in anorexia nervosa (Besjes et al., 2025), but guidelines suggest that this should not occur until it is apparent that the symptoms are not merely due to starvation, and that they persist during restoration of weight (Hay et al., 2014). Particular care in prescribing is required in patients under 18 years old, and because of the high risks of medical complications and side effects in under-weight patients.

A range of novel interventions are being tested for anorexia nervosa (Brockmeyer et al., 2018), including brain stimulation and even neurosurgery. There are preliminary indications these may be of benefit in severe, intractable cases, but there are significant ethical issues, and they are not recommended in current guidelines (Murray et al., 2022).

Management

Starting treatment

Success largely depends on establishing a good relationship with the patient. It should be made clear that

Box 13.6 Treatment of anorexia nervosa based on NICE (2017) guidelines

General factors

Establish therapeutic relationship and provide psychoeducation.
 Monitor weight, mental and physical health, and risk factors.
 Be multidisciplinary and coordinated between services.
 Involve family members or carers (where appropriate).
 Review physical and mental health at least annually.

Psychological treatment for adults

Consider one of: CBT-ED (see text), MANTRA (Maudsley Anorexia Treatment for Adults), or SSCM (specialized supportive clinical management).
 If the above are not acceptable or ineffective, consider eating disorder-focused focal psychodynamic therapy.

Psychological treatment for children and adolescents

Consider focused family therapy.
 If unacceptable or ineffective, consider CBT-ED or adolescent-focused psychotherapy for anorexia nervosa.

Other issues

Monitor and treat physical health complications (see Box 13.4).
 Monitor growth in adolescents.
 Consider bone mineral density measurement and thence osteoporosis treatment.
 Consider day patient or in patient admission for severe or resistant cases.
 Do not use medication routinely.

Reproduced from National Institute for Health and Care Excellence (NICE). (2017). Eating disorders: Recognition and treatment. NICE guideline NG69. Copyright (2017), with permission from National Institute for Health and Care Excellence.

achieving an adequate weight is essential to reverse the physical and psychological effects of starvation. It is important to agree a specific dietary plan, while emphasizing that weight control is only one aspect of the

problem, and help should be offered with the accompanying psychological issues, as well as dealing with any medical complications. Educating patients and their families about the disorder and its treatment is important. Early intervention is associated with better outcomes, and so every effort should be made to engage with the patient as promptly as possible. Conversely, in longstanding cases of anorexia nervosa, treatment goals may be more modest, and include helping patient and carers cope with a serious and chronic illness.

Setting of treatment

Most cases may be treated on an outpatient or day-patient basis, ideally within a specialist eating disorder service. There is no good evidence that inpatient care is more effective, but demand is increasing, especially for adolescent inpatient eating disorder units. Admission is indicated primarily in two situations: to a medical or paediatric ward, if there are serious and imminent physical health risks (Box 13.4), or to psychiatric inpatient care, if there is acute suicidal ideation or if sustained attempts at outpatient or day-patient treatment have failed.

Compulsory treatment for anorexia nervosa is controversial both legally and ethically (see Chapter 4). It is rarely used, although refeeding (sometimes with a nasogastric tube) is accepted as a treatment for eating disorders under the Mental Health Act. For children and adolescents unable or unwilling to give informed consent, either the Mental Health Act or the Children Act (1989) may be used. The views of the parents as well as the patient are critical in decisions of this kind (Nicholls and Barrett, 2015).

Restoring weight

A reasonable aim is an increase of 0.5 kg a week, which will usually require an extra 500–1000 calories a day. More rapid refeeding is potentially dangerous, in part due to hypophosphataemia, and should follow the MEED guidance (Royal College of Psychiatrists, 2022). The target weight should be above a minimum healthy level (a BMI above 18.5); it is not a good idea to agree a compromise target lower than this except in severe and refractory cases. It is good practice to monitor the patient's physical state regularly, and to prescribe vitamin supplements if indicated. It is also important to assess and modify other weight-reducing strategies that the patient may employ, such as compulsive exercise or laxative misuse.

Previously, for those receiving inpatient treatment, the patient would usually stay in hospital until the agreed target weight was reached and maintained, and strict behavioural regimes were often used. Currently,

much shorter admissions, with a more collaborative therapeutic approach, are the norm, followed by intensive day-patient care. While in hospital, eating should be supervised by a nurse, who has several roles—to reassure the patient that they will not lose control over their weight, to be clear about the agreed targets, to monitor nutrition, and to ensure that the patient does not induce vomiting or take laxatives.

Bulimia nervosa

The term *bulimia* refers to binge eating, defined earlier. As mentioned, binge eating occurs in some cases of anorexia nervosa, and is also the hallmark of binge eating disorder (see below). The syndrome of bulimia nervosa was first described by Russell (1979) in an influential paper in which he named the condition and described the key clinical features in 30 patients. The prevalence of bulimic behaviours and the associated harms soon became apparent, and the syndrome was first included in DSM-III.

The central features of bulimia nervosa are as follows:

- A preoccupation with eating, with an irresistible and recurrent urge to overeat, manifesting in repeated 'binges' when large amounts of food are consumed in a short time, accompanied by a sense of loss of control and of guilt. The core feature is the loss of control rather than the quantity eaten.
- The use of extreme measures to control body weight, especially self-induced vomiting and use of laxatives, as well as periods of starvation or excessive exercise.
- Overvalued ideas concerning shape and weight, of the type seen in anorexia nervosa.

ICD-11 requires that these behaviours have occurred at least once a week for one month (three months in DSM-5) and that the patient does not meet criteria for anorexia nervosa (which, if present, takes precedence). Bulimia nervosa also needs to be distinguished from binge eating disorder, described below. However, as shown in Figure 13.1, many patients move between these categories. DSM-5 classifies the severity of bulimia nervosa according to the frequency of 'inappropriate compensatory behaviours', ranging from mild (one to three per week) to extreme (14 or more per week). ICD-11 does not have these elements.

Patients with bulimia nervosa are usually of normal weight (BMI 18.5–25), not least since patients who are substantially underweight usually fulfil criteria for a diagnosis of anorexia nervosa.

There is often a feeling of a profound loss of control overeating. Episodes of binge eating may be precipitated

by stress, or by the breaking of self-imposed dietary rules, or may occasionally be planned. During the episodes, large amounts of food are rapidly consumed, on average over 2000 kcal (e.g. a loaf of bread, pot of jam, cake, and biscuits). This voracious eating usually takes place when the person is alone. At first it brings relief from tension, but this is soon followed by guilt and disgust, and the person induces vomiting or engages in another compensatory behaviour.

Depressive symptoms are very common, and usually secondary to the eating disorder. A high proportion meet criteria for major depression. The depressive symptoms usually remit as the eating disorder improves.

For review of bulimia nervosa, see Treasure et al. (2020).

Physical consequences

Bulimia nervosa can impact on physical health, mostly due to repeated vomiting or use of purging agents such as laxatives. With repeated vomiting, reflux symptoms are common, and tooth enamel can erode, increasing the risk of decay and sensitivity to temperature. *Russell's sign* describes callouses on the knuckles caused by repeatedly putting fingers down the throat to induce vomiting. Parotid and other salivary glands can become enlarged. More serious physical health problems can occur, but are rare (Box 13.7).

The physical complications are best treated by cessation of the causative behaviour, but symptomatic treatments are also available, for example, proton pump inhibitors for oesophageal reflux, or spironolactone for peripheral oedema. However, medication for these purposes should be used with caution, and only in severe cases.

For review, see Gibson et al. (2019).

Epidemiology

As with anorexia nervosa, the prevalence and incidence of bulimia nervosa are uncertain. In the community, the prevalence is around 1% among women aged between 16 and 40 years in western societies (Kessler et al., 2013). It is at least ten times less common in men. The dramatic increase in presentation and diagnosis seen in the United Kingdom in the early 1990s has been followed by stability or a modest decline. For review, see van Eeden et al. (2021).

Onset, course, and prognosis

Bulimia nervosa usually has an onset in late adolescence or early adulthood (i.e. several years later than anorexia nervosa). It often follows a period of concern about body shape and weight, and 25% of patients

Box 13.7 Physical complications of bulimia nervosa

General

- Menstrual irregularities
- Non-specific gastrointestinal symptoms

Related to vomiting

- Cardiac arrhythmias
- Oesophageal reflux
- Oesophageal tears
- Gastric rupture
- Epistaxis and subconjunctival haemorrhage
- Damage to teeth (perimyolysis)
- Parotid gland enlargement (sialadenosis) and raised salivary amylase
- Metabolic alkalosis
- Peripheral oedema

Related to stimulant laxatives

- Metabolic abnormalities as described for vomiting
- Hyponatraemia
- Chronic constipation
- Melanosis coli

have a history of anorexia nervosa (see Figure 13.1). There is commonly an initial period of dietary restriction which, after a variable length of time, but usually within three years, breaks down, with increasingly frequent episodes of overeating. As the overeating becomes more frequent, the body weight returns to a more normal level. At some stage, self-induced vomiting or laxative abuse are adopted to compensate for the overeating. However, this may result in even less control of eating.

Despite the original assertion by Russell (1979) that bulimia nervosa was an 'ominous variant', its outcome is clearly better than anorexia nervosa. Nevertheless, even five to ten years later between one third and a half of individuals still have a clinical eating disorder, although in many of these cases it will take an atypical form. No convincing predictors of course or outcome have been identified, although childhood obesity and low self-esteem may be associated with a worse prognosis.

The mortality rate is approximately double that of the general population, in part through cardiovascular risk, but the excess is significantly less than that for anorexia nervosa (Semchishen et al., 2026).

Aetiology

Like anorexia nervosa, bulimia nervosa appears to be the result of exposure to general risk factors for psychiatric disorder, including a family history (in part, reflecting a genetic predisposition), especially depression and substance misuse, and a range of adverse childhood experiences. No risk genes have been identified. It was thought that childhood sexual abuse was especially common, but evidence suggests that the rate is no higher than among those who develop other types of psychiatric disorder (Fairburn, 1999).

Epidemiological studies also suggest that, unlike those with anorexia nervosa, patients with bulimia nervosa have increased exposure to factors that specifically promote dieting, such as childhood obesity, parental obesity, and early menarche. Perfectionism appears to be less of a risk factor than in anorexia nervosa.

The neurobiological mechanisms appear to be broadly similar to those described above for anorexia nervosa. There are also models of bingeing, which conceptualize it as ‘food addiction’ (Smith and Robbins, 2013), or note its similarities to impulsive/compulsive behaviours (Pearson et al., 2015), but these remain speculative.

Treatment of bulimia nervosa

There has been more research into the treatment of bulimia nervosa than into that of anorexia nervosa, and more effective psychological and pharmacological treatments have been identified. For some people, guided self-help may be sufficient, but in most cases formal treatment is indicated. For a review, see Svaldi et al. (2019).

Psychotherapy

Both CBT and interpersonal therapy are effective in bulimia nervosa. Of those who complete treatment (about 20% drop out), 60% will have stopped binge eating, and there is an 80% reduction overall. Psychological aspects improve in parallel. Early response is a strong predictor of outcome. The effect is maintained, with low relapse rates seen over 12 months.

The most striking evidence comes from a specifically tailored CBT-based approach (‘CBT-ED’), mentioned above in the description of anorexia nervosa treatment. In a relatively large study, CBT-ED reduced or abolished core behavioural and psychological symptoms of eating disorders, and showed sustained efficacy (Fairburn et al., 2009). The treatment was delivered in two forms—a ‘simple’ intervention which focused solely on the eating disorder, and a ‘complex’ intervention in which personality, mood, and interpersonal issues were also addressed. The authors suggested that the simple treatment is the

default version and widely applicable to bulimia nervosa (and other eating disorders), whereas the latter could be limited to individuals with additional psychopathology. Subsequent studies confirmed the efficacy of CBT-ED and showed its superiority over psychoanalytic psychotherapy and interpersonal psychotherapy. CBT-ED is the recommended treatment for bulimia nervosa in adults (National Institute for Health and Care Excellence, 2017); as with anorexia nervosa, family-based treatment may have some advantages over individual therapy in adolescents with bulimia nervosa, and is recommended in the NICE guidelines.

Medication

Antidepressants are effective, producing a reduction of about 50% in the frequency of binge eating, and cessation in 20% of cases. The onset of therapeutic effect is more rapid than in depression, but a higher dose may be needed (e.g. fluoxetine up to 60 mg daily). However, long-term data are less encouraging and show poor compliance. Antidepressants should be viewed as second-line treatment, used if an effective psychological treatment is unavailable or unsuccessful. Topiramate is effective in suppressing binge eating, but due to side effects its use in bulimia nervosa is not advocated.

Management

The management of bulimia nervosa is less intensive, and in some respects easier, than that of anorexia nervosa because the patient is more likely to wish to recover, and a good working relationship can often be established. Furthermore, there is no need to manage the complications of starvation nor to restore weight. However, it is necessary to assess the patient’s physical state and to measure electrolyte status in those who are vomiting frequently or misusing laxatives (Box 13.7).

As with many common disorders, a ‘stepped-care’ approach appears to be the best way of providing appropriate care for large numbers of people with varying degrees of severity of disorder.

Step 1. Identify the small minority (less than 5%) of individuals who need urgent specialist care because of severe depression, physical complications, or substance abuse that requires treatment in its own right.

Step 2. Offer guided cognitive behavioural self-help, using a self-help book or online programme, and with the guidance of a non-specialist facilitator. Treatment usually takes about four months and requires eight to ten meetings with the facilitator. Guided self-help is appropriate for primary care, and appears to lead to good progress in about a third of patients.

Step 3. Patients who do not show benefit within four to six weeks of commencing Step 2 require CBT-ED. In a minority of cases, where concomitant depressive symptoms are severe or persistent, it is worthwhile adding an antidepressant drug such as fluoxetine in doses of up to 60 mg daily.

Step 4. Patients who do not improve with CBT-ED require comprehensive specialist reassessment. In some cases, measures to provide more intensive cognitive therapy or an antidepressant drug may be useful. It is important to review the initial treatment with the patient, with the aim of agreeing a treatment approach that the patient finds acceptable.

Binge eating disorder

Binge eating disorder is characterized by recurrent binge-eating episodes in the absence of the other diagnostic features of bulimia nervosa; in particular, there are no compensatory behaviours such as vomiting or purging. In ICD-10 such patients would have been classified as eating disorder, unspecified. Patients may have depressive symptoms and some dissatisfaction with their body weight and shape; however, this is usually less severe than in bulimia nervosa. Nevertheless, binge eating disorder is of comparable severity to bulimia nervosa in terms of personal and public health and increased mortality. The prevalence is estimated at 0.6–1.8% in women and 0.3–1.7% in men, with obesity and other health comorbidities being common. The condition often goes undetected and untreated.

The risk factors for, and mechanisms underlying, binge eating disorder are thought to be similar to those for bulimia nervosa. About 25% of patients who present for treatment for obesity have features of binge eating disorder. The condition generally affects an older age group than bulimia nervosa, and up to 25% of those who present for treatment are men.

Binge eating disorder has a relatively high spontaneous remission rate and larger effect sizes than other eating disorders for response to CBT-ED, interpersonal therapy, and antidepressants. Lisdexamfetamine is also effective, and licensed for binge eating disorder in the United States.

For review of binge eating disorder, see Giel et al. (2022).

Other eating disorders

Other specified feeding or eating disorder

As noted earlier, many patients with eating disorders do not meet criteria for anorexia nervosa or bulimia

nervosa. While many are now accounted for in the category of binge eating disorder, others are not. The category of ‘other specified feeding or eating disorder’ largely comprises atypical or mixed forms of the three aforementioned disorders.

Avoidant/restrictive food intake disorder (ARFID)

ARFID occurs in young children, and occasionally in adolescents and adults, and refers to an eating behaviour leading to insufficient dietary intake to the point where there is nutritional deficiency, failure to thrive, and impaired psychosocial functioning. The child may lack interest in food, or be worried about its sensory characteristics, or its adverse consequences. However, there is no disturbance of body image or preoccupation with weight. There is an overlap with autistic spectrum disorder. There are no established treatments.

Rumination–regurgitation disorder

This disorder is characterized by the repeated and intentional regurgitation of swallowed food back into the mouth, after which it is reswallowed (ruminated) or spat out. It occurs repeatedly and is not accounted for by a medical condition such as an oesophageal stricture or pyloric stenosis. There are no established treatments, but behavioural techniques are usually applied (McElroy et al., 2020). For review see Murray et al. (2019).

Pica

Pica is the repeated eating of non-nutritional and ‘non-food’ substances, after the age (about two years) at which the distinction from food is usually made (Leung and Hon, 2019). A wide range of substances may be ingested, such as hair, paper, stones, or raw food ingredients (e.g. salt, flour). Pica is particularly common in children and adults with intellectual disability, and is also seen in several other psychiatric disorders, including autism and schizophrenia. It is also associated with pregnancy and iron deficiency. In each of these situations, a diagnosis of pica should only be made if the behaviour is sufficiently persistent or severe to require clinical attention.

Pica can lead to medical or surgical emergencies due to poisoning from ingested substances (e.g. lead), obstruction, nutritional deficiencies, anaemia, or parasitosis. The differential diagnosis includes anorexia nervosa, factitious disorder, and personality disorder. Behavioural approaches are usually used; SSRIs or atypical antipsychotics are sometimes tried in refractory cases.

Sleep–wake disorders

There is increasing recognition of the prevalence and clinical significance of sleep problems, and of circadian rhythm disruptions more broadly, in psychiatric (Walker et al., 2020) and medical (Logan and McClung, 2019) disorders. The impact of poor sleep on brain health is also now widely accepted (Gottesman et al., 2024). Reflecting these issues, there has been a corresponding growth of the specialty of sleep medicine. However, psychiatrists are sometimes asked to see patients whose main problem is either difficulty in sleeping (*dyssomnia*, which includes insomnia and hypersomnia) or behaviours associated with sleep (*parasomnias*). Many other patients have disruptions to their sleep–wake cycle as one of their symptoms, and the psychiatrist needs to be able to distinguish these from primary sleep disorders.

Sleep–wake problems are important for several reasons.

- They may represent primary disorders (the focus of this section).
- They may be symptoms of psychiatric disorder, especially mood and neurodegenerative disorders, or occur in a range of medical disorders, or be a side effect of treatment.
- They may be causes of psychological symptoms and contribute to the onset of several psychiatric disorders (Freeman et al., 2020).
- Better sleep can improve mental health (Freeman et al., 2017).
- Persistent sleep–wake difficulties are significantly associated with a range of adverse physical health outcomes.

For a review of sleep and circadian disorders, see Meyer et al. (2022). For a review of the biology of sleep, see Krause et al. (2017).

Classification

Table 13.2 lists the main categories of the ICD-11 and DSM-5 classifications of sleep–wake disorders. These disorders are grouped together in ICD-11 in chapter 7, and are thus separate from the chapter wherein psychiatric disorders are classified in chapter 6.

A more complex classification of sleep disorders intended for use by specialists in sleep medicine is the third edition of the International Classification of Sleep Disorders (ICSD 3). It has 80 types of sleep disorder.

Epidemiology

Sleep problems and sleep–wake disorders are frequent, but there is a wide variation in estimates, depending on the definition and the population studied. For example, one in three people report insomnia symptoms in the past year, and the point prevalence of an insomnia diagnosis is 6–15% (Krystal et al., 2019). Excessive sleepiness is reported in 5% of adults, and about 15% have a chronic sleep–wake disorder of some kind. Risk factors for specific sleep–wake disorders are mentioned in the relevant section below.

Assessment

Assessment requires a full psychiatric and medical history, together with detailed enquiries about the sleep complaint (see Box 13.8). In some cases specialist investigation, including polysomnography, is necessary. For a review, see Anderson (2020).

Insomnia

Insomnia is a condition of unsatisfactory quality and/or quantity of sleep, with difficulty falling asleep, remaining asleep, or waking early. For diagnosis of an insomnia disorder, there must also be associated daytime impairment such as fatigue, irritability, or problems in functioning. DSM-5 requires the insomnia symptoms to have been present at least three nights a week for three months. This is also required for the ICD-11 diagnosis of chronic insomnia; ICD-11 also has a category of short-term insomnia when a similar degree of disturbance has been present for less than three months.

Transient insomnia occurs at times of stress or as ‘jet lag’. Short-term insomnia is often associated with personal problems—for example, illness, bereavement, relationship difficulties, or stress at work. Sleep-onset insomnia can be a relatively common complaint among adolescents. Insomnia in clinical practice is usually chronic, and secondary to other disorders, notably painful physical conditions, depressive disorders, and anxiety disorders; it also occurs with excessive use of alcohol or caffeine, and in dementia. It can be provoked by prescribed drugs. In about 15% of cases of insomnia, no cause is found.

For a review of insomnia and its management, see Perlis et al. (2022) and Morin and Buysse (2024).

Table 13.2 Classification of sleep–wake disorders

ICD-11 Sleep–wake disorders	DSM-5 Sleep–wake disorders
Insomnia disorders	Insomnia disorders
Hypersomnolence disorders	Hypersomnolence disorders
Narcolepsy	Narcolepsy
Kleine–Levin syndrome	
Sleep-related breathing disorders	Breathing-related sleep disorders
Circadian rhythm sleep–wake disorders	Circadian rhythm sleep–wake disorders
Parasomnia disorders	Parasomnias
Other specified sleep–wake disorders	Other specified sleep–wake disorders
Sleep–wake disorders, unspecified	Sleep–wake disorders, unspecified
Sleep-related movement disorders	

Source: *International Classification of Diseases, Eleventh Revision (ICD-11)*. Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)*. Copyright (2013), American Psychiatric Association.

Box 13.8 Assessment of sleep disturbance

Screening questions

Do you sleep well enough and long enough?
Are you very sleepy during the day? Do you nap in the daytime?
Is your sleep disturbed at night?

Sleep history

Detailed history of the sleep complaint and a typical sleep–wake cycle
Factors that improve or worsen sleep
Effect on mood and functioning
Past and present treatment
History from bed partner

Sleep diary

Systematic two-week or longer record

Possible investigations

Video recording
Actigraphy (wrist-worn)
Polysomnography
HLA typing (if narcolepsy suspected)
CSF orexin (hypocretin) levels (if narcolepsy suspected)

Assessment of insomnia

Usually the diagnosis of insomnia is made on the basis of the account given by the patient, or parent/carer in the case of children. Assessment should focus on the nature of the sleep problem and its daytime consequences, as well as screening for psychiatric and other medical disorders and use of alcohol and drugs. Sleep studies are rarely undertaken.

Treatment of insomnia

Both non-pharmacological and pharmacological approaches can be used to treat insomnia. Hypnotic drugs were widely prescribed in the past, but non-pharmacological approaches are now recommended as first-line treatment. These include a range of cognitive and behavioural components (Box 13.9), including *sleep hygiene*, which refers to a series of steps that are useful for all patients with insomnia and can be applied in primary care (Box 13.10). For more serious or persistent insomnia, a range of additional specific non-pharmacological treatments have been shown to be effective. For example, *stimulus control* focuses on the principle that one only goes to bed when sleepy, and, if not asleep within 20 minutes, one should get up and engage in a relaxing activity before returning to bed. Adaptations of CBT for insomnia for delivery via the internet show promise (Freeman et al., 2017).

Box 13.9 Treatments for insomnia**Cognitive and behavioural interventions**

Sleep hygiene
 Sleep restriction
 Stimulus control
 Cognitive behaviour therapy
 Relaxation therapy

Pharmacological treatments

Short-acting benzodiazepines (e.g. temazepam)
 ‘Z drugs’ (zolpidem, zopiclone)
 Dual orexin antagonists (e.g. daridorexant, lemborexant)
 Low-dose sedative antidepressants (e.g. trazodone, mirtazapine)
 Melatonin or melatonin agonists
 Histamine H1 antagonists

Medication has a limited role to play. Although it is sometimes justifiable and useful to give a hypnotic for short-term use up to about two weeks (e.g. in the early stages of SSRI treatment), demands for prolonged medication should be resisted. Apart from the risks of tolerance, dependence, and ‘hangover’ effects, withdrawal of hypnotics can lead to insomnia that is as distressing as the original sleep disturbance. Melatonin is widely used in some countries, where it is available without prescription. Low doses of sedative antidepressants, such as trazodone or mirtazapine, are an alternative pharmacological treatment. Daridorexant is an orexin antagonist, the first of a new class of drug for the treatment of insomnia; their place in clinical practice is not yet established.

For evidence-based guidelines on the treatment of insomnia and other sleep disorders, see Wilson et al. (2019). For a systematic review of pharmacological treatments, see De Crescenzo et al. (2022).

Hypersomnolence disorders and excessive daytime sleepiness

Excessive daytime sleepiness is common (with a reported prevalence of 3–5%) and underdiagnosed. Many cases are secondary to loss of night-time sleep. Box 13.11 lists the principal causes. For review, see Gandhi et al. (2021).

Box 13.10 Principles of sleep education (sleep hygiene)**Sleep environment**

Familiar and comfortable
 Dark
 Quiet

Encourage

Bedtime routines
 Consistent time for going to bed and waking up
 Going to bed only when tired
 Thinking about problems *before* going to bed
 Regular exercise

Avoid

Late-evening exercise
 Caffeine-containing drinks late in the day
 Using mobile devices or watching TV in bed
 Excessive alcohol and smoking
 Excessive daytime sleep
 Large late meals
 Too much time in bed lying awake

Box 13.11 Causes of excessive daytime sleepiness

Insufficient night-time sleep
 Obstructive sleep apnoea
 Depression
 Narcolepsy
 Medical disorders (e.g. hypothyroidism, Prader–Willi syndrome)
 Shift-work sleep disorder
 Use of sedative medications

Narcolepsy

Narcolepsy is an important cause of chronic sleepiness, affecting about 1 in 2,500 people. It usually begins with the sudden onset of persistent daytime sleepiness between the ages of 10 and 20 years, although diagnosis is often delayed by several years. A useful diagnostic clue is that, unlike daytime sleepiness caused by insufficient sleep or that which is often observed in teenagers, people with

narcolepsy typically feel refreshed after a full night's sleep. In addition, narcolepsy is characterized by a disturbance in rapid eye movement (REM) sleep, which can occur at any time and lead to unusual states in between sleep and wakefulness (such as hypnopompic and hypnagogic hallucinations). The most striking of these REM-like states is *cataplexy*: sudden episodes of partial or complete paralysis of voluntary muscles, sometimes causing the person to fall to the ground. These features do not occur in all patients, with cataplexy being limited to type 1 narcolepsy (see below). It can be precipitated by laughter and other sudden or unexpected positive emotions. Additional clinical features of narcolepsy include a tendency to obesity (owing to a low metabolic rate), and depression is common.

For review of narcolepsy, see Bassetti et al. (2019).

Aetiology and pathology of narcolepsy

Considerable progress has been made in understanding narcolepsy, leading to the delineation of two types. Type 1 narcolepsy is characterized by cataplexy and by marked reduction of a peptide called orexin-A (also called hypocretin-1) in the cerebrospinal fluid (CSF). This occurs because of a severe and selective loss of the hypothalamic neurons that make this peptide, which is known to be involved in regulation of wakefulness. Type 2 narcolepsy is not associated with cataplexy, and CSF orexin-A levels are normal; its diagnosis is therefore more challenging, and its biological basis less well understood.

The main genetic association to narcolepsy is *HLA-DQB1*06:02*. More than 98% of patients with type 1 narcolepsy (and 50% of those with type 2 narcolepsy), have this genotype, compared to 15–30% of the general population. *HLA-DQB1*06:02* has been estimated to increase the risk of narcolepsy by a factor of 200. Other genes involved in immune function are also implicated in narcolepsy risk. A small percentage of cases are familial or secondary to another disorder.

There is increasing evidence that narcolepsy may be an autoimmune disorder, triggered by infections or other stimuli. For example, its onset is most common in late spring, and may follow streptococcal and other infections. Strikingly, there was an increase in cases of narcolepsy in children following the 2009–2010 H1N1 influenza pandemic, related at least in part to a specific influenza vaccine used at the time. The various genetic and environmental triggers are thought to lead to an immune response involving a T-cell population that damages the orexin-A producing cells.

Assessment of narcolepsy

Narcolepsy usually presents to neurologists or paediatricians. The differential diagnosis is from other causes of

excessive daytime sleepiness (Box 13.11). A full history, especially a sleep history, is the main assessment tool. The Epworth Sleepiness Scale is often used. In specialist settings the *multiple sleep latency test* (in which the latency to daytime naps, and the occurrence of REM sleep during them) is carried out. Measurement of HLA status and orexin-A levels in CSF is sometimes undertaken, but is not routine.

A psychiatrist may be involved in the care of a person with narcolepsy for several reasons:

- To assist in the diagnostic assessment.
- If cataplexy is triggered by strong emotions, as is often observed.
- If the patient has comorbid psychiatric disorder, or is distressed by the psychological and social consequences of their symptoms, especially cataplexy.
- To advise regarding the use of psychotropic medication for treatment of narcolepsy.

Treatment of narcolepsy

Patients need considerable help in adjusting to this disabling chronic illness. They should be encouraged to follow a regular routine, with good quality sleep at night and planned naps during the day. If stressful events or other factors (e.g. fatigue) appear to provoke cataplexy, efforts should be made to avoid them or to arrange the patient's lifestyle so as to minimize their impact. Particular caution should be taken with driving and other activities where sudden drowsiness or cataplexy may be dangerous.

Most patients require treatment with stimulant drugs. For mild or moderate cases, the non-amphetamine stimulant modafinil reduces daytime sleepiness and increases alertness. Amphetamines are more potent but have more side effects and risk of abuse. Venlafaxine, clomipramine, and other antidepressants can be used to decrease cataplexy. Night-time sleep can be improved by sodium oxybate (the sodium salt of GABA, also known as GBH), with beneficial effects on daytime functioning, but its side effects and potential for abuse restrict its use to severe cases. Pitolisant is a recently introduced H3 histamine receptor antagonist/inverse agonist which is effective in treating sleepiness and cataplexy and with fewer side effects than sodium oxybate.

For guidelines on the treatment of narcolepsy, see Bassetti et al. (2019).

Kleine–Levin syndrome

This very rare secondary sleep disorder consists of episodes of severe hypersomnia, increased appetite, apathy,

and hypersexuality, often lasting for days or weeks, with long intervals of normality between them. Cognitive impairment is also a common feature. The syndrome usually affects adolescent boys. The symptoms suggest a hypothalamic disorder, but its aetiology is unknown other than a possible contribution from *TRANK1* and *LMOD3* genes. There are no established diagnostic tests nor treatments, but lithium, stimulants, and a range of other drugs are sometimes used. For a review, see Qasrawi and BaHammam (2023).

Breathing-related sleep disorder

This syndrome consists of daytime drowsiness together with periodic respiration, recurrent apnoea, and excessive snoring at night. Psychiatrists usually encounter this syndrome in people referred because of tiredness and associated symptoms secondary to the sleep disorder, such as irritability and attentional deficits. It is usually associated with upper airways obstruction, hence the term *obstructive sleep apnoea*. The prevalence is about 4% in the male population. The typical patient is a middle-aged overweight man who snores loudly. Treatment consists of relieving the cause of the respiratory obstruction and encouraging weight loss. Continuous positive pressure ventilation using a face mask is often effective. Obstructive sleep apnoea is a risk factor for stroke. For a review, see Gottlieb and Punjabi (2020).

Some children with a history of problems with tonsils, adenoids, and/or ear infections can have frequent arousals during sleep, diminishing the overall quality of sleep from obstructed airways, leading to daytime difficulties with irritability and attentional problems.

Circadian rhythm sleep disorder (sleep–wake schedule disorders)

There are several forms of circadian rhythm sleep disorder, of which jet lag is the most well known. The others include disorders associated either with advanced or delayed sleep phases, and shift-work sleep disorder. The latter is a common and increasing problem, the consequences of which are widely underestimated. Fatigue and transient difficulties in sleeping accompany regular changes of shift, or the irregular alternating of night work and days off may lead to chronic problems of poor sleep, fatigue, impaired concentration, and an increased risk of accidents as well as adverse physical and social

effects. Melatonin may be useful for jet lag and some other circadian rhythm disorders, but should be initiated by specialists.

Puberty triggers a change towards an evening preference in a sizeable proportion of adolescents (up to 40%), exacerbated by a range of social and behavioural factors. The most prevalent sleep complaint therefore tends to be an inability to wake up for school, with significant impairment in a small subset of adolescents, and leading to public health calls for a later start time for the school day (Alfonsi et al., 2020).

For a review of circadian rhythm sleep disorders, see Meyer et al. (2022).

Sleep-related movement disorders

Restless legs syndrome

This syndrome is a distressing and painful condition that can result in severe insomnia and periodic limb movements during sleep. It is common, with 2.5% of the population having significant symptoms. Risk factors include female gender, pregnancy, ageing, low iron status, and parkinsonism. The syndrome can be mistaken for a psychiatric symptom or disorder (e.g. anxiety, akathisia), and it can be comorbid, especially with depression. For a review, see Manconi et al. (2021).

Parasomnias

Parasomnias are abnormal behaviours or physiological events occurring in association with sleep or sleep–wake transitions. Some types of parasomnia are especially common in children whereas others occur mostly in older adults and may be associated with neurodegenerative disorders. Parasomnias are broadly divided into those associated with REM sleep and those occurring during non-REM sleep. For a review, see Khurana and Carvalho (2020).

Parasomnias related to rapid eye movement (REM) sleep

REM sleep behaviour disorder

REM sleep behaviour disorder is a parasomnia that should be considered when behavioural problems, particularly agitation or aggression, occur during the night.

It is thought to occur when the normal atonia of REM sleep is lost, so that dreams are acted out. It is more common in the elderly, particularly men. It is often associated with, and can precede the onset of, neurodegenerative disorders, particularly Parkinson's disease and Lewy body dementia (Galbiati et al., 2019). Clonazepam and donepezil may be effective.

Nightmares (dream anxiety disorder)

A nightmare is an awakening from REM sleep to full consciousness with detailed dream recall. Children often experience nightmares, with a peak frequency around the age of 5 or 6 years. Nightmares may be stimulated by frightening experiences during the day, and frequent nightmares usually occur during a period of anxiety. Other causes include post-traumatic stress disorder, fever, psychotropic drugs, side effects of other drugs (e.g. malarone), and alcohol detoxification.

Sleep paralysis

Sleep paralysis is an inability to perform voluntary movements during the transitions between REM sleep and wakefulness, either at sleep onset (hypnagogic) or during awakening (hypnopompic). The episodes are often accompanied by extreme fear (Sharpless and Kilkova, 2019).

Disorders of arousal from non-REM sleep

Night terror disorder

Night terrors are much less common than nightmares. They are sometimes familial. The condition begins and usually ends in childhood, but occasionally persists into

adult life. A few hours after going to sleep, the person, while in stage 3–4 non-REM sleep, sits up and appears terrified. They may scream, and they usually appear confused. There are marked increases in heart rate and respiratory rate.

After a few minutes they slowly settle and return to normal calm sleep. There is little or no dream recall. A regular bedtime routine and improved sleep hygiene have been shown to be helpful. Benzodiazepines and imipramine have been shown to be effective in preventing night terrors, but their prolonged use should be avoided.

Sleepwalking disorder

Sleepwalking is an automatism that occurs during deep non-REM sleep, usually in the early part of the night. It is most common between the ages of 5 and 12 years, and 15% of children in this age group walk in their sleep at least once. Occasionally, the disorder persists into adult life. Sleepwalking may be familial. Most children do not actually walk, but sit up and make repetitive movements; those that do walk around usually do so with their eyes open. They do not respond to questions, and are difficult to wake. They can usually be led back to bed. Most episodes last for a few seconds or minutes.

As sleepwalkers can occasionally harm themselves, they need to be protected from injury. Doors and windows should be locked and dangerous objects removed. Adults with severe problems should be given advice about safety, avoidance of sleep deprivation, and any other circumstances that might make them excessively sleepy (e.g. drinking alcohol before going to bed).

Disorders of sexual function and paraphilic disorders

In the past, as reflected in earlier editions of this book, sexual matters were prominent in psychiatry. In contemporary psychiatry, there has been a marked shift in the psychiatric relevance of, and a lesser involvement in, issues related to sexual function, preference, and gender identity. Psychiatry's residual involvement in this area is now primarily in specialized clinics, wherein the psychiatrist's main role is to identify and manage any comorbid mental illness or distress that has arisen. Much less emphasis is placed on the 'psychiatric' basis of sexual disorder itself. Nevertheless, a

brief sexual history continues to be an integral part of assessment (Chapter 3), and knowledge of how psychiatric and medical disorders and their treatments can impair sexual function remains important. The psychiatric aspects of childhood sexual abuse are discussed in Chapter 16, and forensic aspects of sexual offences covered in Chapter 18.

These changes have come about for several reasons:

- Sexual factors were prominent in psychodynamic theories of causation and, perhaps related to this historical

view, more attention was paid to the patient's sexual history, attitudes, and behaviour than is now usual.

- Disorders of sexual function and preference were often attributed to mental illness and therefore in the domain of psychiatry. Reflecting this, sex therapy was more widely practised by psychiatrists.
- *Gender identity disorder* was a psychiatric diagnostic category in ICD-10. Views about gender identity are changing rapidly, and in ICD-11 the neutral term *gender incongruence* sits in a separate grouping of all conditions related to sexual health. In DSM-5, the term employed is *gender dysphoria*, with the requirement that there is significant associated distress or functional impairment.

This chapter reflects these developments, while recognizing that current classifications use categories and concepts that retain elements of the earlier views, and that there can be markedly different beliefs between individuals, countries, and cultures. This particularly applies to issues related to gender identity. Hence, we discuss the main features of the three categories of conditions related to sex and gender:

- *Sexual dysfunction* denotes impaired or unsatisfying sexual enjoyment or performance. Such conditions are common. They are subdivided, according to the stage of the sexual response that is mainly affected, into disorders of sexual desire, disorders of sexual arousal, and disorders of orgasm. There are also categories for conditions associated with painful sex, notably dyspareunia and vaginismus.
- *Paraphilic disorders (paraphilias)* are abnormalities of sexual preference. They are not commonly encountered in clinical practice but have forensic implications.
- *Gender incongruence* (gender dysphoria).

For a review of sexual dysfunctions and paraphilias, see Graham and Bancroft (2020). For a review of gender dysphoria, see Elaut and Heylens (2020).

Classification

As shown in simplified form in Table 13.3, ICD-11 and DSM-5 take similar approaches to classification in this area. Both require that the problem causes significant distress or functional impairment, and that it has been present for several months. Each disorder can be further described in terms of severity, duration, and whether it is generalized or situational. In ICD-11, suspected aetiological factors can be coded, such as illnesses and drugs, as well as cultural and relationship issues.

Note that in ICD-11, only paraphilic disorders are considered to be psychiatric disorders; sexual dysfunction and gender incongruence are part of a separate category of sexual health conditions.

Sexual dysfunction

In men, sexual dysfunction refers to repeated impairment of normal sexual performance, and less often to impairment of sexual interest and pleasure. In women it more often refers to a repeated unsatisfactory quality to the experience, or to impaired desire or libido. Note also that there is no agreed definition of 'dysfunction' in this context. What is regarded as 'normal' (e.g. in terms of the frequency or duration of sexual activity or interest in it) depends in part on the expectations of the individual(s) concerned.

As explained above, problems of sexual dysfunction are classified into those that affect:

- sexual desire and sexual enjoyment
- the genital response (erectile dysfunction [impotence] in men, and lack of arousal in women)
- orgasm (premature or delayed ejaculation in men, and orgasmic dysfunction in women).

To this list can be added problems that result in pain during sex (vaginismus and dyspareunia in women, and painful ejaculation in men).

It is important to remember that sexual dysfunction may not be disclosed directly, but only revealed during enquiries about another complaint, such as depression or poor sleep, or gynaecological symptoms.

For a comprehensive review of assessment and treatment of sexual dysfunction, see Wincze and Weisberg (2015).

Prevalence of sexual dysfunction

The prevalence of sexual dysfunction is difficult to ascertain since it depends upon which questions are asked, the population surveyed, and the accuracy and completeness of responses. However, it is clear that it is common in both sexes. For example, a systematic review of 135 studies of female sexual dysfunction showed considerable heterogeneity, with an overall estimate in premenopausal women of 41% (McCool et al., 2016). In men, erectile dysfunction affects 20–40% of those aged 60–69, and more than 50% in those over 70 (Shamloul and Ghinem, 2013).

In a UK survey, a sample of over 10,000 people aged 16–74 years were asked about sexual function (Mitchell et al., 2013b). Over 40% of men and 50% of women

Table 13.3 Classification of sexual disorders and gender incongruence/dysphoria

ICD-11*	DSM-5
Sexual dysfunctions	Sexual dysfunctions
Hypoactive sexual desire dysfunction	Sexual interest and arousal disorders
Sexual arousal dysfunctions	Erectile disorder
Male erectile dysfunction	
Orgasmic dysfunction	Female orgasmic disorder
Ejaculatory dysfunction	Premature ejaculation and delayed ejaculation
Sexual pain disorders	Genito-pelvic pain/penetration disorder
Aetiological considerations (e.g. illnesses, drugs, cultural factors)	Substance/medication-induced sexual dysfunction
Other specified sexual dysfunctions	Other specified sexual dysfunction
Paraphilic disorders	Paraphilic disorders
Exhibitionist disorder	Exhibitionist disorder
Voyeuristic disorder	Voyeuristic disorder
Frotteuristic disorder	Frotteuristic disorder
Coercive sexual sadism disorder	Sexual sadism disorder
	Sexual masochism disorder
Pedophilic disorder	Pedophilic disorder
	Fetishistic disorder
	Transvestic disorder
Other paraphilic disorders	Other paraphilic disorders
Gender incongruence	Gender dysphoria
In childhood	In childhood
In adolescence and adulthood	In adolescence and adulthood

*Sexual dysfunction and gender incongruence are in ICD-11 Chapter 17 (Conditions related to sexual health). Paraphilic disorders are in ICD-11 Chapter 6 (Mental, behavioural, or neurodevelopmental disorders) along with other psychiatric categories.

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (DSM-5). Copyright (2013), American Psychiatric Association.

reported a problem with sexual response in the past year, but distress about sex lives was much less common (10% and 11%, respectively). Sexual dysfunction increased with age and was associated with depression and poor physical health. The percentages of adults reporting different types of sexual dysfunction lasting at least three months are shown in Table 13.4.

Causes of sexual dysfunction

The aetiology of most sexual dysfunction is multiply determined, often with physical, psychological, and relationship factors (Graham and Bancroft, 2020). For

a review of female sexual dysfunction, see McCool et al. (2016), and for a review of male dysfunction, see Shamloul and Ghanem (2013).

Biological/medical causes

Sexual dysfunction sometimes arises secondary to physical illness (see Box 13.12). Of the diseases that have a direct effect on sexual performance, *diabetes mellitus* is particularly important, due both to autonomic neuropathy and vascular disease. Hormonal changes that accompany the menopause in women can produce vaginal dryness and dyspareunia. Other important organic

Table 13.4 Reported frequency of sexual dysfunction lasting three months or more among people aged 16–74 years

	Men	Women
Lacked interest in sex	15%	34%
Lacked enjoyment in sex	5%	12%
Felt physical pain due to sex	2%	7%
Difficulty reaching climax	9%	16%
Reaching climax too early	15%	2%
Erectile problems	13%	-
Uncomfortably dry vagina	-	13%
At least one of the above	42%	51%
At least two of the above	14%	22%

Source: Mitchell, K. R., Mercer, C. H., Ploubidis, G. B., Jones, K. G., Datta, J., Field, N., Copas, A. J., Tanton, C., Erens, B., Sonnenberg, P., Clifton, S., Macdowall, W., Phelps, A., Johnson, A. M., & Wellings, K. (2013b). Sexual function in Britain: Findings from the third National Survey of Sexual Attitudes and Lifestyles (Natsal-3). *Lancet*, 382(9907), 1817–1829. Copyright (2013), Elsevier.

causes of sexual dysfunction include neurological disorders and a range of general medical conditions (Basson and Schulz, 2007).

Several commonly used classes of drugs—psychiatric, medical, and recreational—have side effects that involve sexual function (see Box 13.13). The psychotropic medications most likely to negatively affect sexual functioning are SSRIs, antipsychotics (especially those causing prolactin elevation), and mood stabilizers or anticonvulsants that lower bioavailable testosterone (Clayton et al., 2016). Intoxication from, or the excessive use of, alcohol and illicit drugs can also impair sexual performance.

Psychological causes

Psychological contributions to dysfunction include negative body image and performance anxiety (fear of negative evaluation, hypervigilance, or rejection). Many psychiatric disorders may impair sexual function, as discussed in other chapters. There is a strong relationship with depression, in which rates of sexual dysfunction are estimated to be up to 50%, compounded by treatment-emergent sexual dysfunction with rates of 37–62% (Reichenpfader et al., 2014).

Sociocultural influences

Causes of sexual dysfunction can also arise from a person's social context. For example, communication and relationship inequalities can foster sexual dysfunction, and larger sociocultural influences such as sex role or religious proscriptions may also have an impact.

Box 13.12 Medical conditions and surgical procedures commonly associated with sexual dysfunction

Medical conditions

Endocrine

- Diabetes, hyperthyroidism
- Addison's disease, hyperprolactinaemia

Gynaecological

Endometriosis, pelvic infections

Cardiovascular

Hypertension, myocardial infarction, peripheral vascular disease

Respiratory

Asthma, obstructive airways disease

Other

- Prostate cancer
- Arthritis
- Renal failure
- Pelvic autonomic neuropathy, spinal cord lesions, stroke

Surgical procedures

- Mastectomy
- Colostomy, ileostomy
- Prostate surgery
- Prolapse surgery

Furthermore, environmental factors, such as a lack of privacy and disparate work schedules, can be significant contributors to sexual dysfunction.

Assessment of patients with sexual dysfunction

History-taking

The interviewer needs to be particularly sensitive when enquiring about sexual function and dysfunction, and when detecting and dealing with embarrassment experienced by the patient. Whenever possible, the patient and their sexual partner should be interviewed, both separately and together. The major areas of enquiry are listed in Box 13.14. In addition to full characterization of the dysfunction itself, the assessment should search for evidence of any of the medical and psychiatric disorders noted earlier that can lead to sexual dysfunction.

Box 13.13 Drugs that may impair sexual function**Therapeutic agents****Diuretics and antihypertensive agents**

β -blockers, calcium channel blockers, spironolactone

Antidepressants and mood stabilizers

SSRIs, tricyclic antidepressants, monoamine oxidase inhibitors, lithium

Anxiolytics and hypnotics

Benzodiazepines

Antipsychotics**Antihistamines and histamine H_2 -receptor antagonists**

- Diphenhydramine
- Ranitidine

Parkinson's disease medications**Misused substances**

Alcohol, heroin, amphetamine, MDMA, cocaine, marijuana

Physical examination and special investigations

If the general practitioner or another specialist has not already done so, a physical examination should be performed, especially in men (see Box 13.15). Laboratory tests or other investigations should be arranged in appropriate cases—for example, fasting blood sugar, testosterone, and other hormones in men with erectile dysfunction.

Treatment of sexual dysfunction

The main treatments for sexual dysfunction are shown in Box 13.16. The first step, after a detailed assessment, is to provide advice and reassurance, since the problem is often longstanding by the time the patient presents, and it may have led to adverse secondary effects—for example, by increasing anxiety or impairing the quality of the relationship generally. Where possible, the underlying causes should be treated.

Specific interventions include a range of psychological and behavioural treatments, including *sex therapy/sensate focus techniques*. Psychological interventions are an effective treatment options for sexual dysfunction,

Box 13.14 Assessment of sexual dysfunction**Define the problem (ask both partners)**

Origin and course
Why seek help now?
Prior (baseline) sexual function
Sexual drive
Knowledge and fears
Relationship between the partners
Psychiatric disorder
Substance misuse
Medical illness; medical or surgical treatment
Physical examination
Laboratory tests

Box 13.15 Physical examination of men presenting with sexual dysfunction**General examination (directed especially to evidence of diabetes mellitus, thyroid disorder, and adrenal disorder)**

Hair distribution
Gynaecomastia
Blood pressure
Peripheral pulses
Ocular fundi
Reflexes
Peripheral sensation

Genital examination

Penis: congenital abnormalities, foreskin, pulses, tenderness, plaques, infection, urethral discharge
Testicles: size, symmetry, texture, sensation
Prostate

with the strongest evidence for female hypoactive sexual desire disorder and female orgasmic disorder. There are also physical treatments, of which drug therapy with phosphodiesterase type 5 (PDE-5) inhibitors such as sildenafil is by far the most widely used (Shamloul and Ghanem, 2013). The roles of the various treatment

Box 13.16 Treatments for sexual dysfunction

Advice, information, and reassurance

Treatment of underlying cause

Psychological methods

Sexual skills training
Sex therapy (including sensate focus exercises)
CBT
Marital therapy
Systematic desensitization
Educational intervention

Drug treatments

PDE-5 inhibitors

Other physical treatments

Vacuum devices
Dilators

modalities and the evidence for their efficacy vary according to the specific type of sexual dysfunction, as discussed below.

Sex therapy

As noted above, many patients benefit from simple advice and reassurance. Other cases resolve with treatment of the cause of the sexual dysfunction, or the use of medication. For some people, a formal type of behavioural therapy, known as *sex therapy*, may be appropriate. In the United Kingdom, its availability within the National Health Service is limited, but it is also provided by the charity Relate.

Sex therapy owes much to the work of Masters and Johnson (1970), and has four characteristic features.

1. The partners are treated together.
2. They are helped to communicate better about their sexual relationship.
3. They receive education about the anatomy and physiology of sexual intercourse.
4. They complete a series of graded tasks, often called *Sensate Focus*, which are a hierarchical series of structured touching opportunities with a focus as much on what must not yet be attempted as what is to be

done. This prohibition reduces ‘performance anxiety’, resulting in increased confidence and subsequent success through desensitization (Linschoten et al., 2016).

As with the majority of treatments for sexual dysfunction, there are few adequately controlled studies of sex therapy.

Types of sexual dysfunction

In this section we discuss the features and treatment of the more common categories of sexual dysfunction.

Lack or loss of sexual desire (hypoactive sexual desire disorder)

Complaints of diminished sexual desire are much more common among women than among men. There is controversy about the extent to which lack of sexual desire should be considered a disorder. Lack of desire can be categorized as either a lack or a loss, and as either global or situational.

Hypoactive sexual desire disorder can be treated effectively with psychological interventions—especially sexual skills training or CBT. Medication has a very limited role.

Female sexual arousal disorder

This category is combined with hypoactive sexual desire disorder in DSM-5. Lack of sexual arousal in the female is manifested as reduced vaginal lubrication. This may be due to inadequate foreplay, lack of sexual interest, anxiety, or low oestrogen levels (typically related to the menopause). For a review of female sexual dysfunction and its treatment, see Clayton and Juarez (2019).

Male erectile dysfunction (impotence)

This condition is the inability to achieve an erection or to sustain it for long enough for satisfactory coitus. It may be present from the first attempt at intercourse (primary male erectile disorder) or develop after a period of normal function (secondary male erectile disorder). It is much more common among older than younger men.

Primary cases may occur because of a combination of low sexual drive and anxiety about sexual performance. Secondary cases may arise from diminishing sexual drive in the middle-aged or elderly, loss of interest in the sexual partner, anxiety, depressive disorder, or organic disease and its treatment.

The following aspects of the assessment are particularly important.

- Has there been a previous period of normal function?

- Does erection occur during foreplay?
- Does erection occur on waking or in response to masturbation? Erection in these circumstances suggests psychological causes for the failure of erection at other times.
- Is there evidence of alcohol or drug abuse?
- Are there possible effects of any medication?

If the aetiology is uncertain after history-taking, physical examination, and blood tests, several special investigations may be considered, including measurements of tumescence and blood flow using ultrasound.

Any reversible causes should be treated. The first-line specific therapy is a PDE-5 inhibitor—sildenafil (Viagra), vardenafil, and tadalafil—which are effective in about 70% of cases. These inhibit the breakdown of cyclic GMP by a phosphodiesterase located in the vascular smooth muscle of the penis. They do not initiate arousal but enhance the effects of sexual stimulation. The most common side effects are headache, flushing, dyspepsia, and nasal congestion. PDE-5 inhibitors have supplanted earlier physical treatments for erectile dysfunction (i.e. intracavernosal injections and vacuum devices).

For a review of erectile dysfunction and its treatment, see Shamloul and Ghanem (2013).

Female orgasmic disorder

Many women do not regularly reach orgasm during intercourse, but do so in response to clitoral stimulation. Whether failure to reach orgasm regularly should be regarded as a disorder has been questioned. Nevertheless, some women regard it as a problem and ask for advice and help. The best evidence is for sexual skills training. For a review, see Marchand (2021).

Male orgasmic disorder

This term refers to serious delay in, or absence of, ejaculation. It is usually associated with general inhibition about sexual relationships, but it may be caused by drugs, including SSRIs, antipsychotics, and monoamine oxidase inhibitors.

Premature ejaculation

This term refers to habitual ejaculation either before penetration or so soon afterwards that the woman has not experienced pleasure. It is more common among younger men, especially during their first sexual relationships. When it persists, it is often caused by fear of failure.

Psychological causes should be treated initially with advice and reassurance. If the partner is willing, the

‘squeeze technique’ can be useful. When the man indicates that he will soon have an orgasm, his partner grips the penis for a few seconds and then releases it suddenly. Intercourse is then continued. There are also other methods designed to regulate and reduce the amount of sexual stimulation in order to delay orgasm, which can be implemented as part of sex therapy. Sometimes SSRIs, which delay ejaculation, are used in treatment, but continued use is required.

Genitopelvic pain/penetration disorder

Dyspareunia. This refers to pain on intercourse, and implicitly relates to women, although it does occur in men. Such pain can have many different causes. Pain that is experienced after partial penetration may result from impaired lubrication of the vagina, from scars or other painful lesions, or from the muscle spasm of vaginismus. Pain on deep penetration strongly suggests pelvic pathology such as endometriosis, ovarian cysts or tumours, or pelvic infection, although it can be caused by impaired lubrication associated with low sexual arousal. Treatment depends on the aetiology. Hormone replacement therapy is widely used for pain related to vaginal dryness in menopausal and post-menopausal women (Weinberger et al., 2019).

Vaginismus. This refers to spasm of the vaginal muscles, which causes pain when intercourse is attempted, in the absence of an identifiable physical cause. The spasm is usually part of a phobic response to penetration, and may be made worse by an inexperienced partner. Spasms often begin as soon as an attempt to enter the vagina is made. Points of special relevance in the sexual history include the circumstances and objects that provoke spasm, the partner’s sexual technique, and any history of traumatic sexual experience. Treatment employs the general sex therapy techniques described above, with emphasis on an initial ban on attempts at intercourse. Fears are treated by using psychological approaches, and the woman is helped to desensitize herself gradually by inserting first her finger and then dilators of increasing size.

Paraphilic disorders

The paraphilias are abnormalities of sexual preference affecting either the *object* of the person’s sexual interest, or in the preferred sexual *act*. To be diagnosed as a paraphilic disorder, the abnormality needs to be persistent and intense. For sexual preferences involving another person, diagnosis is limited to cases in which the other party is unable to consent due to age or status. Paraphilias

involving solitary sexual acts are only diagnosed if they lead the person to experience marked distress.

The concept of abnormal sexual preference has a social aspect—that is, the behaviour does not conform to some generally accepted view of what is normal. The accepted view is not the same in every society or at every period in history. For centuries, abnormalities of sexual preference were regarded as offences against the laws of religion, rather than conditions that doctors should study and treat. The systematic investigation of these disorders began in the 1870s. Richard von Krafft-Ebing wrote an account of paraphilias in his book *Psychopathia Sexualis*. In England, Havelock Ellis (1859–1939) wrote extensively about these and other sexual disorders, which at the time included masturbation.

The currently recognized subtypes of paraphilic disorders are shown in Table 13.3. All are much more common in men than in women. Some carry major legal implications, and paraphilic disorders are primarily in the domain of forensic psychiatrists. However, any doctor may encounter abnormalities of sexual preference in their clinical work—for example, as a request for help, or an expression of concern by a third party. Controversy over the category of paraphilic disorders continues, and at present focuses on whether sexual preferences of the sexual act, such as fetishism or transvestism, should be considered a disorder. They are included in DSM-5 but were removed from ICD-11 as part of a broader change in how sexual and gender-related disorders are conceptualized (Reed et al., 2019).

For a review of paraphilic disorder, see Beech et al. (2016).

Aetiology of paraphilic disorders

The aetiology of paraphilias remains unknown. Although a range of theories have been proposed, none are supported by good evidence.

- *Behavioural models.* According to these models, sexual preference is shaped by events and reinforcements during development. For example, that fetishism arises from the chance co-occurrence of sexual excitement with the object that later becomes the fetish object.
- *Psychoanalytical models.* For example, Freud argued that fetishism arises when castration anxiety is not resolved in childhood, with each fetish object representing a phallus.

- *Biological models.* These include reports linking paraphilias with abnormal brain activity or structure, or with genetic predisposition.
- *Disease models.* Paraphilias that begin in middle age or later may be secondary to dementia or other organic disorders, or to their treatment. For example, high-dose dopamine agonists that are used to treat Parkinson's disease are associated with a range of abnormal sexual behaviours.

Assessment of abnormalities of sexual preference

Assessment often takes place in the context of possible or actual legal proceedings. It is important therefore to explain the relationship of the interview to any such proceedings, to explain the position with regard to confidentiality and its limits, and to obtain the necessary consent. People who are seen for the assessment of abnormality of sexual preference are often reluctant to disclose the true nature of their desires, or of their sexual behaviour. Whenever possible, consent should be obtained for an interview with any regular sexual partner, and with other informants who may be able to assist with, for example, the assessment of personality.

Assessment includes a psychiatric history, and should incorporate the following steps:

- *Exclude another psychiatric disorder,* especially if the patient first presents with the abnormal sexual preference in middle age or later. Abnormal sexual preference is sometimes secondary to dementia, alcoholism, or mania.
- *Document the sexual behaviour and its effect.* Obtain details of the normal and abnormal sexual behaviour, both currently and in the past. Remember that it is not uncommon for people to have more than one form of abnormal sexual preference. Find out what part the abnormal sexual preference is playing in the patient's life other than as a source of sexual arousal.
- *Assess the patient's motivation.* This will influence decisions about management. Patients' motives for seeking treatment are often mixed. Individuals who have little wish to change their sexual behaviour may consult a doctor because of pressure from a sexual partner, a relative, or the police. Such people may hope to be told that no treatment can help, so as to justify the continuation of their paraphilia. Others seek help when they are depressed and feel guilty about their behaviour.
- *Psychophysiological assessment.* Some specialists use penile plethysmography or polygraphy to assess

sexual interests. These methods are not part of routine assessment.

For a review of assessment of paraphilias, see Seto et al. (2014).

Management of paraphilic disorders

Some aspects of management apply to all kinds of abnormality of sexual preference. Management specific to particular disorders is described below, along with the other aspects of those disorders. The following outline is suggested:

- Complete the assessment to make a provisional diagnosis.
- Assess the immediate risks—for example, whether the patient should be reported to the authorities, or whether their level of distress requires urgent treatment or admission. Often it is a crisis that first leads the person to present—for example, if they have been accused of a sexual offence or caught in a compromising sexual act.
- Decide what treatment is indicated and acceptable to the patient. This includes agreement about the aims of treatment, and the patient's motivation. It may also include incentives or punishments determined by the legal system.

Treatments for paraphilic disorders are of three types, although there is little evidence available:

1. Psychotherapy—individual, couple, or group based on behavioural or psychodynamic theories
2. Pharmacotherapy—gonadotropin-releasing hormone (GnRH) analogues, SSRIs, and synthetic steroid analogues
3. Rehabilitation—counselling, education, social skills training.

An algorithm to assist clinicians in determining appropriate biological treatments for individuals with paraphilic disorders has been developed, which provides guidance on treatments depending on severity of impairment and risk of harm (Thibaut et al., 2020). In some countries, treatment of paraphilia may also be mandated by legal policies (Turner et al., 2019).

Abnormalities of preference for the sexual object

These paraphilias involve preferences for an 'object' other than another adult for the achievement of sexual excitement. The alternative 'object' may be a child (paedophilic disorder), an animal (zoophilia), or an

inanimate object as in fetishistic disorder. The more common paraphilias are mentioned below, noting that fetishism and transvestism are not included in ICD-11 as mentioned earlier.

Paedophilic disorder

Paedophilic disorder has serious criminal as well as clinical implications. Paedophilia is repeated sexual activity (or fantasizing about such activity) with pre-pubertal children as a preferred or exclusive method of deriving sexual excitement. Like most paraphilias, it is much more prevalent in the male population, although female cases are recognized. Paedophilia has to be distinguished from intercourse with young people who have passed puberty but not yet reached the legal age of consent (which differs between countries).

Paedophiles usually choose a child between the age of 6 years and puberty, but some prefer very young children. Some paedophiles approach children within their extended family, or in their professional care, while others befriend unrelated children. Although most paedophiles who are seen by doctors are men of middle age, the condition is usually established early in life, and can also rarely affect women. With younger children, fondling or masturbation is more likely to occur than full coitus, but sometimes young children are injured by forcible attempts at penetration.

Estimates of the prevalence of paedophilic disorder range from 1% to 5% in the male population (Beech et al., 2016). It occurs in all ethnic and socioeconomic groups. As is evident from the existence of child prostitution in some countries, the ready sale of pornographic material depicting sex with children and the number of internet sites that show such material, interest in sexual relationships with children is not rare.

Management is discussed in the forensic chapter under child sexual abuse (Chapter 18). GnRH antagonists may be of use (Landgren et al., 2020).

Fetishistic disorder

In sexual fetishism, the preferred means of achieving sexual excitement are inanimate objects or parts of the human body that do not have direct sexual associations. The objects that evoke sexual arousal are many and varied, the most common being female items of clothing, but for each individual there is usually only a small number of such objects. Contact with the object causes sexual excitement, which may be followed by solitary masturbation or by sexual intercourse that incorporates the fetish. The disorder merges into normal sexual behaviour, but is considered to be abnormal when the

behaviour takes precedence over the usual patterns of sexual intercourse.

Transvestic disorder

Transvestic disorder ranges from the occasional wearing of a few articles of clothing of the opposite sex to complete cross-dressing. It occurs almost entirely in men. Onset usually occurs at around the time of puberty. Most are heterosexual and many have sexual partners, who may or may not know about the behaviour. There have been no reliable follow-up studies or evaluated treatments.

Abnormalities in the preference for the sexual act

The second group of abnormalities of sexual preference involves variations in the behaviour that is engaged in to obtain sexual arousal.

Voyeuristic disorder

Voyeurism refers to observing the sexual activity of others repeatedly as a preferred means of sexual arousal. The voyeur spies on individuals (directly, or using cameras) who are undressing or without clothes, but does not attempt to engage in sexual activity with them. Voyeurism is usually accompanied by or followed by masturbation. There is limited treatment efficacy reported with either pharmacological or psychoanalytical approaches.

Exhibitionistic disorder

Exhibitionistic behaviour—broadly defined as the exposing of the genitals to an unsuspecting stranger—is the most commonly reported of all sex offences (McNally and Fremouw, 2014). Exhibitionistic behaviour is estimated to occur at rates as high as 2–4% in the general population. It is almost entirely a male disorder, and adolescents account for approximately 10% of exhibitionistic offences. It is usually done for the purpose of achieving sexual excitement, without any attempts at further sexual activity with the person.

Over a five-year follow-up, 5–10% of exhibitionistic perpetrators were found to escalate to contact sexual offending. The strongest risk factor for escalation was a general clustering of antisocial behaviour, including a history of sexual and non-sexual convictions (McNally and Fremouw, 2014).

As a generalization, two groups of exhibitionists can be described. The first group includes men of inhibited temperament who struggle against their urges and feel much guilt after the act; they sometimes expose a flaccid penis. The second group includes men who have aggressive traits, sometimes accompanied by features of

antisocial personality disorder; they usually expose an erect penis. They gain pleasure from any distress they cause, and often feel little guilt. Few evidence-based treatments exist, but those that are most effective are along the lines of treatment for sex offenders and can include CBT (Beech et al., 2016).

Frotteuristic disorder

In *frotteurism*, the preferred form of sexual excitement is by rubbing the male genitalia against another person, or by fondling the breasts of an unwilling participant, who is usually a stranger, generally in a crowded place. Frotteuristic activity usually starts in adolescence, with most acts occurring between the ages of 15 and 25 years (Beech et al., 2016).

Gender dysphoria

Transgender is an umbrella term used to describe individuals who have a gender identity that does not align with their assigned gender, and *gender dysphoria* (formerly *gender identity disorder*) occurs when the individual experiences distress due to this discrepancy (Cooper et al., 2020). It is receiving significant attention, and referrals to gender-confirming clinics (formerly known as *sex reassignment clinics*) has risen dramatically, as has the prominence of the subject in the media (Zucker et al., 2015). The management of gender dysphoria should be multidisciplinary, as endocrine, surgical, and psychological input are all often needed. There is, however, a growing argument that gender dysphoria should not necessarily sit within a psychopathological or psychiatric model (Wylie et al., 2016). Such views are seen as outdated and not able to reflect the multitude of different experiences and terms used to describe individuals whose gender identity or gender role behaviour does not match up with societal expectations or stereotypes associated with the (biological) male–female binary (Zucker et al., 2015). Reflecting the changing views, ICD-11 has categories of *gender incongruence of adolescent and adulthood* and *gender incongruence of childhood*, located within the section on sexual health, not mental health. These replace ICD-10's *transsexualism* and *gender identity disorder of childhood*. DSM-5 adopts the term *gender dysphoria* to replace '*gender identity disorder*', placing emphasis on the emotional distress that can result when there is incongruence between one's experienced/expressed gender and one's assigned sex. An increasing number of other terms are also being used to describe aspects of gender and gender dysphoria (see Box 13.17, also Winter et al., 2016). Subtypes have been described that include sexual orientation—whether sexually attracted to males,

Box 13.17 Some terms used relevant to gender dysphoria

Transgender: an umbrella term for people whose gender (identity, expression, general sense of self) is different from the sex they were assigned at birth

Transsexual: someone who has transitioned from one sex to the other through the use of hormones and/or surgical procedures

Transgender woman/trans woman/MtF: someone who was assigned male at birth but living as a woman (male to female)

Transgender man/trans man/FtM: someone who was assigned female at birth but living as a man (female to male)

Sexual orientation: refers to an individual's enduring physical, romantic, and/or emotional attraction to another person. Transgender people may be straight, bisexual, lesbian, gay, or asexual

Gender fluid: refers to an individual who has a non-fixed gender identity that shifts over time or depending on the situation

Gender non-conforming: refers to an individual not conforming to the gender roles that society expects

Non-binary: someone who does not identify with either gender, or has a gender that changes over time, or feels in between genders

Cisgender: someone who identifies with the gender they were assigned at birth

females, both, or neither, as well as age of onset of the gender dysphoria, with early onset usually denoting the preschool period. The implications of these subtypes are unclear, although they have played a historical role in assessments for sex reassignment surgery.

For a review of gender dysphoria in adults, see Zucker et al. (2015), and in adolescents and children, see Skordis et al. (2020). For reviews of transgender people and mental health, see Wylie et al. (2016) and Winter et al. (2016).

Epidemiology

Prevalence rates are difficult to determine. Previous data were based on referrals to specialist centres and on people requesting a change to the gender they had been assigned at birth. This is unlikely to be a true

prevalence rate because of the stigma attached to individuals with gender dysphoria and the limited services available. However, the greater sociocultural acceptability of gender variability seems to be driving an increase in transgender people requesting health care, including for adolescent populations (Paz-Otero et al., 2021). Prevalence estimates also depend on inclusion criteria, age group, and geographic location, and range from a fraction of a per cent to several percentage points (Coleman et al., 2022). A significant increase in referral rates (with data primarily from North America) has also been reported for individuals seeking gender reassignment, in particular birth-assigned female adolescents.

Associated psychopathology

Mental health is a major issue for many individuals with gender dysphoria. A review of 37 studies examining psychiatric disorders in gender dysphoric and gender non-conforming people (Paz-Otero et al., 2021), mostly including adolescent and younger people (under 30 years) found the most frequent pathology being depression (prevalence of up to 50%), and anxiety (37%) with self-harm episodes also common (8%). There have been several recent studies exploring the prevalence of autism spectrum disorders in this population, which in the same review was estimated to be present in 14% (minors) to 36% (older people) (Paz-Otero et al., 2021).

Aetiology

The aetiology of gender dysphoria remains unclear, in part because the principles of 'normative' or 'sex typical' psychosexual development are poorly understood, but it is likely to be heterogeneous and require a complex model. A number of causal mechanisms have been proposed. These are predominantly based on social stresses of living as a gender that is not congruent with the person's preferences, or as a result of social stigma, and can come to the fore during the many biopsychosocial milestones of puberty. There may also be a contribution from genetic predisposition, and from prenatal sex hormones that impact sexual differentiation of the brain (Ristori et al., 2020). Very rare cases may be attributed to chromosomal abnormalities, congenital conditions (such as congenital adrenal hyperplasia), and intersex conditions (hermaphroditism).

Treatment

The treatment of adults with gender dysphoria has been standardized across many high-income countries, reflecting the influence of a number of clinical

guidelines, notably the Standards of Care for the Health of Transsexual, Transgender and Gender-Nonconforming People, Version 8 (SOC-8) (Coleman et al., 2022). The document contains key gender-affirming recommendations for assessment and medical interventions for healthcare professionals. It highlights the importance of being aware of the marked vulnerabilities of transgender and gender-diverse individuals, who experience high rates of stigmatization, ignorance, and with limited access to care which contributes to significant health disparities. A key change from previous guidelines is more guidance on the care of children, emphasizing a conservative approach: for adolescents to only begin hormone suppression after the patient first exhibits physical changes of puberty and, in the case of intersex children, to delay any gonadal surgery until a time when the child can participate in the decision. For adults, the potential harm of some progestogens was noted. The guidelines generally follow consensus (from experts and stakeholders) rather than evidence from clinical trials, which remains limited.

Current treatment recommendations highlight the importance of a patient-centred approach and include the following steps.

Counselling and psychotherapy should be undertaken to discuss the full range of options available, set realistic goals, and to consider the consequences of any contemplated changes, both for the person and for their family. The focus of therapeutic work is to assist the individual to be able to affirm their correct gender and not for this to be necessarily congruent with the sex they were assigned at birth. The agreed consensus is to make the focus of therapeutic involvement the ‘long-term comfort in ... gender identity expression, with realistic chances for success in ... relationships, education and work’ (Coleman et al., 2012).

Real-life experience and social transition in the preferred gender role. Almost all individuals with gender dysphoria will have to live full time in their identified gender role for a period of time (often one year) prior to surgery. During this time, hormone treatment (see below) is prescribed. An individual may then have to demonstrate to an assessment panel that they are better adjusted in the new gender role than in the old one to be considered for surgery. Despite the widespread requirement to live in the new gender role for a year, there is little evidence to support the practice, and it carries psychosocial risks, including loss of employment, impaired relationships, and gender-based

discrimination; thus it has been argued that this step is unnecessary unless the individual themselves would find it helpful (Zucker et al., 2015).

Hormone therapy. Hormone therapy is effective and thought to be reasonably safe in adults with gender dysphoria, although long-term follow-up data are limited. The treatments for males-to-females include testosterone suppression, spironolactone, cyproterone acetate, or gonadotropin-releasing hormone (GnRH) agonists. Androgens may be prescribed to females-to-males, as a result of which the voice deepens, hair increases on the face and body, menstruation ceases, the clitoris enlarges, and sex drive increases. Treatment, at present, is influenced by location and availability of care, with a relatively poor evidence base from which to draw and an absence of long-term studies, although studies of GnRH agonists have shown that treatment delays bone maturation and gain in bone mineral density (Ludvigsson et al., 2023).

Gender-confirming (sex reassignment) surgery. There is a spectrum of gender-affirming surgical procedures. Opinions differ about the merits of sex reassignment, and decisions should be taken with the person and members of the multidisciplinary team. For a male-to-female, possible surgical procedures include mastoplasty, penectomy, orchidectomy, and the creation of a vagina-like structure. For a female-to-male, surgical procedures include mastectomy, ovariectomy, and phalloplasty.

Follow-up studies show that gender-affirming surgery is generally safe and with low complication rates (Akhavan et al., 2021). Gender-affirmation surgery decreases rates of gender dysphoria, depression, and suicidality and improves quality-of-life measures.

Detransitioning

A small number of individuals with gender dysphoria who chose to undergo medical and/or surgical transition subsequently detransition, by discontinuing medications, having surgery to reverse the effects of transition, or both (Littman, 2021). The prevalence of detransition is unclear but might be around 1% (Bustos et al., 2021). Until more is understood about the longer-term outcomes of people with gender dysphoria, the medical and psychological needs of the detransitioning population are important to consider as they might experience additional medical, psychological, and social difficulties.

Further reading

- Fairburn, C. G. (2008). *Cognitive Behavior Therapy and Eating Disorders*. Guilford Press.
- Kryger, M. H., Roth, T., & Goldstein, C. A. (2021). *Principles and Practice of Sleep Medicine*, 7th edn. Elsevier.
- Wincze, J. P., & Weisberg, R. B. (2015). *Sexual Dysfunction: A Guide for Assessment and Treatment*, 3rd edn. Guilford Publications.

CHAPTER 14

Dementia, delirium, and other neuropsychiatric disorders

Introduction 335

Classification 336

Symptoms associated with regional brain pathology 338

Assessment of the 'neuropsychiatric patient' 340

Delirium 342

Amnesia and amnesic disorders 344

Dementia 347

Movement disorders 363

Cerebrovascular disorders 367

Head injury (traumatic brain injury) 368

Epilepsy 370

Brain tumours 375

Other neuropsychiatric syndromes 376

Secondary or symptomatic neuropsychiatric syndromes 379

Introduction

Neuropsychiatry encompasses psychiatric disorders considered to arise from demonstrable abnormalities of brain structure or function. Though not an entirely satisfactory definition, the term is widely used. *Dementia*, *delirium*, and *amnesic syndrome* are the three disorders at its heart, all of which are characterized by acquired cognitive deficits as their cardinal feature. This chapter covers the disorders listed in Box 14.1, but the management of dementia is covered in Chapter 19.

- *Organic psychiatry* is a related but broader term, since it also includes medical disorders with their basis outside the brain (such as endocrine and metabolic disorders); these are covered in Chapter 22. The term also has the implication that all other psychiatric disorders are 'functional', an unhelpful and false dichotomy (see Chapter 2).

Box 14.1 Neuropsychiatric disorders**Neurocognitive disorders**

Delirium

Amnesic syndrome

Dementia

Mild neurocognitive disorder (mild cognitive impairment)

Cerebrovascular disorders**Movement disorders**

Parkinson's disease

Tic disorders

Epilepsy**Head injury****Other neuropsychiatric disorders**

Cancer

Encephalitis

Multiple sclerosis

Neuropsychiatric systemic lupus erythematosus

Secondary or symptomatic neuropsychiatric syndromes**Classification**

ICD-11 and DSM-5 group dementia, delirium, and amnesic syndrome together under a new umbrella category of *neurocognitive disorders*, since they all have acquired deficits in cognitive functioning as their primary feature. Each of these categories is then divided, whenever possible, into diagnoses based on pathology or aetiology (Table 14.1). There are some differences between the two systems, particularly in their approach to amnesic syndromes and minor/mild neurocognitive syndromes, as discussed below. The other neuropsychiatric disorders

listed in Box 14.1 do not have a specific classificatory grouping.

In this chapter we cover the main features of each diagnostic category, starting with the neurocognitive disorders, followed by the other neuropsychiatric disorders. First, however, we introduce how symptoms map onto the regions and networks of the brain, and consider how to approach assessment of a 'neuropsychiatric case'.

Table 14.1 Classification of neurocognitive disorders

ICD-11	DSM-5
Delirium	Delirium
Due to disease classified elsewhere	Delirium due to another medical condition
Due to psychoactive substances and medications*	Substance intoxication delirium
Due to multiple factors	Substance withdrawal delirium
Due to other specified cause	Medication induced delirium
Unknown or unspecified cause	Delirium due to multiple aetiologies
Amnesic disorder	
Due to diseases classified elsewhere	
Due to psychoactive substances including medications*	
Due to other specified cause	
Unknown or unspecified cause	
Dementia	Major neurocognitive disorder (dementia)
Due to Alzheimer disease	Alzheimer's disease
Due to cerebrovascular disease	Vascular disease
Due to Lewy body disease	Lewy body disease
Frontotemporal dementia	Frontotemporal lobar degeneration
Dementia due to psychoactive substances including medications*	Substance/medication induced
Dementia due to diseases classified elsewhere	Prion disease
Neurocognitive disorders, unspecified	Parkinson's disease
	Huntington's disease
	HIV infection
	Due to another medical condition
	Due to multiple aetiologies
	Unspecified
Mild neurocognitive disorder	Minor neurocognitive disorder**
Secondary neurocognitive syndromes	

*Category can be subdivided according to the substance or medication concerned (e.g. hypnotic-induced delirium; amnesic disorder due to alcohol).

**Category can be divided based on the aetiology as for the major neurocognitive disorders

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (DSM-5). Copyright (2013), American Psychiatric Association.

Symptoms associated with regional brain pathology

Before considering the various syndromes, it is helpful to consider the characteristic features associated with lesions in different regions of the brain, and the neuro-anatomical basis of memory. Knowledge of the regional affiliation of neurological and psychopathological findings is relevant when attempting to localize neuropsychiatric conditions. However, the clinical features are not diagnostically specific and the clinicopathological correlations are often modest rather than strong. Moreover, localization of features to specific brain regions is now complemented by a focus on the circuits and networks, both structural and functional, that connect brain regions. For a review, see David (2009a) and Sporns (2020).

Frontal lobe

The frontal lobes, together with their reciprocal connections to other cortical and subcortical regions, have a crucial role in personality and judgement. Patients with a frontal lobe syndrome may present with a variety of clinical syndromes.

- They may be disinhibited, overfamiliar, tactless, and garrulous, make fatuous jokes and puns (*Witzelsucht*), commit errors of judgement and sexual indiscretions, and disregard the feelings of others.
- They may appear inert (abulic) and apathetic, with a paucity of spontaneous speech, movement, and emotional expressions.
- They may engage in obsessive, ritualistic behaviours, with perseveration of thought and gesture.

Measures of formal intelligence are generally unimpaired in frontal lobe disease. However, there may be difficulties in abstract reasoning (e.g. 'How are glass and ice different?') and cognitive estimates are typically inaccurate but precise (e.g. '364 miles from London to New York'). Concentration and attention are reduced, and insight is often markedly impaired. Verbal fluency, assessed using word generation by letter (e.g. number of words beginning with 's' in one minute) and category (e.g. number of animals), is reduced, and unusual (low-frequency) examples may be volunteered. The patient has difficulty switching between tasks (perseveration), carrying out sequenced movements, and understanding rules. Utilization behaviour (e.g. donning several pairs of spectacles) may be evident.

Posterior extension of a dominant frontal lobe lesion may involve Broca's area and produce an expressive (non-fluent) dysphasia. Encroachment on the motor cortex or deep projections may result in a contralateral hemiparesis. Other signs may include ipsilateral optic atrophy or anosmia, a grasp or other primitive reflexes, and, if the process is bilateral or in the midline, incontinence of urine.

Parietal lobe

Lesions of the parietal lobe may cause various neuropsychological disturbances that are easily mistaken for conversion disorder (Chapter 22). Involvement of the non-dominant parietal lobe characteristically gives rise to visuospatial difficulties, with neglect of contralateral space, and constructional and dressing apraxias. Lesions of the dominant lobe may be associated with receptive dysphasia, limb apraxia, body image disorders, right-left disorientation, dyscalculia, finger agnosia, and agraphia. Other signs may include contralateral sensory loss, astereognosis, and agraphesthesia, and (with more extensive lesions) a contralateral hemiparesis or homonymous inferior quadrantanopia.

Persistent unawareness of neurological deficit (anosognosia) is not uncommon, especially with non-dominant parietal lesions. In extreme cases, the patient may deny that a paretic limb belongs to him. This should be distinguished from denial due to a psychological unwillingness to recognize disability and its consequences.

Temporal lobe

The temporolimbic syndromes are characterized by complex and wide-ranging neuropsychiatric clinical pictures. There may be personality change resembling that of frontal lobe lesions, but more often accompanied by specific cognitive deficits and neurological signs. The relatively florid behavioural disturbances that characterize the frontotemporal dementias reflect the combined temporal and frontal involvement, and their interconnections.

Unilateral medial temporal lobe lesions, especially those involving the hippocampus, produce lateralizing memory deficits—left hippocampal damage impairs verbal memory (and semantic impairment and fluent dysphasia), whereas right hippocampal damage affects

non-verbal (spatial) aspects of memory. Some evidence also suggests that left medial temporal lobe lesions are more likely to produce psychotic symptoms, and right-sided lesions produce affective ones.

Occipital lobe

Occipital lobe lesions rarely present to psychiatrists, but they may cause disturbances of visual processing that are easily misinterpreted as being of psychological origin. Such phenomena occasionally accompany migraine or occipital lobe seizures. Complex visual hallucinations may occur with lesions involving visual association areas, sometimes referred to a hemianopic field. These include multiple visual images (polyopia), persistent aftertraces of the features of an image (visual perseveration or palinopsia), and distortions of the visual scene (metamorphopsia). Lesions that impinge anteriorly on the parietal or temporal lobes may produce visual disorientation (inability to localize objects in space under visual guidance) with asimultagnosia (difficulty in perceiving the visual scene as a unity), or prosopagnosia (inability to recognize familiar faces). In patients with suspected occipital lobe pathology, the visual fields should be mapped using perimetry, and neuropsychological tests performed to delineate visual agnosias and other higher-order derangements of visual processing. Some patients who are blind due to occipital lobe damage deny that they are blind (*Anton's syndrome*).

Corpus callosum

Corpus callosum lesions (classically, the 'butterfly glioma') typically extend laterally into both hemispheres. They then produce a picture of severe and rapid intellectual deterioration, with localized neurological signs varying with the degree and direction of extension into adjacent structures. Callosal degeneration is a hallmark of the rare Marchiafava–Bignami syndrome, seen in severe alcohol dependence. Pure callosal lesions (rare and usually iatrogenic, following surgery for intractable epilepsy) can be surprisingly difficult to identify, and require specialized neuropsychological testing to expose a 'disconnection syndrome', reflecting disruption of interhemispheric communication. These unique 'split-brain' patients raise intriguing questions concerning the mechanisms that normally bind the two hemispheres together to generate a consistent, unitary sense of the self (Gazzaniga, 2000).

As well as the corpus callosum, damage to other white matter tracts—both subcortical and periventricular—can have neuropsychiatric and behavioural consequences.

The clinical features of white matter pathology depend on the location of the damage and whether it is focal or diffuse. For a review, see Thiebaut de Schotten et al. (2020).

Subcortical structures and cortico-subcortical circuits

The regional cortical associations with cognitive, affective, and behavioural features reflect the classic 'localisationist' approach to neurology and neuropsychiatry. As noted above, this has largely been replaced by a 'connectionist' approach, in which emphasis is placed on distributed neural systems in which cortical regions are linked with subcortical structures, and with the white matter pathways that connect them.

Mesulam (1998) described five networks:

1. A right-hemisphere spatial awareness network, including the posterior parietal cortex and frontal eye fields.
2. A left-hemisphere language network, including Broca's and Wernicke's areas.
3. A memory–emotion network, including the hippocampus, amygdala, and cingulate cortex.
4. A working memory–executive function network, including the prefrontal cortex and posterior parietal cortex.
5. A face and object recognition network in the temporoparietal and temporo-occipital cortex.

Another influential model is that of Alexander and Crutcher (1990), who proposed four parallel circuits linking different parts of the cerebral cortex with specific basal ganglia and thalamic nuclei. Each circuit mediates different functions. For example, the 'limbic' circuit, which is involved in emotional and motivational processes, links the anterior cingulate cortex and medial prefrontal cortex with the ventral striatum, ventral pallidum, and mediodorsal thalamus. These anatomical pathways are now complemented by a range of 'networks', identified by functional MRI, which reveal a range of coactivated cortical and subcortical regions that are linked to different cognitive states (Matthews and Hampshire, 2016; Sporns, 2020).

Thalamus, basal ganglia, and cerebellum

A variety of cognitive and psychiatric consequences have been described following lesions of subcortical nuclei. These structures, previously considered to be primarily involved in sensory processing (the thalamus) or in motor control (the basal ganglia and cerebellum)

are now all known also to be integrally involved in cognition and behaviour, and lesions therein may present with psychiatric as well as neurological features (DeLong and Wichmann, 2007; Schmahmann and Pandya, 2008), and include disturbances of memory, language, and mood. Reduced initiation of actions is also characteristic of basal ganglia pathology, and impaired consciousness with thalamic lesions.

Rostral brainstem

Behavioural disturbances frequently accompany lesions of the rostral brainstem. The most characteristic features are an amnesic syndrome (see below), hypersomnia, and the syndrome of *akinetic mutism* ('vigilant coma') or stupor.

Memory systems and their neuroanatomy

Clinical, neuropsychological, and brain imaging studies (both structural and functional) support the existence of multiple memory systems in the human brain. These functions may all be affected more or less selectively by brain lesions. The most basic division lies between implicit (i.e. procedural) and explicit (i.e. declarative) memory. The former includes a range of phenomena that are not usually subject to conscious analysis, such as motor skills, conditioned behaviours, and repetition priming. Explicit memory is subclassified into episodic

(memory of autobiographical events) and semantic (knowledge of the world).

The short-term store underpins working memory (e.g. when dialling an unfamiliar telephone number). Distinct anatomical substrates for short-term storage of verbal and visuospatial information, both controlled by a central executive, have been proposed. In neuropsychological terms, 'short-term' refers to immediate recall. By contrast, the concept of 'short-term' memory, as sometimes applied by clinicians to recall over minutes and days, does not correspond to an anatomical substrate.

Specific types of memories, such as faces and topographical information, may engage dedicated subsystems. Episodic memory has both anterograde (new learning) and retrograde (recall of past events) components. It appears to be mediated by a network of cortical and subcortical structures, which include the hippocampus, parahippocampal and entorhinal cortices, amygdala, mammillary bodies, fornix, cingulate, thalamus, and frontobasal cortex, whereas semantic memory may be subserved by a partly independent network overlapping the language areas. Broadly speaking, verbal memories are mediated by the left (dominant) hemisphere and non-verbal memories by the right hemisphere.

For a review of the classification and neuroanatomical basis of memory and its dysfunction, see David (2009a) and Ranganath and Ritchey (2012).

Assessment of the 'neuropsychiatric patient'

The assessment of cognitive function was introduced in Chapters 1 and 3 as part of the general psychiatric assessment, and in this chapter the evaluation of amnesia, delirium, and dementia is discussed. In this section, we introduce key aspects of the initial approach to the patient who has a suspected neuropsychiatric disorder. For a more detailed discussion, see Kipps and Hodges (2005) and David (2009b).

Before embarking upon the assessment, it is worth considering the range of major diagnostic possibilities. One question to be addressed early on is whether there is clouding of consciousness, as this is a hallmark of delirium, and the assessment can proceed to determine its cause. If there is no impaired consciousness, the main diagnostic categories to consider are amnesia,

dementia, or a 'functional' cause of cognitive impairment. The key feature of an amnesic syndrome is a specific deficit in episodic memory; although rare, amnesic syndrome needs to be considered in patients presenting with memory impairment, especially in those with alcohol dependence. A 'functional' cause should always be considered, as memory impairment may occur secondary to many psychiatric disorders; in particular, depression in the elderly may present as *pseudodementia* (see Chapter 19). Distinction between organic and functional causes requires positive evidence to be sought for both forms of disorder (Hallett et al., 2022). It is an important distinction to make, as it has a significant impact upon treatment and prognosis. Once these other causes of cognitive impairment (delirium, amnesic syndrome,

and functional disorder) have been ruled out, a provisional diagnosis of dementia can be made (assuming that the impairment is of sufficient severity), and attention can then turn to determining the type of dementia from which the patient is suffering.

History and mental state examination

Although physical examination and laboratory investigations play a much larger role than elsewhere in psychiatry, the history remains essential: 'the difference between a good neuropsychiatrist and a mediocre one is a good history' (David, 2009b). An informant is especially important, as the presence of impaired cognition or consciousness will necessarily limit the patient's ability to provide a full and accurate history. Key points in the history include the onset, duration, and progression of the impairment—for example, an acute onset suggests delirium or, if it began after a fall, may indicate a subdural haematoma. The neurological, medical, and family history is important, too, as many causes of cognitive impairment are secondary to pre-existing disorders or have a genetic basis.

Physical examination

The physical examination needs to be comprehensive and careful, as signs may not be conspicuous. Particular attention should be paid to the nervous system, as well as to searching for peripheral stigmata of systemic disease and alcohol dependence. Specific signs may provide diagnostic clues (e.g. the Argyll–Robertson pupil of neurosyphilis, optic disc pallor in vitamin B₁₂ deficiency, or cranial nerve involvement in neurosarcoidosis).

Investigations

The choice and extent of investigations will depend on the findings from the history, mental state examination, and physical examination, but usually include a core set of tests, such as the cognitive tests and blood tests used in evaluation of dementias noted below. In difficult or atypical cases and in younger patients, investigations may be extensive, and the opinion and assistance of a neurologist, physician, or neurosurgeon may be required.

Some examples of specialized investigations used in neuropsychiatric evaluation are listed here. Many are described later in the chapter with regard to investigation of dementia.

- *Structural brain imaging* with computed tomography (CT) or magnetic resonance imaging (MRI). Neuroimaging can detect focal and diffuse pathologies, and longitudinal scans can map progressive changes that mirror clinical decline. MRI is superior to CT for most purposes, including evaluation of white matter disease, and the ability to perform volumetric measurements. *Functional brain imaging* with fMRI, magnetic resonance spectroscopy (MRS), or positron emission tomography (PET) are valuable as research tools, but are not in widespread clinical use.
- *Neuropsychological testing* is less widely used than previously, in part because of the increasing availability of brain imaging. However, it can still play a valuable part in characterization of the cognitive impairment (e.g. the cognitive domains that are most affected), and measurement of severity and progression (David, 2009b). Rating scales for the assessment of dementia are considered below.
- *Electroencephalogram (EEG) studies* retain a limited but valuable role in several situations where EEG findings are characteristic—for example, in delirium, prion disease, and detection of non-convulsive status epilepticus. They are also useful in the differential diagnosis of stupor, as a normal EEG would suggest a dissociative state.
- *Cerebrospinal fluid (CSF) examination* after lumbar puncture is essential if an inflammatory or infective process is suspected. It may also become more widely used in the evaluation or prediction of dementia, as different proteins are discovered to have diagnostic or prognostic value as biomarkers.
- *Genetic testing* has a key role in the diagnosis (and prediction) of a limited range of familial disorders in which the mode of inheritance, and the causative gene, are known (e.g. Huntington's chorea).
- *Brain biopsy*, usually of the right frontal lobe, is occasionally considered as a last resort in the diagnosis of suspected prion disease or other rapid, severe early onset dementia. However, the risks of this procedure must always be weighed against the diagnostic and prognostic information that will be obtained from it.

Delirium

Delirium is characterized by a disturbance of attention, orientation, and awareness (sometimes called *clouding of consciousness*) that develops rapidly and includes a range of other symptoms that are typically fluctuating. Disturbance of the sleep–wake cycle is common. Delirium is associated with several adverse outcomes, and its prompt recognition and treatment are important. A number of other terms have also been used to describe delirium, such as *acute confusional state*, *acute brain failure*, and *acute organic syndrome*.

For a review of delirium and its management, see Thom et al. (2019) and Wilson et al. (2020).

Epidemiology

Delirium is much more common at the two ends of life, in childhood and in the elderly, than in adulthood. Other high-risk individuals are those with compromised ‘cerebral reserve’, notably those with pre-existing dementia or a range of other medical factors (Box 14.2) are at higher risk. Delirium is a common disorder in all medical settings, especially for older people.

Delirium has an estimated prevalence of 23% in inpatients (Gibb et al., 2020), with higher rates in intensive care and palliative care settings and following burns or major surgery. Estimates in non-hospital settings are less well established but much lower.

Clinical features

The cardinal feature of delirium is disturbed consciousness. It is manifested as drowsiness, decreased awareness of one’s surroundings, disorientation in time and place, and distractibility. At its most severe the patient may be unresponsive (stuporose), but more commonly the impaired consciousness is quite subtle. Indeed, the first clue to the presence of delirium is often one of its other features, which include mental slowness, distractibility, perceptual anomalies, and disorganization of the sleep–wake cycle (see Box 14.3).

Symptoms and signs vary widely between patients, and in the same patient at different times of day, typically being worse at night. Though the majority of patients are *hypoactive*, with retardation and perseveration, some have *hyperactive delirium* and are restless and irritable, with psychotic symptoms. Repetitive,

Box 14.2 Delirium: predisposing and precipitating factors

Predisposing factors

- Age—extremes of life, especially old age
- Dementia
- Previous episode of delirium
- Physical ill health
- Sensory impairment
- History of cerebrovascular disease
- Alcohol misuse

Precipitating factors

Substance-related

- Prescribed medical drugs (e.g. steroids, digoxin, diuretics)
- Psychotropic medication (e.g. sedatives, opiates, anticholinergics)
- Alcohol—intoxication, withdrawal, delirium tremens
- Polypharmacy

Physiological causes

- Septicaemia
- Infection (e.g. urinary tract, respiratory)
- Hypoxia (e.g. postoperative)
- Pyrexia
- Organ failure (e.g. renal, hepatic, cardiac)
- Metabolic acidosis
- Hypoglycaemia or hyperglycaemia
- Dehydration

Neurological causes

- Post-ictal
- Head injury
- Encephalitis

Other causes

- Invasive devices (e.g. urinary catheter)
- Pain
- Sleep deprivation
- Constipation

Adapted from Inouye, S. K., Westendorp, R. G. J., & Saczynski, J. S. (2014). Delirium in elderly people, *Lancet*, 383(9920), 911–922. Copyright (2014), with permission from Elsevier.

Box 14.3 Clinical features of delirium**Cognitive features**

Clouding of consciousness
 Impaired attention
 Disorientation for time and place
 Impaired memory
 Psychotic symptoms

Perceptual disturbances

Delusions
 Perplexity
 Thought disorder

Behavioural and other symptoms

Agitation
 Irritability
 Labile affect
 Word-finding difficulties

Temporal course

Rapid onset
 Fluctuation over 24-hour period
 Reversal of sleep–wake cycle

purposeless movements are common in both forms. Thinking is slow and muddled, but often rich in content ('dream-like'). Ideas of reference and delusions (often persecutory) are common, but are usually transient and poorly elaborated. Visual perception is often distorted, with illusions, misinterpretations, and visual hallucinations, sometimes with fantastic content. Tactile and auditory hallucinations also occur. Anxiety, depression, and emotional lability are common. The patient may be frightened, or perplexed. Experiences of depersonalization and derealization are sometimes described. Attention and registration are particularly impaired, and on recovery there is usually amnesia for the period of the delirium. A range of delirium screening tools are available, of which the Confusion Assessment Method and 4AT rapid clinical test are most widely used.

Aetiology

The main causes of delirium are listed in Box 14.2. In practice, many cases are multifactorial, and sometimes idiopathic. In elderly people in medical settings, drugs (especially polypharmacy) and infection are probably the most common causes; in other settings and populations,

delirium has a wide range of causes and hence requires urgent and wide-ranging investigations (see below).

The pathophysiological basis of delirium is unclear and likely varies depending on the cause. The severity of clinical disturbance correlates with the degree of slowing of cerebral rhythms on EEG. Current theories include impairment of brain energy metabolism, inflammation, and roles for the neurotransmitters acetylcholine, dopamine and gamma-aminobutyric acid (GABA), with impaired functional connectivity (Wilson et al., 2020).

Management of delirium**Prevention**

There are more effective interventions to prevent delirium than to treat it. This applies to pharmacological and particularly to non-pharmacological strategies, which include a range of validated approaches such as the Hospital Elder Life Program (HELP) and targeting of delirium risk factors (e.g. reorientation, promotion of sleep). Evidence for drug strategies is limited, but low doses of antipsychotics have some efficacy in preventing post-operative delirium, and melatonin is used in various hospital settings (Thom et al., 2019). In practice, however, preventative interventions are either not possible or unsuccessful, and it becomes necessary to treat delirium.

Treatment

Delirium is a medical emergency. Although it is a clinical diagnosis, it is essential to identify and treat the underlying cause, and a range of investigations may be required (Box 14.4). Many are routine (e.g. electrolytes, full blood count, urinalysis), whereas others are carried out depending on the context (e.g. lumbar puncture). Vital signs need regular measurement. As delirium is often caused by drugs (due to side effects or withdrawal effects), these should always be suspected until there is evidence of another cause.

The mainstay of treatment should be non-pharmacological, since there is little evidence that any medication improves outcome, and antipsychotics (which are commonly used) have known adverse effects (Neufeld et al., 2016). General measures should always be used to relieve distress, control agitation, and prevent exhaustion. These include frequent explanation, reorientation, and reassurance. Unnecessary changes in the staff caring for the patient should be avoided. The patient should ideally be nursed in a quiet single room. Relatives should be encouraged to visit regularly.

Box 14.4 Investigations for delirium**Blood tests**

Full blood count
 Urea and electrolytes
 Renal function tests
 Liver function tests
 Calcium
 Random blood glucose
 Blood cultures
 Arterial blood gas
 Syphilis serology

Other tests

Urinalysis
 Chest X-ray
 Drug screen
 Cardiac enzymes
 MRI or CT brain scan
 EEG
 Lumbar puncture

At night, lighting should be sufficient to promote orientation, while not preventing sleep. In many cases, use of these behavioural measures can be sufficient to manage the patient while the cause of the delirium is being investigated and treated.

Nevertheless, despite such interventions, in practice many patients with delirium are treated with medication to control agitation and distress, and to allow adequate sleep (Thom et al., 2019). Antipsychotics are usually first-line medication. They should be limited to cases with severe agitation (endangering the patient) and for psychotic symptoms causing distress (hallucinations or delusions). Haloperidol has conventionally been used in all age groups, starting at a very

low dose, and then carefully titrated to achieve the desired calming effect without excess sedation or side effects. If necessary, the first dose can be given intramuscularly, followed by doses every six hours (typically 1–5 mg per day, although elderly patients may require less). Atypical antipsychotics are now often prescribed instead of haloperidol; the same principles of careful administration and monitoring apply. Some causes of delirium require avoidance of antipsychotics, or particular caution when using them. This includes all patients with coexisting dementia because of their sensitivity to this medication class, as well as delirium associated with alcohol withdrawal (delirium tremens) or with epilepsy, because of the risk of seizures. Other drugs sometimes used to treat delirium include sodium valproate and dexmedetomidine. All drugs should be used with caution in liver failure because of the danger of precipitating hepatic coma.

Outcome

Many cases of delirium recover rapidly. The prognosis is related to the underlying cause, and is worse in the elderly and in those with pre-existing dementia or physical illness. Patients with a hypoactive behavioural profile have a worse outcome than those who are hyperactive. There is an elevated mortality rate following delirium, with an estimated 25% mortality at three months, although published estimates vary markedly; a meta-analysis of elderly patients found that an episode of delirium was independently associated with a twofold increased risk of death in the next two years (Witlox et al., 2010).

The relationship between delirium and dementia is complex and bidirectional. As noted earlier, pre-existing dementia is a major risk factor for delirium, and delirium is associated with a more than doubled risk of subsequent cognitive decline and dementia (Goldberg et al., 2020). Delirium also acts synergistically with underlying dementia neuropathology to accelerate the pace of cognitive decline (Davis et al., 2017).

Amnesia and amnestic disorders

Amnesia is loss of memory, and *amnestic disorders* are those in which episodic memory is specifically and persistently affected, without other significant cognitive impairment or altered consciousness, and with a decline from the previous level of functioning. In ICD-11 and in routine clinical practice, these criteria together distinguish amnestic disorders from dementia and delirium,

in which multiple domains of cognition are affected (Box 14.5). However, as noted earlier, DSM-5 subsumes amnestic disorders within the broader category of neurocognitive disorder and its subdivisions.

Amnestic disorders manifest as an inability to learn new information (anterograde amnesia) and to recall past events (retrograde amnesia). To make the diagnosis,

Box 14.5 Causes of amnesia**Transient**

Transient global amnesia
 Transient epileptic amnesia
 Head injury
 Alcoholic blackouts
 Post-electroconvulsive therapy
 Post-traumatic stress disorder
 Psychogenic fugue
 Amnesia for criminal offence

Persistent (amnesic syndrome)

Korsakoff syndrome
 Herpes encephalitis
 Posterior cerebral artery and thalamic strokes
 Head injury

there should be a significant impairment in social or occupational functioning, and evidence of a general medical condition which can be aetiologically related to the memory impairment. *Korsakoff (or Korsakov) syndrome* is sometimes erroneously referred to synonymously with amnesic disorder, but is in fact a specific form of it, as described below.

For a review of amnesia and amnesic disorders, see Markowitsch and Staniloiu (2012).

Clinical features

The cardinal feature is a profound and circumscribed deficit in episodic memory. The full clinical picture is striking. There is disorientation for time, loss of autobiographical information (often extending back for many years), severe anterograde amnesia for verbal and visual material, and lack of insight into the amnesia. Events are recalled immediately after they occur, but forgotten a few minutes later. Thus the digit span, which tests the short-term memory store, is typically normal. New learning is grossly defective, but retrograde memory is variably preserved and shows a temporal gradient, with older memories being better preserved. Other cognitive functions are relatively intact, although some emotional blunting and inertia are often observed.

The other classic feature, seen particularly in Korsakoff syndrome, is *confabulation*, in which gaps in memory are filled by a vivid and detailed but wholly

fictitious account of recent activities, which the patient believes to be true. The confabulating patient is often highly suggestible.

Aetiology and pathology

Amnesia results from lesions in the medial thalamus, other midline diencephalic structures, or medial temporal lobes (hippocampus and adjacent cortex). Cases due to damage in the medial temporal lobe typically produce the 'purest' amnesia, with little in the way of disorientation or confabulation; the latter features are characteristic of thalamic and diencephalic lesions. More recent studies indicate that the hub of the 'amnesia circuit' is at the junction of the presubiculum and retrosplenial cortex (Ferguson et al., 2019).

Korsakoff syndrome

The commonest cause of amnesic disorder is *Korsakoff syndrome*, named after the Russian neuropsychiatrist (also sometimes spelled Korsakov) who described it in 1889. The alternative term, *Wernicke–Korsakoff syndrome* is also used because the syndrome often follows an acute neurological syndrome called *Wernicke's encephalopathy*, which consists of delirium, ataxia, pupillary abnormalities, ophthalmoplegia, nystagmus, and a peripheral neuropathy. Korsakoff syndrome is usually caused by thiamine deficiency, secondary to alcohol abuse, although it occasionally results from other causes, such as hyperemesis gravidarum and severe malnutrition (Scalzo et al., 2015). The classic neuropathological findings are neuronal loss, gliosis, and microhaemorrhages in the periaqueductal and paraventricular grey matter, the mammillary bodies, and the anterior and mediodorsal thalamus (Krill and Harper, 2012). For a review, see Kopelman (2022).

Other causes of amnesic syndrome include tumours and infarcts in the medial thalamus (diencephalic amnesia) and encephalitis.

Investigation and management

Alertness to the possibility of amnesic syndrome is essential; the patient may not fit the stereotype of chronic alcohol misuse associated with Korsakoff syndrome, and this is a potentially reversible condition. Useful findings from investigations include a reduced red cell transketolase level, which is a marker of thiamine deficiency, and an increased MRI signal in midline structures.

In practice, Korsakoff syndrome should be assumed to be the cause of amnesic syndrome until another aetiology can be demonstrated, and should be treated urgently with thiamine without awaiting the results of investigations. Thiamine is given parenterally in an acute presentation, together with rehydration, general nutritional support, and treatment of supervening alcohol withdrawal. Thiamine replacement should always precede administration of intravenous glucose-containing solutions. Close liaison with physicians and neurologists is important. Apart from thiamine, there are no interventions of proven efficacy for amnesic disorders.

In the longer term, amnesic syndrome may require substantial rehabilitation and support, as the condition markedly impairs normal activities and ability to provide self-care.

Course and prognosis

In the classic series of 245 patients reported by Victor et al. (1971), 96% presented with Wernicke's encephalopathy. Mortality was 17% in the acute stage, and 84% of the survivors developed a typical amnesic syndrome. There was no improvement in 50% of cases, complete recovery in 25%, and partial recovery in the remainder. Favourable prognostic factors were a short history before diagnosis and prompt commencement of thiamine replacement.

The prognosis is poor in cases of amnesic syndrome resulting from viral encephalitis and other causes of irreversible bilateral hippocampal or diencephalic damage. However, amnesic syndrome owing to head injury has

a better outlook. Progressive amnesia suggests a slowly expanding structural lesion, such as a midbrain tumour.

Transient global amnesia

The syndrome of *transient global amnesia* is important in the differential diagnosis of paroxysmal neurological and psychiatric disturbance (see Box 14.6). It occurs in middle or late life. The clinical picture is of sudden onset of isolated, often profound, anterograde amnesia in a clear sensorium, generally lasting for between 15 minutes and 24 hours. Migraine is a risk factor. Functional imaging studies during transient global amnesia have demonstrated localized transient hypoperfusion or hyperperfusion consistent with dysfunction of circuits that mediate episodic memory.

The patient appears bewildered, and requires repeated reorientation, only to ask the same questions moments later. However, there is no disturbance of alertness, and (in contrast to psychogenic fugue) personal identity is retained. Procedural memory is spared—for example, the patient may carry on driving competently during the episode. Apart from the memory disturbance, the neurological examination is entirely normal.

Complete recovery, with amnesia for the period of the episode, is usual. Recurrence is rare, but more common in younger people and those with migraine. However, investigation is always indicated, to exclude other causes of amnesia. Patients with transient global amnesia often present as emergencies, and the syndrome may be misdiagnosed as a dissociative fugue. There is no increased risk of stroke.

For a review, see Ropper (2023).

Box 14.6 Differential diagnosis of transient amnesia and other paroxysmal neuropsychiatric symptoms

Neuropsychiatric

Transient global amnesia
Syncope (cardiogenic, vasovagal, reflex)
Transient ischaemic attacks
Migraine
Epileptic seizure (ictal or post-ictal)
Hypoglycaemia
Pheochromocytoma
Narcolepsy and other parasomnias
Tonic spasms of multiple sclerosis

Treatment-related complications in Parkinson's disease
Drug abuse
Medial temporal lobe tumour

Other

Panic attacks and hyperventilation
Dissociative disorder
Schizophrenia
Bipolar affective disorder
Aggressive outburst in personality disorder

Dementia

Dementia is an acquired global impairment of intellect, memory, and personality, but without impairment of consciousness. ICD-11 requires impairment in two or more cognitive domains (in contrast to amnesic disorders), but DSM-5 does not. Dementia is nearly always progressive. The syndrome of dementia is caused by a wide range of diseases. Alzheimer's disease is the commonest cause, followed by vascular dementia (with which Alzheimer's disease often co-exists) and dementia with Lewy bodies (Box 14.7).

Although dementia is a global or generalized disorder, it often begins with focal cognitive or behavioural disturbances. At this stage, deficits may be too mild or circumscribed to fulfil this definition, and are then called *mild cognitive impairment* or *mild neurocognitive disorder*.

In this section, the main features of the dementia syndrome are described, followed by the principles of assessment. We then discuss the clinical, aetiological, and neuropathological features of the major diseases that produce dementia. Note in Box 14.7 that many

Box 14.7 Causes of dementia

Primary neurodegenerative disorders

Alzheimer's disease*, dementia with Lewy bodies, frontotemporal dementias*,

Parkinson's disease*, prion diseases*, limbic-predominant age-related TDP-43 encephalopathy (LATE), Huntington's disease*

Vascular causes

Vascular dementia*, strokes, subdural haematoma

Inflammatory and autoimmune causes

Systemic lupus erythematosus, other forms of cerebral vasculitis, Behçet's disease, neurosarcoidosis, Hashimoto's encephalopathy, multiple sclerosis

Trauma

Severe head injury, repeated head trauma ('dementia pugilistica')

Infections and related conditions

HIV, iatrogenic and variant CJD (prion disease), neurosyphilis, postencephalitic

Metabolic and endocrine causes

Renal failure, hepatic failure, hypothyroidism, hyperthyroidism,

hypoglycaemia, Cushing's syndrome, hypopituitarism, adrenal insufficiency

Neoplastic causes

Intracranial space-occupying lesions, carcinomatous or lymphomatous meningitis, paraneoplastic limbic encephalitis

Post-radiation

Acute and subacute radionecrosis, radiation thromboangiopathy

Post-anoxia

Severe anaemia, post-surgical (especially cardiac bypass), carbon monoxide poisoning, cardiac arrest, chronic respiratory failure

Vitamin and other nutritional deficiencies

Vitamin B₁₂ deficiency, folate deficiency

Toxins

Alcohol, poisoning with heavy metals, organic solvents, organophosphates

Other

Normal pressure hydrocephalus
Leukodystrophy*

* Can occur in genetically determined forms.

other neuropsychiatric disorders and some medical disorders can also include cognitive impairment; these conditions are considered later in this chapter and in Chapter 22, respectively. Dementia as a result of substance misuse, especially alcohol misuse, is discussed in Chapter 20. The management of dementia, and the relationships between dementia and ageing, are deferred until Chapter 19.

For an overview of dementia, see Gale et al. (2018) and Arvanitakis et al. (2019).

Clinical features of dementia

The presenting complaint is usually of poor memory. Other features include disturbances of behaviour, language, personality, mood, or perception.

The clinical picture is much determined by the patient's premorbid personality, as well as by the underlying cause. People with good social skills may continue to function adequately despite severe intellectual deterioration. Dementia is often exposed by a change in social circumstances or an intercurrent illness. The elderly, socially isolated, or deaf are less likely to compensate for failing intellectual abilities; however, their difficulties may go unrecognized or be dismissed.

Forgetfulness is usually early and prominent, but may sometimes be difficult to detect in the initial stages. Impaired attention and concentration are common and non-specific. Difficulty with new learning is usually the most conspicuous feature. Memory loss is more evident for recent than for more remote material. Disturbed episodic memory is manifested as forgetfulness for recent day-to-day events, with relative preservation of procedural memory (e.g. how to ride a bicycle) and, at least initially, general knowledge about the world at large. By contrast, words and, ultimately, the very objects to which they refer, lose their meaning for patients with marked semantic memory impairment (as in frontotemporal dementia).

Loss of flexibility and adaptability in new situations, with the appearance of rigid and stereotyped routines ('organic orderliness'), and, when taxed beyond restricted abilities, sudden explosions of rage or grief ('catastrophic reaction') may occur. As dementia worsens, patients are less able to care for themselves, and they neglect social conventions. Disorientation for time, and later for place and person, is common. Behaviour becomes aimless, and stereotypes and mannerisms may appear. Thinking slows and becomes impoverished in content and perseverative. False ideas, often of a persecutory kind, gain ground easily. In the

later stages, thinking becomes grossly fragmented and incoherent. This is reflected in the patient's speech, with syntactical and dysnomic errors. Eventually the patient may become mute. Life expectancy is reduced, with death often following bronchopneumonia and a terminal coma.

Behavioural, affective, and psychotic features often accompany the cognitive deficits and are an important part of the clinical picture and as targets for treatment (Tampi and Jeste, 2022). They appear to be part of the underlying biology of the disease process, although in the early stages, while insight is retained, they may also be a psychological response to the realization of cognitive decline. Mood disturbances are particularly common, together with distress, anxiety, irritability, and sometimes aggression. Later, emotional responses become blunted, and sudden, apparently random, mood changes occur. Psychotic symptoms are also a common and fluctuating feature during dementia.

The balance of these core symptoms and signs, together with some additional features, forms the basis for the clinical differentiation between the various causes of dementia, as summarized in Table 14.2 and described in the following sections.

Subcortical and cortical dementia

A distinction is sometimes drawn between subcortical and cortical dementia, based upon their putative neuroanatomical basis. Although the distinction is blurred, clinically and pathologically, the terms have descriptive utility. The key features are summarized in Table 14.3, and examples of the diseases are listed in Box 14.8. The term *subcortical dementia* is seen to refer to a syndrome of slowness of thought, difficulty with complex, sequential intellectual tasks, and impoverishment of affect and personality, with relative preservation of language, calculation, and learning. It contrasts with the spectrum of dysfunction seen in *cortical dementias*, which includes early, prominent impairments of memory, word finding, and visuospatial abilities).

Presenile and senile dementia

Another traditional distinction is made between dementia occurring in those under 65 years of age (presenile or early onset dementia) and dementia beginning later in life (senile or late-onset dementia). It arose in part because of the belief that the major causes were different—Alzheimer's disease in the former, and vascular dementia in the latter. With the realization that Alzheimer's disease is the most common form of dementia in both groups, less attention is now paid to this categorization.

Table 14.2 Clinical features that help to distinguish between major causes of dementia

	Prominent symptoms and signs	Other clinical features
Alzheimer's disease	Memory loss, especially short-term	Relentlessly progressive
	Dysphasia and dyspraxia	Survival 5–8 years
	Early loss of sense of smell	
	Behavioural changes (e.g. wandering)	
Vascular dementia	Personality change	Stepwise progression
	Labile mood	Signs of cerebrovascular disease
	Preserved insight	History of hypertension
		More common in men, and in smokers
Dementia with Lewy bodies	Fluctuating alertness	Frequent adverse reactions to antipsychotics
	Parkinsonism	
	Visual hallucinations	
	Falls and faints	
Frontotemporal dementia	Prominent behavioural change	Onset usually before the age of 70 years
	Expressive dysphasia	Range of clinical subtypes
	Early loss of insight	
	Early primitive reflexes	
Prion disease	Myoclonic jerks	Often early-onset
	Seizures	Rapid onset and progression
	Cerebellar ataxia	Transmissible
	Psychiatric symptoms (vCJD)	
Limbic-predominant age-related TDP-43 encephalopathy (LATE)	Alzheimer-like features	Affects the older old (>85 years)
Normal pressure hydrocephalus	Mental slowing, apathy, inattention	Most common in 50–70-year age group
	Urinary incontinence	Most common reversible dementia
	Gait apraxia	

However, presenile dementia does differ in certain respects from late-onset dementia. Frontotemporal dementia and prion disease are relatively more common, vascular dementia is rarer, and a higher proportion of cases monogenic causes (see Box 14.7). Patients with presenile dementia are more likely to be referred to and investigated by neurologists before psychiatrists become involved in their care. Investigations are more intensive and extensive to give a definite diagnosis and provide a clear prognosis. The clinical course tends to be more rapid than in late-onset cases.

Assessment of dementia

Assessment of a patient who is presenting with a complaint of cognitive impairment involves several stages. A key question to be addressed initially is whether the impairment is due to dementia. This involves ruling out other causes, notably delirium, amnesia, and depression. Other patients will not meet criteria for a major neurocognitive disorder but may have *mild cognitive impairment* (see below), while others will have *subjective cognitive decline*, in which complaints

Table 14.3 Features of cortical and subcortical dementias

	Subcortical dementia	Cortical dementia
Memory impairment	Moderate	Severe, early
Language	Normal	Dysphasias, early
Mathematical skills	Preserved	Impaired, early
Personality	Apathetic, inert	Indifferent
Mood	Flat, depressed	Normal
Coordination	Impaired	Normal
Cognitive and motor speed	Slowed	Normal
Abnormal movements	Common, e.g. tremor	Rare

Box 14.8 Examples of cortical, subcortical, and mixed causes of dementia

Cortical

Alzheimer’s disease
Frontotemporal dementias

Subcortical

Huntington’s disease
Multiple sclerosis

Mixed

Vascular dementia
Dementia with Lewy bodies
Corticobasal degeneration

Assessment of the severity and clinical profile

Screening tests are useful in the assessment of dementia and its severity, and for monitoring progression. Different scales are available to assess cognition, behavioural symptoms, global functioning and activities of daily living, and depression; the latter is useful because depression can coexist with dementia and worsen functioning. Some commonly used screening tests, most of which take ten minutes or less, are listed in Box 14.9. For a review, see Velayudhan et al. (2014).

For cognitive impairment, the benchmark is the Mini-Mental State Examination (MMSE), which has a sensitivity and specificity for dementia of around 80% when a cut-off score of 23 is chosen. The Montreal Cognitive Assessment scale (MoCA) and clock test are valuable and brief alternatives, and the MoCA is superior to MMSE in the detection of mild cognitive impairment (Pinto et al., 2019). The ADAS-cog is specifically designed for suspected Alzheimer’s disease, and is widely used in clinical trials to monitor treatment response. CAMCOG is largely a research tool. Note that these tests are validated for use in the diagnosis of dementia, and are not recommended for ‘case finding’ in the population.

Careful evaluation of the behavioural symptoms of dementia is an integral part of the assessment, and needs to be repeated during the illness, since these symptoms are common, fluctuate, and pose many difficulties for carers.

Assessment of the cause of dementia

Definitive diagnosis of the cause of dementia can usually only be made neuropathologically or, in rare cases,

of poor memory are accompanied by normal performance on standardized cognitive tests (Jessen et al., 2020).

Having established the probable diagnosis of dementia, its characteristics (including severity, symptom and behaviour profile, rate of progression, and risks) are considered, together with assessment of its cause, and identification of potentially reversible causes (Gale et al., 2018). The latter are rare, and in one study consisted mostly of hydrocephalus and space-occupying lesions (Hejl et al., 2002).

Box 14.9 Screening tests for dementia

Cognitive function

Mini-Mental State Examination (MMSE)
 Montreal Cognitive Assessment (MoCA)
 Six-Item Cognitive Impairment Test
 Seven-minute screen
 Clock drawing test
 Hopkins Verbal Learning Test (HVLT)
 Mental Test Score (MTS)
 Alzheimer's Disease Assessment Scale—cognitive subscale (ADAS-cog)
 Cambridge Examination for Mental Disorders of the Elderly, cognitive section (CAMCOG)

Behavioural and psychological features

Neuropsychiatric Inventory
 MOUSEPAD
 BEHAVE-AD
 Cohen–Mansfield Aggression Inventory

Activities of daily living

Bristol Scale
 Alzheimer's Disease Functional Assessment and Change Scale
 Disability Assessment for Dementia

Depression

Cornell Scale
 Geriatric Depression Rating Scale

Global assessment

Clinical Dementia Rating (CDR)

by identification of genetic mutations. However, the differing clinical profiles of the various dementias allow 'probable' diagnoses to be made by experienced clinicians with reasonable accuracy, considering the fact that many patients have a mixed dementia, especially the combination of Alzheimer's disease and vascular dementia. The speed of the cognitive decline is another factor to consider since rapidly progressive dementias—such as prion disease—need more urgent and extensive assessment (Hermann and Zerr, 2022). The use of biochemical, radiological, and genetic investigations only modestly increases the diagnostic accuracy for common

Box 14.10 Investigations for establishing the cause of dementia

Blood tests

Full blood count
 Urea and electrolytes
 Liver function tests
 Calcium and phosphate
 Thyroid function tests
 Vitamin B₁₂ and folate
 Syphilis serology
 Inflammatory markers (e.g. CRP)
 Blood biomarkers (e.g. Aβ42, p-tau, Nf-L)

Other tests

MRI or CT brain scan
 Chest radiograph
 Urinalysis
 Neuropsychological assessment
 Lumbar puncture (if infective or autoimmune cause suspected)
 Genetic testing
 EEG
 PET: fluorodeoxyglucose, amyloid or tau imaging

Adapted from Burns, A., & Illiffe, S. (2009). Dementia. *BMJ*, 338, b75. Copyright (2009), with permission from BMJ Publishing Group Ltd.

dementias, but are important for ruling out rarer and reversible causes.

Box 14.10 summarizes the potential investigations for dementia. An MRI or CT scan is recommended as routine in many guidelines but, in practice, brain scanning, like most investigations, is used in some but not all patients. PET neuroimaging with ligands to detect β-amyloid or tau, and plasma or CSF biomarkers such as β-amyloid, Aβ42 and Aβ40), phosphorylated tau (p-tau217), and neurofilament light chain (NF-L) show real promise for diagnosis and differential diagnosis of Alzheimer's disease and some other dementias (Hansson, 2021). None of these tests are yet routinely available, but this is likely to change soon in the case of the blood biomarkers (Teunissen et al., 2022), with the recent approval in the United States of the first blood test for Alzheimer's disease diagnosis. The test measures the p-tau217/Aβ42 ratio.

In summary, the extent to which investigations listed in Box 14.10 are carried out, as well as the addition of more specialized tests, depends upon the patient's age

and history, the results of the initial tests, the subsequent differential diagnosis, and the healthcare setting.

Assessment of risk in dementia

Patients with dementia are at risk from self-neglect, poor judgement, wandering, and abuse. Their physical health is often a problem. Risks to others may occur because of aggressive or disinhibited behaviour. Fitness to drive is a specific issue to consider, and may be difficult to determine in the early stages (Wilson and Pinner, 2013); current UK regulations allow those with ‘sufficient skills’ and whose ‘progression [of dementia] is slow’ to continue to drive, subject to annual review. Thus risk assessment is part of a full assessment of dementia. A good history from carers and other informants is essential, and an occupational therapist has an important role to play in assessment of functional ability.

Early detection of dementia

Mild cognitive impairment

Increasing attention is being given to the earlier diagnosis of dementia in those with equivocal evidence of worsening memory. This is in part driven by the research focus on developing treatments to delay or prevent progression to dementia. This intermediate category is usually called *mild cognitive impairment* (MCI) though ICD-11 refers to it as *mild neurocognitive disorder* and DSM-5 as *minor neurocognitive disorder*. The key distinction from dementia is that the impairments, though present, are not severe enough to significantly interfere with an individual’s functioning. MCI is thought to represent the prodromal stage of dementia, and so can be further categorized by its putative cause. In people presenting with features of MCI, it is important to consider and exclude depression and other ‘non-dementia’ causes of cognitive impairments, sometimes called *pseudodementia* (Chapter 19).

It is worth noting that the clinical and prognostic significance of MCI remains unclear. Only 5–20% of patients convert to dementia each year, and 40–70% of MCI patients have not progressed to dementia ten years later. For a review of MCI, see Langa and Levine (2014).

Presymptomatic diagnosis and biomarkers

There is increasing evidence that dementia, especially Alzheimer’s disease, can be detected premorbidly, long before overt symptoms of any kind (including MCI) are present. The term *preclinical Alzheimer’s disease* is used to describe people with evidence of Alzheimer’s pathology (from biomarker studies, discussed below) but who are cognitively intact.

Table 14.4 Biomarkers of preclinical Alzheimer’s disease

Clinical	Comments
Neuropsychological deficits	E.g. in verbal episodic memory
Olfactory deficits	
Biochemical	
Cerebrospinal fluid or blood	
Amyloid Aβ42 or Aβ42/Aβ40 ratio	Correlates with cortical amyloid deposition
Phosphorylated tau	Correlates with neurofibrillary changes
Brain imaging	
Hippocampal atrophy	More apparent and progressive than in controls
Fluorodeoxyglucose PET	Normal scan has high negative predictive value
Amyloid PET	Normal scan has high negative predictive value
Tau PET	Not yet in clinical use

Features and biomarkers of the preclinical phase of Alzheimer’s disease are listed in Table 14.4. Longitudinal studies show that there are selective and characteristic neuropsychological impairments detectable up to 20 years before the onset of symptoms. Olfactory deficits are also a simple and valuable marker (Devanand et al., 2015). Hippocampal atrophy, either determined visually or quantitatively, using CT or MRI, is a useful imaging biomarker. The most important recent development is that the blood biomarkers mentioned earlier, especially p-tau217 and the β-amyloid Aβ42/Aβ40 ratio, are predictors of future Alzheimer’s disease dementia, and significantly more accurate than clinicians’ judgement (Palmqvist et al., 2021). For a review of biomarkers in preclinical and early dementia, see Teunissen et al. (2022).

Risk factors for dementia

There are a range of demographic and environmental risk factors for dementia that complement those for specific dementias discussed below (Livingston et al., 2020). The twelve most established environmental factors are

listed in Box 14.11, together with the relative risk each confers for dementia. Notably, together they account for 40% of the worldwide risk, and highlight that potential for prevention of dementia by treating modifiable life-style factors (Kivipelto et al., 2018). A few comments on the risk factors are warranted:

- Depression is not the only psychiatric disorder associated with subsequent risk of dementia. A 30-year population study using hospital records found a relative risk of 4.2, and a five-year earlier age at onset of dementia for people with any psychiatric diagnosis, with rates particularly high for psychotic disorders (Richmond-Rakerd et al., 2022).
- The strong evidence that higher levels of educational attainment, as well as cognitive activity, are protective, supports a ‘use it or lose it’ hypothesis, whereby such attributes are thought to increase cerebral reserve.
- Users of non-steroidal anti-inflammatory drugs, hormone replacement therapy, and statins, have lower rates of dementia and Alzheimer’s disease, and for each class of drug there is also a plausible biological explanation for this effect. However, randomized trials have failed to find any beneficial effect of these drugs in preventing dementia, and hence these drugs should not be considered to be established protective factors.
- Regarding protective factors, there is now strong evidence that vaccination against herpes zoster is associated with a reduced risk of dementia, although caution is still needed in assuming causality (Taquet et al., 2024).
- There are also ethnic differences in dementia risk, with higher rates in Black and lower rates in Asian groups compared to white (Shiekh et al., 2021).

Alzheimer’s disease

In 1907, Alois Alzheimer reported the case of Auguste D, a woman with presenile dementia whose brain exhibited unusual neuropathological features. It was Alzheimer’s colleague, Emil Kraepelin, who named the disease (Maurer et al., 1997). For many years the disease was thought to be rare and limited to presenile forms of dementia, but a seminal report by Roth and colleagues (Blessed et al., 1968) suggested that it was in fact the most common cause of senile dementia, a conclusion confirmed by many subsequent studies. About 60% of dementia is attributable to Alzheimer’s disease occurring either in isolation or together with features of vascular dementia and other neurodegenerative disorders.

Box 14.11 Risk factors for dementia

Age

Female sex

Early life

Less education (1.6)

Mid-life

Hypertension (1.6)

Obesity (body mass index >30) (1.6)

Alcohol (>21 units per week) (1.2)

Hearing loss (1.9)

Traumatic brain injury (1.8)

Later-life

Smoking (1.6)

Depression (1.9)

Social isolation (1.6)

Physical inactivity (1.4)

Diabetes (1.5)

Air pollution (1.1)

Numbers in brackets show relative risk for dementia.

Adapted from Livingston, G., Huntley, J., Sommerlad, A., Ames, D., Ballard, C., Banerjee, S., Brayne, C., Burns, A., Cohen-Mansfield, J., & Cooper, C. (2020). Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. *Lancet*, 396(10248), 413–446. Copyright (2020), with permission from Elsevier.

See Chapter 19 for the epidemiology and treatment of Alzheimer’s disease.

For a review of Alzheimer’s disease, see Scheltens et al. (2021).

Clinical features

The main features of dementia have been described above, and Box 14.12 summarizes the clinical features of Alzheimer’s disease (strictly, ‘dementia of the Alzheimer type’, since formal diagnosis awaits neuropathology).

The first evidence of the condition is often minor forgetfulness, which may be difficult to distinguish from normal ageing. The condition progresses gradually for the first two to four years, with increasing memory disturbance and lack of spontaneity. Memory is lost for recent events first. Language is usually affected early on, with difficulty in finding words or naming objects, and impairments in the ability to construct fluent sentences. Visuospatial skills may be affected, with difficulties in tasks such as copying pictures or learning the way round unfamiliar environments (e.g. when on holiday or in an unfamiliar house). Disorientation in time gives rise to

Box 14.12 Key clinical features of Alzheimer's disease

Core features

Memory impairment (amnesia), with gradual onset and continuing decline
 Aphasia
 Apraxia
 Agnosia
 Anosmia
 Disturbance in executive functioning (e.g. planning, reasoning)

Other features

Depression
 Psychosis
 Behavioural symptoms (e.g. agitation, wandering)
 Personality change

poorly kept appointments and changes in the diurnal pattern of activity.

Depression

The relationship between Alzheimer's disease and depression is complex. Depression is a risk factor for the disease, may be confused with it, or may occur as part of the syndrome. Regarding the latter point, major depression occurs in about 10% of cases, with less marked episodes and symptoms occurring in over 50% of cases. Patients who experience depression have greater decreases in serotonin and noradrenergic markers than other patients with Alzheimer's disease.

Psychotic symptoms

Delusions and hallucinations occur in a significant minority of patients at some stage in the illness. Their prevalence is unclear, as many studies did not distinguish Alzheimer's disease from dementia with Lewy bodies (see below), but estimates suggest rates of 10–50% for delusions and 10–25% for hallucinations. Importantly, the psychotic symptoms are often transient, which has implications for their treatment, especially given the risks of antipsychotic drugs in this population (see Chapter 19).

Behaviour

Changes in behaviour are common, and are of particular concern to carers. The patient may be restless and wake at night, disorientated, and perplexed. Motor activity

may increase in the evening ('*sundowning*'), and eventually the sleep–wake cycle may become completely disorganized. *Aggression* (both verbal and physical) is common, and often takes the form of resistance to help with personal care. Serious physical violence towards others is rare. Both increases and reductions in level of activity are common, involving varying degrees of purposefulness. *Wandering* can refer to a variety of different behaviours, but patients may place themselves at risk by going into unsafe environments. Patients with dementia may undereat or overeat, with associated changes in weight and nutritional state. Changes in sexual behaviour occur, usually with a reduction in drive, although sexual disinhibition occasionally occurs.

Self-care and social behaviour decline, although some patients maintain a good social facade despite severe cognitive impairment, particularly if carers are able to assist with these functions.

Course

In the early stages of Alzheimer's disease, the clinical features are modified by the patient's premorbid personality, and their traits tend to be exaggerated. In the middle and later stages of the illness, the cognitive impairments increasingly predominate, together with the neurological and behavioural features noted above. Incidental physical illness may cause a superimposed delirium, resulting in a sudden deterioration in cognitive function. Median survival from diagnosis is five to seven years, and is slightly less in men than in women.

Investigations

The investigation of suspected Alzheimer's disease follows the same principles as the investigation of dementia in general outlined above. In addition, a comment on genetic testing is warranted. Such testing is indicated in the very rare cases of familial early onset Alzheimer's disease, for which three causative genes are known (see below). Although the apoE4 allele of the apolipoprotein E gene is a major risk factor in all forms of Alzheimer's disease (see below), routine testing for this variant in the differential diagnosis of dementia is not recommended, and there is no place for genetic screening of healthy subjects to predict future dementia.

Neuropathology

On gross examination the brain is shrunken, with widened sulci and enlarged ventricles. Brain weight is reduced. On microscopic examination, the cardinal diagnostic features are *neurofibrillary tangles* and *senile plaques* (also called *amyloid plaques*) in the cerebral cortex and many subcortical regions. The diagnostic

criteria are based upon the abundance and distribution of plaques and tangles—the Consortium to Establish a Registry for Alzheimer’s Disease (CERAD) criteria (Mirra et al., 1991). However, it should be noted that these features are also seen, to some extent, in some non-demented elderly people, and the relationship between clinical dementia and neuropathology is more complex than is sometimes assumed to be the case. A revised definition of Alzheimer’s disease, called the ATN classification, that combines clinical criteria with findings from disease biomarkers (A: amyloid; T: tau; N: neurodegeneration) is currently under debate (Jack et al., 2018).

In addition to the tangles and plaques, there is selective loss of neurons in the hippocampus and entorhinal cortex, proliferation of astrocytes (gliosis), and loss of synapses. The latter is the strongest neuropathological correlate of cognitive impairment. Other histological findings include amyloid deposits in blood vessel walls (so-called *vascular amyloid* or *congophilic angiopathy*), *Hirano bodies* (intracellular crystalline deposits) and *granulovacuolar accumulation* (vacuoles or ‘holes’ within neurons).

For a review of the neuropathology of Alzheimer’s disease, see DeTure and Dickson (2019).

Progression of neuropathology

The disease process starts in the entorhinal cortex, before spreading to the hippocampus, association areas of the parietal lobe, and some subcortical nuclei. Six neuropathological stages based upon β -amyloid deposition are recognized, called *Braak stages*, which correlate with clinical severity (Braak and Braak, 1991), and which complement the evidence that the onset of neuropathology long precedes the symptoms of dementia (Braak and Del Tredici, 2015) and consistent with the preclinical features noted earlier. The spread of pathology along cortico-cortical projections leads to an effective ‘disconnection’ between affected regions.

Senile (amyloid) plaques

Senile plaques are deposits of insoluble proteins, together with degenerating neurites (neuronal processes) and glia. They occur in the space between neurons (the neuropil). Both neuritic and diffuse plaques are recognized, depending on their appearance using silver stains; neuritic plaques have a dense-staining core, whereas diffuse plaques have been likened to cotton wool. The neuritic plaques are pathologically more significant. The protein at the heart of senile plaques is β -amyloid (also called amyloid- β , A β , or β A4), a 39–42-amino acid peptide that is a fragment of the amyloid precursor protein (APP)

gene. Both are central to pathophysiological theories of the disease.

Neurofibrillary tangles

Neurofibrillary tangles occur within the cell body of neurons, especially pyramidal neurons of the cerebral cortex and hippocampus. They are formed of paired helical filaments, comprised of the microtubule-associated protein tau. The normal function of tau is in axonal transport and maintenance of the neuronal cytoskeleton. Tangles are thought to occur because tau becomes hyperphosphorylated, rendering it insoluble. The presence of a tangle causes dysfunction and death of the neuron.

Aetiology and pathogenesis

Genes

In a small number of families with an early onset of illness (before the age of 60 years), there is an autosomal dominant mode of inheritance. Causative mutations have been identified in three genes—amyloid precursor protein (*APP*, on chromosome 21), presenilin 1 (*PSEN1*, on chromosome 14), and presenilin 2 (*PSEN2*, on chromosome 1). Discovery of *APP* as the first ‘Alzheimer gene’ was a seminal event (see Box 14.13). Different mutations are known in each gene, and the age of onset, features, and progression of disease vary depending on the causative mutation.

Despite the importance of these findings, the vast majority of Alzheimer’s disease is not inherited in a Mendelian fashion, and it is often termed ‘sporadic’. However, first-degree relatives of patients with late-onset Alzheimer’s disease have an elevated risk of developing the disorder, and a genetic predisposition is now confirmed by the unequivocal association between polymorphisms in the apolipoprotein E (*apoE*) gene and all forms of Alzheimer’s disease. The heritability is estimated at 60–70% (Scheltens et al., 2021).

As discussed in Box 5.4, the *apoE4* variant of this gene accounts for about 50% of the vulnerability to late-onset Alzheimer’s disease. Its main effect is to promote an earlier onset of disease, by about a decade. *ApoE4* heterozygotes have a threefold greater risk, and homozygotes have an eightfold greater risk of the disease. Other data suggest that *apoE4*-positive cases may show some differences from *apoE4*-negative cases in terms of pathology, course, symptoms, and response to medication. In Caucasians, the *apoE2* variant reduces the risk of Alzheimer’s disease (compared with the commonest *apoE3* form), but this does not hold in all ethnic groups. The (lack of a) role of genetic testing for *apoE4* was mentioned above. It is stressed that *apoE4* is not a determinant of disease; at least one third of patients with Alzheimer’s disease are

Box 14.13 Discovery of *APP* gene mutations in familial Alzheimer's disease

In 1984, the protein that accumulates in senile plaques, cerebral vasculature, and meninges in Alzheimer's disease was discovered to be the β -amyloid peptide. Knowledge of its amino acid sequence allowed the encoding gene, called amyloid precursor protein (*APP*), to be identified and localized to chromosome 21. It was already known that there was a relationship between Alzheimer's disease and Down syndrome (trisomy 21). These observations together made *APP* a 'candidate gene' for Alzheimer's disease in the absence of Down syndrome. Researchers studied families with autosomal dominant presenile Alzheimer's disease to test this hypothesis. The work culminated in the landmark discovery in 1991 of an *APP* mutation (a point mutation changing valine to isoleucine at position 717), which caused Alzheimer's disease (Goate et al., 1991). Since then, different *APP* mutations have been found in other families. *APP* mutations are pathogenic primarily because they affect metabolism of *APP*, promoting the formation of the β -amyloid fragments that accumulate and contribute to the characteristic amyloid pathology described in the text. The other two known genes causing familial Alzheimer's disease are the presenilins (*PSEN1* and *PSEN2*), which provide the catalytic subunit for the enzyme that cleaves *APP*, called γ -secretase. Hence, all three genes share a common influence on *APP* processing. *APP* and *PSEN* mutations explain most cases of familial Alzheimer's disease, but only a tiny fraction (<0.1%) of all cases of Alzheimer's disease. However, β -amyloid is thought to be involved in all forms of Alzheimer's disease, as described in the text.

apoE4-negative, and some *apoE4* homozygotes never develop the disease. It is not known how *apoE4* elevates the risk of Alzheimer's disease, but it is likely to reflect its interaction with β -amyloid metabolism and clearance from the brain, cholesterol metabolism, and other cellular functions (Yamazaki et al., 2019).

There are at least 40 other susceptibility genetic loci for Alzheimer's disease in addition to *apoE* (Andrews et al., 2020). Most of the implicated genes contain common variants with small effects on disease risk (e.g. *CLU*, *PICALM*), while a few genes (e.g. *TREM2*) have rarer variants with larger effect sizes. These genes suggest

microglial cells play a key role, together with pathways involved in immune function and inflammatory responses. For a review, see Scheltens et al. (2021).

Environmental factors

The environmental risk factors for Alzheimer's disease are essentially those listed in Box 14.11 applying to dementia in general. However, it should be noted that neuropathological diagnoses are rarely available in epidemiological studies and so some factors may be more or less relevant to Alzheimer's disease.

The amyloid cascade hypothesis

The *APP* research summarized in Box 14.13 led to the 'amyloid cascade hypothesis', originated by Hardy and Higgins (1992), which became the leading molecular model for the disorder. It proposes that the central pathogenic event is increased formation and deposition of β -amyloid, particularly the 42-amino acid variant. *APP* is a transmembrane protein, which can be cleaved by one of three enzymes, called secretases. Normally, α -secretase activity predominates, and this does not give rise to β -amyloid. However, in Alzheimer's disease, more *APP* is processed via β - and γ -secretase pathways, leading to increased β -amyloid formation. The increased production of β -amyloid leads to disease because it tends to aggregate, becomes insoluble, and is toxic to synapses and neurons. Decreased removal of β -amyloid from the brain also contributes. The other pathological changes of Alzheimer's disease, including neurofibrillary tangles and the involvement of tau, are all seen as downstream in the causal process.

Although the amyloid cascade hypothesis is a convincing explanation of familial Alzheimer's disease, its relevance to sporadic forms of the disease remains controversial for various reasons, not least the failure of treatments targeting β -amyloid to show clearer therapeutic benefits in the disease (Frisoni et al., 2022). It is currently thought that β -amyloid is only one component of a more complex, non-linear, process, which also involves tau (Chang et al., 2021), and a range of other molecules and events (Scheltens et al., 2021).

Inflammatory mechanisms

Neuroinflammation, including microglial activation, was known to be a component of the neuropathology of Alzheimer's disease, but until recently was generally considered to be a response to ongoing neurodegeneration. However, inflammatory, and immune, factors are now thought to be an important part of the disease process itself. The evidence is diverse. It includes interactions with β -amyloid, and the fact that *TREM2*, one of the risk genes for Alzheimer's disease mentioned above,

is a microglial receptor involved in β -amyloid clearance. Alterations in the glymphatic drainage of the brain, cerebral blood vessels, and the blood–brain barrier, may all contribute. For a review, see Long and Holtzman (2019).

The cholinergic hypothesis

Prior to the amyloid hypothesis, the prevailing view was the *cholinergic hypothesis* of Alzheimer's disease, based on findings in the 1970s that there is a severe and widespread loss of acetylcholine in the cerebral cortex. The loss occurs because of pathology and atrophy in the cells of origin, in the nucleus basalis of Meynert. The findings led to the development of the current cholinergic therapies for the disease. However, although cholinergic deficits may well explain some of the symptoms, they do not appear to have a primary causal role. Cholinergic pathology may in fact be of more relevance in dementia with Lewy bodies, and it is possible that the earlier studies of Alzheimer's disease included patients in this category. For a review, see Hampel et al. (2018).

Infectious agents

Another long-standing proposal has been that some infectious agents, particularly reactivation of herpes simplex virus (HSV-1), play a role in the pathogenesis of Alzheimer's disease (Itzhaki et al., 2020). This view has received renewed interest with the increasing evidence for a protective effect of herpes zoster vaccination against dementia (Taquet et al., 2024).

Vascular dementia

The dementia caused by cerebrovascular disease was in the past referred to as 'atherosclerotic psychosis'. Following the separation of distinct syndromes of psychiatric disorder in late life (Roth, 1955), it became apparent that dementia was often associated with multiple infarcts, and Hachinski et al. (1974) suggested the term *multi-infarct dementia*. Subsequent research has shown that these patients are in fact a small subgroup of those with cognitive impairment due to cerebrovascular disease, and the umbrella term of *vascular dementia* (or *vascular cognitive impairment*) is now preferred, within which a range of subtypes is recognized (see Box 14.14). Small-vessel disease affecting subcortical regions is now considered the most common form of vascular dementia. However, the neuropathological findings of vascular dementia are very varied, and include embolus, vasculitis, angiopathy, and haemorrhage, as well as infarcts.

For a review, see Iadecola et al. (2019).

Vascular dementia is usually said to be the second commonest cause of dementia after Alzheimer's disease,

Box 14.14 Subtypes of vascular dementia

Multi-infarct dementia (cortical vascular dementia)
Small-vessel dementia (subcortical vascular dementia; includes Binswanger's disease)
Hypoperfusion dementia
Haemorrhagic dementia
Hereditary vascular dementia (CADASIL: cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy)

of comparable prevalence to dementia with Lewy bodies. However, there is considerable uncertainty over these figures, because of the pathological heterogeneity of vascular changes, the absence of accepted neuropathological criteria for vascular dementia, and their frequent coexistence with Alzheimer-type pathology (Anderle et al., 2025).

Clinical features

Onset is usually in the late sixties or seventies. With large-vessel disease, onset is often relatively acute and follows a stroke, in which case the term *post-stroke dementia* is sometimes used. Emotional and personality changes can appear first, followed by impairments of memory and intellect that characteristically progress in stages. Depression is frequent, and episodes of emotional lability and confusion are common, especially at night. Transient ischaemic attacks or mild strokes may recur from time to time. Insight is often maintained until a late stage. Apathy, behavioural retardation, and anxiety are more common than in Alzheimer's disease, but cognitive impairment itself is more variable and often of lesser degree. Attention, information processing, and executive function are typically most affected, and tests that focus on these domains (such as the MoCA, or vascular dementia assessment scale, VADAS-cog) are more likely to identify deficits than the MMSE.

The course of vascular dementia is classically a step-wise progression, with periods of deterioration that are sometimes followed by partial recovery for a few months. About 50% of patients die from ischaemic heart disease, and others from cerebral infarction. From the time of diagnosis, lifespan varies widely, but most studies show somewhat shorter survival than in Alzheimer's disease.

The diagnosis is difficult to make with confidence unless there is a clear history of strokes or localizing neurological signs. Suggestive features are patchy

Box 14.15 Risk factors for vascular dementia

History of stroke—risk related to number, location, and size
 Increasing age—risk doubles every five years
 Low education
 Hypertension
 Orthostatic hypotension
 Evidence of vascular disease (e.g. carotid bruit, cardiac disease)
 Vascular risk factors (e.g. smoking, hyperlipidaemia, obesity)
 High blood homocysteine
 Late-life depression

psychological deficits, erratic progression, and relative preservation of the personality. On physical examination there may be signs of hypertension and of arteriosclerosis in peripheral and retinal vessels, and there may be neurological signs such as pseudobulbar palsy, rigidity, akinesia, and brisk reflexes. CT or MRI imaging are useful for demonstrating the nature and severity of vascular pathology. CT is sufficient to show established infarcts and large areas of white matter damage, but MRI is more sensitive in revealing the full profile of involvement.

Aetiology

It is assumed that vascular dementia results from the neuronal dysfunction and death, and the damage to white matter tracts, that follow from the cumulative focal areas of ischaemia and necrosis (Iadecola et al., 2019). Unsurprisingly, a history of stroke is a strong risk factor (Kuzma et al., 2018), together with other risk factors for cerebrovascular disease and dementia (Box 14.15). The overlap of these risk factors with those posing risk for Alzheimer's disease is consistent with the frequent co-occurrence of the two pathologies and the emerging concept of mixed dementia (see below). The main genetic form of vascular dementia is the rare autosomal dominant disorder CADASIL, caused by a mutation in the *Notch* gene on chromosome 19. There are also over 50 other genetic loci contributing to risk of cerebral blood vessel disease (Bordes et al., 2022).

Dementia with Lewy bodies

A hitherto unrecognized type of dementia with a relatively distinct pathology and clinical course was described

Box 14.16 Terminology of dementia involving Lewy bodies

Dementia with Lewy bodies: dementia occurring before or concurrently with, or within a year of the onset of, parkinsonism.

Parkinson's disease dementia: dementia occurring a year or more after the onset of established Parkinson's disease

Lewy body dementias: umbrella term that includes both the above categories

Lewy body disease: a pathological diagnosis, based on the presence and distribution of Lewy body-type pathology

Adapted from Walker, Z., Possin, K. L., Boeve, B. F., & Aarsland, D. (2015). Lewy body dementias. *Lancet*, 386(10004), 1683–1697. Copyright (2015), with permission from Elsevier.

in the late 1980s using various terms, including cortical Lewy body disease, diffuse Lewy body disease, and Lewy body variant of Alzheimer's disease. *Dementia with Lewy bodies* is now the consensus term. Box 14.16 summarizes some relevant terminology. The disorder accounts for 4–25% of cases of dementia; the wide range of estimates (and the belated 'discovery' of dementia with Lewy bodies) probably reflects sampling bias in some studies (e.g. underrepresentation in patients coming to autopsy) and the fact that cortical Lewy bodies are difficult to see using routine neuropathological stains.

For a review, see Arnaoutoglou et al. (2019).

Clinical features

The core clinical features, which form the recognized criteria for diagnosis of dementia with Lewy bodies, are summarized in Box 14.17 (McKeith et al., 2017). The fluctuating level of dementia with recurrent delirium-like phases, together with visual hallucinations and parkinsonism, is characteristic, although the motor features are not always apparent. Use of these criteria allows dementia with Lewy bodies to be diagnosed with good specificity and sensitivity, but it remains underdiagnosed in routine clinical practice. The average survival is five to seven years.

Neuropathology

The characteristic histopathological feature is the presence of Lewy bodies in the cerebral cortex. Lewy neurites

Box 14.17 Clinical criteria for dementia with Lewy bodies

Key features

Progressive cognitive decline, especially in attention and visuospatial ability
Pronounced fluctuations in cognition and attention
Recurrent visual hallucinations, usually well-formed and detailed
Motor features of parkinsonism

Supportive features

Repeated falls
Syncope
Transient loss of consciousness
Systematized delusions
Non-visual hallucinations
Adverse reactions to antipsychotics

Diagnosis made less likely in the presence of

Evidence of cerebrovascular disease or stroke
Evidence of other disorder sufficient to account for the clinical picture

are also seen. As in Parkinson's disease, these features are also seen in the substantia nigra, and their biochemical composition is the same in both diseases. Lewy bodies are intraneuronal inclusions, composed mainly of abnormal aggregations of a protein called α -synuclein. They also contain other proteins, including ubiquitin, detection of which is widely used in the routine detection of Lewy bodies. There is also a significant neuropathological overlap with Alzheimer's disease, with dementia with Lewy bodies often exhibiting abundant senile plaques and widespread reductions in choline acetyltransferase in the neocortex; however, neurofibrillary tangles are rare. Brain atrophy is less marked than in Alzheimer's disease, especially in the hippocampus.

Aetiology

Dementia with Lewy bodies is closely related to Parkinson's disease, and both are characterized as 'synucleinopathies', reflecting the abnormal aggregation of α -synuclein protein present in Lewy bodies. There is a genetic predisposition, with an estimated heritability of 36%, and associations with *apoE4* and some of the other genes implicated in Parkinson's disease and Alzheimer's

disease (Guerreiro et al., 2018). Like these disorders, there is also a long prodrome (up to 20 years), with olfactory deficits and sleep disturbance being early features (Fereshtehnejad et al., 2019). However, compared to the other common neurodegenerative disorders, the aetiology and pathophysiology of dementia with Lewy bodies remain poorly understood.

Frontotemporal dementias

Frontotemporal dementias are the second most common form of presenile dementia, accounting for about 40% of cases, and also underlie about 7% of late-life dementias. The term was proposed in 1994 when criteria were developed to better classify Pick's disease (see below) and related disorders, based upon their shared clinical features and regional neuropathology, as described below. The term *frontotemporal lobar degeneration* is the neuropathological counterpart

For review of frontotemporal dementia, see Sivasathiaselan et al. (2019).

Clinical features and subtypes

Frontotemporal dementia encompasses a range of neurodegenerative diseases affecting the frontal and temporal lobes and which are characterized clinically by progressive deficits in behaviour, language, and memory. The diagnostic terminology for frontotemporal dementia is complicated because of the clinical, neuropathological, and genetic heterogeneity; Box 14.18 lists some of

Box 14.18 Diagnostic categories subsumed within frontotemporal dementia

Pick's disease
Lobar atrophy
Frontal lobe degeneration of non-Alzheimer type
Dementia lacking distinctive histology
Semantic dementia
Motor neuron disease with dementia
Progressive non-fluent aphasia
Progressive aphasic syndrome
Frontotemporal dementia with parkinsonism
Corticobasal degeneration*
Progressive supranuclear palsy*
Parkinsonism–dementia complex of Guam*

* Not always included.

Box 14.19 Clinical features of frontotemporal dementias

Behavioural features

Insidious onset, slow progression
Early loss of insight
Early signs of disinhibition and lack of judgement
Mental inflexibility
Stereotyped and imitative behaviour
Hyperorality (e.g. craving for sweet foods)
Distractibility and impulsivity

Language and speech features

Progressive decrease in speech output
Perseveration
Echolalia

Affective features

Depression
Apathy
Emotional blunting
Hypochondriasis

Physical signs

Early primitive reflexes
Early incontinence
Late parkinsonism
Low and labile blood pressure

Box 14.20 Features characteristic of the three main clinical subtypes of frontotemporal dementia

Behavioural variant

Personality change
Disinhibition
Apathy
Selfishness
Self-neglect
Overeating
Behavioural stereotypies
Mental rigidity

Semantic variant primary progressive aphasia

Anomia (especially for nouns)
Word-finding difficulties
Impaired verbal comprehension
Increased pain responses
Prosopagnosia
Irritability

Non-fluent variant primary progressive aphasia

Slow, laboured speech
Misuse of grammar
Speech sound errors
Anomia (especially for verbs)
Irritability

Source: Bang, J., Spina, S., Miller, B. L. (2015). Frontotemporal dementia. *Lancet*, 386(10004), 1672–1682. Copyright (2015), with permission from Elsevier.

the older terms, including Pick's disease, which are now subsumed within the category. Compared to most other dementias, cognitive impairment is a relatively minor feature, with behavioural, psychiatric, and linguistic aspects being more prominent. There is a significant clinical and aetiological overlap with motor neuron disease, and with parkinsonian syndromes. *Progressive supranuclear palsy* (PSP) and *corticobasal syndrome* can be considered as frontotemporal dementias, or categorized separately; supranuclear gaze palsy is characteristic of PSP (Stamelou et al., 2021).

Based upon the clinical profile, two major subtypes of frontotemporal dementia are now recognized: *behavioural-variant frontotemporal dementia* and *primary progressive aphasia*. The latter is then divided into *semantic-variant* and *non-fluent variant* forms. All forms can exhibit the features shown in Box 14.19, but the

clinical features that characterize each subtype are summarized in Box 14.20. About one in eight patients with frontotemporal dementia develop motor neuron disease (and vice versa), and another 20–30% exhibit features of parkinsonism. As discussed below, the presence of these additional neurological elements is related to the neuropathological features and, in familial cases, to the causal gene.

Frontotemporal dementia is difficult to diagnose, since in the early stages it can easily be mistaken for depression or psychosis, and hard to distinguish from other forms of dementia. The history should concentrate

on the past psychiatric history, family history, and pre-morbid personality and functioning. The assessment should also include a careful evaluation of the behavioural features and language abilities. Physical examination of the motor system, looking for muscular weakness and parkinsonism, is also important. Structural brain imaging can play a useful role, showing focal, often asymmetrical, atrophy of the temporal and frontal poles (behavioural and language symptoms being associated with right and left hemisphere involvement, respectively). Functional imaging shows disproportionate hypometabolism and hypoperfusion in these regions.

The average survival of frontotemporal dementia is six to eleven years from symptom onset (and about two to six years from diagnosis); patients with concurrent motor neuron disease have the worst prognosis.

Pick's disease

Arnold Pick described the disease that bears his name in 1892. Pick's disease remains the archetypal frontotemporal dementia, and is sometimes referred to synonymously with it but, if defined properly (including the neuropathological features), it explains only a small proportion of cases. The clinical features that he noted were of aphasia with dementia, such that the case would probably be diagnosed today as a primary progressive aphasia. The pathological findings of Pick's original case were reported by Alzheimer, and were striking, with focal 'knife-blade' atrophy of the frontal and temporal poles, with ballooned neurons (Pick cells) that contained inclusions called Pick bodies. Pick bodies also occur in other areas, notably the hippocampus. They are composed of tau, ubiquitin, and other proteins.

Neuropathology

The classic features of Pick's disease have been summarized above. Other forms of frontotemporal dementia also show gross atrophy of the temporal and frontal lobes, with neuronal loss, gliosis, microvacuolar changes, and often ballooned cells. Different neuropathological subtypes of frontotemporal dementia are now recognized, based primarily upon the identity of the protein which is aberrantly deposited: tau, TDP-43, or FUS. The 'proteinopathy' spreads across neuronal circuits, as described for other neurodegenerative disorders. There is some correspondence between clinical and pathological subtypes of frontotemporal dementia, for example, most cases of semantic variant primary progressive aphasia have TDP-43 pathology, whereas tau is seen in non-fluent primary progressive aphasia (Hodges and Piguet, 2018).

Aetiology

About 10% of cases of frontotemporal dementia are caused by autosomal dominant mutations, with three genes (*C9orf72*, *MAPT* [which encodes tau], and *GRN* [which encodes progranulin]) accounting for about 80% of these cases. In Caucasian populations, *C9orf72* is the most common cause, particularly for frontotemporal dementia with motor neuron disease. The *C9orf72* mutations are hexanucleotide repeats (a repeated sequence of six nucleotides, which occurs up to 20 times in the normal gene, but hundreds of times in those with disease). The *MAPT* mutations disrupt the expression and structure of tau, and the *GRN* mutations are null, preventing production of the encoded progranulin protein. Each of the three genes is associated with a particular neuropathological profile (for example, *C9orf72* cases have TDP-43 neuropathology), but less strongly with the clinical picture, although *C9orf72* cases are associated with prominent psychiatric symptoms, including anxiety and somatically focused delusions.

The pathogenic mechanisms by which the various mutations cause neurodegeneration are uncertain, and may be different for each gene. Genetic testing is now available and should be considered for all patients with frontotemporal dementia with a family history suggestive of autosomal dominant inheritance of neurodegenerative disorders. For a review, see Hodges and Piguet (2018).

The causes of the more common non-familial forms of frontotemporal dementia are not clear, beyond a genetic predisposition that includes the *TMEM106B* gene. PSP and corticobasal syndrome are associated with a specific form of tauopathy (Stamellou et al., 2021).

Prion diseases

Prion diseases are a category of neurodegenerative disorders that is unique for two reasons. First, because of the central causal role of a specific protein, called prion protein (PrP). Second, because they are infectious. They also have characteristic neuropathological features, notably diffuse spongiosis (hence the older term, '*spongiform encephalopathy*'), marked neuronal loss and gliosis, and, in many cases, amyloid plaques. In addition to being acquired via infection (including iatrogenically), other cases are genetic while others are idiopathic. Prion diseases are the most important of the rapidly progressive dementias (Herrmann and Zerr, 2022). Because of these properties, their interest and importance are out of all proportion to their rarity (about 100 cases per year in the United Kingdom). In the United Kingdom, all cases

of suspected prion disease should be referred to the National Surveillance Centre in Edinburgh.

For a review of prion diseases, see Nihat (2020).

Creutzfeldt–Jakob disease

Creutzfeldt–Jakob disease (CJD) is the main prion disease, with an approximate annual incidence of one case per million. A small number of cases are inherited as an autosomal dominant disorder due to mutations in the *PrP* gene (see below). Other cases have been transmitted iatrogenically, via pituitary-derived growth hormone, contaminated neurosurgical instruments or graft material, and possibly by blood transfusion. The incubation time in these cases can exceed 20 years. Because of the potential infectivity, no patients with dementia should be organ donors.

Sporadic CJD affects both sexes equally. Onset is typically between 50 and 65 years of age. It is usually heralded by memory impairment, which may be accompanied by prominent behavioural abnormalities or personality change, prompting initial referral to a psychiatrist. Visual hallucinations, cerebellar signs, involuntary movements, myoclonic jerks, ataxia, and other motor features are frequent. Seizures occur later in the course. There is usually a relentless and rapid progression to death, often within six months.

Variant Creutzfeldt–Jakob disease

Intense interest in prion disease in the United Kingdom followed the description of variant Creutzfeldt–Jakob disease (vCJD) in 1996. It was linked with bovine spongiform encephalopathy (BSE; ‘mad cow disease’), a prion disease which was epidemic in British dairy herds at that time, and it is beyond reasonable doubt that vCJD occurred as a result of eating contaminated bovine products. The incidence of vCJD peaked around the year 2000, with only one new case diagnosed since 2013 and a total of 178 confirmed cases. It is now considered highly unlikely that a human epidemic will occur, as some had feared. There is a polymorphism at codon 129 of the *PrP* gene that encodes either methionine or valine; all vCJD patients except one have been homozygous for methionine (compared with 40% in the population), which suggests that this genetic subgroup is more susceptible to development of vCJD.

Other prion diseases

Kuru, described in the Fore tribe of New Guinea highlanders, was transmitted by ritual cannibalism; the disease has disappeared since this practice was abolished in the

1950s. Transmission of kuru to monkeys in 1970 was the first experimental proof of the infectivity of prion disease.

Other inherited prion diseases are Gerstmann–Sträussler–Scheinker syndrome and fatal familial insomnia. Like familial CJD, both of these are caused by autosomal dominant mutations of the *PrP* gene. Both are extremely rare.

Investigation of suspected prion disease

Several tests are useful in diagnosing prion disease, and have almost entirely superseded the requirement for brain biopsy, which had been the definitive test. The EEG classically shows a triphasic 1–2 Hz discharge. MRI, using diffusion weighted imaging, typically reveals hyperintensities in the pulvinar thalamic nuclei and some other regions. CSF analysis is valuable, showing an increase in 14-3-3 and tau proteins. Potentially definitive diagnostic tests, measuring formation of the pathogenic forms of PrP in CSF or other tissues (see below), are under development. For a review of diagnostic investigations for prion disease, see Zerr (2022).

Aetiology

The name ‘prion’ denotes proteinaceous infectious particles. It was coined in 1982 by Stanley Prusiner, who received the 1999 Nobel Prize for Medicine. PrP is encoded by a gene on chromosome 20. The functions of ‘normal’ PrP (denoted PrP^C) are unknown, but it is expressed by neurons and may serve as a receptor and influence synaptic properties. Prion diseases are caused when PrP^C takes on an abnormal conformation, called PrP^{Sc} (named after scrapie, the prion disease that affects sheep). PrP^{Sc} is both the core molecular marker and presumed causative agent of prion diseases. Compared with PrP^C, PrP^{Sc} is resistant to breakdown by proteases, and tends to self-aggregate and be deposited in the brain. PrP^{Sc} can spread trans-synaptically from neuron to neuron, thereby promoting synaptic and neuronal loss and gliosis, and producing the resulting spongiform appearance of the brain.

In familial cases, the mutated PrP is assumed to be intrinsically more likely to self-aggregate. In forms that are acquired (either iatrogenically or through diet), it is postulated that the normal PrP^C becomes ‘corrupted’ by the acquired PrP^{Sc}, which changes the conformation of PrP^C into PrP^{Sc}, thus propagating ever more PrP^{Sc}. The molecular details of this remarkable process remain obscure. In the case of peripheral acquisition, PrP^{Sc} may spread via the nerves, lymphatics, and blood to reach the brain. Different PrP conformations and

modifications (glycosylation patterns) give rise to various disease ‘strains’. The aetiology of ‘sporadic’ prion disease is not known. It may result from spontaneous mutation or conversion of PrP^C to PrP^{Sc}, or it may arise from occult environmental sources. It is not known why PrP^{Sc} is infectious, when other misfolded, amyloidogenic proteins (such as β -amyloid and α -synuclein) do not appear to be. Prion diseases are not caused by ‘slow viruses’, as was formerly believed.

Limbic-predominant age-related TDP-43 encephalopathy (LATE)

LATE is a recently described dementia, in which the accumulation of TDP-43 (a protein described above in frontotemporal dementia), is seen in very elderly patients (typically, over 85 years), especially in the hippocampus (Nelson et al., 2019). There may be coexisting amyloid plaques and tau pathology. The clinical picture is similar to the dementia of Alzheimer’s disease. It is suggested that LATE may be relatively common, providing the neuropathological basis for 25–30% of clinically diagnosed Alzheimer’s disease in advanced age (Wolk et al., 2019).

Dementia due to HIV disease

See Chapter 22.

Dementia due to alcohol misuse

See Chapter 20.

Emerging concepts of dementia

Classification and clinical practice are based on the dementia syndromes described above. However,

epidemiological and biological research is increasingly calling current diagnostic concepts into question.

Mixed dementia

The frequency and importance of ‘mixed dementia’ are becoming apparent, particularly the coexistence of Alzheimer’s disease and vascular dementia. This occurs in at least 20% of cases (and in the majority of cases if lower thresholds for cerebrovascular disease are used). Indeed, in the over 80s, mixed dementia ‘is the norm not the exception’ (O’Brien and Thomas, 2017). The clinical overlap is now complemented by the increasing evidence that risk factors and disease mechanisms for Alzheimer’s disease overlap with those for vascular dementia (Saito and Ihara, 2016).

Classification of neurodegenerative diseases as proteinopathies

There is further blurring of diagnostic boundaries at the molecular level, in that the same core biochemical processes—misfolding and accumulation of specific proteins, and their spread along neuronal pathways in the brain—appears to be central to many neurodegenerative disorders, including β -amyloid in Alzheimer’s disease, PrP in prion disease, and α -synuclein in Parkinson’s disease (Schechel and Aguzzi, 2018). Equally, it is apparent that there is not a one-to-one correspondence between protein and syndrome, as exemplified by the involvement of tau in both Alzheimer’s disease and some forms of frontotemporal dementia. These factors are leading to the molecular reclassification of dementias as proteinopathies, such as ‘tauopathies’ and ‘synucleinopathies’, etc. (Soto and Pritzkow, 2018)); since the proteins are often related to the synapse, they are also sometimes called ‘synaptopathies’ (Taddei and Duff, 2025). These issues are likely to influence diagnostic and clinical practice as their prognostic and treatment implications become clear.

Movement disorders

Parkinson’s disease, and other disorders in which movement and coordination are the cardinal features, are managed by neurologists. However, they often also have prominent and clinically significant psychiatric features, including cognitive impairment, psychosis, and mood disorders. The psychiatric aspects are considered here.

Parkinson’s disease

Clinical features

The cardinal triad of idiopathic Parkinson’s disease is a rest tremor, cog-wheel rigidity, and bradykinesia. The main psychiatric consequences are cognitive impairment, depression, psychosis, and behavioural problems

Box 14.21 Psychiatric manifestations in Parkinson's disease

Delirium, stupor (especially due to drugs, or intercurrent infection)

Cognitive decline (subcortical dementia, dysexecutive syndrome)

Depression, apathy, mania

Hallucinations (chiefly visual)

Delusions

Sleep attacks, REM sleep behaviour disorder

Sexual disorders

Impulse control disorders, e.g. gambling (largely medication-related)

(Weintraub et al., 2022; Box 14.21). Impaired olfaction is a neglected feature, and can pre-date other symptoms by several years; constipation, sleep disturbances, and depression are also common well before the onset of motor symptoms. The differential diagnosis includes other parkinsonian syndromes, and, of particular relevance to psychiatrists, the parkinsonian side effects of antipsychotic drugs. In clinicopathological terms, Parkinson's disease is part of the spectrum of Lewy body disease, discussed earlier.

For a review, see Bloem et al. (2021).

Dementia

The prevalence of cognitive impairment in Parkinson's disease is about 30%, with up to 80% of patients eventually developing dementia. The diagnosis is '*Parkinson's disease dementia*' if the dementia begins more than one year after the onset of established Parkinson's disease; if dementia occurs less than a year after Parkinson's disease, the diagnosis is *dementia with Lewy bodies* (Box 14.16). Dementia is more common in later-onset cases and those in whom bradykinesia is more prominent than tremor. The clinical picture is of a subcortical dementia, although other profiles of cognitive impairment are seen in Parkinson's disease, notably a 'dysexecutive syndrome' in which planning and working memory are especially affected. For treatment of dementia in Parkinson's disease, see Chapter 19.

Depression

The association of Parkinson's disease with depression is well established; clinically significant depressive

symptoms (including apathy) are present in 35% of cases, and a major depressive disorder in about 20%. Depression is most common in the early and the very advanced stages of the disease. The mechanism is uncertain; depression correlates poorly with degree of disability and disease duration, and may be related to frontal lobe abnormalities and disturbed dopaminergic mechanisms.

Psychotic symptoms

Psychotic symptoms occur at some stage in around 20% of patients. Visual hallucinations are associated with dopaminergic medication, increasing age, disease duration and severity, depression, cognitive impairment, and reduced visual acuity. Delusions are less frequent and are usually paranoid in content. Anti-parkinsonian drugs have been implicated as causes, and whenever possible these should be reduced. If antipsychotic medication is necessary, clozapine or quetiapine should be used.

Other neuropsychiatric manifestations

Excessive somnolence, a disordered sleep-wake cycle, sleep attacks, and REM sleep behaviour disorder are more common in (and before the onset of) Parkinson's disease. As a result, patients with Parkinson's disease must be counselled about the risks of driving, even in the absence of dementia. Pathological gambling and other impulsive behaviours can be problematic, especially in patients treated with dopamine agonists such as pramipexole.

Aetiology

Idiopathic Parkinson's disease is usually a disorder of later life, occurring in around 1% of the population over the age of 55 years, and rising rapidly with age thereafter. There is a significant and unexplained rise in incidence of the disease that has been likened to a pandemic (Bloem et al., 2021). It is more common in men than women, with a ratio of 3:2. The disease results primarily from degeneration of dopaminergic neurons in the zona compacta of the substantia nigra, although the first site of brain pathology is in the IX/X cranial nerve nuclei from where it spreads to other regions. The pathological hallmark is Lewy bodies (α -synuclein inclusions), as described for Lewy body dementia. The early pathology in the IX/X cranial nerve nuclei may reflect an origin of α -synuclein pathology in the gut and transport via the vagus nerve (Kim et al., 2019). The dementia of Parkinson's disease involves more widespread pathology, and includes β -amyloid and tau as well as α -synuclein. Pathophysiologically, the disease process is thought to involve impairment of

Box 14.22 Risk factors for Parkinson's disease

Genetic

Autosomal dominant: e.g. *SCNA*; *LRRK2*; *GBA*
Autosomal recessive: *PRKN*, *PINK1*, *PLA2G6*
Risk polymorphisms: *GBA*; *INPP5F*; and many others

Environmental risk factors

Pesticide exposure
Head injury
Rural living
Drinking well water
Agricultural occupation
Bipolar disorder

Environmental protective factors

Tobacco smoking
Coffee drinking
Physical activity
Use of NSAIDs
High serum urate

Source: data from Kalia, L. V., & Lang, A. E. (2015). Parkinson's disease. *Lancet*, 386(9996), 896–912. Copyright (2015), with permission from Elsevier.

mitochondrial and synaptic functioning, and oxidative stress. Neuroinflammation may also play a role. For review see Rocha et al. (2018).

Risk factors for Parkinson's disease are summarized in Box 14.22; 3–5% of cases, particularly those of early onset, are inherited as autosomal dominant or autosomal recessive disorders. Mutations have been identified in more than 20 genes, including *SCNA*, which encodes α -synuclein (Blauwendraat et al., 2020) and *GBA*. Cases due to *GBA* mutations are characterized by prominent and rapid cognitive decline. The large majority of Parkinson's disease cases are sporadic, reflecting a combination of a modest and complex genetic predisposition (with over 90 genomic loci identified), and a range of environmental factors, notably pesticide exposure and head injury. However, the interpretation of many of the environmental factors is unclear (Bloem et al., 2021). For example, tobacco smoking is associated with a decreased risk, but this appears to be due to people who are developing the disease being more able to quit smoking, rather than a true protective effect. A history

Box 14.23 Features of Huntington's disease

Motor abnormalities

Involuntary movements
Chorea
Fidgeting
Impairment of voluntary movements
Incoordination
Gaze apraxia
Failure to sustain tongue protrusion ('serpentine tongue')
Bradykinesia
Rigidity

Cognitive impairment

Cognitive slowing
Decreased attention
Reduced mental flexibility and planning
Impaired emotional recognition
Poor visuospatial function

Psychiatric and behavioural features

Depression
Irritability
Apathy
Delusions and hallucinations
Early loss of insight

of bipolar disorder is also a risk factor for Parkinson's disease (Harrison and Luciano, 2021).

Huntington's disease

This autosomal dominant disease, also called *Huntington's chorea*, was described by the New England physician George Huntington in 1872. It has a worldwide distribution, with an estimated prevalence of 4–7 per 100,000. Onset is typically in middle life, although adolescent cases are well recognized.

The clinical features of Huntington's disease are shown in Box 14.23. They comprise a triad of movement abnormalities, cognitive impairment, and psychiatric features. The classical choreiform movement abnormality is required for diagnosis, but significant cognitive impairment and psychiatric symptoms frequently occur in the decade before chorea appears (the 'prodrome' of

Huntington's disease), and individuals may therefore present to psychiatrists during this time. Apathy is commonly the first psychiatric manifestation. Most subjects will be aware of their family history, and many will have had predictive genetic testing (see below), and so the psychiatrist will need to deal with the resulting implications and concerns. After presentation, the motor and cognitive symptoms are invariable and progress inexorably, whereas the psychiatric symptoms are much more variable. Depression is seen in over 50% of cases at some stage, and irritability and apathy are also common. Psychotic symptoms, both schizophrenia-like and affective in nature, occur in about 10% of cases.

For a review, see McColgan and Tabrizi (2018).

Neuropathology

The pathological changes mainly affect the caudate nucleus and frontal lobes. The basal ganglia, especially the caudate nucleus, is markedly atrophic and gliotic. The shrinkage of the caudate can be seen using MRI; it begins many years before diagnosis and its extent correlates strongly with the motor and cognitive impairments. There is also some involvement of the frontal cortex, with thinning of the grey matter and pyramidal neuronal loss. Polyglutamine nuclear inclusions are present within some cells, reflecting the causative gene mutation described below. Neurochemically, there is a reduction of the inhibitory transmitter GABA in the caudate nucleus.

Aetiology

Huntington's disease is one of the few single gene, autosomal dominant disorders in psychiatry. The gene, on chromosome 4p, is called *huntingtin* (*HTT*). The mutation, identified in 1993, is a multiple repeat of the codon CAG, which codes for glutamine. It is therefore known as a 'triplet repeat', 'trinucleotide repeat', or 'CAG repeat' mutation, giving rise to a 'polyglutamine' sequence in the *HTT* protein. Normal individuals have fewer than 30 CAG repeats in the gene; disease occurs in those who have more than 36 copies, with full penetrance (i.e. all carriers of the mutation develop the disease) at 40 copies or more. New mutations are very rare; most apparently sporadic cases reflect an incomplete family history, or lack of knowledge of true paternity. The expansion tends to increase in succeeding generations, leading to an earlier age of onset, a feature called *anticipation*, which is characteristic of trinucleotide repeat diseases. The normal function of *huntingtin*, and the pathogenic mechanism by which the mutation causes disease, remain unclear, but a range of molecular pathways leading to cellular dysfunction and death are implicated (Bates

et al., 2015). Diagnostic and predictive genetic testing is widely available. Because of the devastating implications for the sufferer and their descendants, genetic counselling is required.

Dystonias

Dystonias are uncontrolled focal muscle spasms that lead to involuntary movements of the eyelids, face, neck, jaw, shoulders, larynx, hands, and (rarely) other parts of the body. The aetiology is uncertain. In the past, dystonias were regarded as conversion phenomena (i.e. psychogenic); this is now considered to be rare although psychiatric factors may exacerbate symptoms and disability. In contrast, there is evidence that some cases are caused by gene mutations, while others are due to infection, brain injury, neurotoxicity, and a range of other causes. Drugs, especially antipsychotic drugs, are an important cause of dystonias as one of their motoric side effects (Chapter 25). Clinical types of dystonia include blepharospasm, torticollis, writer's cramp, and laryngeal dystonia (Albanese et al., 2025). The most effective treatment is the injection of botulinum toxin directly into the affected muscles. Deep brain stimulation is also being used. For a review, see Balint et al. (2018).

Occupational dystonia

Muscular problems are common among musicians, and may threaten to end their careers. There are many causes, including overuse injury, pressure on peripheral nerves, and focal dystonias. These problems should be assessed by a physician with experience in the field. Performance anxiety is also frequent, and may impair or prevent performance. Beta-blockers alleviate this symptom and are used by many musicians, sometimes without medical supervision. Anxiety management is effective in some cases. Other occupations also have characteristic dystonias (e.g. golfers' 'yips').

Tics

Tics are purposeless, stereotyped, and repetitive jerking movements that most commonly occur in the face and neck. They are much more common in childhood than in adulthood, although a few cases begin at up to 40 years of age. Tic disorders usually begin between the ages of 4 and 8 years and are more common in males. Motor tics usually precede vocal tics, with initial simple motor tics involving the face, head, or neck. Tics can be highly variable and fluctuate, and the maximum severity usually occur between the ages of 8 and 12 years with tics declining in severity throughout the teenage to

adult years. Most sufferers have just one kind of abnormal movement, but a few people have more than one (multiple tics). Like almost all involuntary movements, tics are worsened by anxiety. Unlike dystonias, they can be controlled briefly by voluntary effort, but this results in an increasing unpleasant feeling of tension. Tics in children are associated with Tourette's syndrome (below) and a range of psychiatric disorders, notably obsessive-compulsive disorder, attention deficit hyperactivity disorder, and anxiety disorders. Tics are treated primarily with behavioural interventions, including exposure and response prevention, and habit reversal therapy. These approaches have been formalized in the Comprehensive Behavioral Intervention for Tics (CBIT). A range of medications including atypical antipsychotics and alpha-2 adrenergic agonists are sometimes used (Pringsheim et al., 2019).

Tourette syndrome

This condition was first described by Jean Itard in 1825 and subsequently by Georges Gilles de la Tourette in 1885. It is the most common tic disorder. The syndrome is defined by the onset in childhood of both motor and vocal tics (grunting, snarling, and similar ejaculations), lasting for at least a year. Up to a third of affected individuals exhibit *coprolalia* (involuntary uttering of obscenities). Around 10–40% exhibit echolalia or echopraxia.

There may be stereotyped movements, such as jumping and dancing. The tics are often preceded by premonitory sensations. Associated features include overactivity, difficulties in learning, emotional disturbances, and social problems. Comorbidity with attention deficit hyperactivity disorder and obsessive-compulsive disorder occurs in over 50% of cases.

The prevalence in children and adolescents is estimated at 1% and is four times more common in boys than girls (Yilmaz and Jankovic, 2025). There is a substantial genetic contribution to the syndrome and an overlap with the genetic predisposition to obsessive-compulsive disorder. Prenatal maternal smoking, perinatal hypoxia, and autoimmune abnormalities are also implicated. The neural basis of the syndrome is thought to involve altered dopamine function and aberrant cortico-striatal connectivity.

Mild cases may not require specific treatment. The tics are treated as described above, together with treatment of any comorbid psychiatric disorders. Deep brain stimulation is also being used. There are few good outcome data. Clinical experience suggests that two thirds of patients can expect an improvement or lasting remission by early adulthood, but the outcome is frequently poor.

For a review of tic disorders and Tourette syndrome, see Pringsheim et al. (2019).

Cerebrovascular disorders

Stroke

Cognitive deficits

Strokes may lead to dementia, as described earlier. However, strokes have other neuropsychiatric implications, too. Overt strokes usually present as a neurological emergency, with hemiparesis, dysphasias, and other focal symptoms and signs. Subsequently, survivors may be left with these and other impairments, in addition to a range of psychiatric symptoms, which depend largely on the site and size of the vascular event.

Personality change

Irritability, apathy, lability of mood, and occasionally aggressiveness may occur. Inflexibility in coping with problems is common, and may be observed in extreme form as a catastrophic reaction. These behavioural changes are often as disabling, and as distressing

to carers, as residual hemiplegia or dysphasia. They are probably due more to associated widespread arteriosclerotic vascular disease than to a single stroke, and they may continue to worsen even when the focal signs of a stroke are improving.

After a stroke, some people become abnormally emotional, with mixtures of spontaneous laughter and crying, and the emotional display is frequently at odds with the patient's apparent mood. Antidepressants are considered to be helpful.

Post-stroke depression

Depressed mood is common after stroke and may contribute to the apparent intellectual impairment or impede rehabilitation. The prevalence of depression is about 30% and remains stable for up to ten years post-stroke; there is a cumulative incidence of about 45% for depression within five years of a stroke. The main risk factors are the extent of post-stroke impairment in

activities of daily living, the degree of cognitive impairment, and a past history of depression. Suggestions that the risk of depression is related to the location of the stroke (being greater following a left frontal or left basal ganglia stroke) have not been well confirmed. Post-stroke depression is associated with a reduced quality of life, and increased disability and mortality compared to non-depressed stroke survivors.

Treatment of post-stroke depression depends in part on active rehabilitation. Antidepressants, especially SSRIs, are effective, and they may also enhance recovery after stroke independent of their effects on mood. Medication should, however, be used cautiously, as side effects are frequent and patients are at risk of further cerebrovascular events.

For a review of post-stroke depression, see Villa et al. (2018).

Subarachnoid haemorrhage

In the survivors of subarachnoid haemorrhage, cognitive impairment, personality change, and anxiety

and depression are all common. The long-term outcome is poor, and psychosocial problems are often prominent.

Subdural haematoma

Subdural haematoma may follow a fall in elderly patients, especially in those with a history of alcoholism. However, a history of head trauma is commonly lacking. Acute haematomas may cause coma or fluctuating impairment of consciousness, associated with hemiparesis and oculomotor signs. The psychiatrist is more likely to see the chronic syndromes, in which patients present with headache, poor concentration, vague physical complaints, and fluctuating consciousness, but often with few localizing neurological signs. It is important to consider this possibility as a cause of accelerated deterioration in patients with dementia. Treatment is by surgical evacuation, which may reverse the symptoms.

Head injury (traumatic brain injury)

Head injury and traumatic brain injury (TBI) are, in practice, interchangeable terms. The psychiatrist is likely to see two main groups of patients:

1. A relatively small group with persistent, serious cognitive and behavioural sequelae
2. A larger group with emotional symptoms and personality change.

The severity of non-penetrating (closed) head injury is best assessed by the duration of post-traumatic amnesia (PTA)—that is, the interval between the injury and the return of normal day-to-day memory. This measure has the advantage of being reasonably accurate even when assessed retrospectively (i.e. by asking the patient several months later what they remember of the immediate post-injury period). A PTA of less than one week is associated with a reasonable outcome (e.g. return to work) in the majority of cases, but a PTA of more than one month often results in failure to return to work. Retrograde amnesia (i.e. loss of memory of events prior to the injury) is much less predictive of outcome. An MRI scan is useful for defining the extent of brain injury, but a normal MRI scan does not preclude some degree of brain damage. Levels of the biomarkers Nf-L

and tau in CSF or blood are also useful in predicting outcome.

The vast majority of closed head injuries are due to acceleration and deceleration forces. When loss of consciousness occurs for a few seconds, it is thought to be due to disruption of cholinergic transmission in the brainstem. With more severe injuries, haemorrhagic areas of damage and diffuse axonal injury and shearing in white matter are the two main pathological events (Blennow et al., 2016). Both contribute to coma duration. Other complications include extradural and subdural haemorrhage and anoxia. Deposition of β -amyloid occurs in some cases, which may explain the link between head injury and later development of Alzheimer's disease, and with dementia pugilistica (see below). Apolipoprotein E4 (*apoE4*) genotype may increase the risk of death or cognitive deficits after head injury.

For a review of the psychiatric aspects of head injury, see Howlett et al. (2022).

Acute psychological effects

After severe injury, a phase of delirium may follow awakening from coma. Prolonged delirium may

be accompanied by a transient confabulatory state. Occasionally, delusional misidentification or reduplicative paramnesia is observed—for example, ward staff may be identified as old friends, or the ward may be identified as a duplicate in a distant town. Agitation and disinhibition (often sexual) are often present, and may take days or weeks to resolve.

Chronic psychological effects

Both primary and secondary (due to the effects of brain swelling and raised intracranial pressure) damage determine the neurological and cognitive deficits. The long-term outcome is also influenced by premorbid personality traits, early reactions to the injury, occupational attainment, availability of social support, and compensation issues (van der Naalt et al., 2017). Post-traumatic epilepsy may be a further significant complicating factor in more serious injuries.

Post-concussional syndrome

After mild head injury, 15–30% of patients describe a group of symptoms known as the post-concussional syndrome. The main features are anxiety, depression, and irritability, accompanied by headache, dizziness, fatigue, poor concentration, and insomnia. The duration and severity of these symptoms are highly variable. Most cases resolve without specific medical intervention.

Cognitive impairment

As well as being a risk factor for dementia, head injury can lead to more circumscribed neuropsychological deficits, with memory and executive function being most affected. The patient may show significant impairments in these domains (e.g. organizing and planning activities) without an overall decline in performance in terms of IQ.

Personality change

Personality change is common after severe injuries, particularly if there is frontal lobe damage, when there may be irritability, apathy, loss of spontaneity and drive, disinhibition, and occasionally reduced control of aggressive impulses. ICD-11 has a category of ‘secondary personality change’ to describe this group. Management is difficult, demanding considerable social support and, in some cases, prolonged rehabilitation. Such resources are scarce and often unavailable. Head injury is associated with an increased rate of violence and suicide (Fazel et al., 2014e).

Depression and psychosis

Depression, anxiety, and emotional lability are very common after brain damage, although estimates of prevalence vary widely. Post-head-injury mania is much less common, and can be mistaken for personality change. Head injury may be a risk factor for schizophrenia-like psychosis, but the evidence is inconclusive (Molloy et al., 2011).

Boxing and head injury

In 1969, Roberts drew attention to the tendency for professional boxers to develop a chronic traumatic encephalopathy, sometimes called *punch-drunk syndrome* or *dementia pugilistica* (Roberts, 1969), related to the cumulative extent of head injuries suffered during their boxing careers. The principal early features are executive dysfunction, bradyphrenia, mild dysarthria, and incoordination. The fully developed syndrome may consist of a range of motor features, cognitive deficits, and a variety of behavioural manifestations. Neuropathologically, there is loss of neurons in the cortex, substantia nigra, and cerebellum, together with neurofibrillary tangles and diffuse amyloid plaques. Changes to the rules of boxing, and more careful medical assessments, have reduced the risks, but similar concerns are now being raised with other sports and activities involving regular head trauma (e.g. soccer, rugby, and American football), and the term *chronic traumatic encephalopathy* is increasingly used (Smith et al., 2019).

Treatment of head injury

Understanding the impact of head injury to subsequent neuropsychiatric problems has dramatically increased over the last few decades. There are important consequences for both mild and severe brain injury. For mild TBI there can be common longer-term symptoms of headache, sleep difficulty, and judgement problems. For individuals who survive moderate to severe TBI, cognitive deficits and behavioural changes are frequently reported, alongside personality changes. Early assessment of the extent of neurological signs provides a useful guide to the likely pattern of long-term physical disability. Neuropsychiatric problems should be assessed and their impact anticipated, and a comprehensive social assessment is crucial. The clinical psychologist can sometimes contribute behavioural and cognitive techniques. Practical support is needed for the patient's family and carers. Issues of compensation and litigation should be settled as quickly as possible.

Medication is often used to treat aggression, depression, apathy, psychosis, or concentration problems, but evidence (as for psychological treatments) is lacking. If drugs are considered necessary, always start with a low dose and choose agents that have less potential for seizure generation and anticholinergic or extrapyramidal side effects. Most improvement after a head injury

occurs within the first year while a minority of individuals may suffer from long-lasting symptoms (post-concussion syndrome). Late deterioration should raise suspicion of a progressive or second event, such as a subdural haematoma or hydrocephalus, epilepsy, the development of a psychiatric disorder (e.g. depression), or drug toxicity.

Epilepsy

Epilepsy is the tendency to recurrent seizures, where a seizure consists of a paroxysmal electrical discharge in the brain and its clinical sequelae. Epilepsy is managed primarily by neurologists, and for a full description the reader is advised to refer to a neurology textbook or to Thijs et al. (2019). However, epilepsy has several important psychiatric aspects, reflecting its description as the 'bridge' between psychiatry and neurology:

- the differential diagnosis of episodic disturbances of behaviour (particularly 'atypical' attacks, aggressive behaviour, sleep problems, and 'pseudoseizures')
- the treatment of the psychiatric comorbidities of epilepsy
- seizures caused by psychotropic medication
- the psychological side effects of anticonvulsant drugs.

Types of seizures

The classification of seizures is shown in simplified form in Box 14.24. The principal distinction is between seizures which are generalized from the start and those with a focal onset, called partial seizures. Since focal-onset seizures may become generalized, an accurate description of the onset is essential. It is also necessary to distinguish between types of epilepsy and types of seizure. Older terms such as '*petit mal*' and '*grand mal*' are ambiguous, and are best avoided. It should be remembered that an 'aura' is in fact a partial seizure; most 'absences' and '*petit mal*' are actually complex partial seizures, implying a focal rather than generalized disturbance, as in true absences.

A brief clinical description of the more common types of seizure follows.

Simple partial seizures

The content depends upon the site of the focus. Simple partial seizures include Jacksonian motor seizures and a variety of sensory seizures in which the phenomena are

relatively unformed. Awareness is not impaired. Focal neurological or cognitive dysfunction may persist for a variable period following the seizure.

Complex partial seizures

Complex partial seizures are characterized by altered awareness of self and the environment, and include a wide range of 'psychiatric' features (see Table 14.5), hence the earlier term '*psychomotor epilepsy*'—making them the form of epilepsy of most importance to psychiatrists, and frequently forming part of the differential diagnosis.

Consciousness is not lost, unless secondary generalization occurs. However, during this period subjects appear out of touch with their surroundings, and often have great difficulty in describing their experiences later. The seizures arise most commonly in the temporal lobe, reflecting the additional former term '*temporal lobe epilepsy*'. However, that is not an appropriate term, as complex partial seizures can also begin in the frontal lobes

Box 14.24 Classification of seizure types

Focal onset

Motor or non-motor onset (with or without impaired awareness)

Focal to bilateral tonic-clonic

Generalized onset

Motor

Tonic-clonic

Myoclonic

Non-motor (absence)

Unknown onset

Table 14.5 Clinical features of complex partial seizures

Domain	Features
Consciousness	Altered
Autonomic and visceral	'Epigastric aura', dizziness, flushing, tachycardia, and other bodily sensations
Perceptual	Distorted perceptions, <i>déjà vu</i> , <i>jamaïs vu</i> , visual, auditory, olfactory, gustatory, somatic hallucinations
Cognitive	Disturbances of speech, thought, and memory, derealization, depersonalization
Affective	Fear and anxiety; occasionally, euphoric or ecstatic states
Psychomotor	Automatisms, grimacing, and other bodily movements; repetitive or more complex stereotyped behaviours

and other sites. Seizures that originate in the latter region are particularly likely to be misdiagnosed as functional, as they are frequently accompanied by bizarre posturing and other semi-purposeful, complex motor behaviours. Complex partial seizures of temporal lobe origin are often heralded by an *aura*, which may take the form of olfactory, gustatory, auditory, visual, or somatic hallucinations. Particularly common is the 'epigastric aura'—a sensation of churning in the stomach that rises towards the neck. The patient may also experience odd disturbances of thought or emotion, including an intense sense of familiarity (*déjà vu*) or unfamiliarity (*jamaïs vu*), depersonalization, or derealization, or, rarely, vivid hallucinations of past experiences ('experiential phenomena'). The sequence of events during the seizure tends to be stereotyped in the individual patient, which is an important diagnostic aid. The whole ictal phase usually lasts for up to one to two minutes. After recovery, only the aura may be recalled. Non-convulsive status epilepticus may take the form of a prolonged single seizure, or a rapid succession of brief seizures. In such cases, a protracted period of automatic behaviour may be mistaken for a dissociative fugue or other psychiatric disorder.

Absences

The key feature of an absence attack is loss of awareness that starts suddenly, without an aura, lasts for seconds, and ends abruptly. Simple automatisms (e.g.

eyelid fluttering) often accompany the attack. There are no post-ictal abnormalities. For the purposes of diagnosis and treatment, it is essential to distinguish between absence seizures and complex partial seizures. The latter last longer, automatisms during the episode are more complex, recovery occurs more slowly, and the patient may subsequently recall an aura. Absence attacks in children were previously called 'petit mal', and are classically associated with 3 per second 'spike-and-wave' EEG discharges.

Generalized tonic–clonic seizures

This is the familiar epileptic convulsion with a sudden onset, tonic and clonic phases, and a succeeding period of variable duration (up to many hours) in which the patient may be unrousable, sleepy, or disoriented. Incontinence and tongue-biting or other injuries may occur during the seizure. During the post-ictal phase the patient may present with delirium, which may cause diagnostic uncertainty if the convulsion was not witnessed. Generalized tonic–clonic seizures may be initiated by a partial seizure, implying localized brain disease, which is often overlooked. This is an issue of importance, as primary and secondary generalized seizures differ in significance and management.

Myoclonic, atonic, and other seizure types

There are several types of seizures with predominantly motor symptoms, including myoclonic jerks and drop attacks with loss of postural tone.

Epidemiology

In the United Kingdom, general practice data indicate that the prevalence of epilepsy in adults is about 7 per 1,000. About 1 in 30 people have a seizure at some stage. The inception rate is highest in early childhood, and there are further peaks in adolescence and over the age of 65 years. Epilepsy is usually of short duration, and only becomes a chronic condition in about 20% of subjects. This means that regular attenders at specialist clinics are a minority of all those with epilepsy, and who are especially likely to suffer from its medical and social complications. There is psychiatric comorbidity in up to 50% of people with epilepsy (Gurgu et al., 2021), and there are particular psychiatric and cognitive implications for children with epilepsy.

Aetiology

Genes play an important role. Mutations in over 30 different genes have been identified in families with

autosomal dominant epilepsy, including in ion channel genes. In children with severe developmental epileptic encephalopathies, a *de novo* mutation can be identified in over a third of cases. In ‘sporadic’ epilepsy, there is a genetic predisposition from many risk loci.

Age at onset is an important clue to aetiology. For example, in the newborn, birth injury, congenital brain malformations, and metabolic disorders are common causes. Infantile febrile convulsions, especially status epilepticus, are classically associated with later complex partial epilepsy, via damage to the hippocampus (hippocampal sclerosis). In adults, identifiable causes include cerebrovascular disease, brain tumours, head injury, autoimmune disorders, and neurodegenerative disorders. Patterns of alcohol and other drug use should always be established. Seizure threshold may be lowered by drug therapy, including antipsychotics, tricyclic antidepressants, and bupropion. Sudden withdrawal of substantial doses of any drug with anticonvulsant properties, most commonly diazepam or alcohol, can precipitate seizures.

For a review, see Devinsky et al. (2018).

Making the diagnosis

Epilepsy is essentially a clinical diagnosis that depends upon detailed accounts of the attacks provided by witnesses as well as by the patient. The background history, physical examination, and special investigations are directed towards establishing the aetiology. The extent of investigation is guided by the initial findings, the type of attack, and the patient’s age. Only an outline can be given here; for a full account the reader is advised to refer to a neurology or epilepsy textbook. An EEG can confirm but cannot exclude the diagnosis. It is more useful for determining the type of epilepsy and site of origin. The standard EEG recording may be supplemented by sleep recording, ambulatory monitoring, and split-screen video (telemetry) techniques. Although neuroimaging has an increasing clinical role, no specific brain abnormality (or other cause) is found in the majority of patients with epilepsy.

Epilepsy can be erroneously diagnosed as the cause of paroxysmal neurological and psychiatric symptoms, and it is important to keep in mind the extensive differential diagnosis (see Box 14.6). A clear description of the circumstances surrounding the episode and the mode of onset is fundamental. The most important differential diagnoses of generalized seizures are vasovagal syncope (commonly associated with involuntary movements, a fact not always appreciated) and cardiac

arrhythmias. Hyperventilation (of which the patient is often unaware) and panic attacks frequently produce symptoms similar to complex partial seizures, and may lead to actual loss of consciousness if prolonged. Sudden changes in motor activity, affect, and cognition can occur in schizophrenia.

Factors that together suggest a seizure include abrupt onset, a stereotyped course lasting from many seconds to a few minutes, tongue-biting, urinary incontinence, cyanosis, sustaining injury during the attack, and prolonged post-ictal drowsiness or confusion. However, none of these alone is necessary or sufficient to make the diagnosis. Some forms of frontal lobe epilepsy are particularly likely to be misdiagnosed as psychogenic. If the diagnosis remains uncertain, and attacks are frequent, close observation, including video recording and EEG telemetry and ambulatory monitoring, may be worthwhile.

Psychogenic non-epileptic seizures

Paroxysmal attacks that can be mistaken for epilepsy but that are not caused by seizures or epileptiform activity in the brain are called *psychogenic non-epileptic seizures*; an older term is *pseudoseizures*. They can be very difficult to distinguish from seizures, and make up about 20% of referrals to epilepsy services. As with other ‘functional’ neurological disorders, positive evidence should be sought for the diagnosis, rather than simply an absence of evidence for seizures (Hallett et al., 2022). Features that suggest non-epileptic episodes include identifiable psychosocial precipitants, a past history of physical or sexual abuse, a history of psychiatric disorder, an unusual or variable pattern of attacks, occurrence either only in public or only while alone, and the absence of autonomic signs or change in colour during ‘generalized’ attacks. The patient may be suggestible or betray other evidence of retained awareness during the episode. Complex, purposeful behaviour is more often seen in dissociative states. Video recordings of the attacks (e.g. from smart phones) and ambulatory EEG may be helpful. However, some types of seizure may not be reflected in the surface EEG and, conversely, EEG abnormalities occur in a small percentage of healthy individuals. Post-ictal serum prolactin levels are useful in a minority of cases (they are elevated after a generalized seizure), but should not be relied upon to make the distinction. Complicating the issue, some patients have both seizures and psychogenic non-epileptic seizures. For a review, see Milan-Tomas et al. (2018).

Psychiatric aspects of epilepsy

Psychiatric comorbidity is common in people with epilepsy, with overall rates increased by at least twofold, and higher among those in specialist care. It is a major determinant of their quality of life. Many different types of psychiatric disorder are associated with epilepsy, including cognitive, affective, emotional, and behavioural disturbances. They are usually classified according to whether they occur before, during, after, or between seizures (Box 14.25).

The relationship between epilepsy and psychiatric disorder may reflect any of the following factors:

- an overlapping aetiology or pathophysiology
- the stigma and psychosocial impairments associated with epilepsy
- the side effects of antiepileptic drugs.

Box 14.25 Associations between epilepsy and psychological disturbance

Psychiatric and cognitive disorders associated with the underlying cause

Behavioural disturbances associated with seizures

Pre-ictal: prodromal states and mood disturbance

Ictal: complex partial seizures (affective disturbances, hallucinations, experiential phenomena, automatisms); absence seizures (altered awareness, automatisms)

Post-ictal: impaired consciousness; delirium; psychosis; Todd's paresis (hemiparesis, dysphasia, other focal signs)

Epileptic pseudodementia (non-convulsive status)

Inter-ictal disorders

Cognitive

Psychoses

Sexual behaviour

Depression and emotional disorder

Suicide and deliberate self-harm

Crime and other antisocial behaviour

Personality change

For a review of the psychiatric aspects of epilepsy and their management, see Kanner (2016). For review in children, where 37–77% have co-morbid psychopathology, including depression, anxiety, ADHD, and autism spectrum disorders, see Fazel et al. (2021).

Pre-ictal disturbances

Increasing tension, irritability, and anxiety and depression are sometimes apparent as prodromata for several hours or even days before a seizure, generally increasing in intensity as the seizure approaches.

Ictal disturbances

Ictal psychiatric disturbances (i.e. those directly related to seizure activity) are common and diverse, as noted above. During seizures, transient confusional states, affective disturbances, anxiety, automatisms, and other abnormal behaviours often occur (especially in partial seizures). On occasion, an abnormal mental state may be the only sign of non-convulsive (complex partial or absence) status epilepticus, and the diagnosis is easily overlooked.

Psychosis may occur as an ictal phenomenon. Clues to this possibility include sudden onset and termination of psychiatric disturbance, olfactory or gustatory hallucinations (especially with partial seizures), a relative lack of first-rank symptoms, and amnesia for the period of the disturbance.

Ictal violence is extremely rare, and crimes committed during epileptic automatisms are probably even rarer, an important medicolegal consideration (Brodie et al., 2016).

Post-ictal disturbances

Psychiatric disturbances may occur during the hours following a seizure. Psychotic symptoms are seen in 2% of cases (Clancy et al., 2014), and are associated with bilateral and extratemporal seizure foci, long duration of epilepsy, and structural brain lesions. These transient psychoses are distinct from the inter-ictal psychoses described below. Diverse motor, sensory, cognitive, and autonomic dysfunction can also occur and, as with the psychoses, may occur as part of a delirium, or in clear consciousness.

Post-ictal violence is rare, although it is more common than violence during the seizure, and may be secondary to psychotic experiences. Extreme post-ictal violence may be recurrent, stereotyped, and more likely to occur in men, after a cluster of seizures. There is usually amnesia for the episode.

Inter-ictal disturbances

Cognitive impairment

In the nineteenth century it was widely believed, based on experience with institutionalized populations, that epilepsy was associated with an inevitable decline in intellectual functioning. However, it is now established that relatively few people with epilepsy show cognitive changes. When these do occur, they are likely to reflect concurrent brain damage, unrecognized non-convulsive seizures, or the effects of antiepileptic drugs. A few patients with epilepsy show a progressive decline in cognitive function. In such cases, careful investigation is required to exclude an underlying progressive neurological disorder; this is a particular concern in paediatric practice. For a review, see Sen et al. (2018).

Personality

The historical concept of an ‘epileptic personality’, characterized by egocentricity, irritability, religiosity, quarrelsomeness, and ‘sticky’ thought processes, has been discarded. Community surveys indicate that only a minority of patients have serious personality difficulties, and these probably reflect the adverse effects of brain damage on education, employment, and social life, rather than a specific association with epilepsy. It has been suggested that behavioural abnormalities (including hypergraphia) are particularly associated with medial temporal lobe lesions.

Depression and emotional disorders

Depression and anxiety are common in people with epilepsy, for both biological and psychosocial reasons. The rate is increased almost threefold, with a prevalence of over 20% (Scott et al., 2017). Many subjects meet the criteria for dysthymia rather than major depression, and the term *inter-ictal dysphoric disorder* is sometimes used. The risk factors for depression in epilepsy are summarized in Box 14.26. The relationship between epilepsy and depression is bidirectional, with depression preceding epilepsy, as well as vice versa, and reflecting common pathogenic factors for the two conditions. For a review of depression in epilepsy, see Kanner et al. (2024).

Inter-ictal psychosis

The nature and prevalence of inter-ictal psychosis has long been controversial, including the fundamental question as to whether the two coexist more or less often than expected. Following the landmark study by Slater and Beard (1963), the evidence now supports the view that epilepsy is associated with an increased risk of psychosis, especially a schizophrenia-like presentation, with a meta-analysis reporting a rate of inter-ictal psychosis of

Box 14.26 Risk factors for depression in epilepsy

Biological

- Family history of mood disorder
- Focus in temporal or frontal lobe
- Left-sided focus

Psychosocial

- Perceived stigma
- Fear of seizures
- Pessimistic attributional style
- Decreased social support
- Unemployment

Iatrogenic

- Epilepsy surgery
- Antiepileptic drugs, especially polypharmacy and high serum levels

5.2% (Clancy et al., 2014). Religious and paranoid delusions appear to be particularly common, and affect tends to be preserved. Risk factors include complex partial seizures, especially with the focus in the mesial temporal or frontal lobe, a lesion which is prenatal in origin, and possibly in the left hemisphere more than the right one.

Suicide

Suicide and deliberate self-harm are more common among people with epilepsy than in the general population. Although estimates vary markedly, a meta-analysis found an adjusted odds ratio of 3.7 (Fazel et al., 2013). The rate appears to be higher in those with temporal lobe epilepsy, and after surgical treatment for epilepsy. Suicide risk factors in epilepsy encompass the same range of risk factors as in the general population (see Chapter 21).

Social aspects of epilepsy

The consequences for quality of life correlate with the severity of the seizure disorder, the presence of structural brain pathology, and with any psychiatric comorbidity. The social implications and stigma attached to a diagnosis of epilepsy can be far-reaching, as is the unpredictability of seizure occurrence. When counselling patients and their families, it is important to be sensitive to these

issues and to allay groundless fears and misconceptions. Restrictions on driving are a major burden for many patients, whose livelihood may be at stake. To obtain a UK driving licence, the patient must have had at least one year with no seizures while awake, whether or not they are still taking antiepileptic drugs. Those who have seizures only while asleep may hold a licence if this pattern has been stable for at least three years.

Sexual dysfunction with reduced libido and impaired performance is common in patients with epilepsy, especially where there is a temporal lobe focus. This is thought to be mainly due to antiepileptic medication, although psychosocial factors are also important, and rarely there may be a direct link to the cause of the seizures.

Epilepsy has been said to be associated with crime and violence. However, there is little evidence for either assertion (Fazel et al., 2009).

Treatment of epilepsy

The drug treatment of epilepsy is usually undertaken by general practitioners and neurologists. Here the discussion is restricted to some key points of psychiatric relevance.

- It is important to distinguish between peri-ictal and inter-ictal psychiatric disorders when planning treatment. For peri-ictal disorders, treatment is aimed at control of the seizures.
- Treatment of inter-ictal psychiatric disorders is usually similar to that in non-epileptic patients, but there are relatively few data demonstrating the efficacy of pharmacological or psychological therapies in this population.
- The potential for psychotropic drugs to worsen the seizure disorder should always be considered, though most drugs are safe at therapeutic doses. Indeed, commonly used antidepressants may in fact be anti-convulsant, contrary to common belief. However, antidepressants can cause seizures at toxic doses. Among antipsychotics, first-generation drugs such as haloperidol and fluphenazine have a lower seizure-inducing propensity than olanzapine, quetiapine, or clozapine.
- Some antiepileptic drugs can cause a variety of cognitive and psychiatric symptoms, while others (such as lamotrigine and valproate) are increasingly being used to treat psychiatric disorders.
- There are pharmacokinetic interactions between antiepileptic, psychotropic, and other drugs that can lead to toxicity or subtherapeutic levels. The potential for such interactions should always be checked before prescribing (or withdrawing) medication.

Given these considerations, close liaison between neurologist, psychiatrist, and general practitioner is recommended. For a review of drug treatment of psychiatric disorders in epilepsy, see Kanner (2016). For psychosocial aspects of treatment, see Michaelis et al. (2018).

Brain tumours

Many brain tumours cause psychological symptoms at some stage in their course, and a significant proportion present with such symptoms. Psychiatrists are likely to see patients with slow-growing tumours in 'silent' (especially frontal) areas. These produce psychological effects, but few neurological signs—for example, a frontal meningioma or glioma of the corpus callosum. The nature of the psychological symptoms is influenced by the global effects of raised intracranial pressure, in addition to the tumour location. The rate of tumour growth is also important; rapidly expanding tumours with raised intracranial pressure can present as delirium, whereas slower-growing tumours are more likely to cause chronic cognitive deficits. Focal lesions give rise to a variety of specific neuropsychiatric syndromes; those near the

frontal poles typically manifest initially as personality change. Craniopharyngiomas and other tumours around the hypothalamus are also often associated with personality changes and apathy. Cognitive impairment after treatment of brain tumours in children is particularly problematic, especially in those who have received radiotherapy.

For a review, see Mellado-Calvo and Fleming (2009).

Cognitive impairments in cancer

In addition to the direct effects of brain tumours, neoplasms anywhere in the body can impair cognition, and

Box 14.27 Some causes of cognitive impairment in patients with cancer**Mass effects**

Primary and metastatic brain tumours

Diffuse infiltration

Gliomatosis cerebri

Carcinomatous and lymphomatous meningitis

En plaque meningioma

Paraneoplastic limbic encephalitis

Anti-NMDA-receptor and other autoantibodies

Metabolic derangements

Hypercalcaemia

Hyponatraemia

Acidosis

Hypoglycaemia

Radiotherapy

Acute and subacute radionecrosis

Radiation thromboangiopathy

Accelerated cerebral atherosclerosis

Radiation leukodystrophy

Second malignancies (especially glioma, meningioma)

Chemotherapy

Metabolic encephalopathies

Opportunistic infections secondary to immunosuppression (herpes zoster, progressive multifocal leukoencephalopathy)

produce other neuropsychiatric symptoms, via a range of mechanisms (see Box 14.27). Occasionally, such complications are the presenting feature, but usually they

arise during treatment, at which time the psychiatrist may be asked for a diagnostic and therapeutic opinion. For a review, see Holroyd et al. (2021).

Other neuropsychiatric syndromes**Multiple sclerosis**

Multiple sclerosis is the most common cause of chronic neurological disability in young adults in developed countries. Its consequences for work and relationships may be profound. The disease may be difficult to diagnose early in the course, and physical symptoms are sometimes misinterpreted as psychiatric. Psychiatric symptoms are rarely the presenting feature, but the majority of patients will experience them at some stage, especially fatigue, depression, and cognitive impairment (Box 14.28). Such symptoms probably result both directly from the disease process and from the disabilities associated with it. There is a severalfold increase in the risk of suicide.

Depression is particularly common in multiple sclerosis, with prevalence estimates of up to 50% (Solaro et al., 2018). It does not appear to be closely related to the severity of the clinical syndrome, but may be associated with loss of frontal and temporal grey matter. Depression is also a side effect of beta-interferon, used

to treat multiple sclerosis. Both antidepressant and cognitive behaviour therapy approaches can be effective.

Cognitive impairment is present in at least 40% of patients from community samples. It may be an early manifestation of the illness, and occasionally a rapidly progressive dementia occurs. In most cases, however, intellectual deterioration begins later, is less severe, and progresses slowly. Well-practised verbal skills are often preserved despite deficits in problem-solving, abstraction, memory, and learning. Cognitive impairment correlates with total lesion load and degree of callosal atrophy on MRI, and probably reflects axonal loss rather than demyelination per se. For a review, see Benedict et al. (2020).

Neurosyphilis

Neurosyphilis, a manifestation of the tertiary stage of infection with the spirochaete *Treponema pallidum*, is now rare in western countries. However, increasing numbers of cases have been reported in lower- and

Box 14.28 Psychiatric features of multiple sclerosis

Common

Fatigue
Depression
Cognitive impairment

Less common

Euphoria
Pseudobulbar affect
Psychosis

Source: Wong, E. K. O., Krishnadas, R., & Cavanagh, J. (2013). The interface between neurology and psychiatry: The case of multiple sclerosis. *Adv Psychiatr Treat*, 19(5), 370–377. Copyright (2013), Cambridge University Press

middle-income countries, and in association with HIV. Because of its protean manifestations, the possibility of neurosyphilis should be considered in all ‘neuropsychiatric’ patients, especially those with delirium or dementia, and appropriate blood or CSF serological tests ordered (Marra, 2025). Prior treatment with antibiotics may produce partial and atypical syndromes.

An asymptomatic stage precedes clinical disease with variable latency. Symptomatic neurosyphilis takes three forms.

1. *Meningovascular syphilis* appears within five years of primary infection. It presents with strokes, or with personality changes, emotional lability, and headache. Dementia may occur subsequently, accompanied by psychotic symptoms.
2. *General paresis* (also called *general paralysis of the insane*, or *dementia paralytica*) starts to develop about 20 years after infection. Presentation is with dementia, personality change, dysarthria, and motor symptoms and signs. Affective and psychotic symptoms may be florid, classically with euphoria and grandiose delusions. Discovery of the cause of general paresis was an important landmark in the history of psychiatry, stimulating a search for organic causes of other psychiatric syndromes.
3. *Tabes dorsalis* is a degeneration of spinal cord pathways and is unlikely to present to psychiatrists.

In the early stages, treatment with penicillin reverses the condition, and halts progression later in the illness. If the clinical disease is left untreated, death usually occurs within five years.

For a review, see Ropper (2019).

Encephalitis

Encephalitis is inflammation of the brain parenchyma. The term is sometimes used to apply specifically to a primary viral infection, although encephalitic involvement can also occur as a complication of bacterial meningitis (meningoencephalitis) or a cerebral abscess. Moreover, it is now clear that many cases of encephalitis are not due to infection but have an autoimmune basis.

The most common viral cause of encephalitis is herpes simplex. Without treatment, it was often fatal or left devastating consequences. Although intravenous aciclovir provides an effective treatment, it remains a serious condition. In the acute stage, headache, vomiting, and impaired consciousness are usual, and seizures are common. Presentation may be with delirium. The psychiatrist may be involved in initial diagnosis, but is more likely to see chronic complications, which include prolonged anxiety and depression, a profound amnesic syndrome, personality change, or complex partial epilepsy. Other significant types of encephalitis in adults include those produced by arthropod-borne viruses and, especially in the immune compromised, varicella zoster. For a review of viral encephalitis, see Tyler (2018).

A major recent development is the realization that many cases of encephalitis are not viral, but are autoimmune in origin. In particular, autoantibodies against various neuronal proteins (especially NMDA glutamate receptors and potassium channels) are now recognized as being a relatively common cause of limbic encephalitis, and often present floridly with delirium and psychosis, as well as with seizures and other neurological features (Dalmau and Graus, 2018). Hence, they may present to psychiatrists, and liaison psychiatrists involved in their management. These antibodies not only underlie the encephalitis of paraneoplastic syndromes but can occur in the absence of cancer or other known cause. They can also be triggered by herpes simplex, blurring the distinction between viral and autoimmune forms of encephalitis. Screening for autoantibodies is now routine in investigation of suspected encephalitis, and positive cases are treated using immunosuppression. Whether anti-neuronal antibodies can produce psychiatric syndromes (especially a schizophrenia-like psychosis)

without other evidence of encephalitis remains controversial (see Chapter 11).

An outbreak of *encephalitis lethargica* (also called *post-encephalitic parkinsonism*) was reported in 1917 by Constantin von Economo (Lutters et al., 2018). The condition reached epidemic proportions in the 1920s, and was thought to be related to the influenza pandemic of the time. By the 1930s it had largely disappeared. Parkinsonism was the most disabling complication; personality change with antisocial behaviour, and psychosis, also occurred. In his 1973 book *Awakenings*, Oliver Sacks gave a vivid description of such cases, and the striking but temporary ‘awakening’ brought about in some by L-dopa. Current speculation is that the condition may have been caused by anti-NMDA receptor antibodies. Concern that Covid-19 infection might lead to a similar disorder has, to date, proven unfounded (Taquet et al., 2022).

Normal pressure hydrocephalus

The characteristic clinical triad in this condition (Adams et al., 1965) consists of a striking ‘gait apraxia’ (a broad-based, small-stepped gait with difficulty in initiation) on which supervenes a progressive frontal subcortical syndrome with bradyphrenia and, later, urinary incontinence. Frank dementia is rare. The prevalence is uncertain, but is reported to be up to 6% of cases of late-life dementia in published series. The condition is more common in the elderly, but sometimes occurs in middle life, and in children with congenital abnormalities.

The pathogenesis has been thought to be a block to CSF flow within the ventricular system owing to aqueduct stenosis, or in the subarachnoid space. Impairment of glymphatic drainage of the brain is also implicated. However, often no cause can be discovered. Ventricular pressure is generally normal or low, although episodes of raised pressure may also occur. Ventricular enlargement out of proportion to the degree of cortical atrophy, often with periventricular signal change, is the hallmark finding on brain imaging.

It is important to differentiate this condition from a degenerative dementia, and from depression with pseudodementia. Cases with a short history and prominent gait disturbance with relative sparing of intellect may be amenable to a neurosurgical shunt procedure to improve the circulation of CSF. Outcome is variable, but overall the intervention improves cognitive symptoms in over 60% of patients (Giordan et al., 2018). However, the conventional view that normal pressure

hydrocephalus is distinct from neurodegenerative disorders is challenged by its common co-occurrence with Alzheimer’s disease and the suggestion that in fact ‘pure’ normal pressure hydrocephalus is very rare (Müller-Schmitz et al., 2020). For review, see Johnson and Williams, 2025.

Anoxia, hypoglycaemia, and carbon monoxide poisoning

Anoxia (e.g. due to cardiac arrest), hypoglycaemia, and carbon monoxide poisoning produce similar patterns of brain injury. Cerebral and cerebellar atrophy may occur, and the hippocampus and globus pallidus are particularly vulnerable. Clinical experience suggests that the recovery trajectory is relatively brief compared with TBI, and that neurological sequelae are less disabling. Parkinsonism is not infrequent after a latent interval. The most common cognitive impairment is poor memory.

Carbon monoxide poisoning usually arises from deliberate self-harm by car exhaust fumes, and, more recently, from burning charcoal in the Far East. However, it can also occur accidentally as a result of badly ventilated gas boilers or fires. The prevalence of complications after carbon monoxide poisoning is unclear, with a wide range of figures reported. It is also unclear whether residual symptoms ever occur if the poisoning was insufficient to cause loss of consciousness, although complaints of problems with memory, concentration, fatigue, and headache are common (Ho et al., 2012).

Systemic lupus erythematosus

Systemic lupus erythematosus (SLE) has neuropsychiatric manifestations in about 10% of patients at time of first presentation, and in 20–75% of patients at some stage of the illness (Hanly, 2014). The most common feature is cognitive dysfunction, affecting all domains of memory, as well as poor attention and concentration, and reduced psychomotor speed. Severe cognitive dysfunction occurs in 3–5%. Mood disorder is also common, and psychosis affects up to 8% of cases. SLE is an autoimmune disorder, and anti-double-stranded DNA and other antibodies are implicated, together with an inflammatory component and blood-brain barrier dysfunction. For a review of neuropsychiatric SLE, see Jeltsch-David and Muller (2014).

Secondary or symptomatic neuropsychiatric syndromes

This chapter has considered neuropsychiatry from the perspective of the underlying conditions, such as dementia and epilepsy, and has described the psychiatric symptoms that they may produce. In clinical practice, it is also important to view this relationship the other way round; that is, to recognize that common psychiatric disorders, such as depression, psychosis, and anxiety, can be a manifestation of a recognized underlying disease process within the brain or as part of a medical disorder arising elsewhere in the body (Chapter 22). Psychiatric disorders in this context are variously described as *secondary*, *symptomatic*, or *organic*. For example, ICD-11 has a category of ‘secondary mood syndrome, with depressive symptoms’ and DSM-5 has categories of psychiatric disorders ‘due to another medical condition’; in both systems the presumed underlying disease (e.g. multiple sclerosis) is also coded. Examples of secondary neuropsychiatric disorders are listed in Table 14.6. See also Lovestone (2009).

The clinical features of these secondary disorders are generally indistinguishable from those in the equivalent primary psychiatric disorder. Thus, recognition of a secondary syndrome depends on the associated features. The following suggest that the disorder is secondary to a physical condition:

- Evidence of cerebral disease, damage, or dysfunction, or of physical disease, known to be associated with one of the listed syndromes.
- A temporal relationship (of weeks or a few months) between the development of the underlying disease and the onset of the psychiatric syndrome.
- Recovery from the psychiatric disorder following removal or improvement of the presumed cause.

Table 14.6 Some causes of symptomatic or secondary psychiatric syndromes

Syndrome	Causes
Psychosis	Temporal lobe disorders, Huntington’s disease, basal ganglia lesions, dementias, endocrinopathies, metabolic disorders, cerebral vasculitis, neurosyphilis, limbic encephalitis
Mood disorder	Alzheimer’s disease, stroke, traumatic brain injury, Parkinson’s disease, multiple sclerosis, Huntington’s disease, endocrinopathies, metabolic disorders (especially hypercalcaemia)
Personality change	Frontotemporal dementias, frontal lobe lesions, Huntington’s disease, focal basal ganglia lesions, neurosyphilis, prion disease, traumatic brain injury
Obsessive–compulsive behaviours	Frontotemporal dementias, complex partial seizures, basal ganglia disorders, Rett syndrome

- Absence of evidence suggesting an alternative ‘psychological’ cause of the psychiatric disorder (e.g. in the case of depression, no evidence of a family history of mood disorder, relevant personality traits, previous episodes of mood disorder, recent life events).

Further reading

Ames, D., O’Brien, J. T., & Burns, A. (2017) *Dementia*, 5th edn. Routledge.

Arciniegas, D. B., Yudofsky, S. C., & Hales, R. E. (2018). *Textbook of Neuropsychiatry and Clinical Neurosciences*, 6th edn. American Psychiatric Association Publishing.

David, A. S., Fleming, S., Kopelman, M. D., et al. (Eds.). (2009). *Lishman’s Organic Psychiatry*, 4th edn. Wiley Blackwell.

CHAPTER 15

Personality and personality disorder

Personality 381

History of personality types 382

The origins of personality 383

The assessment of personality 384

Personality disorder 384

Classification of abnormal personalities 386

Descriptions and diagnostic criteria 387

Rates of personality disorder in the clinic and the general population 392

Aetiology 393

The course of personality disorder 396

Treatment 396

Management 397

The management of specific personality disorders 398

Ethical problems 400

Personality

The term personality refers to those enduring qualities of an individual that are shown in their ways of behaving in a wide variety of circumstances. Personality differs from mental disorder. The behaviours that define it have been present throughout adult life, whereas behaviours that define a new onset mental disorder differ from the person's previous behaviour. When we consider the emergence of mental illness, we are drawing on our understanding of a person's previous function, their personality, and usual behaviours. The distinction between manifestations of personality disorder and mental illness is easier to make when behaviour changes over a short period of time (as in a manic disorder), but can be difficult when the changes occur very gradually (as in some cases of schizophrenia).

The importance of personality

Gaining an understanding of and familiarity with a patient's personality is important in psychiatry for a number of reasons. Different personalities predispose to some psychiatric disorders, and they may colour the presentation ('pathoplastic' factors). They may also influence how a patient approaches treatment, and

dictate different strategies for establishing and maintaining a successful therapeutic relationship.

Personality as predisposition

Personality can predispose to psychiatric disorder by modifying an individual's response to stressful events. For example, adverse circumstances are more likely to induce an anxiety disorder in a person who has always worried excessively about everyday problems.

Personality as a pathoplastic factor

Personality can contribute to unusual features of a disorder, particularly when personality traits become exaggerated with illness. For example, rumination and inhibition may be the presentation of depression in an individual with an obsessional personality. The underlying diagnosis can be obscured if the psychiatrist has not made an accurate assessment of personality.

Personality in relation to treatment

Personality is an important determinant of a person's approach to treatment. For example, people with obsessional traits may become frustrated and resistant if treatment does not follow their expectations; and

anxiety-prone people may discontinue medication prematurely because of concerns about side effects even if they are mild. Some people with a severe disorder of personality (particularly antisocial, borderline, impulsive, histrionic, and narcissistic types) have often been excluded from services because of difficult relationships they form with clinicians (Chartonas et al., 2017; Lewis and Appleby, 1988). This is a significant problem, as there is strong evidence that personality disorder is common in clinical populations, and that people with personality disorders have an increased risk for a range

of mental illnesses (Guzzetta and de Girolamo, 2009). There has been a concerted move to remedy this, as outlined later in this chapter and in Chapter 26. Personality traits also affect outcomes of treatment for psychiatric disorders (Bucher et al., 2019).

Patients are often aware of their personalities—they know if they are ‘emotional’ or ‘conscientious’ or ‘anxious’. Acknowledging this and discussing how it may interact with their treatment (and with their life more generally) can be an effective tool in establishing an alliance and maintaining treatment.

History of personality types

A first step in understanding personality is to identify some basic types. Clinicians have generally derived these from a mixture of clinical and common-sense collective experience with several generally recognizable categories, such as a sociable and outgoing type, a solitary and self-conscious type, and an anxious and timid type. Psychologists have adopted a more rigorous scientific approach using personality tests to measure aspects of personality (‘traits’) such as anxiety, energy, flexibility, hostility, impulsiveness, moodiness, orderliness, and self-reliance. These are then subject to statistical investigations to discover how they cluster into ‘factors’.

Although these statistical procedures appear very scientific, their results are determined by the investigators’ original hypotheses (which traits they considered important and included in their analyses, and the order in which they included them, etc.). Like diagnoses generally, these factors and other descriptions about personality are essentially working hypotheses, which are continuously evolving and should not be treated with too much reverence—their value lies in their utility.

Different investigators have derived different personality factors from such traits (Cattell and Mead, 2008). Eysenck (1953) originally proposed only two ‘dimensions’ (high-level factors), which he labelled extraversion–introversion and neuroticism (Eysenck, 1970). Subsequently he added a third dimension, ‘psychoticism’ (Eysenck and Eysenck, 1976), but his use of the term is rather misleading, denoting coldness, aggressivity, cruelty, and antisocial behaviour. A five-factor formulation for personality has persisted, although the terms used for the factors have varied. They can be identified as openness to experience (or novelty-seeking), conscientiousness,

extraversion–introversion, agreeableness (or affiliation), and neuroticism.

Cloninger et al. (1993) developed an alternative scheme with three ‘basic behavioural dispositions’ expressed as four basic temperaments (a seven-factor model). The behavioural dispositions are behavioural activation, behavioural inhibition, and behavioural maintenance. Cloninger considered behavioural activation to be associated with the basic temperament of *novelty-seeking*, behavioural inhibition with *harm avoidance*, and behavioural maintenance with *reward dependence*. The fourth basic temperament is *persistence*. Cloninger’s scheme also includes three character traits, namely *self-directedness*, *cooperativeness*, and *transcendence*. Cloninger’s scheme includes both inherited differences in brain function and the effects of experience.

These factor schemes of personality have survived despite sustained criticism that they are largely dependent on what was included in the original factor analyses (Zuckerman, 2005). Zuckerman has criticized these theories because they assume an alignment between personality traits and brain systems. In recent years there has been some refinement, with greater emphasis on cognitive factors. A cognitive–adaptive theory has been proposed (Penke et al., 2007); Mischel et al. (2004) emphasized the dynamic nature of personality in the cognitive–affective system.

Despite these scientific approaches, clinicians continue to use everyday words to describe the positive and negative features of normal personality. Positive attributes include outgoing, self-confident, stable, and adaptable. Negative attributes include sensitive, jealous, irritable, impulsive, self-centred, rigid, and aggressive. Such terms should be used with care as they can

become stigmatizing or a shorthand for describing a person's presentation rather than giving a fuller and more

dynamic account of how symptoms and behaviours fluctuate in different contexts.

The origins of personality

Genetic influences

Everyday observation suggests that children often resemble their parents in personality. Such similarities could be either inherited or acquired through social learning. Two prominent types of scientific study have been used to study the inheritance of personality.

Studies of twins. More direct evidence has been obtained from personality tests of monozygotic and dizygotic twins. These suggest that the heritability for traits of extraversion and neuroticism is 35–50% (McGuffin and Thapar, 1992). The heritability of other personality traits is broadly similar.

Molecular genetic studies. Genetic linkage and association studies have revealed a number of loci and genes contributing to personality traits, especially neuroticism and impulsivity (Lo et al., 2017). It is clear that the genetic predisposition to personality (and its disorders) arises from the cumulative effect of multiple genes, each of very small effect. Some of these genetic influences are shared with those underlying psychiatric disorders, and this overlap likely explains much of the association between them (such as neuroticism with depression).

Childhood temperament and adult personality

Young infants differ enormously in patterns of sleeping and waking, approach or withdrawal from new situations, intensity of emotional responses, and span of attention. These differences, which are described in more detail in Chapter 16, could form the basis for personality development. They do persist into later childhood, but their relationship to adult personality has been difficult to establish.

Childhood experience and personality development

Childhood experience shapes personality, yet relevant experiences are difficult to quantify or even to record reliably. Few prospective studies have examined transitions from childhood to adult personality, although

some ongoing studies, such as the Dunedin cohort study in New Zealand, are able to link confirmed early experience and adult states (Poulton et al., 2015). There is some evidence showing how individual personality features show continuity from middle childhood to old age (Costa et al., 2019),

The *Freudian theory* of personality development emphasizes events during the first five years of life. Freud proposed that crucial stages of libido development must be accomplished successfully for healthy personality development. Failure or fixation at particular stages explained certain features of adult personality. Freud's explanation can be made to explain almost all personality variation in terms of infantile experience, but this very flexibility makes it impossible to test scientifically. However, it retains enormous intuitive appeal and is widely used both within and outside of psychiatry.

Jung considered personality development to be a life-long process, with 'individuation' as its aim. He introduced the terms 'introvert' and 'extrovert'.

Later analysts rejected Freud's exclusive focus on libido development. *Adler* emphasized a struggle for mastery (overcoming the 'inferiority complex'). The 'neo-Freudians' increasingly emphasized social and peer-group factors in personality development.

Erikson extended Freud's schema, using less off-putting terms, for the individual developmental challenges. Erikson's eight developmental stages continued right into late adult life, and each was presented as a conflict, such as 'trust versus mistrust', 'autonomy versus doubt', that had to be resolved. Erikson emphasizes the importance of adolescence.

Attachment theory

Attachment theory derives from the work of John Bowlby and plays an important role in informing assessment and treatment of early childhood mental health. Although a psychoanalyst, Bowlby's work is based on the importance of available and responsive primary caregivers to allow an infant to develop a sense of security. Bowlby emphasized the importance of *maternal deprivation* in the development of personality

in orphan children just after the Second World War. He proposed that a close, reliable relationship was needed for effective attachment and that a 'secure attachment' promoted health (Bowlby, 1951). The scientific study of attachment was facilitated by the development of research measures for assessing attachment in children and adults, including the Strange Situations Procedure and the Adult Attachment Interview (Galbally et al., 2020). From these studies, four main attachment

styles of secure, avoidant-secure, ambivalent-secure, and later disorganised-insecure have been described (Ainsworth and Bell, 1970). Although attachment styles in early infancy are not necessarily those then observed in adulthood, children who securely attach as infants tend to experience less depression and anxiety as adults, whereas negative life events act to maintain an insecure attachment style (Sutton, 2019).

The assessment of personality

The assessment of personality is discussed in Chapter 3, but two points need to be emphasized. The first is that the assessment of personality used in everyday life cannot be applied reliably in clinical practice. Normally we assume that current behaviour reflects the person's habitual ways of behaving (their personality), and in general this assumption is correct. This is not the case when we assess patients, because current behaviours reflect the effects of current illness, as well as personality, or a change in their personality functioning. A patient's personality can only be judged confidently from reliable accounts of past behaviour, which have been obtained wherever possible from informants as well as from the patient.

Secondly, the assessment instruments developed for personality mentioned earlier are more reliable in healthy individuals and can be misleading in the presence of mental disorder. Personality tests, although useful in research, are seldom used in clinical practice. For a review of personality assessment methods, see (Clark and Harrison, 2001).

The importance of personality assessment

The assessment of personality is important when making decisions about aetiology, diagnosis, and treatment. In aetiology, knowledge of personality helps to explain why certain events are stressful to that patient. In diagnosis, an understanding of personality may explain the presence of unusual features in a disorder that might otherwise cause uncertainty. In treatment, an assessment of personality helps to explain the patient's reaction to their illness and its treatment, and aids the establishment of an effective therapeutic relationship. Personality assessment should be an integral part of every formulation, and not just reserved for those where a personality disorder is suspected.

It is best to record a series of descriptive terms chosen. Examples would be 'sensitive', 'lacking in self-confidence', and 'prone to worry'. Such descriptions help to construct a picture of the unique features of each patient, which is a fundamental element of good clinical practice.

Personality disorder

The concept of abnormal personality

Some personalities are obviously troublesome, for example, paranoid personalities characterized by extreme suspiciousness, sensitivity, and mistrust. However, it is impossible to draw a sharp dividing line between normal and abnormal personalities. Abnormal personalities are in practice recognized because of the *pattern* of their

characteristics, but our current classificatory processes in psychiatry demand that we identify *criteria* for inclusion. However, precisely which criteria should be used to make this distinction remain controversial. Two types of criteria have been suggested: statistical and social.

According to the *statistical criterion*, abnormal personalities are quantitative variations from the normal, and the dividing line is decided by a cutoff score on an appropriate measure. This approach is attractive, as it

parallels that used successfully when defining abnormalities of intelligence, it appears non-judgemental, and it has obvious value in research. However, its usefulness in clinical practice is uncertain.

According to the *social criterion*, abnormal personalities are those that cause the individual to suffer, or to cause suffering to other people. For example, an abnormally sensitive and gloomy personality causes suffering for the individual who has it, and an emotionally cold and aggressive personality causes suffering for others. These criteria are subjective and lack the precision of the first approach, but they serve the needs of clinical practice better and they have been widely adopted.

It is not surprising that it is difficult to frame a satisfactory definition of abnormal personality.

ICD-10 defined personality disorder as:

... severe disturbances in the personality and behavioural tendencies of the individual; not directly resulting from disease, damage, or other insult to the brain, or from another psychiatric disorder; usually involving several areas of the personality; nearly always associated with considerable personal distress and social disruption; and usually manifest since childhood or adolescence and continuing throughout adulthood.

There were specific subtypes including: paranoid, schizoid, antisocial, emotionally unstable, histrionic, anankastic, anxious (avoidant), dependent, 'other specific personality disorder', and unspecified personality disorders. Most research and clinical application is usually found for borderline and antisocial personality disorders (ASPD). Categories are often preferred by clinicians as they can make definitive diagnoses, yet, like much of psychiatry, there are overlaps in diagnostic categories and people may vary in the extent to which they meet all the criteria for one or another disorder. In response to such criticisms of categorical approaches, ICD-11 reviewed the research data and departed from the previous approaches. ICD-11 introduced distinct dimensions (called domain qualifiers) in place of categories (with one exception), and the classification requires judgement about the presence and absence of a personality disorder and its severity:

- The dimensions include *negative affectivity, detachment, disinhibition, dissocial (instead of antisocial), and anankastia*.

- Severity is assessed as *none, personality difficulty, mild personality disorder, moderate personality disorder, severe personality disorder*.

Despite the evidence base for dimensional approaches, the ICD-11 has retained the previous ICD-10 and ICD-9 category of borderline personality disorder (Mulder, 2021) due to its enduring empirical value.

Personality disorder carries stigma as a label, and many people are excluded from services or misjudged as a consequence. Therefore, such labels should be used with care, and explained to patients in terms of their often complex and overlapping nature, and the degree of change over time, and that people with personality difficulties also have advantages, as these traits may be favourable in some situations; for example, those with obsessional traits are often dependable and conscientious. Management plans that play to an individual's strengths are more likely to be helpful for them.

Personality change

In some circumstances during adult life there may be a profound and enduring change in personality that is distinct from the temporary changes that may accompany stressful events or illness. This lasting change may result from:

- injury to or organic disease of the brain
- severe mental disorder, especially schizophrenia
- exceptionally severe stressful experiences (e.g. those experienced by hostages or by prisoners undergoing torture).

Corresponding to similar diagnoses in ICD-10, ICD-11 refers to secondary personality change, meaning pathophysiological consequences of a health condition not classified under mental and behavioural disorders. This judgement is based on history, examination, and laboratory tests. The symptoms are not accounted for by delirium, or other mental and behavioural disorders, and are not psychologically mediated as a response to a severe medical condition. Enduring personality change can occur following complex post-traumatic stress (catastrophic experience), or after a severe psychiatric disorder.

Classification of abnormal personalities

General issues

The use of categories

Personality traits are continuously distributed, but the classification of psychiatric disorders uses categories and requires definitions and boundaries. However, the criterion for inclusion (distress to the person or to others) is very imprecise, so cases that fall just short of it (sub-threshold cases) are frequent. These subthreshold cases often present clinical problems that are similar to those of definite cases.

Comorbidity

It is not only the boundary between normal and abnormal personality that is imprecise and arbitrary; the boundaries between the different types of personality are also ill defined. This leads to an important divergence of practice between clinical psychiatrists and researchers and some psychologists. Many patients have features contained in the criteria for more than one personality disorder (Fyer et al., 1988). Structured instruments for measuring personality disorder, such as the International Personality Disorder Examination (Loranger et al., 1997), are often used by researchers and psychologists. They permit more than one personality diagnosis to be made in a single patient, when the term comorbidity is used. From a clinical perspective it seems more likely that the patient has a single personality disorder with features that overlap two of the arbitrary sets of criteria used in the current systems of diagnosis. We would usually diagnose the single personality disorder that best fits the mixed picture and is most useful in understanding and helping the patient. Comorbidity is used when a patient has both a mental illness and a personality disorder (or more often a mental illness and substance abuse), but makes little clinical sense for personality disorder alone.

Conditions related to personality disorder and classified elsewhere

Cyclothymia and schizotypal disorder were previously classified as personality disorders. In ICD-10 and ICD-11 both of these conditions have been removed and classified instead with the mental disorders (cyclothymia with affective disorders, and schizotypal disorder with schizophrenia). This reflects the fact that these two conditions may begin in adult life and, in the case of schizotypal disorder, evidence from family studies

that links it genetically to schizophrenia (see page 264). Multiple personality disorder is classified with dissociative disorders.

ICD-11 Classification

ICD-11 is radically different from other classifications and was built on a series of field trials. The dissatisfaction with categories of diagnosis, comorbidities, and the lack of fit to complex patient experiences, not to mention stigma and negative stereotypes that the diagnostic labels carry, all led to a rethink by the World Health Organization (WHO). The following is a summary of the key steps to make a diagnosis:

- The clinician is invited to consider the presence of a core problem of function of the self and of interpersonal relationships.
 - Self-function includes:
 - ◆ stability and coherence of identity
 - ◆ maintain a positive overall sense of self-worth
 - ◆ accuracy of one's characteristics, strengths, and limitations
 - ◆ capacity for self-direction.
 - Interpersonal function relates to the following abilities:
 - ◆ interest in engaging others
 - ◆ understand and appreciate other's perspectives
 - ◆ develop and sustain close and mutually satisfying relationships
 - manage conflicts in relationships.
- If there is a personality difficulty, this is not a diagnosis but shows itself in difficulties in interpersonal function, and cognitive and emotional experiences that are intermittent or of low intensity.
- If a personality disorder is present, the clinician considers the severity as either mild, moderate, severe, or unspecified. This is judged on emotional, cognitive, and behavioural manifestations.
- Then the pattern of symptoms is considered against one or more of the following dimensions: negative affectivity (same in DSM-5), detachment, antisocial, anankastia, borderline pattern. DSM-5 includes psychoticism, which appears as schizotypal disorder in ICD-11 under 'schizophrenia or other primary psychotic disorders'. ICD-11 retains the borderline category, which is also present in DSM-5.

Only time will tell whether ICD-11 is able to shift the focus from categories to dimensions and whether this will be adopted by clinicians. Consequently, we present information on the classifications that may still be used in the meantime, including ICD-10. Importantly, the assessment should indicate which system of classification is used and why, for example, if the pattern of presenting symptoms closely matches a particular diagnostic group; or where there are relevant treatment and care options linked to a particular diagnosis.

Classification of personality disorders in ICD-10 and ICD-11.

Table 15.1 shows the classification of personality disorders in ICD-10. The ICD-11 classification is pioneering and does not map easily onto these categories and will need further research and clinical testing (Tyrer et al., 2019). There is debate on whether the new dimensional and severity approach is too basic for clinical value, and whether it will be useful for improving patient outcomes. In the meantime, understanding these different categories and their meaning remains relevant when assessing patients where their records refer to these categories.

Table 15.1 Classification of personality disorders

ICD-10
Paranoid
Schizoid
(Schizotypal; see text)
Antisocial
Emotionally unstable
Impulsive type
Borderline type
Histrionic
—
Anankastic (obsessive–compulsive)
Anxious (avoidant)
Dependent

Source: *The ICD-10 Classification of Mental and Behavioural Disorders: Clinical Descriptions and Diagnostic Guidelines*. Copyright (1992), World Health Organization.

Descriptions and diagnostic criteria

This section contains an account of the abnormal personalities listed in ICD-10. The criteria for diagnosis are lengthy.

All personality disorder diagnoses must meet the basic criteria for personality disorder summarized from the ICD-10 criteria for research as follows:

- The characteristic and enduring patterns of behaviour differ markedly from the cultural norm and in more than one of the following areas: cognition, affectivity, control of impulses and gratification, and ways of relating to others.
- The behaviour is inflexible, and maladaptive or dysfunctional in a broad range of situations.
- Personal distress is caused to others and/or to self.
- The presentation is stable and long lasting, usually beginning by late childhood or adolescence.
- The behaviour is not caused by another mental disorder, or by brain injury, disease, or dysfunction.

There are several diagnostic instruments for personality disorder, but they are mainly of value in research.

Paranoid personality disorder

Such people are suspicious and sensitive (see Box 15.1). They have a marked sense of self-importance, but easily feel shame and humiliation. They are suspicious and constantly on the lookout for attempts by others to deceive or exploit them, which makes it difficult for others to get on with them. They have difficulty making friends and avoid involvement in groups. They are usually mistrustful and often jealous. They may appear secretive and self-sufficient to a fault. They take offence easily and see criticism where none was intended.

Suspicious ideas can be so intense that they are mistaken for persecutory delusions. These *sensitive ideas of reference* are considered further in Chapter 12. Paranoid individuals can be resentful and bear grudges, engaging in litigation, often against all advice.

Schizoid personality disorder

The term ‘schizoid’ was suggested by Kretschmer (1936), who believed that this type of personality is related to

Box 15.1 Features of paranoid personality disorder

Suspicious
Mistrustful
Jealous
Sensitive
Resentful
Bears grudges
Self-important

Box 15.2 Features of schizoid personality disorder

Emotionally cold
Detached
Aloof
Lacking enjoyment
Introspective

schizophrenia. People with this disorder are emotionally cold, detached, aloof, introspective, and prone to engage in excessive fantasy (see Box 15.2). They are unable to express either tender feelings or anger, and show little interest in sexual relationships. When the disorder is extreme, they may appear callous. They are generally ill at ease in company, do not form intimate relationships, and show little family feeling. They are seclusive, following a solitary course through life, and often remain unmarried. They show little sense of enjoyment or pleasure in the activities that most people enjoy, which contributes to their separation from others. They are often interested in intellectual matters, and have a complex inner world of fantasy, albeit without emotional content.

Schizotypal personality disorder

Schizotypal individuals are socially anxious and behave eccentrically. They can experience cognitive and perceptual distortions, and have oddities of speech and inappropriate emotional responses. Schizotypal individuals are not classified as a personality disorder in ICD-10, where the category is placed with schizophrenia and called schizotypal disorder (see Chapter 11).

Schizotypal individuals experience *social anxiety* in company, so have difficulty in forming relationships and lack friends and confidants. They feel different from other people and do not fit in. (This is in contrast to schizoid individuals, who are quite content with this situation.) Their *cognitive and perceptual distortions* include ideas of reference (but not delusions), suspicious ideas, odd beliefs, magical thinking (e.g. belief in clairvoyance, mind reading, and telepathy), and unusual perceptual experiences (e.g. awareness of a 'presence', or experiences bordering on hallucinations). They also show *oddities of speech*, such as unusual constructions, words, and phrasing, as well as vagueness, and a tendency to digression. Their *affective responses are unusual*, and they

appear stiff, odd, and constricted in their emotions. Their *behaviour is eccentric*, with odd mannerisms, unusual choices of clothing, disregard of conventions, and awkward social behaviour.

Antisocial personality disorder (ASPD)

ASPD is not simply another term for *delinquency*. People with this disorder show a callous lack of concern for the feelings of others. They disregard the rights of others, act impulsively, lack guilt, and fail to learn from adverse experiences. Often their abnormal behaviour is made worse by the abuse of alcohol or drugs. Hervey Cleckley's book, *The Mask of Sanity* (Cleckley, 1951) is a vivid and influential description of this type of personality. The disorder is of particular forensic significance because of its inclusion—under the term *psychopathic disorder*—in the Mental Health Act. The criteria for diagnosis differ slightly between classifications; the following criteria are taken from ICD-10.

A central feature is a *callous lack of concern for others*. People with ASPD may be exploitative and even violent, and their sexual activity lacks tenderness. They may inflict cruel or degrading acts on other people, and their partners may be physically or sexually abused. They may have a *superficial charm*, but their *relationships are shallow and short-lived*. They are *irresponsible and depart from social norms*. They do not obey rules and may repeatedly break the law, often committing violent offences. Their offending typically begins in adolescence (Box 15.3).

Such individuals are *impulsive*, rarely plan ahead, and typically have an unstable work record. They take risks, disregarding their own safety and that of other people. They are *irritable*, and when angry sometimes assault others in a violent way. These features of personality are accompanied by a striking *lack of guilt or remorse* and a failure to change their behaviour in response to punishment or other adverse outcomes. They *avoid*

Box 15.3 Features of antisocial personality disorder (ASPD)

Callous
 Transient relationships
 Irresponsible
 Impulsive and irritable
 Lacking guilt and remorse
 Failure to accept responsibility

responsibility, transferring blame on to other people and rationalizing their own failures. They are often deceitful and irresponsible about finances (Box 15.4).

The concept of '*psychopathy*' overlaps with ASPD. Although it has been dropped from the diagnostic classification (in ICD-11, it is classified as 'personality disorder, severity unspecified'), it is still in use in psychology (particularly forensic psychology) and the prison service, and has a very specific meaning. It is generally considered to be a character trait, and is usually assessed using the Hare Psychopathy Checklist-Revised (PCL-R; see <http://www.hare.org/scales/pclr.html>). Hare has critiqued the value and challenges of applying the scale and the levels of expertise required to interpret the findings (Hare, 1998). The scale distinguishes two factors—violent and impulsive behaviour, and charm and exploitation. The terms '*psychopathy*' and '*antisocial personality disorder*' are often unhelpfully interchanged, and psychopathy is probably best avoided by general psychiatrists.

Borderline (or emotionally unstable) personality disorder

The term '*borderline personality disorder*' is a confusing, which reflects its complex history. It was originally used in psychodynamic circles to describe people with a marked '*instability*' of presentation, as opposed to a steady neurotic state. The proposed '*border*' at that time was with psychosis. Kernberg originally described it in 1975 as involving:

- ego weakness
- poor control of impulses
- '*primary process*' thinking despite intact reality testing
- use of certain defence mechanisms, such as projection and denial
- diffuse personal identity.

Box 15.4 Legal implications of psychopathic disorder

The Mental Health Act for England and Wales (1983) defined psychopathic disorder as a persistent disorder or disability of mind (whether or not including significant impairment of intelligence) which results in abnormally aggressive or seriously irresponsible conduct on the part of the person concerned. The requirement of '*treatability*' as a condition of detention was added exclusively to it, out of all the mental disorders. This distinction was abandoned in the 2007 amendment to the Act because it was perceived to present a barrier to effective treatment.

A recent short-lived development in the United Kingdom was the administrative category of '*dangerous and severe personality disorder*' (DSPD). It applied mainly to men with antisocial personality disorder (ASPD) and a history of serious violent or sexual offending that could be linked to their personality disorder. The evolution of effective treatments for this group proved elusive and, in any event, changes in the law to permit indeterminate sentences for prisoners who pose persisting serious risks have made the category redundant.

(see Kernberg's recent summary of the history and emerging evidence on the validity of the diagnosis in Kernberg and Michels, 2009). People with borderline personality disorder experience their lives as being dominated by *strong and fluctuating emotions* that often overwhelm them. They strive for affection and intimacy but are regularly disappointed and may exhaust their partners with the intensity of their emotional demands. They are themselves troubled by the intensity and *unpredictability of their moods* as they plunge into anger or despair. They are often *insecure in their personal identity* and need reassurance and stability, which they may then find constricting and irritating. *Self-harm* is common (either suicidal attempts, or cutting to release tension). Such self-destructive behaviour can be extreme and may dominate the relationship with care services. *Alcohol and drug misuse* are also common, as they attempt to blunt their distressing emotions. Clinical populations are dominated by relatively young women, although the epidemiological studies challenge this distribution.

When this type of personality disorder was introduced into the classification systems, more objective criteria were developed, but it remains unsatisfactory. The

Table 15.2 Abbreviated criteria for emotionally unstable and borderline personality disorders

ICD-10
Emotionally unstable personality disorder
<i>Borderline type</i>
Disturbed or uncertain self-image
Intense and unstable relationships
Efforts to avoid abandonment
Recurrent threats or acts of self-harm
Chronic feelings of emptiness
—
<i>Impulsive type</i>
Impulsive
Liability to anger and violence
Unstable capricious mood
Quarrelsome
Difficulty maintaining a course of action

Source: *The ICD-10 Classification of Mental and Behavioural Disorders: Clinical Descriptions and Diagnostic Guidelines*. Copyright (1992), World Health Organization.

diagnosis was (and still is) almost exclusively applied to young women. This raised the question of whether it was a more culturally determined expression of mood disorder (i.e. depression). The suggestion that it may respond (at least partially) to antidepressants has added to the controversy, as has the greater degree of variation over time than in other personality disorders, and the observation that in most patients it seems to resolve.

Emotionally unstable personality disorder in ICD-10 is divided into a ‘borderline type’ and an ‘impulsive type’. The diagnostic criteria (slightly abbreviated) are shown in Table 15.2.

See Bach et al. (2022) and Skodol et al. (2002) for a comparison and critique of the strengths and weaknesses of different diagnostic systems. Many people who meet the criteria for borderline personality disorder may also meet the criteria for histrionic, narcissistic, and ASPD (Skodol et al., 2002).

For review of borderline personality disorder, see Bohus et al. (2021).

Impulsive personality disorder

This disorder is recognized in ICD-10 as a subtype of emotionally labile personality disorder. The diagnostic

criteria are listed in Table 15.2, and three criteria must be met for the diagnosis. People with impulsive personality disorder cannot control their emotions adequately, and are liable to outbursts of sudden unrestrained anger, which they subsequently regret. They may use physical violence occasionally, causing serious harm. Unlike people with ASPD, they do not generally have other difficulties in relationships.

In ICD-11, the impulse control personality disorder is refined, as discussed above, by severity and core symptoms. ICD-11 includes impulse control disorders defined by repeated failure to resist an impulse, drive, or urge to perform an act that is rewarding (at least in the short term), despite harm either to the individual or others. These disorders include pathological gambling, intermittent explosive disorder, kleptomania, and pyromania, as well as compulsive sexual behaviour disorder (Grant et al., 2014).

Histrionic personality disorder

Histrionic personality disorder is included in both ICD-10; Box 15.5 lists the features that are diagnostic criteria in ICD-10.

Self-dramatization is a striking feature, and may include ‘emotional blackmail’, angry scenes, and demonstrative suicide attempts. These individuals are *suggestible* and easily influenced by others, especially by figures of authority. They *seek attention and excitement*, are easily bored, and have short-lived enthusiasms. They have a *shallow labile affect*. Their emotional life is dramatic and may exhaust others. However, they recover quickly and often cannot understand why others are still upset. They seek intimacy and can be *inappropriately seductive*. Again, there is not separate category of histrionic disorders in ICD-11, rather dimensions of symptoms are invoked in different ways alongside severity ratings. See Bach and First (2018) for example of cross-reads for all ICD-10 categories across diagnostic systems.

Box 15.5 Features of histrionic personality disorder

- Self-dramatization
- Suggestibility
- Shallow labile affect
- Seeks attention and excitement
- Inappropriately seductive
- Over-concern with physical attractiveness

Histrionic individuals are often insecure about their value, and consequently may be *over-concerned with physical attractiveness*. They can appear *self-centred* and vain. They may also have a marked capacity for *self-deception*, being convinced by their own fabrications—however elaborate and improbable these may be—even when other people have seen through them. In its most extreme form, histrionic personality disorder is observed in ‘pathological liars’ and swindlers. Blanche DuBois in Tennessee Williams’s *A Streetcar Named Desire* poignantly portrays the disastrous consequences that such a personality can have both for the individual and for those around them.

Narcissistic personality disorder

This personality disorder is found in DSM-5 but not ICD-10, where it is one of the disorders coded in the residual category ‘other specific personality disorder’.

People with this disorder have a grandiose sense of self-importance and are boastful and pretentious. They are preoccupied with fantasies of unlimited success, power, beauty, or intellectual brilliance. They consider themselves special, and expect others to admire them and offer them special services and favours. They feel entitled to the best, and seek to associate with people of high status. They exploit others and do not empathize with or show concern for their feelings. They envy the possessions and achievements of others, and expect that those individuals will envy them in the same way. They appear arrogant, disdainful, and haughty, and behave in a patronizing or condescending way.

Narcissistic personality disorder carries probably the most pejorative overtones of any of the personality disorder diagnoses. It should be reserved for only the most undeniable examples, and only after you have made sure that the diagnosis is not coloured by personal dislike. On balance it is probably a term to avoid if at all possible, like some older personality terms such as ‘immature personality’ or ‘inadequate personality’, which have no place in modern mental healthcare.

Anxious personality disorder

In contrast to DSM-5, which uses the term ‘avoidant’, ICD-10 uses the term ‘anxious’ personality disorder (Box 15.6). Liability to tension is a criterion only in ICD-10.

People with this disorder are *persistently tense*. They feel insecure and lack self-esteem. They feel *socially inferior*, unappealing, and socially inept. They are *preoccupied with the possibility of rejection*, disapproval, or criticism, and worry that they will be embarrassed or ridiculed.

Box 15.6 Features of avoidant (anxious) personality disorder

- Feels socially inferior
- Preoccupied with rejection
- Avoids involvement
- Avoids risk
- Avoids social activity

They are cautious about new experiences, *avoid risk*, and *avoid social activity*. They have few close friends, but they are not emotionally cold, and indeed crave the social relationships that they cannot manage to attain.

Dependent personality disorder

Box 15.7 lists the diagnostic criteria in ICD-10. People with dependent personality disorder *allow others to take responsibility* for important decisions in their lives. They are *unduly compliant*, but are *unwilling to make direct demands* on other people. They lack vigour and *feel unable to care for themselves*. Lacking self-reliance, they avoid responsibility, and may need *excessive help to make decisions*.

They are often protected by support from a more energetic and determined partner, and may only come to medical attention when the partner leaves or dies.

Anankastic (obsessive–compulsive) personality disorder

ICD-10 uses the term ‘anankastic’, following Kahn (1928), who used the term to avoid the implication

Box 15.7 Features of dependent personality disorder

- Allows others to take responsibility
- Unduly compliant
- Unwilling to make reasonable demands
- Feels unable to care for himself or herself
- Fear of being left to care for himself or herself
- Needs excessive help to make decisions

Box 15.8 Features of anankastic (obsessive–compulsive) personality disorder

Preoccupied with details, rules, etc.
 Inhibited by perfectionism
 Over-conscientious and scrupulous
 Excessively concerned with work and productivity
 Over-conscientious, scrupulous, and inflexible in ethics and morals
 Unable to discard worthless objects
 Reluctant to delegate tasks or work with others
 Miserly
 Rigidity and stubbornness

that this personality disorder is linked to obsessive-compulsive disorder. Box 15.8 lists the diagnostic criteria in ICD-10.

These individuals are *preoccupied with details and rules*, order, and schedules. They have an *inhibiting perfectionism* that makes ordinary work a burden as they are immersed in endless detail. They may lack imagination and fail to take advantage of opportunities. However, they usually have high moral standards, are often *excessively conscientious*, and may be rather judgemental. They can appear humourless and ill at ease when others are enjoying themselves.

These people lack adaptability to new situations. They are generally *rigid and inflexible*, avoiding change and preferring a familiar routine. They can be *stubborn and controlling* and generally thrifty, sometimes to the

point of being miserly. They may hoard objects and money. They may be *pedantic* and *unduly concerned with social conventions*. They can also be particularly troubled by *excessive doubt and caution*, leading to indecision.

Affective personality disorders

Some people have lifelong disorders of mood regulation. They may be persistently gloomy (depressive personality disorder) or habitually in a state of inappropriate elation (hyperthymic personality disorder). A third group alternates between these two extremes (cycloid or cyclothymic personality disorder). These types of personality have been described for many years, and are readily recognized in clinical practice. In ICD-10 they are classified under disorders of mood and not under disorders of personality.

People with *depressive personality disorder* (dysthymia in ICD-10) always seem to be in low spirits. They have a persistently gloomy and pessimistic view of life, and may brood about their misfortunes and worry unduly. They have little capacity for enjoyment and are dissatisfied with their lives. Some are irritable and bad-tempered.

People with *cyclothymic personality* (in ICD-10 this falls under *persistent mood disorders*) alternate between depressive and hypomanic states. This instability of mood is much more disruptive than either of the persisting conditions. People with this disorder are periodically extremely cheerful, active, and productive, at which times they take on additional commitments in their work and social lives. When their mood changes, they become gloomy and defeatist, and with reduced energy levels they now find their newly acquired obligations a burden.

Rates of personality disorder in the clinic and the general population

Clinical population rates

For a psychiatrist, the more important question is how frequently personality disorder will be encountered in their clinical practice. Prevalence rates vary more for personality disorder than for mental illnesses both in clinic and in population surveys because of diagnostic uncertainty. A series of studies have investigated this, and the 16 largest and most rigorous studies were summarized by Zimmerman et al. (2008), whose review showed that

much higher rates at around 40% were found in structured assessments. Rates of personality disorder in those in contact with mental health services have been estimated at 40% for UK community mental health teams (CMHTs) (Newton-Howes et al., 2010) or 50% internationally for outpatient attendees (Beckwith et al., 2014; Fazel and Danesh, 2002). Rates are even higher in prison populations (Fazel and Danesh, 2002). Among primary care consultations in the United Kingdom, the rate is as high as 30% (Beckwith et al., 2014).

However, we know that rates in the clinic and rates in the general population are likely to be very different. Treatment seeking by individuals with personality disorder is powerfully affected by the existence of a comorbid mental illness (Kessler et al., 1999).

General population rates

Epidemiological research into personality disorders in the general population began with the development of standardized instruments using DSM-III criteria. Studies require large samples because the prevalence rates of some personality disorders are low. It is also difficult to identify personality disorder reliably in community surveys without access to informants.

Estimates of the prevalence of the various types of personality disorder are shown in Table 15.3 (Alarcón and Palmer, 2020). Some of these estimates have a substantial range reflecting, among other sources of variation, the use of different assessment instruments. The rates of some disorders vary between men and women, notably ASPD, which is more common among men (estimated ratios range between 2.1 and 7.1). Although borderline and histrionic personality disorders are clinically reported more frequently in women, this is not consistently seen in epidemiological studies. UK community studies show that personality disorder is as common in ethnic minority as in host populations (Crawford et al., 2012; McGilloway et al., 2010), and higher in urban populations (Coid et al., 2006).

The estimates in Table 15.3 are complemented by a recent meta-analysis, which found the global prevalence of any personality disorder in the community to be 7.8% (95% CI 6.1–9.5%) (Winsper et al., 2020). Rates were higher in high compared to low- and medium-income countries.

Table 15.3 Prevalence rates of personality disorders in epidemiological surveys

Personality disorder type (ICD-10)	Prevalence (% median and range)
Any	11.6 (3.9–22.7)
Paranoid	1.6 (0.4–4.4)
Schizoid	0.6 (0.4–4.9)
Antisocial	1.3 (0.2–4.5)
Borderline	1.7 (0.2–5.9)
Histrionic	1.6 (0.3–3.9)
Narcissistic	0.5 (0.1–1.5)
Obsessive-compulsive	2.1 (2.1–7.9)
Avoidant	1.7 (1.0–6.0)
Dependent	0.9 (0.1–11.0)

Source: Alarcón, R. D., & Palmer, B. A. (2020). Personality disorders: Epidemiology and clinical course. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1218–1228). Copyright (2020), Oxford University Press.

Structured assessment instruments have high reliability but may have more questionable validity. Certainly, they seem to generate more positives than most clinicians would expect, and the absolute rates should be treated with some caution. However, the relatively primitive state of personality disorder theory and assessment may explain much of this variation. The surprising findings of remarkably high rates of remission in personality disorder when using these instruments may indicate that they should be treated cautiously. However, there is a growing belief among personality disorder researchers that their lifelong nature has been overstated, as discussed below.

Aetiology

General issues

The causes of personality disorder are uncertain, and this is an area in which hypotheses are many and research findings varied. The two major areas are genetic factors and various kinds of early-life experience. Some personality disorders have been linked aetiologically with the psychiatric disorders that they resemble. The state of

evidence for each of these three general issues will be considered with the individual personality disorders.

The study of the influence of *early-life experiences* is made difficult by the long interval between these experiences and the diagnosis of personality disorder in adult life. Psychodynamic theories linking childhood experience and personality are hard to test, but have been influential.

Most clinicians accept that there are causal links between childhood experience, personality, and personality disorder. It is agreed good practice to assess childhood experiences and to use common-sense judgement to decide whether any of these may have influenced the personality. For example, extreme and repeated rejection by the parents may explain low self-esteem in adult life. Retrospective studies present obvious problems, but prospective studies are now becoming available. For example, independently documented gross physical neglect or abuse or sexual abuse in childhood has been shown to be associated with some personality disorders (Johnson et al., 1999). Different causes have been suggested for the various personality disorders.

Antisocial personality disorder (ASPD)

Genetic causes

The much-quoted early twin studies, for example, Rosanoff et al. (1934), were concerned with probands with repeated conviction for criminal offences rather than ASPD as such, so their relevance is uncertain. More recent twin studies have confirmed the heritability of antisocial behaviour in adults, and have shown that genetic factors are more important in adults than in antisocial children or adolescents where the shared environmental factors dominate (Lyons et al., 1995).

One gene contributing to antisocial behaviour is monoamine oxidase A (MAOA). A common low-activity MAOA genetic variant predisposes to adult antisocial behaviour in men, especially in those who experienced early adversity—an example of a gene–environment interaction (Byrd and Manuck, 2014).

Data linkage studies from Sweden confirm the genetic contribution to sexual offending by demonstrating the association between rates and degree of relationship (Långström et al., 2015).

Cerebral pathology and cerebral maturation

As some brain-injured patients show aggressive behaviour, minor degrees of brain injury have been sought as a cause of antisocial personality. However, there is no convincing evidence to support this idea. Magnetic resonance imaging (MRI) studies have found reduced prefrontal grey matter (Raine et al., 2000), and reduced amygdala volume may be linked to the characteristic lack of empathy (Blair, 2003).

5-Hydroxytryptamine and aggression

Abnormalities in brain 5-hydroxytryptamine (5-HT) neurotransmission have been reported in patients with

impulsive and aggressive behaviour, and low levels of the 5-HT metabolite 5-hydroxyindoleacetic acid (5-HIAA) have been found in the cerebrospinal fluid of subjects who have committed acts of unpremeditated violence (Linnoila and Virkkunen, 1992). In addition, more later studies found abnormalities in 5-HT-mediated neuroendocrine function and brain 5-HT_{2A}-receptor availability *in vivo* in people with high trait aggressiveness (Coccaro et al., 2010; Rosell et al., 2010). It has been suggested that the same abnormalities may be relevant to personality disorders that are characterized by impulsive behaviour, particularly in the presence of prefrontal deficits (Dolan et al., 2002).

Developmental theories

Separation. In a much-quoted early study, Bowlby (1944) studied 44 ‘juvenile thieves’ and concluded that separation of a young child from their mother may be related to the development of antisocial behaviour and failure to form close relationships. Not all separated children are affected adversely, and the effects of separation depend on many factors, including the child’s age, the previous relationship with the mother and father, and the reasons for separation. Rutter showed that *parental marital disharmony* partly accounts for the association between separation and antisocial disorder in sons (Rutter, 1972).

Social learning in childhood. Scott (1960) proposed four ways in which antisocial behaviour could develop through social learning:

1. Growing up in an antisocial family.
2. Lack of consistent rules in the family.
3. Overcoming other problems (e.g. aggressive behaviour to hide feelings of inferiority).
4. Poor ability to sustain attention, and other impediments to learning.

More recent research views antisocial personality as a neurodevelopmental outcome following multiple forms of adversity in the context of social, psychological, and biological adaptations (Glenn, 2019). These studies implicate parental sensitivity, maternal harsh intrusion, and commenting on the emotional state of the child in infancy, alongside attachment styles. The child’s temperament and biological affordances, for example, the autonomic nervous system and hypothalamic–pituitary–adrenal axis, are also relevant.

Childhood behaviour problems and antisocial personality. Early conduct problems have been shown to predict antisocial personality independently of associated adverse family and social factors (Hill, 2003), hence the requirement for a history of conduct disorder for the diagnosis of ASPD in DSM-5, and population studies

have demonstrated how early adolescent aggression predicts ASPD (Whipp et al., 2019).

Paranoid personality disorder

Little is known about the causes of this disorder. Some investigators have reported that paranoid personality disorder is more frequent among first-degree relatives of probands with schizophrenia than among the general population (Kendler et al., 1985), but others have not confirmed this finding (see Chapter 12).

Schizoid personality disorder

The cause of this disorder is unknown. It does not appear to be closely related genetically to schizophrenia (Fulton and Winokur, 1993). The psychoanalyst Melanie Klein outlined what she called the ‘paranoid–schizoid position’ (Klein, 1996). This was an early developmental phase before infants could sustain a clear understanding of their mother as an individual who came and went (the ‘depressive position’, so called because the infant could then miss the mother). Klein’s views were not intended to explain either paranoid or schizoid disorders, although they are often quoted as such, but rather to emphasize the importance of the mother–child relationship more than she thought Freud had done.

Schizotypal personality disorder

This is more frequent among biological relatives of individuals with schizophrenia than among adopted relatives or controls (Kendler et al., 1981). A review of 17 structural imaging studies of people with this personality disorder found brain abnormalities that were similar in most ways to those in people with schizophrenia (Dickey et al., 2002). Similarly, there is strong evidence for a genetic contribution, with heritability scores of 0.35–0.81 in twin studies (Ji et al., 2006; Kendler et al., 2007). These findings suggest that this personality disorder may be a milder form of schizophrenia, or that the two are related in some other way.

Borderline personality disorder

There is now an impressive volume of research into biological factors in borderline personality disorder (New et al., 2008), and some evidence for the importance of

genetic factors. Twin studies show heritability scores of 0.6–0.7 (Torgersen et al., 2012) and relatives of patients with borderline personality disorder are ten times more likely to be treated for borderline personality disorder or ‘similar personality disorder’ (Loranger et al., 1982). No specific genetic loci have been reliably identified but there is genetic overlap between borderline personality disorder and major depression, bipolar disorder, and schizophrenia (Witt et al., 2017). Imaging studies have implicated subtle grey matter changes in prefrontal and limbic regions, as well as functional dysregulation of the cortico-limbic circuitry involved in emotional regulation (Ruocco et al., 2013). Ligand binding and PET imaging studies have suggested that there may be abnormalities in opioid regulation in the same brain regions (Karas et al., 2021). More recent formulations of aetiology have focused on the developmental precursors of borderline personality. These theories point to trait abnormalities in impulse control and emotional regulation interacting with early adverse experiences such as harsh parenting or bullying (Winsper, 2018). For review, see Bohus et al. (2021).

Histrionic personality disorder

There have been few objective studies of the causes of this personality disorder. The genetics have not been studied with standardized methods of assessment, and the few reported investigations have yielded inconsistent findings (McGuffin and Thapar, 1992).

Obsessional personality disorder

Obsessional personality appears to have a substantial genetic aetiology (Reveley and Murray, 1980), although its nature is unknown. Not unexpectedly, people with obsessional personality show impaired executive function and cognitive inflexibility on neuropsychological testing (Marincowitz et al., 2021).

Anxious–avoidant, dependent, and passive–aggressive personality disorders

The genetics of these personality disorders have not been specifically studied, and their causes remain unknown at present.

The course of personality disorder

Personality disorders are defined as lifelong conditions, so little change would be expected with time. There is little reliable evidence about their outcome, other than for borderline personality disorder and, to a lesser extent, ASPD. The results of those studies that have been conducted make surprising reading. Much of the influential work comes from the United States and used repeated assessment according to DSM-IV criteria for personality disorder diagnosis or for personality disorder traits. Overall, this work indicates that personality disorder is much less stable than had previously been believed (Tyrer et al., 2015). For review, see Alarcón and Palmer (2020).

Borderline personality disorder. In the most influential outcome study, the McLean Hospital follow-up (Zanarini et al., 2006), 290 patients with borderline personality disorder were interviewed using a structured interview (SCID II) every two years for ten years. A total of 88% achieved 'remission' over this time period, 39% by two years, a further 22% by four years, a further 22% by six years, and so on. Those who still showed a borderline personality disorder more often had comorbid substance abuse or a history of childhood sexual abuse. A high rate of suicide has been found in some studies (8.5% in the study by Stone et al., 1987), raising the suspicion of an atypical affective disorder.

Aggressive and ASPDs. Clinical impressions indicate that minor improvement may take place slowly, especially in aggressive and antisocial behaviour ('psychopathy burns out').

Persistent personality disorder is associated with a poorer outcome (Tyrer et al., 2016). In a study of 524

subjects, 94 qualified for an ASPD diagnosis in adulthood, 82 of whom were interviewed an average of 30 years later (Robins, 1966); 12% had remitted, 27% had improved but not remitted, and 61% were unimproved or worse. That a subject had improved did not mean the disorder was no longer a problem. Whether ASPD improves significantly over time is the subject of conflicting evidence. Robbins and Regier (1991) reported a high rate of remission in the third and fourth decades of life, whereas Rutter and Rutter (1993) emphasized its stability. The view that such patients 'burn out' may reflect a reduction in help-seeking rather than late maturing.

Much less is known about the longer-term outcome in the other personality disorders. The overwhelming clinical impression is that borderline personality disorder is distinguished from the other disorders by its apparent variability. However, in the longitudinal studies using structured assessments there is also a high level of remission for other personality disorders. For instance, in a collaborative multi-site study of 668 DSM-IV-diagnosed personality disorder patients in the United States (Skodol et al., 2005), remission occurred in 38% of those with obsessive-compulsive personality and 23% of those with schizotypal personality disorder within two years. In a review of borderline disorder patients, 85% recovered for at least two years, while over 90% had periods of remission lasting at least two months (Biskin, 2015).

Mehlum et al. (1991) reported a three- to five-year follow-up of schizotypal personality disorder in which the outcome was worse than borderline personality disorder.

Treatment

There is little evidence on the treatment of personality disorder (Bateman et al., 2015) and most of the available data are concentrated on borderline disorder and ASPDs; the latter data were gathered mainly from incarcerated offenders. It is not surprising that research is limited. This is a patient group that is hard to engage in either treatment or research; the core features, such as functional disability and disturbed relationships, are much harder to operationalize and measure than traditional symptoms; comorbidity complicating the picture is the norm rather than the exception.

Most of the studies that have been conducted use the epiphenomena of the disorders as outcome. For example, in borderline personality disorder, self-harm and readmissions to hospital might be used to assess outcomes, while in ASPD, reduced aggression in the short term and reduced offending in the longer term are used as outcome measures. Despite (or perhaps because of) the variation in the manifestations of personality disorder reported in studies using structured assessments (Skodol et al., 2005), no trials have demonstrated changes to core features such as social functioning or sense of identity.

If treatments reduce some of the disruptive and distressing aspects of these disorders, that is an important and worthwhile achievement. The general approach should be one of engaging the person in a therapeutic relationship, using non-judgemental language, providing support, and adapting the service to provide helpful care while maintaining boundaries.

There are contradictions between different guidelines around the role of pharmacotherapy and the

role distinct contexts of care (Simonsen et al., 2019; Tyrer and Bateman, 2004). Despite the poverty of evidence, national guidelines have been published in the United Kingdom for the management of borderline disorder and ASPDs (for example, see NICE, <https://www.nice.org.uk/guidance/qs88>). The emphasis is firmly on psychosocial interventions with the use of drugs restricted to symptomatic management of crises.

Management

Assessment

There are general principles for the assessment and care of people with personality disorders in a non-stigmatizing and non-judgemental fashion, and at the same time recognizing the role of trauma and adversity, and considering varying levels of self-awareness (Davison, 2002). General indicators of a personality disorder include negative affect and affective dysregulation, impulsivity, interpersonal problems, strong emotional reactions to the clinician and the care system, and poor response to evidence-based treatments for other psychiatric conditions. As well as establishing a diagnosis of personality disorder (see guidance under classification, above), the *strengths and weaknesses* of the individual should be assessed. Strengths are important, because treatment should build on favourable features as well as attempting to modify unfavourable ones. The patient's *circumstances* should be assessed.

One of the most important services that a psychiatrist can provide for a personality-disordered patient is an honest yet sensitive sharing of their understanding of what is going on. Most patients have recognized that they have a problem of some kind and may be quite unrealistic about what psychiatrists have to offer. They may only have come to the psychiatric services because others have suggested it. An opportunity to talk through the strengths and weaknesses of their habitual responses can help an individual to put their characteristic behaviour patterns in perspective and perhaps find less harmful strategies for dealing with difficulties. This process can also avoid the development of unrealistic expectations of what others can do, which could otherwise lead to unproductive dependency, or more commonly feelings of being let down or failed. Regular structured attendance at outpatient clinics when there is little else that can be done can seem benign and compassionate,

and some find it helpful, but it may also carry a strong misleading message both for the patient and for others who relate to them. Yet discharge from services is also very distressing for some, who may have few other relationships.

Supportive therapy

Psychological support has been the mainstay of treatment of people with personality disorder. For some, modest but useful progress can be achieved by having a secure point of reference and a sense of being taken seriously over a period of months or, sometimes, years. Support may be provided by a member of the psychiatric team or, if the person has broken the law, by a probation officer. The dangers of this rather 'paternalistic' approach are increasingly being recognized, and supportive psychotherapy is probably best restricted to fairly senior staff, as it is difficult to do well and has a number of pitfalls.

General aims of management

Although there has been some progress in the treatment of personality disorders, management still consists largely of helping people to find a way of life that conflicts less with their character. Whatever approach is used, the aims should be modest, and considerable time should be allowed to achieve them. A trusting and confiding relationship is the basis of treatment. However, care should be exercised to avoid excessive dependency. Group treatment can sometimes be a way of lessening dependency and increasing social learning.

Often more than one professional is involved in the care of these patients. This can be risky, and close collaboration is needed to avoid inconsistencies of approach. Many of these patients react badly to changes in staff,

which may re-enact painful losses, rejections, or separations in their earlier life. One consistent, albeit limited, relationship may be a much better learning experience than a range of contacts, no matter how skilled they are. There is also the risk of idealizing some members of the team and not others, thus splitting team efforts, so careful communication across the team is essential.

The overall plan should include attempts to help the patient to have less contact with situations that provoke difficulties, and more opportunity to develop assets in their personality. They should be encouraged to take an active part in planning their care, and the reasons for decisions should be explained clearly and discussed fully. Helping the patient avoid adding to their problems by misusing drugs or alcohol is important, although not always possible in the short term. Even if no major improvement is achieved, this approach may stabilize the situation until some fortuitous change in the patient's life brings about improvement.

Choice of psychotherapy. Psychodynamic or cognitive psychotherapy may be explored when the patient is well-motivated and enough stability has been achieved to enable them to focus on the treatment. The approach will depend on the type of personality disorder, on local availability, and on the patient's own preference.

Choice of medication. Medication plays only a small part in the treatment of personality disorders. It should not be the first choice, and when it is prescribed it should be part of a wider plan that embraces psychological and social care. Anxiolytic medication should be used with special caution because of the risk of dependency. Antidepressants are used mainly for associated mood disorder—a common reason for the worsening of the emotional and behavioural problems associated with the personality disorder. The use of antidepressants

in borderline personality disorder is discussed below. Antipsychotics are occasionally useful to reduce aggressive behaviour, but again there is inconsistent evidence to support their value.

Organization of services. People with personality disorder can be cared for by a single practitioner who integrates the psychological, pharmacological, and social aspects of treatment and calls on specialist help when needed. This makes it easier for the practitioner to understand and help with relationship problems, and it emphasizes the essential unity of the person's diverse problems. However, it can be difficult to sustain when the personality disorder is severe. Alternatively, care can be provided by a community team, with a key worker who coordinates the contributions of other team members. Since the publication of the Department of Health paper, *Personality Disorder: No Longer a Diagnosis of Exclusion* (Snowden and Kane, 2003), specialist teams in the United Kingdom have been established specifically for the treatment of personality disorder. Their configuration varies enormously, but proposes a form of therapeutic day programme based on therapeutic community principles and often incorporating principles of mentalization (Bateman and Fonagy, 2010).

Progress

Progress is often achieved as a series of small steps by which the person gradually moves closer to a satisfactory adjustment. Setbacks are common, but they can be used constructively, as it is at these times that the patient is most likely to be willing to confront their problems. Although therapists should try long and hard to help patients with personality disorder, they should recognize that some people may not benefit.

The management of specific personality disorders

The type of personality disorder is not always a good guide to the choice of effective treatment. Nevertheless, some associations have been described, mainly on the basis of clinical experience, and these will be considered here before outlining the approach to borderline and antisocial disorders.

Patients with *paranoid personality disorder* do not engage well with treatment because they are touchy and suspicious. Patients with *schizoid personality disorder* avoid close personal contact, and often drop out after a few sessions of treatment. If they can be persuaded to

continue, they tend to intellectualize their problems and question the value of their treatment. The therapist should try to help these people to become more aware of their problems and respond to them in ways that cause fewer difficulties. For both disorders progress is slow, and the results are limited. Exploratory psychotherapy is unlikely to succeed, and medication is generally unhelpful.

Paranoid personality traits and experiences have become the focus of recent research into the possibility of intervening early in psychosis. Studies in virtual

reality laboratories have been used to identify them and tailored forms of cognitive behaviour therapy are being tested as possible approaches (Freeman and Garety, 2014).

Patients with *obsessional personality disorder* do not respond well to psychotherapy. Unskilled treatment can lead to excessive morbid introspection, which leaves the person feeling worse rather than better. Treatment should be directed towards avoiding situations that increase difficulties, and developing better ways of coping with stressful situations. Patients often seek help during an associated depressive disorder, and it is important to identify and treat this comorbid condition.

Patients with *avoidant (anxious) personality disorder* generally have low self-esteem, and fear disapproval and criticism. They can be helped by a therapeutic relationship in which they feel valued and able to reconsider their perception of themselves. As with obsessional personality disorder, the possibility of a comorbid depressive disorder that requires treatment should be considered.

Patients with *dependent personality disorder* are usually helped more by problem-solving counselling, in which they are encouraged to take increasingly more responsibility for themselves. Medication should be avoided unless there is an associated depressive disorder.

Borderline personality disorder

Little is achieved by arguing about the nosological coherence of borderline personality disorder—its contested history and boundaries and, least of all, its misleading name. Borderline is the personality disorder that most often calls for a response from psychiatry and psychiatrists. It can present with crises, depression, and self-harm. The care has to be negotiated—patients will inevitably exert a considerable influence on the choice of treatment, and all psychiatrists, irrespective of their interests, are likely to have to manage people with borderline disorders alone or comorbid with other conditions.

While the main body of research is into sophisticated daycare units based on either mentalization or dialectic behaviour therapy (DBT; see below), not all patients can have access to these, nor will all patients accept this approach. As a consequence, virtually every form of psychotherapy or social therapy has responded with adaptations and strategies (Bateman et al., 2015).

In many parts of the world psychoanalytical approaches are still pursued in outpatient or office-based practice, although focused more firmly on transference issues (Levy et al., 2019). Cognitive behaviour

therapists focus more on schemas (the underlying assumptions patients make rather than exclusively on automatic thoughts; see Beck et al., 2015). Short-term integrative therapies such as cognitive analytical therapy (Ryle and Karr, 2020) have specified adaptations for borderline patients. The two most influential adaptations have been in behaviour therapy, with the development of DBT (Linehan, 2018) and in psychodynamic therapy, with mentalization (Bateman and Fonagy, 2010; Bateman et al., 2015).

Dialectical behaviour therapy

DBT is an intense form of psychotherapy that includes individual sessions, group sessions, and, optimally, daycare. It was developed by the psychologist Marsha Linehan to help people reduce harmful behaviours, such as self-harm, suicidal thinking, and substance abuse. The approach aims to increase emotional and cognitive regulation by awareness of triggers and assessing coping skills to apply. Linehan has more recently acknowledged that her life was dominated by such behaviours and that finding some effective treatment was vital for her own functioning. It includes an eclectic mix of therapies, with cognitive behaviour therapy to increase emotional regulation plus mindfulness to increase awareness, and some pretty blunt confrontation. The treatment is outlined in Chapter 24.

Mentalization

The residential therapeutic communities developed after the Second World War to help individuals with personality disorder were aimed at those with ASPD but contained many with what would now be recognized as borderline personality disorder. They are described in Chapter 24 and, because of an absence of evidence for their effectiveness and their cost, are no longer a feature of mental health care (although they are still influential in prisons and drug treatment units). However, day units based broadly on therapeutic community principles have evolved for the treatment of borderline personality disorder. What distinguishes them is their focus on ‘mentalization’. Essentially, this involves helping the patient become more fully aware of their thoughts and feelings rather than acting on them without reflection.

Mentalization day hospitals vary, but all usually include both individual and group work, and their approach is summarized in Box 15.9. For the many patients unable or unwilling to attend such units, principles of ‘generalist’ management is summarized in Box 15.10.

Box 15.9 Five common characteristics of evidence-based treatments for borderline personality disorder

1. Structured (manual-directed) approaches to prototypic borderline personality disorder problems
2. Patients are encouraged to assume control of themselves (i.e. sense of agency)
3. Therapists help connections of feelings to events and actions
4. Therapists are active, responsive, and validating
5. Therapists discuss cases, including personal reactions, with others

Box 15.10 Proposed characteristics for a generalist approach to treating borderline personality disorder

1. Treatment providers have previous experience with borderline personality disorder
2. Supportive (i.e. encouraging, advisory, and educational)
3. Focus on managing life situations (not on the in-therapy interactions)
4. Non-intensive (i.e. once per week, with additional sessions as needed)
5. Interruptions are expected; consistent regular appointments are optional
6. Psychopharmacological interventions are integrated; group or family interventions are encouraged when necessary

Other psychotherapeutic approaches

There is also evidence that transference-focused therapy, schema-focused therapy, and cognitive analytic therapy can all be helpful (see Chapter 24; see also the new clinical update from Bax et al., 2023).

Antisocial personality disorder (ASPD)

Most treatment programmes are delivered by forensic psychologists in the prison service and comprise schema-based cognitive behaviour therapy, often in groups. Few psychiatrists feel they have much to offer in the treatment of ASPD beyond identifying it and exploring its consequences with the patient and family.

Pharmacotherapy

NICE guidance (see <https://www.nice.org.uk/guidance/qs88>) advises against the use of medication unless a clear comorbid disorder (e.g. major depression) is present. A meta-analytical review by Lieb et al. (2010) concluded that there was some evidence, albeit limited, that antipsychotic drugs and anticonvulsant mood stabilizers could have beneficial effects on aspects of borderline personality disorder, such as affective instability, impulsive dyscontrol, and cognitive perceptual symptoms. Nevertheless, it does not appear likely that current drug treatments are able to modify core borderline symptomatology, including fear of abandonment, feelings of emptiness, and identity disturbance (Gartlehner et al., 2021)

Ethical problems

Stigma and patient involvement

From the early ideas of moral insanity to the present time, the concept of personality disorder has been linked to a moral judgement. People who are told that they have a disordered personality believe there is something inadequate about who they are. This can be entrenched by some professionals' attitudes, and by the difficulty that people with personality disorder experience in accessing psychiatric and other services. Once

the diagnosis has been made it tends to stick and to affect the person's subsequent care, as well as the way that other people think of them over the longer term. For this reason, many clinicians are reluctant to make the diagnosis.

This situation is changing, and the policy document, *Personality Disorder: No Longer a Diagnosis of Exclusion* highlighted the need for all psychiatrists to engage with personality-disordered individuals. The existence of a personality disorder is associated with increased rates of

other psychiatric disorders. Even if psychiatrists do not believe that they can change the patient's personality, that is no reason not to treat the associated problems. This renewed interest has also led to the establishment of specific psychotherapeutic teams and services for

personality-disordered patients. These services are very varied in their form (reflecting the absence of strong evidence for specific treatments), but, as mentioned above, many of them are centred around a psychodynamic day service (Bateman and Krawitz, 2013).

Further reading

Geddes, J. R., et al. (Eds). (2020). Section 19: Personality disorders. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press. (The six chapters in this section provide a comprehensive review of personality disorder.)

Schneider, K. (1950). *Psychopathic Personalities* (trans. M. W. Hamilton). Cassell. (A classic text of great importance in the development of ideas about personality disorder.)

CHAPTER 16

Child and adolescent psychiatry

Introduction 403

Normal development 404

Classification of psychiatric disorders in children
and adolescents 407

Epidemiology 409

Aetiology 410

Psychiatric assessment of children and
their families 415

Physical examination 417

Psychiatric treatment for children and
their families 419

Review of disorders 423

Ethical and legal problems in child and
adolescent psychiatry 464

Appendix: Basic elements in child and adolescent
psychiatric assessments 465

Introduction

A significant proportion of 'adult' mental illness starts before the age of 18 years. There are also specific psychiatric disorders of childhood. These factors attest to the importance of properly assessing child development, identifying early signs of mental illness, and instituting effective treatment. The period of childhood is relatively brief yet the developmental tasks a child needs to perform are considerable, including learning a broad range of skills, developing physically and cognitively, and managing their social and peer environment with adequate communication skills and emotional control. Mental illness can impact on the developmental tasks of childhood and potentially negatively impact on a child's developmental trajectory for the rest of their life, for example, a child might have such low self-esteem they are unable to make friends at school, or perform badly in school achievement tests and do not pursue higher education.

In recent years there has been a shift in the understanding of the importance of adolescence as a unique period of brain development. Adolescence is a time of social, psychological, and biological development during which relationships with others, especially peers, become

paramount (Foulkes and Blakemore, 2018). Substantial structural and functional changes take place in the brain during this period that is now increasingly defined as lasting from the ages of 12 to 24 years (Sawyer et al., 2018). This chapter focuses on children and adolescents up until the age of 18 years and will be referred to under 'child' in this chapter unless specific age groups are being described.

The practice of child and adolescent psychiatry differs from that of adult psychiatry in a number of important ways.

- *Children are usually dependent on an adult to help them access services.* They seldom initiate the consultation and are usually brought by a parent, or another adult, who has concerns about some aspect of the child's cognitive, emotional or behavioural development. Whether a referral is sought depends on the attitudes and tolerance of these adults, and how they perceive the child's behaviour. Healthy children may be brought to the doctor by over-anxious and solicitous parents or teachers, while in other circumstances severely disturbed children may be left without involving formal services.

- *The child's difficulties may reflect the problems of other people*—for example, parental physical or mental illness.
- *The child's stage of physical and cognitive development must be considered* when deciding what is abnormal. For example, repeated bedwetting may be normal in a 3-year-old child but is abnormal in a 7-year-old. In addition, the child's response to life events changes with age. Thus separation from a parent will affect children in different ways across their childhood.
- *Children are generally less able to express themselves in words.* For this reason, evidence of disturbance often comes from observations of behaviour made by parents, teachers, and others. These informants may give differing accounts, in part because children's behaviour varies with their circumstances, and in part because the various informants will likely have different beliefs about the parameters of normal and abnormal behaviour. For this reason, informants should be asked for specific examples of any problem they describe, and asked about the circumstances in which it has been observed. Furthermore, physical manifestations of psychological difficulties, often termed somatization, are more common and developmentally appropriate in younger populations where it can be harder for the child to find the words to describe or even understand what they may be feeling.
- The vast majority of children attend schools that are found across the world with variable sizes, resources, philosophies, and success. They can play an important role in the overall development of children and can also be areas of great academic and social stress for children. Furthermore, schools offer a location for accessing children for prevention and treatment of mental illness.
- *The emphasis of treatment is different.* Medication is used considerably less in the treatment of children than in the treatment of adults, and is usually started by a specialist rather than the family doctor. Instead, there is more emphasis on psychological interventions, working with the parents and the whole family, reassuring and retraining children, and coordinating the efforts of others who can help children, especially at school. Thus multidisciplinary working is even more important in child psychiatry than in adult psychiatry. Consequently, treatment is usually provided by a team that might include a psychiatrist, psychiatric nurses, a clinical psychologist, and potentially also those trained in other therapies, social care, or occupational therapy.
- Ethical considerations can make participation of children in research complicated, and so the evidence base, particularly on pharmacological interventions, is limited for children.

The first part of this chapter discusses a number of general issues concerning psychiatric disorder in childhood, including its frequency, causes, assessment, and management. The second part of the chapter contains information about the principal syndromes encountered in the practice of child and adolescent psychiatry. The chapter does not provide a comprehensive account of child psychiatry. It is an introduction to the main themes for psychiatrists who are undertaking specialist general training with more detail available from specialist texts such as one of those listed under Further reading. Disorders of intellectual development (DID) among children is considered in Chapter 17 but the reader should remember that many aspects of the study and care of children with DID are similar to those described in this chapter.

Normal development

The practice of child psychiatry calls for some knowledge of the normal process of development from a highly dependent infant into an independent adult. In order to judge whether any observed emotional, social, or intellectual functioning is abnormal, it has to be compared with the corresponding normal range for that age group. This section gives a brief and simplified account of the main aspects of development that concern the psychiatrist, drawing largely on Western developmental frameworks and therefore not able to provide a fully cross-cultural perspective. A textbook of paediatrics

should be consulted for details of these developmental phases (see Lissauer and Carroll, 2021).

The first year of life

This is a period of rapid motor, sensory and social development as well as the first stages of language development. Three weeks after birth, the baby smiles at faces, selective smiling appears by six months, fear of strangers by eight months, and anxiety on separation from the mother shortly thereafter.

Bowlby (1980) emphasized the importance of a general process of attachment of the infant to the parents, and of more selective emotional bonding. Although bonding with the mother is most significant, important attachments are also made to the father and other people who are close to the infant. Other studies have shown the reciprocal nature of this process and the role of early contacts between the mother (or other carers) and the newborn infant in initiating bonding. Attachment and bonding are discussed further on p. 383–384 and in Chapter 15 on personality and personality disorders.

By the end of the first year, the neurotypical child will have likely formed a close and secure relationship with their primary caregiver/s. There is often an ordered pattern of sleeping and feeding, and weaning onto solid food has usually been accomplished. The infant will be able to point at objects and things they want. Children will have begun to learn about objects outside themselves, simple causal relationships, and spatial relationships. By the end of the first year, they enjoy making sounds and may say ‘mama’, ‘dada’, and perhaps one or two other words.

Year two

This too is a period of rapid development. Children begin to want to please their parents/caregivers, and appear anxious when they perceive disapproval. They explore their environment more through play and with increasing mobility, as most will be walking by the time they are 18 months of age, they also begin to learn to control their behaviour. By now, attachment behaviour should be well established. *Temper tantrums* may occur, particularly if exploratory wishes are frustrated. These tantrums do not last long, and should lessen as the child learns to accept constraints. They start to point at things that they want others to observe out of interest. By the end of the second year the child should be able to put two or three words together as a *simple sentence*.

Preschool years (2–5 years of age)

This phase brings a further increase in intellectual abilities, especially in the complexity of language. Social development occurs as children learn to live within their social environment, often including their family and other places of care, such as at nursery or preschool. They begin to identify with their parents/caregivers, and their social life develops rapidly as they learn to interact with siblings, other children, and adults. Temper tantrums can continue, but diminish and often disappear before the

child starts school. Attention span and concentration increase steadily. At this age, children are very curious about the environment and ask many questions.

In children aged 2–5 years, fantasy life can be rich and vivid. It can form a temporary substitute for the real world, enabling desires to be fulfilled regardless of reality. Special ‘transitional’ or attachment objects such as teddy bears or comfort blankets can become important to the child, offering them psychological support.

Children begin to learn about their own *identity*. They realize the differences between males and females in appearance, clothes, behaviour, and anatomy.

During these years the most common mental health presentations to primary care services include difficulties in feeding, sleeping, and constipation, as well as clinging to their primary caregivers (separation anxiety), temper tantrums, oppositional behaviour, and minor degrees of aggression. These are often managed in primary care or in paediatric services but can be referred to mental health services if they are persistent and interfere with the child’s ability to develop physically or interact socially in their family or with peers.

Middle childhood

By the age of 5 years, children usually know their gender identity and their position in the family. They learn to cope with school, and to read, write, and begin to acquire numerical concepts. School staff, especially a class teacher, become important other adults in children’s lives. At this stage children gradually learn what they can achieve and what their limitations are. Conscience and standards of social behaviour develop further.

Common problems in this age group are predominantly anxiety related and include fears, nightmares, minor difficulties in relationships with peers, as well as behavioural problems such as conduct disorder (CD).

Adolescence

Adolescence is the growing-up period between childhood and maturity and also a sensitive period in the development of the brain. Adolescence is a time of significant neuroplasticity—where the nervous system adapts its structure and function in response to environmental demands, experiences and physiological changes (Fuhrmann et al., 2015). Among the most obvious features are the physical changes of puberty. The age at which these changes occur is quite variable, usually between 11 and 13 years in girls, and between 13 and 17 years in boys. The production of sex hormones precedes these changes, starting in both sexes between the ages of

8 and 10 years. An adolescent has increased awareness of personal identity and individual characteristics. At this age, young people become self-aware, are concerned to know who they are, and begin to consider where they want to go in life. They can look ahead, consider alternatives for the future, and feel hope and despair. Some experience emotional turmoil and feel alienated from their family, but such experiences are not universal.

Peer group influences become more prominent and close friendships often develop, especially among girls. Membership of a group is common, and this can help the adolescent in moving towards autonomy. Adolescence brings a greater sense of their own gender identity as well as an increase in sexual interest and activity. At first, tentative approaches are made to individuals to whom they are attracted. Gradually these become more direct and confident. How far and in what way sexual feelings are expressed depends greatly on the societal norms, the behaviour of their peer group, and the attitudes of the family.

Common problems among children aged from 12 to 16 years include mood disorders sometimes presenting with self-harm, a range of different anxiety problems, difficulties in relationships with peers, behavioural difficulties including CD and truancy, as well as starting to experiment with alcohol and illicit substances. Schizophrenia and bipolar disorder may have their onset in this age group, but are uncommon.

Developmental psychopathology

In child psychiatry it is important to adopt a developmental approach for three reasons.

1. *The stage of development determines whether behaviour is normal or pathological.* For example, as noted above, bedwetting is normal at the age of 3 years but abnormal at the age of 7 years.
2. *The effects of life events differ as the child develops.* For example, infants aged under six months can move to a new carer with little disturbance, but children six months to three years of age show great distress when separated from a familiar carer, because an attachment relationship has been formed. After the age of 3 years, attachment bonds are still strong but the child's ability to understand and to use language can reduce the effect of a change of caretaker, especially if arranged sensitively.
3. *Psychopathology may change as the child grows older.* Anxiety disorders in childhood tend to improve as

the child develops, depressive disorders often recur and continue into adult life, CDs can continue into adolescence as aggressive and delinquent behaviour, and also as substance abuse—a problem that is less common in younger children. These changes and continuities are sometimes related as much to changes in the environment as to developmental changes in the child.

The psychopathology of individual disorders will be discussed later in the chapter. Here some general issues regarding childhood psychopathology are summarized.

The influence of genes. Most of the major syndromes of childhood and adolescence have a genetic contribution, mostly from common variants of small effect but also some rare variants of large effect, as described below. These genetic influences often show a significant overlap between syndromes, and are thought to interact with epigenetic and environmental influences (Kiser et al., 2015).

The influence of the environment. Factors in the environment may predispose to or precipitate disorders, and they may maintain them. They may also protect from the effects of other causative agents—for example, the risk of depression in adulthood as a consequence of poor parental care in childhood is reduced by the experience of a caring relationship with another person (Blakemore, 2019). This experience is not protective against all causative factors—for example, a caring relationship might not reduce the risk of depression in adulthood following child abuse. Furthermore, genetic and environmental factors interact, making certain exposures more likely to occur, or affecting their consequences. For example, genes influence the personality traits of irritability and impulsiveness that can contribute to repeated relationship difficulties.

The dividing line between normal and abnormal. The common childhood disorders are at the extreme end of a continuum with normal behaviour. Despite this, we study them using a categorical system because decisions about interventions and involvement with mental health services often require 'yes or no' answers. To form a category, a cut-off point has to be decided, and this is usually rather arbitrary, and can be influenced by the environment in which the child is assessed, their physical state, and the other individuals in their presence. Children who fall just below the cut-off point—and hence outside the diagnostic group—can nevertheless have problems and need help. For example, some adolescents who fall below the threshold for depressive disorder have significant psychosocial impairment, and even mild depressive symptomatology can be

associated with poor academic performance (Wickersham et al., 2021).

Continuities and discontinuities. Some symptoms and behaviour problems in childhood are associated with longer-term problems continuing into adulthood and might make that adult up to six times more likely to experience an adverse outcome. For example, untreated ADHD or CD can be associated with increased prevalence of substance misuse and suicidality when that child becomes a young adult (Copeland et al., 2015).

Parent–child interactions. Maternal behaviour affects the child, but children also elicit behaviours from their parents. For example, a parent might find it harder to be responsive to an infant who does not respond to cuddling and play than to a sibling who is affectionate towards them. This lack of responsiveness may then affect the infant, thus increasing attachment difficulties.

Many of these issues are also relevant to psychopathology in adults, and are considered further in Chapter 16.

Classification of psychiatric disorders in children and adolescents

DSM-5 and ICD-11 contain schemes for classifying the psychiatric disorders of childhood. Compared to their predecessors, both have shifted to a more lifespan approach in how many of the disorders are described with an appreciation of developmental change over time. Impairment of function is a key principle for the diagnosis of disorders. Disorders of adolescence are classified partly with this scheme, and partly with the categories used in adult psychiatry.

Seven main groups of childhood psychiatric disorders are generally recognized by clinicians. The terms used in this book for the seven groups are listed below, with some alternatives shown in parentheses:

1. Adjustment reactions
2. Autism spectrum disorders (ASDs)
3. Specific developmental disorders
4. Conduct (antisocial or externalizing) disorders
5. Attention-deficit hyperactivity disorders (ADHDs)
6. Emotional (internalizing) disorders
7. Symptomatic disorders.

Many child psychiatric disorders cannot be classified in a satisfactory way by allocating them to a single category. Therefore more stress has been placed on psychiatric comorbidity, and allowance for multiple diagnoses. This represents a move away from the previous multi-axial framework where five to six different axes were described in the formulation (including: clinical psychiatric syndromes; specific delays in development; intellectual level; other somatic medical conditions; challenging social situations; level of adaptive functioning).

More emphasis is also placed in DSM-5 on *clinical case formulation* where a clinical summary of the social, psychological, and biological factors that contribute to the development of a mental disorder are incorporated, as well as a measure of clinical significance, which is a proxy for individual distress and/or impairment (Harris, 2014).

The DSM-5 and ICD-11 classifications for child psychiatric disorders are shown in Box 16.1 and Box 16.2 where they are separated by degree of similarity and difference between the two diagnostic systems (First et al., 2021). (In this book, disorders of intellectual development in childhood are considered in Chapter 17). Both schemes are complicated, so only the main categories as relevant to this chapter and not covered in other areas of this book, are shown in the table. In DSM-5 many of the disorders of childhood have been removed from a more generic ‘infancy, childhood, and adolescence section’ that was in DSM-IV and integrated into the chapter relevant to that specific disorder. The chapters are then organized and presented in chronological fashion with diagnoses most applicable to infancy and childhood listed first. The neurodevelopmental disorders that are categorized in ICD-11 are of particular relevance to child psychiatry and are not covered in as much detail in other chapters of this book. The neurodevelopmental disorders include disorders of intellectual development (in Chapter 17); developmental speech or language disorders; ASDs; developmental learning disorders; developmental motor coordination disorder; ADHD; stereotyped movement disorder and a remainder category of ‘other neurodevelopmental disorders’. For more information on the two classification systems see Stein (2020) and Garralda (2016).

Box 16.1 ICD 11 Categories of note for children (and not primarily covered in other chapters)

Neurodevelopmental disorders

Disorder of intellectual development (Chapter 17)
 Developmental speech or language disorders
 Autism spectrum disorders (ASD)
 Developmental learning disorders
 Developmental motor coordination disorders
 Attention-deficit hyperactivity disorders (ADHD)
 Stereotyped movement disorder
 Other neurodevelopmental disorders

Disruptive behaviour and dissocial disorders

Oppositional defiant disorder
 Conduct-dissocial disorder
 Other specified disruptive behaviour or dissocial disorders
 Disruptive behaviour or dissocial disorders, unspecified

Source: *International Classification of Diseases, Eleventh Revision (ICD-11)*, Copyright (2022), World Health Organization. <https://icd.who.int>

Box 16.2 Differences in DSM-5 and ICD-11 classification of primarily childhood psychiatric disorders (not covered in other chapters)

Disorders with essentially identical definitions in the two diagnostic systems

- Developmental speech fluency disorder in ICD-11/childhood onset speech fluency disorder in DSM-5
- Avoidant restrictive food intake disorder
- Enuresis
- Encopresis
- Intermittent explosive disorder

Disorders with minor differences between the two diagnostic systems due to greater degree of specification in the DSM-5

Conduct-dissocial disorder in ICD-11/conduct disorder in DSM-5

Disorders with minor definitional differences between the two diagnostic systems

- ASD
- Developmental learning disorder in ICD-11/specific learning disorder in DSM-5
- ADHD (differences also in terms of degree of specification)
- Reactive attachment disorder

Disorders or diagnostic entities with major differences between the two diagnostic systems

Developmental language disorder in ICD-11/
 Language disorder *plus* social (pragmatic) communication disorder in DSM-5

Source: *International Classification of Diseases, Eleventh Revision (ICD-11)*, Copyright (2022), World Health Organization, <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)*, Copyright (2013), American Psychiatric Association.

Epidemiology

Behavioural and emotional disorders are common in childhood. Estimates vary according to the diagnostic criteria and other methods used. A landmark study, carried out more than 50 years ago on the Isle of Wight in the UK, studied all the 10- and 11-year-olds attending state schools on the island—a total of 2193 children (Rutter et al., 1970).

Overall rates of psychiatric disorder. There are many barriers to estimating an exact prevalence, including differing definitions of mental ill health and diversity of methodological approaches (Kieling et al., 2011). Globally, meta-analytic evidence suggests that the point prevalence of mental health disorders among children and adolescents aged 6–18 years is 13.4% (95% confidence interval [CI] 11.3–15.9; (Polanczyk et al., 2015)). The National Comorbidity Study in US adolescents showed how for adolescents, anxiety disorders are the most common condition (31%), followed by behaviour disorders (19%), mood disorders (14%), and substance use disorders (11%) (Merikangas et al., 2010). The overall prevalence of disorders with severe impairment and/or distress was 22% (11% mood; 8% anxiety; 9% behaviour). The median age of onset for disorders was earliest for anxiety (6 years) followed by 11 years for behaviour, 13 years for mood, and 15 years for substance use disorders. In the UK, the Mental Health of Children and Young People (MHCYP) survey provides high-quality evidence regarding the prevalence of ‘probably mental disorders’ among English children and adolescents. The 2017 findings are largely comparable to findings from the meta-analysis of global prevalence described above, with an estimated 11.6% of children and adolescents aged 5–17 years estimated to have at least one diagnosable mental health disorder (Sadler et al., 2018).

In determining time trends, before the Covid-19 pandemic there was concern that the prevalence of child and adolescent mental health difficulties might be increasing. A 2015 review of studies, including representative population samples of children and adolescents, conducted from 1970 onwards found evidence for increases in emotional problems (including symptoms of anxiety and depression) and conduct problems across

several countries, particularly for adolescents (Collishaw, 2015). However, these trends were not consistent across all studies (with some finding improvements over time). Following the Covid-19 pandemic, there has been additional concern raised about the negative impact this might have had on the mental health of children and adolescents. The online follow-up surveys (Newlove-Delgado et al., 2021; Vizard et al., 2020) to the 2017 MHCYP survey provide high-quality evidence on post-pandemic changes. Compared with the changes in prevalence from 1999 to 2017, these two follow-up surveys demonstrated a marked rise. From 2017 to July 2020, the prevalence of ‘probable’ mental health disorder among 5- to 16-year-olds increased from 10.8% to 16.0%. From 2017 to 2021, the analogous increase for 6- to 16-year-olds (noting the one-year difference in age range than was used in 2020) was 11.6% to 17.4% (Newlove-Delgado et al., 2021). This deterioration in mental health was not universal and indeed some children and adolescents reported improved mental wellbeing during this time (Soneson et al., 2022). It is important to consider the potential for longer-term impacts of the pandemic and associated lockdowns on the mental health of children and adolescents.

Variations with gender and age. Before puberty, disorders are more frequent overall among boys than among girls after puberty, disorders are more frequent among girls. Particular disorders also vary in frequency according to gender and age (see Table 16.1).

Other sources of variation. Studies have categorized risk factors that increase the likelihood of mental, emotional, and behavioural disorders into six domains (Greenberg et al., 2015):

1. Child: e.g. self-regulation problems
2. Family: e.g. quality of parent-child attachment
3. Peer: e.g. isolation
4. Demographic: e.g. family income and family structure/parental relationship
5. School: e.g. school climate
6. Community/policy: e.g. neighbourhood policy and crime rates.

Table 16.1 Comparative frequency of childhood psychiatric disorders and presentations in boys and girls

More frequent in boys	Equal	More frequent in girls
ASDs	Depression before puberty	Depression (post-pubertal)
Specific developmental disorders	School refusal*	Anxiety disorders Specific phobia
ADHD	Selective mutism	Eating disorders
ODD and CDs		Self-harm
Juvenile delinquency*		
Nocturnal enuresis—in older children		
Tic disorders		
Suicide		

* Juvenile delinquency and school refusal are descriptive terms, not diagnoses.

Aetiology

In discussing the causes of child psychiatric disorders, the principles are similar to those described in Chapter 5 on the aetiology of adult disorders. Most childhood disorders are likely to emerge from a complex interaction between emerging neurodevelopmental vulnerabilities, in part genetically influenced, and aspects of the child's pre- and post-natal environment. There is also a developmental aspect: children mature psychologically and socially as they grow up, and their disorders reflect this maturation. In the following paragraphs, four interacting groups of factors are considered briefly: heredity, temperament, physical impairment with special reference to brain damage, and environmental factors in the family and society. All aspects of aetiology are discussed more fully in *Rutter's Child and Adolescent Psychiatry* textbook (Thapar et al., 2015).

Genetic factors

Children with psychiatric disorders tend to cluster within families, with higher rates observed among the parents and other relatives of affected children. Environmental factors account for a substantial part of this association. However, population-based twin studies indicate that there is a significant genetic contribution to some psychiatric disorders, especially to ASDs, hyperactivity, and even about 40% heritability of anxiety disorders.

Genetic studies are considered further when the aetiology of particular childhood psychiatric disorders is described later in this chapter. Here, we consider some general matters concerned with the interpretation of the results of genetic studies. Examples of genetic findings in child psychiatry are given at relevant points in this chapter, especially in the sections on autism and hyperactivity, where genetic research has produced some relevant findings (see p. 434). Some general points about the interpretation of genetic studies are shown in Box 16.3.

Temperament

There is an association between temperamental styles and subsequent childhood psychopathology, with temperament measured between ages 0–6 years able to predict a number of life outcomes. These include cognitive ability, educational attainment, BMI, civic engagement, and divorce (Wright and Jackson, 2022).

Brain disorders

Although serious physical disease of any kind can predispose to psychological problems in childhood, those affecting the brain, such as traumatic injury, epilepsy, infections, tumours, and neurodegenerative or other

Box 16.3 Interpreting the results of genetic studies of childhood psychiatric disorders

Phenotype and genotype. Although standard diagnostic criteria are valuable in research, the results of this research show that diagnostic categories generally do not map closely onto the genetic factors in aetiology. For example, the phenotype of ASDs includes some kinds of developmental language disorder, while the phenotype of Tourette syndrome (Chapter 14) seems to extend to obsessional thinking and behaviour.

Comorbidity. Many children have symptoms that qualify for more than one psychiatric diagnosis—for example, depressive disorder and CD, or depressive disorder and eating disorder. It is not clear whether this overlap arises because one disorder predisposes to another, or because they are two manifestations of the same genetic predisposition.

Polygenic effects and environmental influences. The hereditary factors of importance in child psychiatry are largely polygenic. As in adulthood, these multiple genes interact with psychosocial factors, and genetic investigations may include estimates of environmental factors.

Indirect effects. Genes may exert their effects through factors such as intelligence and temperament, which in part determine whether certain situations are experienced as stressful.

Shared and non-shared environment. Analyses of population genetic data divide the variance into genetic, shared environmental, and non-shared environmental factors (see p. 105). Twins, full siblings, and half-siblings are increasingly used as comparison groups in studies and although these sibling comparison groups are assumed to be similar in many aspects of ‘shared’ environment this assumption has limitations, especially in the interpretations of the impact of the non-shared environment. For example, a shared environment can affect different children in different ways, and this effect then appears in the analyses as ‘non-shared environment.’

conditions, are important risk factors (Fazel et al., 2021; Verkuijl et al., 2015).

Children with brain injury are more likely to develop psychiatric disorder if they encounter adverse

psychosocial influences of the kind that provoke psychiatric disorder in children without brain damage.

Maturation changes and delayed effects

The effects of brain lesions are more complex in childhood than in adult life because the brain is still developing. This has two consequences:

- *Greater capacity to compensate.* The immature brain is more able than the adult brain to compensate for localized damage—it has greater *plasticity*. For example, even complete destruction of the left hemisphere in early childhood can be followed by normal development of language.
- *Delayed effects.* Early damage may not be manifested as a disorder until a later stage of development when the damaged area takes up some key function. It is well established that brain injury at birth may not result in seizures until many years later. It has been suggested that there may be similar delays in the behavioural consequences of brain injury.

The consequences of head injury in childhood

Head injury is a common cause of neurological and psychological impairment in childhood and increasing evidence is being gathered regarding the different chronic negative impacts of head injury. For example, following a concussion there can be cognitive, physical, and affective symptoms, such as difficulty concentrating, headache, and irritability that can occur in up to 30% of children with mild head injury (Ewing-Cobbs et al., 2018). Parents have reported up to 40% of children still acting differently one year after the injury with some emotional and cognitive symptoms emerging over time.

Epilepsy as a cause of childhood psychiatric disorder

The relationship between epilepsy and psychiatric disorder in adults is considered on pp. 370. In childhood there is a strong association between recurrent seizures and psychiatric disorder. As in adult life, the causal relationship may be of four kinds:

1. the brain lesion causing the epilepsy may also cause the psychiatric disorder;
2. the psychological and social consequences of recurrent seizures may cause the disorder;
3. the effects of epilepsy on school performance may cause the disorder; and/or
4. the drugs used to treat epilepsy may cause the disorder through their side-effects.

The *site and the type of epilepsy* appear to be generally less important, apart from the fact that temporal lobe epilepsy seems to be more likely to be associated with psychological disorder. The age of onset of seizures also determines the child's response to epilepsy. For a review of the influence of brain injury and epilepsy on psychopathology, see Heyman et al. (2015) .

Environmental factors

The effect of life events

The concept of *life events* (see p. 100) is useful in child psychiatry as well as adult psychiatry. Life events may predispose to or provoke a disorder, or protect against one. Events can be classified by their severity, their social characteristics (e.g. family problems, or the death of a parent), or their general significance—for example, exit events (i.e. separations) or entrance events (i.e. additions to a family by the birth of a sibling). The way in which stressful life events contribute to childhood psychiatric disorders is not well understood, but as in adults, the psychological impact of such events is influenced by factors such as temperament, cognitive style, and previous life experience.

It is important to note that life events often occur in a setting of chronic stress, which in itself may play a causal role in the precipitation of psychiatric disorder. Also, in children as in adults, there are important differences in the liability of individuals to experience negative life events. This may reflect societal factors such as poverty and discrimination, but the behaviour of individuals can also shape life experience in important ways. For example, children with CD have a substantially increased risk of experiencing negative life events (Scott and Gardner, 2015).

Family influences

As children progress from complete dependence on others to independence, they need a stable and secure family background with a consistent pattern of emotional warmth, acceptance, help, and constructive discipline. *Prolonged separation* from or *loss* of parents can have a profound effect on psychological development in infancy and childhood. Poor relationships in the family may have similar adverse effects, and overt conflict between the parents seems to be especially important.

Maternal deprivation and attachment

The much quoted work of Bowlby (1951) led to widespread concern about the effects of 'maternal deprivation.' Bowlby suggested that prolonged separation from

the mother was a major cause of juvenile delinquency. Although this suggestion has not been confirmed, it was an important step towards his idea that attachment is a crucial stage of early psychological development. Attachments are formed in the second six months of life, and adaptive attachments are characterized by an appropriate balance between security and exploration. Bowlby proposed that, from their experiences of attachment, infants derive internal models of themselves, of other people, and of their relationships with others. He proposed further that these internal models persist through later childhood into adult life, and that they influence self-concept and relationships. Thus children whose caregivers are loving, sensitive to their child's needs, and consistent in their responses are likely to grow up with self-esteem and able to form loving and trusting relationships.

Bowlby's ideas have been generally supported by studies in which attachment has been measured in infancy and in later years. Attachment has usually been measured at around 6–18 months of age using the Strange Situation Procedure (Ainsworth and Bell, 1970), which evaluates the infant's response to separation from, and subsequent reunion with, the mother or other attachment figure. In this test, the response to the initial separation appears to be determined more by temperament than by attachment, and it is the response to reunion that is used to measure attachment. Other measures of attachment have been developed for older children. Using such tests, attachments have been divided into one secure and three insecure types:

- *Secure*: present in about 60% of infants, and associated with caregiving that is sensitive and responsive to the child's needs.
- *Insecure*:
 - *avoidant*: present in about 15%, and associated with caregiving that is rejecting or intrusive.
 - *disorganized*: present in about 15%, and associated with caregiving that is unpredictable or frightening.
 - *insecure-ambivalent*: present in about 10%, and associated with caregiving that is inconsistent or lacking.

Studies of older children indicate that the type of attachment develops early in life and tends to be stable thereafter, although it is modified to some extent by changing life circumstances. Securely attached children generally do better than those with insecure attachments. Insecure attachments are associated not only with the factors mentioned above, but also with maternal depression, maternal alcoholism, and child abuse.

Of the three types of insecure attachment, the disorganized type is the best predictor of future problems, especially externalizing problems.

The effect of separation. Bowlby's original studies of the effects of separation on young infants have been broadly confirmed, but it has become clear that the effects depend on many factors, including:

- the age of the child at the time of separation
- the previous relationship with the carers
- the reasons for the separation
- how the separation was managed
- the quality of care during the separation.

The impact of adverse caregiving environments: The more extreme caregiving abnormalities include a severe lack of stimulation and interaction with the infant/child; inconsistent experiences of the caregiver by the child and a lack of emotionally invested caregiving. In DSM-5 two separate disorders are described relating to children who have experienced adverse caregiving. These are reactive attachment disorder (RAD) and disinhibited social engagement disorder (DSED), both of which are discussed later in this chapter.

Family risk factors

Family risk factors for psychiatric disorder in childhood are multiple and cumulative. The risk increases in children of families with severe marital or other relationship conflict, low social status, large size or overcrowding, paternal criminality, and parental psychiatric disorder. The risk is also increased in children who are placed in care away from the family.

Protective factors

Protective factors reduce the rate of psychiatric disorder associated with a given level of risk factors. Protective factors include good parenting, strong affectionate ties within the family, including good sibling relationships, sociability, and the capacity for problem solving in the child, and support outside the family from individuals, or from the school or a faith-based organization (Jenkins et al., 2015).

Effects of parental mental disorder

Some common disorders in adults during the childbearing years, such as depression, anxiety, eating disorders, alcoholism, are associated with an increased risk of psychological disturbances in children. The impacts are likely to be different depending on the developmental stage of the child. Perinatal mental illness in particular has been shown to have long-term effects reaching into early adulthood and may manifest as emotional or behavioural difficulties, insecure or disorganized

attachment, and lower levels of cognitive development, as well as child physical growth and development (Stein et al., 2014). Antenatal depression and anxiety have also been associated with later childhood disturbances; however, whether these arise through genetic confounds, through exposure to mental health disorders in the postnatal period, or through direct effect on foetal development remains unclear (see Chapter 22). Children of parents with depression or anxiety have a markedly increased rate of behaviour problems such as CD at school age, and are at increased risk of depression and anxiety disorders. Paternal depression in particular has also been shown to be associated with increased risk of behavioural problems in the early years of life. The causes are complex and are likely to be both genetic and environmental. An important mediator of the effect of parental depression on children is parenting. For example, this has been demonstrated for depressed mothers, where attachment to the infant is impaired by lower levels of warmth and responsiveness. Importantly, the effects of parental mental health disorders on children are not inevitable and are moderated by socioeconomic status, education, and the chronicity of the parental disorder.

Effects of parental separation

There are often two main effects of childhood parental separation/divorce: economic difficulties and exposure to interpersonal conflict. The children of divorced parents have more psychological problems than the children of parents who are not divorced but it is not certain how far these problems precede the divorce and are related to conflict between the parents; the behaviour of one or both parents that contributed to the decision to divorce; and the changes in family relationships that accompany remarriage or cohabitation following the divorce. Distress and dysfunction in the children are greatest in the year after the divorce; after two years these problems are still present but are generally less severe than those of children who remain in conflictual marriages. In a review, parental divorce was associated with multiple negative outcomes among children and although evidence for a causal effect of divorce on children was reasonably strong, effect sizes were small in magnitude. Children showed a substantial degree of variability in outcomes following parental divorce, with some children declining, others improving, and most not changing at all (Amato and Anthony, 2014).

Death of a parent

In children, the response to the death of a parent varies with age, but data from two large cohorts show that

parental death is associated with multiple negative outcomes in children (Amato and Anthony, 2014). Children aged below 4–5 years do not have a complete concept of death as causing permanent separation. Such children react with despair, anxiety, and regression to separation, however caused, and their reaction to bereavement is no different.

Children aged 5–11 years have an increasing understanding of death. They usually become depressed and overactive, and may show disorders of conduct. Some have suicidal thoughts and ideas that death would unite them with the lost parent. Suicidal actions are infrequent. In children over the age of 11 years, the response is increasingly like that of adults (see p. 156).

Bereavement may have long-term effects on development, especially if the child was young at the time of the parent's death, and if the death was sudden or violent. Outcome probably depends largely on the effects of the bereavement on the surviving parent. Most studies of bereavement in children have been concerned with the death of a parent; few have studied the death of a sibling.

Social and cultural factors

Although the family and home is undoubtedly a significant influence on a child's development, wider social influences are also important, particularly as a cause of CD. In the early years of childhood, these social factors act indirectly through their influence on the patterns of family life. As the child grows older and spends more time outside the family, they have a direct effect as well. These factors have been studied by examining the associations between psychiatric disorder and type of neighbourhood and school.

Effects of neighbourhood

Rates of childhood psychiatric disorder are higher in areas of social disadvantage and studies are starting to better elucidate the importance of social cohesion on mental health outcomes in children and adolescents (Kingsbury et al., 2020). There are also increasing numbers of studies looking at the role of green space, opportunities for children to play outside, adequate social amenities for older children and teenagers, exposure to violence, overcrowded living conditions, and lack of community involvement as impacting on mental health outcomes.

Effects of school

Children spend more time in school than in any other institutional structure and the importance of the school environment has long been recognized as important in

child development and promoting mental health (Fazel et al., 2014a). Schools can play an important role in a number of areas for the child—from academic attainment to social and peer relationships, as well as physical and moral development. It has long been known that rates of child psychiatric referral and delinquency vary between schools. These differences persist when allowance is made for differences in the neighbourhoods in which the children live. It seems that children are less likely to develop psychiatric problems in a school in which teachers praise, encourage, and give responsibility to their pupils, set high standards, and organize their teaching well. Factors that do not seem to affect rates of psychiatric disorder include the size of the school and the age of its buildings. Some school-specific factors can impact on children's mental health, for example, bullying and teacher–pupil relationships.

Bullying often takes place within the school context and is prevalent across schools in both high- and low- and middle-income countries; a UK survey showed that over 45% children have been bullied and experienced peer victimization. Both bullies and victims of bullying are at risk for psychiatric problems in childhood and beyond. Olweus (2013) uses three key criteria—intentionality, some repetitiveness and imbalance of power—to define bullying or peer victimization: a student exposed, repeatedly and over time, to negative actions on the part of one or more other students. This definition emphasizes intentionally negative or aggressive acts that are carried out repeatedly and over time. It also assumes a certain imbalance of power or strength with the student who is exposed to negative actions having difficulty defending himself or herself. With the advent of cyber-bullying these criteria remain broadly similar.

A number of meta-analyses have shown convincingly that being involved in bullying, either as a bully, a victim, or a bully–victim with some regularity is not a harmless and passing school problem. Rather, such involvement can result in serious long-term consequences and adjustment problems for all groups, but of a different nature: depression and other internalizing problems for the victims and criminality/antisocial behaviour and other externalizing problems for the bullies (Olweus, 2013).

It seems that around 2–8% of children are bullied once or more a week. They may be divided into a passive group, who are insecure and anxious and withdraw when attacked, and a provocative group, who are themselves aggressive either directly or by getting others into trouble. The bullies are more often boys than girls; boys are more likely to be physically aggressive, whereas girls are

more likely to exclude their victims or campaign against them. Bullying involvement—as a victim, perpetrator, or both—has been associated with anxiety, depression, psychosocial difficulties, and self-harm behaviour. Girls are more likely to be affected than boys in mental health outcomes. A significant association between bullying victimization and negative mental health outcomes is also observed (Eyuboglu et al., 2021).

Note also that many interpersonal challenges with peers are exacerbated by social media contact, and cyber-bullying, which takes place through social media, has been reported as a significant new method through which to victimize other peers (Kowalski et al., 2014). Its

use and effects should be explored with young people, although many victims of cyber-bullying also experience traditional bullying within the school context and levels of cyber-bullying are seen to reduce following strategies to reduce traditional bullying (Eyuboglu et al., 2021).

Relationship with teachers: poor relationships between students and teachers are a predictor of the onset of childhood psychiatric disorder and of low attainment (Fazel et al., 2014b). Furthermore, difficulties with behavioural management can negatively affect teacher–pupil relationships and the classroom environment, and might impact on both teacher and child mental health.

Psychiatric assessment of children and their families

The aims of assessment are to obtain a clear account of the presenting problem, to find out how this problem is related to the child's past development and present life in its psychological and social context, and to plan treatment for the child and the family.

The psychiatric assessment of children differs in several ways from that of adults, which was described in Chapter 3:

- *A more flexible approach.* With children it is often difficult to follow a set routine, so a flexible approach to interviewing is required, although it is still important that information and observations are recorded systematically.
- *Interviewing of family members.* Parents or other carers should be asked to attend the assessment interview, and it is often helpful to have other siblings present.
- *Information from schools.* Time can be saved by asking permission to obtain information from teachers before the child attends the clinic. This information should be concerned with the child's behaviour in both structured and unstructured settings at school and their educational attainments.

Child psychiatrists vary in their methods of assessment. All agree that it is important to see the family together at some stage to observe how they interact. Some psychiatrists do this before seeing the patient alone, while others do it afterwards. It might be easier to better engage some adolescents if initially seen on their own, but this might not help with other adolescents and so must be decided on a case-by-case basis. With younger children the main informants are usually the parents, but children over the age of 6 years should

usually be seen on their own at some stage. In the special case of suspected child abuse, the interview with the child is particularly important.

Interviewing the parents

Parents are likely to be anxious, and some fear that they may be blamed for their child's problems. Time should be taken to put them at ease and explain the purpose of the interview. They should then be encouraged to talk spontaneously about the problems before systematic questions are asked. The methods of interviewing are similar to those used in adult psychiatry (see Chapter 3). The items to be included in the history are listed in the Appendix to this chapter. As in adult psychiatry, an experienced interviewer will keep the complete list in mind, while focusing on those items that are relevant to the particular case. It is important to obtain some developmental history and specific examples of general problems to elicit factual information and to assess feelings and attitudes.

Interviewing and observing the child

Because younger children may not be able or willing to express their ideas and feelings in words, observations of their behaviour and interaction with the interviewer are especially important. With very young children, drawing and the use of toys may be helpful. With older children, it may be possible to follow a procedure similar to that used with adults, provided that care is taken to use words and concepts appropriate to the child's age and background. Standardized methods of observation and

interviewing have been developed mainly for research purposes.

Starting the interview. It is essential to begin by establishing a friendly atmosphere and winning the child's confidence, and asking what they like to be called. It is usually appropriate to begin with a discussion of neutral topics such as pets, favourite games, or birthdays, or current school information—such as which class they are in or which subjects they like—before turning to the presenting problem.

Techniques of interviewing. When a friendly relationship has been established, the child can be asked about the problem, their likes and dislikes, and their hopes for the future. It is often informative to ask what they would request if they were given three wishes or with older children to ask them for three goals they would like to achieve following their contact with mental health services; alternatively, many clinicians might ask about what is going well for that person and what is not going so well. Younger children may be given the opportunity to express their concerns and feelings in paintings or play. Children can generally recall events accurately, although not always in the correct sequence. They can be more suggestible than adults, and when asked a question are used to trying to give the answer that the adult might be expecting—as they might be accustomed to, for example, in school. Therefore it is particularly important to try and use open questions when interviewing.

Behaviour and mental state. The items to be noted are listed in the Appendix to this chapter. Children who are brought to see a psychiatrist may appear silent and withdrawn at the first meeting; this can be normal or part of a presentation of anxiety or depressive disorder, but only additional meetings and informant interviews can help the diagnostic process.

Developmental assessments. By the end of the interview an assessment should have been made of the child's stage of development relative to other children of the same age.

For further information and specific examples of questions, interview schedules can be helpful such as that in the Child and Adolescent Psychiatric Assessment (CAPA) (Angold and Costello, 2000).

Interviewing and observing the family

A family interview can contribute to the assessment of the interactions between family members, but it is not a good way to obtain factual information. The latter is generally more effectively obtained in interviews with the parents or other family members on their own. Of the

various aspects of family interaction, the psychiatrist will usually be most interested in discord and disorganization, which are the features most closely associated with the development of psychiatric disorder. Patterns of communication between family members are also important.

It is usually best to see the family at the first assessment or soon after this, before the interviewer has formed a close relationship with the young patient, or with one of the parents, as this may make it more difficult to interview the other family members.

The interviewer could begin by asking 'Who would be the best person to tell me about the problems?' or by asking the child 'What do you think your father might say about why we are here today?', even if their parent is in the room, and observe responses and interactions. If one family member monopolizes the interview, another member should be asked to comment on what has been said. As an alternative, family members who are present can be asked what they think an absent member would think about the issues.

While observing the family's ways of responding to these and other questions, the interviewer should consider the following:

- Who is the spokesperson for the family?
- Who seems most worried about the problem?
- What are the alliances within the family?
- What is the hierarchy in the family? For example, who is most dominant?
- How well do the family members communicate with one another?
- How do they seem to deal with conflict?

Psychological assessment

Measures of intellectual and educational achievement can often be valuable, depending on the presentation. If developmental milestones and achievement are inconsistent with chronological age, or with the expectations of parents or teachers, this may indicate a generalized or specific disorder of development or may indicate a different psychological disorder. Some of the more commonly used psychometric tools are listed in Table 16.2 (Braconnier and Siper, 2021).

Other information

Important additional informants often include primary care practitioners (including health visitors) and staff at the child's school. Teachers can describe classroom behaviour, educational achievements, and relationships with other children. They may also have insights into

Table 16.2 Notes on some psychological measures in use with children

Intelligence tests	
Stanford–Binet test, 5th edition	Tests verbal and non-verbal intelligence from the age of 2 years to adulthood.
Weschler Intelligence Scale for Children (WISC V)	Provides a profile of specific verbal and performance ability as well as IQ for children 6–14 years of age; widely used and well standardized; cannot be used for IQ below 40. Translated into several languages.
Weschler Pre-School and Primary Scale of Intelligence (WPPSI-IV)	A version of WISC for use with children 3–7 years of age.
British ability scales (BAS3)	One for children 2–6 years of age, the other for children 6–18 years of age, covering seven areas: speed of information processing; reasoning; spatial imagery; perceptual matching; short-term memory; retrieval; application of knowledge. Analysis can be general or specific.
Social development assessments	
Bayley scales of infant and toddler development, Bayley-IV	Standard tasks scored on mental, motor, and behaviour scales and compared with norms. Useful measure of infant development from 1 month to 3½ years of age.
Other developmental assessment	
Denver developmental screening test, 2nd edition	General developmental screening test from birth to 6½ years of age. Identifies delays in personal and social development, fine and gross motor development, and language skills. For use only as a screen to identify those needing further assessment.

the child's home environment. It can be helpful for a member of the mental health team to observe the child at school as well as visit the home. This visit can provide

useful information about material circumstances in the home, the relationship of family members, and the pattern of their life together.

Physical examination

If the child has not been examined recently by a medical practitioner, an appropriate physical examination may be needed to complete the assessment. What is appropriate depends on the nature of the problem, but it will often be looking for signs of conditions that might affect the brain. Therefore the first step is to observe the child's appearance, coordination, and gait, at rest and during play. A basic physical examination may follow, with emphasis on the nervous system. If abnormalities are found or suspected, the opinion of a paediatrician or paediatric neurologist may be needed.

Ending the assessment

At the end of the assessment the psychiatrist should explain to the parents—and to the child, in terms appropriate to their age—the result of the assessment and

they could at this point agree goals for any treatment plan. They should explain how they propose to inform and work with other health care professionals, and seek consent to contact other relevant workers involved with the child, such as school staff or social workers. Throughout, the psychiatrist should encourage questions and discussion.

Formulation

A formulation can be helpful when summarizing key issues. It usually starts with a brief statement of the *current problem*. The *diagnosis* and *differential diagnosis* are discussed next. The diagnosis and any comorbidities can be recorded as per the relevant classification system (as discussed above). *Aetiology* is then considered, with attention to predisposing, precipitating, and perpetuating factors.

The *developmental stages* of the child should be noted. An assessment of the problems and the strengths of the family can also be recorded. Any further assessments should be specified with a *treatment and follow-up plan* drawn up.

Court reports

Psychiatrists may be asked to prepare court reports in relation to children. These reports are usually undertaken by specialists in child psychiatry; therefore only an outline is provided here. If general psychiatrists are required to prepare such a report, they should ensure that they are thoroughly aware of the relevant legislation, ask advice from a colleague with relevant special experience, and read a more detailed account of the requirements for court reports.

Courts concerned with children obtain evidence from several sources, including social workers, education staff, probation officers, community nurses, psychologists, and psychiatrists. In their reports, psychiatrists should focus on matters within their expertise, including:

- the child's age, stage of development, temperament, and the relevance of these to the case;
- whether the child has a psychiatric disorder;
- the child's own wishes about their future (if this is regarding parental separation proceedings), considered in relation to their age and understanding; and
- the parenting skills of the carers, how far they can meet the child's needs, and other relevant aspects of the family.

While focusing on these matters, the psychiatrist should also be prepared to provide information about the following issues:

- the child's physical, emotional, and educational needs;
- the likely effect on the child of any possible change of circumstances (e.g. removal from home, or living with one or other parent after a divorce); and
- any harm that the child has suffered or is likely to suffer.

The *wishes of the child* should be noted in relation to their age and ability to understand the present situation and possible future arrangements, and other relevant factors in the present situation—for example, some abused children maintain strong attachments if, for example, their abuser is also a primary caregiver, which renders these assessments highly complex.

Parenting skills are judged partly on the basis of the history and reports of other people. They are also judged on direct observations of the interactions between the parents and the child, including the parents' attachment to the child, their sensitivity to cues from the child, and their ability to try meet the child's needs, sometimes with a commitment for ongoing parenting skills training or other plans to work with care workers.

The report is similar in structure to a court report for an adult, and is presented under the sections shown in Box 16.4.

Box 16.4 Topics to be addressed in a court report about a child

- The qualifications of the writer.
- Who commissioned the report, and what questions were asked.
- What written information was available, and who was interviewed.
- A summary of the findings from the interview (it is not necessary to repeat information contained in social enquiry reports).
- The writer's interpretation of the information from the interviews and written material.
- In the light of the findings, comments on the options before the court, remembering always that it is for the court and not the psychiatrist to determine which option is selected.
- Developing rapport with children is essential and allowing younger children to bring a toy for comfort can help.
- Age-appropriate methods to take account of developmental levels with younger children who have shorter attention span.
- Allow the child to talk freely, asking as few questions as possible and keeping the questions open ended if possible. Prop use, in particular the use of drawing can be helpful, although anatomical dolls and body diagrams are not recommended.
- Obtain the evidence as soon as possible after the event, and whenever possible before any counselling has taken place.
- Try to complete the account on the first occasion, as subsequent accounts are likely to be less accurate.

Magnusson et al. (2020) suggest the use of the following principles when evidence is obtained from a child:

Psychiatric treatment for children and their families

This section is concerned with the general features of treatment for children. Aspects of treatment that are specific to individual disorders are discussed later in the sections concerned with those disorders.

The role of the primary care team

General practitioners and other members of the primary care team spend much of their time in advising parents about children, but they refer only a small proportion of these children to a mental health services or a paediatrician. General practitioners are more likely to refer children with the following to a paediatrician:

- developmental difficulties
- physical symptoms with a probable psychological cause
- psychological complications of physical illness.

They are more likely to refer emotional, attentional, and CDs to mental health services. Such services typically receive many referrals and self-harm and more severe presentations of eating disorders or depression are usually prioritized.

The psychiatric team

Way of working. Although psychiatrists, psychologists, psychotherapists, family therapists, nurses, occupational therapists, and social workers have special skills, when working in a child mental health team they do not confine themselves to their traditional professional role. Instead, they adopt whatever role is most likely to be helpful in the particular case, as they do when working in a community team caring for adults.

The team usually adopts a *family-centred approach*, and it liaises closely with other professionals involved with the child or the family. These professionals include paediatricians, members of child health and social services, teachers, and other school-based professionals. Since many childhood problems are evident at school, or lead to educational difficulties, the child's teachers usually need to be involved in some way. They may need advice about the best way to manage disturbed behaviour in the classroom, changes that may be needed in the child's school timetable, or remedial teaching that may be required. Occasionally a change of school is indicated.

In the following sections, brief accounts are given of the main kinds of treatment. In the next part of the chapter, information is given about the management of specific disorders. Additional information about treatment in child psychiatry can be obtained from one of the textbooks listed at the end of this chapter.

Medication

Drugs have an important role for a proportion of cases in child psychiatry, although they are often prescribed only after psychological therapies have been tried because of the potential effects of medication on the developing brain and because the evidence base for pharmacological therapies is more limited in child and adolescent populations compared with those over 18 years of age. The main indications for medication are in the treatment of depression, severe anxiety disorders, ADHD, sleep disorders, psychosis, obsessive-compulsive disorders (OCD), and Tourette syndrome. For children, it is important to check dosage against recommendations for the child's age and body weight. Stimulants to treat ADHD, melatonin to treat sleep difficulties, fluoxetine to treat depression and anxiety disorders, and antipsychotics are the most commonly prescribed medications; with evidence that the use of medication for children and adolescents is increasing with boys receiving double the rate of prescriptions as girls (Ball et al., 2022; Klau et al., 2022).

Psychological treatment

Psychological aspects of management

Whatever the plan of management, children benefit from a *warm, secure, and accepting relationship* with the therapist. The security of this relationship helps the child to express their feelings and to find alternative ways of behaving. For younger children, play with toys can help to establish the relationship and provide a medium through which they can express their problems and feelings more effectively than they can with words.

At first, children often perceive the therapist as aligned with their parents, and might expect the therapist to share their attitudes. Children should feel accepted in their own right, and not judged; it is therefore helpful for some children to delay discussion of the presenting problems until the child's confidence has been gained by talking about neutral things that interest them.

More commonly used psychological treatments in psychiatric clinics include behavioural interventions, cognitive-behaviour therapy (CBT), and systemic family therapy with occasional psychodynamic psychotherapy (depending on availability), although the evidence base for many of these is poor for pre-adolescents (Forti-Buratti et al., 2016).

Cognitive-behaviour therapy

Behavioural methods have several applications in child psychiatry. They can be used to encourage new behaviour by positive reinforcement and modelling. This is often done by first rewarding behaviour that approximates to the desired behaviour (*shaping*), and then giving reinforcement in a more discriminating way. Instead, efforts are made to identify and remove any factors in the child's environment that are reinforcing unwanted behaviour. It is often found that undesired behaviour is being *reinforced unwittingly* by extra attention given to the child when the behaviour occurs. If the child is ignored at such times and attended to when their target behaviour is observed, beneficial changes are more likely to take place. More specific forms of behaviour therapy are available for enuresis (see p. 451) and phobias (see p. 666). *Social skills training* in a group or in individual sessions may be used for children who have difficulty in relationships with other children and adults. The methods generally resemble those used with adults (see Chapter 24).

CBT is increasingly indicated for older children and adolescents, and similar to adult studies, has a growing evidence base for both depression (Oud et al., 2019) and anxiety (Sigurvinsdóttir et al., 2020); in individual, group, family, and remote delivery formats; and for other areas of need such as pain, insomnia, and eating disorders. Evidence from studies of depression highlight how effects of therapy improve when components of behavioural activation, challenging thoughts and caregiver involvement are included (Oud et al., 2019).

Acceptance and commitment therapy (ACT) is a contemporary behavioural and cognitive therapy that aims to enhance the psychological flexibility of individuals to respond to thoughts, emotions, and sensations through six core therapeutic processes: acceptance, cognitive defusion, mindfulness, self-as-context, committed action, and valued living (Fang and Ding, 2020). It therefore aims to helping a child understand and accept their inner emotions and assist them to find ways to commit to moving forwards in positive ways. It can be used individually with children and adolescents, in group settings, and with parents. Studies have shown that ACT significantly improves symptoms of anxiety

and depression and, while more effective than treatment as usual, is not superior to traditional CBT.

Family and parenting interventions

There are many interventions for child and adolescent psychiatry that benefit from a family-based component to treatment, and there are also specific family therapy interventions, for example, systemic family therapy. The focus is on helping the family function in a more positive and constructive way by exploring patterns of communication and providing support and education. The symptoms of the child or adolescent are viewed as an expression of malfunctioning of the family, which is the primary focus of treatment. Several approaches have been used, based on systemic, behavioural, psychoanalytical, interpersonal, or structural theories (Carr, 2020). These kinds of therapy are described on pp. 679. In practice, many therapists adopt an eclectic approach.

The best evidence for effectiveness is for structural and behavioural forms of family therapy, as well as for multifamily therapy (Gelin et al., 2018) with a growing evidence base for mood disorders, eating disorders, and substance misuse disorders. For a comprehensive review of family interventions see Eisler and Lask (2015).

Parenting interventions are used for a range of child psychiatric problems and primarily to assist parents of children with behaviour problems that require special parenting skills—for example, the parents of children with CD or ADHD. Parent training can be carried out with an educational approach, in which skills of general importance are taught, or with a behavioural approach, in which the specific problems of the particular parent and child are analysed and corrected. In the behavioural approach, use may be made of video-recorded vignettes showing desirable and undesirable parental responses to children's behaviour. These responses are discussed with the parents of a particular child, or with a group of parents. Whatever approach is adopted, it is important to consider the stage of development of the child, and of the changing needs of children of different ages.

Studies of the behavioural training of parents have now firmly established the effectiveness of this approach in improving parenting skills and parent-child relationships, and reducing antisocial behaviour in children. For example, for groups often considered as 'hard to treat' for reasons of social disadvantage or psychopathology, parenting interventions can improve child behaviour. For families with very complex needs, such as those in the child protection system, the evidence is more promising than for other interventions. Research is now focusing on which families can benefit, and the long-term effects.

For a review of parent training programmes, see Jeong et al. (2021).

Psychodynamic therapies

Psychodynamic psychotherapy with children and adolescents has a long history, based on the methods of Anna Freud and Melanie Klein. Empirical support for its use has been limited and so long-term, analytically based, psychodynamic therapies are now used infrequently in the United Kingdom, although group therapy and creative therapies (e.g. play therapy and art therapy) continue to be provided for selected children with severe and enduring difficulties. A review of short-term psychodynamic psychotherapies has suggested that these may be effective across a range of common mental disorders in children and adolescents (Midgley et al., 2017).

Remote and digital interventions

In recent years there has been a rise in digital technologies that can assist children and adolescents to access remote interventions such as telephone or video calls, online chat messaging or forums, text messages, or emails. There are also computer-assisted psychological therapies, such as CBT, being used in this manner. These interventions have been developing over the last decade but were propelled forwards during the Covid-19 pandemic (Witteveen et al., 2022). The evidence base for many of these interventions is being developed, with few effective digital smartphone apps identified for younger populations (Bear et al., 2022), although other digital interventions for mental health have shown potential effectiveness in supplementing more traditional in-person mental health treatments (Lehtimäki et al., 2021). This is an area where more evidence is being gathered.

Social work

Social workers play an important role in the care of children with psychiatric disorders, and of their families. In the United Kingdom social workers have statutory duties with regard to the protection of children who are at risk within the family, and who require special care or special supervision. They help parents to improve their skills in caring for their children, and to solve problems with finances or accommodation. Social workers carry out family assessments and family therapy, and may also provide individual counselling for the child and members of the family. They can play a key role when the relationships around an adolescent break down, and the adolescent might need to find alternative accommodation and/or educational provision.

Occupational therapy

Occupational therapists can play a valuable part in assessment of the child's development, in psychological treatment, and in devising measures to improve parent-child interaction. They work both in day-patient and inpatient units, and in the community. They work closely with teachers in both assessing and providing therapeutic activities for children, in particular those with neurodevelopmental disorders.

Special education

Special educational provision can be appropriate for children and adolescents with a range of difficulties and occasionally can play an important role in providing a therapeutic space where children are better able to access their different or additional educational needs. Special teaching may be needed to restore confidence, and to remedy delays in writing, reading, and arithmetic, which can be common among children with CDs as well as those with specific developmental disorders. Information from a UK school census (Department for Education, 2022) highlighted that for students educated in special schools for the year 2021/22, 35% of their primary needs were as a result of ASD, 19% had severe learning difficulties, and 12% had social, emotional, and mental health needs. For young people who are educated outside of mainstream education within pupil referral units (PRUs), over 80% have recognized social, emotional, and mental health needs (Department for Education, 2022).

Substitute care

Residential care

There are two main reasons for residential care in children: first, for those with profound intellectual and neurodevelopmental disorders in order to provide for their physical, social, and educational needs; second, for children who have been maltreated or have significant emotional and behavioural problems with symptoms that result from or are maintained by a severely unstable home environment or extreme parental rejection. The rest of this section discusses the needs of this second group. For this second group, residential care has a long history in the provision of services and is an umbrella term capturing various forms of residentially based living arrangements, from small group homes to large institutions, across three service systems—social services/child welfare, mental health, and juvenile justice. These care settings can provide therapeutically

planned behavioural health interventions to unrelated youth with a wide range of problems in a 24-hour structured and multidisciplinary care environment. Children who are considered for residential placement often present with severely disordered behaviour and might have a number of psychiatric diagnoses including CDs or emerging personality disorders, as well as breakdown in their educational provision.

Removal of a child from their primary caregivers should be considered only after every practical effort has been made to improve the circumstances of the family. Residential care may be provided in a foster home, a children's home (in which a group of about ten children live in circumstances as close as possible to those of a large family), or a residential school. Residential care, other than fostering, is seldom arranged for children under five years of age, because of the need to try and ensure attachment to a primary caregiver. In general, children in residential care are a population particularly vulnerable to physical and sexual abuse and have high rates of psychosocial problems in later childhood and in adult life; thus, there is international consensus about the need to avoid prolonged stays in residential care. Residential care is associated with adult criminal convictions, depression, and low self-efficacy. Evidence of the abuse of children placed within care are reminders of the need to ensure good training and supervision of the staff of children's homes and residential schools.

For a review of residential care, see Ijzendoorn et al. (2015).

Fostering and adoption

Foster care may be of three kinds:

1. *Short-term emergency care*—for example, when a caregiver is ill or when the parents of a high-needs autistic child need respite.
2. *Medium-term care*, which may be followed by a return home—for example, if the caregiver is receiving treatment for problems that led to neglect or abuse of the child.
3. *Long-term care*, in which the child remains until they are adopted or able to live independently.

Children in foster care have better outcomes than those in residential care but children in long-term foster care have more problems than children who have been adopted. It is difficult to determine to what extent these problems are related to experiences before fostering, and to what extent they are due to the lesser security of fostering as opposed to adoption. Problems seem to be greater when the fostered child has experienced

pre-natal exposure to substances, is older, and when children in the fostering family are of the same age as the fostered child. Children in foster care usually retain some contact with their biological parents, but it is not helpful to the child to have sporadic and distressing contacts. For a review of fostering, see Ijzendoorn et al. (2015).

Adoption is the process whereby the parenting of a child is transferred from that person's biological or legal parent/s to another person, and it is usually intended to mark a permanent change in status, legally recognized as such. Adoption practices can be open or closed: if open then there remains an opportunity for contact with biological parents and closed adoption where the identities of adoptive parents, biological kin, and adoptees are not disclosed; however, the transmittal of non-identifying information such as medical history can be allowed. It is generally understood that a lack of openness can be detrimental to the psychological wellbeing of adoptees.

Adoptions can occur between related family members or unrelated individuals. Unrelated adoptions may occur through the following routes:

- *private domestic adoptions*: under this arrangement, charities and other organizations bring together, from within the same country, prospective adoptive parents and children to be placed
- *foster care adoption*: this is an adoption of a child who is initially placed in foster care
- *International adoption*: this is the placing of a child for adoption outside that child's country of birth. Different countries have different laws regarding international adoptions. Research following international adoptions from severely deprived institutional settings has helped elucidate the long-term effects of early deprivation and how to try and address the emotional, cognitive, social and physical sequelae.
- *Embryo adoption*: this is the donation of embryos (usually remaining after another couple have completed in vitro fertilization (IVF) treatments) to another individual or couple, followed by the placement of those embryos in the woman's uterus to achieve pregnancy.

Given the range of different types of adoption and also different reasons leading to a child being adopted, as well as experiences prior to and age at adoption, the psychological impacts of adoption can either be minimal or multiple. The areas most commonly raised are those around loss and grief for biological parents; identity development, which can be more difficult for an adopted child; and poor self-esteem. For a review of

adoption in children and its psychological impact, see Cohen and Farnia (2015).

Intensive treatments

Intensive community services and inpatient care are essential components of the provision of child psychiatry, particularly for those children with severe physical or risk needs such as those with eating disorders, psychosis and at risk of self-harm, suicide or harm to others. Intensive community services form an increasingly important level of care between outpatient and inpatient services. Child psychiatric inpatient units require easy access to paediatric advice, adequate space for socializing, easy access to schooling, and an informal design that still allows close observation. There should be some provision for caregivers to stay with their younger children.

Admission for inpatient treatment is usually arranged for any of three reasons:

1. *Severity*: when the disorder is too severe to treat in any other way—for example, more violent behaviours, extreme mood disorders (both depression and mania), difficult-to-treat psychosis, severe pervasive developmental disorder, and life-threatening anorexia nervosa.
2. *Observation*: when the diagnosis is uncertain.
3. *Separation*: inpatient treatment can provide a necessary period away from a severely disturbing home

environment and an assessment of the child is needed outside of this environment.

Psychiatric inpatient treatment for children and adolescents is sometimes beneficial but predictors of who benefits and in what circumstances remain unknown (Kennedy et al., 2020). There is some evidence that compliance with treatment and female gender are predictors of positive change as well as positive peer relationships during the admission. The high costs of inpatient units have contributed to expansion of other types of intensive community-based and often early-intervention services which are often welcomed by families and young people. *Day-hospital treatment* for children provides many of the advantages of inpatient care without removing them from home. Unless there is any danger that the child may be abused, remaining at home has the advantage that relationships with other family members are maintained. Day care can relieve the family from some of the stressful effects of managing a child with severe behavioural disturbance or eating problems.

Increasingly, intensive community-based support is provided for children and adolescents with severe problems. Such support requires a well-staffed and experienced team of professionals with an ‘assertive-outreach’ approach (see p. 765). Although such services are valuable, they do not remove the need for inpatient provision.

Review of disorders

The review of disorders begins with the problems encountered in *preschool children*. *Specific and pervasive developmental disorders* are described next. An account is then given of *the main psychiatric disorders of childhood*, in the order in which they appear in the major systems of classification. Other psychiatric disorders of childhood are described next, before a brief account of the disorders of adolescence (which are generally similar to those of either childhood or adulthood). The chapter ends with a discussion of *child abuse*.

Psychiatric disorders of preschool children and their families

It has already been noted that in the preschool years children are learning several kinds of social behaviour

although it should be noted that children are also shaped by culturally regulated customs, child-rearing practices, and belief systems. Preschool children are acquiring physical developmental milestones with increasing independence in movement, feeding, and sphincter control. They are learning how to conduct themselves at mealtimes, to go to bed at an appropriate time, and to control their emotions. Most of these behaviours are learned within the family as well as in childcare settings, depending on the circumstances of each particular child. The psychiatric problems of preschool children centre around these behaviours, and they often reflect factors in the family and caregiving environment as well as factors in the child. Many psychological problems at this age are brief, and can be thought of as variants in normal development. Most are treated in primary care and by paediatricians. The more serious problems may be referred to child mental health teams.

Prevalence

Epidemiological surveys have found significant rates of psychological problems in preschool children. The estimated population rate for any psychiatric disorder was 7.1%. The most common disorders were ADHD (1.9%), oppositional defiant disorder (ODD) (1.8%), CD (0.7%), anxiety disorders (1.5%), and depressive disorders (2.0%). Comorbidity among disorders was common. Emotional and behavioural disorders were more common in children whose parents did not live together and in those of low socioeconomic status. Boys more often had ADHD and depressive disorders than girls (Wichstrøm et al., 2012).

Some common problems of preschool children

Temper tantrums and disruptive behaviours

Temper tantrums are among the most common childhood behaviour problems and approximately 5–7% of children between one and three years of age have temper tantrums lasting at least 15 minutes three or more times per week. They are extreme episodes of frustration or anger with mood and behaviour returning to normal between tantrums. Only persistent or very severe tantrums are abnormal. Children often have temper tantrums in order to get something they want, to avoid something they do not want to do, or to seek parental attention. They are more likely to occur when the child is tired, hungry, ill, or frustrated. The immediate effect can be an unwitting reinforcement by excessive attention and inconsistent discipline on the part of the parents. When tantrums persists it can be because parents have emotional problems of their own, or because the relationship between them is unsatisfactory.

Temper tantrums can be difficult to stop once they start and so identifying triggers and potential causes and preventing tantrums is important. During a tantrum it is usually best to try and remain calm and to respond kindly with firm and consistent setting of limits. In treatment it is first necessary to discover why the parents have been unable to set limits in this way. They might need help with any problems of their own and advised how to respond to the tantrums.

Sleep problems

The most common sleep difficulty is *wakefulness at night*, which is most frequent between the ages of 1 and 4 years and reported by around 20–30% of parents. About 20% of children of this age take at least an hour to get to sleep or are wakeful for long periods during the night. Bedtime struggles and middle-of-the-night wakings are

not only a source of sleep disruption for children and their parents but can also be a source of conflict and negative emotion among family members. When wakefulness is an isolated problem and not very distressing to the family, it is enough to reassure the parents that it is likely to improve. If sleep disturbances are severe or persistent, two possible causes should be considered. First, the problems may have been made worse by various comorbidities (emotional disorders, autism, ADHD) or physical illness. Secondly, they may have been exacerbated by the parents' excessive concern and inability to reassure the child.

If no medical or psychiatric disorder is detected, the reasons for the parents' concerns should be sought and dealt with as far as possible. Some parents overstimulate their child in the evening, or unwittingly reinforce crying in the night by taking the child into their own bed. A behavioural approach to these problems is generally helpful. Other sleep problems such as *nightmares* and *night terrors* are common among healthy toddlers, but they seldom persist for long. They are discussed on p. 321. For a review of sleep problems in children, see Williamson et al. (2019).

Feeding problems

Normal feeding and eating depends on the successful integration of a range of physical functions and interpersonal relationships during early development. Therefore a minor disruption in one or more of these areas can result in a feeding problem in infants and early childhood. Common disturbances include delayed development of feeding/eating skills, difficulty managing or tolerating foodstuffs, and reluctance or refusal to eat based on taste, texture, and other sensory factors. Minor food fads and food refusal are common in preschool children, but do not usually last long. In a minority, however, the behaviour is severe or persistent, although not accompanied by signs of poor nourishment. When this happens it might be because the parents are over-attentive and this might unwittingly reinforce the child's behaviour. Feeding problems become 'Feeding disorders of infancy and childhood' when the problems are persistent (over one month in duration) and associated with significant failure to gain weight (Bryant-Waugh et al., 2010).

Treatment is directed to the parents' management of the problem. They should be encouraged to ignore the feeding problem and refrain from offering the child special foods or otherwise attempting to do anything unusual to persuade him or her to eat. Instead, the child should be offered a normal meal and left to decide whether or not to eat it.

Pica

Pica is the diagnostic term for the eating of items that are generally regarded as inedible, such as soil, paint, or paper. It is often associated with other behaviour problems and inappropriate to the child's developmental level. Cases should be investigated carefully because some are due to brain damage, ASD, or intellectual disability. It can also co-occur with other mental disorders, for example, schizophrenia, but for a diagnosis of pica it must present as severe enough to warrant independent clinical attention. Otherwise, treatment consists of common-sense precautions to keep the abnormal items of diet away from the child. Pica usually diminishes as the child grows older. For further discussion see Pica section in Chapter 13.

Disorders associated with adverse caregiving

As explained above, children's *attachments to their parents* vary in their security, and they may vary between caregivers—for example, being insecure with one parent but secure with the other. Attachment disorders are more extreme variations from the norm, and do not correspond exactly to any of the types of insecure attachment described earlier (see p. 412). They are *pervasive*, affecting all relationships. They start before the age of 5 years and are associated with grossly abnormal caregiving.

In DSM-5 two separate disorders are described relating to children who have experienced adverse caregiving. The first is *reactive attachment disorder* (RAD), where there is an absence of attachment behaviour secondary to social neglect. These children have the developmental capacity to form attachments, and these disorders are very responsive to enhanced caregiving. For example, in children from Romanian orphanages who had experienced extreme deprivation, their signs of RAD would diminish or disappear once children were placed in families and received adequate care, with best outcomes if they had been removed before two years of age (Zeanah et al., 2015). The second is *disinhibited social engagement disorder* (DSED), in which the child lacks appropriate reticence with unfamiliar adults and violates socially sanctioned boundaries. Compared with RAD, children with DSED are usually more functionally impaired, have difficulties with close relationships, more need for special education, and have a poorer outcome.

Assessment and treatment of the problems of preschool children

Assessment. Usually the information is largely obtained from the parents. The assessor seeks to discover whether the problem is primarily in the child or related to

difficulties in primary caregiver, other potential attachment figures, or the entire family. The problem behaviour is assessed, together with the child's general level of development, and the functioning of the family.

Treatment. Apart from the particular points already mentioned under the specific disorders, treatment includes advice for the parents (and if necessary for other family members) about relevant aspects of child rearing. There is little systematic evidence about the value of treatment, although parenting interventions are the best-established approaches, and have some benefit even where the underlying condition has a strong genetic basis. It may be helpful to arrange for the child to spend part of the day away from the family in a play-group or nursery school, provided that the care offered is of high quality. Specially adapted cognitive approaches can be helpful for emotional and anxiety disorders.

Specific developmental disorders

DSM-5 and ICD-11 both contain categories for specific developmental disorders, which are circumscribed developmental delays that are not attributable to another disorder or to lack of opportunity to learn. It is debatable whether these conditions should be classified as mental disorders at all, since many children who meet the criteria have no other signs of psychopathology. Developmental learning, communication, and motor problems can affect children throughout development, influencing their lives and personalities and can therefore have long-term implications for children's emotional and regulatory development, affecting their transitions into adulthood. Some of these disorders, when detected and treated early, can be remediated successfully. Others are chronic, and compensatory skills must be taught. In ICD-11, these developmental disorders are divided into specific developmental disorders of *speech or language, learning, and motor coordination disorders*. In DSM-5, the same disorders are called *learning disorder, communication disorders, and motor skill disorder*, respectively.

Specific learning disorders

Reading disorder

In DSM-5, reading disorder is grouped with other learning disorders, where previously the disorders of reading, writing, or arithmetic were considered selective or specific impairments. However, it is increasingly apparent that they can frequently co-occur with each other

and with other communication impairments, attention disorder, and developmental coordination disorders. Significant advances have been made in understanding the nature and causes of children's reading difficulties, especially through better elucidation of the cognitive models involved. There are two different, but commonly occurring reading disorders in children: dyslexia and reading comprehension disorder.

Dyslexia

Dyslexia refers to a difficulty in learning to decode leading to poor appreciation of the spelling patterns of words and their pronunciation. Therefore, children with dyslexia can have difficulties accurately reading aloud and with spelling. It is usually diagnosed because the child has serious *delay in learning to read*, evident from the early years of schooling, and sometimes preceded by delayed acquisition of basic language skills. *Writing and spelling are impaired*, and in older children these problems may be more obvious than the reading problems. Errors in reading and spelling include omissions, substitutions, or distortions of words, slow reading, long hesitations, and reversals of words or letters. There may be associated emotional problems, but development in other areas is not affected. Of note the DSM-5 uses the term specific learning disorder (SLD) to describe a number of different disorders of language and not the term dyslexia; the SLDs are considered a type of neurodevelopmental disorder.

Estimates of its presence in English-speaking countries is 3–10% in school-aged children with more boys referred than girls and higher numbers in urban versus rural populations and in those with English as a second language. Typically the spelling difficulties in people with dyslexia are more resistant to remediation than their reading difficulties. Those with specific reading difficulties are more often male and more likely to have minor neurological abnormalities, and to come from socially disadvantaged homes. For example, sensory impairments are common; some auditory processing tasks, especially frequency discrimination, present in up to 30–40% of children with dyslexia.

Reading comprehension impairment

Children with *reading comprehension impairment* show a contrasting pattern where they are able to read aloud accurately but have difficulty understanding what they have read. These children present less frequently to services as they often go unnoticed. It might be quite a common disorder, as some studies have suggested up to 5% of secondary school pupils have this. They might experience higher-level language difficulties, including

problems with inferencing and with figurative language use. Much of the clinical profile has similarities with that seen in ADHD.

Disorders of communication, speech, and language

Communication is the transmission of information (a message) using a common signalling system and language is a form of communication that humans have developed to assist in communication. Language is a complex formal system that makes use of sounds and words in rule-governed ways to convey an infinite number of meanings. Language too can take a variety of forms: it can be written or conveyed using manual signs. However, the most widely used form for expressing language is through speech (Norbury and Paul, 2015). The process by which language is acquired is complex, and is still not fully understood.

Children vary widely in their achievement of speech and language. Half of all children use words with meanings by about 12 months of age, and 97% do so by 21 months of age. Half of all children form words into simple sentences by 23 months of age. Vocabulary and complexity of language develop rapidly during the preschool years. Specific language impairment appears to be a relatively common disorder with estimates of around 7% in kindergarten children; and there are high levels of comorbidity with reading disorders, motor coordination, and general learning disorders and ADHD. They are also associated with psychiatric problems because they can both be a sign of a common brain abnormality, and also because language disorder impairs social interaction and education.

Causes of speech and language disorder

The causes of language impairment remain poorly understood. The children appear to have pronounced problems in word learning (vocabulary knowledge), grammar learning, and phonology. A minority might have another primary disorder, which is important to detect, such as *intellectual disability*, hearing problems, *cerebral palsy*, and ASDs. *Social deprivation* can cause mild delays in speaking or add to the effects of the other causes. There is a significant genetic contribution, including mutations in a gene called *FOXP2* causing some cases of severe language disorder (Peterson and Pennington, 2015).

Classification

The classification differs in some ways between ICD-11 and DSM-5. ICD-11 classifies speech and language disorders as neurodevelopmental disorders with a

differentiation between primary and secondary disorders. DSM-5 has these disorders under ‘communication disorders.’

Language disorder

Language disorders are persistent difficulties in the acquisition and use of language across modalities and may involve comprehension or production deficits in one or more of the following: i) vocabulary (word knowledge and use); ii) sentence structure (the ability to put words and word endings together to form sentences based on rules of grammar); and iii) discourse (the ability to use vocabulary and connect sentences to explain or describe a topic or have a conversation). The disorders in language therefore follow these three areas: *disorders of vocabulary; grammar and morphology, and discourse*.

Prevalence. The prevalence of language disorder depends on the method of assessment; a rate of 1–5% of children has been proposed (Norbury and Paul, 2015).

Prognosis. Children with language comprehension deficits can be resistant to change and are likely to require ongoing support for their language. It is reported that about 50% of the children who meet the DSM-5 criteria develop normal speech by adulthood, while the rest have long-lasting difficulties. The prognosis is worse when the language disorder is severe, and when there is a comorbid condition, such as CD.

Treatment. This mainly involves specialist speech and language therapists and special education provision, for example, focusing on developing vocabulary and narrative skills. Psychiatrists may be involved when there is a comorbid disorder, and may need to advise the parents about the child’s rights for special education.

For a review of language disorder, see Norbury and Paul (2015).

Assessment of speech and language disorders

Early investigation is essential both to determine the nature and severity of the speech and language disorder and to exclude disorders of intellectual development, deafness, cerebral palsy, and pervasive developmental disorder. The speech-producing organs should be examined. It is particularly important to detect deafness at an early stage.

Parents can give some indication of the child’s speech and language skills, especially if they complete a standardized inventory. With younger children it may be necessary to rely on this information, but children from the age of about 3 years can be tested by a standard test of language appropriate to the child’s age. If possible, such a test should be carried out by a speech therapist or a psychologist specializing in the subject.

Social (pragmatic) communication disorder

The introduction of this category of disorder in DSM-5 is predominantly as a result of changes to the diagnostic criteria for ASD (see below). This category is used for children with persistent difficulties in the social use of verbal and non-verbal communication (Saul et al., 2023). Four key areas of deficit need to be present for a diagnosis: using communication for social purposes (greetings); changing communication to match context or needs of the listener; following rules for conversation (turn taking); and understanding what is not explicitly stated and non-literal or ambiguous meanings. These children must not meet diagnostic criteria for ASDs (no repetitive behaviours and restrictive interests), although similarities might be evidence between the two groups and they also must not have intellectual and language impairments that cause these deficits (Norbury and Paul, 2015). As this is a new disorder, there is not yet clear evidence for interventions, although speech and language therapy focusing on communication needs is the usual intervention provided (Saul et al., 2023).

Autism spectrum disorders

ASDs are similarly categorized in ICD-11 and DSM-5 with two main categories: *difficulties in initiating and sustaining social communication and social interaction* and by *restricted interests and repetitive behaviours* (RRBs). As both components are required for diagnosis of ASD, social communication disorder is diagnosed if no RRBs are present. These features occur in a wide range of situations. Usually development is typical from infancy, and most cases are manifested before the age of 5 years. ASD now encompasses the previous DSM-IV and ICD-10 autistic disorder, Asperger syndrome, childhood disintegrative disorder, and pervasive developmental disorder not otherwise specified.

Clinical characteristics

ASD is used to denote impairments in reciprocal social communication and a tendency to engage in repetitive stereotyped patterns of behaviours, interests, and activities. ASD is a neurodevelopmental disorder that arises from atypical brain development so core features are often present in early childhood although not always apparent.

In both DSM-5 and ICD-11, three main categories of abnormality are important for diagnosis:

1. abnormalities of social development
2. abnormalities of communication
3. restriction of interests and behaviour.

The clinical presentation is remarkably diverse and is variable both between individuals, and in the same individual at different ages (see Box 16.5).

In childhood some might have an apparent history of early regression or lack of progress. Some of the earliest social communication symptoms are difficulties in joint attention, eye contact, lack of intention to communicate with others, lack of socially imitative play, and fascination with sensory stimuli.

Abnormalities of social development. The child has difficulty forming reciprocal, emotional relationships with people and might not respond to their caregivers' affectionate behaviour by smiling or cuddling. Instead, they appear to dislike being picked up or kissed. They are sometimes no more responsive to their parents than to strangers, and do not show interest in other children, although this can be more because of sensory overload than a lack of interest in these interactions. There can be little difference in their behaviour towards people and inanimate objects. A characteristic sign is gaze avoidance—that is, the absence of eye-to-eye contact.

Abnormalities of communication. Speech may develop late or never appear. Occasionally, it develops normally until about the age of 2 years and then disappears in part or completely. This lack of speech is a manifestation of a severe cognitive defect. As children with more severe ASD grow up, about 50% acquire some useful speech, although serious impairments usually remain, such as the misuse of pronouns and the inappropriate repeating of words spoken by other people (echolalia). Some children are talkative, but their speech can be a repetitive monologue rather than a conversation with another person.

The cognitive defect also affects *non-verbal communication and play* as the child might not take part in the imitative games of the first years of life, and later they do not use toys in an appropriate way. They show little imagination or creative play.

Restriction of interests and behaviour. *Obsessive desire for sameness* is a term often applied to the child with ASD stereotyped behaviour, and their distress if there is a change in the environment. For example, some children insist on eating the same food repeatedly, on wearing

Box 16.5 Clinical clues for ASD

In preschool children

- **Limited facial expressions**—May have a flat or restricted range of facial expressions, especially for social interactions.
- **Unusual attachment to objects**—May develop intense attachments to unusual objects (e.g. a piece of string, a specific toy).
- **Emotional dysregulation**—Prone to intense meltdowns, often triggered by minor changes in routine or sensory overload.
- **Unusual gait or motor skills**—May have an awkward walk, clumsiness, or delays in motor coordination (e.g. difficulty with running or jumping).

In school-aged children

- **Rigid thinking patterns**—Struggles with abstract concepts, jokes, sarcasm, or figurative language.
- **Unusual emotional responses**—Laughs or cries at unexpected times or inappropriately to social situations.
- **Hyperfocus on routines or rules**—May enforce strict personal or external rules (e.g. correcting others' behaviour).

- **Monotone or unusual prosody**—Speech may be overly robotic, sing-song, or lack natural inflection.
- **Sensory-seeking behaviours**—May seek excessive movement (rocking, spinning), stare at lights, or crave deep pressure.

In adolescents

- **Literal interpretation of language**—Difficulty understanding sarcasm, metaphors, or indirect social cues.
- **Challenges with emotional expression**—Struggles to identify or express emotions, may appear emotionally detached.
- **Masking or camouflaging**—May mimic peers' behaviours to fit in, leading to exhaustion or anxiety.
- **Increased risk of anxiety and depression**—More prone to mental health difficulties, often linked to social challenges.
- **Executive functioning difficulties**—Struggles with planning, organization, or prioritizing tasks (e.g. managing schoolwork).
- **Unusual food preferences or aversions**—Strong preferences for certain textures, tastes, or routines around eating.

the same clothes, or on engaging in repetitive games. Some are fascinated by spinning toys. Odd behaviour and mannerisms are common. Some children carry out odd motor behaviours such as whirling round and round, twiddling their fingers repeatedly, flapping their hands, or rocking. Others do not differ obviously in motor behaviour from neurotypical children.

Other features. Children with ASD may suddenly show anger or fear without apparent reason. They may be overactive and distractible, sleep badly, or soil or wet themselves. Some injure themselves deliberately. They have a higher prevalence of epilepsy than the general population, with a median overall prevalence of 9%, although with a substantial range observed in studies (Lukmanji et al., 2019).

Intelligence level. Cognitive difficulties are very common, and some disorder of intellectual development is identified in 25–50% of individuals with ASD with the most common pattern being poor language and social comprehension but with relative strengths ‘splinter skills’ in visuospatial abilities. ‘High functioning individuals’ with ASD was a term previously used to describe individuals who could read, write, speak, and manage life skills without much assistance but this is not a diagnostic category; more appropriate descriptions include ‘mild autistic traits’ to ‘needing help with day-to-day functioning’. Some children show areas of ability despite impairment of other intellectual functions, and in some cases, they have exceptional but restricted powers of memory or mathematical skill.

Although there is a tendency for core behaviours to improve over time, some may persist and cause difficulties in the longer term. Those affected can have difficulties with independent living, motor coordination, sensory sensitivities, sleep and eating problems, mental health difficulties, and behaviours that place themselves and others at risk.

Epidemiology

The analysis of the literature on ASD prevalence studies published since 2014 confirms a high variability of prevalence estimates worldwide. This variability is accompanied by methodological differences among the performed studies that concern how cases are detected, which population is involved, and, to a lesser measure, how cases are defined. Interestingly, the longitudinal analysis of data across years within the same geographical area confirms the increase of prevalence estimates that has repeatedly drawn the attention of researchers in the last twenty years. Studies from Australia, Canada, Oman, and the United States and some European countries show a substantial increase of ASD prevalence

estimates over the years. However, the consistency of the increase over countries is masked by the high variability of the prevalence estimates over the continents, with a range from 0.8/1000 in the North Sirajganj district of Bangladesh to 93/1000 in Japan (Chiarotti and Venerosi, 2020). In the United Kingdom, the 2017 prevalence estimate of ASD was 1.2%, with higher prevalence in lower-income households (Sadler et al., 2018) and a rate of 2.8% ever having been diagnosed with ASD in the US National Survey of Children’s Health in 2016 (Xu et al., 2019). It cannot be denied that there has also been an increase in public awareness of ASD, a broadening of the diagnostic criteria, and more clinicians having a better understanding of the disorder. The *prevalence* of ASD is much higher in boys than in girls—as high as 5–6 to 1 ratio—but it is unclear if this is more about poor diagnostic classification in girls than a true gender difference. Of children with ASD, 31% were classified as having IQ scores in the range of intellectual disability (IQ ≤ 70), 23% in the borderline range (IQ = 71–85), and 46% in the average or above-average range of intellectual ability (IQ > 85). The proportion of children classified in the range of intellectual disability differed by race/ethnicity with, for example, significantly greater prevalence among non-Hispanic white children than non-Hispanic Black children.

Risk factors and possible aetiology

ASDs have a strong genetic basis. The heritability of ASDs in the population is around 90%. The rate of ASDs is about 25 times higher in siblings of affected children than in the general population. The genetic architecture of ASD is complex. Like most psychiatric syndromes, the genetic predisposition in most cases results from the combination of multiple common polymorphisms of small effect. There are also copy number variants that are rare but confer a larger risk, and some cases that are caused by specific gene mutations or chromosomal abnormalities (Box 16.6). These cases can be identified by genetic testing (Ellison-Wright and Boardman, 2015). Evidence is steadily accumulating that the highly heterogeneous and functionally diverse set of ASD genes identified so far converge on a smaller set of specific molecular pathways or brain circuits, possibly contributing to aberrant synaptic *pruning*. For a review of the genetics of ASD, see de la Torre-Ubieta (2016) and Eyring and Geschwind (2021).

Neuroimaging studies have also provided useful insights into the neural substrates underlying ASD and the pathological changes that occur in the brain. They indicate that the brain matures along an atypical trajectory. This leads to differences in neuroanatomy,

Box 16.6 Genetic architecture of autism spectrum disorders

Inheritance patterns

- **Complex (polygenic)**—Polygenic liability arising from multiple common genetic variants.
- **Autosomal recessive**—(e.g. Smith–Lemli–Opitz syndrome, Sanfilippo syndrome).
- **Autosomal dominant**—(e.g. Timothy syndrome, Tuberous sclerosis complex, PTEN hamartoma tumour syndrome).
- **X-linked**—(e.g. Fragile X syndrome, Rett syndrome).
- **De novo mutations**—Arise spontaneously in the affected individual, commonly seen in high-impact ASD risk genes (e.g. *SCN2A*, *CHD8*, *ADNP*).

Genetic mechanisms

- **Single nucleotide polymorphisms (SNPs)**—Associated with polygenic risk.
- **Copy number variations (CNVs)**—(e.g. 16p11.2 deletion/duplication, *NRXN1* deletion, 22q11.2 deletion syndrome, Phelan–McDermid syndrome [22q13 deletion]).
- **Trinucleotide repeat expansions**—(e.g. Fragile X syndrome, Myotonic dystrophy type 1).
- **Epigenetic dysregulation**—Methylation abnormalities affecting gene expression (e.g. *MECP2*-related disorders).
- **Mitochondrial dysfunction**—Some mitochondrial DNA mutations have been implicated in ASD-related neurodevelopmental disorders.

functioning, and connectivity within the wider neural systems that probably mediate autistic symptoms and traits (Ecker et al., 2015).

Various risk factors have been identified, for example, a sibling with autism (reflecting the genetic predisposition), birth defects including cerebral palsy, gestational age less than 35 weeks, and maternal use of valproate in pregnancy. As noted earlier, ASD can also occur in a range of genetic disorders such as fragile X, muscular dystrophy, neurofibromatosis, and tuberous sclerosis; since autistic features are not invariable in these conditions, they may also be viewed as being as risk factors for ASD, rather than specific genetic causes of it.

Other potential risk factors include obstetric complications, although it is not clear whether these are a true risk factor or represent an outcome associated with

a primary abnormality in the foetus. Other environmental causes being explored include immune dysfunction, with studies showing a variety of peripheral and central immune changes, including T-cell dysfunction and microglial and astroglial activation. Altered gene expression is also implicated, with differences in two modules of co-expressed genes being observed in the brain in autism. The first module, related to synaptic function and neuronal projection, was under-expressed, whereas the second module, which was enriched for immune genes and glial markers, was overexpressed. These data support the view that synaptic dysfunction as well as immune dysregulation are important in the pathogenesis of ASD. For a review of the non-genetic risk factors see Kim (2019).

Socioeconomic status might influence ASD as risk is increased for children with mothers born abroad, with the risk for low-functioning autism peaking when migration occurred around the time of pregnancy. Different mechanisms can be proposed to explain these results, such as the high level of maternal stress or low immunity regarding common infections. Four reviews have suggested that both maternal age and paternal age are risk factors for autism, with paternal age perhaps playing a more significant role; this might be because of a high frequency of point mutations, gene imprinting effects, genetic discredited changes in sperm cells, *de novo* copy number variations (CNVs), or epigenetic influences that may increase risk of autism in their offspring (Couteur et al., 2015).

Finally, there have been a number of other discredited speculations by the public, media, and scientific community about possible causes of ASD, without scientific data to support them, including immunological causes (infections or vaccinations such as MMR), and exposure to mercury in tooth fillings.

Cognitive and emotional processes

Identifying and understanding the cognitive profile of ASD is likely to play an important role in understanding the underlying brain systems and structures involved in this very heterogeneous disorder. The more prominent theoretical models include: Theory of Mind (ToM) executive dysfunction and weak central coherence (WCC).

Theory of mind. This theory attempts to identify a basic psychological disorder in ASD. By the age of 4 years, neurotypical children are able to form an idea of what others are thinking. As an example, consider a child who watches while another child is first shown the location of a hidden object and then sent out of the room while the object is moved to a new hiding place.

The child who has remained in the room will conclude that the child who left temporarily will expect the object to be in the original position when he returns to the room. A child with ASD tends to lack this appreciation of what another child is likely to be thinking. In the example, a child with ASD is likely to say that the child who left the room will think that the object has been moved to its new place. It is not certain how specific to ASD is this difficulty in appreciating what others know and expect, nor how central it is to the psychopathology. In any case, its cause is not known.

Impairment of frontal lobe executive functions involved in planning and organization, resulting in perseveration and poor self-regulation, as well as impaired ability to extract high-level meaning from diverse sources of information has been hypothesized to contribute to the ASD profile. However, this might be more related to atypical cognitions in ASD rather than executive function deficits. The *WCC theory* tries to explain how in autism a 'local processing bias' explains the enhanced performance of children with autism on some neurocognitive tasks and could explain the 'islets of ability' that are commonly observed in individuals with ASD. For a review of cognitive phenotypes in autism, see Charman et al. (2011).

Differential diagnosis

ASD represents a broad phenotype with differing levels of impairment, but the presentation must be distinguished from the following:

- neurodevelopmental disorders (e.g. language disorders).
- *Developmental speech or language disorder* (see p. 425), which differs from ASD in that the child usually responds normally to people and has good non-verbal communication.
- *disorders of intellectual development* in which responses to other people are more 'normal' than those of a child with ASD. A child with ASD has more impairment of language relative to other skills than is found in an intellectually impaired child of the same age.
- conditions in which there is developmental regression (Rett syndrome).
- *deafness*, which can be excluded by appropriate tests of hearing.

Furthermore, it is important to be aware that certain disorders are more likely to occur with ASD, such as ADHD, depression, anxiety, and CD/challenging behaviours.

Prognosis

Longitudinal studies have found ASD to be a stable diagnosis, with a systematic review demonstrating that the diagnostic stability of ASD diagnosis appears in 53–100% of children studied (Woolfenden et al., 2012). There is wide variation in outcome with between <5 and 25% having a very good outcome from childhood to adult. For example, around 10–20% of children with childhood autism begin to improve between the ages of about 4 and 6 years, and are eventually able to attend a mainstream school and obtain work. A further 10–20% can live at home, but cannot work and need to attend a special school or training centre and remain very dependent on their families and/or support services. The remainder, at least 60%, improve little and are unable to lead an independent life; many need long-term residential care. Those who improve may continue to show language problems, emotional coldness, and odd behaviour. As noted already, a minority develop epilepsy in adolescence.

Overall for many individuals there is gradual reduction in autistic symptoms and improvement in adaptive ability over time but there is marked variability in individual outcomes. The major predictors of better outcomes are higher IQ and presence of useful speech at the age of 5 years. Periods of transition can be particularly difficult, such as becoming an adolescent and then becoming an adult, where at times functioning can be seen to deteriorate. This may coincide with development of comorbid anxiety and mood disorders.

Assessment

Assessment should be concerned with more than the diagnosis of autism. Therefore an ASD-specific developmental history, ASD observational assessments, and standardized individual assessments should be undertaken. The following additional factors need to be considered:

- cognitive level
- language ability
- communication skills, social skills, and play
- repetitive or otherwise abnormal behaviour
- stage of social development in relation to age, mental age, and stage of language development
- associated medical conditions (e.g. comorbid epilepsy)
- psychosocial factors, including the needs of the family.

Individual ASD assessments (through direct interaction and observation) are often made with the use

of an ASD specific tool such as the Autism Diagnostic Observation Schedule (ADOS); the Diagnostic Interview for Social and Communication Disorders (DISCO); the Development, Dimensional and Diagnostic Interview (3di). or the Child Autism Rating Scale (CARS).

Management

The key aspect of management for a child with ASD is to have an individualized approach informed by the specific strengths and difficulties of that child and also with their changing developmental needs and goals. The management will often need to consider a number of key areas which include reducing the core symptoms and behaviours of ASD; enabling the individual to achieve their own potential; treating any comorbid or co-occurring problems including any difficulties encountered by caregivers and supporting the family through education and specific evidence-based approaches. These are summarized below but see Kevelson (2021) for a more detailed description.

Management of abnormal behaviour: psychosocial and behavioural interventions. Early intensive behavioural intervention (EIBI) is the most studied pre-school intervention and along with a number of other, predominantly behaviourally informed interventions have been shown to improve short-term outcomes for preschool children (Tonge et al., 2014). These interventions include strategies for parents, teachers, and carers to improve joint attention skills and reciprocal communication often through interactive play and action routines. There are fewer studies on older children, and the quality of evidence is poorer, but social skills training and peer-mediated social communication interventions show some promise. Overall, the social-communication programmes may help address social isolation (NICE, 2013b) but there is little evidence, to date, for interventions to treat the core features of restricted, repetitive behaviours or sensory sensitivities.

Other interventions: It is important to note that many different therapeutic interventions have been tried, but as with those addressing the core features above, have insufficient evidence to support their use, at present. These include sensory integration therapy, auditory integration therapy, visual therapies, music therapies and restricted diets or dietary supplementation. For example, the gluten-free casein-free (GFCF) diet is the most frequently implemented restrictive dietary intervention but there remains a lack of evidence for its use. There is, however, some evidence that vocational programmes can increase employment success for some individuals with ASD. There are no drugs licensed to treat the core ASD syndrome.

Identification and treatment of co-occurring mental health problems. As described above, children with ASD have higher rates of other mental health problems including ADHD, ODD, OCD, anxiety disorders, and mood disorders. Most clinical guidelines recommend implementing evidence-based interventions for these disorders, with necessary modifications to meet the needs of the person with ASD. For example, CBT can help children with anxiety disorders if they have good verbal and cognitive capacity, as can family and school-based interventions for those with ADHD. The response rate for stimulant medication for ADHD is however lower for a child with ASD when compared to a child without ASD. There is, similarly, little good evidence to support the use of SSRIs for OCD and mood and anxiety disorders, although these might be commonly used in practice. There is some evidence to support the use of antipsychotic medication to ameliorate associated symptoms of aggression and irritability, but these should be used with caution, starting at low doses and for short-term use only. As with most management of psychological problems, a focus on the factors that can increase risk of developing problems is likely to be important. For children with ASD these will include understanding sensory sensitivities, changing circumstances, physical or mental illness, exploitations, and bullying at school.

Educational issues. Depending on the level of cognitive and verbal skills of the child, there has been a move across the globe to try and include children with ASD within mainstream schools although some children with ASD will require special educational environments and if the difficulties are so severe that the child cannot stay in the family, residential schooling is necessary. In some cases, the educational and residential needs of children with ASD can be provided through the services for those with disorders of intellectual development. Older adolescents may need vocational training. There is no evidence for the effectiveness of different educational environments, but for those children with better functioning, their education will need to ensure that both culturally valued skills are learned as well as specific skills and social understanding.

Family support. The family of a child with ASD may experience increased parental and family stress, psychological difficulties, and poorer quality of life. There are, however, few evidence-based interventions that have been shown to help and the general principles of interventions should be the management of behaviours that challenge (NICE, 2013b). The family might need prompt assessment of their child's needs and easy access to appropriate educational and other provisions. Although doctors may be able to do little specifically to

help the child, they must not withdraw from the family, who need continuing support as well as support for their efforts to help the child themselves, and obtain help from educational and social services. Some parents request genetic counselling, and it seems that the risk of a further child with ASD is about 3%. Many parents find it helpful to join a voluntary organization in which they can meet other parents of children with ASD and discuss common problems.

For a review of the prevalence, diagnosis, assessment and treatment of ASDs, see Kvelson et al. (2021), Hodges et al. (2020) and Bougeard et al. (2021).

Rett syndrome

Rett disorder (or Rett syndrome) is a rare X-linked condition that occurs almost exclusively in girls. The reported prevalence is about 1 per 10,000 girls. After a period of normal development in the first months of life, head growth slows and over the next two years there is arrest of cognitive development and loss of purposive skilled hand movements. Stereotyped movements develop with hand-clapping and hand-wringing movements. Ataxia of the legs and trunk may develop. Interest in the social environment diminishes in the first few years of the disorder, but may increase again later. Expressive and receptive language development is severely impaired and there is psychomotor retardation. Some patients develop severe intellectual disability.

The disorder is caused by sporadic mutations in the *MeCP2* gene, located on the X chromosome. These mutations interfere with the normal roles of *MeCP2* in epigenetic regulation of brain development (Sandweiss et al., 2020).

Attention deficit/hyperactivity disorder (ADHD)

ADHD is the second most common psychiatric disorder of childhood. Although about a third of children are described by their parents as overactive, and 5–20% of school children are so described by teachers, these reports encompass a continuum of behaviour ranging from normal high spirits to a severe and persistent disorder. This overactivity often varies in different situations. The norms of childhood behaviour therefore encompass a spectrum of behaviour that is often culturally determined and therefore it is important to ensure that ADHD denotes the functionally impairing, persistent end of the spectrum. ADHD is the term that is most commonly used and comes from the DSM-5 criteria where symptoms need to be present from the main categories

of inattention and hyperactivity and impulsivity. For a summary of ADHD, see Verkuil et al. (2015) and Nigg et al. (2020).

Clinical features

The core features of ADHD are inattention, hyperactivity, and impulsivity. These features are pervasive, occurring across situations, although they can vary somewhat in different circumstances, so that parents and teachers may give rather different accounts of the child's behaviour.

Children with the disorder are often *reckless, and prone to accidents*. They may have *learning difficulties*, which result in part from *poor attention and lack of persistence with tasks*. Many develop minor forms of antisocial behaviour as the condition continues, particularly disobedience, temper tantrums, and aggression. These children are often socially disinhibited and can cause disruption both at home and in the classroom. They can be susceptible to bullying or can be easily influenced to do 'silly' things. Mood fluctuates, but low self-esteem and depressive mood are common.

Restlessness, overactivity, and related symptoms often start before school age. Sometimes the child was overactive as a baby, but more often significant problems begin when the child starts to walk; they are constantly on the move, interfering with objects and exhausting their parents.

Diagnostic criteria

In both ICD-11 and DSM-5 the cardinal features for the diagnosis of ADHD are impaired attention, hyperactivity, and impulsiveness starting in childhood and lasting for at least six months to a degree that is maladaptive and inconsistent with the developmental level of the child. ADHD was previously classified as hyperkinetic disorder in ICD-10. According to ICD-11, ADHD can present as predominantly inattentive, predominantly hyperactive-impulsive, or combined, with both symptoms being clinically significant and neither predominating the clinical presentation. Furthermore the symptoms must be presenting in more than one setting in order for a diagnosis to be made. They both require an element of symptomatic interference with the child's level of social, academic, and occupational functioning. Both require symptoms to have started before the age of 12 years. DSM-5 also has more information on diagnoses in older children and, for example, if over 16, then fewer symptoms are required for a diagnosis. There are relatively recent data to suggest there is a late-onset variant of the condition beginning in adolescence or early adulthood (Sonuga-Barke et al., 2022).

Comorbidity and associated conditions. About 50% of children with ADHD meet diagnostic criteria for other conditions as well, principally ODD, CD, depressive disorder, or anxiety disorder. Sleep problems are also extremely common. Specific learning difficulties and poor motor coordination are also commonly present. Estimations vary substantially for prevalence rates of ASDs in children with ADHD but have been estimated at 20–50%.

Epidemiology

Estimates of the prevalence of ADHD vary according to the criteria for diagnosis.

The 2010 Global Burden of Disease Study found that worldwide point prevalence rates of childhood ADHD were 2.2% in males and 0.7% in females (Erskine et al., 2013). However, the diagnosis rates of and treatment approaches to ADHD vary worldwide. For example, two studies, from the United States and United Kingdom, both explored parent-reported ADHD diagnosis ‘ever’ made. The US study reported a 42% increase in ADHD diagnoses from 2003–2011, with 8.8% having a current diagnosis of ADHD. The UK study reported a prevalence of 1.7%, with no evidence of an increase between 1999. Overall, 6.1% of children in the United States receive drugs for ADHD, in contrast with an estimated 0.7% in the United Kingdom (Verkuijl et al., 2015).

Aetiology

ADHD is due to both heritable and non-heritable factors, with evidence suggesting a disorder of higher cognitive executive function, related to abnormalities of neurotransmission in the prefrontal cortex and associated subcortical structures.

Neurological findings. Signs suggesting neurodevelopmental impairment or delay are found in children with hyperkinetic disorder—for example, clumsiness, language delay, and abnormalities of speech. Although these signs are generally associated with birth complications, they could result from factors acting at an earlier stage of development of the brain. The disorder can also occur as a consequence of severe traumatic brain injury.

Neuroimaging studies. Recent studies have shown a reduction in volume and cortical thickness in certain areas of the brain associated with ADHD, especially the grey matter of the basal ganglia, and are possibly related to a delay in brain maturation. Other studies have shown functional and structural abnormalities in areas involved in self-organization, including the prefrontal and striatal regions, as well as the cerebellum. There is also evidence for white matter disruption and disordered anatomical and functional connectivity between these various brain regions. For a review, see Pereira-Sanchez et al. (2021).

Genetic studies. Investigations of family members, monozygotic and dizygotic twins, and adopted children all suggest that genetic factors are important, with heritability estimates of about 70–80%. Compared with controls, probands with ADHD have more first-degree relatives with the same disorder. Adoption studies show that the biological parents of children with ADHD are more likely to have had the same or a related disorder than are the adoptive parents. Molecular genetic studies indicate that both common and rare variants are involved (see Chapter 5) and include genes influencing early brain development (Demontis et al., 2022). Some of the genetic risk likely interacts with environmental factors, such as maternal smoking. For a review, see Grimm et al. (2020).

Other factors. A number of other factors have been shown to be associated with ADHD. These include early psychosocial adversity (demonstrated by studies of children raised in extreme deprivation), maternal alcohol and substance misuse during pregnancy, low birth weight, prematurity, nutritional deficiencies, and exposure to environmental toxins.

Prognosis

Inattention and impulsiveness often persist into adult life while overactivity usually lessens gradually as the child grows older, especially when it is mild and not present in every situation and often ceases by puberty. About 50% of the cases diagnosed in childhood retain the full diagnosis in adolescence. The prognosis for any associated learning difficulties is less good, while antisocial behaviour has the worst outcome. When *the overactivity is severe*, and is accompanied by *learning failure* or *low intelligence*, the prognosis is poor and the condition may persist into adulthood as antisocial disorder and drug misuse, rather than as continued hyperactivity.

By adulthood most patients will no longer meet the full criteria for ADHD or hyperkinetic syndrome, but the majority will retain some functional impairment. Although this would suggest that the incidence of the disorder should be low in adulthood, surveys have indicated rates of about 4%, suggesting the presence of additional cases that were not detected during childhood or might commence in adulthood as late-onset ADHD. Furthermore, the diagnostic criteria in the DSM-5 has reduced the number of symptoms required for adults and so potentially increased the rate of detection of ADHD in adults. Adults with ADHD can experience more opportunities to ‘live with’ the disorder as they no longer need to attend school with its associated institutional structure and can potentially choose career paths more suited to their work patterns and needs. However, many describe

procrastination, poor motivation, and mood lability, while irritability, inattention, and poor organization can lead to problems with both work and social relationships. Comorbid mood disorders and substance misuse are common. Studies of psychostimulants and atomoxetine show a similar clinical response to that seen in children.

Treatment

Psychosocial interventions. An overactive child can exhaust their caregivers, who might need support as they might have had feelings of being judged by other parents or teachers for not being able to control their child's behaviour for a number of years. Educational interventions to help teachers understand how best to support their students with ADHD are also important, as the majority of treatment approaches are to focus on reducing impairment and improving quality of life (Sonuga-Barke et al., 2022) and additional special education provision may be needed. Psychosocial interventions are often recommended in guidelines as a first step but with more severe cases this is usually combined with stimulant medication. Although the evidence base for most psychosocial interventions is limited, parent training, social skills training, cognitive training, and specific classroom interventions have all been studied. Behavioural interventions often include parent-, child-, and school-based elements and seem to be most effective in combination with medication.

Parent training for ADHD is commonly used although it has a weak evidence base. It is usually group based and focuses on making unwanted behaviours for the child explicit, clarifying family rules, and anticipating potentially difficult times in the day, such as transition times. CBT for children with ADHD might have some benefit for those with comorbid anxiety or depressive disorders. Interventions to assist children with ADHD in the classroom, including advice for teachers, can provide some positive effects.

Dietary advice. A minority of children with ADHD might benefit from free fatty acid supplementation and some parents who suspect their children's behaviour is affected by the food they eat have observed improvements following the elimination of certain foods after following a strict elimination diet. The restriction of artificial food colourants can also help a minority of children with ADHD.

Medication. Drugs can improve the three core symptoms of inattention, overactivity, and impulsivity, and should be tried in children and adolescents with moderate to severe impairment, and where parent training and psychological approaches alone have not proved effective. There are two main groups of clinically effective drugs for

ADHD: stimulants and non-stimulant drugs. *Stimulants* increase available central dopamine and noradrenaline, and it is thought that these actions underlie their therapeutic effects. The most commonly prescribed medication is *methylphenidate*, which can be prescribed in short- or long-acting preparations. Dexamphetamine is also used, often now in the form of lisdexamfetamine (see page 742). Initial dosage should be related to body weight, but titrated primarily according to clinical response and tolerability.

The potential short-term benefits of stimulant treatment are decreased restlessness, reduced aggressiveness, and (sometimes) improved attention. These effects do not usually diminish with time, but it has not been shown in controlled trials that they are associated with better long-term outcome. However, epidemiological studies of adults with ADHD have shown longer-term benefits of stimulant treatment in terms of increased employment and reduction in risks of injuries, traffic accidents and crime, provided medication continues to be taken (Swanson, 2019).

The *side-effects* of stimulants include irritability, depression, insomnia, and poor appetite with resultant height and weight impacts. Tics may be temporarily made worse. As medication may be needed for many months, and some children take it for years, careful monitoring is essential. Some parents are understandably reluctant to agree to long-term drug treatment for their children. In such cases it can be helpful to give the drugs for a trial period so that the benefits and disadvantages can be assessed for a particular child. The drug may be stopped from time to time (drug holidays) in an attempt to minimize side-effects and to confirm that medication is still needed. Baseline assessment should include cardiovascular history and examination, with ongoing monitoring of blood pressure and heart rate.

Other drugs include the noradrenaline reuptake inhibitor *atomoxetine*, which has also been licensed for the treatment of hyperkinetic syndrome and ADHD. Common adverse effects of atomoxetine include nausea, abdominal pain, loss of appetite, and sleep disturbance. Rarely (about 1 in 50,000 patients treated), severe liver damage can occur. Although atomoxetine has the theoretical advantage of lacking psychostimulant properties, experience with its use is necessarily limited compared with that for methylphenidate and studies show that although atomoxetine is effective, methylphenidate has better outcomes. It may be more suitable in patients with comorbid tic disorders. It can also be used in those who have not responded to stimulant treatment, or who are poorly tolerant of it.

Another non-stimulant drug, more recently approved for the management of ADHD, is guanfacine, which is a noradrenergic α_2 -adrenoceptor agonist. It is indicated for children and adolescents for whom stimulants are ineffective or poorly tolerated. Common side effects of guanfacine include somnolence, headache, fatigue, and abdominal pain. Like other α_2 -adrenoceptor agonists, guanfacine can lower blood pressure and cause fainting, so careful dose titration in the early stages of treatment is required (Huss et al., 2018).

In clinical trials, short-term symptomatic and functional benefits of stimulants have been shown in about two thirds of children, but the long-term benefits are uncertain. It therefore seems best to reserve drug treatment for moderate to severe ADHD, particularly where symptoms cause significant functional impairment.

For clinical guidelines for drug treatment of ADHD see Caye et al. (2019) and for an update on medication for ADHD see Steingard et al. (2019).

ADHD in adults

General adult psychiatrists are seeing a marked increase in referrals for suspected ADHD in people who had not been diagnosed in childhood or adolescence. As noted above, this may reflect a late-onset subtype of ADHD, but more likely it reflects cases that had not been recognized as ADHD previously and had therefore not presented for treatment. The Adult ADHD Self-Report Rating Scale (ASRS) is a recommended screening tool. Bipolar disorder is both a differential and a comorbid diagnosis, and comorbidity with substance use disorders is common. Management of adult ADHD follows similar principles as in children and adolescents, with both non-stimulant and stimulant medications being used, and with good evidence that treatment is associated with improved outcomes. For review, see Kooij et al. (2019).

Oppositional and conduct disorders

Oppositional and conduct disorders are characterized by persistent patterns of behaviour that deviate from socially accepted norms and often intrude on other people's expectations or rights. They form the largest single group of psychiatric disorders in older children and adolescents and are a highly impairing psychiatric disorder. For review, see Fairchild et al. (2019).

Clinical features

The essential feature of both ODD and CD is persistent abnormal conduct which is more serious than ordinary

childhood mischief. The abnormal behaviours centre around *defiance*, *aggression*, and behaviours that violate social expectations or the rights of others. The upset, disruption, and costs inflicted on the family, peer group, schools, and wider society can be considerable.

In the preschool period, oppositional defiant behaviour usually manifests as defiant and aggressive behaviour in the home, often with overactivity. The behaviours include disobedience, temper tantrums, physical aggression towards siblings or adults, and destructiveness. In later childhood, CD is manifested in the home as stealing, lying, and disobedience, together with verbal or physical aggression. Later, the disturbance often becomes evident outside as well as inside the home, especially at school, or as truancy, delinquency, vandalism, and reckless behaviour, or as alcohol or drug abuse. Antisocial behaviour among teenage girls includes spitefulness, emotional bullying of peers, and running away.

In children older than seven years of age, persistent stealing is abnormal. Below that age, children seldom have a real appreciation of other people's property. Many children steal occasionally, so that minor or isolated instances need not be taken seriously. A small proportion of children with CD present with sexual behaviour that incurs the disapproval of adults. In younger children, masturbation and sexual curiosity may be frequent and obtrusive. Frequent unprotected sex and pregnancy may be a particular problem in adolescent girls. Fire-setting is rare, but obviously dangerous (see p. 501).

To constitute ODD and CDs, these behaviours have to be more persistent than a reaction to changing circumstances, such as adjusting to the arrival in the family of a new step-parent. There is no sharp dividing line between these disorders and ordinary bad behaviour; instead, there is a continuum on which diagnostic criteria define a cut-off point. The cut-off defines the most severe cases that have the worst outcome and are most in need of help. Much of this help is social or educational, but psychiatrists have an important role in identifying comorbid disorders and arranging multidisciplinary care.

Classification

Both ICD-11 and DSM-5 include ODD/CD, which are characterized by a repetitive pattern of antisocial behaviour. They are highly heterogeneous disorders with different symptom clusters and the diagnostic criteria in CD, for example, include subtypes based on age of onset of symptoms (childhood versus adolescent-onset CD) and the presence or absence of limited prosocial emotions (such as empathy). In ICD-11, ODD is a subtype of CD and is usually ascribed to younger children and does not include delinquent acts. In DSM-5 it is a separate entity.

ODD is usually a diagnosis given to younger children while CD is typically given to older children and teenagers. ODD requires four from a list of eight symptoms to be present for at least six months and CD requires the presence of three symptoms from a list of 15, with a duration of at least 12 months. The criteria are closely similar in the two systems of classification.

Of note, DSM-5 has two other disorders that may overlap, Intermittent explosive disorder (which can only be diagnosed after the age of 6 years) and a new disorder called disruptive mood dysregulation disorder (DMDD). Although DMDD disorder has been explicitly introduced to prevent over-diagnosis and medication for bipolar disorder, it appears to have considerable overlap with ODD.

Prevalence

The overall prevalence of ODD and CD together varies between 5% and 10%, with much variability attributed to diagnostic criteria and methods used in studies. ODD/CD makes up about half of all child and adolescent psychopathology, and ODD is more common in younger children while CD is more common in adolescents. There is fairly good agreement on rates of 5–8% in general longitudinal surveys where multiple informant have been used (Scott and Garnder, 2015). ODD is about twice as common in boys than girls, but this ratio rises for CD with rates that are 3–7 times more common in boys than girls. There is also a strong socioeconomic factor to prevalence with those experiencing socioeconomic disadvantage having five times greater prevalence than those in the highest groups and rates are also highest in children who have experienced maltreatment, residential care, foster care, and those with intellectual disability. Furthermore, prevalence has shown to have increased considerably over the second half of the twentieth century.

Aetiology

Many different factors influence the development of ODD/CD and although some risks are intercorrelated, exposure to multiple risk factors has a cumulative effect, worsening outcomes. The importance of the parenting/family environment and the complex interplay this might have with individual temperaments and differential responses to environments and interventions is an important area of current research. The aetiological factors can be divided into individual-level influences and family level influences, although as indicated above, the relationship between the two can be close.

Individual-level influences

Genetic factors. It has become clear that for some subtypes, there is a strong genetic contribution to CD/ODD.

CD clusters in families. There is a highly heritable trait of liability to externalizing disorders in general (CD, ODD, and ADHD), but shared environmental factors are also important in the aetiology of CD. For example, genetic contribution is higher for antisocial behaviour in the presence of inattention and hyperactivity, callous unemotional traits, or high levels of physical aggression but where these factors are absent, the genetic contribution is low. There is evidence that a variant of the *monoamine oxidase-A (MAOA) gene* predisposes to CD, but only when combined with adverse factors in the child's environment, such as child maltreatment. If genetic factors are involved, it is not known whether they exert their effect by influencing temperament, or in some other way. For a review, see Fairchild et al. (2019).

Pregnancy and perinatal complications. Maternal alcoholism during pregnancy has been associated with CD, but might be mainly because of the effect on the developing child's IQ.

Temperament. Several studies have shown associations between early temperament and later conduct problems, with traits such as negative emotionality, poor emotional regulation, inattention, and restlessness usually identified. It seems likely that a 'difficult' temperament becomes more likely to lead to disruptive behaviour problems when it interacts with a harsh and inconsistent parenting style.

Brain functioning. There is some evidence of abnormalities in brain regions involved in motivation and affect, with limbic structures and the amygdala as well as the orbitofrontal cortex affected. Children with conduct problems have been consistently shown to have poor executive functions, compromising their ability to successfully achieve goals through appropriate, effective actions. Children with brain damage and epilepsy are more prone to CD, as they are to other psychiatric disorders.

Language, IQ and educational attainment. Low IQ and low school achievement are important predictors of ODD/CD and delinquency. In a study of twins aged 13 years, low IQ of the child predicted conduct problems independent of social class and the IQ of the parents (Goodman, 1995). Children who struggle to assert themselves verbally may attempt to gain control of social exchanges using aggression and low IQ can contribute to academic difficulties, which can impact on self-esteem and a higher likelihood of school being perceived as an unrewarding environment. Thus low school attainment then adds to the probability of ODD/CD developing.

Family-level influences

Quality of parenting, along with gender, is the strongest predictor of antisocial behaviour. In most studies CDs

are commonly found in children from families experiencing instability, insecurity, or low emotional availability, often living in socioeconomically deprived areas. Antisocial behaviour is frequent among children from families with disrupted caregiving arrangements, those from homes in which family relationships are poor, and those who have been in residential care in their early childhood. CDs are also related to adverse factors in the wider social environment of the neighbourhood and school (Scott and Gardner, 2015).

Child maltreatment and exposure to interparental conflict and violence. There is ample evidence that ODD/CD is associated with less-than-optimal parenting practices, characterized by harsh inconsistent discipline, low warmth and involvement, and high criticism (Scott and Gardner, 2015). Associations between physical abuse and conduct problems are well established and although some child conduct presentations might increase the likelihood of experiencing corporal punishment (smacking, spanking) when it comes to injurious physical maltreatment of children, the effects this has on their later likelihood of CD are strong. In addition, children exposed to domestic violence between adults are more likely themselves to become aggressive.

Other influences beyond the family. There are important peer influences, with studies showing how children with ODD/CD have more negative interchanges with other children. They are more likely to be rejected by peers and to associate with deviant peers. Peer influences on antisocial behaviour have most evidence in adolescence. There are also important school effects as children with ODD/CD are more likely to attend schools with high delinquency rates with high levels of distrust between teachers and students, low pupil commitment to school, and unclear rules. As mentioned above, not only do children with ODD/CD come disproportionately from poor families, but they also come from neighbourhoods that are more likely to be inner-city areas with high disorganization and high residential mobility.

Prognosis

The long-term outcome of CD varies considerably with the nature and extent of the disorder. In an important study, Robins (1966) found that almost 50% of people who had attended a child guidance clinic for CD in adolescence showed some kind of antisocial behaviour in adulthood. Subsequent studies have shown that this is most likely to take the form of violent offending, heavy drug usage, teenage pregnancy, leaving school without qualifications, as well as receiving state benefits. No cases of sociopathic disorder were found in adulthood among those with diagnoses other than CD in adolescence.

Follow-up of conduct-disordered children cared for in children's homes and of controls led to similar conclusions. About 40% of the conduct-disordered children had antisocial personality disorder in their twenties, and many of the rest had persistent and widespread social difficulties below the threshold for diagnosis of a personality disorder. Where CDs first *present* in adolescence, the prognosis is better, with about 80% no longer demonstrating significant antisocial behaviour in adulthood (Scott and Gardner, 2015) (see Box 16.7). As individuals get older their offending does reduce considerably, especially by the time they are in their forties, but their overall 'life failure' rates, as determined by multiple indices, remains high. Their general health outcomes are also worse.

Management

Mild CDs often subside without treatment other than common-sense advice to the parents. For more severe disorders, treatment for the child is often combined with treatment and social support for the family although psychological treatments have been shown to have a small effect in reducing parent-, teacher-, and observer-rated CD problems. There is not enough evidence to support one specific psychological treatment over another (Bakker et al., 2017). Any coexisting disorders (e.g. ADHD, depression) should be treated. There is no convincing evidence that any treatment affects the overall long-term prognosis. Nevertheless, some short-term benefits can often be achieved, and in some cases adverse family factors can be modified in a way that could improve prognosis. Some families are difficult to help by any means, especially where there is severe material deprivation, chaotic relationships, and limited access to educational or social resources.

Parent training programmes. These programmes are the mainstay of treatment of ODD/CD and use behavioural principles (see p. 420). Parents are taught how the child's antisocial behaviour may be reinforced unintentionally by their attention to it, and how it may be provoked by interactions with members of the family. Parents are also taught how to reinforce normal behaviour by praise or rewards, and how to set limits on abnormal behaviour—for example, by removing the child's privileges, such as an hour less time to play a game. Parents in some programmes are also trained how to read with their children, thus tackling other influences for the child's behaviour. As aids to learning, parents are provided with written information and video recordings showing other parents applying behavioural procedures. For a review of parent training programmes, see Scott and Gardner (2015).

Box 16.7 Factors that predict poor outcome in children with conduct disorder

Individual factors (child-specific predictors)

- **Early onset**—Symptoms emerging **before the age of 8 years** are strongly associated with persistent antisocial behaviour into adulthood.
- **Severity and persistence**—Frequent, diverse, and severe antisocial behaviours (e.g. aggression, property destruction, serious rule violations) indicate a worse prognosis.
- **Comorbid ADHD and impulsivity**—**Hyperactivity, attention problems, and poor impulse control** increase risk of persistent conduct problems.
- **Low cognitive functioning**—**Lower IQ, executive dysfunction, and verbal skill deficits** contribute to poor social problem-solving and emotion regulation.
- **Limited pro-social emotions (callous-unemotional traits)**—Lack of empathy, guilt, and emotional responsiveness is associated with a more **severe, persistent, and treatment-resistant** form of CD.
- **Pervasiveness**—Conduct problems occurring **across multiple settings** (home, school, community) indicate a more entrenched disorder.

Family factors

- **Parental psychopathology**—Parental **criminality, antisocial behaviour, substance abuse, and severe mental health conditions** increase risk.
- **High conflict and maltreatment**—**Hostile, aggressive, or neglectful parenting**, exposure to domestic violence, and child maltreatment worsen prognosis.

- **Inconsistent or harsh discipline**—Unpredictable, overly punitive, or permissive parenting styles exacerbate conduct issues.
- **Low socioeconomic status (SES)**—Socioeconomic disadvantage, financial instability, unemployment, and chronic stress in the home environment contribute to poorer long-term outcomes.
- **Poor parent-child attachment**—Weak emotional bonds, lack of parental warmth, and insecure attachment patterns are linked to more severe externalizing behaviours.

Environmental and social factors

- **Exposure to violence and crime**—Growing up in neighbourhoods with high crime rates increases antisocial modelling.
- **School failure and exclusion**—Early academic struggles, school disengagement, and exclusionary discipline practices (e.g. suspension, expulsion) predict worsening conduct problems.
- **Peer influences**—Association with **delinquent or deviant peer groups** (especially in adolescence) reinforces antisocial behaviour.
- **Social disadvantage and systemic inequality**—Structural disadvantage, discriminatory policies, lack of access to social services, and instability in housing and community resources worsen risk.
- **Neurobiological risk factors**—Structural and functional differences in prefrontal cortex, amygdala, and reward-processing systems have been linked to greater aggression and poor emotional regulation.

Anger management. Young people who are habitually aggressive have been shown to misperceive hostile intentions in other people who are not in fact hostile. They also tend to underestimate the level of their own aggressive behaviour, and choose inappropriate behaviours rather than more appropriate verbal responses. *Anger management programmes* seek to correct these ideas by teaching how to inhibit sudden inappropriate responses to angry feelings—for example, the child says to himself ‘Stop! What should I do?’ They also learn how to *reappraise the intentions* of other people and use socially acceptable forms of self-assertion.

Other methods. Remedial teaching should be arranged if there are associated reading difficulties. Medication is

of value if there is a comorbid disorder such as ADHD. There is increasing use of atypical antipsychotics to target reactive aggression, although the evidence that this is effective is only modest and might best be considered when there is poor emotional regulation characterized by prolonged rages; if used, medication should only be considered as a short-term intervention.

Interventions in school. There is growing interest in how schools can try and develop interventions to help address the effects of ODD/CD in their classrooms. A number of interventions have been developed that are mainly preventive programmes, including promoting social and emotional learning. They are often delivered by teachers to all children and include proactive

strategies focusing on positive behaviours with effective instructions given to children alongside effective behavioural management. As yet, the evidence base for these interventions as well as the use of teaching assistants in classrooms is poor.

Forensic child psychiatry and juvenile delinquency

Forensic child psychiatry is predominantly associated with children's interactions with the juvenile justice system and can include the assessment and management of mental disordered offenders, or those at risk of offending, and issues such as mental competency, fitness to testify, victimization, and child custody disputes. This section focuses on the young offender, with the term juvenile delinquent often used to denote a young person who has broken the law and been found guilty of an offence that would be categorized as a crime if committed by an adult. In most countries, the term applies only to a young person who has attained the age of criminal responsibility (at present ten years of age in the United Kingdom, but varying widely in other countries). Thus delinquency is not a psychiatric diagnosis but a legal category. The term 'children in conflict with the law' is gaining increasing popularity. However, juvenile delinquency may be associated with psychiatric disorder, especially CD. For this reason, it is appropriate to interrupt this review of the syndromes of child psychiatry in order to consider juvenile delinquency.

The majority of adolescent boys, when asked to report their own behaviour, admit to offences against the law, and about 20% are convicted at some time. Most offences are against property. In England in 2009, young people 10–17 years of age committed about 200,000 detected offences of sufficient gravity to result in a court disposal. The largest category (about 42,000 offences) was for theft and handling stolen goods, although violence against the person was a close second (about 38,000 offences). Over the 2000–2017 period, the proportion of proven offences committed by girls remained much lower than that of boys—around 30% of the total—though the gender gap has narrowed somewhat over time (Arnull et al., 2023). Chronic juvenile delinquency has been shown to lead to increased life 'failure'—a composite measure based on offending, housing stability, marital stability, employment, substance use, and mental health functioning at both ages 32 and 48 (Drury, 2019).

Delinquency is sometimes equated with CD. This is wrong, for although the two categories overlap, they are not the same. Many delinquents do not have CD (or any

other psychiatric disorder). Equally, many of those with CD do not offend. Nevertheless, in an important group, persistent law-breaking is preceded and accompanied by abnormalities of conduct, such as *truancy*, *aggressiveness*, and *attention seeking*, and by *poor concentration*.

Causes

The causes of juvenile delinquency are complex and overlap with the causes of CD. The causes are reviewed briefly here and many of the findings have come from the Cambridge Study in delinquent development, which has followed up 411 males from age 8 to 50 years; for a more detailed account, see Young and Church (2015).

Social factors. Delinquency is related to socioeconomic disadvantage, poverty, insecure or poor-quality housing, bullying, and poor education. There are marked differences in delinquency rates between adjacent neighbourhoods which differ in these respects. Rates also differ between schools. Many social theories have been put forwards to explain the origins of crime, but none offers a completely adequate explanation.

Family factors. Many studies have found that crime and interpersonal violence runs in families. For example, about 50% of boys with criminal fathers are convicted, compared with 20% of those with fathers who are not criminals.

Factors in the child. *Genetic factors* appear to be less significant among the causes of delinquency than in the more serious criminal behaviour of adulthood (see p. 496). There are important associations between delinquency and a range of neurodevelopmental disorders (Hughes, 2012) with prevalence rates in those with delinquent behaviours of 23–32% of intellectual disability, 43–57% of specific learning disability, 15% of ASDs, and ADHD in 11–30% (Hughes, 2012; Young and Church, 2015).

Assessment

When the child is seen as part of an ordinary psychiatric referral and the delinquency is accompanied by a psychiatric syndrome, the latter should be assessed in the usual way. It is important to be aware that there is a high prevalence of mental disorder among juvenile offenders, and it is important to screen for anxiety, depression, and suicidal risk in particular. A thorough assessment is often needed as a juvenile offender can present with a broad range of needs and risks, requiring information from collateral sources, clear notes, clear risk assessments, and management plans; a number of structured risk assessment instruments have been developed with varying utility, generally focusing on relationships between risk and re-offending. Sometimes the child psychiatrist

is asked to see a young person specifically to prepare a court report. In these circumstances, as well as making enquiries among the parents and teachers, it is essential to consult any social worker or probation officer who has been involved.

Psychological testing of intelligence and educational achievements can be useful. Other factors to be considered are listed in Box 16.8. The form of the report is similar to that described earlier in this chapter, and in Chapter 4. It should include a summary of the history and present mental state together with recommendations about treatment.

Interventions for juvenile offenders

This section considers interventions that are intended to reduce the chances of further offending, often falling under the risk–need–responsivity (RNR) model, which focuses on the prediction of risk and classification of

offenders for treatment. For juvenile offenders with psychiatric disorders, including CDs, mood disorders, ADHD, substance misuse and dependence, and disorders of intellectual development, the treatment of these disorders should be a core aspect of care and are treated according to principles described elsewhere in this chapter. The range of interventions that have been developed for juvenile offending can be conceptualized within a public health framework and are listed in Box 16.9, and described in more detail in Young and Church (2015).

Psychiatrists who treat juvenile offenders need to understand the legal system in the country in which they work. The legal responses usually include a warning not to offend again, a fine, the requirement that the parent or guardian take proper control, supervision by a social worker, a period at a special centre, or an order committing the child to the care of the local authority. Since delinquent behaviour is common, mainly not serious,

Box 16.8 Assessment of young offenders

1. Current offence analysis

- **Nature and seriousness**—The type of offence (e.g. violent, sexual, property-related) and its severity, including harm caused.
- **Characteristics of the victim**—Age, vulnerability, relationship to the offender, and any patterns in victim selection.
- **Motive and intent**—Underlying reasons for the offence (e.g. financial gain, peer influence, impulsivity, revenge, lack of empathy).
- **Role in group offending**—Whether the young person acted alone or within a group; leader or follower dynamics if others were involved.
- **Attitude towards the offence and victim**—Level of remorse, minimization or justification of actions, victim-blaming tendencies.

2. Previous offending and associated behaviours

- **Offence history**—Number, nature, escalation, and whether offences were detected, charged, or resulted in convictions.
- **Patterns of offending**—Recurring themes (e.g. gang involvement, opportunistic vs. premeditated crimes, victim targeting).
- **Substance misuse**—Type, frequency, and impact on behaviour and decision making.

- **History of violence**—Aggression towards peers, adults, or authority figures, including triggers and situational factors.
- **Self-harm and mental health concerns**—Presence of self-injurious behaviour, suicide risk, or emotional dysregulation.
- **Cruelty to others**—Acts of physical or emotional harm towards children, animals, or vulnerable individuals.
- **Fire-setting**—Incidents of arson or fascination with fire, including underlying motivations (e.g. thrill-seeking, revenge, sensory interest).

3. Strengths and protective factors

- **Individual resilience**—Personal strengths, coping skills, ability to express emotions, and any pro-social aspirations.
- **Family support**—Presence of a supportive caregiver, stable home environment, or other positive role models.
- **Community and peer influences**—Engagement in structured activities (e.g. sports, arts), positive peer networks, and access to mentoring or diversion programmes.
- **Education and employment**—School attendance, learning abilities, and vocational interests.

Box 16.9 Interventions for youth offending

Primary prevention

Parenting programmes
Preschool programmes
Day-care programmes
School programmes
Cognitive skills training
Peer programmes
Community programmes
Situational crime prevention

Secondary prevention

Family-focused therapies
Mentoring
Therapeutic foster care
Safeguarding of children
Treatment of parental substance misuse

Tertiary prevention

Cooperation between police, social services, and mental health
Intensive supervision and surveillance
Assertive treatment of mental disorder
Restorative justice
Victim support

and often a temporary phase, it is generally appropriate to treat first offences with minimal intervention coupled with firm disapproval. The same applies to minor offences that are repeated. A more vigorous response is required for more serious, recurrent delinquency. For this purpose a *community-based programme* is usually preferred, with the main emphasis on improving the family environment, reducing harmful peer group influences, helping the offender to develop better skills for solving problems, and improving educational and vocational accomplishments. In the United Kingdom, these are usually conducted by young offender teams. When this approach fails, custodial care may be considered.

The main aim of the law as it applies to children and adolescents is treatment rather than punishment. There has been extensive criminological research to determine the effectiveness of the measures used. The general conclusions are not encouraging, although not surprising since, as explained above, delinquency is strongly

related to factors external to the child, including family disorganization, antisocial behaviour among the parents, and socioeconomic deprivation.

Of the many approaches to the treatment of juvenile offending, a few are described below, but the evidence-base remains poor for most of the interventions.

- *Structured programs following cognitive behavioural models as well as problem-solving skills training.* These can focus on teaching social skills, anger management, and problem solving. An additional component, attributional retraining, helps to correct cognitive distortions whereby delinquent youths readily perceive threat and hostility even in neutral interpersonal situations.
- *Multisystemic therapy.* This consists of an intensive set of integrated interventions across multiple systems in the young person's life, following nine treatment principles, including family therapy; helping the young person to find non-delinquent friends; personal development (including assertiveness training); improving family problem-solving skills; liaison with teachers; and coordination of other involved agencies.
- *Multidimensional treatment foster care.* The young person lives in a foster home, away from delinquent friends, for about six months and learns better life skills. There are high levels of clinical support, daily contact, and points systems for rewards alongside clear consequences for negative behaviour. At the same time, the family are taught the skills needed to respond to the young person more effectively.

Studies of these and other approaches indicate the need to match the type of treatment to the needs of the particular offender. Some seem to respond better to authoritative supervision, others to more permissive counselling. However, it is not yet possible to provide any satisfactory guidelines for matching treatment and offender. See Young and Church (2015) for a review of forensic aspects of child and adolescent psychiatry and Piquero et al. (2016) for a review of parent-training interventions on antisocial behaviour.

Anxiety disorders

Anxiety disorders are the most common mental disorders of childhood. Fear and anxiety play important roles in functioning and have major evolutionary significance and therefore it is important for a clinician to distinguish between normal anxiety and anxiety disorders in childhood. This distinction is usually when the symptoms are impairing function and/or causes marked avoidance

and significant distress. Both ICD-11 and DSM-5 classify childhood and adult anxiety disorders together in an *anxiety disorders* section, although in ICD-11 childhood separation anxiety disorder is listed separately with 'emotional disorders with onset specific to childhood'. Both diagnostic schemes place OCD and post-traumatic stress disorder (PTSD) in different categories from anxiety disorders.

Prevalence

Anxiety disorders are the most common mental disorders in childhood and more frequent in girls than boys. They are associated with significant impairment with phobias and separation anxiety disorders most common in childhood and social anxiety becoming more prominent in adolescence. Surveys of the general population suggest overall prevalence rates of anxiety disorders in children around 5–10% with age and cultural factors possibly influencing prevalence rates in individual studies that can range from 1–24%. (Pine and Klein, 2015).

Anxiety at different ages

The nature and manifestations of anxiety change as the child grows older. Infants pass through a stage of fear of strangers. During the preschool years, separation anxiety and fears of animals, imaginary creatures, and the dark are common. In early adolescence these fears are replaced by anxiety about social situations and personal adequacy. Anxiety disorders in childhood resemble these normal anxieties and follow the same developmental sequence, although they are more severe and more prolonged. Phobias and separation anxiety disorder usually start in early childhood, and social anxiety disorder starts in adolescence. In later adolescence, rates of generalized anxiety disorder and panic disorder start to increase. Comorbidity is common in childhood anxiety disorders, especially with other types of anxiety disorders.

Aetiology

The main factors that have been studied in the aetiology of anxiety disorders concern the interlinked issues of the child's temperament, genetic influences, and the family social environment, including parenting behaviours. Family aggregation and genetic studies indicate raised vulnerability to anxiety in offspring of adults with the disorder (e.g. the temperamental style of behavioural inhibition, or information processing biases). Environmental factors are also important; these include adverse life events and exposure to negative information or modelling. Both mothers and fathers are likely to be key, although not unique, sources of such influences,

particularly if they are anxious themselves. Some parenting behaviours associated with child anxiety, such as overprotection, may be elicited by child characteristics, especially in the context of parental anxiety, and these may serve to maintain an anxiety disorder in the child. Emerging evidence emphasizes the importance of taking the nature of child and parental anxiety into account and of considering bidirectional influences. For a review of aetiology, assessment, and treatment see Creswell et al. (2020).

Prognosis

Longitudinal surveys of anxiety disorders highlight how most affected children will be free of anxiety disorders in adulthood. However, anxiety disorders in childhood do predict later major depression in some samples, and in studies of adults with anxiety disorders, the majority had childhood anxiety disorders; therefore, they do increase risk of later disorders. Anxiety may also have greater long-term impact in females.

Management

Early detection and treatment of childhood anxiety disorders can prevent substantial impairment over the course of a child's development and accumulation of functional disability. Early treatment also may prevent later development of adult psychiatric illness and prevent impaired development and accumulated functional disability. Family education and training can reduce reinforcement of anxiety and avoidance, but parents with their own anxiety disorders or other psychopathology may need separate treatment. Existing studies support a number of pharmacological and psychotherapeutic treatments for childhood anxiety disorders. Both CBT and SSRIs are effective treatments for separation, generalized, and social anxiety disorders in children and adolescents, although non-pharmacological interventions are usually used in the first instance. Combination treatment with SSRIs and CBT has been found to be more effective than either treatment alone.

The most frequently evaluated psychological treatment for anxiety disorders in children and young people is CBT, which typically involves the application of exposure to enable children and young people to confront feared situations, typically in a graded fashion, in order to develop new learning about what really happens when they enter anxiety-provoking situations (Creswell et al., 2020). In CBT programmes, exposure is typically accompanied by cognitive restructuring procedures, to help children identify and challenge negative automatic thoughts. Some programmes also include other forms of skills training, such as

relaxation, social skills, and problem-solving training. It has consistently been concluded, across a number of meta-analyses, that CBT shows clear benefits over waitlist controls, although longer-term outcomes are less conclusive.

Separation anxiety disorder

Separation anxiety disorder is a fear of separation from people to whom the child is attached that is clearly greater than normal separation anxiety of toddlers or preschool children, or persists beyond the usual preschool period, and is associated with significant problems of social functioning. The onset is before the age of 6 years. The diagnosis is not made when there is a generalized disturbance of personality development.

Clinical picture

While separation anxiety is part of normal early child development, children with this disorder are excessively anxious when separated from parents or other attachment figures, and unrealistically concerned that harm may befall these persons or that they will leave the child. They may refuse to sleep away from these persons or, if they agree to separate, may have disturbed sleep with nightmares. They cling to their attachment figures by day, demanding attention. Anxiety is often manifested as physical symptoms of stomachache, headache, nausea, and vomiting, and may be accompanied by crying, tantrums, or social withdrawal. Separation anxiety disorder is one cause of school refusal (see School refusal).

Epidemiology

Community surveys suggest that rates of separation anxiety disorder are about 5% among 7- to 11-year-old boys and girls (Pine and Klein, 2015).

Aetiology

Separation anxiety disorder is sometimes precipitated by a *frightening experience*. This may be brief (e.g. admission to hospital) or prolonged (e.g. conflict between the parents). In some cases separation anxiety disorder develops in children who react with excessive anxiety to a large number of everyday stressors and who are therefore said to have an *anxiety-prone temperament*. Sometimes the condition appears to be a response to anxious or over-protective parents.

Prognosis

The disorder often improves with time, but may worsen again when there is a change in the child's routine, such as a move of school. Some cases progress to generalized or other anxiety disorders in adult life.

Treatment

The whole range of possible aetiological factors should be considered, including stressful events, previous actual separation, an anxiety-prone temperament, and the behaviour of the parents. Stressors should be reduced if possible, and the children should be helped to talk about their worries. It is more important to involve the family, helping them to understand how their own concerns or overprotection affect the child, and to find ways of making the child feel more secure. Anxiolytic drugs may be needed occasionally when anxiety is extremely severe, but they should be used for short periods only. When separation anxiety is worse in particular circumstances, the child may benefit from the behavioural techniques used for phobias.

Phobic anxiety disorder

This diagnosis for children corresponds to specific phobia for adults (see p. 170). Minor phobic symptoms are common in childhood. They usually concern animals, insects, the dark, school, and death. The prevalence of more severe phobias varies with age. Severe and persistent fears of animals usually begin before the age of 5 years, and nearly all have declined by the early teenage years but a minority, probably about 10%, persists into adult life.

Most childhood phobias improve without specific treatment provided that the parents adopt a firm and reassuring approach. For phobias that do not improve, simple behavioural treatment can be combined with reassurance and support. The child is encouraged to encounter feared situations in a graded way, as in the treatment of phobias in adult life (page 171).

Post-traumatic stress disorder

Although not included in ICD-11 among the anxiety disorders with onset usually in childhood, PTSD can occur in children. The clinical picture resembles that of the same disorder in adult life (see p. 140), with disturbed sleep, nightmares, flashbacks, and avoidance of reminders of the traumatic events. Developmental factors can play an important role in the presentation of PTSD and so children may also present with behavioural problems, developmental regression, physical symptoms, and more generalized fears.

Aetiology

As in adults, the cause is an encounter with an exceptionally severe stressor—for example, those encountered by children experiencing natural and man-made disasters, war, physical or sexual abuse, violence to self and others, and serious or life-threatening injury or illness.

As with adults, cases after vehicle and other accidents are more common than those after the less-frequent major disasters. Estimates of PTSD in children who have been exposed to potentially traumatic events are of around 16% (Alisic et al., 2014). The prevalence of PTSD appears to vary with a number of factors, such as the gender and age of the child or adolescent, type of trauma, frequency and severity of exposure, and the amount of time since the traumatic event. Children and adolescents who have undergone complex trauma defined by repeated or chronic trauma, which is often of an interpersonal nature and begins early in life, may demonstrate different responses to children and adolescents who have undergone other forms of trauma and therefore may respond differently to different therapies for PTSD.

Prognosis

Exposure to trauma in childhood is associated with a broad range of adverse psychiatric outcomes in adulthood. Longitudinal studies have shown how 15–30% of children with PTSD can continue to suffer from the disorder in adulthood as well as generalized anxiety disorders, substance misuse disorders, depression, and suicidality (Yule et al., 2015).

Management

Treatment resembles that for adults (see p. 146). The most studied treatments include trauma-focused CBT (TF-CBT), for which there is the greatest evidence base, as well as eye-movement desensitization and reprocessing (EMDR), narrative exposure therapy (NET), and supportive counselling (Smith et al., 2019).

Obsessive–compulsive disorder

OCDs are increasingly recognized in childhood, and although previously thought to be rare, are now seen to be a common cause of distress for children and adolescents. As with the disorder in adults, OCD is characterized by the presence of obsessions (unwanted, repetitive, or intrusive thoughts), and compulsions (unnecessary repetitive behaviours or mental activities). OCD is a complex disorder as there are clear continuities and discontinuities of the disorder with normal development. Of note, several related forms of *repetitive behaviour* are common, particularly between the ages of 4 and 10 years. These repetitive behaviours include preoccupation with numbers and counting, the repeated handling of certain objects, and hoarding. Typically developing children commonly adopt rituals such as avoiding cracks in the pavement or touching lampposts. These behaviours cannot be called compulsive because the child does not struggle against them. The preoccupations and rituals

of OCD are more extreme than these behaviours of healthy children, and take up an increasing amount of the child's time—for example, re-checking schoolwork many times, or frequently repeated hand washing. OCD is also associated with certain neurological disorders, such as Tourette disorder and there is increasing evidence that there might also be some subgroups of OCD unique to child populations (Rapoport et al., 2015).

Clinical features

OCD typically develops in late childhood and early adolescence. Obsessional disorders in childhood generally resemble those in adulthood (see p. 183). Most children have a combination of obsessions and rituals and commonly present with washing *rituals*, followed by repetitive actions and checking. Obsessional thoughts are most often concerned with contamination, accidents, or illness affecting the patient or another person, and concerns about orderliness and symmetry. The content of symptoms often changes as the child grows older. The obsessional symptoms may be provoked by external cues such as unclean objects. Children with obsessional symptoms usually try to conceal them, especially outside the family. Obsessional children often involve their parents by asking them to take part in the rituals or give repeated reassurance about the obsessional thoughts.

Aetiology

OCD in childhood reflects a combination of biological and psychological factors, with both genetic and environmental influences, as in adult-onset OCD.

Genetic factors are suggested as OCD is a highly heritable disorder, particularly when it has childhood onset. The concordance rate in monozygotic twins (about 80%) is significantly more than that of dizygotic twins (about 40%). Family studies find a tenfold increase in rates of OCD among relatives of children with OCD, compared to a doubling of rates of OCD among relatives of those with adult-onset OCD. The genetic contribution to OCD is increased when there is a comorbid tic disorder (Rapoport et al., 2015).

Neural basis. The conceptual model of OCD links the endless repetitive thoughts and actions of OCD with uncontrolled activity of parallel, discrete loops within the brain. These loops are postulated to connect the basal ganglia, prefrontal cortex, and thalamus. Another model of OCD views the disorder as a result of the over-activation of a system designed to monitor performance completion, leading to a constant feeling that the action is 'not just right' which then creates a need to correct perceived mistakes.

Autoimmune factors. The association with Sydenham's chorea, which is thought to be an autoimmune disorder following group A beta-haemolytic streptococcal infection, has led to the description of a subtype of acute onset childhood-onset OCD with a similar aetiology. The condition is known as *paediatric autoimmune neuropsychiatric disorder associated with streptococcal infection* (PANDAS) and is at times a debated diagnosis (Wilbur et al., 2019).

Epidemiology

The overall prevalence of OCD in young people up to 18 years of age is around 1–3%, with males having an earlier age of onset and therefore higher prevalence rates in child as compared to adult populations. Rates are much lower in young children and rise exponentially with age.

Associated disorders

Severe and persistent obsessional thoughts and compulsive rituals in childhood are often accompanied by other anxiety and depressive symptoms. The patterns of comorbidity among childhood-onset OCD are similar to adult-onset with tic disorders (30%), major depression (26%), and specific developmental disabilities (17%) the most common. Simple phobias, ADHD, CD, and separation anxiety disorders also occur.

Prognosis

OCD in childhood has a *generally poor prognosis*, persisting in almost half of cases and 50% of adults with OCD reporting that their symptoms started in childhood. The poorest outcomes are predicted by earlier age of onset, comorbid psychiatric illness, and poor initial response to treatment. OCD patients with tics or Tourette disorder have a waxing and waning course and although tics often can improve after the age of 10 years, OCD symptoms can continue for several more years.

Management

The first step is to inform the child, the parents, and the teachers about the disorder and allow time for discussion of the implications. When obsessional symptoms occur as part of an anxiety or depressive disorder, treatment is directed to the primary disorder. True obsessional disorders of later childhood are treated along similar lines to the same disorder in adults (see p. 187), with *cognitive behavioural methods, medication, or a combination of the two*. The cornerstone of psychological therapies for children with OCD is exposure and response prevention (ERP). A hierarchy of increasingly intense anxiety-provoking situations that trigger obsessional thinking is constructed and subjects are exposed gradually to these situations and encouraged to refrain from engaging in

compulsive behaviours. They are taught relaxation and anxiety management techniques to help them in the process. Several randomized controlled trials (RCTs) have evaluated the efficacy of CBT for youth with OCD using manualized treatment protocols that emphasize either cognitive or behavioural components, with most treatments involving both aspects. Irrespective of these distinctions, CBT significantly reduces obsessive-compulsive symptom severity relative to waitlist, placebo, treatment-as-usual, and relaxation training (RT) comparison conditions, with maintenance up to seven years after acute treatment (McGuire et al., 2015). CBT is often also a preferred treatment option for parents of children with OCD, and is recommended as a first-line monotherapy for youth with mild-to-moderate OCD symptom severity and together with SSRIs for more severe cases.

Medication management with SSRIs are also empirically supported treatments for children and adolescents with OCD (McGuire et al., 2015). SSRIs evaluated in RCTs for child and adolescent OCD include sertraline, fluoxetine, and fluvoxamine and all significantly reduce obsessive-compulsive symptom severity relative to placebo and waitlist control conditions, with escitalopram perhaps being the most effective (Tao et al., 2022). Long-term benefit from acute SSRI treatment has been observed for up to 12 months with maintenance medication, although symptom reemergence can occur with discontinued use.

Somatoform and related disorders

Children with a psychiatric disorder often complain of *somatic symptoms* that do not have an underlying structural cause. These complaints include abdominal pain, headache, and limb pains. Most of these children are treated in primary care. The minority who are referred to specialists are more likely to be sent to paediatricians than to child psychiatrists.

Chronic fatigue syndrome (CFS)

This condition, which is described on p. 623, usually occurs after puberty. It is associated with severe mental and physical fatigue that is not alleviated by rest, coupled with significant disability. Its prevalence has been estimated at around 0.1–2% of child and adolescent populations, with two thirds estimated to be female and those with CFS having high rates of psychiatric comorbidity, predominantly anxiety and depressive disorders. The strongest evidence regarding aetiology in younger populations is that the Epstein-Barr virus is associated with

an increased risk of CFS. More recently, post-infectious fatigue syndromes following SARS-CoV-2 infection have been increasingly reported in children and adolescents. There is some evidence for certain personality traits, including excessive conscientiousness, rigidity, fearful behaviour, and sensitivity being reported more frequently in young people with CFS (Lievesley et al., 2014).

The principles of treatment resemble those for the treatment of the disorder in adults (see p. 624). CBT, similar to that used for adults, has been found to be effective in adolescents with an understanding of the maintaining power of inactivity for fatigue and related behavioural vicious circles that follow from this being most helpful (Garralda and Rask, 2015). Often it is best for treatment to be carried out jointly by a paediatrician and a psychiatrist or clinical psychologist and a good collaborative relationship with the family is essential. Although the longer-term prognosis for fatigue is probably quite good, the short-term effects on school performance can have serious consequences.

Recurrent abdominal pain

Childhood recurrent abdominal pain is one of the most common medical complaints of childhood with no structural pathology identified in the majority of cases. It affects approximately 10% of children, and is a common reason for presentation to primary care services and referral to a paediatrician. In many cases, the abdominal pain is associated with headache, limb pains, and sickness, some are related to anxiety, some to 'masked' depressive disorder, and some to stressful events. Treatment is similar to that for other emotional disorders. Follow-up suggests that 25% of cases severe enough to require investigation by a paediatrician develop adult depression, generalized anxiety disorder, or unexplained physical symptoms.

For a review of child somatoform and related disorders see Garralda and Rask (2015).

Mood disorders

Depression

Major unipolar depression is a significant global health problem, with the highest incident risk being during adolescence. A depressive illness during this period is associated with negative long-term consequences including suicide, additional psychiatric comorbidity, interpersonal relationship problems, poor educational performance, and poor employment attainment well into adult life (Cousins and Goodyer, 2015).

It is normal for healthy children to experience low mood or feel depressed in response to distressing

circumstances—for example, when a parent is seriously ill or a grandparent has died. Some, including those experiencing grief, may also lose interest, concentrate poorly, and eat and sleep badly. In adolescence, depressive mood may be less immediately obvious than anger, alienation from parents, withdrawal from social contact with peers, underachievement at school, substance abuse, and deliberate self-harm. This section covers clinical depression and not normal variations in mood or depressive symptoms that are part of another disorder, such as anxiety or CD. Although there are many similarities with adult depression, the treatment implications can be very different. For a review of depression in young people, see Thapar et al. (2022).

Clinical picture

The clinical picture of major depression in childhood and adolescence is in most ways similar to that in adults but can be easily missed. Irritability can be a common feature in adolescents. Young children might not express guilt in an adult way, and may have difficulty in describing feelings of sadness. Furthermore their sleep may not be disturbed in the ways found among adults with depressive disorder and they might complain of symptoms such as unexplained abdominal pains, headache, anorexia, and enuresis. However, while these latter symptoms can be the first to be brought to medical attention, their investigation should include the possibility of a depressive disorder. When this is done, history taking and mental state examination will reveal persistent sadness, anhedonia, irritability, and other symptoms typical of depressive disorder.

Epidemiology

Depressive symptoms are more common in adolescence than in childhood. Estimates of the prevalence of major depressive disorder give figures of less than 1% in pre-pubertal children with an equal male to female ratio. The picture changes dramatically following puberty with a one-year prevalence of around 4% with females at great risk with a 2:1 ratio of females to males, and evidence from a UK population survey of highest rates in older white female adolescents (16- to 17-year-olds) (Sadler et al., 2018). These prevalence estimates can vary according to instrument used, population studied, and duration investigated. Following the Covid-19 pandemic, there have been concerns about the increase in depressive symptoms with pooled prevalence estimating that the rates of depression had doubled in the first year of the pandemic up to with 1 in 4 youth said to be experiencing clinically elevated depression symptoms (Racine et al., 2021).

Aetiology

There are many individual, family, and social risks for child and adolescent depression that are strongly correlated and might also impact on continuing and later adversities. The causes of depressive disorder in childhood appear to be similar to those of depressive disorder in adulthood; however, in adolescence, major pubertal, brain, and cognitive maturation processes take place and are likely to have an influence on the aetiology and/or impact of depression. The increased risk of depression in women correlates with endocrinological/pubertal changes rather than chronological age (Joinson et al., 2011). The heritability of major depression in adolescents is greater than that in younger children. As in children, genes could act indirectly—for example, through interaction with life events in the family or elsewhere. Negative cognitive biases and increased cortisol secretion have also been identified as risk factors. For a description of the different models of depression and its neurobiology see Brent and Maalouf (2015) and Table 16.3.

Genetic factors. Genetic factors appear similar in nature and magnitude as in adult depression (see

Chapter 9). No individual genes have been identified. Longitudinal studies of young people show a number of distinct trajectories for the development of depression. The onset of clinical symptoms by the age of 16 years is associated with the presence of an increased polygenic risk score for adult major depression (Rice et al., 2019).

Neurobiology. This is described in detail in Brent and Maalouf (2015). Current models include the potential role of cortisol in the HPA-stress system, its link with the prefrontal cortex, and the increased sensitivity of female brains to psychosocial stressors.

Other causes. Negative *life events* often precede the onset of depressive disorder in children, as they do in adults. *Temperament* also seems to be important, especially the tendency to react intensely to environmental stimuli.

Management

General measures. Any distressing circumstances and stressors should be reduced as far as possible, while the child is helped to talk about their feelings. The possibility of depression in the parents should be considered,

Table 16.3 Aetiology of depressive disorder in childhood

Vulnerability factors for depression		Environmental risk factors	
<i>Genetic</i>	See text below	<i>Parental depression and family discord</i>	Negative impact of maternal depression possibly mediated by family discord and expressed emotion directed towards the child.
<i>Cognitive distortion and rumination</i>	Biased attention to negative emotional cues along with rumination can influence the onset of depression.	<i>Child maltreatment</i>	Potent risk factor for the onset and recurrence of depression.
<i>Emotion regulation</i>	If difficulties in emotional regulation, less able to shift attention away from negative cues in environment and possibly related to growth of children's depressive symptoms.	<i>Peer victimization</i>	Higher rates of depression in both bullies and their victims; can have long-lasting effects.
<i>Behavioural disorders and irritability</i>	Increased rates in children with ADHD as well as their parents.	<i>Sexual minority status</i>	3× higher risk of depression.
<i>Sleep</i>	Sleep deprivation leads to increased mood lability and greater attention to negative emotional cues.	<i>Bereavement</i>	3× increased risk; greatest risk immediately after the bereavement and if a past history of depressive symptomatology.
<i>Comorbid medical illness</i>	Poor physical health associated with depression.	<i>Resilience</i>	Higher IQ

Adapted from Brent, D., & Maalouf, F. (2015). Depressive disorders in childhood and adolescence. In A. Thapar, D. S. Pine, J. F. Leckman, S. Scott, M. J. Snowling, & E. Taylor (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 874–892). Copyright, with permission from John Wiley and Sons.

and treated if necessary. With a school-aged child, the management plan should involve the teachers to help them to understand the effect of depression on the child's performance, and to help to identify and reduce any stressors at school, including bullying. The nature of the disorder is explained to the parents and, in terms appropriate to their age, to the child. This explanation, and the approach to possible stressors, should take full account of culture and ethnicity. A stepped care model has been described in the 2019 NICE guidance for treatment of depression, with recommendations that the majority of children and young people be managed on an outpatient or community basis (Luxton and Kyriakopoulos, 2022). General advice on the benefits of regular exercise, sleep hygiene, anxiety management, nutrition, and the benefits of a balanced diet should be given.

Psychological treatment. If depression does not improve with the above measures, a specific psychological treatment should be considered, based on an individual assessment of the patient's profile (Luxton and Kyriakopoulos, 2022). Cognitive therapy, interpersonal therapy, and brief family therapy can be used, although most of the evidence for their efficacy is from trials with older children and adolescents.

Medication. When a child has failed to respond to psychosocial measures and psychological treatment, or if the depressive disorder is severe at the outset, antidepressant medication can be considered. For depressive disorders, SSRIs, and particularly fluoxetine, have been shown to have clinical benefit for the treatment of moderate to severe depression (Boaden et al., 2020). Fluoxetine has the strongest evidence base for child and adolescent populations (Luxton and Kyriakopoulos, 2022). Clinicians should warn young people and their carers about the possible emergence or exacerbation of suicidality on starting SSRIs, and therefore prescribing decisions in these circumstances are usually made by a specialist in child psychiatry. The potential benefits and adverse effects are examined in each case, and discussed with the carers and, in appropriate terms, with the child. The information should include warning about the delay between starting medication and improvement, and dangerous side-effects, including *worsening ideas of and behaviour around self-harm*. If an antidepressant is prescribed, it should be for a child or adolescent with a confirmed diagnosis of moderate or severe depression, with provision made for closer monitoring for any emergent adverse drug reactions, especially in the first few weeks and it should, if possible, be combined with some form of psychological treatment (Luxton and Kyriakopoulos, 2022).

In the Treatment for Adolescents with Depression (TADS) study sponsored by the US National Institute for Mental Health (NIMH), 439 depressed adolescents were randomly allocated to fluoxetine, pill placebo, CBT, or a combination of fluoxetine and CBT. After 12 weeks of treatment, response rates for pill placebo (34%) and CBT (43%) were significantly less than for fluoxetine alone (61%) and the combination treatment (71%). By 18 weeks, the response to CBT (65%) was similar to that found with fluoxetine alone (69%), while the response to combination treatment was now 85%. As noted above, fluoxetine treatment was associated with an *increased risk of suicidal ideation and self-harm* relative to placebo, but this increase was not seen in patients in whom fluoxetine was combined with CBT (March and Vitiello, 2009). The TADS study supports the view that the best treatment for moderate to severe depression in adolescence is a *combination of fluoxetine and CBT*.

Psychological treatments. A network meta-analysis has shown that *interpersonal therapy* is likely to be the most effective psychotherapy or CBT in combination with fluoxetine for depression in children and adolescents (as described above) (Zhou et al., 2020).

Prognosis

As with adult depression, child and adolescent depression tends to be *chronic and recurrent*, although 60–90% will remit with one year; 50–70% will develop subsequent depressive disorders within five years. There is an increased risk of depression in adulthood and chronic depression in childhood is also associated with *educational underachievement* and *interpersonal difficulties*.

Bipolar disorder

In the last two decades there has been a marked increase in knowledge about childhood-onset bipolar disorder, and it can be one of the most impairing childhood psychiatric illnesses with significant morbidity and mortality. Mania is generally thought to be extremely uncommon before puberty, although there have been significant recent diagnostic changes giving rise to estimates of bipolar disorder in paediatric populations of about 1%.

The diagnostic criteria for making a diagnosis are similar to those in adults and are described in Chapter 10. The DSM-5 criteria have helped clarify some of the more complex issues involved in diagnosis. These include a clearer distinction between the prominence of irritability in childhood presentation of mania compared to other disorders such as generalized anxiety disorder, ASD, and ADHD. In the new classification of childhood bipolar disorder, irritability has to be present

at the same time as some of the 'B symptoms' of mania, such as decreased need for sleep. The second key area that helps distinguish mania from the overactivity of ADHD involves the presence of these five symptoms: elation; grandiosity; hypersexuality; flight of ideas/increased goal-directed activity; and decreased need for sleep (Leibenluft & Dickstein, 2015).

The most significant departure in the DSM-5 classification was in the introduction of a category of DMDD (as described on p. 437), which is now a key differential diagnosis to consider. DMDD, as distinguished from childhood bipolar disorder, involves chronic, severe, irritability characterized by frequent temper outbursts out of proportion to the situation and child's developmental level, and a persistently irritable or angry mood. A longitudinal perspective is very important to ensure the presence of different episodes to help confirm a bipolar diagnosis compared to DMDD.

Children with bipolar disorder tend to cycle more frequently than adults and are more likely to have an impairing illness with symptoms much of the time. Their treatment requires a multimodal approach to meet the complex needs of the disorder with targeted pharmacotherapy and psychotherapy. Most of the treatment data comes from adult studies. For mania, second-generation antipsychotic medications have the strongest efficacy data in youth with bipolar disorder; however, as with in the treatment of psychosis, there are significant side-effects such as substantial weight gain and metabolic syndrome. Risperidone, aripiprazole, and quetiapine have all been licensed by the US Food and Drug Administration to treat mania or mixed states in 10–17-year-olds. There is some support for the use of lithium in childhood bipolar disorder, and weakest support for the therapeutic efficacy of antiepileptic drugs. The treatment of depression associated with bipolar disorder in childhood is poorly studied, although the data suggests that adolescents with bipolar disorder spend considerable time suffering from symptoms of depression and anxiety. There is some suggestion that the risk of manic induction on antidepressant treatment is higher in prepubertal children than in older adolescents.

For a full review of bipolar disorder in childhood see Demeter et al. (2022).

School refusal

School refusal is not a psychiatric disorder but a pattern of behaviour that can have many causes. It is convenient to consider it at this point in the chapter because of its association with *anxiety* and *depressive disorder*. School refusal represents one presentation within the broader problem of persistent school absence. Physical illness is the most common. A small number of children

are deliberately and repeatedly kept at home by parents to help in the home or for other reasons. Others are *truants* who could go to school but choose not to, often as a form of rebellion, and usually without their parents' knowledge. Finally, school refusers stay away from school but with parents' knowledge, experience emotional distress at the prospect of attending school, and their parents often try to get them to attend school (Maynard et al., 2015). Since the pandemic there has been an increase in the numbers persistently absent (10% or more of possible school sessions) from school.

Epidemiology

Temporary absences from school are extremely common, but the prevalence of school refusal is around 1–2% of the general population and up to 15% of those referred to UK mental health services.

Clinical picture

At times, the first sign to the parents that something is wrong is the child's sudden and complete refusal to attend school. More often there is an increasing reluctance to set out, with signs of unhappiness and anxiety when it is time to go. These children often complain of somatic symptoms of anxiety such as headache, abdominal pain, diarrhoea, sickness, or vague complaints of feeling ill. These complaints occur on school days but not at other times. Some children appear to want to go to school, but become increasingly distressed as they approach it. The final refusal can arise in several ways. It may follow a period of gradually increasing difficulty of the kind just described. It may appear after an enforced absence for another reason, such as a respiratory tract infection. It may follow an event at school such as a change of class. It may occur when there is a problem in the family such as the illness of a grandparent to whom the child is attached. Whatever the sequence of events, the child is extremely resistant to efforts to return them to school, and their evident distress makes it hard for the parents to insist that they go.

Almost 50% of school refusers referred to clinics have a diagnosable anxiety disorder, including separation anxiety disorder, specific phobias, social phobia, generalized anxiety disorder, and panic disorder with agoraphobia.

Aetiology

School refusal is a complex problem that is determined by a broad range of risk factors that interact with each other and change over time. The risk factors involve individual, family, school, and community factors. Individual factors include behavioural inhibition, fear of failure, low self-efficacy, sensory overstimulation (for example in children with ASD), and physical illness.

The family factors can include separation and divorce, parental mental illness, overprotective parenting style, and dysfunctional family interactions. School factors include bullying, physical education lessons, transition to secondary school; community factors can include increasing pressure to achieve academically and inadequate support services (Maynard et al., 2015). Separation anxiety is particularly important in younger children. In older children there may be a true school phobia, for example, a specific fear of certain aspects of school life, including travel to school. Some fear bullying, or failure to do well in class. Other children have no specific concerns, but feel generally inadequate and depressed. Some older children have a depressive disorder.

Prognosis

In the absence of treatment most youth with school refusal continue to display problematic school attendance and emotional distress leading to short- and long-term adverse consequences. Studies of the longer prognosis of school refusal show that of those with the more severe problems, between a third and a half have emotional problems and others will have poorer social adjustment and limited social contacts. The impact on families can also be great.

Treatment

Except in the most severe cases, arrangements should be made for an early return to school and restoration of regular attendance. There should be discussion with the schoolteachers, who should be asked about the child's problems, asked how they can help the child to catch up with missed education, and advised how to manage any difficulties that may arise when the child returns. By the time that help is sought, parents often have difficulty in pressuring the child to go to school. Psychosocial treatments for children with school refusal typically incorporate both cognitive and behavioural interventions. There are a number of manualized CBT interventions and all involve individual treatment, some level of parental involvement, consultation with school staff, and between-session tasks. Graded exposure to school attendance is commonly advocated and many manuals incorporate problem solving training with the young person and family work on communication and problem solving. For a review of psychosocial interventions, see Maynard et al. (2015).

Other childhood psychiatric disorders

Functional enuresis

Functional enuresis is the repeated involuntary voiding of urine occurring after an age at which continence is

usual (see below), in the absence of any identified physical disorder. Enuresis may be nocturnal (bed-wetting) or diurnal (daytime wetting), or both. Most children achieve daytime and night-time continence by three or four years of age. Nocturnal enuresis is often referred to as primary if there has been no preceding period of urinary continence. It is called secondary if there has been a preceding period of urinary continence.

Nocturnal enuresis can cause great unhappiness and distress, particularly if the parents blame or punish the child. This unhappiness may be made worse by limitations imposed by enuresis on activities such as staying with friends or going on school trips. Nocturnal enuresis occasionally results from physical conditions, but more often appears to be caused by delay in the maturation of the nervous system, either alone or in combination with environmental stressors. There is some evidence for a *genetic contribution*; about 70% of children with enuresis have a first-degree relative who has been enuretic. Stressful events are associated with the onset of secondary enuresis. Rigid or other particular kinds of training have not been shown to improve outcomes.

Primary care and paediatric services usually treat these disorders. Any physical disorder should be treated. If the enuresis is functional, an explanation should be given to the child and the parents that the condition is common, and the child is not to blame. The caregivers should be encouraged to reward success without drawing attention to failure, and not to focus attention on the problem. Many younger enuretic children improve spontaneously soon after being given an explanation of this kind, but those over six years of age are likely to need more active measures. The next step is usually advice about restricting fluid before bedtime, voiding at daytime intervals and/or before bedtime, and taking the child during the night to the toilet, and the use of star charts to reward success. Enuresis alarms and medication (desmopressin, either as a nasal spray or as a tablet) are further interventions, usually implemented by specialist services.

Faecal soiling

At the age of 3 years, 6% of children are still soiling themselves with faeces at least once a week; at the age of 7 years the figure is 1.5%. By the age of 11 years, the figure is only 1% once a month or more. Soiling is three times more frequent in boys than in girls (Butler, 2008). The term *encopresis* is used, but in two senses. In its wider sense it is a synonym for faecal soiling. In its narrower sense it denotes the repeated deposition of formed faeces in inappropriate places, including the underclothes. Because of this ambiguity, the term faecal soiling is used here. Children who soil their clothes for any reason may

feel ashamed, deny what has happened, and try to hide the dirty clothing.

Faecal soiling has several causes that include constipation with overflow; fear of using the toilet; failure to learn bowel control; stress-induced regression; and rebellion. Treatment depends on the cause and is often conducted in paediatric settings. Constipation is generally managed with a combination of laxatives and behavioural interventions aimed at re-establishing normal bowel habits, including short periods (less than 5 minutes) of toilet sitting after meals, self-initiated toileting, and self-management of cleanliness. A child who is fearful of the toilet should be reassured sympathetically. Inadequate toilet training may be improved using behavioural techniques, including achievable targets, and star charts or other rewards, together with help for the parents. Stress-induced regression usually disappears when the child has been helped to overcome the trauma. Soiling as rebellion is more difficult to treat, as it is generally part of wider social and psychological difficulties, which may require intensive and prolonged help.

Whatever the cause, it is unusual for encopresis to persist beyond the middle teenage years, although associated problems may continue. When treated, most cases improve within a year.

Selective mutism

In this condition, sometimes called *elective mutism*, a child consistently refuses to speak in certain social situations, although they do so normally in others. Usually speech is normal in the home but lacking at school. There is evidence that although these children are able to speak in some situations, they do have lower scores on standardized measures of language than their peers. These children often have a comorbid anxiety disorder, usually social phobia, and so in DSM-5 this condition has now been placed with the anxiety disorders. The condition usually begins between three and five years of age, after *normal speech has been acquired*. Although reluctance to speak is not uncommon among children starting school, clinically significant elective mutism is rare, probably occurring in about 1 per 1,000 children.

Assessment is difficult because the child often refuses to speak to the clinician, so that diagnosis depends to a large extent on the parents' and other informant accounts. When questioning the parents, it is important to ask whether speech and comprehension are normal at home. Treatment approaches aim to lower the anxiety that a child has for speaking in certain situations and increase the contexts in which the child may speak comfortably. In general behavioural treatments, CBT and/or play therapy will be the first choice of intervention.

Selective mutism is persistent, with a remission rate of only 58% at 13 years after first referral, and high rates of phobia and social anxiety as well as increased risk for depression and substance misuse in unresolved cases (Norbury and Paul, 2015).

Stammering

Stammering (or *child onset fluency disorder* in DSM-5) is a disturbance of the rhythm and fluency of speech. It may take the form of repetitions of syllables or words, or of blocks in the production of speech. This may be accompanied by excessive tension, struggle behaviour, and secondary mannerisms such as eye blinking or arm movements. Stammering is four times more frequent in boys than in girls. It is usually a brief problem in the early stages of language development, although prognosis after the age of 7 years is more guarded. However, 0.3–1% of children suffer from stammering after they have entered school.

The cause of stammering is not known, although many theories exist. It seems unlikely that all cases have the same causes. Over 60% have a concurrent speech and language disorder, of which the most common is speech sound disorder. Genetic factors, brain damage, and anxiety may all play a part in certain cases, but do not seem to be general causes. Stammering is not usually associated with a psychiatric disorder, even though it can cause embarrassment and distress. Most children improve whether treated or not. Many kinds of psychiatric treatment have been tried, including psychotherapy and behaviour therapy, but none has been shown to be effective. The usual treatment is speech therapy, including four classes of intervention: fluency shaping; stuttering modification; use of electronic devices; and parent-implemented techniques. For a summary see Norbury and Paul (2015).

Tic disorders

Tic disorders, including *Tourette syndrome*, are considered on pp. 366.

Psychosis and early-onset schizophrenia

Psychotic-like symptoms are relatively common in children, but are rarely clinically significant. Prevalence is higher in younger (aged 9–12 years) compared to older (aged 13–18 years) children, although rare before the age of 9 years. For example, in a systematic review of population-based studies, the prevalence of psychotic symptoms was 17% among children aged 9–12 years and 7.5% among adolescents aged 13–18 years (Kelleher et al., 2012). In a UK population cohort, up to 14% of 12-year-olds had 'non-clinical psychotic symptoms',

with higher levels in those from low socioeconomic status families and with low IQ. Their experience of the psychotic symptoms seemed to be similar to those of adult psychosis but only 3% met strict phenomenological criteria for adult psychotic disorders. Polanczyk et al. (2010) confirmed that a significant minority of 12-year-olds in the community self-report hallucinations (most commonly) and delusions. In addition, these symptoms were associated with many of the same risk factors and correlates as adult schizophrenia, including genetic, social, neurodevelopmental, home-rearing, and behavioural risks. Therefore, psychotic symptoms in childhood can be a marker of an impaired developmental process. Psychotic symptoms generally occur in the context of other childhood psychiatric problems, indicating that it is worthwhile to ask all preadolescent psychiatric patients about hallucinations and delusions. Even if the psychotic symptoms are not themselves impairing, they are associated with important risk factors, such as chaotic household, maternal negativity, and physical maltreatment, and with behavioural problems, such as early tobacco use and self-harm, that should be a focus of attention.

For those at the most severe end of the spectrum, early-onset schizophrenia, starting in childhood or adolescence, is accepted to be clinically and biologically continuous with the adult disorder. The most prominent differences are that the early-onset form of schizophrenia is often more severe, has a greater association with early neurodevelopmental difficulties, and has poorer treatment response. There may also be a higher proportion of cases attributable to copy number variants (see Chapter 11). The whole range of symptoms that characterizes schizophrenia in adulthood may occur, and in both DSM-5 and ICD-11 the criteria for diagnosis in children are the same as those used in adults; there is no separate category of childhood schizophrenia. It is highlighted in DSM-5 that if other communication disorders of childhood are present where disorganized speech and negative symptoms are present that a diagnosis of schizophrenia should be made only if prominent hallucinations or delusions are also present for at least one month. Schizophrenia in adolescence is more common in boys than in girls. Before symptoms of schizophrenia appear, many of these children experience non-specific behavioural changes, social withdrawal, and declining school performance. The prognosis may be good for a single acute episode with florid symptoms, but for the majority, early-onset schizophrenia runs a chronic course with especially poor prognosis when the onset is insidious (Fraguas et al., 2014).

Effective treatments are less well established for early-onset schizophrenia than for adult populations and include antipsychotic medications combined with psychoeducational, psychotherapeutic, and educational interventions. For children, adolescents, and young adults, the balance of risk and benefit of antipsychotics appears less favourable than in adults (Stafford et al., 2015). Studies suggest that many youth with early-onset schizophrenia do not respond adequately to available *antipsychotic drugs* and are vulnerable to adverse events, particularly metabolic side-effects (McClellan and Stock, 2013). In recent years there has been a growing emphasis on early detection and intervention to intervene in those with 'at-risk mental states' but especially to minimize the duration of untreated psychosis in child and adolescent populations, as these populations often have long periods of untreated psychotic symptoms.

Gender dysphoria

These disorders are discussed in detail in Chapter 13 and represent severe or persistent distress association with an incongruence between one's gender identity and biological sex. This field, and the classification of these disorders, has changed dramatically over the last few years. Gender dysphoria is a diagnosis in DSM-5 and categorized under disorders related to sexuality and gender identity in the ICD-11 described in Chapter 13. Children with gender dysphoria often experience a range of psychiatric comorbidities, with a high prevalence of mood and anxiety disorders, trauma, eating disorders, autism spectrum conditions, suicidality, and self-harm. There is significant variation across the world in how children with gender dysphoria are assessed and treated (Bloom et al., 2021).

Suicidal behaviour and self-harm

Self-harm and suicide are major public health problems in adolescents with rates of self-harm being high in the teenage years and suicide being the second most common cause of death in young people worldwide. Important contributors to self-harm and suicide include genetic vulnerability as well as psychiatric, psychological, familial, social, and cultural factors. The effects of media and contagion are also important. Prevention of self-harm and suicide needs both universal measures aimed at young people in general and targeted initiatives focused on high-risk groups. There is little evidence of effectiveness of either psychosocial or pharmacological treatment, with particular controversy surrounding the usefulness of antidepressants. Restriction of access to means for suicide is important.

These issues are discussed in Chapter 21.

Psychiatric aspects of physical illness in childhood

The associations between physical and psychiatric disorders in children resemble those in adults (see Chapter 22). There are four main groups to consider (Fazel et al., 2021), and these include children with:

1. Life-limiting illnesses and conditions with high disease burden, such as cancer, where there is an increased risk of a comorbid psychiatric disorder.
2. Disorders involving the brain, including epilepsy and traumatic brain injuries, which are associated with high psychopathology (e.g. the risk of depression for a child with epilepsy is 2–5 times higher).
3. Common chronic physical disorders (e.g. asthma and type 1 diabetes) where studies have shown a correlation between psychiatric disorders and lower adherence to treatment with resultant poorer health outcomes.
4. Disorders that commonly present to paediatric services but where psychiatric input is often required, such as chronic pain, somatic symptom disorder, and severe eating disorders.

The impact of childhood physical illness

Children with somatic illnesses serious enough to warrant hospitalization are at increased risk for emotional disorders, with estimates of psychological distress in children and adolescents varying but higher rates of anxiety and mood disorders than would be expected, with estimates of those with chronic illnesses range from 10–30% (Fazel et al., 2021). Those that impact on the central nervous system, such as epilepsy and cerebral palsy, pose additional risks; other factors associated with poor psychological adjustment to chronic medical conditions include physical disability and brain dysfunction; pain frequency; younger age; poverty; single-parent family; and increased psychological symptoms in the parents. There is evidence that a child's psychiatric illness may affect the physical disease process not only by influencing adherence and lifestyle, but also by producing psychophysiological changes. For instance, children with diabetes mellitus are at greater risk for depressive disorders, and those who have comorbid depression are at increased risk for treatment non-adherence and repeated hospitalization as well as disease-related complications.

Non-adherence with medical treatments is common across conditions, ranging from estimates of 50–80%; and the factors contributing to poor adherence are many but include the medication regimen, family systems,

denial, psychiatric illness, and relationships with medical providers. *Asthma* is the most common chronic illness of childhood, and it is important when treating a child with asthma to understand the factors that affect adherence, to delineate common psychosocial stresses that may act as triggers for asthma attacks, and to teach stress management skills to improve adherence and decrease morbidity.

At the other extreme, childhood *cancer* is rare but also presents numerous challenges to adaptation and development, and although many childhood cancer survivors adapt well following treatment, leukaemia survivors can have a 1.5 times increased risk of later depression when compared to their siblings. Psychological support is particularly beneficial at the more stressful times, which are often at the time of diagnosis, the onset of treatment, and treatment completion, as well as for some dealing with any late effects of cancer and its treatment—such as any effects on sexual maturation that might have been affected by chemotherapy or relapse. Challenges to psychosocial adaptation often include persistent pain; persistent debilitating side effects of treatment such as nausea; alteration in body image; school disruptions; compromised peer relationships; concerns about relapse; and family stresses and conflicts.

There is evidence that CBT has positive effects in the treatment of symptoms of depression and anxiety in children who have chronic physical illness (Bennett et al., 2015). There are growing psychological medicine or psychiatry-liaison services working in general hospital settings, and mental health services integrating with the general or specialist medical teams can provide enhanced models of care for children with medical illnesses and work closely as part of the team contributing to the management of often complicated needs (Fazel et al., 2021).

Effect on parents

The effects on parents can be significant, especially when the child's physical illness is *chronic or disabling*. Their response depends on factors such as the nature of the disorder, the temperament of the child, the parents' emotional resources, and the circumstances of the family. The parents may experience a sequence of emotional reactions like those of bereavement, and their marital and social lives may be affected. The support of a multidisciplinary team and, if needed, respite options for care are important for families. It is important also to ensure that the siblings of children with medical illnesses have their questions and worries answered and be aware that there may be times when their family are having to prioritize the needs of their unwell sibling. Although some

studies have shown more emotional and behavioural disturbances in siblings than would be expected by chance, many siblings manage well, and might develop enhanced abilities to cope with stress and to show compassion for others.

Children in hospital

It is important to prepare children for admission by explaining in appropriate terms what will happen, and by introducing to them the members of staff who will care for them and if possible, to visit the hospital or treatment area in advance. Once admitted, many hospitals appreciate the importance of ensuring a family member can stay with their child, and depending on each particular situation, let them stay overnight in the same room with them or sleep close to the hospital. The wider family and friends should be able to visit frequently and, where appropriate, primary carers can take part in the care of the child. Many hospitals have educational facilities attached or associated with them as children who have prolonged hospital stay, and are well enough, might benefit from educational input, not only to distract them while in hospital but also to make the adjustment back into school easier as they will not have to catch up on quite as much missed work. For issues to consider in the very unwell or when a child dies in hospital, see Pinsky et al. (2015).

Special considerations for adolescent populations

Adolescence and young adulthood are the years in which an individual establishes the social, cultural, emotional, educational, and economic resources to maintain their health and wellbeing across the life course (Patton et al., 2016). It is a dynamic period of brain development with significant changes taking place within neural systems. Early adolescence (10–14 years of age) is biologically dominated by puberty and the effects of the rapid rise in pubertal hormones on body morphology and sexual and brain development. Adolescence is a time of remodelling of the brain's reward system. Psychologically it is characterized by low resistance to peer influences, low risk perception often leading to increases in risk-taking behaviour, and poor self-regulation. School and family environments are critical social contexts during this period. In late adolescence (15–19 years of age) pubertal maturation continues and the brain continues to be extremely developmentally active, particularly in the development of the prefrontal cortex and the increased connectivity between brain networks. During this phase

many adolescents develop greater executive functioning and also increased autonomy. When making decisions, adolescents seek out and are more influenced by exciting, arousing, and stressful situations compared with adults (Blakemore, 2019).

In many low- and middle-income settings adolescent males and females might start to work, get married, and have children, while in high-resource settings, education often continues into mid or late adolescence, and responsibilities come later. Adolescents are expected both to conform to the rules of society and to become more independent and develop restraints on their own behaviour. Many of the problems of adolescence relate to conflicts between these two expectations, or to the rejection of the rules of society. Although such problems are sometimes conspicuous, they are not inevitable.

Epidemiology of psychiatric disorder in adolescence

Although psychiatric disorders are only slightly more common in adolescence than in the middle years of childhood, the pattern of disorder is closer to that of adults. In the previously mentioned MHCYP UK surveys (Sadler et al., 2018), up to 24% of adolescent girls and 10% of boys had a psychiatric disorder. Emotional disorders were the most frequent, with some evidence of increase during the first Covid lockdown (Vizard et al., 2020).

Clinical features of psychiatric disorders of adolescence

There are no specific disorders of adolescence. Nevertheless, special experience and skill can help to apply the general principles of psychiatric diagnosis and treatment to patients at this time of transition between childhood and adult life. It can be difficult to distinguish psychiatric disorder from the normal emotional reactions of the teenage years. The clinical features of adolescent depression, anxiety, psychosis, and other disorders are discussed under their relevant sections in this chapter. A few specific disorders not included elsewhere and important to mention are discussed next.

Eating disorders

Problems with *eating* and *weight* are common in adolescence. They are discussed in Chapter 13 and closely resemble the same conditions in adulthood. It is particularly important to involve the parents and perhaps other family members in the treatment of an adolescent patient with eating disorder. Systemic family therapy has been shown to be of value for anorexia nervosa in adolescents. For a review, see Eddy et al. (2021).

Alcohol and substance use disorders

Substance use disorders are among the most prevalent psychiatric disorders in those 9–21 years of age and in children and adolescents are similar to those in adulthood (Chapter 20). The number of people with substance use disorders rises rapidly in adolescence, peaks around the age of 20 years, and has a declining incidence after 25 years of age. However, adolescent substance use disorders often persist into adulthood and those persisting further become leading causes of adult deaths. There is considerable country-to-country variability in substance use, with vaping, alcohol use, cannabis, and nitrous oxide relatively prevalent in the United Kingdom in 2022. Substance use patterns also shift over time as well, as new, abusable substances and prescription medications appear and become more readily available in-person and from online sources.

Excessive drinking can be common among some demographics of adolescents, especially among those with CD. Most adolescent heavy drinkers seem to reduce their drinking as they grow older, but a few progress to more serious drinking problems in adulthood. Prevention programmes have been developed, but there is little evidence for their effectiveness.

Occasional *drug taking* can be common in adolescence, and is often a group activity. Vaping, cigarette smoking, and the use of cannabis are the most frequent. Solvent abuse is largely confined to adolescence, and is usually of short duration. Abuse of drugs such as ecstasy, amphetamines, barbiturates, opiates, and cocaine is less common but more serious, as most drug-dependent adults will have experimented with these drugs during adolescence. There is a strong association between CD in childhood and drug taking in adolescence. A US adolescent cohort reported that one in five adolescents had used cannabis frequently between the ages of 13 and 17 years (higher among males) (Shanahan et al., 2021).

Many adolescents experiment with drugs for short periods and do not become regular users. Those who persist in taking drugs are more likely to come from discordant families or broken homes, to have failed at school, and to be members of a group of persistent drug users. Feelings of alienation and low self-esteem may also be important.

Assessment of adolescents

When interviewing adolescents, special considerations can help to decide which approach might be best for that particular person and whether an approach similar

to that for children or adults would be most appropriate at that time. Always remember that a proportion of adolescents attending a mental health service do so somewhat unwillingly, and also that many will have difficulty in expressing their feelings in adult terms. Spending time establishing a relationship with adolescent patients can help, and listening and respecting their perspective is important.

It might help to build rapport if the adolescent is seen before interviewing caregivers as this can help the adolescent feel autonomous. Later, other members of the family may be interviewed, and sometimes the family may be seen as a whole. As well as the usual psychiatric history, particular attention should be paid to information about the adolescent's functioning at home, in school, or at work, and about their relationships with peers.

Services for adolescents

Although small, the proportion of adolescents in the population who are seen in psychiatric clinics is growing; note that some might prefer to be seen in child services, and others in adult services. As many might struggle in this transition, there are some examples of adolescent services developing, catering to the 0–25 or 12–25 year age group (Fusar-Poli, 2019).

There are variations in the organization of services for adolescents and the treatments that they provide, but most combine individual and family psychological treatment with the possibility of drug treatment for severe disorders. Increasingly, services are accepting outpatient referrals not only from doctors but also from school staff, social workers, the courts, and even families and young people directly (Rocks et al., 2020). Inpatient facilities are usually limited in extent, so it is important to agree with social services as to what kinds of problems need admission to a health service unit, and which should be cared for in residential facilities provided by social services (in the United Kingdom). Reasons for admission to a health service inpatient unit include the following:

- severe mental illness requiring that the person's mental state be observed carefully, investigations carried out, or treatment monitored closely including acute psychosis, severe eating disorders; or
- behaviour that is dangerous to the self or to others, and that is due to psychiatric disorder.

When dangerous behaviour relates to personality and circumstances and not to illness, secure residential accommodation is likely to be more appropriate.

Child maltreatment

The WHO definition states that child maltreatment includes ‘all forms of physical and/or emotional ill treatment, sexual abuse, neglect or negligent treatment or commercial or other exploitation, resulting in actual or potential harm to the child’s health, survival, development or dignity in the context of a relationship of responsibility, trust or power.’

In recent decades the concept of child maltreatment and abuse has therefore been widened to include the overlapping categories of *physical abuse (non-accidental injury)*, *emotional abuse*, *sexual abuse*, and *neglect*. (Table 16.4).

Most of the literature on child abuse refers to high-income countries; however, abuse of children is prevalent throughout the world, although it can take different forms such that in low- and middle-income

Table 16.4 How to recognize child maltreatment

Emotional abuse	● Behavioural signs: Extreme compliance, withdrawal, excessive aggression, persistent fearfulness, difficulty trusting adults. ● Self-soothing behaviours: Body-rocking, head-banging, repetitive hand movements. ● Developmental delays not explained by other causes. ● Low self-esteem. ● Dissociation: Emotional numbness, lack of response to distressing situations. ● Self-harm and suicidal ideation: Cutting, burning, hair-pulling, suicidal thoughts or attempts. ● Inconsistent or inappropriate emotional responses: extreme emotional outbursts over minor events.
Physical abuse	● Injury patterns: Unexplained bruises, abrasions, cuts, eye injuries, subdural haemorrhages. ● Bruises in atypical areas: Ears, neck, torso, buttocks, upper arms, back of legs. ● Patterned injuries: Loop marks (cords), belt marks, bite marks, grip bruises, cigarette burns, immersion burns (stocking-glove pattern). ● Fractures: Spiral fractures, rib fractures (especially in infants), multiple fractures at different stages of healing. ● Head trauma signs: Retinal haemorrhages, petechiae, skull fractures, persistent vomiting, loss of consciousness. ● Oral injuries: Torn frenulum, bruising inside the mouth, unexplained dental trauma. ● Child’s explanation does not match injuries or changes in stories.
Sexual abuse	● Anogenital injuries: Tears, bruises, fissures, unexplained erythema. ● Unexplained rectal or genital symptoms: Dysuria, frequent UTIs, rectal bleeding, pain. ● Sexually transmitted infections in a prepubertal child. ● Pregnancy in a minor without a clear history of consensual age-appropriate sexual activity. ● Sexualized behaviour: Inappropriate sexual knowledge or talk, touching others inappropriately. ● Regressive behaviours: Bedwetting, soiling, thumb-sucking (re-emerging in older children).
Neglect	● Physical signs: Failure to thrive (low weight, malnutrition, developmental delays), chronic poor hygiene, severe dental decay, persistent infestations (lice, scabies). ● Medical neglect: Uncontrolled asthma, untreated infections. ● Lack of supervision with frequent accidents or injuries due to lack of adult care. ● Inappropriate clothing for weather or consistently dirty clothing. ● Failure to provide basic needs: Food, shelter, emotional warmth, access to education.
Other forms of maltreatment, including child exploitation	● Online exploitation: Grooming, coercion into participating in sexual or violent content. ● Labour exploitation: Forced domestic work, unpaid labour under abusive conditions. ● Domestic violence exposure. ● Unnecessary medical procedures, failure to improve despite interventions.

countries other causes and risk factors are involved. For example, there might be greater numbers of children in low-income countries exposed to physical violence and corporal punishment; growing up without a primary caregiver for reasons ranging from parental illness to migrant labour where parents have to leave home in order to find employment; and child labour and trafficking.

A UK community survey found that nearly 1 in 5 children experience some forms of maltreatment and that children who have experienced one form of maltreatment are two to three times more likely to experience other forms of maltreatment and to be victimized by other perpetrators over time (Danese and McCrory, 2015). The term *safeguarding* is increasingly used to denote the consideration of and care needed for children either at risk of or who have a past or current history of child maltreatment and how to ensure they come to least harm and are protected. During the period of national lockdown which occurred in the Covid-19 pandemic, families were living in virtual confinement and subject to massive economic disarray. This created the conditions likely to produce a rise in children's experience of abuse and neglect, which was observed in a number of studies (Rodriguez et al., 2021).

The term *foetal abuse* is sometimes applied to behaviours that are detrimental to the foetus, including physical assault and the taking by the mother of substances that are likely to cause foetal damage. *Fabricated and induced illness* (also referred to as *Factitious Disorder Imposed on Another (FDIA)* or *Munchausen syndrome by proxy*) is the name given to apparent illness in children which has been fabricated by the parents, and to conditions induced by parents—for example, by giving medication or other substances when not indicated (see p. 625).

Aetiology

There are many interacting causes for child maltreatment (see Box 16.10). Although several risk factors show a statistical association with child maltreatment, no single risk factor is either necessary or sufficient for maltreatment to occur. The most useful way to frame the many risks is to consider them in a multilevel model with individual, family, and community factors considered alongside appreciating the bidirectional nature of the child–adult interaction and how this might influence maltreatment.

The environment. Child maltreatment is more frequent in neighbourhoods in which family violence is common; schools, housing, and employment are unsatisfactory, and there is little feeling of community.

Box 16.10 Risk factors for child maltreatment

The parent(s)

- Young age
- Limited educational attainment or cognitive difficulties
- Social isolation/no one to provide help
- Breakdown of relationship with partner
- Poor parenting skills: lack of awareness of child's needs, harsh punishment, little reward; unrealistic expectations of child
- Experience of maltreatment as a child
- Criminal record
- Difficulties with emotional regulation
- Psychiatric problems: depression, substance abuse, personality disorder

The child

- Factors leading to weak attachment to the parents:
- Premature, health problems including disability
- Separation from mother during early life (e.g. in neonatal unit)
- Difficult temperament, frequent or prolonged crying

The environment

- Neighbourhood adversity: high levels of family and community violence, schools with high levels of absence and disruption, high unemployment
- Large family size or single parent
- Non-biological or transient carers in the home
- Poverty
- Little feeling of community cohesion
- Isolation including periods of lockdown

The parents. Factors associated with child maltreatment include young age of parents, impaired capacity for empathy, low education, psychiatric disorder and/or substance misuse, social isolation, family dysfunction, and a criminal record. If a parent has a psychiatric disorder, it is most often a personality disorder. Many parents give a history of having themselves suffered abuse or deprivation in childhood, and the parental relationship is marked by conflict and sometimes violence. Although physical abuse is more common in families with other forms of social pathology, it is certainly not limited to such families.

The children. Risk factors include younger age, especially under the age of 5 years, and other factors associated with increased child vulnerability, including prematurity, physical and intellectual disabilities, chronic illness, and those with either a more difficult temperament or a more dependent relationship with carers.

Physical abuse (non-accidental injury)

Estimates of the prevalence of physical abuse of children vary according to the criteria used. Most studies report prevalence rates of physical abuse in childhood in the range 5–15%, depending on how this is defined with lower rates of significant violence.

Clinical features

Parents may bring an abused child to the doctor with an injury said to have been caused accidentally. Alternatively, relatives, neighbours, school staff, or other people may become concerned and report the problem to police, social workers, or voluntary agencies. The most common forms of injury are multiple bruising, burns, abrasions, bites, torn upper lip, bone fractures, and subdural or retinal haemorrhages. Some infants are smothered, often with a pillow, and the parents might report an apnoeic attack. Suspicion of physical abuse should be aroused by the pattern of the injuries, a previous history of suspicious injury, a vague or inconsistent account of the way in which injuries came about, an account that is inconsistent with the nature and extent of injury, delay in seeking help, and incongruous parental reactions. The psychological characteristics of abused children vary, but include fearful responses to the parents, other evidence of anxiety or unhappiness, and social withdrawal. Such children often have low self-esteem, may avoid adults and children who make friendly approaches, and may be aggressive.

Assessment and management

Health care professionals, especially those working in primary care and emergency departments, should always be alert to the possibility of physical abuse and non-accidental injury. They need to be particularly aware of the risks to children who have some of the characteristics described above, or who are being cared for by parents with the predisposing factors listed. It is important for professionals to receive regular child protection training in order to be aware of the different ways these children might present, especially as the majority are thought to be missed by health services, and also of the heightened needs of these children.

Concerns about possible abuse can be raised by anyone working with the child. If in hospital this is usually

done through discussions with the safeguarding team and if in the community directly with social care, unless there are acute health concerns where primary care often are involved. Safeguarding assessments are often conducted by paediatricians who will take a careful history, documenting the information in as much detail as possible, and also perform a thorough physical examination. For further information about the physical examination, see a textbook of paediatrics. In the hospital emergency department, inpatient admission should be arranged for all children in whom non-accidental injury is suspected.

If possible, social services should become involved early and the doctor's concerns should be discussed with the parents, and in any case, they should be told that admission is necessary to allow further investigations. If the parents refuse admission, it may be necessary in England and Wales to apply to a magistrate for a Place of Safety Order; similar action may be appropriate in other countries. During admission, assessment must be thorough and can include formal photographs of injuries (of a quality that could be used in court if needed, so this can often be through the medical illustration departments of the hospital and with designated equipment) and full skeletal survey radiography. Radiological examination may show evidence of previous injury or, rarely, of abnormalities such as osteogenesis imperfecta. A computed tomography (CT) scan may be needed if subdural haemorrhage is suspected.

Once it has been decided that non-accidental injury is probable, the first priority is to establish the child's safety and that of any other children in the household. The procedures involved will vary according to the administrative arrangements in different countries. In the United Kingdom, social services is responsible for child protection and should be notified; police departments also are often involved. It may be decided to put the child's name on a child abuse register, thereby making the social services department responsible for visiting the home and checking the problem regularly.

In some cases, if the risk of returning the child to their family home is too great, separation might be required. If the parents do not agree to separation, a *care order* can be sought by the social services department. If the abuse is severe, prolonged, or permanent, separation may be necessary, and the parents may face criminal charges. Because there are known cases of injury or death in children who have been returned to their parents, it is vitally important that the most careful assessment is undertaken before physically abused children are returned to their family. Countries vary in the requirements and procedures for reporting and monitoring possible physical abuse in children, and readers

should familiarize themselves with the arrangements that exist in the area in which they work. Often in the first instance, children will be placed with other family members, such as grandparents.

Prognosis

Children who have been subjected to physical abuse are at high risk of further problems. For example, the *risk of further severe injury is probably around 10–30%*, and occasionally the injuries are fatal. Therefore, a priority in the intervention is to try to prevent any recurrence of maltreatment. Abused children are likely to have subsequent high rates of physical disorder, delayed development, and learning difficulties. There are also increased rates of behavioural and emotional problems in later childhood and adulthood, even when there has been earlier therapeutic intervention. As adults, many former victims of abuse have difficulties in rearing their own children. The outcome is better for abused children who can establish a good relationship with an adult (often from outside the family), and who have dispositions that include sociability and academic competence.

Emotional abuse

The term emotional abuse usually refers to *persistent neglect or rejection sufficient to impair a child's development*. However, the term is sometimes applied to gross degrees of overprotection, verbal abuse, or scapegoating, which impair development. Emotional abuse often accompanies other forms of child abuse.

Emotional abuse has various effects on the child, including failure to thrive physically, impaired psychological development, and emotional and CDs. Diagnosis depends on observations of the parents' behaviour towards the child, which may include frequent belittling or sarcastic remarks about them during the interview. One or both parents may have a disorder of personality, or occasionally a psychiatric disorder. The parents should be interviewed separately and together to discover any reasons for the abuse of this particular child—for example, he may fail to live up to their expectations, or may remind them of another person who has been abusive to one of them. Each parent's mental state should be assessed.

Management

In treatment, the parents should be offered help with their own emotional problems and with their day-to-day interactions with the child. It is often difficult to persuade parents to accept such help. If they reject help and if the effects of emotional abuse are serious, it may

be necessary to involve social services and to consider the steps described above for the care of children suffering physical abuse. The child is likely to need individual help.

Child neglect

Child neglect is the failure to provide necessary care. It can take several forms, including emotional deprivation, neglect of education, physical neglect, lack of appropriate concern for physical safety, and denial of necessary medical or surgical treatment. These forms of neglect may lead to physical or psychological harm, including poor academic performance and disturbed behaviour.

Child neglect is more common than physical abuse, and it may be detected by various people, including relatives, neighbours, teachers, doctors, or social workers. Child neglect is associated with adverse social circumstances, and can lead to the need for a child to be taken into foster care or other living arrangements.

Growth faltering

Paediatricians recognize that some children show inadequate growth when no clear medical cause is identified. In such cases, insufficient nutritional intake and emotional deprivation often play a major role. These children typically fall below the 5th centile for weight and height, and their rate of weight gain is less than expected for age. Different terms have been used to describe this condition, from non-organic failure to thrive (NOFTT) and deprivation dwarfism to psychosocial short stature syndrome (PSSS) in older children; the terms growth faltering and psychosocial growth failure are now more widely adopted.

Clinical picture

Growth faltering is caused by the deprivation of food and close affection. In the children who are seen in the psychiatric service, there is usually evidence of problems in the parent–child relationship since the child's early infancy; these include rejection and, in extreme cases, expressed hostility towards the child. There may be physical or sexual abuse as well. The infant who is failing to thrive may present with recent weight loss, persistently low weight, or reduced height. There may be cognitive and developmental delay. The infant may be irritable and unhappy, or, in more severe cases, lethargic and resigned. There is a clinical spectrum ranging from infants with mild feeding problems to those with all of the severe features described above. If food and care are provided, the infant usually grows and develops quickly.

These children might also have abnormally short stature, unusual eating patterns, retarded speech

development, and temper tantrums. Although short in stature, the child may be of normal weight. Emotional and behavioural disorders occur, and there may be cognitive and developmental delay with impairment of language skills. These children have low self-esteem and are commonly depressed. There is usually a history of deprivation or of psychological maltreatment. Away from the deprived environment, these children often eat ravenously.

Management

When treating either syndrome, the first essential is to ensure the child's safety, which may require admission to hospital. Subsequently, some children can be managed at home, but others need to be moved to alternative living arrangements including foster care. Some parents can be helped to understand their child's needs and to plan for them; other parents are too hostile to be helped. If help is feasible, it should be intensive and should probably focus on changing patterns of parenting. It is unusual for the parents to be psychiatrically ill. Home-based intervention programmes may improve treatment effects, particularly in cognitive and behavioural domains.

Prognosis

With both syndromes, the prognosis for early cases is relatively good (Rudolf and Logan, 2005). However, some children have to be placed permanently in foster care because family patterns are resistant to change. In a positive emotional setting the abnormal behaviour is usually lost quickly, and mental development follows physical growth.

Sexual abuse

The term *sexual abuse* refers to the involvement of children in sexual activities that they do not fully comprehend and to which they cannot give informed consent, and which violate generally accepted cultural rules. The term covers various forms of sexual contact with or without varying degrees of violence. The term also covers some activities that do not involve physical contact, such as posing for pornographic photographs or films. The abuser is commonly known to the child and is often a member of the family (*incest*). A minority of children are abused by groups of paedophiles (sex rings). There is increased exposure to risk through social media.

Epidemiology

Child sexual abuse is highly prevalent around the globe and differentiating between types of abuse is important to obtain meaningful estimates of prevalence. A review

collating results from other studies estimated prevalence rates of sexual abuse and separated abuse into four different types. For non-contact abuse (inappropriate sexual solicitation, indecent exposure) the rates were estimated at 17% for males and 31% for females; for contact abuse (fondling, touching, kissing) the rates were 6% for males and 13% for females, for forced intercourse (oral, vaginal, anal, and attempted) the rates were 3% for males and 9% for females, and mixed sexual abuse (more than one type of abuse occurring) for 8% males and 15% of females (Barth et al., 2013). Females therefore have a two- or threefold risk compared to males to be sexually abused in childhood and about one in ten women experience this. Children with disabilities are more likely to be victims of sexual abuse.

The majority of children know their abuser, usually because they are in their family, or occasionally because they have been befriended as part of a grooming process. The majority of abusers are male (85–95%) and stepfathers are over-represented. The extent of sexual abuse by women is not known, a few carry out abuse in conjunction with a man, and those that abuse alone are more likely to abuse boys. Sexual abuse by children and adolescents, mostly boys, has become widely recognized and is no longer considered an acceptable variant of childhood or adolescent sexual development. Sixty per cent of contact sexual abuse reported by children in the United Kingdom was perpetrated by other children and young people under the age of 18 (Glaser, 2015). Many adult abusers report the onset of their abusive activities in adolescence.

Clinical features

The presentation of child sexual abuse depends on the type of sexual act and the relationship of the offender to the child. Children are more likely to report abuse when the offender is a stranger. It might come to light when a child talks about it to a friend, relative, or teacher or it is suspected on the basis of less specific indicators described below. Spontaneous and intentional disclosures are likely to be credible. Specific indicators include age-inappropriate sexualized behaviour, rarer genital physical signs, sexually transmitted diseases, and pregnancy in a young girl especially when the identity of the father is not clear. Non-specific indicators include sudden onset of unexplained difficulties in a previously untroubled child such as distractibility, educational deterioration, social isolation, aggressiveness, low self-esteem, marked unhappiness, disturbed sleep and nightmares, fearfulness, and separation anxiety. Depression, running away, deliberate self-harm, and drug and alcohol misuse may arise later (Glaser, 2015). There may also

be physical symptoms in the urogenital or anal area, pregnancy, and behavioural or emotional disturbance. When abuse occurs within the family, marital and other family problems are common.

Effects of sexual abuse

Childhood sexual abuse is a significant risk factor for mental health disorders in childhood, adolescence, and adulthood as well as having effects on physical health.

Effects in childhood and adolescence. Not all children will develop difficulties, and some will only develop them later, but a wide range of difficulties which may not appear immediately has been found with considerable effect sizes: depression, anxiety, complex PTSD, self-harm, low self-esteem, CD, bulimia, promiscuity, and sexual victimization of other children. It may lead to unwanted pregnancy. The worst prognosis is for children who are younger than seven years of age when the abuse takes place and also those who have negative self-attributions about the abuse, including self-blame and shame, children who have experienced penetration in the abuse, and those who have depressed parents.

Effects in adulthood. These are significant and have been estimated to contribute around 13% of all adult psychopathology. For example, in an Australian study, 23% of adult survivors of childhood sexual abuse required long-term mental health services as compared with 7% of a comparison population (Cutajar et al., 2010). The effects include depressed mood, low self-esteem, self-harm, difficulties in relationships, and sexual maladjustment in the form of either hypersensitivity or sexual inhibition. The effects of abuse are generally greater when the abuse has involved physical violence and penetrative intercourse. Some of the long-term effects may be related to the events surrounding the disclosure of the abuse, including any legal proceedings, and to other problems in the family, such as neglect of the children and sexual deviance or substance abuse. Nevertheless, even when these other factors are controlled for, sexual abuse in childhood seems to be associated with psychiatric disorder later in life, especially with *depressive disorders*, *anxiety disorders*, and *personality disorders* (Glaser, 2015).

Aetiology

Sexual abuse of children occurs in all socioeconomic groups, and studies have shown no ethnic differences in rates and although socioeconomic status is unrelated to the incidence of childhood sexual abuse, there is an overrepresentation of low socioeconomic groups in clinic samples (Glaser, 2015). There are several preconditions

that make sexual abuse more likely. In the abuser, these include deviant sexual motivation, impulsivity, a lack of conscience, and a lack of external restraints (e.g. cultural tolerance); in the child, they include a vulnerability by virtue of age, disability, social isolation, and previous sexual abuse which can be identified and preyed upon by abusers.

Assessment

It is important to be ready to detect sexual abuse and to give serious attention to any complaint by a child of being abused in this way. When abuse has been established, it is important to assess whether it is likely to continue if the child remains at home and, if so, how dangerous it is likely to be. It is also important not to make the diagnosis without adequate evidence, which requires social investigation of the family as well as psychological and physical examination of the child.

The child should be interviewed sympathetically and encouraged to describe what has happened. Drawings or toys may help younger children to give a description, but great care must be taken to ensure that they are not used in a way that suggests to the child events that have not taken place. Young children can recall events accurately, but they are more suggestible than adults. Interviewing is difficult, and whenever possible it should be carried out by a child psychiatrist or social worker with special experience.

In the United Kingdom, responsibility for child protection rests with social services, to whom suspicions of abuse and actual abuse must be reported. Multidisciplinary involvement, often including the police and the judicial system, is usually needed to establish whether abuse has taken place and what the appropriate response should be. As with the physical abuse, the first priority is to protect the child and other children who might be at risk. The arrangements vary in different countries, and readers should find out how they apply in the country in which they are working.

Management

The initial management and the measures to protect the child are similar to those for physical abuse (see Box 16.11), including a decision about separating the child from the family. There are particular difficulties involved in intervening with families in which sexual abuse has occurred. These include a marked tendency to deny the seriousness of the abuse and of other family problems, and in some cases deviant sexual attitudes and behaviour of other family members, possibly including other children. Not all children require therapy; however, a minimum requirement is for the child to have an

Box 16.11 Steps in the assessment and management of child sexual abuse

1. Recognition, disclosure, and referral

- Identify **signs and indicators** of possible sexual abuse (physical, behavioural, or verbal disclosures).
- Refer cases of suspected abuse to **child protection services** (e.g. social services, police, and safeguarding teams).
- Consider **non-verbal cues** and behavioural changes in children who may be unable or unwilling to disclose.

2. Immediate safety and protection measures

- Assess whether the child is at **immediate risk**
- Liaise with **multi-agency safeguarding teams**.
- **Ensure medical needs are addressed**—Consider immediate forensic examination if the abuse has occurred recently.

3. Structured multi-agency investigation

- Conduct an **inter-agency strategy discussion** (police, social services, healthcare, and education professionals).
- **Child interview**—A **forensically sound interview** should be conducted by trained professionals (e.g. Achieving Best Evidence in criminal proceedings (ABE) guidelines in the UK).
- **Medical examination**—Performed by a trained paediatrician, focusing on physical evidence, health needs, and psychological impact.
- **Family assessment**—Evaluate family dynamics, caregiver responses, and potential for further safeguarding concerns.

4. Initial multi-agency child protection meeting

- Convene a **multi-agency meeting** to assess risk and determine the need for a **child protection plan**.

- Consider whether **other children in the household** are at risk.

5. Investigation and protection planning

- **Develop a protection plan** outlining safeguarding measures, child welfare needs, and parental responsibilities.
- Consider the need for **psychological and therapeutic support** for the child and non-offending family members.
- Engage **specialist child protection services** for ongoing risk management and intervention.

6. Implementation and ongoing review

- **Monitor the effectiveness** of the child protection plan, with scheduled reviews and reassessments.
- Adjust interventions based on **the child's wellbeing and risk factors**.
- Provide **continuous support for the child and non-offending caregivers**.

7. Legal and court proceedings

- Work with **family courts** to determine custody, restraining orders, or removal of the child from harmful environments.
- Ensure **child-friendly legal processes**, including special measures to support testimony (e.g. pre-recorded video statements, intermediary support).

8. Therapeutic support and long-term recovery

- Specialist trauma-focused therapy (e.g. TF-CBT, EMDR) for children experiencing psychological distress.
- **Support for non-offending caregivers**, including psychoeducation on supporting an abused child.
- Address **long-term impact** (e.g. attachment issues, self-esteem, PTSD symptoms) through ongoing mental health and social care interventions.

appropriate narrative about their abuse and to have access to an identified person who believes the child and is able to listen to the child supportively and uncritically. Several meta-analyses have found large to moderate effect sizes for psychotherapy, group therapy, and CBT combined with supportive, psychodynamic, or play therapy.

The treatment of child and adolescent abusers requires coordinated multidisciplinary support with child protection services. CBT individually and in groups has been used, as has structured group work.

For a review of the treatment of child sexual abuse, see Glaser (2015).

Ethical and legal problems in child and adolescent psychiatry

As well as the ethical and legal problems related to the treatment of adults and discussed in Chapter 4 and elsewhere, the following issues are particularly likely to arise in the care of children with psychiatric disorders.

Best interests principle

In general, *the interests of the child take precedence over those of the parents*. This principle is most obvious, and most easily followed, in cases of child abuse. In other cases, the decision is more difficult—for example, when a depressed mother is neglecting her child and is likely to become more depressed if substitute care is arranged. Such problems can usually be resolved by discussion with the parents, and between the professionals who are caring for the child and for the parent.

Confidentiality

The care of children often involves collaboration between medical and social services, and sometimes with teachers. Different agencies may have different policies about the confidentiality of records, and clinicians should consider these differences when deciding what information to disclose.

Consent

In each country, the law decides an age below which the child is deemed not to have capacity and so parents give consent on behalf of the child. Below this age, the child's assent should be obtained, if possible, as this will aid treatment, but if the child refuses the parents can decide. This problem arises, for example, when an adolescent under the legal age of consent refuses treatment for anorexia nervosa. The parents can also refuse treatment; however, their right to do so is linked to their duty to protect the child. In cases in which the parents' refusal appears not to be in the interests of the child, countries generally have provisions for a decision by a court of law.

Some of the complexities of English law can be mentioned briefly to illustrate the problems that have to be resolved in all legal systems. A fuller account of these and other issues is given by Hale and Fortin (2015). Readers are advised to find out for their respective countries the age at which a 'minor' can consent to their own treatment and the rights of parents when it comes to treatment.

In English law, the age from which people are judged legally capable of giving or refusing consent is 18 years. However, English law recognizes that most of those aged between 16 and 18 years have the capacity to give consent to receive treatment, and allows them to do so without the need for consent by a parent. The position is less clear when a 16- or 17-year-old does not consent to treatment, but it is probably the case that the parents' consent is sufficient. The decision in each case is likely to depend on the consequences of refusal; the more severe these are, the more likely it is that a court would accept that the child's refusal can be overridden by the parents. Further complexities arise with children under 16 years of age, some of whom are competent to give consent to certain treatments. There is no general assumption of competence at this age, and it has to be established in the individual case, but if it is established, the minor can give consent.

The question arose most notably with regard to the provision of contraception without the additional consent of a parent. In *Gillick v. West Norfolk AHA* it was ruled that, in English law, a sufficiently competent minor could consent without the need to obtain the consent of the parent (now called *Gillick competence* and related *Fraser guidelines*). It is probable, however, that with certain more invasive and risky treatments, the consent of a parent could be legally necessary, as well as clinically desirable. If a minor under the age of 16 years refuses treatment, this can be overruled by the parents if refusal is likely to result in harm. A final complexity concerns the definition of a parent. This problem arises, for example, when the person accompanying the child is not the person recognized by the law as having parental responsibility. For example, in English law, a father who is not married to the mother does not automatically have legal responsibility.

Consent for research poses similar problems for patients below the legal age of consent. In most countries, parents consent on behalf of their children, and they may find it difficult to balance the risks to the child against the benefits, which might not be for their child but for other children who might be treated for the same condition in the future. It is important that children, however, give their assent to the research and that they are able to discuss the issues and any concerns they might have, including with a person other than the individual who is requesting the consent (e.g. a nurse in their usual clinic).

Further reading

Thapar, A., Pine, D., Cortese, S., Creswell, C., Ford, T. J., Leckman, J. F., & Stringaris, A. (2025). *Rutter's Child and Adolescent Psychiatry*, 7th edn. Wiley-Blackwell.

Dulcan, M. K. (Ed.). (2021). *Dulcan's Textbook of Child and Adolescent Psychiatry*. American Psychiatric Publications. (A concise introduction to the subject.)

Appendix: Basic elements in child and adolescent psychiatric assessments

The format and extent of an assessment will depend on the nature of the presenting problem and the age of the child. The role of the parental history and information from other sources including education and other carers will be important in child psychiatry. The following scheme is adapted from Leckman and Taylor (2015), which should be consulted for further information.

1. Reason for referral: why this child and why now? Nature, frequency, and severity of presenting problem(s). Situations in which it occurs. Provoking and ameliorating factors. Stresses thought by parents to be important. Need to use this initial time to build therapeutic relationship and also gauge opinions on potential treatment options.
2. Child's strengths, talents, and interests; environmental supports
3. Presence of other comorbid problems or complaints:
 - a) Physical. Headaches, stomachache. Hearing, vision. Seizures, faints, or other types of episodes.
 - b) Eating, sleeping, or elimination problems.
 - c) Relationships with other children. Special friends. Experience of bullying or negative experiences with peers (including on social media).
 - d) Level of activity, attention span, concentration.
 - e) Mood, energy level, sadness, misery,
 - f) Response to frustration. Temper tantrums.
 - g) Antisocial behaviour. Aggression, stealing, truancy.
 - h) Educational attainments, attitude to school attendance.
 - i) Sexual interest and behaviour.
 - j) Any other symptoms, tics, etc.
4. Family history, current structure, and functioning (construct genogram).
 - a) Parents. Ages, occupations. Current physical and emotional state. History of physical or psychiatric disorder. Whereabouts of grandparents. Quality of parental relationship. Sharing of attitudes over child's problems. Quality of parent-child relationship. Positive interaction: mutual enjoyment. Parental level of criticism, hostility, rejection.
 - b) Siblings. Ages, presence of problems.
 - c) Home circumstances: poverty, space, access to internet, sleeping arrangements.
 - d) Overall pattern of family relationships. Alliance, communication. Exclusion, scapegoating. Intergenerational confusion.
5. Developmental, educational and medical history including possible traumatic events and current level of development (can include review of birth and paediatric records, education evaluations).
 - a) Pregnancy issues, delivery and state at birth, early mother-child relationship, post-partum depression, early feeding patterns. Milestones. Obtain exact details only if outside range of normal.
 - b) Language: comprehension, complexity of speech.
 - c) Spatial ability.
 - d) Motor coordination, clumsiness.
 - e) Past illnesses and injuries. Hospitalizations.
6. Personal history.
 - a) Early temperamental characteristics. Easy or difficult, irregular, restless baby and toddler.
 - b) Separations lasting a week or more. Nature of substitute care.

- c)** Description of gender preference and sexual orientation.
 - d)** Schooling history. Ease of attendance. Educational progress. Bullying. Exclusions.
 - e)** Use of social media/screen time: how much, which sites, distressing experiences online.
 - f)** Vaping history, alcohol consumption, other non-prescribed medications or substances such as cannabis.
 - g)** Exposure to violence or abuse—can ask about social services or police involvement in their homes, if they have been threatened or subject to any emotional, physical, sexual abuse, or neglect.
- 7. Mental state examination**
- a)** Appearance. Signs of dysmorphism. Nutritional state. Evidence of neglect, bruising, cuts, etc.
 - b)** Activity level. Involuntary movements. Capacity to concentrate.
 - c)** Mood. Expression of signs of sadness, misery, anxiety, tension.
 - d)** Rapport, capacity to relate to clinician. Eye contact. Spontaneous talk. Inhibition and disinhibition.
 - e)** Relationship with parents. Affection shown. Resentment. Ease of separation
 - f)** Habits and mannerisms.
 - g)** Presence of delusions, hallucinations, thought disorder.
 - h)** Level of awareness. Evidence of minor epilepsy.
- 8. Physical and neurological examination of child, only indicated if concern of presence of any genetic, medical, metabolic, endocrine, or neurological disorders that may impact the child's course of illness and treatment; or conduct as a baseline pre-treatment.**
- a)** Note any facial asymmetry
 - b)** Eye movements. Ask the child to follow a moving finger and observe eye movement for jerkiness, incoordination.
 - c)** Finger-thumb apposition. Ask the child to press the tip of each finger against the thumb in rapid succession. Observe clumsiness, weakness.
 - d)** Copying patterns. Drawing a man.
 - e)** Observe grip and dexterity in drawing.
 - f)** Jumping up and down on the spot.
 - g)** Hopping.
 - h)** Hearing. Capacity of child to repeat numbers whispered two metres behind them.

CHAPTER 17

Disorders of intellectual development

Introduction 467

Terminology 467

Epidemiology 469

Clinical features 469

Physical disorders 470

Psychiatric disorders 471

Other clinical aspects 474

Aetiology 476

Assessment and classification 483

The care of people with disorders of intellectual development 486

Treatment of psychiatric disorder and behavioural problems 489

Ethical and legal issues 490

Introduction

This chapter is concerned with a general outline of the features, epidemiology, and aetiology of disorders of intellectual development (DID), the organization of services and, more specifically, with the psychiatric disorders that affect these individuals. These disorders often present in infancy or the early childhood years, although in some cases are not diagnosed until the child

is older than five years of age, when standardized measures of developmental skills become more reliable and valid (Moeschler et al., 2014).

Many of the psychiatric problems of children with DID are similar to those of neurotypical children and therefore much is of relevance in their management is covered in Chapter 16.

Terminology

Over the years, there has been significant change in the terms that have been applied to people with DID. In the nineteenth and early twentieth centuries, derogatory terms such as ‘idiot’ for those with severe intellectual impairment, and ‘imbecile’ for those with moderate impairment was used. The special study and care of such people was known as the field of *mental deficiency*. When these words came to carry stigma, they were replaced by the terms *mental subnormality*

and *mental retardation*. The terms *mental handicap* and *learning disability* have also been widely used, but now the term *intellectual disability* (ID) is generally preferred and has been adopted in the DSM-5. ICD-10 used the term mental retardation; however, progress has been made in the adoption of the term ‘disorders of intellectual development (DID)’ in ICD-11. This shows a useful shift in terminology that places emphasis on the condition, classification, and defining features in the context

of advances in our understanding of their aetiology and brain functioning and development. Of note, in the United States the term learning disability is generally applied to dyslexia and similar forms of specific learning disability, rather than to DID.

DID are sometimes divided into *syndromic* and *non-syndromic* forms. *Syndromic intellectual disability* refers to the presence of additional medical or behavioural features (which allow a 'syndrome' to be diagnosed); if not, the term *non-syndromic intellectual disability* is used.

The development of ideas about DID

A fundamental distinction, first made by Esquirol in 1845, is drawn between general intellectual impairment starting in early childhood (*ID*) and intellectual impairment developing later in life (*dementia*). Complementing these two categories, *acquired brain injury* (ABI) is the term used to describe individuals who develop an ID after birth, either because of trauma (e.g. following a head injury) or non-traumatic (e.g. because of a brain tumour).

Early in the twentieth century, Binet's tests of intelligence provided quantitative criteria for categorizing intellectual ability. These tests also made it possible to identify the range of abilities that might not be obvious without the objective criteria testing provided. Unfortunately, these measures did not focus on adaptive functioning and so might have contributed to assumptions at the time that individuals with lower scores were socially incompetent and required institutional care.

Similar views were reflected in the legislation of the time. For example, in England and Wales the Idiots Act of 1886 made a simple distinction between idiocy (more severe) and imbecility (less severe). In 1913, the Mental Deficiency Act added a third category for people who 'from an early age display some permanent mental defect coupled with strong vicious or criminal propensities in which punishment has had little or no effect.' As a result of this legislation, people of normal or near-normal intelligence were admitted to hospital for long periods simply because their behaviour deviated from the values of society.

The inclusion of adaptive functioning and social criteria enables more information to be gathered of relevance to everyday care and the shift in DSM-5 to place ID in the broader category of neurodevelopmental disorders was an important step forwards.

DSM-5 defines ID as involving 'impairments of general mental abilities that impact adaptive functioning in three domains, or areas'. These domains determine how well an individual can cope with everyday life and include communication and social skills, personal independence, and school/work functioning. The disorder is considered chronic and often co-occurs with other mental conditions like depression, attention-deficit hyperactivity disorder, and autism spectrum disorder. Having defined the disorder with a focus on adaptive functioning and ability to meet developmental and sociocultural standards, it is subdivided by severity levels according to adaptive functioning into mild, moderate, severe, and profound categories placing some weight on standardized IQ testing. The categories broadly map onto the previous ICD-10 subtypes established according to IQ: mild (IQ 50–70); moderate (IQ 35–49); severe (IQ 20–34); and profound (IQ below 20).

ICD-11 has introduced the new categorization of DID (6A00) as part of 'neurodevelopmental disorders'. These are then classified as mild, moderate, severe, and profound with two further options of provisional and unspecified. Those that are mild are described as generally achieving independent living, while those with severe disorders typically require daily support in a supervised environment.

The original World Health Organization (WHO) classification of impairment, disability, and handicap has shifted to using more constructive terminology, with *activity* replacing 'disability', and *participation* replacing 'handicap.' The impairment is of the central nervous system, and the disability (or limitation of activity) is in learning and acquiring new skills. The extent to which impairment leads to disability (limits activity) depends, in part, on experiences in the family and at school, and on the correction of associated problems such as deafness. The final stage, namely handicap (or limited participation), depends on the degree of disability and on other factors such as the support that is provided. The DSM-5 aligns most closely with the WHO recommendations for an integrated approach bring neurodevelopmental and social approaches together.

Educationalists can use other terms, and these also differ between countries. In the United Kingdom, the term commonly used is *special educational needs*, whereas in the United States, mild, moderate, and severe cognitive impairment is most commonly used, with 90% of the children with ID falling in the mild category, which approximately equates to an IQ lying between 55 and 70.

Epidemiology

Level of IQ has been used as a key criterion in defining DID, with IQ tests designed to be normally distributed with a mean of 100 and a standard deviation of 15. In both DSM-5 and ICD-11 ID is considered to be approximately two standard deviations or more below the population, which equals an IQ score of about 70 or below, approximately 2.3% of the population (Simonoff et al., 2015). Placing the diagnostic criteria on both IQ and adaptive functioning allows for considerable individual differences in presentation and diagnosis as these differences can be within syndromes and dependant on family functioning and levels of engagement with activities of daily living; therefore, the boundaries between the categories of mild, moderate and severe ID are not firmly fixed. Current estimates place approximately 1–2% of the population as having DID (Mazza et al., 2020).

Variability in apparent prevalence rates is due partly to experimental factors such as methods of ascertainment and definition, but also reflects genuine differences across time and place. In recent years, the *incidence* has fallen substantially, because of the improvement in preventable prenatal and perinatal causes of severe intellectual disabilities. However, the

prevalence has not fallen, and in fact is expected to rise by over 10% by 2020. The preserved or increasing prevalence despite reduced incidence reflects two factors. First, people with DID, particularly those with Down syndrome, are living longer. This has also affected the age distribution of people with severe ID, so that the numbers of adults have increased. Secondly, improvements in maternal and neonatal care, especially those born extremely premature (below 32 weeks' gestation) are resulting in more children surviving this early life stress and growing up with DID, especially those in the severe and profound categories. Furthermore, only a proportion of the milder cases are likely known to service providers as it is estimated that less than 50% of all such people receive special provision, possibly leading to underestimates regarding need. Those in contact with services can be described as reflecting 'administrative prevalence' are greater in number from lower socioeconomic groups and in greater contact during childhood when more patients need services.

The clinical features of the individual syndromes that cause DID and the epidemiology of psychiatric disorders for those with DID are covered later in the chapter.

Clinical features

The most frequent manifestation of DID is uniformly low performance on different intellectual tasks, including learning, short-term memory, the use of concepts, and problem solving (see Table 17.1). Specific abnormalities may lead to particular difficulties. For example, lack of visuo-spatial skills may cause practical difficulties, such as inability to dress, or there may be disproportionate difficulties with language or social interaction, both of which are strongly associated with behaviour disorders. Among children with DID, the common behaviour problems of childhood tend to occur when they are older and more physically developed than neurotypical children, and their difficulties might last for longer. We describe the different features according to IQ and extrapolate what this might mean for functioning, although we are aware of the limitations of this approach, as described above.

Mild ID (IQ 50–70)

People with mild ID account for about 85% of those with DID. Usually their appearance is unremarkable and any sensory or motor deficits are slight. Most people in this group develop more or less normal language abilities and social behaviour during the preschool years, and their learning difficulties may never be formally identified. In adulthood, most people with mild ID can live independently in ordinary surroundings, although they may need help in parenting and coping with family responsibilities, housing, and employment, or when under unusual stress.

Moderate ID (IQ 35–49)

People in this group account for about 10% of those with DID. Many have better receptive than expressive

Table 17.1 Features of mild, moderate, and severe/profound intellectual disability

	Mild	Moderate	Severe/profound
IQ range	69–50	49–35	< 35
Percentage of cases	85%	10%	5%
Ability to self-care	Independent	Need some help	Limited
Language	Reasonable	Limited	Basic or none
Reading and writing	Reasonable	Basic	Minimal or none
Ability to work	Semi-skilled	Unskilled, supervised	Supervised basic tasks
Social skills	Normal	Moderate	Few
Physical problems	Rare	Sometimes	Common
Specific cause discovered	Sometimes	Often	Usually

language skills, which is a potent cause of frustration and behaviour problems. Speech might be primarily using simpler phrases, and is often better understood by people who know them well. Many make use of signing systems such as Makaton sign language. Activities of daily living such as dressing, feeding, and attention to hygiene can be acquired over time, but other activities of daily living, such as the use of money and road sense, generally require support. Similarly, supported employment and residential provision are usually needed.

Severe ID (IQ 20–34)

It is difficult to estimate IQ accurately when the score is below 34 because of the difficulty in administering the tests in a valid manner to individuals in this group. Estimates suggest that people with severe ID account for about 3–4% of those with DID. In the preschool

years their development is usually greatly slowed. Eventually many of them can be helped to look after themselves under close supervision, and to communicate in a simple way—for example, by using objects of reference. As adults they can undertake simple tasks and engage in some social activities, but they need supervision and often benefit from a clear structure to their lives.

Profound ID (IQ below 20)

People in this group account for 1–2% of those with DID. Development across a range of domains tends to be around the level expected of a 12-month-old infant. Accordingly, people with profound DID are a highly vulnerable group who require considerable support and supervision, even for the least complicated activities of daily living.

Physical disorders

People with DID experience a greater variety and complexity of physical health problems than the rest of the population, but may not complain of feeling unwell, nor be able to articulate their symptoms, and conditions may be noticed only because of changes in behaviour. The prevalence of obesity among youth with DID is almost double that of the general population and it is unclear if this is because of physiological or metabolic differences, mealtime behaviours, and/or infrequent physical activity (Segal et al., 2016). Clinicians should

be aware of the associations between certain DID and physical illness (e.g. Down syndrome; see p. 477), and that people with DID are more likely to die prematurely. A UK review of deaths of over 200 people with IDs showed that nearly a quarter were younger than 50 years of age when they died, and the median age at death was 64 years of age—for males 13 years younger and for females 20 years younger than the median age of death for the general population. Contributory factors to the premature deaths included poor health care and living

in inappropriate accommodation (Heslop et al., 2014). Following progress in understanding and improved resources, with the exception of people with severe and multiple disabilities or Down syndrome, the life expectancy of this group is now more closely approximating with that of the general population. Middle and old age, which until 30 years ago were not recognized in this population, are now important parts of the life course of these individuals. Older adults with DID form a small but significant and growing proportion of older people in the community (Coppus, 2013).

Sensory and motor disabilities and incontinence are often prominent physical disorders in people with DID. People with severe ID (especially children) usually have several of these problems. Only one third are continent, ambulant, and without severe behaviour problems. Around 25% are highly dependent on other people. Among those with mild ID, similar problems occur, but less frequently. Nevertheless, they are important because they determine whether special educational programmes are needed. Sensory disorders add an important additional obstacle to normal cognitive development. Motor disabilities include spasticity, ataxia, and athetosis. Ear infections and dental caries are common in this population.

Epilepsy is a frequent and clinically important condition in DID. Around 22% of people with ID have a history of epilepsy, compared with 5% in the general population (Robertson et al., 2015). The prevalence increases with the severity of the disorder, with lifetime history of epilepsy estimated to be 12% in people with Down syndrome, 15% in mild to moderate ID, and 30% in severe and profound ID. The risk appears higher among those with additional neurological diagnoses, such as cerebral palsy and those with autism spectrum disorder. Epilepsy is also more commonly associated with fragile X syndrome, tuberous sclerosis, Angelman syndrome, and Rett syndrome, while certain epilepsy syndromes, such as infantile epileptic spasms syndrome (IESS, previously known as West syndrome) and Lennox–Gastaut syndrome, are more common among people with DID. The proportion of epilepsy cases that are drug-resistant is also greater among people with DID. Of particular concern is the incidence of *sudden unexplained death in epilepsy* (SUDEP), which is estimated to be 1 in 295 per year in the ID population, compared with 1 in 1,000 per year in epilepsy patients in the general population.

For a review of the management of epilepsy in DID, see Kerr et al. (2009), but see also Kerr et al. (2023).

Psychiatric disorders

In the past, psychiatric disorder among individuals with DID was often viewed as different from that seen in neurotypical populations, but they are similar at times, differently modified by their developmental stage, and may not be easily recognized or communicated. Therefore, when diagnosing psychiatric disorder among people with DID, more emphasis may have to be given to behaviour and less to reports of psychological phenomenology. Comorbid psychiatric disorders in DID affect overall management with implications for the individual's life, in their caregiver's burden, in psychopharmacological and psychological treatment, and in service providers (Mazza et al., 2020).

For reviews of psychiatric disorders in DID, see Mazza et al., (2020).

Epidemiology and features of psychiatric disorder in people with ID

Studies of individuals with DID have published rates of psychiatric disorder that vary widely because of

methodological limitations in studies with case ascertainment, detection, and definition. Early studies primarily had the selection bias of including mainly people from institutions, although more recent studies have included people from the community. A 2020 systematic review of co-occurring psychiatric disorders in adults and adolescents with ID brought together 22 existing studies (Mazza et al., 2020). They concluded that prevalence rates were high with individuals having at least double the risk of the general population. The pooled prevalence of any co-occurring psychiatric disorders was 33.6% (95% CI: 25.2%–43.1%). Prevalence was lower in population-based studies in studies that used ICD criteria for the psychopathology. The prevalence was higher in mild, moderate, and severe ID than in profound ID. Mood disorders were most common with a prevalence of 6.7%, followed by anxiety disorders at 5.5%, and schizophrenia with a prevalence of 4.8; unspecified psychotic disorder was 3.9% and personality disorders had a pooled prevalence of 2.8%, with high heterogeneity in all analyses.

In general, the aetiology of psychiatric disorders in DID is similar to that in the general population in that

they result from a complex and non-deterministic combination of biological, psychological, and social factors. However, there are some specific factors that add additional risk and complexity to the incidence and detection of psychiatric disorders. Although the direction of effects can be difficult to unpick, DID and psychiatric disorders might have shared genetic and other risk factors; they might result from medical adversities that affect the brain (e.g. epilepsy, cerebral palsy); they might come from environments of increased family psychosocial adversity; and the ID itself might impact on the individual's capacity to communicate, understand their environment, and respond flexibly to various situations (Simonoff, 2015). Furthermore, the complexity of the presentation of psychiatric disorders in those with DID reflect the often behavioural, medical, and neuropsychiatric components of an individual's specific DID contributes to why a psychiatrist is often a key member of the multidisciplinary team helping to manage the needs of someone with a DID experiencing difficulties.

Psychotic disorders

The epidemiological studies available suggest an increased prevalence of psychotic disorders among individuals with ID. Schizophrenia affects around 4% of individuals with DID, compared with less than 1% in the general population (Mazza et al., 2020). The overlap between the two conditions largely reflects shared genetic factors. Clinically, delusions may be less elaborate than in most other patients with schizophrenia, hallucinations may be less complex, and thought disorder can be difficult to identify. When IQ is less than 45, it is difficult to make the diagnosis with any certainty. Furthermore, some of the symptoms of underlying brain damage, such as stereotyped movements and social withdrawal, may be incorrectly attributed to schizophrenia, so a comparison of current with previous behaviour is always needed.

The diagnosis of schizophrenia should be considered as one of several possibilities when intellectual or social functioning worsens without evidence of an organic cause, and especially if any new behaviours that are odd and out of keeping emerge. When there is continuing doubt, a trial of antipsychotic drugs is sometimes appropriate. The principles of treatment of schizophrenia in people with ID are the same as those for general populations of patients (see Chapter 11); however, as with all pharmacological interventions in those with DID, it is important to be aware that often lower therapeutic doses are often needed and titration should be slower with close monitoring of side effects.

Mood disorder

The rate of depressive disorders is estimated to lie around 6–7%, slightly higher than that of the general population (Mazza et al., 2020). However, people with DID can be less likely to complain of mood change or to express depressive cognitions. Diagnosis often has to be made on the basis of an appearance of sadness, changes in appetite and sleep, and behavioural changes of retardation or agitation. Severely depressed patients with adequate verbal abilities may describe hallucinations or delusions. Of note, depressive disorder occurs commonly in individuals with Down syndrome during late adolescence and early adulthood, independent of life events (Collacott et al., 1998). Mania can be diagnosed on the basis of hyperactivity and behavioural signs of excitement, irritability, or nervousness.

The differential diagnosis of mood disorder in individuals with DID can be broad and range from physical health problems such as thyroid dysfunction, which is especially prevalent in people with Down syndrome, to psychological responses to grief.

The rate of suicide in people with moderate and more severe DID is lower than in the general population. The rate of self-harm is less certain, because it can be difficult to determine extent of intent with some self-injurious behaviour.

See Chapter 9 for principles of treatment of mood disorders.

Anxiety disorders and related conditions

Adjustment disorders are common among individuals with DID, occurring when there are changes in the routine of their lives. Anxiety disorders are also frequent, especially at times of stress, and social anxiety is often seen in individuals with fragile X syndrome (Cordeiro et al., 2010). Obsessive-compulsive disorders are also found. Conversion and dissociative symptoms are sometimes florid, taking forms that can be interpreted in terms of the patient's understanding of illness. Somatoform disorders and other causes of functional somatic symptoms can result in persistent requests for medical attention. Treatment usually focuses on psychoeducation to bring about adjustments in the patient's environment, and give reassurance. Interventions are often directly aimed at thoughts and beliefs with many studies reporting positive outcomes. Counselling, at an appropriate level of complexity, can also be helpful (Appleton et al., 2019; Dagnan et al., 2018).

Eating disorders

Overeating and unusual dietary preferences are frequent among people with DID. Abnormal eating behaviours, including *pica* (Chapter 13), are not uncommon, but classical eating disorders appear to be less common than in the general population. Overeating and obesity are features of Prader–Willi syndrome (see below).

Delirium and dementia

Delirium. This may occur as a response to infection, medication, and other precipitating factors. As in neurotypical individuals, delirium in individuals with DID is more common in childhood and in old age than at other ages. Disturbed behaviour due to delirium is sometimes the first indication of physical illness. Delirium may also occur as a side effect of drugs (especially anti-epileptics, antidepressants, and other psychotropic medication).

Dementia. As the life expectancy of individuals with DID increases, dementia in later life is becoming more common. Alzheimer's disease is particularly common among people with Down syndrome (see p. 477), but dementia also is also more common in other elderly intellectually disabled people, with a prevalence of 18.3% in those over the age of 65 years, making it two to three times more common in this population (Strydom et al., 2009). Prevalence rates do not differ between mild, moderate, and severe ID groups. Dementia in individuals with DID may initially present with seizures, or with the usual progressive decline in intellectual and social functioning, which has to be distinguished from conditions such as depression and delirium.

Disorders that are usually first diagnosed in childhood and adolescence

Many of the disorders in this category are more frequent in children with individuals with DID than in the general population, and they are more likely to continue into adulthood. It is important to be aware that relatively specific developmental disorders of scholastic skills, speech, and language and motor function may occur alongside more global difficulties.

Autism spectrum disorder (ASD) and attention-deficit hyperactivity disorder (ADHD)

Hyperactive behaviour and autistic-like behaviour are frequent symptoms in individuals with DID. In addition, the diagnoses of ASD and ADHD are more common for those with DID than among the general population. There is a particular comorbidity between DID and autistic spectrum disorders, probably reflecting an overlap in

their aetiology, especially with regard to genetic factors (Mefford et al., 2012).

Abnormal movements

Stereotypies, mannerisms, and rhythmic movement disorders (including head banging and rocking) occur in about 40% of children and 20% of adults with severe DID. Repeated self-injurious behaviours are less common but important. There is a specific association with Lesch–Nyhan syndrome, in which the biting away of the corner of a lip is common. Prader–Willi syndrome is strongly associated with a pattern of self-injury where individuals pick at their skin.

Behaviour that challenges

The term *behaviour that challenges* (or *problem behaviour*) is used to describe problematic behaviour that is relatively specific to DID and is associated with neurodevelopmental disorders such as ASD. It describes behaviour of an intensity or frequency sufficient to impair the physical safety of a person with DID, to pose a danger to others, or to make participation in the community difficult (Simonoff, 2015). Those who present with behaviour that challenges are often marginalized, stigmatized, disempowered, and excluded from mainstream society as it often provokes punitive or restrictive responses (Banks et al., 2007). It is probable that around 20% of children and adolescents with DID and 15% of intellectually disabled adults have some form of behaviour that challenges. The causes of such behaviour are listed in Box 17.1. Whenever possible, the primary cause should be understood and treated. It is essential that the behaviour is understood in the context of that individual and their environment and therefore any intervention must address the person, the environment, and the interaction between the two. The person might be expressing unhappiness in their current environment, and the complexities of assessment might require a comprehensive and multidisciplinary assessment including: a functional assessment of behaviour, underlying medical and organic factors, psychological/psychiatric factors, communication, and social/environmental factors.

Intervention should be delivered in a person-centred context and a framework of positive behavioural support. This can include proactive and reactive strategies, psychotherapy, communication, positive programming, physical and/or medical, and psychopharmacological. Behavioural treatment is the most widely used treatment. Behaviour that challenges is more likely to be encountered in the family home, in small-scale community settings, or in environments that are poorly organized and unable to respond well to the needs of

Box 17.1 Causes of behaviour that challenges

Physical: pain, discomfort, malaise

Psychiatric and neuropsychiatric disorder: mood disorders, psychosis, anxiety, obsessive-compulsive disorders, ADHD, dementia

Psychological trauma: reaction to abuse or loss; wish to escape an unpleasant situation

Communication difficulties: hearing loss, unclear communication, insufficient vocabulary or means of expression, difficulties understanding communication of others

Phenotype-related behaviours: Prader–Willi syndrome, Lesch–Nyhan syndrome, Williams syndrome

Under- or over-stimulation

Desire for attention or other reward

Side effects of medication

the person. Therefore the response of those in their environment is key and as these interventions are often provided in the places where the behaviour usually appears, competency-based training and professional support is often required for carers; in severe cases, for example, in a residential unit, such support might also be needed.

Sleep disorders

Impaired sleep is common in individuals with DID with prevalence estimates ranging from 9–34% (van de Wouw et al., 2012). In a meta-analysis of sleep duration and sleep quality, in 93% of comparisons between groups,

sleep was found to be of poorer quality in the group of people with DID (Surtees et al., 2018). Serious sleep difficulties such as obstructive sleep apnoea, excessive daytime sleepiness, and parasomnias have a reported prevalence of 9%, and can be a source of considerable distress. Furthermore, sleep disorders may be associated with subsequent challenging behaviours and a worsening of cognitive impairment. The high rate of sleep disorders is accounted for by five factors:

1. coexisting damage to brain structures that are important for the sleep–wake cycle
2. epileptic seizures that start during sleep
3. epilepsy-related sleep instability that disrupts sleep architecture
4. structural abnormalities in the upper respiratory tract causing sleep apnoea (particularly common among people with Down syndrome)
5. poor sleep hygiene.

The management of sleep disorders is complicated by the heterogeneity of the underlying causes of DID, along with the nature of associated comorbidities. Treatment follows the usual principles of identifying and treating the cause, as well as non-pharmacological interventions such as improving sleep hygiene and some components of CBT (Korb et al., 2021) (see Chapter 24). Even modest adaptations to daily routine and the sleeping environment can benefit adults with intellectual disabilities, who are more likely to lack adequate daily exercise and regular exposure to natural light compared with the general population. The evidence for the pharmacological management of sleep disorders in adults with DID is not well established. The medication that has received the most attention is melatonin, probably because of its favourable side effect profile, and some studies have shown it to be effective.

Other clinical aspects

Sex, relationships, and parenthood

Most individuals with DID develop sexual interests in the same way as individuals without DID. Although people with ID are encouraged to live as normally as possible in other ways, sexual expression can be discouraged by parents and carers, and sexual feelings may not even be discussed.

In the past, sexual activity of individuals with DID was discouraged because it was feared that they might produce children with DID. It is now known that many kinds of severe ID are not inherited, and that those that are inherited are often associated with infertility. Another concern is that people with DID will not be good parents. Studies have shown that although low parental intellectual capacity can negatively impact child outcomes, other studies indicate that child

development approaches population norms (Collings and Llewellyn, 2012). Parents will be best able to care for a child successfully if they are strongly supported, especially with behavioural interventions such as teaching parenting skills. For children raised by parents with DID, social exclusion, bullying, and stigma are commonplace, and they are at high risk of being removed from parental care.

It is especially important to consider issues of capacity and ability to consent to sexual relationships if individuals with DID become involved in intimate relationships. These issues should be considered carefully in each case, and contraception made available where appropriate.

Sexuality

There is little published literature on the sexual wellbeing of individuals with DID and they might experience restrictions in their access to sexual health information and services. Protective attitudes may be linked to the fact that individuals with DID have been targeted for abuse and parents may fear that sexual knowledge might exacerbate this vulnerability. Some individuals with DID can have a childlike curiosity about other people's bodies, which can be misunderstood as sexual. Some expose themselves without fully understanding the significance of their actions. This is usually best dealt with by behavioural interventions or with specialized group therapy depending on which interventions are most accessible to them. Promising approaches have been developed regarding sex and relationship education, starting early and using innovative methods, and to sexual health services, adapted to allow additional time, address specific fears, and ensure accessible communication (Talbot et al., 2010).

Maltreatment and abuse

There is a strong association between disability (physical and intellectual) and child maltreatment, with individuals significantly more likely to experience abuse than their non-disabled peers, more so if the child has communication difficulties and behavioural disorders (Stalker and McArthur, 2012). There is also evidence to suggest that abuse is often under-reported in children with DID. This might be for a number of reasons, including difficulty communicating, perceived threat, fears of being separated from family, and tolerance of abuse in order to be accepted or receive rewards or affection. When abuse has occurred, it may lead to psychological problems concurrently and later in life that are similar to those experienced by any other victim of such abuse—for example, depression and post-traumatic stress disorder. However, due to difficulty in emotional expression

and communication, such disorders may present primarily with challenging behaviours in situations with which the individual associates past abuse.

Growing old

Several problems arise more frequently as individuals with DID live longer other than just the emergence of dementia and physical health problems. When parents are the carers, they may find care increasingly difficult to deliver as they grow old. Understandably, such parents are often concerned about how their deaths will affect their child's future, yet can find it difficult to arrange alternative care while still alive.

The older person with DID also faces special challenges. If their parents die first, they face problems of bereavement with potentially little preparation for these events. The isolation felt by many bereaved individuals may be increased because other people are not sure how to offer comfort, and because the person with DID may be excluded from the ritual of mourning and other support that might be needed at this time. Bereaved people with DID should be helped to come to terms with the loss, using the principles that apply generally to grief counselling, but choosing appropriately simple forms of communication, such as life story work, memory boxes, photographs, storytelling, artwork, and films to help them explore loss in constructive ways. Studies exploring how best to support older individuals with DID have indicated the importance of funding, provision, and organization of services/personnel, education, and cooperation among different support systems in order to best help them. The provision of good housing or activities, support when these provisions change, provision of (mental) health care, dementia care, end-of-life care, life story work, future planning, and support for activities of daily living are crucial (Schepens et al., 2019).

Effects on the family

When a newborn child is found to have DID, many parents can be distressed. Feelings of rejection are common, but seldom last for long, and can be replaced by feelings of loss of the hoped-for neurotypical child. Frequently the diagnosis of DID is not made until after the first year of life, and the parents then have to change their hopes and expectations for the child. They can experience prolonged depression, guilt, shame, or anger, and have difficulty in coping with the many practical and financial problems. A few reject their children, some become overinvolved in their care, sacrificing other important

aspects of family life, while others seek repeatedly for a cause to explain the DID. Most families eventually achieve adjustment, although the temptation to over-indulge the child can remain. However well they adjust psychologically, the parents might still be faced with the prospect of prolonged dependence, frustration, and social difficulties. If the child also has a physical disability, these difficulties can be increased.

There have been several studies on the effect of a child's DID on the family. Although caregiving can often be a rewarding experience, caregivers in the family of those with DID also face considerable responsibility and accompanying stress (Mitter et al., 2019). Not only do they have to cope with both the physical and emotional needs accompanying the affected individual's disability, but also the stigma of their child being undervalued by others, as well as often continuous and/or repeated battles with services, can adversely affect quality of life. Consequently, they may feel dejected and helpless, resulting in lowered self-esteem and impaired family relationships. Mothers of children with DID have been found to be especially susceptible to poor mental health, reporting more depressive symptoms, higher levels of malaise, depression, and anxiety than do mothers of typically developing children. Affiliate stigma may contribute to the negative impact of being a parent of an individual with DID. As a means of coping with some of the effects of affiliate stigma, caregivers may react by withdrawing socially, or even distancing themselves from their relative with DID to avoid association (Mitter et al., 2019). Services that facilitate improved functioning in activities of daily living can play an important role in also assisting the families in accessing suitable family support that has the potential to positively influence family quality of life. Teams specializing in services for children with DID sometimes also offer a support service or support groups for siblings.

Considerations in Covid

The impact of the Covid-19 pandemic disproportionately affected individuals with DID, with many being especially vulnerable to the physical, mental, and

social effects of the pandemic. Individuals with Down syndrome were at ten times greater risk than the general population from death from contracting the virus. Cognitive impairments may have limited their understanding of information about protecting themselves from the virus, leaving those with DID to rely on carers to be vigilant on their behalf during quarantine. Restrictions on usual activities were likely to induce mental stress, leading to an escalation in challenging behaviours, risk of placement breakdown, and increased the use of psychotropic medication. People with DID were vulnerable to exploitation by others where the usual community supports no longer functioned to protect them (Perera et al., 2020).

Criminal justice system

Individuals with mild DID have higher rates of criminal behaviour than the general population (see also Chapter 18). The causes of this excess are multiple, but influences in the family and social environment are often important. Impulsivity, suggestibility, vulnerability to exploitation, and desire to please are other reasons for involvement in crime, alongside an increased likelihood in being detected. Compared with the general population, individuals with DID who commit offences are more likely to be detected and, once apprehended, may be more likely to confess. Among the more serious offences, arson and sexual offences (usually exhibitionism) are said to be particularly common.

Of note, individuals with DID may be suggestible and may therefore give false confessions, so particular care should be taken when questioning them about an alleged offence. In the United Kingdom, police interrogation should accord with the Police and Criminal Evidence Act, which requires the presence of an appropriate adult to ensure that the person with ID understands the situation and the questions. Once convicted, psychiatric supervision in specialized forensic learning disability units and specialized education may be needed. Those with DID are also more likely to be the victims of crime and are perceived to be poor witnesses.

Aetiology

In a classic study of the 1,280 mentally retarded people living in the Colchester Asylum, Penrose (1938) found that most cases were due not to a single cause but to a

hypothesized interaction of multiple genetic and environmental factors. This conclusion still broadly applies, especially for mild DID. Conversely, a specific cause for

severe DID is increasingly often found. Table 17.2 lists the main categories for DID aetiology, and some examples of each.

Genetic factors are a major cause of DID. IQ is heritable (with estimates of 30–50%; Deary et al., 2009), and ID is in this respect just the tail end of the normal distribution of intelligence in the population (indeed, by definition, 2% of people are expected to have an IQ of less than 70). This aspect of genetic predisposition to intelligence reflects the cumulative effects and interactions of a large number of genes, most of which have yet to be identified. In addition, a specific chromosomal or genetic defect can be the necessary and sufficient cause of a person's DID. These include Down syndrome and fragile X syndrome, the two most common causes of DID. It has recently been estimated that over 1,000 genes underlie intellectual ability.

The crude division of DID into cases where a specific cause can be found, and those where it reflects multiple, largely 'non-specific' causes, parallels earlier categorization into either 'subcultural' (later sociocultural) or 'pathological' mental retardation (as it was then called). At a mechanistic level, the types of genetic abnormality that cause DID are diverse (see Box 17.2) and can be because of chromosomal abnormalities, X-linked disorders, and autosomal dominant and recessive disorders, as well as copy number variants (duplications deletions) of chromosomal regions (see Chapter 5 for an introduction to genetics.) The clinical features of the major genetic causes of DID are listed in Table 17.3, with additional details given in the text for Down syndrome and fragile X syndrome. For a review of the genetic causes of ID, see Vissers et al. (2016).

Environmental factors are divided into pre-, peri-, and postnatal factors, reflecting the time at which they are believed to have occurred. The relative importance of environmental factors varies according to setting. For example, they are more significant in low- and middle-income countries (LMICs) and where healthcare provision or general health are poorer, and they may be affected by local factors (e.g. areas of low iodine predispose to congenital hypothyroidism). Social as well as biological factors should be considered in the environmental category. Some non-genetic causes of ID are summarized in Table 17.4.

Note also that aetiology of DID is sometimes defined in terms of an accompanying physical characteristic, such as hydrocephalus, microcephaly, or cerebral dysgenesis. Finally, the descriptive category of *inborn errors of metabolism* encompasses multiple different syndromes, mostly due to a single gene disorder affecting an enzyme that is important in a particular biochemical pathway.

They include urea cycle disorders and lysosomal storage disorders, which in turn include mucopolysaccharidoses (e.g. Hurler syndrome) and sphingolipidoses (e.g. Tay-Sachs disease), and others (e.g. phenylketonuria) (Scaglia, 2014).

Overall, it is estimated that prenatal (genetic and environmental) factors cause 50–70% of DID, with 10–20% originating perinatally, and 5–10% originating postnatally; the proportions, and the causes within each category, depend on the population being studied.

Down syndrome

Down syndrome, first described by Langdon Down in 1866, is a common cause of DID, occurring in 1 in about every 650 live births. It is most commonly caused by an additional chromosome 21 (95% of cases) or chromosomal translocation causing partial trisomy (in approximately 4% of cases) or chromosomal mosaicism (in approximately 1% of cases). It is more frequent in children of older women, occurring in about 1 in 2,000 live births to mothers aged 20–25 years, and 1 in 30 live births to those aged 45 years. Improvements in the antenatal screening process (maternal serum testing and, later, measurement of foetal nuchal translucency) has led to an increase in the number of terminations of pregnancy due to early detection of a fetus with Down syndrome.

The clinical picture consists of a number of features, any one of which can occur in an individual without Down syndrome. Four features together are generally accepted as strong evidence for the syndrome. The most characteristic signs are listed in Box 17.3. IQ is generally between 20 and 50, but in 15% of individuals it is above 50. Mental abilities usually develop fairly quickly in the first six months to a year of life, but then increase more slowly. Children with Down syndrome are often described as affectionate and easy-going, but there is wide individual variation. Emotional and behaviour problems are less prominent.

In the past, the infant mortality of Down syndrome was high, but with improved medical care survival into adulthood is usual, and about 25% of people with Down syndrome now live beyond 50 years of age. Although longevity in people with Down syndrome has improved appreciably, age-specific risk for mortality still is considerably increased, with many body systems exhibiting signs of premature or accelerated aging. This may be due to both genetic and epigenetic factors (Zigman, 2013).

Aetiology

In 1959, Down syndrome was found to be associated with the chromosomal disorder of trisomy (three

Table 17.2 The types of causes of intellectual disability

Genetic category	Examples
<i>Chromosomal disorders</i>	
• Trisomies	Down syndrome, Edwards syndrome
• Other aneuploidies	Turner syndrome (XO), Klinefelter syndrome (XXY)
• X-linked	Coffin–Lowry syndrome
<i>Mitochondrial disorders</i>	
<i>Trinucleotide repeat expansion disorders - Fragile X syndrome</i>	
<i>Copy number variation deletions</i>	Angelman syndrome (some cases), velocardiofacial syndrome, cri du chat
<i>Single gene disorders</i>	
• Autosomal dominant	Neurofibromatosis, tuberous sclerosis
• Autosomal recessive	Phenylketonuria, Tay–Sachs disease, Hurler syndrome
• X-linked	Rett syndrome
<i>Mitochondrial disorders</i>	Leigh Syndrome, MELAS, MERRF
<i>Complex (non-Mendelian) disorders</i>	
Environmental causes	Examples
<i>Prenatal</i>	
• Infection	Rubella, toxoplasmosis, syphilis
• Toxins	Foetal alcohol syndrome, lead poisoning
• Maternal	Pre-eclampsia, placental insufficiency
• Nutritional	Iodine deficiency, severe malnutrition
• Immune	Rhesus incompatibility
<i>Perinatal</i>	
• Obstetric complications	Brain injury, cerebral palsy
• Complications of prematurity	
<i>Postnatal</i>	
• Infection	
• Injury	
• Severe psychosocial deprivation	
Miscellaneous	Examples
Hydrocephalus	
Microcephaly	
Spina bifida	
Inborn errors of metabolism	Mucopolysaccharidoses, Lesch–Nyhan syndrome
Idiopathic	

Box 17.2 The genetic basis of disorders of intellectual development (DID)

Genetic factors of various kinds underlie DID.

Chromosomal abnormalities

An abnormal number of chromosomes (*aneuploidy*) invariably causes DID. *Trisomy* refers to possession of an extra copy of a chromosome; the classic example is trisomy 21 (Down syndrome). Absence of an autosome is not compatible with life, but an absence (or excessive number) of a sex chromosome is, and produces a variable phenotype in which mild ID is common (e.g. XO, Turner syndrome).

ID is 25% more common in boys, might be due to X-chromosome-linked causes given similar identification in fragile X syndrome, and many others have been identified as well. *X-linked DID* is a collective term used to describe these disorders. Their clinical picture can be particularly complex (e.g. in terms of girls being affected, and the severity of the phenotype).

Copy number variations (CNVs)

CNVs were introduced in Chapter 5. They comprise *deletions* and *duplications* of part of a chromosome. Note that we have two copies of each autosomal chromosome, one from each parent, and only one is affected, hence the more precise term 'hemideletion'. Occasionally, the clinical picture depends upon which parental chromosome is affected, via a process called *imprinting*. For example, in chromosome 15q11-13, the parental origin of the deletion determines whether Prader–Willi syndrome (paternal) or Angelman syndrome (maternal) is present. Some

parts of the genome are 'hotspots' for deletions and rearrangements, such as the q11 band of chromosome 22, which causes velocardiofacial syndrome (Box 11.9). The tips of the chromosomes (telomeres) also seem to be particularly vulnerable, with *subtelomeric deletions* thought to cause 5–10% of 'idiopathic' DID. Cumulatively, CNVs represent the most common cause of DID.

Single gene disorders

These are attributed to a defect limited to a single gene, and where the identity of the gene, and often the causative mutation within it, is known. This category includes most *inborn errors of metabolism*. Single gene disorders can be dominant or, more commonly, recessive, and can be autosomal or X-linked.

Note that these points illustrate that genetic factors do not in fact fall into discrete categories, but are more like a spectrum of causality. Thus, the size of the abnormality can be anything from a whole chromosome, to a CNV, to a mutation in a single nucleotide. Similarly, the impact of the genetic variation can range from being the sole cause of the syndrome through to acting as a minor risk factor. And since many genetic deficits arise *de novo* (i.e. are spontaneous mutations) in the individual (due to an error during gametogenesis, fertilization, or early development), they blur the distinction between genetic (in the sense of inherited) and environmental causes.

For review, see Vissers et al. (2016).

chromosomes instead of the usual two). About 95% of cases are due to trisomy 21. These cases result from failure of disjunction during meiosis, and are associated with increasing maternal age. The risk of recurrence in a subsequent child is about 1 in 100. The remaining 5% of cases of Down syndrome are attributable either to translocation involving chromosome 21 or to mosaicism. The disorder leading to translocation is often inherited, and the risk of recurrence is about 1 in 10. Mosaicism occurs when non-disjunction takes place during any cell division after fertilization. Normal and trisomic cells occur in the same person, and the effects on cognitive development are particularly variable.

Although trisomy 21 is one of the most widespread genetic causes of ID, it still poorly understood. Down syndrome pathology is assumed to be due primarily to

the increased 'dosage' of genes, and thus increased production of the proteins that they encode. This probably also accounts for the frequent early onset of Alzheimer's dementia, with approximately 80% of individuals with Down syndrome, developing Alzheimer's disease pathology by 40 years of age (see Box 14.13) (Ballard et al., 2016).

Fragile X syndrome and X-linked ID

Fragile X syndrome is the second most common specific cause of DID after Down syndrome, and is the most common inherited cause. It occurs in around 1 in 4,000 males and in a milder form in about 1 in 6,000 females. It accounts overall for about 10% of those with DID. The

Table 17.3 Notes on some genetic causes of disorders of intellectual development

Syndrome	Aetiology*	Clinical features	Comments**
Chromosome abnormalities and copy number variants			
Down syndrome	Trisomy 21	See text	
Fragile X syndrome	CGG trinucleotide repeat expansion of FMR1 gene on X chromosome	See text	
Edwards syndrome	Trisomy 18	Growth deficiency, abnormal skull shape and facial features, clenched hands, rocker-bottom feet, cardiac and renal abnormalities.	Rare. Early death common.
Cri du chat	Deletion of chromosome 5p, mostly sporadic.	Microcephaly, hypertelorism, typical cat-like cry, failure to thrive.	1 in 35,000. ID often severe. Hyperactivity. Early death common.
Prader–Willi syndrome	70% due to deletion of paternal 15q11; the rest are due to inheriting two copies of the same region of maternal chromosome 15.	Floppy infants. Small stature, hypogonadal. Excessive appetite, leading to obesity. Daytime sleepiness. Delayed motor milestones and speech difficulties. Skin-picking.	1 in 40,000. ID usually mild or moderate.
Angelman syndrome	Overlaps with Prader–Willi syndrome. Deletions and mutations affect <i>UBE3A</i> gene on 15q11.	Ataxia, hypotonia, and epilepsy. Small head, hook nose, characteristic face and wide mouth. Characteristic ataxic gait and frequent laughter, sociable.	1 in 10,000. Severe to profound ID with absence of speech. Hyperactivity.
22q deletion syndrome (velocardiofacial syndrome, DiGeorge syndrome)	Deletion involving chromosome 22q11; 10% inherited and 90% spontaneous.	Variable. Cardiac deficits, characteristic facial appearance with prominent nasal root/bulbous nasal tip, velopharyngeal insufficiency, short stature, long thin fingers.	1 in 2,000. ID variable, mild ID to borderline IQ typical. Poor social interactions with autistic features and ADHD. 25% develop schizophrenia-like disorders in adolescence (Box 11.9).
Williams syndrome (idiopathic infantile hypercalcaemia)	Deletion of 7q11, affecting elastin gene. Most cases are sporadic.	Distinctive facies with ‘elfin’ appearance, small stature, with harsh, brassy voice. Supra-valvar aortic stenosis common, failure to thrive.	1 in 7,500 to 1 in 18,000. Most have moderate ID, delayed but preserved language skills. Superficially sociable but lacks social flexibility and often anxious.
Inborn errors of metabolism			
Phenylketonuria	Autosomal recessive, causing lack of liver phenylalanine hydroxylase. Most common inborn error of metabolism.	Lack of pigment (fair hair and blue eyes), growth, delay, associated epilepsy, microcephaly, eczema, and hyperactivity.	1 in 10,000. Detectable by neonatal screening of blood or urine; treated by removing phenylalanine from diet early in life.
Galactosaemia	Autosomal recessive, deficiency of one of three enzymes involved in galactose metabolism (GALT).	Presents following introduction of milk into diet; failure to thrive, hepatosplenomegaly, cataracts.	Very rare. Detected by neonatal screening. Intellectual disability usually mild, if treated early (by galactose-free diet).

Table 17.3 Continued

Syndrome	Aetiology*	Clinical features	Comments**
Hurler syndrome (mucopolysaccharidosis type I (MPS I))	Autosomal recessive, affecting α -L-iduronidase, causing accumulation of dermatan sulphate and heparan sulphate.	Coarse facial features, protuberant abdomen, hepatosplenomegaly, cardiac abnormalities.	Rare. Severe disease with reduced life expectancy although outcomes have improved with modern treatment. MPS II (Hunter syndrome) and MPS III (Sanfilippo syndrome) are milder.
Lesch–Nyhan syndrome	X-linked mutation of <i>HGPRT</i> gene, leading to enzyme defect affecting purine metabolism. Excessive uric acid production and excretion.	Severe self-injurious behaviour, compulsive biting of fingers and lips. Development of spastic cerebral palsy and choreoathetoid movements.	Almost all are normal at birth but then develop low-mild ID. Very rare. Diagnosed pre- or postnatally, carrier screening possible. Death in early adult life.
Smith–Magenis syndrome	Deletion of chromosome 17p11.2, affecting the <i>RAI1</i> gene. Usually sporadic.	Broad flat face, brachycephaly, midface hypoplasia, short stature, hoarse voice, peripheral neuropathy.	Severe ID often with speech delay. Engaging personalities but severe self-injurious behaviour, tantrums, and sleep disturbance.
Other single gene disorders			
Neurofibromatosis 1 (von Recklinghausen syndrome)	Autosomal dominant, but 50% of cases are spontaneous mutations. Caused by neurofibromin gene at 17q11.	Neurofibromata, café au lait spots. Astrocytomas and meningiomas.	1 in 3,000. ID in 50%.
Tuberous sclerosis (epiloia)	Autosomal dominant, but variable penetrance and many cases sporadic. Caused by two genes: <i>TSC1</i> (hamartin) on 9q, and <i>TSC2</i> (tuberin) on 16p.	Epilepsy, facial angiofibromas, white skin patches, shagreen skin, retinal phakoma, subungual fibromata, multiple renal and other tumours.	1 in 7,000. ID usually mild. Autism and other psychiatric disorders common.
Duchenne muscular dystrophy	X-linked recessive; dystrophin gene on Xp21.	Males. Progressive muscle weakness, including respiratory and cardiac muscles.	1 in 4,000. 25% have ID. Verbal IQ lower than performance IQ. Depression and anxiety common.
Rett syndrome	MeCP2 gene mutation at Xq28	Females affected (lethal in males). Epilepsy, compulsive hand-wringing movements, agitation.	1 in 10,000. Normal early development, abnormalities beginning end of first year, leading to severe neurodevelopmental disorder with regression, with significant disability in adulthood (see Chapter 16).
Cornelia de Lange syndrome	50% of cases due to mutation in <i>NIPBL</i> gene; other genes include <i>SMC1A</i> , <i>HDAC8</i> , <i>RAD21</i> , and <i>SMC3</i> . All affect the cohesin complex, which plays a role in early growth.	Typical facial features including low anterior hairline and arched eyebrows, long eyelashes. Pre- and post-natal growth retardation. Ptosis, myopia, microcephaly, GOR 60%.	Typically severe ID but variable IQ. Self-injurious behaviour. Communication impaired.

*Numeral refers to chromosome number. p, short arm; q, long arm. **ID: intellectual disability.

Table 17.4 Some non-genetic causes of intellectual disability

Syndrome	Aetiology	Clinical features	Comments
Infection			
Rubella	Viral infection in first trimester.	Cataract, microphthalmia, deafness, microcephaly, congenital heart disease.	If mother infected in first trimester, 80–90% infants are affected.
Cytomegalovirus	Intrauterine infection.	Brain damage; only severe cases are apparent at birth.	60% of survivors have ID.
Toxoplasmosis	Protozoal infection of mother.	Hydrocephaly, microcephaly, cerebral calcification, renal problems, hepatosplenomegaly, epilepsy.	Variable.
Congenital syphilis	Syphilitic infection of mother.	Many die at birth; variable neurological signs, characteristic dental and cutaneous features (Hutchinson's teeth and rhagades).	Uncommon since routine testing of pregnant women.
Other			
Hydrocephalus	Sex-linked recessive; inherited developmental abnormality, e.g. atresia aqueduct, Arnold–Chiari malformation, meningitis, spina bifida.	Rapid enlargement of head. In early infancy, symptoms of raised CSF pressure; other features depend on aetiology.	Mild cases may arrest spontaneously; may be symptomatically treated by CSF shunt; intelligence can be normal.
Microcephaly	Recessive inheritance, irradiation in pregnancy, maternal infections.	Features depend upon aetiology.	ID usually severe.
Cerebral palsy	Perinatal brain damage; strong association with prematurity.	Spastic (common), athetoid and ataxic types; variable in severity.	Majority have below-average intellectual functioning, athetoid are more likely to be of normal IQ.
Hypothyroidism	Congenital hypothyroidism, iodine deficiency or (rarely) atrophic thyroid.	Appearance normal at birth; abnormalities appear at 6 months; growth failure, puffy skin, lethargy.	Now rare in UK; responds to early thyroxine replacement.
Foetal alcohol syndrome		Facial dysmorphology—thin upper lip, smooth philtrum, low-set ears, growth impairment, microcephaly.	1 in 3,000, depending on criteria. ID usually mild, or moderate. Hyperactivity and irritability.
Hyperbilirubinaemia	Haemolysis, rhesus incompatibility, and prematurity.	Kernicterus (choreoathetosis), opisthotonus, spasticity, convulsions.	Prevention by anti-rhesus globulin; treatment by exchange transfusion.

condition was so named because the X chromosome is 'fragile' when lymphocytes from affected individuals are cultured without sufficient folic acid.

As shown in Box 17.4, there are a number of characteristic but highly variable clinical features, none of which is diagnostic, including enlarged testes, large ears, a long face, and flat feet. Psychological features include abnormalities of speech and language, social anxiety, autistic behaviour and other social impairments, disorders of attention and concentration, and hyperactivity.

The inheritance of the condition, and the characteristics of carriers, are unusual and complex. Fragile X syndrome is caused by an amplified CGG repeat sequence in the *FMR1* gene, which leads to silencing of the gene and thus an absence of FMR1 protein, which regulates genes involved in synaptic function and plasticity, and the syndrome is thought to arise because the lack of *FMR1* interferes with the action of these other genes. Some of these genes are now therapeutic targets for fragile X syndrome, notably the metabotropic glutamate receptor mGlu5, but

Box 17.3 Features of Down syndrome

Moderate or severe ID

Placid temperament

Physical features:

- Slanted eyes and epicanthic folds
- Small mouth with furrowed tongue
- Flat nose
- Flattened occiput
- Stubby hands, fingers, and single transverse palmar crease
- Hypotonia with hyperextensibility of joints

Associated medical problems:

- Cardiac anomalies, especially septal defects
- Gastrointestinal abnormalities
- Atlantoaxial instability
- Susceptibility to infection
- Impaired hearing

Increased risk of leukaemia, hypothyroidism, and autoimmune disorders

Increased risk of adolescent and early adult depression

Box 17.4 Features of fragile X syndrome

More common in males

Caused by *FMR1* gene mutation

ID:

- Variable from mild to profound
- Increases late in childhood
- Performance IQ affected more than verbal IQ
- Poor attention and concentration
- Speech repetitive, lacking themes or content ('lit-any speech')

Behavioural features: autistic features common

Physical features:

- Large, protruding ears
- Long face with high-arched palate
- Flat feet
- Lax joints
- Soft skin
- Large testes (after puberty)
- Mitral valve prolapse

Note: All features are particularly variable.

as yet there are no established treatments. Folic acid is ineffective. For a review of fragile X syndrome, see Bagni et al. (2012) and Hagerman et al. (2017).

It is now realized that fragile X is one of a family of X-linked causes of DID, which together occur in up to 1 in

600 boys, account for about 16% of cases of male ID, and underlie the male excess of the condition. However, these disorders can also affect females; an extreme example is Rett syndrome (see Table 17.3 and Chapter 16), which only affects girls, probably because it is lethal to male embryos.

Assessment and classification

Different diagnostic processes have been developed to ensure that individuals with DID are assessed using more validated instruments as assessment needs can be different, especially when the individual has communication and cognitive difficulties affecting their emotional, perceptual, and thought-related experiences. The assessment can be directed towards five main areas:

1. the cause and severity of the disorder
2. intellectual and social skills development
3. associated medical conditions
4. associated psychiatric disorders
5. assessment of current function and needs.

A multi-axial classification has been developed to record some of this information, the DC-LD (*Diagnostic Criteria for Psychiatric Disorders for Use with Adults with Learning Disabilities/Mental Retardation*) (Royal College of Psychiatrists, 2001), namely:

Axis I: Severity of ID

Axis II: Cause of ID

Axis III: Psychiatric disorders:

Level A: Developmental disorders

Level B: Psychiatric illness

Level C: Personality disorders

Level D: Problem behaviours

Level E: Other disorders.

This system tries to consider and record all of the key components and highlights the fact that the assessment is both about characterizing the DID and its consequences, and also about making a specific diagnosis, where possible, as to its aetiology.

Severe DID is usually diagnosed in infancy, especially as it is often associated with detectable physical abnormalities or with delayed motor development. Some people with ID have specific developmental disorders—that is, impairment of specific functions greater than would be expected from the general intellectual level (see Chapter 16). The clinician should be cautious when diagnosing less severe ID on the basis of delays in development. Although routine examination of a child may reveal signs of developmental delay that might suggest possible ID, confident diagnosis often requires specialist assessment. Milder forms of DID often present when the child is of school age, and can include non-attendance at school, low academic achievements, and developmentally delayed interests and behaviour.

Full assessment involves several stages, including history-taking, examination of the mental state, physical examination, genetic and other laboratory investigations, developmental testing, functional behavioural assessment, analysis of the interactions between the individual with a DID, the family and the social support systems, and other aspects of adjustment. Each of these stages is considered in turn. Although this section is concerned mainly with the assessment of children, similar principles apply to assessments later in life.

For reviews of the assessment of DID, see DiStefano et al. (2020) and Eng et al. (2013).

History-taking

In the course of obtaining a full history, particular attention should be given to any family history suggesting an inherited disorder, and to difficulties or abnormalities in the pregnancy or the delivery of the child. Dates of passing developmental milestones should be ascertained. A full account of any behaviour disorders should be obtained. Details of any associated medical conditions, such as congenital heart disease, epilepsy, and cerebral palsy, should be documented.

Mental state examination

The approach should be flexible. Many individuals with DID can have difficulties with attention and concentration. Therefore, the interview may need to be carried

out rather informally while the person is engaged intermittently in some other interest. Questions should be simplified to take account of each person's receptive language and developmental level. Behavioural observations by family, friends, and carers are often more helpful than the observations that the assessor is able to make. Clinical assessment can also be supplemented by using adapted scales for the assessment of psychopathology.

Physical examination

A systematic physical examination should include noting the child's overall appearance, looking for dysmorphic features and other physical signs suggestive of specific disorders. Head circumference should be recorded. The parents' appearance may also be worth noting. Neurological examination is important and should include particular attention to impairments of vision and hearing.

As in the assessment of patients of normal intelligence, it is important to find out the person's previous baseline state before concluding that an item of current behaviour is evidence of a psychiatric disorder. Some long-standing behaviours, such as stereotyped movements or social withdrawal, may resemble and be mistaken for symptoms of schizophrenia or other psychiatric disorders—so-called *diagnostic overshadowing*. However, an increase in a long-standing behaviour may be the first, and sometimes the only, evidence of psychiatric disorder—so-called *baseline exaggeration*.

Genetic and laboratory investigations

As many cases of DID, especially moderate and severe forms, are caused by a specific chromosomal or genetic abnormality, genetic investigations are a major component of assessment and thorough testing often now take place when any difficulties first emerge; these assessments are usually done by paediatric and clinical genetics teams. A clinical geneticist can provide up-to-date information about the conditions for which screening is available and the cytogenetic and molecular genetic methods in use (since both are evolving rapidly) (Vissers et al., 2016), as well as help with the interpretation of test results, and sharing the information with the family and addressing their questions. Depending on the clinical suspicion, and informed by knowledge of the common causes of DID, a range of blood tests to detect inherited metabolic and endocrine disorders should be done, and brain imaging may also be indicated.

Developmental assessment

This assessment is based on a combination of clinical experience and standardized methods of measuring intelligence, language, motor performance, and social skills. Although the IQ is the best general index of intellectual development, it is not reliable in the very young or among people who have severe to profound degrees of intellectual impairments.

Tests used in developmental assessment

Standardized assessment instruments are widely used for screening, diagnosing, and assessing the severity of disorders of psychological development (Carr et al., 2016). Many assessments require special training for correct administration, but some do not. The choice of test is important to ensure it is suitable for the developmental level of the individual being tested. The type of test is also important. There are norm-referenced tests such as the Wechsler Adult Intelligence Scale (WAIS) (for adults with sufficient verbal and cognitive abilities) and other IQ measures; criterion-referenced tests which apply to particular skills without reference to population norms; tests of adaptive behaviour in social settings; and assessments of behavioural functioning. An abbreviated version of the WAIS, called the Wechsler Abbreviated Scales of Intelligence (WASI) (Psychological Corporation, 2004), is now available. Another quick useful method is the use of a Picture Vocabulary Test or another non-verbal test such as Raven's Progressive Matrices. Other commonly used instruments are summarized in Table 17.5 and include general, adaptive functioning, intelligence, language, and other aspects of development. All have good reliability and validity.

Functional behavioural assessment

The functional assessment of behaviour involves an assessment of events before, during, and immediately after the behaviour takes place. It is based on observations reported by family, carers, and members of the clinical team. It is concerned with abilities related to self-care, and social abilities, including communication, sensory motor skills, and social relationships.

Assessment of social interaction and adjustment

This assessment is concerned with the interaction between the individual with DID and the individuals

closely involved in their care. It is also concerned with opportunities for learning new skills, making relationships, and achieving more choice. If the person with ID has reasonable language ability, it is usually possible to obtain much of the information from them. If language ability is less well developed, the account has to be obtained mainly from informants. It is particularly important to obtain a complete description of any change from the usual pattern of behaviour. It is often appropriate to ask parents, teachers, or care staff to keep records of behaviours such as eating, sleeping, and general activity so that problems can be identified and quantified. The assessor should keep in mind the possible causes of psychiatric disorder outlined above, as well as unrecognized epilepsy.

Assessment of needs

As the diagnostic assessment progresses, the likely needs and wishes of the patient require careful consideration, too, as these will form the basis of the care package and management plan to be implemented. The needs assessment brings together the social, emotional, and health needs of the person, and considers their views and wishes as well as those of the other key individuals in their lives. This approach, sometimes termed 'person-centred planning', is now integral to most DID services.

A modification of the Camberwell Assessment of Need, known as CANDID (Camberwell Assessment of Need for Adults with Developmental and Intellectual Disabilities), is available for use.

Follow-up assessments

Once DID has been diagnosed, regular reviews might be required. For children, these are usually carried out by a multidisciplinary child health team, often in community paediatrics services, together with teachers and social workers. The child psychiatrist can work with the team, and has a particular role where the child has emotional, behavioural, or other psychiatric problems. As the child reaches school-leaving age, plans to ensure their ongoing educational, social, occupational, and broader needs are addressed, including preparing for more independent living, if appropriate. The need for ongoing specialist physical and psychological healthcare for adults with DID is important to make sure that their care remains appropriate; a specialist psychiatrist in ID psychiatry can be part of these multidisciplinary reviews, especially if there is a comorbid serious mental illness.

Table 17.5 Commonly used instruments for developmental assessment

Vineland Adaptive Behaviour Scale: measures adaptive behaviour across the life-course and assesses four domains: communication, daily living skills, socialization, and motor skills.	Autism Behaviour Checklist (ABC): questionnaire is completed by the parent or other carer. It is an older scale indicating problems in the area of autism and delay in social development and relatively easy to administer (Oro et al., 2014).
The British Ability Scales and Differential Ability Scales: measure a range of functions and educational attainments (spelling, arithmetic, reading), and can be used to calculate an overall IQ.	The Disability Assessment Schedule (WHO-DAS): focuses on adults and related social, developmental, language disorders, and behavioural problems.
ABAS (Harrison and Oakland, 2000): assesses adaptive behaviours within domains such as communication, functional academics, health and safety, self-care, social, home living, and community use.	British (Peabody) Picture Vocabulary Scale (BPVS): a test of language comprehension, suitable for non-speaking children. The test booklet is largely pictorial and the age range covered is 3–19 years. Although professional training is not required, the test is used mainly by psychologists and speech therapists.
The Portage Guide to Early Education: adopted for use internationally. It provides a broad developmental assessment including socialization, self-help, language, and cognitive and motor domains. Only a brief period of training is required for the assessor. The test requires the active involvement of the parent or other carer.	Reynell Scales of Language Development: assess comprehension and expressive language in the age range 1 month to 6 years. The test is of particular use with non-verbal children.
Griffiths mental development scales: 2–8 years of age. Consists of subscales of locomotor, personal–social, language, hand–eye coordination, performance, and practical reasoning.	Developmental, Dimensional and Diagnostic Interview (3Di): a parental autism interview that measures both symptom intensity and comorbidity across the full range of the autistic spectrum.
Bayley scales of infant development (Bayley III; Bayley, 2005): assesses children in the five key developmental domains of cognition, language, social–emotional, motor, and adaptive behaviour. Validated for 1–42 months.	Wechsler Adult Intelligence Scale (WAIS IV): gives full-scale IQ and verbal comprehension, perceptual reasoning, working memory, and processing speed scales. 16–90 years of age.
Wechsler Intelligence Scale for Children/WISC V: gives a full-scale IQ and measures cognitive functioning within four domains of verbal comprehension, perceptual reasoning, processing speed, and working memory. Validated for ages 6–16.	

The care of people with disorders of intellectual development

A historical perspective

Up until the 1960s, many individuals with moderate to severe DID in high-income settings were cared for in institutions with up to 50,000 people documented in such institutions by 1940 in the United Kingdom. In the 1960s the need for reform was recognized in several high-income countries, prompted in part by the changes in care away from institutions and towards community care that had already been effected in general

psychiatry. Campaigning by groups of parents, and public concern about the generally—and in some instances scandalously—poor conditions in which people with DID were housed, was also important. Surveys of hospitals for the ‘retarded’ showed that the mean IQ of their patients was over 70, and thus many residents had only mild ID and did not need hospital care. At around the same time it was shown that training was beneficial in DID across the severity range, and that there was an advantage to providing residential care in small home-like units.

It is now recognized in many high-income countries that people with DID should be integrated into and included by society, under a principle of *normalization*, as far as possible. However, there have been divergent views about the best way to achieve greater integration. In the United Kingdom, resources have been inadequate, and progress has been slow. In the United States, deinstitutionalization was carried out more quickly, with both successes and failures. Normalization implies that almost all people with DID should live in the community, as much as they are able, and participate in everyday community and societal activities and relationships, make choices, and have full social opportunities. Children are raised whenever possible with their families, and adults are encouraged to live as independently as is feasible. For the few who need special social and health care, accommodation and activities are designed to be as close as possible to those of family life. The concept of normalization has been further developed in the United States and elsewhere, and includes specialist educational and other help to enable people to achieve their full potential. Advocacy groups play an important role for individuals and their family members, and social media has provided many with the opportunity to extend their social contacts and opportunities to network and/or learn and engage more with a range of other individuals and organizations.

Specific services

The main elements in a comprehensive service for individuals with DID are as follows:

- the prevention and early detection of DID
- regular assessment of the person's attainments and difficulties
- advice, support, and practical measures for families
- provision for education, training, occupation, or work
- housing and social support to maximize self-care
- medical, nursing, and other services for those who require these forms of help as outpatients, day patients, or inpatients
- psychiatric and psychological services.

Preventive services

Primary prevention consists of genetic counselling, the education of pregnant women (and the population in general) about behaviours that may put the foetus at risk, and good obstetric and perinatal care. *Secondary prevention* aims to prevent the progression of impairment by either medical or psychological means. The latter

include 'enriching' education and early attempts to reduce behavioural problems. In LMICs, the incidence of DID could be substantially reduced by general measures to improve the health of mothers during pregnancy, and by better perinatal care.

Genetic screening and counselling

These measures begin with an assessment of the different vulnerabilities and risks a child with DID might carry. Such an assessment is based on study of the family history, on knowledge of the genetics of conditions that give rise to DID, and on awareness of the possibilities for genetic screening. The risks of screening are explained to the parents, who are encouraged to discuss them. Most parents seek advice only after having a child with DID to discuss risks with other children. Those who seek advice before starting to have children usually do so because there is a person with DID in the family. Providing a space to ensure families can discuss all possible options about pregnancy and its outcome is important. Therefore, all those involved in screening should be alert to psychological issues and have the appropriate counselling skills. Close liaison with a clinical geneticist is recommended, given the rapid developments in the indications for screening, the methods used, and the significant ethical and counselling issues involved.

Prenatal care

Prenatal care begins even before conception, with immunization against rubella for women who lack immunity, and advice on diet, alcohol, smoking, and folic acid supplementation.

Prenatal diagnosis overlaps with genetic screening, which is becoming available for an increasing number of conditions with the aim of providing information to those at risk of having children with certain DID, reassurance to others, and appropriate treatment of affected infants through early diagnosis. Maternal blood tests, ultrasound scanning, foetoscopy, chorionic villus sampling, and amniocentesis can reveal chromosomal abnormalities, most open neural-tube defects, and about 60% of inborn errors of metabolism. Amniocentesis carries a small but definite risk, and so is usually offered only to women who have previously carried an abnormal foetus, women with a family history of congenital disorder, and those with significant risk that the foetus they are carrying will have a serious condition or abnormality.

Rhesus incompatibility is now largely preventable. Sensitization of a rhesus-negative mother can usually be avoided by giving anti-D antibody. An affected foetus can be detected by amniocentesis and treated if necessary by

exchange transfusion. For pregnant women with diabetes mellitus, special care can improve the outlook for the foetus. Further information about these aspects of care can be found in textbooks of obstetrics and paediatrics.

Postnatal prevention

In the United Kingdom, all newborns are tested for a number of congenital and inborn errors of metabolism, such as phenylketonuria. Screening for elevated levels of lead has been advocated, especially in areas where these are known to be high. Intensive-care units and improved methods of treatment for premature and low birth weight infants can prevent DID in some who would previously have suffered brain damage or died.

Help for families

As noted earlier, having a child with DID can have a major emotional and practical impact on the family, and support is often needed for the family from the time that the diagnosis is first made. It is not enough to give worried parents a full explanation on just one occasion. They may need to hear the explanation several times before they can absorb all of its implications. Adequate time must be allowed to explain the prognosis, indicate what help can be provided, and discuss the part that the parents can play in helping their child to achieve his or her full potential. Community paediatricians, health visitors, and sometimes clinical geneticists are usually involved in this process.

Thereafter, the parents might benefit from continuing support to help them and the child access education, outside school activities, support, respite care if needed, as well as their often-multiple health care needs. Families are likely to need extra help when their child reaches puberty or leaves school. Making the transition from child to adult services can be stressful and planning to support this transition can be valuable. Both respite day and overnight care can be required to relieve carers and to encourage the individual to become more independent.

Education, training, and occupation

One aspect of the policy of normalization is that children with DID might benefit from being educated as far as possible within inclusive preschool settings and mainstream schools. The extent to which this is done varies in different countries and also in different regions of the same country and it is important to ensure that they do not experience marginalization at school with their peers.

Compensatory education is intended to provide optimal conditions for the development of the child with

DID. An early example was the Head Start programme in the United States, which provided extra education for deprived children, including nursery schooling and attempts to teach specific skills. Many of the results were disappointing, but subsequent early intervention studies can be effective, especially if they are family centred. However, many uncertainties remain about the components and the delivery of such help. Research has consistently shown the improved cognitive and social outcomes for children with DID, if educational interventions are commenced early and if possible in inclusive play group or nursery classes (Guralnick, 2005). When school age is reached, the children with milder impairments can be supported within mainstream classes and if needed attend remedial classes. Others might benefit from attending special educational programmes for children with DID. It is still not certain which children might benefit from mainstream schooling, have better opportunities for social integration, and an expectation of progress. However, there may exist a lack of special teaching skills and equipment, as well as the children potentially feeling excluded from their peer group. Also, the methods of teaching that emphasize self-expression might not be optimal for some children with DID who might benefit from special teaching of language and communication. Another advantage of educating children in mainstream schools is that other pupils might better appreciate how to include them into their everyday lives.

Before children with DID leave school, they need reassessment and vocational guidance. The resources and opportunities available to young people with ID have an unfortunate tradition of being inadequate and not meeting the complex needs of those with DID, as well as lack of progress being exacerbated by many accepting limited aspirations set for them by themselves and others (Talbot et al., 2010). Most young people with mild ID are able to take regular employment or enter sheltered employment. Adults with severe disabilities are likely to transfer to adult day centres, which can provide a wide range of activities if the abilities of each attendee are to be developed as much as possible.

Residential care

It is now widely accepted that children with DID are likely to have better outcomes if their parents are supported in caring for them within the family home, if possible. If the care needs cannot be met then finding other arrangements will depend on what is available and possible, and often another family home is sought in the first instance. Adults with DID should be supported in regular or supported housing, or placed with a family

or in other suitable lodgings, such as small residential group homes. Staff often support these individuals to develop their social skills and to live as independently as possible.

Medical services

Individuals with DID need equitable access to general and specialist medical services as do other citizens, but those with DID often require extra support in order to do so. This care is usually obtained from usual primary care services, and health teams often need to allocate additional time and resources for these individuals. Integrated care with identified individuals to coordinate their care is often helpful for the individuals and their families, a role sometimes conducted by specialist nursing staff. Shared care between neurologists and

psychiatrists can improve outcomes for some patients—for example, in the diagnosis and treatment of epilepsy. As a result of their increased health needs, it is generally recommended that annual health checks are offered. This helps to screen for treatable conditions and improve their quality of life.

Psychiatric services

Psychiatric care is an essential part of a comprehensive community service for individuals with DID. In some countries it is provided by generic mental health services, but in the United Kingdom, it is generally provided by staff who specialize in the care of people with ID or by 'learning disability services'. Teams often consist of psychiatrists, specialist nurses, psychologists, and social workers.

Treatment of psychiatric disorder and behavioural problems

Psychiatric disorder in individuals with DID is often first identified by changes in behaviour. It should be remembered, however, that behavioural change can also result from physical illness or from stressful events, both of which should be carefully excluded. In the most severe cases, and especially those with sensory deficits, behavioural disturbance may be due to under-stimulation and frustration due to the inability to communicate wishes and needs. Once the cause is clear, the treatment can be planned. Physical illness should be treated promptly, stressful events reduced if possible, or a more stimulating environment provided when appropriate. If the disturbed behaviour results from a psychiatric disorder, the treatment is similar in most ways to that for general patient populations. Carers are often involved in behavioural assessment and treatment, and it is important to support them adequately.

The most serious and persistent disorders occasionally require hospital admission for more intensive behavioural management, which may be combined with pharmacotherapy.

Medication

Psychotropic medications are frequently prescribed for individuals with DID, often at high doses and these medications are associated with both positive and negative outcomes (Costello et al., 2022). The indications for psychotropic drugs are generally the same as for general

patients, and the full range of such medication should be available for individuals with DID. However, the psychiatrist has particular responsibility for organizing effective ongoing monitoring, including regular physical examination and trying to avoid polypharmacy. The latter is especially important for patients with severe communication impairments who cannot describe adverse effects. Also, neurologically impaired patients may develop adverse effects at lower doses and suffer from over-sedation, delirium, and extrapyramidal symptoms. Antipsychotic and benzodiazepine drugs are often useful in the short-term control of behaviour problems, and generally at much lower doses than would be prescribed in the general population. There are few controlled trials of the efficacy and tolerability of antipsychotic drugs in the longer-term treatment of abnormal behaviour. Nonetheless, clinical experience suggests that severely behaviourally disturbed patients who do not respond to psychosocial interventions sometimes benefit from prolonged use of medication, with frequent monitoring of its effects and adjustment of dosage. For a review, see Costello et al. (2022). Anti-epileptic drug treatment is frequently required, given the prevalence of epilepsy in DID.

Psychological treatment

Although limited understanding of language sets obvious limitations to the use of psychotherapy, it can be

useful in some settings and there has been substantial change in interventions offered and services developed over the last decade, with the result that more therapies are being offered with the opportunity to expand the evidence base. More patients can benefit from a modified or simplified approach and psychological therapies are a key intervention for individuals with DID when experiencing distress. Some of the key psychological therapies currently used include behaviour therapy, cognitive behaviour therapy, psychodynamic psychotherapy, systemic family therapy, and a range of arts therapies (Talbot et al., 2010). Cognitive therapies can be attempted with some patients with higher levels of verbal ability although the most useful psychological treatment takes the form of behavioural therapies. These may be, and are, used for a wide variety of problems (e.g. challenging behaviour in a wide variety of patient abilities). Counselling for parents is an important part of treatment. If more formal family therapy is undertaken, families might appreciate structural approaches, which address the problems and solutions that are relevant to them. Some children with profound DID can be helped by the use of play and sensory stimulation to encourage developmental advances. For a review of these therapies, see Beail (2017).

Behaviour modification

Behavioural methods are helpful to individuals with severe DID as some of the methods do not require language, and can be used to encourage basic skills such as washing, toilet training, and dressing. Often family members and teachers can learn these skills to help maintain progress in the individual's everyday environment. If the problem is an undesired behaviour, it might help to rule out and remove any environmental triggers. Carers should avoid rewarding certain behaviours while also reinforcing alternative adaptive responses. Aggressive behaviour might best be managed by ignoring unwanted behaviours, although a range of strategies might be needed to avoid those that negatively reinforce the behaviours. Rewarding adaptive behaviours with material or social rewards can help, in a similar manner to how these are used with young children. Reward should be given immediately after the desired behaviour has taken place (e.g. using the toilet). For training in skills such as dressing, it is often necessary to provide modelling and prompting in the early stages, and to reduce them gradually later.

Ethical and legal issues

Normalization, autonomy, and conflict of interests

The policy of normalization has encouraged individuals with DID to live as near to normal lives as possible. Normalization can produce unintended effects. For example, it requires that children with DID should be educated in mainstream schools whenever possible. However, in secondary schools, children with special needs were found to be bullied substantially more often than other children. It could be argued that it is in the immediate interests of an individual child to be educated in a non-mainstream school where they are less likely to be bullied, whereas the long-term interests of children with DID as a group may be advanced more by a policy of education in mainstream schools while making strenuous efforts to eradicate bullying.

Normalization also leads to practical ethical questions about sexual activity, contraception, and parenting, as mentioned earlier. The social inclusion of individuals with DID includes being able to participate

in social networks, having friends, holding membership of clubs, and being able to travel.

Consent to treatment

Many individuals with severe DID are unable to give informed consent, and it is essential to be aware of local legislation and practice. In the United Kingdom, these issues are covered by the Mental Capacity Act (see Chapter 4) which includes the Deprivation of Liberty Safeguards (DoLS), under which the practitioner is required to try to explain the nature of the procedure in a way that the patient can understand. If the patient is deemed to lack capacity, steps should be taken to help the person understand through assistive methods. The least restrictive option in keeping with the best interests of the individual with DID should be pursued usually through the involvement of a person closely involved in their care and who knows them well (Kelly, 2015). An advocate may also be involved. If there is doubt and inpatient admission is being sought, it is good practice

to discuss matters with the local Mental Capacity Act office and hold a 'best interests' meeting. Discussing the case with an experienced practitioner is also helpful.

Seriously ill patients who refuse potentially life-saving treatments can prove difficult to manage in general medical settings. If the patient is sufficiently impaired as to not understand the nature of the choice they face and there is a medical emergency, a Mental Capacity Act assessment should be completed by the medical team. If there is time, it may be necessary to refer the case for review in court.

Consent to research

Consent to research requires the ability to understand information, to use the information rationally, to appreciate the consequences of situations, and to decide between alternatives (see Chapter 4). Any individual who

has agreed to take part in research should understand that they can withdraw consent if they wish—a point that should be explained with particular care. In general, no research should be undertaken involving people who cannot consent, unless the same research cannot be successfully carried out without involving these people. It is important to remember that individuals with DID vary in their capacity to provide informed consent. Some may be able to provide verbal rather than written consent. Some may need sign language, pictures, or written information in simple language and a large font size. For some the proposal may have to be discussed repeatedly. It is also good practice to have agreement with the carers and the multidisciplinary team members. In every case the research should follow strictly the relevant research governance criteria and, in the United Kingdom, the Mental Capacity Act (see Chapter 4).

Further reading

Bhaumik, S., Branford, D., Barrett, M., & Gangadharan, S. K. (2015). *The Frith Prescribing Guidelines for Adults with Learning Disability*, 3rd edn. Wiley-Blackwell.

Matson, J. (2020). *Handbook of Intellectual Disabilities. Integrating Theory, Research and Practice*. Springer.

CHAPTER 18

Forensic psychiatry

Introduction 493

General criminology 494

Causes of crime 496

The association between mental disorder and crime 497

Specific offender groups 501

Psychiatric aspects of specific crimes 502

Psychiatric aspects of being a victim of crime 508

The role of the psychiatrist in the criminal courts 509

The treatment of offenders with mental disorder 511

The management of violence in health care settings 514

Risk assessment 515

The psychiatric report 517

Introduction

Chapter 4 covered general legal and ethical issues in the practice of medicine and psychiatry. This chapter is concerned with other aspects of psychiatry and the law covered by the term *forensic psychiatry*, which is used in two ways.

1. Narrowly, it is applied to the branch of psychiatry that deals with the assessment and treatment of mentally disordered offenders.
2. Broadly, the term is applied to all legal aspects of psychiatry, including the civil law and laws regulating psychiatric practice.

Forensic psychiatrists are concerned with both of these issues. In addition, they also assess risk and treat people with violent behaviour who have not committed an offence in law. They have a growing role in the assessment and treatment of victims.

Offenders with mental disorders constitute a minority of all offenders, but they present many difficult problems for psychiatry and the law. These include legal issues, such as the relationship between the mental disorder and the crime, which may affect the court's determination of responsibility, and practical clinical questions, such

as whether an offender needs psychiatric treatment, and finding the appropriate setting for that treatment.

The psychiatrist therefore needs knowledge not only of the law but also of the relationship between particular kinds of crime and particular kinds of psychiatric disorder. Mental health services form only a small part of the social and legal response to criminal activities. The psychiatrist who is working with offenders needs to be able to liaise with others in the criminal justice system, such as lawyers, prison staff, and probation officers. The application of concepts of deviance, guilt, and legality are influenced by people in legal, political, and social systems, and the prevailing policies and laws, as well as by clinical issues.

In reading this chapter it is important to be aware of the very large differences between countries and jurisdictions, with substantial national variations in epidemiology, definitions of crime, legal practice, and the role of the psychiatrist. Readers need to be aware of legal issues and procedures in their jurisdiction. In the following account, the situations and procedures in the United Kingdom (and more especially England and Wales) are used as examples to illustrate general themes. Ethical issues are summarized in Box 18.1.

Box 18.1 Ethical issues in forensic psychiatry

The principal ethical issues relate to 'boundary problems'. The psychiatrist and others involved need to be clear about accountability to legal authorities, etc., rather than to the individual who is being assessed and treated. This applies to a variety of activities:

1. Preparation of medicolegal reports. In the United Kingdom, the responsibility of the psychiatrist is to

the Court rather than to the patient or their legal representative.

2. Assessment of risk. Assessment of the risk to others may be paramount.
3. Voluntary and compulsory treatment. The primary reason for treatment may involve decreasing the risk of harm to others.

General criminology

There is an extensive literature on criminology and many theories of the causes of crime; interested readers are referred to criminology textbooks (see Liebling et al., 2017). Most theories have emphasized the sociological aspects of crime and deviance, particularly social and economic causes of crime in the family, peer group, and subculture. These include poverty, poor schooling, and unemployment. Predisposing social and individual factors are often interrelated, so simple conclusions are rarely possible. However, societal and social influences can be more important than individual factors considered in isolation; for example, genetics, biology, and psychological traits are all relevant. Most correctional and forensic mental health programmes rely on psychological assessment of risk factors to encourage, for example, the development of coping skills and understanding how to challenge offending behaviours (Douglas et al., 2017).

Prevalence

The latest figures (2024) for crime in England and Wales are reported in the Office for National Statistics Bulletin (Office for National Statistics, 2025). The findings suggest that crime against individuals and households has fallen over the last ten years, with some exceptions, such as sexual assault. The latest Crime Survey for England and Wales (CSEW) estimated 9.5 million incidents, which includes theft, robbery, criminal damage, fraud, computer misuse, and violence with or without injury, in the survey year ending September 2024. This was 12% higher than the previous year (8.5 million incidents, mainly because of a 19% rise in fraud (to around 3.9 million incidents). Over the last decade, there has been a gradual decrease in domestic abuse, but an increase in

sexual assault. The CSEW for year ending September 2024 showed no significant changes in these experiences compared with the previous year, except for stalking:

- 4.5% of people aged 16 years and over had experienced domestic abuse in the last year (approximately 2.2 million victims)
- 2.8% (around 1.4 million people) had experienced stalking, a 0.6 percentage point decrease from 3.4% (1.6 million victims)
- 2.2% (approximately 1.0 million people) had experienced sexual assault
- 9.0% had experienced some form of harassment (around 4.4 million people; this estimate is not comparable with the previous year).

Police recorded crime does not provide reliable estimates of general crime trends, but it can offer insights about lower-frequency-high-harm crimes. The following figures reflect reported crimes for the year ending September 2024, compared to the previous year:

- homicides decreased by 4% (to 550 offences) compared with 2023 (572 offences)
- offences involving knives or sharp instruments increased by 4% (to 55,008 offences) compared with 2023 (52,969 offences)
- offences involving firearms decreased by 17% (to 5,372 offences) compared with 2023 (6,499 offences)
- robbery increased by 4% (to 82,347 offences) in comparison with 2023 (78,865 offences)
- shoplifting offences rose by 23% (to 492,914 offences) compared with the previous year (402,220 offences).

Each country will have its own approach to recording and measuring crime, so care is needed when comparing

across countries. Yet, there are attempts to compare homicide rates and organized crimes, for example, using the organized crime index (see <https://ocindex.net/>).

Crime time trends

Between 2003–04 and 2013–14 crime levels fell across England and Wales, from 6 million to around 4 million. There were yearly increases after 2013–14, leading to a peak of offences in the most recent year. In 2021–22, over 6.3 million offences were recorded by the police in England and Wales, the highest since 2002–3. During the same period, Scotland recorded 286,500 crimes and Northern Ireland a further 106,600. Given population growth, the actual crime rate for 2019–20 is lower than the early 2000s (detailed breakdown in Statista, 2022).

Covid-19 and Crime Survey for England and Wales (2022)

Patterns of crime were substantially affected by the coronavirus (Covid-19) pandemic and government restrictions on social contact (Office of National Statistics, 2025). During the pandemic, data were collected by telephone. However, the 2022 estimates are directly comparable with pre-coronavirus pandemic, except the period of data collection was nine months rather than the usual 12-month period, potentially producing lower response rates and more uncertain estimates. The figures from 2022 show, since March 2020, total crime decreased by 8%, including:

- theft, which decreased by 19%
- fraud significantly increased during the pandemic but by June 2022, fraud estimates returned to pre-coronavirus levels. Interestingly, this suggests increases were specific to the coronavirus pandemic.
- computer misuse offences increased during the coronavirus pandemic; however, by June 2022 estimates showed a 27% decrease compared with pre-coronavirus pandemic levels. Thus again, any real increase was likely to be specific to the pandemic.

Police recorded data focus on more severe crimes. Since the year ending March 2020, there were decreases in crime for the year ending June 2022 (reductions as a %):

- burglary (28%)
- robbery (23%)
- vehicle offences (19%)
- knife-enabled crime (9%)
- offences involving firearms (10%)
- homicide (5%).

These may not be long-term trends. Police recorded sexual offences increased by 21% compared with the pre-pandemic year ending March 2020. This continued increase may again reflect the impact of high-profile cases and campaigns on victims' willingness to report incidents.

National differences

There are large national differences in the rates and patterns of crime (Harrendorf et al., 2010). Within Europe, overall crime rates in the United Kingdom were lower than those in the Netherlands but higher than those in the Scandinavian countries. Crimes of assault are more common in the United States and Australia. National statistics are affected by local legislation, the recording of crimes, and by the conduct of legal proceedings.

Gender

Crime is predominantly an activity of young men. In England and Wales, 50% of all indictable offences are committed by males aged under 21 years of age, and 25% by those under 17 years of age. Men represent at least 80% of offenders (Monahan, 1977), and, in most western countries, male prisoners outnumber female prisoners by 30:1. This gender difference is reflected in forensic mental health services, where the majority of patients are male.

Ethnicity

Ethnic minority groups are usually overrepresented in prisoner populations (Hassan et al., 2011). Minority groups are more likely to be poor, unemployed, and living in poor housing, all of which are risk factors for crime. Consequently, they are overrepresented in forensic patient populations. There is evidence of Black offenders being more likely to be remembered and receiving more severe penalties (see Hood, 1992). Thus, the high rates of arrests and convictions in ethnic minority groups may also reflect discrimination at all stages of the criminal justice system—from stop and search through to trial and sentencing. This is especially worrisome if those with mental illness are criminalized, and end up receiving care through the criminal justice system (Bhui et al., 2015).

Victims

Most property crime is committed against strangers, but most serious interpersonal violence (rape, homicide, or child abuse) is committed by offenders known to the victim. Women are exposed to sexual violence from men in both high- and low-income countries.

Women and children are particularly at risk in countries experiencing armed conflict, where about 70% of casualties are non-combatants (Bergen et al., 2005). Victims of crime tend to be disadvantaged groups

living in poorer parts of urban communities, so those with mental illness are more likely to be victims of crime, and the risk is further increased as a consequence of their illness.

Causes of crime

Criminal behaviour (crime) needs to be distinguished from rule-breaking behaviour. Not all rule-breaking or socially unacceptable behaviour is criminal, nor is all criminal behaviour violent. In fact, the vast majority is not, so general theories of the causes of crime will not necessarily address the causes of violence. Aggressive behaviour is not always violent, as it can be constrained by social rules. Violence is aggressive behaviour that transgresses social norms, as in street fighting as opposed to, for example, boxing.

Many factors determine whether an individual is aggressive in a particular situation. These include personality, the immediate social group, the behaviour of the victim, disinhibiting factors such as alcohol or drugs, general environmental factors such as noise and social pressure, physiological factors such as fatigue, hunger, and lack of sleep, and the presence of mental abnormality.

Genetic and physiological factors

Early studies of twins suggested that concordance rates for criminality were substantially greater in monozygotic twins than in dizygotic twins. Results from well over 100 behavioural genetics studies with different designs—including twin studies, studies of twins reared apart, and adoption studies—have converged on the conclusion that antisocial and aggressive behaviour have a considerable genetic basis. Estimates of the variance that is attributable to genetics vary, but several meta-analyses estimate heritability at between 40% and 60%, broadly consistent between genders and ethnicities (Glenn and Raine, 2014).

The gender difference in offending has raised the question of the influence of either the Y chromosome or testosterone levels on offending. However, there is little evidence that chromosomal or hormonal abnormalities are causally associated with criminal behaviour or aggression. However, lower brain 5HT levels are associated with impulsiveness and aggression (see Chapter

15). There is evidence that individuals with sociopathic personalities have longstanding neuropsychological deficits, particularly in executive processes (Moreira et al., 2019).

Psychosocial factors

Individual psychological development interacts with social factors and cultural values to make offending more likely (see Box 18.2). Rule-breaking and antisocial behaviour often start in childhood or early adulthood:

- Follow-up studies of delinquent young people show that early patterns of antisocial behaviour are likely to persist into adulthood.
- Delinquency is associated with harsh parenting styles and poverty.
- Exposure to physical abuse or neglect in childhood significantly increases the risk of violent offending in later life, for both men and women (Rutter, 2005).

The association between childhood adversity and violence may have several mechanisms. Abused and neglected children may have a heightened perception of threat from others. They may show less trust (Neil et al., 2022). They may have decreased capacities to form successful interpersonal relationships owing to possible childhood neglect, a decreased empathy for others, or reduced capacity for self-awareness. Alternatively, they may have a decreased capacity to manage arousal or regulate anger or anxiety arising from excessive exposure to fear experiences. It is also possible that children and parents simply share genetic factors that impair affect regulation or the ability to empathize. However, it is always important to consider the impact of resilience or vulnerability factors, such as temperament (Rutter, 2005). There is an interaction between maltreatment and individual differences in risk and protective genes, leading to different neurobiological circuitry, which can predispose to later psychopathology, including conduct disorders and externalizing symptoms (McCrory et al., 2012). Furthermore, there is increasing debate about the

Box 18.2 Psychosocial risk factors for offending

Individual and family factors

Prenatal and perinatal influences—birth complications, maternal rejection, foetal neurological maldevelopment, maternal nicotine and alcohol consumption
Hyperactivity and impulsivity
Low intelligence
Punitive parenting style—poor supervision, harsh discipline, rejection
Teenage mothers
Parental conflict—separations
Parents in the criminal justice system
Large family size

Social factors

Socioeconomic deprivation
Peer influences
School influences
Community influences

Note: The risk factors were listed among many in a review by Farrington (2009).

Source: Farrington, D. P. (2009). Psychosocial causes of offending. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp.1908–1916). Copyright (2009), Oxford University Press; Glenn, A. L., & Raine, A. (2014). Neurocriminology: Implications for the punishment, prediction and prevention of criminal behavior. *Nature Rev Neurosci*, 15(1), 54–63. Copyright (2014), Nature Publishing Group.

age of criminal responsibility which can start in some countries as low as ten years, and how, with increased understanding of the adolescent brain, there might need to be more nuanced approaches to adolescents, such as progressive legal responsibility according to adolescent brain development.

Psychiatric causes

A small but important group of offenders exhibit criminal behaviour that seems to be partly explicable by specific psychological or psychiatric abnormalities. This group particularly concerns the psychiatrist, and is discussed in the next section.

The association between mental disorder and crime

The association between mental disorder and violence has been repeatedly demonstrated (Fazel et al., 2009) conducted a systematic review of violence and schizophrenia and other psychoses. This meta-analysis of 20 studies, including 18,423 individuals, found that schizophrenia and other psychoses were associated with violence (especially homicide). However, this risk was mediated by comorbid substance abuse, and was the same for individuals with substance abuse without psychosis. The increased risk of violent offending in both men and women with a diagnosis of affective psychosis could be accounted for by alcohol and substance misuse. Alcohol and substance misuse by themselves are associated with a substantially increased risk of violent offending, as is personality disorder.

Research on the prevalence and association of mental disorder in prisoners has recognized limitations:

- Not all criminals are brought to trial and found guilty.
- Not all criminals go to prison (potential sampling bias).

- Mentally disordered offenders may be diverted away from courts and prisons, or they may not be prosecuted.

The proportion of prisoners with mental disorders is higher than the proportion of the general population with such disorders. In a meta-analysis of 62 psychiatric surveys of prisoners, (Fazel and Danesh, 2002) found that about one in seven had a treatable psychiatric illness (see Table 18.1). A larger number suffer from anti-social personality disorder. Hassan et al. (2011) surveyed the mental health of over 3,000 prisoners on admission and found up to 17% with mental illness and over half with drug or alcohol misuse.

The nature of the association

A causal relationship between mental disorder and crime is difficult to show empirically, especially if the type of crime is not defined. The finding that mentally

Table 18.1 Relative risk estimates of violent crime from the Danish Birth Register

Disorder	Male	Female
Schizophrenia spectrum disorder	4.8	13.7
Affective disorders	3.5	4.3
Learning difficulties	1.7	—
Personality disorders	5.7	7.9

Reproduced from Stevens, H., Laursen, T. M., Mortensen, P. B., et al. (2015). Post-illness-onset risk of offending across the full spectrum of psychiatric disorders. *Psychol Med*, 45(11), 2447–2457. Copyright (2015), with permission from Cambridge University Press.

disordered individuals are overrepresented in prison does not necessarily mean that their mental disorder caused them to offend. Nor is criminal law-breaking an indicator in itself of mental disorder, no matter how heinous or bizarre the behaviour.

Only a small minority of all people who commit violent acts have serious mental illness such as psychosis. Swedish national registers linking hospital admissions and criminal convictions over 13 years demonstrated a population-attributable risk fraction of 5.2% (i.e. only 5% of convictions were accounted for by individuals with severe mental illness). This attributable risk fraction was higher in women across all age bands. In women aged 25–39 years, it was 14.0%, and in those aged 40 years or over, it was 19.0% (Fazel and Grann, 2006). In a more recent Swedish register study, Fazel et al. (2014d) showed that, within five years of their first diagnosis of schizophrenia, 10.7% of men and 2.7% of women

were convicted of a violent offence. The vast majority of patients with psychotic illnesses are no more dangerous than members of the general population. There is no evidence that homicidal behaviour is becoming more common in people with mental illness—indeed, it appears to have been fairly stable over time in recent decades (Simpson et al., 2022).

For a review of the association between mental illness and offending, see Whiting et al. (2021) (see also Box 18.3). Generally, people with mental illness are more likely to be victims of violence than perpetrators (see Ghiasi et al., 2022 for a brief review).

Specific psychiatric disorders

Substance dependence and crime

There are close relationships between substance abuse and crime, which have substantially affected legislation, enforcement, and national policies (Grann and Fazel, 2004).

Alcohol and crime are related in three important ways:

1. Alcohol intoxication may lead to charges related to public drunkenness or to driving offences.
2. Intoxication reduces inhibitions and is strongly associated with crimes of violence, including murder.
3. The neuropsychiatric complications of alcoholism may also be linked with crime.

Offences may be committed during alcoholic amnesias (blackouts), which are periods of several hours or days and which the heavy drinker cannot recall, although

Box 18.3 Psychiatric disorder and offending

- People with psychotic disorders are more likely than members of the general public to acquire convictions for violent or other crimes (by factors of approximately 4 and 10, respectively).
- This increased likelihood is altered in strength by local factors such as crime rate and sociodemographic variables.
- Antisocial personality disorder and substance misuse disorders have stronger associations with offending than does psychotic illness.
- A combination of psychiatric disorders (particularly when one of them is substance misuse disorder) may

be more relevant than any single category of psychiatric disorder.

- Most offending by those with psychiatric disorder is minor in nature; violence, when it occurs, is likely to be targeted at a family member.

Source: Thomson, L., & Darjee, R. (2009). Associations between psychiatric disorder and offending. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1917–1926). Copyright (2009), Oxford University Press; Mullen, P.E., Mackenzie, R., Ogloff, J. R., et al. (2006). Assessing and managing the risks in the stalking situation. *J Am Acad Psychiatry Law*, 34(4), 439–450. Copyright (2006), American Academy of Psychiatry and the Law.

at the time they appeared normally conscious to other people and were able to carry out complicated actions. However, the association is complex, and social factors related to drinking may be as important as alcohol itself. For a review, see Johns (2009).

Intoxication with drugs may lead to criminal behaviour, including violent offences. Drug misusers, especially those who are dependent on heroin and cocaine, commit repeated offences against both property and people to fund their drug habit. Some of the offences involve violence. Rates of drug abuse are increased among prisoners, and many succeed in obtaining drugs in prison. Involvement in criminal activity and with other criminals may lead to drug usage. For a review of the relationship between drug dependence and crime, see Johns (2009).

Intellectual disability

There is no evidence that most criminals are of markedly low intelligence. However, reduced intellectual ability is an independent predictor of offending (Holland et al., 2002).

People with disorders of intellectual development may commit offences because they do not understand the implications of their behaviour, or because they are susceptible to influence and exploitation by others. Property offences are the most common, but sexual offences are overrepresented, particularly indecent exposure by males (Perry et al., 2002). The exposer is often known to the victim and therefore the rate of detection is high. There is also an association between intellectual disability and arson (Taylor et al., 2002).

Mood disorder

Depressive disorder is sometimes associated with shoplifting. Severe depressive disorder may rarely lead to homicide. The depressed person may be deluded, for example, that the world is too dreadful to live in, and they consequently kill their family members to spare them. Suicide often follows. A mother suffering from postpartum disorder may sometimes kill her newborn child or her older children. Rarely, a person with severe depressive disorder may commit homicide because of a persecutory belief—for example, that the victim is conspiring against them. Occasionally, ideas of guilt and unworthiness lead depressed patients to confess to crimes that they did not commit.

Manic patients may spend excessive amounts of money on expensive objects, such as jewellery, or cars, that they cannot pay for. They may be charged with fraud or theft, and acting under false pretences. They are

also prone to irritability and aggression, which may lead to offences of violence, although this is seldom severe.

Schizophrenia and related disorders

Psychotic illnesses may be associated with violence, especially when paranoid symptomatology is present, or the patient also has a substance abuse problem. Violence may occur because the offender is frightened, and self-control may be reduced by the presence of the psychotic state. Any state in which paranoid psychotic symptoms feature carries an increased risk of violent behaviour.

Epidemiology

Epidemiological studies have strongly suggested that schizophrenia is associated with an increased risk of both violent and non-violent offending (Fazel et al., 2014). This risk is substantially increased by substance misuse, but the proportion of violent crime attributable to schizophrenia is low. A number of clinical risk factors for violence in schizophrenia have been proposed, including the following:

- fear and loss of self-control associated with non-systematized delusions
- irresistible urges
- instructions from hallucinatory voices (command hallucinations)
- dual diagnosis, particularly substance misuse
- a strong negative affect (e.g. depression, anger, agitation).

Coid et al. (2013) found that six out of 32 types of delusions generated anger in the patient and, of these, three (persecution, being spied on, and conspiracy) resulted in increased risks of violence. Risk assessment is discussed on pages 53, 54. Violent threats made by patients with psychosis should be taken very seriously (especially in those with a history of previous violence). Most serious violence occurs in those already known to psychiatrists, particularly if there is an identifiable victim.

Post-traumatic stress disorder

Post-traumatic stress disorder (PTSD) may be related to offending in three ways:

1. Patients with PTSD may abuse drugs and alcohol.
2. PTSD is associated with increased irritability and decreased affect regulation.
3. Patients with PTSD may rarely experience dissociative episodes involving violence, especially in circumstances resembling their original trauma. This is often hard to determine retrospectively.

PTSD has increasingly been the basis for psychiatric defences to homicide, especially in the United States. It has also been advanced for battered women with a history of prolonged trauma, in whom occasional acts of retaliatory violence are not uncommon.

Morbid jealousy

The syndrome of morbid (pathological) jealousy (see Chapter 12) may be associated with several of the above diagnoses. It has been identified in 12% of 'insane' male and 3% of 'insane' women murderers. It is particularly dangerous because of the risk of the offence being repeated with another partner. It may sometimes be difficult to distinguish morbid jealousy from extreme possessiveness or control over women's behaviour, which may be accompanied by violence.

Organic mental disorders

Delirium is occasionally associated with criminal behaviour, usually because of the associated confusion or disinhibition. Diagnostic problems may arise if the mental disturbance improves before the offender is examined by a doctor.

Dementia is sometimes associated with offences, although crime is otherwise uncommon among the elderly, and violent offences are rare. Violent and disinhibited behaviour may also occur after a head injury (Fazel et al., 2009) and be difficult to distinguish from PTSD.

Epilepsy

The association between epilepsy and crime is complex and poorly understood. It has been widely believed that the risk of epilepsy is greater in prisoners than in the general population, but a meta-analysis of seven studies did not find this (Fazel et al., 2002), and a longitudinal study found that, after adjusting for familial confounding, epilepsy was not associated with increased risk of violent crime (Fazel et al., 2011). Violent behaviour associated with EEG abnormalities in the absence of clinical epilepsy is unlikely to indicate a causal relationship (Fazel et al., 2009). Epileptic automatisms may, very rarely, be associated with violent behaviour and subsequent criminal proceedings. Violence is more common in the post-ictal state than ictally.

Intermittent explosive disorder

This term is used to describe repeated episodes of seriously aggressive behaviour out of proportion to any provoking events and not accounted for by another psychiatric disorder (e.g. antisocial personality disorder,

substance abuse, or schizophrenia). The aggression may be preceded by tension, followed by relief, followed by remorse. The condition is rare, and many psychiatrists doubt whether it is a distinct psychiatric disorder.

Pathological gambling

Pathological gambling, or 'gambling disorder', has a separate category in ICD-11. Gambling is pathological when it is repeated frequently and dominates the person's life, and persists when the person can no longer afford to pay their debts. The person lies, steals, defrauds, or avoids repayment to continue the habit. Family life may be damaged, other social relationships may be impaired, and employment put at risk. Pathological gamblers demonstrate a range of cognitive distortions, such as magnification of gambling skill, superstitions, temporal telescoping, predictive skill, and obviously selective memory. Treatments using cognitive behaviour therapy (CBT) and family therapy have received some support. Gamblers Anonymous is widely used but has not been carefully evaluated. Because of the reclassification of gambling disorder with addictions, various trials have been conducted with opioid receptor antagonists and indicate some modest benefits (Potenza et al., 2019).

Pathological gamblers have an intense urge to gamble, which is difficult to control. They are preoccupied with thoughts of gambling, much as a person dependent on alcohol is preoccupied with thoughts of drink. Increasing sums of money are gambled, either to increase the excitement or in an attempt to recover previous losses. If gambling is prevented, the person becomes irritable and even more preoccupied with the behaviour.

Similarities with the behaviour of people who are dependent on drugs have led to the suggestion that pathological gambling is itself a form of addictive behaviour. Brain imaging studies of pathological gamblers show abnormalities in mesolimbic reward pathways similar to those in drug dependence (Potenza et al., 2019).

Estimates of the lifetime prevalence of gambling disorder in the United States gives an overall figure of around 0.5%. The explosive growth in internet gambling, smartphone gambling apps, and TV spot-betting have undoubtedly increased the risks. There is currently considerable governmental pressure to reduce the spread of online betting and betting shops and so limit the potential losses with gambling machines. It is probably more frequent among males. Most gamblers seen by psychiatrists are adults, but there is concern that young people are increasingly being involved, usually with gambling machines in amusement arcades. For a review of pathological gambling, see Potenza et al. (2019).

Pyromania

Pyromania refers to repeated episodes of deliberate fire-setting, which are not carried out for identifiable reasons such as monetary gain, to conceal a crime, as an act of vengeance, or as a consequence of hallucinations or delusions. Impaired judgement resulting from intoxication, dementia, or mental retardation excludes the diagnosis of pyromania, as does antisocial personality disorder, a manic episode, or (among children or adolescents) a conduct disorder. Firesetting in general is more commonly committed by men than women with about a sixfold male predominance. However, in forensic settings this gender difference is much reduced, suggesting that in women who set fires, mental health factors are more often considered relevant in causation. Pyromania as specific disorder is rare even in psychiatric samples of people who set fires (Tyler and Barnoux, 2022). In this condition, the act of firesetting is preceded by tension or arousal, followed by relief. People with pyromania have a preoccupation with fires and firefighting, and enjoy watching fires. They may plan the offence in advance, taking no account of the danger to other people.

Kleptomania

Kleptomania refers to repeated failure to resist impulses to steal objects that are not needed, either for use or for their monetary value. The impulses are not associated with delusions or hallucinations, or with motives of anger or vengeance. Before stealing there is increased tension, which the stealing relieves. The diagnosis is excluded by associated antisocial personality disorder, a manic episode, or (among children or adolescents) a conduct disorder, or when the stealing results from sexual fetishism. The objects stolen may be of little value and may be hoarded, thrown away, or later returned to the owner. The patient knows that the stealing is unlawful, and may feel guilty and depressed after the immediate pleasurable sensations that follow the act.

Kleptomania occurs more often among women. Associations with anxiety and eating disorders have been described. It may be sporadic, or persist for years despite repeated prosecutions. The diagnosis depends entirely on accused individuals' descriptions of their own motives and is therefore viewed with some scepticism.

Specific offender groups

Female offenders

Women are more law-abiding than men. Men and women are treated differently by the criminal justice system. Shoplifting accounts for 50% of all their convictions, and violent and sexual offences are uncommon. Women are sentenced more leniently for similar offences, and they are more likely to be viewed as 'ill'. Psychiatric disorder is frequent among women admitted to prison, with personality disorder and substance misuse being especially common (Fazel and Danesh, 2002) and self-harm before and during imprisonment is also common (Jenkins et al., 2005).

Young people

National crime statistics indicate that a significant proportion of crime is conducted by those under the age of 18 years, or 'juveniles'. In England and Wales, the rates appear to be declining (Ministry of Justice, 2022). For the year 2020/21, the number of children receiving a caution or sentence had fallen by 17%, with an 82% decrease over ten years. There was a 2% reduction in crime for the

year ending 2022, involving knives and offensive weapons. Reoffending rates have also decreased by almost 4% over ten years. In Scotland, offending rates in young people are also falling (Lightflower et al., 2014).

Where a young person is violent, the victim is often well known to them, commonly a family member. Serious violence by young people and children is rare, and the individuals involved have frequently been the recipients of violence themselves, both within and outside their families (Hamilton et al., 2002). Data from the American National comorbidity study explored the relationship between crime and psychiatric disorders in those under 18 years of age (Coker et al., 2014). They showed that there was a high prevalence of crime, with 18.4% of those with a psychiatric disorder before the age of 18 years reporting a crime and arrest history. Conduct disorder, alcohol use disorders, and drug use disorders had the greatest odds.

Ethnic minorities

Some ethnic groups are overrepresented in both the criminal justice system and forensic psychiatric services.

Non-white prisoners are more likely to receive diagnoses of mental illness rather than personality difficulties. Those with African-Caribbean origin have higher rates

of imprisonment in the United Kingdom, but lower rates of psychiatric morbidity despite an excess of African-Caribbeans in secure hospitals (Coid et al., 2002).

Psychiatric aspects of specific crimes

The following sections are concerned with the types of offences that are most likely to be associated with psychological factors. These are crimes of violence, sexual offences, and some offences against property.

Crimes of violence

Violence among mentally abnormal offenders is more strongly associated with personality disorder than with major mental illness. It is particularly common in people with antisocial personality traits who misuse alcohol or drugs, or who have marked paranoid or sadistic traits. It is often part of a persistent pattern of impulsive and aggressive behaviour, but it may be a sporadic response to stressful events in 'over-controlled' personalities.

Homicide

Stranger homicides by the mentally ill have fallen somewhat since 2006, constituting 7–10% of all such homicides between 2003 and 2022 (Appleby, 2015; Appleby et al., 2025). Thus, the proportion of homicides committed by the mentally ill has fallen in the era of community care.

Normal and abnormal homicide

Mental disorders may count as mitigation for an individual charged with murder, reducing the charge to manslaughter. Homicide can be divided according to the legal outcome into normal (murder or common-law manslaughter) and abnormal (insane murder, suicide murder, diminished responsibility, or infanticide).

Normal homicide accounts for half to two-thirds of all homicides in the United Kingdom, as in other western countries, including the United States, where the rate is much higher. It is most likely to be committed by socially disadvantaged young men. In the United Kingdom, the victims are mainly family members or close acquaintances. In countries with high homicide rates, a greater proportion of killings are associated with robbery or sexual offences. Sexual homicide may result from panic during a sexual offence. Alternatively, it may be a feature of a sadistic killing, sometimes committed by a shy man with bizarre sadistic and other violent fantasies.

Abnormal homicide accounts for a third to a half of all homicides in the United Kingdom. It is usually committed by older people. The victims of abnormal homicide are often family members. The most common psychiatric diagnoses are psychoses, substance use disorder, and personality disorder (Fazel and Grann, 2004). Depressive disorder can also be involved, especially in those who kill themselves afterwards. Homicide by women is much less frequent than that by men, but is nearly always 'abnormal'.

A large proportion of all murderers are under the influence of alcohol at the time of the crime, and drug misuse is also an important factor.

Multiple homicide

Multiple murders are rare, although they attract great public attention. They include:

- individuals without mental illness who kill several people at once, sometimes a family killing, which is often followed by suicide. Paranoid and grandiose character traits are common (Mullen, 2004).
- killings attributable to a psychotic illness in which the killer aims to save himself or his family from a perceived threat.
- serial killings that take place over a period of time. These may be 'normal' (e.g. killings by terrorists) or 'abnormal' (e.g. psychotic, or motivated by sexual sadism or necrophilia).

Homicide followed by suicide

A study of 180 victims and 147 perpetrators of homicides followed by suicides in the United Kingdom reported that 80% of the incidents involved one victim and one perpetrator, 88% exclusively involving members of the same family (Barracough and Harris, 2002). Around 75% of the victims were female, and 85% of the perpetrators were male, whose victims were nearly always current or previous female partners and their children. The victims of female perpetrators were predominantly their children. Data from the United States (Bossarte et al., 2006) report between 65 and 144 homicide/suicide incidents (homicide rate = 0.230/100,000 and 0.238/100,000) in 2003 and 2004, respectively. Most victims

(58%) were a current or former intimate partner of the perpetrator. Among male perpetrators of intimate partner homicide, 30.6% were also suicides. Victims were often children of the perpetrator (13.7%). Most victims (74.6%) were women and perpetrators were mostly men (91.9%). Legal and financial problems were common among perpetrators.

Parents who kill their children

Around 25% of all victims of murder or manslaughter in the United Kingdom are under the age of 16 years, and babies under the age of 1 year are at the highest risk of all age groups (Breslin and Evans, 2004; Stöckl et al., 2017). Most child homicide is by a parent or stepparent, while for adolescents it is by acquaintances. Mothers represent over half of perpetrators. The classification of child murder is difficult, but useful categories are mercy killing, psychotic murder, and, most common, killing resulting from battering or neglect (d'Orban, 1979).

Infanticide

A woman who kills her child may be charged with murder or manslaughter. English law recognizes a special category, infanticide, where the child is under 12 months of age. Infanticide is treated as manslaughter with less harsh penalties. The English legal concept of infanticide is unusual, requiring only that the woman's mind was disturbed as a result of birth or lactation, not that the killing was a consequence of her mental disturbance.

Fewer than five cases a year are recorded in the United Kingdom. Infants are most at risk on the first day of life, and the relative risk decreases steadily to that of the general population by one year. Fathers are slightly more likely to be recorded as the prime suspect over time and receive more severe sentences. Puerperal psychosis is a relatively infrequent cause of homicide, and depression is a factor in some cases. Later infant homicides are usually due to fatal child abuse. Infanticide has been associated with early motherhood.

Family homicide

Homicides can be distinguished by the relationship between perpetrator and victim. Most serious violence takes place within the family (25% of homicide victims are aged under 16 years, 80% killed by their parents). Around 50% of female victims are wives or partners, and the rest are often friends or relatives.

Domestic violence

In the British Crime Survey (March 2022), non-sexual domestic abuse (physical force, emotional abuse, financial abuse, threats and force) was the most common abuse experienced by adults (5.0%), followed by

domestic stalking (1.0%), and domestic sexual assault (0.3%). Domestic violence accounts for about 25% of all violent incidents recorded by the British Crime Survey, about one million incidents annually in England and Wales, about two thirds against women. Most batterers do not have either a diagnosable mental disorder or a criminal history. However, heavy drinking is common.

Some people are violent only within their family, while others are also violent outside. Violence in the family can have long-term deleterious effects on the psychological and social development of the children and on the mental health of the partner. Violence in the family also affects children and elderly relatives (page 458). Any of these forms of violence may rarely result in homicide.

Alertness to possible domestic violence is required not only in accident and emergency departments, but also in primary care and in obstetric and paediatric clinics. Intervention is difficult and raises ethical issues (see Box 18.4).

Violence towards partners

Violence by men towards their female partners is much more frequent than violence in other relationships, such as women towards their male partners. It is physically more serious and is more often reported. Most 'wife batterers' are men with aggressive personalities, while a minority are violent only when psychiatrically unwell, usually with a mood disorder. Other common features among these men are morbid jealousy and heavy drinking. Such men may have been a victim of violence in childhood, and often come from backgrounds in which

Box 18.4 Ethical and legal issues: domestic violence

Confidentiality is especially important because of the risk of retaliation by the abuser.

Careful records are essential, including documentation of the injuries. Written consent should be obtained for photographs.

Specialist advice should be sought about providing practical and other forms of help to those who wish to end the relationship.

Where the risk of serious violence is believed to be very high, disclosure to the police and other authorities to provide protection needs to be carefully planned with the maximum collaboration with the victims.

violence is frequent and tolerated. Behaviour by the victim may contribute to or provoke (but never justify) violence. This is difficult to assess if only the perpetrator is interviewed but, when battering is possibly a 'joint' problem, may be more difficult to stop. Repetitive battering, especially motivated by jealousy, is a real risk factor for homicide. The impacts of intimate partner violence to any children in the household must always be considered.

Sexual offences

Sexually violent offences

In the United Kingdom, sexual offences account for less than 1% of all indictable offences recorded by the police. Although only a small proportion of these are referred to psychiatrists they account for a significant number of assessments. Most sexual offences are committed by men.

Sexual offenders are generally older than other offenders. Reconviction rates are generally lower, but recidivist sexual offenders are extremely difficult to manage (see Box 18.5). In the United Kingdom, Part 1 of the Sex Offenders Act 1997 requires those convicted or cautioned for relevant sex offences to be kept by police on the Sex Offenders Register.

The most common sexual offences are indecent assault against women, indecent exposure, and unlawful intercourse with girls aged under 16 years. Some sexual offences do not involve physical violence (e.g. indecent exposure, voyeurism, and most sexual offences involving children), but others, such as rape, may involve considerable violence. The nature and treatment of non-violent sexual offences are discussed in Chapter 13; only their forensic aspects are considered here (Gordon and Grubin, 2004).

Sexual abuse of children

The age of consent for sexual intercourse varies in different countries. In England and Wales it is illegal to have any sexual activity with a person aged under 16 years, and this accounts for over 50% of all reported sexual offences. Underreporting is likely, particularly within families. Severity varies from mild indecency to seriously aggressive behaviour, but the large majority do not involve violence.

Offenders may or may not be paedophiles, defined as having a primary sexual interest in prepubertal children. They are almost always male and usually abuse children not previously well known to them. Paedophiles

Box 18.5 Some factors associated with increased risk of reoffending in sex offenders

- Previous criminal history
- Higher number of sexual offences and more than one type of sexual offence
- Being a childhood victim of sexual abuse
- Violent sexual fantasies
- Negative attitudes to women
- Belief that victims consent to or enjoy the act
- Choice of location and occupation to facilitate access to victims
- Use of sadomasochistic or paedophilic pornography
- Substance misuse
- Treatment non-compliance

Source: Gordon, H., & Grubin, D. (2004). Psychiatric aspects of the assessment and treatment of sex offenders. *Adv Psychiatr Treat*, 10(1), 73–80. Copyright (2004), The Royal College of Psychiatrists.

are rarely mentally ill. Victims are often prepared ('groomed') over a long period of time, increasingly via the internet. Some paedophiles may seek work in occupations where they will have access to children who will be left in their care. Cultural and institutional tolerance of paedophilia is negligible, and high profile 'historical abuse' investigations are taking place worldwide, focusing on the church, children's homes, and the entertainment industry.

It is difficult to classify paedophiles, but the following groups have been recognized:

- the timid and sexually inexperienced
- the learning disabled
- those who have experienced normal sexual relationships but prefer sexual activity with children
- a predatory group who may use violence. In rare cases, paedophile sexual activities end in murder.

However, not all child sex offenders are paedophiles thus defined. A significant majority do not have a primary sexual interest in children. Many have 'normal' heterosexual histories and may be involved in such relationships at the time when they offend. Child sex offenders typically are strangers to children, or else have gained opportunistic access through their chosen work or social activities. However, the majority of child sex offending is carried out by men (usually), who have some familial

relationship to the child. Most commonly these are step-fathers or other male members of the extended family circle.

The prognosis is difficult to determine. Among those who receive a prison sentence, the recidivism rate is about one in three. A minority progress to violent sexual offences, so psychiatrists may be asked to give an opinion on their dangerousness.

Assessment

Usually, the perpetrator and the victim are assessed by different people. When trying to decide whether an offence is likely to be repeated or to progress to more serious offences, the psychiatrist should first consider the depositions and the victim's statement to gain available information about the following:

- the duration and frequency of the particular sexual activity in the past (remembering that paedophiles often deny their offending)
- the offender's predominant sexual preferences; exclusively paedophile inclinations and behaviour indicate a greater risk of repetition. Older paedophiles are less likely to be aggressive.

The interviews should determine:

- the offender's previous sexual history
- whether alcohol or drugs played any part in the offence, and, if so, whether the person is likely to continue to use them
- whether there are any feelings of regret or guilt
- any stressful circumstances associated with the offence (and the likelihood that these will continue)
- the degree of access to children
- evidence of any psychiatric disorder or relevant personality features.

Treatment

Treatment is directed towards any associated psychiatric disorder. Direct treatment of the sexual behaviour is difficult. Group therapy run jointly by mental health and probation services may be helpful, as is individual and group support provided by some charities. A recent systematic review showed some effect of CBT to reduce reoffending (Harrison et al., 2020) but, despite enormous investment in prison services, there is little strong evidence of effect. The use of antiandrogens such as cyproterone and medroxyprogesterone has also been advocated, but their use is associated with many adverse effects and ethical issues.

Indecent exposure

This is the legal term for the offence of exposing the genitals to other people. It is applied to all forms of exposure; exhibitionism is by far the most frequent form, but exposure may also occur as an invitation to intercourse, as a prelude to sexual assault, or as an insulting gesture. Exhibitionism is the medical name for the behaviour of those who gain sexual satisfaction from repeatedly exposing to the opposite sex. In England and Wales, indecent exposure is one of the most frequent sexual offences, usually by men aged between 25 and 35 years with no history of psychiatric disorder or other criminal behaviour. Occasionally it is associated with compulsive disorder or substance misuse. Despite this, exhibitionism is listed as a psychiatric disorder in ICD-10. A proportion of offenders are repeat recidivists, and may proceed to more serious sexual violence.

Indecent assault

The term indecent assault refers to a wide range of behaviour, from attempting to touch a stranger's buttocks to sexual assault without attempted penetration. The psychiatrist is most commonly asked to give a psychiatric opinion on adolescent boys and on men who have assaulted children. Although many adolescent boys behave in ways that could be construed as 'indecent', more serious indecent behaviour is associated with aggressive personality, ignorance, lack of social skills, and, occasionally, learning difficulties. Treatment depends on the associated problems.

Stalking

The lay term 'stalking' is usually taken to mean the repeated, unwanted, and intrusive targeting of a particular victim with following and other harassment. It implies an intensive preoccupation with the victim. The scope of behaviour is wide, and includes:

- following the victim
- communication by telephone, mail, and electronic communication
- ordering goods and services in the victim's name
- aggression and threats, including violence, damage to property, and false accusations.

Most stalkers are men, and most victims are women. The victims invariably suffer severe distress. Management requires cooperation between forensic psychiatrists and the criminal justice system in assessing risks, treating any associated psychiatric disorder (e.g. erotomania),

and protecting and treating the victim. (Purcell and Mullen, 2020; also see Chapter 12).

Rape

In English law, a man commits rape if:

- he has unlawful sexual intercourse (whether vaginal or anal) with a woman or man who at the time of the intercourse does not consent to it
- at the time he knows that the victim does not consent to the intercourse, or he is reckless as to whether they consent to it.

Most jurisdictions define rape in terms of the lack of consent of the victim. Not all jurisdictions recognize male rape or sexual assault. Some countries (such as the United States) additionally define lack of consent by age—‘statutory rape’. English law is among the few that recognize rape within marriage.

Rape varies in the degree of violence used, and the extent to which it is mainly to exert control, or exciting in its own right. Most rapists are married or in partnerships, and over half fail to perform sexually during the assault. Rapists frequently have previous non-sexual violent convictions.

Stranger rape is rare, and most rapists know their victims. Stranger rapes are more likely to be physically violent and involve the use of weapons, whereas rapists who know their victims may not need to use physical threats. Most rapes take place in the home.

Epidemiology

Rape and other forms of sexual aggression towards women and increasingly men are probably much more frequent than reported. The 2020 Crime Survey for England and Wales found 49% were victims more than once; 16% reported the assault to the police and of those that told someone but not the police, 40% stated embarrassment as a reason. Over a third (38%) did not think the police could help, and 34% thought it would be humiliating; 44% were victimized by their partner or ex-partner; 9% were victimized on the street, in a car park, park, or another open public space compared with over one third 37% victimized in their own home. Over a half (54%) said the perpetrator used physical force, and 6% said the perpetrator had threatened to kill them. Alleged rapists have only a one in three likelihood of being convicted, mostly when the rape fits the stereotype of stranger rape.

Victims of acquaintance rape assume that they will not be believed.

The prevalence of male rape is unknown and even then, it is likely that many male rapes go underreported, as male victims may be reluctant to come forwards. As in rape of females, it often aims to degrade or dominate the victim. Rapists of males tend to be violent heterosexual men.

Causes

Most explanations of rape are sociocultural in terms of cultural attitudes to women, and social constructions of male and female gender roles. Men who are violent towards women often have rigid and conservative views of their role. Rapists frequently blame their victims for the attack (e.g. ‘She deserved it because she was flirting with X’). That the victim is often an acquaintance, and in 20% of cases may have some engagement in events leading up to the offence, should not be interpreted as consent or voluntary participation. There is little evidence for the frequently expressed view that rape victims encourage the rape or change their minds after having sex.

Rape has been found in population studies to be associated with severe mental illness, and the proportion of all sexual crimes that were committed by patients psychiatrically hospitalized at some point in their lifetimes is 20% (Fazel et al., 2007). Sexual behaviour may be associated with disinhibition as part of a manic or other psychotic illness, substance abuse comorbidity, or paranoid delusions in psychotic states. Evidence of current substance abuse is found in over 50% of rapists. Often both rapist and victim will have been using drugs or alcohol, as many rape scenarios begin in social situations. Some men who commit rape, homicide, or other violent offences have considerable sexual problems or suffer sexual jealousy, which contribute to their dangerousness. A small group of men obtain sexual pleasure from sadistic assaults on unwilling partners (Novak et al., 2007).

Prognosis

In the United Kingdom, most rapists serve only half their sentence in prison, and are then released on licence to be supervised by the probation service. The reconviction rate is 30%. The prison service offers psychological treatment to rapists as part of the Sex Offender Treatment Programme (SOTP) in prisons, but data to date suggest that little improvement occurs in individuals on this programme and indeed

some evidence that treated offenders have a slightly worse outcome than those untreated (Mews et al., 2017). Adding to the evidence base is made difficult by the constant evolution of the SOTP programme, so that effective comparisons and systematic reviews are problematical. In a broader review of treatments for sex offenders, there is evidence for several individual interventions, but this is usually supported by small studies and few with effective randomization (Schmucker and Lösel, 2015).

Child abduction

Child abduction is rare. A child may be abducted by one of the parents, by a man with a sexual motive, or by an older child. Babies are usually abducted by women. Their motives are to achieve comfort, or to manipulate another person, or may occur on impulse in psychiatrically disturbed women. Fortunately, most stolen babies are well cared for and are found quickly.

Shoplifting

The vast majority of shoplifting, like other theft, is carried out by people without any mental disorder. Many adolescents admit to occasional shoplifting. Both observational studies and the reports of huge losses from shops suggest that shoplifting is common among adults (including shop staff).

A minority of shoplifters suffer from psychiatric disorders. Depressive disorders are most common, but various other psychiatric diagnoses can be associated with shoplifting (Lamontagne et al., 2000). Patients with any type of mental illness, especially those with substance abuse problems, may steal because of economic necessity. Patients with disinhibiting conditions may be more likely to steal impulsively, and patients with eating disorders may steal food. In other conditions, shoplifting may result from distractibility—for example, in organic mental disorders, when the person is confused or forgetful, and during panic attacks when the person may run out of the shop without paying.

The assessment of a person charged with shoplifting is similar to that for any other forensic problem. If the accused has a depressive disorder at the time of the examination, the psychiatrist should try to establish

whether the disorder was present at the time of the offence or developed after the charge was brought. The legal question most often posed is whether the accused had the intention to steal, and if a mental condition could have affected that intention.

Arson

Arson is regarded extremely seriously, because it can result in great damage to property and threatens life. Most arsonists are males. Although the courts refer many arsonists for psychiatric assessment, the psychiatric literature on arson is small. Certain groups can be recognized:

- firesetters who are free from psychiatric disorder and who start fires for financial or political reasons, or for revenge; they are sometimes referred to as motivated arsonists.
- so-called pathological firesetters, who suffer from learning difficulties, mental illness, or alcoholism; this group accounts for about 10–15% of arsons
- those with psychotic illness.

At follow-up, 10% were re-convicted for arson, but over 50% for offences of other kinds (Soothill et al., 2004). A person convicted of arson a second time is at much greater risk of committing further offences. The factors associated with an increased risk of repetition include:

- antisocial personality disorder
- learning difficulties
- persistent social isolation
- firesetting for sexual gratification or relief of tension.

The scope for psychiatric intervention is limited. Management of arsonists within hospital requires a secure setting and close observation.

Children also present with firesetting. Sometimes it represents extreme mischievousness in psychologically normal children, at times it is a group activity, and sometimes it arises from psychiatric disturbance, most commonly conduct disorder (Martin et al., 2004). The recurrence rate in the following two years is reported to be less than 10%.

Psychiatric aspects of being a victim of crime

It is only relatively recently that criminology and society have paid attention to the role and needs of victims. General population surveys indicate that being a victim of crime is frequent, and is related to geographical area, gender, age, and social habits. Much violence, especially sexual and domestic assaults, goes unreported. Young men are particularly at risk of personal violence, whereas women are more likely to suffer domestic and sexual violence. In the United Kingdom, around one sixth of assaults on Asians and Afro-Caribbeans are believed to be racially motivated. The response of the victim is important in determining whether an offence is reported to the police and whether charges are brought.

Psychological impact

Childhood abuse and experience of violence during childhood may have major consequences in adulthood. Adult crime victims are at risk of a variety of early and late psychological problems. These include the immediate distress following the crime, and the subsequent distress associated with the investigation and court hearings. PTSD is frequently reported. These consequences are more common and severe immediately after the crime, but they may persist for many years.

Types of crime

Murder

Relatives of victims experience feelings of isolation and shame, and an inability to share their distress that is greater than in other kinds of bereavement. The bureaucracy and delay of legal processes increases anger and feelings of isolation.

Rape

Rape victims suffer long-term psychological effects (Mezey and Robbins, 2009). Recent research has shown very high levels of intrusive thoughts and other post-traumatic symptoms in the week following rape. Serious distress may also be experienced by the partners and families of rape victims. In many countries, including the United Kingdom, crisis intervention centres staffed by multidisciplinary teams have been set up for rape victims. Police practice has significantly improved, and there is now more sensitive handling of rape cases.

Burglary and robbery

Although the consequences are less severe than those following violent crime, they can include adjustment disorder and PTSD. Victims may become excessively preoccupied with security.

Terrorist crimes

There are increasing reports of terrorist crimes, including shootings, bombings, and hostage-taking, and all note severe immediate distress. PTSD and other psychiatric consequences only become persistent in a minority of cases. There is now also a public duty on health and social care professionals to scrutinize those coming to their attention for potential signs of radicalization and extremist beliefs, which may lead to offending. This duty has been challenged by many health professionals on ethical and confidentiality grounds, yet the government guidance is also seen to be entirely consistent with existing forensic practice and supported by the General Medical Council (see Bhui, 2019).

Management

Critical incident debriefing as a routine treatment is not helpful (Raphael and Wilson, 2000). Clinical judgement is needed in assessing the severity and persistence of psychological problems to determine whether victims require specific psychological help.

Support for victims of crime may be available within the community. In the United Kingdom, the Home Office funds the national Victim Support schemes, which routinely contact victims of crime to offer support, and a service is also available to support crime victims appearing at the Crown Court. However, these services rely on volunteers, who may not be able to offer long-term help and who cannot offer specialist psychiatric intervention. Voluntary groups, such as Rape Crisis, offer support to victims of sexual assault. Compensation is available to crime victims from the Criminal Injuries Compensation Board, and psychiatrists may be asked to provide reports in relation to claims for compensation for psychological distress.

Specialist services

Specialist psychiatric assessment and treatment services acquire particular experience of the problems suffered by victims. These may be provided within normal community services, specialized trauma clinics, or designated units such as rape clinics.

Routine psychiatric care

Assessment of routine referrals to psychiatric services should include enquiry about experiences of being a

victim, as this may be important in both aetiology and planning treatment.

The role of the psychiatrist in the criminal courts

Mental state, intention, and responsibility

Most jurisdictions require evidence of guilty intention for an offender to be convicted. Psychiatrists are therefore most often asked to provide opinions about whether a psychiatric illness affects the accused's intent to commit the crime. The underlying principle is that no one should be regarded as culpable unless they were able to control their own behaviour and to choose whether to commit an unlawful act or not. In determining guilt, it is necessary to consider the mental state at the time of the act, and especially intention (*mens rea*). This means the person perceives and intends that their act will produce unlawful consequences. Three other forms of intent need consideration.

1. Recklessness: the deliberate taking of an unjustifiable risk when the consequences can be foreseen but are not avoided.
2. Negligence: bringing about a consequence which a 'reasonable and prudent' person would have foreseen and avoided.
3. Accident (or 'blameless inadvertence').

The key issue is whether the accused had the mental capacity to form the intention, or whether mental disorder might have affected that capacity. Sometimes it will be beyond psychiatric expertise or evidence to answer this question. Asked to give an opinion on these matters, the psychiatrist should liaise closely with the lawyers.

Children

In most jurisdictions, the age of the accused is thought to affect their capacity to form the intent to commit crime. Most jurisdictions exclude children under a certain age from criminal prosecution; for example, in English law, children under ten years of age are excluded because they are deemed incapable of forming criminal intent. The Latin term for this is *doli incapax*. Children between the ages of 10 and 14 years may be convicted if there is evidence of *mens rea* and that the child knew that the offence was legally or morally wrong. The

English age for legal responsibility is lower than in most other European states, and is likely to be raised in the near future.

Competence to stand trial

This issue may arise in relation to any charge. Most jurisdictions require that a defendant must be in a fit condition to defend themselves. In English law, the issue is called 'fitness to plead' and may be raised by the defence, the prosecution, or the judge. It cannot be decided in a magistrates' court, but only by a jury.

It is necessary to determine how far the defendant can:

- understand the nature of the charge
- understand the difference between pleading guilty and not guilty
- instruct counsel
- challenge jurors
- follow the evidence presented in court.

A person may be suffering from severe mental disorder but still be fit to stand trial. If an individual is found not fit to plead, the court will hold a 'trial of the facts' to determine whether the individual carried out the offence. If the offence is not serious, the court may make an order directing the offender to have treatment, often as an outpatient. In cases where the offence is serious, or carries a mandatory penalty (e.g. murder), the court will direct the offender to be detained in hospital indefinitely. If the person should become fit to plead, they may be returned to court for a trial. Detention after being found unfit to plead (or legally insane) operates in the same way as detention accompanied by a restriction order.

Legal insanity (not guilty by reason of insanity)

This defence may also be raised to any charge. It is argued that the defendant lacked *mens rea* for the charge because they were 'legally insane'. This term has nothing to do with diagnostic terms or classifications such as

ICD-10 or DSM-5. Legal insanity is defined in different ways in different jurisdictions. It usually results in the defendant being admitted for treatment in hospital, as opposed to being sent to prison. In some jurisdictions, a 'not guilty by reason of insanity' verdict may result in more lenient sentencing.

In English law, insanity is defined in law by the M'Naughton Rules, after the famous case of Daniel M'Naughton, who in 1843 shot and killed Edward Drummond, the Private Secretary to the then Prime Minister, Sir Robert Peel. In the trial at the Old Bailey, a defence of insanity was presented on the grounds that M'Naughton had suffered from delusions that he was being persecuted by spies. His delusional system gradually focused on the Tory Party, and he decided to kill their leader, Sir Robert Peel. M'Naughton was found not guilty on the grounds of insanity, and was admitted to Bethlem Hospital. Because this was such a contentious decision, rules were provided for guidance. It must be clearly proved that, at the time of committing the act, the accused was: *labouring under such a defect of reason, from disease of the mind, as not to know the nature and quality of the act he was doing, or, if he did know it, that he did not know what he was doing was wrong.*

Several other common law jurisdictions (including some states in the United States and Australia) have used this as the basis for their own definitions of legal insanity. Critics argue that the rules are too narrow, and that few mentally ill offenders would fulfil these criteria. Indeed, M'Naughton himself may not have fulfilled them. In some countries, the insanity defence is widely used, especially in crimes of violence such as homicide. In English law, the alternative defence of diminished responsibility is more usual.

Diminished responsibility

Some jurisdictions include the concept of diminished responsibility. In this an individual's blameworthiness may be reduced (rather than removed) by virtue of having a mental illness. In English law, it is only available in relation to the charge of murder, and is defined as follows: where a person kills or is party to a killing of another, he shall not be convicted of murder if he was suffering from such abnormality of mind (whether arising from a condition of arrested or retarded development of mind or any inherent causes or induced by disease or injury) as substantially impaired his mental responsibility for his acts and omissions in doing or being party to the killing.

There are several difficulties with this definition. 'Abnormality of mind' does not fit any diagnostic category but is basically anything which a 'reasonable man'

would call abnormal. It has been widely interpreted. Successful pleas have been based on conditions such as 'emotional immaturity', 'mental instability', 'psychopathic personality', 'reactive depressed state', 'mixed emotions of depression, disappointment, and exasperation', and 'premenstrual tension'.

The relationship between abnormality of mind and responsibility is also unclear, and psychiatrists have no special expertise in this area. Most legal commentators argue that the finding of responsibility is for the jury to decide, not a matter of expert evidence. Nevertheless, psychiatrists are frequently asked to comment on this issue.

Assessment

Most defendants who are charged with murder undergo extensive psychiatric assessment, often by a specialist in forensic psychiatry but sometimes by a general psychiatrist. Defence lawyers often seek independent psychiatric advice. It is good practice for the doctors involved, whether engaged by prosecution or defence lawyers, to discuss the case together. Disagreement is unusual. Copies of the reports are distributed to the judge and to the prosecution and defence lawyers. Similar arrangements apply to other offences in which a psychiatric opinion is required.

If the psychiatric evidence is accepted by the court, supporting diminished responsibility, the defendant will be convicted of manslaughter rather than murder. In some jurisdictions, diminished responsibility may protect the convicted offender from the death penalty.

In the United Kingdom, offenders who are convicted of manslaughter in this way may be detained in hospital under the Mental Health Act. If the offence is particularly dangerous, and the offender presents a risk to the public, the court may impose a restriction order, requiring the Home Office to agree discharge.

Infanticide is a particular form of manslaughter charge, which can only be brought against women who have killed their newly born children (under one year old). If there is psychiatric evidence to show that the woman was mentally ill at the time of the killing, she will be found guilty of infanticide rather than murder. If the court makes a hospital order, a restriction order is rarely applied. This is a rare example of the English law formally recognizing the existence of psychiatric illness, namely postpartum psychosis, as relevant to the commission of an offence.

Absence of intention (automatism)

It is sometimes argued that the defendant lacked intention altogether (technically, the absence of *mens rea*) and

this is referred to as automatism. The paradigm example is acts committed while sleepwalking. Automatism is difficult to determine retrospectively, and the defence is now rarely used, although it played a significant role in nineteenth-century psychiatry's development. With patients who abuse alcohol or drugs, it may be argued that because they were 'intoxicated' they had no intention to commit the crime. The law on intoxication is complicated, and specialist legal advice should be sought.

Fitness to be punished

In those jurisdictions that have corporal or capital punishment, psychiatrists may be asked to assess offenders to determine whether they are mentally well enough to be punished. In addition to assessment, psychiatrists may be asked to treat offender patients to make them fit to be punished or executed. Clearly it is unethical for psychiatrists to be involved in such procedures.

Other psychiatric issues that may be relevant to the criminal court

Amnesia

Over one third of those charged with serious offences, especially homicide, report some degree of amnesia for the offence and inadequate recall of what happened. It has been argued, unsuccessfully, that loss of memory should be regarded as evidence of unfitness to plead. The factors most commonly associated with claims of amnesia are extreme emotional arousal, alcohol abuse and intoxication, and severe depression. Amnesia has to be distinguished from malingering, but there appear to be instances of true amnesia for offences, just as there is impaired recall by victims and witnesses. Moreover, the factors associated with amnesia are similar in offenders and victims. In the absence of a relevant

neuropsychiatric disorder, the presence of amnesia is unlikely to be accepted as having any legal implications.

False confessions

False confessions to criminal deeds are sometimes made, but their frequency is unknown. Gudjonsson (1992) suggested that there are three main types of false confession:

1. voluntary
2. coerced-compliant
3. coerced-internalized.

Voluntary confessions may arise from a morbid desire for notoriety, from difficulty in distinguishing fact from fantasy, from a wish to expiate guilt feelings, or from a desire to protect another person. Coerced-compliant confessions result from forceful interrogation, and are usually retracted subsequently. Coerced-internalized confessions are made when the technique of interrogation undermines the suspect's own memories and recollections, so that they come to believe that they may have been responsible for the crime. Factors that make a person more likely to make a false confession include a history of substance abuse, head injury, a bereavement, current anxiety, or guilt.

The assessment of possible false confessions is difficult. It requires a thorough review of the circumstances of arrest, custody, and interrogation, as well as an assessment of the personality and the current mental and physical state of the suspect. Clinical psychologists can carry out a neuropsychological assessment and, in some cases, an assessment of suggestibility.

False accusations

Occasionally there are reports of individuals who claim to be the victims of a crime that has not occurred, and who make false accusations. Examples are accusations of rape and also of stalking. Legal and clinical experience suggests such cases are uncommon, and that accusers frequently have severe personality and other problems.

The treatment of offenders with mental disorder

General issues

The assessment needs to include as much information as possible from a variety of sources, including, if possible, the general practice notes. Relatives may not be the most reliable informants, particularly if they are victims of interpersonal violence. Careful attention must be paid

to both mental illness and personality disorders, as well as histories of substance misuse, which are extremely common.

Forensic psychiatric treatment usually involves treating general psychiatric conditions in specialized settings, such as secure treatment units or hospitals. It may also involve involuntary outpatient care. Treatment

planning involves not only the appropriate medications, but also organization of appropriate psychological interventions. This is particularly important for forensic patients with severe personality disorders. Management of such patients requires specialist training for staff and support by forensic psychotherapists. It also depends upon introducing evidence-based psychiatric care into forensic practice. Many forensic patients have personality disorder; the general principles of management are described in Chapter 15.

Settings of treatment

After conviction, an offender may be treated on either a compulsory or voluntary basis. In the United Kingdom, special treatment for mentally abnormal offenders is, in principle, provided by the Home Office (the prison medical service and the probation service) and by the Department of Health (high-security hospitals, specialist forensic services, and general psychiatry services). There has been a lack of research in forensic psychiatry, which means that the evidence base for effective treatment is slender. For a general review of the organization of forensic psychiatric services, see Taylor and Dunn (2009).

Much work with offenders is carried out by general psychiatrists, who assess patients and prepare court reports. General psychiatrists as well as forensic psychiatrists treat offenders who have been given non-custodial sentences. Forensic psychiatrists work in separate units and undertake specialized assessment and court work. In many places there are community forensic services to provide assessment and treatment, and the management of the boundary with general services is subject to ongoing controversy. Forensic psychiatrists may work to provide care for patients who need security in ordinary psychiatric hospitals.

The mentally abnormal in prison

About a third of sentenced prisoners have a psychiatric disorder and 4% have a psychosis (Fazel and Danesh, 2002), although more recent surveys using structured diagnoses find higher rates, with up to 10% classified with psychotic illnesses (Hassan et al., 2011). Most of these disorders can be treated in prison, but a few offenders need transfer to a hospital (see Box 18.6).

The care of mentally disordered prisoners has come under intense scrutiny in recent years. The increase in

Box 18.6 Reasons for hospital transfer of prisoners with psychiatric disorder

- Psychosis
- Failure to improve with medical treatment in prison
- Refusal to have treatment for serious psychiatric illness
- Life-threatening self-harm
- Risk of abuse

prison suicides (Fazel et al., 2011), when that in the general population is falling, plus the transfer of prison health care to the NHS in the United Kingdom, have highlighted the issue. Prison medical services have to provide psychiatric care under extremely difficult conditions, and a substantial increase in psychiatric input to prisons is needed. A few prisons offer psychological treatment, usually for personality disorders and sexual offences, as a main part of their work. Grendon Underwood in England is a long-established specialist prison for such treatments. A recent initiative in the United Kingdom established four special units in prisons and high-security hospitals to treat prisoners/patients with severe personality disorder (see below, dangerous severe personality disorder units [DSPD]). While effectively discontinued, the initiative has generated a pathway within prisons for the care of such individuals. Although there is an undoubted need for psychiatric care within prisons, the benefits should be weighed against any encouragement for courts to send the mentally unwell or neurodivergent to prison, rather than to hospital.

Offenders in hospital

Most jurisdictions allow for the detention of mentally abnormal offenders in secure psychiatric settings. In England and Wales, a convicted offender may be committed to hospital for compulsory psychiatric treatment under a Mental Health Act hospital order. There is also provision in law for a prisoner to be transferred from prison to a psychiatric hospital. Hospital orders may have no time limit, whereas most prison sentences are of fixed length. The length of stay in a psychiatric hospital may be shorter than a prison sentence, or it may

be longer. Prisoners cannot be treated against their will in prisons.

Special hospitals and secure units in the United Kingdom

In the United Kingdom, detention of mentally abnormal offenders may be in a local psychiatric hospital, a medium security unit, or a maximum secure hospital ('special hospital'). In England, the first special provision for the criminally insane was made in 1800, when a criminal wing was established at the Bethlem Hospital. In 1863, Broadmoor, the oldest of the special hospitals, opened under the management of the then Home Office. There are now four high-security special hospitals in England and Wales for patients who require such high levels of security.

The detention of patients in special hospitals is usually for an indeterminate duration. For those with mental illness (mostly schizophrenia), the length of detention is determined by the severity or chronicity of the psychiatric disorder, rather than by the nature of the offence. By contrast, for patients suffering from psychopathic disorder, the main determinant of length of stay is the assessment of the future risk of offending.

The closure of the larger mental hospitals in the United Kingdom has had unforeseen consequences for the care of mentally abnormal offenders. There is less physical security in the new psychiatric wards and hospitals, and less willingness by hospital staff and other patients to tolerate severely disturbed behaviour. Allied with shorter admission practices, it has become increasingly difficult to arrange admission to hospital for offenders, particularly those who are severely disturbed. Two alternative provisions have been developed:

1. Well-staffed secure areas in ordinary psychiatric hospitals in which the less dangerous of these patients can be treated.
2. Special secure units associated with psychiatric hospitals, to provide a level of security intermediate between that of an ordinary hospital and a special hospital (medium-secure units).

Dangerous severe personality disorder units

The notion of DSPD is an administrative category based on clinical diagnosis and risk assessment. It was

introduced in England in 1999 as a response to two pressing problems of public safety. The first was offenders coming to the end of their sentences with a persisting serious risk of violence. The second was that psychiatrists would not detain severely personality-disordered individuals because they judged there to be no evidence of their 'treatability'. Treatability was, uniquely, required in the 1983 Mental Health Act for personality disorder but not for mental illness stemming from the original 1959 Act. In the 2007 Amendment, treatability was replaced by 'medical treatment which is appropriate in his case, taking into account the nature and degree of the mental disorder and all other circumstances of his case.'

Four pilot units were established—two in prisons and two in high secure hospitals. They reflected the context of decreased public tolerance of risk of violent crime, improved standardized risk assessment, establishment of CBT programmes for sexual and violent offenders (McGuire, 2008). The criteria for admission were very prescriptive, requiring a violent crime, a clear personality disorder, and an established link between the personality disorder and the crime.

In practice, the units have proved difficult to staff and utilize fully, and the introduction of indefinite sentencing has removed much of their *raison d'être*. The hospital units are already closed, and the prison units are revising their remit and practice. Forensic mental health care and non-criminal pathways to care are being encouraged in new legislation being scrutinized before parliament at the time of writing (2023).

Treatment in the community

Offender patients may not pose sufficient risk, or be sufficiently ill, to require treatment in hospital. Courts may also use non-custodial sentences in which the offender patient may receive support from the probation service as well as psychiatric treatment. On occasion, psychiatric treatment may be made a condition of probation, with which the offender must agree to comply.

The psychiatric treatment provided for a mentally abnormal offender is similar to that for a patient with the same psychiatric disorder who has not broken the law. It is often difficult to provide psychiatric care for offenders with chronic psychiatric disorders who commit repeated petty offences, move around, and often become homeless. In the past many would have been long-stay patients in a psychiatric hospital.

The management of violence in health care settings

Violent incidents are not confined to patients with forensic problems, but this is a convenient place to consider their management. Although not frequent, violent incidents in hospitals are increasing. The reasons for this increase appear to include the following:

- changes in mental health policies emphasizing dangerousness as a reason for admission
- overcrowding

- lack of sufficiently experienced staff
- increased use of illegal drugs.

All psychiatrists should be familiar with how to manage incidents of violence in inpatient settings. Prior education and training are essential, and the National Institute for Health and Clinical Excellence has provided guidelines (see Figure 18.1). It is important that staff have a clear policy for managing incidents of violence

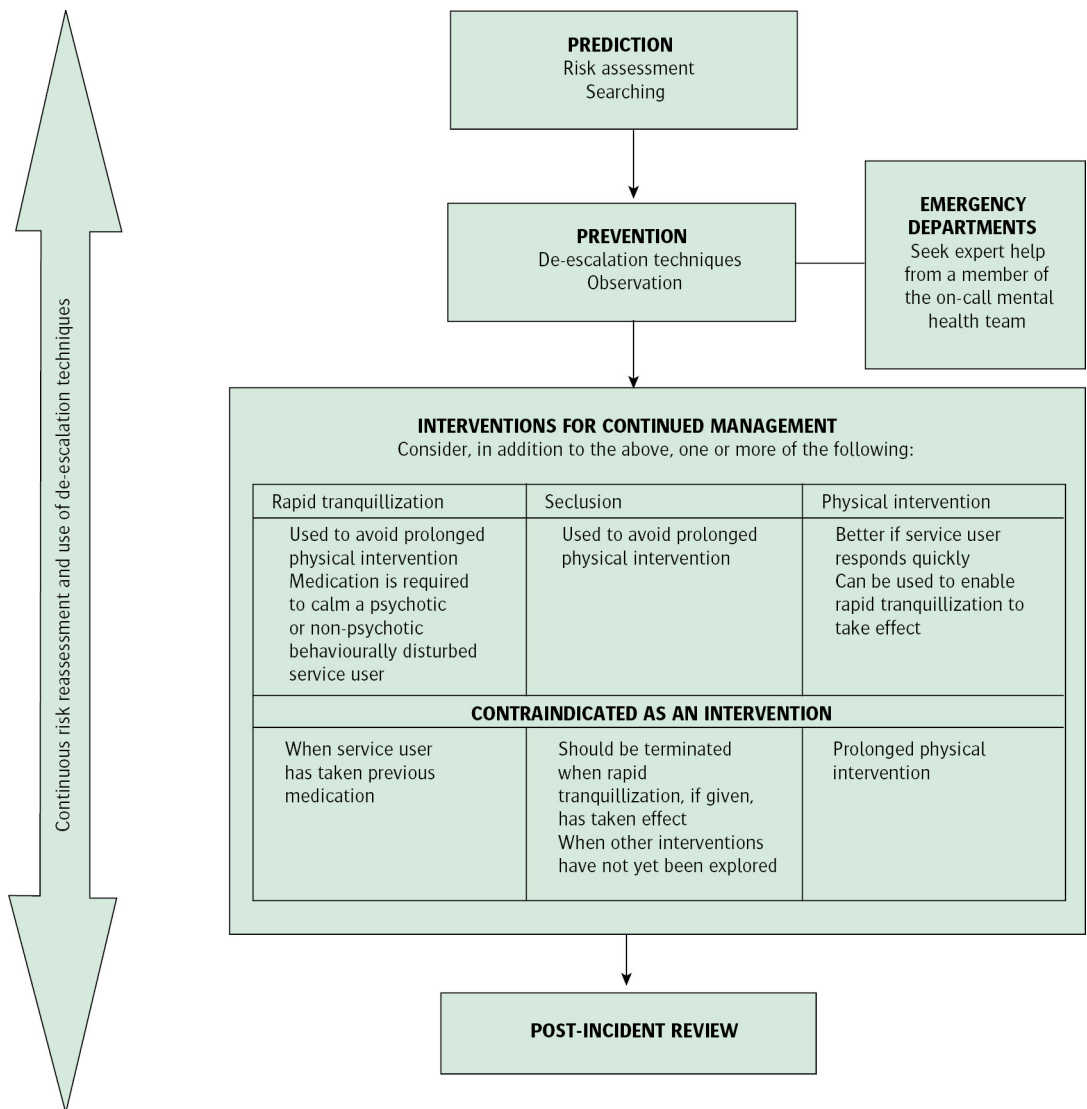


Figure 18.1 Algorithm for the short-term management of disturbed or violent behaviour in health care settings.

Source: National Institute for Health and Care Excellence. (2015). Violence and aggression: Short-term management in mental health, health and community settings. NICE guideline NG10. <https://www.nice.org.uk/guidance/ng10>. Copyright (2015), National Institute for Health and Care Excellence.

Box 18.7 De-escalation techniques used to prevent violence

- If a service user becomes agitated or angry, one staff member should take the primary role in communicating with them. That staff member should assess the situation for safety, seek clarification with the service user, and negotiate to resolve the situation in a non-confrontational manner.
- Use emotional regulation and self-management techniques to control verbal and non-verbal expressions of anxiety or frustration (e.g. body posture and eye contact) when carrying out de-escalation.
- Use a designated area or room to reduce emotional arousal or agitation and support the service user to become calm. In services where seclusion is practised, do not routinely use the seclusion room for this purpose because the service user may perceive this as threatening.

Source: National Institute for Health and Care Excellence. Copyright (2015).

and are trained to implement it. Such a policy calls for attention to the design of wards, arrangements for summoning assistance, and suitable training of staff.

When violence is threatened or actually occurs, staff should be available in adequate numbers, and emergency medication such as intramuscular lorazepam and haloperidol should be unobtrusively available. The emphasis should be on the prevention of violence (see Box 18.7).

Potentially dangerous people can often be calmed by sympathetic discussion or reassurance, preferably given by someone whom they know and trust. It is important

not to challenge the patient but inappropriate to reward violent or threatening behaviour by making concessions. Every effort should be made to allow the patient to withdraw from confrontation without loss of face. The use of medication should be followed by appropriate monitoring (for advice about the use of medications in emergencies, see Chapter 25).

After an incident has occurred, the clinical team should meet to consider the following issues:

- The future care of the patient. For mentally disordered patients, there should be a review of the drugs prescribed and their dosage. When violence occurs in a person with a personality disorder, medication may be required in an emergency, but it is usually best to avoid maintenance medication. Other measures include trying to reduce factors that provoke violence, or providing the patient with more constructive ways of managing tension, such as taking physical exercise or asking a member of staff for help.
- Supportive psychological interventions. These may be required for patients or staff who have been the victims of a violent assault (see the earlier section on victims of crime).
- Whether the police should be informed. It should not be forgotten that such assaults may be criminal. Opinion has moved to a preference for involving the police more often, although they are often reluctant to press charges.
- The possible effect on the whole patient group. Other patients may need support whether or not they were present at the incident.
- The need for changes in the general policy of the ward. A violent incident may enable lessons to be learned that are applicable in a general way to ward policies and procedures.

Risk assessment

The change to community care has made both minor criminality and rare violent offences more conspicuous, and has resulted in increased public disquiet. Psychiatric services need the resources to minimize difficulties and to identify and manage serious threats of violence. For a review of dangerousness and risk, see Buchanan (2008). The psychiatrist may need to assess risk in everyday psychiatric practice and also in forensic work.

In everyday practice, careful risk assessment may be required so that the most appropriate steps can be taken

in the interests of the patient and of other people. Risk of serious harm to others is an important consideration for compulsory detention in hospital.

In forensic work, the court may ask about the defendant's dangerousness, so that a suitable sentence can be passed. The psychiatrist may also be asked to comment on offenders who are being considered for release from institutions. In both situations there is an ethical conflict between the need to protect the community and respect for the rights of the offender.

Box 18.8 Factors associated with dangerousness

Male gender

History

One or more previous episodes of violence*
 Repeated impulsive behaviour
 Evidence of difficulty in coping with stress
 Previous unwillingness to delay gratification
 Antisocial traits and lack of social support
 History of conduct disorder

The offence

Bizarre violence
 Lack of provocation
 Lack of regret
 Continuing major denial

Mental state

Morbid jealousy
 Paranoid beliefs plus a wish to harm others
 Deceptiveness
 Lack of self-control
 Threats to repeat violence
 Attitude to treatment, poor compliance

Circumstances

Provocation or precipitant likely to recur
 Alcohol or drug misuse
 Social difficulties and lack of support

* Note: Of these predictors, the most useful is a history of past violence (Buchanan, 2008).

There have been two broad approaches to risk assessment.

1. Clinical psychiatrists have tried to identify factors associated with dangerousness in an individual patient (see Box 18.8). While general predictors of violence (e.g. past violence, antisocial personality disorder, substance misuse) are helpful, they lack specificity in identifying particular individuals at risk (Fazel et al., 2012)
2. Actuarial methods have been used to predict future criminal behaviour among offenders and psychiatric patients. In general, the low correlations between predicted and observed behaviour have meant that they have been unhelpful for making individual predictions. Recent instruments have an improved predictive accuracy, but may be more useful in predicting those not likely to be violent than those who are (Fazel et al., 2012)

There are no fixed clinical rules for assessing risk, but there are recognized basic principles. A thorough review should be conducted of the history of previous violence, the characteristics of the current offence and the circumstances in which it occurred, and the mental state of the individual (see Figure 18.1 and Figure 18.2). When making the review, it is helpful to consider the following key factors:

- whether any consistent pattern of behaviour can be discerned
- whether any circumstances have provoked violence in the past and are likely to occur again in the future

- whether there is any good evidence that the defendant is willing to change their behaviour
- whether there is likely to be any response to treatment (see Figure 18.1 and Figure 18.2).

Particular difficulties may arise in the assessment of dangerousness in people with antisocial personality or learning disabilities, who may be poorly motivated. Another difficult problem is presented by the person

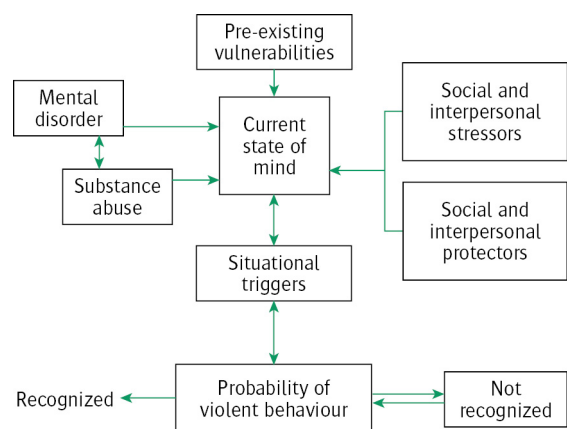


Figure 18.2 Schematic representation of the issues that should be considered when assessing the probability of violent behaviour.

Source: Mullen, P. (2000). Dangerousness, risk, and the prediction of probability. In M. Gelder, J. Lopez-Ibor, Jr., & N. Andreasen (Eds.), *New Oxford Textbook of Psychiatry*, Vol. 2 (2066–2078). Copyright (2009), Oxford University Press.

who threatens to commit a violent act such as homicide. Here the assessment is much the same as for suicide threats but with a lower threshold for intervention. The psychiatrist should ask the individual about their intent, their motivation, and the potential victim, and should make a full assessment of the person's mental state. Some patients who make threats can be helped

by outpatient support and treatment, but sometimes hospital admission is required if the risk is high. It may be necessary to warn potential victims or the police.

It is good practice for the psychiatrist to discuss their evaluation of dangerousness with other colleagues, including other psychiatrists, general practitioners, social workers, and the individual's relatives.

The psychiatric report

A psychiatric report prepared for a major criminal charge is an important document, and should be based on a full psychiatric and social assessment. It is essential that the psychiatrist reads all of the depositions by witnesses, statements by the accused, and any previous medical notes and social reports. Family members should be interviewed. When evidence about previous offences is not admissible (as is the case in English law) the psychiatrist's report should not include these facts. This may cause problems for the psychiatrist, whose opinion is often based in large part on the offender's previous behaviour. The writing of the court report follows

the format described in Box 18.9, and should include an assessment of the person's mental state at the time of the alleged offence, and of their fitness to plead. The involvement of the psychiatrist at various stages of the legal process in England and Wales is shown in Table 18.2.

The role of the psychiatrist in relation to the court

The psychiatrist's role is to draw on their special knowledge to help the court. They should not attempt to

Box 18.9 Some headings for a court report

- *Statement:* Give the psychiatrist's full name, qualifications, and present appointment (and, in England and Wales, whether they are approved under Section 12 of the Mental Health Act).
- *Interview:* Where and when the interview was conducted and whether any third person was present.
- *Sources of information,* including documents that have been examined.
- *Family and personal history of the defendant/plaintiff.* Usually this does not need to be given in great detail, particularly if a social report is available to the court. The focus should be on information that is relevant to the diagnosis and disposal.
- *Present mental state.* Only the salient positive findings should be stated, and negative findings should be omitted. A general diagnosis should be given in the terms used in the old 1983 Mental Health Act (mental illness, mental impairment, or psychopathic disorder). A more specific diagnosis can then be given, but the court will be interested in a broader categorical statement rather than the finer nuances of diagnosis.
- *Mental state at the time of the relevant events.* This is often a highly important issue, especially in criminal cases, and yet it can be based only on retrospective speculation. The assessment can be helped by accounts given by eyewitnesses who saw the offender at the time of the crime or soon afterwards. A current psychiatric diagnosis may suggest the likely mental state at the time of the crime. For example, if the accused suffers from chronic schizophrenia or a chronic organic mental syndrome, the mental state may well have been the same at the time of the crime as at the examination. However, if the accused suffers from a depressive disorder (now or recently) or from an episodic disorder such as epilepsy, it is more difficult to infer the mental state at the material time. Even if it is judged that the defendant was suffering from a mental disorder, a further judgement is needed as to their *mens rea* at the time of the crime.
- *Conclusions.* A summary of the key findings.

Table 18.2 The involvement of psychiatrists in the stages of the UK legal process

Stage I: Arrest	Stage II: Pre-trial	Stage III: At the trial	Stage IV: After the trial
Removal to a place of safety (police station, hospital, for medical examination)	Court report; Remand for inpatient assessment or treatment	Special problems: fitness to plead, diminished responsibility	Treatment under hospital orders or guardianship
Assessment after arrest; Court diversion schemes	Transfer from prison for assessment	Advice about subsequent management	Transfer from prison; Decisions about release; Treatment in the community

tell the court what to do. In the United Kingdom, the duty of the expert medical witness is to the court as a whole; they are not expected to be partisan. It is sometimes hard for psychiatrists to appreciate that they must remain neutral and not provide evidence to order that supports the party instructing them. This is particularly difficult because most psychiatrists use their clinical skills to establish rapport with individuals on whom they are preparing a report, and they experience a desire to advocate for them.

The psychiatrist should be aware that the court will see the report and that it may be read out in open court. Reports that are commissioned and paid for by lawyers are the property of the court.

The assessment

When conducting a psychiatric assessment on a person who is accused of a crime or who has been convicted of an offence, some key points should be kept in mind.

1. Prepare as thoroughly as possible before the interview. Have a clear idea of the purpose of the examination, and particularly about any question of fitness to plead. Obtain details of the present charge and past convictions, together with copies of any statements made by the defendant and witnesses. Study any available reports of the defendant's social history; during the subsequent interview go through this report with the defendant and check its accuracy.
2. Begin by explaining the source of the referral and why it was made. You should explain that their opinion may be given in court and that they are under no obligation to answer any of your questions if they choose not to do so.

3. Make detailed notes, recording any significant comments in the defendant's own words. At some stage in the interview (not necessarily at the start), the alleged crime should be discussed. The defendant may or may not admit guilt, but the psychiatrist is not obliged to comment on this.
4. Take a detailed history of any physical illnesses, paying particular attention to neurological disorders, including head injury and epilepsy.
5. Obtain a careful history of previous psychiatric disorder and treatment. Make a full examination of the present mental state. Special investigations should be requested if appropriate. If the defendant's intelligence level is under question, a clinical psychologist should make a separate assessment.
6. If possible, obtain further information from relatives and other informants. If the defendant is remanded in custody, the staff may have long periods of contact with the prisoner and may be able to provide particularly useful information.

Preparing the report

The preparation of a court report will be affected by the circumstances of the case, and the instructions given by solicitors. Court reports for civil and criminal cases may be very different. A possible outline is shown in Box 18.9. For further details, see O'Grady (2009).

When preparing a court report, the psychiatrist should remember that it will be read by people with a non-medical background. Therefore, the report should be written in simple English and should avoid the use of jargon. If technical terms are used, they should be defined as accurately as possible. The report should be concise.

Advice on medical treatment

One of the psychiatrist's main functions is to give an opinion on whether psychiatric treatment is indicated. The psychiatrist should make sure that any recommendations on treatment are feasible, if necessary by consulting colleagues, social workers, or others. If hospital treatment is recommended, the court should be informed whether or not a suitable placement is available.

The assessment of risk is important here (see pages 53, 54). The psychiatrist should not recommend any form of disposal other than treatment. However, the court often welcomes respectfully worded comments on the suitability of possible sentences, particularly in the case of young offenders.

The psychiatrist appearing in court

The psychiatrist appearing in court should be fully prepared and should have well-organized copies of all reports and necessary documents. It is helpful to speak to the lawyer involved beforehand to clarify any points that may be raised in court. When replying to any questions in court, it is important to be brief, clear, to restrict the answers to the psychiatric evidence, and to avoid speculation. A number of expert witness training programmes are available, which may be useful for psychiatrists who often give expert evidence.

Further reading

Geddes, J. R., et al. (2020). Section 24: Forensic psychiatry. In J. Geddes, N. C. Andreasen, & G. M. Goodwin (Eds.) *The New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1415–1485). Oxford University Press.

Liebling, A., Maruna, S., & McAra, L. (Eds.). (2017). *The Oxford Handbook of Criminology*, 6th edn. Oxford University Press.

CHAPTER 19

Psychiatry of the elderly

Introduction 521

Normal ageing 521

Epidemiology of psychiatric disorders in the elderly 523

Principles and practice of old age psychiatry 525

Clinical features and treatment of psychiatric disorders in the elderly 532

Treatment of dementia 536

Introduction

Older people with mental health problems present particular challenges to the practice of old age psychiatry and the organization of its services. They are often physically as well as mentally frail, and this affects presentation and course. Dementia comprises a substantial part of the clinical practice. On the other hand, the elderly have a rich history to tell and a lifetime's experience of responding to fortune and adversity, and positive aspects of ageing such as resilience and wisdom are important to consider, too (Reynolds et al., 2022). Indeed, a UK survey showed that personal wellbeing ratings are highest in people in their seventh and eighth decades of life (Office for National Statistics, 2016).

When considering psychiatric disorder in the elderly, the clinician must be able to collect and integrate information from a variety of sources, and produce a management plan that takes account of physical and social needs, as well as psychological ones. This plan is likely to involve the cooperation of several professionals. It is in this clinical complexity that much of the challenge and fascination of old age psychiatry lies.

This chapter deals with the psychiatry of old age, with two important exceptions, both of which were covered in Chapter 14:

- delirium
- the clinical features, aetiology, and investigation of dementia.

Normal ageing

Demographics

In 1993, 6% of the world's population was over 65 years of age. As shown in Table 19.1, by 2019 it was 9.1% and it is predicted to increase to 15.9% by 2050. Over the same period, the global population is expected to increase

from 7.7 to 9.7 billion, with marked regional differences in both population growth and the proportion of elderly people. These figures highlight that, throughout the world, there will be an increasing number of elderly people with psychiatric disorders. This applies particularly to the syndromes normally occurring in the elderly,

Table 19.1 Estimates of global population and percentage of elderly people, 2019–2050

	2019	2030	2050
Population (millions)			
World	7,713	8,548	9,735
Europe and Northern America	1,114	1,132	1,136
Sub-Saharan Africa	1,066	1,400	2,118
Eastern and South-Eastern Asia	2,335	2,427	2,411
Percentage of population 65 years or older			
World	9.1	11.7	15.9
Europe and Northern America	18.0	22.1	26.1
Sub-Saharan Africa	3.0	3.3	4.8
Eastern and South-Eastern Asia	11.2	15.8	23.7
For clarity, not all global regions are shown.			

Note: For clarity, not all global regions are shown.
Source: Department of Economic and Social Affairs: Population Division, World Population Ageing 2019. Copyright (2020), United Nations.

notably dementia and delirium, but there will also be more elderly people with pre-existing disorders.

Physical changes in the brain

The weight of the human brain decreases by approximately 5% between the ages of 30 and 70 years, by a further 5% by the age of 80 years, and by another 10–20% by the age of 90 years. As well as these changes, the ventricles enlarge and the meninges thicken. MRI studies show a complex temporal and spatial profile of changes affecting both grey and white matter, with volume reductions prominent in the hippocampus, frontal cortex, and cerebellum. There is some loss of neurons, although this is regionally selective and much less marked than was formerly believed. Instead, loss of synapses appears to be more important. The cytoplasm of some neurons contains a pigment, lipofuscin. The ageing brain also tends to accumulate senile plaques and neurofibrillary tangles, but with a more restricted distribution and smaller numbers than in Alzheimer’s disease.

Ageing itself is thought to reflect genomic changes (e.g. acquired damage to DNA and shortening of the telomeres), mitochondrial and free radical damage, and alterations in some growth factor and signalling pathways. For a review of brain ageing, see Woulfe et al. (2020).

The neuropsychology of ageing

Assessment of cognitive function in the elderly is complicated by the frequent presence of physical ill health, sensory deficits, and by the need carefully to distinguish normal age-related changes from the early phase of dementia.

Longitudinal studies suggest that intellectual function, as measured by standard intelligence tests, shows a significant decline only in later old age. A characteristic pattern of change occurs, with psychomotor slowing and impairment in the manipulation of new information. By contrast, tests of well-rehearsed skills such as verbal comprehension show little or no age-related decline.

Short-term memory, as measured by the digit span test, for example, does not change in the normal elderly. Tests of working memory show a gradual decrease in capacity, so that the elderly perform significantly less well than the young if attention has to be divided between two tasks or if the material has to be processed additionally in some way. The elderly can usually recall remote events of personal significance with great clarity. Despite this, their long-term memory for other remote events shows a decline. Overall, there appears to be a balance between losses in flexible problem solving, and the benefits of accumulated wisdom derived from experience. The term

successful memory ageing is sometimes used to describe elderly people who have preserved cognition (Nyberg and Pudas, 2019).

As well as these cognitive and motor changes, there can be alterations in personality and attitudes, such as increasing cautiousness and rigidity but also attenuation of other traits (Woods and Windle, 2020).

Physical health

In addition to a general decline in functional capacity and adaptability with ageing, chronic degenerative conditions are common. As a result, the elderly consult their family doctors frequently and occupy the large majority of hospital beds. These demands are particularly large in those aged over 75 years. Medical management is made more difficult by the presence of more than one disorder and by sensory and motor

disabilities, as well as by an increased risk of side effects of treatment.

Social circumstances

For most people, ageing brings with it profound changes in social circumstances. Retirement affects not only income but also social status, time available for leisure, and social contacts. Loss of income is a serious problem facing many older people, and financial problems were the most common worry reported in a large European survey of people over 65 years of age.

Social isolation is a fact of life for many older people, especially in higher-income countries. In the United Kingdom, half of all people over 75 years live alone, and 1 in 5 elderly people report being lonely. Loneliness in the elderly is a significant risk factor for adverse mental and physical health outcomes (Reynolds et al., 2022).

Epidemiology of psychiatric disorders in the elderly

Almost sixty years ago, Kay et al. (1964) conducted the first systematic prevalence study of psychiatric disorder among older people in the general population, including those living at home as well as those living in institutions, in an area of Newcastle upon Tyne, in northern England (Table 19.2). Subsequently there have been many large-scale prevalence and incidence studies, in different populations, and the main findings for the major disorders are summarized below. For a review, see Lilford and Hughes, 2020. Key findings include:

- Neurodegenerative disorders become markedly more common with increasing old age, whereas other disorders show lesser or opposite trends.
- Rates of psychiatric disorder are much higher in health and care settings. For example, a third of residents in old people’s homes have significant cognitive impairment. In general hospital wards, between one-third and half of the patients aged 65 years or over have some form of psychiatric illness.
- Gender ratios for some disorders are different from those in younger adults. For example, schizophrenia and bipolar disorder are more common in elderly women than in elderly men.

Table 19.2 Prevalence of psychiatric disorder in people over 65 years of age

Disorder	Prevalence (%)
Dementia (severe)	5.6
Dementia (mild)	5.7
Neurosis and personality disorder	12.5
Manic depression	1.4
Schizophrenia	1.1
Any disorder	26.3

Source: Kay, D. W. K., Beamish, P., & Roth, M. (1964). Old age mental disorders in Newcastle-upon-Tyne: 1: a study in prevalence, *BJPsych*, 110(465), 146–158. Copyright (1964), Cambridge University Press

However, because a variety of methods and diagnostic criteria have been used, and some findings appear to be changing with time, there is still uncertainty about many aspects of the epidemiology of psychiatric disorders in the elderly (Minett et al., 2020).

Dementia

Although there are references in classical literature, dementia in the elderly has been recognized by modern medicine since the French psychiatrist Esquirol described *démence sénile* in 1838. Emil Kraepelin distinguished dementia from psychoses due to other organic causes such as neurosyphilis, and divided it into presenile,

Table 19.3 Prevalence and incidence of dementia in different populations

	Prevalence at age ≥60 years (%)	Incidence per 1000 individuals	Number of people with dementia in 2001	Predicted increase in proportion with dementia during the period 2001–2040
Western Europe	5.4	8.8	4.9 million	Twofold
North America	6.4	10.5	3.4 million	2.7-fold
Latin America	4.6	9.2	1.8 million	3.9-fold
China	4.0	8.0	6 million	3.4-fold
India and South Asia	1.9	4.3	1.8 million	3.1-fold
Africa	1.6	3.5	0.5 million	2.3-fold
Combined values	3.9	7.5	24.3 million	2.3-fold

Adapted from Ferri, C. P., Prince, M., Brayne, C., Brodaty, H., Fratiglioni, L., Ganguli, M., Hall, K., Hasegawa, K., Hendrie, H., & Huang, Y. (2005). Global prevalence of dementia: A Delphi consensus study. *Lancet*, 366(9503), 2112–2117. Copyright (2005), with permission from Elsevier.

senile, and arteriosclerotic forms. In an important study, Roth (1955) showed that dementia in the elderly differed from affective disorders and paranoid disorders in its poorer prognosis.

Reflecting its growing contribution to the global burden of disease, there has been extensive research on the prevalence and incidence of dementia, and several meta-analyses (see Tables 19.3 and 19.4). A systematic review reported a global prevalence of 5–7% in those aged 60 or older, corresponding to about 36 million people; this number is predicted to double every 20 years, with over 70% of cases being in low- or middle-income countries by 2050 (Prince et al., 2013). The prevalence of dementia rises continuously in old age, approximately doubling every five years, reaching 45% in those aged 95 years or older (Alexander et al., 2015). Equally, these figures illustrate that half of the oldest old still remain cognitively intact. Moreover, there is now strong evidence that the age-adjusted incidence of dementia is decreasing considerably (Stallard et al., 2025); the decline is unexplained, and not attributable solely to reductions in vascular risk factors (Wu et al., 2017).

See Chapter 14 for discussion of the causes of dementia and their epidemiology, including modifiable risk factors.

Mood disorder

Despite the focus on dementia in the elderly, depressive disorders are considerably more common. A systematic review of community-based studies found an average prevalence of clinically relevant depression of 13.5% for those aged 55 years or over, of which 9.8% was classed as minor and 1.8% as major. Prevalence was higher in

women and in older people living in adverse circumstances (Beekman et al., 1999). Among the over-75s, the prevalence of major depression is about 7%, and 17% for all depressive disorders (Luppa et al., 2012). The figures show that many depressed older people do not meet the criteria for major depression, but are instead variously labelled as having minor depression, dysthymia, or sub-threshold depression. However, this does not diminish its clinical significance, as these cases have a similar morbidity and course to those of patients who meet the criteria for major depression (Beekman et al., 2002).

Rates of depression in the elderly depend on the setting. Koenig and Blazer (1992) reported rates of 0.4–1.4%

Table 19.4 Prevalence of dementia at different ages in European populations

Age (years)	Prevalence
60–65	0.15 (0–2.3)
65–70	1.2 (0–3.4)
70–75	3.6 (1.3–5.9)
75–80	7.6 (5.2–10.0)
80–85	13.4 (10.6–16.1)
85–90	21.3 (18.1–24.6)
90–95	31.2 (27.7–35.6)
>95	44.7 (39.8–49.6)

Values are percentages, with 95% confidence interval in brackets.

Source: Alexander, M., Perera, G., Ford, L., et al. (2015). Age-stratified prevalence of mild cognitive impairment and dementia in European populations: A systematic review. *J Alzheimers Dis*, 48(2), 355–369. Copyright (2015), Sage Publications.

in the community, 5–10% among medical outpatients, 10–15% among medical inpatients, and 15–20% among nursing home patients. These variations probably reflect the frequent comorbidity of depression in the elderly with medical disorders, especially cardiac and pulmonary disease (Huang et al., 2010). The physical comorbidities likely contribute to the increased mortality rate, especially from cardiac events (Wei et al., 2019). The high prevalence of depressive disorder masks the fact that first episodes become less common after the age of 60 years and rare after the age of 80 years. Similarly, though bipolar disorder remains a prevalent disorder in the elderly, first episodes of mania are rare, and an organic cause (e.g. secondary to steroids) should always be suspected (Sajatovic et al., 2015).

For a review of depression in the elderly, see Thomas (2020).

Anxiety disorders

In general practice, Shepherd (1966) found that, after the age of 55 years, the incidence of new cases of neurosis (i.e. anxiety disorder) declined. However, the frequency of consultations with the general practitioner for neurosis did not fall—presumably as a result of chronic or recurrent cases. Subsequent larger population surveys have largely confirmed the reduction in frequency from the sixth decade onwards, although prevalence and incidence figures differ markedly between studies. For example, 12-month prevalence rates for any anxiety disorder in the 65–85-year-old age group were estimated at 6%

in an Australian survey, but 17% in a European survey (Byrne, 2020). As with depression, anxiety disorder is frequently comorbid with a medical disorder—78% in the Australian survey—and more common in those in residential care or in hospital settings.

Late-onset schizophrenia-like disorders

Schizophrenia has an estimated prevalence of 0.1–0.5% in those aged 65 or older, with 20–25% of these having an onset after the age of 40 years (Cohen et al., 2015). Unlike in younger cases, schizophrenia with onset after the age of 65 is more common in women than men (Lee, 2020). See Chapters 11 and 12 for discussion of schizophrenia-like disorders in the elderly.

Personality disorders

As discussed in Chapter 15, some personality traits and disorders attenuate with age, while others remain or become exaggerated. However, data are varied, with meta-analyses reporting personality disorder in 3–13% of the elderly (van Alphen et al., 2012), somewhat lower than in younger adults. The decline is mainly attributable to a decline in Cluster B personality disorders, whereas obsessive-compulsive and schizoid characteristics may become more prominent (Debast et al., 2014). Schizoid or paranoid traits may become accentuated by the social isolation of old age, sometimes to the extent of being mistaken for a delusional disorder or schizophrenia.

Principles and practice of old age psychiatry

Having surveyed the epidemiological landscape, in this section we outline the principles and practice of old age psychiatry, in terms of service organization, assessment, treatment, and legal issues.

Organization of services

An international consensus statement defined the essential elements of a mental health service for older people as follows (Wertheimer, 1997):

- primary health care team
- specialist old age psychiatry team
- inpatient unit
- rehabilitation
- daycare
- availability of respite care
- range of residential care facilities
- family and social supports
- liaison with geriatric medicine
- education of health care providers about the needs of older people with psychiatric problems
- research, especially into epidemiological issues, and evaluation of services.

The ‘team’ concept is highlighted as being the key to effective provision of care, with multidisciplinary involvement being the norm. However, beyond these broad points of consensus, national policies have

differed considerably in terms of the development of services and how they are implemented. In the United States, emphasis has been placed on care in hospitals and nursing homes. In Europe, Canada, and Australasia, there has been varying emphasis on social policies to provide care in the community and sheltered accommodation. In this section, services in the United Kingdom are described as an example. Health, social, and voluntary services are described separately, although good care depends on close collaboration at all levels, from strategic planning to the coordinated provision of care for each individual patient. Reflecting this need, health and social services, and their budgets, are now usually integrated in the United Kingdom.

A National Dementia Strategy, 'Living Well with Dementia' was launched in England and Wales in 2009 (Department of Health, 2009), with three key steps:

1. Ensure better knowledge about dementia and remove stigma.
2. Ensure early diagnosis, support, and treatment for people with dementia and their families and carers.
3. Develop services to better meet changing needs.

The strategy remains in place, and dementia remains high on the political agenda. There are no research data regarding the extent to which the goals of the strategy have yet been achieved.

For a review of old age psychiatry services, see Denning and George (2020).

Health services

Primary care

The general practitioner has a central role in assessment and management of the problems of mentally ill older people (Robinson and Chew-Graham, 2020). In many instances, the general practitioner, together with other health professionals in the primary care team and community health services, assesses and manages patients without referring them to a specialist. In other cases, management often remains in primary care once a specialist assessment, such as in a memory clinic, has been carried out. However, general practitioners do not detect all of the psychiatric problems of the elderly at an early stage (Mitchell et al., 2011), perhaps due both to time pressures and to lack of awareness of the significance of psychiatric illness among the elderly.

Old age psychiatry services

The organization of psychiatric services varies in different localities, as it reflects the local styles of service providers, local needs, and the extent of provision for

this age group by general psychiatric services, as well as national policies. Nevertheless, there are some general principles of planning:

- The aims should be to maintain the elderly person at home for as long as possible, to respond quickly to medical and social problems as they arise, to ensure coordination of the work of those providing continuing care, and to support relatives and others who care for the elderly person at home.
- There should be close liaison with primary care, other hospital specialists who may be involved, social services, and voluntary agencies.
- A multidisciplinary approach should be adopted with a clinical team that may include psychiatrists, psychologists, community psychiatric nurses, occupational therapists, and social workers. Some members of the team should expect to spend more of their working day in patients' homes and in general practices than in the hospital.
- Issues of mental capacity should always be at the forefront of decision making. In the United Kingdom, the Mental Capacity Act provides the relevant legislation (see Chapter 4).
- It is important that the care provided by different agencies is coordinated. In the United Kingdom the Care Programme Approach (CPA) is intended to improve coordination by making a single person, the key worker or 'care coordinator', responsible for ensuring that appropriate assessment, care planning, and review take place (see Chapter 26).
- The needs of particular elderly groups, such as those with intellectual disability, prisoners, and otherwise vulnerable people, should be considered. For example, some minority groups may have literacy issues, or different views of symptoms such as memory loss, and services need to be tailored to reflect their needs (Giebel et al., 2015).

We now consider the contributions of the various parts of an old age psychiatry service. In the United Kingdom, financial constraints and NHS reorganizations mean that services for older people are changing and will no doubt change further over the lifetime of this book; they also differ in the devolved nations. Although the basic principles and components of care may not change radically, readers will need to familiarize themselves with local conditions.

Domiciliary psychiatric care

Assessment and treatment commonly takes place in the patient's own home. This is more convenient for the

patient, and offers a more relevant and realistic assessment of the difficulties facing the patient and their carers. Assessment is often undertaken by a member of the community mental health team who has a clinical background in nursing, occupational therapy, or social work. This person may become the patient's care coordinator, able to take an overview of their health and social needs, and who can liaise between different agencies, including primary care and specialist services. Care coordinators may assess referrals from general practitioners, monitor treatment in collaboration with general practitioners and the psychiatric services, and take part in the organization of home support for older people with dementia.

Outpatient clinics and memory clinics

Although assessment at home is desirable, outpatient clinics are convenient for the assessment and follow-up of mobile patients. *Memory clinics*, sometimes called *memory services*, have been developed for the specialist assessment and investigation of patients with memory problems (Banerjee and Farina, 2020). In the United Kingdom, these clinics provide information, advice, and support to patients and their carers; they also initiate medication used to treat dementia, which can be prescribed subsequently in primary care.

Day hospitals and day centres

In the past, day hospitals were an important aspect of psychiatric care for the elderly. However, daycare is now usually provided by day centres combining health and social care elements, often with the financial and practical involvement of voluntary organizations such as the Alzheimer's Society. Day centres provide an important and cost-effective component of the services available for the elderly with psychiatric problems. All daycare-based arrangements depend crucially on adequate transport facilities.

Inpatient units

Inpatient teams should be able to provide multidisciplinary assessment and treatment of patients with severe disorders. In addition to psychiatrists and psychiatric nurses, these teams may include occupational therapists, psychologists, speech and language therapists, physiotherapists, social workers, healthcare assistants, and others. There is substantial variation in different areas both in the composition of teams and in whether patients with functional illnesses are cared for separately from or together with those with organic disorders. As in the rest of psychiatry, there are far fewer inpatient beds than in the past, leading to a high threshold for admission, and short durations of stay.

Geriatric medicine

There is inevitably some overlap in the characteristics of patients treated by units for geriatric medicine and those treated in psychiatry units; both are likely to treat patients with dementia. However, in the geriatric medicine setting the primary concern is likely to be with a patient's physical health; in old age psychiatry units the predominant problem usually relates to behaviour, personality, or comorbid mental illness (e.g. depression). Optimal placement depends on the relative predominance of behavioural and physical problems. Medical and psychiatric teams need to cooperate closely if all patients are to receive appropriate treatment. Geriatric medicine units are usually found in acute general hospitals, wherein the increasing availability of liaison psychiatry expertise helps to integrate medical and psychiatric treatments.

Long-term care

In some countries, many elderly psychiatric patients are still treated as inpatients. However, in most countries, including the United Kingdom, long-term care is now provided outside hospital, in residential and nursing homes. There is an ongoing debate about how long-term care for the elderly should be funded. The Dilnot Report in 2011 recommended that, once a certain funding contribution ('cap') had been reached by a particular individual, their social care costs should be free, and in England and Wales, the Care Act (2014) enshrines in law the duty of local authorities to provide appropriate assessment and support for social care needs. However, at the time of writing, plans for how long-term care should be funded in the United Kingdom have not been finalized. In the meantime, the provision, availability, and costs incurred by older individuals with psychiatric (and physical) health needs vary considerably in different regions.

Social services

Domiciliary services

In addition to medical services, domiciliary services include home helps, meals at home, laundry, telephone, and emergency call systems. In the United Kingdom, local authorities both provide and commission these services; they also support voluntary organizations and encourage local initiatives such as good-neighbour schemes and self-help groups. Increasing financial pressures on local authorities, and rising demand from the increasing number of elderly people, mean that all services are under constant threat, and an increasing proportion of these services are now provided by private or voluntary organizations.

Residential and nursing care

Older people may need a variety of social service accommodation, ranging from entirely independent housing through sheltered housing schemes, where there may be some communal provision, often including access to a warden, to residential or nursing homes where there are staff available at all times. There is a need for special housing for the elderly that is conveniently sited and easy to run. Ideally, the elderly should be able to transfer to more sheltered accommodation if they become more disabled, without losing all independence or moving away from familiar places.

In residential homes, the needs of residents for help with personal care can be met by health care assistants, whereas in nursing homes more skilled nursing care is available with a greater proportion of staff with a higher level of training. Some, but not all, nursing homes specialize in the care of older people with mental disorders.

In the United Kingdom, local social services are responsible for providing residential homes and other sheltered accommodation. However, many independent organizations and charities also provide residential homes for older people.

Voluntary and third-sector services

Voluntary agencies play a large and increasing role in the provision of facilities and support for patients, their families, and carers. Previously the task for NHS services was to ensure that their contributions were integrated with health and social service provisions, whereas now these agencies often play the dominant role. Any discussion of 'voluntary' service provision for elderly mentally ill patients must also acknowledge the major part played by informal carers.

Carers

In the present context, carers are unpaid relatives, neighbours, or friends who look after older people at home. These informal carers provide substantially more care to older people than do the statutory services, and their role in the overall provision of care should not be underestimated (Oyeboode and Parveen, 2020). In the United Kingdom, over 700,000 people provide care or support for a person with dementia. Most informal carers of older people are (in descending order of frequency) partners, daughters, or sons. About twice as many women as men are informal carers, and about 50% of all informal carers are themselves elderly.

Carers suffer from high rates of depression and stress, especially those caring for a dementia sufferer (del Pino Casado et al., 2019). Incontinence, behavioural

disturbance at night, and aggression are the most distressing problems for carers, both practically and emotionally. The degree of strain imposed on these committed individuals is often undeclared and easily underestimated.

Accurate assessment of carers' needs is important and, in the United Kingdom, social services now have a statutory responsibility to provide this, where requested. Time should be spent with carers, giving advice and discussing their problems. Such support can help families to avoid some of the frustration and anxiety involved in caring for elderly relatives. Published guides available on the internet are also useful. Other practical help can ease the burden; for example, carer support groups, additional support at home (e.g. laundry or meal services), daycare, respite care, or holidays designed to be suitable for people with dementia. The assessment of need and carer support may often have to be multidisciplinary; both community psychiatric nurses and care managers (social workers) play essential roles in coordinating these services, supporting relatives, and providing direct nursing care.

Interventions to support carers are discussed later in the chapter.

Psychiatric assessment in the elderly

The principles and purpose of psychiatric assessment of older people are no different from those for all other patients (Chapter 3). Assessment is intended to establish a diagnosis, to develop a plan of care, and to inform the prognosis. However, there are differences of emphasis and process (Huntley et al., 2020):

- The assessment is likely to take longer, requiring more than one interview, and with a greater reliance on informants. In the assessment of suspected dementia, an informant is essential to complete the history and to verify the extent to which the impairments are affecting the patient's capabilities and safety. Time will also be required to administer cognitive rating scales.
- Assessment at the patient's home has many advantages. It provides the most realistic evaluation of symptoms and their functional impact within the patient's normal environment. The assessment will include assessment of the wider psychosocial situation, considering not only the patient's needs but also those of carers, dependents, and other individuals who are involved. This is particularly relevant for the patient with dementia.

- The physical examination and laboratory investigations play a greater role, because of the higher prevalence of organic disorders as a cause of psychiatric symptoms in the elderly. The psychiatric assessment may also bring coincidental medical problems to light which require investigation or referral.

The referral

It is important to establish at the outset what prompted the referral, and what the referrer hoped to gain from it. This may well require information to be collected even before the patient is seen, not least to establish the most useful way to approach the assessment. In any one case, each person involved in the patient's care may have different, and possibly conflicting, expectations and needs.

Informants

Many elderly people seen by psychiatrists are unable to give complete or reliable information about themselves. Frequently there is a partner or other close relative living with the patient, but in other cases it may be necessary (with consent) to talk to neighbours, friends, or relatives living away from the patient, to build up a picture of their personality and past history. Most patients will already be well known to their general practitioner or to other health professionals, and it is always worth consulting them.

Where to assess the patient

In the United Kingdom, most old age psychiatrists prefer to assess patients in their own homes. This enables much essential 'real-life' information about the patient's ability to function at home to be gained (e.g. ability to make a cup of tea, or recognize relatives in family photographs). It also avoids the patient appearing disoriented simply because of the disturbing effect of having to travel to a hospital for assessment in an unfamiliar environment. Furthermore, it makes it easier to interview other members of the family, and to assess the level of support from neighbours or outside carers, who may be available at the patient's home during the assessment. For a review of home assessment in old age psychiatry, see Ng and Atkins (2012).

Old age psychiatrists may be asked to assess patients on general hospital wards, for example, because the admission may have revealed, or exacerbated, underlying cognitive (or other) difficulties. Although this provides less information about the patient's circumstances, it makes close liaison with the hospital team possible. Even if the family or other carers cannot be present, it

may be possible to telephone them. Similar principles apply to assessment in residential and nursing homes.

As noted earlier, patients with early cognitive impairments are usually offered assessments in a memory clinic. This offers the advantages of a systematic and detailed assessment, and enables relevant further investigations and follow-up to be arranged as efficiently as possible. It is also the vehicle by which drugs to treat dementia are prescribed in the NHS.

Finally, as in other specialties, the Covid-19 pandemic led to an increased use of remote assessments in old age psychiatry. This is likely to continue, though elderly peoples' reduced access to, and familiarity with, computers compared to younger age groups should be considered.

The assessment

The information that is required about the medical and psychiatric history is the same as in younger patients. However, it may be necessary to piece it together from accounts given by the patient and by other informants. It is particularly important to obtain a clear medical history and to determine the patient's past and present medication.

During the mental state examination, assessment of cognitive function has particular significance, especially for those in whom memory impairment is apparent or suspected. This assessment requires a range of clinical questions and observations, supplemented by one or more of the questionnaires available for this purpose (see Chapter 3), such as the Mini-Mental State Examination (MMSE) or Montreal Cognitive Assessment (MoCA).

As physical problems are extremely common, the psychiatrist will often need to examine the patient (especially if this has not been done recently by the general practitioner), including a neurological examination. Laboratory investigations are important for patients on admission, and in all others with significant psychiatric disorder (for investigation of dementia, see Chapter 14). The Comprehensive Geriatric Assessment (CGA) is one tool to ensure mental and physical health are considered together in an integrated approach (Harwood and Cowan, 2020). Assessments should be adapted to reflect the needs and characteristics of ethnic minority groups (Nielsen et al., 2011).

Finally, the assessment process includes assessments of risk, and of the needs of the carer(s), discussed above, as most older people with psychiatric disorders live in their own homes and are cared for by family members or, occasionally, by good neighbours or friends.

Psychiatric treatment in the elderly

As with assessment, the principles of psychiatric treatment in the elderly resemble those for other adults, but the greater age of these patients requires consideration of three issues that have an impact upon treatment.

1. Elderly patients are likely to have multiple problems. Psychiatric, physical, and social difficulties usually coexist to some extent. 'Treatment' may thus include a broad range of interventions beyond those normally associated with psychiatry—for example, antibiotics for a urinary tract infection, or liaison with a district nurse. These complex treatment needs are reflected in the multidisciplinary nature of services, described above, and require careful planning and integration of treatment provision.
2. Clear boundaries between normality and disease are rare. This poses challenges for treatment thresholds and service provision.
3. Lack of competence (capacity) is common, due to cognitive impairment. There is an increasing focus on how (lack of) competence should be assessed and responded to. The question of capacity underlies a range of ethical and legal issues in which old age psychiatrists become involved (see below).

Physical treatments

The efficacy of psychotropic drugs is generally not affected by age, and elderly patients should not be denied effective drug treatment, especially for depressive disorders. However, prescribing in the elderly requires particular caution, for two main reasons (Bishara, 2020).

1. The incidence of unpleasant or dangerous side effects is high and can produce delirium and other psychiatric disorders. Most problems arise with drugs used to treat cardiovascular disorders (antihypertensives, diuretics, and digoxin) and those that act on the central nervous system (antidepressants, hypnotics, anxiolytics, antipsychotics, and antiparkinsonian drugs). These problems arise because of differences in pharmacokinetics with ageing, and the greater number of drugs that older people are prescribed, which increases the likelihood of harmful interactions.
2. Compliance with treatment may be compromised in those who live alone, who have poor vision, or who are forgetful or confused.

Given these factors, the following points should be considered when prescribing for the elderly person with a psychiatric disorder.

- Start with a low dose, increase slowly, and expect the final dose to be considerably lower than in younger patients. The main exceptions to this are the selective serotonin reuptake inhibitors (SSRIs), for which more comparable doses are used.
- All medication should be reviewed regularly, and kept to a minimum. Before starting a drug in the elderly, it is good practice to state clearly in the notes the reason for the decision (including, for example, the score on a depression rating scale), and the criteria by which the effects of the treatment will be assessed. If good evidence of efficacy is not achieved, medication should be gradually withdrawn.
- The drug regimen should be as simple as possible. Medicine bottles should be labelled clearly, and memory aids, such as packs containing the drugs to be taken on a single day with daily dose requirements, should be provided. If possible, drug-taking should be supervised.

Electroconvulsive therapy

Advanced age is not a contraindication. However, attention should be paid to the physical health of all elderly patients undergoing this treatment, and frail patients should be assessed by an anaesthetist before receiving ECT.

Psychological treatment

In all patients, supportive therapy with clearly defined aims can be helpful, including joint sessions with the patient's partner or carer.

In patients who have no cognitive impairment, the specific psychological therapies that are used in younger individuals remain appropriate for the same range of disorders and with similar expectations of success (Box 19.1). In particular, cognitive and behavioural interventions for mood and anxiety disorders are effective, and are often preferred by the patient. Psychodynamic psychotherapy is less appropriate for the elderly and is seldom used, although cognitive analytic therapy has been found useful.

Even mild cognitive impairment militates against the use of most of the common psychological therapies, as it is likely to prevent the patient from understanding or implementing the treatment). However, a range of specialized psychological and behavioural interventions (e.g. reminiscence therapy) are becoming widely used

in this group, with the benefits targeted as much at the carer as at the patient.

For a review of psychological treatment in the elderly, see Wilkinson and Laidlaw (2020).

Psychosocial treatments

Some patients can achieve independence through measures designed to encourage self-care and domestic skills, and to increase social contacts. For those living at home, a domiciliary occupational therapist may be able to give useful advice on environmental or other modifications that will help the patient to live more independently. More severely impaired patients who are living in institutional care can benefit from an environment in which individual needs and dignity are respected, and each person retains some of their personal possessions. Further psychosocial interventions of this kind are discussed in the section on treatment of dementia.

Legal, financial, and ethical issues

Both the law and ethical principles apply as much to the elderly as to younger people, and these issues are discussed in Chapter 4. However, certain legal and ethical issues that are pertinent in psychiatry have particular relevance to the elderly (see Box 19.2), for several reasons:

- The nature of the common illnesses from which they suffer, especially dementia, which compromises their competence to make decisions and give consent.

Box 19.1 Indications for psychological treatments with older adults

Patient preference, as an alternative to medication (e.g. in anxiety disorder)

To augment the effect of medication (e.g. antidepressant)

Prevention of recurrence (e.g. of depression)

To foster adherence to medication (e.g. in bipolar disorder)

To avoid harms from medication (e.g. antipsychotics in dementia)

To help a distressed caregiver

- Problems arising from increasing age and frailty (e.g. driving).
- The greater likelihood and proximity of death, which gives greater urgency to issues such as testamentary capacity and advance directives (Wheat, 2020).

In the United Kingdom, many of these issues are governed by the Mental Capacity Act (Chapter 4).

Financial affairs

Older people, particularly those with mental disorders, may have difficulty in managing their financial affairs. In the United Kingdom, if the issues are relatively simple—for example, involving only a state pension and benefits—the person can request that someone else, known as an appointee, collects these benefits on their behalf. Alternatively, while the person still has mental capacity to understand what is involved, they can grant a Lasting Power of Attorney in favour of one or more others. The attorney then has authority to carry out any financial transaction on behalf of that person. If the donor of the power subsequently becomes mentally unable to manage their own affairs, the attorney may continue to do so on their behalf, provided the arrangement is registered with the Court of Protection. Sometimes a person becomes mentally incapable of managing their affairs without having made any formal arrangement for someone else to act on their behalf. Under these circumstances, an application supported by medical evidence

Box 19.2 Ethical and legal issues in the elderly

Confidentiality in relation to information from carers
Confidentiality of information about financial circumstances

Consent to treatment

- Capacity to consent to physical and psychological treatment
- Advance directives
- Decisions ‘not to treat’

Damaging behaviour

Management of financial affairs

- Nominating another to take responsibility (Power of Attorney)
- Procedures to enable others to take responsibility

Entitlement to drive a vehicle

Consent to participate in research

of incapacity may be made to the Court of Protection for a deputy to be appointed to manage the patient's affairs.

Under the Mental Capacity Act an attorney can also make proxy health and welfare decisions on behalf of an incompetent person. This requires that the person was competent to donate that Power of Attorney before they became incompetent. The Mental Capacity Act introduced a new power, the Advance Decision to Refuse Treatment (ADRT), which allows an agreed proxy to make decisions that are binding even where they will result in death. Because of the gravity of this power, it must be written and witnessed and confirm the competence of the patient at the time at which it is arranged (see Chapter 4).

For a review, see Wheat (2020).

Driving

A common practical problem is the inability of older patients, especially those with dementia and Parkinson's disease, to drive safely. As with other disorders, doctors generally have an obligation, which overrides confidentiality, to inform the authorities responsible for the provision of driving licences if patients decline to do so themselves. In practice, older people often make the decision to stop driving anyway, or are persuaded to do so by relatives. For a review, see Wilson and Pinner (2013).

Abuse and neglect of the elderly

Abuse and neglect of the elderly by family members, carers, or acquaintances, raises the importance of safeguarding awareness for all healthcare professionals working with the elderly. Its psychiatric relevance arises

in part from the finding that people with dementia are particularly likely to be abused.

Elder abuse

This term refers to actions by a carer or other trusted person that cause harm or create a serious risk of harm to an elderly person (whether or not harm is intended), or to failure by a caregiver to satisfy the elderly person's basic needs or protect them from harm. The term 'elder maltreatment' is also used.

Five forms of elder abuse are recognized—physical, psychological, sexual, financial, and neglect. Prevalence rates of 2–10% are reported, with a higher risk if:

- the abused person has dementia
- the carer and the abused person live together
- the abused person is socially isolated (e.g. lacks close friends)
- the carer has a psychiatric disorder or misuses alcohol
- the carer is heavily dependent (e.g. financially) on the person who is being abused.

Elder abuse is associated with an increased mortality rate compared with that for matched older people, and with other adverse outcomes, ranging from depression to placement in a nursing home.

Although there is now widespread awareness of the problem, and increasing training for healthcare professionals, there is no evidence about the effectiveness of interventions. At present, a range of measures are used, such as increased social support, respite care, relationship counselling, and carer education programmes. Legal approaches (e.g. guardianship, law enforcement agencies) may also be necessary.

For a review of elder abuse, see Patel et al. (2021).

Clinical features and treatment of psychiatric disorders in the elderly

Depressive disorder in the elderly

Clinical features

There is no clear distinction between the clinical features of depressive disorders in the elderly and those in younger people, but some symptoms are more striking in the elderly. Post (1972) reported that a third of

depressed elderly patients had severe retardation or agitation. A meta-analysis confirmed these findings, and also reported that hypochondriasis and somatic symptoms were more common, whereas guilt and loss of libido were less common than in younger adults (Hegeman et al., 2012). Depression itself is sometimes not conspicuous in the elderly and may be masked by other symptoms, particularly hypochondriacal complaints, and so depressive disorder should always be considered when the patient

presents with symptoms of this kind. Psychotic depression in the elderly also differs in some respects from psychotic depression in younger adults, with a greater overall severity and more prominent hypochondriacal delusions (Gournellis et al., 2014).

Depressive pseudodementia

Some retarded depressed patients, especially the elderly, present with '*pseudodementia*'—that is, they have difficulty with concentration and remembering, but careful clinical testing shows that there is no major defect of memory. Differentiation of these patients from those with early dementia is important and can be difficult. Moreover, the distinction is complicated by evidence that depressive pseudodementia is a predictor of subsequent dementia (Connors et al., 2019). Features that suggest depressive pseudodementia include the following:

- The patient's complaint of memory disturbance tends to be greater than, or contrasts with, the informant's account of memory problems in everyday life.
- Depressive symptoms that pre-dated the memory difficulties.
- 'Don't know' responses and poor involvement with neuropsychological tests (these features are characteristic).
- A personal or family history of mood disorder.

Aetiology

The aetiology of depressive disorders of first onset in late life broadly resembles the aetiology of similar disorders at younger ages, with biological, psychological, and social influences (Fiske et al., 2009).

It might be expected that the loneliness and hardship of old age would be important predisposing factors for depressive disorder. Surprisingly, there is no convincing evidence for such an association, perhaps reflecting the fact that the incidence of major depression is lower in the elderly than in younger adults, as discussed above.

The major difference in aetiology for depression in late life is related to vascular factors. First noted by Post (1972), it has become the prominent hypothesis since Alexopoulos et al. (1997) coined the term '*vascular depression*'. There is now diverse evidence for a multifaceted and bidirectional relationship between late-life depression and vascular disease. Meta-analysis shows that cardiovascular disease, diabetes, stroke, and a composite vascular risk score are all significant risk factors for depression in the elderly, but hypertension, smoking, and dyslipidaemia are not (Valkanova and Ebmeier, 2013). Biological mechanisms include findings that the white matter hyperintensities frequently seen in late-life

depression on MRI scans reflects focal areas of ischaemia and infarction, with effects on brain connectivity. Inflammation is also thought to be a contributory factor. For a review, see Empana et al. (2021).

Differential diagnosis

As noted above, it is sometimes difficult to distinguish between depressive pseudodementia and dementia, and this problem is compounded by the fact that depression may occur during the course of dementia. It is essential to obtain a detailed history from other informants and to make careful observations of the patient's mental state and behaviour using the differentiating features listed earlier. Psychological testing can be a useful adjunct, but it requires experienced interpretation, and it usually adds little to skilful clinical assessment. At times, dementia and depressive illness coexist. If there is real doubt, a trial of antidepressant treatment may be appropriate. Other aspects of the differential diagnosis are similar to those for depressive disorder in younger people.

Treatment

The principles of the treatment of depressive disorder are the same as for younger adults, as described in Chapter 9, and should follow current guidelines. Effective treatments thus exist, but often they are not implemented, both because cases are not diagnosed and because interventions are not offered. With elderly patients it is especially important to be aware of the risk of suicide. Any intercurrent physical disorder should be thoroughly treated.

Antidepressants are as effective as in younger people. The drugs should be used cautiously, perhaps starting with half the normal dosage and increasing this slowly in relation to side effects and response. Potential interactions with other drugs that are being taken by the patient must also be considered and may affect the choice of antidepressant. For patients who do not respond to the maximum tolerated dose of an antidepressant, it may be necessary to use a combination of drugs, as in younger patients, or to combine medication with psychological treatment. As with medication, psychological interventions show similar efficacy in the elderly, and are often preferred by patients, but their availability is limited (Simon et al., 2015).

ECT is appropriate for depressive disorder with severe and distressing agitation, suicidal ideas and behaviour, life-threatening stupor, or failure to respond to drugs. Concerns that ECT is dangerous or ineffective in elderly patients are unwarranted; in fact, remission rates are favourable (Rhebergen et al., 2015).

After recovery, expert guidelines suggest that full-dosage antidepressant medication should be continued for at least 12 months for a first episode in the elderly, and proportionately longer for subsequent episodes. Indefinite continuation has also been advocated (Cleare et al., 2015).

For a review, see Thomas (2020).

Course and prognosis

In his classic study, Post (1972) described depression in the elderly as following the rule of threes—a third recover, a third remain the same, and a third deteriorate and become ‘chronic invalids’. Later studies have generally found a similar, if not poorer, outcome. For example, in a six-year longitudinal study of 277 elderly depressed people, Beekman et al. (2002) found that 32% had a severe chronic course, 44% had a fluctuating course, and only 23% achieved substantial remissions.

Factors that are believed to predict a good prognosis for depression in old age include:

- onset before the age of 70 years
- short duration of illness
- good previous adjustment
- the absence of disabling physical illness
- good recovery from previous episodes
- religiosity (self-reported faith, or belonging to a religious group).

Poor outcome is associated with the severity of the initial illness, the presence of organic cerebral pathology, poor compliance with treatment, and severe life events during the follow-up period. There is now convincing evidence that depression in late life, and earlier, is a risk factor for dementia (see Chapter 14), but it is not clear if effective treatment reduces this risk. Finally, as discussed in Chapter 21, depressive disorder in the elderly is strongly associated with suicide (Beghi et al., 2021).

Bipolar disorder in the elderly

In the first prospective study of mania in old age, Broadhead and Jacoby (1990) found that the clinical picture was the same as in younger patients, but that a depressive episode immediately after the manic episode occurs more frequently. Mixed affective episodes may be somewhat more common than in younger patients. As with depression, a history of bipolar disorder appears to increase the risk of dementia (see Chapter 14).

Management of bipolar disorder in the elderly follows the principles described in Chapter 10. Lithium prophylaxis remains valuable, but blood levels should

be monitored with care, particularly regarding declining renal function, and kept at the lower end of the therapeutic range (0.4–0.6 mmol/l). For a review, see Sajatovic et al. (2015).

Anxiety disorders in the elderly

In later life, anxiety disorders are seldom causes for referral to a psychiatrist, although this may be largely because of non-presentation by patients and lack of recognition or referral by general practitioners. It does not reflect their rarity in the population, as shown by the epidemiological data reviewed earlier in the chapter.

Symptoms of anxiety disorders among the elderly are often non-specific, with features of both anxiety and depression. Hypochondriacal symptoms may be prominent. As in younger patients, the possibility that anxiety symptoms are secondary to an underlying depressive episode should always be actively considered before the diagnosis is made. Personality disorder is a predisposing factor, while physical illness and life events such as retirement or a change of accommodation may act as precipitants. A first onset of panic attacks in an elderly person should always prompt a search for an underlying physical or depressive disorder.

The approach to treatment is similar to that for anxiety disorders in younger adults, with a preference for psychological and behavioural interventions. In practice, medication is often prescribed first-line, and should be done cautiously. For a review, see Byrne (2020).

Schizophrenia-like disorders in the elderly

This section focuses on the features that characterize late-onset schizophrenia compared with schizophrenia of younger onset outlined in Chapter 11. It also highlights some issues that affect the management of all patients with schizophrenia as they become elderly (sometimes called ‘graduates’). The complicated terminology in this area has been covered in Chapters 11 and 12, and reflects the longstanding view that these cases are, to some extent, clinically different from earlier-onset cases.

For a review of late-onset schizophrenia, see Lee (2020). For a review of schizophrenia in the elderly, see Cohen et al. (2015).

Clinical features

Although the core symptoms and overall clinical profile are very similar to those of younger-onset cases, the

Box 19.3 Features characteristic of late-onset compared with early-onset schizophrenia

Symptoms that are more common

Visual, tactile, and olfactory hallucinations
Third-person, running commentary, and derogatory auditory hallucinations
Persecutory delusions

Symptoms that are less common

Formal thought disorder
Affective flattening and blunting
Negative symptoms

Other features

Female predominance
No clear genetic loading
Association with sensory deficits and social isolation
Less premorbid educational and psychosocial impairment
Less working memory and verbal learning impairment
Much lower antipsychotic doses required

relative prominence of some features does differ (see Box 19.3). Of these, the extreme rarity of thought disorder and negative symptoms in the elderly is the most striking. A similar generalized cognitive impairment is seen regardless of age of onset, but late-onset cases show somewhat milder deficits, especially in working and verbal memory. However, it should also be noted that a history of schizophrenia is associated with a threefold increase in risk of dementia at age 65 years, with a lesser but still significant increased risk at older ages (Riisgaard et al., 2015).

Differential diagnosis

Compared with younger-onset schizophrenia, there is a greater likelihood that an elderly patient presenting with psychotic symptoms has an organic psychosis of some kind. Therefore all elderly patients who present with a first onset of schizophrenia-like features should be assessed carefully to exclude a delirium, dementia, or organic psychosis due to a neurological or medical disorder such as neurosyphilis, or neoplasm (see Chapter 14). As in younger patients, schizophrenia must also

be distinguished from other psychiatric disorders in which psychotic symptoms occur, notably delusional disorder or affective psychosis. Such distinctions may be particularly difficult if the patient presents with persecutory beliefs but no other clear symptoms, and if there is a history suggestive of paranoid personality traits. For those in whom visual hallucinations are an isolated symptom, consider Charles Bonnet syndrome (see below).

Aetiology

Familial aggregation is much less common than in earlier-onset schizophrenia, which suggests that there is a smaller genetic predisposition. There is no familial association with neurodegenerative or cerebrovascular disorders. Brain imaging changes are similar in nature and magnitude to those described in younger-onset schizophrenia. Unlike late-life depression, rates of white-matter hyperintensities are not increased. A role for sex hormones in the aetiology of late-onset schizophrenia has been proposed, based upon the female predominance (which approaches 3:1), but there is little direct evidence for this. Schizoid or paranoid personality traits are common in older people who develop late-onset psychotic disorders. Deafness is a risk factor.

Treatment

Antipsychotic medication is the mainstay of treatment of late-onset schizophrenia, as in younger patients, but much lower doses (10–20% of a ‘normal’ dose) are often sufficient. This is not just a reflection of the age of the patient, as higher doses are typically needed for earlier-onset cases when they reach the same age. Another reason for cautious use of antipsychotic medication is that the risk of tardive dyskinesia appears to be considerably greater than in younger patients (Liddle, 2013). Atypical antipsychotics probably have a lower risk of tardive dyskinesia than do typical antipsychotics in this age group, but evidence that they increase cerebrovascular and other adverse events in the elderly must be considered (see below). Cognitive and other psychological interventions may reduce disability, as in earlier-onset cases, although there is little direct evidence for this.

Management of schizophrenia in the elderly must also focus due attention on the complex medical and social needs of many patients; services are often fragmentary and inadequate, in part because of the emphasis on younger patients (Cohen et al., 2015).

Other psychiatric syndromes of the elderly

Senile squalor syndrome

Senile squalor syndrome, also known as *Diogenes' syndrome*, is characterized by severe neglect of self and surroundings, domestic squalor, and social withdrawal and isolation. *Syllogomania* (the hoarding of rubbish) also occurs. It may be precipitated by stressful life events, and is associated with a high mortality after hospital admission. Other causes of self-neglect must be excluded, notably dementia, psychosis, or severe depression. The person usually stubbornly refuses all offers of help, and it is difficult to decide when to intervene using compulsory powers.

For a review, see Pavlou and Lachs (2006).

Charles Bonnet syndrome

Charles Bonnet syndrome refers to the occurrence of visual hallucinations without any other features of psychosis, dementia, or delirium. The syndrome is important, as

misdiagnoses, especially of psychosis, resulting in inappropriate treatments have been reported. It is named after a Swiss philosopher who described the case of his own grandfather in 1760. The syndrome is particularly common in older people with failing eyesight, in whom there is a prevalence of 10–15%. It is usually thought to reflect damage to the visual system.

The visual hallucinations are well formed, elaborate, vivid, and perceived in external space. They may be transient or persistent, and either variable in content or stereotyped. They have no personal meaning for the patient. Although they meet most of the criteria for true hallucinations, patients usually retain some insight into their unreality, and can often make the image disappear by closing their eyes.

The condition tends to improve if vision is restored, or when total blindness occurs. There is no specific treatment. The syndrome may be an early indication of dementia, but this is unproven (Russell and Burns, 2014).

For a review, see Hamedani and Pelak (2019).

Treatment of dementia

The clinical, neuropathological and aetiological features of dementia were described in Chapter 14, and its epidemiology was discussed earlier in this chapter. Here we cover its management. A few key points are worth highlighting:

- Management should be multidisciplinary, with the psychiatrist being part of a broad medical and social care team (National Institute for Health and Care Excellence, 2018a).
- Treatment of the behavioural and psychiatric symptoms of dementia has far greater benefits than treatment of the cognitive deficits. Current drugs for dementia are of relatively limited value.
- Non-pharmacological treatment of the behavioural and psychiatric symptoms should always be prioritized before pharmacological interventions, based on efficacy and safety considerations.
- When drugs are used to treat symptoms of dementia, this should be done with particular caution. There are significant risks when using antipsychotic drugs.
- Most patients with dementia have at least two other chronic illnesses.

- Hospitalization rates are increased compared to non-demented people with similar physical health, and hospitalization is associated with cognitive and functional decline.
- Support for carers is important as well as treatment of the patient.

For review of the management of dementia, see Arvanitakis et al. (2019) and Livingston et al. (2020).

Treatment of behavioural and psychiatric symptoms of dementia

A spectrum of behavioural changes and psychiatric symptoms occurs during the course of dementia, as described in Chapter 14 and summarized in Box 19.4. These features affect the majority of sufferers at some stage of the illness, although the overall prevalence and frequency of individual features vary according to the type of dementia. The behavioural and psychological symptoms cause considerable problems for carers, and their management can be difficult. There is increasing

Box 19.4 Behavioural and psychiatric symptoms of dementia

Behaviours

Agitation
Shouting
Wandering
Apathy
Inappropriate sexual behaviour
Impaired sleep

Symptoms

Delusions
Hallucinations
Depression
Anxiety

Contributory factors

Constipation
Pain
Superimposed delirium
Sensory deficits

concern about the inappropriate use of medication, and fortunately an increasing evidence base for non-pharmacological alternatives, although their availability is often limited.

For a review, see Livingston et al. (2020) and Tampi and Jeste (2022).

Before any treatment is considered, careful assessment is required, and attention should be paid to the following points:

- The nature of the problem behaviour(s) should be clearly identified, including its duration, severity, and any suspected causative or modifying factors. Rating scales may be useful, such as the Neuropsychiatric Inventory. Examples of remediable causal factors include urinary tract infections, constipation, and pain. The latter is often overlooked as a cause of agitation, and adequate analgesia may not be provided (Corbett et al., 2012). Pain in dementia may be due to concurrent physical problems (e.g. leg ulcers, arthritis), or it may be part of the disease process.
- A mental state examination is needed to determine whether the behaviour can be explained in terms of an underlying psychopathology (e.g. whether there is evidence of psychotic symptoms, low mood, or

anxiety). If this examination, together with the history and other observations, supports a diagnosis (e.g. of psychosis, depression, or delirium), it will probably influence the treatment plan. On other occasions, however, no such diagnostic evidence may be forthcoming, in which case the behaviour is dealt with on its merits.

- The natural history is that behavioural and psychological symptoms of dementia fluctuate and often remit. Any treatment, especially if drugs are used, should reflect this fact. Regular reassessment and trials of medication withdrawal should be an integral part of the management plan.
- Behavioural changes in dementia are usually problems for the carer rather than for the patient. This reality should be considered when treatment decisions are made, especially as informed consent is often not possible because of the patient's cognitive impairment.

Non-pharmacological treatment of behavioural and psychiatric symptoms

If, following assessment, a behaviour or symptom is deemed to be sufficient to require treatment, a hierarchical approach should be employed. The first approach is to treat any underlying remediable cause, as outlined above. If the problem persists, the next approach should be based on modification of the patient's environment, or a behavioural or psychological intervention, the choice of which will be influenced by the nature of the problem, the severity of the dementia, the setting, and the resources and expertise available. Medication should be reserved for those patients in whom such approaches prove unsuccessful.

Many types of non-pharmacological intervention are used (Box 19.5), as reviewed in Tampi and Jeste (2022). One example is DICE (describe, investigate, create, evaluate) in which structured evaluation of the behavioural and psychiatric symptoms is followed by training and empowerment of caregivers, along with music therapy (Kales et al., 2019). Meta-analyses show effectiveness and a low risk of adverse events for a range of these interventions against different symptoms, including agitation (Leng et al., 2020).

Drug treatment of behavioural and psychological symptoms

High rates of prescribing of psychotropic drugs in dementia, largely to control behavioural problems, have been reported in nursing homes and other care settings. However, the evidence shows that medication has at most a modest effect and, in the case of antipsychotics,

Box 19.5 Non-pharmacological interventions for behavioural and psychological symptoms of dementia

Sensory stimulation

Music therapy
Aromatherapy (e.g. lavender oil)
Bright light therapy
Massage/touch
White noise

Behavioural management

Differential reinforcement
Stimulus control

Social contact

One-to-one interaction
Pet therapy

Person-centred care

Dementia care mapping
Exercise
Structured activity programmes
Environmental modifications
Wandering areas
Carer education and support

is associated with considerable morbidity and mortality, as noted below. Reflecting these increasing concerns, medication for behavioural problems and psychological symptoms in dementia should not be contemplated until:

- a full assessment has been conducted
- any aggravating physical and environmental factors have been addressed
- non-pharmacological strategies have been attempted, given their safety and evidence for efficacy discussed earlier. Fossey et al. (2006) showed that enhanced psychosocial care led to a significant reduction in the prescribing of medication for nursing home residents with dementia.

Antipsychotics

Despite their widespread use, robust evidence for the efficacy of antipsychotics in the behavioural problems associated with dementia is limited to the short-term treatment of aggression. Moreover, their limited benefits

must be set against the significant risk of extrapyramidal side effects, postural hypotension, tardive dyskinesia, metabolic syndrome, and the many other side effects of antipsychotics to which the elderly are prone. The increasing concern about the use of antipsychotics in dementia is enhanced by three other serious potential risks.

1. They are associated with a dose-related increased mortality risk and increased risk of stroke (Yunusa et al., 2019).
2. Antipsychotics can precipitate an irreversible and sometimes fatal syndrome of parkinsonism, impaired consciousness, and autonomic disturbance, notably in patients with dementia with Lewy bodies.
3. Antipsychotics may hasten cognitive decline (Wolf et al., 2017). The mechanism involved is unknown but may relate to antimuscarinic actions.

These concerns apply to atypical and typical antipsychotics. In consequence, current guidelines recommend that antipsychotics should not be used in dementia unless the patient's problems are severe, associated with either psychotic symptoms or serious distress, or the behaviour poses a danger of physical harm. Comparable warnings and restrictions are also in force elsewhere, and have led to a fall in prescribing rates. However, many older people, especially those in nursing homes, continue to be prescribed these drugs, with a pooled prevalence of 27% for antipsychotic use in dementia (Kirkham et al., 2017). Antipsychotics should be used rarely, if ever, in dementia with Lewy bodies. Assessment for vascular risk factors is advised because of the risk of stroke. As older people are more sensitive to antipsychotics, treatment should always start with a low dose, which is increased slowly, and the effects should be monitored regularly. They should be withdrawn gradually. The choice of antipsychotic should be based on the side effect risk profile, with particular attention to extrapyramidal and anticholinergic effects and sedation. Risperidone is commonly used.

Cholinesterase inhibitors and memantine

These drugs were developed to treat the cognitive symptoms of Alzheimer's disease (see below). They are sometimes also used to treat behavioural problems and psychiatric symptoms of dementia, in part because of the increasing concern about antipsychotics. However, they have limited efficacy against these aspects of dementia, although they do have a role in behavioural disturbances occurring in dementia with Lewy bodies (Stinton et al., 2015).

Other drugs

A range of other drugs are sometimes used to treat behavioural and psychiatric symptoms of dementia. These include antidepressants, which have at most a small benefit, and mood stabilizers (sodium valproate and carbamazepine) benzodiazepines, for which there is no good evidence. As with antipsychotics, any modest benefits of these drugs needs to be set against their potential harms; for example, benzodiazepines and trazodone are associated with an increased risk of fall-related injuries (Bronskill et al., 2018).

Drugs of potential use in the future include pimavanserin, a 5-HT_{1A} receptor partial agonist, currently used to treat psychosis in Parkinson's disease and which also shows efficacy in dementia-associated psychosis; brexpiprazole for agitation; and suvorexant for sleep problems (Scheltens et al., 2021).

Drug treatment of Alzheimer's disease

Management of behavioural and psychiatric symptoms continues to be the most important element of therapy of dementia. However, pharmacological treatment of the cognitive impairment of Alzheimer's disease is available using cholinesterase inhibitors and memantine, and a range of other drugs are under investigation. For a review, see Fox et al. (2025). For clinical practice guidelines for the use of anti-dementia drugs, see O'Brien et al. (2017).

Cholinesterase inhibitors

A loss of acetylcholine was the first neurochemical abnormality of Alzheimer's disease to be discovered, and led to the development of cholinesterase inhibitors as the first effective treatments for the disease. They have an important role in clinical practice, and their discovery was responsible for bringing a new therapeutic optimism to old age psychiatry. They work by decreasing the enzymatic breakdown of acetylcholine and thus increase its availability at the synapse; some of the drugs may also enhance activity at nicotinic cholinergic receptors. However, their efficacy is limited, they do not alter the disease process, and the magnitude of their clinical benefit, as well as their cost-effectiveness, has been controversial. In the United Kingdom they are available on the NHS after initial assessment at memory clinics, but they are not remunerated in France.

The drugs currently available are donepezil, rivastigmine, and galantamine. All of these are approved for use in mild to moderate Alzheimer's disease, based upon trial data showing beneficial effects, relative to placebo,

on cognitive function (usually measured on the cognitive portion of the Alzheimer's Disease Assessment Scale, ADAS-Cog) and on global functioning (assessed with the Clinician Interview-Based Impression of Change scale with carer input, CIBIC-Plus). The size of the effect is relatively modest—for example, benefits of about 3 points on the ADAS-Cog (range of scores, 0–70), and 0.4 on the CIBIC-Plus (range of scores, 1–7). As disease progression averages about 7 points on the ADAS-Cog per year, cholinesterase inhibitors may be viewed as delaying progression by about six months. The efficacy of the three drugs is similar.

Side effects of cholinesterase inhibitors are common and can be troublesome. They include nausea, vomiting, diarrhoea, muscle cramps, insomnia, bradycardia, syncope, and fatigue. Their occurrence can be reduced by starting with a low dose, taking the drug with meals, and increasing the dose slowly. If a patient does not respond to or cannot tolerate a cholinesterase inhibitor, a trial of a different one is worthwhile (O'Brien et al., 2017).

Current UK practice is that treatment with a cholinesterase inhibitor should be instituted in memory clinics by a specialist, following a formal assessment of the patient, from which the diagnosis of probable Alzheimer's disease is made, and its severity established as mild or moderate, using the MMSE and other cognitive scales. The response should be assessed two to four months later, and the drug continued only if the MMSE score has not declined, together with evidence of global or behavioural improvement. In those who show this initial response, the drug should continue as long as the MMSE score remains above 12, considering the patient's overall condition and global functioning. Decisions to stop medication are also complicated by some evidence that this is followed by an accelerated cognitive decline, such that within months the patient returns to the expected level of functioning had treatment never been given.

Memantine

Memantine is an N-methyl-D-aspartate (NMDA)-type glutamate receptor antagonist. The rationale for its use in Alzheimer's disease is that neurotoxicity mediated via the NMDA receptor might contribute to the disease process (Danysz and Parsons, 2012).

Randomized controlled trials have shown that memantine benefits cognition and function in moderate to severe Alzheimer's disease; its efficacy is modest, but it is better tolerated than cholinesterase inhibitors (McShane et al., 2019). Common side effects include problems with balance, dizziness, and constipation.

Combining memantine and a cholinesterase inhibitor has equivocal additional benefits and is not routine.

In the NHS, memantine can be prescribed for patients with moderate or severe Alzheimer's disease, and for those with milder Alzheimer's disease who cannot tolerate cholinesterase inhibitors.

Other drugs

As discussed in Chapter 14, exposure to hormone replacement therapy, non-steroidal anti-inflammatory drugs, and statins have all been associated with lower rates of Alzheimer's disease. Each of these drug classes has therefore been used to treat the condition, but without benefit. Vitamin E has uncertain efficacy. Extracts from the leaves of the maidenhair tree, *Ginkgo biloba*, have been reported to have some benefits in dementia, but the evidence is very weak (Laver et al., 2016). None of these interventions are recommended for clinical practice (O'Brien et al., 2017).

Novel treatment approaches

Treatments currently under development for Alzheimer's disease are intended to modify the disease process, and thereby produce greater and more persistent benefits than the drugs that are being used at present.

The hypothesis that β -amyloid is central to the Alzheimer's disease process (see Chapter 14) led to this molecule and its biochemical pathways being the key treatment targets of current research. Three main approaches have been adopted:

1. Drugs to inhibit the enzymes (*secretases*) by which β -amyloid is formed from its precursor molecule.
2. Drugs to prevent aggregation of β -amyloid.
3. Immunization strategies that target β -amyloid.

Three monoclonal antibodies targeting β -amyloid deposits (aducanumab, lecanemab, and donanemab) have recently been licensed in some countries to treat mild Alzheimer's disease. However, their effects are modest, the cost is high, their use requires genotyping and brain imaging, and they can produce serious side effects (amyloid-related imaging abnormalities (ARIA), including cerebral oedema and haemorrhage). For these reasons, they are unlikely to become widely used (Ross et al., 2022), and they are not available on the NHS in the UK. For a review, see Rabinovici and La Joie (2023). The failure of β -amyloid therapies to demonstrate clearer efficacy may indicate that they need to be used at an earlier phase of the illness, or that the strategy is flawed.

The other main therapeutic target being investigated for Alzheimer's disease is tau and its phosphorylation (Chang et al., 2021); there is also interest in anti-inflammatory, antibiotic, and neuroprotective therapies. To date, none of these strategies have produced clear positive results in clinical trials. For a review, see Scheltens et al. (2021).

Drug treatment of other dementias

There are no drugs currently licensed for any other common dementia. However, several of the approaches that are being used in Alzheimer's disease are also being applied in these disorders, and some are mentioned in dementia treatment guidelines (O'Brien et al., 2017).

Vascular dementia

There have been few randomized trials of treatment for vascular dementia. Cholinesterase inhibitors and memantine are of little benefit and not recommended for use. Currently, treatment of vascular dementia remains limited to control of the cardiovascular risk factors, such as cholesterol and blood pressure, that are thought to underlie the condition. However, there is no good evidence that this is effective.

Dementia with Lewy bodies

Cholinesterase inhibitors have moderate effectiveness in dementia with Lewy bodies and in Parkinson's disease dementia (Wang et al., 2015). The evidence is strongest for rivastigmine, which also has some beneficial effects on behavioural symptoms. This is valuable given the particular toxicity of antipsychotic drugs in this patient group. For a review, see Stinton et al. (2015).

Other dementias

In *frontotemporal dementia*, no treatments have been shown to improve cognition; cholinesterase inhibitors are not effective and may worsen the behavioural symptoms (Sivasathiaselan et al., 2019). No treatments are known for the dementia of *Huntington's disease*, but a range of drug interventions for its psychiatric and motor components are available (van Duijn, 2017). Finally, despite much interest in delaying the onset of dementia, there are no treatments shown clearly to be effective for *mild cognitive impairment* (Kasper et al., 2020). Instead, interventions to reduce dementia risk factors are recommended as in the healthy population, such as aerobic exercise mental activity and treatment of hypertension.

(Livingston et al., 2020). Intensive control of blood pressure may reduce progression to mild cognitive

impairment and dementia (SPRINT-MIND Investigators for the SPRINT research group, 2019).

Further reading

Ames, D., Gallagher, D., Loi, S., & Russ, T. (Eds.). (2022). *Guide to the Psychiatry of Old Age*, 2nd edn. Cambridge University Press.

Dening, T., Thomas, A., Stewart, R., & Taylor, J-P. (Eds.). (2020). *Oxford Textbook of Old Age Psychiatry*, 3rd edition. Oxford University Press, Oxford.

CHAPTER 20

The misuse of alcohol and drugs

Introduction 543

Classification of substance use disorders 543

Alcohol use disorders 545

Other substance use disorders 565

Introduction

The presentation of alcohol and drug misuse is not limited to any particular psychiatric or indeed medical specialty. Alcohol and drug use may play an important part in all aspects of psychiatric practice, and is relevant, for example, to the assessment of a patient with acute confusion on a medical ward, a suicidal patient in the accident and emergency department, an elderly patient whose self-care has deteriorated, a troubled adolescent, or a disturbed child who may be inhaling volatile substances.

The phrases ‘substance use disorders’ (DSM-5) or ‘disorders due to substance use’ (ICD-11) are used to refer to conditions arising from the misuse of alcohol, psychoactive drugs, or other chemicals such as volatile substances. In this chapter, problems related to alcohol are discussed first under the general heading of alcohol use disorders. Problems related to drugs and other chemicals are then discussed under the general heading of other substance use disorders.

Classification of substance use disorders

The two classification systems, DSM-5 and ICD-11, use broadly similar categories for substance use disorders, but group them in different ways. Both schemes recognize intoxication and withdrawal states. ICD-11 has separate categories for ‘harmful use’ and ‘dependence’. In contrast, DSM-5 has a single category of ‘use disorder’, which is rated on a continuum from mild to severe depending on the number of features present; there is therefore no separate category for ‘drug dependence’. ICD-11 also recognizes drug-induced psychotic states and amnestic states, while in DSM-5 these are coded with the psychiatric disorders with which they share phenomenology; thus an alcohol-induced psychotic disorder would be classified under ‘schizophrenia spectrum and other psychotic

disorders’. These and some additional categories are shown in Table 20.1.

In both diagnostic systems the first step in classification is to specify the *substance or class of substance* that is involved (see Table 20.2); this provides the primary diagnostic category. Although many drug users take more than one kind of drug, the diagnosis of the disorder is made on the basis of the most important substance used. Where this judgement is difficult or where use is chaotic and indiscriminate, ICD-11 has a category, ‘disorder due to multiple specified psychoactive substances’, which may be employed. Then the relevant disorder listed in Table 20.1 is added to the substance misused. In this system any kind of disorder can, in principle, be attached to any drug, although in practice certain disorders do

Table 20.1 Substance-related disorders

DSM-5	ICD-11
Intoxication	Intoxication
Use disorder	Harmful pattern of use
	Dependence
Withdrawal	Withdrawal
Delirium*	Delirium
Psychotic disorders*	Psychotic disorder
Neurocognitive disorder*	Dementia
Amnestic disorder*	Amnestic syndrome
Mood disorders*	Mood disorder
Anxiety disorders*	Anxiety disorder
Sexual dysfunctions*	
Sleep disorders*	

*Classified under related psychiatric disorder

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (DSM-5). Copyright (2013), American Psychiatric Association.

not develop with individual drugs; for example, hallucinogens do not have a recognized withdrawal syndrome.

Definitions in DSM-5 and ICD-10

Intoxication

Both DSM-5 and ICD-11 provide definitions of *intoxication*. In both systems, intoxication is considered to be a transient syndrome due to recent substance ingestion that produces clinically significant psychological and physical impairment. These changes disappear when the substance is eliminated from the body. The nature of the psychological changes varies with the individual as well as with the drug—for example, some people when intoxicated with alcohol become aggressive, while others become maudlin.

Use disorder, harmful use, and dependence

The terms ‘use disorder’ in DSM-5 and ‘harmful pattern of use’ in ICD-11 refer to maladaptive patterns of substance use that impair health in a broad sense. As noted above in ICD-11, there is a distinction between ‘harmful pattern of use’ and the more serious condition of ‘dependence’, whereas in DSM-5 the concept of ‘use disorder’ covers the range of severity of substance use. The

Table 20.2 Classes of substances

DSM-5	ICD-11*
Alcohol	Alcohol
Stimulants	Stimulants including amphetamines, methylphenidate, and MDMA
Caffeine	Caffeine
Cannabis	Cannabis and synthetic cannabinoids
Cocaine	Cocaine
Hallucinogens (including phencyclidine and MDMA)	Hallucinogens (phencyclidine is listed separately)
Inhalants	Volatile inhalants
Tobacco	Nicotine
Opioids	Opioids
Sedatives, hypnotics, or anxiolytics	Sedatives or hypnotics
Other	Multiple specified psychoactive substances

* The order of entries in the classification has been amended to show parallels with DSM-5.

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>; *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (DSM-5). Copyright (2013), American Psychiatric Association.

latter includes features associated with drug dependence in ICD-11, such as the experience of ‘pharmacological tolerance’ and ‘withdrawal’, a ‘sense of compulsion to take the substance’, and ‘neglect of alternative goals and interests’ (Table 20.3 and Box 20.1). Thus, in ICD-11 it is possible to meet criteria for harmful use without meeting those for dependence, but the presence of dependence implies harmful use.

Tolerance

This is a state in which, after repeated administration, a drug produces a decreased effect, or increasing doses are required to produce the same effect.

Withdrawal state

This refers to a group of symptoms and signs that occur when a drug is reduced in amount or withdrawn, which last for a limited time. The nature of the withdrawal state is related to the class of substance used. Sometimes the pharmacological properties of substances allow for *cross-tolerance*, for example, benzodiazepines can be used to stave off alcohol dependence.

Table 20.3 Criteria for harmful pattern of use and dependence (ICD-11)

Harmful pattern of use	Dependence
● A pattern of continuous, recurrent, or sporadic use of a psychoactive substance that has caused clinically significant damage to a person's physical health or mental health or has resulted in behaviour leading to harm to the health of others. ● Harm to health of the individual occurs due to one or more of the following: 1) behaviour related to intoxication; 2) direct or secondary toxic effects on body organs and systems; or 3) a harmful route of administration. ● Harm to health of others includes any form of physical harm, including trauma, or mental disorder that is directly attributable to behaviour related to substance intoxication on the part of the person to whom the diagnosis of harmful pattern of use applies. ● The pattern of substance use is evident over a period of at least 12 months if substance use is episodic or at least one month if use is continuous. ● Harm to health is not better accounted for by another medical condition or another mental disorder, including another disorder due to substance use (e.g. substance withdrawal).	A pattern of recurrent episodic or continuous substance use with evidence of impaired regulation of substance use that is manifested by two or more of the following: ● impaired control over substance use (i.e. onset, frequency, intensity, duration, termination, context). ● increasing precedence of substance use over other aspects of life, including maintenance of health, and daily activities and responsibilities, such that substance use continues or escalates despite the occurrence of harm or negative consequences (e.g. repeated relationship disruption, occupational or scholastic consequences, negative impact on health). ● physiological features indicative of neuroadaptation to the substance, including: 1) tolerance to the effects of the substance or a need to use increasing amounts of substance to achieve the same effect; 2) withdrawal symptoms following cessation or reduction in use of substance; or 3) repeated use of substance or pharmacologically similar substances to prevent or alleviate withdrawal symptoms. The features of dependence are usually evident over a period of at least 12 months, but the diagnosis may be made if use is continuous (daily or almost daily) for at least three months.

Source: *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>

Alcohol use disorders

Terminology

Alcoholism. In the past, the term *alcoholism* was generally used in medical writing. Although the word is still widely used in everyday language, it is unsatisfactory as a technical term because it has more than one meaning. It can be applied to habitual alcohol consumption that is deemed excessive in amount according to some arbitrary criterion, and it may also refer to damage, whether mental, physical, or social, resulting from such excessive consumption. In a more specialized sense, alcoholism may imply a specific disease entity that is supposed to require medical treatment. However, to speak of an 'alcoholic' often has a pejorative meaning, suggesting behaviour that is morally bad. For most purposes it is better to use four terms that relate to the ICD-11 and DSM-5 classifications outlined above.

1. *Excessive consumption of alcohol.* This refers to a daily or weekly intake of alcohol that exceeds a specified amount (see below). Excessive consumption of alcohol is also known as *hazardous drinking*.

2. *Harmful alcohol use.* This refers to drinking that causes mental, physical, or social harm to an individual. However, it does not include those with formal alcohol dependence.
3. *Alcohol dependence.* This term can be used when the criteria for an ICD-11 dependence syndrome listed in Table 20.3 are met.
4. *Alcohol use disorder.* This term from DSM-5 includes both harmful alcohol use and alcohol dependence.

At this point it is appropriate to examine the moral and medical models of alcohol misuse.

The moral and medical models

According to the *moral model*, if people drink too much, they do so of their own free will, and if their drinking causes harm to them or their family, their actions are morally bad. The corollary of this attitude is that public drunkenness and associated criminal activity should be punished. In many countries this is the official practice;

Box 20.1 Substance use disorder in DSM-5

DSM-5 captures all the symptoms associated with harmful use and dependence in ICD-11. However, there is no separate category for 'dependence' in DSM-5, and the extent of the substance use disorder (in the example below, alcohol) is decided by the number of symptoms endorsed out of a total of 11 (mild, 2–3 symptoms; moderate, 4–5 symptoms; severe, 6 or more symptoms). The symptoms refer to the past year and can be elicited by asking people whether in the past year they have:

1. Had times when they ended up drinking more, or for longer, than they intended?
2. More than once wanted to cut down or stop drinking, or trying to, but couldn't?
3. Spent a great deal of time on activities necessary to obtain alcohol, use alcohol or recover from its effects?
4. Spent a lot of time drinking? Or being sick or getting over other after-effects?
5. Found that drinking—or being sick from drinking—often interfered with taking care of their home or family? Or caused job troubles? Or school problems?
6. Continued to drink even though it was causing trouble with their family or friends?
7. Given up or cut back on activities that were important or interesting to them, or gave them pleasure, in order to drink?
8. More than once got into situations while or after drinking that increased their chances of getting hurt (such as driving, swimming, using machinery, walking in a dangerous area, or having unsafe sex)?
9. Continued to drink even though it was making them feel depressed or anxious or adding to another health problem? Or after having a memory blackout?
10. Had to drink much more than they once did to get the effect they want? Or found that their usual number of drinks had much less effect than before?
11. Found that when the effects of alcohol were wearing off, they had withdrawal symptoms?

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public drunks are fined, and if they cannot pay the fine, they go to prison. Many people now believe that this approach is too harsh and unsympathetic. Whatever the humanitarian arguments, there is little practical justification for punishment, since there is little evidence that it influences the behaviour of dependent drinkers. However, it is possible that social disapproval could play a role in dissuading non-dependent drinkers from excessive consumption.

According to the *medical model*, a person who misuses alcohol is *ill* rather than wicked. Although it had been proposed earlier, this idea was not strongly advocated until 1960, when the physiologist and alcohol researcher, E. Morton Jellinek (1890–1963), published an influential book, *The Disease Concept of Alcoholism*. The disease concept embodies three basic ideas.

1. Some people have a specific vulnerability to alcohol use disorders.
2. Excessive drinking progresses through well-defined stages, at one of which the person can no longer control their drinking.
3. Excessive drinking may lead to physical and mental disease of several kinds.

One of the main consequences of the disease model is that attitudes towards excessive drinking become more humane. Instead of blame and punishment, medical treatment is provided. The disease model also has certain disadvantages. By implying that only certain people are at risk, it diverts attention from some important facts. First, anyone who drinks regularly for a long time may become dependent on alcohol. Secondly, the best way to curtail harmful use of alcohol may be to limit consumption *in the whole population*, and not just among a pre-disposed minority. Thirdly, it may suggest that excessive drinking, at least initially, is not a product of a personal choice, though here one has to remember that personal choice in the use of alcohol and substances is impacted by societal disadvantage and adversity.

Perhaps a useful way to resolve these two approaches is to apply the moral model to excessive drinking in the population in the hope of decreasing the number of people who put themselves at risk of alcohol-related disability as well as potentially risking the lives of others through accidents or violence. However, once dependence has occurred, with its attendant loss of control over drinking, and marked impact on brain and body health, a medical approach may be more appropriate.

Excessive alcohol consumption

In many societies the use of alcohol is sanctioned and even encouraged by sophisticated marketing techniques. Therefore the level of drinking at which an individual is considered to demonstrate excessive alcohol consumption is a somewhat arbitrary concept, which is usually defined in terms of the level of use associated with significant risk of alcohol-related health and social problems. It is usually expressed in units of alcohol consumed per week. For example, in the United Kingdom, government advice published in 2015 suggested that men and women should not regularly drink more than 14 units a week and that those drinking towards the higher end of this range should spread their drinking over three days or more. These limits are lower than those previously recommended (3–4 units a day for men and 2–3 units a day for women).

If the concept of excessive alcohol consumption is to be understood and accepted, it is necessary to explain the units in which it is assessed. In everyday life, this is done by referring to conventional measures

such as pints of beer or glasses of wine. These measures have the advantage of being widely understood, but they are imprecise because both beers and wines vary in strength (see Table 20.4). Alternatively, consumption can be measured as the amount of alcohol (expressed in grams). This measure is precise and useful for scientific work, but is difficult for many people to relate to everyday measures.

For this reason, the concept of a *unit of alcohol* is widely used in health education. A unit can be related to everyday measures because it corresponds to half a pint of beer, one moderately sized glass of table wine, one conventional glass of sherry or port, and one single bar measure of spirits. It can also be related to average amounts of alcohol (see Table 20.4). Thus, on this measure a typical can of beer (450 ml) contains nearly 1.5 units, a bottle of table wine contains about 10 units, a bottle of spirits contains about 30 units, and 1 unit is about 8 g of alcohol. The increase in strength of wine and beer over the past two decades has made many of these calculations difficult. Wine sold in the United Kingdom now contains an average of 12–14% alcohol,

Table 20.4 Alcohol content of some beverages

Beverage	Approximate alcohol content (%)	Grams of alcohol per conventional measure	Units of alcohol per conventional measure (approximate)
Beer and cider			
Ordinary beer	3	16 per pint	2 per pint
		12 per can	1.5 per can
Strong beer	5.5	32 per pint	4 per pint
		24 per can	3 per can
Extra-strong beer	7	40 per pint	5 per pint
		32 per can	4 per can
Cider	4	24 per pint	3 per pint
Strong cider	6	32 per pint	4 per pint
Wine and spirits			
Table wine	10–14	10 per glass	2 per glass
		70 per bottle	12 per bottle
Fortified wines (sherry, port, vermouth)	13–16	8 per measure 120 per bottle	1 per measure 15 per bottle
Spirits (whisky, gin, brandy, vodka)	32–40	8–12 per single measure 240 per bottle	1–1.5 per measure* 30 per bottle

* Somewhat larger measures are used in Scotland and Northern Ireland (12 g).

and sometimes more. To compound matters, the size of wine glass used in most restaurants has increased from 125 ml to 175 ml, so that an average serving in a restaurant or public house is equivalent to a minimum of 2 units.

Epidemiological aspects of excessive drinking and alcohol misuse

Epidemiological methods can be applied to the following questions concerning excessive drinking and alcohol misuse.

- What is the annual per capita consumption of alcohol for a nation as a whole, and how does this vary over the years and between nations?
- What is the pattern of alcohol use of different groups of people within a defined population?
- How many people in a defined population have harmful alcohol use and dependence?
- How do harmful alcohol use and dependence vary with such characteristics as gender, age, occupation, social class, and marital status?

Unfortunately, we lack reliable answers to these questions, partly because different investigators have used different methods of defining and identifying alcohol misuse and ‘alcoholism’, and partly because excessive drinkers tend to be evasive about the amount that they drink and the symptoms that they experience.

Consumption of alcohol in different countries and at different times

In the United Kingdom, the annual consumption of alcohol per adult (calculated as absolute ethanol consumption) doubled between 1960 and 2000, rising from about 4 litres to over 8 litres. Consumption in western Europe has generally been higher than this, particularly in France and Luxembourg. High levels of consumption are also seen in the former Soviet Union and Eastern Europe, while in the Eastern Mediterranean and in Muslim countries consumption is much less.

Recent changes in the United Kingdom can be usefully considered in a historical perspective. For example, in Great Britain, between 1860 and 1900 the annual consumption of alcohol was about 10 litres of absolute alcohol per head of population over 15 years of age (an average of about 19 units of alcohol weekly). It then fell until the early 1930s, reaching about 4 litres per person over 15 years of age per annum. Consumption then increased slowly until the 1950s, when it began to rise more rapidly, reaching a peak in about 9 litres per person in 2005. Over the past 15 years there

has a modest drop in annual consumption; the average annual amount drunk in 2019 was around 7 litres per person (Health Survey for England, 2019).

During the Covid-19 pandemic there was little obvious change in the overall amount of alcohol consumed, though this was difficult to measure because of the closure of many retail outlets. However, there was an increase in alcohol-related deaths mainly driven by an increase in alcohol-related liver disease. This suggests that heavier drinkers were drinking more during this time.

Drinking habits in different groups

Surveys of drinking behaviour generally depend on self-reports, a method that is open to obvious errors. Enquiries of this kind have been conducted in several countries, including the United Kingdom and the United States. Such studies show that the highest reported consumption of alcohol is generally among *young men* who are *unmarried, separated, or divorced*. However, over the past 15 years drinking by women has increased.

Based on population estimates in 2019 in England, 17% of men and 22% of women had drunk no alcohol in the past year while 57% drank at levels with low risk of harm. The remainder—that is about 20% of the adult population—drank at excessive levels that could represent a potentially *harmful pattern of use* (Health Survey for England, 2019)

The prevalence of harmful alcohol use

This can be estimated in three ways—from hospital admission rates, from deaths from alcoholic cirrhosis, and by surveys of the general population.

Hospital admission rates. These give an inadequate measure of prevalence, because a large proportion of excessive drinkers are not admitted to hospital.

- In the United Kingdom during the period 2019–2020 there were just over 1,000,000 hospital admissions where an alcohol-related disease, injury, or condition was the primary reason for hospital admission or a secondary diagnosis.
- In the United Kingdom in 2020 there were around 8,000 deaths directly related to alcohol use (see Alcohol Change, 2022).

Deaths from alcoholic cirrhosis. About 10–20% of people who drink alcohol harmfully develop *cirrhosis of the liver*, and there are correlations in a population between rates of liver cirrhosis and mean alcohol consumption. Therefore deaths from cirrhosis can be used as a means of estimating rates of harmful alcohol use. For example, in the United Kingdom in 2020,

cirrhosis was the cause of over 5,000 of the deaths directly related to alcohol use.

General population surveys. One method of ascertaining the rate of harmful alcohol use in a population is by seeking information from general practitioners, social workers, probation officers, health visitors, and other agencies who are likely to come into contact with heavy drinkers. Another approach is the community survey, in which samples of people are asked about the amount they drink and whether they experience symptoms. For example, the National Survey Comorbidity Replication in the United States found a one-year prevalence rate for DSM-IV alcohol abuse of 3.1%, while the prevalence of alcohol dependence was 1.3% (Kessler et al., 2005b).

A more recent US population study using the more liberal 'alcohol use disorder' criteria of DSM-5 found a one-year prevalence of 13.9% (Grant et al., 2015). The Psychiatric Morbidity Survey in England classified about 20% of adults living in households as excessive (hazardous) drinkers, of whom 3.1% either had a harmful pattern of use or were alcohol dependent (McManus et al., 2016). Over time there are significant individual fluctuations in alcohol consumption, with a substantial proportion of people moving from hazardous drinking towards safer drinking, while others move into the hazardous drinking category.

Surveys based on households are likely to miss certain groups at high risk of alcohol misuse and dependence. For example, in a UK survey among people living in night shelters or sleeping rough, 40% were found to drink more than 50 units of alcohol a week, and about 35% of those sleeping rough were estimated to have severe alcohol dependence (Farell et al., 2003).

Alcohol misuse and population characteristics

Gender

Rates of alcohol misuse and dependence are consistently higher in men than in women, but the ratio of affected men to women varies markedly across cultures. In western countries, about three times as many men as women suffer from alcohol misuse and dependence, but in Asian and Hispanic cultures a higher ratio of men to women are affected. In the Psychiatric Morbidity Survey in England the rate of hazardous drinking among white men (30%) was about twice as high as that among white women (15%), while the rate among South Asian men (4%) was higher than that among South Asian women (2.5%) (McManus et al., 2016). However, perhaps because of changing social attitudes, the gap between men and women with regard to excessive drinking

seems to be narrowing in many western countries. This is of concern because studies indicate that women are more susceptible than men to many of the damaging effects of alcohol (see below).

Age

Early use of alcohol is an important determinant of later use. A cross-sectional survey of over 10,000 US adolescents showed that 15.1% met criteria for lifetime abuse of alcohol with the median age at onset of 14 years (Swendsen et al., 2012). We have seen that the heaviest drinkers are men in their late teens or early twenties. In most cultures the prevalence of alcohol misuse and dependence is lower in those aged over 45 years.

Religion, culture, and ethnicity

The followers of certain religions which proscribe alcohol (e.g. Islam, Hinduism, and the Baptist Church) are less likely than the general population to misuse alcohol. It is also worth noting that the non-Caucasian population in the United Kingdom and the United States are less likely to drink excessively than the Caucasian population, and therefore have a lower rate of alcohol-related disorders. In some instances, the low consumption of alcohol in a particular ethnic group may be due to a *biologically determined lack of tolerance* to alcohol. For example, some Asian populations with a particular variant of the isoenzyme of aldehyde dehydrogenase experience flushing, nausea, and tachycardia owing to accumulation of acetaldehyde when they drink alcohol. Such individuals are likely to be at reduced risk of excess drinking and the consequent development of alcohol-related disorders.

Occupation

The risk of alcohol misuse is much increased among several occupational groups. These include chefs, kitchen porters, bar workers, and brewery workers, who have easy access to alcohol; executives and salesmen who entertain on expense accounts; actors and entertainers; seamen; and journalists. Doctors have been said to have an increased risk of harmful drinking, but whether this is in fact the case has been questioned. However, doctors do have higher rates of prescription drug misuse, and treatment programmes for both this problem and alcohol-related disorders may be difficult to institute.

Alcohol dependence and alcohol withdrawal

Patients are described as *alcohol-dependent* when they meet the ICD-11 criteria for dependence described

in Table 20.3. The presence of *withdrawal phenomena* is not necessary for the diagnosis of dependence, and a substantial minority of individuals who meet the dependence criteria do not experience any withdrawal phenomena when their intake of alcohol decreases or stops. However, about 5% of dependent individuals experience severe withdrawal symptomatology, including *delirium tremens* and *grand mal seizures*.

Course of harmful alcohol use and dependence

Vaillant (2003) described a 60-year follow-up of 194 men who had exhibited *harmful alcohol use* at some point in their adult life. By the age of 70 years, over 50% of the cohort had died and about 20% were abstinent. Around 10% were drinking in a controlled way, while a further 10% continued to misuse alcohol. In a number of individuals, harmful alcohol use had apparently persisted for decades without remission, death, or progression to formal dependence. However, for *alcohol-dependent* men, periods of controlled drinking were invariably followed by a return to a pattern of alcohol dependence. Attendance at Alcoholics Anonymous was the best predictor of good outcome. This study suggests that the prognosis of alcohol misuse is very variable. However, once an individual has become alcohol-dependent the prognosis is poor unless abstinence can be maintained.

Trim et al. (2013) carried out a follow-up study of 129 men aged between 28 and 33 years, who were diagnosed initially with an alcohol use disorder. Over the next 20 years, 60% of the sample showed remission at some point and in 45% the remission was sustained. Predictors of a good response included a diagnosis of *harmful alcohol use* rather than *dependence* as well as a lower drinking frequency and the receipt of a structured treatment for the alcohol problem soon after the onset of the disorder. This supports the view that a diagnosis of *alcohol dependence* carries a poor prognosis, and also suggests a longer-term benefit for specific treatment early in the course of the condition.

The alcohol withdrawal syndrome

Withdrawal symptoms occur across a spectrum of severity, ranging from mild anxiety and sleep disturbance to the life-threatening state known as *delirium tremens*, a specific form of delirium (Chapter 14) that is described further below. The symptoms generally occur in people who have been drinking heavily for years and who maintain a high intake of alcohol for weeks at a time. The symptoms follow a drop in blood alcohol concentration. They characteristically appear on waking,

after the fall in alcohol concentration occurs during sleep. *Dependent drinkers* often take a drink on waking to stave off withdrawal symptoms. In most cultures, *early-morning drinking* is diagnostic of dependency. With the increasing need to stave off withdrawal symptoms during the day, the drinker typically becomes secretive about the amount consumed, hides bottles, or carries them in a pocket. Rough cider and cheap strong beers may be drunk regularly to obtain the most alcohol for the minimum cost.

The earliest and most common feature of alcohol withdrawal is acute tremulousness affecting the hands, legs, and trunk ('the shakes'). The sufferer may be unable to sit still, hold a cup steady, or fasten buttons. They are also agitated and easily startled, and often dread facing people or crossing the road. Nausea, retching, and sweating are frequent. Insomnia is also common. If alcohol is taken, these symptoms may be relieved quickly; otherwise they may last for several days (see Box 20.2).

As withdrawal progresses, *misperceptions* and *hallucinations* may occur, usually only briefly. Objects appear distorted in shape, or shadows seem to move; disorganized voices, shouting, or snatches of music may be heard. Later there may be *epileptic seizures*, and finally, after about 48 hours, *delirium tremens* may develop.

Box 20.2 Symptoms and signs of acute alcohol withdrawal

Mild

Anxiety, insomnia
Tremor in limbs
Sweating
Tachycardia

Moderate

Obvious tremulousness
Panic attacks
Blood pressure raised
Visual hallucinations
Seizures

Severe (delirium tremens)

Recurrent seizures, status epilepticus
Severe confusion and hallucinations
Dehydration
Circulatory collapse
Multi-organ failure

Other alcohol-related disorders

The different types of damage—*physical, psychological, and social*—that can result from alcohol misuse are described in this section. A person who suffers from these disabilities may or may not be suffering from *alcohol dependence* but will certainly meet criteria for *harmful pattern of use* (ICD-11) or *alcohol use disorder* (DSM-5).

Physical damage

Excessive consumption of alcohol may lead to physical damage in several ways. First, it can have a direct toxic effect on certain tissues, notably the brain and liver. Secondly, it is often accompanied by poor diet, which may lead to deficiency of protein and B vitamins. Thirdly, it increases the risk of accidents, particularly head injury. Fourthly, it is accompanied by general neglect, which can lead to increased susceptibility to infection.

Gastrointestinal disorders

Gastrointestinal disorders are common, notably liver damage, gastritis, peptic ulcer, oesophageal varices, and acute and chronic pancreatitis. Damage to the liver, including fatty infiltration, hepatitis, cirrhosis, and hepatoma, is particularly important and is occurring at younger ages. For a person who is dependent on alcohol, the risk of dying from liver cirrhosis is almost ten times greater than the average. However, only about 10–20% of alcohol-dependent people develop cirrhosis.

Nervous system

Alcohol also damages the nervous system. Neuropsychiatric complications are described later; other neurological conditions include *peripheral neuropathy, epilepsy, and cerebellar degeneration*. The last of these is characterized by unsteadiness of stance and gait, with less effect on arm movements or speech. Rare complications are optic atrophy, central pontine myelinolysis, and Marchiafava–Bignami syndrome. The latter condition results from widespread demyelination of the corpus callosum, optic tracts, and cerebellar peduncles. Its main features are dysarthria, ataxia, epilepsy, and marked impairment of consciousness; in the more prolonged forms, dementia and limb paralysis occur. *Head injury* is also common in alcohol-dependent people.

Cardiovascular system and other general medical conditions

Alcohol misuse is associated with *hypertension* and increased risk of *stroke*. Epidemiological studies have sometimes suggested that low to moderate alcohol

consumption can have modest cardiovascular protective effects but this is likely to be explained by methodological confounding (Biddinger et al., 2022). Alcohol use, even at low levels, has been linked to the development of certain cancers, notably of the mouth, pharynx, oesophagus, liver, and breast.

Other physical complications of excessive consumption of alcohol are too numerous to detail here. Examples include anaemia, myopathy, episodic hypoglycaemia, haemochromatosis, cardiomyopathy, vitamin deficiencies, and tuberculosis. They are described in textbooks of medicine, such as the *Oxford Textbook of Medicine* (Firth et al., 2020).

Harmful use of alcohol in women

Studies suggest that women progress more rapidly than men to hazardous drinking, and tend to suffer the medical consequences of alcohol use after a shorter period of exposure to a smaller amount of alcohol. As well as the expected medical complications noted above, female drinkers also show elevated rates of breast cancer and reproductive pathology, including amenorrhoea, anovulation, and possibly early menopause. For a review, see McCaul et al. (2019).

Damage to the fetus

There is evidence that drinking alcohol in pregnancy can lead to a number of foetal abnormalities estimated to affect around 2% of people in high income countries, of which *foetal alcohol syndrome* is the most severe. This consists of a syndrome of *facial abnormality, small stature, low birth weight, low intelligence, and overactivity*. Foetal alcohol syndrome is associated with clearly excessive alcohol use during pregnancy. However, subsequent research has suggested that even moderate or low levels of alcohol consumption at this time can have adverse effects on childhood behavioural, cognitive, and physical development, and the ‘safe limit’ for alcohol consumption during pregnancy (if such a level exists) is not known. Current advice from the UK Department of Health is that women should avoid alcohol during pregnancy. For a review of the prevalence and characteristics of foetal alcohol syndrome, see Wozniak et al. (2019).

Mortality

Some studies have shown that people with a low level of alcohol consumption (about 1 unit daily) have decreased mortality rates compared with non-drinkers; however, it has been suggested that this apparently protective effect is mainly attributable to confounding lifestyle factors (Costanzo et al., 2019).

Not surprisingly, the mortality rate is increased in those who misuse alcohol. Follow-up investigations have studied mainly middle-aged men drinking excessively, in whom overall mortality is at least twice the expected rate, and mortality in women who drink excessively appears to be substantially higher than this. Of course, alcohol must contribute *indirectly* to many more deaths from causes such as accidents. At a global level, the World Health Organization (2014) has estimated that alcohol causes 3.3 million deaths annually (about 5.9% of all deaths).

Psychiatric disorders

Alcohol-related psychiatric disorders fall into four groups:

1. intoxication phenomena
2. withdrawal phenomena
3. neurocognitive syndromes
4. associated psychiatric disorders.

Intoxication phenomena

The severity of the symptoms of alcohol intoxication correlates approximately with the blood alcohol concentration. As noted above, there is much individual variation in the psychological effects of alcohol, but certain reactions, such as lability of mood and belligerence, are more likely to cause social difficulties. At high doses, alcohol intoxication can result in serious adverse effects, such as falls, respiratory depression, inhalation of vomit, and hypothermia.

The *molecular mechanisms* that underlie the acute effects of alcohol are unclear. Key targets appear to be ethanol binding sites located on NMDA and GABA receptors. Like many drugs with 'rewarding' effects, alcohol indirectly increases the release of dopamine and opioids in the brain. The pleasurable effects of alcohol have been attributed to release of dopamine and opioids in the mesolimbic forebrain, while the anxiolytic effects could reflect facilitation of brain gamma-aminobutyric acid (GABA) activity.

Memory blackouts or *short-term amnesia* are frequently reported after heavy drinking. At first the events of the night before are forgotten, even though consciousness was maintained at the time. Such memory losses can occur after a single episode of heavy drinking in people who are not dependent on alcohol. If they recur regularly, they indicate habitual excessive consumption. With sustained excessive drinking, memory losses may become more severe, affecting parts of the day or even whole days.

Withdrawal phenomena

The general withdrawal syndrome has been described earlier under the heading of alcohol dependence. Here we are concerned with the more serious psychiatric syndrome of *delirium tremens*. Delirious states are described generally in Chapter 14, but delirium tremens is discussed here because of its prevalence and mortality, and its somewhat different treatment.

Delirium tremens occurs in people whose history of alcohol misuse extends over several years. Following alcohol withdrawal there is a dramatic and rapidly changing picture of disordered mental activity, with clouding of consciousness, disorientation in time and place, and impairment of recent memory. Perceptual disturbances include misinterpretations of sensory stimuli and vivid hallucinations, which are usually visual but sometimes occur in other modalities. There is severe agitation, with restlessness, shouting, and evident fear. Insomnia is prolonged. The hands are grossly tremulous and sometimes pick up imaginary objects; truncal ataxia may occur. Autonomic disturbances include sweating, fever, tachycardia, raised blood pressure, and dilatation of pupils. Dehydration and electrolyte disturbance are characteristic. Blood testing shows leukocytosis and impaired liver function.

The condition lasts for three or four days, with the symptoms characteristically being worse at night. It often ends with deep prolonged sleep from which the patient awakens with no symptoms and little or no memory of the period of delirium. Delirium tremens carries a *significant risk of mortality* and should be regarded as a medical emergency.

Neurocognitive syndromes

Neurological syndromes include *Korsakoff's psychosis* and *Wernicke's encephalopathy* (see Chapter 14). In the past, the term 'alcoholic dementia' was widely used, but it remains unclear whether this is a discrete category; many cases of dementia in people with a history of alcohol misuse are probably attributable to Alzheimer's disease and other common forms of dementia. However, ICD-11 retains a category of 'dementia due to use of alcohol' and in DSM-5, alcohol is listed as one of the potential causes of 'neurocognitive disorders.' Alcohol does seem likely to be a contributory factor to dementia. For example, a prospective study in 9,000 civil servants indicated that regular alcohol consumption of around 20 units weekly increased the risk of dementia by 17% (Sabia et al., 2018).

People with alcohol use disorders show a spectrum of deficits on cognitive tests in learning, memory, and executive function. In severe disorders these impairments

often have a *multifactorial aetiology*, which can include effects of head injury, vascular changes, and metabolic and nutritional impairments as well as direct alcohol neurotoxicity. The neurotoxicity of alcohol is magnified by concomitant thiamine deficiency, which is common in heavy drinkers. Even young heavy drinkers can exhibit evidence of neuropsychological impairments, for example, in executive function.

Attention has also been directed to the related question of whether chronic excessive alcohol consumption can cause structural brain atrophy. MRI scans in such patients have shown widespread *loss of grey matter* in both cortical and subcortical areas. Lower grey matter volume in striatal, frontal, and thalamic regions is associated with the duration of harmful alcohol use or lifetime alcohol consumption. White matter changes are also common in heavy drinkers, and studies using diffusion tensor imaging indicate that there is demyelination, particularly in the corpus callosum, as well as tracts linking limbic system and prefrontal cortex. Even in people drinking within 'safe limits', alcohol intake is negatively associated with regional grey matter volumes and white matter microstructure (see Daviet et al., 2022).

The changes in brain structure and cognitive impairment that are seen in excessive drinkers remit to some extent with cessation of alcohol use. However, some abnormalities can still be detected after long periods of abstinence. For a review of alcohol-related brain damage, see Sullivan and Pfefferbaum (2019).

Associated psychiatric disorder

Personality deterioration. As the patient becomes increasingly concerned with the need to obtain alcohol, interpersonal skills and attention to their usual interests and responsibilities may deteriorate. These changes in social and interpersonal functioning should not be confused with personality disorder, which should be diagnosed only when the appropriate features have been clearly present prior to the development of alcohol dependence.

Mood and anxiety disorders. The relationship between alcohol consumption and mood is complex. On the one hand, some depressed and anxious patients drink excessively in an attempt to improve their mood. On the other hand, excess drinking may induce persistent depression or anxiety. People with a history of alcohol dependence have a fourfold increased risk of experiencing subsequent *major depression*, and the risk remains elevated even in those who are no longer drinking. The risk of experiencing an *anxiety disorder* is also significantly increased (Sinclair and O'Neil, 2020).

Suicidal behaviour. Suicide rates among people with alcohol use disorders are much higher than those among individuals who do not misuse alcohol, with a tenfold increased risk compared to the general population. After mood disorders, alcohol use disorders are the most common diagnosis in completed suicides (Rizk et al., 2021). Suicide among alcohol misusers is discussed further in Chapter 21. Here it is worth noting that suicide in young men is associated with a high rate of substance misuse, including harmful alcohol use.

Pathological jealousy. Excessive drinkers may develop an overvalued idea or delusion that their partner is being unfaithful. This syndrome of pathological jealousy (also called morbid jealousy) is described in Chapter 12.

Alcoholic hallucinosis. This is characterized by *auditory hallucinations*, usually involving voices uttering insults or threats, which occur in clear consciousness. The patient is usually distressed by these experiences, and appears anxious and restless. The hallucinations are not due to acute alcohol withdrawal, and can indeed persist after several months of abstinence. There has been considerable controversy about the aetiology of the condition. Some follow Kraepelin and Bonhoeffer in regarding it as a rare organic complication of alcoholism; others follow Bleuler in supposing that it is related to schizophrenia. More recent reviewers have concluded that alcoholic hallucinosis is an alcohol-induced organic psychosis, which is distinct from schizophrenia and has a good prognosis if abstinence can be maintained (Narasimha et al., 2019).

In both DSM-5 and ICD-11, 'alcoholic hallucinosis' is subsumed under the heading of 'substance-induced psychotic disorder'.

Social damage

Family problems

Excessive drinking is liable to cause profound social disruption, particularly in the family (Box 20.3). Marital (or other relationship) and family tension is virtually inevitable. The divorce rate among heavy drinkers is high, and the partners of men who drink excessively are likely to become anxious, depressed, and socially isolated. The partners of women who suffer domestic violence frequently drink excessively, and some women who are admitted to hospital because of self-poisoning blame their partner's drinking. The home atmosphere is often detrimental to any children present, because of quarrelling and violence, and a drunken parent provides a poor role model. Children of heavy drinkers are at risk of developing emotional or behavioural disorders, and of performing poorly at school.

Box 20.3 Medical and social consequences of excessive alcohol consumption in the United Kingdom

1. Annual alcohol-related costs of crime and public disorder are £7.3 billion, workplace costs are £6.4 billion, and health costs are £1.7 billion.
2. Up to one third of all accident and emergency attendances involve alcohol.
3. Around 47% of victims of violence believe that their assailant was under the influence of alcohol.
4. Alcohol is involved in 30–60% of child protection cases. Up to 1.3 million children are adversely affected by family drinking.

Source: Prime Minister's Strategy Unit, Alcohol Harm Reduction Strategy for England. Copyright (2004), Cabinet Office.

Work difficulties and road accidents

At work, the heavy drinker often progresses through declining efficiency, lower-grade jobs, and repeated dismissals to lasting unemployment. There is also a strong association between road accidents and alcohol misuse. In England and Wales in 2019, around a quarter of the 1,700 road traffic fatalities involved a driver who had a blood alcohol concentration above the legal limit (Ramnath, 2021).

Crime

Excessive drinking is also associated with crime, mainly petty offences such as larceny, but also with fraud, sexual offences, and crimes of violence, including murder. Studies of prisoners in England and Wales have shown that about a quarter had serious drinking problems before imprisonment (Rebbapragada et al., 2021). It is not easy to determine to what extent alcohol causes the criminal behaviour and to what extent it is part of the lifestyle of the criminal. In addition, there is a link between alcohol misuse and *antisocial personality disorder*.

The causes of excessive drinking and alcohol use disorders

Despite much research, surprisingly little is known about the causes of excessive drinking and alcohol dependence. At one time it was believed that certain people were particularly predisposed, either through

personality or due to an innate biochemical anomaly. Nowadays this simple notion of specific predisposition is no longer held. Instead, alcohol misuse is thought to result from a variety of interacting factors, which can be divided into individual factors and those in society.

Individual factors

Genetic factors

Most genetic studies of alcoholism have investigated individuals with evidence of alcohol dependence. If less severe diagnostic criteria are involved—for example, fairly broadly defined alcohol misuse—the relative genetic contribution is somewhat less. However, it is well established that alcohol use disorders aggregate in families, and twin studies show a higher concordance in monozygotic than dizygotic twins, though the estimated heritability ranges from 40–70% with no clear sex differences (Carvalho et al., 2019).

If a genetic component to aetiology were to be confirmed, it would still be necessary to discover the mechanism. The latter might be biochemical, involving the metabolism of alcohol or its central effects, or psychological, involving personality, for example, sensation-seeking traits. In addition, it is important to note that a predisposition to misuse alcohol and develop dependence will only be expressed if a person consumes excessive amounts of alcohol. Here non-genetic familial factors are likely to play a major role.

Genes for alcohol dependence risk. Candidate gene studies have focused in particular on genes that affect alcohol metabolism because, as we have already seen, people with impaired activity of the alcohol-metabolizing enzyme, aldehyde dehydrogenase, have unpleasant reactions when they consume alcohol, and are therefore at significantly lower risk of alcohol dependence. At a molecular level, a point mutation in the gene for a form of aldehyde dehydrogenase (*ALDH2*) renders the enzyme inactive; this mutation is protective against the development of alcohol use disorders. This polymorphism is carried by about 40% of East Asian individuals but is rare in people of European ancestry.

Other candidate genes have included the *dopamine D₂ receptor* and the *GABA receptor*, and some associations with alcohol dependence have been reported. However, attempts at replication have yielded inconsistent findings. Genome-wide association studies (GWAS) have indicated a polygenic architecture for alcohol use disorders and have confirmed the effect of genes coding for alcohol metabolizing enzymes on the risk of alcohol use disorders (Deak and Johnson, 2021).

Neurobiological factors

Alcohol, like other substances liable to misuse, produces acute rewarding effects in terms of elevated mood and decreased anxiety. These effects are associated with increased release of dopamine and opioids in ventral striatum. Such changes probably explain the widespread use of alcohol but not the transition to compulsive use characteristic of those people with alcohol use disorders. In people with alcohol use disorders, the binding of dopamine D2/3 receptors in striatum is lowered but this is generally seen as a consequence of sustained alcohol use rather than a vulnerability factor. The extent of the dopamine receptor reduction may, however, be associated with craving and liability to relapse after detoxification (Kamp et al., 2019).

A number of studies have sought neurobiological predictors of problematic alcohol use in young people. No difference has been observed in the neural correlates of reward processes in young people with and without a family history of alcohol use disorder. However, in young people without a family history of alcohol use, increased ventral striatal activity in response to reward anticipation predicted heavier future alcohol use (Tschorn et al., 2021).

Family factors

Children tend to follow their parents' drinking patterns. Non-genetic familial factors appear to be important in determining levels of alcohol use. Nevertheless, it is not uncommon to meet people who are abstainers although their parents drank heavily. Generally, parental drinking coupled with favourable parental attitudes towards alcohol and low parental monitoring are associated with an increased risk of alcohol use disorders in young people. In contrast, parental support and high-quality family relationships are protective (Yap et al., 2017).

Personality factors

Some personality disorders, for example, antisocial personality and borderline personality, are associated with high rates of alcohol use disorders (Guy et al, 2018). Certain personality traits are also associated with heavy drinking, in particular impulsivity and novelty seeking (Watts et al., 2021).

Childhood factors

We have seen that genetic factors are involved in the tendency of alcohol dependence to run in families, but the estimated heritability leaves much room for environmental factors. Childhood exposure to traumatic events, particularly emotional, physical, and sexual abuse,

predisposes to alcohol misuse in adulthood. There is also a role for associated factors such as family disruption, poverty, and social isolation (Shin et al., 2015). It is, of course, likely that alcohol dependence in a parent will contribute towards the latter.

Psychiatric disorder

Alcohol misuse is commonly found in conjunction with other psychiatric disorders, and sometimes appears to be secondary to them. For example, some patients with *depressive disorders* use alcohol in the mistaken hope that it will alleviate low mood. Those with anxiety disorders, particularly *panic disorder* and *social phobia*, who use alcohol to relieve anxiety are also at risk. In other cases alcohol misuse appears to *precede* another psychiatric disorder, suggesting that heavy alcohol consumption itself can act as an aetiological factor, perhaps in a person who is predisposed through genetic and other environmental factors to develop the specific psychiatric condition. There can be diagnostic overshadowing, when clinicians are unable to assess underlying psychiatric illness due to acute or chronic alcohol use, where alcohol is used for self-medication. Services that are not prepared to work with comorbidities are likely to not be effective for these more complex presentations.

A national epidemiological survey of alcohol-related disorders carried out in the United States found that 55% of patients with alcohol dependence had a co-existing lifetime psychiatric disorder. The majority of this comorbidity was with *mood disorders* (40%), *anxiety disorders* (32%), and *antisocial personality* (14%) (Kandel et al., 2013). Clinically, alcohol misuse is also seen in patients with *schizophrenia*.

Alcohol consumption in society

There is now general agreement that rates of alcohol dependence and alcohol-related disorders are correlated with the *general level of alcohol consumption* in a society. Previously it had been supposed that levels of intake among excessive drinkers were independent of the amounts taken by moderate drinkers. In 1956, however, the French demographer, Sully Ledermann, challenged this idea, proposing instead that the distribution of consumption within a homogeneous population follows a logarithmic normal curve. If this is the case, an increase in the average consumption must inevitably be accompanied by an increase in the number of people who drink an amount that is harmful.

Although the mathematical details of Ledermann's work have been criticized, there are striking correlations between average annual alcohol consumption in

a society and several indices of alcohol-related damage among its members, for example, rates of liver cirrhosis. For this reason, it is now widely accepted that the proportion of a population that drinks excessively is largely determined by the average alcohol consumption of that population.

What then determines the average level of drinking within a nation? Economic, formal, and informal controls must be considered. The *economic control* is the price of alcohol. There is now ample evidence from the United Kingdom and other countries that the real price of alcohol (i.e. the price relative to average income) profoundly influences a nation's drinking habits. Furthermore, heavy drinkers as well as moderate drinkers reduce their consumption when the tax on alcohol is increased. For example, the introduction in Scotland and Wales of a minimum unit price for alcohol resulted in a 7% increase in price for alcoholic drinks and a similar fall in the overall amount of alcohol consumed. Most of this reduction was in the households that purchased the most alcohol where the potential health gains would be greatest (Beeston et al., 2020).

The main *formal controls* are the licensing laws, but these do not seem to influence drinking behaviours in a consistent way when comparisons are made between different countries. *Informal controls* are the customs and moral beliefs in a society that determine who should drink, in what circumstances, at what time of day, and to what extent. Some communities appear to protect their members from alcohol misuse despite the general availability of alcohol. For example, alcohol-related problems are uncommon among Jewish people even in countries where there are high rates in the rest of the community.

Recognition of harmful alcohol use and dependence

Detection

Only a small proportion of those with harmful alcohol use in the community are known to specialized agencies. Generally, in European settings, only about 20% of patients with formal alcohol dependence receive medical interventions. There are probably several reasons for this, including lack of help-seeking by users, limited availability of specialist alcohol services, and under-identification of alcohol misuse by health professionals. Rates of screening for harmful alcohol use in primary care tend to be low (Carvalho et al., 2019).

Screening questionnaires

Alcohol misuse may go undetected because patients conceal the extent of their drinking. However, doctors and other professionals often do not ask the right questions. It should be a standard practice to ask all patients (medical, surgical, and psychiatric) about their alcohol consumption. Brief screening questionnaires can be helpful—for example, the CAGE questionnaire, which consists of the following four questions.

1. Have you ever felt you ought to *Cut down* on your drinking?
2. Have people *Annoyed* you by criticizing your drinking?
3. Have you ever felt *Guilty* about your drinking?
4. Have you ever had a drink first thing in the morning (an '*Eye-opener*') to steady your nerves or get rid of a hangover?

Two or more positive replies are said to identify alcohol misuse. Some patients will give false answers, but others find that these questions provide an opportunity to reveal their problems.

Overall the CAGE has a good sensitivity but only modest specificity (Mitchell et al., 2014). An alternative is the AUDIT, a ten-item structured interview developed by the World Health Organization for screening for both currently and potentially harmful drinking. It shows good sensitivity and specificity, can identify mild dependence, and is probably the most useful screening questionnaire for clinicians and researchers in primary care (see Box 20.4).

Scores for each question range from 0 to 4, with the first response for each question scoring 0, and the last response scoring 4. For questions 9 and 10, which only have three responses, the scoring is 0, 2, and 4.

People who score in the range 8–15 should receive a brief intervention based on their risk for developing alcohol-related harm. Those who score 16–19 need a brief intervention and regular monitoring, including referral to specialist alcohol services if heavy drinking continues. Those who score in the range 20–40 should receive an early specialist referral and, depending on the severity of physical dependence, community or inpatient detoxification and other treatments (Pilling et al., 2011). It is also possible to use the first three questions of the AUDIT scale as a screen (the AUDIT-C) and proceed to the remainder of the scale if the score on these three items is equal to or greater than 5 (Sinclair and O'Neil, 2020).

Box 20.4 AUDIT questionnaire: screen for alcohol misuse

1. How often do you have a drink containing alcohol?
 - Never
 - Monthly or less
 - 2–4 times a month
 - 2–3 times a week
 - 4 or more times a week
2. How many units of alcohol do you drink on a typical day when you are drinking?
 - 1 or 2
 - 3 or 4
 - 5 or 6
 - 7 to 9
 - 10 or more
3. How often do you have six or more drinks on one occasion?
 - Never
 - Less than monthly
 - Monthly
 - Weekly
 - Daily or almost daily
4. During the past year, how often have you found that you were unable to stop drinking once you had started?
 - Never
 - Less than monthly
 - Monthly
 - Weekly
 - Daily or almost daily
5. During the past year, how often have you failed to do what was normally expected of you because of drinking?
 - Never
 - Less than monthly
 - Monthly
 - Weekly
 - Daily or almost daily
6. During the past year, how often have you needed a drink in the morning to get yourself going after a heavy drinking session?
 - Never
 - Less than monthly
 - Monthly
 - Weekly
 - Daily or almost daily
7. During the past year, how often have you had a feeling of guilt or remorse after drinking?
 - Never
 - Less than monthly
 - Monthly
 - Weekly
 - Daily or almost daily
8. During the past year, have you been unable to remember what happened the night before because you had been drinking?
 - Never
 - Less than monthly
 - Monthly
 - Weekly
 - Daily or almost daily
9. Have you or someone else been injured as a result of your drinking?
 - No
 - Yes, but not in the past year
 - Yes, during the past year
10. Has a relative or friend, doctor or other health worker been concerned about your drinking or suggested that you cut down?
 - No
 - Yes, but not in the past year
 - Yes, during the past year

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General 'at-risk' associations

The next requirement is for the doctor to be suspicious about 'at-risk' associations. In general practice, alcohol misuse may come to light as a result of problems in the patient's primary relationship and family, at work, with finances, or with the law. The alcohol misuser is likely to have many more days off work than the moderate

drinker, and repeated absences on a Monday should arouse strong suspicion. The high-risk occupations (see 'Occupation' above) should also be considered.

Medical 'at-risk' associations

In hospital practice, the alcohol-dependent patient may be noticed if they develop withdrawal symptoms after admission. Florid delirium tremens is obvious, but

milder forms may not be recognized as such, or instead considered to be delirium of another cause—for example, pneumonia or post-operative. In both general and hospital practice, at-risk factors include physical disorders that may be alcohol related. Common examples are gastritis, peptic ulcer, and liver disease, but others, such as neuropathy and seizures, should be borne in mind. Repeated accidents should also arouse suspicion.

Psychiatric at-risk associations include anxiety, depression, erratic moods, impaired concentration, memory lapses, and sexual dysfunction. Alcohol misuse should be considered in all cases of deliberate self-harm and suicidal behaviour.

Drinking history

If any of the above factors raise suspicion of alcohol misuse, the next stage is to take a comprehensive drinking history (see Box 20.5). This should be done sensitively, with understanding that the patient may have difficulty in giving a clear history. The clinician should aim to build up a picture of what and how much the patient drinks throughout a typical day—for example, when and where they have the first drink of the day. Patients should be asked how they feel if they go without a drink for a day or two, and how they feel *on waking*. This can lead on to enquiries about the typical features of dependence and the range of physical, psychological, and social problems associated with it.

To gain an idea of the duration of alcohol problems, key points in the history may include establishing when the patient first began *drinking every day*, when they began drinking *in the mornings*, and when, if ever, they first experienced *withdrawal symptoms*. It is useful to ask about periods of abstinence from alcohol, what factors helped to maintain this state of affairs, and what led to a resumption of drinking. This can lead to enquiries about past attempts at treatment.

Box 20.5 Alcohol use history

- Describe a typical day's drinking. What time is the first drink of the day?
- When did daily drinking start?
- Presence of withdrawal symptoms in the morning or after abstinence.
- Previous attempts at treatment.
- Physical health problems.
- Patient's attitude towards drinking.

It is necessary to obtain a clear understanding of the patient's own view of their drinking behaviour, because there are a number of possible treatment goals. In this situation the *patient's attitude* to their problems plays a key role in deciding which approaches are likely to be most beneficial (see the section on motivational interviewing, below).

Laboratory tests

Several laboratory tests can be used to detect alcohol misuse, although none gives an unequivocal answer. This is because the more sensitive tests can give 'false-positive' results when there is disease of the liver, heart, kidneys, or blood, or if enzyme-inducing drugs such as anticonvulsants or steroids have been taken. However, abnormal values point to the possibility of alcohol misuse. The most useful tests are listed in Box 20.6.

Box 20.6 Laboratory tests for harmful alcohol use

Gamma-glutamyl-transpeptidase (GGT): Estimations of GGT in blood provide a useful screening test. The level is raised in about 70% of alcohol misusers, both men and women, whether or not there is demonstrable liver damage. The heavier the drinking, the greater is the rise in GGT.

Mean corpuscular volume (MCV): MCV is raised above the normal value in about 60% of alcohol misusers, and more commonly in women than in men. If other causes are excluded, a raised MCV is a strong indicator of excessive drinking. Moreover, it takes several weeks to return to normal after abstinence.

Carbohydrate-deficient transferrin: This is a variant of a serum protein that transports iron, and levels are increased in response to heavy drinking. It is probably more specific than GGT.

Breath alcohol concentration: A high concentration does not distinguish between an isolated episode of heavy drinking and chronic misuse. However, if a person is not intoxicated when the breath alcohol concentration is well above the legal limit for driving, they are likely to be unusually tolerant of alcohol. This tolerance suggests persistent heavy drinking. Alcohol is eliminated rather slowly from the blood and can be detected in appreciable amounts for 24 hours after an episode of heavy drinking.

The treatment of alcohol misuse

Early detection and treatment

Early detection of excessive consumption of alcohol and harmful alcohol use is important because treatment of established cases is difficult, particularly when dependence is present. Many cases can be detected early by general practitioners, physicians, and surgeons when patients are seeking treatment for another problem (see Box 20.7).

Brief intervention

General practitioners are well placed to provide early treatment of alcohol problems, and they are likely to know the patient and their family well. It is often effective if the general practitioner gives simple advice in a frank, matter-of-fact way, but with tact and understanding.

Brief interventions generally involve simple education and advice about safe levels of alcohol consumption. Monitoring of alcohol consumption can also be useful in decreasing use. The aim is to promote safer drinking, rather than abstinence. Generally, brief interventions lead to modest reductions in alcohol consumption over the next few years, and are the best initial approach for a person whose alcohol consumption exceeds safe limits. (Sinclair and O'Neill, 2020). It is, however, agreed that brief interventions are not effective for people with severe drinking problems, particularly those who are alcohol-dependent.

Motivational interviewing

Patients with problems of alcohol misuse, particularly those detected by screening methods, may be unsure whether or not to engage in treatment programmes. An appropriate interviewing style, particularly during the first assessment, can help to persuade the patient

to engage in a useful review of their current pattern of drinking.

Confrontation is avoided in *motivational interviewing*, and a less directive approach is taken, during which patients are helped to assess the balance of the positive and negative effects of alcohol on their lives. The clinician can help in this exercise by providing feedback to the patient about the personal risks that alcohol poses both to them and to their family, together with a number of options for change. The aim of motivational interviewing is to persuade the patient to argue their own case for changing their pattern of substance use (see Box 20.8).

For an explanation of the origin of motivational interviewing and a description of the relevant techniques, see Treasure (2004). For a cautionary meta-analysis of the potential benefits, see Foxcroft et al. (2014).

Treatment plans for more established harmful alcohol use and dependence

For patients who have significant alcohol-related disorders, particularly alcohol dependence, treatment generally needs to be more intensive. Any intervention should be preceded by a full assessment, and should include a drinking history and an appraisal of current medical, psychological, and social problems. Other important issues concern the *use of other drugs* (including those bought over the counter) and an *assessment of cognitive function* with, for example, the Mini-Mental State Examination (MMSE).

An intensive and searching enquiry often helps the patient to gain a new recognition and understanding of the problem, and this is the basis of treatment. As noted above, however, it is important to avoid confrontation, as this will only alienate someone who is likely to have very mixed feelings about the prospect of life without alcohol. It is usually desirable to involve the patient's partner in the assessment, both to obtain additional

Box 20.7 Approach to treatment of alcohol misuse

- Raise awareness of problem.
- Increase motivation to change.
- Support and advice.
- Withdraw alcohol (or controlled drinking).
- High-intensity psychological treatments.
- Alcoholics Anonymous.
- Medication (disulfiram, acamprosate, naltrexone).

Box 20.8 Motivational interviewing

- Express empathy.
- Avoid arguing; don't be judgemental.
- Detect and 'roll with' resistance.
- Point out discrepancies in the patient's history.
- Raise awareness about the contrast between the substance user's aims and behaviour.

information and to give the partner an opportunity to unburden their feelings.

A mutually agreed treatment plan should be worked out with the patient (and their partner if appropriate). There should be specific goals, and the patient should be required to take responsibility for realizing these. The goals should deal not only with the drinking problem, but also with any accompanying problems with regard to the patient's health, marriage or other relationship, job, and social adjustment. In the early stages they should be short-term and achievable—for example, complete abstinence for two weeks. In this way the patient can be rewarded by early achievement.

Longer-term goals can be set as treatment progresses. These will be concerned with trying to change factors that precipitate or maintain excessive drinking, such as tensions in the family. When drawing up this treatment plan, an important decision is whether to aim for total abstinence or limited consumption of alcohol (*controlled drinking*).

Total abstinence versus controlled drinking

The disease model of alcoholism proposes that an alcohol-dependent person must become totally abstinent and remain so, as a single drink would lead to relapse. Alcoholics Anonymous have made this a tenet of their approach to treatment.

The issue of abstinence versus controlled drinking remains unresolved. A prevalent view is that controlled drinking may be a feasible goal for people whose alcohol misuse has been detected early, and who are not dependent or physically damaged. Abstinence is a better goal for dependent drinkers and others who have attempted controlled drinking unsuccessfully. If controlled drinking is to be attempted, clinicians need to give the patient clear advice about safe levels (see 'Excessive alcohol consumption' above).

Withdrawal from alcohol

For patients with dependence syndrome, withdrawal from alcohol is an important stage in treatment and should be managed carefully (see Alexiou and King, 2023). In less severe cases, withdrawal may take place at home provided there is adequate support and clinical monitoring (see Box 20.9). This should involve daily assessment by the general practitioner, practice nurse, or specialist alcohol worker to check the patient's physical state and supervise medication (see Box 20.10). In uncomplicated cases, withdrawal symptoms usually resolve within four to six days. However, any patient who is likely to have severe withdrawal symptoms, especially if there is a history of delirium tremens or seizures, should be admitted to an inpatient facility.

Box 20.9 Considerations before alcohol detoxification

1. What are the medical risks?
2. What setting is appropriate?
3. What does the patient want from detoxification?
4. What kind of aftercare is needed to help to maintain abstinence?

Box 20.10 General management of alcohol detoxification

1. Explanation of the process to the patient and their family.
2. The patient should stay off work and rest. They should not drive.
3. The patient should take plenty of fluids but avoid caffeinated drinks.
4. Daily visit by health professional.
5. Maintain abstinence.
6. Reducing course of benzodiazepines over 5–7 days (see text).

The Severity of Alcohol Dependence Questionnaire (SADQ) can be useful for estimating the likely severity of alcohol dependence and thereby gauging the amount of difficulty that the patient might encounter during withdrawal. A SADQ score of more than 30 is an indication for inpatient detoxification. Other indications for residential detoxification are *very high alcohol consumption* (over 30 units daily), *concomitant benzodiazepine misuse*, and *significant medical or psychiatric comorbidity* (National Institute for Health and Clinical Excellence, 2011a).

The extensive research on pharmacological treatment for alcohol withdrawal has been systematically reviewed by Lingford-Hughes et al. (2012). The most important concern is the prevention of major complications of withdrawal, such as seizures or delirium tremens. Treatment with benzodiazepines is usually the most suitable choice.

Benzodiazepines differ in their duration of action (see Chapter 25). Although the longer-acting compounds may carry the risk of oversedation in the elderly or in those with significant liver disease, they also have the advantage of smoother withdrawal and generally

better relief of withdrawal symptoms. *Chlordiazepoxide* is often used. A typical outpatient regimen would start with 20–30 mg four times daily, reducing and stopping over a period of 5–7 days. In more severe cases, particularly in inpatients, higher doses are commonly used. If convulsions occur, the dose of benzodiazepine should be reviewed and increased if necessary. Administration of a benzodiazepine that is well absorbed parenterally (e.g. lorazepam) may be useful to relieve severe acute withdrawal, although oral medication can usually be resumed promptly. In home-based withdrawal it is helpful for a family member or friend to supervise the medication.

It is important that withdrawal is followed by one of the relapse prevention strategies outlined below. These might include psychological interventions, such as individual or group therapy or a self-help group. There are also a number of medications that can be employed at this stage.

Antipsychotic drugs lower the seizure threshold and are less effective than benzodiazepines in preventing delirium. They are sometimes used to reduce agitation during alcohol withdrawal, but use of adequate doses of a benzodiazepine is a safer strategy. However, antipsychotic drugs may be required occasionally in patients with delusions that do not respond to benzodiazepines. In these circumstances, combination of antipsychotic

treatment with benzodiazepines would be expected to attenuate the pro-convulsive effects of the antipsychotic drug.

Vitamin supplements, particularly thiamine, should also be given to prevent the Wernicke–Korsakoff syndrome. Parenteral thiamine should be given to patients at high risk (i.e. those with alcohol dependence and a poor diet). For the treatment of the Wernicke–Korsakoff syndrome, see pages 345, 561.

Psychological treatment

As noted above, *provision of information and advice* about the effects of excessive drinking is an important first stage in treatment. The information given should relate to the specific problems of the individual patient, including both those that have already occurred and those that are likely to develop if drinking continues. The technique of *motivational interviewing* can be useful (see Box 20.8).

More specialized, high-intensity psychological treatments are appropriate if the drinking problem is more serious, particularly in cases of alcohol dependence. Several treatment approaches have been employed, although there is significant overlap between the different therapies (see Box 20.11). Important features of successful psychological approaches include the use of *behavioural techniques*, which require drinkers to take

Box 20.11 High-intensity psychological therapies for alcohol-use disorders

Community reinforcement approach: A broad range of treatments, with the overall aim of changing the drinker's social environment so that sobriety is rewarded. Components include training in communication skills, problem solving, and assertive drink refusal. Pleasurable, alcohol-free activities are prioritized. Relationship therapy and promotion of relapse prevention are also offered.

Cognitive behaviour therapy: Psychoeducation about alcohol use disorders. Analysis of thoughts, feelings, and behaviours that underpin excessive drinking. Identification of triggers to drinking and teaching of skills to deal with craving and mood fluctuations. Development of a relapse prevention plan. Treatment can be provided in individual and group format.

Behavioural couples therapy: Based on social learning theory. Uses behavioural contracting, communication skills training, and behavioural rehearsal with the drinker and their partner. Particularly appropriate in cases where

the drinking problem and relationship difficulties exacerbate each other.

Cue exposure: Based on Pavlovian conditioning theory, and views craving for alcohol as a conditioned response to specific environmental cues. Aims to 'extinguish' conditioned responses and ameliorate craving by presenting relevant cues in the absence of the reinforcing effect of alcohol.

Mindfulness based therapies: In the treatment of alcohol use disorders, mindfulness techniques take a non-judgemental stance towards mood fluctuations and cravings to drink. The promotion of awareness about these mental states and the accompanying triggers prevents automatic responses to distress, including drinking.

Internet-delivered psychological treatments: The above treatments, particularly CBT, can be delivered by the internet either as standalone treatments or to supplement therapist input.

some action themselves, as well as the involvement of a *partner* and the drinker's *wider social network*. There are also important roles for *education* and the improvement of *social and interpersonal skills*, as these relate to alcohol misuse.

It may be helpful, for example, to identify situational or interpersonal triggers that cause the patient to drink excessively, and then to plan and rehearse new methods of coping with these situations. This is an important part of *relapse prevention*. The use of *cue exposure* to alcoholic drinks, without subsequent consumption, may also lessen the risk of subsequent relapse when the patient has to contend with the ready availability of alcohol in social settings. It is important for the patient to appreciate that a *lapse* in drinking behaviour does not have to progress to a full-blown *relapse*. Many patients who misuse alcohol have general deficiencies in problem-solving skills, and appropriate training may help to reduce relapse rates. If the patient is in a relatively stable relationship with a partner, *cognitive behavioural relationship therapy* can produce improvements in both drinking behaviour and relationship adjustment.

For a review of psychological treatments for alcohol use disorders, see National Institute for Health and Clinical Excellence (2011a) and Ray et al. (2019).

Pharmacological treatments to help to maintain abstinence

A number of different kinds of medication have been employed to help people to abstain from drinking or to experience less drinking-related harm. Medications are not regarded as 'stand-alone' treatments and, when used in the management of drinking problems, should be combined with an appropriate psychosocial intervention (see Lingford-Hughes et al., 2012, and Box 20.12).

Disulfiram (*Antabuse*) acts by blocking the oxidation of alcohol so that acetaldehyde accumulates. Some patients find it useful because the anticipation of an unpleasant reaction acts as a deterrent to impulsive drinking. The reaction includes facial flushing, throbbing headache, hypotension, palpitations, tachycardia, and nausea and vomiting. In vulnerable patients, cardiac arrhythmias and collapse may occur.

The main contraindications to the use of disulfiram are a history of heart failure, coronary artery disease, hypertension, psychosis, and pregnancy. The patient should be given clear verbal and written instructions, together with a list of alcohol-containing substances to be avoided. Common side effects are drowsiness, bad breath, nausea, and constipation. Disulfiram is given in a single dose of 800 mg on the first day of treatment, reducing over five days to 100–200 mg daily.

Box 20.12 National Institute for Health and Clinical Excellence (NICE) guidance on the management of harmful drinking and alcohol dependence

1. For harmful drinking and mild dependence offer a high-intensity psychological treatment (see Box 20.9).
2. For those drinking over 15 units of alcohol a day and/or scoring 15–30 or more on the SADQ* test, offer community-based assisted withdrawal, or inpatient treatment if there are safety concerns.
3. After successful withdrawal for patients with moderate or severe alcohol dependence, consider offering acamprosate or oral naltrexone in combination with a high-intensity psychological treatment that promotes relapse prevention.

* Severity of Alcohol Dependence Questionnaire.

Source: National Institute for Health and Care Excellence. (2011a). Alcohol-use disorders: Diagnosis, assessment and management of harmful drinking (high-risk drinking) and alcohol dependence. Clinical guideline CG115. Copyright (2011a), National Institute for Health and Care Excellence.

The efficacy of disulfiram depends on compliance. The treatment is likely to be more successful if a partner or health worker is able to supervise its use, perhaps in the context of a structured treatment such as cognitive behavioural relationship therapy. However, the unsupervised use of disulfiram is not considered to be an effective approach (Luty, 2015).

Acamprosate (*calcium acetyl homotaurinate*) appears to suppress the urge to drink in response to learned cues, and can produce modest but useful reductions in drinking behaviour in alcohol-dependent individuals. It is believed to act by stimulating GABA-inhibitory neurotransmission and decreasing the excitatory effects of glutamate. The effects of acamprosate have not been consistent across studies. For example, a large trial in the United States (COMBINE) showed no increased benefit in measures of drinking behaviour when acamprosate was combined with cognitive behaviour therapy compared with cognitive behaviour therapy alone (Anton et al., 2006). However, a meta-analysis of 19 randomized trials showed that, over a two-year follow-up, acamprosate significantly increased the abstinence rate, with a relative risk of 0.83 (95% CI, 0.77–0.88) (National Institute for Health and Clinical Excellence, 2011a).

The usual dose of acamprosate is two tablets (333 mg each) three times daily with meals. In lighter patients (<60 kg), four tablets daily are recommended. Acamprosate is not metabolized in the liver and is excreted by the kidney. It is therefore unlikely to cause drug interactions. Adverse effects include diarrhoea and, less frequently, nausea, vomiting, and abdominal pain. Skin rashes may occur, as can fluctuations in libido.

Naltrexone is an opioid antagonist believed to block some of the reinforcing effects of alcohol and in this way decrease the likelihood of relapse after detoxification. Not all trials have found a significant benefit over placebo. In the COMBINE study, the best effects on drinking behaviour were shown by participants who received both naltrexone and cognitive behaviour therapy (Anton et al., 2006). A meta-analysis of 27 randomized trials showed that, over a 12-month follow-up, naltrexone significantly increased rates of abstinence, with a relative risk of 0.83 (95% CI, 0.75–0.91), an effect very similar to that of acamprosate (National Institute for Health and Clinical Excellence, 2011a). Common side effects of naltrexone treatment include nausea, vomiting, headache, dizziness, and weight loss.

Nalmefene, like naltrexone, is an orally active opioid antagonist. Nalmefene has been studied in patients who drink heavily but who do not have severe alcohol dependence with withdrawal symptoms. It can therefore be used as an aid to achieve controlled drinking within safe limits. In this group, given with psychosocial support, nalmefene produces modest reductions in total alcohol consumption and number of days of heavy drinking over the following six months. National Institute for Health and Clinical Excellence has recommended nalmefene as a potential treatment for men who drink more than 45 units of alcohol a week and women drinking more than 35 units a week. However, whether the benefits of nalmefene in this population are *clinically significant* has been questioned (Palpacuer et al., 2015). The most common side effects of nalmefene are similar to those of naltrexone, namely nausea, vomiting, dizziness, headache, and insomnia.

Topiramate is an anticonvulsant drug which in randomized trials has been shown to benefit drinking outcomes in terms of reductions in days of heavy drinking and harmful drinking consequences, as well as improving quality of life (Lingford-Hughes et al., 2012). The adverse effects of topiramate, however, can be troublesome, and include paraesthesia, anorexia, headache, abdominal pain, sleep disturbance, and cognitive impairment.

Antidepressant medication may be useful in patients who experience persistent symptoms of major

depression after detoxification. However, tricyclic antidepressants are not recommended because of potentially serious adverse reactions, including cardiotoxicity and death following overdose. There is no evidence that antidepressants are helpful in patients with alcohol use disorders who are not depressed. (Lingford-Hughes et al., 2012).

Other agencies concerned with drinking problems

Alcoholics Anonymous (AA) is a self-help organization founded in the United States by two alcoholic men, a surgeon and a stockbroker. It has since extended to most countries of the world. Members attend group meetings, usually twice-weekly, on a long-term basis. In crisis they can obtain immediate help from other members by telephone. The organization works on the firm belief that abstinence must be complete. At present there are about 1,200 groups in the United Kingdom.

Alcoholics Anonymous is based on a number of fundamental principles, known as the 'Twelve Steps', to which members adhere. It does not appeal to all problem drinkers because the meetings involve an emotional confession of problems, and because of the evangelical, quasi-religious nature of the twelve-step approach. However, the organization is of great value to some problem drinkers, and anyone with a drink problem should be encouraged to try it.

A treatment approach similar to that provided by Alcoholics Anonymous, coupled with encouragement to attend Alcoholics Anonymous meetings, was one of the psychological therapies employed in Project MATCH (see below). In general, all of the treatments that were employed appeared to be roughly equal in efficacy. However, patients with fewer psychiatric problems at entry tended to have a better outcome with the Alcoholics Anonymous type of treatment.

Al-Anon is a parallel organization that provides support for the partners of excessive drinkers. *Alateen* has a similar role for their teenage children.

Non-statutory agencies are voluntary bodies that provide a range of services, including advice for problem drinkers and their families, counselling, and help with occupational and social activities for those who have recovered.

Hostels are intended mainly for homeless problem drinkers. They provide rehabilitation and counselling. Usually abstinence is a condition of residence.

Results of treatment

In a famous controlled trial involving 100 male alcoholics, Edwards et al. (1977) compared simple advice with intensive treatment that included introductions

to Alcoholics Anonymous, medication, repeated interviews, counselling for partners, and, where appropriate, inpatient treatment as well. The advice group received a three-hour assessment together with a single session of counselling with the partner present. The two groups were well matched. After 12 months there was no significant difference between them with regard to drinking behaviour, subjective ratings, or social adjustment.

The effectiveness of structured psychological treatments was studied by Project MATCH, a large randomized controlled trial of three psychological treatments for patients with alcohol problems (Project MATCH Research Group, 1997). The three treatments studied were a cognitive behavioural intervention, a brief motivational enhancement therapy, and a treatment that aimed to help patients to make use of the 'twelve-step' philosophy of Alcoholics Anonymous. The main aim of the study was to attempt to find patient characteristics that predicted response to particular treatments. There were no important differences between the treatments, and few strong correlations were found between effectiveness and patient characteristics. However, in general the outcome was favourable, and the authors concluded that structured psychological treatments of various kinds were helpful in the management of alcohol misuse. In fact, the *brief motivational enhancement therapy* was generally as effective as the two more intensive psychological treatments.

Consistent with this, the UK Alcohol Treatment Trial (UKATT), which was conducted at several UK centres, found that *motivational enhancement therapy* was just as effective as *social network therapy* in decreasing alcohol consumption and alcohol dependence. However, at a one-year follow-up the effects of treatment were fairly modest. For example, relative to baseline levels, average alcohol consumption on a drinking day fell from 27 units to 19 units, and the number of days on which patients abstained from alcohol rose from 30% to 45% (UKATT, 2005).

The guidelines from the National Institute for Health and Clinical Excellence (2011a) recommend that patients with alcohol dependence, once withdrawn from alcohol, should be offered a psychological intervention that matches their needs and social situation (Box 20.11), together with either naltrexone or acamprosate. If neither of these are suitable, and the patient is committed to the goal of abstinence, disulfiram can be offered if supervision is available.

There are likely to be factors within the patient that will predict a good response to a number of different kinds of treatment. There is some disagreement as to what these factors are, but the following generally

predict a better prognosis regardless of which treatment is used:

- Good insight into the nature of the problems.
- Social stability in the form of a fixed abode, family support, and ability to keep a job.
- Ability to control impulsiveness, to defer gratification, and to form satisfactory emotional relationships.

More recent attempts to match patients to treatments have used pharmacogenetic approaches to stratify patients receiving pharmacotherapy. For example, it has been reported that a polymorphism in a glutamate receptor gene (*GRIK1*) can modify the therapeutic response to topiramate. However, positive findings of this nature are the result of post-hoc analyses and require prospective replication before they can be received with any confidence (van den Brink and Kiefer, 2020). For reviews of the effectiveness of a broad range of treatment approaches for alcohol use disorders, see the National Institute for Health and Clinical Excellence (2011a) and van den Brink and Kiefer (2020).

Prevention of alcohol misuse and dependence

In seeking to prevent excessive drinking and alcohol-related disorders, two approaches are possible. The first is to improve the help and guidance available to the individual, as already described. The second is to introduce social changes that are likely to affect drinking patterns in the population as a whole. It is this second group that we are concerned with here. Consumption within a population might be reduced by four methods:

1. *The pricing of alcoholic beverages.* Putting up the price of alcohol would probably reduce consumption within the population. Based on a systematic review, Wagenaar et al. (2010) predicted that doubling alcohol taxes in the United States would reduce direct alcohol-related mortality by 35%, traffic fatalities by 11%, and violent crime by 2%. We have already seen that the introduction in Scotland and Wales of a minimum unit price for alcohol resulted in an increased price of alcoholic drinks and a fall in the overall alcohol consumption.
2. *Controls on advertising.* Controlling or abolishing the advertising of alcoholic drinks might be another preventive measure. It is unclear how far advertising encourages the use of particular brands of alcohol rather than overall consumption. However, in the United Kingdom, annual expenditure on alcohol

advertising and promotion is over £800 million. There is an accepted link between alcohol advertising and consumption, particularly in young people (Stockings et al., 2016). Alcohol marketing now makes extensive use of digital media, which will enhance the impact of advertising in young people.

3. *Controls on sale.* Another preventive measure might be to control sales of alcohol by limiting hours or banning sales in supermarkets. Although relaxation of restrictions has been shown to lead to increased sales of alcohol in some countries, it does not follow that increased restrictions would reduce established rates of drinking. However, there is evidence that a higher minimum age for legal drinking is associated with a decrease in alcohol consumption and fatal

car accidents in young drivers (McCartt et al., 2010). There is concern about the heavy drinking associated with nightlife culture in many UK cities.

4. *Health education.* It is not known whether education about alcohol misuse is effective. Little is known about how attitudes are formed or changed. Although education about alcohol would seem to be desirable, it cannot be assumed that classroom lectures or mass media propaganda alter attitudes. In general, there is little convincing evidence that such approaches are effective. Interventions focusing on the family, designed to improve parenting skills and child–parent relationships, may be more beneficial; for example, by delaying the age at which children start drinking (Stockings et al., 2016).

Other substance use disorders

Under this heading we shall consider the use and misuse of substances other than alcohol. Although these substances include agents such as volatile substances (inhalants), the general term *drug* will be employed because it is in common use. The World Health Organization describes drug misuse as the use of a substance for a purpose not consistent with medical or legal guidelines. This includes the non-medical use of prescription drugs and the recreational use of illegal drugs.

Epidemiology

Illicit drug use

The 2019 National Survey on Drug Use and Health in the United States found that 13% of the population aged 12 years or over had used an illicit substance in the past month (Substance Abuse and Mental Health Services Administration, 2020). Use was highest in unemployed people in the 16–25-year age range. The most commonly used illicit drug was *cannabis*, with 17.5% of the population describing use in the last year. About 8% of the population aged 12 years or older had a substance use disorder in the past year. In about 70% of cases this was an alcohol use disorder.

The Psychiatric Morbidity Survey reported that, in England, 11.3% of men and 6.0% of women had taken an illicit drug in the past year, most commonly cannabis (McManus et al., 2016). Among males, drug use was highest in Black men (14.3%) and lowest in South Asian men (5.9%). In women the highest rate of drug use was

in Black women (9.7%), with South Asian women reporting the lowest rate (0.4%). Highest rates of drug use in the last year were seen in men aged 16–24 years (26.4%) and women aged 16–24 years (17.1%). In 2019, about 3% of the population of England and Wales had used at least one *Class A drug* (heroin, methadone, cocaine, lysergic acid diethylamide [LSD], ecstasy, magic mushrooms, methamphetamine) in the previous year (see Box 20.13).

Drug misuse and dependence

Little is known for certain about the *prevalence* of different types of drug misuse and drug dependence. In the United Kingdom, information comes from two main sources—*criminal statistics* (mainly based on offences involving the use and misuse of illicit drugs) and *community surveys*.

The Psychiatric Morbidity Survey reported that in England the prevalence of drug dependence during the previous 12 months was around 3% (4.3% of men and 1.9% of women). Most of the dependence was on cannabis (3.7% of men and 1.6% of women). The highest frequency was in the 16–24-year age group (McManus et al., 2016). Rates of drug dependence are much higher among the homeless population and those in prison. For example, the British Psychiatric Morbidity Survey found that 24% of people who were sleeping rough met the criteria for drug dependence. In a systematic review, Fazel et al. (2006) found that among prisoners received into custody, around 30% of men and 15% of women reported dependence on drugs before coming into prison.

Box 20.13 Illicit drug use in England and Wales in the past year in adults aged 16–59

Cannabis	7.8%
Cocaine	2.6%
Nitrous oxide	2.4%
Ecstasy	1.4%
Ketamine	0.8%
Amyl nitrite	0.5%
Tranquillizers	0.5%
Magic mushrooms	0.5%
LSD	0.3%
Amphetamines	0.3%
Heroin and methadone	0.2%
Crack	0.1%
Anabolic steroids	0.1%
Volatile substances	0.1%

Source: Crime Survey for England and Wales. Drug misuse in England and Wales: Year ending March 2020. Copyright (2020), Office for National Statistics.

Rates of drug misuse and dependence are higher in disadvantaged areas of large cities. Adolescents are at risk, particularly around school-leaving age. A high proportion of attenders at drug-dependence clinics in large cities are unemployed, with few stable relationships, and leading disorganized lives. However, many young drug users remain in employment and apparently regard their drug taking as part of normal recreational activity for their peer group.

Causes of drug misuse

There is no single cause of drug misuse. It is generally argued that four factors are important:

1. availability of drugs
2. a vulnerable personality
3. an adverse social environment
4. pharmacological factors.

The widespread availability of illicit drugs means that occasional use in a young person can no longer be regarded as abnormal behaviour. However, about 10% of those who experiment with drugs will go on to develop problems with them (Robson, 2009). Once drug taking has started, personal, social, and pharmacological factors play a role in determining misuse and dependence.

Studies of the aetiology of misuse of substances other than alcohol are still at an early stage. For example, it is unclear whether similar risk factors predict misuse of a range of substances, including alcohol, or whether there is some specificity in the mechanisms, which leads certain individuals to misuse certain substances.

Availability of drugs

Drugs involved in misuse can be obtained in three main ways:

1. *Legally without prescription*—nicotine and alcohol are obvious contemporary examples in most countries, and in the nineteenth century much dependence on opioids resulted from taking freely available remedies containing morphia.
2. *By prescription from doctors*—in the first part of the twentieth century much of the known dependence on opioids and barbiturates in western countries was due to drugs obtained from this source; latterly, *benzodiazepine dependence* was often acquired in this way. Currently, there is concern about the misuse of drugs prescribed to relieve chronic pain, such as *opiates* and *gabapentinoids*.
3. *From illicit sources* ('street drugs').

Personal factors

Of those who experiment with drugs, the users who go on to develop problems appear to have some degree of personal vulnerability before they begin taking drugs. They may live in disrupted families and have started taking drugs at a relatively young age. Associated behaviours include a poor school record, truancy, or delinquency. Traits such as sensation seeking and impulsivity are also common. Many of those who misuse drugs report depression and anxiety, but it is seldom clear whether these are the causes or the consequences of drug misuse and dependence. Some give a history of mental illness or personality disorder in the family. Genetic factors are strongly implicated in a variety of drug use *disorders*, but less so in drug use itself. This suggests that drug use in general is largely dependent on factors such as availability and social environment, but that genes contribute to the propensity to develop harmful use and dependence.

In general, twin studies show that substance use disorders (including alcohol) have a heritability of about 50% and like other psychiatric disorders have a polygenic architecture where a large number of genes contribute an individually small effect on risk. GWAS studies have identified some significant single nucleotide

polymorphisms (SNPs) for cannabis and opioid use disorders (Deak and Johnson, 2021).

Social environment

The risk of drug misuse is greater in societies that condone drug use of one kind or another. Within the immediate group, there may be social pressures for a young person to take drugs to achieve status. Thus drug use by individuals is influenced by substance use by their peers or parents. There are also links between drug misuse and indices of social deprivation, such as unemployment and homelessness.

Pharmacological factors

Use, misuse, and dependence

Many individuals use drugs without misusing them, and not all drug misusers become drug-dependent. Therefore it is useful to study the biological mechanisms underlying drug use, misuse, and dependence separately. Drugs are used and misused because they have the ability to serve as *positive reinforcers*—that is, they increase the frequency of behaviours that lead to their use. Drugs act as positive reinforcers because they cause positive subjective experiences such as euphoria or a reduction in anxiety.

Neurobiological mechanisms

An important neurobiological substrate that mediates reinforcing effects is the midbrain dopamine system, the cell bodies of which originate in the ventral tegmental area and innervate the forebrain, particularly the ventral striatum. It has been proposed that these dopamine pathways form part of a physiological reward system, and it is of interest therefore that most drugs of abuse, particularly stimulants and alcohol, increase dopamine release in the ventral striatum. This suggests that activation of midbrain dopamine pathways may be an important property of some drugs that have a propensity to be used and misused.

Although this hypothesis may explain in part the social use of particular drugs, it does not account for the misuse of drugs in some circumstances. Presumably this is a consequence of interactions between the pharmacological properties of the drug, the biological disposition and personality of the user, and the social environment. Current neurobiological formulations of drug dependence suggest that the switch from controlled drug taking to compulsive use (misuse and dependence) is associated with a lessening of *prefrontal 'reflective' processes* and a corresponding increase in *striatal activity*, which underpins more habitual behaviours. These changes might explain the ability of repeated drug administration in

some people to 'hijack' executive behaviour almost exclusively to serve the needs of the drug habit, and the poor judgement and decision making shown by people with substance dependence. Learning and conditioning factors are likely to be important in this context as well as personality factors such as trait impulsivity and novelty seeking.

Neurobiology of tolerance and dependence

The phenomena of tolerance and withdrawal are believed to be a result of *neuroadaptive changes* in the brain. These are part of a homeostatic process that counteracts the acute pharmacological effects that occur when a drug is administered. For example, many drugs that are misused for their anxiolytic and hypnotic properties (e.g. benzodiazepines and alcohol) have, among their acute pharmacological effects, the ability to enhance brain GABA function. During continued treatment with these agents, adaptive changes occur in GABA- and benzodiazepine-receptor sensitivity that tend to offset the effect of the drugs to facilitate GABA neurotransmission. Such an effect could account for the phenomenon of tolerance, with the result that an individual needs to take more of the drug to produce the same pharmacological effect.

If the drug is abruptly discontinued, persistence of the adaptive changes in receptor function could lead to a sudden decline in GABA activity. In fact, many of the clinical features of withdrawal from anxiolytic drugs, such as anxiety, insomnia, and seizures, can be explained on the basis of diminished brain GABA function. Such an effect can also explain the well-known phenomenon of cross-tolerance between anxiolytics and hypnotics and alcohol, which makes it possible, for example, to treat alcohol withdrawal with a benzodiazepine.

Similar kinds of adaptive changes have been proposed to account for the tolerance and withdrawal phenomena that are seen with other drugs of misuse. For example, while acute administration of opioids decreases the firing of noradrenaline cell bodies in the brainstem, tolerance of this effect occurs during repeated treatment, probably because of adaptive changes in the sensitivity of opioid receptors. If opioids are now abruptly withdrawn, there is a sudden increase in the firing of noradrenaline neurons and in the release of noradrenaline in terminal regions. Increased noradrenergic activity may account for several of the clinical features of acute opioid withdrawal, including sweating, tachycardia, hypertension, and anxiety.

Although the positive reinforcing actions of drugs are considered to be the major factor in promoting drug use, withdrawal effects are likely to play an important part once drug taking is established, because they are

invariably unpleasant, and individuals are likely to try to prevent them by taking more of the drug. This ‘negative reinforcement’ may complete the transition from *impulsive* to *compulsive* drug use.

It is also worth noting that for many months following the cessation of a clear-cut withdrawal syndrome, dependent individuals may experience a sudden intense desire to consume the drug. Often particular psychological and social stimuli that were previously associated with drug use may trigger intense craving associated with symptoms resembling a withdrawal state. It has been proposed that a single exposure to the drug during this period may rapidly lead to a full relapse to drug dependence—the so-called *reinstatement effect*. An analogous effect has been shown in previously drug-dependent animals, where a single priming dose of the drug concerned can lead to a full recovery of drug-seeking behaviours that had previously been extinguished.

For a review of the brain mechanisms involved in drug use and misuse and their implications for treatment, see Hayes et al. (2020).

Adverse effects of drug misuse

Drug misuse has many undesirable effects, both for the individual and for society.

Physical health

Drug misuse can lead people to neglect their health, in addition to the direct physical consequences of the substance itself. These are discussed in more detail under the heading of individual drugs.

Intravenous drug use poses particular health risks. This practice is very common with opioid use, but benzodiazepines, amphetamines, and other drugs may be taken in this way. Intravenous drug use has important consequences, which include both local and general effects (see Box 20.14).

The most serious complications of intravenous drug use include *HIV infection* and *hepatitis*. Rates of HIV infection among drug users in the United Kingdom are lower than those found in some other European countries, and some attribute this to the vigorous harm reduction policies in the United Kingdom (methadone prescribing, needle exchanges, and education for drug users). However, hepatitis C is a major concern, as preliminary studies suggest that this infection has a prevalence of 50–80% among UK users who inject drugs. Rates of hepatitis B are lower but still significant (30–50%). Accidental drug overdose can occur by any route, but

Box 20.14 Some consequences of intravenous drug misuse

Local

Vein thrombosis
Infection of injection site
Damage to arteries

Systemic

Bacterial endocarditis
Hepatitis B and C
HIV infection
Accidental overdose

there is a far higher risk of death from heroin overdose when the intravenous route is used.

Substance dependence in women

Substance misuse treatment services, including those for alcohol-related disorders, may not be designed to meet the special needs of women, because existing treatment models have been developed for men. Clinicians need to consider factors such as the following:

- social and parenting responsibilities
- possible history of victimization and sexual abuse
- social barriers to obtaining medical care
- the availability of female therapists.

Drug misuse in pregnancy and the puerperium

It is not uncommon for drug misusers to have limited contact with health care services until late into pregnancy; this can occur because of limited ability to access services, fear of the possible consequences, and the difficulties of addressing dependence issues. Drug use in pregnancy increases the health risks to both mother and foetus. When drugs are taken in early pregnancy, there is a risk of increased rates of foetal abnormality. Opioids may directly decrease foetal growth. When drugs are taken in late pregnancy, the foetus may become dependent on them. The risk of foetal dependence is high with heroin and related drugs, and after delivery the neonate may develop serious withdrawal effects that require skilled care.

If the mother continues to take drugs after delivery, the safeguarding needs of the infant will need to be considered, usually with close monitoring by a care team to support the family and ensure the needs of the infant are

being met. Intravenous drug use may lead to infection of the mother with HIV or other conditions that can affect the foetus. Engagement in treatment and good antenatal care reduce the risks to both mother and baby.

Psychiatric comorbidity

There is a strong association between substance misuse and psychiatric comorbidity (often described as *dual diagnosis*). For example, people with substance misuse often have additional diagnoses of personality disorder, depression, or anxiety. Sometimes symptoms of mood disturbance may be a direct result of drug use. For example, people who use stimulants can experience depression, particularly during periods of acute withdrawal.

In other cases the relationship is more complex, with premorbid psychiatric disorder interacting with substance misuse. Thus patients with primary psychiatric disorders such as schizophrenia, bipolar disorder, and sociopathic personality disorder frequently misuse alcohol and illicit drugs. This practice increases the morbidity of the underlying disorder and heightens the risk of violence and self-harm. There is debate about the best ways of delivering treatment services to such patients, but as yet there is little firm evidence to provide guidance. The general consensus is that an integrated service, where the same clinical team deals with both psychiatric and substance misuse disorders, is likely to be the best approach. However, implementing such a service model poses challenges.

Social consequences of drug misuse

There are three reasons why drug misuse can lead to undesirable social effects.

1. Chronic intoxication may affect behaviour adversely, leading to unemployment, motoring offences, accidents, and family problems, including neglect of children.
2. Illicit drugs are generally expensive, so the user may steal or sell sexual favours to obtain money.
3. Drug misusers often keep company with one another, and those with previously stable social behaviour may be under pressure to conform to a group ethos of antisocial or criminal activity.

The economic costs of drug misuse in the United Kingdom have been estimated to be around £15 billion annually (see Box 20.15). The number of deaths linked to illicit drug use in England and Wales have continued to rise and reached almost 3,000 in 2018. Most drug-related costs to individuals and society are associated with the use of crack cocaine and heroin. Just over a third of the

Box 20.15 Social costs of drug misuse in the United Kingdom

Users: premature death, physical and mental illness, low educational achievement, unemployment.

Families: adverse effect on children, poverty, and deprivation.

Others: victims of dangerous driving, victims of crime, victims of assault.

Community: criminal activity related to drug dealing, environmental impoverishment, health and crime risks to community.

Industry: sickness absence, theft in the workplace, productivity losses, costs of security.

Public sector: health care expenditure, criminal justice expenditure, social services care and benefits.

prison population are serving sentences for drug-related crime and there has been a dismaying increase in the number of young people involved in drug trafficking and its associated violence (see Black, 2020).

Diagnosis of drug misuse

It is important to diagnose drug misuse early, at a stage when dependence may be less established and behaviour patterns less fixed, and the complications of intravenous use may not have developed. We provide some general principles before describing the clinical presentations of the different types of drugs. The clinician who is not used to treating people who misuse drugs should remember that they are in the unusual position of trying to help a patient who may be attempting to deceive them. Patients who are misusing heroin may overstate the daily dose to obtain extra supplies for their own use or for sale to others. Furthermore, many patients take more than one drug but may not say so. It is important to try to corroborate patients' accounts of the amounts that they are taking by asking detailed questions about the duration of drug taking, and the cost and source of drugs, by checking the story for internal consistency, and by external verification whenever possible.

Physical signs

Certain physical signs lead to the suspicion that drugs are being injected. These include needle tracks and thrombosis of veins, especially in the antecubital fossa,

wearing garments with long sleeves in hot weather, and scars. Intravenous drug use should be considered in any patient who presents with subcutaneous abscesses or hepatitis.

Behavioural signs

Behavioural changes may also suggest drug misuse. These include absence from school or work, and occupational decline. Dependent people may also neglect their appearance, isolate themselves from former friends, and adopt new friends in a drug culture. Minor criminal offences, such as petty theft and soliciting, may also be indicators.

Medical presentation

Dependent people may come to medical attention in several ways. Some declare that they are dependent on drugs. Others conceal their dependency, and ask for controlled drugs for the relief of pain such as renal colic or dysmenorrhoea. It is important to be particularly wary of such requests from temporary patients. Others present with drug-related complications, such as cellulitis, pneumonia, hepatitis, or accidents, or for the treatment of acute drug effects, overdose, withdrawal symptoms, or adverse reactions to hallucinogenic drugs. A few are detected during an admission to hospital for an unrelated illness.

Taking a drug history

When taking a drug history (see Box 20.16) from a patient who is misusing drugs, the clinician should ask the patient to describe their drug use the previous day and also to describe a *typical drug-using day* (which drug, how often, and which route of use). A typical week can

be described if drugs are not used every day. The clinician should ask about craving, withdrawal symptoms, and other features of the dependence syndrome, such as increased tolerance of the drug and the priority of drug-seeking over other duties and pleasures. The patient should be asked specifically about *risky behaviour*, such as dangerous injecting (into the groin or neck or infected injection sites), and sharing injection equipment.

A *chronological history* of the development of use of *each drug taken* can then be obtained. Useful milestones can include the first use of the drug, when the patient began to use the drug daily, when withdrawal symptoms were first experienced, and when the patient first injected drugs. A history of sharing injection equipment will be important when estimating the risk of HIV or hepatitis.

If patients have had periods of abstinence, it is useful to ask which influences helped them to achieve this and what factors led to relapse. A history of complications should include adverse effects of the drug itself, as well as complications of the route of administration. Any history of accidental overdose should also be elicited. The patient should be asked about family, occupational, and legal problems. Any past history of treatment should be elicited. There are several possible goals in the treatment of drug misusers. Abstinence is one, but safer drug use (*harm reduction*) may be a more realistic aim for many. It is therefore important to obtain patients' views about their drug use and the changes that they would like to make.

Laboratory diagnosis

Whenever possible, the diagnosis of drug misuse should be confirmed by laboratory tests. Urine testing is most commonly used, as it is easier and less invasive than blood testing. Immunoassay kits are available that are able to measure a wide range of misused agents in a single urine sample with a reasonably long timeframe of detection, ranging from two days for amphetamines to up to three weeks for regular cannabis users. However, saliva has the advantage that samples can readily be obtained under supervision and are therefore less easily tampered with. It should be remembered that these 'point of care' tests are for purposes of screening and, if inconsistent with the clinical history, a confirmatory laboratory test using a chromatographic approach should be requested. One issue is that 'false-positive' urine tests for illicit substances can be produced by prescribed psychotropic medication through cross-reactions. For a review see, Abraham and Luty (2010).

Box 20.16 Drug-using history

- Typical drug-using day or week
- Types and quantities of drugs taken (including alcohol and nicotine)
- Symptoms experienced when drugs are unavailable
- Tolerance and primacy of drug habit
- Risky behaviour
- Developmental history of drug misuse
- Abstinence and relapse triggers
- Medical and social complications
- Psychiatric and forensic history

Prevention, treatment, and rehabilitation: general principles

Prevention

Because treatment is difficult, considerable effort should be given to prevention. For many drugs, important preventive measures such as controlling availability and lessening social deprivation depend on government, not medical, policy. However, the reduction of overprescribing by doctors is important, especially with regard to *benzodiazepines* and other *anxiolytic drugs* and *opiates*.

Although education programmes by themselves do not seem to be effective in prevention, it is important that information about the dangers of drug misuse should be available to young people in the school curriculum and through the media. Another aspect of prevention is the identification and treatment of family problems that may contribute to drug taking. Family therapy and parental skills training can decrease the uptake of illicit drugs in young people (Stockings et al., 2016).

Some young people are particularly vulnerable to drug misuse. This includes those of lower socioeconomic status, those who have been in care, those who are homeless, and those who are not in school due to either truancy or exclusion. Identification of young people at risk of drug misuse should be followed by specific psychosocial interventions that are designed to divert them away from drug use (National Institute for Health and Clinical Excellence, 2017).

Treatment

Motivation and change

When drug misuse has already begun, treatment is more effective if it is given before dependence is established. At this stage, as at later stages, the essential step is to motivate the patient to control their drug taking. This requires a combination of advice about the likely effects of continuing misuse, and help with any concurrent psychological or social problems. The techniques of motivational interviewing may be useful here (Treasure, 2004; see Box 20.8). The *stages of change model* described by Prochaska and DiClemente (1986) can help the clinician to encourage motivation effectively (see Box 20.17).

Aims of treatment

The ultimate aim of treatment of the drug-dependent patient is a good personal and social adjustment in the absence of drug use. However, drug withdrawal (or detoxification) by itself has no effect on long-term

Box 20.17 Stages of change model

Pre-contemplation

Misuser does not believe that there is a problem, although others recognize it.

Contemplation

Individual weighs up the pros and cons and considers that change might be necessary.

Decision

The point is reached where a decision is made to act (or not to act) on the issue of substance misuse.

Action

Individual chooses a strategy for change and pursues it.

Maintenance

Gains are maintained and consolidated. Failure may lead to relapse.

Relapse

Return to previous pattern of behaviour. However, relapse may be a positive learning experience, with lessons for the future

outcome, so this process should be part of a wider treatment programme. If withdrawal cannot be achieved, continued prescribing of certain drugs (e.g. opioids) may be considered as part of a *harm reduction programme*. In addition, psychological treatment and social support are required. The general principles of treatment are outlined below, and later sections of the chapter consider treatment specific to individual drugs.

Treatment setting

In the United Kingdom, drug misusers are treated in a number of different settings, and there is increasing variation in the way that care is delivered locally. Responsibility for commissioning drug and alcohol services now rests with local authorities rather than the health sector. Local authorities, whose budgets are under pressure from other demands, commission care for alcohol and substance misuse from both the NHS and 'third sector' providers. Treatment is largely in community settings with individual and group programmes but therapeutic communities with residential facilities are

also available. Residential care is also sometimes needed for people with complex drug and medical problems who require inpatient detoxification (see below). Some GP practices provide treatment for drug misusers with appropriate support from a local drug service.

Physical complications

The complications of self-injection may require treatment in a general hospital. They include accidental overdose, skin infections, abscesses, septicaemia, HIV, and hepatitis, particularly hepatitis C infection. Drug misusers will also need help with general health problems such as nutrition and dental care. Immunization against hepatitis B may be advisable.

Principles of withdrawal

The withdrawal of misused drugs is sometimes called *detoxification*. For many drugs, particularly opioids, withdrawal may sometimes be most effectively undertaken in hospital (see below). Withdrawal from stimulant drugs and benzodiazepines can often be an outpatient procedure, provided that the doses are not very large and other substances are not taken as well. Nevertheless, the risk of depression and suicide should be borne in mind.

Drug maintenance

Some clinicians undertake to prescribe certain drugs to dependent people who are not willing to give them up. The usual procedure is to prescribe a drug which has a slower action (and is therefore less addictive) than the 'street' drug. For example, methadone is prescribed in place of heroin. When this procedure is combined with help with social problems and a continuing effort to encourage the person to accept withdrawal, it is called *maintenance therapy*.

The rationale of this approach is twofold.

1. Prolonged prescribing will remove the need for the patient to obtain 'street' drugs, and will thereby reduce the associated physical and social damage.
2. Social and psychological help, aided by the natural process of maturing, will give the patient the confidence and skills to be able to give up drugs eventually.

Maintenance drug treatment is used in particular for patients with opioid dependence. If maintenance drug therapy is used, it should be remembered that some drug-dependent people convert tablets or capsules into material for injection, which is a particularly dangerous practice. Furthermore, some attend a succession of GPs in search of supplementary supplies of drugs. They may withhold information about attendance at clinics or pose as temporary residents.

Some patients who receive maintenance drugs achieve a degree of social stability, but others continue heavy drug misuse and deteriorate both medically and socially. Patients who are on maintenance methadone are more likely to be retained in treatment than those in drug-free programmes. This may be important because the length of time spent in treatment, regardless of type, is the best predictor of favourable outcome. (The use of methadone maintenance treatment in the management of opioid dependence is considered in more detail below.)

Harm reduction

The increase in HIV and hepatitis C infection has emphasized the importance of *harm reduction programmes*, of which prescribing maintenance may be one component. The purpose of such programmes is to increase the number of substance misusers who enter and comply with treatment. The aim is to identify intermediate treatment goals which, although not involving total abstinence, nevertheless reduce the risk of drug misuse to the individual and society. For example, even if a patient continues to misuse drugs, the risk of hepatitis and HIV infection can be reduced by providing appropriate education and practical help. Such interventions may result in the drug misuser using safer routes of drug administration or sterile injection equipment. Counselling and screening for hepatitis and HIV may also be worthwhile, and hepatitis B vaccination should be offered to non-immune patients.

Psychosocial treatments

Treatment for drug misuse should always include psychosocial approaches. The usual way to deliver this is through a *key worker* who institutes measures such as *counselling*, *education*, and *motivational interviewing*. Practical help with benefits and information about services can also be provided. All patients should be aware of self-help and mutual aid groups such as Narcotics Anonymous, which can be of great benefit to many individuals. Some patients are helped by treatment in a therapeutic community in which there can be frank discussion of the effects of drug taking on the person's character and relationships within the supportive setting of the group (for an outline of community therapy, see Chapter 24).

More formal psychological treatments, as described for the management of alcohol misuse (see Box 20.11), can also be helpful, and have been reviewed by the National Institute for Health and Clinical Excellence (2007). The aim of such treatment is to increase recreational and personal skills so that the patient becomes

less reliant on drugs and the drug culture as a source of satisfaction. Involvement of the patient's partner and family in structured couple and family therapy is often helpful.

As with alcohol dependence, cognitive behavioural techniques can be used to identify, in advance, situations that act as triggers for drug use. In this way alternative methods of coping can be planned (*relapse prevention*). It has already been mentioned that when a drug misuser is confronted with a situation that contains personal cues for drug use, they can experience acute discomfort associated with a strong desire to use the drug. The technique of cue exposure aims, through repeated exposure, to desensitize the drug misuser to these effects and thus improve their ability to remain abstinent. Although *contingency management* is not currently widely used in the United Kingdom, and is politically sensitive, there is good evidence for its effectiveness in reducing drug misuse, and it appears more effective than other psychosocial approaches such as cognitive behaviour therapy. Contingency management provides a variety of incentives in the form of vouchers, privileges, or modest financial rewards to encourage individuals to modify their drug misuse and increase health-promoting behaviours (Luty, 2014; National Institute for Health and Clinical Excellence, 2007).

Rehabilitation

Many drug takers have difficulty in (re)establishing themselves in normal society. The aim of rehabilitation is to enable the drug-dependent person to leave the drug subculture and to develop new social contacts. Unless this can be achieved, any treatment is likely to fail.

Rehabilitation may be undertaken after treatment in a therapeutic community. Patients first engage in work and social activities in sheltered surroundings, and then take greater responsibility for themselves in conditions that are increasingly similar to those of everyday life. Hostel accommodation is a useful stage in this gradual process. Continuing social support is usually required when the person makes the transition to normal work and living.

Dual-diagnosis patients

As noted above, the management of patients with both substance misuse and serious psychiatric illness, such as schizophrenia or mood disorders, poses several additional challenges. Such comorbidity is associated with an increased risk of violence and suicide, and poorer clinical and social outcomes. Patients with dual diagnoses are particularly difficult to retain in treatment, and frequently present in crisis with many unmet social needs.

It is worth emphasizing that in many settings the co-occurrence of serious mental illness and drug and alcohol misuse is the norm rather than the exception. Treatment is difficult and patients may be rejected by both specialist mental health and substance misuse teams until the 'other' condition is treated. On the other hand, simultaneous management by separate teams can pose problems in communication and coordination. For this reason, integrated mental health and substance use services are regarded as the most appropriate model of care but are often difficult to organize in the context of separate funding streams and commissioning processes (Otasowie, 2021).

The management of patients with *comorbid psychosis and substance misuse* is particularly challenging because of the impact of substances on mental state and the predisposition of some patients with psychosis to misuse substances. Recognizing this complex interaction can help prevent potentially fruitless searches for which disorder 'preceded' the other. Efforts should be directed towards a full medical and social assessment of the patient and the development of a coordinated care plan directed towards identified needs (National Institute for Health and Clinical Excellence, 2016).

Misuse of specific types of drug

Opioids

This group of drugs includes morphine, heroin, codeine, and synthetic analgesics such as fentanyl, pethidine, methadone and oxycodone. The pharmacological effects of opioids are mediated primarily through interaction with specific opioid receptors, with morphine and heroin being quite selective for the μ -opioid-receptor type. The main medical use of opioids is for their powerful analgesic actions; they are misused for their euphoriant and anxiolytic effects.

In the past morphine was misused widely in western countries, but it was largely replaced as a drug of misuse by *heroin*, which has a particularly powerful euphoriant effect, especially when taken intravenously. However, a major source of illicit opioid use currently comes from *medically prescribed analgesics*. There is also a growing trend for the misuse of synthetic opioids such as fentanyl, which is associated with a particularly high mortality rate (Ciccarone, 2019).

Epidemiology

In England and Wales the reported lifetime illicit use of opiates is around 1.0%. This is likely to be an underestimate because in the United Kingdom in 2017 there

were calculated to be about eight people per thousand engaging in high-risk opioid use and around 57,000 people started treatment for an opioid use disorder (Focal Point Annual Report, 2019). In addition, about 12% of the adult population in England are prescribed opioid treatment for various pain syndromes (Osborne, 2021).

Use and misuse

The epidemiological data indicate that many people use heroin without becoming dependent on it. However, there is no doubt that repeated heroin use can lead to the rapid development of dependence and marked physiological tolerance. As well as the intravenous route, opioid users may employ other methods of administration—for example, subcutaneous administration ('skin-popping') or sniffing ('snorting'). Heroin may also be heated on a metal foil and inhaled ('chasing the dragon'). Heroin users may change their customary method of drug administration from time to time. From the perspective of harm reduction, methods that avoid intravenous administration are preferable.

Clinical effects

As well as *euphoria* and *analgesia*, opioids cause *respiratory depression*, *constipation*, *reduced appetite*, and *low libido*. Tolerance develops rapidly, leading to the need for an increasing dosage. Tolerance does not develop equally to all of the effects, and constipation often continues when the other effects have diminished. When the drug is stopped, tolerance diminishes rapidly, so a dose taken after a period of abstinence has greater effects than it would have had previously. This loss of tolerance can result in dangerous—sometimes fatal—respiratory depression when a previously tolerated dose is resumed after a drug-free interval (e.g. after a stay in a residential community or prison).

Withdrawal from opioids

Withdrawal symptoms from opioids may include the following:

- intense craving for the drug
- restlessness and insomnia
- pain in muscles and joints
- running nose and eyes
- abdominal cramps, vomiting, and diarrhoea
- piloerection, sweating, dilated pupils, and raised pulse rate
- disturbance of temperature control.

These features usually begin about six hours after the last dose, reach a peak after 36–48 hours, and then wane.

Withdrawal symptoms rarely threaten the life of someone in reasonable health, although they cause great distress and so drive the person to seek further supplies of the drug.

Methadone

Methadone is approximately as potent, weight for weight, as morphine. It causes cough suppression, constipation, and depression of the central nervous system and of respiration. Pupillary constriction is less marked. The withdrawal syndrome is similar to that of heroin and morphine, and is at least as severe. Because methadone has a long half-life (one to two days), symptoms of withdrawal may begin only after 36 hours and reach a peak after three to five days. For this reason, methadone is often used to replace heroin in patients who are dependent on the latter drug.

The natural course of opioid dependence

Longer-term follow-up studies of opioid misusers have revealed that in most cases the disorder appears to run a chronic relapsing and remitting course, with a significant mortality (10–20%) over ten years. Nevertheless, up to 50% of opioid users have been found to be abstinent at ten-year follow-up, which suggests a trend towards natural remission in survivors (Robson, 2009).

Deaths are not infrequently due to accidental overdose, often related to loss of tolerance after a period of enforced abstinence (see above). Suicide is also a common cause of death. Deaths from HIV infection and hepatitis have become more frequent in recent years. Factors associated with a good outcome include substantial periods of employment and marriage. Abstinence is often related to a change in life circumstances. This is illustrated by the report of 95% abstinence among soldiers who returned to the United States after becoming dependent on opioids during service in the Vietnam War (Robins, 1993).

Prevention

Because dependence develops rapidly and treatment of dependent opioid misusers is unsatisfactory, preventive measures (see 'Prevention' above) are particularly important with regard to this group of drugs.

Treatment of crises

Opioid misusers may present in crisis to a doctor in any of three situations. First, when their *supplies have run out*, they may seek drugs either by requesting them directly or by feigning a painful disorder. Although withdrawal symptoms are very unpleasant, so that the misuser will go to great lengths to obtain more drugs, these symptoms are not usually dangerous to an otherwise healthy

person. Therefore it is best to offer drugs only as the first step of a planned maintenance or withdrawal programme. This programme is described in the following sections. The second form of crisis is *drug overdose*. This requires medical treatment, directed in particular towards any respiratory depression caused by the drug. The availability and use of parenteral naloxone in this situation can be life-saving. The third form of crisis is an *acute complication of intravenous drug usage*, such as local infection, necrosis at the injection site, or infection of a distant organ, often the heart or liver.

Planned withdrawal (detoxification)

The severity of withdrawal symptoms depends on psychological as well as pharmacological factors. Therefore the psychological management of the patient during withdrawal is as important as the drug regimen. The speed of withdrawal should be discussed with the patient to establish a timetable which is neither so rapid that the patient will not collaborate nor so protracted that the state of dependence is perpetuated. During withdrawal, much personal contact is needed to reassure the patient; the relationship that is formed in this way can be important in later treatment. The duration of opioid detoxification is usually about four weeks in a residential setting and up to 12 weeks in a community setting.

When the dose is low, opioids can be withdrawn more quickly while giving symptomatic treatment for the withdrawal effects (Box 20.18). Drugs such as *loperamide* or *metoclopramide* can be useful for gastrointestinal

symptoms. *Non-steroidal analgesics* may help reduce aches and pains (Taylor et al., 2021).

When the daily dose of heroin is high, it may be necessary to prescribe an opioid, reducing the dose gradually. One approach is to use *methadone*, which has a more gradual action. The difficulty lies in judging the correct dose of methadone, because patients may either overstate their use of heroin because they fear the withdrawal symptoms, or understate it in an attempt to avoid censure. In addition, as noted below, the strength of street drugs varies.

Methadone should be given in a liquid form to be taken by mouth. The initial methadone dose is normally 10–40 mg daily, depending on the patient's usual consumption. Users with evidence of opioid tolerance may require dosing at the higher end of this range. If four hours after an initial dose there is evidence of withdrawal symptoms, a supplementary dose may be given, but caution is needed because methadone is a long-acting drug, and accumulation and toxicity could result. Street heroin varies in potency in different places and at different times. Therefore, if possible, advice about the equivalent dose of methadone should be obtained from a doctor with experience of treating drug dependence. The rate of methadone reduction depends on the clinical circumstances and the patient's clinical responses. The most rapid regimen may take about ten days to three weeks, but slower reductions over several months may sometimes be appropriate.

Buprenorphine, a partial agonist at opioid receptors, can also be used to manage opioid withdrawal. Again, tapering doses are used at a rate that has been agreed with the patient. Buprenorphine is often well tolerated, but care needs to be taken when starting treatment, especially in the case of patients who are transferring from methadone. This is because the partial agonist action of the drug may precipitate acute withdrawal in patients who are transferring from higher doses of methadone (more than 30 mg) or heroin. This can usually be avoided by beginning treatment when the patient is already in mild withdrawal.

When opioid drugs are prescribed to drug-dependent patients, there is the possibility of diversion of the medication to the illegal market. This can be avoided by supervision of consumption of each daily dose—for example, by nursing or pharmacy staff. This practice also reduces the risk of overdose, and the daily personal contact probably also has other therapeutic benefits. It is generally regarded as good practice at least while treatment is being established, although daily attendance can be difficult for patients who are in full-time employment. In general, opiate detoxification can take place in

Box 20.18 Pharmacological management of opioid withdrawal

- If short-term, non-opiate treatment is desired, use symptomatic treatments such as loperamide for diarrhoea, metoclopramide for nausea, and mebeverine for stomach cramps. Short-term diazepam can be used for anxiety.
- Buprenorphine can be used for short-term opioid withdrawal.
- Methadone treatment for withdrawal can be successful, but needs to be carried out slowly with a gradual tapering of the dose.

Source: Lingford-Hughes, A. R., Welch, S., Peters, L., et al. (2012). BAP updated guidelines: Evidence-based guidelines for the pharmacological management of substance abuse, harmful use, addiction and comorbidity: Recommendations from BAP. *J Psychopharmacol*, 26(7), 899–952. Copyright (2012), SAGE Publications.

the community, but inpatient management should be considered for patients who have not benefited from previous attempts at detoxification in the community, or who need a high level of nursing and medical care because of comorbid physical or psychiatric problems or polydrug abuse, or those who have significant social problems.

During and after opiate withdrawal it is important to consider and institute a suitable plan of psychosocial intervention, as outlined above (Luty, 2015; National Institute for Health and Clinical Excellence, 2007).

Pregnancy and opioid dependence

The babies of women who misuse opioids are more likely than other babies to be premature and of low birth weight. They may also show withdrawal symptoms after birth, including irritability, restlessness, tremor, and a high-pitched cry. These signs appear within a few days of birth if the mother was taking heroin, but are delayed if she was taking methadone, which has a longer half-life in the body. Low birth weight and prematurity are not necessarily directly related to the drug, as poor nutrition and heavy smoking are common among heroin misusers.

Later effects have been reported, with the children of opioid-dependent mothers being more likely, as toddlers, to be overactive and to show poor persistence. However, these late effects may result from the unsuitable family environment rather than from a lasting effect of the intrauterine exposure to the drug.

The pregnancies of women who are in methadone maintenance programmes have a better outcome than those of women who use opioids but are not in such programmes. Some pregnant women request detoxification, but there are significant risks to the baby if this is carried out in the first or third trimester. The main aim of treatment in the first trimester is to engage the mother in a multidisciplinary care programme that will include stabilization on a suitable dose of methadone, avoidance of illegal drug use, and a high level of antenatal care. Patients who are receiving buprenorphine can remain on this treatment. Detoxification may be carried out in the second trimester, employing small frequent reductions of methadone or buprenorphine (Lingford-Hughes et al., 2012).

Maintenance treatment for opioid dependence

As described above, withdrawal from opioids is the preferred treatment option, but if this is not possible, maintenance treatment, usually with methadone, may reduce the physical and social harm associated with the intravenous use of illegal drug supplies. Instead of heroin,

methadone is prescribed as a liquid preparation formulated so as to discourage attempts to inject it.

Methadone and buprenorphine maintenance treatment. Overall, patients who are receiving methadone maintenance treatment are approximately three times more likely to stay in treatment, and are about two-thirds less likely to use illegal heroin (Lingford-Hughes et al., 2012). In addition, patients in methadone maintenance programmes show less risky injecting behaviours and lower rates of HIV infection.

Methadone doses of 20–40 mg daily have been widely advocated as appropriate for maintenance treatment, but there is evidence that higher doses (60–120 mg daily) are associated with lower rates of illegal opioid use and improved retention in the therapeutic programme. The latter is associated with an improved therapeutic outcome. The best approach is probably to have a flexible dosing policy, bearing in mind the potential toxicity of methadone in patients whose tolerance is unknown or difficult to assess. Generally it is better to start treatment with lower doses initially (not more than 40 mg daily), and to increase the dose over a number of weeks, titrating against the presence of withdrawal symptoms. It should be remembered that methadone levels continue to increase for about five days after the last dosage adjustment.

There is also a growing evidence base for the use of *buprenorphine* as an alternative to methadone for maintenance treatment for opioid users, although there is less consensus on appropriate dosing. In general, doses of 12–16 mg are used in longer-term prescribing. The same principles of treatment apply as in methadone prescribing. In controlled trials, maintenance methadone treatment is superior to buprenorphine in retaining patients in treatment; however, buprenorphine has a greater safety margin, particularly in terms of adverse cardiac effects such as prolongation of the QTc interval. In practice, either buprenorphine or methadone can be offered as maintenance treatment, the decision being guided by safety considerations and patient choice. A prolonged release subcutaneous form of buprenorphine can be administered weekly or monthly and is preferred by some patients (Taylor et al., 2021)

The concerns about diversion of prescribed opioid drugs, and the risk of overdose, have been described above in the section on management of opioid withdrawal. They also apply in maintenance treatment, and daily supervised consumption is commonly continued for a proportion of patients in maintenance treatment. A UK survey showed that supervision of methadone administration produced a fourfold decrease in deaths caused by methadone overdose (Strang et al., 2010).

Naltrexone is a long-acting opioid antagonist that is used to help to prevent relapse in detoxified opioid-dependent patients. Although naltrexone treatment may have a role in certain dependent individuals who are highly motivated, such as doctors recovering from opioid dependence, or patients engaged in structured programmes associated with the criminal justice system, its benefit in the wider community of opioid users is less well established (Herlinger and Lingford-Hughes et al., 2021). A long-acting depot preparation of naltrexone, available in the United States, may have greater efficacy.

Therapeutic community methods

These forms of treatment aim to produce abstinence by effecting a substantial change in the patient's attitudes and behaviour. Drug taking is represented as a way of avoiding pre-existing personal problems, and as a source of new ones. Group therapy and communal living are combined in an attempt to produce greater personal awareness, more concern for others, and better social skills. In most therapeutic communities some of the staff have previously been dependent on drugs, and so are often better able to gain the confidence of patients in the early stages of treatment. For younger populations, the use of a multimodal approach, including family involvement and community support with psychotherapy and pharmacologic interventions is important (National Institute for Health and Care Excellence, 2007).

Anxiolytic and hypnotic drugs

The most frequently misused drugs in this group are now the *benzodiazepines* and *Z drugs* (zopiclone and zolpidem). *Barbiturates* are little prescribed, and their misuse has fallen. Other drugs with sedative properties that are currently misused include *gabapentin* and *pregabalin*. The clinical effects of these drugs are thought to result from their ability, directly or indirectly, to facilitate brain GABA function.

Benzodiazepines and Z drugs

These drugs were in therapeutic use for many years before it became apparent that their prolonged use could lead to tolerance and dependence, with a characteristic withdrawal syndrome. The withdrawal syndrome includes the following:

- *Anxiety symptoms*—anxiety, irritability, sweating, tremor, and sleep disturbance.
- *Altered perception*—depersonalization, derealization, hypersensitivity to stimuli, abnormal body sensations, and abnormal sensation of movement.

- *Other features* (rare)—depression, suicidal behaviour, psychosis, seizures, and delirium tremens.

Epidemiology. Benzodiazepine use has been extremely widespread—for example, it was estimated at one time that about 10% of the population of Europe and the United States were using benzodiazepines as anxiolytics or hypnotics. Over the last few years there has been a decline in the prescription of benzodiazepines for anxiety and insomnia in the United Kingdom. In the 2018 about 3% of the adult population in England were prescribed a benzodiazepine and 2% were prescribed a Z drug (Public Health England 2020).

There is a significant misuse problem of benzodiazepines among younger people, sometimes associated with intravenous administration of very high doses and polysubstance misuse. A significant proportion of people who are dependent on alcohol are also dependent on benzodiazepines. The Crime Survey reported that in England and Wales in 2020, the rate of illicit tranquilizer use in the adult population in the past year was 0.8%. However, the rate of lifetime benzodiazepine use in 15-year-olds was 1.5%. Benzodiazepine misuse in young people often involves high-potency agents such as alprazolam obtained via the internet.

Dependence. Dependence on benzodiazepines usually results from prolonged medical use, but may also come from the availability of benzodiazepines as street drugs because of their euphoriant and calming effects. The withdrawal syndrome closely resembles the anxiety symptoms for which the drugs are usually prescribed. Thus, if symptoms appear after the dose of benzodiazepine has been reduced, the clinician may revert to a higher dosage in the mistaken belief that these symptoms indicate a persistent anxiety disorder. It has been estimated that about one third of patients who take a benzodiazepine at therapeutic doses for more than six months may become dependent (Lingford-Hughes et al., 2012). This dependence usually manifests as withdrawal symptoms when stopping medication rather than significant dose escalation.

Treatment. In mild dependence due to therapeutic prescription, treatment usually consists of gradual withdrawal over a period of several months. The addition of psychological therapies such as group cognitive behavioural therapy can improve cessation rates. Additional medication is of little benefit for most patients but beta-blockers such as propranolol can be helpful for severe physical symptoms of anxiety while antidepressants may be required if a depressive episode emerges.

Withdrawal appears to be more severe from benzodiazepines that have short half-lives and high potency

at the benzodiazepine receptor. For this reason it is often suggested that patients who are taking such compounds should be switched to an equivalent dose of a longer-acting drug such as diazepam before withdrawal is attempted.

Current advice from the National Institute for Clinical Excellence (2022a) is that the dose of benzodiazepine should be lowered very gradually and in decreasing amounts as the daily amount diminishes. In this way, the total withdrawal time for a patient taking 20 mg of diazepam daily might be between 20 to 40 weeks. However, if a patient experiences troublesome withdrawal symptoms, the dose can be maintained or even temporarily increased until the symptoms settle.

Where a patient has been misusing benzodiazepines in very high doses, often in conjunction with opioid misuse, a reduction to a therapeutic dose is a good initial aim. In general, patients should be given no more than 30 mg diazepam daily, which should be enough to prevent withdrawal seizures.

Many patients experience their most troublesome withdrawal symptoms once the benzodiazepine dose has been completely tapered off. Symptoms usually subside over the next few weeks, although the time course can be irregular and some symptoms, such as muscle spasm, may not appear until other features of withdrawal have largely disappeared. A few patients continue to experience withdrawal-like symptoms for months or even years after cessation of benzodiazepines (*prolonged withdrawal syndrome*).

Prevention. The prevention of benzodiazepine dependence lies in the restriction of prescribing. Psychological treatments are effective for anxiety disorders (see Chapter 8), and non-pharmacological approaches to insomnia are also beneficial (see Chapter 13). If benzodiazepines are prescribed, this should be for the short-term relief of symptoms that are severely disabling or distressing. In some patients, who are already long-term users at modest doses, the balance of benefit and risk will favour continued prescribing, but these patients should be regularly reviewed and advised to cut their dose of medication if they can according to the NICE guideline referenced above.

For further information about benzodiazepine dependence and its treatment, see Lingford-Hughes et al. (2012) and National Institute for Clinical Excellence (2022a).

Z drugs such as zopiclone and zolpidem have largely replaced the benzodiazepines as hypnotics. Z drugs are structurally dissimilar to benzodiazepines but nevertheless act via the benzodiazepine receptor. This gives them a similar clinical profile to classical benzodiazepines, but

their pharmacokinetic properties make them suitable as hypnotics.

As with benzodiazepines, the use of Z drugs is not usually associated with rapid dose escalation and drug-seeking behaviour, though these can occur in susceptible individuals. However, many patients remain on long-term Z drug treatment and suffer insomnia and anxiety when they stop medication or reduce the dose suddenly. The National Institute for Clinical Excellence (2022a) also recommends very slow withdrawal as the most appropriate way of managing 'therapeutic dose' Z drug dependence.

Gabapentin and pregabalin (gabapentinoids)

Pregabalin is licensed for the treatment of generalized anxiety disorder (see Chapter 8) while both *gabapentin* and *pregabalin* are used as anticonvulsants and for the treatment of neuropathic pain. Both drugs are structural analogues of GABA, but they do not affect GABAergic mechanisms directly, instead modifying neurotransmitter release through interaction with the $\alpha_2\text{-}\delta$ subunit of voltage-gated calcium channels.

Both drugs produce prominent sedative and anxiolytic effects and there is growing evidence of their misuse and dependence, particularly in people with a history of alcohol and other substance misuse, and in prison settings. Some of the misuse arises from prescribing for licensed indications and subsequent diversion; however, sales on the internet also are an important source. Misuse in combination with opiates is also quite common and gabapentinoids increase the risk of opiate-related deaths (Evoy et al., 2021). In 2019, the UK government placed gabapentin and pregabalin under Schedule 3 of the Misuse of Drugs Regulation and Class C of the Misuse of Drugs Act. This restricts the amount of gabapentin and pregabalin that can be issued on a single prescription.

Sudden cessation of gabapentin and pregabalin can produce a *withdrawal syndrome*. Agitation, confusion, and disorientation are the most frequent symptoms, while sweating, gastrointestinal symptoms, tachycardia, hypertension, and insomnia are also reported. Treatment with gabapentin or pregabalin rapidly alleviates the syndrome, but administration of benzodiazepines does not seem helpful. For a systematic review of pregabalin and gabapentin misuse, see Evoy et al. (2021).

Cannabis

Cannabis is derived from the plant *Cannabis sativa*. It is consumed either as the dried vegetative parts in the form known as marijuana or grass, or as the resin secreted by the flowering shoots of the female plant. Cannabis

contains several pharmacologically active substances, of which the most powerful psychoactive compound is δ -9-tetrahydrocannabinol (THC). It seems likely that the pharmacological effects of cannabinoids are mediated through interaction with specific cannabinoid receptors in the central nervous system. Endogenous ligands for these receptors include 2-arachidonoyl glycerol and arachidonoyl ethanolamide (anandamide) (for a review of the pharmacology of cannabis, see Amin and Ali, 2019). Over the past decade more potent cannabis preparations (known as 'skunk') have become widely available; these have higher levels of THC and a greater risk of adverse effects.

Epidemiology. In some parts of North Africa and Asia, cannabis products are consumed in a similar way to alcohol in western society. In North America and Europe the intermittent use of cannabis by young people is widespread and is rapidly evolving and changing. The Psychiatric Morbidity Survey reported that in households in England the lifetime use of cannabis by adults was about 26%, with 7.2% of those surveyed reporting use in the past year, and 2.6% meeting the criteria for cannabis dependence (McManus et al., 2016).

Clinical effects. The effects of cannabis vary with the dose, the person's expectations and mood, and the social setting. Users sometimes describe themselves as 'high' but, like alcohol, cannabis seems to exaggerate the pre-existing mood, whether that was exhilaration or depression. Users report an increased enjoyment of aesthetic experiences, and distortion of the perception of time and space. There may be reddening of the eyes, dry mouth, tachycardia, irritation of the respiratory tract, and coughing. Cannabis intoxication can lead to dangerous driving.

Adverse effects. No serious adverse effects have been demonstrated among those who use cannabis intermittently in small doses. Although there is no positive evidence of teratogenicity, cannabis has not been proved safe during the first three months of pregnancy. Inhaled cannabis smoke irritates the respiratory tract and is potentially carcinogenic. The most common adverse psychological effect of acute cannabis consumption is *anxiety*. Mild *paranoid ideation* is also not uncommon. At higher doses, *toxic confusional states* and occasionally *psychosis* in clear consciousness may occur. Hyperemesis is also reported.

Cannabis and mental illness. It is well established that cannabis can modify the course of an established psychotic illness, with evidence from a systematic review that users are more likely to experience more *severe positive symptoms*, more *relapses*, and *longer hospitalizations*). Cannabis also worsens the course of bipolar disorder and

probably increases the risk of the illness in those with a familial predisposition (Castle, 2020).

The question of whether cannabis use can predispose to the later *development of schizophrenia* has been more controversial. Andreasson et al. (1987) followed up 45,570 Swedish conscripts for 15 years. They found that the relative risk of developing schizophrenia was 2.5 times higher in individuals who used cannabis, and that the relative risk for heavy users was six times higher. Although these data suggest that cannabis could be a risk factor for the development of schizophrenia, it is also possible that those who are predisposed to develop schizophrenia are also more likely to misuse cannabis.

However, more recent prospective cohort studies that have adjusted for various baseline confounding factors such as history of mental illness, continue to demonstrate an increased risk of new-onset psychotic symptoms, especially in young people who use cannabis. A number of factors appear to increase the risk of persistent or recurrent psychosis following cannabis use. These include a family history of schizophrenia, early age of onset of cannabis use, and consumption of high-potency forms of THC. However, determination of causality is complicated by the fact that there is some overlap in the genes predisposing to cannabis use disorder with those predisposing to schizophrenia. For a review of this area, see D'Souza et al. (2022).

There has also been concern that cannabis use among teenagers might increase the risk of other poor psychosocial outcomes, such as mood and anxiety disorders, cognitive impairment, poor educational performance, job instability, traffic accidents, and uptake of more harmful illegal drugs. Such associations have continued to be reported consistently; however, the direction of causation is not always clear and confounding by factors such as childhood adversity is difficult to exclude completely (Castle, 2020).

Tolerance and dependence. There is evidence that tolerance to the effects of cannabis can occur in individuals who are exposed to high doses for a prolonged period of time, but it is less evident in those who use small or intermittent doses. However, some patients do report inability to control cannabis use despite personal and social harm resulting from it. The *lifetime risk* of cannabis dependence in those who have ever used it is estimated at about 9%, compared to 32% for nicotine, 23% for heroin, 17% for cocaine, and 15% for alcohol (Hall, 2015). Withdrawal from high doses of cannabis gives rise to a syndrome of *irritability*, *nausea*, *insomnia*, and *anorexia*. These symptoms are generally mild in nature. The best way to manage cannabis dependence is unclear, but psychosocial interventions are commonly provided.

Although abstinence may be difficult to achieve, treatment does appear to lower cannabis use and the associated harms.

A discussion of the potential mental health harms of cannabis now needs to consider that in a growing number of countries, cannabis use has been legalized for medical and recreational purposes. Legalization appears to have reduced the illicit cannabis market, but the potency of cannabis preparations has not yet been regulated and continues to rise. Also, a legal cannabis industry has a commercial interest in increasing consumption in the population. There is much uncertainty about the long-term effects of these regulatory changes on public health and how they will interact with the harms of other widely used, licit drugs such as nicotine and alcohol (see Hall et al., 2019).

Stimulant drugs

These drugs include *amphetamines*, and related substances such as methylphenidate. *Cocaine* is also a stimulant drug, but is considered separately in the next section.

Amphetamines

Amphetamines play a significant role in the pharmacological treatment of attention deficit disorder (see Chapter 25) and narcolepsy (see Chapter 13), but have very little other use in medical practice. The psychomotor stimulant effects of amphetamines are believed to result from their ability to release and block the reuptake of dopamine and noradrenaline.

Epidemiology. Amphetamines are currently used rather less than cocaine in the United Kingdom. The Psychiatric Morbidity Survey reported that in households in England the lifetime use of amphetamines was about 6.2%, with 0.5% reporting use in the past year, and 0.1% of those surveyed meeting the criteria for amphetamine dependence (McManus et al., 2016).

In the past, addiction to stimulant drugs arose chiefly from injudicious prescribing. However, most amphetamines are now illegally synthesized and used as a 'street drug', known as 'speed' or 'whizz'. As well as being taken orally or intravenously, amphetamines can also be 'snorted' (taken like snuff). Methamphetamine ('ice') can be smoked or injected, and is said to produce particularly powerful effects, which are prolonged compared to those of amphetamine.

Clinical effects. Apart from their immediate effect on mood, amphetamines cause over-talkativeness, overactivity, insomnia, dryness of the lips, mouth, and nose, and anorexia. The pupils dilate, the pulse rate increases, and the blood pressure rises. With large doses there may

Box 20.19 Some complications of amphetamine and cocaine misuse

Medical

Cardiovascular: hypertension, stroke, arrhythmias, myocardial infarction
 Infective: abscesses, septicaemia, hepatitis, HIV
 Obstetric: reduced foetal growth, miscarriage, premature labour, placental abruption
 Other: weight loss, dental problems, epilepsy, general neglect

Psychiatric

Anxiety, depression, antisocial behaviour, insomnia, paranoid psychosis

be *cardiac arrhythmia*, *severe hypertension*, *cerebrovascular accident*, and, occasionally, *circulatory collapse*. At increasingly high doses, neurological symptoms such as seizures and coma may occur. Acute adverse psychological effects of amphetamines include dysphoria, irritability, insomnia, and confusion. Anxiety and panic can also be present. *Obstetric complications* include miscarriage, premature labour, and placental abruption (see Box 20.19).

Amphetamine-induced psychosis. Prolonged use of high doses of amphetamines may result in repetitive stereotyped behaviour (e.g. repeated tidying). A *paranoid psychosis* that has been likened to paranoid schizophrenia may also be induced by prolonged high doses. The features include persecutory delusions, auditory and visual hallucinations, and sometimes hostile and dangerously aggressive behaviour. Usually, the condition subsides within about a week, but occasionally it persists for months. It is not certain whether these prolonged cases represent true drug-induced psychoses, schizophrenia provoked by the amphetamine, or are merely coincidental. Whatever the nature of the association, it is not uncommon for patients with a history of amphetamine misuse to present to general psychiatric services. The ability of amphetamines to provoke psychosis was one of the observations that led to the dopamine hypothesis of schizophrenia (see Chapter 11).

Tolerance and dependence. From the epidemiology of amphetamine use, it seems that many recreational users do not progress to misuse and dependence. In more persistent users, tolerance to amphetamines leads to higher doses of the drug being taken. A withdrawal syndrome ('*crash*') of varying severity follows cessation of

amphetamine use. In mild cases it consists mainly of low mood and decreased energy. In some cases, particularly in heavy users, depression can be severe, and accompanied by anxiety, tremulousness, lethargy, fatigue, and nightmares. Craving for the drug may be intense, and suicidal ideation may be prominent. Dependence on amphetamines can develop quickly. Dependence on stimulant drugs may be recognized from a history of overactivity and high spirits alternating with inactivity and depression. Whenever amphetamine use is at all likely, a urine sample should be taken for analysis as soon as possible because these drugs are quickly eliminated.

Prevention and treatment. Prevention of amphetamine misuse depends on restriction of the drugs and careful prescribing. Doctors should be wary of newly arrived patients who purport to suffer from narcolepsy.

Treatment of acute overdoses requires sedation and management of hyperpyrexia and cardiac arrhythmias. Most toxic symptoms, including paranoid psychoses, resolve quickly when the drug is stopped. An antipsychotic drug may be needed to control florid symptoms.

The treatment of amphetamine dependence is difficult, as craving for the drug can be intense. Abstinence is the usual goal and to achieve this a full range of social and psychological interventions may be needed (see above). Benzodiazepines may be helpful for managing acute distress caused by a severe withdrawal syndrome. Antidepressants do not appear to be effective in promoting abstinence, although they may be appropriate for treatment of a persistent depressive disorder. Psychosocial approaches such as cognitive behaviour therapy and contingency management continue to be the mainstay of treatment. Where stimulant misuse is associated with attention-deficit hyperactivity disorder (ADHD), treatment with methylphenidate can reduce both illicit stimulant use as well as ADHD symptoms (Winstock and Flechais, 2020).

Cocaine

Cocaine is a central nervous stimulant with effects similar to those of amphetamines (described above). It is a particularly powerful positive reinforcer in animals, and causes strong dependence in humans. These latter effects probably stem from the ability of cocaine to block the reuptake of dopamine into presynaptic dopamine terminals. This leads to substantial increases in extracellular levels of dopamine in the ventral striatum, and consequent activation of the physiological 'reward system'.

Cocaine is administered by injection, by smoking, and by sniffing into the nostrils. The latter practice sometimes causes perforation of the nasal septum. In 'freebasing', chemically pure cocaine is extracted from

the street drug to produce 'crack', which has a very rapid onset of action, particularly when inhaled.

Epidemiology. The Psychiatric Morbidity Survey (McManus et al., 2016) reported that in households in England the lifetime use of cocaine was about 7.9%, with 1.2% of those surveyed reporting use in the past year, and 0.4% meeting the criteria for cocaine dependence.

Clinical effects. The psychological effects of cocaine include excitement, increased energy, and euphoria. This can be associated with grandiose thinking, impaired judgement, and sexual disinhibition. Higher doses can result in visual and auditory hallucinations. *Paranoid ideation* may lead to *aggressive behaviour*. More prolonged use of high doses of cocaine can result in a *paranoid psychosis* with violent behaviour. This state is usually short-lived, but may be more enduring in those with a pre-existing vulnerability to psychotic disorder. *Formication* ('cocaine bugs')—a feeling as if insects are crawling under the skin—is sometimes experienced by heavy cocaine users.

The physical effects of cocaine include increases in pulse rate and blood pressure. Dilatation of the pupils is often prominent. Severe adverse effects of cocaine use include *cardiac arrhythmias*, *myocardial infarction*, *myocarditis*, and *cardiomyopathy*. Cocaine use has also been associated with cerebrovascular disease, including cerebral infarction, subarachnoid haemorrhage, and transient ischaemic attacks. Seizures and respiratory arrest have been reported. *Obstetric complications* include miscarriage, placental abruption, and premature labour.

Tolerance and dependence. In persistent users, tolerance of the effects of cocaine develops and a withdrawal syndrome similar to that seen following withdrawal of amphetamines can occur. After acute cocaine use, the 'crash' consists of dysphoria, anhedonia, anxiety, irritability, fatigue, and hypersomnolence. If the preceding cocaine use has been relatively mild, such symptoms resolve within about 24 hours. After more prolonged use, the symptoms are more severe and extended, and are associated with intense craving, depression, and occasionally severe suicidal ideation. Craving for cocaine can re-emerge after months of abstinence, particularly if the individual is exposed to psychological or social cues previously associated with its use.

Treatment. Acute intoxication may require sedation with benzodiazepines or, in severe cases, an antipsychotic agent. Concurrent medical crises such as seizures or hypertension should be managed in the usual way.

As with amphetamines, the treatment of cocaine dependence is difficult because of the intense craving associated with abstinence from the drug. For moderate cocaine users it may be sufficient to provide

psychological and social support on an outpatient basis. Heavy and chaotic users with strong dependence will need more intensive management, perhaps in a residential setting. There is little evidence to support substitute pharmacological treatments to date, although several medications have been evaluated (Lingford-Hughes et al., 2012). Various psychosocial approaches can be helpful for patients with cocaine dependence. These include cognitive behaviour therapies, contingency management, and programmes that incorporate twelve-step approaches. With regard to individual treatment programmes, it is worth noting that people who misuse cocaine often misuse other drugs, such as opioids and alcohol (see Winstock and Flechais, 2020).

MDMA ('ecstasy')

The recreational use of *3,4 methylenedioxymethamphetamine* (MDMA), also known as 'ecstasy', increased rapidly at the end of the twentieth century but might now be stabilizing to some extent. In 1998, about 5% of 16–24-year-olds reported using ecstasy in the past year, and in 2002 the number had risen to about 7%. In 2016, the Psychiatric Morbidity Survey found a corresponding figure of 6.1% (McManus et al., 2016). Lifetime use of ecstasy in this survey across all ages was about 6.4%. Ecstasy is a synthetic drug that is classified in the DSM-5 substance list as a hallucinogen, but in ICD-11 MDMA has its own category together with the related substance, methylenedioxyamphetamine (MDA). MDMA has both stimulant and mild hallucinogenic properties. It is usually taken in tablet or capsule form in a dose of about 50–150 mg. Given in this way, its effects last for about four to six hours. Like amphetamines, ecstasy increases the release of dopamine, but it also releases 5-hydroxytryptamine (5-HT), which may account for its hallucinogenic and sociotropic properties. The latter has led MDMA to be called by some, an 'empathogen'.

Clinical effects. Ecstasy produces a positive mood state, with feelings of euphoria, sociability, and intimacy. It also produces sensations of newly discovered insights and heightened perceptions. The physical effects of ecstasy include *loss of appetite, tachycardia, bruxism, and sweating*. Tolerance of successive doses of ecstasy develops quickly. Weekend users describe a midweek 'crash' in mood, which may represent withdrawal effects.

Adverse reactions. Rarely, ecstasy can cause severe adverse reactions, and deaths due to hyperthermia and its complications have been reported in healthy young adults. Hyperthermia probably results from the effect of ecstasy in increasing brain 5-HT release, together with the social setting in which the drug is customarily taken (crowded parties with prolonged and strenuous dancing).

Deaths have also been reported due to cardiac arrhythmias, although pre-existing cardiac disease may have played a role. Intracerebral haemorrhage has occurred in ecstasy users, probably as a consequence of hypertensive crises. Cases of toxic hepatitis could reflect impurities in manufacture or an idiosyncrasy in metabolism.

The use of ecstasy has been associated with acute and chronic paranoid psychoses. However, as with other drug-induced psychotic states, it is not clear how far such disorders represent idiosyncratic reactions of vulnerable individuals. There are also reports of 'flashbacks', which are the recurrence of abnormal experiences weeks or months after drug ingestion. Such effects have been reported with other hallucinogens (see below). It is possible that repeated use of ecstasy may increase the risk of adverse psychiatric outcomes such as depression, anxiety, and depersonalization.

In experimental animals, including primates, repeated treatment with ecstasy causes degeneration of 5-HT nerve terminals in the cortex and forebrain. Therefore it is possible that such effects could occur in humans, and some brain imaging studies in heavy MDMA users have shown changes in serotonin transporter binding that could equate to 5-HT terminal damage; however, the findings are not consistent. Whether such a change could be associated with long-term neuropsychological or psychiatric sequelae is disputed, but there is evidence for impairments in executive function in former ecstasy users (Winstock and Flechais, 2020).

Prevention and harm reduction. Although the risk of serious harm following acute ecstasy use appears to be low, it is important to inform potential users about the acute risks and the potential long-term hazard of *neurotoxicity, cognitive impairment, and adverse longer-term psychiatric sequelae*. Consumption of large doses and pre-existing psychiatric disorder are likely to be associated with increased risk of adverse reactions. Education may also help users to avoid heatstroke by encouraging them to take breaks from dancing and to consume sufficient *isotonic* replacement fluid during vigorous exercise. However, the consumption of large amounts of water has sometimes caused death due to *hyponatraemia*.

There has been renewed interest in the possible therapeutic effects of MDMA-assisted psychotherapy particularly for patients with PTSD that has not responded to conventional psychological and pharmacological approaches. Results from placebo-controlled randomized trials have been encouraging (Smith et al., 2022). However, these results should be seen in the context of the difficulty of blinding participants to active MDMA treatment and the fact that the trials of MDMA for PTSD are usually carried out by enthusiasts for this approach.

Hallucinogens

Hallucinogens are sometimes known as psychedelics, but we do not recommend this term because it does not have a single clear meaning. The term psychotomimetic is also used, because the drugs produce changes that bear some resemblance to those of psychotic symptoms. However, the resemblance is not close, so we do not recommend this term either.

The synthetic hallucinogens include LSD, dimethyltryptamine, and methylmethoxyamphetamine derivatives such as 2,5-dimethoxy-4-methylamphetamine (DOM). Of these drugs, LSD is encountered most often in the United Kingdom. Hallucinogens also occur naturally in some species of mushroom ('magic mushrooms'), and varieties containing *psilocybin* are consumed for their hallucinogenic effects. *Mescaline* is found in the peyote cactus. The mode of action of hallucinogenic drugs is generally attributed to partial agonist action at brain 5-HT_{2A} receptors, particularly those located on pyramidal neurons in the neocortex.

Epidemiology. The Psychiatric Morbidity Survey reported that in households in England the lifetime use of LSD was 4.3%, while that of magic mushrooms was 5.9%. The corresponding figures for use in the past year were 0.3% and 0.4%, respectively (McManus et al., 2016).

Clinical effects. The effects of LSD have been well studied, and are described here. The physical actions of LSD are variable. There are initial sympathomimetic effects—heart rate and blood pressure may increase, and the pupils may dilate. However, overdosage does not appear to result in severe physiological reactions.

In predisposed individuals, the hypertensive effects of hallucinogens can cause adverse myocardial and cerebrovascular effects. The psychological effects develop within a period of two hours after LSD consumption, and generally last 8–14 hours. The most remarkable experiences are distortions or intensifications of *sensory perception*. There may be confusion between sensory modalities (*synaesthesia*), with sounds being perceived as visual, or movements being experienced as if heard. Objects may be seen to merge with one another or to move rhythmically. The passage of time appears to be slowed, and experiences seem to have a profound meaning.

A distressing experience may be *distortion of the body image*, with the person sometimes feeling that they are outside their own body. These experiences may lead to panic, with fears of insanity. The patient's mood may be one of exhilaration, distress, or acute anxiety. The physiological and psychological effects of an equivalent dose

of psilocybin are very similar to those of LSD but more short-lived (Holze et al., 2022).

Whenever possible, adverse reactions should be managed by 'talking down' the user, explaining that the alarming experiences are due to the drug. If there is not time for this, an anxiolytic such as diazepam should be given and is usually effective. Tolerance of the psychological effects of LSD can occur, but a withdrawal syndrome has not been described. It is uncertain whether dependence occurs, but, if so, it is likely to be rare.

It has been argued that the use of LSD can cause long-term abnormalities in thinking and behaviour, or even schizophrenia, but the evidence for this is weak. However, *flashbacks* (i.e. the recurrence of psychedelic experiences weeks or months after the drug was last taken) are a recognized event. In DSM-5 this is recognized as 'hallucinogen persisting perception disorder', which describes the reoccurrence of one or more of the perceptual symptoms experienced during intoxication with the hallucinogen. The disorder is rare, and its aetiology not understood. No drug treatment has been shown to be useful and generally symptoms resolve with time and psychological support (Winstock and Rucker, 2020).

Recently, there have been calls for a clinical re-evaluation of psychedelic drugs such as LSD and psilocybin, with claims that their adverse effects, particularly when taken in safe and controlled conditions, have been greatly overstated. There are a growing number of controlled trials of psychedelic-assisted psychotherapy in patients with depressive conditions not responding to conventional pharmacotherapy. The main agent studied in this work has been psilocybin and good short-term responses have been seen in carefully selected patients. As with the MDMA studies described previously, the problem of treatment blinding and controlling for expectancy effects makes interpretation of results difficult (Ling et al., 2022).

Phencyclidine, ketamine, and nitrous oxide

Phencyclidine and ketamine are sufficiently different from the hallucinogens in their actions to require a separate description. They can be synthesized easily, taken by mouth, smoked, or injected. Phencyclidine was developed as a *dissociative anaesthetic*, but its use was abandoned because of adverse effects such as delirium and hallucinations. It is related to the currently used anaesthetic agent ketamine. Ketamine is also misused, and has similar effects to those of phencyclidine. However, ketamine has a shorter half-life than phencyclidine and is less likely than the latter to be associated with grossly disturbed behaviour.

Both phencyclidine and ketamine antagonize neurotransmission at *N-methyl-D-aspartate* (NMDA) receptors, which may account for their hallucinogenic effects. The psychological effects of ketamine in healthy volunteers have been used to model some of the clinical symptoms and cognitive changes that are seen in patients with schizophrenia, and are discussed in Chapter 11. Ketamine has also aroused interest because of its unexpected ability to produce a striking temporary *remission of depression* in treatment-refractory patients (see page 728).

Phencyclidine is widely available in the United States but is little used in the United Kingdom. Most users of phencyclidine also use other drugs, particularly alcohol and cannabis. Ingestion of phencyclidine may be inadvertent, as it is often added to other street drugs to boost their effects. Ketamine is growing in popularity and is mainly used in the 'club scene'. The Psychiatric Morbidity Survey found that the highest lifetime use of ketamine (4.8%) was in those aged 16–24 years and 3.9% of this group had used ketamine in the last year (McManus et al., 2016). Ketamine has a good safety profile when used as an anaesthetic, for which purpose it is given intravenously. Illicit use of ketamine can involve intravenous use or 'snorting'. When taken by either of these routes the drug has short duration of action (about one to two hours).

Clinical effects. Small doses of phencyclidine produce drunkenness, with analgesia of the fingers and toes, and even anaesthesia. Intoxication with the drug is prolonged, the common features being agitation, depressed consciousness, aggressiveness and psychotic-like symptoms, nystagmus, and raised blood pressure. With high doses there may be ataxia, muscle rigidity, convulsions, and absence of response to the environment even though the eyes are wide open. Phencyclidine can be detected in the urine for 72 hours after it was last taken.

If serious overdoses are taken, sympathomimetic crises may occur with hypertensive heart failure, cerebrovascular accident, or malignant hyperthermia. Status epilepticus may appear. Rhabdomyolysis can lead to renal failure. Fatalities have been reported, due mainly to hypertensive crisis but also to respiratory failure or suicide. Other people may be attacked and injured. Chronic use of phencyclidine may lead to aggressive behaviour accompanied by memory loss. Tolerance of the effects of phencyclidine occurs, although withdrawal symptoms are rare in humans. Animal studies suggest that dependence could occur in heavy users.

Low doses of ketamine produce mood elevation, while at higher doses the user experiences sensory and perceptual distortions and out-of-body experiences

similar to those reported with LSD. Frightening hallucinations, thought disorder, and confusion can also occur. These usually remit after a few hours. Patients who are admitted to hospital with ketamine intoxication show evidence of sympathetic overactivity, with chest pain, tachycardia, and palpitations. They may also experience nausea and vomiting, difficulty in breathing, ataxia, mutism, and temporary paralysis. Rarely, severe agitation and rhabdomyolysis can occur.

Treatment of intoxication. Treatment of acute phencyclidine intoxication is symptomatic, according to the features listed above. Haloperidol, diazepam, or both may be given. Caution should be exercised if using benzodiazepines alone, because of the risk of further behavioural disinhibition. Chlorpromazine should be avoided, as it may increase the anticholinergic effects of phencyclidine and worsen the patient's mental state. Hypertensive crisis should be treated with antihypertensive agents such as phentolamine. Respiratory function must be carefully monitored because excessive secretions may compromise the airway in an unconscious patient. Treatment of ketamine intoxication is also supportive and symptomatic. Antipsychotic drugs have little effect on the psychiatric symptoms, but benzodiazepines are reported to be helpful.

Heavy, prolonged use of ketamine can cause progressive inflammation and scarring of the bladder. This results in urinary frequency, dysuria, and haematuria. Prolonged ketamine misuse is also associated with cognitive impairment particularly in visual recognition and spatial working memory. Although serious dependence on ketamine is relatively rare, regular users can experience tolerance to its psychological effects; this can lead to compulsive and harmful use (Winstock and Rucker, 2020).

Nitrous oxide is mentioned here because it also produces transient mood elevation and dissociative experiences through antagonism of NMDA receptors. Nitrous oxide retains a place in clinical medicine as an analgesic and anaesthetic. However, its recreational use in the United Kingdom is restricted by the Psychoactive Substances Act (2016), where possession (apart from approved purposes) has become illegal. Nevertheless, nitrous oxide has continue to be commonly used by young people with 9% of people between 16–24 years of age reporting recreational use in 2020 (Sumnall, 2022).

While the harms of occasional nitrous oxide use are regarded as slight, deaths have been reported through cardiac arrhythmias and asphyxiation. Heavy use of nitrous oxide can lead to functional vitamin B12 deficiency accompanied by myeloneuropathy and peripheral neuropathy. In some of these cases vitamin B12

blood levels may lie within the normal range; however, homocysteine levels are usually raised (Lan et al., 2019).

Volatile substances (solvents, inhalants)

The misuse of *volatile substances* (also known as solvents) is not new, but public concern about widespread misuse first became apparent in the United States in the 1950s. Similar concerns emerged in the United Kingdom in the early 1970s. Although public interest has waned since then, there is a continuing significant level of volatile substance misuse, particularly among adolescents. The pharmacological actions of volatile substances in the central nervous system are unclear, but they may increase the fluidity of neuronal cell membranes and are active at many different kinds of ion channel. This seems to result in increases in brain GABA function and decreases in NMDA receptor activity (Cruz and Bowen, 2021).

Epidemiology. Volatile substance misuse is a worldwide problem. The Psychiatric Morbidity Survey (McManus et al., 2016) reported that in households in England the highest lifetime use of volatile substances was in the 25–34 year age group (2.7%). However, in the past year, use was highest in those aged 16–24 years (0.3%). The survey did not include teenagers under the age of 16 years and a survey of secondary school pupils England in 2018 estimated that 4.2% had used volatile substances in the last year (NHS, 2018). *Amyl nitrite* ('poppers') appears to be more commonly used (see Box 20.14), but has a different pattern of use and is not discussed here. Volatile substance misuse is highly prevalent among the very young homeless populations in South American countries.

Volatile substance use occurs mainly in young men, and is more common in individuals from lower socioeconomic groups. Most of the young people who are known to use volatile substances do so as a group activity, and only about 5% are solitary users. There is some evidence that a subgroup of volatile substance users have antisocial personality disorder and are likely to use and misuse multiple substances. However, the epidemiological data suggest that most individuals who use volatile substances do so only a few times and then abandon the practice (Ives, 2009).

Substances used and methods of use. The volatile substances used are mainly solvents and adhesives (hence the name 'glue-sniffing'), but also include many other substances, such as petrol, cleaning fluid, aerosols of all kinds, agents used in fire extinguishers, and butane. Toluene and acetone are frequently used. In this chapter the term 'volatile substance' is used to denote all of these various substances. The methods of ingestion depend on the substance, and include inhalation from the tops of bottles, beer cans, cloths held over the mouth,

plastic bags, and sprays. Volatile substance use may be associated with the taking of other illicit drugs, or with tobacco or alcohol consumption, which can be heavy.

Clinical effects. The clinical effects of volatile substances are similar to those of alcohol. The central nervous system is first stimulated and then depressed. The stages of intoxication are similar to those of alcohol, namely euphoria, blurring of vision, slurring of speech, incoordination, staggering gait, nausea, vomiting, and coma. Compared with alcohol intoxication, volatile substance intoxication develops and wanes rapidly (within a few minutes, or up to two hours). There is early disorientation, and 40% of cases may develop hallucinations, which are mainly visual and often frightening. This combination of symptoms may lead to serious accidents.

Adverse effects. Volatile substance misuse has many severe adverse effects, of which the most serious is sudden death. These fatalities occur during acute intoxication, and about 100 such deaths occurred annually in the United Kingdom in the 1990s, although there has been a decline subsequently, with 38 deaths being recorded in 2020. Around 50% of the deaths are due to the direct toxic effects of the volatile substance, particularly cardiac arrhythmias and respiratory depression. The rest are due to trauma, asphyxia (due to plastic bag over the head), or inhalation of stomach contents.

Chronic users may show evidence of neurotoxic effects, and severe and disabling peripheral neuropathy has been described in teenager misusers. Other neurological adverse effects, particularly associated with toluene, include impaired cerebellar function, encephalitis, and dementia. Volatile substance misuse can also damage other organs, including the liver, kidney, heart, and lungs. Gastrointestinal symptoms include nausea, vomiting, and haematemesis.

Tolerance and dependence. Dependence can develop if use is regular, but physical withdrawal symptoms are unusual. When such symptoms occur, they usually consist of sleep disturbance, irritability, nausea, tachycardia, and, rarely, hallucinations and delusions. With sustained use over a period of 6–12 months, tolerance can develop.

Diagnosis. The diagnosis of acute volatile substance intoxication is suggested by several features, including glue on the hands, face, or clothes, a chemical smell on the breath, rapid onset and waning of intoxication, and disorientation in time and space. Chronic misuse is diagnosed mainly on the basis of an admitted history of habitual consumption, increasing tolerance, and dependence. A suggestive feature is a facial rash ('glue-sniffer's rash') caused by repeated inhalation from a bag.

Treatment. As noted above, for many users, experimentation with volatile substances is a temporary phase

Table 20.5 Some examples of substances used as ‘recreational’ drugs

Drug	Pharmacology	Intoxication syndrome
Spice (‘Genie’, ‘Yucatan Fire’)	Herbal mixture with addition of synthetic cannabinoids.	Changes in mood and perception, cognitive impairment, psychosis, scleral injection, hunger.
Salvia (‘Magic Mint’, ‘Mystic Sage’)	Plant extract from <i>Salvia divinorum</i> . Active substance is salvinorin A, an opiate kappa receptor agonist.	Alterations in mood and somatic sensations. Striking changes in perception of external reality, confusion, amnesia, anxiety.
Mephedrone, (‘Meow-Meow’), MDPV (‘Vanilla Sky’, ‘Purple Wave’)	Synthetic cathinone derivatives, related to active substance in <i>khat</i> . Cause release of monoamine neurotransmitters, including dopamine.	Euphoria, increased sociability, agitation, psychosis, seizures, tachycardia, chest pain, hyponatraemia.
Gamma hydroxybutyrate (‘GHB’, Fantasy’)	Increase in dopamine, opioids, and GABA.	Increased sociability, disinhibition, amnesia, sedation, coma.

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that does not appear to lead to persistent misuse or dependence. Advice and support may be sufficient for such individuals. However, a significant subgroup of those who misuse volatile substances also misuse other substances, such as alcohol and opioids. Such individuals are more likely to have an antisocial personality disorder and to have experienced a chaotic and abusive family life. Treatment of this group is difficult, and a full range of psychological and social treatments is likely to be needed (see above). There is no specific pharmacotherapy for volatile substance misuse, but associated psychiatric disorders such as depression may require treatment in their own right.

Prevention. Prevention of volatile substance misuse may best be directed at the large numbers of young people who experiment with volatile substances as a result of curiosity or peer pressure. Policies include the restriction of sales of volatile substances to children and adolescents. Education, particularly concerning the risk of severe injury and death (which can, of course, occur in occasional or first-time users), seems to be worthwhile. Wider social measures, such as the provision of improved recreational facilities, have also been advocated.

For a review of volatile substance misuse, see Cruz and Bowen (2021).

Legal highs

The phrase ‘legal high’ has been used to refer to a substance with psychotropic properties whose sale or use

is not banned (or not yet banned) by legislation. The recreational use of such substances appears to be increasing in the United Kingdom and Europe. In 2018, 1.4% of secondary school children in England had taken a legal high in the previous year (NHS, 2018).

Because the development and uptake of legal highs evolves quickly, with the aim of circumventing usual regulatory channels, their effects, particularly in terms of intoxication, may not be well recognized. Generally, legal highs are designed to mimic effects of more traditional drugs of abuse. As one substance is banned, manufacturers search for replacements with a similar pharmacology that have not yet been caught in the regulatory framework. Because of this, any description of available compounds is always likely to be dated, but a few examples are given in Table 20.5. The passage of the UK Psychoactive Substances Act (2016) was designed to lessen the use of legal highs by making it an offence to produce or supply psychoactive substances for human consumption. There are exceptions for approved medical and scientific use.

Whether the Act has succeeded in its aims has been debated. There seems to have a reduction in the use of legal highs in the general population but not in children or the homeless. The use of synthetic cannabinoids continues to have a high prevalence in prisons. There is also the problem that legal highs are now supplied by criminal networks or through the internet rather than more visible retail outlets (Iacobucci, 2018).

Further reading

- Geddes, J. R., Andreasen, N. C., & Goodwin, G. M. (Eds.). (2020). Section 8: Substance use disorders. In *The New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press.
- National Institute for Health and Care Excellence (NICE). (2011a). Alcohol-use disorders: Diagnosis, assessment and management of harmful drinking (high-risk drinking) and alcohol dependence. Clinical guideline CG115.<https://www.nice.org.uk/guidance/cg115>
- Robson, P. (2009). *Forbidden Drugs*. Oxford University Press. (A very readable account of illegal drugs, the reasons for their use, and treatment of misuse.)

CHAPTER 21

Suicide and self-harm

Introduction 589

Suicide 590

The assessment of suicidal risk 596

The management of suicidal patients 597

Reducing suicide rates 598

Non-fatal self-harm 599

The management of self-harm 603

Introduction

Suicide is defined as a death caused by intentional self-directed injury, with more than 700,000 individuals estimated to die from suicide each year across the globe (Naghavi, 2019; Pathirathna et al., 2022). Suicide and self-harm rates vary systematically by age, sex, and means of suicide or self-harm. The contributing risk and protective factors are complex, with consideration needed across the range of individual, family, community, and societal influences. Suicide in adults is among the ten leading causes of death in most countries for which information is available and the third leading cause of death in young people aged 10–19 years (see Figure 21.1) (Naghavi, 2019; Torok et al., 2019). Therefore, although suicide itself is not a psychiatric illness or disease, suicidal behaviour is an important public health concern and mental health conditions are associated with increased risk of suicide but other social determinants of health play a role (San Too et al., 2019).

There are indications that there have been large decreases in overall global rates of suicide fatalities, with some estimates placing this at a reduction of one third since the early 2000s. This trend is not reflected everywhere with the incidence of suicide varying more than tenfold between countries (Turecki et al., 2019). The subject is important to all healthcare practitioners who may encounter people who are at risk of dying by suicide, and who may also at times be involved in helping family members or others after a suicide. The importance of the

subject is reflected in many national and international policy initiatives for reduction of suicide rates from school and workplace initiatives to targeting improved access to support for higher-risk populations and legislation to limit access to means (Hofstra et al., 2020).

For every suicide it is estimated that more than 20 non-fatal episodes of self-harm occur (Turecki et al., 2019). This ratio varies from country to country, often depending on the lethality of the most commonly used suicide method (Lim et al., 2021). Different definitions of self-harm are often used; therefore, to facilitate comparisons it can be useful to distinguish between self-injurious behaviours with inferred or actual intent to die and non-suicidal self-injury without a current intent to die. Depression, substance misuse, and other mental health problems are estimated to be present in half of all suicides and are also more common in people who self-harm. Furthermore, the rate of suicide in the year following an episode of self-harm is substantially higher than that of the general population (Geulayov et al., 2019). The rate of suicide has been shown to be raised following hospital attendance for self-harm, especially in the year following discharge. It is raised in those with either self-injury or self-poisoning, and for attempted hanging. For these reasons, psychiatrists need to be particularly well informed about the nature of suicidal behaviour, and about strategies aimed at their prevention. For an overview of relevant aspects of suicide and suicide risk, see Turecki et al. (2019).

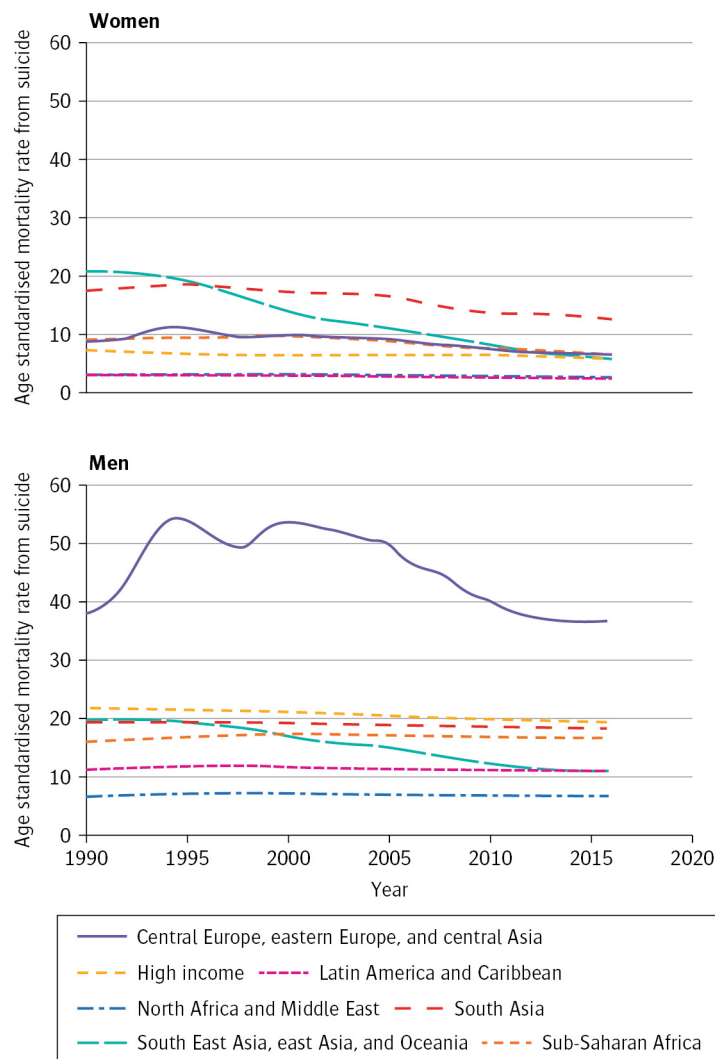


Figure 21.1 Global suicide rates, by country income level and gender (1970–2021).

Citation: Global Suicide Rates by Income Level and Gender (1970–2020). Visualization generated by ChatGPT, based on data trends from the World Health Organization (WHO) and the Institute for Health Metrics and Evaluation (IHME), 2024.

Suicide

The act of suicide

Suicide has been defined as an act with a fatal outcome, deliberately initiated and performed in the knowledge or expectation of its fatal outcome. Overall, approximately 8% of suicidal acts in the United States have been estimated to be fatal, with the highest rates for males and for those over the age of 65 years (Conner et al., 2019). Different methods are used by people who

die by suicide. In England and Wales, according to the Office for National Statistics in 2022, hanging, strangulation, and suffocation (all grouped together) were the most commonly used method for suicide for both males and females, accounting for 58% of all suicides (2022). The second most common method was by poisoning (drug overdose) and accounted for 21% of all suicides; of note, poisoning is the most common method of all self-harm (both fatal and non-fatal) (Conner et al., 2019).

In the United States, firearms are the most frequently used suicide method, much higher than most other Western countries, and when used are the most lethal of all methods (Conner et al., 2019; Pritchard et al., 2020).

Suicidal ideation is considerably more frequent than suicide attempts, and may be active (i.e. with a plan) or passive (i.e. primarily with thoughts about wishing to die). Those with active suicidal ideation are more likely to die by suicide (Turecki et al., 2019). Precautions against discovery are often taken, for example, choosing an isolated location or a time when no one is around. However, in many cases one or more individuals might have received some communication warning them that this is something the individual is considering. Incidence estimates vary depending on the definitions and wording used in the study, as definitions can range from thoughts that life is not worth living to making concrete plans to end one's life. Suicidal ideas have been expressed by more than two thirds of those who die by suicide, and clear suicidal intent by more than one third. The last clinical contact for the majority of people who die by suicide is either in an emergency department or in primary care. Data from the National Confidential Inquiry into Suicide and Safety in Mental Health (Appleby et al., 2019) suggest that 25–29% of people who die by suicide have been in recent contact with mental health services, and half have been in contact with mental health, emergency care, or primary care services.

The epidemiology of suicide

The accuracy of the statistics

Accurate statistics about suicide are difficult to obtain because information about the exact cause of a sudden death is not always available. Completed suicides can be reported as such on death certificates; a 1989 study estimated the sensitivity of suicide on death certificates to be between 55% and 99%, and hypothesized that validity was highly variable by location (Swain et al., 2019). Factors that influence the accuracy of suicide capture on death certificates include method of suicide, patient history, and differential reporting of 'undetermined death'. For example, in England and Wales, official figures depend on the verdicts reached in coroners' courts. A verdict of suicide is recorded by a coroner only if there is clear evidence that the injury was self-inflicted and that the deceased intended to kill themselves. If there is any doubt about either point, a verdict of accidental death or an open verdict is often recorded. Open verdicts were previously more commonly used to describe 'events of undetermined intent', often recorded when the method of self-harm was less active (e.g. drowning

compared with hanging) and when the deceased was younger. However, studies examining 'open' verdicts have estimated that 55–86% were suicides (Swain et al., 2019). It is therefore accepted that official statistics underestimate the true rates of suicide.

Around the world, awareness of the different systems used in countries to capture suicide rates can help make sense of cross-country comparisons alongside measurement of social, economic, cultural, and environmental factors. For example, consideration needs to be given to the rates of depression and access to treatment, socioeconomic pressures, restrictions on the means of suicide, and the manner in which suicide might be reported and documented. Belgium has doctor-assisted suicide laws that might be impacting on their relatively high rates while proposed low rates in certain regions in India might be because of the social stigma attached to suicide, especially among females (Arya et al., 2021; Axe et al., 2022). Attempts to overcome these problems include highlighting the categories where most 'hidden suicide' might be recorded: ill-defined or unknown causes of death, events of undetermined intent (EUIs), and accidental deaths (Snowdon and Choi, 2020).

Changes in suicide rates

Temporal trends in suicide rates vary by region, with global rates falling, as estimated by the World Health Organization (WHO), primarily driven by a reduction of rates in China; however, this drop has masked increases in countries such as the United States and Brazil where rates rose by 35% between 2000 and 2016 (Turecki et al., 2019). In general, key influences on secular trends in suicide include periods of economic recession (which are generally associated with increases), changes in access to commonly used, high-lethality suicide methods, periods of war (which are generally associated with falls in suicide), and media reporting of celebrity suicides (which may lead to transient increases in suicide rates) (Turecki et al., 2019). The impact of the Covid-19 pandemic on suicide rates is as yet unclear (Appleby, 2021). Early evidence from the lockdown periods showed no consistent evidence of a rise in suicide rates but many evolving risk factors subsequent to the lockdown periods, including economic effects and rise in community distress might explain an increased frequency of suicidal thoughts in certain higher-risk groups (John et al., 2020).

Geographic variations

The incidence of suicide varies more than tenfold between countries, with incidence relatively low in the Middle East and some South and Central American

countries and high in Eastern Europe, Russia, India, and South Korea (Turecki et al., 2019). The largest number of suicides occur in low- and middle-income countries, where treatment options are limited and the availability and cultural preference for high-lethality suicide methods can influence suicide rates (for example, with toxic pesticides more lethal than psychotropic medications). Almost half of all suicides in the globe take place in India and China (Turecki et al., 2019), reflecting their larger populations while the countries with the highest suicide rates per head of population are Lesotho, Guyana, Eswatini, South Korea, and Kiribati (World Health Organization, 2022). The country with the highest suicide rates in Europe is Belgium. For 15- to 19-year-olds, the highest suicide rates have been observed in New Zealand, Finland, Ireland, Guyana, and Sri Lanka (Glenn et al., 2020).

Demographic characteristics

Globally, suicide is about three times as common in men as in women, although the rates of suicide attempts is higher in women. This could reflect a male preference for more lethal methods and the fact that in many cultures, males might find it harder to access available supports. The highest rates of suicide in both men and women are in the older adult population, often associated with physical illness and disability (Beghi et al., 2021). Suicide rates are lower among the married than among those who have never been married, and increase progressively through widowers, widows, and the divorced. Rates are higher in the unemployed and those that are incarcerated, especially among those on remand (Zhong et al., 2021). Risk factors for suicidal thoughts and behaviours are listed in Box 21.1.

Box 21.1 Risk factors for suicidal thoughts and behaviours

Individual risk factors

- Biological
 - Male sex (completion); female (attempts)
 - Serotonin dysfunction
 - Family history of suicidal behaviour
- Sociodemographic
 - Gender (non-binary and transgender status)
 - Lesbian, gay, bisexual, or other sexual orientation minority identity
 - Older age
 - High-risk professions
 - Firearm ownership
 - Incarceration
 - Job/financial problems or loss
- Psychiatric and/or other illness
 - Psychiatric disorder (particularly major depressive disorder, bipolar disorder, PTSD, schizophrenia)
 - Neurodiverse individuals (autism spectrum disorder and ADHD)
 - Emotionally unstable personality disorders (EUPDs)
 - Substance and alcohol use/abuse
 - Other serious health conditions and/or functional impairment (including chronic pain)
- Psychological
 - Previous suicide attempt
 - Current suicidal thinking/suicidal intent/plans
 - Access to/presence of lethal means
 - Preparatory behaviours

- Hopelessness or feeling of burdensomeness
- Perfectionism
- Rejection/thwarted belonging
- Impulsive or aggressive tendencies

Interpersonal risk factors

- Bullying
- Suicide in family member/close contact
- Isolation and/or loss of social ties
- High conflict, abusive or violent relationships

Community risk factors

- Barriers to healthcare access
- Suicide cluster in community
- Poor social integration
- Exposure to community violence
- Discrimination and racism

Societal risk factors

- Stigma associated with psychiatric illness and help seeking
- Access to lethal means of suicide
- Exposure to accounts of suicide
- Poverty

Adapted from Cramer, R. J., & Kapusta, N. D. (2017). A social-ecological framework of theory, assessment, and prevention of suicide. *Front Psychol*, 8, 1756. Copyright (2017), Frontiers Media S.A. Reproduced under the Creative Commons Attribution License (CC BY).

Suicide is a complex phenomenon determined by genetic, cultural, and behavioural factors and investigations into the causes of suicide have increased dramatically over the last decades. Studies are trying to better understand and address suicidal behaviour from both an individual and a population perspective. Suicide is rarely caused by a single circumstance or event; a range of factors need to be considered. Understanding genetic, epigenetic, and neurobiology as well as information about the health and wellbeing of the deceased at the time of suicide are all relevant. Prospective studies of suicide are difficult to conduct because of its relative rarity leading to a reliance on epidemiological studies to examine associations between social and demographic factors and rates of suicide in different populations at different times to inform our understanding of social circumstances.

Use has been made of ‘*the psychological autopsy*’ where retrospective case reviews have pieced together the circumstances that surround a suicide by examining records and interviewing doctors, relatives, and friends who knew the deceased well. The most prominent theory of pathways to suicidal behaviour is that of the stress–diathesis model, which depicts suicidal behaviour as a consequence of the interactions between acute stressors or ‘proximal’ risk factors and a diathesis or ‘distal’ factors (Mann and Rizk, 2020). Examples of acute stressors include exacerbations of psychiatric illnesses or financial and interpersonal problems. The diathesis relates to a set of suicide-related traits that moderates the likelihood of suicidal behaviour in response to stressors. Some of the risk factors for suicide and suicidal behaviours are discussed in more detail below.

Biological factors

Studies suggest that there is a heritable component to suicidal behaviour, with national registry-based studies and twin and adoption studies all suggesting a heritability of probably around 17–36% (once the heritability of comorbid psychiatric disorders is controlled for) (Turecki et al., 2019). No single gene or group of genes has been identified. Of note, the offspring of individuals who have attempted suicide have a fivefold greater risk of attempting suicide than the general population, and suicide risk is doubled if the individual has lost a parent to suicide. For a comprehensive summary of the biological aspects of suicidal behaviour, see Mann and Rizk (2020).

Gender and sexuality

Approximately 4% of all patient suicides in the United Kingdom over a ten-year period were by patients who identified as lesbian, gay, or bisexual, and 1% were in a

trans group. The trans patients were younger and over half lived alone and the majority had experienced childhood abuse (Appleby et al., 2023). A population cohort of over 6 million Danish-born individuals over four decades showed that transgender individuals had higher rates of suicide attempt (adjusted incidence rate ratio of 7.7) and suicide mortality (aIRR 3.5) compared with non-transgender individuals (Erlangsen et al., 2023).

Psychiatric disorder

The most consistent finding of studies of proximal factors is that the large majority (90% in North America, 30–70% in East Asia) of those who die from suicide have some form of psychiatric disorder at the time of death (Turecki et al., 2019). The increased risk of suicide is one of the main reasons why people with psychiatric disorders have a 15–20 years’ shorter life expectancy, compared with the general population (Nordentoft et al., 2020). In addition, risk of suicide varies according to the level of mental healthcare received, with those admitted to inpatient care having a 44-fold higher risk than the general population. The level of risk is greatest in the first week of admission and then the first days and months after discharge. The most common conditions implicated include:

- **Major depressive disorder.** The number, duration, and intensity of episodes for major depressive disorder are important to consider for risk of suicidal behaviour. About 4% of individuals with depression will die by suicide and those who do are more likely to be male, have a family history of psychiatric disorder, have a past history of attempted suicide, and to have experienced a sense of hopelessness (Nordentoft et al., 2020).
- **Bipolar disorder.** In people with bipolar disorder, higher risk for suicide attempts are in early-stage illness (estimated 10–15% higher), have mixed-state episodes, and experience irritable dysphoric states (Turecki et al., 2019).
- **Schizophrenia.** In a meta-analysis, the lifetime suicide risk in schizophrenia was 5.6% with symptoms of psychosis conferring additional risk alongside comorbid depressive disorders, substance misuse, and agitation and motor restlessness (Nordentoft et al., 2020) (see Chapter 11).
- **Substance misuse.** Substance use or misuse is found in a large proportion of people who die by suicide, with alcohol misuse observed in up to 40% and use of illicit substances in up to 25%, mainly sedative hypnotic drugs and cannabis (Turecki et al., 2019). It has been proposed that high impulsivity is a likely to be a key mediator in substance use associated with suicidal

behaviours. Among those who are alcohol dependent, suicide is more likely when the person is male, older, has a long history of drinking, and has a history of depression and of previous suicidal attempts. Suicide risk is also increased among those whose drinking has caused physical complications, relationship problems, difficulties at work, or arrests for drunkenness offences.

- Suicide risk is also increased in those with a *personality disorder* (see Chapter 15); *Eating disorders* (see Chapter 13), and individuals experiencing *chronic pain*, *severe somatic disorders*, and other chronic, terminal, or severe health conditions. Suicides are increasing in patients given a diagnosis of personality disorder, particularly among women; historically, they have been excluded from services and better models of safe and compassionate care are needed (Appleby et al., 2023).

For a review, see Nordentoft et al. (2020).

Psychological risk factors

The personality traits most robustly associated with suicidal behaviour are anxiety and impulsive-aggressive traits and might explain the higher rates in attention-deficit hyperactivity disorder, borderline personality disorder, oppositional defiant disorder, and conduct disorder. These traits might also explain the familial aggregation of suicidal behaviour (Turecki et al., 2019). There is some evidence that sensitivity to social rejection, defeat, and entrapment—traits sometimes found in perfectionism—is also associated with higher rates of suicidal behaviour. Diminished problem solving, impaired memory, and decreased positive future thinking have been associated with suicidal behaviour (Richard-Devantoy et al., 2015). Furthermore, there is a fivefold increased risk of suicide attempts in high-achieving students when compared to low-achieving students.

Psychological factors that contribute to suicide have been derived mainly by extrapolation from studies of non-fatal self-harm. Research has indicated that hopelessness, impulsivity, dichotomous thinking, cognitive constriction, problem-solving deficits, and overgeneralized autobiographical memory are all associated with suicidal behaviour. The construct on ‘unbearable psychological pain’ is believed to be a key driver as well as knowledge of and comfort with potentially lethal means (O’Connor and Nock, 2014).

Interpersonal risk factors and early life adversity

Early-life experiences, often falling under the description ‘adverse childhood experiences’ (ACEs), including

neglect, sexual abuse, and physical abuse, are strongly associated with suicidal behaviour later in life (Turecki et al., 2019). Hypotheses as to why this occurs include modification of the stress response and specifically changes in the hypothalamic–pituitary–adrenal (HPA) axis (see Chapter 7).

Societal risk factors

The most important social and economic factors associated with suicide include job loss, economic turmoil, and recent changes in socioeconomic condition. These were first described by Emile Durkheim in the nineteenth century showing how social factors, especially isolation and marginalization, were central to understanding suicide rates (Mueller et al., 2021). There is evidence that the economic recession of 2008, with its following austerity and rising unemployment, has been associated with a rise in the male suicide rate even in relatively protected countries such as the United Kingdom (Barr et al., 2012), with reports of even greater effects in those most affected, such as Greece.

Another social factor that appears to affect rates of suicide is *media coverage of suicide*. Suicide and attempted suicide rates were shown to increase after television programmes and films depicting suicide (Nordentoft et al., 2020). Media guidelines have been developed recommending that reports of suicide should not list details regarding method and should avoid depicting the event as a heroic act. Understanding the role of social media in influencing suicidal behaviour, especially in younger populations, is an area of growing research as there is some evidence of an independent association between problematic use of social media and the internet and suicidal attempts in young people (Sedgwick et al., 2019).

Special groups

Suicide among those in contact with psychiatric services

Given the strong association between mental disorder and suicide, it is not surprising that many people who die by suicide are in contact with psychiatric services. The National Confidential Inquiry into Suicide and Safety in Mental Health, which was set up in 1997, collects information on all people who die by suicide while in contact with the mental health services (Appleby et al., 2023). Between 2010 and 2020, there were 18,403 suicide deaths in the United Kingdom by patients (i.e. people in contact with mental health services within 12 months of suicide), an average of 1,673 deaths per

year, comprising 27% of all suicides (Appleby et al., 2019). Of patients who died over this period of time, approximately half were in post-discharge care. The most common non-acute care settings were community health services and alcohol and drug services.

Their findings along with the UK National Institute for Health and Care Excellence (NICE) guidance (NICE, 2019) has led to the following suggestions to try and reduce the numbers of suicide attempts:

- *Support patients intensively* during the first few weeks after discharge from hospital and help tackle loneliness and comorbid alcohol and drug misuse. Involve families where possible and actively try and re-establish contact if there is loss of contact.
- *Improve the inpatient experience* and enhance the therapeutic relationship to reduce risk. Ensure inpatient areas are safe with removal of all accessible ligature points.
- *Multi-agency planning* and, for those under 18 years of age, more links with educational settings. Ensure good management of any comorbid disorders, such as anxiety and autism spectrum disorders.
- *Suicide-related internet use.* Clinicians need to be aware that suicide-related internet use is a feature of suicide by mental health patients of all ages. It can take a number of forms, including information promoting suicide methods. Enquiry about exposure to internet risks should be a routine part of risk assessment.

Assisted or 'rational' suicide

Despite the findings reviewed above, not all deaths by suicide involve someone with a mental illness. For example, there is an increasing number of countries (e.g. Switzerland, the Netherlands, Belgium, Spain) that have made changes in their law to allow 'assisted dying' for those with a long-term illness to take their own life with help from friends, family, or medical practitioners. An established and well-documented decision is generally required, together with confirmation that the individual does have capacity. The law is under debate in many countries, including in the United Kingdom. Lack of personal autonomy and perceived dignity due to infirmity and chronic pain are the most common reasons given by those who pursue this path.

High-risk occupational groups

A number of high-risk occupational groups have been identified. In both the United Kingdom and the United States, men working in the low-skilled construction industry have the highest rates of suicide, three times

the UK national average. In the United States, over half of the workers in construction and extraction who died by suicide had substance misuse or mental health problems in the past three years (Peterson et al., 2020). For males working in skilled trades, the highest risk was among building trades, particularly plasterers, painters, and decorators, with more than double the risk of suicide than the male national average (Office for National Statistics, 2017). For females the highest-risk profession was for those working in artistic, literary, and media occupations. For healthcare professionals, females carry a particularly high risk, lying 24% above the national average, primarily influenced by the high suicide risk among female nurses. Male and female carers had a risk of suicide that was twice the national average. The highest paid occupational groups, including managers, directors, and senior officials, had the lowest risk of suicide, for example, corporate managers and directors had a 70%-lower suicide rate for both sexes.

Suicide pacts

The majority of people who die by suicide do so alone. In suicide pacts, which are rare events, two (or occasionally more) people mutually agree that at the same time each will take their own life. Such pacts that actually result in death are uncommon. Social media and digital communication has been shown to play a role in facilitating suicide pacts and the planning process (Pourmand et al., 2019). Suicide pacts have to be distinguished from cases where murder is followed by suicide (especially when the first person dies but the second is revived), or where one person aids another person's suicide without intending to die himself or herself.

The psychological causes of these pacts are not known with certainty but one small study showed how psychiatric disorders were the main reason to make a suicide pact (Galante et al., 2022). Usually the two people have a particularly close relationship but are socially isolated from others. Often a dominant partner initiates the suicide (Brown and Barraclough, 1997). *Mass suicide* is occasionally reported. For example, 918 followers of the People's Temple cult, including 276 children, died at Jonestown, Guyana in 1978; 39 members of the Heaven's Gate cult in California died in 1997; and an estimated 400 people, who had been ordered to starve themselves to death, died in the Shakahola massacre in Kenya in 2023. These tragic events are generally initiated by a charismatic leader who has strong convictions and is sometimes deluded. Sometimes there is evidence to suggest murder followed by suicide within such groups.

The assessment of suicidal risk

General issues

Every doctor should be able to assess the risk of suicide. There are two requirements. The first is a willingness to make direct but tactful enquiries about the patient's intentions. There is no convincing evidence that asking a patient about suicidal inclinations makes suicidal behaviour more likely (Polihronis et al., 2022). On the contrary, someone who has already thought of suicide is likely to feel better understood when the issue is raised by the healthcare professional that they are seeing.

The second requirement is to be alert to factors that place the individual at higher risk of suicide. However, prediction has a low sensitivity and specificity as, even if the risk is correctly assessed as high, it is difficult to predict when the suicide will take place. NICE guidance has been published on assessment of self-harm which is relevant for assessing suicidal attempts (NICE, 2022d).

Assessment of risk

The most obvious warning sign is a direct statement of intent. It cannot be repeated too often that there is no truth in the idea that people who talk of suicide do not do it. On the contrary, two thirds of those who die by suicide have told someone of their intentions. However, a difficulty arises with people who talk repeatedly of suicide. In time their statements may no longer be taken seriously, although some who repeatedly make such threats do eventually kill themselves. Just before the act, there may be a subtle change in their way of talking about dying, sometimes in the form of oblique hints instead of former more open statements.

Risk is assessed further by considering factors that have been shown to be associated with suicide. Examples of factors that point to greater risk include:

- marked hopelessness
- a history of *previous suicide attempts*: around 40–60% of those who die by suicide have made a previous attempt
- social isolation
- older age
- *depressive disorder*, especially with severe mood change with insomnia, anorexia, and weight loss
- *alcohol dependence*, especially with physical complications or severe social damage

- drug dependence
- *schizophrenia*, especially among young men with recurrent severe illness, depression, intellectual deterioration, or a history of a previous suicide attempt
- chronic painful illness.

Conducting the history

When the risk factors for suicide have been assessed, the rest of the history should be evaluated. The interview should be conducted in an unhurried and sympathetic way that allows the patient to admit any despair or self-destructive intentions. It is usually appropriate to start by asking about current problems and the patient's reaction to them. Enquiries should cover losses, both personal (e.g. bereavement or divorce) and financial, as well as loss of status. A detailed history of social media use and information accessed from online sources is important. Information about conflict with other people and social isolation should also be elicited. Physical illness should always be enquired after, particularly any painful or terminal conditions in the elderly.

When assessing previous personality, consider that the person's self-description may be coloured by depression, and usual assessment principles should be followed. An informant can help better understand changes in mood, impulsive or aggressive tendencies, and possibly broader attitudes towards death.

Mental state examination

The assessment of mood should be particularly thorough, and cognitive function must not be overlooked. The interviewer should then assess suicidal intent. It is usually appropriate to begin by asking whether the patient thinks that life is too much for them, or whether they no longer want to go on and how they feel about the future, to garner an understanding of whether they are feeling hopeless. These questions can lead to more direct enquiries about thoughts of suicide, specific plans, and preparations for suicide, such as hoarding tablets. It is important to remember that severely depressed patients occasionally have homicidal ideas—they may believe that it would be an act of mercy to kill other people, often their partner or a child, to spare them intolerable suffering. Such homicidal ideas should always be taken extremely seriously.

Assessment of risk among inpatients

The suicide of a hospital inpatient is always of particular concern, and it is important to be able to identify those

at risk. Unfortunately, the general guiding risk factors noted above are present in so many of those admitted to hospital that they differentiate poorly between degrees of risk.

The management of suicidal patients

General issues

Having assessed the suicidal risk, the clinician should make a treatment plan and try to persuade the patient to accept it. The first step is to decide whether the patient should be admitted to hospital or treated as an outpatient or day patient. This decision depends on the intensity of the suicidal intention, the severity of any associated psychiatric illness, and the availability of social support outside hospital. If outpatient treatment is chosen, the patient should be given advice as to how to contact any urgent help—for example, from a crisis service—so that they can obtain aid if they feel worse. Frustrated attempts to find help can be the last straw for a patient with suicidal inclinations.

If the immediate risk of suicide is judged to be high, inpatient care is likely to be required unless there is an effective crisis management team available, or reliable relatives who wish to care for the patient themselves, understand their responsibilities, and are able to fulfil them. Such a decision requires a thorough knowledge of the patient and their problems, as well as of the relatives. If hospital treatment is indicated and the patient refuses it, admission under a compulsory order will likely be necessary. In most countries imminent suicide risk is sufficient rationale for detention to a psychiatric inpatient facility for observation, but the duration and conditions will vary. Readers should be aware of and follow the legal requirements of the jurisdiction in which they work.

Management in the community

The management of patients who have been identified as being at risk of suicide but do not require admission involves continuing assessment of the suicidal risk, and agreed plans for appropriate treatment and support (see Box 21.2). Where they are available and the patient consents to this, relatives or other carers should be involved. The key worker should liaise closely with other members of the community team to ensure that there will be a prompt and appropriate response if the patient or the carers ask for additional help. If medication is

required—for example, to treat a depressive disorder—the drug that is the least dangerous in overdose should be considered. The choice should be discussed with the general practitioner, and small quantities often prescribed. When appropriate, medication should be kept safely by the carer. Both patients and carers need to know how to obtain immediate help in an emergency.

Management in hospital

The primary needs are to provide a safe environment, to treat any underlying psychiatric disorder, and try and prevent patients from harming themselves—hence the importance of appropriate staffing. A policy (see Box 21.3) should be agreed with all staff members when the patient is admitted.

Ward arrangements. Ward design should minimize the availability of means of self-harm. This includes

Box 21.2 Care of the potentially suicidal patient in the community

- Full assessment of patient and proposed carers
- Organization of social support to counteract periods of loneliness and isolation
- Regular review of the suicide risk and the arrangements
- Safe psychiatric medications given in adequate dosage using less-toxic drugs and monitoring side effects to ensure prescribed medication is acceptable and thus facilitate adherence
- Small quantity prescriptions—this needs to be balanced with risk of not requesting repeat prescriptions and so close monitoring and support needed
- Involvement of relatives or friends/companions in the safe storage of tablets
- Arrangements for immediate access to extra help for patient and carers.

Box 21.3 Care of the suicidal patient in hospital

General requirements

Safe ward environment.
An adequate number of well-trained staff.
Good working relationships among staff, and between staff and patients.
Agreed policies for the observation, assessment, and review of patients.

On admission

Assess risk.
Agree the level of observation.
Remove any objects that might be used for suicide.
Discuss and agree plans with the patient.
Agree a policy for visitors (number, duration of visit, and what they need to know).

During admission

Regular review of risk and plans
Agreed plans for the level of supervision.
Clear communication of assessments and plans between staff, especially when shifts change.
Agree action to be taken if the patient leaves the ward without notice or permission.

At discharge

Agree date and plan for aftercare in advance of discharge.
Discuss and agree the plan with the patient and those involved in their care.
Prescribe in adequate but non-dangerous amounts.
Arrange follow-up, and agree action to be taken if the patient does not attend.

preventing access to open windows and other places where jumping could lead to serious injury or death, removing potential ligature points from which hanging could take place (e.g. by boxing in pipework), preventing access to ward areas in which self-injury would be easier to enact, and removing potentially dangerous personal possessions such as razors and belts. If the risk is high, special staffing arrangements may be needed to ensure that the patient is never alone.

The *management policy* (see Box 21.3) should be:

- *reviewed* carefully at frequent intervals until the danger passes
- *explained to and agreed with each new shift of staff*, especially when the review has led to changes in the plan
- *explained to and, if possible, agreed with the patient*. If the patient does not agree to necessary parts of the plan, the reasons for these should be carefully explained. If the patient still refuses to collaborate, and the risk is high, compulsory treatment may be required.

The risk of suicide is higher during periods of home leave arranged to test readiness for discharge. It is higher when under the care of a crisis team than as an inpatient

(Appleby et al., 2017). It is also greater during the period immediately before discharge and in the first few weeks after discharge. Therefore, the discharge plan should include early reassessment, effective psychological and social support, and rapid response to any need for extra help. Plans for leave and for discharge should be discussed with the patient and their concerns elicited. If appropriate, the plans should be modified.

However carefully patients are cared for, suicide will happen, despite all the efforts of services; a reminder why efforts to 'eradicate' or why 'zero suicide' policies are controversial. The doctor then has an important role in supporting others, particularly staff members who have worked closely with the patient, as well as family members. The impact on medical teams who have been involved in a patient who then dies by suicide can be substantial, with a report produced by the Royal College of Psychiatrists to help psychiatrists cope with what is likely to be a very difficult time (Brand et al., 2023). Although it is essential to review every suicide carefully to determine whether useful lessons can be learned, this process should never become a search for a scapegoat on whom to attribute blame.

Reducing suicide rates

The most widely used framework to conceptualize prevention strategies takes a risk-factor based approach

and classifies prevention efforts as universal, selective or indicated on the basis of their target group (Turecki

Box 21.4 Suicide prevention**Universal interventions**

Restricting access to the means of suicide
 Encouragement of responsible media reporting
 Suicide awareness campaigns and educational programmes (including school-based interventions)

Selective interventions

Pharmacological interventions
 General practitioner education programmes
 Development of treatment guidelines

Indicated interventions

Psychological interventions
 Social approaches
 Crisis services and telephone 'helplines'

et al., 2019). In population terms, as suicide is a rare event, controlled trials of interventions to reduce suicide would require many thousands of participants, few trials have therefore been conducted. However, there is some evidence from observational studies to suggest public health measures that could affect the rate of suicide in the population (see Box 21.4). For more information, see Turecki et al. (2019). The NICE guidance (2019) encourages a focus on five main areas:

- To establish multi-agency suicide prevention partnerships with clear governance and accountability structures.
- To reduce access to methods of suicide based on local information.
- To work with local media to encourage journalists and editors to follow best practice when reporting on suicide and suicidal behaviour.
- To discuss with adults presenting with suicidal thoughts or plans whether they would like their

family, carers, or friends to be involved in their care and are made aware of the limits of confidentiality.

- To give information and tailored support to people bereaved or affected by a suspected suicide.

For those with identified suicidal behaviour, the following can be considered:

Treatment of underlying mental illness. This includes careful support if any in-patient care is provided (as above) and appropriate long-term medication is used, for example, there is robust evidence that lithium prophylaxis in mood disorders reduces suicide rates (Smith and Cipriani, 2017), and some evidence that clozapine may reduce suicide attempts among people with schizophrenia or schizoaffective disorder.

Prescribing less-toxic antidepressants. Selective serotonin reuptake inhibitors (SSRIs) are much less toxic in overdose than tricyclic antidepressants. On the other hand, SSRIs have been reported to cause the emergence or worsening of suicidal ideas in young people, possibly because they can cause agitation and insomnia initially but in general this is not evident in adult populations (Nischal et al., 2012). Results from the latest cohort studies do not suggest an increased risk of suicidality in adults taking SSRIs; in fact, they are associated with a reduced risk (Lagerberg et al., 2022; see also Chapter 25).

Behavioural and psychological treatments. Overall, psychosocial interventions, irrespective of type, have been shown to reduce suicide attempts (Turecki et al., 2019). These include longer-term interventions such as CBT, dialectic behavioural therapy, interpersonal psychotherapy, and acceptance and commitment therapy. Brief interventions for suicidal behaviour include safety planning, crisis response planning, and short interventions programmes.

A number of risk assessment tools have been generated to facilitate suicide risk prediction, but studies have shown that these tools have low positive predictive values. More promising results have been generated using machine learning and natural language processing (Velupillai et al., 2019).

Non-fatal self-harm

Until the 1950s, little distinction was made between people who killed themselves and those who survived after an apparent suicidal act; however, studies identified that the intention of a proportion of survivors had not been to die. As a result, several terms—*deliberate self-poisoning*, *parasuicide*, and *deliberate self-harm*—were introduced to

describe episodes of intentional self-harm that did not lead to death and may or may not have been motivated by a desire to die (see Box 21.5). In this chapter, the term '*non-fatal self-harm*' will be used to describe such incidents.

Some people who had no intention of dying succumb to the unintended effects of an overdose. Others

Box 21.5 Terminology for self-harm

Currently used terminology

Suicidal behaviour

Refers to acts such as suicide and attempted suicide. This may also include suicide-related communications, both verbal and non-verbal, and expressing suicidal intent.

Suicide attempt

Refers to an act carried out by an individual with at least some intent to end their life. A suicide attempt is an act of self-harm with the intended outcome being death. Intent is the defining factor differentiating a suicide attempt from non-fatal self-harm, which is also used to refer to situations where the intent is to cope with the experience of distress.

Suicidal ideation

Refers to thinking about, considering or planning to kill oneself. Suicidal thoughts range in intensity and frequency from fleeting to more concrete, well-thought-out plans for killing oneself, or complete preoccupation with self-destruction.

Non-suicidal self-injury

Non-suicidal self-injury (NSSI) is the deliberate, self-inflicted destruction of body tissue without suicidal intent and for purposes not socially sanctioned. It includes behaviours such as cutting, burning, biting, and scratching skin. In DSM-5, NSSI is included as a condition for further study and is also listed as a possible feature of borderline personality disorder (BPD), described as suicidal behaviour, gestures, threats, or self-mutilating behaviour (Zetterqvist, 2015).

Self-harm

Self-harm refers to an intentional act of self-poisoning or self-injury, irrespective of the motivation or apparent purpose of the act and is an expression of emotional distress. Can also refer to *self-injury* or *self-abuse*.

Self-poisoning or self-injury

Self-harm by these methods regardless of suicidal intent.

Self-mutilation

Serious and/or repetitive bodily mutilation without suicidal intent. Can also be referred to as *cutting*, *self-injurious behaviour*, or *self-wounding*.

Less commonly used and previously used terminology

Deliberate self-harm

Previous use in United Kingdom for all episodes that survived, regardless of intent; however, NICE (2022d) guidelines have removed 'deliberate' and use 'self-harm' as 'deliberate' does not accurately describe what the individual is experiencing and can be stigmatizing.

Parasuicide

Episodes survived, with or without suicidal intent (especially in Europe), or episodes without intent. Repetitive bodily harm may be excluded.

Reproduced from Skegg, K. (2005). Self-harm. *Lancet*, 366(9495), 1471–1483. Copyright (2005), with permission from Elsevier.

who intended to die are revived. Importantly, many people might be ambivalent at the time of the act, uncertain whether they wished to die or survive. It should be remembered that, among people who have engaged in self-harm, the suicide rate in the subsequent 12 months is substantially greater than in the general population. For example, in the longstanding Multicentre Study of Self-Harm in England, when examining suicide following hospital presentation for non-fatal self-harm, the incidence of suicide was highest in the year following discharge from hospital, and previous incidents of self-harm that had involved both self-injury and self-poisoning were associated with higher risk (Geulayov et al., 2019). Living in areas of low socioeconomic deprivation increased subsequent risk. Therefore, self-harm should not be regarded lightly, and early follow-up is important.

Box 21.5 gives some of the definitions of the more commonly used terms in the study of suicide and self-harm. There are conceptual and methodological inconsistencies in the nomenclature, as well as the measurement and definitions of terms used (Reeves et al., 2022). This has led to ongoing challenges for researchers, educators, and clinicians

The act of self-harm

Methods of self-poisoning

The majority of self-poisoning episodes in the United Kingdom involve prescribed or over-the-counter medication, while a minority involve illicit drugs, other household substances, or pesticides. In the United Kingdom, about 90% of the cases of self-harm that are referred to general hospitals involve a drug overdose, and most of

them present no serious threat to life. The type of drug used varies somewhat with age, local prescription practices, and the availability of drugs. The most commonly used drugs are the non-opiate analgesics, such as paracetamol and aspirin.

Paracetamol is particularly dangerous because it damages the liver and may lead to the delayed death of patients—a risk that some patients, for example adolescents, might not be aware of. Antidepressants (both tricyclics and SSRIs) are taken in about 25% of episodes (Geulayov et al., 2018; Geulayov et al., 2019). Alcohol is frequently involved in self-harm acts, and its use has been noted to have increased in a UK longitudinal study from 2000 to 2009 (Ness et al., 2015); 36% of self-harm presentations also recorded ‘alcohol misuse’ with a larger number (58%) having some involvement with alcohol.

Methods of deliberate self-injury

Self-injury accounts for about 10% of all non-fatal self-harm presenting to general hospitals in the United Kingdom, it is increasing in incidence, and it is also often not seen by healthcare professionals. The most common method of self-injury is laceration, usually cutting of the forearms or wrists; it accounts for about 80% of the self-injuries that are referred to a general hospital.

There are three forms of deliberate self-laceration:

1. Deep and dangerous wounds inflicted with serious suicidal intent, more often by men.
2. Self-mutilation by psychotic patients (sometimes in response to hallucinatory voices) or people with severe disorders of intellectual development.
3. Superficial wounds that do not endanger life, more often inflicted by women.

Only the last group will be described here.

Usually, the act of laceration is preceded by increasing tension and irritability, which diminish afterwards. After the act, the patient often feels shame and disgust. Some of these individuals report that they lacerated themselves while in a state of detachment from their surroundings, and that they experienced little or no pain. The lacerations (those most likely to present to hospital) are usually multiple, made with glass or a razor blade, and inflicted on the forearms or wrists. Some people also injure themselves in other ways (e.g. by burning with cigarettes, or inflicting bruises). People who cut themselves superficially do not always seek help from medical services, and many of these people are young women, often with problems of low self-esteem, and sometimes impulsive or aggressive behaviour, unstable moods, difficulty in interpersonal relationships, and problems with alcohol and drug misuse.

The epidemiology of self-harm

Self-harm is the main risk factor for suicide and the epidemiology of self-harm might best be conceptualized as an iceberg (Geulayov et al., 2018, Kapur et al., 2020). Individuals with thoughts of self-harming and those who are self-harming but not presenting to clinical services make up the largest group and lowest tier of the proposed ‘iceberg’. It has been estimated that only 6% of episodes of self-harm present to hospitals (Kapur et al., 2020). There are substantial variations in rates across and between countries, and response rates might even vary between centres in one country. Variations in rates of self-harm can follow similar patterns to variations in suicide rates. Moving up the iceberg, from thoughts of self-injury to self-harm in the community and then to self-harm coming to medical attention, the outcomes become more severe, culminating in suicide at the tip of the iceberg. The WHO has estimated that the estimated global annual rate of suicide attempts is around 4 per 1000 adult, with 12 month prevalence rates for suicidal behaviour similar for men and women in high- and low-income countries, but higher for women than men in middle-income countries (Kapur et al., 2020). In a UK household survey, 4.9% of adults reported a lifetime history of self-harm without suicidal intent (McManus et al., 2009).

In England, hospital presentations of self-harm have been collated since 2000 from three hospital sites across the country. These ‘multicentre study’ findings have highlighted how for self-harm the rates are higher for females, and the peak age for presentation was 15–24 years of age (41% of all female episodes) (Kapur et al., 2020). Self-poisoning accounted for 75% of the presentations, with paracetamol ingested in 35% of episodes. Box 21.1 lists the risk factors for self-harm, many of which are shared with suicidal behaviours. Box 21.6 lists the reasons often given for self-harm.

The outcome of self-harm

Repetition of self-harm

It is estimated that 15–25% of those who present to hospital following self-harm will engage in further self-harm within a year (Hawton et al., 2020). In the weeks after self-harm, patients can report changes for the better if properly supported. Those with psychiatric symptoms might report that these have become less intense. This improvement may result from help provided by professionals, or from improvements in the person’s relationships, attitudes, and behaviour. Some people do

Box 21.6 Reasons given for self-harm

Responding to distress

- Managing distress (affect regulation)
- Interpersonal influence (changing or responding to how others think or feel; help seeking)
- Punishment (usually of self)
- Managing dissociation
- Averting suicide

Self-harm as a positive experience

- Gratification (self-harm is comforting or enjoyable)
- Sensation seeking
- Experimenting with something new
- Developing a sense of personal mastery

Defining the self

- Defining boundaries
- Validation
- Self as belonging or fitting in (to a group or subculture)
- Having a personal language

Source: Edmondson, A. J., Brennan, C. A., & House, A. O. (2016). Non-suicidal reasons for self-harm: A systematic review of self-reported accounts. *J Affect Disord*, 191, 109–117. Copyright (2016), Elsevier.

not improve and harm themselves again, in some cases fatally. A systematic review of 129 hospital-based studies identified a number of factors that had a consistent association with repetition (Larkin et al., 2014), many of which matched risk factors for the initiation of self-harm, although the sensitivity of individual variables varied greatly between studies. These included previous self-harm, personality disorder, hopelessness, history of psychiatric treatment, schizophrenia, alcohol abuse/dependence, drug abuse/dependence, and living alone. The review highlighted that risk of repetition more often came from long-standing vulnerabilities rather than characteristics of the index episode. Self-cutting was associated with a higher risk of repetition than self-poisoning.

Results of treatment after self-harm

The needs of individual patients can vary greatly, following self-harm, and so often a broad range of interventions are required. Treatments include both psychosocial and

pharmacological approaches, which might be combined (Hawton et al., 2022). Although randomized controlled studies have shown decreases in psychopathology and improvements in social problems after various treatments, there is less evidence that treatment prevents the repetition of self-harm. A systematic review and meta-analysis of psychological and psychosocial approaches to reduce repetition of self-harm examined data from 45 trials (Hetrick et al., 2016) and a further review focused on three Cochrane reviews focusing on different aspects of self-harm treatment (Saunders and Smith, 2016). In general, the following conclusions (Hawton et al., 2020; Saunders and Smith, 2016) can be drawn from the available combined studies, although a relative paucity of studies and data has been noted:

- **Pharmacological interventions:** Lithium has the strongest evidence for the prevention of suicidal behaviour. There is evidence from studies for the specific anti-suicidal effect of lithium when used to treat affective disorders, although there is a lack of data to support its use for the prevention of recurrence of self-harm, as existing studies have small numbers of participants. It is likely that lithium's mood-stabilizing effect will be of relevance to self-harm interventions. There is emerging evidence that ketamine, for example, given as an infusion in emergency departments, can lead to a rapid reduction in suicidal ideation. Buprenorphine, in low-dose sublingual preparations, has also been associated with a reduction in suicidal ideation. In adolescents, there is evidence that a combination of fluoxetine and CBT may have greater benefit in the treatment of suicidal ideation than either alone in the treatment of adolescents with depression.
- **Brief cognitive behaviour therapy-based interventions:** There is evidence that CBT-based psychotherapy (either problem-solving therapy or CBT) is associated with fewer individuals repeating self-harm, but there are many limitations to the available evidence-base.
- **Dialectical behaviour therapy (DBT):** DBT has been associated with fewer subsequent episodes of self-harm for patients with a history of multiple episodes of self-harm, especially those with features associated with borderline personality disorder.
- **Contact interventions:** These interventions maintain contact with patients remotely (through mail/postcards) and have limited evidence to support their use. More recent systematic reviews into internet and mobile-based psychological interventions have also been inconclusive (Arshad et al., 2020), because while some have demonstrated reductions in suicidal

ideation these are not evident when the interventions have been compared with active control groups.

Special groups

Mothers of young children

Mothers of young children require special consideration because of the association between self-harm and child abuse. It is important to ask about the mother's feelings towards her children, and to enquire about their welfare. In the United Kingdom, information about the children can usually be obtained from the general practitioner, who may ask their health visitor to investigate the case. Caring for young children can put strains on already shaky relationships, and there is evidence of an increased risk of partner violence against vulnerable young mothers (Vaughn et al., 2015).

Children and adolescents

Presentations of children and adolescents to clinical services following self-harm are increasingly common, for example, in the United Kingdom, presentations to hospital for children and adolescents who have self-harmed have increased substantially in both boys and girls in the last few decades (Cybulski et al., 2021). Rates have been estimated at around 17% in some meta-analyses, with approximately 50% reporting one or two incidents (Gillies et al., 2018). It is difficult to obtain the exact rates because many incidents might not reach medical services, although community self-report data have also shown rates to be around 12% in a sample from 2020 (during the Covid-19 pandemic) (Geulayov et al., 2022a). Self-harm is more common among girls and the most commonly report reasons for self-harm are wanting relief from thoughts or feelings (Gillies et al., 2018).

Self-harm in adolescents is associated with loneliness, history of maltreatment, traumatic stress, and peer

victimization (Asarnow and Mehlum, 2019; Geulayov et al., 2022b). Among adolescents it is also associated with alcohol and drug misuse (especially among males), violence and being the victim of violence, mood disorder, and personality disorder (Clarke et al., 2019).

For most children and adolescents, there is a reduction in self-harm into adulthood, but a significant minority continue to have social and psychiatric problems, as well as repeat acts of self-harm. A poor outcome is associated with poor psychosocial adjustment, a history of previous self-harm, and family dysfunction.

Treatment usually involves the family, and is directed to the causal problems and to the coping skills of the child or adolescent, usually overseen by child and adolescent mental health teams. A Cochrane review of interventions for self-harm in children and adolescents (Witt et al., 2021) reported that there is little evidence of beneficial effects for individual cognitive behaviour therapy (CBT)-based psychotherapy, mentalization-based therapy for adolescents (MBT-A), group-based psychotherapy, enhanced assessment approaches, compliance enhancement approaches, family interventions, or remote contact interventions. There is some evidence of effectiveness for dialectical behaviour therapy (DBT-A) for adolescents.

Overall, few trials have been conducted and those that have are generally small, meaning that possible beneficial effects of some of these therapies cannot be ruled out. In a study of use and acceptability of interventions following adolescent self-harm in a community sample in England, Geulayov et al. (2022a) demonstrated that one third had not sought any help, the majority of those seeking support did so from informal sources of support (family, friends), and that around 12% sought support from formal child and adolescent mental health services and around 8% from online resources. Informal sources of support were perceived as the most helpful and online sources as the least helpful.

The management of self-harm

The organization of services

Aspects of provision of mental health services can affect the rates of self-harm and suicide in clinical populations (Hawton et al., 2020). The care of patients with self-harm involves a variety of services, including primary care teams, ambulance services, emergency department staff, and social services with continuity of care likely to

be a particularly important factor in preventing suicide in those at risk (see Boxes 21.3 and 21.9 for principles of care). All of these staff require training to enable them to respond appropriately and to make the necessary decisions, including how to assess immediate risk, to obtain informed consent, and to assess capacity to consent and in what circumstances necessary care can be given without consent. The organization of the medical and

surgical response to these patients is beyond the scope of this book; for such information, see National Institute for Clinical Excellence (2022d).

Arrangements for psychosocial assessment vary and a debate has long persisted about who should conduct the assessment. Whatever arrangement is made, it is important to ensure that all patients receive psychosocial assessment as well as an assessment of the medical effects of the self-harm (see below). A code of practice detailing the arrangements for psychosocial assessment for each hospital site is needed and has to be agreed by general medical and psychiatric services and known to those who work there.

Provision should be available for the special needs of patients from ethnic minorities and other groups experiencing potential barriers to accessing care, such as prisoners and people bereaved by suicide. Children and adolescents should, whenever possible, be assessed by staff who have been trained in the assessment and care of young people and who are familiar with the problems of confidentiality and consent that can arise in such cases. If possible, patients over the age of 65 years should be assessed by staff who are familiar with the special needs of the elderly, especially important as there is a greater risk of suicide in this age group.

The assessment of patients after self-harm

General aims

Assessment is concerned with three main issues:

1. The immediate risks of suicide.
2. The subsequent risks of further self-harm.
3. Current medical or social problems.

The assessment should be conducted in a way that encourages patients to undertake a constructive review of their problems and of the ways in which they could deal with them themselves. It is important to encourage self-help, because many of these people are unwilling to be seen again by a psychiatrist.

Usually the assessment is carried out in an accident and emergency department or a ward in a general hospital, where there may be little privacy. Whenever possible, the interview should be conducted in a side room so that it will not be overheard or interrupted. If the patient has taken an overdose or taken alcohol and/or other substances, the interviewer should first make sure that they have recovered sufficiently to be able to give a satisfactory history. If consciousness is still impaired, the interview should be delayed. Information should

also be obtained from relatives or friends, the primary care team, and any other relevant individuals (such as a social worker) who might know or be trying to assist the patient. Wide enquiry is important, as sometimes information from other sources may differ substantially from the account given by the patient.

Specific enquiries

The interview should address five questions.

1. What were the patient's intentions when they self-harmed?
2. Do they now intend to die?
3. What are their current problems?
4. Is there a psychiatric disorder?
5. What helpful resources are available?

Each of these questions will now be considered in turn.

What were the patient's intentions when they self-harmed?

Patients sometimes misrepresent their intentions. For this reason the interviewer should reconstruct, as fully as possible, the events that led up to the act of self-harm to find the answers to five subsidiary questions.

- *Was the act planned or carried out on impulse?* The longer and more carefully the plans have been made, the greater is the risk of a fatal repetition.
- *Were precautions taken against being found?* The more thorough the precautions, the greater is the risk of a further fatal overdose. Of course, events do not always take place as the patient expected—for example, a family/household member may return earlier than usual so that the patient is discovered alive. In such circumstances, it is the patient's reasonable expectations that are of importance in predicting the future risk.
- *Did the patient seek help?* Serious intent can be inferred if there were no attempts to obtain help after the act.
- *Was the method thought to be dangerous?* If drugs were used, what were they and what amount was taken? Did the patient take all of the drugs available? If self-injury was used, what form did it take? Not only should the actual dangerousness of the method be assessed, but also the dangerousness anticipated by the patient, which may be inaccurate. For example, some people wrongly believe that paracetamol overdoses are harmless, or that benzodiazepines are dangerous.
- *Was there a 'final act' (e.g. writing a suicide note or making a will)?* If so, the risk of a further fatal attempt is greater.

Box 21.7 Factors suggesting high suicidal intent following an episode of self-harm

Changing behaviour

- Preparations made in anticipation of death (e.g. making a will, organizing insurance)
- Communicating intent to others beforehand
- Withdrawing from friends, saying goodbye

Attempt itself

- Extensive premeditation and preparations made for the act (e.g. purchasing the means of suicide, saving up tablets)
- Act carried out in isolation and precautions taken to avoid discovery and not seeking help after the act
- Leaving a note
- Timed so that intervention is unlikely

Mental state examination

- Continuing suicidal ideation
- Current depression
- Feeling hopeless and empty
- Feeling they are a burden to others

Source: Hawton, K., Saunders, K. E. A., & Pitman, A. (2020). Prevention of suicide and treatment following self-harm. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1303–1314). Copyright (2020), Oxford University Press; Choo, C. C., et al. (2019). Clinical assessment of suicide risk and suicide attempters' self-reported suicide intent: A cross-sectional study. *PLoS ONE*, 14(7), e0217613. Copyright (2019), PLOS.

By reviewing the answers to these questions, the interviewer can form a judgement of the patient's intentions at the time of the act (see Box 21.7).

Do they now intend to die?

The interviewer should ask directly whether the patient is pleased to have recovered or wishes that they had died. If the act suggested serious suicidal intent and if the patient now denies such intent, the interviewer should try to find out by tactful questioning whether there has been a genuine change of resolve.

What are their current problems?

Many patients will have experienced a mounting series of difficulties in the weeks or months leading up to the act. Some of these difficulties may have been resolved by the time that they are interviewed, sometimes as a result of the act of self-harm—for example, a partner

who had threatened to leave the patient may subsequently decide to stay. The more serious or ongoing the problems that remain, the greater is the risk of a fatal repetition. This risk is particularly strong if the problems include loneliness or ill health. The review of problems should be systematic and should cover the following:

- intimate relationships with the partner or another person
- relationships with children and other relatives
- employment, finances, and housing
- legal problems
- social isolation, bereavement, and other losses
- physical health
- social media.

Drug and alcohol problems can be considered either at this stage of the assessment or later when the psychiatric state is reviewed.

Is there a psychiatric disorder?

It should be possible to answer this question from the history and from a brief but systematic examination of the mental state. Particular attention should be directed to depressive disorder, substance misuse disorders, anxiety disorder, and personality disorder. Schizophrenia and dementia should also be considered, especially as there are increasing rates of self-harm in older adults, especially those with dementia. Adjustment disorders are diagnosed in many individuals in response to major life changes and stresses (e.g. bereavement, relationship break-up, migration). The presence of an obvious precipitating event does not rule out the presence of a psychiatric disorder.

What helpful resources are available?

These include capacity to solve problems, material resources, and the help that others are likely to provide. The best guide to a patient's abilities to solve future problems is their record of dealing with difficulties in the past—for example, the loss of a job or a broken relationship. The availability of help should be assessed by asking about friends and confidants, and about any support the patient is receiving or can be expected to receive from the general practitioner, social workers, or third-sector and charitable agencies.

Is there a continuing risk of suicide?

The interviewer now has the information required to answer this important question. The answers to the first five questions outlined above are reviewed:

- Did the patient originally intend to die?
- Do they intend to do so now?
- Are the problems that provoked the act still present?
- Is there a psychiatric disorder?
- Is additional support available, and is the patient likely to accept it?

Having reviewed the answers to these questions, the interviewer compares the patient's characteristics with those that are found in people who have died by suicide.

Is there a risk of further self-harm?

The predictive factors are summarized in Box 21.7. All of the points should be considered before deciding on the risk.

What treatment is required, and will the patient agree to it?

If the risk of suicide is judged to be high, the procedures are those outlined in the first part of this chapter. Around 5–10% of patients who self-harm require admission to a psychiatric unit; most need treatment for depression or alcoholism, and a few for schizophrenia. Some need a period of respite from overwhelming stress.

If admission to hospital is not indicated, a plan of management has to be agreed with the patient and any potential carers. If the patient refuses the offer of help, their care should be discussed with the general practitioner before they are allowed to return home (see below and Box 21.8). For most patients it is useful to provide an emergency telephone number that allows immediate access to advice or an urgent appointment should there be a further crisis.

Management after the assessment

Patients who have harmed themselves might be reluctant to accept further help, and it is important to conduct the assessment interview in a way that fosters a therapeutic relationship. Plans should be agreed with the patient, so that the plan is mutually acceptable. Patients' needs fall into three groups:

1. A small minority need admission to a psychiatric unit for treatment.
2. About one third have a psychiatric disorder that requires treatment in primary care, or from a psychiatric team in the community.

Box 21.8 Management of treatment refusal following self-harm

Step	Key considerations		
1. Immediate engagement	Use calm, non-judgemental communication. Discuss proposed treatments, alternatives, and the person's concerns. Involve trusted relatives/friends where appropriate.	4. If has capacity but refuses	Explore motivations (e.g. ambivalence, despair, cultural factors). Offer alternatives and support. If refusal persists, clearly explain the consequences. Document discussion and clinical reasoning.
2. Assess mental capacity	Assume capacity unless evidence suggests otherwise. Assess capacity (see Chapter 4, Box 4.2) as soon as possible and for the decision at hand; incapacity should be confirmed by an appropriately trained clinician, with senior input where needed. Reassess if intoxication is suspected and also if new treatments or plans are introduced.	5. Legal and ethical escalation	Consider if criteria are met for detention under the Mental Health Act (or equivalent). Involve senior clinicians, legal/ethics advisors, or safeguarding where needed.
3. If lacking capacity	Act in the person's best interests under the mental capacity act (or, in rare emergencies, common law). Provide necessary, proportionate, and least restrictive life-saving care. Document reasoning.	6. Discharge planning	If leaving against advice, provide crisis numbers, written safety plans, and re-engagement options. Inform the GP or follow-up team. Aim to make contact the next day (or for most appropriate team to lead on follow-up contact).
		7. Documentation essentials	Record capacity assessment, patient's decision, treatment offered, risks discussed, legal framework applied, and follow-up plans. Use structured templates if available.

For further guidance, see National Institute for Clinical Excellence (2022d).

Box 21.9 Principles of clinical care following self-harm

1. **Continuity and trained care:** Minimize staff changes to promote continuity of care. Ensure that clinical observation is conducted by appropriately trained personnel, avoiding the use of untrained staff wherever possible.
2. **Accessible Care Plans (ACPs):** Ensure care plans are available to all relevant healthcare professionals across primary, secondary, and multidisciplinary teams and are regularly reviewed.
3. **Visibility and engagement:** Clinical staff should be accessible and actively engage with individuals receiving care by being present in communal areas, checking in regularly, and maintaining visibility, particularly during shift handovers and busy periods.
4. **Safe and respectful environment:** Assess environmental risks while respecting the individual's autonomy. Use the least restrictive interventions necessary for safety.
5. **Clinical risk management:** Remove objects that could be used for self-harm in a clinical setting, ensuring the individual is informed about the reasons for this and provided with appropriate emotional support.
6. **Orientation and support:** Introduce individuals to the care setting as early as possible and provide clear information on how to access support.
7. **Rapid escalation of concerns:** Staff must be aware of how to identify and promptly escalate concerns about an individual's safety and wellbeing.

For further guidance, see National Institute for Health and Care Excellence (2025). Scenario: Acute management following an act of self-harm.

3. The remainder need help with various psychosocial problems, and assistance with improving their ways of coping with stressors. This help is needed even when the risk of immediate suicide or non-fatal repetition is low, as continuing problems increase the risk of later repetition. Apart from practical help, problem solving is usually the best approach, starting with the problems identified during the assessment interview.

The principles of clinical care are summarized in Box 21.9 and of general care in the community in Box 21.10. If the episode of self-harm is associated with problems in a specific interpersonal relationship, it is often helpful to interview the other person involved if the patient agrees, first alone and then together with the patient, if appropriate and possible. This joint interview can lead to an improved dialogue between relevant individuals, and this may help to decrease the acute risk of repetition.

Some special problems of management

Patients who refuse assessment

A number of individuals who present to hospital following self-harm do not receive an assessment, and for some of these it is because they refuse an assessment, while for others it might be because of intoxication or long waiting times. Patients who do not receive a

psychosocial assessment are at elevated risk of further non-fatal self-harm and suicide (Quinlivan et al., 2022). It is essential to gather as much information as possible from other sources before allowing the patient to leave. The problem should be discussed with the primary health care and community mental health team if they are involved. If the risk of suicide is judged to be high, it may be necessary to keep the patient until an assessment is made for detention under a compulsory order.

Frequent repeaters

Some patients take overdoses repeatedly at times of stress. Often the behaviour seems intended to reduce tension or gain support through contact with services. However, when overdoses are taken repeatedly, relatives and friends can become unsympathetic or even overtly hostile, and staff of hospital emergency departments may feel frustrated. These patients should be assessed for a possible personality disorder as well as insoluble social problems. Unfortunately, neither counselling nor intensive psychotherapy is usually effective in these situations. All of those involved in the patient's management should agree a care plan, which should seek to reward constructive behaviour. An opportunity for continuing support by one individual should be arranged. However, whatever help is arranged, the risk of eventual death by suicide is high. Services for those with borderline personality disorders (Chapter 15) can potentially offer support to these patients.

Box 21.10 General principles of community care after self-harm

- Monitoring of mental state
 - Monitor patient for further suicidal or self-harm thoughts
 - Treat psychiatric illness effectively (try to avoid medication that could be lethal in overdose)
 - Attend to substance misuse
- Availability of support
 - Identify support available in a crisis
 - Enlist support of family and friends where possible—in agreement with the patient and encouraging a collaborative approach
 - Assertive follow-up in an empathic relationship
- Psychological support
 - Help patient to identify and work towards solving problems
 - Encourage adaptive expression of emotion
 - Affirm the values of hope and of caring for oneself

Repeated self-cutting

The management of self-laceration presents many problems. Patients often have difficulty in expressing their

feelings in words, so formal psychotherapy may not be helpful. Simple efforts to gain the patient's confidence and increase their self-esteem are more likely to succeed. Assessment should include a behavioural analysis of the sequences of events that lead to self-cutting. This may help to formulate ways in which treatment could either interrupt the chain of events that leads to self-cutting, or replace the cutting with an alternative method of obtaining relief.

Medication appears to have a limited role, although atypical antipsychotic agents such as risperidone or olanzapine in low dose may be valuable as a short-term measure to reduce tension. Some people who repeatedly cut themselves have significant personality difficulties. Treatment should be directed towards these problems, although it may be difficult and prolonged. Admission to a psychiatric unit is occasionally necessary, but carries risks, and is advised against unless absolutely necessary (National Institute for Health and Clinical Excellence, 2022d). If the patient is admitted, a clear and detailed policy is needed, as self-cutting can be difficult to manage in an inpatient unit, and may be imitated by other patients. Complex needs services for personality disordered patients can potentially provide appropriate interventions.

Further reading

Geddes, J. R., Andreasen, N. C., & Goodwin, G. (Eds.). (2020). Section 21. Suicide. In: *The New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1279–1316). Oxford University Press.

Wasserman, D. (Ed.). (2021). *Oxford Textbook of Suicidology and Suicide Prevention*, 2 edn. Oxford University Press.

CHAPTER 22

Psychiatry and medicine

Introduction 609

Psychiatry, medicine, and mind–body dualism 609

Bodily distress disorders 611

Dissociative disorders 613

Psychiatric disorders in medical settings 616

Psychological medicine services 628

Psychiatric aspects of obstetrics and
gynaecology care 638

Introduction

This chapter considers the relevance of psychiatry to the rest of medical practice. It covers two related, but distinct, areas. The first is the application of psychiatric knowledge and expertise in a general medical setting. The second covers the psychiatric conditions associated with specific medical disorders.

The chapter is organized as follows:

1. Psychiatry, medicine, and mind–body dualism.
2. Bodily distress disorders (formerly somatoform disorders).

3. Dissociative disorders.
4. Psychiatric disorders in medical settings.
5. Psychological medicine services and consultation–liaison psychiatry.
6. Psychiatric aspects of obstetrics and gynaecology care.

This chapter should be read in conjunction with the chapters on stress-related disorders (Chapter 7) and neuropsychiatry (Chapter 14); aspects of psychiatry and medicine in children are discussed in Chapter 16.

Psychiatry, medicine, and mind–body dualism

Patients usually attend their doctors because of symptoms that are causing distress and/or dysfunction, that is, when they have an *illness*. Medical assessment is directed to making a diagnosis of the illness, and this diagnosis is used to guide the plan of management. The diagnosis is conventionally classified as either medical or psychiatric. Many presentations are not easy to classify in this binary way, and most mental and physical disorders and presenting symptoms co-exist, and may even share causal pathways and thus therapeutic approaches.

Medical diagnosis

Most medical diagnoses are based on symptoms and physical signs and the results of biological investigations that together indicate the presence of bodily pathology (abnormal structure and/or function), which is referred to as *disease*. Not all medical diagnoses are arrived at in this way (e.g. migraine). Ultimately a medical diagnosis is a label for a condition that is conventionally treated by medical doctors and listed in classifications of disease such as ICD-11.

Psychiatric diagnosis

Psychiatric diagnosis was discussed in Chapter 2, and the following brief account is intended to remind the reader of that discussion and to draw out special concerns when making psychiatric diagnosis in the presence of other long-term conditions. A psychiatric diagnosis, in many ways like most medical diagnoses, is essentially a label for a condition that is conventionally treated by doctors, sometimes psychiatrists, but often by other healthcare professionals, too. Psychiatric diagnoses are made on judgements about functioning (in society and in relationships), as well as recognizable patterns of symptoms that suggest a specific mental illness. Psychiatric diagnoses are listed in the classifications of diseases, along with medical diagnoses but, although some have associated physical pathology, many do not. As we learn more about what causes psychiatric disorders, more biological, physiological, or neurochemical mechanisms are being identified. Yet, overall, we are still unable to identify mental illnesses on the basis of biological tests. Even where some evidence of useful diagnostic markers has emerged, these are still not routinely used in clinical practice. Because psychiatric diagnoses generally lack known organic pathology, they are generally referred to as disorders, rather than diseases. Yet, the term ‘disorder’ is also contested as it risks stigma.

In the past, psychiatric diagnoses were regarded as ‘mental’ in nature, in contrast to the ‘physical’ nature of medical diagnoses. Most psychiatric disorders usually present with disturbed mental states or behaviours. Some do present with primarily somatic symptoms, and these situations are considered later in this chapter.

Medicine, psychiatry, and dualism

Underlying this division of illnesses into physical and mental is the assumption that a parallel distinction can be made in healthy people—that there is ‘mind-body dualism’, an idea commonly attributed to the philosopher Descartes. So-called Cartesian dualism had, and

continues to exert, a profound influence on western medical thinking (Miresco and Kirmayer, 2006). While this division has some utility, it can also be problematic.

Limitations of dualism

Dualism is at best an oversimplification. It can be convincingly argued that there are no conditions that are purely physical or psychological, whether in health or illness. The associated assumption that psychological symptoms indicate psychopathology and physical symptoms indicate physical pathology leads to the categories shown in Table 22.1. Two of these, disease and disorder, have been considered; the other two—comorbidity and somatization—are considered next.

Comorbidity means the co-occurrence of two disorders. The term has been extended in the table to describe the co-occurrence of prominent mental symptoms and bodily pathology since these patients are usually given both a psychiatric and a physical diagnosis (Kisely and Goldberg, 1996). In practice neither of these diagnoses may lead to effective treatment because a focus on either may lead to neglect of the other. An example is the widespread neglect of depression in patients with medical disease (Moussavi et al., 2007; Walker et al., 2014). More recently, the term *multi-morbidity* has been used to capture those with at least two or at least three medical conditions, one of which can be a mental disorder, an infectious disease, or a physical non-communicable disease.

Practical consequences of dualism

One consequence of dualism is the professional and organizational separation of medicine and psychiatry, and dualism in training, services, and practice. This makes it difficult to provide integrated care for patients with comorbidity, multimorbidity, and bodily distress disorder (BDD). Once such services and professionals operate within their specific domains of specialization, it is likely that the dualism is further reinforced, rather

Table 22.1 Traditional ‘dualistic’ categories of mental and physical illness

	Psychological symptoms	Physical symptoms
Bodily pathology	Comorbidity	Medical disease
Psychopathology	Psychiatric disorder	Somatization

than one service or clinician taking a holistic view. One organizational remedy has been the establishment of psychological medicine (also known as liaison psychiatry) services to general hospitals (see below). More recently, in the United Kingdom, there are efforts to develop integrated interventions and services whereby psychiatric and medical services are co-located, even if provided by different specialists (Fazel et al., 2021).

An integrated approach

New scientific knowledge, such as the demonstration of a neural basis to many psychiatric disorders, has shown that crude dualistic thinking is untenable. Evidence for the effect of psychiatric disorder on the outcome of medical conditions such as myocardial infarction (Meijer et al., 2011) and cancer (Walker et al., 2014) has pointed to the same conclusion. Such integration is strengthened by evidence that psychological interventions can be effective in reducing fatigue in chronic fatigue syndrome (CFS; White et al., 2011), where the underlying medical condition is uncertain; or in multiple sclerosis (van Kessel et al., 2008), where it is clear. The implication is that psychiatric disorders may be no more distinct from medical conditions than the higher nervous system is from the rest of the body (Sharpe and Carson, 2001).

The psychiatrist working in medical settings needs to be aware of these problems and help by ensuring that

Box 22.1 The biopsychosocial formulation

Biological factors

Disease
Physiology

Psychological factors

Cognition
Mood
Behaviour

Social factors

Interpersonal
Social and occupational
Relating to the health care system

both the biological and psychological and social aspects of illness are considered in every case. This 'biopsychosocial' approach was proposed many years ago (Engel, 1977) and is mostly adopted throughout this book. The factors to consider in a biopsychosocial formulation are shown in Box 22.1. They can be divided further into *pre-disposing*, *precipitating*, and *perpetuating* causes (as introduced in Chapter 3). The last group of causes is the usual target for treatment, while the first two are relevant to prevention.

Bodily distress disorders

Some patients have somatic symptoms but no evidence of bodily pathology. It is unclear whether their illness should be categorized as medical (with presumed but unidentified somatic pathology) or as psychiatric (with assumed psychopathology). In the past these conditions were generally given the medical diagnosis of *functional illness* (function is abnormal but there is no pathology). In psychiatric practice these conditions would have been called a *somatoform disorder*. Historically, the diagnosis of somatoform disorder implied that: (i) the somatic symptoms were caused by psychopathology; and (ii) there was a hypothetical process—somatization—by which the psychopathology caused the bodily symptoms.

In ICD-11 this is now called a 'bodily distress disorder' (BDD). Many of the later sections deal fundamentally with BDD, for example, the sections on *psychiatric*

disorders presenting in medical settings, *psychological medicine services*, and *psychiatric aspects of specific medical conditions*. In DSM-5, the term 'somatoform disorder' has also been abandoned because of significant overlap between different somatoform conditions and the perceived lack of utility of the boundary between medically explained and medically unexplained symptoms (Sharpe, 2013). As mentioned above, ICD-11 offers a new simplified category of BDD, which replaces ICD-10 categories of somatoform disorders (F45.0) and neurasthenia (F45.2). BDD is characterized by distress and excessive attention to the bodily symptoms manifest by repeated contact with healthcare agencies. Specifically, the new classification removes the distinction between 'medical explained' and 'medically unexplained' symptoms. The term 'somatic' is avoided in ICD-11, and hypochondriasis is

now placed under ‘obsessive–compulsive and related disorders’ (see Chapter 8).

Pain as a bodily distress disorder

This term was used to denote patients with chronic pain that did not appear to be caused by any physical or specific psychiatric disorder. ICD-11 includes pain syndromes within the overarching category of BDDs, rather than separate pain syndromes.

Epidemiology

Pain is widely reported in surveys of the general population. Most people report pain that is transient, but a minority describe persistent or recurrent pain leading to disability (Gureje et al., 1998). Pain is the most common symptom among people who consult doctors. Acute pain usually has an organic cause, but psychological factors can affect the subjective response to pain whatever the main cause. Patients with multiple pains are especially likely to have associated psychiatric disorder (Gureje et al., 1998). Pain is particularly associated with depression, anxiety, panic, and somatoform disorders. In general practice, pain is a common presenting symptom of emotional problems. In psychiatric practice, it has been reported that pain is experienced by about one fifth of inpatients and over half of outpatients.

Aetiology

Chronic pain occurs in many conditions, including neurological or musculoskeletal disorders. It often has both physical and psychological causes (Bushnell et al., 2013). A ‘pain-prone disorder’ has been suggested as a variant of depressive disorder, but there is little evidence for this. It is more likely that in these cases the pain arises in the context of personal and social factors, and that beliefs about pain are important in maintaining it (Linton, 2000). Chronic pain may impose great burdens on the patient’s family; also, the attitude of family members and other caregivers can influence the perception of pain, its course, and the response to treatment.

Assessment

The assessment of a patient complaining of pain of unknown cause should include:

- Thorough investigation of possible physical causes. When the results of this investigation are negative, it should be remembered that pain may be the first

symptoms of a physical illness that cannot be detected at an early stage.

- Full description of the pain and the circumstances in which it occurs.
- History-taking of development of social and communication skills and current levels of functioning to ascertain if autism spectrum disorder (ASD) might be present.
- Search for symptoms of a depressive or other psychiatric disorder.
- Description of pain behaviours—for example, the presentation of symptoms, requests for medication, and responses to pain.
- Exploration of the patient’s beliefs about the causes and implications of the pain.

Treatment

The management of chronic pain should be individually planned, comprehensive, and involve the patient’s family. Skill is required to maintain a working relationship with patients unwilling to accept an approach that uses psychological treatments as part of the treatment of pain. Any associated physical disorder should be treated and adequate analgesics provided. The treatment of pain associated with a psychiatric disorder is the treatment of the primary condition.

Psychological care is directed to assessing:

- The presence of any associated mental disorder. This assessment should be made on positive findings and not solely because a specific organic cause has not been identified. If depressive illness is present, it should be treated vigorously. Tricyclic antidepressant medication is effective in some patients with chronic pain even without evidence of a depressive disorder.
- Whether psychological techniques are indicated to modify the pain or any associated behaviours. Cognitive behaviour therapy (CBT) aims to encourage the use of distraction, relaxation, and other ways of coping with the pain, and to reduce social reinforcement of pain-related behaviour. In a meta-analysis these treatments had a significant though modest effect (Morley et al., 2013). Mindfulness approaches are becoming more widely used (Pardos-Gascón et al., 2021).

Multidisciplinary pain clinics bring together expertise in somatic and psychological treatments for pain. See Grandhe et al. (2016) for a review of treatment approaches to chronic pain.

Dissociative disorders

The older term *somatoform disorder* denoted a group of conditions (listed in Table 22.2) characterized by physical symptoms occurring without an adequate physical cause. In DSM-5 the new term *somatic symptom disorder* emphasizes the burden of symptoms, rather than their unclear aetiology (Sharpe, 2013). In ICD-11, these symptoms are listed in the separate category of BDD.

A further difference between classifications concerns the status of *conversion disorder* and its relationship to dissociative disorders. In DSM-5, conversion disorder is classified with somatic symptom disorders and has the alternative name of *functional neurological symptom disorder*. This emphasizes the essential importance of the neurological examination and recognition that the relevant psychological factors may not be demonstrable at the time of diagnosis. ICD-11 classifies these under '*dissociative neurological symptom disorder*'.

Table 22.2 Categories of somatoform disorders in disorders in ICD-11, ICD-10

ICD-11	ICD-10
Bodily distress disorder (BDD)	Somatization disorder
BDD	Undifferentiated somatoform disorder
Obsessive–compulsive disorder	Hypochondriacal disorder
BDD	Somatoform autonomic dysfunction
BDD	Persistent pain disorder
BDD	Other somatoform disorders
BDD	No category
Dissociative neurological symptom disorder, under BDD	No category
Dissociative disorders, unspecified	Neurasthenia
Factitious disorders	

Source: *International Classification of Diseases, Tenth Revision* (ICD-10). Copyright (1994), World Health Organization; *International Classification of Diseases, Eleventh Revision* (ICD-11). Copyright (2022), World Health Organization. <https://icd.who.int>

ICD-11 has a *dissociative disorder* category that includes *dissociative identity disorder*, *dissociative amnesia*, and *depersonalization/derealization disorder*.

Dissociative disorders refer to involuntary disruption of continuity of the usual integrated function of identity, sensations, perceptions, thoughts, memories, and control over bodily movements or behaviour. There are sub-categories for neurological symptoms, amnesia, trance disorder, possession, dissociative identity disorder, partial dissociative identity disorder, depersonalization–derealization, and other specified disorders. The relevance to psychological medicine is that these can present with altered bodily function and may be seen by physicians in the first instance, with the concern that some brain pathology might explain the presentation. Dissociation is a crucial concept in the development of psychiatry and the history of the term is reviewed briefly in Box 22.2.

Types of dissociative disorders

The types of dissociative disorders recognized in ICD-11 are shown in Box 22.3. Some consider that these disorders are well established and common while some doubt the evidence base for the category and for helpful treatments (Casey, 2021).

Box 22.2 A brief history of dissociation

The term dissociation is associated particularly with the French philosopher and psychiatrist Janet (1859–1947), who worked for a time with Charcot at the Salpêtrière in Paris. For a while Janet's ideas were influential; however, his theories were overshadowed by those of Freud. More recently, in the 1970s interest in dissociation revived as a result of studies of the effects of psychological trauma, especially among Vietnam War veterans. These studies suggested a common origin in the mechanism of dissociation, a new name of dissociative symptoms (see Box 22.3), and a new diagnostic category of dissociative disorders.

Box 22.3 Dissociative disorders in ICD-11

- Dissociative neurological symptom disorder
- Dissociative amnesia
- Trance disorder
- Possession trance disorder
- Dissociative identity disorder
- Partial dissociative identity disorder
- Depersonalization–derealization disorder
- Other specified dissociative disorders

Dissociative neurological disorders

These are defined by three criteria:

1. Involuntary disruption or discontinuity in the normal integration of motor, sensory, or cognitive functions, lasting at least several hours.
2. Clinical findings are not consistent with a recognized disease of the nervous system (e.g. stroke) or another medical condition (e.g. head injury).
3. The symptoms are not explained by other disorders.

The subtypes include visual, auditory, sensory, speech, paresis or weakness, abnormal gait, abnormal movement, cognitive symptoms, non-epileptic attacks, and other sensory experiences.

Dissociative amnesia

The essential feature is an inability to recall important personal memories, usually of a stressful nature, that is too extensive for normal forgetfulness. Dissociative amnesia occurs alone and during the course of other dissociative disorders and of post-traumatic stress disorder, acute stress disorder, and somatization disorder. The diagnosis is made only when these other conditions are not present.

Dissociative amnesia must be distinguished from amnesia having a medical cause. It has been described in two forms:

1. Circumscribed amnesia for a single recent traumatic event.
2. Inability to recall long periods of childhood. Among patients who present in this way, some have concurrent organic disease.

See Staniloiu and Markowitsch (2014) for a review.

Trance and possession disorder

Trance and possession states are characterized by a temporary loss of the sense of personal identity and of full awareness of the surroundings. Such states are induced temporarily in willing participants in religious or other ceremonies, and recreationally in 'raves', often aided by drug use. When they arise in this way, the states are not recorded as disorders. A dissociative trance involves narrowing or loss of awareness of immediate surroundings with profound unresponsiveness or insensitivity to environmental stimuli. Such states are not part of an accepted cultural or religious practice.

Some cases resemble multiple personality disorder in that the person behaves as if taken over by another personality for a brief period. The person may feel taken over by a deity or spirit in the context of religious ceremonies.

Dissociative identity disorder

The disorder has been highly controversial and, along with the associated 'recovered memory syndrome', generates fierce debate and legal challenges. ICD-11 defines dissociative identity disorder as follows:

- Disruption of identity characterized by the presence of two or more distinct personalities, involving marked discontinuities in the sense of self and agency. Each personality state includes its own pattern of experiencing, perceiving, conceiving, and relating to self, the body, and the environment.
- At least two distinct personality states recurrently take executive control of the individual's consciousness and functioning.
- There are related alterations in sensation, perception, affect, cognition, memory, motor control, and behaviour. There are typically episodes of amnesia inconsistent with ordinary forgetting, which may be severe.
- The condition is not accounted for by other conditions and generates distress in self or family or in other public settings.

ICD-11 also includes a partial dissociative identity disorder, whereby there is one dominant personality that sustains everyday functions, and it can be disrupted by a non-dominant personality. The condition is probably rare, but it has been reported more frequently in certain periods, notably around the end of the nineteenth century. These variations over time probably reflect the changing interests of doctors and cultural factors rather than true changes in prevalence. Over the past

40 years there has been another increase in the number of reported cases, especially in the United States. It is not certain whether this change is real, and epidemiological studies do not provide the answer because they report such widely varying prevalence rates.

Patients who meet diagnostic criteria for dissociative disorders often meet the criteria for other diagnoses, including schizophrenia, personality disorder, and substance abuse. Many also have symptoms of anxiety and depression. The relationship between dissociative identity disorder and these other conditions would be clarified by long-term follow-up studies, but no systematic studies of this kind have been reported.

Two issues have dominated the discussion of *aetiology*:

1. *The role of severe trauma.* Clinical experience and research find that many of those with the disorder describe severe physical or sexual abuse taking place in childhood. It has been suggested that dissociation begins as a psychological defence mechanism that reduced distress at the time of the original trauma but has unfortunate lasting consequences.
2. *Iatrogenic factors.* It has been argued that the widespread publicity given to some people with multiple personality, the credulity of some therapists, and the use of hypnosis, have been responsible for at least some instances of the disorder.

For a review, see Dorahy et al. (2014).

Depersonalization disorder

Depersonalization disorder is characterized by an unpleasant state in which external objects or parts of the body are experienced as changed in their quality and feel unreal or remote. Transient symptoms of depersonalization are quite common as a minor feature of other syndromes (see Chapter 1), and the symptoms of depersonalization disorder occur occasionally in association with other psychiatric disorders (see differential diagnosis, below). However, primary depersonalization disorder is rare. How rare is uncertain because estimates of the prevalence vary so widely that no useful conclusion can be reached (Reutens et al., 2010).

Clinical picture

The central features are a feeling of being unreal and an unreal quality to perceptions. Emotions seem dulled and actions feel mechanical. Paradoxically, this lack of feeling is experienced as extremely unpleasant. Insight is retained into the subjective nature of their experiences. These symptoms may be intense and are accompanied

by *déjà vu* and by changes in the experience of passage of time. Some patients complain also of sensory distortions affecting a single part of the body (usually the head, the nose, or a limb), a feeling which some describe as if made of cotton wool.

Two thirds of the patients are women. The onset is often sudden, in adolescence or early adult life, with the condition starting before the age of 30 years in about half the cases. Once established, symptoms may persist for years, though with periods of partial or complete remission.

Differential diagnosis

Before diagnosing depersonalization disorder, any primary disorder must be excluded, especially temporal lobe epilepsy, schizophrenia, depressive disorder, another dissociative disorder, and anxiety disorders (Sierra et al., 2012). Most patients who present with depersonalization will be found to have one of these other conditions.

Aetiology

The causes of a primary depersonalization disorder are not known. Apart from the possible association with schizoid personality disorder, no definite constitutional factors have been identified. Jay et al. (2014) suggest that depersonalization is due to excessive inhibition of the insula by ventrolateral prefrontal cortex. At the same time, autonomic responses to unpleasant stimuli are reduced, suggesting inhibition of emotional processing, which could lead to a subjective sense of emotional numbing.

Prognosis

The rare primary depersonalization disorder has not been followed systematically. Clinical experience indicates that cases lasting longer than a year have a poor long-term outcome.

Treatment

When depersonalization is secondary to another disorder, treatment should be directed to the primary condition. Supportive interviews can help the patient to function more normally despite the symptoms, and any stressors should be addressed. No specific pharmacological or psychological treatments are well established. In this, as in other conditions that are difficult to treat, it is important to resist the temptation to give ineffective treatments to appear to be doing something for the patient. It is better to give adequate time for supportive care. For review, see (Hunter et al., 2017)

Other dissociative syndromes in ICD-11

Dissociative fugue

Dissociative fugue is no longer a separate diagnosis in DSM-5 but a specifier for dissociative amnesia. It is extremely rare, with a loss of memory coupled with wandering away from the person's usual surroundings. These people usually deny all memory of their whereabouts during the period of wandering, and some deny knowledge of personal identity. Many dramatic case histories have been published. The disorder must be distinguished from organic disorders, including epilepsy and substance intoxication. As in DSM-5, in ICD-11 dissociative fugue is not a separate category but there are codes for amnesia with and without dissociative fugue, fugue reaction to exceptional stress, and post-ictal fugue. (Box 22.3).

Dissociative stupor

In dissociative stupor, patients show the characteristic features of stupor. They are motionless and mute, and they do not respond to stimulation, but they are aware of their surroundings. Dissociative stupor is rare. It is essential to exclude other possible conditions, namely schizophrenia, depressive disorder, mania, and organic brain diseases.

Ganser's syndrome

Ganser syndrome is a very rare condition with four features:

- giving 'approximate answers' (see below) to questions designed to test intellectual functions
- psychogenic physical symptoms
- hallucinations
- apparent clouding of consciousness.

The syndrome was described in 1898 by Ganser in prisoners, but it is not confined to them. The term 'approximate answers' denotes answers (to simple questions) that are plainly wrong but are clearly related to the correct answer in a way that suggests that the latter is known. For example, when asked to add two and two a patient might answer five. The obvious advantage to be gained from the condition, coupled with the approximate answers, suggests malingering. However, the condition is often maintained so consistently that unconscious mental mechanisms have been thought to play a part. It is important to exclude an organic brain disease or schizophrenia; the former should be considered particularly carefully when muddled thinking and visual hallucinations are part of the clinical picture. In ICD-11 it is not separately named but falls under other dissociative syndromes.

For a review, see Dwyer and Reid (2004).

Psychiatric disorders in medical settings

Although common in all settings, the type and presentation of psychiatric disorders differ from one medical setting to another. The most common diagnoses in each setting are shown in Table 22.3. There are refinements to this, for example, acute psychoses are more often seen in emergency departments while chronic psychoses are more likely to be seen in medical and surgical wards or in primary care, although they may also present to emergency presentations.

Epidemiology

General practice. The most common psychiatric disorders in general practice are depression, anxiety, and stress-related disorders. BDDs (ICD-11), formerly known as somatoform disorders in ICD-10 and somatic symptom disorder in DSM-5, are also common and co-exist with depression, anxiety, and stress-related disorders.

Emergency departments. Deliberate self-harm often triggers presentation to emergency services. Although a minority of these patients have persistent psychiatric disorders, many have stress-related disorders, and some have depressive disorders. Intoxication and delirium related to alcohol and drugs are also common reasons for presentation, particularly in inner-city hospitals and among people involved in accidents. Some patients with BDD (and a few with factitious disorders) are frequent attenders at emergency departments (Wooden et al., 2009).

Medical and surgical outpatient clinics. About a third of people attending medical and surgical outpatient clinics have a psychiatric disorder (Maiden et al., 2003). Half of these have depressive and anxiety disorders, and most of the remainder have somatic symptom disorders. In both groups, adequate recognition and treatment of the psychiatric disorder should be an integral part of

Table 22.3 The relative prevalence of common psychiatric disorders in medical settings other than psychiatry

	General practice	Emergency department	Medical/surgical outpatients	Medical/surgical inpatients
Depression/anxiety	++	++	+++	+++
Delirium	–	+	–	+++
Alcohol misuse	++	+++	++	+++
Psychosis	+	+	–	–
BDD (formerly somatization)	+++	+	+++	++

Notes: – rare; + uncommon; ++ common; +++ very common.

management, since it has been shown to improve outcome: depression is a common cause of apparent worsening of a medical condition.

Medical and surgical wards. About 20% of medical and surgical inpatients have a depressive or anxiety disorder coexisting with their medical disease, 10% have a significant alcohol misuse problem, and at least a quarter of elderly inpatients have an episode of delirium (Saxena and Lawley, 2009). Some patients with severe BDD (somatic symptom disorders) undergo multiple investigations and even surgery before the diagnosis is made.

Presentation of psychiatric disorders in medical settings

Although psychiatric disorders commonly present in medical settings with psychological symptoms or behavioural disturbance, other less obvious presentations are common. These are as: (i) multiple physical complaints referred to as somatic symptoms; (ii) a medical management problem and/or difficulties with adherence to treatment; (iii) an apparent exacerbation of a medical condition.

A significant minority of patients seen in general practice and in hospital outpatient clinics have bodily symptoms that cannot be explained by organic pathology, and some of these might have a psychiatric disorder (Steinbrecher et al., 2011).

Bodily symptoms due to depressive and anxiety disorders. Depression is associated with bodily symptoms, such as fatigue, weight loss, and pain, which may lead to referral to a medical specialty. Anxiety is associated with symptoms of autonomic arousal, such as palpitations, and with breathlessness and sensory symptoms. A World Health Organization (WHO) collaborative study of patients presenting to primary care in 14 countries found a strong

association between bodily symptoms and depressive and anxiety disorders in all centres, despite different cultures and health services. Also, there was a linear relationship between number of bodily symptoms and the presence of depression and anxiety disorder (Simon et al., 1999). Among the anxiety disorders, panic disorder is an especially important cause of bodily symptoms, such as chest pain, dizziness, and tingling.

Psychiatric disorder presenting as apparent worsening of a medical condition

An exacerbation of complaints about symptoms or disability associated with a chronic medical condition is sometimes caused by a comorbid depressive disorder. *Refusal of treatment* may also be the first pointer to a psychiatric disorder, warranting careful assessment of social, psychological, and biological influences.

Comorbidity

Comorbidity of psychiatric disorders with other medical conditions is important. For example, an eating disorder may greatly complicate the treatment of diabetes, and depression is a risk factor for increased mortality and morbidity following myocardial infarction.

Epidemiology

Psychiatric disorder is present in as many as a third of patients with serious acute, recurrent, or progressive medical conditions. It is difficult to ascertain the exact proportion because standard criteria for the diagnosis of psychiatric disorder include some symptoms that can

also be caused by medical illness, for example, fatigue and poor sleep. Although modifications have been suggested to make the criteria more appropriate for use with people with physical as well as psychiatric disorder, none is wholly satisfactory. It is best to start with standard criteria and then use knowledge of the medical condition to decide which of the symptoms that point to psychiatric disorder could have originated in this other way. However, this approach requires skilled interviewers and is difficult to achieve on a large scale.

Suicidal thinking and behaviour should be assessed; these are more common in the medically ill than in the general population. There are associations with cancer, multiple sclerosis, and a number of other conditions (Webb et al., 2012).

Common associations between psychiatric and other medical illness are shown in Box 22.4. Epidemiological surveys consistently reveal extensive comorbidity in health conditions (Jacobi et al., 2004).

The importance of psychiatric comorbidity in the medically ill

A comorbid psychiatric disorder can greatly affect the impact and outcome of medical conditions, for example, in patients with ischaemic heart disease (Surtees et al., 2008), diabetes (Holt and, 2015), and cancer (Walker et al., 2013). A study of people with severe mental illness showed multimorbidity clusters that include substance misuse, atopy, and affective cardiovascular and complex multimorbidity, where each cluster includes seven to nine long-term conditions (Ma et al., 2022). Anxiety and depression are also risk factors for non-compliance with medical treatment. Multiple physical conditions create a health and financial burden on individuals, including

loss of work and reduced quality of life, as well as the healthcare system, and society.

The causes of psychiatric comorbidity in the medically ill

Psychiatric disorder may be present in medical patients for three main reasons:

1. By chance, as both are common.
2. The psychiatric disorder may have caused the medical condition, for example, alcohol dependence causing cirrhosis of the liver.
3. The medical condition may have caused the psychiatric disorder, either through an action of the disease or its treatment on the brain, or as a reaction to the psychological impact of the medical condition or side effects of its treatment, or as a result of the social effects of the medical condition or its treatment (for example, loss of employment). These factors interact with the person's premorbid vulnerability.

Biological mechanisms. In addition to delirium and dementia, several other psychiatric disorders may be caused by effects of the medical condition on the brain. Conditions that may act in this way include acute infection, endocrine disorders, and some forms of malignancy. This resulting psychiatric condition is referred to as an 'organic' mental disorder (see Chapter 2). The principal medical conditions that may act in this way are listed in Box 22.5.

Box 22.4 Psychiatric disorders that are common in long-term conditions

Adjustment disorder
Severe depression
Anxiety disorders
Acute stress disorder
Post-traumatic stress disorder
Bodily distress disorder
Substance misuse
Eating disorder
Sleep disorder
Factitious disorder
Sexual disorders

Box 22.5 Medical conditions that may cause psychiatric disorder directly

Depression

Carcinoma
Infections
Neurological disorders (see Chapter 14)
Diabetes, thyroid disorder, Addison's disease
Systemic lupus erythematosus

Anxiety

Hyperthyroidism
Hyperventilation
Pheochromocytoma
Hypoglycaemia
Drug withdrawal
Some neurological disorders (see Chapter 14)

Table 22.4 Some drugs with psychological side effects

Drug	Side effect
Antiparkinsonian agents	
Anticholinergic drugs	Delirium, visual hallucinations
L-Dopa	Delirium, depression, psychotic symptoms
Cardiovascular drugs	
Methyldopa	Tiredness, weakness, depression
Clonidine	Drowsiness, fatigue, mood disturbances
Sympathetic blockers	Impotence, mild depression
Digitalis	Delirium, mood disturbance
Diuretics	Weakness, apathy, and depression (owing to electrolyte depletion)
Analgesics	
Salicylamide	Confusion, agitation, amnesia
Other examples	
Isoniazid	Acute organic syndrome and mania
Steroids	Irritability, mania, depression
Mefloquine	Psychotic symptoms, abnormal dreams, insomnia, depression, anxiety

Medical treatment may also cause psychiatric disorder by its effect on the brain. Table 22.4 lists some commonly used drugs that may produce psychiatric disorder as a side effect. Other treatments associated with psychiatric disorder include radiotherapy, cancer chemotherapy, and operations such as mastectomy.

Psychological and social mechanisms. The most common means by which a medical condition can cause psychiatric disorder is by its psychological impact. Certain types of medical conditions are particularly likely to provoke serious psychiatric consequences. These include life-threatening acute illnesses and recurrent progressive conditions. Psychiatric disorder is more common in chronic medical illness when there are distressing symptoms such as severe pain, persistent vomiting, and breathlessness, and where there is significant disability (Katon et al., 2007).

Patients at risk for both acute and persistent psychiatric disorder in the course of medical illness include those who:

- have developed psychological problems in relation to stress in the past
- have suffered other recent adverse life events
- are in difficult social circumstances.

The reactions of family, friends, employers, and doctors may affect the psychological impact of a medical

condition. They may reduce the consequences by their support, reassurance, and other help, or they may increase it by their excessive caution, contradictory advice, or lack of sympathy.

Prevention of psychiatric disorder in the medically ill

There are three main strategies.

1. Identify those at risk including those with pre-existing illness and those undergoing treatments associated with new onset or exacerbation of psychiatric disorder.
2. Minimize the negative effect of illness by providing good medical and nursing care.
3. Detect, and treat effectively, the early stages of any psychiatric disorder.

Effectiveness of psychiatric treatments in the medically ill

Clinical experience indicates that psychiatric treatments are generally effective in patients who are also medically ill. There was, however, limited evidence from randomized controlled trials because comorbid patients

are usually excluded—but there are now more trials of antidepressants in medically ill patients and a clearer indication of their value (Taylor et al., 2011). Clinical experience also suggests that psychological treatments such as CBT are effective in medically ill patients, and again there is now some evidence from clinical trials (van Kessel et al., 2008). There is better evidence for the benefits of ‘collaborative care’, in which medical and psychiatric treatment are coordinated. This is especially in the management of chronic medical conditions (Katon, 2003), where there is evidence of considerably reduced treatment costs, primarily because of shorter inpatient stays (Koopmans et al., 2005).

Management of psychiatric disorder in the medically ill

Assessment is as for psychiatric disorder as in other circumstances, but with the additional need to:

- Be well informed about the medical condition and its treatment.

Agree the type and pace of treatment with each patient and openly discuss uncertainty (a driver of anxiety), changes to social functioning (losses that can lead to depression), and other impacts of medical diseases.

- Distinguish anxiety and depressive disorders that might benefit from treatment, from normal emotional responses to physical illness and its treatment. Symptoms that seldom occur in normal distress (such as hopelessness, guilt, loss of interest, and severe insomnia) help to make the distinction.
- Be aware that medical conditions and their treatment may cause symptoms such as fatigue and loss of appetite, which are commonly used to diagnose psychiatric disorder.
- Explore the patient’s understanding of the medical condition and its treatment.

General considerations

The nature of the medical condition and its treatment should be explained clearly and the opportunity provided for patients to express their worries and fears. The treatment for the associated psychiatric disorder is conducted using the methods appropriate for the same disorder in a physically healthy person. However, careful consideration should be given to possible interactions between the proposed psychiatric intervention and the medical condition and its treatment, and whether the patient can engage with and manage any adverse effects of psychiatric treatment. Treatment can often be

given by the general practitioner or medical specialist, but more complex cases may require a psychiatrist, for example, if there are concerns about risks to self or others, or questions about capacity to consent, or behavioural disturbance.

Medication

Anxiolytic and sedative drugs can give valuable short-term relief when distress is severe. The indications for antidepressants are generally the same as those for patients who are not physically ill, but in the presence of mild to moderate depression with physical illness, the National Institute for Health and Care Excellence (NICE) recommends CBT with antidepressants (Pilling et al., 2009). The side effects and possible interactions of the psychotropic drugs with other medication should be considered carefully before prescribing.

Psychological treatments

Explanation and advice are part of the treatment of every patient. CBT may be chosen to reduce distress, increase adherence to treatment, reduce disproportionate disability, and to modify lifestyle risk factors.

Bodily symptoms not explained by physical pathology

Bodily symptoms that are not clearly associated with physical pathology are common in the general population and in patients in all medical settings (Mayou and Farmer, 2002). Although most of these symptoms are transient, a minority are persistent and disabling, and a cause of frequent medical consultations. Some conditions that may give rise to physical symptoms are shown in Box 22.5.

As noted, the most common association of physical symptoms with psychiatric disorder is with anxiety and depressive disorders, and some conditions meet the criteria for a somatic symptom disorder. Some physical symptoms can be explained by hypervigilance, health beliefs about physical causes, lower pain thresholds, illness behaviours for secondary gain, and psychological elaboration of underlying physical pathology. A few of these patients have a factitious disorder, and a very small number indeed are malingering. However, many of those who present to doctors with unexplained somatic symptoms do not meet the diagnostic criteria for any of these conditions. Even in these undiagnosed cases, however, psychological and social factors are often important as causes of the symptoms and as reasons for seeking help, and psychological measures are often helpful.

Terminology

Many terms have been used to describe medically unexplained somatic symptoms, and none is wholly satisfactory. These terms include hysteria, hypochondriasis, somatization, somatoform symptoms, functional somatic symptoms, and functional overlay. Although important in terms of understanding past diagnostic systems, we adopt the ICD-11 nomenclature and seek to avoid unnecessarily dualistic terms.

Epidemiology

Medical symptoms unrelated to physical disorders are common in the general population and in people attending primary care; they are more frequent in women than in men (Guthrie, 2008). Although most of these symptoms are recognized as not serious and although most are short in duration, a sizeable minority lead to distress, functional disability, and role impairment with more time off work, consulting doctors, and taking medication (Harris et al., 2009).

Aetiology

Although unexplained by physical pathology, something is known about the causes of these medical symptoms. Most arise from misinterpretation of the significance of normal bodily sensations, which are interpreted as a sign of disease. Concern leads to the focus of attention on the sensations and this leads to even greater concern, apprehension, and anxiety, which exacerbate and maintain the symptom. For example, awareness of increased heart rate when excited or anxious may lead to worry about heart disease, restriction of daily activities, and repeated requests for investigation and reassurance. Good communication by clinical staff can help to counter the effects of such misinterpretations.

Box 22.6 lists sources of bodily sensations that may be misinterpreted as symptoms of disease.

Box 22.6 Physiological sources affecting bodily sensations

Sinus tachycardia and benign arrhythmias
Effects of fatigue
Hangover
Effects of overeating
Effects of prolonged inactivity
Autonomic effects of anxiety
Effects of lack of sleep

Predisposing factors include:

- *Beliefs about illness* shape a person's response to sensations and symptoms. These beliefs may be related to personal experience of illness in earlier life, to involvement in the illness of family or friends, or to portrayals of illness in the media (see Box 22.7).
- *Adverse experiences in childhood.* Adults with medical symptoms in the absence of physical illness commonly report adversities in childhood, for example, poor parenting and various forms of abuse.
- *Social circumstances.* Physical symptoms are associated with poor socioeconomic circumstances and acute or chronic adversity.
- *Personality.* Some people may have concerns about bodily health going back to adolescence or earlier, and these can be viewed as part of their personality.

Perpetuating factors include:

- *Behavioural factors*, such as repeatedly seeking information about illness, and inactivity.
- *Emotional factors*, especially chronic anxiety and depression.
- *The reactions of others*, including doctors as well as family and friends. Doctors may inadvertently prolong the problem by failing to give clear, relevant information that considers the patient's individual fears and other concerns.

Prevention

Until we know more of the causes of physical symptoms in the absence of physical illness, there is no solid basis for longer-term prevention. While a reduction in predisposing factors in the population, such as childhood abuse and lack of parenting, might have some effect, it

Box 22.7 Experiences that may affect the interpretation of bodily sensations

Childhood illness and other adverse childhood experiences, including encounters with medical services, and absence from school.
Neurodevelopmental disorders including autism.
Illness among family and friends.
Physical illness in adult life.
Accounts of illness in the media.
Other sources of information about illness, including cultural idioms of distress and health beliefs.

is equally important and more practical to address factors such as poor communication by doctors (Ring et al., 2004) and inadequate treatment of depression and anxiety in patients who present with physical complaints. See Hotopf (2004) for a fuller discussion of prevention.

Assessment

An adequate medical assessment is essential. At the end of this the physician should explain:

- The purpose and results of all investigations carried out, and why it has been concluded that there is no medical cause for the symptoms.
- That the symptoms are nevertheless accepted as real and that it makes sense to seek other causes.

If the patient is then referred to a psychiatrist, the latter should be informed about the results of the investigations and of the way these have been explained to the patient.

The psychiatrist explores the nature and significance to the patient of the unexplained symptoms, and completes the usual psychiatric history and a mental state examination, with particular attention to:

- previous concerns about illness
- current beliefs about illness including cultural influences on explanatory models
- personality
- psychological problems
- detection of a depressive or anxiety disorder.

Social problems including housing, poverty, contextual, geographical, and political pressures.

Information should be sought from other informants as well as from the patient. Some services have screening

questionnaires for depression that are completed by the patients, either on their own or with the help of a member of the medical team

Management

The individual care plans should take account of the patient's special concerns, the type of symptoms, and any associated psychiatric disorder.

The management of chronic fatigue syndrome (CFS) and some other 'functional' medical syndromes is discussed below. The management of BDDs (formerly somatoform) is described later in the chapter.

After completing the basic procedures outlined in Box 22.8, many patients will be reassured. Those who are not reassured may seek repeated investigation and reassurance. Therefore, when all medically necessary investigations have been completed, the clinician should explain thoroughly why no further investigation is required. After these initial discussions, it is seldom helpful to engage in repeated arguments about the causes. Most patients are willing to accept at least that social and psychological, as well as biological, factors may influence their symptoms, and this acceptance can provide a basis for psychological management. Keeping a diary to explore the relationship of symptoms to psychosocial causes can be more convincing than further explanation.

Much can usually be achieved by the primary care team, and by the physicians using these basic procedures. However, chronic and recurrent problems may need to be referred to a psychiatrist or clinical psychologist for further treatment of associated psychiatric disorder, possibly involving cognitive therapy or dynamic psychotherapy. The psychiatrist can also help to coordinate any continuing medical care, especially when this

Box 22.8 Approaches to bodily symptoms in the absence of an identified physical illness

Emphasize that the symptoms are real and familiar to the clinician.

Explain the role of psychosocial factors in all medical conditions.

Offer and discuss a psychosocial explanation of the symptoms.

Allow adequate time for the patient and partner/family to ask questions.

Agree a treatment plan to include:

- Treatment of any minor medical problem contributing to the symptoms.
- Treatment of any associated psychiatric disorder (commonly anxiety or depression).
- If appropriate, improve fitness by graded activity.
- If appropriate, diary keeping to explore the relationship between symptoms and possible psychosocial causes, such as external life events.

involves several specialties and professions. In many cases one joint assessment (physician plus psychologist or psychiatrist) clarifies diagnoses (and what the patient does not have) to reduce further rounds of clinics and excessive help-seeking behaviour to find another diagnosis.

Further treatment should be based on the formulation of the person's individual problems. The treatment plan might include, for example, antidepressant medication, anxiety management, and cognitive therapy to change the inaccurate beliefs about the origin and significance of symptoms.

Antidepressant medication has been shown to benefit several kinds of medically unexplained symptoms, including fibromyalgia and irritable bowel syndrome. Facial pain may be helped by low-dose tricyclic antidepressants. This benefit is especially (but not exclusively) in patients with marked depressive symptoms (Somashekar et al., 2013).

CBT is moderately effective in the treatment of non-cardiac chest pain, irritable bowel, and CFS. Treatment delivered by psychologists has greater benefit than that given by primary care physicians (Gerger et al., 2015).

Prognosis

The prognosis for less complex cases of recent onset is good, but for chronic, multiple, or recurrent conditions it is much less so. For such cases the realistic goal may be to limit disability and requests for unnecessary medical investigation. Doctors need to recognize that they may also contribute to this process because doctors are as active in proposing biological explanations and tests as patients are in asking for them (Bensing and Verhaak, 2006). Promoting psychosocial explanations is one pathway to promoting long term self-management.

For fuller information about the management of medically unexplained symptoms, see Isaac and Paauw (2014).

Chronic fatigue syndrome

Many terms have been used to describe this syndrome, whose most prominent feature is chronic disabling fatigue: the terms include *postviral fatigue syndrome*, *neurasthenia*, and *myalgic encephalomyelitis* (ME). The descriptive term *chronic fatigue syndrome* (CFS) is now preferred. The diagnosis requires that the illness must have lasted for least six months and that other causes of fatigue have been excluded (Fukuda et al., 1994).

CFS has a long history. In the nineteenth century the symptoms were diagnosed as neurasthenia, and in ICD-10 the syndrome can still be recorded in this way, an

option that is chosen widely in China and some other countries. The diagnostic criteria for CFS overlap with those for several other psychiatric disorders, including depression, anxiety, somatoform disorders, and fibromyalgia. CFS remains a highly contested disorder with a powerful lobby against any psychiatric interpretation. The persistence of this conflict exemplifies the problems with defining these disorders.

The US Institute of Medicine has proposed renaming CFS as '*systemic exertion intolerance disease*', with the aim of improving acceptance and understanding of the patient experience and stimulating research into the pathophysiology of key symptomatology ICD-11 places benign myalgic encephalomyelitis and CFS separately, within the category of 'diseases of the nervous system', and specifically with the label of post-viral fatigue syndrome. The WHO concludes there is sufficient evidence of an infectious aetiology but insufficient consensus on reliable diagnostic patterns and aetiology. Persistent fatigue was a constant feature of all studies reviewed.

Clinical features

The central features are fatigue at rest and prolonged exhaustion after minor physical or mental exertion. These features are accompanied by muscular pains, poor concentration, and other symptoms from the list in Box 22.9. People with this condition are commonly most concerned to avoid activity. Frustration, depression, and loss of physical fitness are common.

Epidemiology

Surveys of the general population indicate that persistent fatigue is reported by up to a quarter of the population at any one time. The complaint is common among people attending primary care and medical outpatient clinics. However, only a small proportion of people who complain of excessive fatigue meet the former diagnostic criteria for CFS. Estimates of prevalence are around 0.5% of the general population with a marked (20–40-fold) predominance of females (Prins et al., 2006). CFS in children is considered in Chapter 16.

Aetiology

The aetiology of CFS is controversial. Suggested factors include:

- *Biological causes.* Several biological causes have been proposed, including chronic infection, immune dysfunction, muscle disorder, neuroendocrine dysfunction, and ill-defined neurological disorders. Despite strong claims for several specific causative agents, evidence is lacking. For example, *Lyme disease*

Box 22.9 Case definition of chronic fatigue syndrome

Inclusion criteria

Clinically evaluated, medically unexplained, fatigue of at least six months' duration that is:

- of new onset (not lifelong)
- not a result of ongoing exertion
- not substantially alleviated by rest
- a substantial reduction in previous level of activities.

The occurrence of four or more of the following symptoms:

- Subjective memory impairment
- Sore throat
- Tender lymph nodes
- Muscle pain
- Joint pain
- Headache
- Unrefreshing sleep
- Post-exertional malaise lasting more than 24 hours.

Exclusion criteria

- Active, unresolved, or suspected disease
- Psychotic, melancholic, or bipolar depression
- Psychotic disorders
- Dementia
- Anorexia or bulimia nervosa
- Alcohol or other substance misuse
- Severe obesity

Source: Fukuda, K., Straus, S. E., Hickie, I., et al. (1994). The chronic fatigue syndrome: A comprehensive approach to its definition and management. International Chronic Fatigue Syndrome Study Group. *Ann Intern Med*, 121(12), 953–959. Copyright (1994), The American College of Physicians.

(Lyme borreliosis) has been said to produce chronic fatigue, myalgia, and impaired concentration (Stanek et al., 2012).

- *Psychological and behavioural factors.* These appear to be important, especially concerns about the significance of symptoms, the resulting focus of attention on symptoms, and avoidance of physical mental and social activities that worsen them. Many patients are depressed and/or anxious. However, not all are, and the intense opposition to a psychological explanation in some patients is very striking.
- *Social factors.* Stress at work is sometimes important. Belief that there must be a physical cause may be influenced by the stigma attached to a psychiatric

diagnosis, and by some of the information provided by patient groups and practitioners.

An alternative approach to aetiology is to consider predisposing, precipitating, and perpetuating factors:

- *Predisposing factors* include a history of major depressive disorder and perhaps personality characteristics such as perfectionism.
- *Precipitating factors* include viral infection and life stresses.
- *Perpetuating factors* may include neuroendocrine dysfunction, emotional disorder, attribution of the whole disorder to physical disease, coping by avoidance, chronic personal and social difficulties, and misinformation from the media and other sources.

Course and prognosis

For established cases, the long-term outcome has been considered poor. However, it must be remembered that most studies of prognosis relate to patients referred to specialist centres who may have experienced long-term fatigue before the referral. Also, most studies refer to prognosis before modern treatments (see below) were widely used. A recent long-term study found a sixfold increase in suicide, but no overall excess mortality, in CFS (Roberts et al., 2016).

Treatment

Many treatments have been suggested but very few are of proven efficacy. There have been several randomized controlled trials of CBT and of graded exercise regimes, which have shown the benefits of these procedures over standard medical care alone, and over relaxation therapy.

The PACE study (White et al., 2011) and its follow-up (Sharpe et al., 2015) indicate a strong advantage to CBT and graded exercise rather than 'living within the dictates of the disorder', as recommended by many CFS groups. The findings have been challenged and led to NICE removing the intervention from its guidelines, although there remains controversy (Flottorp et al., 2022).

Management

Assessment. It is important to enquire carefully about depressive symptoms, and other mental health-related complaints, as part of a comprehensive biopsychosocial assessment. Although extensive medical investigations are unlikely to be rewarding, the psychiatrist can usefully collaborate with a physician when assessing patients, especially if the symptoms related to a specific

body system or medical differential diagnosis. The formulation should refer to any relevant aetiological factors (see above).

Starting treatment. The six basic steps are to:

1. Acknowledge the reality of the patient's symptoms and the disability associated with them.
2. Provide appropriate information about the nature of the syndrome to the patient and to the family.
3. Present the aetiological formulation as a working hypothesis to be tested, not argued over.
4. Treat identifiable depression and anxiety.
5. Encourage gradual return to normal functioning by overcoming avoidance and regaining the capacity for physical activity.
6. Help with any occupational and other practical problems.

Medication. After establishing a clear and mutually agreed therapeutic care plan, and when there is a definite evidence of a depressive disorder, antidepressant drugs can be prescribed in usual doses. Clinical experience suggests that selective serotonin reuptake inhibitor (SSRI) drugs are best tolerated. Antidepressant drugs may also be useful in reducing anxiety, improving sleep, and reducing pain. Low-dose tricyclic antidepressants can have a role here.

Psychological treatment. On the simplest level, psychological approaches include education about the condition, and the correction of misconceptions about cause and treatment. Cognitive behaviour methods include addressing misconceptions about the nature of the condition and excessive concerns about activity; and encouraging a gradual increase in activity. Associated personal or social difficulties can be addressed using a problem-solving approach (Harvey and Wessely, 2009).

Irritable bowel syndrome

Irritable bowel syndrome is characterized by abdominal pain or discomfort, with or without an alteration of bowel habits, persisting for longer than three months in the absence of any demonstrable disease.

The condition occurs in as many as 10% of the general population (Wilson et al., 2004) the majority of whom do not consult a doctor. The cause of the syndrome is uncertain, although there appears to be a disturbance of bowel function and sensation. Depressive and anxiety disorders are common among people who attend gastroenterology clinic with irritable bowel syndrome, especially among those who fail to respond to treatment. People with mild symptoms usually respond

to education, reassurance about the absence of serious pathology, change in diet, and, when required, antispasmodics or laxative drugs. More severe and chronic symptoms may need additional treatment. CBT has been shown to be of benefit, and so have tricyclic antidepressants, although SSRIs do not seem particularly useful (Hookway, 2015), except where more prominent anxiety and/or depression is present.

Fibromyalgia

The term fibromyalgia refers to a syndrome of generalized muscle aching, tenderness, stiffness, and fatigue, often accompanied by poor sleep. A physical sign of multiple specific tender points has been described, but it is probably non-specific. Women are affected more than men, and the condition is more common in middle age. The aetiology is uncertain, but there is a marked association with depression and anxiety. A network meta-analysis shows limited evidence for the efficacy of either pharmacological or psychological therapy, but a combination of antidepressant, aerobic exercise, and CBT shows some promise (Nüesch et al., 2013).

Factitious disorder

ICD-11 broadly defines factitious disorder as the falsification or feigning of physical or psychological signs or symptoms, or induction of injury or disease, associated with identified deception. The category is divided further into that imposed on the self and that imposed on another, or unspecified. ICD-11 specified deceptive behaviour is not motivated by external reward such as disability payments or evading criminal proceedings. Malingering, where there are external rewards, is not included in the diagnosis. In factitious disorder, symptoms are feigned to enable the person to adopt a sick role and obtain medical care (in malingering, symptoms are feigned to obtain other kinds of advantage—see below). The term *Munchausen's syndrome* denotes an extreme form of this disorder (described below).

The prevalence of factitious disorder is not known with certainty, but it accounts for about 1% of referrals to consultation liaison services. The disorder usually begins before the age of 30 years. 'Illness deception' in physical disorder, covering both factitious disorder and malingering, is considered relatively common, albeit mild, in occupational health services, where one study (Poole, 2010) found it in 8% of consecutive attendees. The cause is uncertain, in part because many affected individuals give histories that are inaccurate. There is often a history of parental abuse or neglect, of chronic illness in early

life, with many encounters with the medical services, sometimes long periods in hospital, and sometimes alleged medical mismanagement. Previous substance misuse, mood disorder, and personality disorder are other common features. Many patients have worked in occupations related to medicine. This is variable but the condition is usually long lasting. Few patients accept psychological treatment, but some improve during supportive medical care. In some cases, there is evidence of other disturbed behaviours, including abuse of children and (on the part of those working in health professions) harm to patients.

Assessment. When factitious disorder is suspected, the available information should be reviewed carefully, including the history given by informants as well as that provided by the patient. A psychiatrist may be able to assist in this assessment, and in cases of doubt further specialist medical investigation may be needed. Additional evidence may be obtained by careful observation of the patient, but the ethical and legal aspects of any proposal to make covert observations should be considered most carefully.

Starting treatment. When the diagnosis has been made, the doctor should explain to the patient the findings and discuss their implications. This should be done in a way that conveys an understanding of the patient's distress and makes possible a discussion of potential psychosocial causes. Although some patients admit at this point that the symptoms are self-inflicted, others persistently deny this. One should not simply assume in this case that the patient is being deliberately misleading. Examining these patients often yields a sense of self-deception where the process lies somewhere between conscious (fabulist) and unconscious processes. In these cases, management should still be directed to helping

the patient identify and overcome associated psychological and social difficulties, in the hope that improvement in such problems may be followed eventually by a lessening of the factitious disorder. Staff who have been caring for the patient while the investigations were being carried out may feel deceived and become angry when they discover that the patient has deceived them. Other staff may resist the diagnosis if they have been persuaded and enlisted by the patient to make the case for them. Such feelings make management more difficult, and the psychiatrist should play a part in resolving them through discussion, and by explaining the nature and severity of the patient's psychosocial problems. All closely involved staff should be involved in agreeing to a treatment plan that defines what future medical care, and what help with the associated problems, is needed, both for the patient and/or the family. Special risks and difficult ethical problems may arise when the patient is a healthcare worker—see Box 22.10. See Bass and Halligan (2014) for a review of factitious disorder.

Munchausen's syndrome

Richard Asher (1951) suggested the term *Munchausen's syndrome* for an extreme form of factitious disorder, in which patients attend hospital with a false but plausible and often dramatic history suggesting serious acute illness. Often the person is found to have attended and deceived the staff of many other hospitals, and to have given a different false name at each. Many of these people have scars from previous (negative) exploratory operations.

People with this disorder may obstruct efforts to obtain information about them, and some interfere with diagnostic investigations. When further information is obtained, especially that about previous hospital attendance, the patient often leaves. The aetiology

Box 22.10 Healthcare workers with factitious disorder

Highly publicized cases of serious physical harm to patients caused by a small number of healthcare workers with factitious disorder, or by those producing factitious disorder by proxy, have aroused great public concern. It is essential, therefore, in the management of these patients to consider the risk to others if the patient continues to work in healthcare.

These infrequent cases can present difficult medico-legal problems. If the diagnosis is in doubt, it may be judged necessary to seek additional evidence by searching the patient's belongings for items (such as needles

or medication) that could have been used to simulate symptoms. In general, it is unethical to search patients' belongings without first explaining the reason and obtaining permission. If the patient refuses to be searched, and there is a serious risk to others should the diagnosis be missed, it is usually appropriate to obtain advice from one or more experienced professionals, including a medicolegal opinion. Advice may be needed also when judging risk, when the risk is serious, and when the patient refuses to allow discussion with the employers in deciding whether to breach confidentiality.

and long-term outcome of this puzzling disorder are unknown.

Fabricated or induced illness

Fabricated or induced illness (FII; sometimes known as *factitious disorder by proxy* or *Munchausen disorder by proxy*) is a rare form of child abuse. A parent or carer, usually the child's mother, exaggerates or gives false accounts of symptoms in their child, and may fake or deliberately induce signs of illness and seek repeated medical investigations and needless treatment for the child (Meadow, 1985).

The signs reported most commonly are neurological signs, bleeding, and rashes. Some children collude in the production of the symptoms and signs. The perpetrators (usually mothers) often have a severe personality disorder, including pathological lying, and many suffer from somatic symptom disorder themselves. Early abusive experiences are common (Bass and Jones, 2011). Hazards for children include disruption of education and social development. The prognosis is usually poor for both children and perpetrators, and there is a significant mortality. Occasional cases of murder of children by professional carers have been described as an extreme form of this disorder.

Factitious disorder by proxy is a child protection issue. Hence, possible cases need to be investigated by a multi-professional team, including social care and police. The diagnosis should be made with caution, and only after detailed investigation. In the United Kingdom, a number of high-profile legal cases have highlighted the danger of diagnosis on insufficient positive evidence and without adequate exclusion of other causes of the child's symptoms (Bass and Glaser, 2014).

Malingering

Malingering is not a medical diagnosis but a description of behaviour. The term denotes the deliberate simulation or exaggeration of symptoms for the purpose of obtaining some gain, such as financial compensation. The distinction from factitious disorder and from somatic symptom disorder can be difficult because it requires an accurate understanding of the person's motives (Bass and Halligan, 2014).

Malingering is infrequent and is most often encountered among prisoners, the military, and people seeking compensation for accidents. Several kinds of clinical picture have been described:

- malingered medical conditions and disability;
- malingered psychosis, seen in those wishing to obtain admission to hospital for shelter or to prolong their

stay in hospital, and in criminal defendants trying to avoid standing trial or to influence sentencing;

- malingered or exaggerated post-traumatic stress disorder;
- malingered cognitive deficit.

Ganser's syndrome (see page 616) is thought by some to be a form of malingering.

Assessment. This depends on careful history-taking and clinical examination, and a watch for discrepancies in the person's behaviour. Lawyers or insurers sometimes use surveillance by video or other means to detect the behaviour but, for ethical reasons, clinicians seldom do. Psychological tests have been used to aid detection, but none is of proven validity.

Management. When malingering is certain, the patient should be informed tactfully and their situation discussed in a non-judgemental way. They should be encouraged to deal more appropriately with any problems that led to their behaviour and, in appropriate cases, offered some brief face-saving intervention as a way to give up the symptoms. These dilemmas are considered in a comprehensive review of the literature (Eastwood and Bisson, 2008).

Body dysmorphic disorder

Body dysmorphic disorder was first described by Morselli in 1886 as 'a subjective description of ugliness and physical defect which the patient feels is noticeable to others'. The preoccupation with the imagined defect in appearance is usually an overvalued idea, but individuals 'can receive an additional diagnosis of delusional disorder (Chapter 12); there is also overlap with hypochondriasis and obsessive-compulsive disorder. Indeed, in both DSM-5 and ICD-11, body dysmorphic disorder is classified with 'obsessive-compulsive and related disorders'.

Patients with dysmorphophobia are convinced that some part of their body is too large, too small, or misshapen. To other people the appearance is normal, or there is a trivial abnormality. In the latter case, it may be difficult to decide whether the preoccupation is disproportionate. The common concerns are about the nose, ears, mouth, breasts, buttocks, or penis, but any part of the body may be involved. Patients may be constantly preoccupied with and tormented by their mistaken beliefs. It seems to them that other people notice and talk about the supposed deformity. They may blame all their other difficulties on it: if only their nose/breasts were a better shape, they would be more successful in their work, social life, or sexual relationships. Time-consuming

behaviours which aim to re-examine, improve, or hide the perceived defect are frequent. Social impairment is considerable. There is substantial comorbidity, especially with major depression and social phobia.

The condition usually begins in adolescence. It is chronic, though it often fluctuates over time. It is probable that there is some improvement over many years but there have been no long-term prospective studies.

The severe cases described in the psychiatric literature are infrequent, but less-severe forms of dysmorphophobia are more common, especially among those seeking plastic surgery or consulting dermatologists. As with the more severe cases, many meet diagnostic criteria for other disorders.

This should include questions about the nature of the preoccupations with appearance and of the ways in which this has interfered with personal and social life. Diagnosis can be difficult because some sufferers fail to reveal the precise nature of their symptoms because of embarrassment. This failure may result in misdiagnosis such as social phobia, panic disorder, or even a psychotic disorder.

When body dysmorphic disorder is secondary to a psychiatric disorder such as major depression, the latter should be treated in the usual way. The treatment of primary body dysmorphic disorder is often difficult. It

is essential to establish a working relationship in which the patient feels that the psychiatrist is sympathetic, understands the severity of the problems, and is willing to help. Since many patients will be requesting surgery, it is important to explain the lack of success of this approach and suggest that there are other effective treatments. There is some modest support with SSRIs and for CBT (Phillips, 2004; Wilhelm et al., 2014). Counselling and practical help should be provided for any occupational, social, or sexual difficulties that accompany the condition. Although some patients are helped by this approach coupled with continued support, many are not.

Cosmetic surgery is often successful for patients with conditions other than dysmorphophobia. However, surgery is usually contraindicated for people who have body dysmorphic disorder because many remain dissatisfied after the operation. Selection for surgery therefore requires careful assessment of the patient's expectations. Those with the most unrealistic hopes generally have poorer prognoses. Assessment is difficult, and collaboration between plastic surgeons, psychiatrists, and psychologists is valuable. Unintended rebuffs by surgeons or psychiatrists can increase the difficulties of management.

Psychological medicine services

Psychiatric services for general hospitals are named and organized in several ways.

Psychological medicine. This refers to the integration of a psychiatric team into the main body of an acute hospital and its respective specialties, with shared recording systems, often joint working on ward rounds and within teams (Sharpe, 2014). These psychological medicine services can take a proactive approach to identification of patients with mental health needs rather than wait for a referral.

Consultation–liaison psychiatry (sometimes known as C–L psychiatry or as *liaison psychiatry*) is the traditional term. In consultation work, the psychiatrist gives opinions on patients referred by physicians and surgeons; in liaison work the psychiatrist is a member of a medical or surgical team and offers advice about any patient to whose care he feels able to contribute.

Behavioural medicine is the term for similar arrangements provided by clinical psychologists rather than psychiatrists.

Psychosomatic medicine is a term used widely in the past in some countries, and which is still current in Germany. It has been revived in the United States as the name for consultation–liaison psychiatry as a recognized subspecialty (Gitlin et al., 2004).

The composition of services

Consultation–liaison units vary in their size and organization. Some are staffed entirely by psychiatrists, and others by a multidisciplinary team of psychiatrists, nurses, social workers, and clinical psychologists. A few services have inpatient beds for patients who are both medically and psychiatrically ill. In some North American hospitals with large consultation–liaison services, up to 5% of all admissions are referred to psychiatrists. In the United Kingdom and many other countries, a smaller proportion of inpatients are referred, most being emergencies, especially those with deliberate self-harm (see Chapter 21). Most of the literature on consultation–liaison

psychiatry has focused on inpatients, but most of the patients referred are from outpatient clinics and emergency departments.

Consultation–liaison psychiatry is increasingly concerned with the provision of psychiatric and psychological assessment, and the collaborative management of patients with either medically unexplained symptoms or psychiatric disorders comorbid with chronic medical illness. A major challenge is to find ways of extending these overstretched services to similar patients treated in primary care.

For further information about consultation and liaison psychiatry and psychosomatic medicine, see Leentjens et al. (2011).

Some emergencies

General approach

The successful management of any psychiatric emergency depends importantly on the initial clinical interview. The aims are those of any assessment interview: to establish a good relationship with the patient; elicit information from the patient and other informants; and observe the patient's behaviour and mental state. A relaxed, sympathetic, and structured approach helps to calm the situation enough for the doctor to reach an understanding of the patient's concerns and suggest a plan to which the patient will agree. In an emergency, it may not be possible to conduct a full assessment interview, but the assessment should be as systematic and as complete as the circumstances permit.

The anxious patient

Panic attacks. The somatic symptoms of a panic attack are frequent reasons for an emergency presentation. Common features include non-cardiac chest pain, tingling in the extremities, and the effects of hyperventilation (see pages 164, 179). Most patients can be talked through an episode of panic, and hyperventilation responds to rebreathing into a paper bag. Occasionally a small dose of a benzodiazepine is needed to control the anxiety. Follow-up treatment for panic disorder may be required.

Severe generalized anxiety may complicate any medical presentation. Attendance at an emergency department can be a frightening experience and the consequent anxiety may be made worse by the response of unaware staff. It is usually possible to reduce anxiety by explaining what is happening in a sympathetic manner. Occasionally a small dose of a benzodiazepine may be necessary to control the anxiety.

See also Chapter 3 for assessment of anxious patients.

The angry or aggressive patient

It can be very upsetting to clinicians and other carers when the person they are trying to help responds with anger. When this happens, it is essential to appear calm and avoid doing or saying anything that may increase the person's anger, as well as being mindful of the physical safety of patient, staff and others (see below). The clinician should try to find out and understand why the person is angry. Sometimes it is helpful to comment on the anger and to ask directly why the person is so upset. It is always unwise either to show anger in return or to be unduly submissive. It may be necessary to apologize for the problem that has caused the anger, for example, if the patient has been kept waiting for a long time.

If the patient is potentially violent, it is essential to arrange for adequate but unobtrusive help to be available. Physical contact (including physical examination) should not be attempted unless the purpose has been clearly understood by and agreed with the patient. If restraint cannot be avoided, it should be accomplished quickly by an adequate number of people using the minimum of force. Staff should not attempt single-handed restraint. Extreme caution is, of course, required if the patient could be in possession of an offensive weapon.

See also Chapter 18 for discussion of prevention and management of violence in psychiatric settings. These guidelines change in response to evidence, so readers should be aware of updates in their local settings. Interventions are delivered in the context of organizational policies and regular staff training in de-escalation and safe restraint.

Emergency drug treatment of disturbed or violent patients

Rapid tranquillization may be recommended if all other forms of management are unsuccessful. Diazepam (5–10 mg) may be useful for a patient who is frightened. Oral administration should be preferred. For more severely disturbed patients, intramuscular lorazepam (2 mg) or promethazine (50 mg) can be used; intramuscular haloperidol (2–5 mg) can also be employed for those who are psychotic and who have previously been exposed to antipsychotics. Intramuscular olanzapine (10 mg) is an alternative to haloperidol but should not be given with parenteral benzodiazepines. For further details, see Chapter 25 (page 710; see also pages 514, 515). For treatment of delirium, see Chapter 14.

When the patient has become calm, medication may be continued in smaller doses, usually three to four times a day and preferably by mouth, using syrup if the patient will not swallow tablets. The dosage depends on the patient's weight and on the initial response to the drug. Careful observations by nurses of the physical state and behaviour are necessary during this treatment.

Problems in consent to treatment

General principles relating to consent to treatment are discussed in Chapter 4. Psychiatrists can often be asked to advise on issues around the competency of patients and are sometimes asked to advise about patients who are refusing to accept medical or surgical treatment. There are several reasons why patients may be unwilling to accept treatment recommended to them. They may not have understood the information they have been given, they may be frightened or angry, or, occasionally, they may have a mental illness that interferes with their ability to make an informed decision.

A conscious mentally competent adult has the right to refuse treatment even after a full and rational discussion of the reasons for carrying it out. When the patient's condition is such that he cannot give consent, then, in the United Kingdom and many other countries, the doctor in charge of the patient has the right to give immediate treatment in life-threatening emergencies. The medicolegal issues are summarized in Box 22.11 and are also discussed in Chapter 4.

See Boland et al. (2000) for a review of the management of psychiatric syndromes in intensive care units.

Psychiatric aspects of surgical treatment

Pre-operative problems

Patients who are about to undergo surgery are often anxious, and those who are most anxious before an operation are likely to be anxious afterwards. Most studies of psychological preparation for surgery have shown that it can reduce post-operative distress, especially when the preparation includes measures to improve coping.

Psychiatrists may be asked to assess patients before surgery. Common reasons for such a request include:

- clarification of the role of emotional factors in the patient's physical complaints
- uncertainty about the patient's cognitive state and capacity to provide informed consent
- help in the management of current psychiatric problems
- help in predicting the patient's response to surgery and their capacity to cooperate with post-operative treatment and rehabilitation, including needle phobias and fears about medical procedures.

Psychiatric problems in the post-operative period

Delirium

Delirium is common after major surgery, especially in the elderly. Whether delirium develops depends on the type of surgery, the type of anaesthetic, the presence of post-operative physical complications, and the type of medication. Delirium is associated with increased mortality and longer stay in hospital (see Chapter 14).

Box 22.11 Medicolegal and ethical issues: patients who refuse to accept advice about emergency treatment

- In life-threatening emergencies where it is not possible to obtain the patient's consent (consciousness is impaired, or there is evidence of psychiatric disorder which cannot be immediately assessed), opinions should be obtained from medical and nursing colleagues, and, if possible, from the patient's relatives. Detailed records should be kept of the reasons for the decision. It is essential for all doctors to know the law about these matters in the country in which they are practising.
- If a patient has a *mental disorder that impairs the ability to give informed consent*, it may be appropriate to use legal powers of compulsory assessment and treatment of the mental disorder. In the United Kingdom, the powers for compulsory treatment of a mental disorder do *not* give the doctor a right to treat concurrent physical illness against the patient's wishes. However, after successful treatment of the psychiatric disorder, the patient may decide to give informed consent for the treatment of the physical illness. See also Chapter 4.

Pain

Psychiatrists are sometimes asked to advise on the management of patients with unusually severe post-operative pain. Patients who are given greater control over the timing of analgesia usually experience less pain and make less use of analgesia. Some are helped by anxiety management, and others by help in resolving anger arising, for example, from disagreements with the staff who are caring for them.

Long-term psychological problems after surgery

Adjustment problems are particularly common after mastectomy and laryngectomy, and after surgery that has not led to the expected benefit. Psychiatrists can sometimes contribute to the management of such problems, especially when the surgery is part of the treatment of a relapsing, chronic, or progressive disorder. The psychiatrist may help to discover psychosocial factors that are impeding adjustment, and to help the patient resolve or come to terms with the problems. Some patients require antidepressant medication.

Limb amputation

Limb amputation has different psychological consequences for young and for elderly people. Young adult amputees, such as those losing a leg in a road accident or in military action, characteristically show denial at first, and may later experience depression and phantom limb pains, which resolve slowly. Children and adolescents seem to have a similar outcome. Older people usually undergo amputation after prolonged problems associated with vascular disease. Such patients do not commonly report severe distress immediately after the operation, but they often develop phantom limb pain. Some have difficulty with the prosthesis and show a degree of functional incapacity disproportionate to their physical state.

Plastic surgery

People with physical deformities often suffer from embarrassment and distress, and this may markedly restrict their lives as children and as adults. When such patients are psychiatrically healthy, plastic surgery usually gives good results. Even when there is no major objective defect to match the concerns about appearance, cosmetic surgery to the nose, face, and breasts is usually successful. Nevertheless, psychological assessment can contribute to assessment before plastic surgery: the outcome is likely to be poor in patients who have delusions about their appearance, dysmorphophobia, or greatly

unrealistic expectations, or have been dissatisfied with previous surgery. See von Soest et al. (2009) for a review.

Bariatric surgery

There is a high prevalence of psychiatric disorders among candidates for bariatric surgery. There is also evidence that psychiatric symptoms decrease post-surgery, although these improvements may not persist (Müller et al., 2013; Sarwer and Heinberg, 2020).

Organ transplantation

These operations have considerable associated psychiatric consequences related to the nature of the surgery, and the need for continuing intensive medical care.

Selection for transplantation. The selection process can be stressful, and so can the wait for a suitable organ, though the latter tends to be less so for kidney transplants because there is the alternative of continuing dialysis during the waiting period. There are few psychological contraindications for these operations, mostly relating to inability to cope with the demands of the necessary long-term post-operative care. After operation, transplantation is associated with the same psychiatric and emotional problems that occur after other major surgery, especially anxiety, delirium, and depression. There are also some problems specific to a particular type of transplant, for example, liver transplantation has the highest rate of pre-operative and post-operative neuropsychiatric complications.

These problems are sufficiently frequent and serious to justify psychiatric liaison with transplant units to support staff and train them to recognize and respond appropriately to patients who have such problems (Heinrich and Marcangelo, 2009).

Diabetes mellitus

Diabetes is a chronic condition requiring prolonged medical supervision and informed self-care, and many physicians emphasize the psychological aspect of treatment. It is associated with an increased prevalence of psychiatric disorder. Moreover, type 2 diabetes is two to three times more common in patients with severe mental disorder, related both to shared disease mechanisms and the effects of antipsychotic and antidepressant drugs (Holt and Mitchell, 2015), highlighting the need for psychiatrists to be aware of the features and management of diabetes. For a general review of psychiatric aspects of diabetes see Doherty (2015) and Ducat et al. (2014).

Psychological factors and diabetic control

Psychological factors are important in established diabetes because they influence its control, and it is now generally accepted that good control of blood glucose is the most important single factor preventing long-term complications. Psychological factors can impair control in two ways. First, stressful experiences can lead to endocrine changes. Second, many diabetics do not adhere well to their treatment regime, especially at times of stress, and this is an important cause of 'brittle' diabetes.

Problems of being diabetic

Psychological and social problems may be caused by restrictions of diet and activity, the need for self-care, and the possibility of serious physical complications such as vascular disease and impaired vision. Although most diabetic people overcome or adapt to such problems, an important minority have difficulties. Adherence to regimes for testing, diet, and insulin is often unsatisfactory so that glycaemia control is less than optimal. Problems of this kind are particularly likely in adolescents with type 1 diabetes.

Other problems

Associated eating disorder. Control of diabetes is more difficult when the diabetic person has an eating disorder, particularly when insulin is part of the treatment regimen, since it can be misused to lose weight.

Associated medical complications. Psychosocial problems are more common in diabetics who have severe medical complications such as loss of sight, renal failure, and vascular disease. Diabetic neuropathy is often very painful and can restrict activities.

Sexual problems are common among diabetics. Two kinds of impotence occur in diabetic men. The first is psychogenic impotence of the kind found in other chronic debilitating diseases. The second kind is more common in diabetes and may pre-date other features of the disease. It is thought to be associated with pelvic autonomic neuropathy and vascularity, while endocrine factors may play a part.

Pregnancy is a difficult time for diabetic women, with problems in the control of diabetes and increased risks of miscarriage and foetal malformations.

Organic psychiatric syndromes in diabetic patients

Delirium. The first evidence of developing diabetic coma is sometimes an episode of disturbed behaviour, starting either abruptly or insidiously. The cause of the behaviour becomes clearer as other prodromal symptoms develop, including thirst, headaches, abdominal pain, nausea,

and vomiting. The pulse is rapid, and blood pressure is low. Dehydration is marked and acetone may be smelt on the breath.

Dementia. Diabetes is a risk factor for dementia (see Chapter 14).

Psychiatric aspects of diabetes management

Diabetes has been the object of considerable psychiatric research, involving a range of psychological treatments and particularly in adolescent and brittle diabetics. Possible measures include treatment for depressive disorder; blood glucose awareness training to improve the ability to recognize and act on fluctuations in blood glucose concentrations; weight management programmes; CBT approaches to improve self-care; help with psychosocial problems; and treatment of sexual dysfunction. Tricyclic antidepressants may be helpful in relieving the pain of neuropathy. See further reading for a review of psychological care of patients with insulin-dependent diabetes, and for guidance on psychiatric care in diabetes (Doherty, 2015; Plack et al., 2010).

Other endocrine disorders

Many endocrine disorders, most conspicuously thyroid and adrenal dysfunction, have been associated with psychiatric complications. Box 22.12 summarizes some of the more common associations, together with aspects of steroid therapy.

See Doherty and Dinneen (2021) for a review of the psychiatric aspects of endocrine disorder.

Other metabolic and autoimmune disorders

Psychiatric aspects of systemic lupus erythematosus and autoimmune encephalitis are covered in Chapter 14. For other metabolic and medical disorders, such as the porphyrias, sarcoidosis, and myasthenia gravis, see Harrison and Kopelman (2009).

Cardiac disorders

It has long been suggested that emotional disorder predisposes to ischaemic heart disease. Recent research has concentrated on several possible risk factors, including chronic emotional disturbance, social and economic disadvantage, overwork and other chronic stress, and the 'type A personality'. The latter consists of hostility, excessive competitive drive, ambitiousness, a chronic sense of urgency, and a preoccupation with deadlines (Friedman and Rosenman, 1959). Although type A behaviour has

Box 22.12 Psychiatric aspects of other endocrine disorders and steroid therapy

Hyperthyroidism (thyrotoxicosis): May present with psychiatric symptoms such as anxiety, irritability, emotional lability, and difficulty in concentrating. These, together with hyperactivity, fatigue, and tremor, may make differential diagnosis from anxiety disorder difficult. Treatment of thyroid dysfunction usually results in improvement of psychiatric symptoms.

Hypothyroidism (myxoedema): Cognitive impairment and other psychiatric disorders are common. Mood disorder may take a rapid-cycling form. Replacement therapy may reverse the mood symptoms, but neuropsychiatric problems may persist. See Dugbartey (1998) for a review.

Hyperadrenalism (Cushing's syndrome): Depressive symptoms are common. Their severity is not closely related to plasma cortisol concentrations, and personality and stressful circumstances may play a part. Psychological symptoms usually improve quickly when

the medical condition is controlled. Paranoid symptoms also occur, especially in those with the most severe illness. See Pivonello et al. (2015).

Steroid therapy: Affective symptoms, especially euphoria or mild mania, are frequent. Paranoid symptoms are less common. The severity of the mental disorder is not closely associated with dosage. Symptoms usually improve when the dosage is reduced but, when severe, lithium prophylaxis should be considered for patients who need to continue steroid treatment after a mood disorder has been controlled. Withdrawal of corticosteroids may cause lethargy, weakness, and joint pain.

Anabolic steroids: Anabolic-androgenic steroids are widely used by athletes. Significant mood disturbances and increased aggression have been reported (Oberlander and Henderson, 2012).

been widely accepted as an independent risk factor for ischaemic heart disease, recent evidence has cast doubt on this conclusion. However, there is evidence that cardiac patients with 'type D' personality (high levels of distress and negative affect) have a poorer outcome (Denollet et al., 2010).

Angina

Angina is often precipitated by emotions such as anxiety, anger, and excitement. It is a frightening symptom, and some patients become overcautious despite reassurance and encouragement to resume normal activities. Angina is sometimes accompanied by atypical chest pain and breathlessness caused by anxiety or hyperventilation, and it is important to identify this rather than increasing medical anti-angina therapy.

Myocardial infarction

Patients often respond to the early symptoms of myocardial infarction with denial and consequently delay seeking treatment. In the first few days in hospital, acute organic mental disorders and anxiety symptoms are common.

Recent research has concentrated on the replicated finding that depression, anxiety, and social isolation are important risk factors for both a lower quality of life (Dickens et al., 2006) and for a 22% increased risk of death after myocardial infarction (Meijer et al., 2011).

Some survivors of cardiac arrest suffer cognitive impairment. When such impairment is mild, it may show up later as apparent personality change or as behavioural symptoms, which may be attributed wrongly to an emotional reaction to the illness.

When patients return home from hospital, they commonly report non-specific symptoms such as fatigue, insomnia, and poor concentration, as well as excessive concern about somatic symptoms and an unnecessarily cautious attitude to exertion. Most patients overcome these problems and return to a fully active life. A few continue with emotional distress and social disability out of proportion to their physical state, often accompanied by atypical somatic symptoms. Such problems are more common in patients with longstanding psychiatric or social problems, overprotective families, and myocardial infarction with a complicated course.

The finding that depression and social isolation are associated with increased mortality after myocardial infarction has led to research to evaluate interventions to treat them. In the large trials so far completed, nursing support to combat social isolation was ineffective. CBT, medication, and exercise all show some efficacy in treating depression after myocardial infarction, but antidepressants must be used with caution, and tricyclic antidepressants are best avoided. Reductions in cardiac events and mortality have not been demonstrated reliably (Shapiro, 2015).

Non-cardiac chest pain

During the American Civil War, Da Costa described a condition which he called 'irritable heart'. This syndrome consisted of a conviction that the heart was diseased, together with palpitations, breathlessness, fatigue, and inframammary pain. This combination has also been named 'disorderly action of the heart', 'effort syndrome', and 'neurocirculatory asthenia'. The symptoms were originally thought to indicate a disorder of the functioning of the heart.

Non-cardiac chest pain is very common among patients in primary care and in cardiac outpatient clinics. Most patients are reassured by a thorough assessment, but a significant minority continue to complain of physical and psychological symptoms and to limit their everyday activities. Follow-up studies of patients with chest pain and normal coronary angiograms have consistently found subsequent mortality and cardiac morbidity to be little greater than those without cardiac disease, but persistent disability to be common.

Many causes have been suggested for atypical cardiac symptoms, including pain originating in the chest wall, oesophageal reflux and spasm, microvascular angina, mitral valve prolapse, and psychiatric disorder. In most patients, chest pain appears to be due to minor non-cardiac physical causes or to hyperventilation, which are misconstrued as heart disease and associated with anxiety. The aetiology is as for other medically unexplained symptoms. The most common psychiatric concomitant is panic disorder; less common are depressive disorder and hypochondriasis.

Management should follow the principles described in Chapter 8, with a particular emphasis on the treatment of hyperventilation, graded increase in activity, and discussion of beliefs about the cause of the pain. CBT may be effective in the management of anxiety with hyperventilation. Depressive disorder should be treated with antidepressant medication.

See George et al. (2016) for a review.

Sensory disorders

Deafness

Deafness may develop before speech is learned (prelingual deafness) or afterwards. Profound early deafness interferes with speech and language development, and with emotional development. Prelingually deaf adults often maintain their own social groups and communicate by sign language. When they have problems, it appears that these are more often diagnosed as behavioural problems and social maladjustment than emotional

disorder. These problems are managed best by those with special knowledge of the effects of deafness.

Deafness of later onset has less severe effects than those just described. However, the acute onset of profound deafness can be extremely distressing, while milder restriction of hearing may cause depression and considerable social disability.

Kraepelin was the first to suggest that deafness is an important factor in the development of persecutory delusions. Subsequent evidence supports an association between deafness and paranoid disorders in the elderly (see Chapter 12). There is also a link between deafness and dementia, and correction of hearing loss may help prevent cognitive decline (see Chapter 12).

Tinnitus

Tinnitus is very common, but few patients seek treatment, and most can live a normal life. Persistent tinnitus may be associated with low mood. Some patients are helped by devices that mask tinnitus with a more acceptable sound. Antidepressant medication may improve mood and reduce the intensity of the tinnitus. Cognitive and behavioural methods may enable people to accept their tinnitus and to minimize their social handicaps and perhaps protect against the increased risk of suicide in this disorder. For review, see Langguth et al. (2013).

Blindness

Although it imposes many difficulties, blindness in early life need not lead to abnormal psychological development in childhood or to unsuccessful later development. In previously sighted people, the later onset of blindness often causes considerable distress. Initial denial and subsequent depression are common, as are prolonged difficulties in adjustment. See Berman and Brodaty (2006).

Infections

Psychological factors may affect the course of recovery from an acute infection (Hotopf et al., 1996). In a classic early study, psychological tests were completed by 600 people who subsequently developed Asian influenza. Delayed recovery from the influenza was no more common among people whose initial illness had been severe, but it was more frequent among those who had obtained more abnormal scores on the psychological tests before the illness (Imboden et al., 1961). More recent findings of research on viral illness in general practice and infectious mononucleosis have reported similar conclusions.

In addition, some infectious diseases (e.g. hepatitis A, influenza, and brucellosis) are frequently followed by periods of depression and fatigue. The possible role of

infections, such as Lyme disease, as a cause of chronic fatigue was discussed on pages 623, 624. *Viral encephalitis* is discussed in Chapter 22.

HIV infection

HIV infection affects the brain at an early stage, and the disease has a chronic progressive course associated with a wide range of psychiatric consequences. With modern combined antiretroviral therapy, most patients with AIDS lead relatively healthy lives for substantial periods; indeed, their life expectancy is no longer significantly reduced. The nature of the physical symptoms, their progressive course, and the reactions of other people all explain why emotional distress is common in people with HIV infection. A further reason is that some of those at high risk for HIV (e.g. drug abusers) may have other psychological problems. Effects on the family may be considerable, especially where the partner and or one or more of the children are infected. Women might first find out about their diagnosis when pregnant and so will have to navigate the difficult adjustment of having a potentially life-limiting condition at the same time as preparing to become a parent and their concerns about how the infection might affect their parenting as well as the physical and psychological health of their child. Psychiatrists can contribute to the care of AIDS patients by providing counselling and specialist treatment for neuropsychiatric and other psychiatric complications.

Reactions to testing. Although HIV antibody testing is worrying for most of those who undergo it, the distress is usually short lived whatever the outcome of the test. People who have persistent and unjustified worries about having AIDS require psychiatric help of the kind appropriate for other illness fears.

Psychiatric problems, including adjustment disorder, depressive disorder, and anxiety disorder, are frequent at the time of diagnosis, although they may occur at any stage of the disease. People with previous psychological problems, longstanding social difficulties, or lack of social support are especially vulnerable.

Suicide and deliberate self-harm may occur in people who are concerned about the possibility of HIV infection as well as in people with proven disease. Among the latter, the risk is greater in those with advanced symptoms. However, it is not certain how much greater the risk of suicide and deliberate self-harm is in AIDS patients than in the general population.

Neuropsychiatric disorders occur secondary to the complications of immune suppression and as a result of the direct effects of HIV on the brain (Saylor et al., 2016). Minor cognitive disorders are frequent. HIV-associated dementia (AIDS-dementia complex), HIV

encephalopathy, and subacute encephalitis occur late in the illness in around a third of patients. There is usually an insidious onset with progression to profound dementia. HIV infection can also result in neurological symptoms and dementia in those who do not have AIDS. Delirium may occur when there is an opportunistic infection or cerebral malignancy.

Social consequences are considerable because of the public fears of the condition and stigma. Cultural differences in acceptance or rejection, and in the availability of family and other support, are major determinants of quality of life.

Problems in relation to illicit drug use are considered in Chapter 20. The disorganized lives of some HIV-positive drug users and their personal and social problems can make treatment of HIV difficult.

Ethical problems are related to confidentiality. They include the importance of maintaining confidentiality, disclosure to third parties who are at risk of infection, disclosure to insurers and to employers, and protection of the public from risk of transmission from HIV-infected healthcare workers.

Covid-19

Covid-19 has had implications for mental health both in terms of effects of the SARS-CoV-2 virus, and of the pandemic itself.

Psychiatric sequelae of SARS-CoV-2 infection

Large-scale cohort studies using health records show that infection with SARS-CoV-2 leads to a modestly increased risk of most psychiatric disorders in adults when compared to otherwise similar individuals diagnosed with other respiratory infections or other health events (Harrison and Taquet, 2023). The absolute increase in risk is small and, for common disorders like anxiety and depression, it is transient, lasting only a few weeks or months. However, for neurocognitive disorders and psychosis, the excess risk of receiving a diagnosis persists for up to two years (Taquet et al, 2022).

The psychiatric risks of Covid-19 affect people who had not previously been diagnosed with a psychiatric disorder and increase risk of recurrence in those who had. The risks are greater in, but not limited to, those who were hospitalized due to infection. Men and women show broadly similar risks. There is an age gradient, whereby children are not at increased risk of a psychiatric diagnosis overall (though they are at risk of disorders such as psychosis and seizures) and older adults have the greatest increased risk (Taquet et al., 2022). The risks are somewhat lower in people who experience breakthrough Covid-19 infection post-vaccination, and in

those with a second Covid-19 infection. Despite being a milder acute infection, the Omicron strain appears to confer similar psychiatric risks in survivors as previous SARS-CoV-2 strains (Taquet et al., 2022).

The mechanisms linking Covid-19 to subsequent psychiatric disorder are unknown. They likely include the acute infection acting as a psychosocial stressor (especially for disorders where the risk is transient) but also endotheliopathy, microthrombosis, and inflammation (especially for disorders with persisting risks post-Covid-19). There are no specific treatments for psychiatric disorders attributed to Covid-19.

It should also be noted that people with a pre-existing psychiatric diagnosis are more at risk of Covid-19 infection, independent of physical comorbidities, and have a poorer outcome, including greater mortality. The mechanisms are unknown.

Brain fog and long Covid

In addition to the risk of specific psychiatric (and medical) disorders following Covid-19, there is a broad range of symptoms and difficulties that people report, collectively known as *long-Covid* or *post-acute Covid-19 syndrome* if they occur or persist at least three months after infection (Nalbandian et al., 2021). Relevant to psychiatry, common features of long-Covid include complaints of poor attention, memory and concentration (collectively known as *brain fog*) as well as fatigue and anxiety/depressive symptoms. Estimates of the prevalence of long Covid vary widely, depending on the population surveyed and the method and timing of ascertainment, but a significant proportion of people complain of at least one feature. The explanation for long Covid symptoms and their relationship to CFS and other post-infectious syndromes with similar features remains controversial (Monje and Iwasaki, 2022).

Mental health effects of the Covid-19 pandemic

These long-term sequelae of SARS-CoV-2 infection are in addition to the effects of the Covid-19 pandemic on population mental health. Evidence indicates that the pandemic was associated with clear but relatively modest increases in mental health symptoms, and the Global Burden of Disease study estimated that it led to approximately 25% more depressive and anxiety disorders. Mental health was disproportionately affected in women, children, adolescents and young adults, health-care workers, and those with preexisting medical and psychiatric disorders. The pandemic also exacerbated existing sociocultural disparities in mental health. For review, see Penninx et al. (2022).

Cancer

The psychological consequences of cancer are similar to those of any other serious physical illness:

- *Delay in seeking medical help* because of fear or denial.
- *Response to the diagnosis*, which may be anxiety, shock, anger, disbelief, or depression. Sometimes the response is severe enough to meet the criteria for a psychiatric disorder, usually an adjustment disorder or sometimes a depressive disorder. The risk of suicide is increased, particularly in the year after diagnosis (Hem et al., 2004).
- *Later consequences*. Major depression occurs throughout the course of cancer and appears to be more frequent in those suffering pain. Prevalence estimates for depression in cancer range from 5% to 15%, with higher rates in palliative care settings (Walker et al., 2014).
- *The progression and recurrence* of cancer are often associated with increased psychiatric disturbance, which may result from a worsening of physical symptoms such as pain and nausea, from fear of dying, or from the development of an organic psychiatric syndrome.
- *Delirium and dementia* may arise from brain metastases, which originate most often from carcinoma of the lung, but also from tumours of the breast and alimentary tract, and from melanomas. Occasionally, brain metastases produce psychiatric symptoms before the primary lesion is discovered (see also Chapter 14).
- *Neuropsychiatric problems (paraneoplastic syndromes)* are sometimes induced by certain kinds of cancer in the absence of metastases, notably by ovarian teratomas, lung tumours, and Hodgkin's lymphoma. The mechanism is thought to be an autoimmune response or secretion of psychoactive substances by cancer cells (see Chapter 14).

Treatment for cancer may cause psychological disorder. Emotional distress is particularly common after mastectomy and other mutilating surgery. *Radiotherapy* causes nausea, fatigue, and emotional distress. *Chemotherapy* often causes malaise and nausea, and anxiety about chemotherapy may cause anticipatory nausea before the treatment. The latter may be helped by behavioural treatments in addition to antiemetic medication. Chemotherapy is also associated with cognitive impairment, which may persist after the conclusion of treatment (Mounie et al., 2020)

Family and other close relatives of cancer patients may experience psychological problems, which may persist even if the cancer is cured. Nevertheless, many

patients and relatives make a good adjustment to cancer. The extent of their adjustment depends partly on the information they receive.

Treatment for psychological consequences of cancer

In the past, doctors were reluctant to tell patients that the diagnosis was cancer, but most patients prefer to know the diagnosis and how it will affect their lives. However, the information must be communicated well, otherwise there may be problems in achieving a psychological adjustment. Depression and anxiety disorders are often missed in these patients, and systematic screening has been recommended (Andersen et al., 2014), but is often not carried out in practice (Shaw et al., 2016).

Depression in cancer patients can be effectively treated (Walker et al., 2013). A significant improvement in depressive symptoms, and an improved quality of life, was reported using an integrated collaborative care model that includes psychological therapy and antidepressants (Sharpe, 2014).

Childhood cancer

Childhood cancer presents special problems. Communication of the diagnosis is particularly difficult, but it is generally better to explain the diagnosis in terms appropriate to the child's stage of development. The child often reacts to the illness and its treatment with behaviour problems. Many parents react at first with shock and denial and may take months to accept the full implications of the diagnosis. Some mothers develop an anxiety or depressive disorder, and other family members may be affected. In the early stages of the illness parents are usually helped by advice about practical matters, and later by discussions of their feelings, which often include guilt. Adult survivors of cancer in childhood or adolescence appear to be at risk of social difficulties.

Genetic counselling

Genetic counselling about the reproductive risks of hereditary disease is mainly given to couples contemplating, planning, or expecting a child. It is now the responsibility of medical geneticists, not least because of rapid developments in the range and methods of testing available. However, psychiatrists do need to keep up to date with the increasing number of clinical situations in which genetic testing, and thence counselling, may be indicated (see Chapter 5).

Counselling includes:

- providing information about risks
- helping people to cope with concerns about the diagnosis
- enabling patients to take informed decisions about family planning.

Occasionally medical geneticists may request the collaboration or advice of psychiatrists, particularly when the patient is significantly distressed by the advice or information given, and if as a result they are unable to concentrate on making decisions.

Accidents

Psychiatric disorder predisposing to accidents. This is often through cognitive impairment occurring, for example, in alcohol or drug intoxication, delirium, or dementia, and less so in depression and some personality disorders.

Psychiatric disorder caused by accidents includes adjustment disorder, anxiety and depressive disorders, and post-traumatic stress disorder. Avoidance of situations associated with the accident is common and may be severe enough to meet diagnostic criteria for phobic anxiety disorder. *Head injury* may cause specific cognitive disorders and personality change (see pages 368, 369).

Associations with particular kinds of accident

Criminal assault can have especially severe and persistent consequences for victims (Kilpatrick and Acierno, 2003). Victims' problems are discussed further on pages 503–506.

Road traffic accidents are the leading cause of death in people under 40 years of age and a major cause of physical morbidity. Psychiatric factors contributing to road accidents include the misuse of alcohol and drugs, psychiatric illness, suicidal and risk-taking behaviour, and the side effects of some prescribed psychotropic drugs. Psychiatric consequences include acute stress disorder, anxiety and depressive disorders, post-traumatic stress disorder, phobias of travel, and disorders caused by brain injury. Some of these conditions are transient, but others persist and may cause considerable disability. Most of those affected do not seem to have been psychologically vulnerable before the accident (Chossegros et al., 2011).

Occupational injury. The psychiatric consequences of occupational injury resemble those of other accidents. It is sometimes alleged that hopes of compensation or other benefits help to maintain the symptoms and disability (see 'Compensation neurosis').

Spinal cord injury. Around a quarter of patients admitted to a spinal injury unit suffer from psychiatric problems

requiring treatment. Depression is common in the period immediately after a spinal cord injury but usually improves with time. Nevertheless, suicide appears to be more common among these patients than in the general population.

Burns. In children, burns are associated with overactivity, learning disability, child abuse, and child neglect. In adults, burns are associated with alcohol and drug misuse, deliberate self-harm, and dementia.

Post-traumatic stress disorder is common among those with severe burns, and persistent anxiety and depression occur in more than a third (Palmu et al., 2011). The outcome is worse for patients with burns affecting the face. Such patients are likely to withdraw from social activities. These patients need considerable support from the multidisciplinary team at a burns unit. For a review, see Jeschke et al (2020).

Compensation neurosis

The term 'compensation neurosis' (or accident neurosis) refers to psychologically determined physical or mental symptoms occurring when there is an unsettled claim for compensation. It was believed that such psychological factors were important in claims for persistent physical disability after occupational injuries and road accidents. Compensation claims were thought to prolong symptoms and that settlement was followed by recovery, although evidence has failed to substantiate this extreme view (Margoshes, 2000).

In fact, many accident victims do not claim compensation, and few become involved in prolonged litigation. However, it does appear that time off work and disability are affected by the type of accident, social factors, and the prospect of compensation, social security, or other benefits.

See Malt (2009) for a review of the psychiatric aspects of accidents, burns, and other trauma.

Psychiatric aspects of obstetrics and gynaecology care

Pregnancy

Psychiatric disorder is more common in the first and third trimesters of pregnancy than in the second. In the *first trimester* unwanted pregnancies are associated with anxiety and depression. In the *third trimester* there may be fears about the impending delivery or doubts about the normality of the foetus. Psychiatric symptoms in pregnancy are more common in women with a history of previous psychiatric disorder and probably also in those with serious medical problems affecting the course of pregnancy, such as diabetes. It is sometimes said that psychiatric disorders are less common in pregnant than non-pregnant women, but this is not borne out by systematic reviews of depression (Gavin et al., 2005) and anxiety (Goodman et al., 2014) disorders. Maternal depression during pregnancy is associated with an increased rate of premature delivery and some other adverse perinatal outcomes (Jarde et al., 2016), highlighting the need for effective screening to improve detection, and thence treatment (O'Connor et al., 2016a).

Management of depression in pregnancy involves careful assessment, consideration of treatment options, and providing the woman with accurate information about risks and benefits (Box 22.13). In particular, care must be taken when using antidepressants, as with all

psychotropic drugs, during pregnancy (Chisolm and Payne, 2016). Nevertheless, if the woman has a history of recurrent moderate to severe depression and has been effectively maintained on antidepressant medication, it is generally better to continue this treatment during pregnancy because the risk of relapse is high if medication is withdrawn, and because of the risks to the foetus of untreated maternal depression (Jarde et al., 2016). Women with a history of less-severe depression can often withdraw antidepressant medication successfully (Yonkers et al., 2011).

Women not taking antidepressant medication who become depressed during pregnancy should be provided with psychoeducation and psychological treatment in a stepped care manner, depending on the severity of their condition. If the depression is severe, or if a moderate depressive episode fails to respond to psychological management, antidepressant medication should be offered, usually an SSRI (Vigod et al., 2016). Indeed, generally the risks of SSRI treatment in pregnancy are low and are outweighed by those of untreated depression. However, self-limiting neonatal withdrawal syndromes have been described, although the incidence across studies varies widely. SSRI use in later pregnancy may also be associated with a small increase (1–2 per 1,000 live births) in the risk of pulmonary hypertension of

Box 22.13 Depression in pregnancy

- Offer all pregnant women information about depression, its significance for her and her baby, and the recommended treatment options.
- Women with a history of recurrent depression of moderate to severe intensity who are successfully maintained on antidepressant treatment may be advised to continue treatment.
- Women who experience the onset of a mild–moderate depressive episode should be offered education and psychological treatment using a stepped care approach.
- Women with severe depression or those with moderate depression not responding to psychological treatment should be offered antidepressant medication, usually an SSRI other than paroxetine.
- The absolute risks of harm to the foetus from antidepressants in pregnancy are low and need to be placed in the context of the risks of untreated depression in the mother.

the newborn, a potentially fatal condition (Huybrechts et al., 2015). Paroxetine has been associated with cardiac defects and should therefore be avoided in pregnancy; other antidepressants appear to be safe in this respect (Huybrechts et al., 2014).

For review of depression in pregnancy, see Vigod et al. (2016).

Hyperemesis gravidarum

About half of all pregnant women experience nausea and vomiting in the first trimester. In the past it was sometimes suggested that these symptoms, as well as the severe condition of hyperemesis gravidarum, have a psychological basis, but there is no evidence for this.

Pseudocyesis

Pseudocyesis is a rare condition in which a woman believes that she is pregnant when she is not, and develops amenorrhoea, abdominal distension, and other changes similar to those of early pregnancy. The condition is more common in younger women. Pseudocyesis usually resolves quickly once diagnosed, but some patients persist in believing that they are pregnant. Recurrence is common. Only rarely is the condition associated with a psychiatric disorder.

Couvade syndrome

In this syndrome, the partner of the pregnant woman reports that he is experiencing some of the symptoms of pregnancy, complaining of nausea, morning sickness, and often of toothache. These complaints generally resolve after a few weeks.

Termination of unwanted pregnancy

In the past, psychiatrists in the United Kingdom were often asked to see pregnant women who were seeking

a therapeutic abortion on the grounds of mental illness. The provisions of current legislation in the United Kingdom and many other countries now make it generally more appropriate for decisions to be made by the family doctor and the gynaecologist, without involving a psychiatrist. Nevertheless, psychiatric opinions are still sought at times, not only about the grounds for termination of pregnancy but also for an assessment of the likely psychological effects of termination in a particular patient (Morris et al., 2012). Most of the evidence suggests that the psychological consequences of termination are usually mild and transient (Fergusson et al., 2009).

In vitro fertilization

The process of *in vitro* fertilization (IVF) is often protracted and, not surprisingly, associated with emotional distress (Rockliff et al., 2014). However, postpartum depression is no more common than in women following spontaneous pregnancy (Gressier et al., 2015).

Spontaneous abortion

Approximately a fifth of diagnosed pregnancies do not progress beyond 20 weeks, mainly because of foetal defects. Distress is common after miscarriage, although most women improve rapidly (Broen et al., 2005).

Antenatal death

Antenatal death (stillbirth) causes an acute bereavement reaction and, for some women, long-term psychiatric problems, as well as concern about future pregnancy. It is believed by many midwives that parents should be helped to mourn by encouraging them to hold the baby, to name it, and to have a proper funeral. However, the evidence regarding this issue is weak (Hennegan et al., 2015). Parents may need some continuing support, and the next pregnancy may be a particularly worrying time.

The risk of stillbirth is higher in women with a psychiatric history, but this may reflect factors such as antenatal care attendance rather than the psychiatric disorder itself (King-Hele et al., 2009).

Caesarean section

Caesarean section has been said to have adverse psychological consequences for parents and infants. Most of the research has failed to test this association because it has not separated the effects of surgery from other adverse factors. However, it would seem sensible to pay particular attention to parental support and to initial bonding.

See Brockington (2009) for a review of the psychiatric aspects of pregnancy.

Postpartum mental disorders

These disorders can be divided into minor mood disturbance (maternity blues), puerperal psychosis, and chronic depressive disorders of moderate severity.

Minor mood disturbance—‘maternity blues’

Among women giving birth to a normal child, between half and two thirds experience brief episodes of irritability, lability of mood, and episodes of crying. Lability of mood is particularly characteristic, taking the form of rapid alternations between euphoria and misery. The symptoms reach their peak on the third or fourth day postpartum. Patients often speak of being ‘confused’, but tests of cognitive function are normal. Although frequently tearful, patients may not be feeling depressed at the time but tense and irritable.

‘Maternity blues’ is more frequent among primigravida. The condition is not related to complications at delivery or to the use of anaesthesia. ‘Blues’ patients have often experienced anxiety and depressive symptoms in the last trimester of pregnancy; they are also more likely to give a history of premenstrual tension, fears of labour, and poor social adjustment.

Both the frequency of the emotional changes and their timing suggest that maternity blues may be related to readjustment in hormones after delivery, although this has not been established. No treatment is required because the condition resolves spontaneously in a few days.

Postpartum psychosis (puerperal psychosis)

In the nineteenth century, puerperal and lactational psychoses were thought to be specific entities distinct from other mental illnesses. Later psychiatrists such as Bleuler and Kraepelin regarded the puerperal psychoses as no

different from other psychoses, and neither ICD-11 nor DSM-5 have a specific diagnostic category for puerperal psychosis. Nevertheless, the term continues to be used widely by clinicians and patients.

For a review, see Jones (2020).

Epidemiology

The incidence of postpartum psychoses has been estimated at 1–2 per 1,000 births. This incidence is substantially above the expected rate for psychoses in non-puerperal women of the same age. Puerperal psychoses are more frequent in primiparous women and in those who have suffered previous major psychiatric illness. In particular, women with a history of bipolar disorder have a 20% risk of puerperal relapse. However, about half of women who develop puerperal psychosis do not have any form of ‘high-risk’ history (Jones et al., 2014). Puerperal illnesses are reported to be more common in developing than in developed countries; the excess may reflect cases with an organic aetiology.

Aetiology

There is a genetic predisposition, which overlaps with that of bipolar disorder, but apart from a possible susceptibility locus on chromosome 16, no specific genes or genetic pathways have been identified. The early onset of puerperal psychoses has led to speculation that they might be caused by the dramatic hormonal changes that follow delivery. There is, however, no evidence that hormonal changes in women with puerperal psychoses differ from those in other women in the early puerperium. Hence, if endocrine factors do play a part, they would seem to act only as precipitating factors in predisposed women. Immunological factors have also been implicated. The sleep deprivation associated with childbirth may play a role since disrupted sleep can precipitate mania in vulnerable individuals.

Clinical features

The onset of puerperal psychosis is often within two to three days of delivery, and usually within the first one to two weeks. Sudden onset and rapid deterioration is typical. Three types of clinical picture are observed: delirium, mood disorder, and schizophreniform disorder. Delirium was common in the past but is now much less frequent since the incidence of puerperal sepsis was reduced by antibiotics. Nowadays affective psychoses predominate, either bipolar disorder with mania or mixed features, or depressive psychosis. Schizophreniform disorders presenting for the first time occur but are rare. The clinical features of these syndromes are similar to those of the corresponding non-puerperal syndrome. Insomnia and

overactivity are often early features. Perplexity and confusion are common.

Management

Assessment should be prompt and pay attention to the potential risks to mother and baby (Spinelli, 2009). As well as the usual psychiatric assessment it is essential to ascertain the mother's ideas concerning the baby. Severely depressed or psychotic patients may have delusional ideas that the child is malformed, imperfect in some way, or evil. These false ideas may lead to attempts to kill the child to spare it from future suffering, or to protect others. Infanticide is extremely rare, but suicide, though also uncommon, is a leading cause of maternal death.

Treatment

This usually requires inpatient care, often under the Mental Health Act. If available, admission is often best to a mother and baby unit, where the child can remain with the mother to minimize adverse effects on maternal bonding. Once the admission is arranged, all contacts between mother and baby should be supervised, at first by nursing staff and thereafter reviewed in the light of clinical progress. Child protection considerations should always be an integral part of management.

As with other psychoses, antipsychotics are a mainstay of treatment, with antidepressants also used if depressive symptoms are prominent. Adjunctive benzodiazepines are helpful for insomnia and sleep disturbance. Lithium may also be worth considering if there is a clear bipolar component, though it is advised that women taking lithium should not breastfeed. Where pharmacological treatment and psychological support do not resolve symptomatology, the early use of electroconvulsive therapy (ECT), which can be rapidly effective in puerperal psychosis, should be considered. For review, see Bergink et al. (2015).

Prognosis

Over 75% of women have a good outcome after puerperal psychosis. Recovery usually occurs within a few months. After a puerperal psychosis there is a high (at least 35%) risk of relapse after the next delivery (Wesseloo et al., 2016), and almost 70% of women will have a non-puerperal recurrence, usually of bipolar disorder (Blackmore et al., 2013). Given these risks, women with a history of puerperal psychosis should be offered preconception care and regular psychiatric support during and immediately after each pregnancy. In this instance, the benefits of maintenance medication (antipsychotics and mood stabilizers) often outweigh

the risks. However, sodium valproate should be avoided, as in all women of childbearing age. For review, see Perry et al. (2021).

Postnatal depression

Less severe depressive disorders are much more common than the puerperal psychoses. Systematic reviews suggest a point prevalence of 5% in the first three months after delivery, rising to 19% if minor depression is included (Gavin et al., 2005). It is estimated that about a third of these cases start during pregnancy. Higher figures have been reported in low- and middle-income countries. Tiredness, irritability, and anxiety are often more prominent than depressive mood change, and there are often phobic and obsessional symptoms concerning fears about harming the baby. It is important to distinguish such obsessional thinking from delusional beliefs related to puerperal psychosis, as discussed above.

Most patients recover after two to six months, but up to 30% still have some depressive symptoms in the year following childbirth. There is a high risk (about 40%) of subsequent depressive relapse, either postnatally or at other times.

For review, see Howard et al. (2014a).

Clinical observation suggests that postnatal disorders are often precipitated in vulnerable mothers by the psychological adjustment required after childbirth, as well as by the loss of sleep and the hard work involved in the care of the baby. There is little evidence of a specific biological basis different from depression in other situations. The main risk factors are a previous history of depression (especially when accompanied by obstetric complications) and indications of social adversity. Low levels of partner or other support, relationship difficulties, and domestic violence are also risk factors. Unintended pregnancy is associated with a twofold increase in postnatal depression (Abajobir et al., 2016).

Despite the medical and other care given to women after childbirth, many postpartum depressions are undetected or, if detected, untreated. Therefore, those providing care to mother and baby need to be alert to the possibility of depression. Screening appears to decrease the prevalence of postnatal depression (O'Connor et al., 2016a), and the Edinburgh Postnatal Depression Scale performs reasonably well as a screening tool.

Many women can be treated effectively in primary care, with non-directive counselling and help with solving practical problems (Clarke et al., 2013). A small proportion needs specific psychological interventions or antidepressant medication, and a few with severe

or complex problems require referral to psychiatric services. Evidence shows efficacy of psychological and pharmacological therapies comparable to depressive episodes in other situations. Exposure of the infant to drug via breast milk needs to be considered but is generally low with SSRIs such as sertraline. For a review, see Larsen et al. (2015).

Other psychiatric disorders in the postnatal period

There has been much less emphasis on other disorders in the postnatal period, but research indicates that anxiety disorders, eating disorders, and post-traumatic stress disorder can all present at this time and require treatment. For review, see Howard et al. (2014a) and Howard et al. (2014b).

Effects of maternal mental health on the child

Maternal psychiatric disorders, especially depression, during pregnancy and in the postnatal period have significant effects on the development of the child (Goodman et al., 2011). For example, both antenatal and postnatal depression in the mother are associated with an increased risk in the child for emotional problems, behavioural difficulties, disorganized attachment, and depression. Postnatal depression also impacts on intellectual development in the child, although the long-term effects on cognitive functioning are uncertain. Although most studies have focused on mothers, there is increasing evidence that paternal depression shows similar associations with child development.

The mechanisms linking parental mental health with child health and development are complex. They likely include an interplay of genetic, epigenetic, biological, and social factors, acting both as mediators and moderators of risk.

Several interventions to reduce the adverse effects of parental psychiatric disorder on the health of children have been tested. Treatment of maternal perinatal disorders has yielded a mixed pattern of results, with not all studies showing benefits for the child even when the mother has been treated successfully. Greater effects on the child have emerged from specific parenting interventions, including home-visiting programmes to improve the quality of interactions and the attachment between mother and infant. There are no data regarding effects of interventions in fathers.

For further discussion, see Chapter 16. For review, see Stein et al. (2014).

Psychiatric aspects of gynaecology

Premenstrual syndrome and premenstrual dysphoric disorder

These categories denote a group of psychological and physical symptoms starting a few days before and ending shortly after the onset of a menstrual period. The psychological symptoms include anxiety, irritability, mood lability, food cravings, and depression; the physical symptoms include breast tenderness, abdominal discomfort, and a feeling of distension. Premenstrual syndrome (PMS) is not included in current classifications of psychiatric disorder, but premenstrual dysphoric disorder (PMDD) is included in DSM-5 within the 'depressive disorders' chapter. Both PMS and PMDD should be distinguished from the much more frequent occurrence of similar symptoms that are not strictly premenstrual in timing.

For review, see Ryu and Kim (2015).

Epidemiology

The frequency of PMS in women of reproductive age is estimated at 30–40%, and 3–8% for PMDD. Such estimates should be interpreted with caution. First, there is a problem of definition. Mild and brief symptoms are frequent premenstrually, and it is difficult to decide when they should be classified as PMS. Second, information about symptoms is often collected retrospectively by asking women to recall earlier menstrual periods, and this is an unreliable way of establishing the time relationships. Prospective ratings are more likely to be accurate. Third, the description of premenstrual symptoms is subjective and may be influenced by knowledge that the enquiry is concerned specifically with the premenstrual syndrome. The aetiology is uncertain. Biological explanations have been based on ovarian hormones (excess oestrogen, lack of progesterone), pituitary hormones, and disturbed fluid and electrolyte balance. None of these theories has been proved. Various unproven psychological explanations have been based on possible associations of the syndrome with neuroticism or with attitudes towards menstruation.

Management

PMS has been treated with progesterone as well as with oral contraceptives, bromocriptine, diuretics, and psychotropic drugs. There is no convincing evidence that

any of these is effective, and treatment trials suggest a high placebo response (up to 65%). Psychological support and encouragement may be as helpful as medication.

For PMDD, SSRIs and oral contraceptives are considered the treatment of choice, with the latter primarily improving the physical symptoms. CBT and lifestyle modifications also have a role (Lustyk et al., 2009).

See O'Brien et al. (2011) for a review of premenstrual syndromes and their treatment.

The menopause

In addition to the physical symptoms of flushing, sweating, and vaginal dryness, menopausal women often complain of headache, dizziness, and depression. It is not certain whether depressive symptoms are more common in menopausal women than in non-menopausal women (Rössler et al., 2016). Nevertheless, among patients who consult general practitioners because of emotional symptoms, a disproportionately large number of women are in the middle-aged group that spans the menopausal years.

Depressive and anxiety symptoms at the time of the menopause could have several causes. Hormonal changes have often been suggested, notably deficiency of oestrogen, but a causal role has not been established. Psychiatric symptoms at this time of life could equally

reflect changes in the woman's role as her children leave home, her relationship with her husband alters, and her own parents become ill or die.

There are no specific treatments for depression associated with the menopause, and it should be treated in the usual fashion. For review, see Llana et al. (2012).

Hysterectomy

The alleged relationship between hysterectomy and psychiatric disorder has been controversial. A systematic review concluded that hysterectomy is associated with an improvement in depressive symptoms and no effect on anxiety symptoms (Darwish et al., 2014). Moreover, patients who are free from psychiatric symptoms before hysterectomy seldom develop them afterwards (Gath et al., 1982).

Sterilization

Considerations similar to those for hysterectomy apply to these procedures. Although retrospective studies suggested that sterilization leads to psychiatric disorder and sexual dysfunction, prospective enquiry has contradicted this. Indeed, sexual relationships are more likely to improve than worsen, and definite regrets are unusual.

For review of psychiatric aspects of obstetrics and gynaecology, see Brockington (2009).

CHAPTER 23

Public mental health and global mental health

Introduction 645

Public mental health 645

Prevention 646

Life course 646

Why prevention? 647

Social determinants 647

Cost effectiveness 649

Global mental health: concepts and mission 650

The global burden of mental illness 651

Introduction

Most of the world's population has limited access to specialist mental health services. In addition, there are important mental health promotion and prevention strategies, including early intervention, that will likely have an impact on the mental health of a population

in high and well as low and middle-income countries. This chapter introduces some of the interventions and strategies that can be delivered by mental health professional from a public health and global health perspective.

Public mental health

Public mental health can be defined as a science and art to promote and protect mental health and emotional wellbeing, prolong life, and improve quality of life through the organized efforts of society. Contrary to clinical specialties, which focus on individuals, public mental health takes a population-based approach and emphasizes collective responsibility for mental health, as well for the protection and prevention of mental distress and psychiatric disorders.

The approach involves the state acting in partnership with public agencies in health and social care, employers, teachers, and local government to better respond to

the multiple social and societal drivers of poor health. Population interventions have been used to prevent violence, knife crime, smoking, and obesity. Extending this to prevention of mental illness and to minimize disabilities is a natural step, since psychiatric disorders are caused and influenced by many factors, for example, early childhood experiences, poverty, peer interactions, school climate, and employment. All of these can be considered as part of a complex 'exposome' that interacts with individual, group, and area-based vulnerabilities and dispositions to shape the trajectories of health and illness over the life course.

Definitions

Public health

The science and art of promoting and protecting health and wellbeing, preventing ill health and prolonging life through the organized efforts of society (UK faculty of public health).

Population

All people living in a defined geographical area using local government boundaries, or regions through which health social care is organized, or national populations.

Prevention

Primary prevention seeks to prevent the onset of new conditions by targeting all populations.

Secondly prevention includes early detection and intervention.

Tertiary prevention seeks to reduce disability and improve recovery of those receiving specialist care.

Primary prevention (Fazel and Soneson, 2023) can also be organized by the following schema:

Universal: everyone receives the intervention.

Selective: high-risk populations receive the intervention.

Indicated: those with early symptoms receive the intervention.

Prevention

The main rationale for public health interventions is to shift risk factors in the population, either to reduce exposure to noxious agents (e.g. poisons, pollution, energy-dense foods) or to encourage healthier behaviours (e.g. stop smoking, take enough exercise and sleep, safe housing). By shifting these risk factors, fewer people are exposed to the risk factor and so fewer will develop illness, or where mental health is concerned, fewer will develop symptoms. Yet, this assumes a normal or near-normal distribution (for a good explanation, see Adams, 2004). For some behaviours, for example, sleep duration, there is a J-shaped relationship between duration of sleep and health outcomes where people who sleep seven to eight hours at night have better health than those who sleep more or less than this (Chaput et al., 2020).

Universal policies applied to entire populations can be efficient and have small impact on many more people than clinical interventions applied when people are already unwell. However, those least likely to benefit are those living in precarity and poverty and who are already marginalized. Shifting their health-risk behaviours or the quality of their housing may be difficult, so they may not benefit from universal policies, and one might propose those with severe mental illness are among that group. Therefore, universal policies, especially those seeking behaviour change, are likely to widen inequalities rather than reduce them; another approach is 'proportionate universalism' whereby marginalized groups receive modified or targeted universal interventions designed to meet their needs and to engage them, when universal interventions would not work.

Life course

A life course approach means that we consider health promotion and prevention from conception to death. Thus life course interventions to prevent mental illness include protecting children from harm in utero (e.g. maternal smoking or extreme nutritional problems), harms around childbirth (e.g. asphyxia or infections), during the early and school years (e.g. treating parental

psychiatric disorder, preventing bullying, supporting families), and promoting mental health among the workforce and ensuring healthy lifestyles (e.g. managing work stress, demand-control, effort-reward). There also interventions to promote healthy ageing and reduce the risk of long-term conditions and premature mortality, for example, enhancing physical and cognitive activity.

Why prevention?

At least 50% of all lifelong mental illness is evident by 14 years of age (Kessler et al., 2005); therefore, prevention and early intervention is critical. Mental health research has provided advances in understanding neurodevelopmental impacts of early adverse life events, and this evidence is informing the development of appropriate policies and services, alongside preventive and therapeutic interventions. Cumulative adversity in childhood is associated with increased risks of adult mental illnesses, substance and alcohol misuse, suicide and self-harm, and shortened life expectancy (Bellis et al., 2023; Gondek et al., 2021). Indeed, adult psychiatric disorders may be better understood to be, at least in part, the consequence of cumulative life course trauma, including early childhood and contemporary social disadvantage. Poverty, unemployment, discrimination, poor housing, fear of crime, and violence all impact on mental health. Specific adversities include sexual, physical, and emotional abuse and neglect, racism, multiple loss events, experience of the care system or incarceration, and parents with mental illness or who misuse substances; there exist many others. Those with neurodiversity will likely experience more adversities, and with more intensity, as will those living in precarity. Therefore, improving trauma-informed care in health and social services, as well as in general public services such as welfare, schools, and policing will potentially impact on mental health outcomes, for example, by facilitating more acceptable and accessible support (Arnold et al., 2021; Bellis et al., 2023; Lorey and Fegert, 2022).

In terms of prevention and early intervention, much attention has been devoted to psychoses and those at risk of or at ultra-high risk of psychosis (see Chapter 11 on ultra-high risk of psychosis, or UHR). The UHR state is associated with high rates of experienced trauma, victimization such as bullying, substance use, comorbid mental disorders, suicidal ideation, and self-harm (Joa et al., 2021). The changes in legislation around cannabis use may also risk higher levels of psychosis in those already at risk (Hjorthøj et al., 2021). The conversion from UHR to actual psychosis is about 20%. Those living in urban areas, with multiple adversities, people from ethnic minorities (especially Black African and Caribbean ethnic groups), and migrants in general, are more likely to transition from high-risk states into psychosis. Thus, prevention and intervention need to address societal and social determinants, early detection and intervention in primary and secondary care, as well as more public health campaigns to ensure the general population is better prepared. This is especially important for those with family histories of poor mental health and where there has been a family member with psychosis. Children of parents with psychosis may experience neglect or fear when their parent is unwell, or detained, and ensuring they are cared for is essential. Yet, little is currently stated in policy and practice on this. Children of parents with psychosis could, for example, be one group targeted for selective prevention (Fazel and Soneson, 2023).

Social determinants

The World Health Organization (WHO) defines social determinants of health as ‘the non-medical factors that influence health outcomes’ (World Health Organization, 2022). These factors include the conditions and contexts in which people are born, mature, work, live, and grow older, and the wider systems around them that shape, constrain, or offer opportunities in their daily lives. These conditions include economic, social, political, and health policies, and priorities given to mental illness. Thus, employment practices, poverty, and investment in schools all have broader health impacts. Many argue that every area of government should include a mental health impact assessment, as all such decisions will have a cost or benefit to the mental health of populations.

The WHO provides a list of examples of social determinants that impact on health, the following is a modified version:

- Income protection
- Protecting minimum social standards of safety, housing, amenities, and the environment
- Education and school environment
- Unemployment and job insecurity
- Conditions of work life
- Food insecurity
- Early childhood development
- Social inclusion and non-discrimination

- Conflict—actual and structural in society leading to inequalities
- Access to affordable health services of decent quality
- Isolation and meaningful activity
- Meaning and purpose in everyday life
- Physical activity and opportunities to engage in activities irrespective of physical and social constraints.

This list is important for the universal interventions in the general population, and is especially relevant for people with mental illness, for whom social determinants are risk factors for illnesses; in addition, once mental illnesses develop, more of these risk factors are acquired over the life course. On developing mental illness we notice a social decline in functioning, levels of employment, income, social status, and social interaction; in some instances, there are more interactions with the criminal justice system. Social determinants can substantially influence health outcomes, and influencing wider systems (tackling poverty, safety, justice, activity, etc.) may have more impact on health than interventions in formal health care systems (World Health Organization, 2022a).

Michael Marmot has championed research in health inequalities and repeatedly exposed major difference between rich and poor, and across levels of geographical deprivation; For example, a recent report on the social determinants of health in England (Marmot, 2020) concludes:

- Life expectancy is proportionately reduced for people living in more deprived areas, and especially for women. This social gradient has become steeper over the last decade.
- Marked regional differences in life expectancy exist, especially for those living in relatively more deprived areas.
- Mortality rates are increasing for men and women 45–49 years of age, possibly due to ‘deaths of despair’ (suicide, drugs, and alcohol abuse).
- Child poverty has increased.
- Children’s and youth centres have closed and there is less funding for education.
- A housing crisis with a rising homeless population is increasingly evident.
- People have insufficient money to lead a healthy life.

Marmot concludes that austerity and a lack of investment to reduce poverty and social determinants is largely responsible, and as austerity is a political decision, it should sit alongside other events, for example,

Covid-19 and economic crises. Tackling social determinants is hard as it requires careful coordination between multiple agencies in prevention work that could manifest outcomes in decades and not in short timescales, with many social, government, and public support systems potentially involved.

Population attributable fractions for public health interventions

The population attributable fraction (PAF) is an estimation of how much of a particular condition could be prevented if the candidate risk factor could be eliminated. It assumes causal relationships are plausible and that complete elimination of that risk is possible. It is thus a theoretical construct which gives support for preventive interventions where the assumptions holds. A recent meta-analysis of the literature on preventing modifiable risk factors and their hypothetical expected impact (Dragioti et al., 2022) found the following PAFs for specific outcomes:

- 38% for childhood adversities and schizophrenia spectrum disorders
- 25% for tobacco smoking and opioid use disorders
- 18% for job strain and depression
- 15% for insufficient physical activity and Alzheimer’s disease
- 13% for childhood sexual abuse and depressive disorders
- 12% for clinical high-risk state for psychosis and any non-organic psychotic disorders
- 10% for three metabolic factors and depression
- 10% for cannabis use and schizophrenia spectrum disorders.

They conclude that specific interventions should be assessed for real-world effectiveness. Similar studies of other risks and disorders are also instructive. For example, a meta-analysis of longitudinal studies found women who experienced domestic violence during pregnancy were at a threefold increased unadjusted odds (OR = 3.1, 95% CI 2.7–3.6) of probable depression in the post-natal period (Howard et al., 2013). The findings could not be adjusted for antenatal depression due to a lack of data. The pooled PAF was 12.7% (95% CI 11.8%–13.6%). Therefore, if the association between domestic violence during pregnancy and post-natal depression were causal, experiences of domestic violence during pregnancy contribute significantly to the burden of post-natal

psychiatric disorder, and interventions could reduce this burden, which would also have a positive impact on the development of children of these mothers. There are also raised rates of domestic violence in those presenting to psychiatric and gynaecology departments, inpatient psychiatric services; domestic violence is also a common experience in women receiving diagnoses of PTSD, depression and anxiety (Agnew-Davies, 2016).

For multiply disadvantaged populations, the findings can be dramatic. For example, 49.3% of psychological distress among Aboriginal and Torres Strait Islander adults could be attributable to discrimination in general (39.4–58.8% across strata) and 27.1% to racial discrimination specifically (Thurber et al., 2022). Studies in the United Kingdom have also shown how prevalent discrimination is and how it is associated with common psychiatric disorders and long term medical conditions (Bhui et al., 2005; Hackett et al., 2020).

The role of psychiatrists in managing social determinants

Psychiatrists have generally been embedded in hospital or community mental health settings. Their involvement in public mental health means they need to engage with wider social systems, including local government, the police, housing, food and alcohol industries, schools, and employers to shape relevant policies. This advocacy role is a component of competencies in tackling structural drivers of inequalities (Kirmayer et al., 2018). This preventive work can be done in partnership with other mental health professionals and those already involved

in public health initiatives. However, psychiatrists aware of the social determinants of health can also ensure that people they see in clinics, wards, and community settings have a full assessment of the social factors implicated in illness onset and course. Skills necessary to tackle wider determinants of inequalities are called structural competencies (Metzl and Hansen, 2014). Clinicians also need to facilitate working with other agencies to ensure these determinants are better addressed as they can hinder and complicate recovery. In 2017, Robert Like, a family physician in the United States, developed a checklist for physicians to use when assessing social determinants in clinical care. The mnemonic for this checklist is called THEEUS, each letter standing for one item of the list:

- Transportation (e.g. car, bus, taxi)
- Housing (e.g. home-owner, renter, living arrangements, housing stock)
- Eating (e.g. typical diet/nutrition, adequacy of food supplies, meals on wheels, food deserts)
- Education (e.g. educational attainment, literacy, numeracy, health literacy)
- Safety (e.g. interpersonal, physical, community, environmental)
- Economics (e.g. current and long-term financial assets, budget for food, clothing, medications)
- Utilities (e.g. electricity, gas, water, heating, phone, internet, other)
- Social supports (e.g. family, friends, work, religious, recreational, community)

Cost effectiveness

Understandably, the greatest life course impacts are those applied early in life, and there is good evidence that interventions are cost effective. An evidence base on public health interventions (Royal College of Psychiatrists, 2022) found the strongest evidence for the following seven interventions:

- Interventions during pregnancy and immediately after birth to prevent child mental disorder.
- Interventions to prevent and treat parental mental disorder and parental drug/alcohol misuse.
- Parenting programmes that prevent child mental disorder, substance use, antisocial behaviours, and

unintentional injury, and improve child behavioural outcomes, parenting, and parental mental health.

- Home visiting and parenting programmes to improve child–parent attachment and prevent child adversity.
- School-based interventions to prevent mental disorder and alcohol/tobacco/drug use, reduce child adversity, promote mental wellbeing and resilience, and improve social–emotional skills.
- Workplace-based interventions to reduce employee mental disorder, increase wellbeing, and promote recovery from mental disorder.

- Interventions to reduce smoking, alcohol, drug use, physical inactivity, Covid-19 infection, and promote appropriate care of physical health conditions, including among people with mental disorder.

School-based interventions (health promotion, prevention, and early intervention) are especially promising (Fazel et al., 2014a). There were two priority areas of highest public health relevance, but with a more limited evidence base:

- address socioeconomic inequalities to improve outcomes among marginalized groups
- target marginalized and higher-risk groups to improve access to public mental health interventions.

A report from the Royal College of Psychiatrists (2010) included cost-effectiveness data and shows good evidence of cost-effectiveness for parenting and early childhood prevention for the following outcomes:

- conduct disorder in children
- post-natal depression (health visitors),
- violence and bullying through whole-school programmes
- offending and re-offending
- alcohol problems.

Global mental health: concepts and mission

Global mental health is ‘an evolving field of research and practice that aims to alleviate mental suffering through the prevention, care and treatment of mental and substance use disorders, and to promote and sustain the mental health of individuals and communities around the world’ (Patel and Prince, 2010), primarily in lower-resource settings. It sets out to tackle inequities in mental health status due to national and regional differences in wealth and investment in mental health prevention and care, and the proportion of professionals devoted to such tasks. Patel and Prince argue that the most striking inequity concerns the disparities in provision of care and respect for human rights of persons living with mental disorders between high- and low-income countries.

Low- and middle-income countries (LMICs) represent 80% of the global population but receive less than 20% of global mental health resource allocation. In LMICs, it is estimated that more than 75% of those with anxiety-, mood-, impulse-, or substance misuse-related disorders fail to receive any formal care in the preceding 12 months (Demyttenaere et al., 2004). The absence of treatment for psychosis is even greater, estimated to be around 90%, and where it is available it is often below acceptable and ethical standards. In some countries, people with mental illnesses are still highly stigmatized and service provision is poor, leading to high rates of incarceration, either formally in hospital and the criminal justice system, as well as informally in homes and communities. This is despite great improvements in low-cost treatments and better understanding of the potential role that lay practitioners can play in the provision of care (Kohrt et al., 2018). Resources are markedly

limited for child and adolescent populations, with little provision in health and education services for those with autism spectrum disorders, ADHD, or mood and anxiety disorders, although prevalence rates are similar.

All the social determinants and influences on mental health discussed in the section above (see ‘Public mental health’) apply also to the global health initiatives, yet even more attention to these factors is needed, for two reasons. First, the social determinants and material environment are harsher; second, insufficient resources (financial and human) or infrastructure investment are allocated to the prevention treatment of mental illness, let alone being able to replicate high-intensity and high-cost models of care.

This reality requires that the solutions in LMICs reflect the local geopolitical contexts, levels of investment, and adverse experiences, and make best use of existing resources and assets. Thus the concept of ‘task shifting’ or ‘task sharing’ are potential solutions (Collins, 2020)—meaning local people can either be trained to deliver low-intensity and low-cost interventions, or existing non-mental health professionals might be involved in delivering mental health interventions and prevention (discussed in detail below). Thus, social workers, teachers, support workers, and local government employees might all become valuable assets in the broader mental health agenda and can potentially assist through task-shifting.

Locally embedded and codesigned solutions are also more likely to factor in cultural influences such as complex intersectional identities (age, gender, caste and class, religion, language) and consider how best to

adapt interventions to be useful in the context of these identities. Furthermore, there is poor literacy around what mental illness and mental health are, and there is much stigma, which deters people from seeking help. Alternative explanatory models of illness (e.g. what this condition is called, what causes it, what we can do about it) also need better exploration, and local people are more likely to be able to work within these cultural divergences to optimize interventions. Nonetheless, there are some universal conclusions to be drawn, namely that sufficient social support, nutrition, safe housing and schools, and social conditions are essential; then medication and culturally adapted psychosocial interventions can be helpful and even used in a preventive function.

The *World Mental Health Report* (World Health Organization, 2022b) reinforces calls to provide better infrastructure, increase focus on prevention and health promotion, tackle stigma, and increase political support and investment (see Cuijpers et al., 2023). Clearly, provision of interventions to those with psychiatric disorder (the traditional health model) is still important; however, in accord with the public health model,

the emphasis is on prevention, and improving lives of children and adolescents, including school-based programmes. The enormity of the task to respond to global needs is challenging, yet there are some useful frameworks emerging. Following Rajabzadeh et al. (2021), it seems sensible to promote:

- establishing a Global Mental Health evidence base
- implementation of research into practice
- mental-health-relevant policies, services, and resources
- community participation
- learning from LMICs
- prioritizing LMIC relevant actions.

The overarching approach is to develop public mental health; tackle the treatment gaps (those with mental illness not receiving any intervention); culturally adapt interventions and develop new ones suited for specific sociocultural, geopolitical, and identity contexts; tackle individual and systems levels interventions; and tackle stigma and discrimination while protecting human rights (Barbui et al., 2017).

The global burden of mental illness

With the use of disability-adjusted life years (DALYs), the importance of neurological and psychiatric disorders as a global contributor of disease has become undeniable; the term *neurological, mental health, developmental and substance use disorders* (NMDS) is often used in the context of global health. By 2010, NMDS disorders were estimated to account collectively for 24.6% of global burden of diseases, excluding neurological sequelae in various infectious, systemic, and congenital disorders (Table 23.1) (Silberberg et al., 2015), having increased by around 40% in the preceding two decades (Whiteford et al., 2013).

Lay mental health workers

The Mental Health Gap Action Programme (mhGAP) has focused attention on *task shifting*: the delegation of responsibilities to lower-level cadres who are supported and supervised by more senior professionals, for example, using lay health workers to deliver psychological interventions in LMICs (Chibanda et al., 2015). In South Africa, a study using trained

community lay health workers improved the quality of mother–infant engagement (Cooper et al., 2002). Similarly, in Pakistan, it was possible to integrate a cognitive behaviour therapy intervention in the routine work of community health workers, resulting in better outcomes than usual care in reducing depression (Rahman et al., 2008). In India, a trained lay counsellor-led collaborative stepped care intervention improved recovery from depressive and anxiety disorders among patients attending primary healthcare facilities (Patel et al., 2010). In Chile, a multicomponent stepped-care intervention was more effective than usual care for depression at a public health care level (Araya et al., 2006). Finally, a ‘common elements’ approach to treatment of mood, post-traumatic stress, or anxiety problems involves trained lay workers to deliver treatment using a transdiagnostic approach; the early findings suggest a decrease in clinical symptoms with good retention and indicating good acceptability of treatment (Murray et al., 2014). This approach has been piloted in the United States, Iraq, and on the Thailand–Myanmar border.

Table 23.1 Proportions of age-standardized estimated disability adjusted life years (>0.5%) attributable to mental health, substance-use, and non-communicable neurological disorders by high-income and low- and middle-income countries

Disorder	Absolute DALYs* as global per thousand people (Rank**)	Percentage in high-income countries (developed)	Percentage in LMICs (developing)
Mental health and behavioural disorders			
Mental health (all)	185,200	11.1	6.73
Unipolar depressive disorder	74,260	4.01	2.81
Major depressive disorder	63,200 (2)	3.42	2.39
Anxiety disorders	26,830 (9)	1.60	0.98
Schizophrenia	15,000 (11)	0.85	0.49
Bipolar disorders	12,870 (17)	0.63	0.50
Dysthymia	11,100 (16)	0.6	0.42
Substance-use disorders			
Drug-use disorders	20,000 (19)	1.53	0.67
Alcohol-use disorders	17,640 (22)	1.52	0.56
Neurological disorders: non-communicable			
Cerebrovascular disease and strokes	102,200	5.97	3.79
Neurological disorders (all)	73,800	4.42	2.71
Migraine	22,360 (6)	1.21	0.84
Epilepsy	17,400 (23)	0.44	0.75
Alzheimer's disease and other dementias	11,350 (21)	1.75	0.22
Other neurological disorders	17,870	0.53	0.76

* DALY is defined as a measure of overall disease burden and expressed as the sum of years of potential life lost due to ill health, disability, or premature mortality.

** Number in brackets specifies where the disorder comes in the DALY ranking of all neurological, psychiatric, and substance abuse disorders. For further details of categories and estimates, see Silberberg et al. (2015).

Adapted from Silberberg, D., Anand, N. P., Michels, K., & Kalaria, R. N. (2015). Brain and other nervous system disorders across the lifespan—Global challenges and opportunities. *Nature*, 527(7578), S151–154. Copyright (2015), with permission from Nature Publishing Group, reproduced under the Creative Commons License 4.0.

Integration into primary health care

The challenge of scaling up mental health services in LMICs is less *what* to implement than *how* to implement. There is already robust evidence for a range of cost-effective interventions, but little evidence on how these may be delivered in diverse low-resource settings. A key element is to increase the capacity to treat common mental disorders within primary health care settings. The Programme for Improving Mental Health Care (PRIME) (Lund et al., 2016) was a large multisite

study in five LMICs to develop district mental health care plans. They highlighted some key components, including having shared objectives; the need for participation and engagement with local stakeholders; focusing on community, health facility, and health organization levels; challenges of overburdened primary health care systems; and the limited impact of training without systemic changes in the form of new mental health resources (usually in the form of an experienced mental health professional to provide support), referral pathways, improved medication supply, and reorientation of health facility managers (Hanlon et al., 2016).

Specific populations of concern

HIV/AIDS pandemic

People living with HIV/AIDS have a prevalence of over 30% of psychiatric disorders (depression, anxiety, and related conditions) in LMICs. The rates are higher than for non-HIV-infected individuals (Brandt, 2009), and mental and neurological disorders contribute to increased risk of HIV acquisition as well as to significantly poorer HIV disease outcomes (Chibanda et al., 2015). Depression is the most common mental disorder and is found among 14–32% of people living with HIV (MacLean and Wetherall, 2021). The identification of highly active antiretroviral treatments (ART) and the expansion in access to ART has resulted in greater numbers now living with chronic psychological comorbidities alongside HIV. A large treatment gap, particularly for depression, exists in sub-Saharan Africa, where the ratio of mental health professionals to the population is 1 per 2.5 million for psychologists, 1 per 1 million for mental health nurses, and 1 per 2 million for psychiatrists. There are some promising results for the effectiveness of cognitive behaviour therapy for common mental disorders in HIV/AIDS-affected populations, but the evidence is limited.

In considering the needs of mothers, 90% of the global number of pregnant women with HIV live in sub-Saharan Africa (Stein et al., 2014). Being diagnosed with HIV during pregnancy, which is when many women first learn of their diagnosis as it is part of antenatal screening programmes, increases the risk of depression. As women who are depressed are less likely to adhere to ART, this can compromise both the mother's survival and increase likelihood of vertical transmission of HIV to the foetus or infant.

Perinatal mental illness

Evidence is emerging that poor perinatal maternal mental health, especially depression in the context of socioeconomic disadvantage, is linked to poor infant growth and stunted development (Howard et al., 2014; Stein et al., 2014; see also Chapter 16). Post-natal depression in LMICs is associated with high rates of diarrhoea in children, which could contribute to their poor growth. Furthermore, comorbidity is more common in LMICs, likely increasing the adverse consequences of perinatal mental illness. There is evidence that perinatal depression raises the risks for offspring, even into late adolescence, with increased rates of depression. Of particular concern is the extent to which outcomes for children are moderated by persistent maternal disorder and

socioeconomic disadvantage. Perinatal mental illness might also account for a substantial proportion of maternal death in low-income countries as it is suspected that suicide is not fully reported or recorded. There is evidence that trained non-specialist health workers can provide effective interventions for post-natal depression in LMICs.

Several randomized controlled trials in LMICs showed that psychological interventions delivered by local community health workers can have a positive effect on parenting and child development. A study in Pakistan used a form of cognitive behaviour therapy together with parenting support that began during pregnancy (Rahman et al., 2008). No effect was noted on child growth, the principal outcome, but rates of post-natal depression were reduced, and parents reported decreased rates of children's diarrhoea, increased rates of immunization, and increased play with children. In Jamaica, an intervention targeting child rearing and parenting self-esteem led to improvements in both maternal depressive symptoms and infant global development compared with standard care (Baker-Henningham et al., 2005). As a final example, in a socioeconomically disadvantaged South African community, an intervention focused on helping mothers to attend to their infant's communication and respond to them sensitively led to improvements in the quality of mother-child interaction and increased rates of secure attachment (Cooper et al., 2009).

Children and adolescents

Children and adolescents comprise more than a third of the global population, with 80% of all children and adolescents living in LMICs. Addressing their mental health needs more effectively than is currently the case might alleviate suffering, improve educational attainment, and potentially reduce the burden of psychiatric disorders in adulthood (Kieling et al., 2011; Patton et al., 2016). Schools provide a potentially valuable location for mental health support services given the scarcity of dedicated mental health services for children in LMICs, since schools exist in almost all urban and rural contexts (Fazel et al., 2014b). For example, the School Health Promotion and Empowerment programme (SHAPE) intervention in India successfully trained lay school health counsellors to conduct physical health screening (visual and hearing) as well as mental health interventions. The programme included supervision of school counsellors. Over time, as the counsellor's skills improved, less supervision was needed and further benefits were observed with whole-school interventions

around violence and bullying reduction (Rajaraman et al., 2012).

The Lancet Commission on adolescent health and wellbeing (Patton et al., 2016) highlighted how the lack of mental health provision in LMICs leaves adolescents vulnerable to the harmful consequences of societal economic and cultural changes, as well as noting the equally positive opportunities to improve their health and wellbeing if their most basic health needs are better addressed. Better childhood health and nutrition, improvements to education—especially for girls, delays in age of first pregnancy, and new technologies offer positive opportunities for improved mental health. However, concerning global trends including austerity, especially following the Covid-19 pandemic, unhealthy lifestyles and nutrition, youth unemployment, less family stability, environmental degradation, armed conflict, and mass migration, all pose major threats to child and adolescent wellbeing. Of note, just over half of adolescents grow up in ‘multi-burden countries’, characterized by high levels of health problems, including diseases of poverty (HIV and other infectious diseases, undernutrition, and poor sexual and reproductive health), injury and violence, and non-communicable diseases. The Commission argued that the most powerful actions are intersectoral, multilevel, and multicomponent, and would need to involve young people themselves.

For more information on the background and issues relevant to child and adolescent psychiatry across the globe, see Rahman and Kieling (2015) and Patton et al. (2016).

Humanitarian emergencies

Natural disasters, conflicts, and other humanitarian disasters provide not only a high need but also a critical opportunity to scale-up care to the affected population. The prevalence of mental health and psychosocial problems in these situations is high. Most frequently, mental health researchers in humanitarian settings have

focused on identifying rates of post-traumatic stress disorder (PTSD) and other common mental disorders. However, severe psychiatric disorders (e.g. psychotic disorders), non-specific forms of psychological distress, and psychosocial problems specific to young people are also important (Reed et al., 2012). Evidence also exists of the effect of deteriorated environmental conditions on mental health and wellbeing in humanitarian settings, including undermined networks of social support, loss of opportunities for generating income, and lack of respect for human rights.

Interventions in these settings have been reported in a systematic review (Tol et al., 2011). In 160 reports, the five most reported activities were basic counselling for individuals (39%); facilitation of community support of vulnerable individuals (23%); provision of child-friendly spaces (21%); support of community-initiated social support (21%); and basic counselling for groups and families (20%). Most interventions took place and were funded outside national mental health and protection systems. Two studies showed promising effects of strengthening both community and family support. Studies on refugee children have shown that interventions delivered within the school setting can be successful in helping children overcome difficulties associated with forced migration (Tyrer and Fazel, 2014) as well as specific trauma-focused interventions (Fazel et al., 2015b).

With increasing appreciation of the importance of mental health and psychosocial support (MHPSS), international guidelines have been agreed integrating key components of mental health interventions into humanitarian assistance programmes (Marshall, 2022). The key factors include preparedness steps to be taken before emergencies occur, minimum responses to be implemented during the acute phase of the emergency, and comprehensive responses to be implemented once the minimum responses have been applied.

Further reading

Bhugra, D., Bhui, K., Wong, S. Y. S., & Gilman, S. E. (Eds.). (2018). *Oxford Textbook of Public Mental Health*. Oxford University Press. <https://doi.org/10.1093/med/9780198792994.001.0001>

Wilkinson, R., & Pickett, K. (2019). *The Inner Level: How More Equal Societies Reduce Stress, Restore Sanity and Improve Everyone's Well-being*. Penguin.

Patel, V., & Minas, H. (2013) *Global Mental Health: Principles and Practice*. Oxford University Press.

CHAPTER 24

Psychological treatments

Introduction 655

How psychological treatments developed 656

Classification of psychological treatments 659

Common factors in psychological treatment 659

Counselling and crisis intervention 661

Supportive psychotherapy 663

Interpersonal psychotherapy 663

Cognitive behaviour therapy 664

Individual dynamic psychotherapies 673

Treatment in groups 675

Psychotherapy with couples and families 679

Psychotherapy for children 681

Psychotherapy for older people 681

Treatments of mainly historical and cultural interest 681

Ethical problems in psychological treatment 683

Introduction

This chapter is concerned with various kinds of counselling, psychotherapy, behavioural and cognitive therapies, and some related techniques. The United Kingdom has a separate faculty and specialist training in psychotherapy for psychiatrists. In most countries, psychotherapy is considered a core aspect of a psychiatrist's role, fundamental to their professional identity. It is no longer routine for trainee psychiatrists to be trained fully in one or other form of psychotherapy, but rather they gain an overview and experience in several modalities. Irrespective of early training, it is important to continue learning, and strengthen skills in communications and psychotherapies over the course of a career. Modifications and improvements in therapies are occurring all the time.

Much of what follows relates to psychological therapies woven into the daily practice of psychiatry. The competence in psychotherapies that is expected of psychiatrists is likely to vary in the future, but currently for the UK Royal College of Psychiatrists, psychiatrists in training must develop skills in communications, and undertake a case based learning process (like Balint groups around two hours every two weeks), a short case

(12–20 sessions) and long-term therapy case (20–40 sessions) for core training under supervision; for advanced training, more intensive experience is required over three years, including developing expertise in supervision and teaching and research, and a period of personal therapy is often encouraged (mandatory for psychoanalytic and some other specialist trainings). This personal experience of therapy is not required in non-psychotherapy related specialties. Each modality of therapy will also have its own curriculum, accreditation process, and concepts of what mental illness is and how to intervene, including theoretical perspectives on how interventions work.

The subject is large, and some basic general principles underly this chapter.

- Psychological treatment is not given in isolation, and this chapter complements the chapters on physical treatment, social care, and community services.
- This chapter focuses on the general nature of the treatments; their use in specific disorders is covered in the relevant chapters.

Table 24.1 Examples of types of psychological therapies

Supportive
Counselling
Mentalization
Cognitive behaviour therapy (CBT)
Acceptance and commitment therapy
Trauma-focused therapies (e.g. narrative exposure therapy, EMDR)
Digital therapies (providing education and specific types of intervention, such as CBT)
Family therapy
Group therapy
Coupe therapy
Psychosexual therapies

- Psychological treatments are often combined with medication, considered in more detail with the relevant disorders.
- Many different techniques are considered here. Cognitive behaviour therapy (CBT) is the most widely used approach in current UK practice so will be described in somewhat greater detail.
- Supervised experience is essential before any of these treatments can be used with patients.

For a fuller account of commonly used psychotherapies, see Adshead and Fonagy (2012), Denman (2011), and Gabbard et al. (2005). See also Table 24.1.

Terminology. The word psychotherapy is used in two ways. It can denote all forms of psychological treatment, including counselling and CBT. More traditionally it indicates established psychotherapies that require specific training. These can involve personal therapy and include counselling and CBT. Generally, psychological treatment provides broad support in helping patients make sense of their difficulties and make decisions in their lives. However, psychotherapy can also be used to provide a specific intervention that seeks to address an underlying difficulty and does not seek to be only supportive; indeed, it may be distressing and the patient’s ability to cope must be assessed before recommending the more challenging types of therapy. Similarly, people in unstable social situations or in a social crisis are not always able to make use of the therapy or commit to intensive approaches. For people living in complex and unstable social situations, some supportive work might be required before or alongside the delivery of therapy, yet this is not built into many existing modalities and theories. Social interventions may need to be considered alongside therapy. Although a stable social situation is ideal, it may not be realistic and should be the basis of denying treatment. Equally, the therapeutic approach might warrant adaptation. Careful choice of modality becomes important, for example, people may benefit from creative arts, play, or narrative approaches and find these empowering even at times of difficulty. Finding ways to provide regular care becomes the focus for the therapist and the patient.

How psychological treatments developed

The use of psychological healing is as old as the practice of medicine—parallels have been drawn with the ceremonial healing in temples in ancient Greece. In psychiatry, psychological treatment evolved towards the end of the eighteenth century with developments in hypnosis. The physician Anton Mesmer (1734–1815) challenged the prevailing practice of ‘casting out devils’ in 1775. He proposed that the body could be influenced by magnetism, initially actual magnets, but more importantly the therapist’s force of personality. ‘Mesmerism’ was renamed hypnosis by the Manchester doctor James Braid, who believed it was related to sleep.

Treatment with hypnosis was revived in France by the eminent neurologist Jean-Martin Charcot (1825–1893).

He treated many patients suffering from hysteria and recognized that hypnosis worked by suggestion. Freud visited Charcot to study hypnosis and used it with his patients back in Vienna. He used hypnosis not to modify symptoms directly, but to release the emotions associated with repressed conflicts that he believed to be their cause. Freud’s revolutionary advance was his recognition that patients could recall forgotten events and conflicts without hypnosis. Recall was achieved by the patient lying on a couch and being encouraged to let their mind wander with the therapist out of sight—‘free association’. In time he attended to the intensity of the relationship with his patients—‘transference’. These discoveries formed the basic technique of psychoanalysis

and subsequently of the larger group of dynamic psychotherapies.

Freud published vivid accounts of his new treatment and elaborated his increasingly complex theories, collecting a group of followers. Some of these later broke away, forming their own ‘schools’ of dynamic psychotherapy. These developments will be described briefly. More detailed descriptions are widely available, and for an overview, see Burns and Lundgren (2015). In summary, as introduced in Chapter 5, Freud suggested a psychic structure including the *superego* (internalization of parental figures and morality), the *ego* (sense of self that negotiates with the real world), and the *id* (a reservoir of thoughts and feelings that are strongly characterized by instinct and impulse). Freud proposed the existence of an unconscious; painful or morally unacceptable feelings and wishes were repressed and were not consciously known or felt. Freud especially gave rise to concepts of transference, the patient behaving as if the therapist were a past figure in their life (a reconstruction of that figure in the mind rather than real aspects of that figure). These terms are very precise and should be used with care. Charles Rycroft’s (1995) classical dictionary of psychoanalysis is a useful source of definitions to guide the reader through a complex theoretical language. For a more detailed glossary of concepts, see Laplanche and Pontalis (1988).

Early departures from Freud’s original group

Alfred Adler left Freud’s group in 1910; he stressed social factors in personal development and rejected the libido theory. He also stressed social factors in personal development, introducing the term ‘inferiority complex’. His ‘individual analysis’ focused on current problems and solutions and was highly influential with American analysts. Carl Jung emphasized the inner world of fantasy, and the interpretation of unconscious material, deduced from dreams, paintings, and other artistic productions. Jung believed that part of the content of the unconscious mind was common to all people (the ‘*collective unconscious*’) and was expressed in universal images, which he called *archetypes*.

The neo-Freudians formed in the United States in the 1930s from predominantly Jewish refugees from Nazi Germany. While they accepted that the origins of neurosis were in childhood, they rejected Freud’s emphasis on early infantile sexuality and considered family and social factors more important.

Melanie Klein in London adapted psychoanalytical techniques for use with very young children. She

interpreted their play and originated the ‘object relations’ school of psychoanalysis. ‘Object’ is a confusing term here as it refers both to a person (e.g. the mother) and to parts of that person (e.g. the mother’s breast), as well as (most importantly) to their internal psychological representation. The language in her psychoanalytic writing can be difficult to understand. The language can be excessively dramatic, or harsh, and emphasize strong instinctual feelings of love and hate. Her most famous model of emotional development suggested movement moving through a ‘paranoid–schizoid position’ to the ‘depressive position’ in early infant and child development. Klein has been widely influential, particularly for therapists working with children and with severely ill patients (Spensley, 2014).

Attachment theory originated in the work of John Bowlby, a British analyst. The theory is that infants need a secure relationship with their parents in reality—not just in fantasy. Insecure attachments can lead to difficulty in establishing relationships, and emotional problems later. Bowlby’s ideas had a considerable effect on the care of children, such as the need to minimize periods of parental absence and, for example, allowing parents to visit and stay with their child when admitted to hospital. For a review of the historical development of attachment theory and its relevance for psychotherapy, see Levy and Johnson (2019).

Brief psychodynamic psychotherapy. The Hungarian psychoanalyst Sándor Ferenczi saw the need to develop treatments shorter than psychoanalysis. He did this by setting time limits, making the role of the therapist less passive, and planning the main themes of treatment. These innovations have found their way into the brief dynamic psychotherapy that is used today. Such therapy can be used to help patients suffering from depression, bereavement, or anxiety. This approach requires rapid understanding and active interpretation of transference (described below) by the therapist; however, the time-limited process places some pressures on both patient and therapist. Ultimately, selecting patients suitable for this form of therapy is the key to success, as someone with more complex or enduring problems may not respond or engage.

The trend has continued towards briefer treatment that attends more to the patient’s current problems than to those in the past. For example, *interpersonal therapy* (directed to current interpersonal problems in depressed patients) and *cognitive analytic therapy* (see page 674), use cognitive therapy techniques within a framework of psychodynamic understanding.

Patient selection is critical, and patients can enter a therapy finding they are unable to settle or do not

understand the process or their role and that of the therapist. These transference feelings are to be understood and may reflect the core issues on which therapy should progress; although patients may wish to choose their therapist or switch, there is little provision for this in public services. Much psychotherapy work does take place in the private sector, where there is more choice and the threshold for accessing care is not as high as in public services. In public services, in order to minimize failures of therapy, adequate psychoeducation is commonly offered before therapy, as well as detailed assessment of suitability for a particular therapy, and an assessment of a patient's ability to use and tolerate the approach of the particular recommended therapy, as therapies can sometimes lead to unexpected adverse effects (Crawford et al., 2016; Strauss et al., 2021).

More recently *mentalization* as a first-stage intervention has been proposed to strengthen the core skills and abilities that can help people with their emotions, and is especially useful for people receiving a diagnosis of borderline personality disorder (Bateman and Fonagy, 2010). Sometimes, psychoeducation and mentalization are offered initially, during which an assessment of the person's need for and receptivity to more intensive psychodynamic therapies can be reviewed. If a person is unable to make use of mentalization, it is unlikely they will settle in less structured psychoanalytic approaches, although more structured brief therapies may still be useful. Transference-focused therapy is also an effective alternative for those presenting with a diagnosis of borderline personality disorder (Doering et al., 2010).

The development of cognitive behaviour therapy

Behaviour therapy. Interest in a treatment based on scientific psychology arose in the early twentieth century. Watson and Rayner (1920) in the United States

used learning principles in the treatment of children's fears, and aversion therapy for alcoholism. Maudsley psychologists in the 1930s used learning principles in treatments for phobic disorders. Joseph Wolpe in South Africa published *Psychotherapy by Reciprocal Inhibition* (Wolpe, 1968), describing a treatment for neurotic disorders, based on learning theory and making use of relaxation. In the United States, B. F. Skinner (1954) proposed operant conditioning in the treatment of psychiatric disorders. Wolpe's ideas were adopted in the United Kingdom, and Skinner's ideas were initially more influential in the United States. These approaches converged, and practice in the two countries is now similar.

There has always been an emphasis on evaluation for these new methods. The first clinical trial was reported (Gelder et al., 1967), followed by several trials establishing a strong evidence base for behavioural methods.

Cognitive therapy began with the work of A. T. Beck, a US psychiatrist who was dissatisfied with psychoanalytical psychotherapy for depressive disorders. Beck noted recurring themes in the thinking of depressed patients, and he concluded that these themes were an essential part of the disorder and had to be changed by challenging them in specific ways (see Box 24.1). US psychologists were also dissatisfied with operant conditioning and its 'black box' approach. They proposed that the recurrent thoughts played a part in maintaining distress, with suggestions of how these thoughts might be controlled (Meichenbaum, 1977).

CBT. These cognitive approaches were integrated with behaviour therapy to produce CBT. The strong evidence base, clearly described procedures, and relatively brief treatment time of CBTs have made them the preferred psychological treatment for many disorders (David et al., 2018). As outlined in Table 24.1, CBT is an element of many therapies. The early emphasis on cognition and behaviour attracted some criticism from specialists in other modalities where people presented

Box 24.1 Cognitive errors in depressive disorders

Exaggeration: magnifying small mistakes and thinking of them as major failures.

Catastrophizing: expecting serious consequences from minor problems (e.g. thinking that a relative who is late home has been involved in an accident).

Overgeneralizing: thinking that the bad outcome of one event will be repeated in every similar event in future

(e.g. that having lost one partner, the person will never find a lasting relationship).

Ignoring the positive: dwelling on personal shortcomings or on the unfavourable aspects of a situation while overlooking the favourable aspects.

with complex emotional states that were not easily contained or understood in a narrow paradigm, albeit a strongly researched one. However, CBT is evolving and a third wave of approaches is now commonly available. These attend to issues such as mindfulness, emotions, acceptance, the relationship, values, goals, and

meta-cognition. As a consequence, new modalities are being evidenced as helpful, for example, acceptance and commitment therapy (ACT), dialectical behaviour therapy (DBT), mindfulness-based cognitive therapy (MBCT), functional analytic psychotherapy, and meta-cognitive therapy, among others (Hayes and Hofmann, 2017).

Classification of psychological treatments

Several simple classifications of psychological treatments have been proposed. Two are given here, with a further classification given below, with hints to identify their use in most healthcare systems.

1. Classification by *technique*:

- Eclectic
- Psychodynamic
- Cognitive behavioural
- Other (e.g. systems theory).

2. Classification by *number* of patients taking part:

- Individual therapy
- Couple therapy
- Family therapy
- Small and large group therapy.

These two classifications can be combined—for example, *individual cognitive behavioural* or *psychodynamic group* therapy.

Psychotherapy in public services

Psychological treatment is the principal treatment for some psychiatric disorders, alone or with medication. Counselling, crisis intervention, and CBT are used in this way when they have been shown to be effective in clinical trials. Long-term dynamic psychotherapy can take

years and can be expensive. It is less often available in public services in the United Kingdom. A recent review suggests long-term therapies do benefit and are justified despite cost and duration (Leichsenring et al., 2023). Brief psychodynamic approaches may be effective, for example, in depressive disorders (Malhi et al., 2015). ‘Subthreshold’ conditions are now generally offered counselling, and psychodynamic treatment for them now occurs mainly in private practice. Planning their uses within a public health service has generated a third classification.

1. *Simple psychological aspects of all health care.* This refers to skills and techniques to help individuals to adjust to stressful situations or confront difficult decisions. These are often considered aspects of a good doctor–patient relationship. These are usually supportive and problem solving, and can be delivered by the primary care team and other trained practitioners.
2. *Moderately complex and provided by most mental health professionals.* This includes simpler CBT and brief dynamic psychotherapies. These treatments are usually an identified part of the management plan that includes medication and social measures.
3. *Highly complex and provided by formally trained therapists.* This group includes the more complex psychodynamic and CBTs. These are used to treat more severe or complex disorders, alone or as part of a wider plan of management.

Common factors in psychological treatment

Different psychological treatment methods achieve results that are broadly equal and are greater than placebo. The core features that psychotherapies share may be more important than their differences. Jerome Frank (1974) identified them and they are listed in Box 24.2. Recent work seeks to identify which mechanisms

are influential in persistent depression and which subgroups of depression might be treated by which therapies (see Malhi et al., 2015 for a comprehensive review). Psychotherapies are also as effective as pharmacotherapies in depression (Cuijpers and Gentili, 2017), although half of depressed patients do not respond and only

Box 24.2 Common factors in psychotherapy

The therapeutic relationship: The most important of the common factors in psychotherapy but can become too intense.

Listening: Listening attentively shows concern for the patient's problems and develops the helping relationship in which the patient feels understood.

Release of emotion: Helpful early in treatment, but repeated release is seldom useful. Intense and rapid emotional release is called abreaction.

Restoration of morale: Many patients have suffered repeated failures, and may feel helpless. Improved morale enables the patient to begin to help himself or herself.

Providing information: Patients may remember little of what they have been told about their condition because

of poor concentration. Information should be as simple as possible, expressed clearly, repeated, and perhaps written down.

Providing a rationale: All psychotherapies provide some explanation for the current problem, which improves the patient's confidence. It may be by direct explanation (as in short-term psychotherapy), or fostered indirectly (as in much long-term psychotherapy).

Advice and guidance: These are part of all psychotherapy, whether direct as in brief therapies or indirect as in long-term therapies.

Suggestion: All psychological treatment contains an element of suggestion, which is powerful during the early stages.

a third show remission (Cuijpers et al., 2021). Further advances in psychotherapeutic practice build on early CBT studies, producing third-wave CBT (Hayes and Hofmann, 2017), cognitive analytic therapy (CAT; Ryle et al., 2014), and ACT (Webster, 2011). Each is suited to specific situations and complexities. The evidence base is constantly evolving, and numerous meta-analyses demonstrated respective effectiveness of different modalities as well as the hazards, as psychotherapies are not without adverse effects (Strauss et al., 2021).

Newer therapies seem to identify which mechanisms and which processes are most important for specific patients, carrying specific diagnoses and complex comorbidities, and catering for their beliefs and preferences. Furthermore, given the way care is organized and the need for managing resources, some interventions (mindfulness, mentalization, counselling) are promoted as first line to prepare people for more intensive therapies, if there is not a specific or urgent indication for intervention in other ways. The psychotherapy formulation should indicate what approach is most appropriate for whom.

Transference and countertransference

Therapeutic relationships are inevitably emotional, and they can sometimes become very intense. These powerful emotions were labelled 'transference' and 'countertransference'.

Transference refers to intense feelings in the patient towards the therapist. It can become increasingly intense as treatment progresses. Transference can be positive, with warm feelings, or negative, with critical or hostile feelings. Initially considered an impediment, transference and its resolution are now seen as essential parts of successful treatment. The skills to notice and make use of transference, in the form of transference interpretations, often requires supervision and practice with diligence and discipline. Curiosity and attention to unspoken aspects of the patient's presentation are important. Transference-focused therapies require considerable training, but understanding their nature and how to respond are important for all psychotherapies and for psychiatric practice in general.

Countertransference refers to intense feelings in the therapist towards the patient. The important point is the therapist notices, and tries to make sense of, their feelings as a way of understanding more about the patient's inner world. Apart from managing oneself professionally, and not acting on impulse in response to these strong feelings, the use of countertransference in therapy has become more popular and important. It is a way of understanding how the patient relates and patterns of relational interactions. It is also one of the reasons a personal therapy is advised for advanced practitioners (interested readers are referred to dictionaries of psychoanalysis for precise use and historical understanding of these terms).

Counselling and crisis intervention

Counselling

Broadly speaking, counselling is less formal and less extensive (fewer sessions) than psychotherapy. Counsellors have a shorter training than psychotherapists and are more open and equal in the therapeutic relationship. Often counselling is focused on a specific problem (bereavement, drug abuse) and problem solving and support, rather than an attempt to permanently alter personality or uncover conflicts and release repressed unconscious aspects of the self.

Counselling is more widespread than formal psychotherapy, and counselling skills are a valued part of the professional identity of nurses and social workers. It is widely provided by voluntary sector organizations such as the Samaritans, Cruse, and Relate in the United Kingdom.

Counselling incorporates the non-specific factors shared by psychotherapies (see Box 24.2). The relationship between the counsellor and the person who is being counselled is believed to be the primary therapeutic agent in all counselling, but the relative importance of giving information, allowing the release of emotion, and exploring the situation varies.

In the model of counselling developed from Carl Rogers' *client-centred approach*, the counsellor largely restricts their interventions to helping the client understand their feelings better by reflections back to the emotional content of the client's utterances, often simply repeating the last statement with an interrogative tone. The approach is optimistic in tone and actively encouraging but focused on increasing self-awareness. Rogerian, or client-centred, counselling is widely available in primary care and third-sector agencies but has been largely replaced in secondary mental health care by the more structured and focused procedures.

Approaches to counselling

Problem-solving counselling is highly structured and is particularly suitable for problems related to stressful circumstances or when life problems are exacerbating or maintaining other disorders. Basic counselling is combined with a systematic approach to the resolution of problems. The patient is helped to:

- *identify and list* problems causing distress
- *consider courses of action* to solve or reduce problems
- *select a problem and course of action* that appear feasible and likely to succeed

- *review the results* and select another problem if successful, or another course of action if not.

Interpersonal counselling was developed by Klerman (1984) (Markowitz and Weissman, 2012). Interpersonal therapy has many similarities to the problem-solving approach. Attention is focused on current problems in personal relationships within the family, at work, and elsewhere. These problems are considered under four headings—*loss, interpersonal disputes, role transitions, and interpersonal deficits*. Using a problem-solving approach, the therapist encourages patients to consider alternative ways of coping with these difficulties, and to try these out between sessions. For a review, see Blanco et al. (2009).

Psychodynamic counselling encourages more understanding about unconscious processes by which previous relationships influence current feelings and relationships. Its developmental approach fits well in student health centres.

Counselling for specific purposes

Debriefing

Debriefing for survivors of disasters has become a worldwide phenomenon. Survivors are encouraged to recall the distressing events, with emphasis on emotional release, and responses to the immediate problems. Evidence from clinical trials is discouraging, suggesting that this approach, especially for those who do not want to talk, may prolong ruminations (Mayou et al., 2000) and distract from essential social supports and the traditional advice to 'get back on the horse as soon as possible'. Guidance from the National Institute for Health and Care Excellence (2018b) specifically advises against its routine use. However, group discussions with a narrative approach can assist those who have experienced potentially traumatic events, and those who want some support. Debriefing remains common practice but is offered more on an optional basis.

Counselling for relationship problems

Couples are encouraged to talk constructively about problems in their relationship. The focus is on the need for each partner to understand the point of view, needs, and feelings of the other, and also to identify positive aspects of the relationship and potential strategies to move forwards. A 'safe space' to explore can clarify what further steps are required, and whether more intensive couple therapy, often provided in the voluntary sector

or more intensive approaches delivered through public funded services (where available), are needed.

Bereavement counselling

Bereavement counselling draws heavily on the identified stages of normal grief (see Box 7.8). It combines an opportunity for emotional release (including anger), information about the usual course of grieving, and support in passing through it. At an early stage it may involve sensitive advice about viewing the body and disposing of clothing at an appropriate time. It can also extend to advice on practical matters and living without the deceased person. Bereavement counselling is not always delivered in mental health services, and the voluntary sector and social care services often contribute. Organizational counsellors might also be available to support people who are bereaved.

Counselling in primary care

In primary care, many patients are referred to practice counsellors and (in the United Kingdom) IAPT workers (IAPT stands for Improving Access to Psychological Therapies; see Chapter 26, now called NHS Talking Therapies). IAPT workers have received a relatively limited and focused training in CBT, but not a formal training as a health or social care professional. GP counsellors use various methods of brief treatment, although most often they employ non-directive Rogerian approaches, usually four to six sessions. Although very popular, the effectiveness of counselling in primary care is modest (Bower et al., 2003).

Crisis intervention

Crisis intervention was originally conceived to use the crisis as an entry to longer-term problems but now is used mainly to cope with current crises. It originated in dealing with grief and disasters (Caplan, 1964; Lindemann, 1944; Roberts and Ottens, 2005). Caplan outlined four stages:

1. Emotional arousal and problem-solving efforts.
2. Increasing arousal leading to disorganization of behaviour.
3. Testing alternative ways of coping.
4. Exhaustion and decompensation.

Crisis intervention seeks to limit the reaction to the first stage, or to avoid the fourth stage.

Common problems leading to crisis include:

- *Loss and separation*, such as bereavement or divorce, but also during severe illness.
- *Role changes*, such as marriage, parenthood, or even a new job.

Box 24.3 Crisis intervention

Treatment is immediate, brief, and collaborative

Stage 1

Reduce arousal
Focus on current problems
Encourage self-help

Stage 2

Assess problems
Consider solutions
Test solutions

Stage 3

Consider future coping methods
Crisis intervention methods

- *Relationship problems*, such as those between sexual partners, or between parent and child.
- *Conflicts*, arising from pressing but impossible choices.

The methods used in crisis intervention (see Box 24.3) resemble interpersonal counselling and problem-solving counselling, but with a greater emphasis on reducing arousal. Treatment starts as soon as possible after the crisis and consists of a few sessions over a period of days or, at most, a few weeks. The focus is on current problems, although relevant past events can be considered. High levels of emotional arousal interfere with problem solving, and the first aim of treatment is to reduce arousal. Reassurance and ventilation of emotions are usually effective, but anxiolytic medication is often used. Roberts and Ottens (2005) suggest seven stages of intervention:

1. Plan and conduct a thorough biopsychosocial and lethality/imminent danger assessment.
2. Make psychological contact and rapidly establish the collaborative relationship.
3. Identify the major problems, including crisis precipitants.
4. Encourage an exploration of feelings and emotions.
5. Generate and explore alternatives and new coping strategies.
6. Restore functioning through implementation of an action plan.
7. Plan follow-up and booster sessions.

Supportive psychotherapy

Supportive psychotherapy is one of the most difficult but also one of the most important skills that any psychiatrist must acquire. It is used to help a person to cope with enduring difficulties such as chronic mental or physical illness, and in the care of the dying (pages 219, 397, 663). Supportive therapy is based on the common factors of psychological treatment (see Box 24.2). Its basic elements, listed in Box 24.4, are:

- *The therapeutic relationship.* A trusting and supportive relationship is central in sustaining patients with long-term difficulties. The most common anxiety voiced by staff is the risk of excessive dependence. An effective relationship is most often achieved by an acceptance of the realistic need for dependence. After

Box 24.4 Basic procedures of supportive treatment

Develop a therapeutic relationship
Listen to the patient's concerns
Inform, explain, and advise
Allow the expression of emotion
Encourage hope
Review and develop assets
Encourage self-help

all, the patient is seeing you because they 'need' you, at a time when they may feel vulnerable.

- *Listening.* As in all forms of psychological treatment, the patient must feel that they have their doctor's full attention and sympathy, and that their concerns are being taken seriously.
- *Information and advice* are important, but their timing should be considered carefully. Information should be accurate, but it is not necessary to explain everything during the first session. Indeed, the patient may need to receive information gradually and have time to absorb it. Most patients indicate, directly or indirectly, how much detail they wish.
- *Emotional release* and its acceptance can be helpful.
- *Encouraging hope* is important, but unrealistic reassurance can undermine trust. Reassurance should always be specific, offered only when the patient's concerns have been fully understood. A positive approach builds on encouraging the patient to exploit their assets and opportunities.
- *Persuasion.* Doctors can help patients to take difficult but necessary decisions and actions.

Self-help groups can give valuable support to patients and to carers because it comes from others who have struggled with the same problems. Support groups vary enormously, and it is important have some familiarity with a particular group before recommending it. For an account of supportive treatment, see Bloch (2006).

Interpersonal psychotherapy

Interpersonal psychotherapy (IPT) was developed as a structured psychological treatment for the interpersonal problems of depressed patients (Klerman, 1984). IPT encourages the patient to recognize that depression is an illness and that they are suffering from it. The treatment is highly structured. The number and content of treatment sessions are planned carefully. The initial assessment period lasts from one to three sessions. Interpersonal problems are considered under the following four headings:

1. Bereavement and other loss
2. Role disputes

3. Role transitions
4. 'Interpersonal deficits' (e.g. loneliness).

Each problem is considered using specific situations, and alternative ways of coping are evaluated. Clear goals are set and progress towards them is monitored, with new coping strategies tried out in homework assignments. In the middle phase of treatment, specific methods are used for each of the four kinds of problem listed above. For grief and loss, the methods resemble grief counselling. For interpersonal disputes and role transitions, patients are helped to identify clearly the issues in the dispute, as well as any differences

between their own values and those of the relevant others. They are helped to recognize their own contributions to problems that they ascribe to that person. Interpersonal deficits are addressed by analysing present relationship problems and previous attempts to overcome them, and then discussing alternatives.

In the final two or three sessions, future problems are anticipated and considered.

Several clinical trials have shown that IPT is effective for depressive disorders in adults and adolescents, dysthymia, and bulimia nervosa (Cuijpers et al., 2016). For an account of IPT, see Blanco et al. (2009).

Cognitive behaviour therapy

All psychiatric disorders have cognitive and behavioural components, and both have to change if the patient is to recover. With other treatments, change comes about indirectly—for example, as mood improves with antidepressant therapy, or as conflicts become resolved with psychotherapy. CBT aims to change cognitions and behaviour directly. Unlike dynamic psychotherapy, it is not concerned with the origins of the disorder, but what is maintaining it now.

Behaviour therapy focuses on what provokes symptoms or abnormal behaviour. *Avoidance* is particularly important in phobic and anxiety disorders, as it prevents the normal extinction of the anxiety response. Many behaviours are *maintained by their consequences*. For example, the reduction in anxiety following escape from stressful situations reinforces the phobic avoidance. *Reinforcement* also entrenches symptoms, such as when parents pay more attention to a child's noisy and unruly behaviour.

Cognitive therapy generally focuses on two kinds of abnormal thinking—*automatic thoughts* and *dysfunctional beliefs and attitudes*. Automatic thoughts are often stereotyped and exaggerated responses to stressors such as 'everybody despises me' after mild criticism. They provoke an immediate emotional reaction, usually of anxiety or depression, and undermine self-esteem. Dysfunctional beliefs and attitudes determine the way in which situations are perceived and interpreted.

Three factors are thought to maintain dysfunctional beliefs and attitudes.

1. *Attending selectively* to evidence that confirms them, and ignoring or discounting evidence that contradicts them. For example, patients with social phobias attend more to the critical behaviour of others than to signs of approval.
2. *Thinking illogically* in stereotyped ways (Box 24.5).
3. *Safety-seeking behaviour*, believed to reduce an immediate threat, but perpetuates the fear.

General features of cognitive behaviour therapy

Core features of CBT:

- *The patient is an active partner.* The patient takes an active part in treatment, with the therapist acting as an expert adviser who asks questions, and offers information and guidance.
- *Socratic questioning.* Automatic thoughts are challenged by questioning them—Are there other ways of understanding what is happening? What are the implications of thinking this way?
- *Attention to provoking and maintaining factors.* The patient keeps daily records to identify factors that precede or follow the disorder, and which may be provoking or maintaining it. This kind of assessment is sometimes called the *ABC approach*, the initials referring to Antecedents, Behaviour, and Consequences.
- *Attention to ways of thinking*, revealed by recording thoughts associated with the behavioural or emotional disturbances.

Box 24.5 Examples of illogical thinking

Overgeneralization: Drawing general conclusions from single instances (e.g. 'He does not love me, so no one will ever love me').

Selective abstraction: Focusing on a single unfavourable aspect of a situation and ignoring favourable aspects.

Personalization: Blaming themselves for the actions of other people.

All-or-nothing thinking: Viewing people or situations as 'black-and-white'.

Box 24.6 An agoraphobic patient's record of a behavioural experiment

Situation: Shopping in a crowded supermarket.

My predictions: I shall panic and feel dizzy. Unless I tense my stomach muscles and breathe deeply, I shall faint.

Experiment: When anxious, do not tense stomach muscles or breathe deeply.

Outcome: I did panic quite badly. I felt dizzy, but less so than usual. I did not faint.

What I learned: I did not faint despite a severe panic, and without actions to prevent fainting. I may be wrong thinking I shall faint whenever I panic; tensing and deep breathing may not help. My therapist could be right in thinking that deep breathing makes me feel more dizzy.

What I should do next: Repeat the experiment next time I go shopping.

- *Treatment as investigation* ('collaborative empiricism'). Therapeutic procedures are usually presented as experiments which, even if they fail, help reveal more about the condition.
- *Homework assignments and behavioural experiments.* Patients practise new behaviours between sessions with the therapist, or carry out experiments to test explanations suggested by the therapist (Box 24.6).
- *Highly structured sessions.* At each session, an agenda is agreed, and progress since the last session is reviewed, including any homework. New topics are considered, the following week's homework is planned, and the main points of the session are summarized.
- *Monitoring.* Assessment includes checking daily records kept by the patient, and often includes rating scales.
- *Treatment manuals* describe procedures and how they are applied. Manuals ensure that therapists stick to effective procedures.

Assessment for cognitive behaviour therapy

Topics to be covered

As well as a full psychiatric history, certain additional topics are addressed (see Box 24.7). For each of the

Box 24.7 Topics to be considered during assessment for cognitive behaviour therapy

1. A description of each problem, including behaviour, thoughts, and emotions associated with it
 - Where it occurs most often
 - Common prior events
 - Response to these events
 - What follows the problem
2. Factors that alleviate or worsen the problem
3. Maintaining factors
 - Avoidance
 - Safety behaviours
 - Selective attention
 - Ways of thinking
 - The responses of others

presenting problems, the therapist enquires about antecedents, behaviour, and consequences (the ABC approach). The term '*behaviour*' is used to include *thinking and emotion*, as well as *actions*. By considering the ABC sequence on several occasions, regular patterns of thinking and responding are identified. The therapist focuses on the patient's reasons for their beliefs, which is essential for planning experiences that will change them.

Sources of information

Self-monitoring. Records of thoughts, behaviours, and experiences over days or weeks are made in real time. The record sheet usually has columns for symptoms, thoughts, emotions, and actions, plus the day and time at which they occurred. Events immediately preceding the problem are noted, as well as those occurring at the time and afterwards.

Observations during treatment sessions. The patient may be asked to imagine situations in which problems arise, and to report accompanying thoughts and emotions. Symptoms resembling the disorder may be induced (e.g. panic-like symptoms produced by hyperventilation), and accompanying thoughts and emotions noted.

Special interviewing. Some patients need help to become aware of their maladaptive beliefs. *Laddering* involves a series of questions, each about the answer to the previous question. For example, a patient with an eating disorder might be asked what would happen if she were to gain weight, and she answers that she would lose her friends. To the question 'Why?' she might reply that she would be unlikeable. To a further question

‘Why?’ she might say that only thin people are attractive and popular.

The formulation

The information obtained in these various ways is combined with the usual psychiatric history in a formulation consisting of:

- the type of *events that provoke symptoms*
- any *special features of these events*
- *background factors*
- *maintaining factors*, including avoidance, safety behaviours, and ways of thinking.

The formulation is guided by a cognitive model of the disorder. The therapist discusses the formulation with the patient and may build it up, step by step, perhaps using a diagram on paper or whiteboard. The formulation is modified in light of the discussion.

Behavioural techniques

There are many behavioural techniques, some for a single disorder (such as the enuresis alarm; see page 451), and others such as exposure that can be used for a variety of disorders. Here we describe the more commonly used methods. Evidence for them is considered in the relevant chapters for the particular disorders.

Relaxation training

This usual technique is ‘progressive muscular relaxation’, in which patients are trained to relax individual muscle groups one by one, and to regulate their breathing (Kohl, 2002). This form of therapy is used in anxiety disorders with good effects (Montero-Marin et al., 2018) and can be learned in part from pre-recorded instructions or in a group. However, it has to be practised regularly, and many patients lack the motivation to do this.

Exposure

Exposure is used to reduce avoidance behaviour, especially in the treatment of phobic disorders. For simple phobias it is often sufficient to use exposure alone, but for complex phobic disorders exposure is usually combined with cognitive procedures. Exposure can be carried out *in practice* by confronting the actual situations that provoke anxiety or *in imagination* by imagining the phobic situations vividly enough to induce anxiety. Exposure can be gradual, progressing through a series of increasingly difficult situations (*desensitization*), or abrupt and intensive (*flooding*). In practice, exposure is

usually paced between these two extremes, preferably in practice rather than in imagination. Virtual reality with its ability to provide a realistic but ‘safe’ immersive environment is also emerging as a useful addition to conventional exposure techniques.

Desensitization

In desensitization the patient is helped to:

1. *Construct a hierarchy* of situations that provoke increasing degrees of anxiety. About ten items are chosen with an equal increment of anxiety between them. Sometimes two or more hierarchies can be constructed if no obvious theme links the stressful situations.
2. *Enter or imagine entering the situations* on each step of the hierarchy until achieved without anxiety.
3. *Use relaxation* while entering or imagining the situation to reduce the anxiety response, and enhance the imagery.

Flooding

In flooding, the patient enters the most feared situation from the start, and remains there until the anxiety has diminished. The process is repeated with other near-maximal stimuli. The experience is distressing, and the results are no better than those obtained with desensitization, so flooding is seldom used.

Exposure in routine practice

Sessions last for about 45 minutes of entering a feared situation every day, either alone or with a relative or friend. Usually anxiety diminishes with each exposure. Sometimes, if the anxiety becomes overwhelming, the treatment has to restart from a less-stressful situation. Disengagement from anxiety-provoking situations by thinking of other things is a commonly encountered defence.

Exposure with response prevention

This is a treatment for obsessional rituals. The pressure for rituals decreases if they can be resisted for periods of about an hour.

1. The therapist *explains the rationale for treatment and agrees targets* for exposure, such as to touch a ‘contaminated’ object such as a door handle and not to wash their hands for the next hour. A more advanced target might be to do all the household cleaning without washing their hands until the task is completed. Patients need to feel confident that every task

will be agreed in advance and that they will never be faced with the unexpected.

2. The therapist may demonstrate the necessary exposure, a procedure known as *modelling*.
3. Initially the therapist accompanies and supports the patient during prevention, but later the patient does this on their own.
4. With progress the patient practises entering previously avoided situations that normally provoke rituals. This is called *exposure*.

The obsessional thoughts that accompany rituals usually improve as the rituals are brought under control. Obsessional thoughts that occur without rituals are more difficult to treat. *Habituation training* is a form of mental exposure treatment in which patients dwell on the obsessional thoughts for long periods or listen repeatedly to a recording of the thoughts spoken aloud for an hour or more. Alternatively, *thought stopping* is used (see distraction techniques on page 187).

Social skills training

Some aspects of social behaviour include skills that can be learned, for example, making eye contact or starting a conversation. These skills can be improved through modelling, guided practice, role-play, and video feedback. The training is mainly useful for socially anxious people to improve self-confidence and in the management of adolescents with autism and Autism Spectrum Disorder (Dubreucq et al., 2022).

Assertiveness training

Assertiveness training is a form of social skills training in which patients practise appropriate self-assertion. By a combination of coaching, modelling, and role reversal, patients are encouraged to practise appropriate verbal and non-verbal behaviour, and to judge the level of self-assertion that is appropriate to various situations. Assertiveness training has, surprisingly, been found helpful not just with timid people but also with aggressive individuals, as it teaches a more appropriate way of channelling anger.

Anger management

In this form of social skills training, patients are helped to:

- identify situations that lead to anger
- identify attitudes that lead to anger that is out of proportion
- identify factors that reduce restraints on anger, especially the use of alcohol

- discover and practise alternative ways of dealing with situations that provoke anger—for example, delaying their response until anger can be brought under control ('count to 10').

Self-control techniques

All behavioural treatments aim to increase patients' control over their own behaviour. Self-control techniques attempt to do this directly without the intermediate step of changing thoughts or emotions as in cognitive therapy. Self-control techniques are based on operant conditioning principles, and on the studies by Bandura (1969) of the role of self-reward in social behaviour. Overeating and excessive smoking are examples of target behaviours. Self-control training is usually part of a wider cognitive behaviour programme.

Self-control treatment has three stages:

1. *Self-monitoring*. Daily records are kept of the problem behaviour and the circumstances in which it arises. Keeping a record enhances motivation by demonstrating the severity of the problem. These records are subsequently used to assess progress.
2. *Self-evaluation*. Achievements to be rewarded are agreed with the patient, and progress is monitored by the patient.
3. *Self-reward*. A system of reward points that can be accumulated to earn a material reward such as a week without smoking may be rewarded by going out for dinner.

Enuresis alarms. This behavioural treatment is based on classical conditioning and was developed specifically for nocturnal enuresis (page 451). The original 'pad and bell' comprised two metal plates under the sheets attached to a battery and a buzzer, activated if the child passed urine. Nowadays a small sensor is attached to the pyjamas. The noise of the alarm wakes the child, who must then rise to empty his bladder and, if necessary, change the bed sheet. With repetitions the child begins to wake before his bladder empties involuntarily.

Complex behavioural techniques

Habit reversal

Habit reversal is a complex procedure that is generally used to treat tics, Tourette syndrome, and stuttering. The classical treatment has five components: training in becoming aware of the onset of the behaviour; monitoring the behaviour; training in initiating competing responses that are incompatible with the behaviour; relaxation; and social support.

Eye movement desensitization and reprocessing

This treatment was developed for post-traumatic stress disorder (PTSD). It has three components:

1. *Exposure* using imagined scenes of the traumatic events
2. A *cognitive component* in which the patient attempts to replace negative thoughts associated with the images with positive ones
3. *Saccadic eye movements* induced by asking the patient to follow rapid side-to-side movements of the therapist's finger.

Eye movement desensitization and reprocessing (EMDR) therapy remains controversial, including whether the eye movements contribute to its efficacy (Landin-Romero et al., 2018). A review suggests that it is effective for PTSD, although the quality of the evidence is relatively low (Cuijpers, et al., 2020).

Cognitive techniques

Four methods are commonly used to bring about cognitive restructuring (i.e. change in cognitions).

1. *Distraction*, or focusing attention away from distressing thoughts. This is done by attending to something in the immediate environment (e.g. the objects in a shop window), by engaging in a demanding mental activity (e.g. mental arithmetic), or by producing a sudden sensory stimulus (e.g. snapping a rubber band on the wrist), called *thought stopping*.
2. *Neutralizing*. The emotional impact of anxiety-provoking thoughts can be reduced by rehearsing

a reassuring response (e.g. 'My heart is beating fast because I feel anxious, not because I have heart disease'). Patients may carry a 'prompt card' on which the reassuring thoughts are written.

3. *Challenging beliefs*. The therapist produces evidence that contradicts the patient's beliefs. However, such beliefs persist and, for these, if they are severe, CBT is indicated. CBT therapists do this by arranging behavioural experiments, as in Box 24.6, and by asking questions, as in Box 24.8.
4. *Reassessing the patient's responsibility*. We all overestimate our responsibility for events that have many determinants. Constructing a *pie chart* that shows all of the determinants can help reassess such responsibility. Allocating appropriately sized sectors to other identified factors before entering their own contribution demonstrates that there is less room for the patient's contribution than had been supposed.

Cognitive behavioural treatments

CBT is now the most widely used psychological treatment internationally. It dominates UK provision for a range of disorders—mainly anxiety and depression, but also eating disorders, and it has even been used in psychosis. It has a coherent theory and a set of practices based on a collaborative approach between patient and therapist over a limited (often predetermined) number of sessions. The approaches are succinctly explained by Kennerley et al. (2017).

Box 24.8 Useful questions for challenging beliefs

What is your evidence for this belief?

Is there an alternative way of looking at the situation?

How might other people think if they were in the same situation?

Are you focusing on what you felt rather than on what happened?

Are you forgetting relevant facts? Are you focusing on irrelevant matters?

Are you overestimating how likely this is?

Are you applying higher standards to yourself than you would to others?

Are you thinking in black-and-white terms when you should be considering shades of grey?

Are you overestimating your responsibility for the outcome?

What is the worst that could happen? How bad would this be?

What if the worst should happen? Could you cope?

Are you underestimating what you can do to deal with the situation?

Adapted from Clark, D. M. (2009). Cognitive-behaviour therapy for anxiety disorders. In M. G. Gelder, N. C. Andreasen, J. J. López-Ibor, Jr, & J. R. Geddes (Eds.), *The New Oxford Textbook of Psychiatry*, 2nd edn. Copyright (2000), with permission from Oxford University Press.

CBT for anxiety disorders

In anxiety disorders, cognitive techniques are combined with exposure. The importance of exposure relates to avoidance behaviour, more in the phobic disorders and less in generalized anxiety disorders.

Three kinds of cognition are considered in treatment:

1. *Fear of fear*: general concerns about the effects of being anxious (e.g. losing control).
2. *Fear of symptoms*: concerns about specific symptoms (e.g. fears that palpitations are a sign of heart disease).
3. *Fear of negative evaluation*: concerns that other people will react unfavourably to the patient.

The balance of these cognitions varies in different anxiety disorders. In generalized anxiety disorder, fear of fear and general worry predominate. In social phobia, fears of negative evaluation are particularly important, as are concerns about blushing and trembling. In agoraphobia, fear of fear (especially thoughts that the person will faint, die, or lose control) and fears about the symptoms of a panic attack are central. Treatment involves the techniques outlined above—giving information, questioning the logical basis of the fears, and arranging behavioural experiments.

Information about the physiology of anxiety helps patients to attribute symptoms such as dizziness and palpitations correctly, instead of to physical illness, often heart disease. The *illogical basis of the fears* is discovered by questioning the patient's own evidence for the beliefs. *Behavioural experiments* are devised to test the patient's beliefs and the alternative explanation suggested by the therapist.

Anxiety management is a general treatment for anxiety disorders. It has six stages:

1. *Assessment*. The patient keeps a diary record of:
 - the frequency and severity of symptoms
 - the situations in which they occur
 - avoidance behaviour.
2. *Information* about the physiology of anxiety and any other matters that will correct misconceptions.
3. *Explanation* of the various vicious circles of anxiety.
4. *Relaxation* training as a means of controlling anxiety.
5. *Exposure* to situations that provoke anxiety.
6. *Distraction* to reduce the impact of anxiety-provoking thoughts.

Treatment for panic disorder is focused on the belief that physical symptoms of anxiety are evidence of a serious physical condition, usually heart disease. These beliefs

create a vicious circle in which anxiety symptoms such as tachycardia generate additional anxiety, and this further increases the physical symptoms. Treatment consists of:

1. *Explanation* of how physical symptoms are part of the normal response to stress, and how fear of these symptoms sets up a vicious circle of anxiety.
2. *Record keeping*. Patients record the anxious cognitions that precede and accompany their panic attacks.
3. *Demonstration*. The therapist demonstrates that:
 - physical symptoms can provoke anxious thoughts (e.g. by asking the patient to induce such symptoms by over-breathing or strenuous exercise and noting the accompanying thoughts and fears)
 - these thoughts can induce anxiety (e.g. by asking the patient to focus their mind on the cognitions and observe the effect). This demonstration that physical symptoms lead to anxious thoughts, which in turn lead to anxiety, helps to *validate the vicious circle account* of the aetiology of panic attacks.
4. *Safety-seeking behaviours*. Attention is given to safety behaviours, and to any dysfunctional beliefs that make ordinary situations stressful.
5. *Behavioural experiments* to test the patient's ideas against those proposed by the therapist.
6. *Cognitive restructuring* follows the observation that symptom changes affect the severity of the panic attacks. Repeating the sequence gradually leads to control of the panic attacks.

Treatment for PTSD includes repeated attention to the characteristic intrusive visual images in the condition while in a safe environment. Patients repeatedly imagine the situations depicted in these images, as they would do in systematic desensitization, while anxiety and distress are managed with a relaxation technique such as controlled breathing. Cognitive techniques are also used to alter trauma-induced cognitive distortions and negative self-beliefs. Treatment for PTSD is considered further on pages 146, 147.

Overall, CBT is the psychological treatment of choice for anxiety disorders (van Dis et al., 2020). For a review of the current status and practice, see Clark and Beck (2011).

CBT for depressive disorders

Cognitive therapy for depressive disorders was developed by Beck (1976). It is a complex procedure intended to alter three aspects of the thinking of depressed

patients: negative intrusive thoughts; beliefs and assumptions that render ordinary situations stressful; and errors of logic that allow these beliefs and assumptions to persist despite evidence to the contrary.

Monitoring is of three kinds.

1. Patients identify intrusive thoughts (e.g. 'I am a failure') by writing down their thoughts when their mood is low.
2. Therapists uncover dysfunctional beliefs and assumptions by asking questions such as those shown in Box 24.8. A typical belief of a depressed patient is 'Unless I always try to please other people, they will not like me.'
3. Patients record their activities and identify if it was pleasurable and if it was accompanied by a sense of mastery and achievement.

If the patient is severely depressed, the monitoring of thoughts is deferred, and attention is focused on activities. The resulting 'activity schedule' is used to encourage activities that have been identified as leading to pleasure and mastery. The schedule also helps to bring a sense of order and purpose. At this stage the therapist helps the patient to reduce the need to make decisions, which are difficult for someone who is severely depressed.

If the patient is less severely depressed, treatment begins with an explanation of the cognitive model of depression, and an attempt is made to reduce intrusive thoughts. This is done through distraction and by rehearsing reassuring alternatives (e.g. 'Even though I think my work is bad, my boss praised me yesterday'). To help the patient to concentrate on the positive statement, the alternative can be written on a prompt card. As treatment proceeds, more time is spent in challenging depressive cognitions using the techniques outlined in Box 24.8 combined with behavioural experiments.

The following points are important in relation to cognitive therapy for depression.

- *Reviewing evidence.* Depressed patients are particularly prone to focus on evidence that supports their negative ideas, and to overlook evidence that contradicts them. The therapist should help the patient to give appropriate weight to the positive evidence.
- *Considering alternatives.* Depressed patients often reject positive alternatives to their thoughts and beliefs. The therapist can help the patient to consider alternatives by asking questions such as 'What do you think that another person would think about this situation?' or 'What would you think if another person had done what you have done?' (For additional questions, see

Box 24.8.) Behavioural experiments are used to challenge beliefs and assumptions.

- *Considering consequences.* Patients should be helped to see the consequences of thinking negative thoughts. For example, the thought that everything is hopeless may prevent them from attempting even small changes that could accumulate beneficially.
- *Considering errors of logic* (Box 24.8). The patient should be helped to ask questions such as 'Am I thinking in black-and-white terms?', 'Am I drawing too wide conclusions from this single event?', 'Am I blaming myself for something for which I am not responsible?', or 'Am I exaggerating the importance of events?' These questions are asked in relation to specific ideas, beliefs, and situations.
- *Considering beliefs.* As depression improves, more attention is given to the patient's beliefs, as abnormal beliefs can lead to relapse. Laddering (page 689) can uncover such beliefs. Useful questions include 'In what ways is this idea helpful?', 'In what ways is it unhelpful?', and 'What alternatives are there?'
- *'Mindfulness'.* People who are prone to depression may have a cognitive set in which thoughts and feelings are experienced as events rather than as aspects of the self. Modifying this set reduces the risk of relapse. MBCT is an eight-week programme conducted in groups of 8–15 patients (Segal-Zindall et al., 2013), designed to modify these cognitive routines and reduce the risk of relapse. MBCT in addition to standard care significantly reduces the relapse rate in patients suffering from recurrent major depression (Chiesa and Serretti, 2011), and is recommended by the National Institute for Health and Clinical Excellence (2009). The contribution of the specific 'mindfulness' component of treatment is unclear.

CBT for bulimia nervosa

Currently CBT therapists emphasize the applicability of their approach for all eating disorders and believe the strict distinction between bulimia nervosa and anorexia nervosa is unhelpful and the same approach (CBT-E) should be applied. Evidence for effectiveness is strong for bulimia nervosa but weaker for anorexia nervosa. See Chapter 13 for a review of the evidence for, and contemporary approaches to, psychological treatment for eating disorders.

The treatment of bulimia nervosa with CBT is based on the idea that the central problems are excessive concern about shape and weight, and low self-esteem. This leads to extreme dieting, followed by binge eating and often by self-induced vomiting and abuse of laxatives and diuretics. This vicious circle can be interrupted by:

- restoring a regular pattern of eating three meals a day
- increasing restraint on binge eating
- discussing ideas about shape, weight, and self-esteem.

The therapist attends first to the disordered pattern of eating before attempting to modify cognitions. They *explain the cognitive model* and relate it to the patient's experience. They emphasize the importance of regular meals, the causal role of long periods of fasting, and the adverse effects of repeated vomiting, and of repeatedly taking laxatives and diuretics. The patient *keeps a record* of what they eat, when they eat, and when they induce vomiting or take laxatives and diuretics. The situations that provoke binge eating are recorded. With this information, the patient is better able to control the urge to overeat and, subsequently, the bouts of vomiting. The patient takes a number of precautions to help them to control their eating:

- Meals are eaten in a place separate from that in which food is prepared or stored.
- A limited amount of food is available at each meal—for example, two slices of bread are put on the table, rather than a whole loaf.
- A small amount of food is left on the plate and then thrown away, to mark the end of the meal.
- When shopping for food, a list is made in advance and purchases are strictly limited to the items and quantities on the list.

The therapist strongly discourages frequent checking of weight and of appearance, as both habits maintain the disorder.

Because patients often binge when they are unhappy, lonely, or bored, they are helped to *find other ways of dealing with unpleasant emotions*. For example, they might seek out friends, listen to music, or simply go out for a walk. Vomiting usually stops when the binges are under control. The dangers of abuse of laxatives and diuretics are emphasized once more, and the patient is strongly encouraged to dispose of all such drugs safely.

When eating is under better control, *attention turns to cognitions*. The patient records these together with the eating behaviour. Relevant cognitions are concerned not only with body shape and weight, but also with self-esteem. Examples of these cognitions include the following:

- to be fat is to be a failure
- dieting is a sign of strong will and self-control
- it is necessary to be thin if one is to be happy and successful.

Such beliefs persist because of the *illogical ways of thinking* (see Box 24.5). Identifying these illogical ways of thinking is not to deny that many of them receive strong social endorsement. The fact that many people agree with such cognitions does not mean that the thinking is correct. The questioning used to identify cognitions and illogical thinking resembles that described in Box 24.8.

Treatment also exists in a self-help format that can be effective with well-motivated patients (Box 24.9).

CBT for hypochondriasis

The approach is twofold—first, to identify behaviours that maintain the disorder and, second, to change hypochondriacal ideas directly. The relevant behaviours include:

- repeatedly seeking reassurance, which relieves anxiety briefly but reinforces the concerns in the longer term
- checking bodily functions (e.g. measuring the pulse rate)
- checking bodily structure (e.g. palpating for lumps).

Box 24.9 Self-help for bulimia nervosa

Monitoring

- A daily record of eating, binges, and vomiting
- Weighing no more than once a week

Regular planned meals

- Three normal-sized meals a day
- Three small between-meal snacks

Stop binges

- Eat only the planned amount
- Keep other food out of sight
- Keep limited stocks of food
- Take just enough money when buying food

Control vomiting

Urge to vomit declines as binges stop

Control purging

Reduce laxatives/diuretics, if necessary, in stages

Find alternatives to binge eating

- List distracting activities
- Try them

Reduce life problems

Problem-solving counselling

Hypochondriacal ideas are treated in the same way as anxiety and depressive disorders, using questioning and behavioural experiments. CBT has been found to be moderately helpful and MBCT has also been tested (Williams et al., 2011).

CBT for schizophrenia

Two approaches are used. The first aims to help the patient to *reduce and cope better with stressors* that may be exacerbating the disorder, and to *cope better with hallucinations*. The techniques for dealing with stressors are similar to those described above—namely, identifying and finding ways of dealing with stressful situations. Patients are helped to cope with hallucinations by distancing themselves and repeating statements that neutralize their effects.

The aim of the second approach is to *challenge delusions*. This approach is directed to secondary delusions, especially those that seem to have developed to explain hallucinations. The therapist encourages the patient to regard the delusions as beliefs rather than facts, and to discuss alternatives. To do this effectively requires a detailed formulation of each patient's delusions and other beliefs. When questioning the basis of the delusions, the therapist should avoid challenging them directly, and instead, should try to persuade the patient to consider the consequences of holding the delusion and what the consequences would be of thinking in another way. The therapist then tries to reformulate the delusion as a way of making sense of certain experiences, which can be understood in terms of the knowledge the patient had at the time, but should now be reconsidered.

Turkington et al. have taught the approach to nurses with considerable success, and it is a module in several postgraduate mental health nursing courses (Gournay and Birley, 1998; Turkington et al., 2006; Turkington et al., 2015).

CBT is included as an integral component of treatment in current schizophrenia treatment guidelines. For details, see Chapter 11.

CBT for personality disorder

Beck suggested that the techniques he had developed for the treatment of depressive disorders could be adapted for personality disorders. He described beliefs and ways of thinking that characterize each type of personality disorder in terms of self-view, the views of others, general beliefs, perceived threats, main strategies for coping, and primary affective responses. Beck also suggested a 'schema' characteristic of each type of personality disorder consisting of statements that can be challenged in treatment. For example, the schema for histrionic personality disorder includes the following statements:

- 'Unless I captivate people, I am nothing.'
- 'To be happy, I need other people to admire me.'
- 'I must show people that they have hurt me.'

Schemas are challenged using the general techniques of cognitive therapy (see Box 24.8). There is early evidence of value in the care of people with borderline personality disorder, and the approach can prove helpful in forensic settings.

DBT for borderline personality disorder

Marsha Linehan developed DBT for patients with borderline personality disorder who repeatedly harmed themselves (Linehan et al., 1994). Despite the name, it uses cognitive as well as behavioural techniques. The treatment is highly structured and described in a manual. Therapy is intensive, with individual sessions, skills training in a group, and access by telephone to the therapist between sessions. It is delivered by a small team of therapists and lasts for up to a year.

Individual sessions have four elements:

1. *Cognitive behavioural techniques*, including self-monitoring, and a collaborative style of working with the patient.
2. *Dialectical ways of thinking about problems*, such as seeing causality in terms of 'both/and' rather than 'either/or', and the possibility of reconciling opposites. This approach is particularly valuable to avoid unhelpful confrontations.
3. '*Mindfulness*'—the practice of detachment from experience.
4. *Aphorisms*—that is, phrases that encapsulate the approach (e.g. 'Although I may not have caused all my problems, I still have to solve them').

During the sessions, treatment goals are prioritized, dealing first with life-threatening situations, then with matters that could reduce collaboration with treatment, and after that with behaviours that impair quality of life.

Skills training sessions are usually provided in weekly groups lasting for two hours or more. Patients learn how to control anger and other strong emotions, how to tolerate distress, how to develop interpersonal skills, and mindfulness. The procedures for teaching these skills are described in a manual.

Telephone contact is designed to help patients get through crises by using the skills that have been learned in the sessions. The hours at which contact will be available are agreed in advance between patient and therapist.

DBT has been claimed to give good results with borderline personality disorder, and has become widely

available, although providing the full programme can be difficult. The beneficial effects are primarily on behaviour (i.e. reducing self-harming and suicide attempts) rather than on mood and self-esteem, and the degree of prescription in its provision may be a barrier to better understanding of what is effective within it.

Acceptance and commitment therapy (ACT)

ACT emerged from CBT and reflection on the relational framework, noting how thinking itself can become

a learned behaviour leading to cognitive fusion and experiential avoidance. The key elements are then to show flexibility; acceptance of the situation (non-judgemental awareness of thoughts and feelings); focus on the immediate moment; defusion or distancing from harmful functions of thoughts; transcendence or taking perspective of self as a context in which thoughts and feelings emerge; describe their values or what they wish to achieve as a greater goal; and commit to actions (Webster, 2011). The approach may effectively support the care of anxiety, depression, addictions, and a number of physical health problems (Tjak et al., 2015).

Individual dynamic psychotherapies

Brief insight-oriented psychotherapy

This kind of psychotherapy seeks to understand the origins of a psychiatric disorder by identifying unconscious factors that account for distorted thinking, intense emotions, and troublesome behaviours. There is a recognition that early-life experiences, and the way they are remembered, influence current thoughts, emotions, and behaviours. In its usual form, it aims to produce change, with weekly sessions over six to nine months. The treatment is focused upon specific problems—hence, an alternative term *focal psychotherapy*. The procedures can be summarized as follows.

Starting treatment

The initial assessment is considered very important and is not hurried. It assesses suitability for brief treatment and selects the problems that will be the focus of treatment. This focus and the length of treatment are agreed with the patient, and that not all problems will be resolved is acknowledged. The therapist explains the general aim of linking past and present behaviour patterns, and that the therapist's role is to help the patient to find their own solutions. An atmosphere is created in which the patient feels involved and listened to, and one that is safe to explore ideas and fantasies that were previously hidden.

Subsequent sessions

In subsequent sessions the patient is encouraged to:

- give specific examples of the selected problems, and consider how they thought, felt, and acted at the time
- talk freely about these emotionally painful subjects

- express ideas and feelings even if they seem illogical or shameful
- review their own part in difficulties they usually ascribe to other people
- look for common themes in their problems and their responses
- consider how their present patterns of behaviour began, their original function, and why they persist
- consider alternative ways of thinking and behaving in distressing situations
- try out new responses to their emotions.

The *therapist's role* is to respond to the emotional as well as the intellectual content of the patient's descriptions (e.g. 'It sounds as though you felt angry when this happened'). They help the patient to examine feelings previously denied, and to consider past situations in which similar feelings were experienced. Non-verbal behaviour can point to problems that are not being expressed directly. They maintain the agreed focus and avoid problem areas too complex to address.

Interpretations are key features of dynamic therapy. These are essentially interventions based on hypotheses linking present to past events and behaviours. They can include observations of defence mechanisms, such as blocks in recall during the sessions. Interventions can be verbal or non-verbal and seek to bring unconscious material into conscious awareness.

Transference and countertransference. The therapist has to be alert to intense emotions arising in sessions—transference and countertransference. *Transference* feelings about the therapist are a key to how the patient responded to key figures in childhood. In brief therapy, *countertransference* is usually taken to include all the

therapist's emotional responses to the patient. Insight into countertransference can be difficult so therapists have their own regular supervision.

Ending treatment. The difficulty, and inevitability, of termination is discussed from an early stage. As it approaches, the patient should have a better understanding of their problems and be more confident about dealing with them. Endings can involve a 'flare up' of symptoms, allowing their understanding to be refreshed, and are part of the separation and return to independence. It is common to arrange a couple of follow-up appointments spaced over two or three months.

Indications

The indications for short-term dynamic psychotherapy are based predominantly on clinical experience. It is chosen when the problem:

- can be conceptualized in psychodynamic terms
- is emotional or interpersonal (rather than a specific psychiatric disorder)
- involves low self-esteem, and recurrent problems in forming intimate relationships.

In addition, treatment is more effective when the patient:

- has adequate social support while treatment continues
- is willing to attempt change through their own efforts
- can look honestly at their own motives
- is capable of ceasing self-exploration when the sessions end.

Contraindications include obsessional or hypochondriacal disorders, severe mood disorder, psychoses, and some personality disorders, especially those characterized by acting out. For a review of the theories and methods of brief individual dynamic psychotherapy, see Bloch (2006).

Cognitive analytic therapy (CAT)

CAT is a brief therapy developed by the general practitioner Tony Ryle. Treatment is based on a 'procedural sequence model'. This proposes that behaviour follows a set sequence. Procedural sequences can be faulty in three ways.

- *Traps* are repetitive cycles of behaviour in which the consequences of behaviour perpetuate it. For example, depressed people think in ways that lead to failure, making further depression more likely.
- *Dilemmas* are false choices or unduly narrowed options. For example, people who fear angry feelings

may think that the only alternative to aggression is excessive submission. This allows others to take advantage of them, making them even more angry.

- *Snags* are the anticipation of highly negative consequences of actions such that the action is never carried out and therefore never subjected to a reality check.

The *theory of reciprocal roles* was developed when CAT was used with borderline personality disorder patients. Ryle proposed that people develop internalized 'templates' of social roles, which consist of a role for the self, a role for the other person, and a paradigm of the relationship. Such roles include teacher/pupil, bully/victim, and abuser/abused. When one person adopts one of the roles, the other person feels under pressure to adopt the reciprocal role. These 'templates' can become narrow and inflexible.

Outline of the treatment

Following assessment and an explanation of the treatment, the patient is helped to construct a list of problems, moods, and behaviours, and records them in a diary. Recurrent maladaptive behaviours, role problems, and faulty procedural sequences are identified and formulated, often using diagrams to explain the procedural sequences. Specific examples of general procedures are sought in the diaries, and homework is arranged to try out alternative procedures, for example, asserting oneself appropriately. The origins of maladaptive procedures are also considered from a psychodynamic perspective—present maladaptive behaviour may have arisen from behaviour that was adaptive when the person was younger.

The formulation is summarized in the form of a letter to the patient. The therapist helps the patient become aware of and change their behaviours, procedural sequences, and role problems. Transference and countertransference problems are anticipated, identified, and discussed. Treatment usually lasts either 16 or 24 sessions. When it ends, the therapist writes another letter to the patient summarizing their progress, prognosis, and future action, and the patient is asked to write to the therapist summarizing their experience. Patients often need help to become aware of their problems ('develop an observing I'). This is achieved by modelling the evaluation and analysis of problems using diagrams. For further information about CAT, see Ryle and Karr (2020).

Long-term individual dynamic psychotherapy

Long-term dynamic psychotherapy is a general term for most individual psychotherapy that lasts for longer than nine months. The best known is psychoanalysis, and

most methods are derived from it. Long-term dynamic psychotherapy is costly both in public services and if privately undertaken. There are ongoing evaluations of whether long-term therapies are more effective than short-term therapies. The important question becomes whether short-term therapy is a good as long-term therapy over a longer-term follow-up. Longer-term therapies, where available, are often used for patients judged too complex and unlikely to engage or respond to short-term psychotherapy.

Long-term dynamic psychotherapy aims to increase insight—that is ‘the conscious recognition of unconscious factors on current experiences and behaviour’ (Fonagy and Kächele, 2009). It also involves integrating these insights into current functioning, called ‘working through’. Insight is achieved by bringing feelings and ideas previously outside consciousness into conscious awareness, and then interpreting their significance and linking past experiences to present ones.

The basic tools include free association, and examining the content of fantasies and dreams, and helping patients reflect and make meaning of their experiences. Analysis of transference provides further information about unconscious processes, as does analysis of the countertransference. Recognizing countertransference is a key goal of personal psychotherapy during training. These are not easy concepts to understand or to practise, so should only be undertaken in supervised practice, matching the complexity of the patient to the skills

of the therapist. Listening in supervision groups to how experienced therapists work is one approach before taking on a case in training. Careful selection is important both for the patient and for the therapist in training.

Analysis of resistance, as with analysis of transference, can often markedly increase insight. *Resistance* to accessing unconscious material takes three forms:

1. *Repression*: active blocking access to unconscious material.
2. *Transference resistance*: restricting the intensity of the therapeutic relationship.
3. *Negative therapeutic reactions*: new symptoms that retard progress.

Interpretations are considered the hallmark of this treatment. Interpretations link together seemingly unrelated mental phenomena and involve defences, unconscious processes, transference, or the links between past experience and present patterns. Transference interpretations are particularly emphasized in long-term therapy.

Long-term psychodynamic psychotherapy also differs from brief dynamic psychotherapy in that:

- *It is less structured*—the patient is encouraged to talk and associate ideas freely without a specific focus.
- *The therapist is less active* and guides the patient less.
- *Patients may be seen twice a week* (up to five times a week in classical psychoanalysis).

Treatment in groups

Small-group psychotherapy

This is psychotherapy carried out with a group of usually about eight patients. Small-group psychotherapy is used most often to modify interpersonal problems, as a form of supportive treatment, or to encourage adjustment to the effects of physical or mental illness.

The origins of group psychotherapy

Group therapy's early origins were in American physician Joseph Pratt's supportive and educational groups for patients with pulmonary tuberculosis. Modern group therapy originated in the treatment of war neuroses in the United Kingdom. Group analysis (Foulkes and Lewis, 1945) was based on psychoanalysis with a rather passive group leader using analytical interpretations. Wilfred Bion, a Kleinian analyst, focused

specifically on the unconscious defences of the group as a whole rather than on the problems of individual members (Bion, 1961). In the United States, ‘action groups’ in the 1960s and early 1970s provided a more intense experience. Group techniques are legion, but what they have in common seems more important than the differences.

Classification of small-group therapies

One classification of groups is by their goals (specific versus non-specific) and the activity of the leader (high versus low) (Schlapobersky and Pines, 2009):

1. *Specific goals—high leader activity*: such as structured alcohol and drug programmes and CBT in groups.
2. *Specific goals—low leader activity*: includes problem-solving groups.

3. *Non-specific goals—high leader activity:* includes the many short-term group therapies, including psychodrama.
4. *Non-specific goals—low leader activity:* includes various psychodynamic groups.

Terminology

Groups can be described in terms of their structure, process, and content.

- *Structure* describes the enduring reciprocal relationships between each member of the group and the therapist, and between the members.
- *Process* describes the short-term changes in emotions, behaviours, relationships, and other experiences of the group.
- *Content* refers to the observable events in the group meetings—the themes, responses, discussions, and silences.

Therapeutic factors in group therapy

Group treatments share the therapeutic factors common to all kinds of psychological treatment, including restoring hope, releasing emotion, giving information, providing a rationale, and suggestion (see Box 24.2). Group treatment also shares additional factors, such as shared experience, support for and from group members, socialization, imitation, and interpersonal learning (Yalom and Leszcz, 2005), summarized in Box 24.10.

General indications for group therapy

There is no strong evidence that the results of group therapy differ from those of equivalent individual psychotherapy. Nor is there evidence of differing results for forms of group therapy. Clinical experience suggests that they are somewhat less effective than individual therapies unless they build on shared clinical features.

Group therapy appears most appropriate for patients whose problems are mainly in relationships. Contraindications are similar to those for individual psychotherapy.

Types of small-group psychotherapy

Supportive groups

Many of the therapeutic factors in a group (Box 24.10) work in supportive treatment. In supportive groups, the therapist encourages self-help and ensures that the experiences are used positively. They should also ensure that relationships do not become too intense, protect vulnerable patients when necessary, and ensure that each member is supported by and gives support to other members.

Self-help ('mutual-help') groups

Self-help groups are organized and led by patients or former patients who have learned ways of overcoming

Box 24.10 Therapeutic factors in group therapy

Universality (shared experience). This helps the patient to realize that they are not alone and that others have similar experiences and problems. Hearing about others' experiences is often more convincing and helpful than reassurance from a therapist.

Altruism. Supporting others increases self-esteem of the giver, as well as helping the receiver. Mutual support leads to a sense of belonging to the group.

Group cohesion. Belonging to a group is especially valuable for those who have previously felt isolated. Group cohesion sustains the group through difficult patches.

Socialization. Social skills are acquired in the group from the comments and reactions of members in response to one another's behaviour. Members can try out new behaviours within the safety of the group.

Imitation. This involves learning from observing and adopting the behaviours of other group members. Patients imitate adaptive behaviours, although there is a risk that maladaptive behaviours can also be learned.

Interpersonal learning. This involves learning from the interactions within the group and from practising new ways of interacting. Interpersonal learning is an important component of group therapy.

Recapitulation of the family group. Interactions can become over time increasingly based on past relationships between patients and their parents and siblings. This group transference is encouraged and used in analytical groups.

or adjusting to their difficulties. Group members benefit from the opportunity to talk about their own problems and express their feelings, and from mutual support. Group processes develop very strongly in these groups. Some such groups, such as Alcoholics Anonymous, have strict rules of procedure.

There are countless self-help groups for people who suffer from a wide range of different problems. These include Alcoholics Anonymous (pages 550, 559), Weight Watchers, groups for patients with chronic physical conditions, and groups for the bereaved (Cruse Clubs). Few self-help groups (now often referred to as 'mutual-help groups') have been evaluated.

Therapeutic groups

Interpersonal group therapy

Interpersonal group therapy developed in particular from the work of Yalom and Leszcz (2005), and characterizes most group therapy. Treatment is focused on problems in current relationships, and examines the ways in which these problems are reflected in the group. The past is discussed only in so far as it helps to make sense of present problems.

Patients are prepared for their experience in a group by emphasizing:

- *Confidentiality*: the proceedings of the group are confidential.
- *Reliability*: members must attend regularly and on time, and not leave early.
- *Disclosure*: members are required to disclose their problems.
- *Concern*: members must show concern for the problems of others.
- *Disappointment*: at first members may be disappointed by the lack of rapid change, or frustrated by the need to share the time available for speaking.
- *Keeping apart*: The group members should not meet outside the group, and if they do so this should be reported at the next meeting.
- *Duration*: the length of the group is explained (e.g. 'ten weeks' or 'two years'), together with the need to remain until the end.

Setting up the group

General considerations. About eight members are chosen. They should have some problems in common, and no member should have exclusive problems that set them apart from the rest. Meetings need a room of adequate

size, with the chairs in a circle so that all members can see one another.

Meetings usually last for 60–90 minutes to allow adequate time for every member to take part; they are usually held once a week, and generally continue for 12–18 months. Most groups are '*slow open*', with new members joining only to replace those who leave. Totally *closed* groups are very difficult to maintain, and rare outside residential settings. Groups that accept new members frequently, known as *open* groups, are usually supportive or psychoeducational.

One or two therapists? Most groups are run by one therapist, but many have co-therapists. The advantage of having two therapists is both practical and theoretical. It ensures continuity if one of the therapists has to be away, and it also provides an excellent training opportunity. Theoretically it can also help with countertransference problems. A risk is that co-therapists may compete with or behave defensively towards one another. However, in general, if differences are discussed as they arise, they can provide further insight into the group process and offer healthy modelling of problem exploration and resolution.

Some problems in group therapy

However skilful the therapist, certain problems commonly arise.

- *Formation of subgroups.* Some members may form a coalition based on age, social class, shared values, or other characteristics. The therapist should discourage such groupings while encouraging the group to discuss the reasons for their formation.
- *Members who dominate by talking too much.* The therapist needs to address this at an early stage, before the group rejects the talkative member, and ask the group why they allow one person to absolve the rest from the need to speak about themselves.
- *Members who talk too little.* The therapist should assist silent members to speak and should therefore understand the reasons for silence.
- *Conflict between members.* The therapist should not take sides in conflicts but should encourage the whole group to discuss the issue in ways that lead to understanding of its arising, perhaps a hostile transference.
- *Avoidance of the present.* Members may talk excessively about the past to avoid present conflicts. The therapist can ask questions or use interpretations to bring the discussion back to the present, or actively relate it to current group process.

- *Potentially embarrassing revelations.* Common sense and judgement have to be used to protect vulnerable patients from blurting out potentially devastating information (e.g. about sexual or even criminal activity) early in the group.

Group analysis

This technique has been widely used within the UK health service. It differs from the interpersonal method described above in the greater use of interpretations about transference and unconscious mechanisms and in encouraging 'free-floating discussion' rather than a specific focus. Particular attention is given to transferences to the therapist and between members generating hypotheses about previous relationships to understand current problems.

Encounter groups and psychodrama (action techniques)

In *encounter groups* the interaction between members is actively intensified to provoke change. The encounter can be verbal, using challenging language, but it can also include touching or hugging between the participants. Sessions can last several hours. These groups are attractive to volatile individuals and undoubtedly carry some risks of things getting worse.

Psychodrama groups enact events from the life of one member, in scenes reflecting either current relationships or those of their family of origin. Such enactments often provoke strong feelings reflecting the problems of other group members. Members often swap roles to understand the other person's perspective. The drama is followed by a group discussion. Psychodrama is useful with less-educated and less-verbal patients and is favoured in some prison settings.

Ward groups

Group meetings, often called 'community meetings', are part of the daily programme of many psychiatric wards. The approach originated in therapeutic communities. These are large groups, usually including all of the patients in a treatment unit together with some or all of the staff. At the simplest level, large groups allow patients to examine and deal with the problems of living together. They can confront individual patients about disordered or disruptive behaviour, and provide opportunities for social learning. Care needs to be exercised to avoid bullying, and a predominantly supportive atmosphere is needed for these groups to work. The group is sometimes used as a kind of governing body that formulates rules and seeks to enforce them.

Therapeutic communities

In a therapeutic community, every shared activity is viewed as a potential source of learning and change. Members live, work, and play together, and learn about themselves through the reactions of other members in the course of these activities. Within the safe environment of the community, they are encouraged to experiment with new behaviours and appreciate points of view other than their own. Members take part in frequent group meetings. Maxwell Jones, its founder, called it a *living-learning situation* (Jones, 1968), others a culture of enquiry. The underlying principles of the regimen have been summarized as *democracy*, *reality confrontation*, *permissiveness*, and *communality* (Rapoport, 1960). These translate into the features shown in Box 24.11. Residential therapeutic communities are no longer available in the National Health Service (NHS), but the approach is still predominant in day units and in drug rehabilitation and some offender institutions.

Box 24.11 Principal features of a therapeutic community

Informality. There are few rules, and staff dress and behave informally.

Mutual help. Members support each other and help others to change.

Permissiveness. Members tolerate behaviour that they might not accept elsewhere.

Directness and honesty. Members respond directly to distortions of reality and other kinds of self-deception.

Shared decisions. Members and staff join in the day-to-day decisions about the running of the unit, the behaviour of its members, and often about the admission of new members.

Shared activities. Members provide some of the 'hotel' services in the community, so that each has a job involving responsibilities to other people.

Therapeutic day hospitals

Therapeutic day units are increasingly widespread for the treatment of personality disorder (Bateman and Fonagy, 2008). They have a similar structure to therapeutic communities, with a culture of enquiry, but emphasize ‘mentalization’. They are informed by psychoanalytical theory, but emphasize group work and a supportive

environment. Using nurse therapists, the aim is to help patients to become aware of their strong emotions and reflect and learn to tolerate them, rather than act them out impulsively. There is no emphasis on understanding past experiences as long as the patient can begin to manage their intense emotions.

Psychotherapy with couples and families

Couple therapy

Couple therapy is usually proposed either because conflict in a relationship appears to be the cause of emotional disorder in one of the partners or to save a threatened relationship. The problem is conceived in terms of how the couple interact, and treatment is directed to this interaction. Couples are required themselves to identify the difficulties that they would like to put right.

Several techniques of therapy have developed, based on psychodynamic, behavioural, and systems theory approaches, and on a combination of techniques drawn from these latter two.

Family therapy

Several, sometimes all, members of a family take part in this treatment. Usually both parents are involved, often together with the child whose problems have led the family to seek help. They may be joined by other members of the extended family. The aim of treatment is to improve family communication and functioning, and consequently to help the identified patient. Since success depends on the collaboration of several people, dropout rates are high. Whatever their method, family therapists have the following goals for the family:

- improved communication
- improved autonomy for each member
- improved agreement about roles
- reduced conflict
- reduced distress in the member who is the patient.

The systems approach is very influential, and was developed in the United States by Salvador Minuchin, who, in his structural family therapy, advocated a practical approach to resolving problems. In Italy, the Milan school used hypotheses about the family system to suggest ways of promoting change. These approaches

are described briefly below, together with an eclectic approach. The reader will find more detailed accounts in Bloch and Harari (2009). Family therapy is used in the treatment of some young people with anorexia nervosa after weight has been restored by other means (Chapter 13) (Bloch and Harari, 2009). Special kinds of family treatment have been developed to reduce relapses in schizophrenia (page 279).

Systemic family therapy

Systemic family therapy is concerned with the present functioning of the family, rather than with members’ past experiences. The therapist’s task is to identify the family’s unspoken rules, their disagreements about who makes these rules, and their distorted ways of communicating. The therapist helps the family to understand and modify the rules, and to improve communication.

The *Milan approach* usually consists of five to ten sessions, spaced at intervals of one month or more (‘long brief therapy’) (Barbetta and Telfener, 2021). *Circular questioning* is often used to assess the family. In this technique, one person is asked to comment on the relationships of others, for example, the mother may be asked how her husband relates to their son, and others are asked to comment on her response. The purpose is to discover and clarify confused or conflicting views. A hypothesis is then constructed about the family functioning and presented to the family, who are asked to consider the hypothesis during and between sessions. The family may be asked to try to behave in new ways. Sometimes the therapist provokes change with *paradoxical injunctions* designed to provoke the family into making changes that they cannot make in other ways. Paradoxical injunctions are impossible or counterintuitive suggestions that force the family to confront their hidden or unacceptable motives. A review of ten outcome studies of Milan therapy found symptomatic improvement in about two thirds of patients, and improved functioning in about 50% of the families (Carr, 1991).

Eclectic family therapy

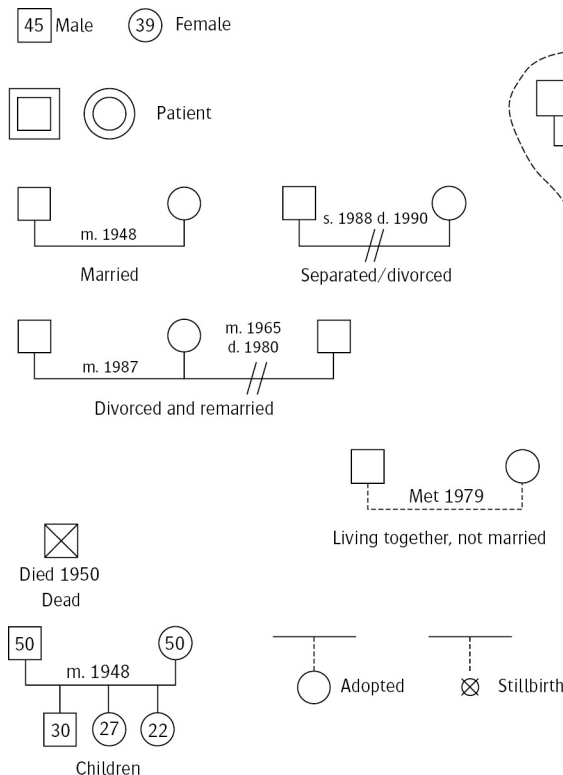
In everyday clinical work, especially with adolescents, it is almost impossible to do without simple short-term interventions to bring about limited changes in the family. The present situation of the family and how the members communicate with one another is usually the focus.

Assessment

Constructing a genogram using conventional symbols (Figure 24.1) is a particularly useful step in family therapy, leading to questions on current and past family life, and the roles of the members. The therapist tries to answer two questions: namely how the family functions and whether family factors are involved in the patient's problems. Bloch and Harari (2009) have proposed a helpful framework in which to consider these questions.

1. How does the family function?

- *structure* recorded in the genogram (e.g. single parent, a step-parent, size and age spread of the sibship)



- *changes and events* (e.g. births, deaths, departures, and financial problems)
- *relationships* (e.g. close, distant, loving, conflictual)
- *patterns of interaction* involving two or more people (e.g. a child who sides with one parent against the other).

2. Are family factors involved in the patient's problems? The family may be:

- *reacting* to the patient's problems (note that there may be other, unrelated, problems)
- *supporting* the patient
- *contributing* to the patient's problems (e.g. the problems of a daughter who cannot leave her lonely mother).

Intervention

Specific goals for change are agreed with the members of the family, who are asked to consider how any changes will affect themselves and others, and what has prevented the family from making the changes. Paradoxical

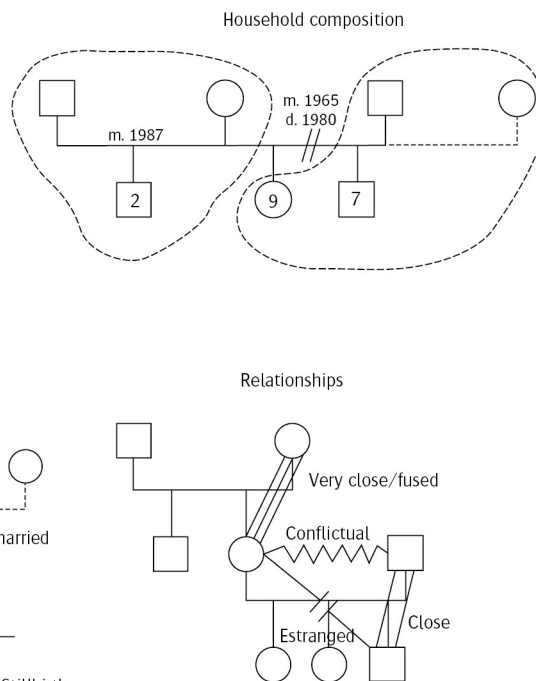


Figure 24.1 Symbols used in the construction of a genogram.

Reproduced from Bloch, S., & Harari, E. (2009). Family therapy in the adult psychiatric setting. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1380–1391). Copyright (2009), with permission from Oxford University Press.

injunctions may be included, but should be made only after the most careful consideration of the range of possible responses. The therapist should remember that

interchanges in the sessions are likely to continue when the family return home, and should try to ensure that this does not lead to further problems.

Psychotherapy for children

The kinds of psychotherapy discussed so far do not lend themselves to the treatment of young children, who lack the necessary verbal skills. In practice many emotional problems of younger children are secondary to those of their parents, and it is often appropriate to direct psychotherapy mainly to them.

Klein believed it was possible to use the child's play as equivalent to the words of the adult in psychotherapy. Anna Freud developed child psychotherapy by a less extreme adaptation of her father's techniques to the needs of the child. She accepted that non-analytical techniques could be helpful, including reassurance, suggestion, the giving of advice, and acting as a role model (an 'auxiliary ego'). For many neurotic disorders, she

advocated analytical techniques to identify the unconscious content and to interpret it to strengthen ego function.

In the United Kingdom, most psychotherapy for children is eclectic; the therapist tries to establish a good relationship with the child and to learn about their feelings and thoughts, partly through talking and listening, and partly through play. Older children can communicate verbally with adults, but younger children can communicate better through play. The therapist can help children to find words that express their thoughts and feelings, and can thus make it easier for them to control and modify them. Child psychotherapy is discussed further in Chapter 16. For a more detailed account, see Pearce (2009).

Psychotherapy for older people

Increasing emphasis is being placed on the provision of psychological treatments as part of the care plan for elderly patients. Provided that they do not have cognitive impairment, elderly patients can take part in any of the treatments described for younger adults. When

using cognitive therapy, it is important to be aware of minor cognitive impairment that, while insufficient to affect general functioning, may impair the patient's engagement in therapy. For a more detailed account, see Cook et al. (2005).

Treatments of mainly historical and cultural interest

In our evidence-based era it may seem inappropriate to list treatments that have fallen into disuse because of lack of effect. However, many of these treatments have entered our culture and have a strong hold on the public imagination. Psychiatrists are likely to be approached by patients who are keen to receive such treatments, and we need to be familiar with them in the same way as we are with outdated but influential concepts (e.g. the 'schizophrenogenic' mother).

Hypnosis is a state in which the person is relaxed and drowsy, and more suggestible than usual. Hypnosis can be induced in many ways. The main requirements are that the subject should be willing to be hypnotized and

convinced that hypnosis will occur. Most hypnotic procedures contain some combination of a task to focus attention (e.g. watching a moving object), rhythmic monotonous instructions, and the use of a graduated series of suggestions (e.g. that the person's arm will rise). The therapist uses the suggestible state either to implant direct suggestions of improvement, or to encourage recall of previously repressed memories.

Hypnosis is used infrequently in psychiatry but is used widely for stopping smoking, dieting, etc. A light trance is used occasionally as a form of relaxation. For this purpose hypnosis has not been shown to be generally superior to relaxation. A deeper trance is used occasionally to

enhance suggestion to relieve symptoms, especially those of conversion disorder. Although sometimes effective in the short term, this method has not been shown to be superior to suggestion without hypnosis. The authors do not recommend the use of hypnosis in clinical psychiatry.

Autogenic training was described by Schultz and Luthe (1959) and was in use mainly in continental Europe as a treatment for physical symptoms caused by emotional disorder. Patients practise exercises to induce feelings of heaviness, warmth, or cooling in various parts of the body, and to slow their respiration. Repeated use of these exercises is supposed to induce changes in autonomic nervous activity, thereby alleviating physical symptoms in stress-related and anxiety disorders, including hypertension. Its results do not differ substantially from those achieved with simple relaxation, nor is there any good evidence that it has a specific therapeutic effect.

Abreaction (the unrestrained expression of emotion) has long been used to relieve mental distress and some psychiatric symptoms. Abreaction is part of many forms of religious healing (see below). It was used to great acclaim during the Second World War, using rapid-acting barbiturates to bring prompt relief from acute war neuroses (predominantly acute stress disorders), notably by Sargant and Slater (1963). In civilian practice, abreaction is less effective, perhaps because fewer disorders are the result of discrete overwhelming stresses.

Meditation and *yoga* are increasingly used by people with minor psychiatric problems as an alternative or adjunct to psychiatric treatment. There are many approaches, each associated with different systems of belief, but sharing common features. They involve relaxation and the regulation of breathing, and directing attention away from the external world and from the stream of thoughts that would otherwise occupy the mind, often by repeating a word or phrase (a mantra). An important feature is setting time aside when calm can be restored. In addition, the espousal of a value system and association with similar-minded individuals (the activities are commonly group-based) may explain some of the reported successes of the methods. Aspects of meditation have been incorporated into MBCT.

Traditional healing still plays an important part in many individuals' lives, and is often a precursor to seeking professional help. It is not restricted, as is often assumed, to ethnic minority groups. Alternative medicines and treatments are ubiquitous in all strata of society. However, the use of traditional healers is generally restricted to minority groups. Traditional healers can be broadly divided into four groups (Jilek, 2000):

- *Herbalists* are concerned mainly with plant remedies, some of which are known to contain active ingredients, while others appear to be placebos.
- *Medicine men and women* use ritual methods of healing, sometimes combined with plant remedies. They are believed to have special powers, often of supernatural origin.
- *Shamans* use methods like those of medicine men and women, but also enter into altered states of consciousness in which they are believed to communicate with spirits or ancestors, and to recover the abducted souls of people made ill by this supposed loss.
- *Diviners* discover and name the cause of illness by interpreting oracles (in either clear or altered consciousness) from the content of dreams, or through some form of communication with ancestors or spirits.

Traditional healers use methods that incorporate the non-specific processes in western psychological treatment (Box 24.2). In addition, they are aware of the value of naming a condition and answering the questions 'Why am I (or my son) afflicted?' This ends uncertainty and relieves blame and guilt. Some traditional healers use therapeutic suggestion, and many involve the family both in the diagnostic process and in the rituals of treatment. Some employ cleansing or purification rituals to eliminate supposed polluting agents. A few healers use sacrificial rites to appease supernatural beings, sometimes combining these with confession and a promise of changed behaviour. They may involve the wider community to reinforce their message. Traditional healing is not necessarily incompatible with modern medicine and may run parallel with it more often than we think.

Cross-cultural psychotherapies

Recent developments in all psychotherapies include attention to the care of migrants, ethnic minorities, and orthodox religious communities. Some general principles of cultural adaptation have been proposed, including philosophical orientation of the therapeutic modality; practical arrangements to improve access to the therapy including health system and societal factors; technical adjustments in terms of skills, frequency and time allocation; and theoretical refinements. (Naeem et al., 2019; Tseng, 1999) Naeem et al. describe the stages of cultural adaptation for CBT, and similar principles can be applied to other therapies (Bhugra and Bhui, 1998).

Ethical problems in psychological treatment

Autonomy

The need for informed consent is as great in psychological treatments as in any other medical procedure. To give such consent the patient must understand the nature of the treatment and its likely consequences. This is a good opportunity to discuss expectations.

Confidentiality

Group psychotherapy presents special problems of confidentiality. Patients should understand fully the requirement to talk of personal matters in the group, but they need to understand equally clearly the requirement to treat as confidential all they hear in the group. Family therapy presents similar problems, especially if the therapist agrees to see one member outside the family session, and is told of a family secret (e.g. an extra-marital affair). Wherever possible the therapist should avoid such individual meetings and arrange for a colleague to see the family member if this is necessary (e.g. if one member is seriously depressed). Similar problems arise in couple therapy.

The answer to the question of when a therapist should reveal confidential material to a third party is the same as in other treatment situations, namely that it is justified when there is a substantial risk to a third party. Confidentiality is often confused with secrecy in psychotherapy (both by patients and by some therapists). Modern mental healthcare uses a model of shared responsibility, and often information must be shared. Only in exceptional cases should patients be promised that 'absolutely no one else will ever know of this'. In modern health systems, electronic records are the norm, meaning the whole treatment team and clinicians in primary and secondary care across providers can access data. Usually the patient is asked to consent to such releases, although the appropriate and relevancy of access must always be justified in the clinical records.

Ethical issues

Patients who are receiving psychological treatment are potentially vulnerable to exploitation. This arises from the experiences that cause them to seek psychotherapy,

but also because of the intense and often dependent relationship with the therapist. As in other branches of medicine, exploitation may be financial or sexual. Financial exploitation is a potential problem in private practice, in which treatment may inadvertently be prolonged for longer than is necessary. The therapist should discuss the role of the negotiating fees within the therapy. The exploitation may be sexual. In the medical and other caring professions, such exploitation is prohibited in professional codes of conduct. Several unethical therapists have defended inappropriate sexual contact as 'therapeutic' and, remarkably, in some cases have succeeded in this defence. However, it is generally accepted that it is never acceptable to form a sexual relationship with a current or previous patient.

Another form of exploitation is the imposition of the therapist's values on the patient. This may be open and direct (e.g. when a therapist imposes the view that termination of pregnancy is morally wrong) or it may be concealed and indirect (e.g. when a therapist expresses no opinion, but nevertheless gives more attention to the arguments against termination than to those for it). Similar problems may arise, for example, in couple therapy when the therapist's values may affect their approach to the question of whether the couple should separate. A controversial issue is that of 'implanting' erroneous explanations. This has been very contested in relation to 'recovered memory' syndrome, where the recall, after many years, of early familial sexual abuse has been attributed to therapists exploiting suggestibility. Although there has been no suggestion that this is deliberate, there is considerable professional doubt about its status, and the consequences are so potentially catastrophic that it requires very careful monitoring.

In group psychotherapy, one patient may be exploited by another. This is one of the reasons for proscribing contact outside the group. One patient may bully or scapegoat another within the sessions, or may seek a sexual relationship outside. The therapist should try to protect vulnerable patients within the sessions. Clearly it is not possible to be as strict in this matter as in a professional relationship, but strongly emphasizing the purpose of the therapy rules is key to minimizing the problem.

Further reading

- Bateman, A., Brown, D. & Pedder, J. (2010). *Introduction to Psychotherapy: An Outline of Psychodynamic Principles and Practice*. Routledge.
- Bloch, S. (2013). *An Introduction to the Psychotherapies*, 4th edn. Oxford University Press. (An introduction to the commonly used psychological treatments, with a chapter on ethics.)
- Frank, J. D., & Frank, J. B. (1993). *Persuasion and Healing*, 3rd edn. Johns Hopkins Press. (A revised version of a landmark account of the non-specific factors in psychotherapy.)
- Gabbard, G., Beck, J. S., & Holmes, J. (Eds.). (2005). *Oxford Textbook of Psychotherapy*. Oxford University Press.
- (A comprehensive set of reviews of the major forms of psychotherapy and their use in clinical practice.)
- Gurman, A. S. (Ed.). (2003). *Family Therapy: Theory, Practice and Research*. Brunner-Routledge. (A comprehensive work of reference.)
- Hawton, K., et al. (1989). *Cognitive Behaviour Therapy for Psychiatric Problems: A Practical Guide*. Oxford (online edn, Oxford Academic, 1 Aug. 2015), <https://doi.org/10.1093/med:psych/9780192615879.001.0001>, accessed 18 Nov. 2025.

CHAPTER 25

Drugs and other physical treatments

Introduction 685

History of physical treatments 685

General considerations 688

The classification of drugs used in psychiatry 693

Anxiolytic drugs 694

Hypnotics 699

Antipsychotic drugs 700

Anticholinergic drugs 711

Antidepressant drugs 711

Mood-stabilizing drugs 729

Psychostimulants 737

Other physical treatments 738

Introduction

This chapter is concerned with the use of drugs and other physical treatments, such as electroconvulsive therapy (ECT) and neurosurgical procedures. Psychological treatments are the subject of Chapter 24. This separation, although convenient when treatments are described, does not imply that the two kinds of therapy are to be thought of as exclusive alternatives when an individual patient is considered; on the contrary, many patients require both. In this book, the ways of combining treatments are considered in other chapters, where the treatment of individual syndromes is discussed. It is important to keep this point in mind when reading this chapter.

This chapter is concerned with clinical therapeutics rather than basic psychopharmacology, which the reader is assumed to have studied already. An adequate knowledge of the mechanisms of drug action is essential if drugs are to be used in a rational way, but a word of caution is appropriate. The clinician should not assume that the therapeutic effects of psychotropic drugs are

necessarily explained by the pharmacological actions that have been discovered so far. In addition, there is less knowledge about the neuropsychological mechanisms through which pharmacological manipulation can ameliorate symptomatology.

This caution does not imply that a knowledge of pharmacological mechanisms has no bearing on psychiatric therapeutics. On the contrary. There have been substantial advances in pharmacological knowledge since the first specific psychotropic drugs were introduced in the 1950s, and it is increasingly important for the clinician to relate this knowledge to their use of drugs. As noted in Chapter 6, evidence-based clinical guidelines are increasingly used to assist practitioners in their use of medication, and it is important to be aware of these. However, in specialist psychiatric practice there is often rather less high-quality evidence to guide prescribing, and a knowledge of pharmacology will help to ensure that safe and reasonable prescribing practices are followed.

History of physical treatments

Physical treatments have been applied to patients with psychiatric disorders since antiquity, although, in retrospect, the most that could be claimed for the best of

these interventions is that they were relatively harmless. Of course, the same holds for the management of patients with general medical disorders, for which

similar treatments, such as bleeding and purging, were often used regardless of diagnosis. It is wise not to be too censorious about the treatment of disorders of which the aetiology is still largely unknown, but to bear in mind that ‘it may well be that in a hundred years current therapies, psychotherapies as well as physical therapies, will be looked upon as similarly uncouth and improbable’ (Kiloh et al., 1988).

Historically, physical treatments can be divided into two main classes:

1. Those that were aimed at producing a direct change in a pathophysiological process, usually by some alteration in brain function.
2. Those that were aimed at producing symptomatic improvement through a dramatic psychological impact.

The latter interventions were often based on philosophical theories about the moral basis of madness. For example, many physicians appear to have followed the proposal of Heinroth (1773–1843) that insanity was the product of evil and personal wrongdoing. Accordingly, restraint with chains and corporal punishment were seen as appropriate remedies. Other physical treatments, such as the spinning chair introduced by Erasmus Darwin (1731–1802), seemed to be designed to produce a general ‘shock to the system’, and perhaps thereby to interrupt the morbid preoccupations of the patient. A less arduous regimen was the use of continuous warm baths, often given in combination with cold packs. This treatment was recommended by clinicians as distinguished as John Conolly (1794–1866) and Emil Kraepelin (1856–1926), and was still in use at the Bethlem Hospital in the 1950s.

Drugs that produce changes in the function of the central nervous system (CNS), such as opiates and anticholinergic agents, have been used in the treatment of mental disorders for hundreds of years. Although some of these drugs may have had calming effects, they were of no specific value in the treatment of psychiatric disorders. Often a physical treatment was used not because of its proven efficacy, but because it was recommended by an eminent and vigorous physician. Also, the assessment of efficacy depended almost entirely on uncontrolled clinical observation.

In 1933, about ten years after the isolation of insulin by Frederick Banting and Charles Best, Manfred Sakel introduced *insulin coma treatment* for psychosis. A suitable dose of insulin was used to produce a coma, which was terminated by either tube feeding or intravenous glucose. A course of treatment could include up to 60 comas. Not surprisingly, serious side effects were common, and

a mortality of at least 1% could be expected, depending on the standard of the clinic and the physical state of the patient. Insulin coma treatment was rapidly taken up throughout Europe, and many specialized treatment units were built. There was a great improvement in the morale of patients and staff because of the belief that this dramatic treatment could cure symptoms of some of the most serious psychiatric disorders.

There were always some practitioners who doubted the efficacy of insulin coma treatment. Their doubts were reinforced by a controlled trial by Ackner and Oldham (1962), who found that, in patients with schizophrenia, insulin coma was no more effective than a similar period of unconsciousness induced by barbiturates. This study was published about the time when chlorpromazine was introduced, and both factors led to a rapid decline in the use of insulin coma treatment. It should be noted that the controlled studies did not exclude the efficacy of insulin treatment in some circumstances, and some continued to maintain that it was effective. Therefore, it is interesting that recent experimental studies have shown that insulin has important functional roles in the brain and influences, for example, synaptic plasticity and turnover of monoamine neurotransmitters (Kellar and Craft, 2020). Perhaps the main lesson to be learned from insulin coma treatment is that the introduction of a new medical treatment should be preceded by adequate controlled trials to determine whether it is therapeutically more effective or safer than current therapies (see Chapter 6). This lesson is particularly important in psychiatry, because the aetiology of most disorders is obscure and outcome may vary widely, even among patients with the same clinical syndrome.

ECT was introduced about the same time as insulin coma treatment. Unlike the latter, ECT has retained a place in current clinical practice. The rationale for convulsive therapy was a postulated antagonism between schizophrenia and epilepsy such that one would exclude the other. This view is erroneous insofar as schizophrenia-like illnesses are actually more common in patients with temporal lobe epilepsy than would be expected by chance (see Chapter 11). Astute clinical observation, in combination with controlled trials, has shown that ECT is effective in the acute treatment of severe mood disorders. Thus, even though the rationale for the introduction of ECT was incorrect and its mode of action remains unclear, controlled trials have confirmed that, in carefully defined clinical situations, ECT is a safe and effective treatment.

The action of *lithium* in reducing mania was a chance finding by Cade (1949), who had been investigating the

effects of urates in animals and had decided to use the lithium salt because of its solubility. Lithium is a toxic agent, so Cade's important observations did not have a significant impact on clinical practice until the following decade, when controlled trials showed that lithium was effective in both the acute treatment of mania and the prophylaxis of recurrent mood disorders.

Other agents that revolutionized psychopharmacology were introduced about this time (see Box 25.1). Their efficacy and their indications were first recognized through clinical observation, and were subsequently confirmed by controlled clinical trials. None of these agents was introduced on the basis of an aetiological hypothesis. Indeed, such aetiological hypotheses as there are in biological psychiatry have been largely derived from knowledge of the mode of action of effective drugs. Thus the dopamine-receptor-antagonist properties of antipsychotic drugs have given rise to the dopamine hypothesis of schizophrenia, while the action of tricyclic antidepressants and monoamine oxidase inhibitors (MAOIs) in facilitating the effects of noradrenaline and 5-hydroxytryptamine (5-HT) has led to the various monoamine hypotheses of mood disorders.

The past few decades have brought a period of consolidation in psychopharmacology, though useful discoveries, often based on 'drug repurposing', continue to be made (Box 25.1). Clinical trials have been widely used to refine the indications of particular drug treatments and to maximize their risk-benefit ratios. New

compounds have become available, but because most of them have been derived from previously described agents, their range of activity is not strikingly different from that of their predecessors. In general, however, the newer agents are better tolerated and sometimes safer—developments that are important for clinical practice.

There may now be grounds for more optimism about the prospects for advances in psychopharmacology. For example, there is rapidly increasing knowledge about chemical signalling in the brain. Numerous neurotransmitters and neuromodulators interact with specific families of receptors, many of which exist in a number of different subtypes. Most of these receptors have been cloned, and selective ligands for them are becoming available. There is increasing knowledge as to how these chemical messengers may modify behaviour through their interactions with specific brain regions and distributed neuronal circuits.

Future compounds that are developed as a consequence of these scientific advances are likely to differ from current drugs in their range of behavioural effects, and could lead to important new developments in psychopharmacology. Given the complex causes of psychiatric disorders, it seems likely that detailed knowledge of aetiology and pathophysiology may lag behind advances in therapeutics. Of course, this disparity is not uncommon in general medicine. It serves to reinforce the importance of *randomized clinical trials* in the assessment of new psychopharmacological treatments.

Box 25.1 Introduction of some physical treatments in psychiatry

1934	Insulin coma treatment (Sakel)	1966	Valproate in bipolar disorder (Lambert et al.)
1936	Frontal leucotomy (Moniz)	1967	Clomipramine in obsessive-compulsive disorder (Fernandez and Lopez-Ibor)
1936	Metrazol convulsive therapy (Meduna)	1988	Clozapine in treatment-resistant schizophrenia (Kane et al.)
1938	Electroconvulsive therapy (Cerletti and Bini)	1996	Transcranial magnetic stimulation in depression (Pascual-Leone et al.)
1941	Amphetamine for hyperactivity in children (Bradley)	1999	Lamotrigine in bipolar depression (Calabrese et al.)
1949	Lithium (Cade)	2001	Atypical antipsychotic augmentation in depression (Shelton et al.)
1952	Chlorpromazine (Delay and Deniker)	2005	Quetiapine in bipolar depression (Calabrese et al.)
1954	Benzodiazepines (Sternbach)	2006	Ketamine in resistant depression (Zarate et al.)
1957	Iproniazid (Crane and Kline)		
1957	Imipramine (Kuhn)		

General considerations

The pharmacokinetics of psychotropic drugs

Before psychotropic drugs can produce their therapeutic effects, they must reach the brain in adequate amounts. The extent to which they do so depends on their absorption, distribution, passage across the blood–brain barrier, metabolism, and excretion. A short review of these processes is given here. The reader who has not studied them before is referred to the chapter on pharmacokinetics in *Seminars in Clinical Psychopharmacology* (Haddad and Nutt, 2020). The following processes are important:

- absorption
- distribution
- metabolism
- excretion.

Absorption

In general, psychotropic drugs are easily absorbed from the gut because most of them are *lipophilic* and are not highly ionized at physiological pH values. Like other drugs, they are absorbed faster from an empty stomach, and in reduced amounts by patients suffering from intestinal hurry or malabsorption syndrome.

Distribution

Psychotropic drugs are distributed in the plasma, where most of them are largely bound to proteins, for example, diazepam and amitriptyline are about 95% bound. They pass easily from the plasma to the brain, across the blood–brain barrier, because they are highly lipophilic. For the same reason, they enter fat stores, from which they are released slowly long after the patient has ceased to take the drug. This means that psychotropic drugs tend to have *large volumes of distribution*.

Metabolism

Most psychotropic drugs are *metabolized in the liver*. This process begins as the drugs pass through the liver in the portal circulation on their way from the gut. This ‘first-pass’ metabolism reduces the amount of available drug, and is one of the reasons why larger doses are needed when a drug such as chlorpromazine is given by mouth than when it is given intramuscularly. The extent of this liver metabolism differs from one person to another. It is altered by certain other drugs which, if taken at the same time, induce liver enzymes (e.g. carbamazepine)

or inhibit them (e.g. some selective serotonin reuptake inhibitors, SSRIs).

Some drugs, such as carbamazepine, induce their own metabolism, especially after being taken for a long time. Not all drug metabolites are inactive—for example, fluoxetine is metabolized to a hydroxy derivative, norfluoxetine, which is also a potent 5-HT reuptake inhibitor. Where drugs give rise to active metabolites, measurements of plasma concentrations of the parent drug alone are a poor guide to therapeutic activity.

Excretion

Psychotropic drugs and their metabolites are excreted mainly through the *kidney*. When kidney function is impaired, excretion is reduced and a lower dose of drug should be given. Lithium is filtered passively and then partly reabsorbed by the same mechanism that absorbs sodium. The two ions compete for this mechanism; hence reabsorption of lithium increases when that of sodium is reduced (e.g. by thiazide diuretics).

Plasma half-life

Plasma concentrations of drugs vary throughout the day, rising immediately after the dose and falling at a rate that differs between individual drugs. The rate at which a drug level declines after a single dose ranges from hours for lithium carbonate to weeks for slow-release preparations of injectable antipsychotic agents. Knowledge of these differences allows more rational decisions to be made about appropriate intervals between doses.

The concept of *plasma half-life* is useful here. The half-life of a drug in plasma is the time taken for its concentration to fall by a half, once dosing has ceased. For most psychotropic drugs, the amount eliminated over time is proportional to the plasma concentration, and in this case, it will take approximately five times the half-life for the drug to be eliminated from plasma. Equally, when dosing with a drug begins, it will take five times the half-life for the concentration in plasma to reach steady state. This can be important when planning treatment. For example, MAOIs should not be given with SSRIs. Therefore, if a patient is taking sertraline, which has an elimination half-life of about 26 hours, it will be important to leave at least five times the half-life (a week is recommended) before starting MAOI treatment. When sertraline treatment begins, the plasma concentrations will continue to rise for about a week before reaching a steady state.

Measurement of circulating drug concentrations

As a result of individual variations in the mechanisms described above, plasma concentrations after standard doses of psychotropic drugs can vary substantially from one patient to another. For example, tenfold differences have been observed with the antidepressant drug nortriptyline. Therefore, it might be expected that measurements of the plasma concentration of circulating drugs would help the clinician. However, with a few exceptions (e.g. lithium and clozapine), this practice is rarely helpful because the plasma drug levels that predict therapeutic response or toxicity within individuals vary considerably. Clearly, for any drug to work it must be present in plasma above a certain minimum concentration, and for some medications ‘target’ levels have been suggested. However, it is still not unusual for some individuals to show a therapeutic response when plasma levels are lower than recommended.

Pharmacogenetics

Pharmacogenetic approaches to prediction of treatment response are a topic of current interest and are most developed for the pharmacological treatment of depression with a number of commercial kits being available. The genes assessed in such tests are predominantly involved in coding for enzymes involved in drug metabolism; they can therefore identify individuals in whom hepatic capacity to metabolize specific antidepressant medications differs significantly from the general population. While such approaches reduce the risk of being prescribed medications liable to drug–gene interactions, their effect to improve therapeutic outcome is currently modest and rather inconsistent (Oslin et al., 2022). There may, however, be value in assessing Duffy-null genotype (*ACKR1* gene) in people of Middle Eastern and African ancestry who are considering treatment with clozapine. This is because *ACKR1* is associated with a benign ethnic neutropenia and knowledge of its presence may prevent exclusion of this particular patient group from clozapine treatment (Royal College of Psychiatrists, 2023).

Pharmacodynamic measures

As an alternative to these assays, it may be possible to measure the pharmacological property that is thought to be responsible for the therapeutic effect of a particular drug. For example, positron emission tomography (PET) can be used to measure directly the degree of brain dopamine-receptor blockade produced by antipsychotic

drugs during treatment. Such information has proved valuable in designing appropriate dosage regimens of antipsychotic drugs (see Table 25.5). However, these pharmacodynamic measures have not yet been able to identify why some patients do not respond to medication. For example, the degree of dopamine-receptor blockade is the same in patients who respond to antipsychotic drugs as in those who do not.

Drug interactions

When two psychotropic drugs are given together, one may interfere with or enhance the actions of the other. Interference may arise through alterations in absorption, binding, metabolism, or excretion (*pharmacokinetic interactions*), or by interaction between the pharmacological mechanisms of action (*pharmacodynamic interactions*).

Pharmacokinetic interactions

Interactions that affect *drug absorption* are seldom important for psychotropic drugs. Interactions due to *protein binding* are also uncommon, even though many psychotropic drugs are highly protein-bound. Interactions that affect *drug metabolism* are of considerable importance. Examples include the inhibition of the metabolism of antipsychotic drugs by some SSRIs, and the stimulation of the metabolism of many psychotropic drugs by carbamazepine, which induces the relevant *cytochrome P450 enzymes* (see below). Interactions that affect *renal excretion* are mainly important for lithium, the elimination of which is decreased by thiazide diuretics.

Cytochrome P450 enzymes. There have been significant developments in the understanding of the microsomal cytochrome P450 enzyme system. These enzymes are located mainly in the liver but also in other tissues, including the gut wall and brain. Their role is to detoxify exogenous substances such as drugs, and their activity can be increased or decreased by concomitant drug administration. This can give rise to clinically important drug interactions. Importantly, several antidepressants, particularly SSRIs, potentially inhibit P450 enzymes (see Table 25.10).

Pharmacodynamic interactions

Pharmacodynamic interactions are exemplified by the serotonin syndrome, in which drugs that potentiate brain 5-HT function by different mechanisms (e.g. SSRIs and MAOIs) can combine to produce dangerous 5-HT toxicity (see below).

As a rule, a single drug can be used to produce all of the effects required of a combination—for example, many antidepressant drugs have useful anti-anxiety

effects. It is desirable to avoid combinations of psychotropic drugs whenever possible, and, if a combination is to be used, it is essential to know about possible interactions. The *British National Formulary (BNF)* provides a useful guide.

Drug withdrawal

Many psychotropic drugs do not achieve useful therapeutic effects for several days or even weeks. After the drugs have been stopped, there is often a comparable delay before their effects are lost. Psychotropic and, indeed, many other classes of drugs produce neuroadaptive changes during repeated administration. Tissues therefore have to readjust when drug treatment is stopped; this readjustment may appear clinically as a *withdrawal* or *abstinence syndrome*. Characteristic abstinence syndromes have been described for antidepressants, antipsychotics, and anxiolytics, while sudden discontinuation of lithium can provoke a 'rebound' mania in patients with bipolar disorder. It is important to be able to distinguish withdrawal syndromes from relapse of the disorder that is being treated. In addition, the risk of abstinence symptoms makes it prudent to withdraw psychotropic drugs slowly wherever possible.

General advice about prescribing

Use well-tried drugs

It is good practice to use well-tried drugs with therapeutic actions and side effects that are thoroughly understood. Clinicians should become familiar with a small number of drugs from each of the main classes. In this way they can become used to adjusting the dosage and recognizing side effects. Well-tried drugs are usually less expensive than new preparations.

Give an adequate dose

Having chosen a suitable drug, the clinician should prescribe it in *adequate doses*. They should not change the drug or add others without a good reason. In general, if there is no therapeutic response to one established drug, there is no likelihood of a better response to another that has very similar pharmacological properties (provided that the first drug has been taken in adequate amounts). However, since the main obstacle to adequate dosage is usually side effects, it may be appropriate to change to a drug with a different pattern of side effects, for example, from an SSRI to mirtazapine, or vice versa.

Use drug combinations cautiously

Occasionally, combinations of psychotropic drugs are given deliberately in the hope of producing interactions that will be more potent than the effects of either drug taken alone in full dosage (e.g. an SSRI with an atypical antipsychotic agent). This practice, if it is to be used, is best carried out by experienced psychiatrists (or under their guidance) because the adverse effects of combinations are less easy to predict than those of single drugs.

Dosing and treatment duration

When a drug is prescribed, it is necessary to determine the dose, the interval between doses, and the likely duration of treatment. The dose ranges for commonly used drugs are indicated later in this chapter. Ranges for others can be found in the manufacturers' literature, the *BNF*, or a comparable work of reference. Within the therapeutic range, the correct dose for an individual patient should be decided after considering the severity of symptoms, the patient's age and weight, and any factors that may affect drug metabolism (e.g. other drugs that are being taken, liver disease).

Next, the interval between doses must be decided. Psychotropic drugs have often been given three times a day, even though their duration of action is such that most can be taken once or twice a day without any undesirable fall in plasma concentrations between doses. Less frequent administration has the advantage that outpatients are more likely to be reliable in taking drugs. In hospital, less frequent drug rounds mean that nurses have more time for psychological aspects of treatment. Some drugs, such as anxiolytics, are required for immediate effect rather than continuous action; they should not be given at regular intervals, but shortly before occasions on which symptoms are expected to be at their worst, or 'as needed' with an individually defined daily limit. The duration of treatment depends on the disorder under treatment; it is considered in the chapters that deal with the relevant clinical syndromes.

What patients want to know

Psychotropic drugs have the aim of changing what people think and feel; unsurprisingly, many patients have misgivings about taking them. It is therefore important to make it clear what the drug is being used for, what therapeutic effects are expected, and when they should start to appear. Other key questions that must be dealt with include the following:

- What effects will I experience on first taking the drug?
- What side effects can I expect?

- What serious side effects should I report immediately?
- For how long should I take the drug?
- Is the drug addictive?
- What will I notice when I stop the drug?

Compliance, concordance, and collaboration

Many patients do not take the drugs that are prescribed for them. Of course, this problem is not restricted to psychiatric practice, but the use of psychotropic drugs raises additional issues in terms of societal stigma and the nature of the adverse effects. Problems with compliance are mainly manifested when treating outpatients, but also occur in hospital, where some patients find ways of avoiding taking the drugs administered by nurses.

If patients are to take medication reliably, they must be convinced of the need to take it, be free from unfounded fears about its dangers, and be aware of how to take it. Each of these requirements presents particular problems when the patient has a psychiatric disorder. Thus, patients with schizophrenia or seriously depressed patients may not be convinced that they are ill, or they may not wish to recover. Deluded patients may distrust clinical staff, and patients with health anxiety may fear dangerous side effects. Anxious patients often forget the prescribed dosage and frequency of their drugs. Therefore, it is not surprising that many psychiatric patients do not take their drugs in the prescribed way. It is important for the clinician to pay attention to this problem. Time spent discussing the patient's concerns is time well spent, for it often increases compliance with treatment. Written instructions and information can be a valuable adjunct, and are now included with drug packaging.

The successful and safe use of medication requires a *collaborative* relationship between patient and doctor. Some have proposed that the terms *concordance* or *adherence* should therefore be preferred to *compliance*, which carries the implicit assumption that the patient's job is to obey instructions. Whatever the term used, it is clearly important to recognize that the use of drug treatment, particularly in psychiatry, requires a thorough understanding of the *patient's attitude* to both their illness and its treatment.

Ethical aspects of drug prescription

The ethical issues in this complex area have been reviewed by Kader and Pantelis (2009).

1. The basis of ethical prescribing is the practitioner's comprehensive knowledge of the risks and benefits

of drug therapies. This will be derived from evidence-based approaches where possible.

2. The doctor–patient relationship is the appropriate framework through which this knowledge is communicated to the patient.
3. The therapeutic partnership between patient and doctor must lead to true informed consent, which includes the right of competent patients to refuse treatment.

Difficulties arise when the evidence base for treatment is lacking and when there is uncertainty about what approach to pursue. Here the clinician has the responsibility to advise treatments that would be supported by peer opinion and to use clinical guidelines. The clinician should also support the right of patients to genuinely effective treatment where this is being hindered by cost constraint and other economic factors. Another difficult problem concerns refusal of treatment when capacity is impaired and the health and safety of the patient or others are at risk. In fact, empirical research suggests that refusal of treatment in these circumstances is often transient and due to current clinical factors. Indeed, most patients whose refusal of treatment is clinically overridden apparently conclude eventually that the decision to treat them was justified (Kader and Pantelis, 2009). It is, of course, important to elucidate and document the reasons for treatment refusal, and to respect the right of competent patients to refuse treatment.

Prescribing for special groups

Children and the elderly

Psychotropic drugs usually lack specific licences for use in young people, and relevant controlled trials are sparse. However, in the case of antidepressants, for example, only fluoxetine has a licence for the treatment of depression in young people, with accompanying advice from NICE that it should be used in combination with a specific psychological treatment in this age group (see Chapter 16). Practitioners need to make themselves aware of local guidelines concerning the use of psychotropic medication in young people, and should seek specialist advice when in doubt.

Clinical trials of most psychotropic medications often exclude older patients, even though conditions such as depression are common in the elderly. Elderly patients are often sensitive to side effects of medication, and may have impaired renal or hepatic function; for these reasons it is important to start treatment with low doses.

Pregnant women

There are special problems with regard to prescribing psychotropic drugs in pregnancy, because of the risk of *teratogenesis*. Information about the teratogenic risk of individual drugs can be obtained from the relevant manufacturer and from the *BNF*, although the available evidence is often sparse or difficult to interpret. In addition, illnesses such as depression are known to have adverse effects on pregnancy outcomes. The practitioner and patient therefore have the difficult task of weighing available information against the risk of managing the illness without medication. In addition, a substantial number of pregnancies are unplanned. For this reason, it is prudent, where possible, to advise women of childbearing age who require psychotropic drugs specifically to avoid pregnancy until the need for the drug treatment is over. For reviews of the use of psychotropic drugs in pregnancy, see National Institute for Health and Care Excellence (2014b) and McAllister-Williams et al. (2017).

Anxiolytics and antidepressants

Anxiolytic drugs are seldom essential in early pregnancy, and psychological treatments can usually be used. If short-term medication is needed, *benzodiazepines* have in general not been shown to be teratogenic. If an *antidepressant drug* is required, SSRIs and tricyclic antidepressants have similar risks. SSRIs do not appear to cause major malformations. However, their use is associated with a small absolute increase in risks of pre-term delivery, post-partum haemorrhage, and persistent pulmonary hypertension of the newborn. Accounting for residual confounding in these epidemiological studies poses difficulties (Lebin and Novick, 2022).

Antipsychotic drugs and mood stabilizers

It is seldom necessary to start *antipsychotic drugs* in early pregnancy. The teratogenic risks of conventional and 'atypical' antipsychotics appear similar. Overall there is little evidence that the use of atypical antipsychotics during pregnancy causes major congenital malformations, though one population-based cohort study suggested that olanzapine might be associated with an increased risk of musculoskeletal malformations (Andrade, 2021).

Lithium treatment early in pregnancy has been associated with cardiac abnormalities in the foetus, particularly *Ebstein's anomaly*. Therefore, women who are considering pregnancy have been recommended to discontinue lithium before conceiving. Similarly, women who become pregnant while taking lithium have usually been advised to stop the treatment. However, recent epidemiological studies have suggested that, although the relative risk of Ebstein's anomaly is increased in infants

who are exposed to lithium in the first trimester, the absolute risk is still fairly low, at 0.05–0.1%. Withdrawal of lithium carries a high risk of relapse in patients with bipolar illness, and the balance of risk to mother and baby may therefore suggest continuation of lithium treatment during pregnancy in some cases (see Fornaro et al., 2020).

Anticonvulsant drugs such as *valproate* and *lamotrigine* are increasingly used as mood stabilizers. However, valproate treatment is clearly associated with an increased risk of neural tube defects as well as other foetal abnormalities. It is estimated that the risk of major congenital abnormalities with valproate use during pregnancy is around one in ten, while four out of ten children will experience neurodevelopmental problems. Current advice is that valproate should *not* be used in women of childbearing age unless no alternative treatment is available, and in these few cases a pregnancy prevention programme must be in place and treatment supported by an appropriate second opinion (National Institute for Health and Care Excellence, 2022c). In contrast, current data suggests that lamotrigine treatment during pregnancy is not associated with an increased risk of major malformations and is less harmful in this respect than carbamazepine which, like valproate, should *not* be used in pregnancy (Pariente et al., 2017).

Neonatal toxicity

Exposure to psychotropic drugs in the later stages of pregnancy can give rise to *neonatal toxicity*, either through the presence of the drug or through a withdrawal syndrome. For example, late exposure to SSRIs is associated with an increased risk of neonatal complications, including jitteriness, hypoglycaemia, poor muscle tone, and respiratory difficulties. The perinatal toxicity associated with *lithium* use includes 'floppy baby syndrome', with cyanosis and hypotonicity, whereas *benzodiazepine* treatment can result in impaired temperature regulation together with breathing and feeding difficulties.

Animal studies suggest that foetal exposure to psychotropic medication can cause *longer-term abnormalities in brain development and behaviour*, and similar effects might occur in humans. However, disentangling such effects from those associated with illness in the mother and problems in the family environment is challenging. For example, there is an established association between SSRI use in pregnancy and increased risk of autistic spectrum disorders in offspring. However, adjusting this association for confounding by indication suggests that it is probably not casual (Dragioti et al., 2019). Similarly, after adjusting for confounding, a large Scandinavian cohort found there was little or no increased risk of

ADHD or autistic spectrum disorder in the offspring of mothers prescribed antipsychotic drugs during pregnancy (Hálfðánarson et al., 2022).

Breastfeeding

Psychotropic drugs should be prescribed cautiously to women who are breastfeeding. *Diazepam* and other *benzodiazepines* pass readily into breast milk, and may cause sedation and hypotonicity in the infant. In general, sedatives and hypnotics are best avoided during breastfeeding. Antipsychotic drugs and antidepressants also enter breast milk, although rather less readily than diazepam. Sertraline and mirtazapine are present in small amounts and breastfeeding can be permitted while the baby is observed for sedation or feeding difficulties. It is usually advised that mothers should express and discard breast milk that has been exposed to peak plasma levels of the drug concerned. If antipsychotic drugs are required, olanzapine and quetiapine can be considered but women taking clozapine should not breastfeed.

Lithium salts enter the milk freely, and serum concentrations in the infant can approach those of the mother, so breastfeeding is best avoided. A general problem is that, even when the concentration of a particular drug in breast milk is low and no detectable clinical effect on the infant can be discerned, it is nevertheless possible that subtle longer-term effects on brain development and behaviour could occur. For this reason, some authorities recommend that women who are receiving psychotropic medication should not breastfeed at

all. A more pragmatic view is provided by the *BNF* and Taylor et al. (2021).

What to do if there is no therapeutic response

1. Is the drug being taken as recommended? The first step is to find out whether the patient has been taking the drug in the correct dose. They may not have understood the original instructions, or may be worried that a full dose will produce unpleasant side effects. Some patients fear that they will become dependent if they take the drug regularly. Other patients may have little wish to take drugs for the reasons discussed above.
2. Is the patient taking any other drug that could affect the metabolism or pharmacological action of the psychotropic agent? Misuse of legal or illegal substances might interfere with the therapeutic actions of psychotropic drugs.
3. Is the diagnosis correct? Review the diagnosis to make sure that the treatment is appropriate before deciding whether to increase the dose or change the medication.

Failure to respond adequately to psychotropic medication is a common reason for psychiatric referral. Specific pharmacological approaches for individual disorders are discussed in the relevant chapters.

The classification of drugs used in psychiatry

Drugs that have effects mainly on mental symptoms are referred to as *psychotropic*. Psychiatrists also use the term *antiparkinsonian agents*, which refers to drugs that are employed to control the side effects of some psychotropic drugs. Anticonvulsant drugs have a role in the treatment of mood disorders.

Psychotropic drugs are conventionally divided into different classes, as shown in Table 25.1, but the therapeutic actions of particular compounds are not confined to one diagnostic category. For example, SSRIs are classified as antidepressants and are effective in the treatment of major depression, but they also produce useful therapeutic effects in anxiety states, obsessive-compulsive disorders, and some eating disorders. Of course, this breadth of effect does not mean that the latter syndromes are forms of depression. It merely highlights

the fact that the neuropsychological consequences of facilitating brain 5-HT function may provide beneficial effects in a variety of psychiatric disorders.

Although there is considerable understanding of the pharmacological actions of psychotropic drugs, little is known about the neuropsychological consequences of these pharmacological actions and about the ways in which neuropsychological changes are translated into clinical benefit in different diagnostic syndromes. At present, therefore, the best plan is to classify drugs according to their major therapeutic use, but to bear in mind that the therapeutic effects of different classes of drugs *may overlap considerably*.

Each of the main groups of drugs will now be reviewed in turn. For each group, an account is given of therapeutic effects, pharmacology, the principal compounds

Table 25.1 Classification of clinical psychotropic drugs

Class of drug	Examples of classes	Class indications (examples)
Antipsychotic	Phenothiazines	Acute treatment of schizophrenia and mania; prevention of relapse in schizophrenia
	Butyrophenones	
	Substituted benzamides	
Antidepressant	SSRIs	Major depression (acute treatment and prophylaxis), anxiety disorders
	Tricyclic antidepressants	
	MAOIs	
Mood stabilizer	Lithium	Prophylaxis of bipolar disorder and recurrent mood disorder
	Anticonvulsants	
Anxiolytic	Benzodiazepines	Anxiety disorders (short-term use)
	Azapirones (buspirone)	
Hypnotic	Benzodiazepines	Insomnia
	Cyclopyrrolones ('Z drugs')	
	Orexin antagonists	
Psychostimulant	Methylphenidate	Attention-deficit hyperactivity disorder
	Modafinil	Narcolepsy

available, pharmacokinetics, unwanted effects, and contraindications. General advice is given about the use of each group in everyday clinical practice, but specific applications to the treatment of individual disorders are found in the chapters that deal with those conditions.

Drugs that have a limited use in the treatment of a single disorder—for example, *disulfiram* for alcohol problems, or *cholinesterase inhibitors* for dementia—are discussed in the chapters that deal with the relevant clinical syndromes.

Anxiolytic drugs

Anxiolytic drugs, such as *benzodiazepines*, have been prescribed widely and often inappropriately. Before prescribing anxiolytic drugs, it is always important to seek the causes of anxiety and to try to modify them. It is also essential to recognize that a degree of anxiety can motivate patients to take steps to reduce the problems that are causing it. Therefore removing all anxiety in the short term is not always beneficial to the patient in the long run. Anxiolytics such as benzodiazepines are most

useful when given for a short time, either to tide the patient over a crisis or to help them to tackle a specific problem.

Tolerance is a particular problem with benzodiazepine-like anxiolytic drugs, and drug dependence can develop. Because the benzodiazepines are still widely used anxiolytics, they will be considered first. *Antidepressants* are increasingly used to treat specific anxiety syndromes, but their therapeutic actions differ in important ways

from benzodiazepine-like drugs. Their indications in the treatment of anxiety disorders are considered here, but their detailed pharmacology is discussed in the section on antidepressant drugs. When reading this section, it is important to bear in mind that psychological treatments are effective in the management of anxiety disorders and have certain advantages over drug treatment, including more sustained efficacy after treatment cessation, as well as fewer adverse effects.

Benzodiazepines

Pharmacology

Benzodiazepines have several actions:

- anxiolytic
- sedative and hypnotic
- muscle relaxant
- anticonvulsant.

Their pharmacological actions are mediated through specific receptor sites located in a supramolecular complex with gamma-aminobutyric acid (GABA_A) receptors. Benzodiazepines enhance GABA neurotransmission, thereby indirectly altering the activity of other neurotransmitter systems, such as those involving noradrenaline and 5-HT.

Compounds available

Many different benzodiazepines are available. They differ both in the potency with which they interact with benzodiazepine receptors and in their plasma half-life (see Table 25.2). In general, *high-potency benzodiazepines* and those with *short half-lives* are more likely to be associated with dependence and withdrawal. Benzodiazepines with short half-lives (less than 12 hours) include lorazepam, temazepam, and lorazepam.

Because of problems with dependence, long-acting benzodiazepines are preferable for the management of anxiety, even if such treatment is to be given intermittently on an 'as-required' basis. Long-acting benzodiazepines include drugs such as diazepam, chlordiazepoxide, and clonazepam. *Diazepam* is rapidly absorbed and can be used both for the continuous treatment of anxiety and for treatment 'as required'. *Alprazolam*, a high-potency benzodiazepine, is effective in the treatment of panic disorder. This therapeutic efficacy is not confined to alprazolam, because equivalent doses of other high-potency agents such as *clonazepam* are also effective.

Flumazenil is a benzodiazepine-receptor antagonist that produces little pharmacological effect by itself, but blocks the actions of other benzodiazepines. Therefore,

Table 25.2 Half-lives of some drugs that act at the GABA–benzodiazepine-receptor complex

Diazepam	20–100 h*
Chlordiazepoxide	5–30 h*
Lorazepam	8–24 h
Temazepam	5–11 h
Zopiclone	4–6 h
Zolpidem	1.5–2 h
Chlormethiazole	4–6 h
	(4–12 h in the elderly)
Chloral	6–8 h

* Active metabolite increases half-life further.

it may be useful in reversing acute toxicity produced by benzodiazepines, but carries a risk of provoking acute benzodiazepine withdrawal. Flumazenil is available only for intravenous use.

Pharmacokinetics

Benzodiazepines are rapidly absorbed. They are strongly bound to plasma proteins but, because they are lipophilic, pass readily into the brain. They are metabolized to a large number of compounds, many of which have therapeutic effects of their own; temazepam and oxazepam are among the metabolic products of diazepam. Excretion is mainly as conjugates in the urine.

Benzodiazepines with short half-lives, such as temazepam and lorazepam, have a 3-hydroxyl grouping, which allows a one-step metabolism to inactive glucuronides. Other benzodiazepines, such as diazepam and clonazepam, are metabolized to long-acting derivatives, such as desmethyldiazepam, which are themselves therapeutically active.

It is now common practice to give benzodiazepines (often in combination with low-dose antipsychotic drugs) to produce a rapid calming effect in psychosis. In this situation, benzodiazepines may be given parenterally, and it is worth noting that the absorption of diazepam following intramuscular injection is poor, and lorazepam should be preferred if this route of administration is used.

Unwanted effects

Benzodiazepines are well tolerated. When they are given as anxiolytics, their main side effects are due to

the sedative properties of large doses, which can lead to *ataxia* and *drowsiness* and *falls* (especially in the elderly), and occasionally to *confused thinking* and *amnesia*. Minor degrees of drowsiness and of impaired coordination and judgement can *affect driving skills* and the operation of potentially dangerous machinery; moreover, people who are affected in this way are not always aware of it. For this reason, when benzodiazepines are prescribed, especially those with a longer action, patients should be advised about these dangers and about the potentiating effects of alcohol. The prescriber should remember that these effects are more common among elderly patients and those with impaired renal or liver function. Use of benzodiazepines (and Z drugs) for more than three months has been linked to an *increased risk of dementia* but residual confounding cannot be excluded and overall a causal association seems unlikely (Osler and Jørgensen, 2020),

Although in some circumstances benzodiazepines reduce tension and aggression, they can also rarely lead to a *release of aggression* by reducing inhibitions in people with a tendency to aggressive behaviour. In this they resemble alcohol. This possible effect should be remembered when prescribing for those judged to be at risk of child abuse, or for any person with a previous history of impulsive aggressive behaviour.

Toxic effects

Benzodiazepines have few toxic effects. Patients usually recover from large overdoses because these drugs do not depress respiration and blood pressure as barbiturates do. Even so, fatal overdoses of benzodiazepines have occasionally been reported.

Drug interactions

Benzodiazepines, like other sedative anxiolytics, potentiate the effects of alcohol and of drugs that depress the CNS. This can cause serious toxicity when opiate users also take high doses of benzodiazepines. Significant respiratory depression has been reported in some patients receiving combined treatment with benzodiazepines and clozapine.

Dependence and withdrawal

It is now generally agreed that *dependence* develops after prolonged use of benzodiazepines. The frequency depends on the drug and the dosage, and has been estimated to be up to 50% of patients who are long-term users. Benzodiazepines are associated with a withdrawal syndrome and tolerance. Although drug-seeking behaviour and dose escalation are not common in people

prescribed benzodiazepines, it certainly can occur, and benzodiazepines are occasionally involved in polydrug misuse and dependence.

The *withdrawal syndrome* associated with benzodiazepines is characterized by several different kinds of symptoms:

- apprehension, anxiety, and insomnia
- tremor
- nausea
- heightened sensitivity to perceptual stimuli and perceptual disturbances
- depression and suicidal thinking
- epileptic seizures (rarely).

Since many of these symptoms resemble those of anxiety disorder, it can sometimes be difficult to decide whether the patient is experiencing a benzodiazepine withdrawal syndrome or a recrudescence of the anxiety disorder for which the drug was originally prescribed. Perceptual disturbances are more likely to indicate benzodiazepine withdrawal.

Withdrawal symptoms generally begin within two to three days of stopping a short-acting benzodiazepine, and within seven days of stopping a long-acting one. The symptoms generally last for three to ten days. Withdrawal symptoms seem to be more frequent after taking drugs with a short half-life than after taking those with a long one. If benzodiazepines have been taken for a long time, it is best to withdraw them gradually over a period of several weeks, according to recent guidance (National Institute for Clinical Excellence 2022a). If this is done, withdrawal symptoms can be minimized or avoided (see Chapter 20).

Benzodiazepines do have the advantage of becoming quickly effective. Therefore current advice is that they should be administered on a short-term basis only (not more than four weeks) to help a patient to cope with functionally disabling anxiety while other treatment measures are instituted.

Azapirones (buspirone)

Indications and pharmacology

The only drug in the azapirone class that is currently marketed for the treatment of anxiety is *buspirone*. It is effective in the treatment of generalized anxiety disorder but is not helpful in the treatment of panic disorder or social anxiety. Unlike the benzodiazepines, the anxiolytic effects of buspirone take several days to develop

and require dose-titration. It is also important to note that buspirone cannot be used to treat benzodiazepine withdrawal. Perhaps for these reasons it has relatively little use.

Pharmacologically, buspirone has no affinity for benzodiazepine receptors, but stimulates a subtype of 5-HT receptor called the 5-HT_{1A} receptor. This receptor is found in high concentration in the raphe nuclei in the brainstem, where it regulates the firing of 5-HT cell bodies. Administration of buspirone lowers the firing rate of 5-HT neurons and thereby decreases 5-HT neurotransmission in certain brain regions. This action may be the basis of its anxiolytic effect.

Pharmacokinetics and adverse effect

Buspirone has poor systemic availability because it has an extensive first-pass metabolism. The side effect profile differs from that of benzodiazepines. For example, buspirone treatment does not cause sedation, but instead is often associated with *light-headedness*, *nervousness*, and *headache* early in treatment. There is little evidence that tolerance and dependence occur during buspirone use, although such a judgement must always be made with circumspection.

Drug interactions

Buspirone is relatively free from significant drug interactions, but combination with MAOIs has been reported to cause raised blood pressure.

Antidepressant drugs used for anxiety

Antidepressant drugs usually ameliorate the anxiety that accompanies depressive disorders. Tricyclic antidepressants have also been shown to be effective in the management of *generalized anxiety disorder*, *panic disorder*, and *post-traumatic stress disorder*, whether or not significant depressive symptoms are present. Similarly, SSRIs are effective in a broad range of anxiety disorders, as well as *obsessive-compulsive disorder* (Baldwin et al., 2014). Both venlafaxine and duloxetine, which are classified as *selective serotonin and noradrenaline reuptake inhibitors* (SNRIs), are licensed for the treatment of generalized anxiety disorder as well as depression.

The therapeutic profile of antidepressant drugs in the treatment of anxiety differs significantly from that of benzodiazepines. The time of onset of effect is much slower with antidepressants and, particularly in panic disorder, there may be an *exacerbation of symptoms*

early in treatment. However, the ultimate therapeutic effect of antidepressants is at least as great as that of benzodiazepines, and they are less likely to produce cognitive impairment (Baldwin et al., 2014). In addition, the use of antidepressants is not associated with tolerance and dependence, although sudden cessation of treatment can cause withdrawal symptoms.

Antipsychotic drugs used for anxiety

Conventional antipsychotic drugs have sometimes been prescribed in low doses for their anxiolytic effects, particularly in patients with persistent anxiety who have become dependent on other drugs, and those with aggressive personalities who respond badly to the disinhibiting effects of benzodiazepines. However, even low-dose antipsychotic treatment, if maintained, is not free from the risk of *tardive dyskinesia*. Newer antipsychotic drugs, such as quetiapine, may also possess anxiolytic effects when given either as a sole treatment or as augmentation treatment in patients who are non-responsive to more standard drug therapies. However, the adverse-effect profile of antipsychotic drugs, together with the lack of well-controlled studies, suggests that such an indication should be restricted to specialist use in patients with anxiety disorders that have not responded to other pharmacological and psychological approaches (Baldwin et al., 2014).

Beta-adrenoceptor antagonists that are used for anxiety

These drugs relieve some of the autonomic symptoms of anxiety, such as tachycardia, almost certainly by a peripheral effect. They are best reserved for anxious patients whose main symptom is palpitation or tremor, particularly in social situations. They are not effective for formal anxiety disorders such as generalized anxiety disorder or panic disorder. An appropriate drug is *propranolol* in a dose of 20–40 mg three times a day. Contraindications are heart block, systolic blood pressure below 90 mmHg or a pulse rate of less than 60 beats/minute, and a history of bronchospasm. Beta-adrenoceptor antagonists precipitate heart failure in a few patients, and should not be given to those with atrioventricular node block, as they decrease conduction in the atrioventricular node and bundle of His. They can exacerbate Raynaud's phenomenon and hypoglycaemia in diabetics.

Pregabalin

Indications and pharmacology

Pregabalin is a derivative of the anticonvulsant drug, *gabapentin*. Like gabapentin, pregabalin has anticonvulsant and analgesic properties. It is licensed for the treatment of *generalized anxiety disorder* but not for other anxiety disorders. Both gabapentin and pregabalin are analogues of GABA; however, neither compound is active at GABA or benzodiazepine receptors. It is believed that their therapeutic effects are mediated through interaction with the $\alpha 2\text{-}\delta$ subunit of voltage-gated calcium channels with a consequent modification of neurotransmitter release.

Controlled trials have shown that, in doses of 150–600 mg, pregabalin is effective in the treatment of generalized anxiety disorder, and is at least as efficacious as other agents used to treat generalized anxiety disorders, including the SSRIs (Baldwin et al., 2014).

Pharmacokinetics and adverse effect

Pregabalin is rapidly absorbed, with peak plasma concentrations occurring within about one hour. Its half-life is about six hours, and it is eliminated unchanged primarily through renal excretion. Dose adjustment is therefore required in patients with impaired renal function. Because of its pattern of elimination, pregabalin treatment is not expected to cause pharmacokinetic interactions with other drugs, but the effects of central sedatives (e.g. benzodiazepines and alcohol) may be potentiated.

In clinical trials of patients with generalized anxiety disorder, discontinuation owing to adverse events with pregabalin was about 12%, compared with 5% with placebo. The most common adverse effects are somnolence and dizziness. Other common unwanted effects of pregabalin include increased appetite, mood changes, confusion, ataxia, tremor, and memory impairment. The most potentially serious reactions are visual disturbances, including vision loss, blurred vision, and other changes of visual acuity. These symptoms mostly improve when pregabalin is discontinued. It also seems likely that withdrawal of pregabalin is associated with discontinuation symptoms, such as insomnia, headache, nausea, diarrhoea, anxiety, sweating, and dizziness.

As well as therapeutic prescribing (pregabalin is licensed for pain and epilepsy as well as generalized anxiety disorder), there is significant misuse of pregabalin particularly in combination with opiates where it increases the risk of opiate-related deaths (Evoy et al., 2021). In 2019, the UK government placed gabapentin and pregabalin under Schedule 3 of the Misuse of Drugs

Regulation and Class C of the Misuse of Drugs act. This restricts the amount of gabapentin and pregabalin that can be issued on a single prescription. In this context it is worth noting that, while gabapentin is not licensed for any specific psychiatric disorder, it is not infrequently used ‘off-label’ to treat anxiety and insomnia, particularly in patients with bipolar disorder. However, evidence for its efficacy in these situations is weak (Hong et al., 2022).

Advice on management

Before an anxiolytic drug is prescribed, the cause of the anxiety should always be sought. In addition, it is helpful to classify the nature of the anxiety disorder, as this can have implications for drug treatment. It is worth remembering that, although medication is undoubtedly helpful in the treatment of anxiety syndromes, *psychological treatments* are as effective and are often preferred by patients. In practice, medication tends to be used when psychological treatments are not readily available or have not been successful.

For most patients with *anxiety symptoms*, attention to life problems, an opportunity to talk about their feelings, and reassurance from the doctor are enough to reduce anxiety to tolerable levels. If an anxiolytic is needed, a benzodiazepine should be given for a short time—not more than three weeks—and withdrawn gradually. It is important to remember that dependency is particularly likely to develop among people with alcohol-related problems. If the drug has been taken for several weeks, the patient should be warned that they may feel tense for a few days when it is stopped.

A compound such as *diazepam* is suitable for both the intermittent treatment of anxiety and continuous treatment throughout the day. The use of diazepam on an ‘as-needed’ basis usually means that lower total doses are consumed and the risk of tolerance and dependence is diminished. For longer-term treatment of severe generalized anxiety disorder, *antidepressant medication* is more appropriate (see Chapter 8).

Antidepressants are often helpful in the treatment of *panic disorder*, although the risk of early symptomatic worsening must be remembered and explained to the patient. The use of small doses early in treatment can be helpful. High-potency benzodiazepines such as *alprazolam* and *clonazepam* are effective in panic disorder, but can cause cognitive impairment and withdrawal problems. However, they may be helpful in patients who do not respond to other treatments. *MAOI treatment* can also be used in treatment-resistant patients (Baldwin et al., 2014).

Hypnotics

Hypnotics are drugs that are used to improve sleep. Many anxiolytic drugs also act as hypnotics, and these have been reviewed in the previous section. Hypnotic drugs are prescribed widely and are often continued for too long. This reflects the frequency of insomnia as a symptom. Effective psychological treatments are available for the management of insomnia, and they appear to have a more sustained duration of action than hypnotics (van der Zwerde et al., 2019).

Pharmacology

The ideal hypnotic would increase the length and quality of sleep without residual effects the next morning. It would do so without altering the pattern of sleep and without any withdrawal effects when the patient ceased to take it. Unfortunately, no drug meets these exacting criteria. It is not easy to produce drugs that affect the whole night's sleep and yet are sufficiently eliminated by morning for there to be no residual sedative effects.

Most prescribed hypnotics *enhance the action of GABA* through interaction with either the benzodiazepine receptor or other adjacent sites located on the GABA macromolecular complex. Antihistamines and low doses of sedating antidepressants such as amitriptyline and trazodone are also used to facilitate sleep. The pineal hormone, melatonin, is involved in organizing circadian rhythms and is sometimes used for the treatment of primary insomnia.

Compounds available

The most commonly used hypnotics are *benzodiazepines* or non-benzodiazepine ligands, which act at or close to the benzodiazepine-receptor site. The latter include *zopiclone* and *zolpidem* (the 'Z drugs'). The actions of these drugs can be reversed by the benzodiazepine-receptor antagonist, flumazenil. Other available hypnotic agents include *chloral hydrate* (or its derivatives), *chlormethiazole*, and *sedating antihistamines* (the latter are often present in 'over-the-counter' preparations). A sustained-release form of the pineal hormone *melatonin* is licensed for the short-term treatment of insomnia in the middle-aged and elderly, while other formulations of melatonin are used to treat sleep disorders in people with learning difficulties and neurodevelopmental disorders.

Of the benzodiazepines, the shorter-acting compounds such as temazepam and lormetazepam are

appropriate as hypnotics because of their relative lack of hangover effects. Other benzodiazepines that were previously marketed as hypnotics, such as flurazepam and nitrazepam, have a long duration of action and produce significant impairments in tests of cognitive function on the day following treatment.

Zopiclone is a cyclopyrrolone. It produces fewer changes in sleep architecture than benzodiazepine hypnotics. The most common side effect is a *bitter after-taste* following ingestion, but behavioural disturbances, including *confusion*, *amnesia*, and *depressed mood*, have been reported. *Zolpidem* is a similar agent, with a shorter half-life.

Clomethiazole edisylate is a hypnotic drug with anti-convulsant properties. It has been used previously to prevent withdrawal symptoms in patients who are dependent on alcohol. For this reason, it is sometimes mistakenly believed to be a suitable hypnotic for alcoholic patients. This belief is erroneous because the drug is as likely as any other hypnotic drug to cause dependency, and can cause dangerous *respiratory depression* when combined with alcohol. It retains a place in the treatment of insomnia in the elderly because of its short duration of action. Unwanted effects include *sneezing*, *conjunctival irritation*, and *nausea*.

Orexin antagonists block receptors for the neuropeptide orexin, which is a signalling molecule involved in the control of wakefulness and appetite. A number of such compounds are licensed for the treatment of insomnia in the United States, while daridorexant is currently licensed for this indication in Europe. Generally, orexin antagonists are effective in the treatment of insomnia but in terms of tolerability do not appear to have clear advantages over other licensed agents (De Crescenzo et al., 2022).

Unwanted effects

As well as the specific side effects of individual compounds noted above, there are a number of general problems associated with the use of all hypnotics. One of the most important is the presence of *residual effects*, which are experienced by the patient on the next day as feelings of being *slow* and *drowsy*. These are accompanied by deficits in daytime performance and an increased risk of falls and other accidents. Such effects are less apparent with the shorter-acting compounds noted above but can still be detected even with drugs such as zolpidem. Other

problems include the *development of tolerance*, whereby the original dose of the drug has progressively less efficacy, and *'rebound' insomnia* on withdrawal, which makes preparations difficult to stop. *Tolerance* is less of a problem with sedating antidepressants, but such drugs have long half-lives accompanied by residual psychomotor effects the next day.

Interactions

The most important interaction of hypnotic drugs is with *alcohol*, where a potentiated effect can be seen. The *interaction between clomethiazole and alcohol* is particularly dangerous, and can result in death from respiratory failure. Hypnotics will also potentiate the effect of other drugs with sedating actions, such as some antidepressant and antipsychotic agents.

Advice on management

Before prescribing hypnotic drugs, it is important to find out whether the patient is really sleeping badly and, if so, why. Many people have unrealistic ideas about the number of hours for which they should sleep. For example, they may not know that duration of sleep often becomes shorter in middle and late life. Others take 'cat naps' in the daytime, perhaps through boredom, and still expect to sleep as long at night. Some people ask for sleeping tablets in anticipation of poor sleep for one or two nights (e.g. when travelling). Such temporary loss of sleep is soon compensated for by increased sleep on subsequent nights, and any supposed advantage in terms of

alertness after a full night's sleep is likely to be offset by the residual effects of the drugs.

The common causes of disturbed sleep include excessive caffeine or alcohol, pain, cough, pruritus, dyspnoea, and anxiety and depression. When any primary cause is present, this should be treated, not the insomnia. Often simple 'sleep hygiene' measures may be helpful (see Chapter 13). If, after careful enquiry, a hypnotic appears to be essential, it should be prescribed for a few days only. The clinician should explain this to the patient, and should warn them that a few nights of restless sleep may occur when the drugs are stopped, but that this restlessness will not be a reason for prolonging the prescription.

The prescription of standard hypnotics for children is not justified, except for the occasional treatment of night terrors and somnambulism. However, *melatonin* has been regarded as helpful in improving sleep in children with neurodevelopmental disorders or in sleep disorders such as sleep-onset insomnia and delayed-phase sleep syndrome (see Chapter 13; Wilson et al., 2019). Hypnotics should also be prescribed with particular care for the elderly, who may become confused and get out of bed in the night, and perhaps injure themselves. Many patients are started on long periods of dependency on hypnotics by the prescribing of 'routine night sedation' in hospital. Prescription of these drugs should not be routine, but should only be a response to a real need, and should be stopped before the patient goes home. Guidance from the National Institute for Health and Care Excellence (2015) reinforces advice to use hypnotic medicines only for *short periods in disabling insomnia*.

Antipsychotic drugs

This term is applied to drugs that reduce psychomotor excitement and control symptoms of psychosis. Alternative terms for these agents are *neuroleptics* and *major tranquillizers*. None of these names is wholly satisfactory. 'Neuroleptic' refers to the side effects rather than to the therapeutic effects of the drugs, and 'major tranquillizer' does not refer to the most important clinical action. The term 'antipsychotic' is used here because it appears in the *BNF*.

The main therapeutic uses of antipsychotic drugs are to reduce *hallucinations*, *delusions*, *agitation*, and *psychomotor excitement* in schizophrenia, mania, or psychosis secondary to a medical condition. The drugs are also used prophylactically to *prevent relapses* of schizophrenia

and other psychoses. The introduction of chlorpromazine in 1952 led to substantial improvements in the treatment of schizophrenia, and paved the way to the discovery of the many psychotropic drugs that are now available.

Pharmacology

Antipsychotic drugs (see Box 25.2) share the property of *blocking dopamine receptors*. This may account for their therapeutic action, a suggestion that is supported by the close relationship between their potency in blocking dopaminergic receptors *in vitro*, and their therapeutic strength. Imaging studies show that acute psychosis is

Box 25.2 A list of some antipsychotic drugs

Phenothiazines

Chlorpromazine
Trifluoperazine

Thioxanthenes

Flupentixol
Clopenthixol

Butyrophenones

Haloperidol

Dibenzodiazepines

Clozapine
Olanzapine

Dibenzothiazepine

Quetiapine

Substituted benzamides

Sulpiride
Amisulpride

Benzisoxazole

Risperidone

Benzisothiazol

Lurasidone

Quinolinone

Aripiprazole
Cariprazine

associated with increased dopamine release in striatal regions, and that the extent of this increase correlates with the subsequent therapeutic effect of antipsychotic drugs (see Chapter 11). A persuasive formulation of antipsychotic drug action suggests that these agents block the ability of increased dopamine release to attribute *abnormal salience* to irrelevant stimuli (Kaar et al., 2020).

Dopamine receptors are of several subtypes. It is the D_2 receptor that is critical for antipsychotic action, and all licensed drugs in the category are antagonists at this receptor, with varying affinities for the D_2 subtype. PET studies suggest that an antipsychotic effect is obtained when D_2 -receptor occupancy lies in the range 60–70%.

Higher levels are associated with extrapyramidal movement disorders and hyperprolactinaemia, but not with greater efficacy (Kaar et al., 2020). Other side effects are attributable to binding to a variety of other receptors (see below).

Distinction between typical and atypical antipsychotic drugs

The term *atypical* antipsychotic agent was introduced to distinguish the newer antipsychotic drugs from conventional *typical* agents, such as chlorpromazine and haloperidol. An alternative term is *second generation*. Although the definition of the term ‘atypical’ varies in the literature, a fundamental property of an atypical antipsychotic is its ability to produce an antipsychotic effect *without causing extrapyramidal side effects*. This definition is problematic, not least because antipsychotics do not fall clearly into two classes in this respect, but lie along a spectrum. For example, low-potency conventional antipsychotic drugs such as chlorpromazine have a relatively low risk of producing extrapyramidal symptoms when prescribed at modest dosages; conversely, extrapyramidal side effects can occur with the atypical antipsychotic risperidone. However, it is true to say that atypical antipsychotic agents have a lower likelihood of causing extrapyramidal side effects within their usual therapeutic range. In addition, the risk of tardive dyskinesia appears to be lower with the newer antipsychotic drugs (Carbon et al., 2018).

Another property that is sometimes attributed to atypical antipsychotic drugs is improved efficacy relative to typical agents. Although this is true in terms of positive psychotic symptoms for the prototypic atypical antipsychotic clozapine, it is not clear how far more recently developed compounds meet this exacting criterion. A multiple treatments network meta-analysis of placebo-controlled trials showed that evidence in this respect is strongest for amisulpride and olanzapine (Huhn et al., 2019). However, pragmatic trials have so far failed to demonstrate important therapeutic differences between conventional and newer antipsychotic drugs, and even clozapine does not have proven efficacy against negative or cognitive symptoms. For these and other reasons, most authorities now believe that the terms ‘atypical’ and ‘typical’ antipsychotic are not useful, and that attention is better directed towards the pharmacological properties of individual drugs and their associated therapeutic profile (Kaar et al., 2020) (see Table 25.3). See Chapter 11 for further discussion.

Table 25.3 Atypical antipsychotics

Drug	EPS*	Prolactin elevation	Weight gain	Adverse effects
Amisulpride	+	+++	+	Insomnia, agitation, nausea, constipation, QTc prolongation
Sulpiride	+	+++	+	Insomnia, agitation, abnormal liver function tests
Clozapine	0	0	+++	Agranulocytosis (white cell monitoring mandatory), myocarditis and myopathy (rare), fatigue, drowsiness, dry mouth, sweating, tachycardia, postural hypotension, nausea, constipation, ileus, urinary retention, seizures, diabetes
Olanzapine	+	+	+++	Somnolence, dizziness, oedema, hypotension, dry mouth, constipation, diabetes, QTc prolongation
Quetiapine	0	0	++	Somnolence, dizziness, postural hypotension, dry mouth, abnormal liver function tests, QTc prolongation, diabetes
Risperidone	+	+++	++	Insomnia, agitation, anxiety, headache, impaired concentration, nausea, abdominal pain, diabetes, QTc prolongation
Aripiprazole	+	0	+	Agitation, insomnia, nausea, vomiting
Cariprazine	+	0	+	Anxiety insomnia, nausea, vomiting
Lurasidone	+	+	+	Somnolence, insomnia, anxiety, agitation, nausea, dyspepsia, abdominal pain

* EPS, extrapyramidal symptoms; 0, not present; +, sometimes; ++, often; +++, can be excessive.

A new class of antipsychotic drugs utilises stimulation of central muscarinic receptors to achieve an antipsychotic effect, probably by diminishing pre-synaptic dopamine release. A combination of the muscarinic agonist, xanomeline, and the peripherally acting muscarinic antagonist, trospium, has been licensed for the treatment of schizophrenia in the United States. Placebo-controlled trials have shown short-term efficacy in psychosis with low levels of movement disorder. Further work is needed to establish the place of muscarinic agonists in the longer-term treatment of schizophrenia.

Pharmacology of typical (conventional) antipsychotics

All of these drugs are effective dopamine-receptor antagonists, but many possess additional pharmacological properties that influence their *adverse-effect profile*.

Phenothiazines

Chlorpromazine is the prototypic phenothiazine. It antagonizes α_1 -adrenoceptors, histamine H_1 -receptors, and muscarinic cholinergic receptors. Blockade of α_1 -adrenoceptors and histamine H_1 -receptors gives chlorpromazine a sedating profile, while α_1 -adrenoceptor blockade also causes hypotension. The anticholinergic activity may cause dry mouth, urinary difficulties, and

constipation, while on the other hand offsetting the liability to cause extrapyramidal side effects. In contrast to chlorpromazine, piperazine compounds such as *trifluoperazine* and *fluphenazine* are more selective dopamine-receptor antagonists, and are therefore less sedating but more likely to cause extrapyramidal effects.

Thioxanthenes and butyrophenones

Thioxanthenes such as *flupentixol* and *clopenthixol* are similar in structure to the phenothiazines. The therapeutic effects are similar to those of the piperazine group. Butyrophenones such as *haloperidol* have a different structure but are clinically similar to the thioxanthenes. They are potent dopamine-receptor antagonists, with few effects at other neurotransmitter receptors. They are not sedating, but have a high propensity to cause extrapyramidal side effects.

Pharmacology of atypical antipsychotic drugs

Selective D_2 -receptor antagonists

Atypical antipsychotic drugs have a diverse pharmacology, but currently two main groupings can be discerned. On the one hand are substituted benzamides such as *sulpiride* and *amisulpride*. These drugs are highly selective D_2 receptor antagonists which, for reasons that are not well understood, seem less likely to produce

extrapyramidal movement disorders. They also lack sedative and anticholinergic properties. However, they cause a substantial increase in plasma prolactin.

5-HT₂-D₂ receptor antagonists

Other atypical antipsychotic drugs possess 5-HT₂ receptor antagonist properties. In other aspects—for example, potency of dopamine D₂-receptor blockade and other neurotransmitter receptor effects, these drugs differ significantly from one another (Kaar et al., 2020). *Risperidone* is a potent antagonist at both 5-HT₂ receptors and dopamine D₂ receptors. It also possesses α_1 -adrenoceptor-blocking properties, which can cause mild sedation and hypotension. *Paliperidone* is an active metabolite of risperidone and has very similar pharmacological properties when given orally.

Olanzapine is a slightly weaker D₂ receptor antagonist than risperidone, but has anticholinergic and powerful histamine H₁ receptor-blocking activity. This gives it strong sedating effects and leads to increased appetite and frequent weight gain. *Quetiapine* is also a histamine H₁ receptor antagonist; it has modest 5-HT₂ receptor-antagonist effects, and rather weaker D₂ receptor-antagonist properties. It has a low propensity to cause movement disorders and, like olanzapine, is highly sedating. *Lurasidone* is a potent 5-HT₂ and D₂ receptor antagonist. It binds only weakly to histamine H₁ receptors, which makes it less likely to cause weight gain and sedation than olanzapine or quetiapine. *Asenapine* binds potently to 5-HT₂ receptors. It also binds significantly to D₂ receptors, as well as to a range of other 5-HT receptors and the α_2 adrenoreceptor, but the contribution of these latter properties to its therapeutic effect is unclear. In the United Kingdom and Europe, asenapine is licensed for the treatment of mania but not schizophrenia. It requires sublingual administration.

Sertindole is a potent 5-HT₂ receptor antagonist with weak D₂ receptor-antagonist effects. It causes clinically significant prolongation of the QTc interval in the electrocardiogram, and its use is currently suspended. *Ziprasidone* is another 5-HT₂ and D₂ receptor antagonist, which differs from the other 5-HT₂/D₂ receptor antagonists described here, because it also binds to 5-HT_{1A} receptors and is a noradrenaline reuptake inhibitor. Somnolence and dizziness are common side effects of ziprasidone, and it causes relatively little weight gain. It tends to increase the QTc interval, and has not been licensed in the United Kingdom.

Aripiprazole is a partial dopamine agonist that also has 5-HT₂ receptor-blocking and 5-HT_{1A} agonist properties. It has an activating profile and can cause pro-dopaminergic side effects such as insomnia, nausea, and vomiting.

Impulse control disorders have been reported occasionally. It is less likely to cause weight gain than quetiapine or olanzapine. *Cariprazine* is also a partial dopamine agonist with similar adverse effects to aripiprazole. It has a particularly high affinity for the dopamine D₃ receptor and appears somewhat more effective than risperidone in treating patients with predominantly negative symptoms of schizophrenia (Németh et al., 2017).

To some extent these latter drugs were designed to reproduce the pharmacological profile of *clozapine*, which was the first antipsychotic agent to show definite benefit in the treatment of patients whose psychotic symptoms had failed to respond to conventional agents. In addition, clozapine has a low liability to cause movement disorders, and is therefore usually regarded as the prototypic atypical antipsychotic drug (Kane et al., 1988).

Clozapine is a weak dopamine D₂ receptor antagonist but has a high affinity for 5-HT₂ receptors. It also binds to a variety of other neurotransmitter receptors, including histamine H₁, α_1 adrenergic and muscarinic cholinergic receptors (Kaar et al., 2020). However, the precise pharmacological basis for its increased efficacy is not well understood. Various studies have attempted to define a therapeutic plasma range for clozapine, with inconsistent results. A reasonable compromise between efficacy and safety is 350–500 $\mu\text{g/l}$. Above 600 $\mu\text{g/l}$ the risk of seizures increases significantly.

While clozapine undoubtedly has superior efficacy, its use is associated with a significant risk of *leukopenia*. This restricts its use to patients who do not respond to or who are intolerant of other antipsychotic drugs. The adverse effects and haematological monitoring of clozapine treatment is discussed below.

Depot antipsychotic drugs

Intramuscular slow-release preparations are used for patients who need to take antipsychotic medication to prevent relapse but do not take oral medication reliably. These 'depot' preparations include the esters of conventional antipsychotic drugs such as *fluphenazine decanoate*, *flupentixol decanoate*, *zuclopenthixol decanoate*, and *haloperidol decanoate*. All are given intramuscularly in an oily medium. *Zuclopenthixol acetate* reaches peak plasma levels within one to two days and has a shorter duration of action than other depots. It is used for the immediate control of acute psychosis, but its superiority to ordinary intramuscular injections is not established. Slow-release injections of atypical antipsychotic drugs such as *risperidone*, *paliperidone* (an active metabolite of risperidone), *aripiprazole*, and *olanzapine* are also available.

Pharmacokinetics

Antipsychotic drugs are well absorbed, mainly from the jejunum. When they are taken by mouth, part of their hepatic metabolism is completed as they pass through the portal system on their way to the systemic circulation (first-pass metabolism). Antipsychotic drugs are *highly protein-bound*.

With the exception of *sulpiride* and *amisulpride*, which are excreted unchanged by the kidney, antipsychotic drugs are extensively metabolized by the liver to produce a range of active and inactive metabolites. This complex metabolism has made it difficult to interpret the clinical significance of plasma concentrations of most antipsychotics; for this reason, with the exception of clozapine, plasma measures are seldom used in everyday clinical work. The half-life of most antipsychotic drugs (around 20 hours) is sufficient to allow once-daily dosing. However, *quetiapine* has a half-life of about three hours, and twice-daily dosing in the treatment of schizophrenia is recommended, although modified-release preparations are available.

The pharmacokinetic profile of *depot preparations* differs substantially from that of standard preparations. For all compounds except zuclopenthixol acetate, it takes several weeks for steady-state drug levels to be reached in the plasma (see Table 25.4). This means that relapse

after treatment discontinuation is likely to be similarly delayed.

Drug interactions

Antipsychotic drugs potentiate the effects of other *central sedatives*. The *hypotensive* properties of chlorpromazine may enhance the effects of *antihypertensive drugs*, including ACE inhibitors.

Some antipsychotic drugs, and particularly *pimozide* and *amisulpride*, can increase the *QTc interval* and should not be given with other drugs that are likely to potentiate this effect, such as antiarrhythmics, tricyclic antidepressants, citalopram, and escitalopram. *Clozapine* should not be given with any agent that is likely to *potentiate its depressant effect on white cell count*, such as carbamazepine, co-trimoxazole, and penicillamine. Some *SSRIs* (notably fluoxetine and paroxetine) slow the hepatic metabolism and increase blood levels of several antipsychotic drugs, including haloperidol, risperidone, aripiprazole, and clozapine.

Unwanted effects

The many different antipsychotic drugs share a broad pattern of unwanted effects that are mainly related to their *antidopaminergic*, *antiadrenergic*, and *anticholinergic*

Table 25.4 Some pharmacokinetic properties of depot antipsychotic drugs

Depot	Time to peak plasma level (days)	Elimination half-life (days)	Dosing interval (weeks)	Typical dose range
Flupentixol decanoate	3–7	14–28	2–4	50mg 4 weekly to 300mg 2 weekly
Haloperidol decanoate	3–9	21	4	50-300mg every 4 weeks
Zuclopenthixol decanoate	3–7	19–31	2–4	200mg every 4 weeks to 600mg weekly
Risperidone microspheres	28-42	3-6	2	20-50mg every two weeks
Olanzapine pamoate	14–28	30	2–4	150mg-300mg every two weeks or 300-405mg every 4 weeks
Paliperidone palmitate (a three-monthly injection of paliperidone palmitate is also available)	12–14	25–50	4	50-150mg every 4 weeks
Aripiprazole monohydrate	5–7	46	Monthly	300–400 mg every 4 weeks

Box 25.3 Some unwanted effects of antipsychotic drugs

Antidopaminergic movement effects

Acute dystonia
Akathisia
Parkinsonism
Tardive dyskinesia

Antiadrenergic effects

Sedation
Postural hypotension
Inhibition of ejaculation

Anticholinergic effects

Dry mouth
Reduced sweating
Urinary hesitancy and retention
Constipation
Blurred vision
Precipitation of glaucoma

Antihistaminic effects

Sedation
Increased appetite
Weight gain

Other effects

Cardiac arrhythmias
Metabolic syndrome and diabetes
Amenorrhoea
Galactorrhoea
Hypothermia
Deep vein thrombosis
Pulmonary embolus
Extrapyramidal effects

properties (see Box 25.3). Details of the effects of individual drugs can be found in the *BNF* or a similar work of reference. Here we give an account of the general pattern, with examples of the side effects associated with a few commonly used drugs.

Movement disorders are related to the antidopaminergic action of the drugs on the *basal ganglia*. As already noted, the therapeutic effects may also derive from the antidopaminergic action, although at mesolimbic sites. The effects on the extrapyramidal system fall into four groups, which are summarized below.

Acute dystonia

This occurs soon after treatment begins, especially in young men. It is most often observed with drugs like haloperidol and high-potency phenothiazines. The main features are torticollis, tongue protrusion, grimacing, and opisthotonos, an odd clinical picture that can easily be mistaken for histrionic behaviour. It can be controlled by an anticholinergic agent given carefully by intramuscular injection.

Akathisia

This is an unpleasant feeling of physical restlessness and a need to move, leading to an inability to keep still. Agitation with suicidal ideation can also occur. Akathisia may wrongly be mistaken for a worsening of psychosis, and more antipsychotic medication may then be inappropriately prescribed. It usually occurs during the first two weeks of treatment with antipsychotic drugs, but may begin only after several months. Akathisia is not reliably controlled by antiparkinsonian drugs. Beta-adrenoceptor antagonists and short-term treatment with benzodiazepines may be helpful. The best strategy is to reduce the dose of antipsychotic drug, if possible.

Parkinsonian syndrome

Antipsychotic-induced parkinsonism is characterized by akinesia, an expressionless face, and lack of associated movements when walking, together with rigidity, coarse tremor, stooped posture, and, in severe cases, a festinant gait. This syndrome often does not appear until a few months after the drug has been taken, and then sometimes diminishes even though the dose has not been reduced. The symptoms can be controlled with antiparkinsonian drugs. However, it is not good practice to prescribe antiparkinsonian drugs prophylactically as a routine, because not all patients will need them. Moreover, these drugs themselves have undesirable effects in some patients, for example, they occasionally cause an acute organic syndrome, and may worsen or unmask concomitant tardive dyskinesia.

Tardive dyskinesia

This is particularly serious because, unlike the other extrapyramidal effects, it does not always recover when the drugs are stopped. It is characterized by chewing and sucking movements, grimacing, choreoathetoid movements, and possibly akathisia. The movements usually affect the face, but the limbs and the muscles of respiration may also be involved. Although the syndrome is seen occasionally among patients who have not taken antipsychotic drugs, it is more common among those

who have taken antipsychotic drugs for a number of years. It is also sometimes seen in patients who are taking dopamine-receptor blockers for other indications (e.g. metoclopramide for chronic gastrointestinal problems).

Epidemiology. Tardive dyskinesia is more common among women, the elderly, and patients who have diffuse brain pathology. A diagnosis of mood disorder is also a risk factor. In about 50% of cases, tardive dyskinesia disappears when the antipsychotic drug is stopped. Estimates of the frequency of the syndrome vary in different series, but it seems to develop in about 20% of patients with schizophrenia who have been treated with long-term conventional antipsychotic medication. Current evidence suggests that the incidence of tardive dyskinesia is lower with atypical antipsychotic agents such as clozapine and olanzapine than with haloperidol (Carbon et al., 2018). However, cases still occur, albeit at a lower level.

Pathophysiology. The cause of tardive dyskinesia is uncertain, but it could be due to supersensitivity to dopamine as a result of prolonged dopaminergic blockade. This explanation is consistent with the observations that tardive dyskinesia may be aggravated by stopping antipsychotic drugs or by the administration of anticholinergic antiparkinsonian drugs (presumably by upsetting further the balance between cholinergic and dopaminergic systems in the basal ganglia).

Treatment. Many treatments for tardive dyskinesia have been tried, but none is universally effective. Therefore, it is important to reduce its incidence as far as possible by limiting long-term antipsychotic drug treatment to patients who really need it. At the same time, a careful watch should be kept for abnormal movements in all patients who have taken antipsychotic drugs for a long time. If dyskinesia is observed, the antipsychotic drug should be stopped if the state of the mental illness allows this.

Although tardive dyskinesia may first worsen after stopping the drug, in many cases it will improve over several months. If the dyskinesia persists after this time or if the continuation of antipsychotic medication is essential, a trial can be made of an atypical agent such as quetiapine. This can sometimes lead to remission of the disorder. Olanzapine and aripiprazole are also possibilities. Clozapine may be the most effective atypical agent in treating tardive dyskinesia but because of its side effect profile is best reserved for patients who do not respond to other atypical agents (unless the patient also has a concomitant treatment-resistant psychotic illness). Other drugs that have been tried include vitamin E, although the evidence for its efficacy is conflicting. Agents that deplete dopamine by inhibiting its uptake into storage vesicles in presynaptic terminals—so-called

VMAT2 inhibitors—can also be helpful in the treatment of tardive dyskinesia. A drug called tetrabenazine is available in the United Kingdom but tends not to be well tolerated (see Taylor et al., 2021).

Tardive dystonia

Although the two conditions can coexist, tardive dyskinesia needs to be distinguished from *tardive dystonia*, which is the long-term persistence of a dystonic movement disorder. Clinically the condition is indistinguishable from the various idiopathic dystonias that present, for example, with blepharospasm or torticollis. The diagnosis of drug-induced tardive dystonia is made on the basis of exposure to dopamine-receptor-blocking agents and a negative family history of dystonia. Treatment is unsatisfactory. Anticholinergic drugs are ineffective, and the condition often persists after withdrawal of the antipsychotic agent. However, clozapine has been reported to be useful, as has local injection of botulinum toxin into the affected muscle group.

Antiadrenergic effects

These include sedation, postural hypotension with reflex tachycardia, nasal congestion, and inhibition of ejaculation. The effects on blood pressure are particularly likely to appear after intramuscular administration, and may appear in the elderly whatever the route of administration.

Anticholinergic effects

These include dry mouth, urinary hesitancy and retention, constipation, reduced sweating, blurred vision, and, rarely, the precipitation of glaucoma.

Antihistaminic effects

H₁ histamine receptor antagonism causes sedation and also increased appetite. The latter can result in excessive weight gain, which contributes towards the development of the metabolic syndrome and type 2 diabetes.

Other effects

Cardiac conduction defects

Cardiac arrhythmias are sometimes reported. ECG changes are more common in the form of *prolongation of the QTc interval* and T-wave changes. It was noted above that the use of *pimozide* has been associated with serious cardiac arrhythmias. Cautious dose adjustment with ECG monitoring is recommended. A similar effect led to the withdrawal of the phenothiazine *thioridazine*. Other antipsychotic drugs with moderate effects to lengthen

the QT interval are amisulpride, quetiapine, olanzapine and chlorpromazine. Antipsychotic drugs are also associated with an increased risk of *venous thrombosis and pulmonary embolus*, which appears to be greater in those who have started treatment more recently. For a review of the cardiac effects of antipsychotic drugs, see Howell et al. (2019).

Depression

Depression of mood has been said to occur, but this is difficult to evaluate because untreated patients with schizophrenia may have periods of depression. It is certainly possible that excessive dopamine-receptor-blockade in the mesolimbic forebrain could be associated with anhedonia and loss of drive, which could then resemble depression or negative symptoms of schizophrenia.

Diabetes and metabolic syndrome

We have already seen that, probably due to histamine H_1 receptor antagonism, patients may *gain weight* when taking antipsychotic drugs, especially chlorpromazine and atypical agents such as olanzapine, quetiapine, and clozapine (see Table 25.3). Weight gain can result in obesity, which increases the risk of type 2 diabetes. Schizophrenia itself has been associated with an increased risk of diabetes, and this risk is probably increased by antipsychotic drugs, partly but not completely through the associated weight gain. Weight gain is probably least likely to be a problem with aripiprazole, lurasidone, and haloperidol (Huhn et al., 2019).

Other pharmacological properties of antipsychotic drugs may increase the risk of diabetes, independently of weight gain, for example, via blockade of peripheral M_3 muscarinic receptors. Similarly, newer antipsychotic drugs such as olanzapine, clozapine, and quetiapine are associated with increased lipid levels. These adverse effects are of great clinical importance because they increase the risk of *cardiovascular disease* and may therefore contribute to the increase in overall mortality seen in patients with schizophrenia. A meta-analysis reported that just over a third of patients taking antipsychotic drugs have the '*metabolic syndrome*' (a combination of central obesity with two out of four of: hypertension, raised fasting glucose, low HDL cholesterol, and raised triglycerides). The presence of this syndrome significantly increases the risk of cardiovascular disease such as myocardial infarction and stroke.

Patients who are receiving treatment with antipsychotic drugs require careful monitoring of metabolic aspects of their general health (Box 25.4). For a review of the pharmacological mechanisms involved in the adverse metabolic effects of antipsychotic drugs, see Singh et al. (2019).

Endocrine changes

Galactorrhoea and *amenorrhoea* are induced in some women by *high prolactin levels*, and it is possible that low libido and sexual dysfunction may also result. There may also be an increased risk of osteoporosis and breast

Box 25.4 Metabolic monitoring and interventions for patients taking antipsychotic drugs

Monitoring

If possible, monitoring should start before treatment or as soon as possible afterwards.

1. Body mass index (once every four weeks for 12 weeks, then at least biannually)
2. Blood glucose (fasting or random: 12 weeks, six months, and then annually—switch to HbA_{1c})
3. Lipid profile (fasting or random: 12 weeks, six months, and then annually)
4. Blood pressure (12 weeks, six months, and annually)
5. Enquire about tobacco smoking and alcohol use at each visit

Possible interventions

1. 'Lifestyle' interventions: diet and exercise, advice about smoking and alcohol intake
2. Switch to antipsychotic drug less likely to cause weight gain (Table 25.3)
3. Adjunctive metformin for weight gain (requires monitoring of renal function and B_{12} levels)
4. Medical management of diabetes, dyslipidaemia, and hypertension according to NICE guidelines

Source: Cooper, S. J., Reynolds, G. P., Barnes, T., et al. (2016). BAP guidelines on the management of weight gain, metabolic disturbances and cardiovascular risk associated with psychosis and antipsychotic drug treatment. *J Psychopharmacol*, 30(8), 717–748. Copyright (2016), SAGE Publications.

cancer (Rahman et al., 2022). Some atypical agents do not increase prolactin levels significantly (see Table 25.3).

Other unwanted effects

In the elderly, *hypothermia* is an important unwanted effect. Some antipsychotic drugs, particularly lower-potency agents, such as chlorpromazine and clozapine, *lower the seizure threshold* and can increase the frequency of seizures in epileptic patients. Prolonged *chlorpromazine* treatment can lead to *photosensitivity* and to accumulation of pigment in the skin, cornea, and lens.

Sensitivity reactions. Phenothiazines, particularly *chlorpromazine*, have been associated with cholestatic jaundice, but the incidence is low (about 0.1%). Blood cell dyscrasias also occur rarely with antipsychotic drugs, but are most common with clozapine. Skin rashes can also occur.

Adverse effects of clozapine

The use of clozapine is associated with a significant risk of *leukopenia* (about 2–3%), which can progress to agranulocytosis. Weekly blood counts for the first 18 weeks of treatment and at two-weekly intervals thereafter are mandatory. After one year, the frequency of blood sampling may be reduced to monthly intervals. With this intensive monitoring, the early detection of leukopenia can be followed by immediate withdrawal of clozapine and by reversal of the low white cell count. This procedure greatly reduces, but does not eliminate, the risk of progression to agranulocytosis. It is usually recommended that clozapine be used as the sole antipsychotic agent in a treatment regimen. Clearly, it is wise to avoid concomitant use of drugs such as *carbamazepine*, which may also lower the white cell count.

Because of its relatively weak blockade of dopamine D₂ receptors, clozapine is unlikely to cause extrapyramidal movement disorders, including tardive dyskinesia. It does not increase plasma prolactin, so galactorrhoea does not occur. However, its use is associated with *hyper-salivation*, *drowsiness*, *postural hypotension*, *weight gain*, and *hyperthermia*. *Seizures* may occur at higher doses. Clozapine is a sedating compound, and cases of respiratory and circulatory embarrassment have been reported during combined treatment with clozapine and benzodiazepines. *Constipation* is common and should be prevented and treated or a dangerous *ileus* may result (Handley et al., 2022). Rarely, fatal *myocarditis* and *myopathy* have been reported. The weight gain is at least partly responsible for an increased risk of *diabetes mellitus*.

The neuroleptic malignant syndrome

This rare but serious disorder occurs in a small minority of patients who are taking antipsychotic drugs, especially

high-potency compounds. Most reported cases have followed the use of antipsychotic agents for schizophrenia, but in some cases the drugs were used for mania, depressive disorder, or psychosis secondary to a medical condition. *Combined lithium and antipsychotic* drug treatment may be a predisposing factor. The overall incidence is probably about 0.2% of patients who are treated with antipsychotic drugs.

The onset is often, but not invariably, during the first ten days of treatment. The clinical picture includes the rapid onset (usually over 24–72 hours) of severe motor, mental, and autonomic disorders, together with *hyperpyrexia*.

- The prominent motor symptom is generalized muscular hypertonicity. Stiffness of the muscles in the throat and chest may cause dysphagia and dyspnoea.
- The mental symptoms include akinetic mutism, stupor, or impaired consciousness.
- Hyperpyrexia develops, with evidence of autonomic disturbances in the form of unstable blood pressure, tachycardia, excessive sweating, salivation, and urinary incontinence.
- In the blood, *creatinine phosphokinase* (CPK) levels may be greatly elevated, and the white cell count may be increased.
- Secondary complications may include pneumonia, thromboembolism, cardiovascular collapse, and renal failure.

The mortality rate of neuroleptic syndrome appears to have been declining over recent years, but can still be of the order of 10%. The syndrome lasts for one to two weeks after stopping an oral neuroleptic, but may last two to three times longer after stopping long-acting preparations. Patients who survive are usually, but not invariably, without residual disability.

The differential diagnosis includes encephalitis, and in some countries heat stroke. Before the introduction of antipsychotic drugs, a similar disorder was reported as a form of catatonia sometimes called acute lethal catatonia. The condition can probably occur with any antipsychotic agent, but in many reported cases the drugs used have been haloperidol or fluphenazine. Cases have also been reported with atypical antipsychotic drugs, including clozapine. The cause could be related to excessive dopaminergic blockade, although why this should affect only a minority of patients cannot be explained.

Treatment is symptomatic. The main priorities are to stop the drug, cool the patient, maintain fluid balance, and treat intercurrent infection. No drug treatment is certainly effective. *Diazepam* can be used for

Table 25.5 Dosage and D₂ receptor blockade of some antipsychotic drugs

Drug	Relative dose (oral)	Maximum BNF dose (mg)	D ₂ -receptor occupancy in vivo (%) [daily dose (mg)]*
Chlorpromazine	100	1,000	80 [200]
Trifluoperazine	5	NA	80 [10]
Haloperidol	2	20	80 [4]
Flupentixol	1	18	74 [10]
Sulpiride	200	2,400	74 [800]
Clozapine	60	900	47 [600]
Risperidone	2	16	75 [4]
Olanzapine	8	20	75 [15]
Aripiprazole	5	30	80 [15]

NA: not available. * Note: Nord and Farde, 2011; Yokoi et al., 2002.

muscle stiffness. *Dantrolene*, a drug used to treat malignant hyperthermia, has also been tried. *Bromocriptine*, a dopamine agonist, is also recommended. Very ill patients require support in an intensive care unit, with intubation and paralysis to maintain respiration and deal with renal failure.

Some patients who developed the syndrome on one occasion have been given the same drug again safely after the acute episode has resolved. Nevertheless, if an antipsychotic has to be used again, it is prudent to restart treatment cautiously with an atypical agent with weaker dopamine-receptor-blocking properties, used at first in low doses. At least two weeks should elapse before antipsychotic drug treatment is reinstated.

Contraindications to antipsychotic drugs

There are few absolute contraindications to antipsychotic medication, and they vary with individual drugs. Before any of these drugs are used, it is important to consult the *BNF* or a comparable work of reference. Contraindications include myasthenia gravis, Addison's disease, glaucoma (where compounds have significant anticholinergic activity), and, in the case of clozapine, any evidence of bone marrow depression. Caution is also required when there is renal disease, liver disease, cardiovascular disorder, epilepsy, or serious infection. Patients with Parkinson's disease sometimes require antipsychotic medication to deal with psychotic states induced by dopaminergic agents; quetiapine or clozapine is advised in these circumstances.

Antipsychotic drugs can produce severe movement disorders and changes in consciousness in some patients with dementia, particularly *Lewy body dementia*. In the elderly, especially those with pre-existing cerebrovascular disease, they are also associated with an increased risk of stroke. Current recommendations are to use antipsychotic drugs with great caution, and, if possible, to avoid their use in these clinical situations. For a discussion, see Chapter 14.

Dosage

Doses of antipsychotic drugs need to be adjusted for the individual patient, and changes should be made gradually. Doses should be lower for the elderly, for patients with brain damage or epilepsy, and for the physically ill. The dosage of individual drugs can be found in the *BNF* or a comparable work of reference, or in the manufacturer's literature.

PET imaging studies

There has been an important trend towards the recommendation of lower doses of typical antipsychotic drugs such as haloperidol. This is based in part on studies with PET, which have demonstrated that adequate dopamine D₂-receptor blockade (in the basal ganglia at least) can be obtained with what were previously considered *low doses* of conventional antipsychotic drugs (e.g. about 5 mg of haloperidol) (see Table 25.5) (Nord and Farde, 2011). For newer antipsychotic drugs, sufficient PET data are usually available to allow clear dosage recommendations.

Low recommended doses produce an adequate antipsychotic effect in the majority of patients. Higher doses may cause further calming, but are likely to be associated with significant adverse effects, some of which may be serious (e.g. cardiac arrhythmias). A prevailing view is that the combination of modest doses of antipsychotic drugs with a benzodiazepine is a safer and more effective means of producing rapid sedation than high doses of antipsychotic drugs.

Antipsychotic drugs and the risk of sudden death

The association of *sudden unexplained death* with antipsychotic drug treatment is a matter of continuing debate. Patients with schizophrenia treated with antipsychotic drugs appear to have higher rates of cardiac arrest and ventricular arrhythmias than controls. They are also more likely to die through choking. This could be due to the illness or to treatment. However, antipsychotic drugs are known to alter *cardiac conduction*, and drugs such as olanzapine can also have hypotensive effects. Epidemiological studies show that the increased risk of sudden death (about twofold) in patients who are taking antipsychotic drugs is similar for the older and newer agents, and is related to increasing dose. Although this association may not be causal, it is clearly prudent to use as low a dose of an antipsychotic drug as the clinical circumstances permit. Particular caution is needed in patients with pre-existing cardiac disease and those taking other medications that might increase the QT_C interval.

An indication of the relative dosage of some commonly used antipsychotic drugs taken by mouth is given in Table 25.5

Pharmacological treatment of acute behavioural disturbance

The role of antipsychotic drugs in the treatment and management of schizophrenia and related disorders is covered in Chapter 11. Here we cover the appropriate use of antipsychotic agents in control of acute behavioural disturbance—sometimes called rapid tranquillization.

Antipsychotic drugs and benzodiazepines are used to control psychomotor excitement, hostility, and other abnormal behaviour resulting from schizophrenia, mania, or organic psychosis. If the patient is very excited and is displaying abnormally aggressive behaviour, the

aim should be to bring the behaviour under control as quickly and safely as possible. Drug treatment in this situation should only be used if psychological or behavioural approaches cannot be employed or have failed to calm the situation.

If the patient is already taking an antipsychotic drug, the addition of oral *lorazepam* (1–2 mg) or *promethazine* (25–50 mg) can be helpful. If the patient is not taking regular antipsychotic medication, the use of *olanzapine* (10 mg), *quetiapine* (100–200 mg), *risperidone* (1–2 mg), or *haloperidol* (5 mg) (the latter combined with *promethazine* 25 mg) can be considered, although the product licence for haloperidol recommends pre-treatment ECG monitoring, which might be difficult to arrange in an emergency situation.

If parenteral treatment is needed, lorazepam (2 mg), promethazine (50 mg), and olanzapine (10 mg) are available as intramuscular preparations. An intramuscular preparation of aripiprazole (9.75 mg) is licensed for emergency use; it may be somewhat less effective than olanzapine, but is less likely to cause hypotension. When using olanzapine intramuscularly it is important to be aware of the contraindications to its use, particularly with regard to patients with cardiovascular disease. *Intramuscular olanzapine should not be given with parenteral benzodiazepines*. Whatever treatment is used, it is important to check for possible respiratory depression, particularly in the elderly and in patients with concomitant physical illness. The benzodiazepine antagonist *flumazenil* should be available. For a summary of the emergency pharmacological management of disturbed behaviour, see Taylor et al. (2021).

There are several other practical points concerning the management of the acutely disturbed patient that can be dealt with conveniently here. Although it may not be easy in the early stages to differentiate between mania and schizophrenia as causes of the disturbed behaviour, it is necessary to try to distinguish them from psychosis secondary to medical conditions and from outbursts of aggression in abnormal personalities. Among medical conditions it is important to consider post-epileptic states, the effects of head injury, transient global amnesia, and hypoglycaemia. If the patient has been drinking alcohol, the danger of potentiating the sedative effects of antipsychotic drugs and benzodiazepines should be remembered. Similarly, antipsychotic drugs that may be more likely to provoke seizures should be used with caution in post-epileptic states.

For further information about the management of violence in healthcare settings, and clinical guidelines, see Chapter 18.

Anticholinergic drugs

Although these drugs have no direct therapeutic use in psychiatry, they are sometimes required to control the *extrapyramidal side effects* of typical antipsychotic drugs.

Pharmacology

Of the drugs that are used to treat idiopathic parkinsonism, currently only the *anticholinergic compounds* are used for drug-induced extrapyramidal syndromes. These drugs are antagonists of *muscarinic cholinergic receptors* both centrally and in the periphery. Some also possess *antihistaminic* properties.

Preparations available

Many anticholinergic drugs are available, and there is little to choose between the various compounds. However, some authorities suggest that the use of agents that are more selective for the M_1 subtype of the muscarinic receptor—for example, *biperiden*—may be associated with fewer peripheral anticholinergic effects. Other preparations that are employed include *procyclidine* and *benzhexol*. *Orphenadrine* and *benztropine* have combined antihistaminic and anticholinergic properties.

Pharmacokinetics

Limited data are available. Anticholinergic drugs appear to be well absorbed and are extensively metabolized in the liver. They are highly protein-bound. Their half-lives are generally in the range of 15–20 hours.

Unwanted effects

In large doses these drugs may cause delirium (*acute organic syndrome*), especially in the elderly. Their anticholinergic activity can summate with those of antipsychotic drugs so that glaucoma or retention of urine in men with enlarged prostates may be precipitated. Drowsiness, dry mouth, and constipation also occur. These effects tend to diminish as the drug is continued.

Orphenadrine may be more toxic than other anticholinergic drugs in overdose, whereas *benztropine* has been associated with heat stroke. All anticholinergic drugs can exacerbate tardive dyskinesia, but are probably not a predisposing factor in its development. Anticholinergic drugs can worsen cognitive function and should be avoided in patients with dementia. It is also possible that long-term use of drugs with anticholinergic properties (this will include drugs such as tricyclic antidepressants and some antipsychotics) can increase the risk of dementia (Coupland et al., 2019).

Drug interactions

Antiparkinsonian drugs can induce *drug-metabolizing enzymes* in the liver, so that plasma concentrations of antipsychotic drugs are sometimes reduced. As noted above, anticholinergic agents can potentiate the effects of other drugs with anticholinergic activity, such as amitriptyline.

Advice on management

As noted previously, anticholinergic drugs should not be given routinely because they may increase the manifestation of tardive dyskinesia. Also, the presence of persistent extrapyramidal side effects is a sign that the dose of medication is excessive for that patient, in terms of D_2 receptor occupancy. Patients who are receiving injectable long-acting antipsychotic preparations may require anticholinergic drugs for only a few days after injection, if at all. There have been reports of misuse of and dependence on anticholinergic drugs, possibly resulting from a mood-elevating effect.

If anticholinergic drugs are required, *biperiden* (2–12 mg daily) or *procyclidine* (5–30 mg daily) are appropriate for routine use. These drugs are usually given three times daily, although their half-lives would suggest that less-frequent dosing should be possible. It is best not to give anticholinergic drugs in the evening because of the possibility of excitement and *sleep disruption*.

Antidepressant drugs

Most currently used antidepressant drugs can be divided into three main classes, depending on their acute pharmacological properties.

1. **Monoamine reuptake inhibitors.** These are compounds that inhibit the reuptake of noradrenaline and/or 5-HT. They include tricyclic antidepressants, SSRIs,

Table 25.6 Groups of antidepressant drugs

Drug	Advantages	Disadvantages
Tricyclic antidepressants	Well studied	Cardiotoxic*, dangerous in overdose
	Efficacy never surpassed	
	Useful sedative effect in selected patients	Anticholinergic side effects**
		Cognitive impairment
SSRIs/SNRIs		Weight gain during longer-term treatment
	Lack cardiotoxicity***: relatively safe in overdose†	Increase agitation and sleep disturbance early in treatment
	Not anticholinergic	Gastrointestinal disturbance, sexual dysfunction, bleeding risk
	No cognitive impairment	Weight gain during long-term treatment
Trazodone	Relatively easy to give effective dose	Can have troublesome withdrawal symptoms
	Lacks cardiotoxicity††, relatively safe in overdose	Daytime drowsiness
Mirtazapine	Useful sedative effect in selected patients	Weight gain common

* Lofepramine is relatively safe in overdose; ** Long-term use of anticholinergic drug may increase risk of dementia; *** Citalopram and escitalopram increase QT interval; † Venlafaxine is more dangerous than SSRIs in overdose; †† Cardiac arrhythmias have rarely been reported with trazodone.

SNRIs, and selective noradrenaline reuptake inhibitors (NARIs).

2. *5-HT₂ receptor antagonists*. These drugs (mirtazapine and trazodone) have complex effects on monoamine mechanisms but share the ability to block 5-HT₂ receptors.
3. *MAOIs*. These are compounds that deactivate monoamine oxidase irreversibly (phenelzine and tranylcypromine) or reversibly (moclobemide).

In the broad range of patients with major depression, these drugs are of equivalent efficacy. The main differences between them are in their *adverse effects and safety* (see Table 25.6). Each of these three classes of drugs is considered in turn after some comments on the possible mechanism of action of antidepressants.

Mechanism of action

The acute effect of most antidepressants is to enhance the functional activity of noradrenaline and/or 5-HT. These actions can be detected within hours of the start of treatment, yet the full antidepressant effects of drug treatment can be delayed for several weeks. For example, it has been suggested that at least four weeks should elapse before an assessment of the effects of an antidepressant drug can be made in an individual patient.

To some extent, this delay in the onset of therapeutic activity may be due to pharmacokinetic factors. For example, the half-life of most antidepressant drugs is around 24 hours, which means that steady-state plasma drug levels will be reached only after five to seven days. However, it seems unlikely that this can completely account for the lag in antidepressant activity.

The delay in onset of obvious therapeutic effect with antidepressant medication led to suggestions that the antidepressant effect of current treatments is a consequence of slowly evolving neuroadaptive changes in the brain, which are triggered by acute potentiation of monoamine function. Studies in experimental animals have implicated various mechanisms that might underlie this effect, including desensitization of inhibitory autoreceptors on 5-HT and noradrenaline cell bodies, increased production of neurotrophins, such as brain-derived neurotrophic factor (BDNF), and increased synaptogenesis and neurogenesis.

Other research has focused on the effects of antidepressant drugs on the neuropsychological mechanisms involved in the processing of emotional information. Emotional processing is known to be negatively biased in depressed patients, and it is therefore of great interest that *single doses* of antidepressant drugs produce positive biases in emotional processing in healthy volunteers and

reverse the negative biases present in depressed patients, in the absence of any changes in subjective mood.

These findings suggest that relevant *neuropsychological* effects of antidepressants can be detected from the beginning of treatment, and the delay in the appearance of obvious therapeutic effects of antidepressant medication may stem from the time taken for changes in emotional processing to be experienced as subjective changes in mood as an individual with this new emotional ‘set’ interacts with their environment. The latter process could well involve the ‘relearning’ of emotional associations, which makes the changes in synaptic plasticity and neurogenesis that have been reported in animal studies of antidepressants of great interest. For a review, see Harmer et al. (2017).

Selective serotonin reuptake inhibitors

SSRIs are now preferred to tricyclic antidepressants in the first-line treatment of depression because they are moderately better tolerated and markedly less toxic in overdose (National Institute for Health and Clinical Excellence, 2022a).

Pharmacological properties

Six SSRIs—citalopram, escitalopram, fluoxetine, fluvoxamine, paroxetine, and sertraline—are available at present for clinical use in the United Kingdom. SSRIs are a structurally diverse group, but they all *inhibit the reuptake of 5-HT* with high potency and selectivity. None of them has an appreciable affinity for the noradrenaline uptake site, and the present data suggest that they have a low affinity for other monoamine neurotransmitter receptors.

Pharmacokinetics

In general, SSRIs are absorbed slowly and reach peak plasma levels after about four to eight hours, although citalopram and escitalopram are absorbed more quickly. The half-lives of citalopram, escitalopram, fluvoxamine,

paroxetine, and sertraline are 20–30 hours, whereas the half-life of fluoxetine is 48–72 hours. The SSRIs are primarily eliminated by hepatic metabolism. Fluoxetine is metabolized to norfluoxetine, which is also a potent 5-HT uptake blocker and has a half-life of seven to nine days. Sertraline is converted to desmethylsertraline, which has a half-life of two to three days and is five to ten times less potent than the parent compound in inhibiting the reuptake of 5-HT. The contribution of desmethylsertraline to the antidepressant effect of sertraline during treatment is unclear.

Efficacy of SSRIs in depression

SSRIs have been extensively compared with placebo and with reference tricyclic antidepressants. The SSRIs are all *superior to placebo* and are *generally as effective as tricyclics* in the treatment of major depression. Most comparative studies have been of moderately depressed outpatients, and there has been concern that SSRIs may be less effective than conventional tricyclic antidepressants for more severely depressed patients, particularly inpatients.

Unwanted effects of SSRIs

Side effects can be grouped as follows (see Table 25.7):

- *Gastrointestinal effects.* Nausea occurs in about 20% of patients, although it often resolves with continued administration. Other side effects include dyspepsia, bloating, flatulence, and diarrhoea. Over longer-term treatment weight gain can occur, most significantly with paroxetine.
- *Neuropsychiatric effects.* These include insomnia, daytime somnolence, agitation, tremor, restlessness, irritability, and headache. SSRIs have also been associated with seizures and mania, although they are less likely than tricyclics to cause the latter effects. Extrapyramidal side effects such as parkinsonism and akathisia have been reported, but are uncommon.
- *Other effects.* Sexual dysfunction, including ejaculatory delay and anorgasmia, is common during SSRI

Table 25.7 Side effects of SSRIs

Gastrointestinal	<i>Common:</i> nausea, appetite loss, dry mouth, diarrhoea, constipation, dyspepsia <i>Uncommon:</i> vomiting, weight loss
Central nervous system	<i>Common:</i> headache, insomnia, dizziness, anxiety, fatigue, tremor, somnolence <i>Uncommon:</i> extrapyramidal reaction, seizures, mania
Other	<i>Common:</i> sweating, delayed orgasm, anorgasmia <i>Uncommon:</i> rash, bleeding, pharyngitis, dyspnoea, serum sickness, hyponatraemia, alopecia

treatment. Sweating and dry mouth are also reported. Cardiovascular side effects are rare with SSRIs, but some reduction in pulse rate may occur, and postural hypotension has been reported. SSRIs have been associated with skin rashes and, rarely, a more generalized allergic reaction with arthritis. SSRIs can cause low sodium states secondary to inappropriate antidiuretic hormone secretion, especially in the elderly. Elevation of liver enzymes can occur but is generally reversible on treatment withdrawal. *SSRIs may increase the risk of upper gastrointestinal bleeding*, particularly when combined with non-steroidal anti-inflammatory drugs or aspirin. Long-term use of SSRIs has been associated with an increased risk of osteoporotic fracture. Part of this risk probably stems from depression itself, but 5-HT mechanisms play a role in bone physiology, and SSRIs are associated with a greater risk of fracture than antidepressants with a low affinity for the 5-HT transporter.

SSRIs and suicidal behaviour

Meta-analyses of placebo- and comparator-controlled trials of SSRIs in adults have found no significant increase in *completed suicide* in SSRI-treated patients, and no difference in rates of *non-fatal suicidal behaviour* between patients taking SSRIs and those taking tricyclics.

Relative to placebo there may be a small risk that SSRIs can increase rates of *self-harm*, but in the largest meta-analysis of adult patients, involving over 700 studies, the number needed to harm with SSRIs (759) was much greater than the number needed to treat (estimated to be between 4 and 7) (Gunnell et al., 2005).

In adolescents and children the risk of self-harm with SSRIs might be greater. In a meta-analysis of 27 placebo-controlled trials in children and adolescents with a variety of diagnoses, including depression, Bridge et al. (2007) found no completed suicides but a small, significant increase in suicidal ideation and self-harm attempts with SSRIs compared with placebo (number needed to harm = 143). However, a subsequent network meta-analysis of randomized controlled trials found an increased risk of suicidality in depressed young people only with venlafaxine and not with SSRIs (Cipriani et al., 2016).

As noted above, SSRIs can cause agitation and restlessness early in treatment, and it is possible that in predisposed individuals this might trigger dangerous behaviour. An epidemiological study of depressed patients in primary care who received a first prescription for an antidepressant found no significant difference in rates of suicidal behaviour or completed suicide in patients taking SSRIs compared with those taking

tricyclic antidepressants (Jick et al., 2004). Noteworthy, however, was the fourfold increase in risk of attempted suicide seen with all antidepressants in the first nine days of treatment, relative to the risk with longer-term treatment (greater than 90 days). In the small number of completed suicides, the relative risk in the first nine days of treatment was increased almost fortyfold.

There are a number of possible reasons for this important phenomenon. For example, depressed people may visit their doctor and start treatment when they are at a particularly low ebb. Thus it is possible that suicidal feelings and behaviour are important factors in bringing people into treatment in the first place. Alternatively, the observation may reflect the traditional view that the greatest risk of suicidal behaviour occurs during the early stages of antidepressant treatment, because improvement in motor retardation precedes resolution of depressed mood. Whatever the explanation, it reinforces advice that patients should be closely monitored when starting antidepressant medication. Interestingly, a similar phenomenon of increased rates of self-harm has been reported in patients with depression in the early stages of psychological treatment (Simon and Savarino, 2007).

Interactions with other drugs

Pharmacodynamic interactions. The most serious interaction reported is where simultaneous administration of SSRIs and MAOIs has provoked a *5-HT toxicity syndrome* (the 'serotonin syndrome'), with agitation, hyperpyrexia, rigidity, myoclonus, coma, and death (for further details, see the section below on 'Monoamine oxidase inhibitors'). Other drugs that increase brain 5-HT function and that must therefore be used with caution in combination with SSRIs include lithium and tryptophan, which have been reported to be associated with mental state changes, myoclonus, and seizures.

Other medical drugs that have been implicated in the serotonin syndrome when combined with SSRIs include tramadol and linezolid. Serotonin toxicity has also been reported when SSRIs are combined with 5-HT receptor agonists such as sumatriptan.

SSRIs may potentiate the induction of *extrapyramidal movement disorders* by antipsychotic drugs, although this effect could be partly due to a pharmacokinetic interaction whereby SSRIs increase plasma levels of certain antipsychotic drugs (see below). The risk of *gastrointestinal bleeding* is increased when SSRIs are combined with aspirin or non-steroidal anti-inflammatory drugs. The risk of clinically significant bleeding is also elevated with combined administration of SSRIs and anticoagulants such as warfarin.

Table 25.8 Inhibition of P450 enzyme by SSRIs

	CYP 1A2	CYP 2D6	CYP 2C9	CYP 2C19	CYP 3A/4
Inhibitors	Fluvoxamine	Fluoxetine	Fluoxetine	Fluvoxamine	Fluvoxamine
	(+++)	(+++)	(+++)	(+++)	(++)
	Duloxetine (+)	Paroxetine	Fluvoxamine	Fluoxetine	Fluoxetine
		(+++)	(+++)	(++)	(+)
		Duloxetine (++)			
		Sertraline (+)		Venlafaxine	
		Citalopram (+)		(+)	
		Escitalopram (+)			
Some substrates (plasma level increased)	Olanzapine	Tricyclic antidepressants	Warfarin	Tricyclic antidepressants	Benzodiazepines
	Clozapine		Tolbutamide		Carbamazepine
	Haloperidol	Venlafaxine	Phenytoin	Diazepam	Quetiapine
	Tricyclic antidepressants	Haloperidol		Propranolol	Clozapine
		Thioridazine		Omeprazole	
	Theophylline	Risperidone			
		Clozapine			
		Olanzapine			

Inhibition: +, mild; ++, modest; +++, strong.

Pharmacokinetic interactions. Some SSRIs, particularly fluvoxamine, fluoxetine, and paroxetine, can produce substantial inhibition of hepatic cytochrome P450 enzymes, and can decrease the metabolism of several other drugs, thereby elevating their plasma levels (see Table 25.8). Examples where clinically important reactions have been reported include tricyclic antidepressants, antipsychotic agents (including clozapine and risperidone), anticonvulsants, and warfarin. *Citalopram*, *escitalopram*, and *sertraline* cause fewer reactions of this nature.

The clinical use of SSRIs in depression

Whether meta-analyses reveal clinically important differences in efficacy between SSRIs has been disputed. However, there are significant differences in pharmacokinetic profile, which have a bearing on the likelihood of a withdrawal syndrome and the potential for drug interactions (see Table 25.9).

Fluvoxamine appears to have *higher dropout rates* from trials, and may be somewhat less well tolerated. *Fluoxetine* has the *most activating effect*, and also has a distinctive pharmacokinetic profile in relation to its *long-acting metabolite*, which has a half-life of about one week.

On the one hand, this results in potential for troublesome *drug interactions* several weeks after fluoxetine has been stopped. For example, at least five weeks should elapse between stopping fluoxetine and starting an MAOI. On the other hand, this slow tapering of plasma concentration results in fluoxetine being the least likely of the SSRIs to cause a *withdrawal syndrome*.

Escitalopram is the active isomer of citalopram, and is marketed as being more effective than the parent compound, although whether any difference between the two drugs is of clinical significance is unclear. Both citalopram and escitalopram have been associated with prolongation of the QTc interval and their maximum allowable doses lowered accordingly. *Paroxetine* appears to have the most troublesome withdrawal syndrome of the SSRIs (Table 25.9).

When treating depressive disorder, dosing with SSRIs is straightforward because the recommended starting dose is often sufficient to produce significant therapeutic benefit. Generally there seems little to be gained in the treatment of depression for SSRI doses to exceed the minimum daily effective dose (Furukawa et al., 2019)

Patients who are starting SSRIs should be warned about the likely side effects, including nausea and

Table 25.9 Differences between SSRIs

Drug	Risk of pharmacokinetic interaction*	Discontinuation syndrome	Other
Citalopram	+	+	Increases QTc interval
Escitalopram	+	+	Increases QTc interval
Fluoxetine	+++	0	Increased risk of agitation, slower onset of action
Fluvoxamine	+++	++	Less well tolerated
Paroxetine	+++	+++	Weight gain, mildly anticholinergic
Sertraline	++	++	May have mild dopaminergic effects

* Based on inhibition of cytochrome P450 enzymes. See Table 25.8. +, present; ++, common; +++, marked.

some restlessness during sleep; a forewarned patient is more likely to continue with medication. A number of patients become more anxious and agitated early during SSRI treatment; therefore it is important to explain that such effects are sometimes experienced during treatment but do not mean that the underlying depression is worsening. If the patient persists with treatment, such anxiety and agitation usually diminish, but *short-term treatment with a benzodiazepine or Z drug* may be helpful, particularly if sleep disturbance is a problem. Small doses of *trazodone* (50–150 mg) may also help sleep, although there are occasional reports of serotonin toxicity with this combination.

Patients should be *reviewed frequently* during the first few weeks of treatment, when support and advice are helpful both to maintain morale and to ensure compliance with medication. Often the clinician can detect improvements in rapport and initiative early in treatment. It can then be useful to discuss these changes with the patient.

When patients respond to SSRIs there is good evidence that *continuing treatment for at least six months* lowers the rate of relapse. In addition, placebo-controlled studies have shown that SSRIs are effective in the prophylaxis of recurrent depressive episodes. SSRIs should not be stopped suddenly, as patients can experience *withdrawal reactions* (insomnia, nausea, agitation, and dizziness; Box 25.5) after the cessation of treatment, particularly with paroxetine. Liquid preparations of SSRIs can facilitate a slow withdrawal, as can a switch to fluoxetine. Whether withdrawal problems with SSRIs have been underemphasized is controversial (Horowitz and Taylor, 2022; Kalfas et al, 2025).

Box 25.5 Some common SSRI withdrawal symptoms

Systemic

Chills
Lethargy
Sweating
Headache

Gastrointestinal

Nausea, vomiting
Diarrhoea
Abdominal pain
Anorexia

Neurological

‘Electric shock’-like sensations (‘brain zaps’)
Dizziness
Vertigo
Tinnitus

Psychiatric

Irritability
Anxiety, agitation
Low mood
Emotional lability
Sleep disturbance

Source: Jha, M. K., Rush, A. J., Trivedi, M. H. (2018). When discontinuing SSRI antidepressants is a challenge: Management tips. *Am J Psychiatry*, 175(12), 1176–1184. Copyright (2018), American Psychiatric Association.

Tricyclic antidepressants

Though rarely used first line in the treatment of depression, tricyclic antidepressants continue to be useful agents because of their efficacy in severely ill depressed patients and those with treatment-refractory illness (Cowen and Anderson, 2015). At low doses tricyclics are still used widely for the treatment of neuropathic pain syndromes.

Pharmacology

Tricyclic antidepressants have a three-ringed structure with an attached side chain. A useful distinction is between compounds that have a terminal methyl group on the side chain (*tertiary amines*) and those that do not (*secondary amines*). In general, compared with the secondary amines, tertiary amines (e.g. amitriptyline, clomipramine, and imipramine) have a higher affinity for the 5-HT uptake site and are more potent antagonists of α_1 -adrenoceptors and muscarinic cholinergic receptors. Therefore, in clinical use, tertiary amines are more sedating and cause more anticholinergic effects than secondary amines (e.g. desipramine and nortriptyline).

Tricyclic antidepressants inhibit the reuptake of both 5-HT and noradrenaline. They also have *antagonist activities at a variety of neurotransmitter receptors*. In general, these receptor-blocking actions have been thought to cause adverse effects (see Table 25.10), although some investigators have argued that the ability of some tricyclic antidepressants to antagonize brain 5-HT₂ receptors may also mediate some of their therapeutic effects. Tricyclics have quinidine-like *membrane-stabilizing effects*, and this may explain why they impair cardiac conduction and cause high toxicity in overdose.

Pharmacokinetics

Tricyclic antidepressants are well absorbed from the gastrointestinal tract, and peak plasma levels occur two to

four hours after ingestion. Tricyclics are subject to significant first-pass metabolism in the liver and are highly protein-bound. The free fraction is widely distributed in body tissues. In general, the elimination half-life of tricyclics is such that it is unnecessary to give them more than once daily.

Tricyclics are metabolized in the liver by hydroxylation and demethylation. It is noteworthy that demethylation of tricyclics with a tertiary amine structure gives rise to significant plasma concentrations of the corresponding secondary amine, which is itself an active antidepressant. There can be substantial (ten- to forty-fold) differences in plasma tricyclic antidepressant levels between individual subjects when fixed-dose regimens are employed. Plasma measures have been suggested to be helpful for safety reasons when higher doses of tricyclics are employed, but ECG monitoring is probably more useful (Taylor et al., 2021).

Compounds available

These include amitriptyline, clomipramine, desipramine, dosulepin, doxepin, imipramine, lofepramine, nortriptyline, and trimipramine. Clomipramine and lofepramine are sufficiently distinct from amitriptyline and imipramine to warrant separate mention.

Clomipramine is the most potent of the tricyclic antidepressants in inhibiting the reuptake of 5-HT. However, its secondary amine metabolite, desmethylclomipramine, is an effective noradrenaline reuptake inhibitor. Unlike other tricyclic antidepressants, clomipramine is also useful in ameliorating the symptoms of *obsessive-compulsive disorder* (whether or not there is a coexisting major depressive disorder).

Lofepramine is a tertiary amine, which is metabolized to desipramine. However, during lofepramine treatment, desipramine levels are probably too low to contribute significantly to the therapeutic effect. Lofepramine is a fairly selective inhibitor of noradrenaline reuptake, and

Table 25.10 Some adverse effects of tricyclic antidepressants

Pharmacological action	Adverse effect
Muscarinic-receptor blockade (anticholinergic)	Dry mouth, tachycardia, blurred vision, glaucoma, constipation, urinary retention, sexual dysfunction, cognitive impairment
α_1 -Adrenoceptor blockade	Drowsiness, postural hypotension, sexual dysfunction, cognitive impairment
Histamine H ₁ receptor blockade	Drowsiness, weight gain
Membrane-stabilizing properties	Cardiac conduction defects, cardiac arrhythmias, epileptic seizures

has fewer anticholinergic and antihistaminic properties than amitriptyline. Lofepamine has been widely compared with other tricyclic antidepressants, and in general its antidepressant efficacy appears to be equivalent, though there is a lack of placebo-controlled trials.

The most important feature of lofepramine is that, unlike conventional tricyclic antidepressants, it is much less cardiotoxic in overdose. This means that it is likely to be safer than other tricyclics for patients with cardiovascular disease, although caution is still recommended.

Side effects. These are numerous and important (see Tables 25.5 and 25.9).

- *Anticholinergic effects.* These include dry mouth, disturbance of accommodation, difficulty in micturition (leading to retention), constipation (leading rarely to ileus), postural hypotension, tachycardia, and increased sweating. Retention of urine, especially in elderly men with enlarged prostates, and worsening of glaucoma are the most serious of these effects; dry mouth and accommodation difficulties are the most common.
- *Psychiatric effects.* These include tiredness and drowsiness with amitriptyline and other sedative compounds, insomnia with desipramine and lofepramine, and acute organic syndromes. Mania may be provoked in patients with bipolar disorders and generally tricyclics are not recommended for patients with bipolar illness (see Chapter 10).
- *Cardiovascular effects.* Tachycardia and postural hypotension occur commonly. The electrocardiogram frequently shows prolongation of PR and QTc intervals, depressed ST segments, and flattened T-waves. Ventricular arrhythmias and heart block develop occasionally, more often in patients with pre-existing heart disease.
- *Neurological effects.* These include fine tremor (commonly), incoordination, headache, muscle twitching, epileptic seizures in predisposed patients, and, rarely, peripheral neuropathy.
- *Other effects.* Allergic skin rashes, cholestatic jaundice, and, rarely, agranulocytosis; weight gain and sexual dysfunction are also common.
- *Withdrawal effects.* Tricyclic antidepressants should be withdrawn slowly if at all possible. Sudden cessation may be followed by nausea, anxiety, sweating, gastrointestinal symptoms, and insomnia with vivid dreaming.

Toxic effects

In overdosage, tricyclic antidepressants produce a large number of effects, some of which are extremely serious.

Therefore urgent expert treatment in a general hospital is required, but the psychiatrist should know the main signs of overdosage. These can be listed as follows. The cardiovascular effects include *ventricular fibrillation*, *conduction disturbances*, and *low blood pressure*. Heart rate may be increased or decreased depending partly on the degree of conduction disturbance. Sedation and coma lead to *respiratory depression*. The resulting hypoxia increases the likelihood of cardiac complications. Aspiration pneumonia may develop. In practice most patients need only supportive care, but cardiac monitoring is important, and arrhythmias require urgent treatment by a physician in an intensive care unit.

Interactions with other drugs

- Tricyclic antidepressants antagonize the hypotensive effects of α_2 -adrenoceptor agonists such as clonidine, but can be safely combined with thiazides and angiotensin-converting enzyme (ACE) inhibitors.
- The ability of tricyclics to block noradrenaline reuptake can lead to hypertension with systemically administered noradrenaline and adrenaline.
- Tricyclics should not be used in conjunction with antiarrhythmic drugs, particularly amiodarone. Tricyclics increase the QTc interval and should not be given with other psychotropic or general medical drugs that can produce a similar effect, for example, pimozide, erythromycin, clarithromycin, diphenhydramine, and tamoxifen. Caution is required when tricyclics are co-administered with citalopram and escitalopram.
- Plasma levels of tricyclics can be increased by numerous other drugs, including cimetidine, sodium valproate, calcium-channel blockers, and SSRIs. Tricyclics may increase the action of warfarin. Interactions of tricyclic drugs with MAOIs are considered later.

Contraindications

Contraindications include agranulocytosis, severe liver damage, glaucoma, prostatic hypertrophy, uncontrolled epilepsy, and significant cardiovascular disease. Tricyclics must be used cautiously in epileptic patients and in the elderly.

Clinical use of tricyclic antidepressants

It is probably sufficient to be familiar with one sedating compound (e.g. amitriptyline) and one less sedating drug (e.g. nortriptyline). Other tricyclics can then be reserved for special purposes. For example, lofepramine can be used for patients who present a risk of overdose, while clomipramine can be reserved for patients

in whom a depressive disorder is related to obsessive-compulsive disorder.

The prescribing of amitriptyline can be taken as an example. At the outset it is important to explain to the patient that, although side effects may be noticed early in treatment, any significant improvement in mood may be delayed for a week or more, and therefore it is important to persist. Early signs of improvement may include better sleep and a lessening of tension.

The usual practice of starting with a *low dose* of amitriptyline and building up is probably wise, because side effects are generally milder and the patient is more likely to develop tolerance to them. The starting dose will depend to some extent on the patient's age, weight, physical condition, and history of previous exposure to tricyclics; daily doses of 25–50 mg for an outpatient and 50–75 mg for an inpatient would be reasonable. The whole dose can be given at night about one to two hours before bedtime, because the sedative effects of the drug will aid sleep.

The dose of amitriptyline to be aimed for is about 125 mg daily or above. With careful monitoring and encouragement, this dose can usually be reached over a period of two to four weeks. Whether lower doses of tricyclics are effective in less severe depressive states in primary care is still debated.

In some patients, *side effects* limit the rate of dosage increase, but if there is clinical improvement it is reasonable to settle for lower doses. In general, side effects should not be greater than the patient can comfortably tolerate. For patients who show little or no improvement, it is usually advisable to continue amitriptyline for four weeks at the maximum tolerated dose before deciding that the drug is ineffective.

In the ECG it is important to note any evidence of impaired cardiac conduction, for example, *lengthening of the QTc interval* and the appearance of bundle branch block or arrhythmias. Because of the half-life of amitriptyline, each dose increase will take about a week to reach steady state. If the patient has not improved, and if they cannot tolerate an increase in dose or fail to respond to higher doses, other treatments should be considered. Some possible strategies are outlined in Chapter 9 (see page 226).

Maintenance and prophylaxis

If patients respond to amitriptyline, they should be maintained on treatment for *at least six months*, as continuation therapy greatly reduces the risk of early relapse. The same dose of amitriptyline should be maintained if possible, but if side effects become a problem the dose can be lowered until tolerance is again satisfactory.

It is often not clear when antidepressant drug treatment should be withdrawn, because in some patients, depression is a recurrent disorder. Long-term prophylactic treatment may then be justified. Obviously, the risk of recurrence increases with the number of episodes that the patient suffers, but other clinical and biochemical predictors of relapse are not well established (see Chapter 9).

Monoamine oxidase inhibitors

MAOIs were introduced just before the tricyclic antidepressants, but their use has been less widespread because of both troublesome *interactions with foods and drugs* and uncertainty about their *therapeutic efficacy*. However, in adequate doses MAOIs are useful antidepressants, often producing clinical benefit in depressed patients who have not responded to other medication or ECT. In addition, MAOIs can be useful in *refractory anxiety states* (van den Eynde et al., 2023).

These beneficial effects have to be weighed against the need to adhere to *strict dietary and drug restrictions* to avoid reactions with tyramine and other sympathomimetic agents. In practice this means that conventional MAOIs are used only in patients who failed to respond to multiple other treatments.

Pharmacological properties

MAOIs inactivate enzymes that oxidize noradrenaline, 5-HT, dopamine, and tyramine, and other amines that are widely distributed in the body as transmitters, or are taken in food and drink or as drugs. Monoamine oxidase (MAO) exists in a number of forms that differ in their substrate and inhibitor specificities. From the point of view of psychotropic drug treatment, it is important to recognize that there are two forms of MAO, type A and type B, which are encoded by separate genes. In general, MAO-A metabolizes intraneuronal noradrenaline and 5-HT, whereas both MAO-A and MAO-B metabolize dopamine and tyramine.

Compounds available

Phenelzine is the most widely used and widely studied compound. *Isocarboxazid* is reported to have fewer side effects than phenelzine, and can be useful for patients who respond to the latter drug but suffer from its side effects of hypotension or sleep disorder. *Tranylcypromine* differs from the other compounds in combining the ability to inhibit MAO with an *amphetamine-like stimulating effect*, which may be helpful in patients with anergia and retardation. However, some patients have become dependent on the stimulant effect of tranylcypromine.

Moreover, compared with phenelzine, tranylcypromine is more likely to give rise to hypertensive crises, although it is less likely to damage the liver. For these reasons, tranylcypromine should be prescribed with particular caution.

Moclobemide differs from the other compounds in *selectively binding* to MAO-A, which it inhibits in a *reversible* way. This results in a lack of significant interactions with foodstuffs, and a quick offset of action (see below).

Pharmacokinetics

Phenelzine, isocarboxazid, and tranylcypromine are rapidly absorbed and widely distributed. They have short half-lives (about two to four hours), as they are quickly metabolized in the liver by acetylation, oxidation, and deamination. People differ in their capacity to acetylate drugs. For example, in the United Kingdom, approximately 60% of the population are ‘fast acetylators’, who would be expected to metabolize hydrazine MAOIs more quickly than ‘slow acetylators’. Some studies have shown a better clinical response to phenelzine in ‘slow acetylators’, but this finding has not been consistently replicated. However, it may underlie the observation that the best response rate with MAOIs occurs in studies that have used *higher dose ranges*, presumably because even patients who metabolize MAOIs quickly will receive an adequate dose.

Phenelzine, isocarboxazid, and tranylcypromine bind irreversibly to MAO-A and MAO-B by means of a covalent linkage. This means that the enzyme is permanently deactivated and MAO activity can be restored only when new enzyme is synthesized. Thus, despite their short half-lives, irreversible MAOIs cause a *long-lasting inhibition of MAO*.

In contrast to these compounds, moclobemide binds reversibly to MAO-A. This compound has a short half-life (about two hours), and therefore its inhibition of MAO-A is brief, declining to some extent even during the latter periods of a three times daily dosing regimen. Full MAO activity is restored within 24 hours of stopping moclobemide, whereas with the irreversible MAOIs, a period of two weeks or more may be needed for synthesis of new MAO.

Efficacy of MAOIs in depression

For many years MAOIs were in relative disuse because several studies, in particular a large controlled trial by the Medical Research Council (Clinical Psychiatry Committee, 1965), found phenelzine to be no more effective than placebo in the treatment of depressive disorders. It seems likely that the doses of MAOIs were too low in these early investigations; in the Medical

Box 25.6 Adverse effects of MAOIs

Central nervous system

Insomnia, drowsiness, agitation, headache, fatigue, weakness, tremor, mania, confusion

Autonomic nervous system

Blurred vision, difficulty in micturition, sweating, dry mouth, postural hypotension, constipation

Other

Sexual dysfunction, weight gain, peripheral neuropathy (pyridoxine deficiency), oedema, rashes, hepatocellular toxicity (rare), leukopenia (rare)

Research Council study the maximum dose of phenelzine was 45 mg daily, in contrast to the current practice of using doses of up to 90 mg daily if side effects permit. Subsequent studies have shown that in this wider dose range MAOIs are *superior to placebo* and are *generally equivalent to tricyclic antidepressants* in their therapeutic activity (Cleare et al., 2015).

Unwanted effects

These include dry mouth, difficulty in micturition, postural hypotension, confusion, mania, headache, dizziness, tremor, paraesthesia of the hands and feet, constipation, and oedema of the ankles. Hydrazine compounds can give rise to hepatocellular jaundice (see Box 25.6).

Interactions with foodstuffs

Some foods contain *tyramine*, a substance that is normally inactivated by MAO in the liver and the gut wall. When MAO is inhibited, tyramine is not broken down and is free to exert its *hypertensive effects*. These effects are due to release of noradrenaline from sympathetic nerve terminals with a consequent elevation in blood pressure. This may reach dangerous levels and may occasionally result in *subarachnoid haemorrhage*. Important early symptoms of such a crisis include a severe and usually throbbing headache.

The incidence of hypertensive reactions is about 10% in patients who are taking MAOIs, even in those who have received dietary counselling. Therefore, regular reminders about dietary restrictions may be helpful, particularly in patients on longer-term treatment (see Box 25.7). There have been reports of a wide range of foods

Box 25.7 Foods to be avoided during MAOI use

- All cheeses except cream, cottage, and ricotta cheeses
- Red wine, sherry, beer, and liquors
- Pickled or smoked fish
- Brewer's yeast products (e.g. Marmite, Bovril, and some packet soups)
- Broad bean pods (e.g. Italian green beans)
- Beef or chicken liver
- Fermented sausage (e.g. bologna, pepperoni, salami)
- Unfresh, overripe, or aged food (e.g. pheasant, venison, unfresh dairy products)

being implicated in hypertensive reactions with MAOIs, but many of these have cited single cases and therefore are of uncertain validity. Another complication is that the tyramine content of a particular food item may vary, as may the susceptibility of an individual patient to a hypertensive reaction. If a forbidden food has been consumed on one occasion without adverse effects, this does not preclude a future reaction.

It is notable that about 80% of all reported reactions between foodstuffs and MAOIs, and nearly all of the deaths, have followed the *consumption of cheese*. Hypertensive reactions will often settle without specific treatment. However, if hypotensive treatment is needed, parenteral administration of an α_1 adrenoceptor antagonist, such as *phentolamine* is the most appropriate (see van den Eynde et al., 2023). The use of oral *nifedipine* has also been advocated. Whatever treatment is given, blood pressure must be monitored carefully.

Moclobemide and tyramine reactions

Tyramine is metabolized by both MAO-A and MAO-B. Experimental studies have shown that the hypertensive effect of oral tyramine is potentiated much less by moclobemide than by non-selective MAOIs. In patients who are taking moclobemide in doses of up to 900 mg daily, the dose of tyramine required to produce a significant pressor response is above 100 mg. Even a five-course meal with red wine would be unlikely to result in a tyramine intake of more than 40 mg.

Tyramine has relatively little effect in patients who are receiving moclobemide because MAO-B (present in the gut wall and the liver) is still available to metabolize much of the tyramine ingested. Another factor may be

that the interaction between moclobemide and MAO-A is reversible, thus allowing displacement of moclobemide from MAO when tyramine is present in excess.

Interactions with drugs

Patients who are taking MAOIs must not be given drugs whose metabolism depends on enzymes that are affected by the MAOI. These drugs include *sympathomimetic amines* such as *adrenaline*, *noradrenaline*, and *amphetamine*, as well as *phenylpropanolamine* and *ephedrine* (which may be present in proprietary cold cures). *L-Dopa* and *dopamine* may also cause hypertensive reactions. Local anaesthetics often contain a sympathomimetic amine, which should also be avoided. Some *opiates* (particularly *pethidine*, *tramadol*, and *fentanyl*), as well as *cocaine* and *insulin* can be involved in dangerous interactions. Sensitivity to *oral antidiabetic* drugs is increased, with a consequent risk of hypoglycaemia. The ability of MAOIs to cause postural hypotension can increase the hypotensive effects of other agents. Finally, the metabolism of carbamazepine, phenytoin, and other drugs that are broken down in the liver may be slowed.

The serotonin syndrome. A number of drugs that potentiate brain 5-HT function can produce a severe *neurotoxicity syndrome* when combined with MAOIs. In some cases this can be fatal. The main features of this syndrome are listed in Box 25.8. It is worth noting that some of these symptoms resemble the neuroleptic malignant syndrome with which 5-HT neurotoxicity is occasionally confused. In view of the interactions between dopamine and 5-HT pathways, it is possible that similar mechanisms may be involved.

Current clinical data indicate that combination of MAOIs with SSRIs, *venlafaxine*, *duloxetine*, *vortioxetine*,

Box 25.8 Clinical features of the serotonin syndrome

Neurological

Myoclonus, nystagmus, headache, tremor, rigidity, seizures

Mental state

Irritability, confusion, agitation, hypomania, coma

Autonomic

Hyperthermia, sweating, shivering, tachycardia, hypertension, diarrhoea, cardiac arrhythmias

imipramine, and *clomipramine* is contraindicated. The combination of MAOIs with *L-tryptophan* has also been reported to cause 5-HT toxicity. Adverse reactions have been reported between the 5-HT_{1A} receptor agonist, *buspirone*, and MAOIs. In addition, the use of 5-HT_{1D} receptor agonists, such as *sumatriptan*, should be avoided. If used with caution, the combination of *lithium* with MAOIs appears to be safe, and it can be effective in patients with resistant depression.

If a serotonin syndrome develops, all medication should be stopped and supportive measures instituted. In theory, drugs with 5-HT-receptor-antagonist properties such as *cypheptadine* or *propranolol* may be helpful, but controlled studies have not been carried out. For a review of the serotonin syndrome, see Ellahi (2015)

Combination of MAOIs with tricyclic antidepressants. The combined use of MAOIs and tricyclic antidepressants fell into disuse because of the severe reactions associated with the 5-HT syndrome. Current views are that combination therapy is safe provided that the following rules are observed:

- Clomipramine and imipramine are not used. The most favoured tricyclics in combination with MAOIs are amitriptyline and trimipramine.
- The MAOI and tricyclic are started together at low dosage, or the MAOI is added to the tricyclic (adding tricyclics to MAOIs is more likely to provoke dizziness and postural hypotension).

The advantages and disadvantages of combined tricyclic and MAOI therapy have not been fully established. On the one hand, patients who are taking tricyclics with MAOIs are less likely to suffer from *MAOI-induced insomnia*, but on the other hand, they are more likely to experience *postural hypotension* and troublesome *weight gain*. The combination is said to be useful in patients with *resistant depression*. Low doses of *trazodone* (50–150 mg) are also used to ameliorate MAOI-induced insomnia; present experience suggests that this combination is generally well tolerated, although there are occasional reports of adverse effects that could represent serotonin toxicity.

Contraindications

These include liver disease, phaeochromocytoma, congestive cardiac failure, and conditions that require the patient to take any of the drugs that react with MAOIs.

Clinical use of MAOIs in depression

The availability and cost of MAOIs varies widely between countries. In the United Kingdom, *phenelzine* is

generally available and its use can be taken as an example. Treatment should start with 15 mg daily, increasing to 30 mg daily in divided doses (with the final dose taken not later than 3.00 pm) in the first week. Patients should be given *clear written instructions about foods to be avoided* (see below), and should be warned to take no other medication unless it has been specifically checked with a pharmacist or doctor who knows that the patient is taking MAOIs. As always, patients should be warned about the delay in therapeutic response (up to six weeks) and about common side effects (sleep disturbance and dizziness).

In the second week, the dose of *phenelzine* can be increased to 45 mg daily. At this stage a greater increase to 60 mg may produce a quicker response, but is also associated with more adverse effects. Accordingly, if feasible, it is better to find out whether an individual patient will respond to lower doses (about 45 mg) before increments are made (up to 90 mg daily). If the patient does not respond to 45 mg, the dose can be increased by 15 mg weekly if side effects permit.

The response to MAOIs can often be sudden; over the course of a day or two the patient suddenly feels better. If there are signs of *overactivity* or *excessive buoyancy* in mood, the dose can be reduced and the patient monitored for signs of developing hypomania. Side effects that are likely to be particularly troublesome are insomnia and postural hypotension. Insomnia is best managed by lowering the dose of MAOI, if feasible. Otherwise, the addition of a benzodiazepine or *trazodone* (50–150 mg at night) can be helpful, although the latter drug can sometimes increase problems of dizziness and postural hypotension.

Postural hypotension can be a disabling problem with MAOIs. Again, dose reduction is worth considering. Various measures have been suggested—for example, the use of support stockings, an increase in salt intake, or even the use of a mineralocorticoid. Of course, the latter two measures have their own adverse effects.

Withdrawal from MAOIs

Patients who respond to MAOIs have often suffered from disabling depression for many months or even years. For such patients the usual practice is to continue therapy for at least a year and probably longer. With MAOIs, it is wise to lower the dose if the patient can tolerate the reduction without relapsing. Sudden cessation of MAOIs can lead to *anxiety* and *dysphoria*. Even gradual withdrawal can be associated with increasing anxiety and depression.

Clinical experience indicates that it is more difficult to stop MAOI than tricyclic antidepressant treatment.

An explanation for this difference may be that MAOIs produce a more severe discontinuation syndrome than other antidepressants; another possible explanation is that MAOIs are given to patients with chronic disabling disorders who frequently relapse. It is emphasized that, because of the time taken to synthesize new MAO, at least two weeks should elapse between the cessation of irreversible MAOI treatment and the easing of dietary and drug restrictions.

Moclobemide

In their freedom from tyramine reactions and their quick offset of activity, reversible type A MAOIs, such as *moclobemide*, have clear advantages over conventional MAOIs. However, the therapeutic efficacy of moclobemide, particularly in more severely depressed patients, is not as well established. Also, it is not clear that moclobemide is effective for patients with the various forms of atypical depression and drug-resistant depression for which conventional MAOIs can be useful (Cleare et al., 2015).

The starting dose of moclobemide is 150–300 mg daily, which can be increased to 600 mg over a number of weeks. Treatment-resistant patients may require higher doses, but above levels of 900 mg daily it is prudent to institute the usual MAOI dietary restrictions. Moclobemide is better tolerated than tricyclic antidepressants or irreversible MAOIs, but side effects such as nausea, insomnia, and agitation occur in about 20–30% of patients.

Drug interactions of moclobemide

Moclobemide should not be combined with SSRIs, *venlafaxine*, *vortioxetine*, or *clomipramine* because a serotonin syndrome may result. Caution is needed with *sumatriptan*. Like the irreversible MAOIs, moclobemide may react adversely with some *opiates*. Similarly, moclobemide may potentiate the pressor effects of *sympathomimetic amines*; therefore combined use should be avoided. Moclobemide should not be combined with *L-dopa* because of the risk of hypertensive crisis.

Other antidepressant drugs

Other antidepressant drugs are available for use in the United Kingdom and other countries. Their mechanism of action is such that they cannot easily be grouped with SSRIs, tricyclic antidepressants, or MAOIs. These drugs also have differing adverse-event profiles. Therefore they are discussed individually below.

Agomelatine

Agomelatine is a more recently licensed antidepressant which is a melatonin receptor agonist and a somewhat weaker antagonist at 5-HT_{2C} receptors. The mechanism of antidepressant action of agomelatine is not established, but could be mediated through a melatonin-like action on circadian rhythms. It is also possible that 5-HT_{2C}-receptor blockade might lead to increased dopamine release in the prefrontal cortex, though whether agomelatine blocks 5-HT_{2C} receptors at clinically used doses is unclear.

Pharmacokinetics

Agomelatine is rapidly absorbed, reaching maximum levels within one to two hours of ingestion. However, it has a high first-pass metabolism, with a bioavailability of only 5–10%. It has a short half-life of about two hours and no metabolites likely to contribute to its therapeutic action.

Efficacy

Controlled trials indicate that agomelatine is superior to placebo and comparable in efficacy to SSRIs. Agomelatine may also have efficacy in adolescents with depression (Arango et al., 2022). In clinical trials, agomelatine was dosed once-daily at 9.00 pm, which is sufficiently in advance of the endogenous night-time melatonin peak to produce effects on circadian rhythm. Therefore it seems sensible when prescribing agomelatine to use the same treatment schedule.

Unwanted effects

Generally, agomelatine is one of the best tolerated antidepressants and may be useful in the management of depressed patients who suffer significant adverse effects with more conventional agents (Cipriani et al., 2018). The most common adverse effects of agomelatine are *nausea* and *dizziness*. Agomelatine is not sedating, but some patients experience *somnolence*, and *insomnia* has also been reported. Other possible side effects include *anxiety* and *fatigue*, *diarrhoea*, and *constipation*. *Sexual dysfunction* is less frequent than with SSRIs. The most serious potential adverse effect of agomelatine is an increase in liver enzymes (ALT and AST), with a rate of 1.4% and 2.5% in agomelatine-treated patients (25 mg and 50 mg, respectively) compared with 0.6% in patients taking placebo. For this reason, treatment with agomelatine should be preceded by measurement of liver function tests, which should be repeated after approximately at 3, 6, 12, and 24 weeks. Any clinical suspicion of impaired hepatic function should be followed by urgent liver

function tests, and treatment with agomelatine should be stopped if the results are abnormal.

Drug interactions

The main interaction of agomelatine is with drugs that inhibit the hepatic microsomal enzymes CYP1A2 and CYP2C9/19. This is because these enzymes metabolize agomelatine, and higher blood levels of agomelatine are likely to increase the risk of hepatic dysfunction. Therefore agomelatine should not be given with potent CYP1A2 inhibitors such as fluvoxamine and ciprofloxacin. Co-administration of agomelatine with more moderate inhibitors (oestrogens, propranolol, and grepafloxacin) should be employed with caution.

Mirtazapine

Mirtazapine is a quadricyclic compound with complex pharmacological actions. It is a fairly potent antagonist at several 5-HT-receptor subtypes, particularly 5-HT₂ and 5-HT₃ receptors. Mirtazapine is also a competitive antagonist at histamine H₁ receptors and α_1 - and α_2 -adrenoceptors. The latter action leads to an increase in noradrenaline cell firing and release. Mirtazapine is not a muscarinic cholinergic antagonist and is not cardiotoxic. Because of these various actions, mirtazapine has a *sedating profile*, but it is not anticholinergic and is relatively *safe in overdose*.

Pharmacokinetics

Mirtazapine is well absorbed, with peak plasma levels being reached between one and two hours. The half-life is about 16 hours, and the daily dose can be given at night. Mirtazapine is extensively metabolized by the liver, and has only minor inhibitory effects on cytochrome P450 isoenzymes.

Efficacy

Mirtazapine has demonstrated clinical efficacy in both *placebo-controlled and comparator trials* with SSRIs and tricyclic antidepressants in moderate to severely depressed patients. The effective dose is usually between 10 mg and 45 mg daily with the greatest efficacy in randomized trials being seen with 30 mg (Furukawa et al., 2019).

Unwanted effects

The common adverse effects of mirtazapine are attributable to its potent antihistaminic actions, and include *drowsiness* and *dry mouth*. *Increased appetite* and *body weight* are also common. Thus far leukopenia does not appear to be more common with mirtazapine than with other antidepressants. However, the data sheet

recommends that physicians be vigilant for possible signs that might reflect low white cell count.

Drug interactions

Mirtazapine may potentiate other centrally acting sedatives. There is a theoretical risk that mirtazapine could reverse the therapeutic effect of α_2 -adrenoceptor agonists such as clonidine.

Trazodone

Trazodone is a triazolopyridine derivative with complex actions on 5-HT pathways. Studies *in vitro* suggest that trazodone has some weak 5-HT-reuptake-inhibiting properties, which are probably not manifested during clinical use; for example, repeated administration of trazodone does not lower platelet 5-HT content.

Trazodone has antagonist actions at 5-HT₂ receptors, but its active metabolite, *m*-chlorophenylpiperazine (*m*-CPP), is a 5-HT receptor agonist. Therefore the precise balance of effects on 5-HT receptors during trazodone treatment is difficult to determine, and may depend on relative blood levels of the parent compound and metabolite. Trazodone also blocks postsynaptic α_1 -adrenoceptors. Overall it has a distinct *sedating profile*.

Pharmacokinetics

Trazodone has a short half-life (about 4–14 hours). It is metabolized by hydroxylation and oxidation, with the formation of a number of metabolites, including *m*-CPP. During treatment, plasma levels of *m*-CPP may exceed those of trazodone itself.

Efficacy

Several controlled studies have shown that trazodone in doses of 150–600 mg is *superior to placebo* in the treatment of depressed patients. Trazodone also appears to have equivalent antidepressant activity to reference compounds such as imipramine. Many of these studies were carried out in moderately depressed outpatients, and the efficacy of trazodone relative to other antidepressants in more severely ill patients is not well established.

Some have maintained that the efficacy of trazodone is improved if treatment is started at low doses (50 mg) and increased slowly to 300 mg over two to three weeks. Despite the short half-life of trazodone, once-daily administration of the drug is often sufficient. The drug is usually given in the evening to take advantage of its sedative properties. Doses above 300 mg daily are usually better given in divided amounts. Lower doses (50–150 mg) are sometimes used in combination with SSRIs and MAOIs to ameliorate the sleep-disrupting effects of the latter agents.

Unwanted effects

The major unwanted effect of trazodone is *excessive sedation*, which can result in significant cognitive impairment. *Nausea* and *dizziness* are also reported, particularly if the drug is taken on an empty stomach. The α -adrenoceptor-antagonist properties of trazodone may *lower blood pressure* to some extent, and postural hypotension has been reported. Trazodone is less cardiotoxic than conventional tricyclics, but there are reports that *cardiac arrhythmias* may be worsened in patients with cardiac disease. Nevertheless, trazodone is less toxic in overdose than tricyclic antidepressants.

The most serious side effect of trazodone is *priapism*. This reaction is seen rarely (about 1 in 6,000 male patients). It can cause considerable problems, requiring the local injection of noradrenaline agonists such as adrenaline, or even surgical decompression. Long-term sexual dysfunction has sometimes resulted. It is recommended that male patients be warned of this potential side effect and advised to seek medical help urgently if persistent erection occurs.

Drug interactions

As with all sedative antidepressants, trazodone may potentiate the sedating effects of alcohol and other central tranquilizing drugs.

Venlafaxine

Venlafaxine is a phenylethylamine derivative which produces a *potent blockade of 5-HT reuptake*, with lesser effects on noradrenaline reuptake, which probably only become apparent at higher doses. In this respect the pharmacological properties of venlafaxine resemble those of clomipramine to some extent. However, unlike clomipramine and other tricyclic antidepressants, venlafaxine has a negligible affinity for other neurotransmitter receptor sites, and so lacks sedative and anticholinergic effects. Because of these pharmacological properties, venlafaxine has been classified as a *selective serotonin and noradrenaline reuptake inhibitor (SNRI)*.

Pharmacokinetics

Venlafaxine is well absorbed, achieving peak plasma levels about 1.5–2 hours after oral administration. The half-life of venlafaxine is three to seven hours, but it is metabolized to desmethylvenlafaxine, which has essentially the same pharmacodynamic properties as the parent compound and a half-life of 8–13 hours. The extended-release formulation of venlafaxine (venlafaxine XL) reaches a peak plasma level after about six hours. This gives a longer apparent half-life (about 15 hours),

but the drug is still quickly eliminated. Once-daily dosing is possible with this preparation.

Efficacy

Venlafaxine has been studied in both inpatients and outpatients with major depression and compared with placebo and active comparators. Current studies suggest that it is more effective than placebo and of at least equal efficacy to other available antidepressant drugs, including tricyclic antidepressants. Some meta-analyses suggest that venlafaxine is more effective than SSRIs, particularly for more severely depressed patients, but the data are inconsistent (Cleare et al., 2015).

Venlafaxine has a wider dosage range than SSRIs, from 75 to 375 mg daily in two divided doses, or up to 225 mg of the extended-release preparation given as a single dose. The usual starting dose of venlafaxine is 75 mg daily, which may be sufficient for many patients. Upwards titration can be considered in cases where there is insufficient response.

Unwanted effects

The adverse-effect profile of venlafaxine resembles that of SSRIs, with the most common adverse effects being *nausea*, *headache*, *somnolence*, *dry mouth*, *dizziness*, and *insomnia*. *Anxiety* and *sexual dysfunction* may also occur. Venlafaxine occasionally causes *postural hypotension* but, in addition, dose-related *increases in blood pressure* can occur. Blood pressure monitoring is advisable in patients who are receiving more than 150 mg venlafaxine daily. Overdoses of venlafaxine have been associated with cardiac arrhythmias, so it is probably prudent to avoid using venlafaxine in patients with significant cardiac disease. Like SSRIs, venlafaxine can lower plasma sodium levels. Venlafaxine may be somewhat less well tolerated than SSRIs (Cleare et al., 2015).

Venlafaxine appears to be *more toxic than SSRIs* in acute overdose with a toxicity intermediate between that of conventional tricyclic antidepressants and SSRIs. Similarly to some SSRIs, sudden discontinuation of venlafaxine has been associated with troublesome symptoms of *fatigue*, *nausea*, *abdominal pain*, and *dizziness*. It is recommended that patients who have received venlafaxine for six weeks or more should have the dose reduced gradually over a period two to four weeks, and longer if possible.

Drug interactions

Unlike the SSRIs, venlafaxine appears to produce relatively modest effects on hepatic drug-metabolizing enzymes, and therefore should be less likely to inhibit the metabolism of co-administered drugs. Venlafaxine

is metabolized partly by CYP3A4, and caution is recommended where venlafaxine is co-prescribed with CYP3A4 inhibitors such as ketoconazole and clarithromycin. Like other drugs that potentially inhibit the uptake of 5-HT, venlafaxine should not be given concomitantly with MAOIs because of the danger of a toxic serotonin syndrome. It is also recommended that 14 days should elapse after the end of MAOI treatment before venlafaxine is started, and that at least seven days should elapse after venlafaxine cessation before MAOIs are given. As with SSRIs, venlafaxine should be used with caution in combination with other 5-HT promoting drugs (for example, lithium, triptans, and tramadol).

Duloxetine

Like venlafaxine, duloxetine is also classified as an SNRI. It is about five times more potent in inhibiting the reuptake of 5-HT than in inhibiting that of noradrenaline. It has little effect on other neurotransmitter receptors.

Pharmacokinetics

Duloxetine is well absorbed, with maximum blood levels occurring about six hours post ingestion. It has a half-life of about 12 hours. It is extensively metabolized to therapeutically inactive compounds.

Efficacy

Duloxetine given in a single dose of 60 mg daily has greater antidepressant efficacy than placebo, and is equivalent in therapeutic activity to SSRIs. In a network meta-analysis of randomized trials, the efficacy of duloxetine was very similar to that of venlafaxine (Cipriani et al., 2018).

Unwanted effects

The adverse-effect profile of duloxetine is similar to that seen with other 5-HT-promoting antidepressants, and includes *nausea, dry mouth, dizziness, gastrointestinal disturbances, insomnia, somnolence, and sexual dysfunction*. Like venlafaxine, duloxetine can increase blood pressure; there are few data on toxicity in overdose. As would be expected from its pharmacology and half-life, abrupt cessation of duloxetine is associated with troublesome dizziness, insomnia, anxiety, and headache.

Drug interactions

Duloxetine produces a moderate inhibition of CYP2D6 and, to a lesser extent, of CYP1A2. It is therefore likely to increase blood levels of other drugs that are metabolized by these enzymes (see Table 25.8). On the basis of its pharmacology, duloxetine should not be given concomitantly with MAOIs because of the danger of a

toxic serotonin syndrome. For the same reason it seems advisable that 14 days should elapse after the end of MAOI treatment before duloxetine is started, and that at least seven days should elapse after duloxetine cessation before MAOIs are given. Combination with drugs such as lithium, triptans, and tramadol should be approached with caution.

Vortioxetine

Vortioxetine is a more recently licensed antidepressant which blocks the reuptake of 5-HT and also antagonizes a variety of 5-HT receptors, including the 5-HT_{1D}, 5-HT₃, and 5-HT₇ receptors, while acting as an agonist at 5-HT_{1A} receptors. Hence, vortioxetine produces complex effects on 5-HT neurotransmission but probably overall leads to an increase in brain 5-HT function, particularly at post-synaptic 5-HT_{1A} receptors. At higher doses (20 mg daily) vortioxetine produces a similar occupancy of brain 5-HT transporter sites as standard therapeutic doses of SSRIs.

Pharmacokinetics

Vortioxetine is slowly absorbed, with peak plasma concentrations being reached at 7–11 hours. Vortioxetine is metabolized principally through CYP2D6 to an inactive metabolite. The half-life is about 65 hours. Poor CYP2D6 metabolizers (about 5–7% of the Caucasian population) have about a twofold increase in plasma vortioxetine levels.

Efficacy

Short-term studies show superior efficacy of vortioxetine relative to placebo at doses of 10–20 mg, with the 20mg dose showing the highest efficacy (Iovieno et al., 2021). In one randomized study of patients who had failed to respond to first-line SSRI treatment, a switch to vortioxetine was superior to agomelatine (Montgomery et al., 2014). A relapse prevention study over one year showed a benefit of vortioxetine relative to placebo. It is claimed that vortioxetine has a particularly helpful effect on depression-associated cognitive impairment, but systematic studies in comparison to SSRIs have yet to be carried out. It is not yet established how far any cognitive improvement produced by vortioxetine improves 'real-world' functioning (Nierenberg et al., 2019).

Unwanted effects

The main adverse effects of vortioxetine are *nausea, vomiting, diarrhoea, and constipation*. *Dizziness, abnormal dreaming, and pruritis* are also reportedly common. At lower doses (10–15 mg) the incidence of sexual dysfunction appears less than that of other 5-HT reuptake blockers, but at 20 mg the advantage for vortioxetine is lost.

Drug interactions

Data are currently rather limited, but similar precautions as those taken with SSRIs are recommended. Accordingly, concomitant treatment with vortioxetine and MAOIs or within two weeks of stopping a conventional MAOI is contraindicated. As with other 5-HT reuptake blockers, caution is needed when vortioxetine is given to patients at increased risk of bleeding through, for example, anticoagulant treatment or therapy with non-steroidal anti-inflammatory agents. Similarly, caution should be exercised when co-administering vortioxetine with other 5-HT potentiating agents such as tryptophan, lithium, triptans, and tramadol. Vortioxetine levels are likely to be increased by inhibitors of CYP2D6, such as ritonavir, bupropion, and quinidine.

Reboxetine

Reboxetine is a morpholine and is structurally related to fluoxetine. It is an *NARI*, with no clinically significant effects on other neurotransmitter receptors.

Pharmacokinetics

After oral administration, reboxetine reaches peak plasma levels after about two hours. Its half-life is around 13 hours, and twice-daily administration is recommended. Reboxetine is metabolized by the liver, where it is a substrate for cytochrome P450 CYP3A.

Efficacy

Reboxetine reportedly shows efficacy in both *placebo-controlled trials* and against active comparators such as SSRIs. However, different meta-analyses have come to differing conclusions, and overall in moderately depressed outpatients reboxetine may be less effective than some other classes of antidepressants, including SSRIs (Cipriani et al., 2018). It is claimed that reboxetine produces better improvement in social function in depressed patients than SSRIs, but this possibility has not been confirmed. The usual dose of reboxetine is 4 mg twice daily, with a maximum dose of 12 mg daily.

Unwanted effects

Despite its low affinity for muscarinic receptors, reboxetine produces adverse effects characteristic of cholinergic-receptor blockade, presumably through interactions of noradrenergic and cholinergic pathways. The most common side effects are *dry mouth, constipation, sweating, and insomnia*. *Urinary hesitancy, impotence, tachycardia, and vertigo* are also occasionally described.

Drug interactions

Limited information is available. It is recommended that reboxetine should not be given with other agents that

might *potentiate noradrenaline function*, such as MAOIs, or that might increase blood pressure, such as ergot derivatives. Plasma reboxetine levels might be increased by drugs that inhibit cytochrome P450 3A4, such as some antifungal agents, fluvoxamine, and macrolide antibiotics.

St John's wort

St John's wort is an extract from the plant *Hypericum perforatum*. It has been used in medicine for centuries for numerous indications, including burns, arthritis, snakebite, and depression. The active principles are probably derived from six major product groups, including *hypericins* and *hyperforins*. The pharmacology of St John's wort is complex, but animal experimental and some human studies indicate that it potentiates aspects of monoamine neurotransmission.

Efficacy

There have been numerous trials of St John's wort, although these are difficult to interpret because the preparations and dosages have been difficult to standardize. In addition, the trials have been conducted in mild to moderately depressed subjects. More recent meta-analyses suggest that standardized extracts of hypericum (in doses of 600–1800 mg) are more effective than placebo, and about equal in efficacy to other antidepressants, in patients with clearly diagnosed major depression (Apaydin et al., 2016). However, there is a lack of longer-term efficacy data, and the available hypericum preparations are not standardized.

Adverse effects and drug interactions

St John's wort is well tolerated, with the most common side effects being *gastrointestinal disturbance, dizziness, and tiredness*. Cases of mania during treatment have been described. Photosensitivity is also rarely reported. Hypericum extracts may *induce hepatic enzymes*, and there are reports that treatment with St John's wort was associated with lowered levels of theophylline, cyclosporin, digoxin, and ethinyloestradiol. Finally, St John's wort may cause *serotonin neurotoxicity* when combined with SSRIs and other 5-HT-potentiating drugs.

Bupropion

Bupropion has significant use as an antidepressant in the United States. In the United Kingdom it is licensed as an adjunct to smoking withdrawal but not as an antidepressant. It is structurally and pharmacologically distinct from other antidepressants, being a unicyclic aminoketone derivative. Bupropion modestly enhances both dopamine and noradrenaline function in the brain, probably via reuptake blockade.

Efficacy and adverse effects

Numerous controlled trials have shown that the antidepressant effect of bupropion is superior to that of placebo and equivalent to that of SSRIs. In some respects the adverse-effect profile of bupropion is similar to that of the SSRIs, with insomnia, agitation, tremor, and nausea being most frequently reported. However, mania and psychosis can also occur, as can raised blood pressure. Bupropion is less likely than SSRIs to cause sexual dysfunction and weight gain, which gives it an important advantage in some patients.

The main concern with the use of bupropion has been the increased risk of seizures. In its original formulation the risk of seizures at higher doses (0.4%) was about four times greater than that associated with SSRIs (about 0.1%). The risk appears to be less with the slow-release formulation (bupropion SR), which has been marketed for smoking cessation. At doses of 300 mg or less, the risk of seizures with bupropion SR appears to be about the same as that seen with SSRIs. This is the maximum dose recommended for smoking cessation, and is the standard dose used when treating depression. Bupropion should not be used in patients with a history of seizures or eating disorder.

Drug interactions

Bupropion should not be given with other drugs that might lower the seizure threshold, such as tricyclic antidepressants, antipsychotic drugs, and antimalarial drugs. Administration with MAOIs is also contraindicated. Bupropion has been combined safely with lithium, and in the United States is used by specialists to augment ineffective SSRI treatment. Bupropion inhibits *CYP2D6*, and drugs metabolized by this pathway (including some other antidepressants and antipsychotics) should be prescribed only with caution.

Ketamine and esketamine

Ketamine is an antagonist at the glutamate N-methyl-D-aspartate (NMDA) receptor and is licensed in the United Kingdom for the induction and maintenance of general anaesthesia. Ketamine is a racemic mixture of *S* and *R* chiral forms. Esketamine is composed of the pure *S* isomer and is available as an intranasal spray for the treatment of depressed patients not helped sufficiently by conventional medications.

Like many treatments in psychopharmacology, the discovery of ketamine occurred somewhat serendipitously when Berman and colleagues (2000) found that administration of a sub-anaesthetic dose of intravenous ketamine produced a clinically important reduction of symptoms in a small group of depressed patients.

Strikingly, this antidepressant effect outlasted the duration of ketamine administration by several days.

The mechanism of action of ketamine to relieve depression is not clearly understood. Most current theories focus on the ability of ketamine to block NMDA receptors located on GABA inhibitory neurons. The resulting disinhibitory effect leads to an increased release of presynaptic glutamate which, through activation of glutamatergic α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptors, produces prolonged increases in synaptic plasticity and neurogenesis. However, the antidepressant effect of ketamine in depressed patients can be prevented by pre-treatment with the opiate antagonist, naltrexone. This suggests that opiate receptor activation may also be involved in its antidepressant action (Williams et al., 2018).

Pharmacokinetics and drug interactions

The peak blood level of ketamine is found at the end of the intravenous infusion and about 30 minutes following intranasal esketamine. The bioavailability of intranasal esketamine is a little under 50% (a proportion of the intranasal spray tends to be inadvertently swallowed). Ketamine and esketamine are extensively metabolized in the liver via *CYP2B6* and *CYP3A4*. A variety of metabolites are produced, some of which, for example, hydroxynorketamine, may themselves have antidepressant activity. The half-life of intravenous ketamine is two to three hours, while that of esketamine is between 7–12 hours.

Ketamine and esketamine are usually administered in combination with antidepressant medication (for esketamine, this is a condition of its license). Thus far, interactions between ketamine and esketamine with ongoing antidepressant treatment have not appeared problematic. It is uncertain whether concomitant benzodiazepines might diminish the antidepressant effect of ketamine treatment and similar comments apply to the use of lamotrigine, which has been reported in some studies to block the dissociative effects of ketamine. It is wise not to administer ketamine with agents that have pressor effects, for example, psychostimulants.

Efficacy

Several placebo-controlled trials have demonstrated the acute efficacy of intravenous ketamine and intranasal esketamine in patients with treatment-resistant depression, though in more highly treatment-resistant patients, response rates are lower. In the latter group, and particularly in older patients, ECT appears more

effective (Ekstrand et al., 2022; McIntyre et al., 2021). The usual dose of intravenous ketamine is 0.5mg/kg infused over 40 minutes, though doses of up to 1.0mg/kg have been given and appear safe. Esketamine dosing varies between 28–84mg. It is difficult to ‘blind’ participants to ketamine and esketamine administration and it is uncertain how far expectancy effects might contribute to the clinical antidepressant response.

The duration of the antidepressant effect following a single intravenous ketamine varies but is usually between three to seven days. While some depressed patients respond clinically to the first or second ketamine infusion, in others the response can be delayed and usually patients receive a course of four to six infusions, with infusions being administered once or twice weekly. After the cessation of ketamine infusions most patients who have responded relapse within the next month, indicating the importance of effective maintenance treatment. Esketamine has been used in the acute treatment of resistant depression with twice weekly administration initially followed by maintenance treatment, at a frequency of once weekly to once fortnightly. This regime has demonstrated effective relapse prevention over several months. Oral ketamine has also been employed as a maintenance treatment after intravenous infusions but is subject to extensive first pass metabolism, which

greatly diminishes its bioavailability. This makes appropriate dosing difficult to determine.

Adverse effects

Acutely, ketamine and esketamine cause dissociative symptoms, perceptual disturbances, and anxiety. Dizziness and drowsiness are also seen frequently. Ketamine and esketamine produce cardiac stimulatory effects associated with tachycardia and increases in blood pressure. A pretreatment ECG is recommended, and blood pressure should be monitored during ketamine and esketamine administration.

Over longer-term treatment there has been concern about the possibility of urinary symptoms (nocturia, dysuria, haematuria, urgency, and incontinence) which occur in about 30% of people who regularly use ketamine recreationally. However, thus far this has not been reported in people receiving ketamine and esketamine for the treatment of depression. There is also the recognized abuse potential of ketamine. While this has not been an issue in clinical trials, it underlines the importance of careful patient selection and documentation of treatment. For this, as well as the safety reasons outlined above, ketamine and esketamine need to be administered in appropriate, well-regulated, clinical settings. For a review see, McIntyre et al. (2021).

Mood-stabilizing drugs

Several agents are grouped under this heading, such as *lithium* and a number of anticonvulsant drugs, including *carbamazepine* and *sodium valproate*. These three drugs are effective in the *prevention of recurrent affective illness* and also in the *acute treatment of mania*. In addition, lithium has useful antidepressant effects in some circumstances, but the antidepressant activity of carbamazepine and sodium valproate is less well established. Another anticonvulsant, *lamotrigine*, shows promise in the acute treatment and prevention of bipolar depression, but does not appear to be effective against manic states.

Lithium

Placebo-controlled trials have shown that lithium is effective in a number of conditions, including the following:

- the acute treatment of mania
- the prophylaxis of unipolar and bipolar mood disorder

- augmentation therapy in resistant depression
- the prevention of aggressive behaviour in patients with learning disabilities.

Mechanism of action

Animal studies have shown that lithium has important effects on the intracellular signalling molecules or ‘second messengers’ that are activated when a neurotransmitter or agonist binds to a specific receptor. At clinically relevant doses, lithium inhibits the formation of *cyclic adenosine monophosphate* (cAMP) and also attenuates the formation of various *inositol lipid-derived mediators*. Through these actions lithium could exert profound effects on a wide range of neurotransmitter pathways, many of which use the above messenger systems. More recent interest has focused on the ability of lithium to promote cell survival and increase synaptic plasticity, perhaps through inhibition of the activity of the enzyme glycogen synthase kinase-3 (GSK-3).

Pharmacokinetics

Lithium is rapidly absorbed from the gut and diffuses quickly throughout the body fluids and cells. It is removed from the plasma by renal excretion and by entering cells and other body compartments. Therefore, there is rapid excretion of lithium from the plasma, and a slower phase reflecting its removal from the whole-body pool.

Like sodium, lithium is filtered and partly reabsorbed in the kidney. When the proximal tubule absorbs more water, lithium absorption increases. Therefore *dehydration* causes the plasma lithium concentration to rise. Because lithium is transported in competition with sodium, more is reabsorbed by the kidney when sodium concentrations fall. This is the mechanism whereby *thiazide diuretics* can lead to toxic concentrations of lithium in the blood.

Dosage and plasma concentrations

Because the therapeutic and toxic doses are close together, it is essential to measure plasma concentrations of lithium during treatment. Measurements should first be made after about seven days, then about every two weeks, and then, provided that a satisfactory steady state has been achieved, once every six weeks. Subsequently, lithium levels are often very stable, and serum measures can be carried out at intervals of approximately three to six months unless there are clinical indications for more frequent monitoring.

After an oral dose, serum lithium levels rise by a factor of 2 or 3 within about four hours. For this reason, concentrations are normally measured approximately 12 hours after the last dose, usually that given at night. It is important to follow this routine, as published information about lithium levels refers to the concentration 12 hours after the last dose, and not to the 'peak' reached in the four hours after that dose. If an unexpectedly high concentration is found, it is important to establish whether the patient has inadvertently taken a morning dose before the blood sample was taken.

Previously, the accepted range for mood prophylaxis was 0.7–1.2 mmol/l measured 12 hours after the last dose. However, current trends are to maintain lithium at lower serum levels because this decreases the burden of side effects. Nolen et al. (2019) concluded that, in the prophylaxis of bipolar disorder, the minimum efficacious serum level of lithium was 0.4 mmol/l, but in most patients the best therapeutic response was obtained with levels in the range 0.6–0.8 mmol/l. Higher levels (up to 1.0 mmol/l, where tolerance permitted) benefited some patients with more persistent manic symptomatology.

In the treatment of acute mania, serum concentrations below 0.8 mmol/l appear to be ineffective, and a range of 0.8–1.0 mmol/l is probably required. Serious toxic effects appear with lithium concentrations above 2.0 mmol/l, although early symptoms may appear at concentrations above 1.2 mmol/l.

A number of delayed-release preparations of lithium are now available, but their pharmacokinetics *in vivo* do not differ significantly from those of standard lithium carbonate preparations. Liquid formulations of lithium citrate are available for patients who have difficulty in taking tablets. Lithium may be administered once or twice daily. Frequency of administration does not appear to affect urine volume. In general, it is more convenient to take lithium as a single dose at night, but patients who experience gastric irritation on this regimen may be helped by divided daily doses.

Unwanted effects

A mild *diuresis* due to sodium excretion occurs soon after the drug is started. Other common effects include *tremor of the hands*, *dry mouth*, a *metallic taste*, *feelings of muscular weakness*, and *fatigue* (Table 25.11).

Some degree of *mild thirst* and *polyuria* is common in patients taking lithium, probably because lithium blocks the effect of antidiuretic hormone on the renal tubule. This may not be of clinical significance, but up to one-third of patients can show progression to a *diabetes insipidus-like syndrome* with pronounced polyuria and polydipsia. This may necessitate withdrawal of lithium treatment, although the use of lower serum lithium levels may cause the syndrome to remit.

Some patients, especially women, *gain weight* when taking the drug. Persistent fine tremor, mainly affecting the hands, is common, but coarse tremor suggests that the serum concentration of lithium has reached toxic levels. Most patients adapt to the fine tremor; for those who do not, propranolol up to 40 mg three times daily may reduce the symptom. Both *hair loss* and *coarsening of hair texture* can occur.

Thyroid gland enlargement occurs in about 5% of patients who are taking lithium. The thyroid shrinks again if thyroxine is given while lithium is continued, and it generally returns to normal a month or two after lithium has been stopped. Lithium interferes with thyroid production, and *hypothyroidism* occurs in up to 20% of women patients, with a compensatory rise in thyroid-stimulating hormone.

Tests of thyroid function should be performed every six months to help to detect these changes, but these intermittent tests are no substitute for a continuous watch for suggestive clinical signs, particularly *lethargy*.

Table 25.11 Some adverse effects of lithium, carbamazepine, and valproate

	Lithium	Carbamazepine	Valproate
Neurological	Tremor, weakness, dysarthria, ataxia, impaired memory, seizures (rare)	Dizziness, weakness, drowsiness, ataxia, headache, visual disturbance	Tremor, sedation
Renal/fluid balance	Increased urine output with decreased urine-concentrating ability. Thirst, diabetes insipidus, oedema	Acts to increase urine-concentrating ability, low sodium states, oedema	Increased plasma ammonia
Gastrointestinal	Altered taste, anorexia, nausea, vomiting, diarrhoea, weight gain	Anorexia, nausea, constipation, hepatitis	Anorexia, nausea, vomiting, diarrhoea, weight gain, hepatitis (rare), pancreatitis (rare)
Endocrine	Decreased thyroxine with increased TSH. Goitre, hyperparathyroidism (rare)	Decreased thyroxine with normal TSH	Menstrual disturbances
Haematological	Leukocytosis	Leukopenia, agranulocytosis (rare)	Low platelet count, abnormal platelet aggregation
Dermatological	Acne, exacerbation of psoriasis	Erythematous rash	Hair loss
Cardiovascular	ECG changes (usually clinically benign)	Cardiac conduction disturbances	

and substantial *weight gain*. If hypothyroidism develops and the reasons for lithium treatment are still strong, thyroxine treatment should be added. Lithium has also been associated with elevated serum calcium levels in the context of *hyperparathyroidism*. This is occasionally associated with severe depression, making distinction from the underlying mood disorder difficult. Blood calcium levels should be monitored every six months or more frequently if indicated.

Reversible ECG changes also occur. These may be due to displacement of potassium in the myocardium by lithium, as they resemble those of hypokalaemia, with T-wave flattening and inversion or widening of the QRS. They are not usually of clinical significance. However, sinus node block and QTc prolongation are occasionally seen during lithium treatment, particularly with higher lithium levels (Mehta and Vannozzi, 2017). Other changes include a reversible *leukocytosis* and occasional papular or maculopapular *rashes*.

Effects on *memory* are sometimes reported by patients, who complain in particular of everyday lapses of memory, such as forgetting well-known names. It is possible that this impairment of memory is caused by the mood disorder rather than by the drug itself, but there is also

evidence that lithium can be associated with impaired performance on certain cognitive tests.

Long-term effects on the kidney. As noted above, lithium treatment decreases *tubular concentrating ability* and can occasionally cause a nephrogenic diabetes insipidus. In addition, there have been reports that over many years of treatment, lithium can sometimes cause an increasing and in some cases *irreversible decline* in tubular function. This may be more likely in patients with higher serum concentrations of lithium, and where concomitant psychotropic medication has been employed.

Several follow-up studies have examined the effect of longer-term lithium maintenance treatment on *glomerular function*. In general, it has been thought that any decline in glomerular function is usually mild and related to lithium intoxication. However, more recent epidemiological studies suggest that lithium treatment is associated with an approximately twofold increased risk of chronic kidney disease, with lithium-treated patients showing a steeper age-related decline in glomerular filtration rate than patients not receiving such treatment. Not surprisingly the risk increases with duration of lithium treatment (Fransson et al., 2022).

Box 25.9 Lithium, renal function, and estimated glomerular filtration rate

The *estimated glomerular filtration rate* (e-GFR) in a healthy young adult is about 100 ml/min, and it falls by about 1 ml/min per year as people get older, so many healthy 75-year-olds will have an e-GFR of 50–60 ml/min.

Chronic kidney disease (CKD)

CKD stage 1: e-GFR >90 ml/min, normal kidney function, but urine findings or structural abnormalities or genetic trait indicate kidney disease

CKD stage 2: e-GFR 60–90 ml/min, mildly reduced kidney function, and other findings (as for stage 1) indicate kidney disease

CKD stage 3: e-GFR 30–59 ml/min, a moderate reduction in kidney function

CKD stage 4: e-GFR 15–29 ml/min, a severe reduction in kidney function

CKD stage 5: e-GFR <15 ml/min, established kidney failure, when dialysis or a kidney transplant may be needed.

Stage 3 CKD is asymptomatic, but is associated with a greater subsequent risk of cardiovascular disease. People with stage 3 CKD require regular cardiovascular monitoring and tests of renal function, including urinalysis every three months.

Indications for referral to a specialist renal physician in a patient taking lithium

Referral is required if any of the following are present:

- e-GFR is decreasing by >4 ml/min annually
- progressive rise in blood creatinine concentration in three or more serial tests
- proteinuria
- haematuria
- symptoms suggestive of chronic renal failure (e.g. tiredness, anaemia)
- e-GFR <30 ml/min.

Source: Morriss, R., & Benjamin, B. (2008). Lithium and eGFR: A new routinely available tool for the prevention of chronic kidney disease. *BJPsych*, 193(2), 93–95. Copyright (2008), Cambridge University Press.

With the current trends towards long-term prophylaxis of mood disorders, it is clearly wise to monitor plasma creatinine levels regularly and to supplement this with estimated glomerular filtration rate (e-GFR; see Box 25.9). It seems likely that the risk of nephrotoxicity will be minimized by maintaining plasma lithium levels at the lower end of the therapeutic range, provided that they are therapeutically effective for the individual patient. Also, careful attention to medical comorbidity in bipolar patients is important.

Toxic effects

These are related to dose. They include *ataxia, poor coordination of limb movements, muscle twitching, slurred speech, and confusion*. They constitute a serious medical emergency, as they can progress through *coma* and *fits* to *death*.

If these symptoms appear, lithium must be stopped at once and a high intake of fluid provided, with extra sodium chloride to stimulate an osmotic diuresis. In severe cases, renal dialysis may be needed. Lithium is rapidly cleared if renal function is normal, so that most cases either recover completely or die. However, cases of

permanent neurological and renal damage despite haemodialysis have been reported.

As noted earlier, lithium can increase *foetal abnormalities*, particularly of the heart, although the magnitude of the individual risk is low. The decision as to whether or not to continue with lithium treatment during pregnancy must therefore be carefully considered. Important factors include the likelihood of affective relapse if lithium is withheld, and the difficulty that could be experienced in managing an episode of affective illness in the particular individual.

If pregnant patients continue with lithium, serum levels should be monitored closely. Ultrasound examination and foetal echocardiography are valuable screening tests as the pregnancy progresses. Patients with a history of *bipolar disorder* have a substantially *increased risk* of relapse in the postpartum period. In such patients it may be worth considering the introduction of lithium shortly after delivery to provide a prophylactic effect. However, significant concentrations of lithium can be measured in the serum of breastfed infants, which may make bottle feeding advisable.

Box 25.10 Some drug interactions of lithium

Pharmacokinetic

Increased lithium levels

- Diuretics (furosemide is safest)
- Non-steroidal anti-inflammatory drugs (aspirin/sulindac is safest)
- ACE inhibitors
- Angiotensin-II-receptor antagonists
- Antibiotics (metronidazole)

Decreased lithium levels

- Theophylline
- Sodium bicarbonate

Pharmacodynamic

- *5-HT neurotoxicity*
SSRIs (can be used safely with care)
- *Extrapyramidal side effects enhanced*
Antipsychotic agents, metoclopramide, domperidone
- *Enhanced neurotoxicity*
Carbamazepine, phenytoin, calcium-channel blockers, methylidopa

Drug interactions

Because of the narrow therapeutic index of lithium, *pharmacokinetic* drug interactions are of major clinical importance (Box 25.10). *Pharmacodynamic* interactions may involve potentiation of *5-HT-promoting agents*, leading to a serotonin syndrome. In addition, therapeutic serum levels of lithium can be associated with *neurotoxicity* in the presence of certain other centrally acting agents, for example, calcium channel blockers and carbamazepine. It is recommended that lithium be used with caution with drugs that can prolong the QTc interval and with agents liable to cause hypokalaemia since the latter can also increase the risk of cardiac arrhythmias.

ECT and surgery. It is possible that the continuation of lithium during ECT may lead to neurotoxicity. If feasible, lithium treatment should be suspended or serum levels reduced during ECT, because the customary overnight fast beforehand may leave patients relatively dehydrated the following morning. If possible, lithium treatment should be discontinued before *major surgery*, because the effects of *muscle relaxants* may be potentiated. However,

the risk of acute withdrawal and 'rebound' mania must be considered.

Lithium withdrawal

In some studies, abrupt *lithium withdrawal* has been associated with the rapid onset of mania. Undoubtedly there is an increased risk of recurrent mood disorder after lithium discontinuation, probably because lithium is an effective prophylactic agent and because it is used for disorders that have a high risk of recurrence. However, there is probably also a lithium withdrawal syndrome with 'rebound' mania, although this may be restricted to patients with bipolar disorder.

The risk of rapid relapse is reduced if lithium is *discontinued slowly* over a period of several weeks. Even patients who have remained entirely well for many years may experience a further episode of affective disorder after lithium discontinuation. Most of these individuals will respond to the reintroduction of lithium

Contraindications

These include renal failure or recent renal disease, current cardiac failure or recent myocardial infarction, and chronic diarrhoea sufficient to alter electrolytes. Lithium should not be prescribed if the patient is judged to be unlikely to observe the precautions required for its safe use. This includes a propensity to discontinue it suddenly against advice.

The management of patients on lithium

Preparation. A careful routine of management is essential because of the effects of therapeutic doses of lithium on the thyroid and kidney, and the toxic effects of excessive dosage. The following routine is one of several that have been proposed, and can be adopted safely. Successful treatment requires attention to detail, so the steps are described below at some length.

Before starting lithium, a *physical examination* should be performed, including the measurement of blood pressure. It is also useful to weigh the patient and calculate the BMI.

Blood should be taken for estimation of *electrolytes*, *calcium*, *creatinine*, *e-GFR*, and a *full blood count*. *Thyroid function tests* are also necessary and a baseline ECG should be carried out if there is any history of cardiac disease or significant risk factors. A pregnancy test is advisable in women of childbearing age.

If these tests show no contraindication to lithium treatment, the clinician should check that the patient is not taking any drugs that might interact with lithium. A careful explanation should then be given to the

patient. They should understand the possible early toxic effects of an unduly high blood concentration, and also the circumstances in which this can arise, for example, during intercurrent gastroenteritis, renal infection, or the dehydration secondary to fever. They should be advised that if any of these arise, they should stop the drug and seek medical advice.

It is usually appropriate to include another member of the family in these discussions. Providing *printed guidelines* on these points is often helpful (either written by the doctor, or in one of the forms provided by pharmaceutical firms). In these discussions a sensible balance must be struck between alarming the patient by overemphasizing the risks, and failing to give them the information that they need to take a collaborative part in the treatment.

Starting treatment. Lithium should normally be prescribed as the carbonate, and treatment should begin and continue with a *single daily dose* unless there is gastric intolerance, in which case divided doses can be given. If the drug is being used for prophylaxis, it is usual to begin with 200–400 mg daily in a single dose. The lower end of the range is appropriate when patients are taking concomitant medication such as SSRIs that might interact with lithium.

Blood should be taken for lithium estimations every week or two, adjusting the dose until an appropriate concentration is achieved. A lithium level of 0.4–0.8 mmol/l (in a sample taken 12 hours after the last dose) may be adequate for prophylaxis, as explained above. If this is not effective, the previously accepted higher range of 0.8–1.0 mmol/l can be tried if side effects permit and the predominant symptomatology is manic. When judging the response, it should be remembered that several months may elapse before lithium achieves its full effect.

Continuation treatment. As treatment continues, lithium estimations should be carried out about *every three to six months*. It is important to have some means of reminding patients and doctors about the times at which repeat investigations are required. Computerized databases may be helpful in this respect. *Every six months*, blood samples should be taken for electrolytes, urea, creatinine, e-GFR, calcium, and thyroid function tests. If two consecutive thyroid function tests one month apart show evidence of *hypothyroidism*, lithium should be stopped or L-thyroxine prescribed. Troublesome *polyuria* is a reason for attempting a reduction in dose, whereas severe persistent polyuria is an indication for specialist renal investigation, including tests of concentrating ability. A persistent *leukocytosis* is not uncommon and is apparently harmless. It reverses soon after the drug is stopped.

When lithium is given, the clinician must keep in mind the interactions that have been reported with psychotropic and other drugs (see above). It is also prudent to be extra vigilant for toxic effects if ECT is being given. If the patient requires an anaesthetic for any reason, the anaesthetist should be told that they are taking lithium because, as noted above, there is some evidence that the effects of muscle relaxants may be potentiated.

Lithium is usually continued for at least a year, and often for much longer. The need for the drug should be reviewed regularly, considering any persistence of mild mood fluctuations, which suggest the possibility of relapse if treatment is stopped. Continuing medication is more likely to be needed if the patient has previously had several episodes of mood disorder within a short time, or if previous episodes were so severe that even a small risk of recurrence should be avoided.

Some patients have taken lithium continuously for 15 years or more, but there should always be compelling reasons for continuing treatment for more than five years. As noted above, lithium should be withdrawn slowly, over several months if possible. The patient should be advised not to discontinue lithium suddenly on their own initiative.

Sodium valproate

Sodium valproate was first introduced as an anticonvulsant but has also been used as a mood stabilizer and in the management of mania.

There have been several controlled studies indicating that valproate is effective in the *acute management of mania*. It may also have efficacy in the acute treatment of bipolar depression (Bahji et al., 2020). The evidence that valproate is effective in *longer-term prophylaxis* of bipolar disorder is less strong than for that of lithium. In a randomized trial, Bowden et al. (2000) showed a marginal benefit for valproate over lithium and placebo in bipolar patients over a 1-year follow-up. However, in the large pragmatic BALANCE trial, valproate was less effective than lithium (BALANCE Study Group, 2010). A network meta-analysis of randomized controlled trials reported beneficial effects of valproate in the prevention of mania but not depression (Kishi et al., 2021). There have been numerous case studies and open series that have reported useful prophylactic effects of valproate in patients who were *unresponsive to lithium*, including those with rapid-cycling mood disorders.

Mode of action. Valproate is a simple branch-chain fatty acid with a mode of action that is unclear. However, there is some evidence that it can slow down the breakdown of the *inhibitory neurotransmitter GABA* through

inhibition of GABA transaminase. It may also block voltage gated sodium channels. These actions could account for the anticonvulsant properties of valproate, but whether they also underlie the psychotropic effects is unclear.

Pharmacokinetics

Valproate is rapidly absorbed, with the peak plasma concentrations occurring about two hours after ingestion. It is widely and rapidly distributed and has a half-life of 8–18 hours. Valproate is metabolized in the liver to produce a wide variety of metabolites, some of which have anticonvulsant activity. Unlike carbamazepine, valproate does not induce hepatic microsomal enzymes and, if anything, it tends to delay the metabolism of other drugs.

Dosing and plasma concentrations

Valproate can be started at a dose of 200–400 mg daily, which may be increased once- or twice-weekly to a range of 1–2 g daily. Plasma levels of valproate do not correlate well with either the anticonvulsant or the mood-stabilizing effects, but it has been suggested that efficacy in the treatment of acute mania is usually apparent when plasma levels are *greater than 50 µg/ml*.

Unwanted effects

Common side effects of valproate include *gastrointestinal disturbances, tremor, sedation, and tiredness*. Other troublesome side effects include *weight gain and transient hair loss*, with changes in hair texture on regrowth.

Patients who are taking valproate may show some elevation in *hepatic transaminase enzymes*. Provided that this increase is not associated with hepatic dysfunction, the drug can be continued while enzyme levels and liver function are carefully monitored. However, there have been several reports of fatal *hepatic toxicity* associated with the use of valproate; most of these cases have occurred in children taking multiple anticonvulsant drugs. Valproate must be withdrawn immediately if vomiting, anorexia, jaundice, or sudden drowsiness occur.

Valproate may also cause *thrombocytopenia*, and may inhibit platelet aggregation. *Acute pancreatitis* is another rare but serious side effect, and *increases in plasma ammonia levels* have also been reported. Other possible side effects include *oedema, amenorrhoea, and rashes*. Valproate has been associated with *polycystic ovary disease*, and is an important teratogen in both men and women. As discussed earlier, the National Institute for Health and Clinical Excellence (2022c) has recommended that valproate should not be initiated for the treatment of bipolar disorder in women and men under the age of 55 unless two specialists independently

document that there is no other effective or tolerated treatment. In these cases, a pregnancy prevention programme must be in place.

Drug interactions

Valproate potentiates the effects of central sedatives. It has been reported to increase the side effects of other anticonvulsants (without necessarily improving anticonvulsant control). It may increase plasma levels of lamotrigine, phenytoin, nimodipine, and tricyclic antidepressants. Co-administration of valproate and quetiapine may increase the risk of neutropenia. The interaction of valproate with lamotrigine necessitates careful adjustment of lamotrigine dosage (see below). Valproate may decrease *olanzapine* concentrations.

Clinical use

The efficacy of valproate in the acute and continuation treatment of mania has been established by several controlled trials (Goodwin et al., 2016), and it is licensed for this indication in the United Kingdom in the form of the semi-sodium preparation. In the treatment of acute mania it has a quicker onset of action than lithium and carbamazepine because it can be dosed at high levels initially. For example, a therapeutic effect can be apparent within one or two days of employing a loading dose of valproate of 20 mg/kg. Valproate may also be more effective than lithium in the management of patients with *mixed affective states*.

In terms of prophylaxis, valproate appears to be more effective in the prevention of manic episodes. It has often been used in combination with lithium in bipolar patients who have shown a partial response to lithium, and this combination appears to be safe. For patients who continue to show episodes of mood disturbance, valproate can also be combined with atypical antipsychotic drugs. Guidelines recommend that body weight, blood count, and liver function tests are assessed prior to treatment and thereafter at six-monthly intervals.

Lamotrigine

Lamotrigine is a triazine derivative that blocks voltage-dependent *sodium channels* and reduces excitatory neurotransmitter release, particularly that of *glutamate*. It is licensed in the United Kingdom as a monotherapy and adjunctive treatment for epilepsy and also for the prevention of depressive episodes in patients with bipolar 1 disorder.

Lamotrigine is not licensed for the treatment of acute mood disorders in the United Kingdom but placebo-controlled trials have shown that lamotrigine has efficacy

as monotherapy in the *acute treatment of bipolar depression* (Geddes et al., 2009). Lamotrigine was also effective in patients with bipolar depression when added to quetiapine (Geddes et al., 2016). Lamotrigine does not seem to have significant acute or prophylactic antimanic actions. A network meta-analysis confirmed the efficacy of lamotrigine in preventing depressive but not manic episodes in patients with bipolar disorder (Kishi et al., 2021). Lamotrigine may be effective as an augmentation treatment in unipolar depression but the evidence is less strong (Goh et al., 2019).

Pharmacokinetics

Following oral administration, lamotrigine is rapidly absorbed, with peak plasma levels occurring after about 1.5 hours. The drug is extensively metabolized by the liver but does not induce cytochrome P450 enzymes. Its half-life is about 30 hours. A plasma therapeutic range has not been identified.

Adverse effects

Skin eruptions, usually maculopapular in nature, occur in about 3% of patients and may be associated with fever. They are most common in the first few weeks of treatment, and their incidence can be reduced by careful initial dosing (see below). Other side effects include *nausea, headache, diplopia, blurred vision, dizziness, ataxia, and tremor*. Rarely, very serious adverse effects, such as *angioedema, Stevens–Johnson syndrome, and toxic epidermal necrolysis*, have been reported.

Drug interactions

Plasma levels of lamotrigine can be lowered by drugs that induce hepatic-metabolizing enzymes, such as *carbamazepine*. Combination of carbamazepine and lamotrigine can also cause neurotoxicity. Lamotrigine levels are increased by concomitant administration of *valproate*.

Clinical use

Lamotrigine has a useful role to play in the acute and prophylactic management of the depressive pole of bipolar disorder. Because lamotrigine appears to be ineffective in the treatment or prevention of mania, it is often used in combination with other drugs such as lithium. However, in bipolar patients where the burden of illness is almost exclusively depressive, it can be used as a monotherapy (Goodwin et al., 2016).

When initiating lamotrigine treatment, to minimize the risk of rash it is important to follow the dosage recommendations in the *BNF* (25 mg daily for the first two weeks, followed by 50 mg daily for the next two weeks). The usual therapeutic dose in bipolar disorder is in the

range 50–300 mg daily. If the patient is also taking valproate, dosage initiation must be even more cautious (for guidance, see the *BNF*).

Carbamazepine

Carbamazepine was originally introduced as an *anticonvulsant*, and was found to have useful effects on mood in certain patients. Subsequently it was found to be beneficial in some bipolar patients, including those who had proved *refractory to lithium*. There is reasonable evidence that carbamazepine is effective in the management of *acute mania* and also in the *prophylaxis of bipolar disorder*, although overall its efficacy is probably less than that of lithium (Goodwin et al., 2016).

Mode of action. Like certain other anticonvulsants, carbamazepine blocks *neuronal sodium channels*. It is unclear whether this action plays a role in the mood-stabilizing effects. In both humans and animals, carbamazepine facilitates 5-HT neurotransmission, an action that it shares with lithium.

Pharmacokinetics

Carbamazepine is slowly but completely absorbed and widely distributed. It is extensively metabolized, and at least one metabolite, carbamazepine epoxide, is therapeutically active. The half-life during long-term treatment is about 20 hours. Carbamazepine is a strong *inducer of hepatic microsomal enzymes*, and can lower the plasma concentrations of numerous other drugs.

Dosage and plasma concentrations

The dosage of carbamazepine in the treatment of mood disorders is similar to that used in the treatment of epilepsy, within the range 400–1600 mg daily. Treatment is usually given in divided doses twice-daily, because this practice may improve tolerance. No clear relationship has been established between plasma carbamazepine concentrations and therapeutic response, but it seems prudent to monitor levels (about 12 hours after the last dose) and to maintain them in the usual anticonvulsant range as a precaution against toxicity.

Unwanted effects

Side effects are common at the beginning of treatment. They include *drowsiness, dizziness, ataxia, diplopia, and nausea*. Tolerance to these effects usually develops quickly. A potentially serious side effect of carbamazepine is *agranulocytosis*, although this complication is very rare (estimated to be about 1 in 20,000 patients).

A *relative leukopenia* is more common, with the white cell count often falling during the first few weeks of

treatment, although it usually remains within normal levels. Rashes occur in about 5% of patients, and, rarely, severe exfoliative dermatitis develops. Elevations in liver enzymes may also occur, and, rarely, *hepatitis* has been reported. Carbamazepine can cause disturbances of *cardiac conduction*, and is therefore contraindicated in patients with pre-existing abnormalities of cardiac conduction. Carbamazepine is an established *human teratogen* and should not be used in women of child-bearing age.

Carbamazepine lowers plasma *thyroxine concentrations*, but thyroid-stimulating hormone levels are not elevated and clinical hypothyroidism is unusual. Carbamazepine has also been associated with *low sodium states*. The unwanted effects of carbamazepine are compared with those of lithium and valproate in Table 25.11.

Drug interactions

Carbamazepine increases the metabolism of many other drugs, including tricyclic antidepressants, benzodiazepines, antipsychotic drugs, oral contraceptive agents, thyroxine, warfarin, other anticonvulsants, and some antibiotics. A similar mechanism may underlie the decline in plasma carbamazepine levels that occurs after the first few weeks of treatment.

Carbamazepine levels may be increased by SSRIs and erythromycin. The pharmacodynamic effects and plasma levels of carbamazepine may be increased by some calcium-channel blockers, such as diltiazem and verapamil. Conversely, carbamazepine may decrease the effect of certain other calcium-channel antagonists, such as felodipine and nifedipine. Neurotoxicity has been reported when carbamazepine and lithium have been combined even in the presence of normal lithium levels. The manufacturers of carbamazepine recommend that combination of carbamazepine with MAOIs should be avoided. However, there are case reports of these drugs being used safely together. It is possible that some MAOIs may increase plasma carbamazepine levels.

Clinical use of carbamazepine

Current clinical guidelines for bipolar disorder (see, for example, Goodwin et al., 2016) place carbamazepine

rather low down the list of therapeutic options, after drugs such as lithium, valproate, and lamotrigine. To some extent this reflects the liability of carbamazepine to cause significant drug interactions, as well as its perceived poorer tolerability. Also, there is less controlled evidence concerning its use. With this in mind, the usual indications for carbamazepine are as follows:

- The prophylactic management of bipolar illness in patients for whom lithium and valproate treatment is ineffective or poorly tolerated.
- The treatment of patients with frequent mood swings and mixed affective states for which carbamazepine may be more effective than lithium.
- Added to lithium treatment in patients who have shown a partial response to the latter drug; in these circumstances it is important to remember that this combination can cause neurotoxicity.
- In the acute treatment of mania, again usually as an alternative or addition to lithium and valproate.

If clinical circumstances permit, it is preferable to start treatment with carbamazepine slowly at a dose of 100–200 mg daily, increasing in steps of 100–200 mg twice weekly. Patients show wide variability in the blood levels at which they experience adverse effects; accordingly, it is best to titrate the dose against the side effects and the clinical response.

Because of the risk of a *lowered white cell count*, it is prudent to monitor the blood count before treatment and after three and six months of treatment. Some guidelines also recommend routine monitoring liver function tests and plasma electrolytes; however, clinical vigilance is probably the best safeguard. Patients should be instructed to seek help urgently if they develop a fever or other sign of infection. When patients have responded to the addition of carbamazepine to lithium, it is possible subsequently to attempt a cautious withdrawal of lithium. However, the current clinical impression is that, for many patients, the maintenance of mood stability requires continuing treatment with both drugs.

Psychostimulants

This class of drugs includes mild stimulants, of which the best known is *caffeine*, and more powerful stimulants, such as *amphetamine* and *methylphenidate*. *Cocaine* is a powerful psychostimulant, with a particularly high

potential for inducing dependence (see Chapter 20). It is useful as a local anaesthetic, but has no other clinical indications. All potent psychostimulants *increase the release* and *block the reuptake* of dopamine and

noradrenaline. A newer compound, *modafinil*, produces increases in alertness and decreases sleepiness, and, although its mechanism of action is uncertain, it may also involve dopaminergic mechanisms (Sousa and Dinis-Oliveira, 2020).

Indications

The main current indication for psychostimulant treatment with methylphenidate and amphetamine is attention-deficit hyperactivity disorder (ADHD) in children, and in adults where ADHD symptomatology has persisted from childhood. Various preparations are available and *slow-release forms of methylphenidate* are often preferred to diminish the need to take additional doses in the school day and to lessen rapid-onset stimulant effects which are believed to increase the risk of dependence. A licensed form of amphetamine, *lisdexamphetamine*, consists of an inactive complex of dexamphetamine and the amino acid, lysine; this prodrug is then taken up by red cells from which the active dexamphetamine is slowly released, giving the preparation a prolonged action profile.

Another indication for stimulants is the treatment of *narcolepsy*. Some specialists, mainly in the United States, believe that psychostimulants may have a role in combination with other antidepressant drugs for patients with

refractory depressive disorder, but the evidence for this effect is somewhat equivocal. *Modafinil* is licensed for the treatment of sleepiness associated with narcolepsy. Modafinil has also been used with apparent benefit to treat sleepiness and fatigue in depressed patients receiving SSRIs (Nuñez et al., 2022).

Unwanted effects

These include restlessness, insomnia, headache, dry mouth, poor appetite, dizziness, tremor, palpitations, and cardiac arrhythmias. In young people growth retardation can occur. Toxic effects from large doses include disorientation and aggressive behaviour, hallucinations, convulsions, and coma. Persistent abuse can lead to a paranoid state similar to paranoid schizophrenia. *Amphetamines* and *methylphenidate* can cause severe hypertension in combination with MAOIs and, to a lesser extent, with tricyclic antidepressants. They are contraindicated in cardiovascular disease and thyrotoxicosis. *Modafinil* has been associated with dry mouth, nausea, abdominal pain, headache, tachycardia, palpitations, cardiac arrhythmias, anxiety, and insomnia. It should not be used in patients with significant cardiovascular disease and an ECG is advised before starting modafinil treatment.

Other physical treatments

Electroconvulsive therapy

History

Convulsive therapy was introduced in the 1930s on the basis of the mistaken idea that epilepsy and schizophrenia do not occur together. It seemed to follow that induced fits should lead to improvement in schizophrenia. However, when the treatment was tried it became apparent that the most striking changes occurred not in schizophrenia but in *severe depressive disorders*, in which it brought about a substantial reduction in chronicity and mortality.

The first psychiatrist to employ seizures as therapy was Ladislav von Meduna who, in 1934, used medications such as camphor and pentylenetetrazole to produce grand mal fits. However, in 1938 Cerletti and Bini found that electrical stimulation, applied through bitemporal electrodes, produced fits more efficiently

and quickly than drugs, and, as time went by, electrical stimulation became the rule. The subsequent addition of brief anaesthesia and muscle relaxants made the treatment safer and more acceptable.

Indications

This section summarizes the indications for ECT. Further information about the efficiency of the procedure will be found in the chapters that deal with the individual psychiatric syndromes.

ECT is a rapid and effective treatment for *severe depressive disorders*. In the Medical Research Council trial (Clinical Psychiatry Committee, 1965) it acted faster than imipramine or phenelzine, and was more effective than imipramine in women and more effective than phenelzine in both sexes. The indications for ECT have been reviewed by the National Institute for Clinical Excellence (2003) (see Box 25.11). Indications suggested

Box 25.11 Indications for ECT (NICE)

It is recommended that electroconvulsive therapy is used only to achieve rapid and short-term improvement of severe symptoms after an adequate trial of other treatment options has proven ineffective and/or when the condition is considered to be potentially life-threatening in individuals with:

1. Severe depressive illness
2. Catatonia
3. A prolonged or severe manic episode.

Reproduced from National Institute for Health and Care Excellence. (2003). Guidance on the use of electroconvulsive therapy. Technology appraisal guidance TA59. <https://www.nice.org.uk/guidance/ta59> Copyright (2003), with permission from National Institute for Health and Care Excellence

by the Royal College of Psychiatrists (2005) are shown in Box 25.12.

Both sets of guidelines agree with the impression of many clinicians, and with the recommendations of this book, that ECT should be used mainly when it is essential to bring about improvement quickly. Therefore the strongest indications are an immediate *high risk of suicide, depressive stupor, or danger to physical health* because the patient is not drinking enough to maintain adequate renal function.

Box 25.12 Indications for ECT (Royal College of Psychiatrists, 2005)

In severe depressive illness, ECT may be the treatment of choice when the illness is associated with:

- life-threatening illness because of refusal of foods and fluids
- a high suicide risk.

ECT may be considered for the treatment of severe depressive illness associated with:

- stupor
- marked psychomotor retardation
- depressive delusions and hallucinations.

ECT may be considered as second- or third-line treatment of depressive illness that is not responsive to antidepressant drugs.

ECT may be considered for the treatment of mania:

The NICE guidelines for the use of ECT *in depression* were modified slightly in 2009 to the following:

- Consider ECT for acute treatment of severe depression that is life-threatening and when a rapid response is required, or when other treatments have failed.
- Do not use ECT routinely for people with moderate depression but consider it if their depression has not responded to multiple drug treatments and psychological treatment.

ECT can lead to a rapid resolution of *mania*, but is generally reserved for patients who do not respond to drug treatment, or for those whose manic illness is particularly severe, requiring high doses of antipsychotic drugs (Goodwin et al., 2016).

On the basis of clinical case studies, it has long been held that ECT is useful for the treatment of *acute catatonic states*. Controlled studies have also shown that ECT is effective in patients with acute schizophrenia with predominantly positive symptoms. In these studies ECT is effective not only for affective symptoms but also for positive symptoms such as *delusions* and *thought disorder* (Brandon et al., 1985). In general, however, ECT adds little to the effects of adequate doses of antipsychotic drugs, although it probably produces a greater rate of symptomatic improvement in the short term. ECT may also have a role in the treatment of patients with schizophrenia whose positive symptoms do not respond to antipsychotic medication, including clozapine (Petrides et al., 2015; Zheng et al., 2016).

- that is associated with life-threatening physical exhaustion
- that has not responded to appropriate drug treatment.

ECT may be considered for the treatment of acute schizophrenia as a fourth-line option for treatment-resistant schizophrenia after treatment with two antipsychotic drugs and then clozapine has proved ineffective.

ECT may be indicated in patients with catatonia where treatment with a benzodiazepine (usually lorazepam) has proved ineffective.

Reproduced from Scott, A. F. (2005). College Guidelines on electroconvulsive therapy: An update for prescribers, *Adv Psychiatr Treat*, 11(2), 150–156. Copyright with permission from Cambridge University Press.

Mode of action

Role of the seizure

Presumably the specific therapeutic effects of ECT must be brought about through physiological and biochemical changes in the brain. The first step in identifying the mode of action must be to find out whether the therapeutic effect depends on the seizure, or whether other features of the treatment are sufficient, such as the passage of the current through the brain, and the use of anaesthesia and muscle relaxants.

Clinicians have generally been convinced that the patient does not improve unless a convulsion is produced during the ECT procedure. This impression has been confirmed by several double-blind trials which, taken together, show that ECT is strikingly more effective than a full placebo procedure that includes anaesthetic and muscle relaxant (UK ECT Review Group, 2003) (see Figure 25.1).

This evidence does not necessarily support the view that a full seizure is the sufficient and necessary therapeutic component of ECT, and recent studies have shown that this notion is incorrect. Modern ECT machines deliver brief pulses of electrical current that enable a seizure to be induced by administration of relatively low doses of electrical energy. With this mode of administration, both *electrode placement* and *electrical dosage* can have profound effects on the therapeutic

efficacy of ECT. In particular, it appears that the *amount by which the applied electrical dose exceeds the seizure threshold of the individual patient* is an important determinant of both efficacy and cognitive side effects of ECT. Furthermore, the seizure threshold varies greatly (about 15-fold) between individuals.

This situation has important implications for the practical management of ECT when the clinician's aim is to find the best balance between *therapeutic efficacy* and *cognitive side effects* (see below). From a theoretical viewpoint, however, it can be concluded that an important determinant of ECT efficacy is how far the applied electrical energy exceeds the seizure threshold *of the individual patient*.

Neurobiological effects of ECT

Electrical seizures in animals produce many biochemical and electrophysiological changes, and therefore it is difficult to identify the processes that are important in the antidepressant effect of ECT. It is of interest that some of the changes in brain monoamine pathways found in rodents after ECT resemble those found after repeated antidepressant treatment. For example, both ECT and antidepressants increase the expression of *dopamine D₂ receptors* in the nucleus accumbens, which could be associated with improvements in motivational behaviour. In the case of ECT, this may involve interaction with *glutamate pathways*.

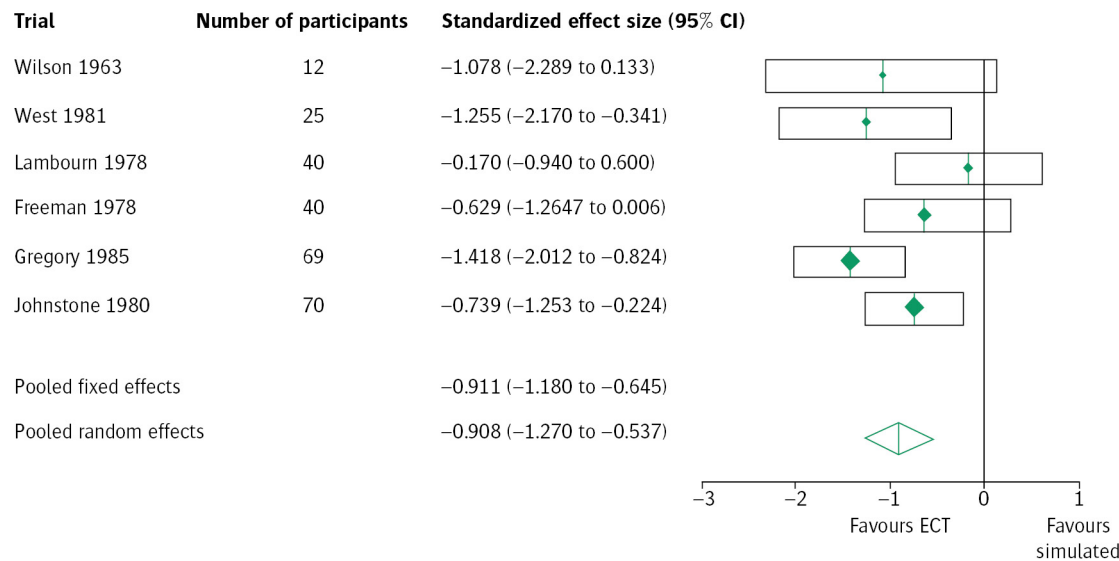


Figure 25.1 Meta-analysis of randomized, placebo-controlled studies of electroconvulsive therapy (ECT) in depression. Reproduced from The UK ECT Review Group. (2003). Efficacy and safety of electroconvulsive therapy in depressive disorders: A systematic review and meta-analysis. *Lancet*, 361(9360), 799–808. Copyright (2003), with permission from Elsevier.

Like antidepressant medication, electrical seizures in animals result in changes in neurotrophins such as *BDNF*, and increases in *neurogenesis* in the hippocampus. Consistent with this, imaging studies in patients undergoing ECT have demonstrated substantial increases in hippocampal volume (Wilkinson et al., 2017). The hippocampus is likely to play an important role in the pathophysiology of depression, and changes in hippocampal activity could therefore underpin some of the therapeutic effects of ECT. Of course, altered hippocampal function could also play a role in the amnesic side effects of ECT. Finally, in terms of network theories of depression (see Chapter 9), ECT appears to decrease the hyperconnectivity of the default mode network that is characteristic of depressive states. For a review, see Cowen (2019).

Physiological changes during ECT

If ECT is given without anticholinergic premedication, the pulse first slows and then rises quickly to 130–190 beats/minute, falling to the original resting rate or beyond towards the end of the seizure, before a final less marked tachycardia that lasts for several minutes. Marked increases in blood pressure are also common, and the systolic pressure can rise to 200 mmHg. Cerebral blood flow also increases by up to 200%.

Unilateral or bilateral ECT

Overall, bilateral ECT has a superior efficacy to unilateral ECT. However, bilateral treatment is associated with more cognitive impairment (UK ECT Review Group, 2003). When right unilateral ECT is dosed to about six times the seizure threshold, its efficacy is similar to that of bilateral ECT, and it retains some cognitive advantages over bilateral treatment (Kolshus et al., 2017).

These data suggest that the appropriate electrode placement for ECT should balance efficacy against cognitive disturbance according to the needs of the individual patient. If the need for improvement is particularly urgent, bilateral ECT should be considered, dosed to about two and a half times the seizure threshold.

For a meta-analytical review of treatment factors that affect the efficacy of ECT, see UK ECT Review Group (2003) (see also Table 25.12).

Unwanted effects after ECT

ECT can cause a brief *retrograde amnesia* as well as loss of memory for up to 30 minutes after the fit. *Brief disorientation* can occur, particularly with bilateral electrode placement. *Headache* can also occur. Some patients complain of confusion, nausea, and vertigo for a few hours

Table 25.12 Efficacy of ECT in depression in relation to antidepressant treatment and different stimulus parameters

Treatment comparison	HAM-D difference (with 95% CI)	Effect size*
ECT superior to antidepressant medication	5.2 (1.4–8.9)	0.46
Bilateral ECT superior to unilateral ECT	3.6 (2.2–5.2)	0.32
Higher ECT dose superior	4.1 (2.4–5.9)	0.57

HAM-D, Hamilton Rating Scale for Depression. * For explanation of effect sizes, see Chapter 6.

Source: UK ECT Review Group (2003).

after the treatment, but with modern methods these unwanted effects are mild and brief.

A few patients complain of *muscle pain*, especially in the jaws, which is probably attributable to the muscle relaxant. Occasional damage to the teeth, tongue, or lips can occur if there have been problems in positioning the gag or airway. Poor application of the electrodes can lead to *small electrical burns*.

All of these physical consequences are rare if a good technique of anaesthesia is used and the fit is modified adequately. Other complications of ECT are rare and mainly occur in people suffering from physical illness. They include *cardiac arrhythmia*, *pulmonary embolism*, *aspiration pneumonia*, and *cerebrovascular accident*. Prolonged apnoea is a rare complication of the use of muscle relaxants. Rarely, *status epilepticus* may occur in predisposed individuals or in those taking medication that prolongs seizure duration.

Since the introduction of ECT, there has always been concern as to whether it may cause *brain damage*. When ECT is given to animals in the usual clinical regimen, there is no neuropathological evidence that brain damage occurs. Imaging studies in humans show that, as would be expected, ECT changes aspects of brain structure and function but there is currently no evidence of brain damage in terms of cortical atrophy, ventricular enlargement, or white matter hyperintensities (Jolly and Singh, 2020). ECT does not increase the subsequent risk of dementia or stroke (Kirov et al., 2021).

Cognitive problems after ECT

As already mentioned, the *immediate effects* of ECT include loss of memory for events shortly before the

treatment (*retrograde amnesia*), and impaired retention of information acquired soon after the treatment (*anterograde amnesia*). These effects depend on both electrode placement (unilateral versus bilateral) and electrical dose, but electrode placement appears to be the more important factor.

A systematic review showed that when patients were studied a short time (1–28 days) after the end of a course of ECT, compared to pre-ECT measures deficits were apparent in autobiographical memory, verbal fluency, and verbal memory. However, other aspects of cognition such as working memory and verbal learning were not affected, and executive function showed a small improvement. Six months following the course of ECT, cognitive tests were generally improved compared to pre-ECT measures and there was no objective change in autobiographical memory, compared to pre-ECT status (Landry et al., 2021).

Nevertheless, there remains concern about deleterious long-term effects of ECT on autobiographical memory and in some patients such an effect appears to be persistent. Loss of autobiographical memory tends to affect memories that are closer in time to the ECT course. Risk factors appear to be the use of bilateral rather unilateral electrode placements, a higher number of treatments, and female sex. Concomitant lithium treatment may also be a risk factor.

Complaints of *subjective difficulty* with memory are quite common after ECT though the frequency of this varies considerably between different studies (Jones and McCollum, 2019). Generally, complaints of subjectively poor memory correlate with symptoms of depression rather than objective performance on cognitive tasks (Kirov et al., 2021).

It seems reasonable to conclude that, when used in the usual way, *ECT is not normally followed by persisting anterograde memory disorder*, nor with problems in *working memory* and *executive function*. However, ECT, particularly when applied bilaterally, may cause a persisting retrograde amnesia for personal memories. Although this is not a significant problem for most patients, some, for reasons that are unclear, have much greater autobiographical memory loss, to the extent that their lives and sense of self are adversely affected. For a review of the cognitive effects of ECT, see Finnegan and McLoughlin (2019).

The mortality of ECT

The death rate attributable to ECT has been estimated to be less than 2 per 100,000 treatments. This is similar to that seen with general anaesthesia for minor surgical conditions (Tørring et al., 2017). The risks are related to

the anaesthetic procedure, and are greatest in patients with cardiovascular disease. When death occurs it is usually due to *ventricular fibrillation* or *myocardial infarction*. Generally, the all-cause mortality of patients who receive ECT is somewhat less than that of matched depressive controls over the following year, but this may be due to the additional medical attention received by patients prior to and during ECT (Jørgensen et al., 2020). Whether ECT reduces suicide rates in the year following treatment is not clear, with different studies coming to differing conclusions.

Contraindications

The contraindications to ECT are any medical illnesses that increase the risk of anaesthetic procedure by an unacceptable amount, for example, respiratory infections, serious heart disease, and serious pyrexial illness. Other contraindications are diseases that are likely to be made worse by the *changes in blood pressure and cardiac rhythm* that occur even in a well-modified fit. These include serious heart diseases, recent myocardial infarction, cardiac failure, cerebral or aortic aneurysm, deep vein thrombosis, and raised intracranial pressure.

In Mediterranean and Afro-Caribbean patients who might have *sickle-cell trait*, additional care is needed to ensure that oxygen tension does not fall. Extra care is also required with diabetic patients who take insulin. Although risks rise somewhat in old age, so do the risks of untreated depression and drug treatment.

Technique of administration

In this section we outline the technical procedures used at the time of treatment. Although the information in this account should be known, it is important to remember that ECT is a *practical procedure* that must be learned by apprenticeship as well as by reading.

ECT clinic. ECT should be given in pleasant safe surroundings. Patients should not have to wait where they can see or hear treatment being given to others. There should be waiting and recovery areas separate from the room in which treatment is given, and adequate emergency equipment should be available, including a sucker, endotracheal tubes, adequate supplies of oxygen, and facilities to carry out full resuscitation. The ECT machine should provide a brief pulse square wave output together with a printable EEG. The machine should be capable of providing a range of easily-set stimuli (Dunne et al., 2019).

The nursing and medical staff who give ECT should receive *special training and accreditation*.

Arrival of the patient. The first step in giving ECT is to put the patient at ease and to check their identity. The

case notes should then be seen to make sure that there is a *valid consent form* and that *the patient continues to consent to treatment*. The drug sheet should be checked to ensure that the patient is not receiving any drugs that might complicate the anaesthetic procedures. It is also important to check for evidence of drug allergy or adverse effects of previous general anaesthetics. The drug sheet should be available for the anaesthetist to see. A full physical evaluation should have been carried out by the patient's treating doctor. Specialist advice should be sought if there may be medical contraindications to ECT.

Electrode placement. A decision about electrode placement should have been made by the treating doctor prior to treatment. When using unilateral treatment, it is important to apply the electrodes to the *non-dominant hemisphere*. In right-handed people, the left hemisphere is nearly always dominant; in left-handed people, either hemisphere may be dominant. Therefore, if there is evidence that the patient is not right-handed, it is usually better to use bilateral electrode placements.

Anaesthetic procedures. Prior to administering the anaesthetic it is necessary to make sure that the patient has taken nothing by mouth for at least five hours, and then, with the anaesthetist, to remove any dentures and check for loose or broken teeth. Finally, the record of any previous ECTs should be examined for evidence of delayed recovery from the relaxant (due to deficiency in pseudocholinesterase) or other complications.

The anaesthetic should be given by a trained anaesthetist (although this cannot always be achieved in developing countries). Suction apparatus, a positive-pressure oxygen supply, and emergency drugs should always be available. A tilting trolley is also valuable. As well as the psychiatrist and the anaesthetist, at least one nurse should be present.

Anaesthesia for ECT was previously induced with *methohexitone*, a short-acting barbiturate. However, this agent became increasingly difficult to obtain, and *propofol* is the most widely used alternative. The injection of propofol can be painful, and it can also *decrease seizure length* and *delay or abolish convulsions*. *Etomidate* may be preferred in patients in whom it is difficult to induce a seizure.

The induction agent is followed immediately by a *muscle relaxant* (often suxamethonium chloride) from a separate syringe, although the same needle can be used. The anticholinergic agent *glycopyrrolate* can be used to prevent ECT-induced increases in parasympathetic tone and bradycardia. The anaesthetist is responsible for the choice of drugs, and should also ensure that the lungs are well oxygenated before a mouth gag is inserted.

Application of ECT. While the anaesthetic is being given, the psychiatrist checks both the *dose of electricity* and the *electrode placement* that has been prescribed for the patient. The skin is cleaned in the appropriate areas and moistened electrodes are applied. (If good electrical contact is to be obtained, it is also important that grease and hair lacquer are removed by ward staff before the patient is sent for ECT.) Although dry electrodes can cause skin burns, it is also important to remember that excessive moisture causes shorting and may prevent a seizure response.

Although enough muscle relaxant should have been given to ensure that convulsive movements are minimal, a nurse or other assistant should be ready to restrain the patient gently if necessary. The electrodes are then secured firmly. For unilateral ECT, the first electrode is placed on the non-dominant side, 3 cm above the midpoint between the external angle of the orbit and the external auditory meatus. The second electrode is placed *at least 10 cm away from the first one*, vertically above the meatus of the same side (see Figure 25.2). A wide separation of the electrodes is thought to maximize the efficacy of unilateral ECT. The stimulus is then given.

For *fixed-dose right unilateral ECT* the initial dose should be set at 400 millicoulombs and subsequently adjusted according to clinical efficacy and cognitive side effects. However, dose titration offers advantages in terms of individualizing the dose for each patient. In dose titration the seizure threshold for the individual patient is determined during the first ECT session by starting with a very low dose and increasing it until a fit occurs (see Dunne et al., 2019 for a typical regime).

For *bilateral ECT*, electrodes are placed on opposite sides of the head, each 3 cm above the midpoint of the line joining the external angle of the orbit to the external

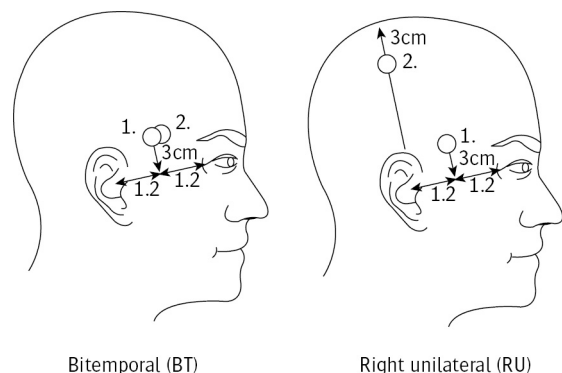


Figure 25.2 Electrode placements for electroconvulsive therapy (ECT).

auditory meatus—usually just above the hairline (see Figure 25.2). When using bilateral electrode placement, the present practice is to administer a dose of electricity that is only modestly (about 1.5 to 2.5 times) above the seizure threshold for the individual. The seizure threshold for ECT is best determined by dose titration as described above. Otherwise an appropriate dose can be estimated from the fact that two-thirds of the population have seizure thresholds in the range 100–200 millicoulombs. In addition, seizure thresholds are higher in men than in women, and increase with age. For example, a reasonable starting dose for a male patient under 40 years of age who is to receive bilateral ECT would be 150 millicoulombs.

Subsequently this dose could be adjusted, depending on the length of the seizure, the cognitive side effects, and the clinical response. The dose might be increased if the seizure was short or absent, or if there was no improvement after several treatments. Conversely, troublesome post-ECT cognitive disturbance would indicate that the dose of electricity should be reduced or that unilateral placement of the electrodes should be used.

It is important to note that seizure duration may *decrease* during a course of ECT, because repeated treatment tends to increase the seizure threshold. Thus if a dose of electricity initially produced a seizure of satisfactory duration, it may subsequently need to be increased.

The seizure. It is essential to observe carefully for evidence of seizure. If satisfactory muscle relaxation has been achieved, the seizure takes the following form. First, the muscles of the face begin to twitch and the mouth drops open. Then the upper eyelids, thumbs, and big toes jerk rhythmically for about half a minute. It is important not to confuse these convulsive movements with muscle twitches due to the depolarization produced by suxamethonium.

EEG monitoring is now generally used to check whether a seizure has been induced, but the records can be difficult to interpret because of the muscle artefact produced by direct stimulation of the frontalis muscle. A seizure can be detected by the appearance in the EEG of high frequency spike waves followed by slower spike and wave complexes with a frequency of around 3 Hz (Scott and Waite, 2013).

There is no direct correlation between treatment outcome and duration of seizure activity, but it is recommended that the dose of electricity is adjusted so as to achieve a seizure duration of *between 20 and 50 seconds*. This duration is recommended because short seizures are likely to be therapeutically ineffective, whereas long seizures are more likely to be associated with cognitive

disturbance. Prolonged seizures >90 seconds should be terminated with anticonvulsant medication such as propofol or diazepam.

Psychotropic drugs may alter the seizure threshold and seizure duration. For example, most antidepressant and antipsychotic drugs lower the seizure threshold, whereas benzodiazepines, valproate, and lamotrigine have the opposite effect. Adverse effects, including increased cognitive impairment, have been reported when ECT has been given with *lithium*, and it may be prudent to suspend lithium treatment or omit the lithium dose the evening before treatment. SSRIs and *venlafaxine* have been associated with prolonged seizures during ECT and it is sensible to lower the dose of venlafaxine below 200mg daily if possible. Some anaesthetists prefer not to anaesthetize patients who are taking MAOIs, but ECT can in fact be given safely to patients who are receiving MAOI treatment. The usual practice in the United Kingdom has been to continue antidepressant medication during a course of ECT. This generally appears safe and may even enhance the therapeutic effect of ECT (Anderson, 2019).

Recovery phase. After the seizure, the lungs are oxygenated thoroughly with an airway in place. The patient remains in the care of the anaesthetist and under close nursing observation until breathing resumes and consciousness is restored. During recovery, the patient should be turned on to their side and cared for in the usual way for any patient recovering from an anaesthetic after a minor surgical procedure. A qualified nurse should be in attendance to supervise the patient and reassure them. Meanwhile, the psychiatrist makes a note of the date, type of electrode placement, drugs used, and amount of current, together with a brief description of the fit and any problems that have arisen. When the patient is awake and orientated, they should rest for an hour or so on their bed or in a chair.

If ECT is given to a day-patient, it is especially important to make certain that no food or drink has been taken before they arrive at the hospital. They should rest for several hours and should not leave until it is certain that their recovery is complete; they should leave in the company of a responsible adult, and certainly not riding a bicycle or driving a car.

Failed stimulations. The most important problem, apart from those relating to the anaesthetic procedure, is failure to produce a clonic convulsion (a tonic jerk produced by the current must not be mistaken for a seizure). If it is certain that no seizure has appeared, checks should be made of the machine, the electrodes, and contact with the skin. The possibility of shorting due to

excess moisture on the scalp should also be considered. If all of these are excluded, the patient may have either an unusually high resistance to the passage of current through the extracranial tissues and skull, or a high convulsive threshold. The charge can then be increased and a further stimulus given, provided the anaesthetist agrees.

Frequency and number of treatments

ECT is usually given twice a week. In general, ECT given three times weekly has *little therapeutic advantage* over a twice-weekly regimen, and may produce more cognitive impairment (UK ECT Review Group, 2003).

Decisions about the length of a course of ECT must depend on clinical experience, as relevant information is not available from clinical trials. *A course of ECT usually consists of between six and a maximum of twelve treatments.* Progress should be reviewed after each treatment. There is usually little response until two or three treatments have been given, after which increasing improvement occurs. If the response is more rapid than this, fewer treatments may be given. If there has been no response after six to eight treatments, the course should usually be abandoned, as it is unlikely that more ECT will produce a useful change.

Clinical change and memory should also be assessed after each treatment. It is better to use reliable, validated scales for this purpose. For clinical change, the Clinical Global Impression of Improvement is suitable and the patient's subjective report should also be noted. Weekly measurement of depressive symptoms can be made on the Hamilton Rating Scale for Depression or the Montgomery-Åsberg Depression Rating Scale. Measurement of the various aspects of memory in a routine clinical setting is more difficult. NICE (2003) has recommended the Mini Mental State Examination, which will capture confusion and short-term memory impairment but not, for example, autobiographical memory loss. Significant cognitive impairment should lead to a reappraisal of the electrical dose and electrode placement.

Prevention of relapse

It is important that, whereas ECT may produce striking benefit in depressed patients, there is *a high relapse rate* unless continuation therapy with antidepressant medication is undertaken. Sometimes the choice of antidepressant drug can be difficult because, if a patient does not respond to an adequate dose of an antidepressant drug prior to ECT, continuing the same drug after the course of ECT is completed *may not provide a useful*

prophylactic effect. Thus, if a patient has required ECT because of non-response to antidepressant medication, it is good practice to consider a different class of antidepressant drug or lithium carbonate in the continuation and prophylactic phases of drug treatment. A randomized study of 200 patients found post-ECT prophylaxis with a combination of lithium and nortriptyline as effective as maintenance ECT in sustaining remission in the six months following a successful course of treatment. Despite this, just over 50% of the patients in each group relapsed (Kellner et al., 2006).

A few patients respond well to ECT but continually relapse even when maintained on multiple drug therapy. In these circumstances, some practitioners have given *continuation* followed by *maintenance ECT* at a reduced frequency (e.g. fortnightly or monthly) with good effect (Dunne et al., 2019). If continuation or maintenance ECT are considered, it is important to arrange ongoing assessments of cognitive function and subjective reports of memory difficulties.

Consent to ECT

Before a patient is asked to agree to ECT, it is essential to provide a full *explanation of the procedure* and indicate its *expected benefits* and *possible risks*, especially the effects on memory. Appropriately written information sheets should be a standard part of the consent procedure (National Institute for Clinical Excellence, 2003).

Many patients expect severe and permanent memory impairment after treatment, and some even expect to experience unmodified fits. Once the doctor is sure that the patient understands what they have been told, the patient is asked to sign a standard form of consent. It should be made clear to them that consent is being sought for the whole course of ECT, and not just for one treatment, although it is also essential to make sure that they understand that consent can be withdrawn at any time. It is also important to ensure that patients continue to give consent prior to each treatment throughout the course. Part of the consent process should involve the assessment of *capacity* (see Chapter 4). Patients cannot give informed consent if they lack capacity, and advance directives declining ECT should be respected unless treatment is considered life-saving and can be administered under the Mental Health Act.

In the United Kingdom, if a patient refuses consent, or is unable to give it because they are in a stupor or for some other reason, and if the procedure is essential, further steps must be considered. The first is to decide whether there are grounds for involving the appropriate section of the Mental Health Act (see Chapter 4). The

section does not allow anyone to give consent on behalf of the patient, but it does establish formally that they are mentally ill and in need of treatment. In England and Wales, the opinion of a second independent consultant is required by the Mental Health Act 1983. Readers working outside the United Kingdom should familiarize themselves with the relevant legal requirements.

Patient attitude to treatment

Surveys carried out by clinicians have generally reported positive experiences of ECT, with the majority of patients reporting that the procedure was tolerable and helpful. The presence of friendly and helpful clinic staff was regarded as important in reducing anxiety about treatment. The major complaint about ECT concerned autobiographical memory loss, which in some patients was persistent and severe. Service user-led research on ECT generally elicits less-positive responses about treatment, with about half the patients reporting that they would not have ECT again. Again, the presence of persistent memory loss is usually the main complaint, but some patients also reported a lack of information about ECT and its adverse effects and also experiences of coercion. Not surprisingly, patients who believe that ECT has helped them are more positive about treatment. However, many patients remain dissatisfied with their experience of undergoing ECT (Waite, 2019).

This situation underlines the need for careful selection of patients for ECT, as well as adequate time being taken to explain the procedure and its possible adverse effects as fully and carefully as possible.

Ethical aspects of ECT

As with drug treatment, the key issue for a competent patient is the concept of *full informed consent*. In general, ECT is widely regarded as a safe and effective treatment—a view that is supported by randomized controlled trials. However, it is important that patients are warned about the possibility of loss of remote memories outlined above. In addition, in the case of medication-resistant patients, the issue of relapse should be discussed.

ECT in the case of non-competent patients raises particularly difficult issues. Usually, in these circumstances, ECT is needed to treat a patient whose life is placed at risk as a result of their illness. The legal safeguards for patients are outlined above, but the clinician has a duty to explain to the patient and their family the reasons for the action taken, and to follow carefully the ethical principles outlined in Chapter 4. A compelling personal account of these issues has been provided by Gergel (2022).

Other neurostimulation techniques

Transcranial magnetic stimulation

The use of transcranial magnetic stimulation (TMS) is based on the principle that, if a conducting medium such as the brain is adjacent to a magnetic field, a current will be induced in the conducting medium. In TMS, an electromagnetic coil is placed on the scalp. The passage of high-intensity pulses of current in the coil produces a powerful magnetic field—typically about 2 Tesla—which results in current flow in neural tissue and neuronal depolarization. Neuropsychological effects of TMS are particularly likely when pulses of current are delivered rapidly, so-called *repetitive TMS* (rTMS). If the stimulation occurs more quickly than once per second (1 Hz), it is called *fast rTMS*. The use of appropriately shaped coils allows reasonably localized stimulation of the specific cortical areas.

Uses of TMS

TMS has been used for many years in clinical neurophysiology to explore, for example, the integrity of the motor cortex after stroke. In research settings, TMS is used to localize the cortical substrates of specific neuropsychological functions. For example, short-term verbal recall can be disrupted by rTMS administered over the left temporal cortex.

Clinically, rTMS has been used to relieve *depressive states*. Initially, studies used fast rTMS applied to the left prefrontal cortex. However, other investigations have employed different kinds of electromagnetic coil, stimulation parameters, and sites of coil application, making comparison between studies difficult. Typically in the treatment of depression, 40 pulses of stimulation are delivered over four seconds followed by a gap of 26 seconds before the next 40 pulses. Around 2,400–3,000 pulses are delivered over a single session, which may last about 30 minutes. To achieve a clinical antidepressant effect, TMS is usually repeated daily on weekdays for two to four weeks. How well the response is maintained after this is unclear.

Recent developments have employed intermittent ‘theta-burst’ (50 Hz) stimulation (iTBS) which permits a shorter duration of daily treatment (three to ten minutes) and is as effective as standard TMS. A further development is the provision of ten iTBS treatments each day for five days with coil placement guided by pre-treatment fMRI connectivity measures. This protocol showed a rapid and complete antidepressant response in a majority of patients with treatment-resistant depression in a sham treatment, controlled trial (Cole et al., 2022).

Meta-analyses suggest that, overall, TMS is more beneficial in the treatment of depression than sham treatment, but while it is better tolerated than ECT, it is somewhat less effective (Chen et al., 2017). TMS has also been employed in the treatment of other disorders, such as obsessive-compulsive disorder, schizophrenia, eating disorders, and depersonalization syndrome. Findings in these conditions are still preliminary and the difficulty of providing a convincing sham control makes assessment difficult. TMS treatment for depression has been approved by NICE and is increasingly used in the United Kingdom for patients resistant to pharmacological treatment. TMS is usually given with adjunctive antidepressant medication.

Adverse effects of TMS

The use of single-pulse TMS in neurophysiological studies has not raised significant safety concerns. The major hazard with rTMS is the risk of inducing *seizures*. This is greater with fast rTMS than with slow rTMS. Current safety protocols, which adjust the amount of magnetic stimulation in relation to the motor threshold of the individual, appear to have greatly reduced the likelihood of fits, although patients with risk factors (e.g. a family history of epilepsy) are generally excluded from TMS studies of healthy volunteers.

Minor side effects are more common, and include *muscle tension headaches*, and sufficient noise is generated by the equipment to cause short-term changes in *hearing threshold*. This can be prevented by the use of earplugs (by both patients and investigators). rTMS appropriately localized also has the potential to disrupt cognitive function, but so far, such changes have been temporary, and have not persisted after the stimulation was terminated. There are insufficient data to establish whether there might be any long-term sequelae to the brain or other organs from the high-intensity magnetic field generated during TMS. No specific hazard has been revealed by current follow-up studies.

Transcranial direct current stimulation

A more recent stimulatory technique for treating psychiatric disorders is provided by transcranial direct current stimulation (TDCS). TDCS involves an application to the scalp of a small current of 1 or 2 milliamperes via two electrodes which function as anode and cathode. Current is typically applied for around 30 minutes. The weak electrical field thus generated is capable of altering neuronal excitability and plasticity. The anode placement is usually over left dorsolateral prefrontal cortex. Treatment sessions last for 20–30 minutes and are repeated daily for two weeks or more. This can be

followed by less frequent maintenance treatments. TDCS is generally well tolerated. Temporary tingling and burning sensations may be experienced at the site of the electrodes. Headache is sometimes reported. There are occasional reports of skin burns under the electrode, which may be due to faulty technique.

TDCS has been applied mainly to the treatment of depression, although even for this single indication a variety of electrode placements and current strengths have been used. In a randomized, sham-controlled study of 200 unipolar depressed patients with a mild level of treatment resistance, TDCS was superior to sham treatment but less effective than escitalopram. Two patients who received active TDCS developed mania, though generally this seems a rare side effect of TDCS treatment (Brunoni et al., 2017).

In an individual patient data meta-analysis, TDCS was found to have superior antidepressant effects to sham treatment though the benefit of TDCS appeared greater during the maintenance phase of treatment, suggesting that its full effects may be delayed (Moffa et al., 2020). A network meta-analysis of non-surgical brain stimulations treatments for depression showed that TDCS was less effective than TMS (Mutz et al., 2019). However, TDCS kits are commercially available and can be used in the home; this may provide an advantage for some patients. Generally, TDCS would seem to have a place in the first line treatment of depression in patients where pharmacotherapy is not suitable or desired.

Vagal nerve stimulation

Vagal nerve stimulation (VNS) is an established treatment for patients with refractory epilepsy, in whom it was noted to improve mood. It has therefore subsequently been applied to the treatment of patients with resistant depression.

As well as providing an efferent parasympathetic outflow from the brain, the vagus nerve conveys afferent sensory fibres that synapse in the nucleus tractus solitarius and subsequently project to the forebrain. Stimulating the vagus nerve therapeutically involves an operative procedure during which a pulse generator is implanted subcutaneously in the left anterior chest wall. Bipolar electrodes, connected to the generator, are placed around the left vagus nerve in the cervical region. These electrodes are intermittently stimulated by the generator, with the stimulation parameters being regulated by a telemetric 'wand', which is connected to a personal computer. Modification of the stimulus parameters is used to balance adverse effects (see below) with therapeutic effects. Typically stimulation occurs for about 30 seconds followed by a five-minute 'off' period.

VNS in depression

Open studies have suggested that over 12 months of treatment about 50% of patients show a good clinical response to VNS. Bottomley and colleagues (2020) carried out an assessment of 22 studies of VNS in depressed patients with resistant depression. Overall, there was evidence of increasing benefit with VNS in terms of both diminished depressive symptoms and improvement in quality-of-life measures over two years of treatment. However, in only two of these studies was there a randomized element of treatment comparison and these did not show significant benefits of VNS. The latter studies were, however, relatively short-term and it appears that the response to VNS may accumulate gradually over many months.

Adverse effects of VNS

During periods of VNS, hoarseness is common and throat pain, cough, and dyspnoea can occur. These adverse effects decline with time, and withdrawal from treatment is uncommon. Hypomania has rarely been reported, but its relationship to VNS treatment is uncertain. Various non-invasive, transcutaneous approaches to vagal nerve stimulation are becoming available; this may offer easier avenues to this form of treatment in the future (see Austelle et al., 2022).

The efficacy of VNS is not clearly established by controlled trials and some aspects of the implantation are irreversible. It has therefore been recommended that VNS should currently be regarded as an investigational procedure and offered to patients only in the setting of an ethically approved research protocol (Matthews and Matthews, 2019).

Deep brain stimulation

Deep brain stimulation (DBS) is established in the treatment of Parkinson's disease, where implantation of electrodes in the subthalamic nucleus significantly alleviates the motor symptoms of advanced disease. DBS was also noted sometimes to improve mood and obsessive symptomatology in patients with Parkinson's disease and this, together with increasing knowledge of the neural circuitry underlying psychiatric disorders, has led to studies of its effects in patients with refractory depression and obsessive-compulsive disorder.

DBS involves bilateral implantation of electrodes into designated target locations using stereotactic guidance. The location of the electrodes is confirmed with MRI scanning. The electrodes are connected to a generator implanted into the abdomen, and the stimulation settings are adjusted according to clinical response and adverse effects. The fact that the settings can be

modified and the generator switched on or off makes DBS more susceptible to controlled trial than neurosurgical treatment.

It seems likely that stimulation frequency is a key factor in determining the clinical efficacy of DBS. In Parkinson's disease, for example, stimulation starts to reduce tremor at a frequency of approximately 50 Hz and reaches a plateau at about 200 Hz. Low-frequency stimulation is believed to activate neurons, while high-frequency stimulation probably causes inhibition. Whereas improvement in rigidity and tremor in Parkinson's disease can be seen within minutes of DBS commencing, amelioration in depression and obsessive-compulsive symptomatology develops gradually over a period of weeks. This might be a consequence of neuroplastic changes induced by DBS, or might perhaps be due to changes in the oscillatory properties of the relevant neural networks.

DBS in obsessive-compulsive disorder

In studies of patients with intractable obsessive-compulsive disorder, electrodes have been implanted in the anterior limb of the internal capsule, with the aim of influencing the brain regions targeted by neurosurgery for obsessive-compulsive disorder (see below). Subsequent studies have also examined the effect of DBS in the ventral striatum, nucleus accumbens, and bed nucleus of stria terminalis. These sites are involved in the regulation of cortico-striato-thalamo-cortical circuitry thought relevant to the pathophysiology of OCD (see Chapter 8).

Gadot and colleagues (2022) conducted a meta-analysis of 34 studies (including nine randomized controlled trials) of DBS in 255 patients with refractory obsessive-compulsive disorder. Clinical response at two years was seen in 66% of the sample, with an overall reduction of about 50% in obsessive compulsive symptomatology and similar improvements in ratings of depression. There was no difference in response rates between randomized and non-randomized studies but treatments targeting the nucleus accumbens seemed somewhat less effective.

DBS in depression

DBS in treatment-refractory depressed patients has also involved a number of different brain regions, based on abnormalities revealed by functional neuroimaging as well as theoretical considerations of the neural underpinnings of symptoms such as anhedonia. Studies have involved mainly the subgenual cingulate cortex and the ventral striatum, but other targets, including the median forebrain bundle and lateral habenula, have also been

used. A meta-analysis of 12 randomized controlled trials in 186 patients found a significant benefit of DBS to diminish depressive symptomatology with an effect size of 0.75. There was no difference in effect between the different target sites (Hitti et al., 2020). However, it should be noted that two well-conducted randomized trials with sham stimulation periods of 16 and 26 weeks failed to show specific benefit of DBS treatment (Matthews and Matthews, 2019).

Adverse effects of DBS

Typical adverse effects associated with DBS include throbbing or buzzing sensations, nausea, and jaw tingling. Orofacial muscle contractions, dyskinesias, dysarthria, and dysphagia are also reported. Mood changes may occur but are difficult to relate specifically to DBS. However, some patients with no history of bipolarity have experienced hypomania. More profound mood swings, including severe depression and suicidal ideation, may occur if the stimulator fails suddenly. Some patients have reported a feeling of being cognitively 'clouded', which has responded to changes in stimulus parameters. To date there is no evidence that DBS causes enduring cognitive impairment.

As with any neurosurgical procedure, implantation of electrodes for DBS carries risks of haemorrhage, infection, and seizure. Such complications appear to be rare; in one series of 141 patients undergoing DBS for Parkinson's disease, the overall mortality attributable to the procedure was 0.7% (Vergani et al., 2010). However, in the meta-analysis of OCD studies outlined above, almost a third of patients experienced at least one serious adverse effect. These included hardware complications, seizures, infections, intracranial haemorrhage, and *de novo* obsessions about the stimulation parameters. Six patients attempted suicide and there was one completed suicide.

Ethical aspects of DBS

DBS is considered to be less invasive than neurosurgery in that the stimulation can be stopped and there is no lesioning of tissue. However, the procedure does carry risks, and there is a lack of long-term efficacy and safety data. The situation is further complicated by involvement of commercial factors, because the stimulator equipment is expensive and manufactured by industry.

This highlights once again the need for a scrupulous fiduciary relationship between the patient and the treating clinical team, as well as full informed consent to the procedure by a competent patient. It also suggests that some DBS trials should be sponsored by non-commercial organizations, and that all clinical data should be made publicly available so that there is no risk of publication

bias. In view of the troubled history of direct neurosurgical interventions for psychiatric disorders, it has been proposed that DBS should only be used to treat psychiatric disorders in the context of an approved research protocol with full independent ethical review.

Neurosurgery for psychiatric disorders (psychosurgery)

History of procedure

Psychosurgery refers to the use of *neurosurgical procedures* to modify the symptoms of psychiatric illness by operating on either the nuclei of the brain or the white matter. Psychosurgery began in 1936 with the work of Egas Moniz, whose operation consisted of an extensive cut in the white matter of the frontal lobes (*frontal leukotomy*). This extensive operation was modified by Freeman and Watts (1942), who made smaller coronal incisions in the frontal lobes through lateral burr holes.

Although their 'standard leucotomy' was far from standardized anatomically, and although it produced unacceptable side effects (see below), the procedure was widely used in the United Kingdom and other countries. There was enthusiasm for the initial improvements observed in patients, but this was followed by growing evidence of adverse effects, including *intellectual impairment, emotional lability, disinhibition, apathy, incontinence, obesity, and epilepsy*.

These problems led to a search for more restricted lesions capable of producing the same therapeutic benefits without these adverse consequences. Some progress was made, particularly with the incorporation of *stereotactic techniques*, but at the same time advances in pharmacology made it possible to use drugs to treat the disorders for which surgery was intended.

Current approaches

The term *psychosurgery* is now often replaced by the phrase *neurosurgery for mental disorders*. This change in terminology is intended to emphasize that:

- The techniques now involve placement of localized lesions in specific cerebral sites.
- The treatment is for specific psychiatric conditions (treatment-resistant major depression and obsessive-compulsive disorder), not for primary behavioural disturbance.

Indications

In the United Kingdom the indications for neurosurgery for mental disorders are restricted to major depression

Box 25.13 Stereotactic procedures used in psychosurgery

Subcaudate tractotomy

Lesion made beneath the head of each caudate nucleus, in the rostral part of the orbital cortex

Anterior cingulotomy

Bilateral lesions within the cingulate bundles

Limbic leucotomy

Subcaudate tractotomy combined with cingulotomy

Anterior capsulotomy

Bilateral lesions in the anterior limb of the internal capsule

and obsessive–compulsive disorder that is chronic, treatment-refractory, and disabling. Neurosurgery is not indicated where these conditions arise as a consequence of organic brain disease or a pervasive developmental disorder.

Types of operation

Nowadays the older ‘blind’ operations have been replaced by stereotactic procedures that allow the lesions to be placed more accurately (see Box 25.13). In the United Kingdom, current procedures are limited to anterior capsulotomy and anterior cingulotomy. The lesions are produced either by radiofrequency thermocoagulation, gamma radiation, or focused ultrasound. Lesions are made bilaterally.

Effectiveness

In the absence of controlled trials, assessment has been in the form of long-term *follow-up studies*. A report by the Royal College of Psychiatrists (2000) found a ‘marked improvement’ rate in 63% of patients with major depression and in 58% of patients with obsessive–compulsive disorder (see Table 25.13). These response rates are similar to those of more recent reviews (Matthews and Matthews, 2019).

Adverse effects

With modern procedures, severe adverse effects are rare. After the operation, headache and nausea are common, and confusion occurs in about 10% of patients. These adverse effects typically last for a few days, but can

Table 25.13 Outcome of psychosurgery in the United Kingdom, 1961–1997

	Depression (n = 727)	Obsessive– compulsive disorder (n = 478)
Marked improvement	63%	58%
Lesser improvement	22%	27%
No response	14%	14%
Worse	1%	1%

Source: *Neurosurgery for Mental Disorder. Report from the Neurosurgery Working Group of the Royal College of Psychiatrists* (Council Report CR89). Copyright (2000), Royal College of Psychiatrists.

persist for up to a month. Long-term cognitive impairment does not seem to occur (see Box 25.14).

Clinical use

Neurosurgical procedures for mental disorders should not be performed until the effects of several years of vigorous multimodal treatment have been observed. If this rule is followed, the operation will hardly ever be used. In fact, fewer than 20 psychosurgical operations a year are currently performed in the United Kingdom—about 25% of the annual rate in the early 1980s. If the surgery is to be considered at all, it should only be for chronic intractable obsessional disorder and severe chronic depressive disorders in older patients. There is no clear justification for neurosurgery for anxiety disorders or

Box 25.14 Adverse effects of stereotactic psychosurgery

Acute effects

Operative mortality (less than 0.1%)
Haemorrhage, hemiplegia (less than 0.3%)
Transient confusion, lethargy (common)

Long-term effects

Epilepsy (1–2%)
Weight gain (10%)
Frontal lobe syndrome (very rare)
Personality changes (usually mild)

schizophrenia. For a review, see Matthews and Matthews (2019).

Ethical issues

The brain is the organ of judgement and decision making, but it is regarded as ethically permissible to operate, for example, on a brain tumour, if a patient gives their consent. The situation with regard to psychosurgery is different because the tissue that is lesioned is not overtly diseased. However, surgeons do sometimes operate on tissue that is not diseased. The practical problems with regard to psychosurgery (which distinguish it from ECT) are as follows:

- the lack of randomized studies to demonstrate that it is effective
- the irreversibility of the procedure
- the potentially serious nature of some of the adverse effects.

For these reasons it seems ethically appropriate that psychosurgery should be offered only to competent patients who are able to give their full informed consent. Determining competence may, of course, be difficult in a patient with chronic severe mood disorder. In the United Kingdom, this has led to the development of safeguards under Section 57 of the 1983 Mental Health Act, which require that:

1. The patient gives their full informed consent.
2. A multidisciplinary panel appointed by the Mental Health Act Commission confirms that the patient's consent is valid.
3. The doctor on the multidisciplinary panel certifies that the treatment should be given. Before doing so, the doctor must consult two people, one a nurse and the other neither a nurse nor a doctor, who have been concerned with the patient's treatment.

Bright light treatment

The use of *phototherapy*, or *artificial bright light*, as a psychiatric treatment was first studied systematically by Rosenthal et al. (1984). These workers used bright light to treat patients with the newly identified syndrome of *seasonal affective disorder*. Since then, phototherapy has become the mainstay of treatment of *winter depression*, particularly in patients with *atypical depressive features* such as hyperphagia and hypersomnia.

Mechanism of action

The light–dark cycle is believed to be one of the most important zeitgebers regulating circadian and seasonal rhythmicity in mammals. Initially, phototherapy was believed to ameliorate the symptoms of winter depression by extending the photoperiod. This was based on the view that patients with winter depression were particularly sensitive to the effects of short winter days, and that bright light treatment produced a day length equivalent to that of summer.

More recent formulations have suggested that the antidepressant effect of bright light may be attributable to a *phase advance in circadian rhythm*. This is supported by controlled trials which show that, in most patients, morning phototherapy is more effective than evening phototherapy. However, other studies have shown that bright light given at midday is also therapeutically effective. It is difficult to devise a truly plausible placebo condition for bright light treatment, and some have argued that the antidepressant effects of bright light may be mediated in large measure by placebo effects, particularly patient expectation.

Pjrek and colleagues (2020) found in a meta-analysis of 19 randomized studies that bright light was significantly superior to a dim light control in reducing depressive symptomatology in patients with seasonal affective disorder, with an effect size of 0.37 at study endpoint. In winter depression the therapeutic effects of bright light treatment, fluoxetine, and cognitive behaviour therapy appear generally equivalent, although cognitive therapy had a better long-term outcome (Cleare et al., 2015).

Equipment

A conventional light box contains fluorescent tubes mounted behind a translucent plastic diffusing screen. The tubes provide an output that can vary between 2,500 and 10,000 lux. Light sources that produce 10,000 lux are more expensive, but may allow a *reduced duration of exposure* (30 minutes, compared with 120 minutes) to secure a therapeutic effect.

Phototherapy has also been administered using *head-mounted units* or *light visors*. These instruments are attached to the head and project light into the eyes, allowing the subject to remain mobile while receiving treatment. Although light visors are more convenient to use than light boxes, results from placebo-controlled trials of light visors have proved disappointing and their use is not currently recommended.

Therapeutic efficacy and indications

The major indication for light therapy is seasonal affective disorder in which patients experience winter depressions. Bright light treatment is more effective than placebo in patients with winter depression, particularly if patients experience:

- increased sleep
- carbohydrate cravings
- an afternoon slump in energy.

Patients with typical melancholic symptoms—for example, weight loss and insomnia—do less well with phototherapy as a sole treatment, even where the disorder is seasonal in nature. However, there are studies to suggest that morning light treatment may be helpful to reduce depressive symptoms in non-seasonal depression and also augment the therapeutic effect of antidepressant medication. However, the quality of literature has been criticized (Dong et al., 2022). The evidence that bright light therapy (given at midday) can be helpful in bipolar depression is somewhat stronger. In a meta-analysis of seven randomized controlled trials, Lam and colleagues (2020) found a significant benefit of bright light treatment in reducing depressive symptomatology in bipolar patients, with an effect size of 0.43. There was no increase in manic switches. An umbrella review found little evidence for the effectiveness of phototherapy in other psychiatric conditions, such as dementia, sleep disorders, and eating disorders (Reis et al., 2022)

Adverse effects

In general, phototherapy is well tolerated, although mild side effects occur in up to 45% of patients early in treatment. These include *headache*, *eye-strain*, *blurred vision*, *eye irritation*, and *increased tension*. Insomnia can also occur, particularly with late evening treatment. Rare adverse events that have been reported include *manic mood swings* and *suicide* attempts, the latter putatively through light-induced alerting and energizing effects prior to mood improvement. Whether these rare events are actually adverse reactions to light is uncertain. There is no evidence that phototherapy employed in recommended treatment schedules causes ocular or retinal damage.

Clinical use of phototherapy

Since the best-established indication for phototherapy is winter depression, the following account describes the

use of bright light treatment in winter depression. One of the major practical difficulties in phototherapy is the time needed to administer the treatment. For this reason, a 10,000-lux light box may be preferred because the daily duration of therapy can be reduced to 30 minutes. It seems likely that cool white light and full-spectrum light have equivalent clinical efficacy, but because cool white light is free of ultraviolet light, it is theoretically safer.

The evidence suggests that bright light treatment of winter depression is most effective when administered in the early morning. However, treatments given at other times of day may prove beneficial and can be more convenient for individual patients. In an initial trial, therefore, it is best to recommend early morning treatment but to advise the patient that the exact timing of therapy can eventually represent a balance of therapeutic efficacy and practical convenience. Treatment should not be given late in the evening because of the possibility of *sleep disruption*.

Early morning phototherapy should start within a few minutes of awakening. Patients should allow an initial duration of treatment of 30 minutes with a 10,000-lux light box or two hours with 2,500 lux equipment. They should seat themselves about 30–40 cm away from the light box screen. They should not gaze at the screen directly, but face it at an angle of about 45° and glance across it once or twice each minute.

The antidepressant effect of light treatment can appear within a few days, but in controlled trials longer periods (up to three weeks) can be necessary before the therapeutic effects of bright light exceed those of placebo treatment. As noted above, mild side effects are common in the early stages of treatment, but usually settle without specific intervention. If they are persistent and troublesome, the patient can sit a little further away from the light source or reduce the duration of exposure. Exposure should also be reduced if elevated mood occurs.

Once a therapeutic response has occurred, it is necessary to continue phototherapy up to the point of natural remission, otherwise relapse will occur. However, it may be possible to lower the daily duration of treatment. Phototherapy may also be started in advance of the anticipated episode of depression to provide a *prophylactic effect*; however, this indication is not currently supported by good evidence.

Further reading

- Ferrier, I. N., & Waite, J. (Eds.). (2019). *The ECT Handbook*, 4th edn. RCPsych Publications.
- Haddad, P. M., & Nutt, D. J. (2020). *Seminars in Clinical Psychopharmacology*. Cambridge University Press.
- National Institute for Health and Care Excellence (2003). Guidance on the use of electroconvulsive therapy. Technology appraisal guidance TA59. <https://www.nice.org.uk/guidance/ta59>
- Stahl, S. M. (2021). *Stahl's Essential Psychopharmacology*. Cambridge University Press.
- Taylor, D., Barnes, T. R. E., & Young, A. H. (2021). *The Maudsley Prescribing Guidelines*, 14th edn. John Wiley.

CHAPTER 26

Psychiatric services

Introduction 755

The history of psychiatric services 755

The components of a mental health service 759

Services for psychiatric disorder in primary care 769

Specialist services for acute psychiatric disorder 772

Psychiatric services that provide long-term care 772

Services for people with particular needs 776

Some difficulties with community care 777

International service principles 778

Introduction

The last two chapters dealt with the treatment of individual patients. This chapter is concerned with the provision of psychiatric care for populations. It deals mainly with the needs of and provisions for people aged 18–65 years ('adults of working age').

Services for children are described in Chapter 16, services for the elderly in Chapter 19, and services for patients with intellectual disability in Chapter 17. The organization of psychiatric services in any country inevitably depends on the organization of general medical services in that country. This chapter will refer specifically to services in the United Kingdom, but the principles embodied apply widely.

The chapter begins with an account of the historical development of psychiatric services. This is followed by descriptions of the commonly available psychiatric services and of the problems encountered with these provisions. Recent developments in the United Kingdom are seeking to ensure people are well supported on transition between adolescent and adult services, as well as to older age services if these are required. Clinicians need to consider which service best meets care needs and levels of maturation, rather than base services on arbitrary age ranges. The chapter ends with a consideration of some innovations designed to overcome these problems.

The history of psychiatric services

What we now recognize as psychiatry dates from the end of the eighteenth century. Before that there were hardly any special provisions for the mentally ill in Europe, other than in Spain, where some hospitals were present from the Islamic Middle Ages (Chamberlain, 1966). Although mental disturbances were often considered as illnesses in antiquity, during the Middle Ages

they became regarded as spiritual problems. Other than prayer, and forms of exorcism and containment, little was provided.

In Britain until the middle of the eighteenth century, most mentally ill people lived with their families or, if very disturbed, were in prison. Private 'madhouses' (later to be called private asylums) were developed mainly for

those who could pay, but accepted some paupers supported by their parishes (Parry-Jones, 1971). At about the same time, a few hospitals or wards were established through private benefaction and public subscription. The Bethel Hospital in Norwich was founded in 1713. In London, the lunatic ward at Guy's Hospital was established in 1728. In 1751, St Luke's Hospital was founded as an alternative to the overcrowded Bethlehem Hospital, which had existed as an institution since 1247, but only latterly had become exclusively a 'madhouse'.

Moral management

At the end of the eighteenth century the Enlightenment led to a more empirical, less theological, understanding of madness. This rational approach led to increased public concern in many countries about the poor standards of private and public institutions, including efforts to improve the care of the mentally ill. In Paris in 1793, Philippe Pinel iconically released patients from their chains and introduced other changes to make the care of patients more humane. In England, similar reforming ideas were proposed by William Tuke, a Quaker philanthropist who founded the Retreat in York in 1792. The Retreat provided pleasant surroundings and adequate facilities for occupation and recreation. Treatment was based on 'moral' (i.e. social) management, consisting of kindness, calm, and activity. Tuke and Pinel both acknowledged that, although mad, patients still could respond to kindness and consideration, in contrast to the previous authoritarian approach aimed at simply controlling patients. Medical treatments such as bleeding and purging were peripheral to the routine care of the insane. Tuke's grandson, Samuel, described their system in *A Description of the Retreat* (1813), and the Retreat was replicated internationally (Raad and Makari, 2010). It became clear that many mentally ill patients could exert self-control and did not require physical restraint and punishment.

The asylum movement

Until the early nineteenth century most mentally ill people received no care and lived as vagrants or as inmates of workhouses and gaols. Public concern about the welfare of the insane in workhouses and gaols and some private madhouses led to the County Asylum Act of 1808. This provided for the building of mental hospitals in English counties. However, it was the Lunatics Act 1845 that required the building of one in every county. At first the new asylums were small and provided good treatment in spacious surroundings. Moral management

was championed. The 'non-restraint movement', which had started with the work of Gardiner Hill at the Lincoln Asylum in 1837, was established by John Conolly at the Middlesex County Asylum, Hanwell, in 1845. This removed the use of all physical restraints, a practice that has characterized mental healthcare in the United Kingdom ever since.

Unfortunately, these early attitudes yielded to a more restrictive approach as the asylums expanded and became overcrowded. Initial optimism about the curability of psychiatric disorder faded with the accumulation of chronic cases and an increased focus on organic and hereditary causes. The original York Retreat housed 30 patients, and the first 16 British asylums about 100 patients each. By 1840 they averaged 300 patients, 540 patients by 1870, and 960 patients by 1900. Some patients with chronic illness were housed in detached annexes or houses in the grounds of the asylum. Other hospitals returned chronic patients to the community either by boarding them out with a family (a form of care that had long been practised successfully at Gheel in Belgium), or by returning them to workhouses. The Lunacy Commissioners, whose role was to oversee the care of the mentally ill, were concerned that these arrangements could lead to abuse, and were opposed to them. Nevertheless, nineteenth-century asylums, even when overcrowded, provided a standard of care for the mentally ill that was lacking elsewhere. They were protected from exploitation, and provided with shelter, food, and general health care, all without cost.

A change to a stricter custodial regime was endorsed by the bureaucratic and rigid Lunacy Act 1890, which imposed restrictions on admission and discharge from hospital. The Lunacy Act had resulted from years of unwarranted committal, but it handicapped British psychiatric development for nearly half a century. For accounts of psychiatric hospitals in the United Kingdom and the United States in the nineteenth century, see Rothman (1971) and Jones (1972).

Arrangements for early treatment

The opening of the Maudsley Hospital in 1923 provided an outpatient service and voluntary inpatient treatment in surroundings in which teaching and research were carried out. In the years between the wars the impetus for change increased. The Mental Treatment Act 1930 repealed many of the restrictions on discharge of patients that had been imposed by the 1890 Act. It allowed county asylums to accept voluntary patients and changed the term 'asylum' to 'mental hospital' and the term 'lunatic' to voluntary 'mental patient'. The

1930 Act also encouraged local authorities to set up outpatient clinics and to establish facilities for aftercare. Therapeutic optimism, revived by von Jauregg's successful malaria treatment for cerebral syphilis, increased further with the introduction of insulin coma treatment (later abandoned) and electroconvulsive therapy. Efforts were made to improve conditions in hospitals, to unlock previously locked wards, and to encourage occupational activities. Similar changes took place in other countries.

In most countries, these reforms were halted by the Second World War. Psychiatric hospitals became understaffed as doctors and nurses were recruited to the war effort, and became overcrowded as some had to be evacuated for the care of the war injured (Crammer, 1990). In Germany, many psychiatric patients died in a eugenics programme—a particularly shameful episode in the history of a country that dominated psychiatric progress in the nineteenth and early twentieth centuries (Burleigh, 2000).

Social psychiatry and the beginning of community care

Several forces led to further changes in psychiatric hospitals directly after the Second World War. Social attitudes had become more liberal and sympathetic towards the disadvantaged. Wartime experience of 'battle neuroses' had encouraged interest in the early treatments, and in the use of groups and social rehabilitation. In the United Kingdom, the establishment of the National Health Service (NHS) led to a general reorganization of medical services, including psychiatry. The introduction of chlorpromazine in 1952 made it easier to manage the disturbed behaviour of patients with psychosis. It became possible to open wards that had been locked (although some hospitals had done so before such drugs became available), to engage patients in social activities, and to discharge more of them into the community.

Despite these changes, services continued to be concentrated at single sites, often remote from centres of population. In the United States, Goffman vividly exposed the detrimental effects of such 'total institutions'. He described how their impersonal, inflexible, and authoritarian regimes eroded identity and, through this 'institutionalization', generated chronicity (Goffman, 1961). In the United Kingdom, Wing and Brown (1970) demonstrated how large mental hospital environments characterized by 'social poverty' led to 'clinical poverty'. Vigorous social rehabilitation was introduced to improve conditions in hospitals and to counteract the effects of years of institutional living. Occupational and industrial therapies were used to prepare chronically

disabled patients for the move from hospital to sheltered accommodation or to ordinary housing (Bennett, 1983). Many long-stay patients were very responsive to these vigorous new methods, and optimism grew that newly admitted patients could be similarly helped.

Away from the hospitals, day units were established to provide continuing treatment and rehabilitation, and hostels were opened to provide sheltered accommodation. The numbers of patients in psychiatric hospitals fell substantially in the United Kingdom and in other countries. The changes started later in the United States, with the introduction of Medicaid in 1965, but then progressed very rapidly. Despite these changes, services were still based in large mental hospitals that were often far from patients' homes, and the provision of community facilities remained inadequate.

Hospital closure

Early successes in discharging institutionalized patients led to expectations that asylums would soon be replaced by small psychiatric units in general hospitals, supported by community facilities. In most countries, the programme of hospital closure took place, albeit very gradually. A notable exception was Italy. In 1978, the Italian Parliament passed Law 180, which prohibited admissions to mental hospitals forthwith. It aimed to abolish Italy's neglected mental hospitals altogether over a period of three years and replace them with a comprehensive system of community care. Psychiatric 'diagnostic' units limited to admissions of seven days were to be set up in general hospitals, and community services developed for each catchment area. This Italian revolution was led by Franco Basaglia in hospitals in north-east Italy, through the movement he founded. This movement—*Psichiatria Democratica*—combined a left-wing political view that patients in psychiatric hospitals were the victims of oppression by the capitalist system with the conviction that severe mental illness was induced more by social conditions than by biological causes. These sentiments and debates on the role of political systems and social forces continues and is not only of historical relevance.

Basaglia's charismatic personality and qualities of leadership (and a wife who became an Italian senator) helped him to succeed with his reforms. Others found them more difficult, and their impact varied. Where the reforms were financed adequately and were implemented by enthusiastic staff (particularly in the wealthy north), the new provisions were successful. However, in Rome and the south facilities were often inadequate and problems encountered (Fioritti et al., 1997). The state

mental hospital system, however, came to a complete end in 1989 (Burti, 2001).

In the United Kingdom and elsewhere the pace of change was slower, but similar problems arose. Many patients needed ongoing intensive support, and many required repeated readmissions to hospital—so-called revolving-door patients. Rehabilitation services had to adjust their expectations and provide continuing care. The early expectations of *deinstitutionalization* had clearly been overoptimistic. The policy of ‘community care’ was introduced to develop more adequate community provision. However, the international comparisons show that community services do indeed reflect social and political values and priorities and are not always based on evidence of effectiveness of interventions and recovery alone, but rather on a complex interplay of ethics and values and political priorities. Rather than see this as an oversight, accepting this reality argues for more psychiatrists to study subjects like philosophy, politics, and economics, and to be more aware of policy and the importance of political engagement (Jenkins, 2013).

The rise of community care

The development of community care falls quite neatly into two phases—before 1980 and from 1980 onwards (Burns, 2014). In the first phase, as hospitals closed, community psychiatric services acquired three responsibilities. The first was to provide treatment for those individuals with severe mental illness who previously would have remained in hospital for many years. The second was to treat severe acute psychiatric disorders as far as possible without lengthy admission to hospital, and as near as possible to the patient’s home. The third was to assist primary care services in the detection, prevention, and early treatment of the less severe psychiatric disorders. Services were to be comprehensive, to deliver continuity of care, and to be provided by multidisciplinary teams.

The framework for this ‘sectorized’ service was laid by the 1959 Mental Health Act (MHA) in the United Kingdom and a similar approach was initiated in France. The UK Act required hospitals to provide outpatient follow-up for their own discharged patients and also required social services involvement in both compulsory admissions and community support. A highly localized service was therefore a practical necessity for effective joint working.

These general principles were applied rather differently in the United Kingdom and in the United States.

In the former, emphasis was placed initially on provision for patients discharged from long-term hospital care. In the latter, community mental health centres (CMHCs) were established, with more emphasis on prevention and early treatment to avoid admission. These CMHCs emphasized crisis intervention and the treatment of acute, often relatively minor, psychiatric disorders. Unlike the UK teams, they had no outreach facility, so patients with chronic and severe mental illness (who seldom actively seek care) became neglected. A rather ‘antimedical’ approach in CMHCs led to difficulties in recruiting and retaining psychiatrists (Talbot et al., 1987). Dissatisfaction with the centres grew as people discharged from long-term hospital care found their way into private hospitals or prisons, or joined the homeless population of large cities (Goldman and Morrissey, 1985).

In the United Kingdom and parts of Europe, some commonly agreed principles evolved in sectorized care:

- *Minimizing inpatient care.* Hospital admissions were to be brief, and as far as possible patients were to be admitted to psychiatric units in general hospitals rather than to psychiatric hospitals. Whenever practicable, people were to be treated as outpatients or day patients.
- *Providing rehabilitation early on.* The aim was to protect and possibly improve residual functioning and prevent further deterioration.
- *Multidisciplinary teams.* Care was to be provided by teams, usually consisting of psychiatrists, community nurses, clinical psychologists, and social workers, often working in collaboration with members of voluntary groups.
- *Legal reform.* New laws were introduced (e.g. the 1983 MHA and its 2007 amendment in England and Wales) to limit the use of compulsory treatment, to encourage alternatives to inpatient care, and to strengthen the rights of the individual.
- *Independent Review of MHA.* At the time of writing, new legislation is under scrutiny in parliament, in the form of a revised MHA. The new proposals emphasize recovery, less stigmatizing language, and there are clauses to exclude learning disability and autism unless there is a comorbid mental illness. The new Act also seeks to provide more protection following detention in the form of appeals and reviews. The review has met with a number of external challenges, including the Covid-19 pandemic, a change of government, and then the death of Queen Elizabeth II, who announced reforms in the

Queen's Speech in 2022. These real-world influences mean that parliamentary time has been limited, and so the Act may take some time to be implemented. As an example of how mental health care can be influenced by other non-evidence-based issues, the war in Ukraine, rising costs of fuel, and continued austerity measures inevitably have taken priority such that further actions on the MHA and its implementation are further delayed in the UK. Similar experiences in other countries mean psychiatrists and mental health professionals need to retain a focus on evidence, on professional values, and highlight deficiencies in care systems, and where there are risks of public harms (Iemmi, 2022) for discussion of global health and political influence; and Chapter 23 on public mental health and global mental health).

As experience increased after the reforms had begun, the following additional features were introduced:

- *Care packages* based on an assessment of each patient's needs.
- *Case management* by a named, clinically trained care worker who led and coordinated the work of others involved in care.
- *Outreach* to take services to vulnerable people who might otherwise find it difficult to engage with the care that is offered, and to arrange follow-up.
- *User involvement*. Service users were increasingly involved in planning both their own treatment and the services for the population.
- *Risk assessment* carried out regularly, as part of the care plan.

The components of a mental health service

From the 1960s to the mid-1980s, mental health services in the United Kingdom and, indeed, most parts of the world evolved, slowly, based on professional consensus. In the 1980s, interest in evidence-based medicine spread to mental health services research. This approach was international and increased the pace of change dramatically. The fundamental components of most care, namely inpatient wards, day hospitals, and community mental health teams (CMHTs), had not been subject to research, but were well established and have proved durable.

The publication of a groundbreaking trial of an assertive community treatment team ushered in the second phase of development—the era of evidence-based mental health services (Stein and Test, 1980). The more recent highly specialized teams (assertive outreach teams, crisis resolution/home treatment teams, and early-onset (psychosis) services) have been driven more by policy than research, which has tended to follow. Since the last edition of this book, we have witnessed a dramatic change in fortunes for some of these new teams as the evidence has accumulated. What follows is a description of the UK services that addresses research and international evidence where this is relevant or helpful. The NHS has entered a phase of profound change, and the predictability and uniformity of services are likely to change markedly during the lifetime of this book. As noted earlier, issues specifically related to psychiatric services for children, those with learning disabilities,

and the elderly, are discussed in Chapters 16, 17, and 19, respectively.

Inpatient wards

No comprehensive service can survive without access to 24-hour nursing supervision for acute episodes of severe illness (Talbot et al., 1987; Thornicroft and Tansella, 2013). Surprisingly, inpatient wards are often overlooked in descriptions of comprehensive services. They serve patients at risk from neglect or suicide and those lacking insight, and where support is needed for containing behaviours and for helping frightened patients to recover. Wards usually accommodate 10–20 patients. It is rarely possible to staff and run stand-alone units of less than three to four such wards effectively (30–60 beds). The famed Italian Law 180 restricted wards to 15 patients, and this is a common international goal (Burns et al., 1998). Ward size is a trade-off between privacy and domesticity on the one hand and effective supervision on the other. Single rooms are preferable and are now the norm in many new-build units. They afford maximum privacy and, although initially expensive, they improve flexibility and have been shown to reduce conflict significantly. However, they do raise concerns about visibility and supervision and safety for very distressed people in need of intensive hospital care.

The number of beds required for acute psychiatric disorder is difficult to determine exactly. However, it

is generally accepted that bed numbers in the United States, the United Kingdom, and parts of Europe are now inadequate (Priebe et al., 2005; Raad and Makari, 2010). Severe bed shortage appears to be a contributor to the recent increase in compulsion (Keown et al., 2011) as it depends on:

- The level of morbidity, including substance misuse, in the population served by the unit.
- The willingness and ability of families to care for acutely ill relatives, which in turn depends on the quality of accommodation and the availability of other family members.
- The extent and availability of crisis services.
- Facilities for treatment of acute psychiatric disorder outside hospital, such as well-staffed hostels.
- Facilities for early discharge of patients from hospital after the acute phase of the disorder; it is generally easier to discharge early than to avoid admission.

The design of inpatient units for acutely ill patients should strike a balance between the patients' needs for privacy and the staff's requirement to observe them, although single, ensuite accommodation is increasingly the norm. Units need to be sufficiently large to ensure adequate staffing for emergencies yet avoid an anonymous or 'institutional' feel. Three 15- to 20-bedded wards is considered to be the minimum necessary for safe effective care, and units of over 100 patients are considered probably too large. Ward design should minimize the possibility of suicide by hanging or jumping. There is a need for secure areas for the most disturbed patients, areas where patients can be alone, and areas where they can interact with others. There should be provision for occupational therapy, the practice of domestic skills, and recreation. Outdoor space is desirable. Single-sex accommodation is now mandatory in many countries including the United Kingdom.

The siting of acute inpatient units. Inpatient care for acute psychiatric disorders is generally provided as part of a general hospital complex. This location reduces stigma and provides easy access to general medical services when required. The disadvantages of such locations include a low threshold for disruptive behaviour in general hospital settings and inadequate space for occupational and recreational activities suitable for psychiatric care. Many of these problems can be overcome if the psychiatric unit occupies a separate building within the general hospital complex.

Collaboration with community teams. Patients are at increased risk, particularly for suicide, when they move

from inpatient to community care. About 25% of suicides among psychiatric patients in England and Wales occur within three months of discharge from inpatient psychiatric care, with a peak in the first one to two weeks, and contact in the week following discharge is essential (Bickley et al., 2013).

Single-sex accommodation

Concern for the safety and privacy of female patients has led to government requirements for single-sex wards. Protecting women and protecting dignity at times of distress is important.

Smaller, more flexible, units such as 'crisis houses' that offer 24-hour care have been a useful complement to inpatient wards, where appropriate, either before admission for care, or to avert admission, or as a means of offering 'step-down' discharge. Ward design and leadership to manage clinical care are increasingly crucial as improved community care concentrates involuntary and disturbed inpatients in them.

Continuity of responsibility

Responsibility by the same clinical team for both community and inpatient care is unusual except in the United Kingdom and Italy, where it has delivered strikingly shorter stays and improved bed management. However, this is not universal in the United Kingdom, where there have been splits between community and inpatient care. One argument for this is that inpatient care requires proficiency in specific skills (acute emergency, legislation, ethics, physical care, legalized coercive care) while community care requires more relational approaches and engagement with community providers for health and social care. This dilemma is partly born of the need to have well-trained psychiatrists (and mental health professionals) who are adept at working in both spaces and can manage the tensions.

Furthermore, continuity is difficult in dispersed populations, and when each team has fewer inpatients. A separate inpatient team ensures better inpatient standards, but with the risk of diverging approaches (e.g. a 'crisis and medical model' inpatient team and a 'relational, psychotherapeutic' community team). Separation has recently become common practice in the United Kingdom at the same time that many European services are trying to overcome it, for reasons that are unclear, but which may include the problems of wards having to cope with several admitting teams.

Both the move to single-sex accommodation and specialized teams result in each team often having only one or two patients on a ward. Diffused responsibility and

organizing multiple ward rounds have promoted the change, but not without cost. It appears to be in conflict with the expectation of continuity of clinical responsibility in the 1983 MHA, and particularly in the 2007 amendment permitting compulsory treatment in the community through the new provision of Supervised Community Treatment Orders (SCTOs, which are invariably referred to as Community Treatment Orders, or CTOs; see Chapters 4 and 26). It is unclear whether the inpatient consultant or the community consultant should make the decision. CTOs are likely to continue, despite dissatisfaction with their effectiveness in trials, and marked variation across geographies and are indeed debated in the proposed revisions to the MHA in the UK parliament (McBride et al., 2020).

Longer inpatient care

Acute inpatient wards admit patients for weeks or a couple of months, with rapid discharge anticipated. Across Europe the rate of compulsory admission has been rising for the past decade (Barnett et al., 2019; Smith et al., 2020). The reasons are unclear but bed shortages are certainly one factor (Keown et al., 2011), as is an increasing focus on risk assessment; amendments to the MHA in 2007 and austerity measures also led to reduced community support (Smith et al., 2020). Some patients require longer, more secure care service because of illness severity or for legal reasons. Modern rehabilitation wards are thus generally restricted to patients with persistently unacceptable behaviour.

Diagnosis-specific wards

Diagnosis- or disorder-specific wards are relatively rare in the United Kingdom. In Scandinavia and Central Europe, separate wards for alcohol and substance abuse are long established. Wards for specific, specialized problems such as anorexia nervosa or resistant schizophrenia offer highly specific regimes. These are generally regional, and are a complement to acute admission wards, not an alternative. More mother and baby units for perinatal mental illness can ensure mother and baby are not separated when a mother suffers with a severe mental illness. And there remains a shortage of inpatient child and adolescent care where essential, although community models are generally preferred.

Daycare

Daycare is provided in day hospitals and day centres, and there is no clear distinction between them. Day hospitals are generally provided by health services, include medical and nursing staff, and can offer treatments (e.g. the

prescription and monitoring of medication, psychotherapies). Day centres are provided by social services or voluntary organizations. They only rarely provide specific treatments or employ clinically trained staff. Acute day hospitals in Europe and partial hospitalization services in the United States have been energetically proposed as acceptable and economical alternatives to inpatient care (Marshall, 2003; Marshall et al., 2011). However, services vary according to local context. A drop-in day centre may provide psychiatric assessment and treatment in areas of high social mobility and homelessness. Broadly speaking, day centres provide long-term social support and day hospitals provide focused interventions and treatments (Catty et al., 2005). Some studies propose day care offers little advantage over usual outpatient care (Marshall et al., 2001; Shek et al., 2009). Day care is used in four main ways:

1. *Acute day hospital care* for patients with acute illness who would otherwise be admitted to hospital. Such care is generally more appropriate for those who can be with their families in the evening and at weekends. Early trials of acute day hospital treatment (involving 1,568 people) found it feasible for 25% of those currently admitted to inpatient care, with comparable rates of improvement and readmission, and some found it cheaper than inpatient care (Marshall, 2003). However, such acute day hospitals were very different from routine day hospitals, and failed to generalize, needing either a charismatic leader or a research programme to sustain them. They are currently experiencing a resurgence as the base for crisis response and home treatment (CR/HT) teams but have not been researched individually.
2. *Transitional daycare* to allow earlier discharge from hospital.
3. *Day treatment programmes* that provide more intensive treatment for people who are not responding to outpatient care. There is evidence for the value of this approach in personality disorder (Bateman and Fonagy, 2013), and it is increasingly being used for severe eating disorders.
4. *Daycare centres* for the support of long-term service users. Such care needs to be planned carefully with an active treatment programme that is specific to each patient's needs, otherwise attenders easily become dependent and 'stuck'.

Day hospitals and day centres vary considerably and their clientele and activities overlap (Catty et al., 2005). CR/HT teams are often integrated with day hospitals.

Supported accommodation and residential care

The extent of required inpatient care depends on capacity and access to other types of supported accommodation. Patients remain well outside hospital only with adequate support and stable, affordable accommodation. Supervision may be needed to ensure self-care, continued medication, and to anticipate and defuse crises. This can be provided by voluntary agencies, social services, or health services. Voluntary agencies tend to be more efficient at providing long-term residential care (Knapp et al., 1999), but are reluctant to accept risky patients. A mixed economy works best.

Outpatient clinics

Psychiatrists assess patients, advise them and their referers, and provide treatments. In state-funded systems ‘office-based’ pure practice is rare, as most professionals work in outpatient clinics or mental health teams. Psychiatrists and psychologists may still operate independently within outpatient clinics, but with easy access to enhanced resources and second opinions. Outpatient clinics may operate either alongside CMHTs or as part of them. They work better for severe illness when fully integrated with CMHTs (Wright et al., 2004). Clinics provide an efficient format for assessments and monitoring of treatment progress. However, recent trends advocate alternate models given the number of missed appointments and the need to offer more flexibility for patients (Killaspy, 2006).

Multidisciplinary community mental health teams

Most community mental health services are made up of multidisciplinary professional groups working together to care for an identified number of patients. These are referred to generically as CMHTs. Nurses have long worked outside hospitals (e.g. midwives, health visitors) and have made psychiatric home visits since the 1950s. Social workers, psychologists, and occupational therapists are increasingly based within CMHTs rather than in their own separate departments. The staffing of these teams varies internationally, but they all hold regular review meetings. These reviews incorporate their varied professional perspectives and allocate tasks according to staff skills and patient needs. They were developed in France and the United Kingdom in the delivery of sectorized psychiatry, refined in Italy, and further elaborated in North America and Australia.

The generic sector CMHT

Who it is for

The CMHT is the fundamental building block of modern community mental health services. It originated as mental hospital catchment areas (which often covered a whole city or county) divided into sectors of 50,000–100,000 inhabitants to facilitate ongoing care. It provides assessment and care for patients who have been discharged from psychiatric units and for outpatients who require more support than can be provided in primary care. Current sector size in Western Europe ranges from 20,000 to 50,000 members of the population, determined both by resources (the size shrinking as investment increases) and by the local configuration. As more specialized teams are established, the CMHT’s remit may be narrowed and its sector size increased.

CMHTs prioritize individuals with severe mental illnesses (SMIs), such as psychoses and severe affective disorders. Many inner-city CMHTs only see such patients. However, diagnosis is not all—complications due to social adversity, personality difficulties, or substance abuse can make secondary mental health care necessary even for apparently ‘minor’ disorders. Several tools have been developed for clarifying a threshold for acceptance (Slade et al., 2000), but these are of limited use and most teams rely on clinical assessments.

Staffing and management

CMHTs range from seven or eight to over 20 full-time staff. They emphasize skill sharing and a degree of generic working, and have evolved an informal, democratic style. Senior psychiatrists initially provided clinical leadership but, with increased staff numbers and treatment complexity, ‘team managers’ now coordinate the workload. There is little consensus on their role, which can range from the purely administrative to determining clinical priorities and supervising staff. Clinically active team managers usually have reduced caseloads. If clinical leadership and team management are separated (which is common if there is a strong medical presence), the roles need to be well defined and relationships clarified. CMHTs appear to help engagement, but leadership and team performance are critical (Krompa et al., 2022).

Assessments

The key to good care is accurate assessment, and CMHTs vary in how they provide this. In general, psychiatrists are responsible for an initial assessment (often in an outpatient clinic) and involve the team members in

treatment. Increasingly, other team members have taken a role in assessments, either individually or jointly, and sometimes with the psychiatrist. This issue generates strong opinions, but there has been surprisingly little research into it. With highly developed primary care, non-medical assessments may be effective, but otherwise medical time should prioritize assessments. For the most severely ill patients, home-based assessments pay considerable dividends (Burns et al., 2007; Stein and Test, 1980). Crisis care and home treatment teams now routinely assess patients in order to minimize admissions to hospital, and work closely with CMHTs (Stulz et al., 2020).

Case management

Most CMHT staff act as clinical case managers (Holloway et al., 1995), taking lead responsibility for the delivery, coordination, and review of care for an agreed number of patients. Currently, clinical case managers provide direct care to build a trusting relationship and provide treatment and continuity for patients with complex needs. They also use the full range of team resources. The caseloads of staff members are explicitly limited (usually 15–25), and reviews are recorded and systematic. In the United Kingdom this has been formalized as the Care Programme Approach (CPA) in a policy document from the Department of Health in 2010 (Goodwin and Lawton-Smith, 2010). The policy required a named case manager (called a *care coordinator*), and a document called a *care plan*, which records the patient's needs or problems, the proposed interventions for them, and who is responsible for each of them, plus an agreed date for *review*. Not everyone felt this was a useful or efficient way of organizing care, and this debate continues (Simpson et al., 2003). The proposals for a new MHA, for example, propose a care and treatment plan be a statutory requirement and that patients be given the right to advance choice directives and independent advocacy. All these efforts seek to ensure all patients receive a high-quality service.

Care plans are often supplemented with a risk assessment and a contingency (crisis) plan. Optimally this should not need more than two pages, but it is often weighed down with administrative details that obscure its clinical purpose. Such simple structured paperwork can be adapted to any service. It fulfils a vital coordinating role in complex care and serves as a natural focus for clinical reviews. The value of this document (as with the risk assessment and contingency plan) lies in its brevity. Too much information is as risky as too little, and the level of detail needs to be clinically (not managerially) determined.

Team meetings

CMHTs always have one, and often two, regular meetings per week for both clinical and administrative business. They last between one and two hours, and the level of structure varies.

Allocation of referrals. Deciding who will assess new patients need occupy only ten minutes at the beginning of the routine meeting or can simply be delegated to a senior member to decide. Referrals can be allocated according to who is first available, or by matching the clinical problem against available skills and training. It is remarkable how many CMHTs fail to have an agreed timetable for new assessments (as is routine in outpatient clinics). Much time is saved if each member has agreed regular assessment 'slots'. There is then no need to 'coordinate diaries' for joint assessments, and patients can be directly booked into available slots if the member is absent at allocation. Time spent discussing allocations (and particularly having a 'referral meeting') before assessment is unprofitable, and most well-functioning teams delegate the task.

Patient reviews. Reviews are needed for *new patients*, *routine monitoring*, *crises*, and *discharge*. They can range from simply reporting the problem and proposed treatment in uncomplicated cases through to detailed, structured, multidisciplinary case conferences. These may include other services (e.g. general practitioner, housing, child protection). *New patient reviews* are particularly important for providing a broad, experienced overview, particularly if assessments are distributed across the team. They also ensure rational and fair allocation to caseloads. *Routine monitoring* is often overlooked yet is probably the most important review for team efficiency. It shapes and redirects treatment and identifies patients who are ready for discharge. The burden on individual staff members is regularly monitored. Routine monitoring is a statutory requirement of the CPA and is good practice in all case management. *Crisis reviews* are unscheduled but allow case managers to seek advice when they are unsure. *Discharge reviews* provide an excellent opportunity for audit and learning within the team.

Managing waiting lists and caseloads. Effective CMHTs need to guarantee prompt access. *Routine assessments* should take place within two to four weeks. Striving for a shorter waiting time is rarely productive and waiting times of much over four weeks have a rapidly rising rate of failed appointments. *Urgent assessments* (which include most psychotic episodes) need to be seen within a week, and usually within a couple of days. *Emergency assessments* are for those associated with immediate risk (e.g. hostile behaviour or suicidal intent) and ideally

need to be seen the same day. CR/HT teams offer a rapid assessment of new presentations, including referral into the most appropriate care pathway and intervention.

Communication and liaison

Team meetings ensure internal communication, but CMHTs also need good links with the wider network of professional colleagues. Most routine communication takes place by letter, phone, or during clinical encounters. More recently, secure email and video conferences are commonly used, and were essential during the Covid-19 pandemic and thereafter. Indeed, many services have now established routine digital as well as face-to-face consultations and reviews.

More structured liaison is advisable with primary care and general hospitals. General hospital links may be between specific CMHTs and wards, or CMHTs may provide input to patients from their sectors. Dedicated liaison psychiatry services are routine in well-resourced services.

General practice liaison. GP liaison systems originated in Balint groups but are now more likely to involve shared care or co-location of CMHTs in GP health centres (Burns and Bale, 1997). An effective and sustainable system involves regular (usually monthly) timetabled meetings between the two teams, or a 'link' CMHT member attending the GP health centre. It is important always to be clear about responsibilities; blurring of boundaries is risky. Working with a joint case register on a regular basis is also useful to ensure mental health and physical health care needs are addressed optimally, and the roles of primary care and mental health teams are clear for each patient.

Liaison with other agencies. The same principles apply to liaison with other agencies (social services, housing, and charitable and voluntary sector providers). Whether regular meetings are cost-effective will depend on the volume of shared work. Professional confidentiality and information sharing are more sensitive. For people with multiple disadvantages, liaison with other agencies becomes a critical part of care, as policies and practices may exclude people with specific needs; for example, housing needs, substance misuse, or perceptions of risk and potential violence.

Mental health services research

The past 30 years have witnessed a change in how mental health services are planned and evaluated. *Mental health services research* (MHSR) attempts to subject services and systems to the same evidence-based medicine approach of rigorous testing (e.g. controlled studies,

randomized controlled trials, meta-analyses) widely accepted with specific treatments. This has had two profound effects. First, it has internationalized thinking about optimal services. The language of science means that findings from other parts of the world can be applied locally. Second, it has shifted the locus of control in service development away from clinicians and towards academics and policymakers. However, narrow models of evaluations, for example trials versus evaluation and qualitative research, may not best capture the complex and systemic impacts of service interventions.

There are undoubted benefits to MHSR. For example, some ineffective practices (e.g. traditional 'brokerage' case management) have been identified and abandoned. However, there are also drawbacks, particularly the risk of overinterpretation of results and their application without recognition of the modifying effects of local contexts. Three fundamental conceptual problems have dogged MHSR:

1. A tendency to equate a service with a treatment rather than with the *platform* for delivering treatments.
2. The assumption that the 'control' services in randomized controlled trials are inert and consistent, when in fact they are active and very variable between 'comparator' services.
3. A tendency to ignore the 'pioneer' effect—shiny new demonstration services with charismatic leaders have an inherent advantage.

MHSR methodology is improving, but these pitfalls in interpreting studies always need to be considered. Indeed, the UK government now has a commissioning stream devoted to health and social care delivery research, which seeks to accommodate multiple research methods and philosophies.

Mental Health National Service Framework 1999

Mental health services in England and Wales were radically reformed following the publication of the National Service Framework (NSF) in 1999 (Appleby, 2004) and the NHS Plan in 2000 (see Holloway, 2002). These drew heavily on MHSR and pioneering services in North Birmingham and Australia. They initially recommended the complete replacement of the United Kingdom's sectorized generic CMHTs with four separate 'functional' teams—*assertive outreach*, *CR/HT*, *early intervention*, and *primary care liaison* teams. The sheer impracticality of such a massive reorganization (for which there simply were not enough trained staff) and challenges to the

strength of the evidence modified the proposal, and CMHTs were retained and primary care liaison teams abandoned.

An ambitious programme to establish the first three types of functional teams was put in place, starting with the creation of 300 assertive outreach teams (AOTs). These three teams are described below. The research evidence is briefly addressed but, as demonstrated by that on AOTs, this can change radically over time. Although the NSF has ended, there are lessons to be learned about missed opportunities and where skills and staff shortages undermined the impact. Despite the NSF ending, some elements have been retained, for example, AOTs and CR/HT teams.

Assertive outreach teams

The most extensively replicated and researched specialist CMHT is the AOT. This is based on the *ACT model*, whose landmark study by Stein and Test (published in 1980 and mentioned earlier) ushered in evidence-based mental health service planning. ACT delivered improved clinical and social outcomes with substantially reduced hospitalization at slightly lower overall costs. Two Cochrane reviews with meta-analyses, one of case management showing an increase in hospitalization and one of ACT showing a reduction so leading to the adoption of the model in the United Kingdom (Marshall and Lockwood, 1998; Marshall et al., 2001). AOTs (see Box 26.1) are costly, requiring one full-time case manager for about ten patients. Consequently, they are reserved for the most difficult ('hard to engage' or 'revolving-door') psychotic patients, who have frequent, often dangerous, relapses.

The AOT approach is based on proactive outreach—visiting patients at home and persisting with visits even

when patients are reluctant. It exploits team-working, with daily meetings and several members working with most patients, rather than exclusive individual relationships. This is needed for reasons of safety (it may not be safe to visit some patients alone) and also because of the complexity of patients' needs. The approach is very practical. Staff take patients shopping, sort out their accommodation, even deliver their medicines daily if necessary. These activities go well beyond traditional professional boundaries. Advocates of AOT insisted that good outcomes depended on a strict replication of Stein and Test's approach and even developed scales to measure this ('model fidelity').

Who and what are AOTs for

Assertive outreach research has been conducted exclusively with psychotic patients, generally those with poor medication compliance and often with comorbid alcohol or drug abuse and offending behaviour. AOTs generally take only patients who cannot be stabilized despite CMHT support.

Enthusiasm for AOTs led to a preoccupation with their structure and process rather than with what they do. Despite the extensive early evidence of a reduction in hospitalization (Marshall and Lockwood, 1998), this has never been achieved against a CMHT control and never in the United Kingdom. Careful examination of AOT studies (Burns et al., 2007) demonstrated that this was because its practice was almost identical to standard UK CMHTs apart from the very small caseload and 24-hour availability, and these appeared to have no significant effect. In the United Kingdom, AOT teams are being abandoned and their caseloads resorbed into CMHTs. Internationally, however, ACT has become the accepted term for any multidisciplinary CMHT that offers flexible outreach.

Functional assertive community treatment teams

Functional assertive community treatment (FACT) teams are a response developed in the Netherlands to the shifting evidence base for AOT (Firn et al., 2018). FACT is a hybrid between an ACT team and a generic CMHT and uses 'zoning' to specify high-risk patients who need intensive monitoring and input. Zoning involves using a 'traffic light' classification of patients. Those in the red zone need weekly to daily contact and are considered at high risk.

Those in amber are routine patients needing weekly to monthly contact, often by a single case manager and not at great immediate risk of relapse. Those in the green zone are not in need of regular contact but

Box 26.1 Core components of assertive community treatment

- Assertive follow-up
- Small caseloads (1:10–1:15)
- Regular (daily) team meetings
- Frequent contact (weekly to daily)
- *In vivo* practice (treatment in home and neighbourhood)
- Emphasis on engagement and medication
- Support for family and carers
- Provision of services using all team members
- Crisis stabilization 24 hours a day, 7 days a week.

only occasional assessment and are usually allowed direct self-referral. Zoning is a welcome structuring of relatively longstanding CMHT practice. Only time and research will tell whether FACT teams are significantly different from CMHTs.

Crisis teams

Crisis intervention therapy in the early 1970s built on Caplan's belief that crises (when psychological defences were breached) presented fertile opportunities for psychotherapy breakthroughs (Caplan, 1964). More recently this psychodynamic focus has faded. Crisis teams have been promoted most where local services are poorly developed (they may be the *only* community services) or social mobility is high (e.g. centres of major cities), with many transient and homeless patients for whom continuity of care cannot be established.

Crisis teams are characterized by rapid response and accessibility. Defining the duration of a crisis (and hence the window for intervention) has generated endless unprofitable debate. Most teams will see patients immediately, certainly on the same day. Their clinical aims (i.e. the assessment and management of acute episodes of mental illness) and staffing (mainly nurses and social workers) are similar to those of CMHTs. Given the acuteness of the problems, medical input is probably essential, despite attempts in some services to dispense with it. Crisis teams may be based in mental health services alongside CMHTs, or in the emergency rooms of general hospitals with 24-hour availability. Important functions include distinguishing acute mental illness from severe personal crises (especially those involving self-harm) and accessing beds for both voluntary and involuntary admissions. Hospital-based crisis teams are often run in conjunction with liaison services (see Chapter 22).

Crisis resolution/home treatment teams

CR/HT teams, which were introduced by the NSF, were developed in Australia (Hoult, 1986). Their primary purpose is to avoid unnecessary admissions and to reduce the duration of inpatient care. They also are, in part, a response to increased consumer demand for access out of hours. They draw heavily on assertive outreach practice, with limited, shared caseloads, flexible working, extended access, and an emphasis on outreach. Reduction in hospitalization was proposed to offset much of their cost (Smyth and Hoult, 2000). They aim to treat patients who would otherwise be in hospital, and to focus on the severely mentally ill. They offer intensive visiting (usually daily for a limited period) and considerable practical support, and they work with patients'

social networks. Most aim to limit themselves to six weeks of involvement. Such intensive teamworking requires highly effective communication, and the teams meet daily (in some cases twice daily). Information transfer is burdensome, and liaison with CMHTs is complex, requiring absolute clarity about local arrangements for clinical responsibility.

Variations in practice and sustainability

Crisis teams are particularly difficult to research rigorously, as the informed consent required for randomized controlled trials is almost impossible to obtain. However, there is some evidence that CR/HT teams may reduce the need for hospital care (Johnson et al., 2005), although this is not clear-cut (Glover et al., 2006). As with AOTs, the potential for impact will depend on local provision. Although the UK model is well specified (including who it should and should not care for; see Box 26.2), in reality, practice varies considerably. Unless local services are particularly undeveloped, a full 24-hour service is very wasteful, and an on-call facility to the emergency room and police station at night usually suffices. In practice, contact frequencies are generally lower, patients stay with the service longer than anticipated, and restricting the service to patients who would otherwise be in hospital is rarely achieved. CR/HT teams are inevitably referred individuals who are

Box 26.2 Remit of UK crisis resolution/home treatment teams

Commonly, adults (16- to 65-years-old) with severe mental illness (schizophrenia, manic depressive disorders, severe depressive disorder) with an acute psychiatric crisis of such severity that, without the involvement of the CR/HT team, hospitalization would be necessary.

The service is not usually appropriate for individuals with:

- mild anxiety disorders
- primary diagnosis of alcohol or other substance abuse
- brain damage or other organic disorders, including dementia
- learning disabilities
- exclusive diagnosis of personality disorder
- recent history of self-harm but not suffering from a psychotic or serious depressive illness
- crisis related solely to relationship issues.

recurrently in crisis (often with alcohol and relationship problems), who cannot be refused care but would not ‘otherwise be in hospital’. Recent evidence from the National Confidential Inquiry into Suicide suggests that deaths by suicide have fallen during inpatient stays but are higher in the early period of discharge from hospital, including when people are cared for by CR/HT teams. Thus, suicide prevention is a critical ingredient in the evolving model.

Medical input and the team’s independence from CMHTs both vary markedly. Some teams operate completely parallel to CMHTs, some share extensively (often acting as an adjunct to CMHT care—e.g. making evening medication drops), and some insist on CMHT involvement. Local circumstances (and staff availability) will ultimately determine the configuration. Most CR/HT teams are now routinely involved at the threshold for admission and discharge. At its most extreme some NHS Trusts insist that no patient can be either admitted or discharged except through the CR/HT team. Other NHS Trusts take a more pragmatic approach—particularly when patients are well known to CMHTs.

Clarity about responsibility and leadership is crucial. Patient evaluation of CR/HT teams varies. Many appreciate the promptness of response and the level of support. Others complain about intrusiveness and having to explain themselves to a series of different staff who are often insufficiently trained to make the necessary decisions.

Traditional crisis services may have a relatively limited lifespan. A World Health Organization (WHO) review found that many early crisis services had ceased to exist within a decade (Cooper, 1979). Many had successfully improved local access and consolidated as a more general service (including, but not restricted to, crisis work). Others had become overwhelmed with inappropriate referrals or patients who could not be referred on, and eventually closed. A similar pattern has been observed (but not documented) in crisis houses. Whether CR/HT teams will persist in the era of clinical commissioning is an open question.

Crisis houses and respite care

The excessive formality involved in hospital admission may prolong stays and dissuade patients from seeking help. ‘Crisis houses’ allow admission with a minimum of fuss, and often with reduced supervision. They are usually small (four to eight beds), in a domestic setting, and take patients for a period of days, or occasionally a week or two. They are favoured for vulnerable women and early intervention services (EIS). They do not replace inpatient care but are generally very welcome for

a minority of patients. However, without careful targeting and integration into local services they can become chaotic (and potentially dangerous) or blocked, and few survive without an active local champion. Furthermore, not all services have a good range of crisis houses, although a flourishing charitable and voluntary sector do offer alternative support in less formal settings.

Early intervention services

EIS are the third functional team introduced by the NSF. As discussed in Chapter 11, a long duration of untreated psychosis (DUP) in schizophrenia is associated with poorer outcomes. EIS teams were developed mainly from Australian and UK models (Birchwood et al., 1998; Edwards et al., 2000). They aim to shorten DUP, protect current social functioning and networks, and improve outcome by providing prompt effective treatment. Despite a detailed prescription, UK teams vary considerably and prioritize different features of the approach. Some minimize the importance of diagnosis in favour of ease of access, while others restrict themselves to schizophrenia to maximize skills. The duration of involvement ranges from 18 to 36 months, with proposals of up to 60 months. Some have a narrow age band (a ‘youth service’), while others take all first-episode patients irrespective of age. Three quite different activities are reported in the EIS literature, which may have varying emphasis in different teams (see Box 26.3).

Case management

The core of EIS is a specialized CMHT that case-manages first-episode psychosis patients. Teams prioritize the protection of social networks and functioning (by keeping the patient at college or work, and emphasizing family interventions, etc.). They assume that there will be a return to premorbid functioning, and hospitalization is avoided (crisis and respite houses are preferred) and obtaining welfare benefits is minimized. An assertive

Box 26.3 Components of early intervention teams

- *Case management*—ongoing care of identified patients
- *Early identification*—awareness-raising campaigns for psychoses
- *High-risk and prodromal patient identification and treatment.*

outreach approach with practical and shared case management is used during the acute phase, with less intensive follow-up thereafter. 'Zoning' is commonly used, as in FACT teams, with 'acute' patients seen at least weekly, recovering patients seen weekly to monthly, and a third category for those who are still open to the team but who are not in active treatment. Teams differ in terms of whether they insist on keeping patients registered when they have recovered and do not want contact. They also may or may not provide inpatient care.

EIS case managers aim to prevent patients and their families from lowering their expectations prematurely. As mentioned, many avoid obtaining benefits for patients, encouraging as rapid a return to work or college as possible. Self-management of illness is strongly endorsed, with an emphasis on learning relapse signals and devising individually appropriate responses.

Early recognition and high-risk intervention

Some EIS teams are actively involved in psychosis public awareness campaigns and lecturing in schools and colleges. A minority attempt to identify 'ultra-high-risk' patients with prodromal symptoms of schizophrenia and offer them treatment (both psychological and pharmacological) to prevent 'transition' to psychosis. This approach remains highly controversial and is generally restricted to research teams. The evidence for the efficacy of EIS interventions is outlined in Chapter 11.

Experience with EIS teams was much less than with AOTs when they were introduced in the NSF, and they included confusing and contradictory prescriptions. Even more than with AOTs, the difficulties of making them work in dispersed rural and semi-rural settings have posed real challenges.

A settled single model has not yet evolved for EIS teams. However, they are now well established nationally.

Forensic and rehabilitation teams

Community-focused services face particular difficulties with treatment-resistant patients, especially those with socially unacceptable or offending behaviour. Such patients do not fit well into open wards. Specialized forensic teams provide care where offending behaviour and danger to others predominate. They provide secure inpatient care (in addition to their traditional roles with prisons and the courts), which is increasing internationally (Priebe et al., 2005). Some forensic teams provide

community services (intensive case management of dangerous patients) but integrating them with general services can be controversial. The focus on risk assessment and extensive joint visiting are the main ways in which they differ from AOTs.

Rehabilitation services have reached out from long-stay wards to more community placements. A significant number of patients remain profoundly disabled despite receiving optimum treatments. Rehabilitation teams focus on the long-term management of disability rather than on symptoms and episode-based care. The threshold from general services to rehabilitation services varies enormously. Rehabilitation teams generally serve patients who cannot survive without some form of supervised accommodation even between relapses. These include a shrinking handful of long-stay patients, but, increasingly, a very disturbed 'new long-stay' population with comorbid substance abuse and behavioural disturbances. Inpatient provision is needed for this group, and a range of hostel care can promote stability, but is difficult to sustain. 'Hospital hostels', where patients can live in hostel-type domestic accommodation, yet be subject to the MHA, have been successfully established by some rehabilitation services.

Diagnosis-specific teams

Highly specialized teams for specific disorders (e.g. eating disorders, autism, personality disorders, bipolar disorder) focus on developing specific skills. They provide specialized treatments, usually at regional level. They are strongly supported both by their professionals and by the families of sufferers. However, the opportunity costs of establishing such teams in a health service with finite resources require careful consideration. There are moves to ensure CMHTs and primary care liaison address all conditions, so that more expensive specialist teams are not essential or used in a targeted manner.

Other components of a mental health service

The main components (teams and facilities) of specialized mental health services have been outlined above. It will be clear from that description that some of these components (e.g. inpatient wards, generic-sector CMHTs) are fairly widespread and durable, while others (e.g. most of the specialized teams) are more variable in their availability and practice. The way in which the

functions and obligations of mental health services are distributed between these different teams will vary. The most important of these components are outlined and described here. They can be summarized as follows:

1. *Primary mental health care teams*, which support primary care and conduct initial assessments.
2. *Outpatient care*.
3. *Specialist psychological treatments*, which are often integrated into the specialist or primary mental health care teams.
4. *Crisis/home-based treatment teams*, which treat acute mental disorder in the patient's home.
5. *Accident and emergency liaison services*, which are often part of the crisis service, that assess and treat people who present to accident and emergency departments.
6. *General hospital liaison services* for medically ill people who have mental health problems.
7. *Assertive outreach teams* for severely mentally ill people who are difficult to engage or high users of services.
8. *Early intervention services* for people during their first episode of psychosis and the following three years.
9. *Employment services*, including supported employment to help find paid employment, training, and education.
10. *Vocational rehabilitation*, which includes sheltered workshops and transitional employment.
11. *Day hospitals and daycare services*, including drop-in centres.
12. *Self-help and service user groups and advocacy services*.
13. *Inpatient wards*.
14. *Forensic services*, with wards for high, medium, and low levels of security, and court-diversion schemes.
15. *Residential services*, providing hostels and supported accommodation, including that for acutely ill people or crises.
16. *Services for specific groups*, such as people with eating disorders, and mother and baby units.
17. *Social and welfare services*, for help with obtaining benefits, advocacy, community support workers, and home help/meals.
18. *Services for special groups*, with outreach to marginalized groups, such as the homeless and refugees.

Services for psychiatric disorder in primary care

Classification

ICD-10 and ICD-11 were developed for use in psychiatry. They are too detailed for routine use in primary care, and their fine distinctions are seldom helpful when selecting treatment in this setting. The WHO has therefore developed a simpler classification for use in primary care. In separate guidance, diagnoses are linked to a care plan.

Identification

The most important psychiatric service in primary care is to detect disorders. This is not a simple matter, as many patients present with both physical and psychological problems, and many of the latter go undetected (see Figure 26.1). Many people with a psychiatric disorder complain only of physical symptoms, whether of a coincidental minor physical illness or a consequence of the psychiatric disorder (e.g. palpitations in anxiety, or tiredness in depression). Some describe physical symptoms because they fear the doctor may not take emotional problems seriously. It is not just the patient who

is responsible for this, as doctors often do focus on the physical symptoms. *Conspicuous morbidity* is the term for cases that are detected, and *hidden morbidity* for the rest. Hidden morbidity is generally less severe than conspicuous morbidity.

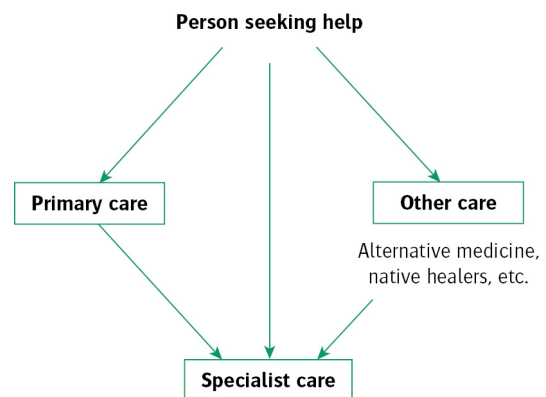


Figure 26.1 Pathways to care in the United Kingdom.

How effectively primary care team members identify an undeclared psychiatric disorder depends on:

- their ability to gain the patient's confidence, thus enabling them to disclose the psychiatric problems of which they are aware;
- their skill in assessing whether physical symptoms are caused by psychiatric illness, which requires a good knowledge of physical medicine as well as psychiatry; and
- their familiarity with the family and frequency of seeing patients over a long period of time for multiple reasons.

Psychiatric disorders treated in primary care

Most psychiatric disorders in primary care attenders can be treated successfully there. Examples include most adjustment disorders, the less severe anxiety and depressive disorders, somatization, and some cases of alcohol misuse. For a more detailed account, see Goldberg and Bridges (1988), and Strathdee (1987).

Patients presenting with physical symptoms. In primary care, many patients with a psychiatric disorder present with physical symptoms. Such patients were called *somatizers*, but the terminology has moved to '*medically unexplained symptoms*'. Many of them will reveal associated psychiatric symptoms and accept a psychiatric diagnosis when interviewed appropriately. Others deny any psychiatric symptoms and reject a psychiatric diagnosis, however skilfully they are examined, and can prove very difficult to treat. Where the concern about physical illness is unshakeable and no other psychiatric disorder is detected, the diagnosis of hypochondriasis is made. Repeated investigations fail to reassure but can be impossible to resist. Most frequent attenders are female and middle-aged (Koch et al., 2007). Around 80% have a present or past psychiatric disorder, usually depression, somatization, or generalized anxiety, and around 60% have a concurrent physical illness (Gili et al., 2011). Since most refuse to be referred to a psychiatric team, the general practitioner must manage them. One approach is to help to make links between symptoms and stressful life events, leaving out psychological processes, and explore how these stresses might be reduced.

Disorders referred from primary care to psychiatric services

Table 26.1 shows that, on average, about one in four of patients with a psychiatric disorder who are identified by GPs are treated by the psychiatric services. This referred group includes patients with severe depressive disorders,

Table 26.1 Pathways to care with rates of psychiatric disorder among adults at each level of care

	Cases per 1000 per annum
In the community	260–315
Attending primary care	230
Detected in primary care [†]	102
Attending psychiatric services	24
Inpatient	6

[†] Conspicuous psychiatric morbidity.

Adapted from Goldberg, D., & Huxley, P. (1992). *Common Mental Disorders: A Biosocial Model*. Copyright (1992), with permission from Routledge.

bipolar disorders, schizophrenia, and dementia. GPs are more likely to refer patients with other disorders when:

- the diagnosis is uncertain
- the condition is severe
- there is a significant suicide risk
- the condition is chronic
- the necessary treatment cannot be provided by a member of the primary care team
- previous treatment in primary care has been unsuccessful
- psychiatric services are accessible and responsive
- the patient is willing to attend.

Treatments provided by the primary care team for acute disorders

Acute problems are often treated with counselling, used alone or combined with medication. Some practices also provide simple behavioural treatments. Many primary care teams include a counsellor. The availability of a counsellor has not been shown to reduce the prescribing of psychotropic drugs in the practice (Mynors-Wallis et al., 2000; Sibbald et al., 1996).

Improving access to psychological therapies

Psychotherapy has traditionally been a less prominent aspect of health care in the United Kingdom compared to the United States or continental Europe. Psychotherapists there are either accessed directly or via

GPs and are paid by insurance or privately. Improving access to psychological therapies, or IAPT (Clark, 2011), is an initiative to reverse this by providing easy access to basic psychotherapy (almost exclusively cognitive behavioural therapy). IAPT is a highly prescribed process, but local flexibility has been encouraged regarding who provides it and where IAPT has two components—a stepped-care treatment of anxiety and depressive disorders, and a supported employment service (see below and Chapter 24 on psychological treatments). The supported employment service aims to help patients to return to work when their anxiety and depression symptoms have receded (reducing unemployment owing to psychiatric disorders was the other political stimulus for IAPT).

The first step in IAPT is self-help using telephone and internet programmes. A form of brief cognitive behaviour therapy is offered to those with residual symptoms, and referral on to specialized services only takes place after both self-help attempts and cognitive behaviour therapy. A striking feature of IAPT is the level of structure—therapists use detailed manuals, and symptomatic status is recorded after every session using the Patient Health Questionnaire (PHQ-9) (Kroenke et al., 2001) for depressive symptoms and the Generalized Anxiety Disorder Assessment (GAD-7) (Spitzer et al., 2006)) for anxiety symptoms.

The format of IAPT is highly variable and is (as yet) unevaluated. The ease of access and clear commitment to psychological treatments have won widespread approval. Those who provide the service are positive about its outcome, but to date there has been little evidence of diversion from psychiatric services nor independent research evidence of its impact.

Treatments provided by the primary care team for chronic disorders

The respective roles of the GP and the community team should be clearly defined in relation to any patient with chronic mental disorder and reviewed regularly. For example, with some patients with chronic schizophrenia, the GP might be responsible for their physical health, assess their general progress, administer and encourage compliance with medication, and support the family. Kendrick et al. (1995) found that, even with additional training, GPs are not very reliable at making comprehensive assessments of patients with long-term psychiatric illness. However, subsequent research demonstrated that this was possible, and financial incentives were introduced to primary care, which has improved performance in this area. Physical monitoring of psychosis patients has become high priority, with the recognition of their dramatically reduced life expectancy

(about 20 years) and the high rates of metabolic syndrome associated with the atypical antipsychotics. It has become more routine that the psychiatric team assesses these patients and agrees a plan with the GP.

Work in primary care by the psychiatric team

There are several ways in which a psychiatric team can work with the primary care team, and these have changed over time (Burns and Bale, 1997)

Advising and training GPs and their staff (Balint groups)

Balint groups are valued by a small group of highly motivated GPs; while some do persist, they are mainly of historical interest. The psychiatrist does not see patients, but gives advice based on the GP's assessment and is psychoanalytically inspired (Balint, 1964); www.balint.co.uk).

Assessing and referring

The psychiatrist assesses patients when the GP is uncertain about diagnosis or treatment. They may do this on their own, or jointly with the GP, after which referral can proceed in the usual way. This approach is now very rarely used, as liaison with GPs has improved.

Assessing and treating ('shifted outpatients')

There are intermittent initiatives to encourage psychiatrists to work mainly in primary care, seeing most patients there or at home, rather than in the hospital outpatient clinic. The approach rarely catches on although it often has vocal local champions. Clinical psychologists and psychiatric nurses can also work in primary care, providing assessment, counselling, or behavioural treatment. Patients no longer need to visit a psychiatric clinic, but there may be little learning between the psychiatric and primary care teams. This approach was popular in the 1980s but was shown to be generally ineffective (Wooff and Goldberg, 1988), as clinical focus shifted and skills were lost. There is more positive evidence for social workers located in primary care.

Shared care

This approach aims for 'seamless care', with enthusiastic advocates (Essex et al., 1990) and several well-publicized demonstration services. However, these demonstration services rarely survived the departure of their advocates, and shared care has not caught on generally. The term is ambiguous, covering everything from GPs monitoring lithium levels to completely shared responsibility. In the care of severely mentally ill patients, absolute clarity

of responsibility has repeatedly been shown to be crucial (hence the CPA), and the risk of confusion does not seem to be worth taking.

Liaison meetings

Probably the most useful and time-efficient method is to arrange regular (usually monthly) face-to-face meetings between members of the psychiatric team and the primary care team (Burns and Bale, 1997) This

can range from meetings of the entire teams to relying on identified 'link' members. The important point is that it is face to face and takes place routinely—not just over crises. Joint patients are discussed, together with possible referrals. Trust is established, along with a more realistic appraisal of what each team can do. Such liaison meetings are increasingly required by commissioners.

Specialist services for acute psychiatric disorder

Patients referred to specialist care

Patients who are treated by psychiatric services are a subgroup of people with mental disorders. In some countries patients can go directly to a specialist, so that those treated by the psychiatric services may not be very different from those treated in primary care. Where the GP acts as the 'gatekeeper' to specialist services, as in the United Kingdom, the number and types of patient reaching the psychiatric services depend on:

- the willingness of GPs to treat psychiatric disorders
- the treatment skills and resources of the primary care team
- patients' willingness to attend for specialist psychiatric advice
- the GP's criteria for referral to the psychiatric services
- the psychiatric services' criteria for accepting referrals.

In the United Kingdom, most patients in contact with the psychiatric services have severe and chronic anxiety disorders, severe mood disorder, schizophrenia and other psychoses, or dementia. Patients who are particularly likely to be cared for by specialists are those who are suicidal, those who are potentially dangerous to others, and those with dual diagnoses of mental disorder and substance misuse or personality disorder.

Generic versus specialized services

Community teams have traditionally been generic, dealing with all kinds of adult psychiatric problems. More recent specialized separate teams for crisis management and continuing care are described above. The potential advantages of each of these arrangements are summarized in Table 26.2.

Psychiatric services that provide long-term care

Characteristics of patients who require long-term care

Diagnosis

With the exception of the elderly (who are considered in Chapter 19), most psychiatric patients who require long-term care have schizophrenia, chronic mood disorders, or personality disorders associated with aggressive behaviour or substance misuse. Patients who need care in hospital for longer than one year, other than forensic patients, are referred to as 'new long-stay'. This distinguishes them from the 'old long-stay', who had been residents in hospital closure programmes.

Problems

There are several ways of classifying the problems of patients who need long-term psychiatric care. They can be divided into seven groups, three of which are contained in the WHO's classification of disablement (page 23), on the basis of the following problems:

- *Symptoms* such as persistent hallucinations or suicidal ideas.
- *Unacceptable behaviours*, such as shouting obscenities, and threatening or carrying out violent acts.
- *Impairments* that interfere with the functioning of a psychological or physical system—for example, poor memory or lack of drive.

Table 26.2 Generic versus specialist services

Advantages of generic services	Drawbacks of specialist services
Simple model, easy to implement	Complex model, difficult to implement
Traditional, tried and tested	New, relatively untried in clinical practice
Comprehensive	Problem of transitions between teams or falling between different team criteria
Less expensive	More expensive
Continuity of care is easier	Continuity is more difficult to achieve
Drawbacks of generic services	Advantages of specialist services
Evidence base is weak	Evidence base better developed
Less efficient—too many different tasks	More efficient with a smaller range of tasks
Less staff satisfaction	More staff satisfaction
Difficult to deliver complex treatments	Easier to deliver complex treatments
Heterogeneous patients, outcomes difficult to evaluate	Homogeneous patients, outcomes easier to assess

- *Disabilities* that interfere with the activities of the whole person, such as inability to dress.
- *Handicaps*, which are social disadvantages directly resulting from disability—for example, inability to work, or to care for children.
- *Other social disadvantages* that are not directly related to disability, but which are a consequence of the *stigmatizing attitudes* of other people—for example, unemployment, poverty, and homelessness.
- *Adverse personal reactions* to illness and social disadvantage—for example, low self-esteem, hopelessness, denial of illness, or the misuse of drugs.

Wing and Furlong (1986) proposed a list of patient characteristics that make it difficult to treat the individual in the community, which, despite its age, is still useful (see Box 26.4). Patients with severe and persistent problems of this kind may need care in a well-staffed hostel, or in a hospital that can provide appropriate rehabilitation and security.

Requirements of a community service providing long-term care

To treat patients in the community, they need all the elements of care previously provided in hospital. The following provisions are required:

1. Suitable and well-supported carers.
2. Appropriate accommodation.

Box 26.4 Patient characteristics that make community care difficult

- Risk of harm to self and others
- Unpredictable behaviour and liability to relapse
- Substance misuse
- Poor motivation and poor capacity for self-management or for performance of social roles
- Lack of insight into the need for treatment
- Low public acceptability.

3. Suitable occupation.
4. Arrangements that enlist the patient's collaboration with treatment.
5. Regular reassessment, including physical health.
6. Effective collaboration among carers (formal and informal).
7. Continuity of care and rapid response to crises.

Complicated and expensive arrangements are required to make these seven elements available as readily in the community as they would be in hospital. Lack of such arrangements may leave patients homeless, without constructive occupation, inadequately treated, and without a carer. Failure of community care may also leave carers unsupported and family life disrupted.

When community care began, it was concerned mainly with patients who had become institutionalized and compliant after many years in hospital. These patients could often be managed in the community without much difficulty. Younger patients who have spent less time in hospital are often less compliant and more difficult to manage.

Carers

When people with severe mental illness live at home, *family* and *friends* are the main carers. They provide much of the help that would have been given by nurses had the patient remained in hospital. For example, they may have to encourage the patient to get up in the morning, maintain personal hygiene, eat regular meals, and occupy himself or herself constructively. Carers also encourage collaboration with treatment. If patients have many problem behaviours, prolonged involvement in their care is stressful. Carers may then need support and advice, and sometimes periods of respite.

Volunteers play an important part in many systems of community care. Trained volunteers can help to support patients and their families, and some charitable organizations employ professional carers, such as hostel staff.

Community psychiatric nurses are the backbone of community care by supporting patients and carers, evaluating patients, supervising drug therapy, and encouraging social interaction. Monitoring is particularly important for patients whose mental state is unstable.

Accommodation

Patients who have been discharged from hospital need food and shelter. Many live with their families, and some care for themselves in rented accommodation. Others require more extensive help, which can be provided in three ways.

In lodgings. Some families take in patients as lodgers and provide them with extra care. This practice works well in several countries but has not been widely adopted in the United Kingdom.

In group homes. Many patients live in group homes of four or five patients together. The houses may be owned by social or health services or by a charity. This model works well for people with schizophrenia as well as for those with social handicaps but few positive symptoms. People in group homes receive regular support and supervision, usually from a community nurse or social worker. They ensure the arrangements are working well and encourage residents to take on as much responsibility as possible.

In hostels. For some people, hostels are half-way houses from which they move on to more independent

living. Others need to remain in hostels for a period of years. Although most hostel residents live fairly independent lives, a few of the most disabled require additional care. Hostels have permanent staffing. Levels of supervision vary, and overnight staff may sleep in, or do night shifts, as in a hospital ward.

Occupation

Most patients would like to work, although rates of unemployment are higher in mental illness than any other disabled group. There are several approaches to increasing work and employment:

- *Vocational rehabilitation* is the longest established approach. Extensive training based on a detailed assessment of disabilities and following a structured programme is provided to develop the skills and confidence needed for competitive employment. Training may take place in sheltered workshops, with transitional jobs in paid posts 'owned' by the rehabilitation scheme. Progress to open employment is disappointing.
- *The clubhouse model* is a specialized form of vocational rehabilitation that is based around a building which is jointly administered and maintained by people with mental illness (Beard et al., 1978). Originating in the United States, it has established itself internationally but is not widespread and may be fading out.
- *Supported employment* was originally developed for people with learning disabilities. Individuals are placed in competitive work without any extended preparation, but with support on the job (Bond and Drake, 2014). It has been shown in several randomized controlled trials (Mueser et al., 2016) to be the most effective way of obtaining open employment. Most of the studies have been conducted in the United States, but a European replication (Burns et al., 2007; Knapp et al., 2013) confirmed equal efficacy (over 50% of psychotic patients became employed) and cost-benefit. It is government policy in the United Kingdom and several European countries and is part of the IAPT programme, but provision remains patchy.
- *Sheltered work* allows people to work productively but more slowly than would be acceptable elsewhere. Schemes include horticulture projects and craft workshops.
- *Voluntary activities.* Some patients who cannot meet the requirements for sheltered work benefit from undertaking voluntary activities.
- *Occupational therapy.* Those unable to contemplate work can benefit from occupational therapy to reduce

boredom, under-stimulation, and lack of social contacts. Occupational activities are often provided in a day hospital or a day centre.

Reassessment

Patients living outside hospital require the same regular reassessments. The patient's mental state and of collaboration with treatment is assessed by the key worker, often a community nurse, with less frequent reassessment by a psychiatrist. Patients can forget appointments, and a recall system prompts re-establishing contact. Physical health reviews are essential because such patients often neglect physical illness or may not accept offered help.

Risk assessment. It is more difficult to anticipate threatening or dangerous behaviour in community care than in hospital. An important part of assessment is a regular evaluation of the patient's risk to himself or herself, and to other people. Such assessment involves *static risk factors*, such as the patient's past behaviours, and *dynamic risk factors*, such as the current level of substance misuse or of psychosocial stressors. A risk management plan is an important component of the overall care plan. The scientific value of structured risk assessment has been disputed (see Chapter 18). However, they demonstrate that due consideration has been given of possible adverse consequences.

Continuity of care

Community care staff need to gain the confidence of their patients and to know them well. Neither can be achieved if there are frequent changes in staff. Continuity of care is important, and staff should be extra vigilant when care passes between workers.

Response to crises

Staff need to respond quickly to crises. Families and the staff of voluntary organizations accept patients more readily if they know that help will be available quickly in an emergency. Furthermore, re-admission to hospital may be avoided by prompt action. A crisis plan agreed in advance with the patient and carers is particularly helpful. Staffing levels need to be adequate to allow a quick response, preferably by staff who know the patient.

Working with family and volunteers

Community care is costly, and in most countries public funds are limited, so arrangements often depend on input from families and voluntary groups, who should be involved in the planning of services, and there should be agreement about their responsibilities. Without such an agreement, family may feel responsible for over-demanding tasks, and professionals concerned that

volunteers are taking on tasks beyond their capabilities. Similarly, it is best practice to involve families and voluntary groups in the evaluation of services.

Other components of a community service

Rehabilitation and recovery

Rehabilitation denotes procedures for helping patients to reach and maintain their best level of functioning. It is based on a careful assessment of disabilities and deficits and relies on structured gradual training. It may be provided in an inpatient unit, a day hospital, or a specialized rehabilitation centre. Rehabilitation procedures include medical, psychological, occupational, social, and residential. Most patients require medication to control symptoms of schizophrenia or chronic mood disorders. Psychological methods include supportive therapy, behavioural programmes, and social skills training (see Chapter 24). Work helps to structure the day and provides an opportunity for interaction with other people. It can improve self-esteem, and payment is an added incentive. For many the aim is to return to ordinary work, but those who cannot achieve this are trained for activities such as gardening, crafts, and cooking. These provide a sense of achievement and a shape to the day. Whenever practicable, patients should be encouraged to join the same social groups attended by healthy people.

The recovery movement

'Recovery' has taken on a quite specific meaning in recent years beyond the absence of symptoms. The 'recovery movement' (Slade et al., 2014) emphasizes the primacy of patient determination in treatment planning and goals. What matters is the quality of life, not the absence of symptoms. Clearly it is eminently sensible to recognize that the goal of treatment is to enable patients to live the life they wish and to make the best adaptation to their residual problems. However, the approach raises some problems for practitioners. Firstly, much of the rhetoric unnecessarily caricatures psychiatric practice as narrow and mechanical.

A tenet of the recovery movement is that psychiatrists are overly pessimistic about outcomes, and it calls on the profession to be more hopeful and 'upbeat'. Most professionals believe they owe their patients honesty. Lastly, some recovery enthusiasts insist that mental health professionals should support the patient's treatment choices even if they believe them wrong. The recovery movement is a good reminder that social and personal outcomes are what matter to patients and

families. Hopefully this will persist when the excesses have faded.

Inpatient care

People with long-term mental illness require inpatient care for acute treatment at times of relapse, for intensive rehabilitation, and in some instances for long stay. The basic requirements for such an

inpatient unit are broadly similar to those for patients with acute illness, except that accommodation can be more domestic. Because the pace and intensity of treatment is usually slower for this group, it is advisable to have a separate facility. Patients with severely disturbed behaviour need secure areas, with outdoor space and occupational and social activities in the same building.

Services for people with particular needs

Clinical work and research have identified several groups of patients whose needs may require special attention.

Members of ethnic minorities and migrants

Members of ethnic minorities are less likely to use routine services. These problems are not primarily due to a limited command of English.

The problems of minorities in the United Kingdom have been studied, in particular among those of South Asian and African-Caribbean origin. People of South Asian origin consult their GPs more frequently about most conditions than do members of the general population, but consult them less about psychiatric symptoms (Halvorsrud et al., 2018). African-Caribbean patients with psychoses lack family support and often bypass GP referral and come to services via the police or criminal justice system. When such patients do present to primary care with mental disorders, they are more likely to be referred to specialist care (Bhui et al., 2003; Halvorsrud et al., 2018).

Under-recognition and over-admission of psychiatric disorder

In the 1980s and 1990s much was made of professionals' overdiagnosis of psychosis in African and African-Caribbean patients based on a lack of cultural sensitivity (Lipsedge and Littlewood, 1997). The AESOP study has dispelled this contention, confirming that the rates are indeed higher and the outcomes, if anything, marginally worse rather than better as they would be as the consequence of misdiagnosis (Fearon et al., 2006). Much effort is being expended on considering cultural adaptations to services, improving access, and offering a broader and more flexible range of options for people living in precarity. Establishing trust and early intervention as well as public health literacy about mental health problems are

part of the solution. Partnerships between voluntary and statutory sector agencies is one mechanism to improve the quality of care (Moffat et al., 2009; Sass et al., 2009). Diagnosis as a form of stigma is a relevant concern for many people, irrespective of age, gender, and ethnicity, and improving classifications of diagnosis for all to be more closely linked with recovery and precision medicine offers some hope.

The drivers of a higher incidence of psychosis and differing pathways to care are found to be sociological, for example, adverse experiences, deprivation, discrimination, requiring more flexibility and cultural adaptation of interventions (see Chapter 24). Culturally competent care and more effective therapeutic communications are also critical elements of professional skills and service design (Bhui et al., 2007; Bhui et al., 2012; Bhui et al., 2015). The ICD-11 and DSM-5 both provide guidance. The DSM-5 includes a cultural interview, which sets out the relevant domains of assessment, including information on racism and social and structural assessments (Aggarwal et al., 2020).

Homeless mentally ill people

When hospitals were closed, it was feared that many discharged patients would become homeless, but when closures were planned carefully, few of them became so (Harrison et al., 1994; Leff, 1993). Nevertheless, surveys have found high rates of chronic psychiatric disorder among the residents of hostels for the homeless (Fazel et al., 2014). The most common disorders were alcohol and drug dependence. Psychotic illness and major depression each ranged from 3% to 42%. Alcohol dependence has increased over recent decades. Services for the homeless mentally ill vary enormously, and there is neither agreement on their need nor on their characteristics (Hwang and Burns, 2014). Specialist homeless teams have been part of the NHS and other systems

globally, although they are rarely sustained and require joint funds from social care, health care, and public health agencies, given the complexities of problems faced by homeless people, including premature mortality. Homeless people often experience multiple health problems, and have multiple adverse experiences early on in their lives, and require personalized care and advocacy, and challenges to organizational policies and practice that exclude them (Bhui et al., 2006).

Refugees

Refugees have the general problems of members of ethnic minority groups (described above), together with specific problems secondary to the experiences that led them to seek refuge in another country. These experiences include persecution, physical injury, torture, rape, or the witnessing of these. The consequences include general medical as well as psychiatric conditions. The latter are mainly post-traumatic stress disorder and depressive disorder.

The specifics of care needed by refugees should be set in the context of many of the principles of global mental

health, dealt with further in Chapter 23. At the very least they need integrated medical and psychiatric care from a team that is sensitive to cultural factors. Integrated care is important because general and psychiatric disorders occur together, and because refugees often come from cultures in which distress and psychiatric disorders are generally manifested by physical symptoms. In addition, aid workers are sometimes reluctant to refer refugees to a solely psychiatric service, which they think inappropriate for what appears to be a normal response to overwhelmingly stressful circumstances. As with the homeless mentally ill, acceptability is much greater with integrated services.

Staff who provide services for refugees should be experienced in the treatment of post-traumatic conditions and should be aided by interpreters. Female members of staff are needed to help female refugees, especially those who have experienced rape. Other staff should have experience in treating children and adolescents (Fazel et al., 2015). For a review of the psychiatric problems of refugees, see Bhui (2022), Lindert et al. (2016), Murray et al. (2010), and Chapter 7.

Some difficulties with community care

The burden on relatives

If members of the family are to take responsibility for patients, by housing them, encouraging adaptive behaviour, supervising their medication, and reporting signs of relapse, they need to be well informed, adequately supported, and able to obtain help in an emergency. Such support is time consuming and is rarely available unless community care is well resourced and the needs of the family are given high priority. In the United Kingdom a carer assessment is obligatory with long-term disorders, yet many carers report poor communication from psychiatric services. Staff experience continuing conflict over their duty of confidentiality to the patient and their wish to support and inform relatives. There is no easy answer to this dilemma. Carers report the greatest difficulties in coping with negative symptoms and socially embarrassing or aggressive behaviour, and anticipatory anxiety about such behaviour and long-term prognosis are regularly reported as more burdensome than practical help. Care plans should always consider the possible effects of the patient's illness on any children in the home.

Problems with the distribution of resources

When resources are limited, there is a conflict between the needs of patients with acute disorders and those with chronic disorders. The conflict is most evident in primary care, where the former group of patients is generally more demanding of care and more responsive to it.

Problems with the coordination of services

In most countries, long-stay hospital care for patients with a chronic psychiatric disorder is provided by a single agency (a hospital authority). Community care requires coordinated action by several agencies, each of which usually has other responsibilities—for example, social services departments have responsibilities for mentally healthy children and frail elderly people as well as for psychiatric patients. This broad responsibility generates endless problems regarding the allocation of resources. Regular negotiation is the only possible response.

International service principles

In countries that have few specialist psychiatric resources, it is essential to decide on priorities. The WHO (1984) has identified four priorities:

1. Rapid response to psychiatric emergencies.
2. Provisions for chronic severe psychiatric disorder.
3. Care for psychiatric disorders associated with general medical illness.
4. Care for any high-risk groups that are found in the country (e.g. drug misusers).

The WHO advocates the training of *auxiliary workers* who can supplement the efforts of fully trained staff. Thus, in countries with few trained psychiatrists, resources may be used most effectively by improving the skills of nurses who can perform the first-line management

of psychiatric disorders, supported by GPs (see also Chapter 23).

The WHO has released a new report on global mental health in 2022, taking stock of the global burden of mental disorders and societal and personal impacts (WHO, 2022). The report emphasizes:

1. More capacity and effective interventions suited to low- and middle-income countries;
2. Better integration of community, primary, and secondary care;
3. Prevention efforts and addressing the social determinants of mental illnesses; and
4. The need for public mental health efforts to be integrated with care systems, taking account of contexts (see Bhugra and Bhui, 2018).

Further reading

Various authors. (eds) (2020). Section 23: Service provision. In: J Geddes, NC Andreasen, GM Goodwin (Eds.), *The New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1363–1411). Oxford University Press.

Goldberg, D., & Huxley, P. (1992). *Common Mental Disorders: A Biosocial Model*. Routledge. (This book is still valuable for

its account of the development and utility of the ‘filter’ model, which relates epidemiology to the provision of services.)

Thornicroft, G., & Szmukler, G. (2001). *Textbook of Community Psychiatry*. Oxford University Press. (A comprehensive work of reference.)

References

- Abajobir, A. A., Maravilla, J. C., Alati, R., & Najman, J. M. (2016). A systematic review and meta-analysis of the association between unintended pregnancy and perinatal depression. *J Affect Dis*, 192, 56–63.
- Abdallah, C. G., Averill, L. A., Akiki, T. J., et al. (2019). The neurobiology and pharmacotherapy of posttraumatic stress disorder. *Annu Rev Pharmacol Toxicol*, 59, 171–189. <https://doi.org/10.1146/annurev-pharmtox-010818-021701>
- Abderhalden, C., Needham, I., Dassen, T., et al. (2008). Structured risk assessment and violence in acute psychiatric wards: Randomised controlled trial. *BJPsych*, 193(1), 44–50.
- Abraham, A., & Luty, J. (2010). Testing for illicit drug use in mental health services. *Adv Psychiatric Treat*, 16(5), 369–379.
- Abrahamyan Empson, L., Baumann, P. S., Söderström, O., et al. (2020). Urbanicity: The need for new avenues to explore the link between urban living and psychosis. *Early Interv Psychiatry*, 14(4), 398–409. <https://doi.org/10.1111/eip.12861>
- Ackner, B., & Oldham, A. (1962). Insulin treatment of schizophrenia: A three-year follow-up of a controlled study. *Lancet*, 279(7228), 504–506.
- Adams, J., & White, M. (2004). When the population approach to prevention puts the health of individuals at risk. *Int J Epidemiol*, 34(1), 40–43. doi:10.1093/ije/dyh232
- Adams, R., Fisher, C., Hakim, S., et al. (1965). Symptomatic occult hydrocephalus with normal cerebrospinal-fluid pressure: A treatable syndrome. *N Engl J Med*, 273(3), 117–126.
- Adshead, G., & Fonagy, P. (2012). How does psychotherapy work? The self and its disorders. *Adv Psychiatric Treat*, 18(4), 242–249. <https://doi.org/10.1192/apt.bp.111.009274>
- Aggarwal, N. K., Jarvis, G. E., Gómez-Carrillo, A., et al. (2020). The Cultural Formulation Interview since DSM-5: Prospects for training, research, and clinical practice. *Transcult Psychiatry*, 57(4), 496–514. <https://doi.org/10.1177/1363461520940481>
- Agnew-Davies, R. (2016). Domestic violence and women's mental health. In D. J. Castle & K. M. Abel (Eds.), *Comprehensive Women's Mental Health* (pp. 161–173). Cambridge University Press. <https://doi.org/DOI:10.1017/CBO9781107045132.015>
- Agorastos, A., Haasen, C., & Huber, C. G. (2012). Anxiety disorders through a transcultural perspective: Implications for migrants. *Psychopathol*, 45(2), 67–77.
- Ahmad, A., Anderson, K. N., & Watson, S. (2021). Sleep and circadian rhythm disorder. In A. H. Young & M. F. Juruena (Eds.), *Bipolar Disorder: From Neuroscience to Treatment* (pp. 133–147). Springer.
- Ainsworth, M. D., & Bell, S. M. (1970). Attachment, exploration, and separation: Illustrated by the behavior of one-year-olds in a strange situation. *Child Devel*, 41(1), 49–67. <https://doi.org/10.2307/1127388>
- Akhavan, A. A., Sandhu, S., Ndem, I., & Ogunleye, A. A. (2021). A review of gender affirmation surgery: What we know, and what we need to know. *Surgery*, 170(1), 336–340. <https://doi.org/https://doi.org/10.1016/j.surg.2021.02.013>
- Alameda, L., Trotta, G., Quigley, H., et al. (2022). Can epigenetics shine a light on the biological pathways underlying major mental disorders? *Psychol Med*, 52(9), 1645–1665. <https://doi.org/10.1017/s0033291721005559>
- Alarcón, R. D., & Palmer, B. A. (2020). Personality disorders: Epidemiology and clinical course. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1218–1228). Oxford University Press. <https://doi.org/10.1093/med/9780198713005.003.0119>
- Albanese, A., Bhatia, K. P., Fung, V. S. C., et al. (2025). Definition and classification of dystonia. *Movement Dis*, 40, 1248–1259.
- Albert, N., & Weibell, M. A. (2019). The outcome of early intervention in first episode psychosis. *Int Rev Psychiatry*, 31(5–6), 413–424. <https://doi.org/10.1080/09540261.2019.1643703>
- Alcohol Change. (2022). Alcohol statistics. <https://alcoholchange.org.uk/alcohol-facts/fact-sheets/alcohol-statistics>
- Alexander, D. A. (2005). Early mental health intervention after disasters. *Adv Psychiatric Treat*, 11(1), 12–18.
- Alexander, G. E., & Crutcher, M. D. (1990). Functional architecture of basal ganglia circuits: Neural substrates of parallel processing. *Trends Neurosci*, 13(7), 266–271.
- Alexander, M., Perera, G., Ford, L., et al. (2015). Age-stratified prevalence of mild cognitive impairment and dementia in European populations: A systematic review. *J Alzheimers Dis*, 48(2), 355–359.
- Alexiou, A., & King, T. (2023). Alcohol withdrawal. *BMJ*, 381, 951.
- Alexopoulos, G. S., Meyers, B. S., Young, R. C., et al. (1997). 'Vascular depression' hypothesis. *Arch Gen Psychiatry*, 54(10), 915–922.
- Alfonsi, V., Scarpelli, S., D'Atri, A., et al. (2020). Later school start time: The impact of sleep on academic performance and health in the adolescent population. *Int J Environ Res Public Health*, 17(7), 2574.
- Alisic, E., Zalta, A. K., Van Wesel, F., et al. (2014). Rates of post-traumatic stress disorder in trauma-exposed children and adolescents: Meta-analysis. *BJPsych*, 204(5), 335–340.
- Allain, N., Leven, C., Falissard, B., et al. (2017). Manic switches induced by antidepressants: An umbrella review comparing randomized controlled trials and observational studies. *Acta Psychiatr Scand*, 135(2), 106–116.
- Allegrini, A. G., Cheesman, R., Rimfeld, K., et al. (2020). The p factor: Genetic analyses support a general dimension of psychopathology in childhood and adolescence. *J Child Psychol Psychiatry*, 61(1), 30–39.
- Alloy, L. B., Tetone, M. K., Ng, T. H., & Bart, C. P. (2020). Stress in bipolar disorder. In K. L. Harkness, & E. P. Hayden (Eds.), *The Oxford Handbook of Stress and Mental Health* (pp. 127–154). Oxford University Press.
- Als, T. D., Kurki, M. I., Grove, J., et al. (2023). Depression pathophysiology, risk prediction of recurrence and comorbid psychiatric disorders using genome-wide analyses. *Nature Med*, 29(7), 1832–1844.

- Altman, E. G., Hedeker, D., Peterson, J. L., & Davis, J. M. (1997). The Altman self-rating mania scale. *Biol Psychiatry*, 42(10), 948–955.
- Amato, P. R., & Anthony, C. J. (2014). Estimating the effects of parental divorce and death with fixed effects models. *J Marriage and Family*, 76(2), 370–386.
- American Psychiatric Association. (2013a). *Diagnostic and Statistical Manual of Mental Disorders: DSM-5*. American Psychiatric Association.
- American Psychiatric Association. (2013b). *Principles of Medical Ethics*. American Psychiatric Association.
- Ames, D., Gallagher, D., Loi, S., & Russ, T. (Eds.). (2022). *Guide to the Psychiatry of Old Age*, 2nd edn. Cambridge University Press.
- Ames, D., O'Brien, J., & Burns, A. (2017). *Dementia*, 5th edn. Routledge.
- Amin, M. R., & Ali, D. W. (2019). Pharmacology of medical cannabis. *Adv Exp Med Biol*, 1162, 151–165.
- an der Heiden, W., & Häfner, (2011). Course and outcome. In D. Weinberger & P. Harrison (Eds.), *Schizophrenia* (pp. 104–141). Wiley-Blackwell.
- Anderle, S., Dixon, M., Quintela-Lopez, T., et al. (2025). The vascular contribution to cognitive decline in ageing and dementia. *Nature Rev Neurosci*, 26, 591–606.
- Andersen, B. L., DeRubeis, R. J., Berman, B. S., et al. (2014). Screening, assessment, and care of anxiety and depressive symptoms in adults with cancer: An American Society of Clinical Oncology guideline adaptation. *J Clin Oncol*, 32(15), 1605–1619. <https://doi.org/10.1200/jco.2013.52.4611>
- Anderson, I. M. (2019). Interactions between ECT and prescribed medication In I. Ferrier & J. Waite (Eds.), *The ECT Handbook* (pp. 218–233). Cambridge University Press.
- Anderson, K. (2020). Diagnosis of sleep and circadian rhythm disorders. In J. R. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1124–1136). Oxford University Press.
- Andrade, C. (2021). Major congenital malformations associated with exposure to second-generation antipsychotic drugs during pregnancy. *J Clin Psychiatry*, 82(5), 37288.
- Andréasson, S., Engström, A., Allebeck, P., & Rydberg, U. (1987). Cannabis and schizophrenia: A longitudinal study of Swedish conscripts. *Lancet*, 330(8574), 1483–1486.
- Andrews, S. J., Fulton-Howard, B., & Goate, A. (2020). Interpretation of risk loci from genome-wide association studies of Alzheimer's disease. *Lancet Neurol*, 19(4), 326–335.
- Angold, A., & Costello, E. J. (2000). The child and adolescent psychiatric assessment (CAPA). *J Am Acad Child Adolesc Psychiatry*, 39(1), 39–48.
- Angst, F. (2009). Course and prognosis of mood disorders. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 665–669). Oxford University Press.
- Angst, J., Hengartner, M. P., Gamma, A. et al. (2013). Mortality of 403 patients with mood disorders 48 to 52 years after their psychiatric hospitalisation. *Eur Arch Psychiatry Clin Neurosci*, 263, 425–434.
- Anton, R. F., O'Malley, S. S., Ciraulo, D. A., et al. (2006). Combined pharmacotherapies and behavioral interventions for alcohol dependence: The COMBINE study: A randomized controlled trial. *JAMA*, 295(17), 2003–2017. doi: 10.1001/jama.295.17.2003
- Apaydin, E. A., Maher, A. R., Shanman, R., et al. (2016). A systematic review of St. John's wort for major depressive disorder. *Syst Rev*, 5(1), 1–25.
- Appleby, L. (2004). *The National Service Framework for Mental Health—Five Years On*. Department of Health.
- Appleby, L. (2015). *National Confidential Inquiry into Suicide and Homicides by People with Mental Illness: Making Mental Health Care Safer: Annual Report and 20-year Review*. University of Manchester. <https://documents.manchester.ac.uk/display.aspx?DocID=37580>.
- Appleby, L. (2021). What has been the effect of Covid-19 on suicide rates? *BMJ*, 372, n834. <https://doi.org/10.1136/bmj.n834>
- Appleby, L., Hunt, I. M., & Kapur, N. (2017). New policy and evidence on suicide prevention. *Lancet Psychiatry*, 4(9), 658–660.
- Appleby, L., Kapur, N., Shaw, J., et al. (2019). *National Confidential Inquiry into Suicide and Safety in Mental Health. Annual Report, England, Northern Ireland, Scotland and Wales*. University of Manchester.
- Appleby, L., Kapur, N., Shaw, J., et al. (2023). *National Confidential Inquiry into Suicide and Safety in Mental Health. Annual Report 2023: England, Northern Ireland, Scotland and Wales. UK Patient and General Population Data, 2010–2020*. University of Manchester.
- Appleby, L., Kapur, N., Turbull, P., et al. (2025). *The National Confidential Inquiry into Suicide and Safety in Mental Health. Annual Report: UK patient and general population data, 2012–2022*. University of Manchester.
- Appleton, H., Roberts, J., & Simpson, K. (2019). How is anxiety identified and diagnosed in individuals with autism spectrum disorder and intellectual disability? A scoping review. *J Ment Health Res Intellect Disabil*, 12(3–4), 152–175.
- Arango, C., Buitelaar, J. K., Fegert, J. M., et al. (2022). Safety and efficacy of agomelatine in children and adolescents with major depressive disorder receiving psychosocial counselling: A double-blind, randomised, controlled, phase 3 trial in nine countries. *Lancet Psychiatry*, 9(2), 113–124.
- Arango, C., & Fraguas, D. (2016). Should psychiatry deal only with mental disorders without an identified medical aetiology? *World Psychiatry*, 15, 22–23.
- Araya, R., Flynn, T., Rojas, G., et al. (2006). Cost-effectiveness of a primary care treatment program for depression in low-income women in Santiago, Chile. *Am J Psychiatry*, 163(8), 1379–1387.
- Arnautoglou, N. A., O'Brien, J. T., & Underwood, B. R. (2019). Dementia with Lewy bodies—from scientific knowledge to clinical insights. *Nat Rev Neurol*, 15(2), 103–112.
- Arnold, K. T., Porter, K. M. P., Frattaroli, S., et al. (2021). Multilevel barriers and facilitators to sustainability of a universal trauma-informed school-based mental health intervention following an efficacy trial: A qualitative study. *School Ment Health*, 13, 174–185. <https://doi.org/10.1007/s12310-020-09402-w>
- Arnall, E., Park, J., & Heimer, K. (2023). Girls in the juvenile justice system in England and Wales, 2002–2017. *J Youth Stud*, 26(1), 19–42.
- Arshad, U., ul Ain, F., Gauntlett, J., et al. (2020). A systematic review of the evidence supporting mobile- and internet-based psychological interventions for self-harm. *Suicide Life Threat Behav*, 50(1), 151–179. <https://doi.org/> <https://doi.org/10.1111/sltb.12583>

- Arvanitakis, Z., Shah, R. C., & Bennett, D. A. (2019). Diagnosis and management of dementia. *JAMA*, 322(16), 1589–1599.
- Arya, V., Page, A., Armstrong, G., et al. (2021). Estimating patterns in the under-reporting of suicide deaths in India: Comparison of administrative data and Global Burden of Disease Study estimates, 2005–2015. *J Epidemiol Community Health*, 75(6), 550–555.
- Aronson, R., Offman, H. J., Joffe, R. T., & Naylor, C. D. (1996). Triiodothyronine augmentation in the treatment of refractory depression: a meta-analysis. *Arch Gen Psychiatry*, 53(9), 842–848.
- Asarnow, J. R., & Mehlum, L. (2019). Practitioner review: Treatment for suicidal and self-harming adolescents—Advances in suicide prevention care. *J Child Psychol Psychiatry*, 60(10), 1046–1054. <https://doi.org/10.1111/jcpp.13130>
- Asher, R. (1951). Munchausen's syndrome. *Lancet*, 1(6650), 339–341. [https://doi.org/10.1016/s0140-6736\(51\)92313-6](https://doi.org/10.1016/s0140-6736(51)92313-6)
- Ashok, A. H., Marques, T. R., Jauhar, S., et al. (2017). The dopamine hypothesis of bipolar affective disorder: The state of the art and implications for treatment. *Mol Psychiatry*, 22(5), 666–679.
- Attia, E., Steinglass, J. E., Walsh, B. T., et al. (2019). Olanzapine versus placebo in adult outpatients with anorexia nervosa: A randomized clinical trial. *Am J Psychiatry*, 176, 449–456. <https://doi.org/10.1176/appi.ajp.2018.18101125>
- Austelle, C. W., O'Leary, G. H., Thompson, S., et al. (2022). A comprehensive review of vagus nerve stimulation for depression. *Neuromodulation*, 25(3), 309–315.
- Avis, K. A., Stroebe, M., & Schut, H. (2021). Stages of grief portrayed on the internet: A systematic analysis and critical appraisal. *Front Psychol*, 12, 772696. <https://doi.org/10.3389/fpsyg.2021.772696>
- Axe, C. J., Doherty, A. M., & Jones, D. A. (2022). Investigating the relationship between euthanasia and/or assisted suicide and rates of non-assisted suicide: Systematic review. *BJPsych Open*, 8(4), e108, Article e108. <https://doi.org/10.1192/bjo.2022.71>
- Ayala, E. S., Meuret, A. E., & Ritz, T. (2009). Treatments for blood-injury-injection phobia: A critical review of current evidence. *J Psychiatric Res*, 43(15), 1235–1242.
- Bach, B., & First, M. B. (2018). Application of the ICD-11 classification of personality disorders. *BMC Psychiatry*, 18(1), 351. <https://doi.org/10.1186/s12888-018-1908-3>
- Bach, B., Kramer, U., Doering, S., et al. (2022). The ICD-11 classification of personality disorders: A European perspective on challenges and opportunities. *Borderline Personal Disord Emot Dysregul*, 9(1), 12. <https://doi.org/10.1186/s40479-022-00182-0>
- Bagayogo, I., Interian, A., & Escobar, J. I. (2013). Transcultural aspects of somatic disorders in the context of depressive disorders. In R. Alarcón (Ed.), *Cultural Psychiatry* (pp. 64–74). Karger.
- Bagni, C., Tassone, F., Neri, G., & Hagerman, R. (2012). Fragile X syndrome: Causes, diagnosis, mechanisms, and therapeutics. *J Clin Invest*, 122(12), 4314–4322.
- Bahji, A., Ermacora, D., Stephenson, C., et al. (2020). Comparative efficacy and tolerability of pharmacological treatments for the treatment of acute bipolar depression: A systematic review and network meta-analysis. *J Affect Disord*, 269, 154–184.
- Bahji, A., Hawken, E., Sepehry, A., et al. (2019). ECT beyond unipolar major depression: Systematic review and meta-analysis of electroconvulsive therapy in bipolar depression. *Acta Psychiatr Scand*, 139(3), 214–226.
- Bahji, A., Vazquez, G. H., & Zarate, C. A. Jr. (2021). Comparative efficacy of racemic ketamine and esketamine for depression: A systematic review and meta-analysis. *J Affect Disord*, 278, 542–555. <https://doi.org/10.1016/j.jad.2020.09.071>
- Baigent, C., Peto, R., Gray, R., et al. (2010). Large-scale randomized evidence: Trials and meta-analyses of trials. In D. Warrell, T. Cox, & J. Firth (Eds.), *Oxford Textbook of Medicine*, 5th edn. (pp. 31–47). Oxford University Press.
- Baio, J. (2012). Prevalence of autism spectrum disorders: Autism and Developmental Disabilities Monitoring Network, 14 sites, United States, 2008. *Morbid Mortal Week Rep*, 61(SS03), 1–19.
- Baker-Henningham, H., Powell, C., Walker, S., & Grantham-McGregor, S. (2005). The effect of early stimulation on maternal depression: A cluster randomised controlled trial. *Arch Dis Childhood*, 90(12), 1230–1234.
- Bakker, M. J., Greven, C. U., Buitelaar, J. K., & Glennon, J. C. (2017). Practitioner review: Psychological treatments for children and adolescents with conduct disorder problems—a systematic review and meta-analysis. *J Child Psychol Psychiatry*, 58(1), 4–18. <https://doi.org/10.1111/jcpp.12590>
- Balance Study Group. (2010). Lithium plus valproate combination therapy versus monotherapy for relapse prevention in bipolar I disorder (BALANCE): A randomised open-label trial. *Lancet*, 375(9712), 385–395.
- Baldwin, D., & Sinclair, J. (2015). Recurrent brief depression: This too shall pass? In D. Bhugra & G. Malhi (Eds.), *Troublesome Disguises: Managing Challenging Disorders in Psychiatry* (pp. 100–113). Wiley-Blackwell.
- Baldwin, D. S., Anderson, I. M., Nutt, D. J., et al. (2014). Evidence-based pharmacological treatment of anxiety disorders, post-traumatic stress disorder and obsessive-compulsive disorder: A revision of the 2005 guidelines from the British Association for Psychopharmacology. *J Psychopharmacol*, 28(5), 403–439.
- Balint, B., Mencacci, N. E., Valente, E. M., et al. (2018). Dystonia. *Nat Rev Dis Primers*, 4(1), 25.
- Balint, M. (1964). *The Doctor, His Patient and the Illness*, Rev and Enlarged 2nd edn. Pitman Medical Publishing Company.
- Ball, W. P., Black, C., Gordon, S., et al. (2022). Inequalities in children's mental health care: Analysis of routinely collected data on prescribing and referrals to secondary care. *medRxiv*. doi: 10.1186/s12888-022-04438-5
- Ballard, C., Mobley, W., Hardy, J., et al. (2016). Dementia in Down's syndrome. *Lancet Neurol*, 15(6), 622–636.
- Ballenger, J. (2009). Panic disorder and agoraphobia. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 750–764). Oxford University Press.
- Bandura, A. (1969). *Principles of Behavior Modification*. Holt, Rinehart, & Winston.
- Banerjee, S., & Farina, N. (2020). Memory assessment services. In T. Denning, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 339–346). Oxford University Press.
- Bang, J., Spina, S., & Miller, B. L. (2015). Frontotemporal dementia. *Lancet*, 386(10004), 1672–1682.

- Banks, R., Bush, A., Baker, P., et al. (2007). *Challenging Behaviour: A Unified Approach. Clinical and Service Guidelines for Supporting People with Learning Disabilities Who Are at Risk of Receiving Abusive or Restrictive Practices. College Report CR 144*. Royal College of Psychiatrists, British Psychological Society and Royal College of Speech and Language Therapists.
- Barbetta, P., & Telfener, U. (2021). The Milan approach, history, and evolution. *Fam Process*, 60(1), 4–16. <https://doi.org/10.1111/famp.12612>
- Barbui, C., Cipriani, A., Patel, V., et al. (2011). Efficacy of antidepressants and benzodiazepines in minor depression: Systematic review and meta-analysis. *BJPsych*, 198(1), 11–16.
- Barbui, C., Purgato, M., Abdulmalik, J., et al. (2021). Efficacy of interventions to reduce coercive treatment in mental health services: Umbrella review of randomised evidence. *BJPsych*, 218(4), 185–195. <https://doi.org/10.1192/bjp.2020.144>
- Barbui, C., Purgato, M., Churchill, R., et al. (2017). Evidence-based interventions for global mental health: Role and mission of a new Cochrane initiative. *Cochrane Database Syst Rev*, 4(4), D000120. <https://doi.org/10.1002/14651858.ED000120>
- Barnes, T. R. (1989). A rating scale for drug-induced akathisia. *BJPsych*, 154(5), 672–676.
- Barnes, T. R., Drake, R., Paton, C., et al. (2020). Evidence-based guidelines for the pharmacological treatment of schizophrenia: Updated recommendations from the British Association for Psychopharmacology. *J Psychopharmacol*, 34(1), 3–78. <https://doi.org/10.1177/0269881119889296>
- Barnes, T. R., Shingleton-Smith, A., & Paton, C. (2009). Antipsychotic long-acting injections: Prescribing practice in the UK. *BJPsych*, 195(S52), s37–s42.
- Barnett, P., Mackay, E., Matthews, H., et al. (2019). Ethnic variations in compulsory detention under the Mental Health Act: A systematic review and meta-analysis of international data. *Lancet Psychiatry*, 6(4), 305–317. [https://doi.org/10.1016/S2215-0366\(19\)30027-6](https://doi.org/10.1016/S2215-0366(19)30027-6)
- Barr, B., Taylor-Robinson, D., Scott-Samuel, A., et al. (2012). Suicides associated with the 2008-10 economic recession in England: Time trend analysis. *BMJ*, 345, e5142. doi: 10.1136/bmj.e5142
- Barracough, B., & Harris, E. C. (2002). Suicide preceded by murder: The epidemiology of homicide-suicide in England and Wales 1988–92. *Psychol Med*, 32(4), 577–584. <https://doi.org/10.1017/s0033291702005500>
- Barth, J., Bermetz, L., Heim, E., et al. (2013). The current prevalence of child sexual abuse worldwide: A systematic review and meta-analysis. *Int J Pub Health*, 58(3), 469–483.
- Baruk, H. (1959). Delusions of passion. In S. Hirsch & M. Shepherd (Eds.), *Themes and Variations in European Psychiatry* (pp. 375–384). Wright.
- Bass, C., & Glaser, D. (2014). Early recognition and management of fabricated or induced illness in children. *Lancet*, 383(9926), 1412–1421. [https://doi.org/10.1016/s0140-6736\(13\)62183-2](https://doi.org/10.1016/s0140-6736(13)62183-2)
- Bass, C., & Halligan, P. (2014). Factitious disorders and malingerers: Challenges for clinical assessment and management. *Lancet*, 383(9926), 1422–1432.
- Bass, C., & Jones, D. (2011). Psychopathology of perpetrators of fabricated or induced illness in children: Case series. *BJPsych*, 199(2), 113–118. <https://doi.org/10.1192/bjp.bp.109.074088>
- Bassetti, C. L. A., Adamantidis, A., Burdakov, D., et al. (2019). Narcolepsy—clinical spectrum, aetiopathophysiology, diagnosis and treatment. *Nat Rev Neurol*, 15, 519–539. <https://doi.org/10.1038/s41582-019-0226-9>
- Basson, R., & Schultz, W.W. (2007). Sexual dysfunction 1: Sexual sequelae of general medical disorders. *Lancet*, 369, 409–424.
- Bateman, A., Brown, D. & Pedder, J. (2010). *Introduction to Psychotherapy: An Outline of Psychodynamic Principles and Practice*. Routledge.
- Bateman, A., & Fonagy, P. (2008). 8-year follow-up of patients treated for borderline personality disorder: Mentalization-based treatment versus treatment as usual. *Am J Psychiatry*, 165(5), 631–638. <https://doi.org/10.1176/appi.ajp.2007.07040636>
- Bateman, A., & Fonagy, P. (2010). Mentalization based treatment for borderline personality disorder. *World Psychiatry*, 9(1), 11–15. <https://doi.org/10.1002/j.2051-5545.2010.tb00255.x>
- Bateman, A., & Fonagy, P. (2013). Impact of clinical severity on outcomes of mentalisation-based treatment for borderline personality disorder. *BJPsych*, 203(3), 221–227. <https://doi.org/10.1192/bjp.bp.112.121129>
- Bateman, A. W., Gunderson, J., & Mulder, R. (2015). Treatment of personality disorder. *Lancet*, 385(9969), 735–743. [https://doi.org/10.1016/s0140-6736\(14\)61394-5](https://doi.org/10.1016/s0140-6736(14)61394-5)
- Bateman, A. W., & Krawitz, R. (2013). *Borderline Personality Disorder: An Evidence-based Guide for Generalist Mental Health Professionals*. Oxford University Press.
- Bates, G. P., Dorsey, R., Gusella, J. F., et al. (2015). Huntington disease. *Nature Rev Disease Primers*, 1(1), 1–21.
- Bax, O. K., Chartonas, D., Parker, J., et al. (2023). Personality disorder. *BMJ*, 382, e050290. <https://doi.org/10.1136/bmj-2019-050290>
- Bayley, N. (2005). *Bayley Scales of Infant and Toddler Development, Third Edition (Bayley-III®)*. APA PsycTests.
- Beail, N. (2017). *Psychological Therapies and People Who Have ID: A Report from the Royal College of Psychiatrists and British Psychological Society*. British Psychological Society.
- Bear, H. A., Nunes, L. A., DeJesus, J., et al. (2022). Determination of markers of successful implementation of mental health apps for young people: Systematic review. *J Med Int Res*, 24(11), e40347.
- Beard, J. H., Malamud, T. J., & Rossman, E. (1978). Psychiatric rehabilitation and long-term rehospitalization rates: The findings of two research studies. *Schizophr Bull*, 4, 622–635. <https://doi.org/10.1093/schbul/4.4.622>
- Beards, S., Gayer-Anderson, C., Borges, S., et al. (2013). Life events and psychosis: A review and meta-analysis. *Schizophr Bull*, 39(4), 740–747.
- Beauchamp, T., & Childress, J. (2019). *Principles of Biomedical Ethics*, 8th edn. Oxford University Press.
- Beck, A., Ward, C., Mendelson, M., et al. (1961). An inventory for measuring depression. *Arch Gen Psychiatry*, 4, 561–585.
- Beck, A. T. (1976). *Cognitive Therapy and the Emotional Disorders*. International Universities Press.
- Beck, A. T., Davis, D. D., & Freeman, A. (2015). *Cognitive Therapy of Personality Disorders*, 3rd edn. Guilford Press.
- Beck, J. G., & Sloan, D. M. (2023). *The Oxford Handbook of Traumatic Stress Disorders*, 2nd edn. Oxford University Press.

- Beech, A. R., Miner, M. H., & Thornton, D. (2016). Paraphilias in the DSM-5. *Annu Rev Clin Psychol*, 12, 383–406. <https://doi.org/10.1146/annurev-clinpsy-021815-093330>.
- Beckwith, H., Moran, P. F., & Reilly, J. (2014). Personality disorder prevalence in psychiatric outpatients: A systematic literature review. *Personal Ment Health*, 8(2), 91–101. <https://doi.org/https://doi.org/10.1002/pmh.1252>
- Beekman, A. T., Copeland, J., & Prince, M. J. (1999). Review of community prevalence of depression in later life. *BJPsych*, 174(4), 307–311.
- Beekman, A. T., Geerlings, S. W., Deeg, D. J., et al. (2002). The natural history of late-life depression: A 6-year prospective study in the community. *Arch Gen Psychiatry*, 59(7), 605–611.
- Beeston, C., Robinson, M., Giles, L., et al. (2020). Evaluation of minimum unit pricing of alcohol: A mixed method natural experiment in Scotland. *Int J Environment Res Pub Health*, 17(10), 3394.
- Beghi, M., Butera, E., Cerri, C. G., et al. (2021). Suicidal behaviour in older age: A systematic review of risk factors associated to suicide attempts and completed suicides. *Neurosci Biobehav Rev*, 127, 193–211. <https://doi.org/https://doi.org/10.1016/j.neubiorev.2021.04.011>
- Beidas, R. S., Bittenheim, A. M., & Mandell, D. S. (2021). Transforming mental health care delivery through implementation science and behavioral economics. *JAMA Psychiatry*, 78(9), 941–942. <https://doi.org/10.1001/jamapsychiatry.2021.1120>
- Beierl, E. T., Böllinghaus, I., Clark, D. M., et al. (2020). Cognitive paths from trauma to posttraumatic stress disorder: A prospective study of Ehlers and Clark's model in survivors of assaults or road traffic collisions. *Psychol Med*, 50(13), 2172–2181. <https://doi.org/10.1017/s0033291719002253>
- Bell, V., Halligan, P. W., & Ellis, H. D. (2006). Diagnosing delusions: A review of inter-rater reliability. *Schizophr Res*, 86(1–3), 76–79.
- Bellis, M. A., Wood, S., Hughes, K., et al. (2023). *Tackling Adverse Childhood Experiences (ACEs): State of the Art and Options for Action*. PHWN Trust.
- Benedict, R. H., Amato, M. P., DeLuca, J., & Geurts, J. J. (2020). Cognitive impairment in multiple sclerosis: Clinical management, MRI, and therapeutic avenues. *Lancet Neurol*, 19(10), 860–871.
- Benjamin, S., & Lauterbach, M. D. (2018). The neurological examination adapted for neuropsychiatry. *CNS Spectr*, 23(3), 219–227. <https://doi.org/10.1017/s1092852918000925>
- Bennett, D. (1983). The historical development of rehabilitation services. In F. N. Watts & D. H. Bennett (Eds.), *Theory and Practice of Psychiatric Rehabilitation* (pp. 15–42). Wiley.
- Bennett, S., Shafan, R., Coughtrey, A., et al. (2015). Psychological interventions for mental health disorders in children with chronic physical illness: A systematic review. *Arch Dis Child*, 100(4), 308–316. <https://doi.org/10.1136/archdischild-2014-307474>
- Bensing, J. M., & Verhaak, P. F. (2006). Somatisation: A joint responsibility of doctor and patient. *Lancet*, 367(9509), 452–454. [https://doi.org/10.1016/s0140-6736\(06\)68155-5](https://doi.org/10.1016/s0140-6736(06)68155-5)
- Berdenis van Berlekom, A., Muflihah, C. H., Snijders, G., et al. (2020). Synapse pathology in schizophrenia: A meta-analysis of postsynaptic elements in postmortem brain studies. *Schizophr Bull*, 46(2), 374–386. <https://doi.org/10.1093/schbul/sbz060>
- Bergen, R. K., Edleson, J. L., & Renzetti, C. M. (2005). *Violence Against Women: Classic Papers*. Pearson/Allyn & Bacon.
- Bergink, V., Burgerhout, K. M., Koorengel, K. M., et al. (2015). Treatment of psychosis and mania in the postpartum period. *Am J Psychiatry*, 172(2), 115–123.
- Berlin, I. (1958). *Two Concepts of Liberty*. Clarendon Press.
- Berman, K., & Brodaty, H. (2006). Psychosocial effects of age-related macular degeneration. *Int Psychogeriatr*, 18(3), 415–428. <https://doi.org/10.1017/s1041610205002905>
- Berman, R. M., Cappiello, A., Anand, A., et al. (2000). Antidepressant effects of ketamine in depressed patients. *Biol Psychiatry*, 47(4), 351–354.
- Berrios, G. E. (1992). Phenomenology, psychopathology and Jaspers: A conceptual history. *Hist Psychiatry*, 3(11), 303–327.
- Berrios, G. E. (1996). *The History of Mental Symptoms: Descriptive Psychopathology since the Nineteenth Century*. Cambridge University Press.
- Berrios, G. E., & Kennedy, N. (2002). Erotomania: A conceptual history. *Hist Psychiatry*, 13(52), 381–400.
- Besjes, M. J., Mares, S. H. W., van Elburg, A. A., Spijker, J. (2025). The efficacy of pharmacological treatment of depression in anorexia nervosa and underweight patients: a systematic review. *Eur Eating Disord Rev* (in press).
- Betz, L. T., Penzel, N., Rosen, M., et al. (2023). Disentangling heterogeneity of psychosis expression in the general population: Sex-specific moderation effects of environmental risk factors on symptom networks. *Psychol Med*, 53(5), 1860–1869. <https://doi.org/10.1017/s0033291721003470>
- Bhugra, D., & Bhui, K. (1998). Psychotherapy for ethnic minorities: Issues, context and practice. *Br J Psychother*, 14(3), 310–326. <https://doi.org/https://doi.org/10.1111/j.1752-0118.1998.tb00385.x>
- Bhugra, D., Bhui, K., Wong, S. Y. S., & Gilman, S. E. (Eds.). (2018). *Oxford Textbook of Public Mental Health*. Oxford University Press. <https://doi.org/10.1093/med/9780198792994.001.0001>
- Bhui, K. (2019). Uncertainties in practice: Terrorism and mental illness: Commentary on... Towards an integrated pragmatic approach in terrorism and mental illness. *BJPsych Adv*, 25(2), 110–112. <https://doi.org/10.1192/bja.2018.70>
- Bhui, K. (2022). A Refugee Rose of competencies and capabilities for mental healthcare of refugees. *BJPsych Open*, 8(2), e45. <https://doi.org/10.1192/bjo.2022.11>
- Bhui, K., Ascoli, M., & Nuamh, O. (2012). The place of race and racism in cultural competence: What can we learn from the English experience about the narratives of evidence and argument? *Transcult Psychiatry*, 49(2), 185–205. <https://doi.org/10.1177/1363461512437589>
- Bhui, K., Aslam, R. H. W., Palinski, A., et al. (2015). Interventions to improve therapeutic communications between Black and minority ethnic patients and professionals in psychiatric services: Systematic review. *BJPsych*, 207(2), 95–103. <https://doi.org/10.1192/bjp.bp.114.158899>
- Bhui, K., Dein, S., & Pope, C. (2021). Clinical ethnography in severe mental illness: A clinical method to tackle social

- determinants and structural racism in personalised care. *BJPsych Open*, 7(3), e78. <https://doi.org/10.1192/bjo.2021.38>
- Bhui, K., Shanahan, L., & Harding, G. (2006). Homelessness and mental illness: A literature review and a qualitative study of perceptions of the adequacy of care. *Int J Soc Psychiatry*, 52(2), 152–165. <https://doi.org/10.1177/0020764006062096>
- Bhui, K., Stansfeld, S., Hull, S., et al. (2003). Ethnic variations in pathways to and use of specialist mental health services in the UK. Systematic review. *BJPsych*, 182, 105–116. <https://doi.org/10.1192/bjp.182.2.105>
- Bhui, K., Stansfeld, S., McKenzie, K., et al. (2005). Racial/ethnic discrimination and common mental disorders among workers: Findings from the EMPiRIC study of ethnic minority groups in the United Kingdom. *Am J Public Health*, 95(3), 496–501. <https://doi.org/10.2105/ajph.2003.033274>
- Bhui, K., Ullrich, S., Kallis, C., & Coid, J. W. (2015). Criminal justice pathways to psychiatric care for psychosis. *BJPsych*, 207(6), 523–529. <https://doi.org/10.1192/bjp.bp.114.153882>
- Bhui, K., Warfa, N., Edonya, P., et al. (2007). Cultural competence in mental health care: A review of model evaluations. *BMC Health Serv Res*, 7, 15. <https://doi.org/10.1186/1472-6963-7-15>
- Bickley, H., Hunt, I. M., Windfuhr, K., et al. (2013). Suicide within two weeks of discharge from psychiatric inpatient care: A case-control study. *Psychiatr Serv*, 64(7), 653–659. <https://doi.org/10.1176/appi.ps.201200026>
- Biddinger, K. J., Emdin, C. A., Haas, M. E., et al. (2022). Association of habitual alcohol intake with risk of cardiovascular disease. *JAMA Netw Open*, 5(3), e223849–e223849.
- Bienvenu, O. J., Davydow, D. S., & Kendler, K. (2011). Psychiatric ‘diseases’ versus behavioral disorders and degree of genetic influence. *Psychol Med*, 41(1), 33–40.
- Bion, W. R. (1961). *Experiences in Groups and Other Papers*. Tavistock Publications <https://doi.org/10.4324/9780203359075>.
- Birchwood, M., Todd, P., & Jackson, C. (1998). Early intervention in psychosis: The critical period hypothesis. *BJPsych*, 172(S33), 53–59.
- Birnbaum, R., & Weinberger, D. R. (2017). Genetic insights into the neurodevelopmental origins of schizophrenia. *Nat Rev Neurosci*, 18(12), 727–740. <https://doi.org/10.1038/nrn.2017.125>
- Bishara, D. (2020). Psychopharmacology. In T. Denning, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 207–216). Oxford University Press.
- Biskin, R. S. (2015). The lifetime course of borderline personality disorder. *Can J Psychiatry*, 60(7), 303–308. <https://doi.org/10.1177/070674371506000702>
- Black, C. (2020). Review of drugs: Phase 1 report—Executive summary. https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/897786/2SummaryPhaseOne\foreword200219.pdf
- Blackman, G., Neri, G., Al-Doori, O., et al. (2023). Prevalence of neuroradiological abnormalities in first-episode psychosis: A systematic review and meta-analysis. *JAMA Psychiatry*, 80(10), 1047–1054. <https://doi.org/10.1001/jamapsychiatry.2023.2225>
- Blackmore, E. R., Rubinow, D. R., O'Connor, T. G., et al. (2013). Reproductive outcomes and risk of subsequent illness in women diagnosed with postpartum psychosis. *Bipolar Disord*, 15(4), 394–404.
- Blackmore, R., Boyle, J. A., Fazel, M., et al. (2020). The prevalence of mental illness in refugees and asylum seekers: A systematic review and meta-analysis. *PLoS MED*, 17(9), e1003337. <https://doi.org/10.1371/journal.pmed.1003337>
- Blair, R. J. R. (2003). Neurobiological basis of psychopathy. *BJPsych*, 182(1), 5–7. <https://doi.org/10.1192/bjp.182.1.5>
- Blakemore, S.-J. (2019). Adolescence and mental health. *Lancet*, 393(10185), 2030–2031.
- Blanco, C., Markowitz, J. C., & Weissman, M. M. (2009). Interpersonal psychotherapy for depression and other disorders. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1318–1326). Oxford University Press.
- Blashfield, R. K., Keeley, J. W., Flanagan, E. H., & Miles, S. R. (2014). The cycle of classification: DSM-I through DSM-5. *Ann Rev Clin Psychol*, 10, 25–51.
- Blauwendraat, C., Nalls, M. A., & Singleton, A. B. (2020). The genetic architecture of Parkinson's disease. *Lancet Neurol*, 19(2), 170–178.
- Blazer, D. G. (2020). Hearing loss: The silent risk for psychiatric disorders in late life. *Clin Geriatr Med*, 36(2), 201–209. <https://doi.org/10.1016/j.cger.2019.11.002>
- Blennow, K., Brody, D. L., Kochanek, P. M., et al. (2016). Traumatic brain injuries. *Nat Rev Dis Primers*, 2(1), 1–19.
- Blessed, G., Tomlinson, B. E., & Roth, M. (1968). The association between quantitative measures of dementia and of senile change in the cerebral grey matter of elderly subjects. *BJPsych*, 114, 797–811.
- Bleuler, M. (1911). *Dementia Praecox or the Group of Schizophrenias* (English edition published in 1950 ed.). International University.
- Bleuler, M. (1974). The long-term course of the schizophrenic psychoses. *Psychol Med*, 4, 244–254.
- Bloch, S. (1986). *An Introduction to the Psychotherapies*, 2nd edn. Oxford University Press.
- Bloch, S. (2006). *An Introduction to the Psychotherapies*, 4th edn. Oxford University Press.
- Bloch, S., & Harari, E. (2009). Family therapy in the adult psychiatric setting. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1380–1391). Oxford University Press.
- Bloem, B. R., Okun, M. S., & Klein, C. (2021). Parkinson's disease. *Lancet*, 397(10291), 2284–2303.
- Bloom, T. M., Nguyen, T. P., Lami, F., et al. (2021). Measurement tools for gender identity, gender expression, and gender dysphoria in transgender and gender-diverse children and adolescents: A systematic review. *Lancet Child Adolesc Health*, 5(8), 582–588. [https://doi.org/10.1016/S2352-4642\(21\)00098-5](https://doi.org/10.1016/S2352-4642(21)00098-5)
- Boaden, K., Tomlinson, A., Cortese, S., & Cipriani, A. (2020). Antidepressants in children and adolescents: Meta-review of efficacy, tolerability and suicidality in acute treatment. *Front Psychiatry*, 11, 717. <https://doi.org/10.3389/fpsy.2020.00717>
- Bohus, M., Stoffers-Winterling, J., Sharp, C., et al. (2021). Borderline personality disorder. *Lancet*, 398(10310), 1528–1540. [https://doi.org/10.1016/S0140-6736\(21\)00476-1](https://doi.org/10.1016/S0140-6736(21)00476-1)
- Boland, R. J., Goldstein, M., & Haltzman, S. D. (2000). Psychiatric management of behavioral syndromes in intensive

- care units. In B. S. Fogel and D. B. Greenberg (Eds.), *Psychiatric Care of the Medical Patient*, 3rd edn. (pp. 1389–1398). Oxford University Press.
- Bolton, D. (2022). Looking forward to a decade of the biopsychosocial model. *BJPsych Bull*, 46(4), 1–5. <https://doi.org/10.1192/bjb.2022.34>
- Bond, G. R., & Drake, R. E. (2014). Making the case for IPS supported employment. *Admin Policy Ment Health*, 41(1), 69–73. <https://doi.org/10.1007/s10488-012-0444-6>
- Bonsack, C., & Borgeat, F. (2005). Perceived coercion and need for hospitalization related to psychiatric admission. *Int J Law Psychiatry*, 28(4), 342–347. <https://doi.org/10.1016/j.ijlp.2005.03.008>
- Border, R., Johnson, E. C., Evans, L. M., et al. (2019). No support for historical candidate gene or candidate gene-by-interaction hypotheses for major depression across multiple large samples. *Am J Psychiatry*, 176(5), 376–387.
- Bordes, C., Sargurupremraj, M., Mishra, A., & Dabette, S. (2022). Genetics of common cerebral small vessel disease. *Nat Rev Neurol*, 18(2), 84–101.
- Borgan, F., Kokkinou, M., & Howes, O. (2021). The cannabinoid CB(1) receptor in schizophrenia. *Biol Psychiatry Cogn Neurosci Neuroimaging*, 6(6), 646–659. <https://doi.org/10.1016/j.bpsc.2020.06.018>
- Bornoalova, M. A., Cummings, J. R., Hunt, E., et al. (2014). Understanding the relative contributions of direct environmental effects and passive genotype-environment correlations in the association between familial risk factors and child disruptive behavior disorders. *Psychol Med*, 44(4), 831–844.
- Bosnjak Kuharic, D., Kekin, I., Hew, J., et al. (2019). Interventions for prodromal stage of psychosis. *Cochrane Database Syst Rev*, 2019(11), CD012236. <https://doi.org/10.1002/14651858.CD012236.pub2>
- Bossarte, R. M., Simon, T. R., & Barker, L. (2006). Characteristics of homicide followed by suicide incidents in multiple states, 2003–04. *Inj Prev*, 12(Suppl 2), ii33–ii38. <https://doi.org/10.1136/ip.2006.012807>
- Bottomley, J. M., LeReun, C., Diamantopoulos, A., et al. (2020). Vagus nerve stimulation (VNS) therapy in patients with treatment resistant depression: A systematic review and meta-analysis. *Comp Psychiatry*, 98, 152156.
- Bougeard, C., Picarel-Blanchot, F., Schmid, R., et al. (2021). Prevalence of autism spectrum disorder and co-morbidities in children and adolescents: A systematic literature review. *Front Psychiatry*, 12, 744709.
- Boutros, N. N., Arfken, C., Galderisi, S., et al. (2008). The status of spectral EEG abnormality as a diagnostic test for schizophrenia. *Schizophr Res*, 99(1–3), 225–237. <https://doi.org/10.1016/j.schres.2007.11.020>
- Bowden, C. L., Brugger, A. M., Swann, A. C., et al. (1994). Efficacy of divalproex vs lithium and placebo in the treatment of mania. *JAMA*, 271(12), 918–924.
- Bowden, C. L., Calabrese, J. R., McElroy, S. L., et al. (2000). A randomized, placebo-controlled 12-month trial of divalproex and lithium in treatment of outpatients with bipolar I disorder. *Arch Gen Psychiatry*, 57(5), 481–489.
- Bower, P., Rowland, N., & Hardy, R. (2003). The clinical effectiveness of counselling in primary care: A systematic review and meta-analysis. *Psychol Med*, 33(2), 203–215. <https://doi.org/10.1017/s0033291702006979>
- Bowlby, J. (1980). *Attachment and Loss*. Vol. 3: *Loss, Sadness and Depression*. Basic Books.
- Bowlby, J. (1944). Forty-four juvenile thieves: Their characters and home-life. *Int J Psychoanal*, 25, 19–53.
- Bowlby, J. (1951). Maternal care and mental health. *Bull World Health Organ*, 3(3), 355–533.
- Braak, H., & Braak, E. (1991). Neuropathological staging of Alzheimer-related changes. *Acta Neuropathol*, 82(4), 239–259.
- Braak, H., & Del Tredici, K. (2015). The preclinical phase of the pathological process underlying sporadic Alzheimer's disease. *Brain*, 138(10), 2814–2833.
- Bracken, P., Fernando, S., Alsaraf, S., et al. (2021). Decolonising the medical curriculum: Psychiatry faces particular challenges. *Anthropol Med*, 28(4), 420–428. <https://doi.org/10.1080/13648470.2021.1949892>
- Bracken, P., Thomas, P., Timimi, S., et al. (2012). Psychiatry beyond the current paradigm. *BJPsych*, 201(6), 430–434.
- Braconnier, M. L., & Siper, P. M. (2021). Neuropsychological assessment in autism spectrum disorder. *Curr Psychiatry Rep*, 23(10), 1–9.
- Brand, F., Carbonnier, A., Croft, A., et al. (2023). *If a Patient Dies by Suicide: A Resource for Mental Health Professionals*. Royal College of Psychiatrists.
- Brandon, S., Cowley, P., McDonald, C., et al. (1985). Leicester ECT trial: Results in schizophrenia. *BJPsych*, 146(2), 177–183.
- Brandt, R. (2009). The mental health of people living with HIV/AIDS in Africa: A systematic review. *African J AIDS Res*, 8(2), 123–133.
- Breedvelt, J. J. F., Brouwer, M. E., Harrer, M., et al. (2021). Psychological interventions as an alternative and addition to antidepressant medication to prevent depressive relapse: Systematic review and meta-analysis. *BJPsych*, 219(4), 538–545. <https://doi.org/10.1192/bjp.2020.198>
- Brent, D., & Maalouf, F. (2015). Depressive disorders in childhood and adolescence. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 874–892). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch63>
- Breslin, R., & Evans, H. (2004). Key child protection statistics. Child homicides and deaths. www.nspcc.org.uk/inform/Statistics/KeyCPstats/4.asp
- Brewin, C. (2009). Recovered memories and false memories. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 713–716). Oxford University Press.
- Bridge, J. A., Iyengar, S., Salary, C. B., et al. (2007). Clinical response and risk for reported suicidal ideation and suicide attempts in pediatric antidepressant treatment: A meta-analysis of randomized controlled trials. *JAMA*, 297(15), 1683–1696.
- Broadhead, J., & Jacoby, R. (1990). Mania in old age: A first prospective study. *Int J Geriatr Psychiatry*, 5(4), 215–222.
- Brockington, I. (2009). Obstetric and gynaecological conditions associated with psychiatric disorder. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1114–1127). Oxford University Press.

- Brockmeyer, T., Friederich, H. C., & Schmidt, U. (2018). Advances in the treatment of anorexia nervosa: a review of established and emerging interventions. *Psychol Med*, 48, 1228–1256. <https://doi.org/10.1017/S0033291717002604>
- Brodie, M. J., Besag, F., Ettinger, A. B., et al. (2016). Epilepsy, antiepileptic drugs, and aggression: An evidence-based review. *Pharmacol Rev*, 68(3), 563–602.
- Broen, A. N., Moum, T., Bødtker, A. S., & Ekeberg, Ø. (2005). The course of mental health after miscarriage and induced abortion: A longitudinal, five-year follow-up study. *BMC Med*, 3, 1–14.
- Bronskill, S. E., Campitelli, M. A., Iaboni, A., et al. (2018). Low-dose trazodone, benzodiazepines, and fall-related injuries in nursing homes: A matched-cohort study. *J Am Geriatr Soc*, 66(10), 1963–1971.
- Broome, M. R., Saunders, K. E., Harrison, P. J., & Marwaha, S. (2015). Mood instability: Significance, definition and measurement. *BJPsych*, 207(4), 283–285. <https://doi.org/10.1192/bjp.bp.114.158543>
- Brown, A. S., & Susser, E. S. (2008). Prenatal nutritional deficiency and risk of adult schizophrenia. *Schizophr Bull*, 34(6), 1054–1063.
- Brown, C., & Lloyd, K. (2001). Qualitative methods in psychiatric research. *Adv Psychiatr Treat*, 7(5), 350–356.
- Brown, G. (2009). Medical sociology and issues of aetiology. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 268–275). Oxford University Press.
- Brown, G. W., & Birley, J. L. (1968). Crises and life changes and the onset of schizophrenia. *J Health Soc Behav*, 9(3), 203–214.
- Brown, G. W., & Harris, T. O. (1993). Aetiology of anxiety and depressive disorders in an inner-city population. 1. Early adversity. *Psychol Med*, 23(1), 143–154.
- Brown, G. W., Monck, E. M., Carstairs, G. M., & Wing, J. K. (1962). Influence of family life on the course of schizophrenic illness. *Br J Soc Prevent Med*, 16, 55–68.
- Brown, M., & Barraclough, B. (1997). Epidemiology of suicide pacts in England and Wales, 1988–92. *BMJ*, 315(7103), 286–287. doi: 10.1136/bmj.315.7103.286
- Bruch, H. (1974). *Eating Disorders: anorexia nervosa and the person within*. Routledge & Kegan Paul.
- Brunoni, A. R., Moffa, A. H., Sampaio-Junior, B., et al. (2017). Trial of electrical direct-current therapy versus escitalopram for depression. *N Engl J Med*, 376(26), 2523–2533.
- Bryant-Waugh, R., Markham, L., Kreipe, R. E., & Walsh, B. T. (2010). Feeding and eating disorders in childhood. *Int J Eat Disord*, 43(2), 98–111. <https://doi.org/10.1002/eat.20795>
- Buchanan, A. (2008). Risk of violence by psychiatric patients: Beyond the ‘actuarial versus clinical’ assessment debate. *Psychiatr Serv*, 59(2), 184–190. <https://doi.org/10.1176/ps.2008.59.2.184>
- Bucher, M. A., Suzuki, T., & Samuel, D. B. (2019). A meta-analytic review of personality traits and their associations with mental health treatment outcomes. *Clin Psychol Rev*, 70, 51–63. <https://doi.org/10.1016/j.cpr.2019.04.002>
- Buckley, P. F., Miller, B. J., Lehrer, D. S., & Castle, D. J. (2009). Psychiatric comorbidities and schizophrenia. *Schizophr Bull*, 35(2), 383–402.
- Bundy, H., Stahl, D., & MacCabe, J. (2011). A systematic review and meta-analysis of the fertility of patients with schizophrenia and their unaffected relatives. *Acta Psychiatr Scand*, 123(2), 98–106.
- Burleigh, M. (2000). Eugenic utopias and the genetic present. *Totalit Move Polit Relig*, 1(1), 56–77.
- Burns, A., & Illiffe, S. (2009). Dementia. *BMJ*, 338, b75.
- Burns, T. (2014). Community psychiatry's achievements. *Epidemiol Psychiatr Sci*, 23(4), 337–344. <https://doi.org/10.1017/S2045796014000560>
- Burns, T., & Bale, R. (1997). Establishing a mental health liaison attachment with primary care. *Adv Psychiatr Treat*, 3(4), 219–224.
- Burns, T., Catty, J., Becker, T., et al. (2007). The effectiveness of supported employment for people with severe mental illness: A randomised controlled trial. *Lancet*, 370(9593), 1146–1152. [https://doi.org/10.1016/S0140-6736\(07\)61516-5](https://doi.org/10.1016/S0140-6736(07)61516-5)
- Burns, T., Catty, J., Dash, M., et al. (2007). Use of intensive case management to reduce time in hospital in people with severe mental illness: Systematic review and meta-regression. *BMJ*, 335(7615), 336. <https://doi.org/10.1136/bmj.39251.599.259.55>
- Burns, T., Gargan, L., Walker, L., et al. (1998). Not just bricks and mortar: Report of the Royal College of Psychiatrists Working Party on the size, staffing, structure, siting, and security of new acute adult psychiatric in-patient units. *Psychiatr Bull*, 22(8), 465–466.
- Burns, T., & Lundgren, E. (2015). *A Very Short Introduction to Psychotherapy*. Oxford University Press.
- Burns, T., Rugkåsa, J., Molodynski, A., et al. (2013). Community treatment orders for patients with psychosis (OCTET): A randomised controlled trial. *Lancet*, 381, 1627–1633. doi:10.1016/S0140-6736(13)60107-5
- Burns, T., Yeeles, K., Koshiaris, C., et al. (2015). Effect of increased compulsion on readmission to hospital or disengagement from community services for patients with psychosis: Follow-up of a cohort from the OCTET trial. *Lancet Psychiatry*, 2(10), 881–890. [https://doi.org/10.1016/s2215-0366\(15\)00231-x](https://doi.org/10.1016/s2215-0366(15)00231-x)
- Burns, T., Yeeles, K., Molodynski, A., et al. (2011). Pressures to adhere to treatment (‘leverage’) in English mental health-care. *BJPsych*, 199(2), 145–150. <https://doi.org/10.1192/bjp.bp.110.086827>
- Burti, L. (2001). Italian psychiatric reform 20 plus years after. *Acta Psychiatr Scand Suppl*, 2001(410), 41–46. <https://doi.org/10.1034/j.1600-0447.2001.1040s2041.x>
- Burton, A., Marougka, S., & Priebe, S. (2010). Do financial incentives increase treatment adherence in people with severe mental illness? A systematic review. *Epidemiol Psychiatr Soc*, 19(3), 233–242.
- Bushnell, M. C., Čeko, M., & Low, L. A. (2013). Cognitive and emotional control of pain and its disruption in chronic pain. *Nat Rev Neurosci*, 14(7), 502–511.
- Bustos, V. P., Bustos, S. S., Mascaro, A., et al. (2021). Regret after gender-affirmation surgery: A systematic review and meta-analysis of prevalence. *Plast Reconstr Surg Glob Open*, 9(3), e3477.
- Butler, R. J. (2008). Wetting and soiling. In M. J. Rutter, D. Bishop, D. S. Pine, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 5th edn. (pp. 916–929). John Wiley & Sons, Ltd.
- Butler, R. W., & Braff, D. L. (1991). Delusions: A review and integration. *Schizophr Bull*, 17(4), 633–647.

- Byrd, A. L., & Manuck, S. B. (2014). MAOA, childhood maltreatment, and antisocial behavior: Meta-analysis of a gene-environment interaction. *Biol Psychiatry*, 75(1), 9–17. <https://doi.org/10.1016/j.biopsych.2013.05.004>
- Byrne, G. (2020). Anxiety disorders in older people. In T. Denning, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 655–670). Oxford University Press.
- Cade, J. F. (1949). Lithium salts in the treatment of psychotic excitement. *Med J Aust*, 2(10), 349–352.
- Callicott, J. H., Mattay, V. S., Verchinski, B. A., et al. (2003). Complexity of prefrontal cortical dysfunction in schizophrenia: More than up or down. *Am J Psychiatry*, 160(12), 2209–2215.
- Campbell, E. J. M., Scadding, J. G., & Roberts, R. S. (1979). The concept of disease. *BMJ*, 2(6193), 757.
- Cannon, M., Clarke, M. C., & Cotter, D. R. (2014). Priming the brain for psychosis: Maternal inflammation during fetal development and the risk of later psychiatric disorder. *Am J Psychiatry*, 171(9), 901–905.
- Cannon, M., Jones, P. B., & Murray, R. M. (2002). Obstetric complications and schizophrenia: Historical and meta-analytic review. *Am J Psychiatry*, 159(7), 1080–1092.
- Caplan, G. (1964). *Principles of Preventive Psychiatry*. Basic Books.
- Carbon, M., Kane, J. M., Leucht, S., & Correll, C. U. (2018). Tardive dyskinesia risk with first-and second-generation antipsychotics in comparative randomized controlled trials: A meta-analysis. *World Psychiatry*, 17(3), 330–340.
- Cardno, A. G., & Gottesman, I. I. (2000). Twin studies of schizophrenia: From bow-and-arrow concordances to star wars Mx and functional genomics. *Am J Med Genet*, 97(1), 12–17.
- Carmi, L., Fineberg, N., Ben Arush, O., & Zohar, J. (2020). Obsessive compulsive disorder. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 987–994). Oxford University Press.
- Carnemolla, S. E., Hsieh, J. W., Sipione, R., et al. (2020). Olfactory dysfunction in frontotemporal dementia and psychiatric disorders: A systematic review. *Neurosci Biobehav Rev*, 118, 588–611. <https://doi.org/10.1016/j.neubiorev.2020.08.002>
- Carr, A. (1991). Milan systemic family therapy: A review of ten empirical investigations. *J Fam Ther*, 13(3), 237–263. <https://doi.org/10.1046/j.1991.00425.x>
- Carr, A. (2020). Evidence for the efficacy and effectiveness of systemic family therapy. In K. S. Wampler, R. B. Miller, & R. B. Seedall (Eds.), *The Handbook of Systemic Family Therapy* (pp. 119–146). Wiley.
- Carr, A., Linehan, C., O'Reilly, et al. (Eds.). (2016). *The Handbook of Intellectual Disability and Clinical Psychology Practice*. Routledge.
- Carroll, A. (2009). Are you looking at me? Understanding and managing paranoid personality disorder. *Adv Psychiatr Treat*, 15(1), 40–48.
- Carvalho, A. F., Heilig, M., Perez, A., et al. (2019). Alcohol use disorders. *Lancet*, 394(10200), 781–792.
- Casey, P. (2021). Editors comment: The debate on dissociative identity disorder. *BJPsych Adv*, 27(2), 102–103. <https://doi.org/10.1192/bja.2020.95>
- Caspi, A., Sugden, K., Moffitt, T. E., et al. (2003). Influence of life stress on depression: Moderation by a polymorphism in the 5-HTT gene. *Science*, 301(5631), 386–389.
- Castle, D. (2020). Cannabis and mental illness. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 519–524). Oxford University Press.
- Castle, D., Scott, K., Wessely, S., & Murray, R. M. (1993). Does social deprivation during gestation and early life predispose to later schizophrenia? *Soc Psychiatry Psychiatr Epidemiol*, 28, 1–4.
- Cattell, H. E. P., & Mead, A. D. (2008). The Sixteen Personality Factor Questionnaire (16PF). In G. J. Boyle, G. Matthews, & D. H. Saklofske (Eds.), *The SAGE Handbook of Personality Theory and Assessment: Volume 2 — Personality Measurement and Testing* (pp. 135–159). SAGE Publications Ltd. <https://doi.org/10.4135/9781849200479>
- Catty, J., Goddard, K., & Burns, T. (2005). Social services day care and health services day care in mental health: Do they differ? *Int J Soc Psychiatry*, 51(2), 151–161.
- Caye, A., Swanson, J. M., Coghill, D., & Rohde, L. A. (2019, Mar). Treatment strategies for ADHD: An evidence-based guide to select optimal treatment. *Mol Psychiatry*, 24(3), 390–408. <https://doi.org/10.1038/s41380-018-0116-3>
- Cermolacce, M., Sass, L., & Parnas, J. (2010). What is bizarre in bizarre delusions? A critical review. *Schizophr Bull*, 36(4), 667–679.
- Chabert, J., Allauze, E., Pereira, B., et al. (2022). Glutamatergic and N-acetylaspartate metabolites in bipolar disorder: A systematic review and meta-analysis of proton magnetic resonance spectroscopy studies. *Int J Mol Sciences*, 23(16), 8974.
- Chamberlain, A. S. (1966). Early mental hospitals in Spain. *Am J Psychiatry*, 123(2), 143–149. <https://doi.org/10.1176/aip.123.2.143>
- Chamberlain, S. R., & Baldwin, D. S. (2021). Monoamine oxidase inhibitors (MAOIs) in psychiatric practice: How to use them safely and effectively. *CNS Drugs*, 35(7), 703–716.
- Chan, R. C., Xu, T., Heinrichs, R. W., et al. (2010). Neurological soft signs in schizophrenia: A meta-analysis. *Schizophr Bull*, 36(6), 1089–1104.
- Chang, C.-W., Shao, E., & Mucke, L. (2021). Tau: Enabler of diverse brain disorders and target of rapidly evolving therapeutic strategies. *Science*, 371(6532), eabb8255.
- Chaput, J. P., Dutil, C., Featherstone, R., et al. (2020). Sleep duration and health in adults: An overview of systematic reviews. *Appl Physiol Nutr Metab*, 45(10 Suppl. 2), S218–S231. <https://doi.org/10.1139/apnm-2020-0034>
- Charman, T., Jones, C. R., Pickles, A., et al. (2011). Defining the cognitive phenotype of autism. *Brain Res*, 1380, 10–21.
- Chartonas, D., Kyratsous, M., Dracass, S., et al. (2017). Personality disorder: Still the patients psychiatrists dislike? *BJPsych Bull*, 41(1), 12–17. <https://doi.org/10.1192/pb.bp.115.052456>
- Chatterton, M. L., Stockings, E., Berk, M., et al. (2017). Psychosocial therapies for the adjunctive treatment of bipolar disorder in adults: Network meta-analysis. *BJPsych*, 210(5), 333–341.
- Chavanne, A. V., & Robinson, O. J. (2021). The overlapping neurobiology of induced and pathological anxiety: A meta-analysis of functional neural activation. *Am J Psychiatry*, 178(2), 156–164.
- Chawla, N., Anothaisintawee, T., Charoenrungrueangchai, K., et al. (2022). Drug treatment for panic disorder with or without agoraphobia: Systematic review and network meta-analysis of randomised controlled trials. *BMJ*, 376, e066084.

- Chekroud, A. M., Zotti, R. J., Shehzad, Z., et al. (2016). Cross-trial prediction of treatment outcome in depression: A machine learning approach. *Lancet Psychiatry*, 3(3), 243–250. [https://doi.org/10.1016/s2215-0366\(15\)00471-x](https://doi.org/10.1016/s2215-0366(15)00471-x)
- Chen, J., Cai, Y., Cong, E., et al. (2014). Childhood sexual abuse and the development of recurrent major depression in Chinese women. *PLoS ONE*, 9(1), e87569.
- Chen, J.-J., Zhao, L.-B., Liu, Y.-Y., et al. (2017). Comparative efficacy and acceptability of electroconvulsive therapy versus repetitive transcranial magnetic stimulation for major depression: A systematic review and multiple-treatments meta-analysis. *Behav Brain Res*, 320, 30–36.
- Chen, L. L., Abbaspour, A., Mkom, G. F., et al. (2021). Gut microbiota in psychiatric disorders: A systematic review. *Psychosom Med*, 83(7), 679–692. <https://doi.org/10.1097/psy.00000000000000959>
- Cheslack-Postava, K., & Brown, A. S. (2022). Prenatal infection and schizophrenia: A decade of further progress. *Schizophr Res*, 247, 7–15. <https://doi.org/10.1016/j.schres.2021.05.014>
- Chiarotti, F., & Venerosi, A. (2020). Epidemiology of autism spectrum disorders: A review of worldwide prevalence estimates since 2014. *Brain Sci*, 10(5), 274. <https://doi.org/10.3390/brainsci10050274>
- Chibanda, D., Cowan, F. M., Healy, J. L., et al. (2015). Psychological interventions for common mental disorders for people living with HIV in low- and middle-income countries: Systematic review. *Trop Med Int Health*, 20(7), 830–839.
- Chiesa, A., & Serretti, A. (2011). Mindfulness based cognitive therapy for psychiatric disorders: A systematic review and meta-analysis. *Psychiatry Res*, 187(3), 441–453. <https://doi.org/10.1016/j.psychres.2010.08.011>
- Chieze, M., Clavien, C., Kaiser, S., & Hurst, S. (2021). Coercive measures in psychiatry: A review of ethical arguments. *Front Psychiatry*, 12, 790886. <https://doi.org/10.3389/fpsy.2021.790886>
- Chisolm, M. S., & Payne, J. L. (2016). Management of psychotropic drugs during pregnancy. *BMJ*, 352, h5918.
- Choo, C. C., Harris, K. M., Chew, P. K., & Ho, R. C. (2019). Clinical assessment of suicide risk and suicide attempters' self-reported suicide intent: A cross-sectional study. *PLoS ONE*, 14(7), e0217613.
- Chossegros, L., Hours, M., Charnay, P., et al. (2011). Predictive factors of chronic post-traumatic stress disorder 6 months after a road traffic accident. *Accid Anal Prev*, 43(1), 471–477.
- Chouinard, G., Ross-Chouinard A., Annable L., & Jones B. D. (1980). The extrapyramidal symptom rating scale. *Can J Neurol Sci*, 7, 233–244.
- Chowdhury, A. N., & Brahma, A. (2020). Update on Koro research methodology. *Indian J Psychiatry*, 62(1), 102–104.
- Christodoulou, G. N. (1991). The delusional misidentification syndromes. *BJPsych*, 159(S14), 65–69.
- Ciccarone, D. (2019). The triple wave epidemic: Supply and demand drivers of the US opioid overdose crisis. *Int J Drug Pol*, 71, 183–188.
- Cipriani, A., Barbui, C., Salanti, G., et al. (2011). Comparative efficacy and acceptability of antimanic drugs in acute mania: A multiple-treatments meta-analysis. *Lancet*, 378(9799), 1306–1315.
- Cipriani, A., Furukawa, T. A., Salanti, G., et al. (2018). Comparative efficacy and acceptability of 21 antidepressant drugs for the acute treatment of adults with major depressive disorder: A systematic review and network meta-analysis. *Lancet*, 391(10128), 1357–1366.
- Cipriani, A., Furukawa, T. A., Salanti, G., et al. (2009). Comparative efficacy and acceptability of 12 new-generation antidepressants: A multiple-treatments meta-analysis. *Lancet*, 373(9665), 746–758.
- Cipriani, A., Salanti, G., Furukawa, T. A., et al. (2018). Antidepressants might work for people with major depression: Where do we go from here? *Lancet Psychiatry*, 5(6), 461–463. [https://doi.org/10.1016/s2215-0366\(18\)30133-0](https://doi.org/10.1016/s2215-0366(18)30133-0)
- Cipriani, A., Williams, T., Nikolakopoulou, A., et al. (2018). Comparative efficacy and acceptability of pharmacological treatments for post-traumatic stress disorder in adults: A network meta-analysis. *Psychol Med*, 48(12), 1975–1984. <https://doi.org/10.1017/s003329171700349x>
- Cipriani, A., Zhou, X., Del Giovane, C., et al. (2016). Comparative efficacy and tolerability of antidepressants for major depressive disorder in children and adolescents: A network meta-analysis. *Lancet*, 388(10047), 881–890.
- Clancy, M. J., Clarke, M. C., Connor, D. J., et al. (2014). The prevalence of psychosis in epilepsy: A systematic review and meta-analysis. *BMC Psychiatry*, 14(1), 1–9.
- Clark, D. A., & Beck, A. T. (2010). *Cognitive Therapy of Anxiety Disorders: Science and Practice*. Guildford Press.
- Clark, D. M. (2011). Implementing NICE guidelines for the psychological treatment of depression and anxiety disorders: The IAPT experience. *Int Rev Psychiatry*, 23(4), 318–327.
- Clark, E., & Koutsogianni, M. (2022). Genetic investigations pathway for people with intellectual disability, autism and/or epilepsy. *BJPsych Adv*, 28(6), 353–362. <https://doi.org/10.1192/bja.2022.44>
- Clark, L., & Harrison, J. (2001). Assessment instruments. In W. Livesley (Ed.), *Handbook of Personality Disorders: Theory, Research, and Treatment* (pp. 277–306). Guildford Press.
- Clarke, K., King, M., & Prost, A. (2013). Psychosocial interventions for perinatal common mental disorders delivered by providers who are not mental health specialists in low- and middle-income countries: A systematic review and meta-analysis. *PLoS Med*, 10(10), e1001541.
- Clarke, S., Allerhand, L. A., & Berk, M. S. (2019). Recent advances in understanding and managing self-harm in adolescents. *F1000Res*, 8, 1794. doi: 10.12688/f1000research.19868.1
- Clayton, A. H., Alkis, A. R., Parikh, N. B., & Votta, J. G. (2016). *Sexual Dysfunction Due to Psychotropic Medications*. Psychiatric Clinics of North America.
- Clayton, A. H., & Valladares Juarez, E. M. (2019). Female sexual dysfunction. *Med Clin North Am*, 103, 681–698. <https://doi.org/10.1016/j.mcna.2019.02.008>
- Cleare, A., Pariente, C. M., Young, A. H., et al. (2015). Evidence-based guidelines for treating depressive disorders with antidepressants: A revision of the 2008 British Association for Psychopharmacology guidelines. *J Psychopharmacol*, 29(5), 459–525.
- Cleckley, H. M. (1951). The mask of sanity. *Postgrad Med*, 9(3), 193–197.

- Clinical Psychiatry Committee. (1965). Clinical trial of the treatment of depression. *BMJ*, 1, 881–886.
- Cloitre, M. (2020). ICD-11 complex post-traumatic stress disorder: Simplifying diagnosis in trauma populations. *BJPsych*, 216(3), 129–131. <https://doi.org/10.1192/bjp.2020.43>
- Cloninger, C. R., Svrakic, D. M., & Przybeck, T. R. (1993). A psychobiological model of temperament and character. *Arch Gen Psychiatry*, 50(12), 975–990. <https://doi.org/10.1001/archpsyc.1993.01820240059008>
- CNV and Schizophrenia Working Groups of the Psychiatric Genomics Consortium. (2017). Contribution of copy number variants to schizophrenia from a genome-wide study of 41,321 subjects. *Nature Genet*, 49, 27–35.
- Coccaro, E. F., Lee, R., & Kavoussi, R. J. (2010). Aggression, suicidality, and intermittent explosive disorder: Serotonergic correlates in personality disorder and healthy control subjects. *Neuropsychopharmacol*, 35(2), 435–444. <https://doi.org/10.1038/npp.2009.148>
- Cochrane, A. L. (1979). 1931–1971: A critical review with particular reference to the medical profession. In G. Teeling-Smith, & N. Wells (Eds.), *Medicines for the Year 2000* (pp. 1–11). Office of Health Economics.
- Cohen, C. I., Meesters, P. D., & Zhao, J. (2015). New perspectives on schizophrenia in later life: Implications for treatment, policy, and research. *Lancet Psychiatry*, 2(4), 340–350.
- Cohen, N. J., & Farnia, F. (2015). Adoption. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 273–284). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch22>
- Coid, J., Petruckevitch, A., Bebbington, P., et al. (2002). Ethnic differences in prisoners. 1: Criminality and psychiatric morbidity. *BJPsych*, 181, 473–480. <https://doi.org/10.1192/bjp.181.6.473>
- Coid, J., Yang, M., Tyrer, P., et al. (2006). Prevalence and correlates of personality disorder in Great Britain. *BJPsych*, 188, 423–431. <https://doi.org/10.1192/bjp.188.5.423>
- Coid, J. W., Ullrich, S., Kallis, C., et al. (2013). The relationship between delusions and violence: Findings from the East London first episode psychosis study. *JAMA Psychiatry*, 70(5), 465–471. <https://doi.org/10.1001/jamapsychiatry.2013.12>
- Coker, K. L., Smith, P. H., Westphal, A., et al. (2014). Crime and psychiatric disorders among youth in the US population: An analysis of the National Comorbidity Survey-Adolescent Supplement. *J Am Acad Child Adolesc Psychiatry*, 53(8), 888–898. <https://doi.org/10.1016/j.jaac.2014.05.007>
- Cole, E. J., Phillips, A. L., Bentzley, B. S., et al. (2022). Stanford neuromodulation therapy (SNT): A double-blind randomized controlled trial. *Am J Psychiatry*, 179(2), 132–141.
- Coleman, E., Radix, A. E., Bouman, W. P., et al. (2022). Standards of care for the health of transgender and gender diverse people, Version 8. *Int J Transgend Health*, 23(Suppl1), S1–S259. <https://doi.org/10.1080/26895269.2022.2100644>
- Collacott, R., Cooper, S., Branford, D., & McGrother, C. (1998). Behaviour phenotype for Down's syndrome. *BJPsych*, 172(1), 85–89.
- Collin, G., Turk, E., & van den Heuvel, M. P. (2016). Connectomics in schizophrenia: From early pioneers to recent brain network findings. *Biol Psychiatry Cogn Neurosci Neuroimaging*, 1(3), 199–208.
- Collings, S., & Llewellyn, G. (2012). Children of parents with intellectual disability: Facing poor outcomes or faring okay? *J Intellect Dev Disabil*, 37(1), 65–82.
- Collins, P. Y. (2020). What is global mental health? *World Psychiatry*, 19(3), 265–266. <https://doi.org/10.1002/wps.20728>
- Collins, P. Y., Patel, V., Joestl, S. S., et al. (2011). Grand challenges in global mental health. *Nature*, 475(7354), 27–30. <https://doi.org/10.1038/475027a>
- Collishaw, S. (2015). Annual research review: Secular trends in child and adolescent mental health. *J Child Psychol Psychiatry*, 56(3), 370–393. <https://doi.org/10.1111/jcpp.12372>
- Conner, A., Azrael, D., & Miller, M. (2019). Suicide case-fatality rates in the United States, 2007 to 2014: A nationwide population-based study. *Ann Intern Med*, 171(12), 885–895. <https://doi.org/10.7326/m19-1324>
- Connors, M. H., Quinto, L., & Brodaty, H. (2019). Longitudinal outcomes of patients with pseudodementia: A systematic review. *Psychol Med*, 49(5), 727–737.
- Constantin, D., Dinu, E. A., Rogozea, L., et al. (2020). Therapeutic interventions for adjustment disorder: A systematic review. *Am J Ther*, 27(4), e375–e386. <https://doi.org/10.1097/mjt.0000000000001170>
- Cook, J., Gallagher-Thompson, D., & Hepple, J. (2005). Psychotherapy with older adults. In G. Gabbard, J. Beck, & J. Holmes (Eds.), *Oxford Textbook of Psychotherapy* (pp. 381–392). Oxford University Press.
- Cooper, J. E. (1979). *Crisis Admission Units and Emergency Psychiatric Services. Report on a Study*. World Health Organization.
- Cooper, J. E., Kendall, R. E., Gurland, B. J., et al. (1972). *Psychiatric Diagnosis in New York and London: A Comparative Study of Mental Hospital Admissions*. Oxford University Press.
- Cooper, K., Russell, A., Mandy, W., & Butler, C. (2020). The phenomenology of gender dysphoria in adults: A systematic review and meta-synthesis. *Clin Psychol Rev*, 80, 101875. <https://doi.org/10.1016/j.cpr.2020.101875>
- Cooper, P. J., Landman, M., Tomlinson, M., et al. (2002). Impact of a mother–infant intervention in an indigent peri-urban South African context. *BJPsych*, 180(1), 76–81.
- Cooper, P. J., Tomlinson, M., Swartz, L., et al. (2009). Improving quality of mother–infant relationship and infant attachment in socioeconomically deprived community in South Africa: Randomised controlled trial. *BMJ*, 338, b974. <https://doi.org/10.1136/bmj.b974>
- Cooper, S. J., Reynolds, G. P., Barnes, T., et al. (2016). BAP guidelines on the management of weight gain, metabolic disturbances and cardiovascular risk associated with psychosis and antipsychotic drug treatment. *J Psychopharmacol*, 30(8), 717–748.
- Cooper, Z., & Dalle Grave, R. (2017). Eating disorders: Transdiagnostic theory and treatment. In S. G. Hofmann & G. J. G. Asmundson (Eds.), *The Science of Cognitive Behavioral Therapy* (pp. 337–357). Elsevier.
- Copeland, W. E., Wolke, D., Shanahan, L., & Costello, E. J. (2015). Adult functional outcomes of common childhood psychiatric problems: A prospective, longitudinal study. *JAMA*

- Psychiatry*, 72(9), 892–899. <https://doi.org/10.1001/jamapsychiatry.2015.0730>
- Coppus, A. (2013). People with intellectual disability: What do we know about adulthood and life expectancy? *Dev Disabil Res Rev*, 18(1), 6–16.
- Corbett, A., Husebo, B., Malcangio, M., et al. (2012). Assessment and treatment of pain in people with dementia. *Nat Rev Neurol*, 8(5), 264–274.
- Cordeiro, L., Ballinger, E., Hagerman, R., & Hessel, D. (2010). Clinical assessment of DSM-IV anxiety disorders in fragile X syndrome: Prevalence and characterization. *J Neurodev Disord*, 3(1), 57.
- Correll, C. U., Rubio, J. M., & Kane, J. M. (2018). What is the risk-benefit ratio of long-term antipsychotic treatment in people with schizophrenia? *World Psychiatry*, 17(2), 149–160. <https://doi.org/10.1002/wps.20516>
- Correll, C. U., Solmi, M., Croatto, G., et al. (2022). Mortality in people with schizophrenia: A systematic review and meta-analysis of relative risk and aggravating or attenuating factors. *World Psychiatry*, 21(2), 248–271. <https://doi.org/10.1002/wps.20994>
- Costa, P. T. J., McCrae, R. M., & Löckenhoff, C. E. (2019). Personality across the life span. *Ann Rev Psychol*, 70(1), 423–448. <https://doi.org/10.1146/annurev-psych-010418-103244>
- Costanzo, S., de Gaetano, G., Di Castelnuovo, A., et al. (2019). Moderate alcohol consumption and lower total mortality risk: Justified doubts or established facts? *Nutr Metab Cardiovasc Dis*, 29(10), 1003–1008.
- Costello, A., Hudson, E., Morrissey, S., et al. (2022). Management of psychotropic medications in adults with intellectual disability: A scoping review. *Ann Med*, 54(1), 2486–2499. <https://doi.org/10.1080/07853890.2022.2121853>
- Coupland, C. A., Hill, T., Denning, T., et al. (2019). Anticholinergic drug exposure and the risk of dementia: A nested case-control study. *JAMA Int Med*, 179(8), 1084–1093.
- Cousins, L., & Goodyer, I. M. (2015). Antidepressants and the adolescent brain. *J Psychopharmacol*, 29(5), 545–555. <https://doi.org/10.1177/0269881115573542>
- Couteur, A., & Szatmari, P. (2015). Autism spectrum disorder. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 661–682). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch51>
- Cowen, P. J. (2015). Neuroendocrine and neurochemical processes in depression. In R. J. DeRubeis & D. R. Strunk (Eds.), *The Oxford Handbook of Mood Disorders* (pp. 190–200). Oxford University Press.
- Cowen, P. J. (2019). Mechanism of action of ECT. In I. Ferrier & J. Waite (Eds.), *The ECT Handbook* (pp. 13–23). Cambridge University Press.
- Cowen, P. J., & Anderson, I. M. (2015). New approaches to treating resistant depression. *BJPsych Adv*, 21, 315–323.
- Coyle, J. T., Ruzicka, W. B., & Balu, D. T. (2020). Fifty years of research on schizophrenia: The ascendance of the glutamatergic synapse. *Am J Psychiatry*, 177(12), 1119–1128. <https://doi.org/10.1176/appi.ajp.2020.20101481>
- Craddock, N., & Mynors-Wallis, L. (2014). Psychiatric diagnosis: Impersonal, imperfect and important. *BJPsych*, 204(2), 93–95.
- Cramer, R. J., & Kapusta, N. D. (2017). A social-ecological framework of theory, assessment, and prevention of suicide. *Front Psychol*, 8, 1756. <https://doi.org/10.3389/fpsyg.2017.01756>
- Crammer, J. (1990). *Asylum History: Buckinghamshire County Pauper Lunatic Asylum-St. John's*. Springer Science & Business.
- Crawford, M. J., Rushwaya, T., Bajaj, P., et al. (2012). The prevalence of personality disorder among ethnic minorities: Findings from a national household survey. *Pers Ment Health*, 6(3), 175–182. <https://doi.org/10.1002/pmh.1186>
- Crawford, M. J., Thana, L., Farquharson, L., et al. (2016). Patient experience of negative effects of psychological treatment: Results of a national survey†. *BJPsych*, 208(3), 260–265. <https://doi.org/10.1192/bjp.bp.114.162628>
- Creswell, C., Waite, P., & Hudson, J. (2020). Practitioner review: Anxiety disorders in children and young people—assessment and treatment. *J Child Psychol Psychiatry*, 61(6), 628–643. <https://doi.org/10.1111/jcpp.13186>
- Cruz, S. L., & Bowen, S. E. (2021). The last two decades on pre-clinical and clinical research on inhalant effects. *Neurotoxicol Teratol*, 87, 106999.
- Cuijpers, P., Donker, T., Weissman, M. M., et al. (2016). Interpersonal psychotherapy for mental health problems: A comprehensive meta-analysis. *Am J Psychiatry*, 173(7), 680–687. <https://doi.org/10.1176/appi.ajp.2015.15091141>
- Cuijpers, P., & Gentili, C. (2017). Psychological treatments are as effective as pharmacotherapies in the treatment of adult depression: A summary from randomized clinical trials and neuroscience evidence. *Res Psychother*, 20(2), 273. <https://doi.org/10.4081/ripppo.2017.273>
- Cuijpers, P., Javed, A., & Bhui, K. (2023). The WHO World Mental Health Report: A call for action. *BJPsych*, 222(6), 227–229. <https://doi.org/10.1192/bjp.2023.9>
- Cuijpers, P., Karyotaki, E., Ciharova, M., et al. (2021). The effects of psychotherapies for depression on response, remission, reliable change, and deterioration: A meta-analysis. *Acta Psychiatr Scand*, 144(3), 288–299.
- Cuijpers, P., Noma, H., Karyotaki, E., et al. (2020). A network meta-analysis of the effects of psychotherapies, pharmacotherapies and their combination in the treatment of adult depression. *World Psychiatry*, 19(1), 92–107. <https://doi.org/10.1002/wps.20701>
- Cuijpers, P., Quero, S., Noma, H., et al. (2021). Psychotherapies for depression: A network meta-analysis covering efficacy, acceptability and long-term outcomes of all main treatment types. *World Psychiatry*, 20(2), 283–293.
- Cuijpers, P., Sijbrandij, M., Koole, S., et al. (2014a). Psychological treatment of generalized anxiety disorder: A meta-analysis. *Clin Psychol Rev*, 34(2), 130–140.
- Cuijpers, P., Sijbrandij, M., Koole, S. L., et al. (2014b). Adding psychotherapy to antidepressant medication in depression and anxiety disorders: A meta-analysis. *Focus*, 12(3), 347–358.
- Cuijpers, P., Sijbrandij, M., Koole, S. L., et al. (2013). The efficacy of psychotherapy and pharmacotherapy in treating depressive and anxiety disorders: A meta-analysis of direct comparisons. *World Psychiatry*, 12(2), 137–148.
- Cuijpers, P., Veen, S. C. V., Sijbrandij, M., et al. (2020). Eye movement desensitization and reprocessing for mental health problems: A systematic review and meta-analysis. *Cogn*

- Behav Ther*, 49(3), 165–180. <https://doi.org/10.1080/16506073.2019.1703801>
- Cutajar, M. C., Mullen, P. E., Oglloff, J., et al. (2010). Suicide and fatal drug overdose in child sexual abuse victims: A historical cohort study. *Med J Aust*, 192(4), 184–187.
- Cuthbert, B. N., & Insel, T. R. (2013). Toward the future of psychiatric diagnosis: The seven pillars of RDoC. *BMC Med*, 11(1), 1–8.
- Cybulski, L., Ashcroft, D. M., Carr, M. J., et al. (2021). Temporal trends in annual incidence rates for psychiatric disorders and self-harm among children and adolescents in the UK, 2003–2018. *BMC Psychiatry*, 21, 1–12.
- D'Souza, D. C., DiForti, M., Ganesh, S., et al. (2022). Consensus paper of the WFSBP task force on cannabis, cannabinoids and psychosis. *World J Biol Psychiatry*, 23(10), 719–742.
- Da Costa, R. T., Sardinha, A., & Nardi, A. E. (2008). Virtual reality exposure in the treatment of fear of flying. *Aviat Space Environ Med*, 79(9), 899–903.
- Dagnan, D., Jackson, I., & Eastlake, L. (2018). A systematic review of cognitive behavioural therapy for anxiety in adults with intellectual disabilities. *J Intellect Disabil Res*, 62(11), 974–991.
- Dalmau, J., & Graus, F. (2018). Antibody-mediated encephalitis. *N Engl J Med*, 378(9), 840–851.
- Dalton, L., Rapa, E., Ziebland, S., et al. (2019). Communication with children and adolescents about the diagnosis of a life-threatening condition in their parent. *Lancet*, 393(10176), 1164–1176. [https://doi.org/10.1016/s0140-6736\(18\)33202-1](https://doi.org/10.1016/s0140-6736(18)33202-1)
- Dane, A., & Bhatia, K. (2023). The social media diet: A scoping review to investigate the association between social media, body image and eating disorders amongst young people. *PLoS Glob Public Health*, 3(3), e0001091. <https://doi.org/10.1371/journal.pgph.0001091>
- Danese, A., & McCrory, E. (2015). Child maltreatment. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 364–375). John Wiley & Sons, Ltd.
- Danysz, W., & Parsons, C. G. (2012). Alzheimer's disease, β -amyloid, glutamate, NMDA receptors and memantine—searching for the connections. *Br J Pharmacol*, 167(2), 324–352.
- Darwish, M., Atlantis, E., & Mohamed-Taysir, T. (2014). Psychological outcomes after hysterectomy for benign conditions: A systematic review and meta-analysis. *Eur J Obstet Gynecol Reprod Biol*, 174, 5–19.
- Dauwan, M., Begemann, M. J., Heringa, S. M., & Sommer, I. E. (2016). Exercise improves clinical symptoms, quality of life, global functioning, and depression in schizophrenia: A systematic review and meta-analysis. *Schizophr Bull*, 42(3), 588–599.
- David, A. S. (2009a). Basic concepts in neuropsychiatry. In A. S. David, S. Fleminger, M. D. Kopelman, S. Lovestone, & J. D. C. Mellers (Eds.), *Lishman's Organic Psychiatry*, 4th edn. (pp. 3–27). Wiley-Blackwell.
- David, A. S. (2009b). Clinical assessment. In A. S. David, S. Fleminger, M. D. Kopelman, S. Lovestone, & J. D. C. Mellers (Eds.), *Lishman's Organic Psychiatry*, 4th edn. (pp. 103–163). Wiley-Blackwell.
- David, D., Cristea, I., & Hofmann, S. G. (2018). Why cognitive behavioral therapy is the current gold standard of psychotherapy. *Front Psychiatry*, 9, 4. <https://doi.org/10.3389/fpsy.2018.00004>
- David, A. S., Fleminger, S., Kopelman, M. D., et al. (Eds.). (2009). *Lishman's Organic Psychiatry: A Textbook of Neuropsychiatry*, 4th edn. Wiley-Blackwell.
- Davies, C., Cipriani, A., Ioannidis, J. P. A., et al. (2018). Lack of evidence to favor specific preventive interventions in psychosis: A network meta-analysis. *World Psychiatry*, 17(2), 196–209. <https://doi.org/10.1002/wps.20526>
- Davies, C., Segre, G., Estradé, A., et al. (2020). Prenatal and perinatal risk and protective factors for psychosis: a systematic review and meta-analysis. *Lancet Psychiatry*, 7(5), 399–410. [https://doi.org/10.1016/s2215-0366\(20\)30057-2](https://doi.org/10.1016/s2215-0366(20)30057-2)
- Davies, G., Welham, J., Chant, D., et al. (2003). A systematic review and meta-analysis of northern hemisphere season of birth studies in schizophrenia. *Schizophr Bull*, 29(3), 587–593.
- Davies, S., Esler, M., & Nutt, D. (2010). Anxiety-bridging the heart/mind divide. *J Psychopharmacol*, 24(5), 633–638.
- Daviet, R., Aydogan, G., Jagannathan, K., et al. (2022). Associations between alcohol consumption and gray and white matter volumes in the UK biobank. *Nature Comm*, 13(1), 1175.
- Davis, D. H., Muniz-Terrera, G., Keage, H. A., et al. (2017). Association of delirium with cognitive decline in late life: A neuropathologic study of 3 population-based cohort studies. *JAMA Psychiatry*, 74(3), 244–251.
- Davison, S. E. (2002). Principles of managing patients with personality disorder. *Adv Psychiatric Treat*, 8(1), 1–9. <https://doi.org/10.1192/apt.8.1.1>
- Dawson, J. (2005). *Community Treatment Orders: International Comparisons*. Otago University.
- Deary, I. J., Johnson, W., & Houlihan, L. M. (2009). Genetic foundations of human intelligence. *Hum Genet*, 126(1), 215–232. doi: 10.1007/s00439-009-0655-4
- de Bartolomeis, A., Barone, A., Vellucci, L., et al. (2022). Linking inflammation, aberrant glutamate-dopamine interaction, and post-synaptic changes: Translational relevance for schizophrenia and antipsychotic treatment: A systematic review. *Mol Neurobiol*, 59(10), 6460–6501. <https://doi.org/10.1007/s12035-022-02976-3>
- De Crescenzo, F., D'Alò, G. L., Ostinelli, E. G., et al. (2022). Comparative effects of pharmacological interventions for the acute and long-term management of insomnia disorder in adults: A systematic review and network meta-analysis. *Lancet*, 400, 170–184.
- de la Torre-Ubieta, L., Won, H., Stein, J. L., & Geschwind, D. H. (2016). Advancing the understanding of autism disease mechanisms through genetics. *Nature Med*, 22(4), 345–361.
- de Stefano, A., & Ducci, G. (2008). Involuntary admission and compulsory treatment in Europe: An overview. *Int J Ment Health*, 37(3), 10–21. <https://doi.org/10.2753/IMH0020-741370301>
- de Zwart, S. M. C., Brouwer, R. M., Tsouli, A., et al. (2019). Running in the family? Structural brain abnormalities and IQ in offspring, siblings, parents, and co-twins of patients with schizophrenia. *Schizophr Bull*, 45(6), 1209–1217. <https://doi.org/10.1093/schbul/sby182>
- Deak, J. D., & Johnson, E. C. (2021). Genetics of substance use disorders: A review. *Psychol Med*, 51(13), 2189–2200.
- Dean, R. L., Marquardt, T., Hurdacas, C., et al. (2021). Ketamine and other glutamate receptor modulators for depression in

- adults with bipolar disorder. *Cochrane Database Syst Rev* 10(10), CD011611.
- Deb, S., Nancarrow, T., Limbu, B., et al. (2020). UK psychiatrists' experience of withdrawal of antipsychotics prescribed for challenging behaviours in adults with intellectual disabilities and/or autism. *BJPsych Open*, 6(5), e112. <https://doi.org/10.1192/bjo.2020.97>
- Debast, I., van Alphen, S. P., Rossi, G., et al. (2014). Personality traits and personality disorders in late middle and old age: Do they remain stable? A literature review. *Clin Gerontol*, 37(3), 253–271.
- Del Matto, L., Muscas, M., Murru, A., et al. (2020). Lithium and suicide prevention in mood disorders and in the general population: A systematic review. *Neurosci Biobehav Rev*, 116, 142–153.
- DeLong, M. R., & Wichmann, T. (2007). Circuits and circuit disorders of the basal ganglia. *Arch Neurol*, 64(1), 20–24.
- del Pino Casado, R., Rodriguez Cardosa, M., López-Martínez, C., & Orgeta, V. (2019). The association between subjective caregiver burden and depressive symptoms in carers of older relatives: A systematic review and meta-analysis. *PLoS ONE*, 14(5), e0217648.
- Demeter, C. A., Townsend, L. D., Wilson, M., & Findling, R. L. (2022). Current research in child and adolescent bipolar disorder. *Dialogues Clin Neurosci*, 10(2), 215–228.
- Demontis, D., Walters, G. B., Athanasiadis, G., et al. (2022). Genome-wide analyses of ADHD identify 27 risk loci, refine the genetic architecture and implicate several cognitive domains. *medRxiv*. doi: 10.1038/s41588-022-01285-8
- Demyttenaere, K., Bruffaerts, R., Posada-Villa, J., et al. (2004). Prevalence, severity, and unmet need for treatment of mental disorders in the World Health Organization World Mental Health Surveys. *JAMA*, 291(21), 2581–2590. <https://doi.org/10.1001/jama.291.21.2581>
- Dening, T., & K, G. (2020). Principles of service provision in old age psychiatry. In T. Dening, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 281–306). Oxford University Press.
- Dening, T., & Thomas, A. (Eds.). (2013). *Oxford Textbook of Old Age Psychiatry*, 2nd edn. Oxford University Press.
- Denman, C. (2011). The place of psychotherapy in modern psychiatric practice. *Adv Psychiatric Treat*, 17(4), 243–249. <https://doi.org/10.1192/apt.bp.109.007807>
- Denollet, J., Schiffer, A. A., & Spek, V. (2010). A general propensity to psychological distress affects cardiovascular outcomes: Evidence from research on the type D (distressed) personality profile. *Circ Cardiovasc Qual Outcomes*, 3(5), 546–557. <https://doi.org/10.1161/circoutcomes.109.934406>
- Department for Education. (2022). Special educational needs in England. <https://explore-education-statistics.service.gov.uk/find-statistics/special-educational-needs-in-england/2022-23>
- Department of Health and Social Care. (2009). Living Well with Dementia: A National Dementia Strategy. www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsPolicyAndGuidance/DH_094058
- Derby, I. M. (1933). Manic-depressive 'exhaustion' deaths: An analysis of 'exhaustion' case histories. *Psychiatr Q*, 7, 436–449.
- DeTure, M. A., & Dickson, D. W. (2019). The neuropathological diagnosis of Alzheimer's disease. *Mol Neurodegener*, 14(1), 1–18.
- Devanand, D., Lee, S., Manly, J., et al. (2015). Olfactory deficits predict cognitive decline and Alzheimer dementia in an urban community. *Neurology*, 84(2), 182–189.
- Devereaux, P., Bhandari, M., Montori, V. M., et al. (2002). Double blind, you have been voted off the island! *Evid Based Ment Health*, 5(2), 36.
- Devinsky, O., Vezzani, A., O'Brien, T. J., et al. (2018). Epilepsy. *Nat Rev Dis Primers*, 4(1), 18024.
- Devoe, D. J., Han, A., Anderson, A., et al. (2023). The impact of the COVID-19 pandemic on eating disorders: A systematic review. *Int J Eat Disord*, 56(1), 5–25.
- Dickens, C. M., McGowan, L., Percival, C., et al. (2006). Contribution of depression and anxiety to impaired health-related quality of life following first myocardial infarction. *BJPsych*, 189, 367–372. <https://doi.org/10.1192/bjp.bp.105.018234>
- Dickey, C. C., McCarley, R. W., & Shenton, M. E. (2002). The brain in schizotypal personality disorder: A review of structural MRI and CT findings. *Harv Rev Psychiatry*, 10, 1–15. <https://doi.org/10.1080/106732202016201>
- DiStefano, C., Sadhwani, A., & Wheeler, A. C. (2020). Comprehensive assessment of individuals with significant levels of intellectual disability: Challenges, strategies, and future directions. *Am J Intellect Dev Disabil*, 125(6), 434–448.
- Djulfbegovic, B., & Guyatt, G. H. (2017). Progress in evidence-based medicine: A quarter century on. *Lancet*, 390(10092), 415–423. [https://doi.org/10.1016/s0140-6736\(16\)31592-6](https://doi.org/10.1016/s0140-6736(16)31592-6)
- Dobrescu, S. R., Dinkler, L., Gillberg, C., Rastam, M., Gillberg, C., & Wentz, E. (2020). Anorexia nervosa: 30-year outcome. *Br J Psychiatry*, 216, 97–104. <https://doi.org/10.1192/bjp.2019.113>
- Doering, S., Hörz, S., Rentrop, M., et al. (2010). Transference-focused psychotherapy v. treatment by community psychotherapists for borderline personality disorder: randomised controlled trial. *BJPsych*, 196(5), 389–395. <https://doi.org/10.1192/bjp.bp.109.070177>
- Doherty, A. M. (2015). Psychiatric aspects of diabetes mellitus. *BJPsych Adv*, 21(6), 407–416. <https://doi.org/10.1192/apt.bp.114.013532>
- Doherty, A. M., & Dinneen, S. F. (2021). An introduction to psychiatry in endocrinology. In A. M. Doherty, A. M. Egan, & S. Dinneen (Eds.), *Mental Health, Diabetes and Endocrinology* (pp. 1–6). Cambridge University Press. <https://doi.org/DOI:10.1017/9781911623625.002>
- Dolan, M., Deakin, W. J., Roberts, N., & Anderson, I. (2002). Serotonergic and cognitive impairment in impulsive aggressive personality disordered offenders: Are there implications for treatment? *Psychol Med*, 32(1), 105–117. <https://doi.org/10.1017/s0033291701004688>
- Dold, M., Li, C., Gillies, D., & Leucht, S. (2013). Benzodiazepine augmentation of antipsychotic drugs in schizophrenia: A meta-analysis and Cochrane review of randomized controlled trials. *Eur Neuropsychopharmacol*, 23(9), 1023–1033.
- Done, D. J., Crow, T. J., Johnstone, E. C., & Sacker, A. (1994). Childhood antecedents of schizophrenia and affective illness: Social adjustment at ages 7 and 11. *BMJ*, 309(6956), 699–703.
- Dong, C., Shi, H., Liu, P., et al. (2022). A critical overview of systematic reviews and meta-analyses of light therapy for non-seasonal depression. *Psychiatry Res*, 314, 114686.

- Dong, D., Wang, Y., Chang, X., et al. (2018). Dysfunction of large-scale brain networks in schizophrenia: A meta-analysis of resting-state functional connectivity. *Schizophr Bull*, 44(1), 168–181. <https://doi.org/10.1093/schbul/sbx034>
- Dorahy, M. J., Brand, B. L., Şar, V., et al. (2014). Dissociative identity disorder: An empirical overview. *Aust N Z J Psychiatry*, 48(5), 402–417.
- d'Orban, P. T. (1979). Women who kill their children. *BJPsych*, 134, 560–571. <https://doi.org/10.1192/bjp.134.6.560>
- Douglas, T., Pugh, J., Singh, I., et al. (2017). Risk assessment tools in criminal justice and forensic psychiatry: The need for better data. *Eur Psychiatry*, 42, 134–137. <https://doi.org/10.1016/j.eurpsy.2016.12.009>
- Downs, J., & Kelly, C. (2025). Improving understanding, recognition and treatment for men with anorexia nervosa. *Br J Hosp Med*, 86, 1–13.
- Dragioti, E., Radua, J., Solmi, M., et al. (2022). Global population attributable fraction of potentially modifiable risk factors for mental disorders: A meta-umbrella systematic review. *Mol Psychiatry*, 27(8), 3510–3519. <https://doi.org/10.1038/s41380-022-01586-8>
- Dragioti, E., Solmi, M., Favaro, A., et al. (2019). Association of antidepressant use with adverse health outcomes: A systematic umbrella review. *JAMA Psychiatry*, 76(12), 1241–1255.
- Dressing, H., Kuehner, C., & Gass, P. (2006). The epidemiology and characteristics of stalking. *Curr Opin Psychiatry*, 19(4), 395–399.
- Drevets, W. C., Wittenberg, G. M., Bullmore, E. T., & Manji, H. K. (2022). Immune targets for therapeutic development in depression: towards precision medicine. *Nat Rev Drug Discov*, 21(3), 224–244. <https://doi.org/10.1038/s41573-021-00368-1>
- Driessen, E., Hegelmaier, L. M., Abbass, A. A., et al. (2015). The efficacy of short-term psychodynamic psychotherapy for depression: A meta-analysis update. *Clin Psychol Rev*, 42, 1–15.
- Drury, A. (2019). What becomes of chronic juvenile delinquents? Multifinality at midlife. *Youth Violence Juv Justice*, 18(2), 119–134.
- Dubreucq, J., Haesebaert, F., Plasse, J., et al. (2022). A systematic review and meta-analysis of social skills training for adults with autism spectrum disorder. *J Autism Dev Disord*, 52(4), 1598–1609. <https://doi.org/10.1007/s10803-021-05058-w>
- Ducat, L., Philipson, L. H., & Anderson, B. J. (2014). The mental health comorbidities of diabetes. *JAMA*, 312(7), 691–692. <https://doi.org/10.1001/jama.2014.8040>
- Duffy, R. M., & Kelly, B. D. (2015). Psychiatric assessment and treatment of survivors of torture. *BJPsych Adv*, 21(2), 106–115.
- Dugbartey, A. T. (1998). Neurocognitive aspects of hypothyroidism. *Arch Intern Med*, 158(13), 1413–1418. <https://doi.org/10.1001/archinte.158.13.1413>
- Dunne, R., O'Neill-Kerr, A., McLoughlin, D. M., & Waite, J. (2019). Practical aspects of ECT. In I. Ferrier & J. Waite (Eds.), *The ECT Handbook* (pp. 183–201). Cambridge University Press.
- Dwyer, J., & Reid, S. (2004). Ganster's syndrome. *Lancet*, 364(9432), 471–473. [https://doi.org/10.1016/s0140-6736\(04\)16772-x](https://doi.org/10.1016/s0140-6736(04)16772-x)
- Eastwood, S., & Bisson, J. I. (2008). Management of factitious disorders: A systematic review. *Psychosom*, 77(4), 209–218. <https://doi.org/10.1159/000126072>
- Eaton, W. W., Bienvenu, O. J., & Miloyan, B. (2018). Specific phobias. *Lancet Psychiatry*, 5(8), 678–686.
- Ecker, C., Bookheimer, S. Y., & Murphy, D. G. M. (2015). Neuroimaging in autism spectrum disorder: Brain structure and function across the lifespan. *Lancet Neurol*, 14(11), 1121–1134, Article 147. [https://doi.org/10.1016/S1474-4422\(15\)00050-2](https://doi.org/10.1016/S1474-4422(15)00050-2)
- Eddy, K. T., Harshman, S. G., Becker, K. R., et al. (2019). Radcliffe ARFID Workgroup: Toward operationalization of research diagnostic criteria and directions for the field. *Int J Eat Disord*, 52, 361–366. <https://doi.org/10.1002/eat.23042>
- Eddy, K. T., Kinasz, K., Murray, H. B., & Le Grange, D. (2021). Eating and feeding disorders. In M. K. Dulcan (Ed.), *Dulcan's Textbook of Child and Adolescent Psychiatry* (pp. 411–436). American Psychiatric Publications.
- Edmondson, A. J., Brennan, C. A., & House, A. O. (2016). Non-suicidal reasons for self-harm: A systematic review of self-reported accounts. *J Affect Disord*, 191, 109–117.
- Edwards, G., Orford, J., Egert, S., et al. (1977). Alcoholism: A controlled trial of 'treatment' and 'advice'. *J Stud Alcohol*, 38(5), 1004–1031.
- Edwards, J., McGorry, P. D., & Pennell, K. (2000). Models of early intervention in psychosis: An analysis of service approaches. In M. Birchwood, D. Fowler, & C. Jackson (Eds.), *Early Intervention in Psychosis: A Guide to Concepts, Evidence and Interventions* (pp. 281–314). Wiley.
- Ehlers, A., Wild, J., Warnock-Parkes, E., et al. (2023). Therapist-assisted online psychological therapies differing in trauma focus for post-traumatic stress disorder (STOP-PTSD): A UK-based, single-blind, randomised controlled trial. *Lancet Psychiatry*, 10(8), 608–622. [https://doi.org/10.1016/s2215-0366\(23\)00181-5](https://doi.org/10.1016/s2215-0366(23)00181-5)
- Ehrlich, T. J., Ryan, K. A., Burdick, K. E., et al. (2022). Cognitive subgroups and their longitudinal trajectories in bipolar disorder. *Acta Psychiatr Scand*, 146(3), 240–250.
- Eisler, I., & Lask, J. (2015). Family interventions. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 510–520). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch39>
- Ekstrand, J., Fattah, C., Persson, M., et al. (2022). Racemic ketamine as an alternative to electroconvulsive therapy for unipolar depression: A randomized, open-label, non-inferiority trial (KetECT). *Int J Neuropsychopharmacol*, 25(5), 339–349.
- Elaut, E., & Heylens, G. (2020). Gender dysphoria. In J. R. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1191–1198). Oxford University Press.
- Elias, A., Thomas, N., & Sackeim, H. A. (2021). Electroconvulsive therapy in mania: A review of 80 years of clinical experience. *Am J Psychiatry*, 178(3), 229–239.
- Ellahi, R. (2015). Serotonin syndrome: A spectrum of toxicity. *BJPsych Adv*, 21(5), 324–332.
- Ellison-Wright, Z., & Boardman, C. (2015). Diagnosis and management of ASD in children and adolescents. *Progress Neurol Psychiatry*, 19(6), 28–32. <https://doi.org/10.1002/pnp.407>
- Empana, J.-P., Boutouyrie, P., Lemogne, C., et al. (2021). Microvascular contribution to late-onset depression: Mechanisms, current evidence, association with other brain diseases, and therapeutic perspectives. *Biol Psychiatry*, 90(4), 214–225.
- Eng, B., Addison, P., & Ring, H. (2013). A guide to intellectual disability psychiatry assessments in the community. *Adv Psychiatr Treat*, 19(6), 429–436.

- Engel, G. L. (1977). The need for a new medical model: A challenge for biomedicine. *Science*, 196(4286), 129–136.
- Engels, G., Francke, A. L., van Meijel, B., et al. (2014). Clinical pain in schizophrenia: A systematic review. *J Pain*, 15(5), 457–467.
- Ennis, J., & Majid, U. (2021). 'Death from a broken heart': A systematic review of the relationship between spousal bereavement and physical and physiological health outcomes. *Death Stud*, 45(7), 538–551. <https://doi.org/10.1080/07481187.2019.1661884>
- Erlangsen, A., Jacobsen, A. L., Ranning, A., et al. (2023). Transgender identity and suicide attempts and mortality in Denmark. *JAMA*, 329(24), 2145–2153. <https://doi.org/10.1001/jama.2023.8627>
- Erskine, H. E., Ferrari, A. J., Nelson, P., et al. (2013). Research review: Epidemiological modelling of attention-deficit/hyperactivity disorder and conduct disorder for the Global Burden of Disease Study 2010. *J Child Psychol Psychiatry*, 54(12), 1263–1274.
- Escott-Price, V., Smith, D. J., Kendall, K., et al. (2019). Polygenic risk for schizophrenia and season of birth within the UK biobank cohort. *Psychol Med*, 49(15), 2499–2504. <https://doi.org/10.1017/S0033291718000454>
- Esquirol, E. (1845). *Mental Maladies: A Treatise on Insanity*. Lea and Blanchard.
- Essen-Möller, E. (1971). Suggestions for further improvement of the international classification of mental disorders. *Psychol Med*, 1(4), 308–311.
- Essex, B., Doig, R., & Renshaw, J. (1990). Pilot study of records of shared care for people with mental illnesses. *BMJ*, 300(6737), 1442–1446.
- Evoy, K. E., Sadrameli, S., Contreras, J., et al. (2021). Abuse and misuse of pregabalin and gabapentin: A systematic review update. *Drugs*, 81, 125–156.
- Ewing-Cobbs, L., Cox, C. S., Clark, A. E., et al. (2018). Persistent postconcussion symptoms after injury. *Pediatrics*, 142(5), e20180939.
- Eyring, K. W., & Geschwind, D. H. (2021). Three decades of ASD genetics: Building a foundation for neurobiological understanding and treatment. *Hum Mol Genet*, 30(20), R236–R244. <https://doi.org/10.1093/hmg/ddab176>
- Eysenck, H. J. (1953). *The Structure of Human Personality*. Methuen.
- Eysenck, H. J. (1970). *The Structure of Human Personality (Psychology Revivals)*. Routledge.
- Eysenck, H. J., & Eysenck, S. B. G. (1976). *Psychoticism as a Dimension of Personality*. Hodder & Stoughton.
- Eyuboglu, M., Eyuboglu, D., Pala, S. C., et al. (2021). Traditional school bullying and cyberbullying: Prevalence, the effect on mental health problems and self-harm behavior. *Psychiatry Res*, 297, 113730.
- Fabrega, H., Jr. (2000). Culture, spirituality and psychiatry. *Curr Opin Psychiatry*, 13(6), 525–530.
- Fairburn, C. (1999). Risk factors for anorexia nervosa: Three integrated case-control comparisons. *Arch Gen Psychiatry*, 56, 468–476.
- Fairburn, C. G., Cooper, Z., Doll, H.A., et al. (2009). Transdiagnostic cognitive-behavioral therapy for patients with eating disorders: A two-site trial with 60-week follow-up. *Am J Psychiatry*, 166, 311–319.
- Fairburn, C. G., Cooper, Z., Doll, H. A., et al. (2013). Enhanced cognitive behaviour therapy for adults with anorexia nervosa: A UK-Italy study. *Behav Res Ther*, 51(1), R2–8. doi:10.1016/j.brat.2012.09.010
- Fairburn, C. G., & Harrison, P. J. (2003). Eating disorders. *Lancet*, 361(9355), 407–416.
- Fairchild, G., Hawes, D. J., Frick, P. J., et al. (2019). Conduct disorder. *Nat Rev Dis Primers*, 5(1), 43. <https://doi.org/10.1038/s41572-019-0095-y>
- Fang, S., & Ding, D. (2020). A meta-analysis of the efficacy of acceptance and commitment therapy for children. *J Context Behav Sci*, 15, 225–234.
- Fanous, A. H., & Kendler, K. S. (2004). The genetic relationship of personality to major depression and schizophrenia. *Neurotox Res*, 6, 43–50.
- Farhat, L. C., Olfson, E., Nasir, M., et al. (2020). Pharmacological and behavioral treatment for trichotillomania: An updated systematic review with meta-analysis. *Depress Anxiety*, 37(8), 715–727.
- Faris, R. E. L., & Dunhman, H. W. (1939). *Mental Disorders in Urban Areas*. Chicago University Press.
- Farrell, M., Howes, S., Taylor, C., et al. (2003). Substance misuse and psychiatric comorbidity: an overview of the OPCS National Psychiatric Morbidity Survey. *Int Rev Psychiatry*, 15(1–2), 43–49. doi: 10.1080/0954026021000045930
- Favril, L., Yu, R., Uyar, A., et al. (2022). Risk factors for suicide in adults: Systematic review and meta-analysis of psychological autopsy studies. *Evid Based Ment Health*, 25(4), 148–155. <https://doi.org/10.1136/ebmental-2022-300549>
- Fazel, M. (2015a). Methylphenidate for ADHD. *BMJ*, 351: h5875.
- Fazel, M., Hoagwood, K., Stephan, S., & Ford, T. (2014a). Mental health interventions in schools 1: Mental health interventions in schools in high-income countries. *Lancet Psychiatry*, 1(5), 377–387. [https://doi.org/10.1016/S2215-0366\(14\)70312-8](https://doi.org/10.1016/S2215-0366(14)70312-8)
- Fazel, M., Patel, V., Thomas, S., & Tol, W. (2014b). Mental health interventions in schools in low-income and middle-income countries. *Lancet Psychiatry*, 1(5), 388–398. [https://doi.org/10.1016/S2215-0366\(14\)70357-8](https://doi.org/10.1016/S2215-0366(14)70357-8)
- Fazel, M., Reed, R., & Stein, A. (2015b). Refugee, asylum-seeking and internally displaced children and adolescents. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 573–585). John Wiley & Sons, Ltd.
- Fazel, M., & Soneson, E. (2023). Current evidence and opportunities in child and adolescent public mental health: A research review. *J Child Psychol Psychiatry*, 64(12), 1699–1719. doi: 10.1111/jcpp.13889
- Fazel, M., Townsend, A., Stewart, H., et al. (2021). Integrated care to address child and adolescent health in the 21st century: A clinical review. *JCPP Advances*, 1(4), e12045.
- Fazel, S., Bains, P., & Doll, H. (2006). Substance abuse and dependence in prisoners: A systematic review. *Addiction*, 101(2), 181–191.
- Fazel, S., & Danesh, J. (2002). Serious mental disorder in 23000 prisoners: A systematic review of 62 surveys. *Lancet*, 359(9306), 545–550. [https://doi.org/10.1016/S0140-6736\(02\)07740-1](https://doi.org/10.1016/S0140-6736(02)07740-1)

- Fazel, S., Geddes, J. R., & Kuschel, M. (2014c). The health of homeless people in high-income countries: Descriptive epidemiology, health consequences, and clinical and policy recommendations. *Lancet*, 384(9953), 1529–1540. [https://doi.org/10.1016/S0140-6736\(14\)61132-6](https://doi.org/10.1016/S0140-6736(14)61132-6)
- Fazel, S., & Grann, M. (2004). Psychiatric morbidity among homicide offenders: A Swedish population study. *Am J Psychiatry*, 161(11), 2129–2131. <https://doi.org/10.1176/appi.ajp.161.11.2129>
- Fazel, S., & Grann, M. (2006). The population impact of severe mental illness on violent crime. *Am J Psychiatry*, 163(8), 1397–1403. <https://doi.org/10.1176/ajp.2006.163.8.1397>
- Fazel, S., Grann, M., Kling, B., & Hawton, K. (2011). Prison suicide in 12 countries: An ecological study of 861 suicides during 2003–2007. *Soc Psychiatry Psychiatr Epidemiol*, 46(3), 191–195. <https://doi.org/10.1007/s00127-010-0184-4>
- Fazel, S., Gulati, G., Linsell, L., et al. (2009). Schizophrenia and violence: Systematic review and meta-analysis. *PLoS Med*, 6(8), e1000120. <https://doi.org/10.1371/journal.pmed.1000120>
- Fazel, S., Lichtenstein, P., Grann, M., & Langstrom, N. (2011). Risk of violent crime in individuals with epilepsy and traumatic brain injury: A 35-year Swedish population study. *PLoS MED*, 8(12), e1001150. <https://doi.org/10.1371/journal.pmed.1001150>
- Fazel, S., Philipson, J., Gardiner, L., et al. (2009). Neurological disorders and violence: A systematic review and meta-analysis with a focus on epilepsy and traumatic brain injury. *J Neurol*, 256, 1591–1602.
- Fazel, S., Singh, J. P., Doll, H., & Grann, M. (2012). Use of risk assessment instruments to predict violence and antisocial behaviour in 73 samples involving 24827 people: Systematic review and meta-analysis. *BMJ*, 345, e4692. <https://doi.org/10.1136/bmj.e4692>
- Fazel, S., Sjöstedt, G., Långström, N., & Grann, M. (2007). Severe mental illness and risk of sexual offending in men: A case-control study based on Swedish national registers. *J Clin Psychiatry*, 68(4), 588–596. <https://doi.org/10.4088/jcp.v68n0415>
- Fazel, S., Vassos, E., & Danesh, J. (2002). Prevalence of epilepsy in prisoners: Systematic review. *BMJ*, 324(7352), 1495. <https://doi.org/10.1136/bmj.324.7352.1495>
- Fazel, S., Wolf, A., Långström, N., et al. (2013). Premature mortality in epilepsy and the role of psychiatric comorbidity: A total population study. *Lancet*, 382(9905), 1646–1654.
- Fazel, S., Wolf, A., Palm, C., & Lichtenstein, P. (2014d). Violent crime, suicide, and premature mortality in patients with schizophrenia and related disorders: A 38-year total population study in Sweden. *Lancet Psychiatry*, 1(1), 44–54. [https://doi.org/10.1016/S2215-0366\(14\)70223-8](https://doi.org/10.1016/S2215-0366(14)70223-8)
- Fazel, S., Wolf, A., Pillas, D., et al. (2014e). Suicide, fatal injuries, and other causes of premature mortality in patients with traumatic brain injury: A 41-year Swedish population study. *JAMA Psychiatry*, 71(3), 326–333.
- Fearon, P., Kirkbride, J. B., Morgan, C., et al. (2006). Incidence of schizophrenia and other psychoses in ethnic minority groups: Results from the MRC AESOP Study. *Psychol Med*, 36(11), 1541–1550. <https://doi.org/10.1017/S0033291706008774>
- Feighner, J. P., Robins, E., Guze, S. B., et al. (1972). Diagnostic criteria for use in psychiatric research. *Arch Gen Psychiatry*, 26(1), 57–63.
- Feldman, H., & Schmack, K. (2025). From reward learning to the Bayesian brain: 20 years of aberrant salience. *Am J Psych*, 182, 801–803.
- Fereshtehnejad, S.-M., Yao, C., Pelletier, A., et al. (2019). Evolution of prodromal Parkinson's disease and dementia with Lewy bodies: A prospective study. *Brain*, 142(7), 2051–2067.
- Ferguson, M. A., Lim, C., Cooke, D., et al. (2019). A human memory circuit derived from brain lesions causing amnesia. *Nature Comm*, 10(1), 3497.
- Fergusson, D. M., Horwood, L. J., & Boden, J. M. (2009). Reactions to abortion and subsequent mental health. *BJPsych*, 195(5), 420–426.
- Fernandez de la Cruz, L., & Mataix-Cols, D. (2020). Hoarding disorder. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press.
- Ferrari, M., Yap, K., Scott, N., et al. (2018). Self-compassion moderates the perfectionism and depression link in both adolescence and adulthood. *PLoS ONE*, 13(2), e0192022.
- Ferri, C. P., Prince, M., Brayne, C., et al. (2005). Global prevalence of dementia: A Delphi consensus study. *Lancet*, 366(9503), 2112–2117.
- Fineberg, N., & Robbins, T. W. (2021). *The Neurobiology and Treatment of OCD: Accelerating Progress*. Springer.
- Fineberg, N., Drummond, L., Reid, J., et al. (2020). Management and treatment of obsessive-compulsive disorder. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1011–1023). Oxford University Press.
- Finnegan, M., & McLoughlin, M. (2019). Cognitive side effects of ECT. In I. Ferrier & J. Waite (Eds.), *The ECT Handbook* (pp. 109–128). Cambridge University Press.
- Fioritti, A., Lo Russo, L., & Melega, V. (1997). Reform said or done? The case of Emilia-Romagna within the Italian psychiatric context. *Am J Psychiatry*, 154(1), 94–98. <https://doi.org/10.1176/ajp.154.1.94>
- Firm, M., Alonso-Vicente, M., Hubbeling, D., et al. (2018). The replacement of assertive outreach services by reinforcing local community teams: A replication study reporting comparative outcomes and patient reported experience. *J Ment Health*, 27(2), 157–163. <https://doi.org/10.1080/09638237.2017.1340601>
- First, M. B., Gaebel, W., Maj, M., et al. (2021). An organization- and category-level comparison of diagnostic requirements for mental disorders in ICD-11 and DSM-5. *World Psychiatry*, 20(1), 34–51.
- Firth, J., Conlon, C., & Cox, T. (Eds.) (2020). *Oxford Textbook of Medicine*, 6th edn. Oxford University Press.
- Fiske, A., Wetherell, J. L., & Gatz, M. (2009). Depression in older adults. *Ann Rev Clin Psychol*, 5, 363–389.
- Florida, A., Velasco, E. R., Monari, S., et al. (2023). Glucocorticoid-based pharmacotherapies preventing PTSD. *Neuropharmacol*, 224, 109344. <https://doi.org/10.1016/j.neuropharm.2022.109344>
- Flottorp, S. A., Brurberg, K. G., Fink, P., et al. (2022). New NICE guideline on chronic fatigue syndrome: More ideology than science? *Lancet*, 399(10325), 611–613. [https://doi.org/https://doi.org/10.1016/S0140-6736\(22\)00183-0](https://doi.org/https://doi.org/10.1016/S0140-6736(22)00183-0)

- Focal Point Annual Report. (2019). Focal Point Annual Report. <https://www.gov.uk/government/publications/united-kingdom-drug-situation-focal-point-annual-report/united-kingdom-drug-situation-focal-point-annual-report-2019#opioids>
- Folstein, M. F., Folstein, S. E., & McHugh, P. R. (1975). 'Minimal state': A practical method for grading the cognitive state of patients for the clinician. *J Psychiatr Res*, 12(3), 189–198.
- Fonagy, P., & Kächele, H. (2009). Psychoanalysis and other long-term dynamic psychotherapies. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1337–1350). Oxford University Press.
- Fornaro, M., Maritan, E., Ferranti, R., et al. (2020). Lithium exposure during pregnancy and the postpartum period: A systematic review and meta-analysis of safety and efficacy outcomes. *Am J Psychiatry*, 177(1), 76–92.
- Forneris, C. A., Gartlehner, G., Brownley, K. A., et al. (2013). Interventions to prevent post-traumatic stress disorder: A systematic review. *Am J Prevent Med*, 44(6), 635–650.
- Forstner, A. J., Awasthi, S., Wolf, C., et al. (2021). Genome-wide association study of panic disorder reveals genetic overlap with neuroticism and depression. *Mol Psychiatry*, 26(8), 4179–4190.
- Forti-Buratti, M. A., Saikia, R., Wilkinson, E. L., & Ramchandani, P. G. (2016). Psychological treatments for depression in pre-adolescent children (12 years and younger): Systematic review and meta-analysis of randomised controlled trials. *Eur Child Adolesc Psychiatry*, 25(10), 1045–1054. doi: 10.1007/s00787-016-0834-5
- Fossey, J., Ballard, C., Juszczak, E., et al. (2006). Effect of enhanced psychosocial care on antipsychotic use in nursing home residents with severe dementia: Cluster randomised trial. *BMJ*, 332(7544), 756–761.
- Foulkes, L., & Blakemore, S.-J. (2018). Studying individual differences in human adolescent brain development. *Nature Neurosci*, 21(3), 315–323.
- Foulkes, S. H., & Lewis, E. (1945). A study in the treatment of groups on psycho-analytic lines. *Br J Med Psychol*, 20(2), 175–184. <https://doi.org/10.1111/j.2044-8341.1945.tb00751.x>
- Fox, N. C., Belder, C., Ballard, C., et al. (2025). Treatment for Alzheimer's disease. *Lancet*, 406, 1408–1423.
- Foxcroft, D. R., Coombes, L., Wood, S., et al. (2014). Motivational interviewing for alcohol misuse in young adults. *Cochrane Database Syst Rev* 8, CD007025.
- Frances, A. J., & Nardo, J. M. (2013). ICD-11 should not repeat the mistakes made by DSM-5. *BJPsych*, 203(1), 1–2.
- Frank, J. D. (1974). Common features of psychotherapies and their patients. *Psychother Psychosom*, 24(4/6), 368–371. <http://www.jstor.org/stable/45114354>
- Franquillo, A. C., Guccione, C., Angelini, G., et al. (2021). The role of personality in schizophrenia and psychosis: A systematic review. *Clin Neuropsychiatry*, 18(1), 28–40. <https://doi.org/10.36131/cnforiteditore20210103>
- Fransson, E., Werneke, U., Harju, V., et al. (2022). Kidney function in patients with bipolar disorder with and without lithium treatment compared with the general population in northern Sweden: Results from the LISIE and MONICA cohorts. *Lancet Psychiatry*, 9(10), 804–814.
- Freeman, D., & Garety, P. (2014). Advances in understanding and treating persecutory delusions: A review. *Soc Psychiatry Psychiatr Epidemiol*, 49(8), 1179–1189. <https://doi.org/10.1007/s00127-014-0928-7>
- Freeman, D., Rosebrock, L., Waite, F., et al. (2023). Virtual reality (VR) therapy for patients with psychosis: Satisfaction and side effects. *Psychol Med*, 53(10), 4373–4384. <https://doi.org/10.1017/s0033291722001167>
- Freeman, D., Sheaves, B., Goodwin, G. M., et al. (2017). The effects of improving sleep on mental health (OASIS): a randomised controlled trial with mediation analysis. *Lancet, Psychiatry*, 4, 749–758. [https://doi.org/10.1016/S2215-0366\(17\)30328-0](https://doi.org/10.1016/S2215-0366(17)30328-0)
- Freeman, D., Sheaves, B., Waite, F., Harvey, A. G., & Harrison, P. J. (2020). Sleep disturbance and psychiatric disorders. *Lancet Psychiatry*, 7, 628–637. [https://doi.org/10.1016/S2215-0366\(20\)30136-X](https://doi.org/10.1016/S2215-0366(20)30136-X)
- Freeman, W., & Watts, J. (1942). *Psychosurgery*. Thomas.
- Freud, S. (1917). Mourning and melancholia. In A. Freud & J. Strachey (Eds.), *The Standard Edition of the Complete Psychological Works*, Vol. 14 (pp. 243–258). Hogarth Press.
- Friedman, M., & Rosenman, R. H. (1959). Association of specific overt behavior pattern with blood and cardiovascular findings; Blood cholesterol level, blood clotting time, incidence of arcus senilis, and clinical coronary artery disease. *J Am Med Assoc*, 169(12), 1286–1296. <https://doi.org/10.1001/jama.1959.03000290012005>
- Frigli, V., Smith, M., Andrews, T. M., et al. (2022). Incidence of fractures in people with intellectual disabilities over the life course: A retrospective matched cohort study. *EClinicalMed*, 52, 101656.
- Frisoni, G. B., Altomare, D., Thal, D. R., et al. (2022). The probabilistic model of Alzheimer disease: The amyloid hypothesis revised. *Nature Rev Neurosci*, 23(1), 53–66.
- Frith, C. (1996). Neuropsychology of schizophrenia: What are the implications of intellectual and experimental abnormalities for the neurobiology of schizophrenia? *Br Med Bull*, 52(3), 618–626.
- Fuhrmann, D., Knoll, L. J., & Blakemore, S. J. (2015). Adolescence as a sensitive period of brain development. *Trends Cogn Sci*, 19(10), 558–566. <https://doi.org/10.1016/j.tics.2015.07.008>
- Fukuda, K., Straus, S. E., Hickie, I., et al. (1994). The chronic fatigue syndrome: A comprehensive approach to its definition and study. International Chronic Fatigue Syndrome Study Group. *Ann Int Med*, 121(12), 953–959.
- Fulford, K. W. (2011). Bringing together values-based and evidence-based medicine: UK Department of Health Initiatives in the 'personalization' of care. *J Eval Clin Pract*, 17(2), 341–343. <https://doi.org/10.1111/j.1365-2753.2010.01578.x>
- Fulford, K. W. M. (1989). *Moral Theory and Medical Practice*. Cambridge University Press.
- Fuller-Thomson, E., Lacombe-Duncan, A., Goodman, D., et al. (2020). From surviving to thriving: Factors associated with complete mental health among childhood sexual abuse survivors. *Soc Psychiatry Psychiatr Epidemiol*, 55(6), 735–744. <https://doi.org/10.1007/s00127-019-01767-x>
- Fulton, M., & Winokur, G. (1993). A comparative study of paranoid and schizoid personality disorders. *Am J Psychiatry*, 150(9), 1363–1367. <https://doi.org/10.1176/ajp.150.9.1363>

- Furukawa, T. (2004). Meta-analyses and megatrials: Neither is the infallible, universal standard. *Evi Based Mental Health*, 7, 34–35.
- Furukawa, T. A., Cipriani, A., Cowen, P. J., et al. (2019). Optimal dose of selective serotonin reuptake inhibitors, venlafaxine, and mirtazapine in major depression: A systematic review and dose-response meta-analysis. *Lancet Psychiatry*, 6(7), 601–609.
- Furukawa, T. A., Suganuma, A., Ostinelli, E. G., et al. (2021). Dismantling, optimising, and personalising internet cognitive behavioural therapy for depression: A systematic review and component network meta-analysis using individual participant data. *Lancet Psychiatry*, 8(6), 500–511.
- Fusar-Poli, P. (2019). Integrated mental health services for the developmental period (0 to 25 years): A critical review of the evidence. *Front Psychiatry*, 10, 355.
- Fusar-Poli, P., McGorry, P. D., & Kane, J. M. (2017). Improving outcomes of first-episode psychosis: An overview. *World Psychiatry*, 16(3), 251–265. <https://doi.org/10.1002/wps.20446>
- Fusar-Poli, P., Papanastasiou, E., Stahl, D., et al. (2015). Treatments of negative symptoms in schizophrenia: Meta-analysis of 168 randomized placebo-controlled trials. *Schizophr Bull*, 41(4), 892–899.
- Fusar-Poli, P., & Radua, J. (2018). Ten simple rules for conducting umbrella reviews. *Evid Based Ment Health*, 21(3), 95–100. <https://doi.org/10.1136/ebmental-2018-300014>
- Fusar-Poli, P., Salazar de Pablo, G., Correll, C. U., et al. (2020). Prevention of psychosis: Advances in detection, prognosis, and intervention. *JAMA Psychiatry*, 77(7), 755–765. <https://doi.org/10.1001/jamapsychiatry.2019.4779>
- Fusar-Poli, P., Salazar de Pablo, G., Rajkumar, R. P., et al. (2022). Diagnosis, prognosis, and treatment of brief psychotic episodes: A review and research agenda. *Lancet Psychiatry*, 9(1), 72–83. [https://doi.org/10.1016/s2215-0366\(21\)00121-8](https://doi.org/10.1016/s2215-0366(21)00121-8)
- Fyer, M. R., Frances, A. J., Sullivan, T., et al. (1988). Comorbidity of borderline personality disorder. *Arch Gen Psychiatry*, 45(4), 348–352. <https://doi.org/10.1001/archpsyc.1988.01800280060008>
- Gabbard, G., Beck, J., & Holmes, J. (Eds.). (2005). *Oxford Textbook of Psychotherapy*. Oxford University Press.
- Gadot, R., Najera, R., Hirani, S., et al. (2022). Efficacy of deep brain stimulation for treatment-resistant obsessive-compulsive disorder: systematic review and meta-analysis. *J Neurol Neurosurg Psychiatry*. doi: 10.1136/jnnp-2021-328738. Online ahead of print.
- Gaebel, W., Stricker, J., & Kerst, A. (2020). Changes from ICD-10 to ICD-11 and future directions in psychiatric classification. *Dialogues Clin Neurosci*, 22(1), 7–15. <https://doi.org/10.31887/DCNS.2020.22.1/wgaebel>
- Galbiati, A., Verga, L., Giora, E., Zucconi, M., & Ferini-Strambi, L. (2019). The risk of neurodegeneration in REM sleep behavior disorder: A systematic review and meta-analysis of longitudinal studies. *Sleep Med Rev*, 43, 37–46. <https://doi.org/10.1016/j.smrv.2018.09.008>
- Galante, N., Gentile, G., Tambuzzi, S., & Zoja, R. (2022). Suicide pacts in the Milan district (Italy): A retrospective autopsy-based study with literature review. *J Forens Legal Med*, 86, 102319. <https://doi.org/https://doi.org/10.1016/j.jflm.2022.102319>
- Galappathie, N., & Khan, S. T. (2016). End-of-life care in psychiatry: ‘One chance to get it right’. *BJPsych Bull*, 40(1), 38–40.
- Galbally, M., Stein, A., Hoegfeldt, C. A., & van Ijzendoorn, M. (2020). From attachment to mental health and back. *Lancet Psychiatry*, 7(10), 832–834. [https://doi.org/10.1016/s2215-0366\(20\)30337-0](https://doi.org/10.1016/s2215-0366(20)30337-0)
- Gale, S. A., Acar, D., & Daffner, K. R. (2018). Dementia. *Am J Med*, 131(10), 1161–1169. <https://doi.org/10.1016/j.amjmed.2018.01.022>
- Gallagher, J. B., III. (2018). Social class and schizophrenia: A review of early and recent findings in the United States. *Curr Psychiatry Rev*, 14(3), 187–194. <https://doi.org/http://dx.doi.org/10.2174/1573400514666181015115442>
- Galling, B., Vernon, J. A., Pagsberg, A. K., et al. (2018). Efficacy and safety of antidepressant augmentation of continued antipsychotic treatment in patients with schizophrenia. *Acta Psychiatr Scand*, 137(3), 187–205. <https://doi.org/10.1111/acps.12854>
- Gandhi, K. D., Mansukhani, M. P., Silber, M. H., & Kolla, B. P. (2021). Excessive Daytime Sleepiness: A Clinical Review. *Mayo Clin Proc*, 96, 1288–1301. <https://doi.org/10.1016/j.mayocp.2020.08.033>
- Ganesh, S., & D’Souza, D. C. (2022). Cannabis and psychosis: Recent epidemiological findings continuing the ‘causality debate’. *Am J Psychiatry*, 179(1), 8–10. <https://doi.org/10.1176/appi.ajp.2021.21111126>
- García-Gutiérrez, M. S., Navarrete, F., Sala, F., et al. (2020). Biomarkers in psychiatry: Concept, definition, types and relevance to the clinical reality. *Front Psychiatry*, 11, 432. <https://doi.org/10.3389/fpsyt.2020.00432>
- Gardner, M. J., Thomas, H. J., & Erskine, H. E. (2019). The association between five forms of child maltreatment and depressive and anxiety disorders: A systematic review and meta-analysis. *Child Abuse Negl*, 96, 104082. <https://doi.org/10.1016/j.chiabu.2019.104082>
- Gardner, W., Hoge, S. K., Bennett, N., et al. (1993). Two scales for measuring patients’ perceptions for coercion during mental hospital admission. *Behav Sci Law*, 11(3), 307–321. <https://doi.org/10.1002/bsl.2370110308>
- Garety, P., & Freeman, D. (2013). The past and future of delusions research: From the inexplicable to the treatable. *BJPsych*, 203(5), 327–333.
- Garraalda, M. (2016). ICD-11—Comparison With DSM-5 and implications for child and adolescent psychiatric disorders. In M. Hodes & S. Gau (Eds.), *Positive Mental Health, Fighting Stigma and Promoting Resiliency for Children and Adolescents* (pp. 15–35). Elsevier.
- Garraalda, M. E., & Rask, C. U. (2015). Somatoform and related disorders. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter’s Child and Adolescent Psychiatry*, 6th edn. (pp. 1035–1054). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch72>
- Gartlehner, G., Crotty, K., Kennedy, S., et al. (2021). Pharmacological treatments for borderline personality disorder: A systematic review and meta-analysis. *CNS Drugs*, 35(10), 1053–1067. <https://doi.org/10.1007/s40263-021-00855-4>
- Gartlehner, G., Gaynes, B. N., Hansen, R. A., et al. (2008). Comparative benefits and harms of second-generation antidepressants: Background paper for the American College of Physicians. *Ann Int Med*, 149(10), 734–750.

- Gath, D., Cooper, P., & Day, A. (1982). Hysterectomy and psychiatric disorder: I. Levels of psychiatric morbidity before and after hysterectomy. *BJPsych*, 140(4), 335–342.
- Gaupp, R. (1974). The scientific significance of the case of Ernst Wagner. In S. Hirsch & M. Shepherd (Eds.), *Themes and Variations in European Psychiatry: An Anthology* (pp. 121–133). John Wright and Sons.
- Gavin, N. I., Gaynes, B. N., Lohr, K. N., et al. (2005). Perinatal depression: A systematic review of prevalence and incidence. *Obstet Gynecol*, 106(5 Part 1), 1071–1083.
- Gazzaniga, M. S. (2000). Cerebral specialization and interhemispheric communication: Does the corpus callosum enable the human condition? *Brain*, 123(7), 1293–1326.
- Geddes, J. R., Andreasen, N. C., & Goodwin, G. (Eds.). (2020). *New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press.
- Geddes, J. R., Calabrese, J. R., & Goodwin, G. M. (2009). Lamotrigine for treatment of bipolar depression: Independent meta-analysis and meta-regression of individual patient data from five randomised trials. *BJPsych*, 194(1), 4–9.
- Geddes, J. R., Carney, S. M., Davies, C., et al. (2003). Relapse prevention with antidepressant drug treatment in depressive disorders: A systematic review. *Lancet*, 361(9358), 653–661.
- Geddes, J. R., Gardiner, A., Rendell, J., et al. (2016). Comparative evaluation of quetiapine plus lamotrigine combination versus quetiapine monotherapy (and folic acid versus placebo) in bipolar depression (CEQUEL): A 2 × 2 factorial randomised trial. *Lancet Psychiatry*, 3(1), 31–39.
- Gelder, M. (1986). Neurosis: Another tough old word. *BMJ*, 292(6526), 972.
- Gelder, M. G., Marks, I. M., Wolff, H. H., & Clarke, M. (1967). Desensitization and psychotherapy in the treatment of phobic states: A controlled inquiry. *BJPsych*, 113(494), 53–73. <https://doi.org/10.1192/bjp.113.494.53>
- Gelin, Z., Cook-Darzens, S., & Hendrick, S. (2018). The evidence base for multiple family therapy in psychiatric disorders: A review (part 1). *J Fam Ther*, 40(3), 302–325.
- Gelkopf, M., Mazor, Y., & Roe, D. (2021). A systematic review of patient-reported outcome measurement (PROM) and provider assessment in mental health: Goals, implementation, setting, measurement characteristics and barriers. *Int J Qual Health Care*, 34(Suppl 1), ii13–ii27. <https://doi.org/10.1093/intqhc/mzz133>
- Geoffroy, P. A., Schroder, C. M., Reynaud, E., & Bourgin, P. (2019). Efficacy of light therapy versus antidepressant drugs, and of the combination versus monotherapy, in major depressive episodes: A systematic review and meta-analysis. *Sleep Med Rev*, 48, 101213. <https://doi.org/10.1016/j.smrv.2019.101213>
- George, N., Abdallah, J., Maradey-Romero, C., et al. (2016). Review article: The current treatment of non-cardiac chest pain. *Aliment Pharmacol Ther*, 43(2), 213–239. <https://doi.org/10.1111/apt.13458>
- Gergel, T. (2022). ‘Shock tactics’, ethics and fear: An academic and personal perspective on the case against electroconvulsive therapy. *BJPsych*, 220(3), 109–112.
- Gerger, H., Hlavica, M., Gaab, J., et al. (2015). Does it matter who provides psychological interventions for medically unexplained symptoms? A meta-analysis. *Psychother Psychosom*, 84(4), 217–226. <https://doi.org/10.1159/000380914>
- Gergov, V., Milic, B., Löffler-Stastka, H., et al. (2022). Psychological interventions for young people with psychotic disorders: A systematic review. *Front Psychiatry*, 13, 859042. <https://doi.org/10.3389/fpsy.2022.859042>
- Geschwind, D. H., & Flint, J. (2015). Genetics and genomics of psychiatric disease. *Science*, 349(6255), 1489–1494. <https://doi.org/10.1126/science.aaa8954>
- Geulayov, G., Borschmann, R., Mansfield, K. L., et al. (2022a). Utilization and acceptability of formal and informal support for adolescents following self-harm before and during the first COVID-19 lockdown: Results from a large-scale English schools survey. *Front Psychiatry*, 13, 881248.
- Geulayov, G., Casey, D., Bale, L., et al. (2019). Suicide following presentation to hospital for non-fatal self-harm in the multi-centre study of self-harm: A long-term follow-up study. *Lancet Psychiatry*, 6(12), 1021–1030.
- Geulayov, G., Casey, D., McDonald, K. C., et al. (2018). Incidence of suicide, hospital-presenting non-fatal self-harm, and community-occurring non-fatal self-harm in adolescents in England (the iceberg model of self-harm): A retrospective study. *Lancet Psychiatry*, 5(2), 167–174.
- Geulayov, G., Mansfield, K., Jindra, C., et al. (2022b). Loneliness and self-harm in adolescents during the first national COVID-19 lockdown: Results from a survey of 10,000 secondary school pupils in England. *Curr Psychol*, 15, 1–12.
- Ghiasi, N., Azhar, Y., & Singh, J. (2022). *Psychiatric Illness And Criminality*. StatPearls.
- Gibb, K., Seeley, A., Quinn, T., et al. (2020). The consistent burden in published estimates of delirium occurrence in medical inpatients over four decades: A systematic review and meta-analysis study. *Age Ageing*, 49(3), 352–360.
- Gibson, D., Workman, C., & Mehler, P. S. (2019). Medical Complications of Anorexia Nervosa and Bulimia Nervosa. *Psychiatr Clin North Am*, 42, 263–274. <https://doi.org/10.1016/j.psc.2019.01.009>
- Giebel, C. M., Zubair, M., Jolley, D., et al. (2015). South Asian older adults with memory impairment: Improving assessment and access to dementia care. *Int J Geriatr Psychiatry*, 30(4), 345–356.
- Giel, K. E., Bulik, C. M., Fernandez-Aranda, F., et al. (2022). Binge eating disorder. *Nat Rev Dis Primers*, 8, 16. <https://doi.org/10.1038/s41572-022-00344-y>
- Gili, M., Luciano, J. V., Serrano, M. J., et al. (2011). Mental disorders among frequent attenders in primary care: A comparison with routine attenders. *J Nerv Ment Dis*, 199(10), 744–749. https://journals.lww.com/jonmd/Fulltext/2011/10000/Mental_Disorders_Among_Frequent_Attenders_in.4.aspx
- Gillies, D., Christou, M. A., Dixon, A. C., et al. (2018). Prevalence and characteristics of self-harm in adolescents: Meta-analyses of community-based studies 1990–2015. *J Am Acad Child Adolesc Psychiatry*, 57(10), 733–741.
- Giordan, E., Palandri, G., Lanzino, G., et al. (2018). Outcomes and complications of different surgical treatments for idiopathic normal pressure hydrocephalus: A systematic review and meta-analysis. *J Neurosurg*, 131(4), 1024–1036.
- Gitlin, D. F., Levenson, J. L., & Lyketsos, C. G. (2004). Psychosomatic medicine: A new psychiatric subspecialty. *Acad Psychiatry*, 28(1), 4–11. <https://doi.org/10.1176/appi.ap.28.1.4>

- Glaser, D. (2015). Child sexual abuse. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 376–388). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch30>
- Glenn, A. L. (2019). Early life predictors of callous–unemotional and psychopathic traits *Infant Ment Health J*, 40(1), 39–53. <https://doi.org/https://doi.org/10.1002/imhj.21757>
- Glenn, A. L., & Raine, A. (2014). Neurocriminology: Implications for the punishment, prediction and prevention of criminal behaviour. *Nat Rev Neurosci*, 15(1), 54–63. <https://doi.org/10.1038/nrn3640>
- Glenn, C. R., Kleiman, E. M., Kellerman, J., et al. (2020). Annual research review: A meta-analytic review of worldwide suicide rates in adolescents. *J Child Psychol Psychiatry*, 61(3), 294–308. <https://doi.org/https://doi.org/10.1111/jcpp.13106>
- Glover, G., Arts, G., & Babu, K. S. (2006). Crisis resolution/home treatment teams and psychiatric admission rates in England. *BJPsych*, 189(5), 441–445.
- Goate, A., Chartier-Harlin, M. C., Mullan, M., et al. (1991). Segregation of a missense mutation in the amyloid precursor protein gene with familial Alzheimer's disease. *Nature*, 349, 704–706.
- Godfrey, K. E., Gardner, A. C., Kwon, S., et al. (2018). Differences in excitatory and inhibitory neurotransmitter levels between depressed patients and healthy controls: A systematic review and meta-analysis. *J Psychiatr Res*, 105, 33–44.
- Godlewska, B. R., Minichino, A., Emir, U., et al. (2021). Brain glutamate concentration in men with early psychosis: A magnetic resonance spectroscopy case-control study at 7T. *Transl Psychiatry*, 11(1), 367. <https://doi.org/10.1038/s41398-021-01477-6>
- Goff, D. C., Falkai, P., Fleischhacker, W. W., et al. (2017). The long-term effects of antipsychotic medication on clinical course in schizophrenia. *Am J Psychiatry*, 174(9), 840–849. <https://doi.org/10.1176/appi.ajp.2017.16091016>
- Goffman, E. (1961). *Asylums: Essays on the Social Situation of Mental Patients and Other Inmates*. AldineTransaction.
- Goh, K. K., Chen, C.-H., Chiu, Y.-H., & Lu, M.-L. (2019). Lamotrigine augmentation in treatment-resistant unipolar depression: A comprehensive meta-analysis of efficacy and safety. *J Psychopharmacol*, 33(6), 700–713.
- Goldberg, D. (2010). The classification of mental disorder: A simpler system for DSM-V and ICD-11. *Adv Psychiatr Treat*, 16(1), 14–19.
- Goldberg, D., & Huxley, P. (1980). *Mental Illness in the Community*. Tavistock.
- Goldberg, D., Rickels, K., Downing, R., & Hesbacher, P. (1976). A comparison of two psychiatric screening tests. *BJPsych*, 129(1), 61–67.
- Goldberg, D. P., & Bridges, K. (1988). Somatic presentations of psychiatric illness in primary care setting. *J Psychosom Res*, 32(2), 137–144.
- Goldberg, D. P., & Hillier, V. F. (1979). A scaled version of the General Health Questionnaire. *Psychol Med*, 9(1), 139–145.
- Goldberg, E., & Morrison, S. (1963). Schizophrenia and social class. *BJPsych*, 109(463), 785–802.
- Goldberg, P. (1972). *The Detection of Psychiatric Illness by Questionnaire*. Maudsley Monograph.
- Goldberg, T. E., Chen, C., Wang, Y., et al. (2020). Association of delirium with long-term cognitive decline: A meta-analysis. *JAMA Neurol*, 77(11), 1373–1381.
- Goldman, H. H., & Morrissey, J. P. (1985). The alchemy of mental health policy: Homelessness and the fourth cycle of reform. *Am J Public Health*, 75(7), 727–731. <https://doi.org/10.2105/ajph.75.7.727>
- Gondek, D., Patalay, P., & Lacey, R. E. (2021). Adverse childhood experiences and multiple mental health outcomes through adulthood: A prospective birth cohort study. *SSM Ment Health*, 1, 100013. <https://doi.org/https://doi.org/10.1016/j.ssmmh.2021.100013>
- Gonzalez-Burgos, G., Cho, R. Y., & Lewis, D. A. (2015). Alterations in cortical network oscillations and parvalbumin neurons in schizophrenia. *Biol Psychiatry*, 77(12), 1031–1040.
- Gonzalez-Rodriguez, A., & Seeman, M. V. (2022). Differences between delusional disorder and schizophrenia: A mini narrative review. *World J Psychiatry*, 19, 683–692.
- Goodkind, M. (2013). Functional neurocircuitry and neuroimaging studies of anxiety disorders. In D. Charney, J. Buxbaum, P. Sklaar, & E. Nestler (Eds.), *Neurobiology of Mental Illness* (pp. 606–620). Oxford University Press.
- Goodman, J. H., Chenausky, K.L., Freeman, M. P. (2014). Anxiety disorders during pregnancy: A systematic review. *J Clin Psychiatry*, 75:e1153–e1184. doi: 10.4088/JCP.14r09035.
- Goodman, S. H., Rouse, M. H., Connell, A. M., et al. (2011). Maternal depression and child psychopathology: A meta-analytic review. *Clin Child Fam Psychol Rev*, 14, 1–27.
- Goodman, W. K., Storch, E. A., & Sheth, S. A. (2021). Harmonizing the neurobiology and treatment of obsessive-compulsive disorder. *Am J Psychiatry*, 178(1), 17–29.
- Goodwin, G., Haddad, P., Ferrier, I., et al. (2016). Evidence-based guidelines for treating bipolar disorder: Revised third edition recommendations from the British Association for Psychopharmacology. *J Psychopharmacol*, 30(6), 495–553.
- Goodwin, G., & Browning, M. (2020). Imaging of depressive disorders. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press.
- Goodwin, N., & Lawton-Smith, S. (2010). Integrating care for people with mental illness: The Care Programme Approach in England and its implications for long-term conditions management. *Int J Integr Care*, 10, e040.
- Goossen, B., van der Starre, J., & van der Heiden, C. (2019). A review of neuroimaging studies in generalized anxiety disorder: 'So where do we stand?' *J Neural Trans*, 126(9), 1203–1216.
- Gordon, H., & Grubin, D. (2004). Psychiatric aspects of the assessment and treatment of sex offenders. *Adv Psychiatr Treat*, 10(1), 73–80. <https://doi.org/10.1192/apt.10.1.73>
- Gordovez, F. J. A., & McMahon, F. J. (2020). The genetics of bipolar disorder. *Mol Psychiatry*, 25(3), 544–559. <https://doi.org/10.1038/s41380-019-0634-7>
- Gottesman, I. I. (1991). *Schizophrenia Genesis: The Origins of Madness*. WH Freeman/Times Books/Henry Holt & Co.
- Gottesman, I. I., & Shields, J. (1967). A polygenic theory of schizophrenia. *Proc Nat Acad Sci*, 58(1), 199–205.
- Gottesman, R. F., Lutsey, P. L., Benveniste, H., et al. (2024). Impact of sleep disorders and disturbed sleep on brain health: A scientific statement from the American Heart Association. *Stroke*, 55, e61–e76. <https://doi.org/10.1161/STR.0000000000000453>

- Gottlieb, D. J., & Punjabi, N. M. (2020). Diagnosis and management of obstructive sleep apnea: A review. *J Am Med Assoc*, 323, 1389–1400. <https://doi.org/10.1001/jama.2020.3514>
- Gottschalk, M. D. (2020). Genetics of stress-related disorders. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press.
- Gournay, K., Birley, J. (1998). Thorn: A new approach to mental health training. *Nurs Times*, 94, 54–55.
- Gournellis, R., Oulis, P., & Howard, R. (2014). Psychotic major depression in older people: A systematic review. *Int J Geriatr Psychiatry*, 29, 784–796.
- Graham, C. A., & Bancroft, J. (2020). The sexual dysfunctions and paraphilias. In J.R. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1181–1190). Oxford University Press.
- Grandhe, R., Souzdanitski, D., & Gritsenko, K. (2016). New chronic pain treatments in the outpatient setting. *Curr Pain Head Rep*, 20, 1–7.
- Granholm, E., & Harvey, P. D. (2018). Social skills training for negative symptoms of schizophrenia. *Schizophr Bull*, 44(3), 472–474. <https://doi.org/10.1093/schbul/sbx184>
- Grann, M., & Fazel, S. (2004). Substance misuse and violent crime: Swedish population study. *BMJ*, 328(7450), 1233–1234. <https://doi.org/10.1136/bmj.328.7450.1233>
- Grant, B. F., Goldstein, R. B., Saha, T. D., et al. (2015). Epidemiology of DSM-5 alcohol use disorder: Results from the National Epidemiologic Survey on Alcohol and Related Conditions III. *JAMA Psychiatry*, 72(8), 757–766.
- Grant, J. E., Atmaca, M., Fineberg, N. A., et al. (2014). Impulse control disorders and 'behavioural addictions' in the ICD-11. *World Psychiatry*, 13(2), 125–127. <https://doi.org/10.1002/wps.20115>
- Grant, J. E., Chamberlain, S. R., Redden, S. A., et al. (2016). N-acetylcysteine in the treatment of excoriation disorder: A randomized clinical trial. *JAMA Psychiatry*, 73(5), 490–496.
- Gray, J., Feldon, J., Rawlins, J., et al. The neuropsychology of schizophrenia. *Behav Brain Sci*, 14(1), 1–20.
- Gray, R., Bressington, D., Ivanecka, A., et al. (2016). Is adherence therapy an effective adjunct treatment for patients with schizophrenia spectrum disorders? A systematic review and meta-analysis. *BMC Psychiatry*, 16, 1–12.
- Green, E., Rees, E., Walters, J., et al. (2016). Copy number variation in bipolar disorder. *Mol Psychiatry*, 21(1), 89–93.
- Green, M. F., Horan, W. P., & Lee, J. (2015). Social cognition in schizophrenia. *Nat Rev Neurosci*, 16(10), 620–631.
- Green, M. F., Horan, W. P., & Lee, J. (2019). Nonsocial and social cognition in schizophrenia: Current evidence and future directions. *World Psychiatry*, 18(2), 146–161. <https://doi.org/10.1002/wps.20624>
- Greenberg, M. T., & Riggs, N. R. (2015). Prevention of mental disorders and promotion of competence. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 215–226). John Wiley & Sons, Ltd.
- Greenhalgh, T. (2019). *How to Read a Paper: The Basics of Evidence-based Medicine and Healthcare*. John Wiley & Sons.
- Gressier, E., Letranchant, A., Cazas, O., et al. (2015). Postpartum depressive symptoms and medically assisted conception: A systematic review and meta-analysis. *Hum Rep*, 30(11), 2575–2586.
- Griesinger, W. (1867). *Mental Pathology and Therapeutics*, 2nd edn. (trans. C. Lockhart Robertson & J. Rutherford). New Sydenham Society.
- Griffiths, K. M., Carron-Arthur, B., Parsons, A., & Reid, R. (2014). Effectiveness of programs for reducing the stigma associated with mental disorders. A meta-analysis of randomized controlled trials. *World Psychiatry*, 13(2), 161–175.
- Grimm, O., Kranz, T. M., & Reif, A. (2020). Genetics of ADHD: What should the clinician know? *Curr Psychiatry Rep*, 22, 1–8.
- Group, P. M. R. (1997). Matching alcoholism treatments to client heterogeneity: Project MATCH posttreatment drinking outcomes. *J Stud Alcohol*, 58, 7–29.
- Grzenda, A., & Widge, A. S. (2020). Electroencephalographic biomarkers for predicting antidepressant response: New methods, old questions. *JAMA Psychiatry*, 77(4), 347–348. <https://doi.org/10.1001/jamapsychiatry.2019.3749>
- Gudjonsson, G. (1992). *The Psychology of Interrogations, Confessions and Testimony*. John Wiley & Sons.
- Guerreiro, R., Ross, O. A., Kun-Rodrigues, C., et al. (2018). Investigating the genetic architecture of dementia with Lewy bodies: A two-stage genome-wide association study. *Lancet Neurol*, 17(1), 64–74.
- Gunnell, D., Saperia, J., & Ashby, D. (2005). Selective serotonin reuptake inhibitors (SSRIs) and suicide in adults: Meta-analysis of drug company data from placebo controlled, randomised controlled trials submitted to the MHRA's safety review. *BMJ*, 330(7488), 385.
- Guralnick, M. J. (2005). Early intervention for children with intellectual disabilities: Current knowledge and future prospects. *J Appl Res Intellect Disabil*, 18(4), 313–324.
- Gureje, O., Lewis-Fernandez, R., Hall, B. J., & Reed, Geoffrey M. (2019). Systematic inclusion of culture-related information in ICD-11. *World Psychiatry*, 18(3), 357–358. <https://doi.org/10.1002/wps.20676>
- Gureje, O., von Korf, M., Simon, G. E., & Gater, R. (1998). Persistent pain and well-being: A World Health Organization Study in Primary Care. *JAMA*, 280(2), 147–151. <https://doi.org/10.1001/jama.280.2.147>
- Gurgu, R. S., Ciobanu, A. M., Danasel, R. I., & Panea, C. A. (2021). Psychiatric comorbidities in adult patients with epilepsy (A systematic review). *Exp Ther Med*, 22(2), 1–25.
- Guthrie, E. (2008). Medically unexplained symptoms in primary care. *Adv Psychiatr Treat*, 14(6), 432–440. <https://doi.org/10.1192/apt.bp.106.003335>
- Guy, N., Newton-Howes, G., Ford, H., et al. (2018). The prevalence of comorbid alcohol use disorder in the presence of personality disorder: Systematic review and explanatory modelling. *Personal Ment Health*, 12(3), 216–228.
- Guze, S. B. (1989). Biological psychiatry: Is there any other kind? *Psychol Med*, 19(2), 315–323.
- Guzzetta, F., & de Girolamo, G. (2009). Epidemiology of personality disorders. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 881–886). Oxford University Press. <https://doi.org/10.1093/med/9780199696758.003.0111>
- Hachinski, V. C., Lassen, N. A., & Marshall, J. (1974). Multi-infarct dementia: A cause of mental deterioration in the elderly. *Lancet*, 304(7874), 207–209.

- Hackett, R. A., Ronaldson, A., Bhui, K., et al. (2020). Racial discrimination and health: A prospective study of ethnic minorities in the United Kingdom. *BMC Public Health*, 20(1), 1652. <https://doi.org/10.1186/s12889-020-09792-1>
- Haddad, P. M., & Correll, C. U. (2018). The acute efficacy of antipsychotics in schizophrenia: A review of recent meta-analyses. *Ther Adv Psychopharmacol*, 8(11), 303–318. <https://doi.org/10.1177/2045125318781475>
- Haddad, P. M., & Nutt, D. J. (2020). *Seminars in Clinical Psychopharmacology*. Cambridge University Press.
- Hagerman, R. J., Berry-Kravis, E., Hazlett, H. C., et al. (2017). Fragile X syndrome. *Nat Rev Dis Primers*, 3, 17065. <https://doi.org/10.1038/nrdp.2017.65>
- Hahlweg, K., & Baucou, D. H. (2023). Family therapy for persons with schizophrenia: Neglected yet important. *Eur Arch Psychiatry Clin Neurosci*, 273(4), 819–824. <https://doi.org/10.1007/s00406-022-01393-w>
- Hailes, H. P., Yu, R., Danese, A., & Fazel, S. (2019). Long-term outcomes of childhood sexual abuse: An umbrella review. *Lancet Psychiatry*, 6(10), 830–839. [https://doi.org/10.1016/s2215-0366\(19\)30286-x](https://doi.org/10.1016/s2215-0366(19)30286-x)
- Hale, B., & Fortin, J. (2015). Legal issues in the care and treatment of children with mental health problems. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 239–249). John Wiley & Sons, Ltd.
- Hálfádnarson, Ó., Cohen, J. M., Karlstad, Ø., et al. (2022). Antipsychotic use in pregnancy and risk of attention/deficit-hyperactivity disorder and autism spectrum disorder: A Nordic cohort study. *BMJ Ment Health*, 25(2), 54–62.
- Hall, J., & Bray, N. J. (2022). Schizophrenia genomics: Convergence on synaptic development, adult synaptic plasticity, or both? *Biol Psychiatry*, 91(8), 709–717. <https://doi.org/10.1016/j.biopsych.2021.10.018>
- Hall, W. (2015). What has research over the past two decades revealed about the adverse health effects of recreational cannabis use? *Addiction*, 110(1), 19–35.
- Hall, W., Stjepanović, D., Caulkins, J., et al. (2019). Public health implications of legalising the production and sale of cannabis for medicinal and recreational use. *Lancet*, 394(10208), 1580–1590.
- Hallett, M., Aybek, S., Dworetzky, B. A., et al. (2022). Functional neurological disorder: New subtypes and shared mechanisms. *Lancet Neurol*, 21(6), 537–550.
- Halmi, K. A. (2020). Epidemiology and primary prevention of feeding and eating disorders In J. R. Geddes (Ed.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1059–1064).
- Halvorsrud, K., Nazroo, J., Otis, M., et al. (2018). Ethnic inequalities and pathways to care in psychosis in England: A systematic review and meta-analysis. *BMC Med*, 16(1), 223. <https://doi.org/10.1186/s12916-018-1201-9>
- Halvorsrud, K., Nazroo, J., Otis, M., et al. (2019). Ethnic inequalities in the incidence of diagnosis of severe mental illness in England: A systematic review and new meta-analyses for non-affective and affective psychoses. *Soc Psychiatry Psychiatr Epidemiol*, 54(11), 1311–1323. <https://doi.org/10.1007/s00127-019-01758-y>
- Hamedani, A. G., & Pelak, V. S. (2019). The Charles Bonnet syndrome: A systematic review of diagnostic criteria. *Curr Treat Options Neurol*, 21, 1–13.
- Hamilton, C. E., Falshaw, L., & Browne, K. D. (2002). The link between recurrent maltreatment and offending behaviour. *Int J Offender Ther Comp Criminol*, 46(1), 75–94. <https://doi.org/10.1177/0306624X02461006>
- Hamilton, M. (1959). The assessment of anxiety states by rating. *Br J Med Psychol*, 32(1), 50–55.
- Hamilton, M. (1967). Development of a rating scale for primary depressive illness. *Br J Soc Clin Psychol*, 6(4), 278–296.
- Hampel, H., Mesulam, M.-M., Cuello, A. C., et al. (2018). The cholinergic system in the pathophysiology and treatment of Alzheimer's disease. *Brain*, 141(7), 1917–1933.
- Handley, S., Every-Palmer, S., Ismail, A., & Flanagan, R. (2022). Clozapine-induced gastrointestinal hypomotility: Presenting features and outcomes, UK pharmacovigilance reports, 1992–2017. *BJPsych*, 220(6), 355–363.
- Hanlon, C., Fekadu, A., Jordans, M., et al. (2016). District mental healthcare plans for five low-and middle-income countries: Commonalities, variations and evidence gaps. *BJPsych*, 208(s56), s47–s54.
- Hanly, J. G. (2014). Diagnosis and management of neuropsychiatric SLE. *Nat Rev Rheumatol*, 10(6), 338–347.
- Hansen, H. G., Speyer, H., Starzer, M., et al. (2023). Clinical recovery among individuals with a first-episode schizophrenia: An updated systematic review and meta-analysis. *Schizophr Bull*, 49(2), 297–308. <https://doi.org/10.1093/schbul/sbac103>
- Hansson, O. (2021). Biomarkers for neurodegenerative diseases. *Nat Med*, 27(6), 954–963.
- Hardy, J. A., & Higgins, G. A. (1992). Alzheimer's disease: The amyloid cascade hypothesis. *Science*, 256, 184–185.
- Hare, R. D. (1998). The Hare PCL-R: Some issues concerning its use and misuse. *Legal Criminol Psychol*, 3(1), 99–119. <https://doi.org/https://doi.org/10.1111/j.2044-8333.1998.tb00353.x>
- Harmer, C. J., Duman, R. S., & Cowen, P. J. (2017). How do antidepressants work? New perspectives for refining future treatment approaches. *Lancet Psychiatry*, 4(5), 409–418.
- Harrendorf, S., Heiskanen, M., & Malby, S. (Eds.). (2010). *International Statistics on Crime and Justice*. HEUNI/UNODC.
- Harris, A. M., Orav, E. J., Bates, D. W., & Barsky, A. J. (2009). Somatization increases disability independent of comorbidity. *J Gen Intern Med*, 24(2), 155–161. <https://doi.org/10.1007/s11606-008-0845-0>
- Harris, C., & Barraclough, B. (1998). Excess mortality of mental disorder. *BJPsych*, 173(1), 11–53.
- Harris, J. C. (2014). New classification for neurodevelopmental disorders in DSM-5. *Curr Opin Psychiatry*, 27(2), 95–97.
- Harrison, G., Hopper, K., Craig, T., et al. (2001). Recovery from psychotic illness: A 15-and 25-year international follow-up study. *BJPsych*, 178(6), 506–517.
- Harrison, G., Mason, P., Glazebrook, C., et al. (1994). Residence of incident cohort of psychotic patients after 13 years of follow up. *BMJ*, 308(6932), 813–816. <https://doi.org/10.1136/bmj.308.6932.813>
- Harrison, J. L., O'Toole, S. K., Ammen, S., et al. (2020). Sexual offender treatment effectiveness within cognitive-behavioral programs: A meta-analytic investigation of general, sexual, and violent recidivism. *Psychiatr Psychol Law*, 27(1), 1–25. <https://doi.org/10.1080/13218719.2018.1485526>
- Harrison, N., & Kopelman, M. (2009). Endocrine diseases and metabolic disorders. In A. David, S. Fleminger, M. Kopelman, et al. (Eds.),

- Lishman's *Organic Psychiatry: A Textbook of Neuropsychiatry* (pp. 617–688). Wiley Blackwell.
- Harrison, P., & Oakland, T. (2000). *Adaptive Behavior Assessment System*. Harcourt Assessment.
- Harrison, P. J. (1999). The neuropathology of schizophrenia: A critical review of the data and their interpretation. *Brain*, 122(4), 593–624.
- Harrison, P. J. (2015). Recent genetic findings in schizophrenia and their therapeutic relevance. *J Psychopharmacol*, 29(2), 85–96.
- Harrison, P. J., & Bannerman, D. M. (2023). *GRIN2A (NR2A)*: A gene contributing to glutamatergic involvement in schizophrenia. *Mol Psychiatry*, 28(9), 3568–3572. <https://doi.org/10.1038/s41380-023-02265-y>
- Harrison, P. J., Colbourne, L., & Harrison, C. H. (2020b). The neuropathology of bipolar disorder: Systematic review and meta-analysis. *Mol Psychiatry*, 25(8), 1787–1808.
- Harrison, P. J., & Luciano, S. (2021). Incidence of Parkinson's disease, dementia, cerebrovascular disease and stroke in bipolar disorder compared to other psychiatric disorders: An electronic health records network study of 66 million people. *Bipolar Disord*, 23(5), 454–462.
- Harrison, P. J., Mould, A., & Tunbridge, E. M. (2022). New drug targets in psychiatry: Neurobiological considerations in the genomics era. *Neurosci Biobehav Rev*, 139, 104763. <https://doi.org/10.1016/j.neubiorev.2022.104763>
- Harrison, P. J., & Taquet, M. (2023). Neuropsychiatric disorders following SARS-CoV-2 infection. *Brain*, 146, 2241–2247. <https://doi.org/10.1093/brain/awad008>
- Harrison, P. J., Tunbridge, E. M., Dolphin, A. C., & Hall, J. (2020a). Voltage-gated calcium channel blockers for psychiatric disorders: Genomic reappraisal. *BJPsych*, 216(5), 250–253. <https://doi.org/10.1192/bjp.2019.157>
- Harvey, A. G., & McGlinchey, E. L. (2015). Sleep interventions: A developmental perspective. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 999–1015). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch70>
- Harvey, S. B., & Wessely, S. (2009). Chronic fatigue syndrome: Identifying zebras amongst the horses. *BMC Med*, 7, 1–4.
- Harwood, R., & Cowan, R. (2020). Physical assessment. In T. Denning, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 164–182). Oxford University Press.
- Hasan, A., von Keller, R., Friemel, C. M., et al. (2020). Cannabis use and psychosis: A review of reviews. *Eur Arch Psychiatry Clin Neurosci*, 270(4), 403–412. <https://doi.org/10.1007/s00406-019-01068-z>
- Hasin, Y., Seldin, M., & Lusi, A. (2017). Multi-omics approaches to disease. *Genome Biol*, 18(1), 83. <https://doi.org/10.1186/s13059-017-1215-1>
- Hasler, G., Nugent, A. C., Carlson, P. J., et al. (2008). Altered cerebral γ -aminobutyric acid type A-benzodiazepine receptor binding in panic disorder determined by [11C] flumazenil positron emission tomography. *Arch Gen Psychiatry*, 65(10), 1166–1175.
- Hassan, L., Birmingham, L., Harty, M. A., et al. (2011). Prospective cohort study of mental health during imprisonment. *BJPsych*, 198(1), 37–42. <https://doi.org/10.1192/bjp.bp.110.080333>
- Hawton, K., Lascelles, K., Pitman, A., et al. (2022). Assessment of suicide risk in mental health practice: Shifting from prediction to therapeutic assessment, formulation, and risk management. *Lancet Psychiatry*, 9(11), 922–928.
- Hawton, K., Saunders, K. E. A., & Pitman, A. (2020). Prevention of suicide and treatment following self-harm. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1303–1314). Oxford University Press.
- Hay, P., Chinn, D., Forbes, D., et al. (2014). Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for the treatment of eating disorders. *Aust N Z J Psychiatry*, 48(11), 977–1008.
- Hayes, A., Herlinger, K., Paterson, L., & Lingford-Hughes, A. (2020). The neurobiology of substance use and addiction: Evidence from neuroimaging and relevance to treatment. *BJPsych Adv*, 26(6), 367–378.
- Hayes, S. C., & Hofmann, S. G. (2017). The third wave of cognitive behavioral therapy and the rise of process-based care. *World Psychiatry*, 16(3), 245–246. <https://doi.org/10.1002/wps.20442>
- Health Survey for England (2019). Health Survey for England Retrieved from <https://digital.nhs.uk/data-and-information/publications/statistical/health-survey-for-england/2019>
- Hecker, E. (1871). Die Hebephrenie. *Virchows Archiv für Pathologie und Anatomie*, 52, 394–429.
- Hegeman, J., Kok, R., Van der Mast, R., & Giltay, E. (2012). Phenomenology of depression in older compared with younger adults: Meta-analysis. *BJPsych*, 200(4), 275–281.
- Heijnen, W. T., Birkenhäger, T. K., Wierdsma, A. I., & van den Broek, W. W. (2010). Antidepressant pharmacotherapy failure and response to subsequent electroconvulsive therapy: A meta-analysis. *J Clin Psychopharmacol*, 30(5), 616–619.
- Heinrich, T. W., & Marcangelo, M. (2009). Psychiatric issues in solid organ transplantation. *Harv Rev Psychiatry*, 17(6), 398–406. <https://doi.org/10.3109/10673220903463259>
- Hejl, A., Høgh, P., & Waldemar, G. (2002). Potentially reversible conditions in 1000 consecutive memory clinic patients. *J Neurol Neurosurg Psychiatry*, 73(4), 390.
- Hem, E., Loge, J. H., Haldorsen, T., & Ekeberg, Ø. (2004). Suicide risk in cancer patients from 1960 to 1999. *J Clin Oncol*, 22(20), 4209–4216. <https://doi.org/10.1200/jco.2004.02.052>
- Hennegan, J. M., Henderson, J., & Redshaw, M. (2015). Contact with the baby following stillbirth and parental mental health and well-being: A systematic review. *BMJ Open*, 5(11), e008616.
- Henssler, J., Alexander, D., Schwarzer, G., et al. (2022). Combining antidepressants vs antidepressant monotherapy for treatment of patients with acute depression: A systematic review and meta-analysis. *JAMA Psychiatry*, 79(4), 300–312. <https://doi.org/10.1001/jamapsychiatry.2021.4313>
- Herlinger, K., & Lingford-Hughes, A. (2021). Addressing unmet needs in opiate dependence: Supporting detoxification and advances in relapse prevention. *BJPsych Adv*, 27(6), 362–372.
- Hermann, P., & Zerr, I. (2022). Rapidly progressive dementias—etiologies, diagnosis and management. *Nat Rev Neurol*, 18(6), 363–376.
- Heslop, P., Blair, P. S., Fleming, P., et al. (2014). The Confidential Inquiry into premature deaths of people with intellectual disabilities in the UK: A population-based study. *Lancet*, 383(9920), 889–895.

- Heston, L. L. (1966). Psychiatric disorders in foster home-reared children of schizophrenic mothers. *BJPsych*, 112(489), 819–825.
- Hetrick, S. E., Robinson, J., Spittal, M. J., & Carter, G. (2016). Effective psychological and psychosocial approaches to reduce repetition of self-harm: A systematic review, meta-analysis and meta-regression. *BMJ Open*, 6(9), e011024. <https://doi.org/10.1136/bmjopen-2016-011024>
- Hettema, J. M., Prescott, C. A., Myers, J. M., et al. (2005). The structure of genetic and environmental risk factors for anxiety disorders in men and women. *Arch Gen Psychiatry*, 62(2), 182–189.
- Hessler, H. S. (2016). Management of sleep disorders in neurodevelopmental disorders and genetic syndromes. *Curr Opin Psychiatry*, 29(2), 138–143. <https://doi.org/10.1097/YCO.0000000000000230>
- Heyman, I., Skuse, D., & Goodman, R. (2015). Brain disorders and psychopathology. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 389–402). John Wiley & Sons, Ltd.
- Hill, J. (2003). Early identification of individuals at risk for antisocial personality disorder. *BJPsych Suppl*, 44, S11–14. <https://doi.org/10.1192/bjp.182.44.s11>
- Hill, K., Mann, L., Laws, K., et al. (2004). Hypofrontality in schizophrenia: A meta-analysis of functional imaging studies. *Acta Psychiatr Scand*, 110(4), 243–256.
- Hirano, Y., & Uhlhaas, P. J. (2021). Editorial: Current MEG research in psychiatry. *Front Psychiatry*, 12, 647085. <https://doi.org/10.3389/fpsy.2021.647085>
- Hirschfeld, R. M., Williams, J. B., Spitzer, R. L., et al. (2000). Development and validation of a screening instrument for bipolar spectrum disorder: The Mood Disorder Questionnaire. *Am J Psychiatry*, 157(11), 1873–1875. <https://doi.org/10.1176/appi.ajp.157.11.1873>
- Hitchcock, C., Rodrigues, E., Rees, C., et al. (2019). Misremembrance of things past: Depression is associated with difficulties in the recollection of both specific and categorical autobiographical memories. *Clin Psychol Sci*, 7(4), 693–700. <https://doi.org/10.1177/2167702619826967>
- Hitti, F. L., Yang, A. I., Cristancho, M. A., & Baltuch, G. H. (2020). Deep brain stimulation is effective for treatment-resistant depression: A meta-analysis and meta-regression. *J Clin Med*, 9(9), 2796.
- Hjorthøj, C., Posselt, C. M., & Nordentoft, M. (2021). Development over time of the population-attributable risk fraction for cannabis use disorder in schizophrenia in Denmark. *JAMA Psychiatry*, 78(9), 1013–1019. <https://doi.org/10.1001/jamapsychiatry.2021.1471>
- Ho, R. C. M., Cheng, W., Chua, A. N. C., & Mak, A. (2012). Neuropsychiatric aspects of carbon monoxide poisoning: Diagnosis and management. *Adv Psychiatr Treat*, 18(2), 94–101.
- Hodges, J. R., & Piguet, O. (2018). Progress and challenges in frontotemporal dementia research: A 20-year review. *J Alzheimers Dis*, 62(3), 1467–1480.
- Hodges, H., Fealko, C., & Soares, N. (2020). Autism spectrum disorder: Definition, epidemiology, causes, and clinical evaluation. *Transl Pediatr*, 9(Suppl 1), S55.
- Hofstra, E., van Nieuwenhuizen, C., Bakker, M., et al. (2020). Effectiveness of suicide prevention interventions: A systematic review and meta-analysis. *Gen Hosp Psychiatry*, 63, 127–140. <https://doi.org/10.1016/j.genhosppsych.2019.04.011>
- Holland, T., Clare, I. C., & Mukhopadhyay, T. (2002). Prevalence of criminal offending by men and women with intellectual disability and the characteristics of offenders: Implications for research and service development. *J Intellect Disabil Res*, 46(Suppl 1), 6–20. <https://doi.org/10.1046/j.1365-2788.2002.00001.x>
- Hollingshead, A. B., & Redlich, F. C. (1958). *Social Class and Mental Illness: Community Study*. John Wiley & Sons.
- Holloway, F. (2002). The NHS Plan: 2 years on. *Psychiatr Bull*, 26(8), 281–283. <https://doi.org/10.1192/pb.26.8.281>
- Holloway, F., Oliver, N., Collins, E., & Carson, J. (1995). Case management: A critical review of the outcome literature. *Eur Psychiatry*, 10(3), 113–128. [https://doi.org/10.1016/0767-399X\(96\)80101-5](https://doi.org/10.1016/0767-399X(96)80101-5)
- Holmes, T., & Rahe, R. (1967). The social adjustment rating scale. *J Psychosom Res*, 11, 213–218.
- Holroyd, K. B., Rubin, D. B., & Vaitkevicius, H. (2021). Neurologic Complications in Patients with Cancer. *Semin Neurol*, 41(5), 588–605.
- Holt, R. I., & Mitchell, A. J. (2015). Diabetes mellitus and severe mental illness: Mechanisms and clinical implications. *Nat Rev Endocrinol*, 11(2), 79–89.
- Holze, F., Ley, L., Müller, F., et al. (2022). Direct comparison of the acute effects of lysergic acid diethylamide and psilocybin in a double-blind placebo-controlled study in healthy subjects. *Neuropsychopharmacol*, 47(6), 1180–1187.
- Hong, J. S., Atkinson, L. Z., Al-Juffali, N., et al. (2022). Gabapentin and pregabalin in bipolar disorder, anxiety states, and insomnia: Systematic review, meta-analysis, and rationale. *Mol Psychiatry*, 27(3), 1339–1349.
- Hood, R. (1992). *Race and Sentencing: A Study in the Crown Court: A Report for the Commission for Racial Equality*. Oxford University Press.
- Hookway, C., Buckner, S., Crosland, P., & Longson, D. (2015). Irritable bowel syndrome in adults in primary care: Summary of updated NICE guidance. *BMJ*, 350, h701.
- Hope, T., Savulescu, J., & Hendrick, J. (2016). *Medical Ethics and Law: The Core Curriculum*, 2nd edn. Churchill Livingstone.
- Horby, P., Lim, W. S., Emberson, J. R., et al. (2021). Dexamethasone in hospitalized patients with Covid-19. *N Engl J Med*, 384(8), 693–704. <https://doi.org/10.1056/NEJMoa2021436>
- Horowitz, M. A., & Taylor, D. (2022). Distinguishing relapse from antidepressant withdrawal: Clinical practice and antidepressant discontinuation studies. *BJPsych Adv*, 28(5), 297–311.
- Horwitz, T., Lam, K., Chen, Y., et al. (2019). A decade in psychiatric GWAS research. *Mol Psychiatry*, 24(3), 378–389. <https://doi.org/10.1038/s41380-018-0055-z>
- Hotopf, M. (2004). Preventing somatization. *Psychologic Med*, 34(2), 195–198. <https://doi.org/10.1017/S003329170300151X>
- Hotopf, M., Noah, N., & Wessely, S. (1996). Chronic fatigue and minor psychiatric morbidity after viral meningitis: A controlled study. *J Neurol Neurosurg Psychiatry*, 60(5), 504–509. <https://doi.org/10.1136/jnnp.60.5.504>

- Hoult, J. (1986). Community care of the acutely mentally ill. *BJPsych*, 149(2), 137–144.
- Howard, L. M., Molyneaux, E., Dennis, C.-L., et al. (2014). Non-psychotic mental disorders in the perinatal period. *Lancet*, 384(9956), 1775–1788.
- Howard, L. M., Oram, S., Galley, H., et al. (2013). Domestic violence and perinatal mental disorders: A systematic review and meta-analysis. *PLoS MED*, 10(5), e1001452.
- Howard, L. M., Piot, P., & Stein, A. (2014). No health without perinatal mental health. *Lancet*, 384(9956), 1723–1724.
- Howell, S., Yarovova, E., Khwanda, A., & Rosen, S. D. (2019). Cardiovascular effects of psychotic illnesses and antipsychotic therapy. *Heart*, 105(24), 1852–1859.
- Howes, O., McCutcheon, R., & Stone, J. (2015). Glutamate and dopamine in schizophrenia: An update for the 21st century. *J Psychopharmacol*, 29(2), 97–115.
- Howes, O. D., Cummings, C., Chapman, G. E., & Shatalina, E. (2023). Neuroimaging in schizophrenia: An overview of findings and their implications for synaptic changes. *Neuropsychopharmacol*, 48(1), 151–167. <https://doi.org/10.1038/s41386-022-01426-x>
- Howes, O. D., Kambeitz, J., Kim, E., et al. (2012). The nature of dopamine dysfunction in schizophrenia and what this means for treatment: Meta-analysis of imaging studies. *Arch Gen Psychiatry*, 69(8), 776–786.
- Howes, O. D., McCutcheon, R., Agid, O., et al. (2017). Treatment-resistant schizophrenia: Treatment Response and Resistance in Psychosis (TRRIP) Working Group Consensus Guidelines on Diagnosis and Terminology. *Am J Psychiatry*, 174(3), 216–229. <https://doi.org/10.1176/appi.ajp.2016.16050503>
- Howes, O. D., & Shatalina, E. (2022). Integrating the neurodevelopmental and dopamine hypotheses of schizophrenia and the role of cortical excitation–inhibition balance. *Biol Psychiatry*, 92(6), 501–513. <https://doi.org/10.1016/j.biopsych.2022.06.017>
- Howes, O. D., Whitehurst, T., Shatalina, E., et al. (2021). The clinical significance of duration of untreated psychosis: An umbrella review and random-effects meta-analysis. *World Psychiatry*, 20(1), 75–95. <https://doi.org/10.1002/wps.20822>
- Howlett, J. R., & Stein, M. B. (2016). Prevention of trauma and stressor-related disorders: A review. *Neuropsychopharmacol*, 41(1), 357–369.
- Huang, C.-Q., Dong, B.-R., Lu, Z.-C., et al. (2010). Chronic diseases and risk for depression in old age: A meta-analysis of published literature. *Ageing Res Rev*, 9(2), 131–141.
- Huckins, L. M., Signer, R., Johnson, J., Wu, Y. K., Mitchell, K. S., & Bulik, C. M. (2022). What next for eating disorder genetics? Replacing myths with facts to sharpen our understanding. *Mol Psychiatry*, 27, 3929–3938. <https://doi.org/10.1038/s41380-022-01601-y>
- Hughes, N., Williams, H., Chitsabesan, P., et al. (2012). *Nobody Made the Connection: The Prevalence of Neurodisability in Young People Who Offend*. Office of the Children's Commissioner.
- Huhn, M., Nikolakopoulou, A., Schneider-Thoma, J., et al. (2019). Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: A systematic review and network meta-analysis. *Lancet*, 394(10202), 939–951. [https://doi.org/10.1016/s0140-6736\(19\)31135-3](https://doi.org/10.1016/s0140-6736(19)31135-3)
- Hui, T., Kandola, A., Shen, L., et al. (2019). A systematic review and meta-analysis of clinical predictors of lithium response in bipolar disorder. *Acta Psychiatr Scand*, 140(2), 94–115.
- Hulme-Dickens, S. (2013). The mental state examination. *Adv Psychiatr Treat*, 19(2), 97–98.
- Hunt, G. E., Large, M. M., Cleary, M., et al. (2018). Prevalence of comorbid substance use in schizophrenia spectrum disorders in community and clinical settings, 1990–2017: Systematic review and meta-analysis. *Drug Alcohol Depend*, 191, 234–258. <https://doi.org/10.1016/j.drugalcdep.2018.07.011>
- Hunter, A., Murray, R., Asher, L., & Leonardi-Bee, J. (2020). The effects of tobacco smoking, and prenatal tobacco smoke exposure, on risk of schizophrenia: A systematic review and meta-analysis. *Nicotine Tob Res*, 22(1), 3–10. <https://doi.org/10.1093/ntr/nty160>
- Hunter, E. C., Charlton, J., & David, A. S. (2017). Depersonalisation and derealisation: Assessment and management. *BMJ*, 356, j745. doi: 10.1136/bmj.j745
- Hunter, R., & MacAlpine, I. (Eds.). (1963). *Three Hundred Years of Psychiatry*. Oxford University Press.
- Huntley, J., Thomas, A., & Stewart, R. (2020). Psychiatric assessment of older people. In T. Denning, A. Thomas, R. Stewart, & J.-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 145–152). Oxford University Press.
- Huss, M., Dirks, B., Gu, J., et al. (2018). Long-term safety and efficacy of guanfacine extended release in children and adolescents with ADHD. *Eur Child Adolesc Psychiatry*, 27, 1283–1294.
- Huybrechts, K. F., Bateman, B. T., Palmsten, K., et al. (2015). Antidepressant use late in pregnancy and risk of persistent pulmonary hypertension of the newborn. *JAMA*, 313(21), 2142–2151.
- Huybrechts, K. F., Palmsten, K., Avorn, J., et al. (2014). Antidepressant use in pregnancy and the risk of cardiac defects. *N Engl J Med*, 370(25), 2397–2407.
- Hwang, S. W., & Burns, T. (2014). Health interventions for people who are homeless. *Lancet*, 384(9953), 1541–1547. [https://doi.org/10.1016/S0140-6736\(14\)61133-8](https://doi.org/10.1016/S0140-6736(14)61133-8)
- Hyde, T. M., & Ron, M. A. (2011). The secondary schizophrenias. In D. R. Weinberger & P. J. Harrison (Eds.), *Schizophrenia*, 3rd edn. (pp. 165–184). Wiley-Blackwell.
- Iacobucci, G. (2018). Ban on legal highs fails to curb use among children. *BMJ*, 363, k4927.
- Iadecola, C., Duering, M., Hachinski, V., et al. (2019). Vascular cognitive impairment and dementia: JACC scientific expert panel. *J Am Coll Cardiol*, 73(25), 3326–3344.
- Iemmi, V. (2022). Establishing political priority for global mental health: A qualitative policy analysis. *Health Policy Plan*, 37(8), 1012–1024. <https://doi.org/10.1093/heapol/czac046>
- Ijzendoorn, M. H., Bakermans-Kranenburg, M. J., & Scott, S. (2015). Residential and foster care. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 261–272). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch21>
- Imboden, J. B., Canter, A., & Cluff, L. E. (1961). Convalescence from influenza. A study of the psychological and clinical determinants. *Arch Intern Med*, 108, 393–399. <https://doi.org/10.1001/archinte.1961.03620090065008>
- Ingvar, D. H., & Franzén, G. (1974). Abnormalities of cerebral blood flow distribution in patients with chronic schizophrenia. *Acta Psychiatr Scand*, 50, 425–462.

- Inouye, S. K., Westendorp, R. G. J., & Saczynski, J. S. (2014). Delirium in elderly people. *Lancet*, 383(9920), 911–922.
- Ioannou, M., Wartenberg, C., Greenbrook, J. T., et al. (2021). Sleep deprivation as treatment for depression: Systematic review and meta-analysis. *Acta Psychiatr Scand*, 143(1), 22–35.
- Iovieno, N., Papakostas, G. I., Feeney, A., et al. (2021). Vortioxetine versus placebo for major depressive disorder: A comprehensive analysis of the clinical trial dataset. *J Clin Psychiatry*, 82(4), 34253.
- Isaac, M. L., & Paaauw, D. S. (2014). Medically unexplained symptoms. *Med Clin North Am*, 98(3), 663–672. <https://doi.org/10.1016/j.mcna.2014.01.013>
- Itzhaki, R. F., Golde, T. E., Heneka, M. T., & Redhead, B. (2020). Do infections have a role in the pathogenesis of Alzheimer disease? *Nature Rev Neurol*, 16, 193–197.
- Ives, R. (2009). Disorders relating to the use of volatile substances. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 502–506). Oxford University Press.
- Jablensky, A. (2016). Psychiatric classifications: Validity and utility. *World Psychiatry*, 15(1), 26–31.
- Jablensky, A., Kirkbride, J. B., & Jones, P. B. (2011). Schizophrenia: The epidemiological horizon. In D. Weinberger & P. Harrison (Eds.), *Schizophrenia* (pp. 185–225). Wiley-Blackwell.
- Jablensky, A., Sartorius, N., Ernberg, G., et al. (1992). Schizophrenia: manifestations, incidence and course in different cultures: A World Health Organization ten-country study. *Psychol Med*, 20, 1–97.
- Jack, C. R., Jr, Bennett, D. A., Blennow, K., et al. (2018). NIA-AA research framework: Toward a biological definition of Alzheimer's disease. *Alzheimers Dement*, 14(4), 535–562.
- Jacobi, F., Wittchen, H.-U., Hölting, C., et al. (2004). Prevalence, co-morbidity and correlates of mental disorders in the general population: Results from the German Health Interview and Examination Survey (GHS). *Psychol Med*, 34(4), 597–611.
- Jafferany, M., & Patel, A. (2019). Skin-picking disorder: A guide to diagnosis and management. *CNS Drugs*, 33(4), 337–346.
- Jäger, M., & Rössler, W. (2009). Informeller Zwang zur Verbesserung der Behandlungsbereitschaft psychiatrischer Patienten. [Informal coercion to enhance treatment adherence among psychiatric patients]. *Neuropsychiatr*, 23(4), 206–215.
- Jaggar, M., Fanibunda, S. E., Ghosh, S., et al. (2019). The neurotrophic hypothesis of depression revisited: New insights and therapeutic implications. In J. Quevedo, A. F. Carvalho, & C. A. Zarate (Eds.), *Neurobiology of Depression* (pp. 43–62). Elsevier.
- James, A., & Worrall-Davies, A. (2015). Provision of intensive treatment: Intensive outreach, day units, and in-patient units. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 648–660). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch50>
- James, M., Painter, J., Buckingham, B., & Stewart, M. W. (2018). A review and update of the Health of the Nation Outcome Scales (HoNOS). *BJPsych Bull*, 42(2), 63–68. <https://doi.org/10.1192/bjb.2017.17>
- Jamison, K. (1997). *An Unquiet Mind*. Picador.
- Jankovic, J., Richards, F., Berrisford, G., & Priebe, S. (2020). Advance statements in adult mental healthcare: An update. *BJPsych Adv*, 26(2), 109–115. <https://doi.org/10.1192/bja.2019.62>
- Jarde, A., Morais, M., Kingston, D., et al. (2016). Neonatal outcomes in women with untreated antenatal depression compared with women without depression: A systematic review and meta-analysis. *JAMA Psychiatry*, 73(8), 826–837.
- Jaspers, K. (1913). *Allgemeine Psychopathologie*. Springer.
- Jaspers, K. (1963). *General Psychopathology* (trans. from the 7th German edition by J. Hoenig and M. W. Hamilton). Manchester University Press.
- Jaspers, K. (1968). The phenomenological approach in psychopathology. *BJPsych*, 114(516), 1313–1323.
- Jauhar, S., Johnstone, M., & McKenna, P. J. (2022). Schizophrenia. *Lancet*, 399(10323), 473–486. [https://doi.org/10.1016/S0140-6736\(21\)01730-x](https://doi.org/10.1016/S0140-6736(21)01730-x)
- Jauhar, S., Laws, K. R., & McKenna, P. J. (2019). CBT for schizophrenia: A critical viewpoint. *Psychol Med*, 49(8), 1233–1236. <https://doi.org/10.1017/S0033291718004166>
- Jay, E. L., Sierra, M., van den Eynde, F., et al. (2014). Testing a neurobiological model of depersonalization disorder using repetitive transcranial magnetic stimulation. *Brain Stimul*, 7(2), 252–259. <https://doi.org/10.1016/j.brs.2013.12.002>
- Jeltsch-David, H., & Muller, S. (2014). Neuropsychiatric systemic lupus erythematosus: Pathogenesis and biomarkers. *Nat Rev Neurol*, 10(10), 579–596.
- Jenkins, J., Madigan, S., & Arseneault, L. (2015). Psychosocial adversity. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 330–340). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch26>
- Jenkins, R. (2013). How to convince politicians that mental health is a priority. *World Psychiatry*, 12(3), 266–268. <https://doi.org/10.1002/wps.20073>
- Jenkins, R., Bhugra, D., Meltzer, H., et al. (2005). Psychiatric and social aspects of suicidal behaviour in prisons. *Psychol Med*, 35(2), 257–269. <https://doi.org/10.1017/S0033291704002958>
- Jeong, J., Franchett, E. E., Ramos de Oliveira, C. V., et al. (2021). Parenting interventions to promote early child development in the first three years of life: A global systematic review and meta-analysis. *PLoS MED*, 18(5), e1003602. <https://doi.org/10.1371/journal.pmed.1003602>
- Jeschke, M. G., van Baar, M. E., Choudhry, M. A., Chung, K. K., Gibran, N. S., Logsetty, S. (2020). Burn injury. *Nat Rev Dis Primers*, 6, 11. doi: 10.1038/s41572-020-0145-5.
- Jessen, F., Amariglio, R. E., Buckley, R. F., et al. (2020). The characterisation of subjective cognitive decline. *Lancet Neurol*, 19(3), 271–278.
- Jha, M. K., Minhajuddin, A., Gadad, B. S., et al. (2017). Can C-reactive protein inform antidepressant medication selection in depressed outpatients? Findings from the CO-MED trial. *Psychoneuroendocrinol*, 78, 105–113.
- Ji, W., Hu, Y., Huang, Y., et al. (2006). A twin study of personality disorder heritability [In Chinese]. *Zhonghua Liu Xing Bing Zue Za Zhi*, 27(2), 137–141.
- Jiang, J., Zhao, Y. J., Hu, X. Y., et al. (2017). Microstructural brain abnormalities in medication-free patients with major depressive disorder: A systematic review and meta-analysis of diffusion tensor imaging. *J Psychiatry Neurosci*, 42(3), 150–163. <https://doi.org/10.1503/jpn.150341>
- Jick, H., Kaye, J. A., & Jick, S. S. (2004). Antidepressants and the risk of suicidal behaviors. *JAMA*, 292(3), 338–343.

- Jilek, W. G. (2000). Traditional non-Western folk healing as relevant to psychiatry. In M. Gelder, J. Lopez-Ibor, Jr., & N. Andreasen (Eds.), *New Oxford Textbook of Psychiatry* (pp. 1061–1066). Oxford University Press.
- Joa, I., Bjørnstad, J., Johannessen, J. O., et al. (2021). Early detection of ultra high risk for psychosis in a Norwegian catchment area: The two-year follow-up of the prevention of psychosis study. *Front Psychiatry*, 12, 573905. <https://doi.org/10.3389/fpsyt.2021.573905>
- Joas, E., Karanti, A., Song, J., et al. (2017). Pharmacological treatment and risk of psychiatric hospital admission in bipolar disorder. *BJPsych*, 210(3), 197–202. <https://doi.org/10.1192/bjp.bp.116.187989>
- Johannsen, M., Damholdt, M. F., Zachariae, R., et al. (2019). Psychological interventions for grief in adults: A systematic review and meta-analysis of randomized controlled trials. *J Affect Disord*, 253, 69–86. <https://doi.org/10.1016/j.jad.2019.04.065>
- John, A., Eyles, E., Webb, R. T., et al. (2020). The impact of the COVID-19 pandemic on self-harm and suicidal behaviour: Update of living systematic review. *F1000Res*, 9, 1097. <https://doi.org/10.12688/f1000research.25522.2>
- Johns, A. (2009). Offending, substance misuse, and mental disorder. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1926–1928). Oxford University Press
- Johnson, J. G., Cohen, P., Brown, J., et al. (1999). Childhood maltreatment increases risk for personality disorders during early adulthood. *Arch Gen Psychiatry*, 56(7), 600–606. <https://doi.org/10.1001/archpsyc.56.7.600>
- Johnson, L. A., Johnson, R. L., & Portier, R.-B. (2013). Current 'legal highs'. *J Emerg Med*, 44(6), 1108–1115.
- Johnson, M. B., & Hyman, S. E. (2022). A critical perspective on the synaptic pruning hypothesis of schizophrenia pathogenesis. *Biol Psychiatry*, 92(6), 440–442. <https://doi.org/10.1016/j.biopsych.2021.12.014>
- Johnson, M. D., & Williams, M. A. (2025). Idiopathic normal-pressure hydrocephalus. *N Engl J Med*, 393, 2243–2253.
- Johnson, S., Nolan, F., Pilling, S., et al. (2005). Randomised controlled trial of acute mental health care by a crisis resolution team: The north Islington crisis study. *BMJ*, 331(7517), 599.
- Johnstone, E., Frith, C., Crow, T., et al. (1976). Cerebral ventricular size and cognitive impairment in chronic schizophrenia. *Lancet*, 308(7992), 924–926.
- Johnstone, E. C., MacMillan, J. F., & Crow, T. J. (1987). The occurrence of organic disease of possible or probable aetiological significance in a population of 268 cases of first episode schizophrenia. *Psychol Med*, 17(2), 371–379.
- Joinson, C., Heron, J., Lewis, G., et al. (2011). Timing of menarche and depressive symptoms in adolescent girls from a UK cohort. *BJPsych*, 198(1), 17–23.
- Joinson, C., Sullivan, S., von Gontard, A., & Heron, J. (2015). Early childhood psychological factors and risk for bedwetting at school age in a UK cohort. *Eur Child Adolesc Psychiatry*, 25(5), 519–528. <https://doi.org/10.1007/s00787-015-0756-7>
- Jolly, A. J., & Singh, S. M. (2020). Does electroconvulsive therapy cause brain damage: An update. *Indian J Psychiatry*, 62(4), 339.
- Jonas, K., Lian, W., Callahan, J., et al. (2022). The course of general cognitive ability in individuals With psychotic disorders. *JAMA Psychiatry*, 79(7), 659–666. <https://doi.org/10.1001/jamapsychiatry.2022.1142>
- Jones, C., Hacker, D., Meaden, A., et al. (2018). Cognitive behavioural therapy plus standard care versus standard care plus other psychosocial treatments for people with schizophrenia. *Cochrane Database Syst Rev*, 11(11), Cd008712. <https://doi.org/10.1002/14651858.CD008712.pub3>
- Jones, E., & Wessely, S. (2003). 'Forward psychiatry' in the military: Its origins and effectiveness. *J Trauma Stress*, 16(4), 411–419.
- Jones, E., & Wessely, S. (2014). Battle for the mind: World War 1 and the birth of military psychiatry. *Lancet*, 384(9955), 1708–1714.
- Jones, I. (2020). Postpartum psychosis: An important clue to the etiology of mental illness. *World Psychiatry*, 19(3), 334.
- Jones, I., Chandra, P. S., Dazzan, P., & Howard, L. M. (2014). Bipolar disorder, affective psychosis, and schizophrenia in pregnancy and the post-partum period. *Lancet*, 384, 1789–1799.
- Jones, K. (1972). *A History of the Mental Health Services*. Routledge and Kegan Paul.
- Jones, M. (1968). *Social Psychiatry in Practice*. Penguin.
- Jones, P., Murray, R., Rodgers, B., & Marmot, M. (1994). Child developmental risk factors for adult schizophrenia in the British 1946 birth cohort. *Lancet*, 344(8934), 1398–1402.
- Jones, S. V., & McCollum, R. (2019). Subjective memory complaints after electroconvulsive therapy: Systematic review. *BJPsych Bull*, 43(2), 73–80.
- Jørgensen, M. B., Rozing, M. P., Kellner, C. H., & Osler, M. (2020). Electroconvulsive therapy, depression severity and mortality: Data from the Danish National Patient Registry. *J Psychopharmacol*, 34(3), 273–279.
- Judd, L. L., Akiskal, H. S., Schettler, P. J., et al. (2002). The long-term natural history of the weekly symptomatic status of bipolar I disorder. *Arch Gen Psychiatry*, 59(6), 530–537.
- Juruena, M. F., Cleare, A. J., & Young, A. H. (2021). Neuroendocrine stress system in bipolar disorder. In A. H. Young & M. F. Juruena (Eds.), *Bipolar Disorder: From Neuroscience to Treatment* (pp. 149–171). Springer Nature.
- Kaar, S. J., Natesan, S., Mccutcheon, R., & Howes, O. D. (2020). Antipsychotics: Mechanisms underlying clinical response and side-effects and novel treatment approaches based on pathophysiology. *Neuropharmacol*, 172, 107704.
- Kader, L., & Pantelis, C. (2009). Ethical aspects of drug treatment. In S. Bloch & S. Green (Eds.), *Psychiatric Ethics* (pp. 339–400). Oxford University Press.
- Kahlbaum, K. (1863). *Die Gruppierung der Psychischen Krankheiten*. Kafemann.
- Kahn, E. (1928). Die psychopathischen Persönlichkeiten. In O. Bumke (Ed.), *Handbuch der Geisteskrankheiten*. (pp. 227). Springer.
- Kahn, R., & Sommer, I. (2015). The neurobiology and treatment of first-episode schizophrenia. *Mol Psychiatry*, 20(1), 84–97.
- Kahn, R. S. (2020). On the origins of schizophrenia. *Am J Psychiatry*, 177(4), 291–297. <https://doi.org/10.1176/appi.ajp.2020.20020147>
- Kaiser, R. H., Andrews-Hanna, J. R., Wager, T. D., & Pizzagalli, D. A. (2015). Large-scale network dysfunction in major depressive disorder: A meta-analysis of resting-state functional connectivity. *JAMA Psychiatry*, 72(6), 603–611.

- Kales, H. C., Lyketsos, C. G., Miller, E. M., & Ballard, C. (2019). Management of behavioral and psychological symptoms in people with Alzheimer's disease: An international Delphi consensus. *Int Psychogeriatr*, 31(1), 83–90.
- Kalia, L. V., & Lang, A. E. (2015). Parkinson's disease. *Lancet*, 386(9996), 896–912.
- Kambeitz, J. P., & Howes, O. D. (2015). The serotonin transporter in depression: Meta-analysis of in vivo and post-mortem findings and implications for understanding and treating depression. *J Affect Disord*, 186, 358–366.
- Kamp, F., Proebstl, L., Penzel, N., et al. (2019). Effects of sedative drug use on the dopamine system: A systematic review and meta-analysis of in vivo neuroimaging studies. *Neuropsychopharmacol*, 44(4), 660–667.
- Kandel, D., Griesler, P. C., Hu, M.-C., & Melas, P. A. (2013). Epidemiology of substance use disorders. In D. Charney, J. Buxbaum, P. Sklaar, & E. Nestler (Eds.), *Neurobiology of Mental Illness* (pp. 772–787). Oxford University Press.
- Kandel, E. R. (1998). A new intellectual framework for psychiatry. *Am J Psychiatry*, 155(4), 457–469.
- Kane, J., Honigfeld, G., Singer, J., & Meltzer, H. (1988). Clozapine for the treatment-resistant schizophrenic: A double-blind comparison with chlorpromazine. *Arch Gen Psychiatry*, 45(9), 789–796.
- Kane, J. C., Ventevogel, P., Spiegel, P., et al. (2014). Mental, neurological, and substance use problems among refugees in primary health care: Analysis of the Health Information System in 90 refugee camps. *BMC Med*, 12(1), 1.
- Kanner, A. M. (2016). Management of psychiatric and neurological comorbidities in epilepsy. *Nat Rev Neurol*, 12(2), 106–116.
- Kanner, A. M., Shankar, R., Margraf, N. G., et al. (2024). Mood disorders in adults with epilepsy: A review of unrecognized facts and common misconceptions. *Ann Gen Psychiatry*, 23, 11.
- Kappelman, N., Perry, B. I., & Khandaker, G. M. (2022). Prenatal and childhood immuno-metabolic risk factors for adult depression and psychosis. *Harv Rev Psychiatry*, 30(1), 8–23. <https://doi.org/10.1097/hrp.0000000000000322>
- Kapur, N., Steeg, S., & Moreton, A. (2020). Self-harm: Epidemiology and risk factors. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1289–1295). Oxford University Press.
- Kapur, S. (2003). Psychosis as a state of aberrant salience: A framework linking biology, phenomenology, and pharmacology in schizophrenia. *Am J Psychiatry*, 160(1), 13–23.
- Karas, K. H., Baharikhoo, P., & Kolla, N. J. (2021). Borderline personality disorder and its symptom clusters: A review of positron emission tomography and single photon emission computed tomography studies. *Psychiatry Res Neuroimag*, 316, 111357. <https://doi.org/10.1016/j.psychresns.2021.111357>
- Kasper, S., Bancher, C., Eckert, A., et al. (2020). Management of mild cognitive impairment (MCI): The need for national and international guidelines. *World J Biol Psychiatry*, 21(8), 579–594.
- Katon, W., Lin, E. H., & Kroenke, K. (2007). The association of depression and anxiety with medical symptom burden in patients with chronic medical illness. *Gen Hosp Psychiatry*, 29(2), 147–155.
- Katon, W. J. (2003). Clinical and health services relationships between major depression, depressive symptoms, and general medical illness. *Biol Psychiatry*, 54(3), 216–226.
- Kaufman, D. M., Geyer, H. L., & Milstein, M. J. (Eds.). (2017). *Kaufman's Clinical Neurology for Psychiatrists*, 8th edn. Elsevier.
- Kay, D., Beamish, P., & Roth, M. (1964). Old age mental disorders in Newcastle-upon-Tyne: 1: A study in prevalence. *BJPsych*, 110(465), 146–158.
- Kay, D., & Roth, M. (1961). Environmental and hereditary factors in the schizophrenias of old age ('late paraphrenia') and their bearing on the general problem of causation in schizophrenia. *J Ment Sci*, 107(449), 649–686.
- Kay, S. R., Fiszbein, A., & Opler, L. A. (1987). The positive and negative syndrome scale (PANSS) for schizophrenia. *Schizophr Bull*, 13(2), 261–276.
- Kaye, W. H., Wierenga, C. E., Bailer, U. F., Simmons, A. N., & Bischoff-Grethe, A. (2013). Nothing tastes as good as skinny feels: the neurobiology of anorexia nervosa. *Trends Neurosci*, 36, 110–120. <https://doi.org/10.1016/j.tins.2013.01.003>
- Keepers, G. A., Fochtmann, L. J., Anzia, J. M., et al. (2020). The American Psychiatric Association Practice Guideline for the treatment of patients with schizophrenia. *Am J Psychiatry*, 177(9), 868–872. <https://doi.org/10.1176/appi.ajp.2020.177901>
- Kessler, R. C., Berglund, P. A., Chiu, W. T., et al. (2013). The prevalence and correlates of binge eating disorder in the World Health Organization World Mental Health Surveys. *Biol Psychiatry*, 73, 904–914. <https://doi.org/10.1016/j.biopsych.2012.11.020>
- Kellar, D., & Craft, S. (2020). Brain insulin resistance in Alzheimer's disease and related disorders: Mechanisms and therapeutic approaches. *Lancet Neurol*, 19(9), 758–766.
- Kelleher, I., Connor, D., Clarke, M. C., et al. (2012). Prevalence of psychotic symptoms in childhood and adolescence: A systematic review and meta-analysis of population-based studies. *Psychol Med*, 42(09), 1857–1863. <https://doi.org/doi:10.1017/S0033291711002960>
- Kellner, C. H., Knapp, R. G., Petrides, G., et al. (2006). Continuation electroconvulsive therapy vs pharmacotherapy for relapse prevention in major depression: A multisite study from the Consortium for Research in Electroconvulsive Therapy (CORE). *Arch Gen Psychiatry*, 63(12), 1337–1344.
- Kelly, B. D. (2015). Best interests, mental capacity legislation and the UN Convention on the Rights of Persons with Disabilities. *BJPsych Adv*, 21(3), 188–195.
- Kendall, K., Van Assche, E., Andlauer, T., et al. (2021). The genetic basis of major depression. *Psychol Med*, 51(13), 2217–2230.
- Kendell, R. (1975). *The Role of Diagnosis in Psychiatry*. Blackwell.
- Kendler, K., & Walsh, D. (1995). Schizophreniform disorder, delusional disorder and psychotic disorder not otherwise specified: Clinical features, outcome and familial psychopathology. *Acta Psychiatr Scand*, 91(6), 370–378.
- Kendler, K. S. (1982). Demography of paranoid psychosis (delusional disorder): A review and comparison with schizophrenia and affective illness. *Arch Gen Psychiatry*, 39(8), 890–902.
- Kendler, K. S. (2001). Twin studies of psychiatric illness: An update. *Arch Gen Psychiatry*, 58(11), 1005–1014.
- Kendler, K. S., Gardner, C. O., & Prescott, C. A. (2002). Toward a comprehensive developmental model for major depression in women. *Am J Psychiatry*, 159(7), 1133–1145.
- Kendler, K. S., Gruenberg, A. M., & Strauss, J. S. (1981). An independent analysis of the Copenhagen sample of the Danish adoption study of schizophrenia. II. The relationship between

- schizotypal personality disorder and schizophrenia. *Arch Gen Psychiatry*, 38(9), 982–984. <https://doi.org/10.1001/archpsyc.1981.01780340034003>
- Kendler, K. S., Hettema, J. M., Butera, F., et al. (2003). Life event dimensions of loss, humiliation, entrapment, and danger in the prediction of onsets of major depression and generalized anxiety. *Arch Gen Psychiatry*, 60(8), 789–796.
- Kendler, K. S., Kuhn, J., & Prescott, C. A. (2004). The interrelationship of neuroticism, sex, and stressful life events in the prediction of episodes of major depression. *Am J Psychiatry*, 161(4), 631–636.
- Kendler, K. S., Masterson, C. C., & Davis, K. L. (1985). Psychiatric illness in first-degree relatives of patients with paranoid psychosis, schizophrenia and medical illness. *BJPsych*, 147(5), 524–531.
- Kendler, K. S., Myers, J., Torgersen, S., et al. (2007). The heritability of cluster A personality disorders assessed by both personal interview and questionnaire. *Psychol Med*, 37(5), 655–665.
- Kendrick, T., Burns, T., & Freeling, P. (1995). Randomised controlled trial of teaching general practitioners to carry out structured assessments of their long term mentally ill patients. *BMJ*, 311(6997), 93–97.
- Kennedy, J., Hemby, P., Green, D., et al. (2020). Predictors of change in global psychiatric functioning at an inpatient adolescent psychiatric unit: A decade of experience. *Clin Child Psychol Psychiatry*, 25(2), 471–482.
- Kennerley, H., Kirk, J., & Westbrook, D. (2017). *Introduction to Cognitive Behaviour Therapy: Skills & Applications*, 3rd edn. Sage.
- Keown, P., Weich, S., Bhui, K. S., & Scott, J. (2011). Association between provision of mental illness beds and rate of involuntary admissions in the NHS in England 1988–2008: Ecological study. *BMJ*, 343, d3736. <https://doi.org/10.1136/bmj.d3736>
- Keren, H., O'Callaghan, G., Vidal-Ribas, P., et al. (2018). Reward processing in depression: A conceptual and meta-analytic review across fMRI and EEG studies. *Am J Psychiatry*, 175(11), 1111–1120. <https://doi.org/10.1176/appi.ajp.2018.17101124>
- Kernberg, O. F., & Michels, R. (2009). Borderline personality disorder. *Am J Psychiatry*, 166(5), 505–508. <https://doi.org/10.1176/appi.ajp.2009.09020263>
- Kerr, M., Goodwin, G., & Hanna, J. (2023). Epilepsy, intellectual disability and the epilepsy care pathway: Improving outcomes. *BJPsych Adv*, 129(5), 295–304.
- Kerr, M., Scheepers, M., Arvio, M., et al. (2009). Consensus guidelines into the management of epilepsy in adults with an intellectual disability. *J Intellect Disabil Res*, 53(8), 687–694.
- Kessing, L. (2020). Epidemiology of mood disorders In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press.
- Kessler, R. C. (2004). The epidemiology of dual diagnosis. *Biol Psychiatry*, 56(10), 730–737.
- Kessler, R. C., Berglund, P., Demler, O., et al. (2005a). Lifetime prevalence and age-of-onset distributions of DSM-IV disorders in the National Comorbidity Survey Replication. *Arch Gen Psychiatry*, 62(6), 593–602. <https://doi.org/10.1001/archpsyc.62.6.593>
- Kessler, R. C., Chiu, W. T., Demler, O., & Walters, E. E. (2005b). Prevalence, severity, and comorbidity of 12-month DSM-IV disorders in the National Comorbidity Survey Replication. *Arch Gen Psychiatry*, 62(6), 617–627.
- Kessler, R. C., Petukhova, M., Sampson, N. A., et al. (2012). Twelve-month and lifetime prevalence and lifetime morbid risk of anxiety and mood disorders in the United States. *Int J Methods Psychiatr Res*, 21(3), 169–184.
- Kessler, R. C., Zhao, S., Katz, S. J., et al. (1999). Past-year use of outpatient services for psychiatric problems in the National Comorbidity Survey. *Am J Psychiatry*, 156(1), 115–123. <https://doi.org/10.1176/ajp.156.1.115>
- Kessler, U. (2019). Electroconvulsive therapy in bipolar disorder depression. In I. N. Ferrier & W. J. (Eds.), *The ECT Handbook*, 4th edn. (pp. 38–43). Cambridge University Press.
- Kety, S. S., Wender, P. H., Jacobsen, B., et al. (1994). Mental illness in the biological and adoptive relatives of schizophrenic adoptees: Replication of the Copenhagen study in the rest of Denmark. *Arch Gen Psychiatry*, 51(6), 442–455.
- Kevelson, S. D., Rahman, J., & Veenstra-VanderWeele, J. (2021). Autism spectrum disorders. In Dulcan, M. K. (Ed.), *Dulcan's Textbook of Child and Adolescent Psychiatry*, (pp. 135–156). American Psychiatric Publications.
- Khachadourian, V., Zaks, N., Lin, E., et al. (2021). Advanced paternal age and risk of schizophrenia in offspring: Review of epidemiological findings and potential mechanisms. *Schizophr Res*, 233, 72–79. <https://doi.org/10.1016/j.schres.2021.06.016>
- Khurana, D. S., Carvalho, K. S. (2020). Parasomnia. In: Sedky, K., Nazir, R., Bennett, D. (Eds.), *Sleep Medicine and Mental Health*. Springer, Cham (pp. 235–258).
- Kieling, C., Baker-Henningham, H., Belfer, M., et al. (2011). Child and adolescent mental health worldwide: Evidence for action. *Lancet*, 378(9801), 1515–1525. [https://doi.org/10.1016/S0140-6736\(11\)60827-1](https://doi.org/10.1016/S0140-6736(11)60827-1)
- Killaspay, H. (2006). Psychiatric out-patient services: Origins and future. *Adv Psychiatr Treat*, 12(5), 309–319.
- Kiloh, L., Smith, J., & Johnson, G. (1988). *Physical Treatments in Psychiatry*. Blackwell Scientific Publications.
- Kilpatrick, D. G., & Acierio, R. (2003). Mental health needs of crime victims: Epidemiology and outcomes. *J Trauma Stress*, 16(2), 119–132.
- Kilpatrick, D. G., Friedman, M. J., & Gilmore, A. K. (2020). Classification and descriptive psychopathology of post-traumatic stress disorder and other stressor-related disorders. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press.
- Kim, H. K., Blumberger, D. M., & Daskalakis, Z. J. (2020). Neurophysiological biomarkers in schizophrenia-P50, mismatch negativity, and TMS-EMG and TMS-EEG. *Front Psychiatry*, 11, 795. <https://doi.org/10.3389/fpsy.2020.00795>
- Kim, J. Y., Son, M. J., Son, C. Y., et al. (2019). Environmental risk factors and biomarkers for autism spectrum disorder: An umbrella review of the evidence. *Lancet Psychiatry*, 6(7), 590–600.
- Kim, S., Kwon, S.-H., Kam, T.-I., et al. (2019). Transneuronal propagation of pathologic α -synuclein from the gut to the brain models Parkinson's disease. *Neuron*, 103(4), 627–641. e627.
- King, S., Stone, J. M., Cleare, A., & Young, A. H. (2019). A systematic review on neuropsychological function in bipolar disorders type I and II and subthreshold bipolar disorders—something to think about. *CNS Spect*, 24(1), 127–143.
- King-Hele, S., Webb, R., Mortensen, P. B., et al. (2009). Risk of stillbirth and neonatal death linked with maternal mental

- illness: A national cohort study. *Arch Dis Child Fetal Neonatal Ed*, 94(2), F105–F110.
- Kingsbury, M., Clayborne, Z., Colman, I., & Kirkbride, J. B. (2020). The protective effect of neighbourhood social cohesion on adolescent mental health following stressful life events. *Psychol Med*, 50(8), 1292–1299.
- Kipps, C., & Hodges, J. (2005). Cognitive assessment for clinicians. *J Neurol Neurosurg Psychiatry*, 76(Suppl 1), i22–i30.
- Kirkbride, J. B., Errazuriz, A., Croudace, T. J., et al. (2012a). Incidence of schizophrenia and other psychoses in England, 1950–2009: A systematic review and meta-analyses. *PLoS ONE*, 7(3), e31660.
- Kirkbride, J. B., Susser, E., Kundakovic, M., et al. (2012b). Prenatal nutrition, epigenetics and schizophrenia risk: Can we test causal effects? *Epigenomics*, 4(3), 303–315.
- Kirkham, J., Sherman, C., Velkers, C., et al. (2017). Antipsychotic use in dementia: Is there a problem and are there solutions? *Can J Psychiatry*, 62(3), 170–181.
- Kirmayer, L. J., Kronick, R., & Rousseau, C. (2018). Advocacy as key to structural competency in psychiatry. *JAMA Psychiatry*, 75(2), 119–120. <https://doi.org/10.1001/jamapsychiatry.2017.3897>
- Kirov, G., Jauhar, S., Sienaert, P., et al. (2021). Electroconvulsive therapy for depression: 80 years of progress. *BJPsych*, 219(5), 594–597.
- Kirov, G., Rees, E., & Walters, J. (2015). What a psychiatrist needs to know about copy number variants. *BJPsych Adv*, 21(3), 157–163.
- Kisely, S., & Goldberg, D. (1996). Physical and psychiatric comorbidity in general practice. *BJPsych*, 169(2), 236–242.
- Kiser, D. P., Rivero, O., & Lesch, K. P. (2015). Annual research review: The (epi) genetics of neurodevelopmental disorders in the era of whole-genome sequencing—unveiling the dark matter. *J Child Psychol Psychiatry*, 56(3), 278–295.
- Kishi, T., Ikuta, T., Matsuda, Y., et al. (2021). Mood stabilizers and/or antipsychotics for bipolar disorder in the maintenance phase: A systematic review and network meta-analysis of randomized controlled trials. *Mol Psychiatry*, 26(8), 4146–4157.
- Kishimoto, T., Hagi, K., Kurokawa, S., et al. (2021). Long-acting injectable versus oral antipsychotics for the maintenance treatment of schizophrenia: A systematic review and comparative meta-analysis of randomised, cohort, and pre-post studies. *Lancet Psychiatry*, 8(5), 387–404. [https://doi.org/10.1016/S2215-0366\(21\)00039-0](https://doi.org/10.1016/S2215-0366(21)00039-0)
- Kivipelto, M., Mangialasche, F., & Ngandu, T. (2018). Lifestyle interventions to prevent cognitive impairment, dementia and Alzheimer disease. *Nat Rev Neurology*, 14(11), 653–666.
- Klau, J., Bernardo, C. D. O., Gonzalez-Chica, D. A., et al. (2022). Trends in prescription of psychotropic medications to children and adolescents in Australian primary care from 2011 to 2018. *Aust N Z J Psychiatry*, 56(11), 1477–1490.
- Klein, D. F. (1964). Delineation of two drug-responsive anxiety syndromes. *Psychopharmacol*, 5(6), 397–408.
- Klein, M. (1996). Notes on some schizoid mechanisms. *J Psychother Pract Res*, 5(2), 160–179.
- Klerman, G., & Weissman, M. M. (1984). *Interpersonal Psychotherapy of Depression: A Brief, Focused, Specific Strategy*. Basic Books.
- Knapp, M., Hallam, A., Beecham, J., & Baines, B. (1999). Private, voluntary or public? Comparative cost-effectiveness in community mental health care. *Policy Politics*, 27(1), 25–41.
- Knapp, M., Patel, A., Curran, C., et al. (2013). Supported employment: Cost-effectiveness across six European sites. *World Psychiatry*, 12(1), 60–68. <https://doi.org/10.1002/wps.20017>
- Koch, H., van Bokhoven, M. A., Riet, G. T., et al. (2007). Demographic characteristics and quality of life of patients with unexplained complaints: A descriptive study in general practice. *Q Life Res*, 16(9), 1483–1489.
- Koenig, H. G., & Blazer, D. G. (1992). Epidemiology of geriatric affective disorders. *Clinics Geriatr Med*, 8(2), 235–251.
- Kohl, F. (2002). Die progressive Muskel-Relaxation nach E. Jacobson (PME). Ein modernes Entspannungsverfahren. [Progressive muscle relaxation according to E. Jacobson. A modern relaxation technique.] *Med Monatsschr Pharm*, 25(3), 77–87.
- Kohrt, B. A., Asher, L., Bhardwaj, A., et al. (2018). The role of communities in mental health care in low-and middle-income countries: A meta-review of components and competencies. *Int J Environ Res Public Health*, 15(6), 1279.
- Kolbe, M., Schmutz, S., Seelandt, J. C., et al. (2021). Team debriefings in healthcare: Aligning intention and impact. *BMJ*, 374, n2042. <https://doi.org/10.1136/bmj.n2042>
- Kolshus, E., Jelovac, A., & McLoughlin, D. (2017). Bitemporal v. high-dose right unilateral electroconvulsive therapy for depression: A systematic review and meta-analysis of randomized controlled trials. *Psychol Med*, 47(3), 518–530.
- Koning, J. P., Tenback, D. E., van Os, J., et al. (2010). Dyskinesia and parkinsonism in antipsychotic-naïve patients with schizophrenia, first-degree relatives and healthy controls: A meta-analysis. *Schizophr Bull*, 36(4), 723–731.
- Kooij, J., Bijlenga, D., Salerno, L., et al. (2019). Updated European Consensus Statement on diagnosis and treatment of adult ADHD. *Eur Psychiatry*, 56(1), 14–34.
- Koopmans, G. T., Donker, M. C., & Rutten, F. H. (2005). Length of hospital stay and health services use of medical inpatients with comorbid noncognitive mental disorders: A review of the literature. *Gen Hosp Psychiatry*, 27(1), 44–56.
- Kooyman, I., & Walsh, E. (2011). Societal outcomes in schizophrenia. In D. Weinberger & P. Harrison (Eds.), *Schizophrenia* (pp. 644–665). Wiley-Blackwell.
- Kopelman, M. D. (2022). What is the Korsakoff syndrome? A paper in tribute to Prof Alwyn Lishman. *Cogn Neuropsychiatry*, 27(4), 296–313.
- Korb, L., O'Regan, D., Conley, J., et al. (2021). Sleep: The neglected life factor in adults with intellectual disabilities. *BJPsych Bull*, 47(3), 139–145. <https://doi.org/10.1192/bjb.2021.122>
- Kowalski, R. M., Giumetti, G. W., Schroeder, A. N., & Lattanner, M. R. (2014). Bullying in the digital age: A critical review and meta-analysis of cyberbullying research among youth. *Psychol Bull*, 140(4), 1073.
- Kraepelin, E. (1904). *Clinical Psychiatry: A Textbook for Students and Physicians*. Macmillan.
- Kraepelin, E. (1919). *Dementia Praecox and Paraphrenia*. Livingstone.
- Kraepelin, E. (1921). *Manic Depressive Insanity and Paranoia*. Livingstone.
- Krause, A. J., Simon, E. B., Mander, B. A., et al. (2017). The sleep-deprived human brain. *Nat Rev Neurosci*, 18, 404–418. <https://doi.org/10.1038/nrn.2017.55>

- Kreitman, N. (1961). The reliability of psychiatric diagnosis. *J Ment Sci*, 107(450), 876–886.
- Kretschmer, E. (1927). *Der sensitive Beziehungswahn*. Reprinted and trans. in S. R. Hirsch & M. Shepherd (Eds.). (1974). *Themes and Variations in European Psychiatry*, Chapter 8. Wright.
- Kretschmer, E. (1936). *Physique and Character*, 2nd edn. WJH Sprott and KP Trench.
- Kril, J. J., & Harper, C. G. (2012). Neuroanatomy and neuropathology associated with Korsakoff's syndrome. *Neuropsychol Rev*, 22, 72–80.
- Kroenke, K. (2021). PHQ-9: Global uptake of a depression scale. *World Psychiatry*, 20(1), 135–136. <https://doi.org/10.1002/wps.20821>
- Kroenke, K., Spitzer, R. L., & Williams, J. B. (2001). The PHQ-9: Validity of a brief depression severity measure. *J Gen Int Med*, 16(9), 606–613.
- Krompa, G. M., O'Mahony, E., Tan, J., et al. (2022). The effectiveness of community mental health teams in relation to team cohesion, authentic leadership and size of the team: A study in the north west of Ireland. *Community Ment Health J*, 58(7), 1393–1402.
- Kruizinga, J., Liemburg, E., Burger, H., et al. (2021). Pharmacological treatment for psychotic depression. *Cochrane Database Syst Rev*, 12(12), Cd004044. <https://doi.org/10.1002/14651858.CD004044.pub5>
- Krynicky, C. R., Upthegrove, R., Deakin, J. F. W., & Barnes, T. R. E. (2018). The relationship between negative symptoms and depression in schizophrenia: A systematic review. *Acta Psychiatr Scand*, 137(5), 380–390. <https://doi.org/10.1111/acps.12873>
- Krystal, A. D., Prather, A. A., & Ashbrook, L. H. (2019). The assessment and management of insomnia: an update. *World Psychiatry*, 18, 337–352. <https://doi.org/10.1002/wps.20674>
- Kubler-Ross, E. (1969). *On Death and Dying*. Macmillan.
- Kullberg, M.-L., Maciejewski, D., van Schie, C. C., et al. (2020). Parental bonding: Psychometric properties and association with lifetime depression and anxiety disorders. *Psychol Assess*, 32(8), 780.
- Kuo, S. S., & Pogue-Geile, M. F. (2019). Variation in fourteen brain structure volumes in schizophrenia: A comprehensive meta-analysis of 246 studies. *Neurosci Biobehav Rev*, 98, 85–94. <https://doi.org/10.1016/j.neubiorev.2018.12.030>
- Kuźma, E., Lourida, I., Moore, S. F., et al. (2018). Stroke and dementia risk: A systematic review and meta-analysis. *Alzheimers Dement*, 14(11), 1416–1426.
- Kvale, G., Berggren, U., & Milgrom, P. (2004). Dental fear in adults: A meta-analysis of behavioral interventions. *Comm Dent Oral Epidemiol*, 32(4), 250–264.
- Lagerberg, T., Fazel, S., Sjölander, A., et al. (2022). Selective serotonin reuptake inhibitors and suicidal behaviour: A population-based cohort study. *Neuropsychopharmacol*, 47(4), 817–823. <https://doi.org/10.1038/s41386-021-01179-z>
- Lai, Y., Wang, T., Zhang, C., et al. (2020). Effectiveness and safety of neuroablation for severe and treatment-resistant obsessive-compulsive disorder: A systematic review and meta-analysis. *J Psychiatry Neurosci*, 45(5), 356–369.
- Lally, J., Ajnakina, O., Stubbs, B., et al. (2017). Remission and recovery from first-episode psychosis in adults: Systematic review and meta-analysis of long-term outcome studies. *BJPsych*, 211(6), 350–358. <https://doi.org/10.1192/bjp.bp.117.201475>
- Lam, R. W., Teng, M. Y., Jung, Y.-E., et al. (2020). Light therapy for patients with bipolar depression: systematic review and meta-analysis of randomized controlled trials. *Can J Psychiatry*, 65(5), 290–300.
- Lamers, F., Milanese, Y., De Jonge, P., et al. (2018). Metabolic and inflammatory markers: Associations with individual depressive symptoms. *Psychol Med*, 48(7), 1102–1110.
- Lamontagne, Y., Boyer, R., Hetu, C., & Lacerte-Lamontagne, C. (2000). Anxiety, significant losses, depression, and irrational beliefs in first-offence shoplifters. *Can J Psychiatry*, 45(1), 63–66. <https://doi.org/10.1177/070674370004500109>
- Lan, S.-Y., Kuo, C.-Y., Chou, C.-C., et al. (2019). Recreational nitrous oxide abuse related subacute combined degeneration of the spinal cord in adolescents—A case series and literature review. *Brain Devel*, 41(5), 428–435.
- Landgren, V., Malki, K., Bottai, M., Arver, S., & Rahm, C. (2020). Effect of gonadotropin-releasing hormone antagonist on risk of committing child sexual abuse in men with pedophilic disorder: A randomized clinical trial. *JAMA Psychiatry*, 77, 897–905. <https://doi.org/10.1001/jamapsychiatry.2020.0440>
- Landi, P., Marazziti, D., Rutigliano, G., & Dell'Osso, L. (2016). Insight in psychiatry and neurology: State of the art, and hypotheses. *Harv Rev Psychiatry*, 24(3), 214–228.
- Landin-Romero, R., Moreno-Alcazar, A., Pagani, M., & Amann, B. L. (2018). How does eye movement desensitization and reprocessing therapy work? A systematic review on suggested mechanisms of action. *Front Psychol*, 9, 1395. <https://doi.org/10.3389/fpsyg.2018.01395>
- Landry, M., Moreno, A., Patry, S., et al. (2021). Current practices of electroconvulsive therapy in mental disorders: A systematic review and meta-analysis of short and long-term cognitive effects. *J ECT*, 37(2), 119–127.
- Langa, K. M., & Levine, D. A. (2014). The diagnosis and management of mild cognitive impairment: a clinical review. *JAMA*, 312(23), 2551–2561.
- Langdon, R., Connaughton, E., & Coltheart, M. (2014). The Fregoli delusion: A disorder of person identification and tracking. *Topics Cog Sci*, 6(4), 615–631.
- Langguth, B., Kreuzer, P. M., Kleinjung, T., & De Ridder, D. (2013). Tinnitus: Causes and clinical management. *Lancet Neurol*, 12(9), 920–930. [https://doi.org/10.1016/s1474-4422\(13\)70160-1](https://doi.org/10.1016/s1474-4422(13)70160-1)
- Långström, N., Babchishin, K. M., Fazel, S., et al. (2015). Sexual offending runs in families: A 37-year nationwide study. *Int J Epidemiol*, 44(2), 713–720. <https://doi.org/10.1093/ije/dyv029>
- Laplanche, J., & Pontalis, J. (1988). *The Language of Psycho-Analysis*. Karnac.
- Larkin, C., Di Blasi, Z., & Arensman, E. (2014). Risk factors for repetition of self-harm: A systematic review of prospective hospital-based studies. *PLoS ONE*, 9(1), e84282.
- Larner, A. J., Welch, K. A., & Carson, A. J. (2021). Neurological examination: What do psychiatrists need to know? *BJPsych Adv*, 27(5), 313–319.
- Larsen, E., Damkier, P., Pedersen, L., et al. (2015). Use of psychotropic drugs during pregnancy and breast-feeding. *Acta Psychiatr Scand*, 132, 1–28.
- Laursen, T. M. (2019). Causes of premature mortality in schizophrenia: A review of literature published in 2018. *Curr Opin Psychiatry*, 32(5), 388–393. <https://doi.org/10.1097/ycp.0000000000000530>

- Laver, K., Dyer, S., Whitehead, C., et al. (2016). Interventions to delay functional decline in people with dementia: A systematic review of systematic reviews. *BMJ Open*, 6(4), e010767.
- Lawrie, S. M. (2025). Brain structure in schizophrenia: what do we know and what next? *Lancet Psychiatry*, 12, 852–862.
- Lawrie, S. M., O'Donovan, M. C., Saks, E., et al. (2016a). Improving classification of psychoses. *Lancet Psychiatry*, 3(4), 367–374.
- Lawrie, S. M., O'Donovan, M. C., Saks, E., et al. (2016b). Towards diagnostic markers for the psychoses. *Lancet Psychiatry*, 3(4), 375–385.
- Laws, K. R., & Conway, W. (2019). Do adjunctive art therapies reduce symptomatology in schizophrenia? A meta-analysis. *World J Psychiatry*, 9(8), 107–120. <https://doi.org/10.5498/wjp.v9.i8.107>
- Leadership Alliance for the Care of Dying People. (2014). *One Chance to Get It Right: Improving People's Experience of Care in the Last Few Days and Hours of Life*. LACDP.
- Lebin, L. G., & Novick, A. M. (2022). Selective serotonin reuptake inhibitors (SSRIs) in pregnancy: An updated review on risks to mother, fetus, and child. *Curr Psychiatry Rep*, 24(11), 687–695.
- Leckman, J. F., & Taylor, E. (2015). Clinical assessment and diagnostic formulation. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 403–418). John Wiley & Sons, Ltd.
- Leclerc, Y., Sheehan, D. V., Weiller, E., et al. (1997). The Mini International Neuropsychiatric Interview (MINI). A short diagnostic structured interview: Reliability and validity according to the CIDI. *Eur Psychiatry*, 12(5), 224–231.
- LeDoux, J. E. (2014). Coming to terms with fear. *Proc Natl Acad Sci USA*, 111(8), 2871–2878. <https://doi.org/10.1073/pnas.1400351111>
- Lee, E. (2020). Late-onset schizophrenia. In T. Denning, A. Thomas, R. Stewart, & J.-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 671–684). Oxford University Press.
- Lee, R., Leighton, S. P., Thomas, L., et al. (2022). Prediction models in first-episode psychosis: Systematic review and critical appraisal. *BJPsych*, 220(4), 179–191. <https://doi.org/10.1192/bjp.2021.219>
- Leentjens, A. F., Rundell, J. R., Diefenbacher, A., et al. (2011). Psychosomatic medicine and consultation-liaison psychiatry: Scope of practice, processes, and competencies for psychiatrists or psychosomatic medicine specialists: A consensus statement of the European Association of Consultation-Liaison Psychiatry and Psychosomatics and the Academy of Psychosomatic Medicine. *Psychosomatics*, 52(1), 19–25. <https://doi.org/10.1016/j.psych.2010.11.022>
- Lees, R., Hines, L. A., D'Souza, D. C., et al. (2021). Psychosocial and pharmacological treatments for cannabis use disorder and mental health comorbidities: A narrative review. *Psychol Med*, 51(3), 353–364. <https://doi.org/10.1017/s0033291720005449>
- Leff, J. (1993). All the homeless people—where do they all come from? *BMJ*, 306(6879), 669–670. <https://doi.org/10.1136/bmj.306.6879.669>
- Leff, J., Kuipers, L., Berkowitz, R., & Sturgeon, D. (1985). A controlled trial of social intervention in the families of schizophrenic patients: Two-year follow-up. *BJPsych*, 146(6), 594–600.
- Lehtimäki, S., Martic, J., Wahl, B., et al. (2021). Evidence on digital mental health interventions for adolescents and young people: Systematic overview. *JMIR Ment Health*, 8(4), e25847.
- Leibenluft, E., & Dickstein, D. P. (2015). Bipolar disorder in childhood. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 858–873). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch62>
- Leichsenring, F., Abbass, A., Heim, N., et al. (2023). The status of psychodynamic psychotherapy as an empirically supported treatment for common mental disorders—An umbrella review based on updated criteria. *World Psychiatry*, 22(2), 286–304. <https://doi.org/10.1002/wps.21104>
- Leichsenring, F., & Leweke, F. (2017). Social anxiety disorder. *N Engl J Med*, 376(23), 2255–2264.
- Lely, J. C. G., Smid, G. E., Jongedijk, R. A., et al. (2019). The effectiveness of narrative exposure therapy: A review, meta-analysis and meta-regression analysis. *Eur J Psychotraumatol*, 10(1), 1550344. <https://doi.org/10.1080/20008198.2018.1550344>
- LeMoult, J., & Gotlib, I. H. (2019). Depression: A cognitive perspective. *Clin Psychol Rev*, 69, 51–66.
- Leng, M., Zhao, Y., & Wang, Z. (2020). Comparative efficacy of non-pharmacological interventions on agitation in people with dementia: A systematic review and Bayesian network meta-analysis. *Int J Nurs Stud*, 102, 103489.
- Lennox, B. R. (2022). Challenging the psychiatry-neurology divide: The case of autoimmune encephalitis. *Nat Rev Neurol*, 18(5), 253–254. <https://doi.org/10.1038/s41582-022-00636-7>
- Leonhard, K., Korff, I., & Schultz, H. (1962). Die Temperamente und den Familien der monophasischen und biphasischen Psychosen. *Psychiatrie und Neurologie*, 143, 416–434.
- Lester, G., Wilson, F. B., Griffin, L., & Mullen, P. E. (2004). Unusually persistent complainants. *BJPsych*, 184(4), 352–356.
- Leucht, S., Cipriani, A., Spinelli, L., et al. (2013). Comparative efficacy and tolerability of 15 antipsychotic drugs in schizophrenia: A multiple-treatments meta-analysis. *The Lancet*, 382(9896), 951–962.
- Leucht, S., Crippa, A., Sifakis, S., et al. (2020). Dose-response meta-analysis of antipsychotic drugs for acute schizophrenia. *Am J Psychiatry*, 177(4), 342–353. <https://doi.org/10.1176/appi.ajp.2019.19010034>
- Leucht, S., Helfer, B., Gartlehner, G., & Davis, J. M. (2015). How effective are common medications: A perspective based on meta-analyses of major drugs. *BMC Med*, 13, 253. <https://doi.org/10.1186/s12916-015-0494-1>
- Leucht, S., Hierl, S., Kissling, W., et al. (2012). Putting the efficacy of psychiatric and general medicine medication into perspective: Review of meta-analyses. *BJPsych*, 200(2), 97–106.
- Leung, A. K. C., & Hon, K. L. (2019). Pica: A common condition that is commonly missed—An update review. *Curr Pediatr Rev*, 15(3), 164–169. <https://doi.org/10.2174/1573396315666190313163530>
- Levkovitz, Y., Tedeschini, E., & Papakostas, G. I. (2011). Efficacy of antidepressants for dysthymia: A meta-analysis of placebo-controlled randomized trials. *J Clin Psychiatry*, 72(4), S964.
- Levy, K. N., Draijer, N., Kivity, Y., et al. (2019). Transference-focused psychotherapy (TFP). *Curr Treat Options Psychiatry*, 6(4), 312–324. <https://doi.org/10.1007/s40501-019-00193-9>

- Levy, K. N., & Johnson, B. N. (2019). Attachment and psychotherapy: Implications from empirical research. *Can Psychol*, 60(3), 178–193.
- Lewis, A. (1953). Health as a social concept. *Br J Sociol*, 4(2), 109–124.
- Lewis, A. (1970). Paranoia and paranoid: A historical perspective. *Psychol Med*, 1(1), 2–12.
- Lewis, A. J. (1934). Melancholia: A clinical survey of depressive states. *J Ment Sci*, 80(329), 277–378.
- Lewis, G., & Appleby, L. (1988). Personality disorder: The patients psychiatrists dislike. *BJPsych*, 153, 44–49. <https://doi.org/10.1192/bjp.153.1.44>
- Lewis, G., Ioannidis, K., van Harmelen, A. L., et al. (2018). The association between pubertal status and depressive symptoms and diagnoses in adolescent females: A population-based cohort study. *PLoS ONE*, 13(6), e0198804. <https://doi.org/10.1371/journal.pone.0198804>
- Lewis, G., Marston, L., Duffy, L., et al. (2021). Maintenance or discontinuation of antidepressants in primary care. *N Engl J Med*, 385(14), 1257–1267. <https://doi.org/10.1056/NEJMoa2106356>
- Lezak, M., Howieson, D. B., Bigler, E. D., & Tranel, D. (Eds.). (2012). *Neuropsychological Assessment*, 5th edn. Oxford University Press.
- Lichtenstein, P., Yip, B. H., Björk, C., et al. (2009). Common genetic determinants of schizophrenia and bipolar disorder in Swedish families: A population-based study. *Lancet*, 373(9659), 234–239.
- Liddle, P., Friston, K., Frith, C., et al. (1992). Patterns of cerebral blood flow in schizophrenia. *BJPsych*, 160(2), 179–186.
- Liddle, P. F. (1987). The symptoms of chronic schizophrenia: A re-examination of the positive-negative dichotomy. *BJPsych*, 151(2), 145–151.
- Liddle, P. F. (2013). Tardive dyskinesia in schizophrenia. *BJPsych*, 203, 6–7.
- Lieb, K., Völm, B., Rücker, G., et al. (2010). Pharmacotherapy for borderline personality disorder: Cochrane systematic review of randomised trials. *BJPsych*, 196(1), 4–12. <https://doi.org/10.1192/bjp.bp.108.062984>
- Liebling, A., Maruna, S., & McAra, L. (Eds.). (2017). *The Oxford Handbook of Criminology*, 6th edn. Oxford University Press
- Lievesley, K., Rimes, K. A., & Chalder, T. (2014). A review of the predisposing, precipitating and perpetuating factors in chronic fatigue syndrome in children and adolescents. *Clin Psychol Rev*, 34(3), 233–248. <https://doi.org/10.1016/j.cpr.2014.02.002>
- Lightflower, C., Orr, D., & Waswani, N. (2014). *Youth Justice in Scotland. Fixed in the Past or Fit for the Future?* CYCJ. <http://www.cycj.org.uk/wp-content/uploads/2014/09/Youth-Justice-in-Scotland.pdf>
- Lilford, P., & Hughes, J. C. (2020). Epidemiology and mental illness in old age. *BJPsych Adv*, 26(2), 92–103.
- Lim, J. S., Buckley, N. A., Chitty, K. M., et al. (2021). Association between means restriction of poison and method-specific suicide rates: A systematic review. *JAMA Health Forum*, 2(10), e213042–e213042. <https://doi.org/10.1001/jamahealthforum.2021.3042>
- Lindemann, E. (1944). Symptomatology and management of acute grief. *Am J Psychiatry*, 101, 141–148. <https://doi.org/10.1176/ajp.101.2.141>
- Lindert, J., Carta, M. G., Schäfer, I., & Mollica, R. F. (2016). Refugees mental health—A public mental health challenge. *Eur J Public Health*, 26(3), 374–375. <https://doi.org/10.1093/eurpub/ckw010>
- Linehan, M. M. (2018). *Cognitive-Behavioral Treatment of Borderline Personality Disorder*. Guilford Publications.
- Linehan, M. M., Tutek, D. A., Heard, H. L., & Armstrong, H. E. (1994). Interpersonal outcome of cognitive behavioral treatment for chronically suicidal borderline patients. *Am J Psychiatry*, 151(12), 1771–1776. <https://doi.org/10.1176/ajp.151.12.1771>
- Ling, S., Ceban, F., Lui, L. M., et al. (2022). Molecular mechanisms of psilocybin and implications for the treatment of depression. *CNS Drugs*, 36(1), 17–30.
- Lingford-Hughes, A. R., Welch, S., Peters, L., & Nutt, D. (2012). BAP updated guidelines: Evidence-based guidelines for the pharmacological management of substance abuse, harmful use, addiction and comorbidity: recommendations from BAP. *J Psychopharmacol*, 26(7), 899–952.
- Linnoila, V. M., & Virkkunen, R. (1992). Aggression, suicidality, and serotonin. *J Clin Psychiatry*, 53 Suppl, 46–51.
- Linschoten, M., Weiner, L., & Avery-Clark, C. (2016). Sensate focus: A critical literature review. *Sexual and Relationship Therapy*, 31, 230–247.
- Linscott, R., & Van Os, J. (2013). An updated and conservative systematic review and meta-analysis of epidemiological evidence on psychotic experiences in children and adults: On the pathway from proneness to persistence to dimensional expression across mental disorders. *Psychol Med*, 43(6), 1133–1149.
- Linton, S. J. (2000). A review of psychological risk factors in back and neck pain. *Spine*, 25(9), 1148–1156.
- Lipsedge, M., & Littlewood, R. (1997). *Aliens and Alienists: Ethnic Minorities and Psychiatry*, 3rd edn. Routledge.
- Lissauer, T., & Carroll, W. (2021). *Illustrated Textbook of Paediatrics E-Book*. Elsevier Health Sciences.
- Littman, L. (2021). Individuals treated for gender dysphoria with medical and/or surgical transition who subsequently detransitioned: A survey of 100 detransitioners. *Arch Sex Behav*, 50(8), 3353–3369.
- Liu, J., Cao, L., Li, H., et al. (2022). Abnormal resting-state functional connectivity in patients with obsessive-compulsive disorder: A systematic review and meta-analysis. *Neurosci Biobehav Rev*, 135, 104574.
- Livingston, G., Huntley, J., Sommerlad, A., et al. (2020). Dementia prevention, intervention, and care: 2020 report of the Lancet Commission. *Lancet*, 396(10248), 413–446.
- Llaneza, P., García-Portilla, M. P., Llaneza-Suárez, D., et al. (2012). Depressive disorders and the menopause transition. *Maturitas*, 71(2), 120–130.
- Lo, M.-T., Hinds, D. A., Tung, J. Y., et al. (2017). Genome-wide analyses for personality traits identify six genomic loci and show correlations with psychiatric disorders. *Nat Genet*, 49(1), 152–156. <https://doi.org/10.1038/ng.3736>
- Lock, J. (2015). An update on evidence-based psychosocial treatments for eating disorders in children and adolescents. *J Child Adolesc Psychol*, 44, 707–721. <https://doi.org/10.1080/15374416.2014.971458>
- Logan, R. W., & McClung, C. A. (2019). Rhythms of life: Circadian disruption and brain disorders across the

- lifespan. *Nat Rev Neurosci*, 20, 49–65. <https://doi.org/10.1038/s41583-018-0088-y>
- Lomax, J., & Meyrick, J. (2022). Systematic review: Effectiveness of psychosocial interventions on wellbeing outcomes for adolescent or adult victim/survivors of recent rape or sexual assault. *J Health Psychol*, 27(2), 305–331. <https://doi.org/10.1177/1359105320950799>
- Long, J. M., & Holtzman, D. M. (2019). Alzheimer disease: An update on pathobiology and treatment strategies. *Cell*, 179(2), 312–339.
- Loranger, A., Janca, A., & Sartorius, N. (1997). *Assessment and Diagnosis of Personality Disorders: The ICD-10 International Personality Disorder Examination (IPDE)*. Cambridge University Press.
- Loranger, A. W., Oldham, J. M., & Tulis, E. H. (1982). Familial transmission of DSM-III borderline personality disorder. *Arch Gen Psychiatry*, 39(7), 795–799. <https://doi.org/10.1001/archpsyc.1982.04290070031007>
- Lorey, K., & Fegert, J. M. (2022). Incorporating mental health literacy and trauma-informed law enforcement: A participative survey on police officers' attitudes and knowledge concerning mental disorders, traumatization, and trauma sensitivity. *Psychol Trauma*, 14(2), 218–228. <https://doi.org/10.1037/tra0001067>
- Lovestone, S. (2009). Other disorders of the nervous system. In A. David, S. Fleminger, & M. Kopelman (Eds.), *Lishman's Organic Psychiatry*, 4th edn, (pp. 845–905). Wiley-Blackwell.
- Ludvigsson, J. F., Adolfsson, J., Höistad, M., et al. (2023). A Systematic Review of Hormone Treatment for Children with Gender Dysphoria and Recommendations for Research. *Acta Paediatrica*.
- Lukmanji, S., Manji, S. A., Kadhim, S., et al. (2019). The co-occurrence of epilepsy and autism: A systematic review. *Epilepsy Behav*, 98, 238–248.
- Lund, C., Tomlinson, M., & Patel, V. (2016). Integration of mental health into primary care in low- and middle-income countries: The PRIME mental healthcare plans. *BJPsych*, 208(s56), s1–s3. <https://doi.org/10.1192/bjp.bp.114.153668>
- Luppa, M., Sikorski, C., Luck, T., et al. (2012). Age-and gender-specific prevalence of depression in latest-life-systematic review and meta-analysis. *J Affect Disord*, 136(3), 212–221.
- Lustyk, M. K. B., Gerrish, W. G., Shaver, S., & Keys, S. L. (2009). Cognitive-behavioral therapy for premenstrual syndrome and premenstrual dysphoric disorder: A systematic review. *Arch Womens Ment Health*, 12, 85–96.
- Lutters, B., Foley, P., & Koehler, P. J. (2018). The centennial lesson of encephalitis lethargica. *Neurology*, 90(12), 563–567.
- Luty, J. (2014). Drug and alcohol addiction: New challenges. *Adv Psychiatr Treat*, 20(6), 413–421.
- Luty, J. (2015). Drug and alcohol addiction: Do psychosocial treatments work? *BJPsych Adv*, 21(2), 132–143.
- Luxton, R., & Kyriakopoulos, M. (2022). Depression in children and young people: Identification and management NICE guidelines. *Arch Dis Child Educ Pract Ed*, 107(1), 36–38.
- Lyons, M. J., True, W. R., Eisen, S. A., et al. (1995). Differential heritability of adult and juvenile antisocial traits. *Arch Gen Psychiatry*, 52(11), 906–915. <https://doi.org/10.1001/archpsyc.1995.03950230020005>
- Ma, R., Romano, E., Ashworth, M., et al. (2022). Multimorbidity clusters among people with serious mental illness: A representative primary and secondary data linkage cohort study. *Psychol Med*, 53(10), 4333–4344. <https://doi.org/10.1017/S00332917200109X>
- Ma, Y. (2015). Neuropsychological mechanism underlying anti-depressant effect: A systematic meta-analysis. *Mol Psychiatry*, 20(3), 311–319. <https://doi.org/10.1038/mp.2014.24>
- MacLean, J. R., & Wetherall, K. (2021). The association between HIV-stigma and depressive symptoms among people living with HIV/AIDS: A systematic review of studies conducted in South Africa. *J Affect Disord*, 287, 125–137. <https://doi.org/10.1016/j.jad.2021.03.027>
- Maercker, A., Cloitre, M., Bachem, R., et al. (2022). Complex post-traumatic stress disorder. *Lancet*, 400(10345), 60–72. [https://doi.org/10.1016/s0140-6736\(22\)00821-2](https://doi.org/10.1016/s0140-6736(22)00821-2)
- Magnusson, M., Ernberg, E., Landström, S., & Akehurst, L. (2020). Forensic interviewers' experiences of interviewing children of different ages. *Psychol Crime Law*, 26(10), 967–989.
- Maiden, N., Hurst, N., Lochhead, A., et al. (2003). Quantifying the burden of emotional ill-health amongst patients referred to a specialist rheumatology service. *Rheumatol*, 42(6), 750–757.
- Maj, M. (2005). 'Psychiatric comorbidity': An artefact of current diagnostic systems? *BJPsych*, 186(3), 182–184.
- Malhi, G. S., Bassett, D., Boyce, P., et al. (2015). Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for mood disorders. *Aust N Z J Psychiatry*, 49(12), 1087–1206. <https://doi.org/10.1177/0004867415617657>
- Malt, U. (2009). Psychiatric aspects of accidents, burns and other trauma. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1105–1113). Oxford University Press.
- Manconi, M., Garcia-Borreguero, D., Schormair, B., et al. (2021). Restless legs syndrome. *Nat Rev Dis Primers*, 7, 80. <https://doi.org/10.1038/s41572-021-00311-z>
- Mann, J. J., & Rizk, M. M. (2020). A brain-centric model of suicidal behavior. *Am J Psychiatry*, 177(10), 902–916.
- March, J. S., & Vitiello, B. (2009). Clinical messages from the Treatment for Adolescents with Depression Study (TADS). *Am J Psychiatry*, 166(10), 1118–1123.
- Marchand, E. (2021). Psychological and behavioral treatment of female orgasmic disorder. *Sex Med Rev*, 9, 194–211. <https://doi.org/10.1016/j.xsmr.2020.07.007>
- Marcks, B. A., Weisberg, R. B., Dyck, I., & Keller, M. B. (2011). Longitudinal course of obsessive-compulsive disorder in patients with anxiety disorders: A 15-year prospective follow-up study. *Compr Psychiatry*, 52(6), 670–677.
- Marder, S. R., & Cannon, T. D. (2019). Schizophrenia. *N Engl J Med*, 381(18), 1753–1761. <https://doi.org/10.1056/NEJMra1808803>
- Marder, S. R., & Galderisi, S. (2017). The current conceptualization of negative symptoms in schizophrenia. *World Psychiatry*, 16(1), 14–24. <https://doi.org/10.1002/wps.20385>
- Margoshes, B., & Webster, B. (2000). Why do occupational injuries have different health outcomes? In T. Mayer, R. Gatchel, & P. Polatin, (Eds.), *Occupational Musculoskeletal Disorders: Function, Outcomes, and Evidence* (pp. 47–61). LW&W.
- Marincowitz, C., Lochner, C., & Stein, D. J. (2021). The neurobiology of obsessive-compulsive personality disorder: A systematic review. *CNS Spectr*, 27(6), 664–675.
- Markowitsch, H. J., & Staniloiu, A. (2012). Amnesic disorders. *Lancet*, 380(9851), 1429–1440.

- Markowitz, J. C., & Weissman, M. M. (2012). Interpersonal psychotherapy: Past, present and future. *Clin Psychol Psychother*, 19(2), 99–105. <https://doi.org/10.1002/cpp.1774>
- Marmot, M. (2020). Health equity in England. The Marmot Review ten years on. *BMJ*, 368, m693.
- Marneros, A. (2020). Delusional disorders. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press.
- Marneros, A., Pillmann, F., & Wustmann, T. (2012). Delusional disorders—are they simply paranoid schizophrenia? *Schizophr Bull*, 38(3), 561–568.
- Marra, C. M. (2025). Neurosyphilis. *Brain*, 148, 4222–4240.
- Marshall, C. (2022). The Inter-Agency Standing Committee (IASC) guidelines on Mental Health and Psychosocial Support (MHPSS) in emergency settings: A critique. *Int Rev Psychiatry*, 34(6), 604–612.
- Marshall, M. (2003). Acute psychiatric day hospitals. *BMJ*, 327(7407), 116–117. <https://doi.org/10.1136/bmj.327.7407.116>
- Marshall, M., Crowther, R., Almaraz-Serrano, A. M., & Tyrer, P. (2001). Day hospital versus out-patient care for psychiatric disorders. *Cochrane Database Syst Rev*, 2001(3), CD003240.
- Marshall, M., Crowther, R., Sledge, W. H., et al. (2011). Day hospital versus admission for acute psychiatric disorders. *Cochrane Database Syst Rev*, 2011(12), CD004026.
- Marshall, M., & Lockwood, A. (1998). Assertive community treatment for people with severe mental disorders. *Cochrane Database Syst Rev*, 2000(2), CD001089.
- Marteau, T. M., Oliver, A., & Ashcroft, R. E. (2008). Changing behaviour through state intervention. *BMJ*, 337, a2543. [doi:10.1136/bmj.a2543](https://doi.org/10.1136/bmj.a2543)
- Martin, G., Bergen, H. A., Richardson, A. S., et al. (2004). Correlates of firesetting in a community sample of young adolescents. *Aust N Z J Psychiatry*, 38(3), 148–154. <https://doi.org/10.1080/j.1440-1614.2004.01318.x>
- Martinho, F. P., Duarte, G. S., & do Couto, F. S. (2020). Efficacy, effect on mood symptoms, and safety of deep brain stimulation in refractory obsessive-compulsive disorder: A systematic review and meta-analysis. *J Clin Psychiatry*, 81(3), 4067.
- Masters W. H. & Johnson V. E. (1970). *Human Sexual Inadequacy*. Churchill, London.
- Matson, L. M., Adler, A. B., Quartana, P. J., et al. (2022). Management of acute stress reactions in the military: A stepped care approach. *Curr Psychiatry Rep*, 24(12), 799–808. <https://doi.org/10.1007/s11920-022-01388-3>
- Matthews, K., & Matthews, C. (2019). Neurosurgery for mental disorders. In I. Ferrier & J. Waite (Eds.), *The ECT Handbook* (pp. 140–148). Cambridge University Press.
- Matthews, P. M., & Hampshire, A. (2016). Clinical concepts emerging from fMRI functional connectomics. *Neuron*, 91(3), 511–528.
- Maudsley, H. (1879). *The Pathology of the Mind*. Macmillan.
- Maurer, K., Volk, S., & Gerbaldo, H. (1997). Auguste D and Alzheimer's disease. *Lancet*, 349(9064), 1546–1549.
- Mavridis, D., Giannatsi, M., Cipriani, A., & Salanti, G. (2015). A primer on network meta-analysis with emphasis on mental health. *Evid Based Ment Health*, 18(2), 40.
- Maynard, B. R., Heyne, D., Brendel, K. E., et al. (2015). Treatment for school refusal among children and adolescents: A systematic review and meta-analysis. *Res Soc Work Pract*, 28(1), 1049731515598619.
- Mayou, R., & Farmer, A. (2002). Functional somatic symptoms and syndromes. *BMJ*, 325(7358), 265–268.
- Mayou, R. A., Ehlers, A., & Hobbs, M. (2000). Psychological debriefing for road traffic accident victims. Three-year follow-up of a randomised controlled trial. *BJPsych*, 176, 589–593. <https://doi.org/10.1192/bjp.176.6.589>
- Mazza, M. G., Rossetti, A., Crespi, G., & Clerici, M. (2020). Prevalence of co-occurring psychiatric disorders in adults and adolescents with intellectual disability: A systematic review and meta-analysis. *J Appl Res Intellect Disabil*, 33(2), 126–138.
- McAllister-Williams, R. H., Baldwin, D. S., Cantwell, R., et al. (2017). British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. *J Psychopharmacol*, 31(5), 519–552.
- McBride, O., Moon, G., Canaway, A., et al. (2020). Use of community treatment orders and their outcomes: An observational study. *Health Soc Care Deliv Res*, 8(9), hsd08090.
- McCartney, M., Nevitt, S., Lloyd, A., et al. (2021). Mindfulness-based cognitive therapy for prevention and time to depressive relapse: Systematic review and network meta-analysis. *Acta Psychiatr Scand*, 143(1), 6–21. <https://doi.org/10.1111/acps.13242>
- McCart, A. T., Hellinga, L. A., & Kirley, B. B. (2010). The effects of minimum legal drinking age 21 laws on alcohol-related driving in the United States. *J Safety Res*, 41(2), 173–181.
- McCaul, M. E., Roach, D., Hasin, D. S., et al. (2019). Alcohol and women: A brief overview. *Alcohol Clin Exp Res*, 43(5), 774–779.
- McClellan, J., & Stock, S. (2013). Practice parameter for the assessment and treatment of children and adolescents with schizophrenia. *J Am Acad Child Adolesc Psychiatry*, 52(9), 976–990. <https://doi.org/http://dx.doi.org/10.1016/j.jaac.2013.02.008>
- McClure, M. M., Graff, F., Triebwasser, J., et al. (2019). Guanfacine augmentation of a combined intervention of computerized cognitive remediation therapy and social skills training for schizotypal personality disorder. *Am J Psychiatry*, 176(4), 307–314. <https://doi.org/10.1176/appi.ajp.2018.18030349>
- McColgan, P., & Tabrizi, S. J. (2018). Huntington's disease: A clinical review. *Eur J Neurol*, 25(1), 24–34.
- McCool, M. E., Zuelke, A., Theurich, M. A., et al. (2016). Prevalence of female sexual dysfunction among premenopausal women: A systematic review and meta-analysis of observational studies. *Sex Med Rev*, 4(3), 197–212.
- McCrory, E., De Brito, S. A., & Viding, E. (2012). The link between child abuse and psychopathology: A review of neurobiological and genetic research. *J Royal Soc Med*, 105(4), 151–156. <https://doi.org/10.1258/jrsm.2011.110222>
- McCutcheon, R., Beck, K., Bloomfield, M. A., et al. (2015). Treatment resistant or resistant to treatment? Antipsychotic plasma levels in patients with poorly controlled psychotic symptoms. *J Psychopharmacol*, 29(8), 892–897.
- McCutcheon, R. A., Krystal, J. H., & Howes, O. D. (2020). Dopamine and glutamate in schizophrenia: Biology, symptoms and treatment. *World Psychiatry*, 19(1), 15–33. <https://doi.org/10.1002/wps.20693>
- McElroy, S. L. (2020). Management and treatment of feeding and eating disorders. In J.R. Geddes, N. Andreasen, & G. Goodwin

- (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1087–1095). Oxford University Press.
- McGilloway, A., Hall, R. E., Lee, T., & Bhui, K. S. (2010). A systematic review of personality disorder, race and ethnicity: Prevalence, aetiology and treatment. *BMC Psychiatry*, 10, 33. <https://doi.org/10.1186/1471-244X-10-33>
- McGrath, J., Saha, S., Chant, D., & Welham, J. (2008). Schizophrenia: A concise overview of incidence, prevalence, and mortality. *Epidemiol Rev*, 30(1), 67–76.
- McGuffin, P., & Thapar, A. (1992). The genetics of personality disorder. *Br J Psychiatry*, 160(1), 12–23.
- McGuire, J. (2008). A review of effective interventions for reducing aggression and violence. *Philos Trans R Soc Lond B Biol Sci*, 363(1503), 2577–2597. <https://doi.org/10.1098/rstb.2008.0035>
- McGuire, J. F., Piacentini, J., Lewin, A. B., et al. (2015). A meta-analysis of cognitive behavior therapy and medication for child obsessive-compulsive disorder: Moderators of treatment efficacy, response, and remission. *Depress Anxiety*, 32(8), 580–593. <https://doi.org/10.1002/da.22389>
- McIntyre, L. L., Kunze, M., Barton, H., & Luehring, M. (2021). Early intervention for children with intellectual and developmental disabilities. In L. M. Glidden, L. Abbeduto, L. L. McIntyre, & M. J. Tassé (Eds.), *APA Handbook of Intellectual and Developmental Disabilities: Clinical and Educational Implications: Prevention, Intervention, and Treatment* (pp. 3–22). American Psychological Association.
- McIntyre, R. S., Berk, M., Brietzke, E., et al. (2020). Bipolar disorders. *Lancet*, 396(10265), 1841–1856.
- McIntyre, R. S., Rosenblat, J. D., Nemeroff, C. B., et al. (2021). Synthesizing the evidence for ketamine and esketamine in treatment-resistant depression: An international expert opinion on the available evidence and implementation. *Am J Psychiatry*, 178(5), 383–399.
- McKeith, I. G., Boeve, B. F., Dickson, D. W., et al. (2017). Diagnosis and management of dementia with Lewy bodies: Fourth consensus report of the DLB Consortium. *Neurology*, 89(1), 88–100.
- McKenna, P. (1984). Disorders with overvalued ideas. *BJPsych*, 145, 579–585.
- McLaughlin, J., & Young, T. (2022). Neurology in the psychiatric patient: How to think about differentials in altered mental status and diagnoses not to miss. *BJPsych Adv*, 28(4), 226–238.
- McLean, C. P., Levy, H. C., Miller, M. L., & Tolin, D. F. (2022). Exposure therapy for PTSD: A meta-analysis. *Clin Psychol Rev*, 91, 102115. <https://doi.org/10.1016/j.cpr.2021.102115>
- McMahon, F. J. (2022). Putting genetics to work in the psychiatric clinic. *Am J Psychiatry*, 179(3), 182–188. <https://doi.org/10.1176/appi.ajp.2021.22010024>
- McManus, S., Bebbington, P. E., Jenkins, R., & Brugha, T. (2016). *Mental Health and Wellbeing in England: The Adult Psychiatric Morbidity Survey 2014*. NHS Digital.
- McManus, S., Meltzer, H., Brugha, T., et al. (2009). *Adult Psychiatric Morbidity in England: Results of a Household Survey*. Health and Social Care Information Centre.
- McNally, M. R., & Fremouw, W. J. (2014). Examining risk of escalation: A critical review of the exhibitionistic behavior literature. *Aggression and Violent Behavior*, 19, 474–485.
- McShane, R., Westby, M. J., Roberts, E., et al. (2019). Memantine for dementia. *Cochrane Database Syst Rev*, 3(3): CD003154.
- Meadow, R. (1985). Management of Munchausen syndrome by proxy. *Arch Dis Child*, 60(4), 385–393. <https://doi.org/10.1136/adc.60.4.385>
- Mechanic, D. (1978). *Medical Sociology*, 2nd edn. Free Press.
- Mefford, H. C., Batshaw, M. L., & Hoffman, E. P. (2012). Genomics, intellectual disability, and autism. *N Engl J Med*, 366(8), 733–743.
- Mehlum, L., Friis, S., Irion, T., et al. (1991). Personality disorders 2–5 years after treatment: A prospective follow-up study. *Acta Psychiatr Scand*, 84(1), 72–77. <https://doi.org/10.1111/j.1600-0447.1991.tb01423.x>
- Mehta, N., & Vannozzi, R. (2017). Lithium-induced electrocardiographic changes: A complete review. *Clin Cardiol*, 40(12), 1363–1367.
- Mehta, S., Jamieson, T., & Ackery, A. D. (2019). Helping clinicians and patients navigate electronic patient portals: Ethical and legal principles. *CMAJ*, 191(40), E1100–e1104. <https://doi.org/10.1503/cmaj.190413>
- Meichenbaum, D. (1977). *Cognitive-Behavior Modification: An Integrative Approach*. Plenum Press.
- Meier, S. M., & Deckert, J. (2019). Genetics of anxiety disorders. *Curr Psychiatry Rep*, 21, 1–8.
- Meijer, A., Conradi, H. J., Bos, E. H., et al. (2011). Prognostic association of depression following myocardial infarction with mortality and cardiovascular events: A meta-analysis of 25 years of research. *Gen Hosp Psychiatry*, 33(3), 203.
- Mellado-Calvo, N., & Fleminger, S. (2009). Cerebral tumours. In A. David, S. Fleminger, & M. Kopelman (Eds.), *Lishman's Organic Psychiatry*, 4th edn, (pp. 281–308). Wiley-Blackwell.
- Menke, A. (2019). Is the HPA axis as target for depression outdated, or is there a new hope? *Front Psychiatry*, 10, 101.
- Merikangas A. K., & Merikangas, K. R. (2020). Genetic epidemiology of depression in the molecular era. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 789–796). Oxford University Press.
- Merikangas, K. R., He, J. P., Burstein, M., et al. (2010). Lifetime prevalence of mental disorders in U.S. adolescents: results from the National Comorbidity Survey Replication—Adolescent Supplement (NCS-A). *J Am Acad Child Adolesc Psychiatry*, 49(10), 980–989. <https://doi.org/10.1016/j.jaac.2010.05.017>
- Mesholam-Gately, R. I., Giuliano, A. J., Goff, K. P., et al. (2009). Neurocognition in first-episode schizophrenia: A meta-analytic review. *Neuropsychol*, 23(3), 315.
- Mesholam-Gately, R. I., Varca, N., Spitzer, C., et al. (2021). Are we ready for a name change for schizophrenia? A survey of multiple stakeholders. *Schizophr Res*, 238, 152–160. <https://doi.org/10.1016/j.schres.2021.08.034>
- Mesulam, M.-M. (1998). From sensation to cognition. *Brain*, 121(6), 1013–1052.
- Metzl, J. M., & Hansen, H. (2014). Structural competency: Theorizing a new medical engagement with stigma and inequality. *Soc Sci Med*, 103, 126–133. <https://doi.org/10.1016/j.socscimed.2013.06.032>
- Mews, A., Di Bella, L., & Purver, M. (2017). *Impact Evaluation of the Prison-based Core Sex Offender Treatment Programme*. Ministry of Justice.

- Meyer, I. H. (2013). Prejudice, social stress, and mental health in lesbian, gay, and bisexual populations: Conceptual issues and research evidence. *Psychol Bull*, 129(5), 674–697.
- Meyer, N., Faulkner, S. M., McCutcheon, R. A., et al. (2020). Sleep and circadian rhythm disturbance in remitted schizophrenia and bipolar disorder: A systematic review and meta-analysis. *Schizophr Bull*, 46(5), 1126–1143.
- Meyer, N., Harvey, A. G., Lockley, S. W., Dijk, D. J. (2022). Circadian rhythms and disorders of the timing of sleep. *Lancet* 400, 1061–1078.
- Mezey, G., & Robbins, I. (2009). The impact of criminal victimization. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1984–1990). Oxford University Press.
- Michaelis, R., Tang, V., Goldstein, L. H., et al. (2018). Psychological treatments for adults and children with epilepsy: Evidence-based recommendations by the International League Against Epilepsy Psychology Task Force. *Epilepsia*, 59(7), 1282–1302.
- Middleton, H., & Moncrieff, J. (2019). Critical psychiatry: A brief overview. *BJPsych Adv*, 25(1), 47–54. <https://doi.org/10.1192/bja.2018.38>
- Midgley, N., O’Keeffe, S., French, L., & Kennedy, E. (2017). Psychodynamic psychotherapy for children and adolescents: An updated narrative review of the evidence base. *J Child Psychother*, 43(3), 307–329.
- Miklowitz, D. J., Efthimiou, O., Furukawa, T. A., et al. (2021). Adjunctive psychotherapy for bipolar disorder: A systematic review and component network meta-analysis. *JAMA Psychiatry*, 78(2), 141–150.
- Milán-Tomás, Á., Persyko, M., Del Campo, M., et al. (2018). An overview of psychogenic non-epileptic seizures: Etiology, diagnosis and management. *Can J Neurol Sci*, 45(2), 130–136.
- Millard, S. J., Bearden, C. E., Karlsgodt, K. H., & Sharpe, M. J. (2022). The prediction-error hypothesis of schizophrenia: New data point to circuit-specific changes in dopamine activity. *Neuropsychopharmacol*, 47(3), 628–640. <https://doi.org/10.1038/s41386-021-01188-y>
- Miller, J. N., & Black, D. W. (2019). Schizoaffective disorder: A review. *Ann Clin Psychiatry*, 31(1), 47–53.
- Minett, T., Brayne, C., & Stephan, B. (2020). Epidemiology of old age psychiatry: An overview of concepts and main studies. In T. Denning, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 45–76). Oxford University Press.
- Minichino, A., Rutigliano, G., Merlino, S., et al. (2019). Unmet needs in patients with brief psychotic disorders: Too ill for clinical high-risk services and not ill enough for first episode services. *Eur Psychiatry*, 57, 26–32. <https://doi.org/10.1016/j.eurpsy.2018.12.006>
- Ministry of Justice. (2022). *Youth Justice Statistics*. Youth Justice Board for England and Wales. <https://www.gov.uk/government/statistics/youth-justice-statistics-2020-to-2021>
- Minuchin S., Rosman B. & Baker L. (1978). *Psychosomatic Families: Anorexia Nervosa in Context*. Harvard University Press.
- Minzenberg, M. J., Laird, A. R., Thelen, S., et al. (2009). Meta-analysis of 41 functional neuroimaging studies of executive function in schizophrenia. *Arch Gen Psychiatry*, 66(8), 811–822.
- Miresco, M. J., & Kirmayer, L. J. (2006). The persistence of mind–brain dualism in psychiatric reasoning about clinical scenarios. *Am J Psychiatry*, 163(5), 913–918.
- Mirra, S. S., Heyman, A., McKeel, D., et al. (1991). The Consortium to Establish a Registry for Alzheimer’s Disease (CERAD): Part II. Standardization of the neuropathologic assessment of Alzheimer’s disease. *Neurology*, 41(4), 479.
- Mischel, W., Shoda, Y., & Smith, R. (2004). *Introduction to Personality: Toward an Integration*. John Wiley.
- Mitchell, A. J., Bird, V., Rizzo, M., et al. (2014). Accuracy of one or two simple questions to identify alcohol-use disorder in primary care: A meta-analysis. *Br J Gen Pract*, 64(624), e408–e418.
- Mitchell, A. J., Meader, N., & Pentzek, M. (2011). Clinical recognition of dementia and cognitive impairment in primary care: A meta-analysis of physician accuracy. *Acta Psychiatr Scand*, 124(3), 165–183.
- Mitchell, A. J., Vancampfort, D., De Herdt, A., et al. (2013a). Is the prevalence of metabolic syndrome and metabolic abnormalities increased in early schizophrenia? A comparative meta-analysis of first episode, untreated and treated patients. *Schizophr Bull*, 39(2), 295–305.
- Mitchell, J. E., & Peterson, C. B. (2020). Anorexia Nervosa. *N Engl J Med*, 382, 1343–1351. <https://doi.org/10.1056/NEJMcp1803175>
- Mitchell, K. R., Mercer, C. H., Ploubidis, G. B., et al. (2013b). Sexual function in Britain: Findings from the third National Survey of Sexual Attitudes and Lifestyles (Natsal-3). *Lancet*, 382, 1817–1829.
- Mitter, N., Ali, A., & Scior, K. (2019). Stigma experienced by families of individuals with intellectual disabilities and autism: A systematic review. *Res Dev Disabil*, 89, 10–21. <https://doi.org/10.1016/j.ridd.2019.03.001>
- Mizen, L. A. M. (2023). Demystifying genetic jargon in psychiatry. *BJPsych Adv*, 29(2), 103–116. <https://doi.org/10.1192/bja.2021.37>
- Moberg, T., Nordström, P., Forslund, K., et al. (2011). CSF 5-HIAA and exposure to and expression of interpersonal violence in suicide attempters. *J Affect Disord*, 132(1–2), 173–178.
- Moeschler, J. B., Shevell, M., Saul, R. A., et al. (2014). Comprehensive evaluation of the child with intellectual disability or global developmental delays. *Pediatrics*, 134(3), e903–e918.
- Moestrup, L., & Hansen, H. P. (2015). Existential concerns about death: A qualitative study of dying patients in a Danish hospice. *Am J Hosp Palliat Care*, 32(4), 427–436.
- Moffa, A. H., Martin, D., Alonzo, A., et al. (2020). Efficacy and acceptability of transcranial direct current stimulation (tDCS) for major depressive disorder: An individual patient data meta-analysis. *Prog Neuropsychopharmacol Biol Psychiatry*, 99, 109836.
- Moffat, J., Sass, B., McKenzie, K., & Bhui, K. (2009). Improving pathways into mental health care for black and ethnic minority groups: A systematic review of the grey literature. *Int Rev Psychiatry*, 21(5), 439–449. <https://doi.org/10.1080/09540260802204105>
- Moher, D., & et al. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *Ann Int Med*, 151, 264–269.

- Molloy, C., Conroy, R. M., Cotter, D. R., & Cannon, M. (2011). Is traumatic brain injury a risk factor for schizophrenia? A meta-analysis of case-controlled population-based studies. *Schizophr Bull*, 37(6), 1104–1110.
- Monahan, J. (1977). Clinical and actuarial predictions of violence. In D. Faigman, D. Kaye, M. Saks, & J. Sanders (Eds.), *Modern Scientific Evidence: The Law and Science of Expert Testimony*. (pp. 309). West Publishing.
- Monahan, J., Redlich, A. D., Swanson, J., et al. (2005). Use of leverage to improve adherence to psychiatric treatment in the community. *Psychiatr Serv*, 56(1), 37–44. <https://doi.org/10.1176/appi.ps.56.1.37>
- Monje, M., & Iwasaki, A. (2022). The neurobiology of long COVID. *Neuron*, 110, 3484–3496. doi: 10.1016/j.neuron.2022.10.006
- Monroe, S. M., Anderson, S. F., & Harkness, K. L. (2019). Life stress and major depression: The mysteries of recurrences. *Psychol Rev*, 126(6), 791.
- Montero-Marin, J., Garcia-Campayo, J., López-Montoyo, A., et al. (2018). Is cognitive-behavioural therapy more effective than relaxation therapy in the treatment of anxiety disorders? A meta-analysis. *Psychol Med*, 48(9), 1427–1436. <https://doi.org/10.1017/s0033291717003099>
- Montgomery, S., & Asberg, M. (1979). A new depression rating scale designed to be sensitive to change. *BJPsych* 134, 382–389.
- Montgomery, S. A., Nielsen, R. Z., Poulsen, L. H., & Häggström, L. (2014). A randomised, double-blind study in adults with major depressive disorder with an inadequate response to a single course of selective serotonin reuptake inhibitor or serotonin-noradrenaline reuptake inhibitor treatment switched to vortioxetine or agomelatine. *Hum Psychopharmacol*, 29(5), 470–482.
- Moreira, D., Azeredo, A. A. C., & Barbosa, F. (2019). Neurobiological findings of the psychopathic personality in adults: One century of history. *Aggress Viol Behav*, 47, 137–159.
- Morgan, C., Knowles, G., & Hutchinson, G. (2019). Migration, ethnicity and psychoses: Evidence, models and future directions. *World Psychiatry*, 18(3), 247–258. <https://doi.org/10.1002/wps.20655>
- Morgan, C., Lappin, J., Heslin, M., et al. (2014). Reappraising the long-term course and outcome of psychotic disorders: The AESOP-10 study. *Psychol Med*, 44(13), 2713–2726.
- Morgan, M. A., Kelber, M. S., Bellanti, D. M., et al. (2022). Outcomes and prognosis of adjustment disorder in adults: A systematic review. *J Psychiatr Res*, 156, 498–510. <https://doi.org/10.1016/j.jpsychires.2022.10.052>
- Morin, C. M., & Buysse, D. J. (2024). Management of insomnia. *N Engl J Med*, 391, 247–258. <https://doi.org/10.1056/NEJMcp2305655>
- Morley, S., Williams, A., & Eccleston, C. (2013). Examining the evidence about psychological treatments for chronic pain: Time for a paradigm shift? *Pain*, 154(10), 1929–1931.
- Morres, I. D., Hatzigeorgiadis, A., Stathi, A., et al. (2018). Aerobic exercise for adult patients with major depressive disorder in mental health services: A systematic review and meta-analysis. *Depress Anxiety*, 36(1), 39–53. <https://doi.org/10.1002/da.22842>
- Morris, E., O'Donovan, M., Virani, A., & Austin, J. (2022). An ethical analysis of divergent clinical approaches to the application of genetic testing for autism and schizophrenia. *Hum Genet*, 141(5), 1069–1084. <https://doi.org/10.1007/s00439-021-02349-1>
- Morris, K., Savell, K., & Ryan, C. J. (2012). Psychiatrists and termination of pregnancy: Clinical, legal and ethical aspects. *Aust N Z J Psychiatry*, 46(1), 18–27.
- Morrison, A. P., Pyle, M., Gumley, A., et al. (2018). Cognitive behavioural therapy in clozapine-resistant schizophrenia (FOCUS): An assessor-blinded, randomised controlled trial. *Lancet Psychiatry*, 5(8), 633–643. [https://doi.org/10.1016/s2215-0366\(18\)30184-6](https://doi.org/10.1016/s2215-0366(18)30184-6)
- Morrison, A. P., Turkington, D., Pyle, M., et al. (2014). Cognitive therapy for people with schizophrenia spectrum disorders not taking antipsychotic drugs: A single-blind randomised controlled trial. *Lancet*, 383(9926), 1395–1403.
- Morris, R., & Benjamin, B. (2008). Lithium and eGFR: A new routinely available tool for the prevention of chronic kidney disease. *BJPsych*, 193(2), 93–95.
- Mounier, N. M., Abdel-Maged, A. E., Wahdan S. A., Gad, A. M. M., Azab, S. S. (2020). Chemotherapy-induced cognitive impairment (CICI): An overview of etiology and pathogenesis. *Life Sci*, 258.
- Moussavi, S., Chatterji, S., Verdes, E., et al. (2007). Depression, chronic diseases, and decrements in health: Results from the World Health Surveys. *Lancet*, 370(9590), 851–858.
- Mueller, A. S., Abrutyn, S., Pescosolido, B., & Diefendorf, S. (2021). The social roots of suicide: Theorizing how the external social world matters to suicide and suicide prevention. *Front Psychol*, 12, 763.
- Mueser, K. T., Drake, R. E., & Bond, G. R. (2016). Recent advances in supported employment for people with serious mental illness. *Curr Opin Psychiatry*, 29(3), 196–201.
- Mukherjee, S., Sackeim, H. A., & Schnur, D. B. (1994). Electroconvulsive therapy of acute manic episodes: A review of 50 years' experience. *Am J Psychiatry*, 151(2), 169–176.
- Mulder, R. T. (2021). ICD-11 personality disorders: Utility and implications of the new model. *Front Psychiatry*, 12, 655548. <https://doi.org/10.3389/fpsyt.2021.655548>
- Mullen, P., & Maack, L. (1985). Jealousy, pathological jealousy and aggression. In D. P. Farrington & J. Gunn (Eds.), *Aggression and Dangerousness* (pp. 103–126). John Wiley & Sons.
- Mullen, P. E. (2004). The autogenic (self-generated) massacre. *Behav Sci Law*, 22(3), 311–323. <https://doi.org/10.1002/bsl.564>
- Mullen, P. E., Mackenzie, R., Ogloff, J. R., et al. (2006). Assessing and managing the risks in the stalking situation. *J Am Acad Psychiatry Law*, 34(4), 439–450.
- Müller, A., Mitchell, J. E., Sondag, C., & de Zwaan, M. (2013). Psychiatric aspects of bariatric surgery. *Curr Psychiatry Rep*, 15(10), 397. <https://doi.org/10.1007/s11920-013-0397-9>
- Müller-Schmitz, K., Krasavina-Loka, N., Yardimci, T., et al. (2020). Normal pressure hydrocephalus associated with Alzheimer's disease. *Ann Neurol*, 88(4), 703–711.
- Mullins, N., Forstner, A. J., O'Connell, K. S., et al. (2021). Genome-wide association study of more than 40,000 bipolar disorder cases provides new insights into the underlying biology. *Nat Genet*, 53(6), 817–829. <https://doi.org/10.1038/s41588-021-00857-4>
- Muñoz-Negro, J. E., & Cervilla, J. A. (2016). A systematic review on the pharmacological treatment of delusional disorder. *J Clin Psychopharmacol*, 36(6), 684–690.
- Murphy, S., Norbury, R., Godlewska, B., et al. (2013). The effect of the serotonin transporter polymorphism (5-HTTLPR) on

- amygdala function: A meta-analysis. *Mol Psychiatry*, 18(4), 512–520.
- Murray, H. B., Juarascio, A. S., Di Lorenzo, C., Drossman, D. A., & Thomas, J. J. (2019). Diagnosis and Treatment of Rumination Syndrome: A Critical Review. *Am J Gastroenterol*, 114, 562–578. <https://doi.org/10.14309/ajg.0000000000000060>
- Murray, K. E., Davidson, G. R., & Schweitzer, R. D. (2010). Review of refugee mental health interventions following resettlement: Best practices and recommendations. *Am J Orthopsychiatry*, 80(4), 576–585. <https://doi.org/10.1111/j.1939-0025.2010.01062.x>
- Murray, L. K., Dorsey, S., Haroz, E., et al. (2014). A common elements treatment approach for adult mental health problems in low- and middle-income countries. *Cogn Behav Pract*, 21(2), 111–123.
- Murray, S. B., Strober, M., Tadayonnejad, R., Bari, A. A., & Feusner, J. D. (2022). Neurosurgery and neuromodulation for anorexia nervosa in the 21st century: A systematic review of treatment outcomes. *Eat Disord*, 30, 26–53. <https://doi.org/10.1080/10640266.2020.1790270>
- Mutz, J., Vipulanathan, V., Carter, B., et al. (2019). Comparative efficacy and acceptability of non-surgical brain stimulation for the acute treatment of major depressive episodes in adults: Systematic review and network meta-analysis. *BMJ*, 364, 11079.
- Mynors-Wallis, L. M., Gath, D. H., Day, A., & Baker, F. (2000). Randomised controlled trial of problem-solving treatment, antidepressant medication, and combined treatment for major depression in primary care. *BMJ*, 320(7226), 26–30.
- Naem, F., Phiri, P., Rathod, S., & Ayub, M. (2019). Cultural adaptation of cognitive-behavioural therapy. *BJPsych Adv*, 25(6), 387–395. <https://doi.org/10.1192/bja.2019.15>
- Nagata, J. M., Ganson, K. T., & Austin, S. B. (2020). Emerging trends in eating disorders among sexual and gender minorities. *Curr Opin Psychiatry*, 33, 562–567. <https://doi.org/10.1097/YCO.0000000000000645>
- Naghavi, M. (2019). Global, regional, and national burden of suicide mortality 1990 to 2016: Systematic analysis for the Global Burden of Disease Study 2016. *BMJ*, 364, 194. <https://doi.org/10.1136/bmj.194>
- Nair, A., Palmer, E. C., Aleman, A., & David, A. S. (2014). Relationship between cognition, clinical and cognitive insight in psychotic disorders: A review and meta-analysis. *Schizophr Res*, 152(1), 191–200.
- Nalbandian, A., Sehgal, K., Gupta, A., et al. (2021). Post-acute COVID-19 syndrome. *Nat Med*, 27, 601–615. doi: 10.1038/s41591-021-01283-z.
- Narasimha, V. L., Patley, R., Shukla, L., et al. (2019). Phenomenology and course of alcoholic Hallucinoses. *J Dual Diagn*, 15(3), 172–176.
- Nasreddine, Z. S., Phillips, N. A., Bédirian, V., et al. (2005). The Montreal Cognitive Assessment, MoCA: A brief screening tool for mild cognitive impairment. *J Am Geriatr Soc*, 53(4), 695–699. <https://doi.org/10.1111/j.1532-5415.2005.53221.x>
- National Institute for Health and Care Excellence (NICE). (2009). Depression: the treatment and management of depression in adults. Clinical Guideline 90. <https://www.nice.org.uk/guidance/ng222>.
- National Institute for Health and Care Excellence (NICE). (2011a). Alcohol-use disorders: Diagnosis, assessment and management of harmful drinking (high-risk drinking) and alcohol dependence. Clinical guideline CG115. <https://www.nice.org.uk/guidance/cg115>
- National Institute for Health and Care Excellence (NICE). (2014b). Antenatal and postnatal mental health: Clinical management and service guidance. Clinical guideline CG192. <https://www.nice.org.uk/guidance/cg192>
- National Institute for Health and Care Excellence (NICE). (2013b). Autism spectrum disorder in under 19s: Support and management. Clinical guideline CG170. <https://www.nice.org.uk/guidance/cg185>
- National Institute for Health and Care Excellence (NICE). (2017). Eating disorders: Recognition and treatment. NICE guideline NG69. <https://www.nice.org.uk/guidance/ng69>
- National Institute for Health and Care Excellence (NICE). (2011b). Generalised anxiety disorder and panic disorder in adults: Management. Clinical guideline CG113. <https://www.nice.org.uk/guidance/cg113>
- National Institute for Health and Care Excellence (NICE). (2022a). Benzodiazepine and Z drug withdrawal. Health topic. <https://cks.nice.org.uk/topics/benzodiazepine-z-drug-withdrawal/>
- National Institute for Health and Care Excellence (NICE). (2014c). Bipolar disorder: assessment and management. Clinical guideline CG185. <https://www.nice.org.uk/guidance/ng222>
- National Institute for Health and Care Excellence (NICE). (2016). Coexisting severe mental illness and substance misuse: Community health and social care services. NICE Guideline NG58. <https://www.nice.org.uk/guidance/ng58>
- National Institute for Health and Care Excellence (NICE). (2018a). Dementia: Assessment, management and support for people living with dementia and their carers. NICE guideline NG97. <https://www.nice.org.uk/guidance/ng97>
- National Institute for Health and Care Excellence (NICE). (2022b). Depression in adults: Treatment and management. NICE guideline NG222. <https://www.nice.org.uk/guidance/ng222>
- National Institute for Health and Care Excellence (NICE). (2017). Drug misuse prevention: Targeted interventions. NICE guideline NG64. <https://www.nice.org.uk/guidance/ng64>
- National Institute for Health and Care Excellence (NICE). (2007). Drug misuse: Psychosocial interventions. <http://www.nice.org.uk/>
- National Institute for Health and Care Excellence (NICE). (2022c). Epilepsies in children, young people and adults. NICE guideline NG217. <https://www.nice.org.uk/guidance/ng217>
- National Institute for Health and Care Excellence (NICE). (2015). Hypnotics. <http://www.nice.org.uk/>
- National Institute for Health and Care Excellence (NICE). (2018b). Post-traumatic stress disorder. NICE guideline NG116. <https://www.nice.org.uk/guidance/ng116>
- National Institute for Health and Care Excellence (NICE). (2014a). Psychosis and schizophrenia in adults: Prevention and management. Clinical guideline CG178. <https://www.nice.org.uk/guidance/cg178>

- National Institute for Health and Care Excellence (NICE). (2013a). Psychosis and schizophrenia in children and young people: Recognition and management. Clinical guideline CG155. <https://www.nice.org.uk/guidance/cg155>
- National Institute for Health and Care Excellence (NICE). (2025). Scenario: Acute management following an act of self-harm. <https://cks.nice.org.uk/topics/self-harm/management/acute-management-following-an-act-of-self-harm/>
- National Institute for Health and Care Excellence (NICE). (2022d). Self-harm: Assessment, management and preventing recurrence. NICE guideline NG225. <https://www.nice.org.uk/guidance/ng225>
- National Institute for Health and Care Excellence (NICE). (2019). Suicide prevention. Quality standard QS189. <https://www.nice.org.uk/guidance/qs189>
- Neil, L., Viding, E., Armbruster-Genc, D., et al. (2022). Trust and childhood maltreatment: Evidence of bias in appraisal of unfamiliar faces. *J Child Psychol Psychiatry*, 63(6), 655–662. <https://doi.org/10.1111/jcpp.13503>
- Nelson, J. C., Baumann, P., Delucchi, K., et al. (2014). A systematic review and meta-analysis of lithium augmentation of tricyclic and second-generation antidepressants in major depression. *J Affect Disord*, 168, 269–275.
- Nelson, P. T., Dickson, D. W., Trojanowski, J. Q., et al. (2019). Limbic-predominant age-related TDP-43 encephalopathy (LATE): Consensus working group report. *Brain*, 142(6), 1503–1527.
- Nemeroff, C. B. (2020). The state of our understanding of the pathophysiology and optimal treatment of depression: Glass half full or half empty? *Am J Psychiatry*, 177(8), 671–685. <https://doi.org/10.1176/appi.ajp.2020.20060845>
- Nemeroff, C. B., Schatzberg, A. F., Rasgon, N., & Strakowski, S. M. (Eds.). (2022). *The American Psychiatric Association Publishing Textbook of Mood Disorders*, 2nd edn. American Psychiatric Association.
- Németh, G., Laszlovszky, I., Czobor, P., et al. (2017). Cariprazine versus risperidone monotherapy for treatment of predominant negative symptoms in patients with schizophrenia: A randomised, double-blind, controlled trial. *Lancet*, 389(10074), 1103–1113.
- Ness, J., Hawton, K., Bergen, H., et al. (2015). Alcohol use and misuse, self-harm and subsequent mortality: An epidemiological and longitudinal study from the multicentre study of self-harm in England. *Emerg Med J*, 32(10), 793–799. <https://doi.org/10.1136/emered-2013-202753>
- Nesse, R. M. (2023). Evolutionary psychiatry: Foundations, progress and challenges. *World Psychiatry*, 22(2), 177–202. <https://doi.org/10.1002/wps.21072>
- Nestadt, G., & Samuels, J. (2020). Genetics of obsessive-compulsive disorder. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 995–1002). Oxford University Press.
- Neufeld, K. J., Yue, J., Robinson, T. N., et al. (2016). Antipsychotic medication for prevention and treatment of delirium in hospitalized adults: A systematic review and meta-analysis. *J Am Geriatr Soc*, 64(4), 705–714.
- New, A. S., Goodman, M., Triebwasser, J., & Siever, L. J. (2008). Recent advances in the biological study of personality disorders. *Psychiatr Clin North Am*, 31(3), 441–461.
- Newlove-Delgado, T., McManus, S., Sadler, K., et al. (2021). Child mental health in England before and during the Covid-19 lockdown. *Lancet Psychiatry*, 8(5), 353–354.
- Newton-Howes, G., Tyrer, P., Anagnostakis, K., et al. (2010). The prevalence of personality disorder, its comorbidity with mental state disorders, and its clinical significance in community mental health teams. *Soc Psychiatry Psychiatr Epidemiol*, 45(4), 453–460. <https://doi.org/10.1007/s00127-009-0084-7>
- Ng, B., & Atkins, M. (2012). Home assessment in old age psychiatry: A practical guide. *Adv Psychiatr Treat*, 18(6), 400–407.
- NHS. (2018). Smoking, drinking and drug use among young people in England 2018. <https://digital.nhs.uk/data-and-information/publications/statistical/smoking-drinking-and-drug-use-among-young-people-in-england/2018>
- Nicholls, D., & Barrett, E. (2015). Eating disorders in children and adolescents. *BJPsych Advances*, 5, 206–216.
- Nielsen, R., Levander, S., Kjaersdam Telléus, G., et al. (2015). Second-generation antipsychotic effect on cognition in patients with schizophrenia—A meta-analysis of randomized clinical trials. *Acta Psychiatr Scand*, 131(3), 185–196.
- Nielsen, T. R., Vogel, A., Riepe, M. W., et al. (2011). Assessment of dementia in ethnic minority patients in Europe: A European Alzheimer's Disease Consortium survey. *Int Psychogeriatr*, 23(1), 86–95.
- Nierenberg, A., Loft, H., & Olsen, C. (2019). Treatment effects on residual cognitive symptoms among partially or fully remitted patients with major depressive disorder: A randomized, double-blinded, exploratory study with vortioxetine. *J Affect Disord*, 250, 35–42.
- Nigg, J. T., Sibley, M. H., Thapar, A., & Karalunas, S. L. (2020). Development of ADHD: Etiology, heterogeneity, and early life course. *Ann Rev Develop Psychol*, 2(1), 559–583. <https://doi.org/10.1146/annurev-devpsych-060320-093413>
- Nihat, A. (2020). Prion disease. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 414–423). Oxford University Press.
- Nijdam-Jones, A., Rosenfeld, B., Gerbrandt, J., et al. (2018). Psychopathology of stalking offenders: Examining the clinical, demographic, and stalking characteristics of a community-based sample. *Crim Justice Behav*, 45(5), 712–731.
- Nimmo-Smith, V., Heuvelman, H., Dalman, C., et al. (2020). Anxiety disorders in adults with autism spectrum disorder: A population-based study. *J Autism Develop Disord*, 50, 308–318.
- Nischal, A., Tripathi, A., Nischal, A., & Trivedi, J. (2012). Suicide and antidepressants: What current evidence indicates. *Mens Sana Monogr*, 10(1), 33–44.
- Nolen, W. A., Licht, R. W., Young, A. H., et al. (2019). What is the optimal serum level for lithium in the maintenance treatment of bipolar disorder? A systematic review and recommendations from the ISBD/IGSLI Task Force on treatment with lithium. *Bipolar Disord*, 21(5), 394–409.
- Norbury, C. F., & Paul, R. (2015). Disorders of speech, language, and communication. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 683–701). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch52>
- Nord, M., & Farde, L. (2011). Antipsychotic occupancy of dopamine receptors in schizophrenia. *CNS Neurosci Therapeut*, 17(2), 97–103.

- Nordahl, H., & Wells, A. (2017). Testing the metacognitive model against the benchmark CBT model of social anxiety disorder: Is it time to move beyond cognition? *PLoS ONE*, 12(5), e0177109.
- Nordentoft, M., Madsen, T., & Erlangsen, A. (2020). Epidemiology and causes of suicide. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1279–1288). Oxford University Press.
- Novak, B., McDermott, B. E., Scott, C. L., & Guillory, S. (2007). Sex offenders and insanity: An examination of 42 individuals found not guilty by reason of insanity. *J Am Acad Psychiatry Law*, 35(4), 444–450. <https://jaapl.org/content/jaapl/35/4/444.full.pdf>
- Nuechterlein, K. H., Green, M. F., Kern, R. S., et al. (2008). The MATRICS Consensus Cognitive Battery, Part 1: Test selection, reliability, and validity. *Am J Psychiatry*, 165(2), 203–213.
- Nüesch, E., Häuser, W., Bernardy, K., et al. (2013). Comparative efficacy of pharmacological and non-pharmacological interventions in fibromyalgia syndrome: Network meta-analysis. *Ann Rheum Dis*, 72(6), 955–962.
- Núñez, N. A., Joseph, B., Pahwa, M., et al. (2022). Augmentation strategies for treatment resistant major depression: A systematic review and network meta-analysis. *J Affect Disord*, 302, 385–400.
- Nyberg, L., & Pudas, S. (2019). Successful memory aging. *Ann Rev Psychol*, 70, 219–243.
- Oberlander, J. G., & Henderson, L. P. (2012). The Sturm und Drang of anabolic steroid use: Angst, anxiety, and aggression. *Trends Neurosci*, 35(6), 382–392. <https://doi.org/10.1016/j.tins.2012.03.001>
- O'Brien, J. T., Holmes, C., Jones, M., et al. (2017). Clinical practice with anti-dementia drugs: A revised (third) consensus statement from the British Association for Psychopharmacology. *J Psychopharmacol*, 31(2), 147–168.
- O'Brien, J. T., & Thomas, A. (2017). Vascular dementia. *Focus*, 15(1), 101–109.
- O'Brien, S., Rapkin, A., Dennerstein, L., & Nevatte, T. (2011). Diagnosis and management of premenstrual disorders. *BMJ*, 342, d2994.
- Ochoa, S., Usall, J., Cobo, J., et al. (2012). Gender differences in schizophrenia and first-episode psychosis: A comprehensive literature review. *Schizophr Res Treatment*, 2012, 916198. <https://doi.org/10.1155/2012/916198>
- O'Connor, R. C., & Nock, M. K. (2014). The psychology of suicidal behaviour. *Lancet Psychiatry*, 1(1), 73–85. [https://doi.org/10.1016/s2215-0366\(14\)70222-6](https://doi.org/10.1016/s2215-0366(14)70222-6)
- O'Connor, E., Rossom, R. C., Henninger, M., et al. (2016a). Primary care screening for and treatment of depression in pregnant and postpartum women: Evidence report and systematic review for the US Preventive Services Task Force. *JAMA*, 315(4), 388–406.
- O'Connor, T. G., Monk, C., & Burke, A. S. (2016b). Maternal affective illness in the perinatal period and child development: Findings on developmental timing, mechanisms, and intervention. *Curr Psychiatry Rep*, 18(3), 24. <https://doi.org/10.1007/s11920-016-0660-y>
- Ødegaard, Ø. (1932) Emigration and insanity. *Acta Psychiatr Neurolog Scandinav. Supplementum*, 4, 1–206.
- Office for National Statistics (ONS). (2025). *Crime in England and Wales: Year ending September 2024*. ONS. <https://www.ons.gov.uk/peoplepopulationandcommunity/crimeandjustice/bulletins/crimeinenglandandwales/yearendingseptember2024>
- Office for National Statistics (ONS). (2020). *Drug Misuse in England and Wales: Year Ending March 2020*. ONS.
- Office for National Statistics (ONS). (2016). *Measuring National Well-Being: At What Age is Personal Well-Being the Highest?* ONS.
- Office for National Statistics (ONS). (2022). *Suicides in England and Wales: 2021 Registrations*. ONS. <https://www.ons.gov.uk/peoplepopulationandcommunity/birthsdeathsandmarriages/deaths/bulletins/suicidesintheunitedkingdom/2021registrations>
- Office for National Statistics (ONS). (2017). *Suicide by Occupation, England: 2011 to 2015*. ONS.
- O'Grady, J. (2009). The expert witness in the criminal court: Assessment, reports and testimony. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 2003–2009). Oxford University Press.
- Okkels, N., Vernal, D., Jensen, S. O. W., et al. (2013). Changes in the diagnosed incidence of early onset schizophrenia over four decades. *Acta Psychiatr Scand*, 127(1), 62–68.
- Olderoid, N. B., Wennerholm, U. B., Pinborg, A., et al. (2018). The effect of paternal factors on perinatal and paediatric outcomes: a systematic review and meta-analysis. *Hum Reprod Update*, 24(3), 320–389. <https://doi.org/10.1093/humupd/dmy005>
- Olfson, M., Druss, B. G., & Marcus, S. C. (2015). Trends in mental health care among children and adolescents. *N Engl J Med*, 373(11), 1079. <https://doi.org/10.1056/NEJMc1507642>
- Olweus, D. (2013). School bullying: Development and some important challenges. *Ann Rev Clin Psychol*, 9, 751–780.
- Orlovská, S., Vestergaard, C. H., Bech, B. H., et al. (2017). Association of streptococcal throat infection with mental disorders: Testing key aspects of the PANDAS hypothesis in a nationwide study. *JAMA Psychiatry*, 74(7), 740–746.
- Oro, A. B., Navarro-Calvillo, M. E., & Esmer, C. (2014). Autistic Behavior Checklist (ABC) and its applications. In V. B. Patel, V. R. Preedy, & C. R. Martin (Eds.), *Comprehensive Guide to Autism* (pp. 2787–2798). Springer.
- Osborne, V. (2021). Opioids in the United Kingdom: Safety and surveillance during COVID-19. *Curr Opin Psychiatry*, 34(4), 357.
- Osler, M., & Jørgensen, M. B. (2020). Associations of benzodiazepines, Z-drugs, and other anxiolytics with subsequent dementia in patients with affective disorders: A nationwide cohort and nested case-control study. *Am J Psychiatry*, 177(6), 497–505.
- Oslin, D. W., Lynch, K. G., Shih, M.-C., et al. (2022). Effect of pharmacogenomic testing for drug-gene interactions on medication selection and remission of symptoms in major depressive disorder: The PRIME Care randomized clinical trial. *JAMA*, 328(2), 151–161.
- Otasowie, J. (2021). Co-occurring mental disorder and substance use disorder in young people: Aetiology, assessment and treatment. *BJPsych Adv*, 27(4), 272–281.
- Otgaard, H., Howe, M. L., & Patihis, L. (2022). What science tells us about false and repressed memories. *Memory*, 30(1), 16–21. <https://doi.org/10.1080/09658211.2020.1870699>

- Oud, M., de Winter, L., Vermeulen-Smit, E., et al. (2019). Effectiveness of CBT for children and adolescents with depression: A systematic review and meta-regression analysis. *Eur Psychiatry*, 57, 33–45.
- Overall, J. E., & Gorham, D. R. (1962). The brief psychiatric rating scale. *Psychol Rep*, 10(3), 799–812.
- Oyeboode, F. (2022). *Sims' Symptoms in the Mind: Textbook of Descriptive Psychopathology*, 7th edn. Elsevier Health Sciences.
- Oyeboode, F. (2021). *Psychopathology of Rare and Unusual Syndromes*. RCPsych Publications.
- Oyeboode, J., & Parveen, S. (2020). Carers and dementia. In T. Denning, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 851–864). Oxford University Press.
- Paez, A. (2018). The 'architect analogy' of evidence-based practice: Reconsidering the role of clinical expertise and clinician experience in evidence-based health care. *J Evid Based Med*, 11(4), 219–226. <https://doi.org/10.1111/jebm.12321>
- Palmer, B. (2014). *Helping People with Eating Disorders: A Clinical Guide to Assessment and Treatment*, 2nd edn. John Wiley and Sons.
- Palmqvist, S., Tideman, P., Cullen, N., et al. (2021). Prediction of future Alzheimer's disease dementia using plasma phosphotau combined with other accessible measures. *Nat Med*, 27(6), 1034–1042.
- Palmu, R., Suominen, K., Vuola, J., & Isometsä, E. (2011). Mental disorders after burn injury: A prospective study. *Burns*, 37(4), 601–609.
- Palpacuer, C., Laviolle, B., Boussageon, R., et al. (2015). Risks and benefits of nalmefene in the treatment of adult alcohol dependence: A systematic literature review and meta-analysis of published and unpublished double-blind randomized controlled trials. *PLoS MED*, 12(12), e1001924.
- Pandis, C., Agrawal, N., & Poole, N. (2019). Capgras' delusion: A systematic review of 255 published cases. *Psychopathol*, 52(3), 161–173.
- Pardos-Gascón, E. M., Narambuena, L., Leal-Costa, C., & van der Hofstadt-Román, C. J. (2021). Differential efficacy between cognitive-behavioral therapy and mindfulness-based therapies for chronic pain: Systematic review. *Int J Clin Health Psychol*, 21(1), 100197. doi: 10.1016/j.ijchp.2020.08.001
- Pariente, G., Leibson, T., Shulman, T., et al. (2017). Pregnancy outcomes following in utero exposure to lamotrigine: A systematic review and meta-analysis. *CNS Drugs*, 31, 439–450.
- Parker, D. A., Trotti, R. L., McDowell, J. E., et al. (2021). Auditory oddball responses across the schizophrenia–bipolar spectrum and their relationship to cognitive and clinical features. *Am J Psychiatry*, 178(10), 952–964. <https://doi.org/10.1176/appi.ajp.2021.20071043>
- Parkes, C. M., & Prigerson, H. G. (2010). *Bereavement: Studies of Grief in Adult Life*, 4th edn. Penguin.
- Parnas, J., Schulsinger, F., Teasdale, T. W., et al. (1982). Perinatal complications and clinical outcome within the schizophrenia spectrum. *BJPsych*, 140(4), 416–420.
- Parry-Jones, W. (1971). *The Trade in Lunacy: A Study of Private Madhouses in England in the Eighteenth and Nineteenth Centuries*. Routledge & Keegan Paul
- Parsons, T. (1951). *The Social System*. Free Press.
- Pasamanick, B., Scarpitti, F. R., Lefton, M., et al. (1964). Home vs hospital care for schizophrenics. *JAMA*, 187(3), 177–181.
- Patel, A. R., & Hall, B. J. (2021). Beyond the DSM-5 diagnoses: A cross-cultural approach to assessing trauma reactions. *Focus*, 19(2), 197–203. <https://doi.org/10.1176/appi.focus.20200049>
- Patel, K., Bunachita, S., Chiu, H., et al. (2021). Elder abuse: A comprehensive overview and physician-associated challenges. *Cureus J Med Sci*, 13, e14375.
- Patel, V., & Prince, M. (2010). Global mental health: A new global health field comes of age. *JAMA*, 303(19), 1976–1977.
- Patel, V., Weiss, H. A., Chowdhary, N., et al. (2010). Effectiveness of an intervention led by lay health counsellors for depressive and anxiety disorders in primary care in Goa, India (MANAS): A cluster randomised controlled trial. *Lancet*, 376(9758), 2086–2095.
- Pathirathna, M. L., Nandasena, H. M. R. K. G., Atapattu, A. M. M. P., & Weerasekara, I. (2022). Impact of the COVID-19 pandemic on suicidal attempts and death rates: A systematic review. *BMC Psychiatry*, 22(1), 506. <https://doi.org/10.1186/s12888-022-04158-w>
- Paton, C., Anderson, I. M., Cowen, P. J., et al. (2020). Prescribing for moderate or severe unipolar depression in patients under the long-term care of UK adult mental health services. *Ther Adv Psychopharmacol*, 10, 2045125320930492. <https://doi.org/10.1177/2045125320930492>
- Patton, G. C., Sawyer, S. M., Santelli, J. S., et al. (2016). Our future: A Lancet commission on adolescent health and wellbeing. *Lancet*, 387(10036), 2423–2478. [https://doi.org/10.1016/s0140-6736\(16\)00579-1](https://doi.org/10.1016/s0140-6736(16)00579-1)
- Pauley, D., Cuijpers, P., Papola, D., et al. (2023). Two decades of digital interventions for anxiety disorders: A systematic review and meta-analysis of treatment effectiveness. *Psychol Med*, 53(2), 567–579.
- Pavlou, M. P., & Lachs, M. S. (2006). Could self-neglect in older adults be a geriatric syndrome? *J Am Geriatr Soc*, 54(5), 831–842.
- Paykel, E. S. (1978). Contribution of life events to causation of psychiatric illness. *Psychol Med*, 8(2), 245–253.
- Paykel, E. S., Scott, J., Teasdale, J. D., et al. (1999). Prevention of relapse in residual depression by cognitive therapy: A controlled trial. *Arch Gen Psychiatry*, 56(9), 829–835.
- Paz-Otero, M., Becerra-Fernández, A., Pérez-López, G., & Ly-Pen, D. (2021). A 2020 review of mental health comorbidity in gender dysphoric and gender non-conforming people. *J Psychiatry Treat Res*, 3(1), 44–55.
- Pearce, J. B. (2009). Counselling and psychotherapy for children. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1764–1769). Oxford University Press.
- Pearson, C. M., Wonderlich, S. A., & Smith, G. T. (2015). A risk and maintenance model for bulimia nervosa: From impulsive action to compulsive behavior. *Psychol Rev*, 122, 516–535. <https://doi.org/10.1037/a0039268>
- Penke, L., Denissen, J. J. A., & Miller, G. F. (2007). The evolutionary genetics of personality. *Eur J Personal*, 21(5), 549–587. <https://doi.org/10.1002/per.629>
- Penrose, L. S. (1938). *A Clinical and Genetic Study of 1280 Cases of Mental Defect*. Special Report Series No. 229. Medical Research Council.

- Penninx, B. W. J. H., Benros, M.E., Klein, R.S. *et al.* (2022). How COVID-19 shaped mental health: from infection to pandemic effects. *Nat Med*, 28, 2027–2037. <https://doi.org/10.1038/s41591-022-02028-2>
- Perälä, J., Suvisaari, J., Saarni, S. I., *et al.* (2007). Lifetime prevalence of psychotic and bipolar I disorders in a general population. *Arch Gen Psychiatry*, 64(1), 19–28.
- Peralta, V., & Cuesta, M. (2016). Delusional disorder and schizophrenia: A comparative study across multiple domains. *Psychol Med*, 46(13), 2829–2839.
- Percie du Sert, O., Unrau, J., Gauthier, C. J., *et al.* (2023). Cerebral blood flow in schizophrenia: A systematic review and meta-analysis of MRI-based studies. *Prog Neuropsychopharmacol Biol Psychiatry*, 121, 110669. <https://doi.org/10.1016/j.pnpbp.2022.110669>
- Pereira-Sanchez, V., Franco, A. R., Vieira, D., *et al.* (2021). Systematic review: Medication effects on brain intrinsic functional connectivity in patients with attention-deficit/hyperactivity disorder. *J Am Acad Child Adolesc Psychiatry*, 60(2), 222–235.
- Perera, B., Laugharne, R., Henley, W., *et al.* (2020). COVID-19 deaths in people with intellectual disability in the UK and Ireland: Descriptive study. *BJPsych Open*, 6(6), e123. <https://doi.org/10.1192/bjo.2020.102>
- Perlis, M. L., Posner, D., Riemann, D., Bastien, C. H., Teel, J., & Thase, M. (2022). Insomnia. *Lancet*, 400, 1047–1060. [https://doi.org/10.1016/S0140-6736\(22\)00879-0](https://doi.org/10.1016/S0140-6736(22)00879-0)
- Perrotelli, A., Giordano, G. M., Brando, F., *et al.* (2021). EEG-based measures in at-risk mental state and early stages of schizophrenia: A systematic review. *Front Psychiatry*, 12, 653642. <https://doi.org/10.3389/fpsy.2021.653642>
- Perry, A., Gordon-Smith, K., Jones, L., & Jones, I. (2021). Phenomenology, epidemiology and aetiology of postpartum psychosis: A review. *Brain Sci*, 11(1), 47. doi: 10.3390/brainsci11010047
- Perry, D. W., Benton, C., Walsh, M., *et al.* (2002). Mental impairment in the West Midlands: 10 years on. *Med Sci Law*, 42(4), 325–333. <https://doi.org/10.1177/002580240204200408>
- Pertusa, A., Gaston, R. L., & Choudry, A. (2019). Hoarding revisited: There is light at the end of the living room. *BJPsych Adv*, 25(1), 26–36.
- Petersen, L., & Sørensen, T. I. (2011). Studies based on the Danish Adoption Register: Schizophrenia, BMI, smoking, and mortality in perspective. *Scand J Public Health*, 39(7 Suppl), 191–195.
- Peterson, C., Sussell, A., Li, J., *et al.* (2020). Suicide rates by industry and occupation—National Violent Death Reporting System, 32 States, 2016. *Morb Mortal Wkly Rep*, 69(3), 57–62. <https://doi.org/10.15585/mmwr.mm6903a1>
- Peterson, D. L., Webb, C. A., Keeley, J. W., *et al.* (2019). The reliability and clinical utility of ICD-11 schizoaffective disorder: A field trial. *Schizophr Res*, 208, 235–241. <https://doi.org/10.1016/j.schres.2019.02.011>
- Peterson, R. L., & Pennington, B. F. (2015). Developmental dyslexia. *Ann Rev Clin Psychol*, 11, 283–307.
- Petrides, G., Malur, C., Braga, R. J., *et al.* (2015). Electroconvulsive therapy augmentation in clozapine-resistant schizophrenia: A prospective, randomized study. *Am J Psychiatry*, 172(1), 52–58.
- Phelan, M., & Blair, G. (2008). Medical history-taking in psychiatry. *Adv Psychiatr Treat*, 14(3), 229–234.
- Phelan, M., Wykes, T., & Goldman, H. (1994). Global functioning scales: A review. *Soc Psychiatry Psychiatr Epidemiol*, 29, 205–211.
- Phillips, K., Menard, W., Pagano, M., *et al.* (2006). Delusional versus nondelusional body dysmorphic disorder: Clinical features and course of illness. *J Psychiatr Res*, 40(2), 95–104.
- Phillips, K. A. (2004). Body dysmorphic disorder: Recognizing and treating imagined ugliness. *World Psychiatry*, 3(1), 12–17.
- Phillips, M., & Drevets, W. C. (2020). Neuroimaging of bipolar disorder. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. Oxford University Press.
- Pichot, P. (1994). Nosological models in psychiatry. *BJPsych*, 164(2), 232–240.
- Pilling, S., Anderson, I., Goldberg, D., *et al.* (2009). Depression in adults, including those with a chronic physical health problem: Summary of NICE guidance. *BMJ*, 339, b4108. <https://doi.org/10.1136/bmj.b4108>
- Pilling, S., Mayo-Wilson, E., Mavranzeouli, I., *et al.* (2013). Recognition, assessment and treatment of social anxiety disorder: Summary of NICE guidance. *BMJ*, 346, f2541.
- Pilling, S., Yesufu-Udechuku, A., Taylor, C., *et al.* (2011). Diagnosis, assessment, and management of harmful drinking and alcohol dependence: Summary of NICE guidance. *BMJ*, 342, d700. doi: 10.1136/bmj.d700
- Pillinger, T., D'Ambrosio, E., McCutcheon, R., & Howes, O. D. (2019). Is psychosis a multisystem disorder? A meta-review of central nervous system, immune, cardiometabolic, and endocrine alterations in first-episode psychosis and perspective on potential models. *Mol Psychiatry*, 24(6), 776–794. <https://doi.org/10.1038/s41380-018-0058-9>
- Pillinger, T., McCutcheon, R. A., Vano, L., *et al.* (2020). Comparative effects of 18 antipsychotics on metabolic function in patients with schizophrenia, predictors of metabolic dysregulation, and association with psychopathology: A systematic review and network meta-analysis. *Lancet Psychiatry*, 7(1), 64–77. [https://doi.org/10.1016/s2215-0366\(19\)30416-x](https://doi.org/10.1016/s2215-0366(19)30416-x)
- Pine, D. S., & Klein, R. G. (2015). Anxiety disorders. In A. Thapar, D. S. Pine, J. F. Leckman, *et al.* (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 822–840). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch60>
- Pinel, P. (1806). *A Treatise on Insanity: In which are Contained the Principles of a New and More Practical Nosology of Maniacal Disorders Than Has Yet Been Offered to the Public* (Vol. 1806). W. Todd.
- Pinsky, E., Rauch, P. K., & Abrams, A. N. (2015). Pediatric consultation and psychiatric aspects of somatic disease. In A. Thapar, D. S. Pine, J. F. Leckman, *et al.* (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 586–598). John Wiley & Sons, Ltd.
- Pinto, T. C., Machado, L., Bulgacov, T. M., *et al.* (2019). Is the Montreal Cognitive Assessment (MoCA) screening superior to the Mini-Mental State Examination (MMSE) in the detection of mild cognitive impairment (MCI) and Alzheimer's disease (AD) in the elderly? *Int Psychogeriatr*, 31(4), 491–504.
- Piquero, A. R., Jennings, W. G., Diamond, B., *et al.* (2016). A meta-analysis update on the effects of early family/parent training programs on antisocial behavior and delinquency. *J Exp Criminol*, 12, 229–248.

- Pivonello, R., Simeoli, C., De Martino, M. C., et al. (2015). Neuropsychiatric disorders in Cushing's syndrome. *Front Neurosci*, 9, 129. <https://doi.org/10.3389/fnins.2015.00129>
- Pjrek, E., Friedrich, M.-E., Cambioli, L., et al. (2020). The efficacy of light therapy in the treatment of seasonal affective disorder: A meta-analysis of randomized controlled trials. *Psychotherapy and Psychosomatics*, 89(1), 17–24.
- Plack, K., Herpertz, S., & Petrak, F. (2010). Behavioral medicine interventions in diabetes. *Curr Opin Psychiatry*, 23(2), 131–138. <https://doi.org/10.1097/ycp.0b013e3283366555>
- Plante, D. T. (2021). The evolving nexus of sleep and depression. *Am J Psychiatry*, 178(10), 896–902.
- Plomin, R., DeFries, J. C., Knopik, V. S., & Neiderhiser, J. M. (2016). Top 10 replicated findings from behavioral genetics. *Perspect Psychol Sci*, 11(1), 3–23. <https://doi.org/10.1177/1745691615617439>
- Plomin, R., & von Stumm, S. (2022). Polygenic scores: Prediction versus explanation. *Mol Psychiatry*, 27(1), 49–52. <https://doi.org/10.1038/s41380-021-01348-y>
- Polanczyk, G., Moffitt, T. E., Arseneault, L., et al. (2010). Etiological and clinical features of childhood psychotic symptoms: Results from a birth cohort. *Arch Gen Psychiatry*, 67(4), 328–338.
- Polanczyk, G. V., Salum, G. A., Sugaya, L. S., et al. (2015). Annual research review: A meta-analysis of the worldwide prevalence of mental disorders in children and adolescents. *J Child Psychol Psychiatry*, 56(3), 345–365. <https://doi.org/10.1111/jcpp.12381>
- Polihronis, C., Cloutier, P., Kaur, J., et al. (2022). What's the harm in asking? A systematic review and meta-analysis on the risks of asking about suicide-related behaviors and self-harm with quality appraisal. *Arch Suicide Res*, 26(2), 325–347.
- Pompili, M., Lester, D., Dominici, G., et al. (2013). Indications for electroconvulsive treatment in schizophrenia: A systematic review. *Schizophr Res*, 146(1–3), 1–9.
- Pompoli, A., Furukawa, T. A., Efthimiou, O., et al. (2018). Dismantling cognitive-behaviour therapy for panic disorder: A systematic review and component network meta-analysis. *Psychol Med*, 48(12), 1945–1953.
- Ponsford, J., Alway, Y., & Gould, K. R. (2018). Epidemiology and natural history of psychiatric disorders after TBI. *J Neuropsychiatry Clin Neurosci*, 30(4), 262–270.
- Poole, C. (2010). Illness deception and work: Incidence, manifestations and detection. *Occup Med*, 60(2), 127–132.
- Poole, R., & Higgs, R. (2017). *Psychiatric Interviewing and Assessment*. Cambridge University Press.
- Popovic, D., Benabarre, A., Crespo, J., et al. (2014). Risk factors for suicide in schizophrenia: Systematic review and clinical recommendations. *Acta Psychiatr Scand*, 130(6), 418–426.
- Post, F. (1972). The management and nature of depressive illnesses in late life: A follow-through study. *BJPsych*, 121(563), 393–404.
- Potenza, M. N., Balodis, I. M., Derevensky, J., et al. (2019). Gambling disorder. *Nat Rev Dis Primers*, 5(1), 51. <https://doi.org/10.1038/s41572-019-0099-7>
- Poulton, R., Moffitt, T. E., & Silva, P. A. (2015). The Dunedin Multidisciplinary Health and Development Study: Overview of the first 40 years, with an eye to the future. *Soc Psychiatry Psychiatr Epidemiol*, 50(5), 679–693. <https://doi.org/10.1007/s00127-015-1048-8>
- Pourmand, A., Robertson, J., Caggiula, A., et al. (2019). Social media and suicide: A review of technology-based epidemiology and risk assessment. *Telemed J E Health*, 25(10), 880–888. <https://doi.org/10.1089/tmj.2018.0203>
- Premraj, L., Kannapadi, N. V., Briggs, J., et al. (2022). Mid- and long-term neurological and neuropsychiatric manifestations of post-COVID-19 syndrome: A meta-analysis. *J Neurol Sci*, 434, 120162. <https://doi.org/10.1016/j.jns.2022.120162>
- Price, J. L., & Drevets, W. C. (2012). Neural circuits underlying the pathophysiology of mood disorders. *Trends Cogn Sci*, 16(1), 61–71.
- Priebe, S., Badesconyi, A., Fioritti, A., et al. (2005). Reinstitutionalisation in mental health care: Comparison of data on service provision from six European countries. *BMJ*, 330(7483), 123–126. <https://doi.org/10.1136/bmj.38296.611215.AE>
- Priebe, S., Bremner, S. A., & Pavlickova, H. (2016). Discontinuing financial incentives for adherence to antipsychotic depot medication: Long-term outcomes of a cluster randomised controlled trial. *BMJ Open*, 6(9), e011673. <https://doi.org/10.1136/bmjopen-2016-011673>
- Priebe, S., Yeeles, K., Bremner, S., et al. (2013). Effectiveness of financial incentives to improve adherence to maintenance treatment with antipsychotics: Cluster randomised controlled trial. *BMJ*, 347, f5847. <https://doi.org/10.1136/bmj.f5847>
- Prince, M., Bryce, R., Albanese, E., et al. (2013). The global prevalence of dementia: a systematic review and metaanalysis. *Alzheimers Dement*, 9(1), 63–75. e62.
- Pringsheim, T., Holler-Managan, Y., Okun, M. S., et al. (2019). Comprehensive systematic review summary: treatment of tics in people with Tourette syndrome and chronic tic disorders. *Neurology*, 92(19), 907–915.
- Prins, J. B., van der Meer, J. W., & Bleijenberg, G. (2006). Chronic fatigue syndrome. *Lancet*, 367(9507), 346–355. [https://doi.org/10.1016/s0140-6736\(06\)68073-2](https://doi.org/10.1016/s0140-6736(06)68073-2)
- Pritchard, C., Parish, M., & Williams, R. J. (2020). International comparison of civilian violent deaths: A public health approach to reduce gun-related deaths in US youth. *Public Health*, 180, 109–113. <https://doi.org/10.1016/j.puhe.2019.11.003>
- Prochaska, J., & DiClemente, C. (1986). Towards a comprehensive model of change. In W. Miller & N. Heather (Eds.), *Treating Addictive Behaviours: Processes of Change* (pp. 3–27). Plenum Press.
- Pu, J., Liu, Y., Zhang, H., et al. (2021). An integrated meta-analysis of peripheral blood metabolites and biological functions in major depressive disorder. *Mol Psychiatry*, 26(8), 4265–4276.
- Public Health England. (2020). Prescribed medicine review: Summary. <https://www.gov.uk/government/publications/prescribed-medicines-review-report/prescribed-medicines-review-summary#introduction>
- Puntis, S., Minichino, A., De Crescenzo, F., et al. (2020). Specialised early intervention teams for recent-onset psychosis. *Cochrane Database Syst Rev*, 11(11), CD013288. <https://doi.org/10.1002/14651858.CD013288.pub2>
- Purcell, R., & Mullen, P. (2020). Stalking and querulous behaviour. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 1142–1460). Oxford University Press.

- Qasrawi, S. O., & BaHammam, A. S. (2023). An update on Kleine-Levin Syndrome. *Curr Sleep Med Rep*, 9, 35–44. <https://doi.org/10.1007/s40675-022-00246-1>
- Quinlivan, L., Gorman, L., Littlewood, D. L., et al. (2022). 'Wasn't offered one, too poorly to ask for one': Reasons why some patients do not receive a psychosocial assessment following self-harm: Qualitative patient and carer survey. *Aust N Z J Psychiatry*, 56(4), 398–407. <https://doi.org/10.1177/00048674211011262>
- Rabinovici, G. D., & La Joie, R. (2023). Amyloid-targeting monoclonal antibodies for Alzheimer disease. *JAMA*, 330(6), 507–509.
- Racine, N., McArthur, B. A., Cooke, J. E., et al. (2021). Global prevalence of depressive and anxiety symptoms in children and adolescents during Covid-19: A meta-analysis. *JAMA Pediatr*, 175(11), 1142–1150. <https://doi.org/10.1001/jamapediatrics.2021.2482>
- Radua, J., Ramella-Cravaro, V., Ioannidis, J. P. A., et al. (2018). What causes psychosis? An umbrella review of risk and protective factors. *World Psychiatry*, 17(1), 49–66. <https://doi.org/10.1002/wps.20490>
- Rahman, A., & Kieling, C. (2015). Global psychiatry. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 201–214). John Wiley & Sons, Ltd.
- Rahman, A., Malik, A., Sikander, S., et al. (2008). Cognitive behaviour therapy-based intervention by community health workers for mothers with depression and their infants in rural Pakistan: A cluster-randomised controlled trial. *Lancet*, 372(9642), 902–909.
- Rahman, T., Sahrman, J. M., Olsen, M. A., et al. (2022). Risk of breast cancer with prolactin elevating antipsychotic drugs: An observational study of US women (ages 18–64 years). *J Clin Psychopharmacol*, 42(1), 7–16.
- Rai, D., Zitko, P., Jones, K., et al. (2013). Country-and individual-level socioeconomic determinants of depression: Multilevel cross-national comparison. *BJPsych*, 202(3), 195–203.
- Raine, A., Lencz, T., Bihle, S., et al. (2000). Reduced prefrontal gray matter volume and reduced autonomic activity in antisocial personality disorder. *Arch Gen Psychiatry*, 57(2), 119–127; discussion 128–119. <https://doi.org/10.1001/archpsyc.57.2.119>
- Rajabzadeh, V., Burn, E., Sajun, S. Z., et al. (2021). Understanding global mental health: A conceptual review. *BMJ Global Health*, 6(3), e004631. <https://doi.org/10.1136/bmjgh-2020-004631>
- Rajaraman, D., Travasso, S., Chatterjee, A., et al. (2012). The acceptability, feasibility and impact of a lay health counsellor delivered health promoting schools programme in India: A case study evaluation. *BMC Health Serv Res*, 12(1), 127. doi:10.1186/1472-6963-12-127
- Ramchandani, P., Stein, A., & Murray, L. (2009). Effect of parental psychiatric and physical illness on child development. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1752–1758). Oxford University Press.
- Ramnath, R. (2021). *Alcohol and Drugs in Road Fatalities: 2021 Report Based on 2019 Data Project Report MIS071*. TRL.
- Ranganath, C., & Ritchey, M. (2012). Two cortical systems for memory-guided behaviour. *Nat Rev Neurosci*, 13(10), 713–726.
- Raphael, B., & Wilson, J. (Eds.). (2000). *Psychological Debriefing: Theory, Practice and Evidence*. Cambridge University Press. <https://doi.org/DOI:10.1017/CBO9780511570148>
- Rapoport, J. L., & Shaw, P. (2015). Obsessive compulsive disorder. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 841–857). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch61>
- Rapoport, R. (1960). *Community as Doctor: New Perspectives on a Therapeutic Community*. Tavistock.
- Ray, L. A., Bujarski, S., Grodin, E., et al. (2019). State-of-the-art behavioral and pharmacological treatments for alcohol use disorder. *Am J Drug Alcohol Abuse*, 45(2), 124–140.
- Raad, R., & Makari, G. (2010). Samuel Tuke's description of the Retreat. *Am J Psychiatry*, 167(8), 898. <https://doi.org/10.1176/appi.ajp.2010.10020265>
- Reardon, J. M., & Williams, N. L. (2007). The specificity of cognitive vulnerabilities to emotional disorders: Anxiety sensitivity, looming vulnerability and explanatory style. *J Anxiety Disord*, 21(5), 625–643.
- Rebbapragada, N., Furtado, V., & Hawker-Bond, G. W. (2021). Prevalence of mental disorders in prisons in the UK: A systematic review and meta-analysis. *BJPsych Open*, 7(S1), S283–S284.
- Reed, G. M., First, M. B., Kogan, C. S., et al. (2019). Innovations and changes in the ICD-11 classification of mental, behavioural and neurodevelopmental disorders. *World Psychiatry*, 18(1), 3–19. <https://doi.org/10.1002/wps.20611>
- Reed, G. M., Drescher, J., Krueger, R. B. et al. (2016). Disorders related to sexuality and gender identity in the ICD-11: revising the ICD-10 classification based on current scientific evidence, best clinical practices, and human rights considerations. *World Psychiatry*, 15, 205–221. <https://doi.org/10.1002/wps.20354>
- Reed, R. V., Fazel, M., Jones, L., et al. (2012). Mental health of displaced and refugee children resettled in low-income and middle-income countries: Risk and protective factors. *Lancet*, 379(9812), 250–265 doi: 10.1016/S0140-6736(11)60050-0
- Rees, E., Walters, J. T., Georgieva, L., et al. (2014). Analysis of copy number variations at 15 schizophrenia-associated loci. *BJPsych*, 204(2), 108–114.
- Reeves, K. W., Vasquez, G., & Weiss, S. J. (2022). Characteristics of suicidal ideation: A systematic review. *Arch Suicide Res*, 26(4), 1736–1756.
- Regier, D. A., Goldberg, D. P., Üstün, B. T., & Reed, G. M. (2020). DSM-5 and ICD-11 classifications. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 51–61). Oxford University Press.
- Reichenpfader, U., Gartlehner, G., Morgan, L. C. et al. (2014). Sexual dysfunction associated with second-generation antidepressants in patients with major depressive disorder: results from a systematic review with network meta-analysis. *Drug Safety*, 37, 19–31. <https://doi.org/10.1007/s40264-013-0129-4>
- Reis, D. J., Hoffberg, A. S., Stearns-Yoder, K. A., & Bahraini, N. H. (2022). Bright light therapy for mental and behavioral illness: A systematic umbrella review. *Chronobiol Int*, 40(2), 204–214. <https://doi.org/10.1080/07420528.2022.2140669>
- Reutens, S., Nielsen, O., & Sachdev, P. (2010). Depersonalization disorder. *Curr Opin Psychiatry*, 23(3), 278–283. <https://doi.org/10.1097/YCO.0b013e3283387ab4>
- Reveley, A., & Murray, R. M. (1980). The genetic contribution to the functional psychoses. *Br J Hosp Med*, 24(2), 166, 169–171.

- Reynolds, C. F., Jeste, D. V., Sachdev, P. S., & Blazer, D. G. (2022). Mental health care for older adults: Recent advances and new directions in clinical practice and research. *World Psychiatry*, 21(3), 336–363.
- Reynolds III, C. F., Dew, M. A., Pollock, B. G., et al. (2006). Maintenance treatment of major depression in old age. *N Engl J Med*, 354(11), 1130–1138.
- Rhebergen, D., Huisman, A., Bouckaert, F., et al. (2015). Older age is associated with rapid remission of depression after electroconvulsive therapy: A latent class growth analysis. *Am J Geriatr Psychiatry*, 23(3), 274–282.
- Riblet, N. B. V., Shiner, B., Young-Xu, Y., & Watts, B. V. (2017). Strategies to prevent death by suicide: Meta-analysis of randomised controlled trials. *BJPsych*, 210(6), 396–402. <https://doi.org/10.1192/bjp.bp.116.187799>
- Rice, F., Riglin, L., Thapar, A. K., et al. (2019). Characterizing developmental trajectories and the role of neuropsychiatric genetic risk variants in early-onset depression. *JAMA Psychiatry*, 76(3), 306–313.
- Richard-Devantoy, S., Berlim, M. T., & Jollant, F. (2015). Suicidal behaviour and memory: A systematic review and meta-analysis. *World J Biol Psychiatry*, 16(8), 544–566. <https://doi.org/10.3109/15622975.2014.925584>
- Richmond-Rakerd, L. S., D'Souza, S., Milne, B. J., et al. (2022). Longitudinal associations of mental disorders with dementia: 30-year analysis of 1.7 million New Zealand citizens. *JAMA Psychiatry*, 79(4), 333–340.
- Riecher-Rössler, A., Butler, S., & Kulkarni, J. (2018). Sex and gender differences in schizophrenic psychoses—A critical review. *Arch Womens Ment Health*, 21(6), 627–648. <https://doi.org/10.1007/s00737-018-0847-9>
- Riisgaard, A. R., Laursen, T. M., Charles, M., et al. (2015). Long-term risk of dementia in persons with schizophrenia: A Danish population-based cohort study. *JAMA Psychiatry*, 72(11), 1095–1101.
- Ring, A., Dowrick, C., Humphris, G., & Salmon, P. (2004). Do patients with unexplained physical symptoms pressurise general practitioners for somatic treatment? A qualitative study. *BMJ*, 328(7447), 1057.
- Ristori J., Cocchetti C., Romani A., et al. (2020). Brain sex differences related to gender identity development: genes or hormones? *Int J Mol Sci* 19, 2123. doi: 10.3390/ijms21062123
- Rizk, M. M., Herzog, S., Dugad, S., & Stanley, B. (2021). Suicide risk and addiction: The impact of alcohol and opioid use disorders. *Curr Addict Rep*, 8, 194–207.
- Robbins, L., & Regier, D. A. (1991). *Psychiatric Disorders in America: The Epidemiologic Catchment Area Study*. Free Press.
- Robins, L. N. (1966). *Deviant Children Grown Up: A Sociological and Psychiatric Study of Sociopathic Personality*. Williams & Wilkins.
- Roberts, A. (1969). *Brain Damage in Boxers*. Pitman.
- Roberts, A. R., & Ottens, A. J. (2005). The seven-stage crisis intervention model: A road map to goal attainment, problem solving, and crisis resolution. *Brief Treat Crisis Interv*, 5, 329–339. <https://doi.org/10.1093/brief-treatment/mhi030>
- Roberts, E., Wessely, S., Chalder, T., et al. (2016). Mortality of people with chronic fatigue syndrome: A retrospective cohort study in England and Wales from the South London and Maudsley NHS Foundation Trust Biomedical Research Centre (SLaM BRC) Clinical Record Interactive Search (CRIS) Register. *Lancet*, 387(10028), 1638–1643. [https://doi.org/10.1016/s0140-6736\(15\)01223-4](https://doi.org/10.1016/s0140-6736(15)01223-4)
- Robertson, J., Hatton, C., Emerson, E., & Baines, S. (2015). Prevalence of epilepsy among people with intellectual disabilities: A systematic review. *Seizure*, 29, 46–62.
- Robins, L. N. (1966). *Deviant Children Grown Up: A Sociological and Psychiatric Study of Sociopathic Personality*. Williams & Wilkins.
- Robins, L. N. (1993). Vietnam veterans' rapid recovery from heroin addiction: A fluke or normal expectation? *Addiction*, 88(8), 1041–1054.
- Robins, L. N., Wing, J., Wittchen, H. U., et al. (1988). The Composite International Diagnostic Interview: An epidemiologic instrument suitable for use in conjunction with different diagnostic systems and in different cultures. *Arch Gen Psychiatry*, 45(12), 1069–1077.
- Robinson, L., & Chew-Graham, C. (2020). Primary care management of older people with mental health problems. In T. Denning, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 317–338). Oxford University Press.
- Robson, P. (2009). *Forbidden Drugs*. Oxford University Press.
- Rocha, E. M., De Miranda, B., & Sanders, L. H. (2018). Alpha-synuclein: Pathology, mitochondrial dysfunction and neuroinflammation in Parkinson's disease. *Neurobiol Dis*, 109, 249–257.
- Rockliff, H. E., Lightman, S. L., Rhidian, E., et al. (2014). A systematic review of psychosocial factors associated with emotional adjustment in in vitro fertilization patients. *Hum Rep Update*, 20(4), 594–613.
- Rocks, S., Fazel, M., & Tsiachristas, A. (2020). Impact of transforming mental health services for young people in England on patient access, resource use and health: A quasi-experimental study. *BMJ Open*, 10(1), e034067.
- Rodolico, A., Bighelli, I., Avanzato, C., et al. (2022). Family interventions for relapse prevention in schizophrenia: A systematic review and network meta-analysis. *Lancet Psychiatry*, 9(3), 211–221. [https://doi.org/10.1016/s2215-0366\(21\)00437-5](https://doi.org/10.1016/s2215-0366(21)00437-5)
- Rodrigues, H., Figueira, I., Lopes, A., et al. (2014). Does D-cycloserine enhance exposure therapy for anxiety disorders in humans? A meta-analysis. *PLoS ONE*, 9(7), e93519.
- Rodriguez, C. M., Lee, S. J., Ward, K. P., & Pu, D. F. (2021). The perfect storm: Hidden risk of child maltreatment during the COVID-19 pandemic. *Child Maltreat*, 26(2), 139–151.
- Rogers, A., Day, J., Randall, F., & Bentall, R. P. (2003). Patients' understanding and participation in a trial designed to improve the management of anti-psychotic medication: A qualitative study. *Soc Psychiatry Psychiatr Epidemiol*, 38, 720–727.
- Rogers, J. P., Oldham, M. A., Fricchione, G., et al. (2023). Evidence-based consensus guidelines for the management of catatonia: Recommendations from the British Association for Psychopharmacology. *J Psychopharmacol*, 37(4), 327–369. <https://doi.org/10.1177/02698811231158232>
- Roosen, L., & Sienaert, P. (2022). Evidence-based treatment strategies for rapid cycling bipolar disorder: A systematic review. *J Affect Disord*, 311, 69–77. <https://doi.org/10.1016/j.jad.2022.05.017>
- Ropper, A. H. (2019). Neurosyphilis. *N Engl J Med*, 381(14), 1358–1363.

- Rosanoff, A. J., Handy, L. M., & Rosanoff, I. A. (1934). Criminality and delinquency in twins. *J Crim Law Criminol*, 24, 923–934.
- Rosell, D. R., Thompson, J. L., Slifstein, M., et al. (2010). Increased serotonin 2A receptor availability in the orbitofrontal cortex of physically aggressive personality disordered patients. *Biol Psychiatry*, 67(12), 1154–1162. <https://doi.org/10.1016/j.biopsych.2010.03.013>
- Rosenheck, R., Mueser, K. T., Sint, K., et al. (2017). Supported employment and education in comprehensive, integrated care for first episode psychosis: Effects on work, school, and disability income. *Schizophr Res*, 182, 120–128. <https://doi.org/10.1016/j.schres.2016.09.024>
- Rosenthal, N. E., Sack, D. A., Gillin, J. C., et al. (1984). Seasonal affective disorder: a description of the syndrome and preliminary findings with light therapy. *Arch Gen Psychiatry*, 41(1), 72–80.
- Ross, E. L., Weinberg, M. S., & Arnold, S. E. (2022). Cost-effectiveness of aducanumab and donanemab for early Alzheimer disease in the US. *JAMA Neurol*, 79(5), 478–487.
- Rössler, W., Ajdacic-Gross, V., Riecher-Rössler, A., et al. (2016). Does menopausal transition really influence mental health? Findings from the prospective long-term Zurich study. *World Psychiatry*, 15(2), 146–154.
- Roth, M. (1955). The natural history of mental disorder in old age. *J Ment Science*, 101(423), 281–301.
- Rothman, D. J. (1971). *The Discovery of the Asylum: Social Order and Disorder in the New Republic*, 2nd edn. Routledge.
- Royal College of Psychiatrists. (2001). *DC-LD: Diagnostic Criteria for Psychiatric Disorders for Use with Adults with Learning Disabilities/Mental Retardation*. Occasional Paper OP 48. Royal College of Psychiatrists.
- Royal College of Psychiatrists. (2000). *Good Psychiatric Practice 2000*. Royal College of Psychiatrists.
- Royal College of Psychiatrists. (2022). Medical emergencies in eating disorders: Guidance on recognition and management. *College Report CR233*. Available at www.rcpsych.ac.uk/CR233.
- Royal College of Psychiatrists. (2010). *No Health without Public Mental Health: The Case for Action*. https://www.rcpsych.ac.uk/docs/default-source/improving-care/better-mh-policy/position-statements/ps04_2010.pdf?sfvrsn=b7316b7_4
- Royal College of Psychiatrists. (2022). *Summary of Evidence on Public Health Interventions*. Royal College of Psychiatrists.
- Royal College of Psychiatrists. (2023). *The Role of Genetic Testing in Mental Health Settings. College Reports: CR237*. [https://www.rcpsych.ac.uk/improving-care/campaigning-for-better-mental-health-policy/college-reports/2023-college-reports/the-role-of-genetic-testing-in-mental-health-settings-\(cr237\)](https://www.rcpsych.ac.uk/improving-care/campaigning-for-better-mental-health-policy/college-reports/2023-college-reports/the-role-of-genetic-testing-in-mental-health-settings-(cr237))
- Royal College of Psychiatrists (2005). *The Third Report of the Royal College of Psychiatrists Special Committee on ECT*. Royal College of Psychiatrists.
- Rudolf, M. C. J., & Logan, S. (2005). What is the long-term outcome for children who fail to thrive? *Arch Dis Child*, 90, 925–931.
- Ruhé, H. G., Mason, N. S., & Schene, A. H. (2007). Mood is indirectly related to serotonin, norepinephrine and dopamine levels in humans: A meta-analysis of monoamine depletion studies. *Mol Psychiatry*, 12(4), 331–359.
- Ruocco, A. C., Amirthavasagam, S., Choi-Kain, L. W., & McMain, S. F. (2013). Neural correlates of negative emotionality in borderline personality disorder: An activation-likelihood-estimation meta-analysis. *Biol Psychiatry*, 73(2), 153–160. [doi:10.1016/j.biopsych.2012.07.014](https://doi.org/10.1016/j.biopsych.2012.07.014)
- Ruscio, A. M., Stein, D. J., Chiu, W. T., & Kessler, R. C. (2010). The epidemiology of obsessive-compulsive disorder in the National Comorbidity Survey Replication. *Mol Psychiatry*, 15(1), 53–63.
- Rush, A. J., Trivedi, M. H., Ibrahim, H. M., et al. (2003). The 16-Item Quick Inventory of Depressive Symptomatology (QIDS), clinician rating (QIDS-C), and self-report (QIDS-SR): A psychometric evaluation in patients with chronic major depression. *Biol Psychiatry*, 54(5), 573–583.
- Russell, G. (1979). Bulimia nervosa: An ominous variant of anorexia nervosa. *Psychol Med*, 9(3), 429–448.
- Russell, G., & Burns, A. (2014). Charles Bonnet syndrome and cognitive impairment: A systematic review. *Int Psychogeriatr*, 26(9), 1431–1443.
- Rutter, M. (2005). Environmentally mediated risks for psychopathology: Research strategies and findings. *J Am Acad Child Adolesc Psychiatry*, 44(1), 3–18. <https://doi.org/10.1097/01.chi.0000145374.45992.c9>
- Rutter, M. (2011). Research review: Child psychiatric diagnosis and classification: Concepts, findings, challenges and potential. *J Child Psychol Psychiatry*, 52(6), 647–660.
- Rutter, M. L. (1972). Relationships between child and adult psychiatric disorders. Some research considerations. *Acta Psychiatr Scand*, 48(1), 3–21. [doi:10.1111/j.1600-0447.1972.tb04346.x](https://doi.org/10.1111/j.1600-0447.1972.tb04346.x)
- Rutter, M. L., & Rutter, M. (1993). *Developing Minds: Challenge and Continuity Across the Life Span*. Basic Books.
- Rutter, M. L., Tizard, J., & Whitmore, K. (Eds.). (1970). *Education, Health and Behaviour: Psychological and Medical Study of Childhood Development*. Longman.
- Rycroft, C. (1995). *A Critical Dictionary of Psychoanalysis*, 2nd edn. Penguin.
- Ryle, A., & Karr, I. B. (2020). *Introducing Cognitive Analytic Therapy: Principles and Practice of a Relational Approach to Mental Health*, 2nd edn. John Wiley.
- Ryle, A., Kellett, S., Hepple, J., & Calvert, R. (2014). Cognitive analytic therapy at 30. *Adv Psychiatr Treat*, 20(4), 258–268. <https://doi.org/10.1192/apt.bp.113.011817>
- Ryu, A., & Kim, T.-H. (2015). Premenstrual syndrome: A mini review. *Maturitas*, 82(4), 436–440.
- Saab, M. M., Murphy, M., Meehan, E., et al. (2022). Suicide and self-harm risk assessment: A systematic review of prospective research. *Arch Suicide Res*, 26(4), 1645–1665.
- Sabia, S., Fayosse, A., Dumurgier, J., et al. (2018). Alcohol consumption and risk of dementia: 23-year follow-up of Whitehall II cohort study. *BMJ*, 362, k2927.
- Sackett, D. (1996). Evaluation of clinical method. In D. Weatherall, J. Ledingham, & D. Warrell (Eds.), *Oxford Textbook of Medicine*, 3rd edn (pp. 15–21). Oxford University Press.
- Sadler, K., Vizard, T., Ford, T., et al. (2018). *Mental Health of Children and Young People in England, 2017*. NHS Digital.
- Saito, S., & Ihara, M. (2016). Interaction between cerebrovascular disease and Alzheimer pathology. *Curr Opin Psychiatry*, 29(2), 168–173.
- Sajatovic, M., Strejilevich, S. A., Gildengers, A. G., et al. (2015). A report on older-age bipolar disorder from the International Society for Bipolar Disorders Task Force. *Bipolar Disord*, 17(7), 689–704.

- Salazar de Pablo, G., Radua, J., Pereira, J., et al. (2021). Probability of transition to psychosis in individuals at clinical high risk: An updated meta-analysis. *JAMA Psychiatry*, 78(9), 970–978. <https://doi.org/10.1001/jamapsychiatry.2021.0830>
- Samara, M. T., Leucht, C., Leeflang, M. M., et al. (2015). Early improvement as a predictor of later response to antipsychotics in schizophrenia: A diagnostic test review. *Am J Psychiatry*, 172(7), 617–629.
- Sampogna, G., Zinno, F., Giallonardo, V., et al. (2020). The de Clérambault syndrome: More than just a delusional disorder? *Int Rev Psychiatry*, 32(5–6), 385–390.
- San Too, L., Spittal, M. J., Bugeja, L., et al. (2019). The association between mental disorders and suicide: A systematic review and meta-analysis of record linkage studies. *J Affect Disord*, 259, 302–313.
- Sandweiss, A. J., Brandt, V. L., & Zoghbi, H. Y. (2020). Advances in understanding of Rett syndrome and *MECP2* duplication syndrome: Prospects for future therapies. *Lancet Neurol*, 19(8), 689–698.
- Santiago, P. N., Ursano, R. J., Gray, C. L., et al. (2013). A systematic review of PTSD prevalence and trajectories in DSM-5 defined trauma exposed populations: Intentional and non-intentional traumatic events. *PLoS ONE*, 8(4), e59236.
- Santomauro, D. F., Melen, S., Mitchison, D., Vos, T., Whiteford, H., & Ferrari, A. J. (2021). The hidden burden of eating disorders: an extension of estimates from the Global Burden of Disease Study 2019. *Lancet Psychiatry*, 8, 320–328. [https://doi.org/10.1016/S2215-0366\(21\)00040-7](https://doi.org/10.1016/S2215-0366(21)00040-7)
- Sareen, J. (2014). Posttraumatic stress disorder in adults: Impact, comorbidity, risk factors, and treatment. *Can J Psychiatry*, 59(9), 460–467.
- Sargant, W., & Slater, E. (1963). *An Introduction to Physical Methods of Treatment in Psychiatry*. Livingstone
- Sarwer, D. B., & Heinberg, L. J. (2020). A review of the psychosocial aspects of clinically severe obesity and bariatric surgery. *Am Psychol*, 75(2), 252–264. <https://doi.org/10.1037/amp0000550>
- Sashidharan, S. P., Mezzina, R., & Puras, D. (2019). Reducing coercion in mental healthcare. *Epidemiol Psychiatr Sci*, 28(6), 605–612. <https://doi.org/10.1017/s2045796019000350>
- Sass, B., Moffat, J., Bhui, K., & McKenzie, K. (2009). Enhancing pathways to care for Black and minority ethnic populations: A systematic review. *Int Rev Psychiatry*, 21(5), 430–438. <https://doi.org/10.1080/09540260802204121>
- Saul, J., Griffiths, S., & Norbury, C. F. (2023). Prevalence and functional impact of social (pragmatic) communication disorders. *J Child Psychol Psychiatry*, 64(3), 376–387.
- Saunders, K. E., & Smith, K. A. (2016). Interventions to prevent self-harm: What does the evidence say? *Evid Based Ment Health*, 19(3), 69–72. <https://doi.org/10.1136/eb-2016-102420>
- Sawyer, S. M., Azzopardi, P. S., Wickremarathne, D., & Patton, G. C. (2018). The age of adolescence. *Lancet Child Adolesc Health*, 2(3), 223–228.
- Saxena, S., & Lawley, D. (2009). Delirium in the elderly: A clinical review. *Postgrad Med J*, 85(1006), 405–413.
- Saylor, D., Dickens, A. M., Sacktor, N., et al. (2016). HIV-associated neurocognitive disorder—Pathogenesis and prospects for treatment. *Nat Rev Neurol*, 12(4), 234–248. <https://doi.org/10.1038/nrneurol.2016.27>
- Scadding, J. G. (1967). Diagnosis: The clinician and the computer. *Lancet*, 2(7521), 877–882.
- Scaglia, F. (2014). *Inborn Errors of Metabolism: From Neonatal Screening to Metabolic Pathways*. Oxford University Press.
- Scalzo, S. J., Bowden, S. C., Ambrose, M. L., et al. (2015). Wernicke–Korsakoff syndrome not related to alcohol use: A systematic review. *J Neurol Neurosurg Psychiatry*, 86(12), 1362–1368.
- Schanda, H., Knecht, G., Schreiner, D., et al. (2004). Homicide and major mental disorders: A 25-year study. *Acta Psychiatr Scand*, 110(2), 98–107.
- Scheckel, C., & Aguzzi, A. (2018). Prions, prionoids and protein misfolding disorders. *Nat Rev Genet*, 19(7), 405–418.
- Scheepens, D. S., van Waarde, J. A., Lok, A., et al. (2020). The link between structural and functional brain abnormalities in depression: A systematic review of multimodal neuroimaging studies. *Front Psychiatry*, 11, 485. <https://doi.org/10.3389/fpsy.2020.00485>
- Scheltens, P., De Strooper, B., Kivipelto, M., et al. (2021). Alzheimer's disease. *Lancet*, 397(10284), 1577–1590.
- Schepens, H., Van Puyenbroeck, J., & Maes, B. (2019). How to improve the quality of life of elderly people with intellectual disability: A systematic literature review of support strategies. *J Appl Res Intellect Disabil*, 32(3), 483–521. <https://doi.org/10.1111/jar.12559>
- Schiffman, J., Walker, E., Ekstrom, M., et al. (2004). Childhood videotaped social and neuromotor precursors of schizophrenia: A prospective investigation. *Am J Psychiatry*, 161(11), 2021–2027.
- Schirmbeck, F., & Zink, M. (2022). Obsessive-compulsive symptoms in psychotic disorders: Pathogenesis and treatment. *Der Nervenarzt*, 93(7), 688–694.
- Schlapobersky, J., & Pines, M. (2009). Group methods in adult psychiatry. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. (pp. 1350–1369). Oxford University Press.
- Schmahmann, J. D., & Pandya, D. N. (2008). Disconnection syndromes of basal ganglia, thalamus, and cerebellar systems. *Cortex*, 44(8), 1037–1066.
- Schmucker, M., & Lösel, F. (2015). The effects of sexual offender treatment on recidivism: An international meta-analysis of sound quality evaluations. *J Exp Criminol*, 11(4), 597–630. <https://doi.org/10.1007/s11292-015-9241-z>
- Schneider-Thoma, J., Chalkou, K., Dorries, C., et al. (2022). Comparative efficacy and tolerability of 32 oral and long-acting injectable antipsychotics for the maintenance treatment of adults with schizophrenia: A systematic review and network meta-analysis. *Lancet*, 399, 824–836.
- Schramm, E., Klein, D. N., Elsaesser, M., et al. (2020). Review of dysthymia and persistent depressive disorder: History, correlates, and clinical implications. *Lancet Psychiatry*, 7(9), 801–812.
- Schultz, J. H., & Luthe, W. (1959). *Autogenic Training: A Psychophysiologic Approach to Psychotherapy*. Grune & Stratton.
- Schulze, T., & McMahon, F. (2018). *Psychiatric Genetics: A Primer for Clinical and Basic Scientists*. Oxford University Press.
- Schwartz, M. A., & Wiggins, O. P. (1987). Diagnosis and ideal types: A contribution to psychiatric classification. *Compr Psychiatry*, 28(4), 277–291. [https://doi.org/10.1016/0010-440x\(87\)90064-2](https://doi.org/10.1016/0010-440x(87)90064-2)

- Schwartz, R. C., & Blankenship, D. M. (2014). Racial disparities in psychotic disorder diagnosis: A review of empirical literature. *World J Psychiatry*, 4(4), 133–140. <https://doi.org/10.5498/wjp.v4.i4.133>
- Scoboria, A., Wade, K. A., Lindsay, D. S., et al. (2017). A mega-analysis of memory reports from eight peer-reviewed false memory implantation studies. *Memory*, 25(2), 146–163. <https://doi.org/10.1080/09658211.2016.1260747>
- Scott, S., & Gardner, F. (2015). Parenting programs. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 483–495). John Wiley & Sons, Ltd.
- Scott, A., & Waite, J. (2013). Monitoring a course of ECT. In J. Waite & A. Eaton (Eds.), *The ECT Handbook* (pp. 60–70). Royal College of Psychiatrists Publications.
- Scott, A. I. (2005). College guidelines on electroconvulsive therapy: An update for prescribers. *Adv Psychiatr Treat*, 11(2), 150–156.
- Scott, A. J., Sharpe, L., Hunt, C., & Gandy, M. (2017). Anxiety and depressive disorders in people with epilepsy: A meta-analysis. *Epilepsia*, 58(6), 973–982.
- Scott, P. D. (1960). The treatment of psychopaths. *BMJ*, 1(5186), 1641–1646. <https://doi.org/10.1136/bmj.1.5186.1641>
- Sedgwick, R., Epstein, S., Dutta, R., & Ougrin, D. (2019). Social media, internet use and suicide attempts in adolescents. *Curr Opin Psychiatry*, 32(6), 534–541. <https://doi.org/10.1097/ycp.0000000000000547>
- Segal, M., Eliasziw, M., Phillips, S., et al. (2016). Intellectual disability is associated with increased risk for obesity in a nationally representative sample of US children. *Disabil Health J*, 9(3), 392–398.
- Segal, T., & Ranjith, G. (2016). Psychiatric assessments on medical wards: A guide for general psychiatrists. *BJPsych Adv*, 22(1), 8–15.
- Segal-Zindall, V. J., Williams, M. G., & Teasdale, J. D. (2013). *Mindfulness-Based Cognitive Therapy for Depression*, 2nd edn. Guildford Press.
- Sehatazadeh, S., Daskalakis, Z. J., Yap, B., et al. (2019). Unilateral and bilateral repetitive transcranial magnetic stimulation for treatment-resistant depression: A meta-analysis of randomized controlled trials over 2 decades. *J Psychiatry Neurosci*, 44(3), 151–163. <https://doi.org/10.1503/jpn.180056>
- Sekar, A., Bialas, A. R., De Rivera, H., et al. (2016). Schizophrenia risk from complex variation of complement component 4. *Nature*, 530(7589), 177–183.
- Selten, J. P., van der Ven, E., & Termorshuizen, F. (2020). Migration and psychosis: A meta-analysis of incidence studies. *Psychol Med*, 50(2), 303–313. <https://doi.org/10.1017/s0033291719000035>
- Semchishen, S. M., Hojlund, M., Lemaire, D., et al. (2026). All-cause and cause-specific mortality in individuals with eating disorders: systematic review and meta-analysis of relative risk and aggravating or attenuating factors. *World Psychiatry*, 26, 125–140.
- Semkovska, M., Quinlivan, L., O'Grady, T., et al. (2019). Cognitive function following a major depressive episode: A systematic review and meta-analysis. *Lancet Psychiatry*, 6(10), 851–861.
- Sen, A. (2009). *The Idea of Justice*. Belknap Press.
- Sen, A., Capelli, V., & Husain, M. (2018). Cognition and dementia in older patients with epilepsy. *Brain*, 141(6), 1592–1608.
- Sendt, K.-V., Tracy, D. K., & Bhattacharyya, S. (2015). A systematic review of factors influencing adherence to antipsychotic medication in schizophrenia-spectrum disorders. *Psychiatry Res*, 225(1–2), 14–30.
- Seto, M. C., Kingston, D. A., & Bourget, D. (2014). Assessment of the paraphilias. *Psychiatr Clin North Am*, 37, 149–161. <https://doi.org/10.1016/j.psc.2014.03.001>
- Shalev, A. Y., Gevonden, M., Ratanatharathorn, A., et al. (2019). Estimating the risk of PTSD in recent trauma survivors: Results of the International Consortium to Predict PTSD (ICPP). *World Psychiatry*, 18(1), 77–87. <https://doi.org/10.1002/wps.20608>
- Shamloul, R., & Ghanem, H. (2013). Erectile dysfunction. *Lancet*, 381, 153–165. [https://doi.org/10.1016/S0140-6736\(12\)60520-0](https://doi.org/10.1016/S0140-6736(12)60520-0)
- Shanahan, L., Steinhoff, A., Bechtiger, L., et al. (2021). Frequent teenage cannabis use: Prevalence across adolescence and associations with young adult psychopathology and functional well-being in an urban cohort. *Drug Alcohol Depend*, 228, 109063.
- Shapiro, P. A. (2015). Management of depression after myocardial infarction. *Curr Cardiol Rep*, 17(10), 80. <https://doi.org/10.1007/s11886-015-0634-4>
- Sharpe, M. (2014). Psychological medicine and the future of psychiatry. *BJPsych*, 204(2), 91–92. <https://doi.org/10.1192/bjp.bp.113.132894>
- Sharpe, M. (2013). Somatic symptoms: Beyond 'medically unexplained'. *Br J Psychiatry*, 203(5), 320–321.
- Sharpe, M., & Carson, A. (2001). 'Unexplained' somatic symptoms, functional syndromes, and somatization: Do we need a paradigm shift? *Ann Int Med*, 134(9, Part 2), 926–930.
- Sharpe, M., Goldsmith, K. A., Johnson, A. L., et al. (2015). Rehabilitative treatments for chronic fatigue syndrome: Long-term follow-up from the PACE trial. *Lancet Psychiatry*, 2(12), 1067–1074.
- Sharpe, M., Walker, J., Hansen, C. H., et al. (2014). Integrated collaborative care for comorbid major depression in patients with cancer (SMaRT Oncology-2): A multicentre randomised controlled effectiveness trial. *Lancet*, 384(9948), 1099–1108.
- Sharpless, B. A., & Klinkova, M. (2019). Clinical features of isolated sleep paralysis. *Sleep Med*, 58, 102–106. <https://doi.org/10.1016/j.sleep.2019.03.007>
- Shaw, J. M., Price, M. A., Clayton, J. M., et al. (2016). Developing a clinical pathway for the identification and management of anxiety and depression in adult cancer patients: An online Delphi consensus process. *Support Care Cancer*, 24(1), 33–41. <https://doi.org/10.1007/s00520-015-2742-5>
- Shaw, R. J., Atkin, K., Bécares, L., et al. (2012). Impact of ethnic density on adult mental disorders: Narrative review. *BJPsych*, 201(1), 11–19.
- Shek, E., Stein, A. T., Shansis, F. M., et al. (2009). Day hospital vs outpatient care for people with schizophrenia. *Schizophr Bull*, 35(6), 1057–1058. <https://doi.org/10.1093/schbul/sbp102>
- Shepherd, M. (1961). Morbid jealousy: Some clinical and social aspects of a psychiatric symptom. *J Ment Sci*, 107(449), 687–753.
- Shepherd, M. (1966). *Psychiatric Illness in General Practice*. Oxford University Press.

- Sheridan Rains, L., Zenina, T., Dias, M., et al. (2019). Variations in patterns of involuntary hospitalisation and in legal frameworks: An international comparative study. *Lancet Psychiatry*, 6(5), 403–417. [https://doi.org/10.1016/S2215-0366\(19\)30090-2](https://doi.org/10.1016/S2215-0366(19)30090-2)
- Shi, Y., Luan, D., Song, R., & Zhang, Z. (2020). Value of peripheral neurotrophin levels for the diagnosis of depression and response to treatment: A systematic review and meta-analysis. *Eur Neuropsychopharmacol*, 41, 40–51. <https://doi.org/10.1016/j.euroneuro.2020.09.633>
- Shiekh, S. I., Cadogan, S. L., Lin, L.-Y., et al. (2021). Ethnic differences in dementia risk: A systematic review and meta-analysis. *J Alzheimers Dis*, 80(1), 337–355.
- Shin, S. H., Lee, S., Jeon, S.-M., & Wills, T. A. (2015). Childhood emotional abuse, negative emotion-driven impulsivity, and alcohol use in young adulthood. *Child Abuse Negl*, 50, 94–103.
- Shiraishi, N., & Reilly, J. (2019). Positive and negative impacts of schizophrenia on family caregivers: A systematic review and qualitative meta-summary. *Soc Psychiatry Psychiatr Epidemiol*, 54(3), 277–290. <https://doi.org/10.1007/s00127-018-1617-8>
- Sibbald, B., Addington-Hall, J., Brennenman, D., & Freeling, P. (1996). Investigation of whether on-site general practice counsellors have an impact on psychotropic drug prescribing rates and costs. *Br J Gen Pract*, 46(403), 63–67.
- Sideli, L., Murray, R. M., Schimmenti, A., et al. (2020). Childhood adversity and psychosis: A systematic review of bio-psychosocial mediators and moderators. *Psychol Med*, 50(11), 1761–1782. <https://doi.org/10.1017/s0033291720002172>
- Sierra, M., & David, A. S. (2011). Depersonalization: A selective impairment of self-awareness. *Conscious Cogn*, 20(1), 99–108.
- Sierra, M., Medford, N., Wyatt, G., & David, A. S. (2012). Depersonalization disorder and anxiety: A special relationship? *Psychiatry Res*, 197(1–2), 123–127. <https://doi.org/10.1016/j.psychres.2011.12.017>
- Sigurvinsdóttir, A. L., Jensínudóttir, K. B., Baldvinsdóttir, K. D., et al. (2020). Effectiveness of cognitive behavioral therapy (CBT) for child and adolescent anxiety disorders across different CBT modalities and comparisons: A systematic review and meta-analysis. *Nord J Psychiatry*, 74(3), 168–180.
- Silberberg, D., Anand, N. P., Michels, K., & Kalaria, R. N. (2015). Brain and other nervous system disorders across the lifespan—Global challenges and opportunities. *Nature*, 527(7578), S151–154. <https://doi.org/10.1038/nature16028>
- Silveira, J., M., & Seeman, M., V (1995). Shared psychotic disorder: A critical review of the literature. *Can J Psychiatry*, 40(7), 389–395.
- Simeone, J. C., Ward, A. J., Rotella, P., et al. (2015). An evaluation of variation in published estimates of schizophrenia prevalence from 1990–2013: A systematic literature review. *BMC Psychiatry*, 15, 1–14.
- Simon, G. E., & Savarino, J. (2007). Suicide attempts among patients starting depression treatment with medications or psychotherapy. *Am J Psychiatry*, 164(7), 1029–1034.
- Simon, G. E., VonKorff, M., Piccinelli, M., et al. (1999). An international study of the relation between somatic symptoms and depression. *N Engl J Med*, 341(18), 1329–1335.
- Simon, S. S., Cordás, T. A., & Bottino, C. M. (2015). Cognitive behavioral therapies in older adults with depression and cognitive deficits: a systematic review. *Int J Geriatr Psychiatry*, 30(3), 223–233.
- Simonoff, E. (2015). Intellectual disability. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 719–737). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch54>
- Simonsen, S., Bateman, A., Bohus, M., et al. (2019). European guidelines for personality disorders: Past, present and future. *Borderline Personal Disord Emot Dysregul*, 6(1), 9. <https://doi.org/10.1186/s40479-019-0106-3>
- Simpson, A., Miller, C., & Bowers, L. (2003). The history of the Care Programme Approach in England: Where did it go wrong? *J Ment Health*, 12(5), 489–504.
- Simpson, A. I., Penney, S. R., & Jones, R. M. (2022). Homicide associated with psychotic illness: What global temporal trends tell us about the association between mental illness and violence. *Aust N Z J Psychiatry*, 56(11), 1384–1388. <https://doi.org/10.1177/00048674211067164>
- Sinclair, J. M., & O'Neill, A. (2020). Update on the management of alcohol use disorders. *BJPsych Adv*, 26(2), 82–91.
- Singh, R., Bansal, Y., Medhi, B., & Kuhad, A. (2019). Antipsychotics-induced metabolic alterations: Recounting the mechanistic insights, therapeutic targets and pharmacological alternatives. *Eur J Pharmacol*, 844, 231–240.
- Singh, T., Poterba, T., Curtis, D., et al. (2022). Rare coding variants in ten genes confer substantial risk for schizophrenia. *Nature*, 604(7906), 509–516. <https://doi.org/10.1038/s41586-022-04556-w>
- Siskind, D., Orr, S., Sinha, S., et al. (2022). Rates of treatment-resistant schizophrenia from first-episode cohorts: Systematic review and meta-analysis. *BJPsych*, 220(3), 115–120. <https://doi.org/10.1192/bjp.2021.61>
- Siskind, D. J., Lee, M., Ravindran, A., et al. (2018). Augmentation strategies for clozapine refractory schizophrenia: A systematic review and meta-analysis. *Aust N Z J Psychiatry*, 52(8), 751–767. <https://doi.org/10.1177/0004867418772351>
- Sivasathiseelan, H., Marshall, C. R., Agustus, J. L., et al. (2019). Frontotemporal dementia: A clinical review. *Semin Neurol*, 39(2), 251–263.
- Skapinakis, P., Caldwell, D. M., Hollingworth, W., et al. (2016). Pharmacological and psychotherapeutic interventions for management of obsessive-compulsive disorder in adults: A systematic review and network meta-analysis. *Lancet Psychiatry*, 3(8), 730–739.
- Skinner, B. F. (1954). *Science and Human Behavior*. Macmillan.
- Skodol, A. E., Gunderson, J. G., Pfohl, B., et al. (2002). The borderline diagnosis I: Psychopathology, comorbidity, and personality structure. *Biol Psychiatry*, 51(12), 936–950.
- Skodol, A. E., Gunderson, J. G., Shea, M. T., et al. (2005). The Collaborative Longitudinal Personality Disorders Study (CLPS): Overview and implications. *J Pers Disord*, 19(5), 487–504. <https://doi.org/10.1521/pedi.2005.19.5.487>
- Skordis, N., Kyriakou, A., Dror, S., et al. (2020). Gender dysphoria in children and adolescents: An overview. *Hormones*, 19(3), 267–276. <https://doi.org/10.1007/s42000-020-00174-1>
- Slade, M., Amering, M., Farkas, M., et al. (2014). Uses and abuses of recovery: Implementing recovery-oriented practices in mental health systems. *World Psychiatry*, 13(1), 12–20.

- Slade, M., Powell, R., Rosen, A., & Strathdee, G. (2000). Threshold Assessment Grid (TAG): The development of a valid and brief scale to assess the severity of mental illness. *Soc Psychiatry Psychiatr Epidemiol*, 35(2), 78–85. <https://doi.org/10.1007/s001270050011>
- Slade, M., & Thornicroft, G. (2020). *Camberwell Assessment of Need*. Cambridge University Press.
- Slater, E., & Beard, A. (1963). The schizophrenia-like psychoses of epilepsy: I. Psychiatric aspects. *BJPsych*, 109(458), 95–112.
- Slater, E., & Shields, J. (1969). Genetical aspects of anxiety. In M. Lader (Ed.), *Studies of Anxiety* (pp. 62–71). Royal Medico-Psychological Association.
- Slee, A., Nazareth, I., Bondaronek, P., et al. (2019). Pharmacological treatments for generalised anxiety disorder: A systematic review and network meta-analysis. *Lancet*, 393(10173), 768–777.
- Small, J. G., Klapper, M. H., Kellams, J. J., et al. (1988). Electroconvulsive treatment compared with lithium in the management of manic states. *Arch Gen Psychiatry*, 45(8), 727–732.
- Smeland, O. B., Frei, O., Dale, A. M., & Andreassen, O. A. (2020). The polygenic architecture of schizophrenia: Rethinking pathogenesis and nosology. *Nat Rev Neurol*, 16(7), 366–379. <https://doi.org/10.1038/s41582-020-0364-0>
- Smid, G. E., Groen, S., de la Rie, S. M., et al. (2018). Toward cultural assessment of grief and grief-related psychopathology. *Psychiatr Serv*, 69(10), 1050–1052. <https://doi.org/10.1176/appi.ps.201700422>
- Smigielski, L., Jagannath, V., Rössler, W., et al. (2020). Epigenetic mechanisms in schizophrenia and other psychotic disorders: A systematic review of empirical human findings. *Mol Psychiatry*, 25(8), 1718–1748. <https://doi.org/10.1038/s41380-019-0601-3>
- Smith, D. G., & Robbins, T. W. (2013). The neurobiological underpinnings of obesity and binge eating: a rationale for adopting the food addiction model. *Biol Psychiatry*, 73, 804–810. <https://doi.org/10.1016/j.biopsych.2012.08.026>
- Smith, D. H., Johnson, V. E., Trojanowski, J. Q., & Stewart, W. (2019). Chronic traumatic encephalopathy—confusion and controversies. *Nat Rev Neurol*, 15(3), 179–183.
- Smith, K. A., & Cipriani, A. (2017). Lithium and suicide in mood disorders: Updated meta-review of the scientific literature. *Bipolar Disord*, 19(7), 575–586.
- Smith, K. W., Scignano, D. J., Hernandez, A. V., & White, C. M. (2022). MDMA-assisted psychotherapy for treatment of posttraumatic stress disorder: A systematic review with meta-analysis. *J Clin Pharmacol*, 62(4), 463–471.
- Smith, P., Dalgleish, T., & Meiser-Stedman, R. (2019). Practitioner review: Posttraumatic stress disorder and its treatment in children and adolescents. *J Child Psychol Psychiatry*, 60(5), 500–515.
- Smith, S., Gate, R., Ariyo, K., et al. (2020). Reasons behind the rising rate of involuntary admissions under the Mental Health Act (1983): Service use and cost impact. *Int J Law Psychiatry*, 68, 101506. <https://doi.org/10.1016/j.ijlp.2019.101506>
- Smoller, J. W., Andreassen, O. A., Edenberg, H. J., et al. (2019). Psychiatric genetics and the structure of psychopathology. *Mol Psychiatry*, 24(3), 409–420. <https://doi.org/10.1038/s41380-017-0010-4>
- Smyth, M., & Hoult, J. (2000). The home treatment enigma. *BMJ*, 320(7230), 305–309. doi: 10.1136/bmj.320.7230.305.
- Snowden, P., & Kane, E. (2003). Personality disorder: No longer a diagnosis of exclusion. *Psychiatr Bull*, 27(11), 401–403. <https://doi.org/10.1192/pb.27.11.401>
- Snowdon, J. (2019). Indian suicide data: What do they mean? *Indian J Med Res*, 150(4), 315.
- Snowdon, J., & Choi, N. G. (2020). Undercounting of suicides: Where suicide data lie hidden. *Glob Public Health*, 15(12), 1894–1901. <https://doi.org/10.1080/17441692.2020.1801789>
- Solaro, C., Gamberini, G., & Masuccio, F. G. (2018). Depression in multiple sclerosis: Epidemiology, aetiology, diagnosis and treatment. *CNS Drugs*, 32(2), 117–133.
- Solmi, M., Croatto, G., Piva, G., et al. (2023). Efficacy and acceptability of psychosocial interventions in schizophrenia: Systematic overview and quality appraisal of the meta-analytic evidence. *Mol Psychiatry*, 28(1), 354–368. <https://doi.org/10.1038/s41380-022-01727-z>
- Solmi, M., Wade, T. D., Byrne, S., et al. (2021). Comparative efficacy and acceptability of psychological interventions for the treatment of adult outpatients with anorexia nervosa: a systematic review and network meta-analysis. *Lancet Psychiatry*, 8, 215–224. [https://doi.org/10.1016/S2215-0366\(20\)30566-6](https://doi.org/10.1016/S2215-0366(20)30566-6)
- Somashekar, B., Jainar, A., & Wuntakal, B. (2013). Psychopharmacotherapy of somatic symptoms disorders. *Int Rev Psychiatry*, 25(1), 107–115. <https://doi.org/10.3109/09540261.2012.729758>
- Soneson, E., Puntis, S., Chapman, N., et al. (2022). Happier during lockdown: A descriptive analysis of self-reported wellbeing in 17,000 UK school students during Covid-19 lockdown. *Eur Child Adolesc Psychiatry*, 32(6), 1131–1146.
- Sonuga-Barke, E. J. S., & Taylor, E. (2015). ADHD and hyperkinetic disorder. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 738–756). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch55>
- Sonuga-Barke, E. J. S., Becker, S. P., Bolte, S., et al. (2022). Annual research review: Perspectives on progress in ADHD science—from characterization to cause. *J Child Psychol Psychiatry*, 64(4), 506–532. <https://doi.org/10.1111/jcpp.13696>
- Soothill, K., Ackerley, E., & Francis, B. (2004). The criminal careers of arsonists. *Med Sci Law*, 44(1), 27–40. <https://doi.org/10.1258/rsmmsl.44.1.27>
- Soto, C., & Pritzkow, S. (2018). Protein misfolding, aggregation, and conformational strains in neurodegenerative diseases. *Nat Neurosci*, 21(10), 1332–1340.
- Sousa, A., & Dinis-Oliveira, R. J. (2020). Pharmacokinetic and pharmacodynamic of the cognitive enhancer modafinil: Relevant clinical and forensic aspects. *Subst Abuse*, 41(2), 155–173.
- Spensley, S. (2014). The Kleinian tradition. *Br J Psychother*, 30(2), 169–180. <https://doi.org/10.1111/bjp.12076>
- Spielberger, C., Gorsuch, R. L., Lushene, R., et al. (1983). *Manual for the State-Trait Anxiety Inventory*. Consulting Psychologists Press.
- Spielmans, G. I., Berman, M. I., Linardatos, E., et al. (2016). Adjunctive atypical antipsychotic treatment for major depressive disorder: A meta-analysis of depression, quality of life, and safety outcomes. *Focus*, 14(2), 244–265.

- Spinelli, M. (2009). Postpartum psychosis: Detection of risk and management. *Am J Psychiatry*, 186, 2.
- Spitzer, R. L., First, M. B., Williams, J. B., et al. (1992). Now is the time to retire the term 'organic mental disorders'. *Am J Psychiatry*, 149(2), 240–244.
- Spitzer, R. L., Kroenke, K., Williams, J. B., & Löwe, B. (2006). A brief measure for assessing generalized anxiety disorder: The GAD-7. *Arch Intern Med*, 166(10), 1092–1097. <https://doi.org/10.1001/archinte.166.10.1092>
- Spitzer, R. L., Williams, J. B., & Gibbon, M. (1987). *Structured Clinical Interview for DSMIV (SCID)*. New York State Psychiatric Institute.
- Sporns, O. (2020). The connectome. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 113–121). Oxford University Press.
- SPRINT Mind Investigators, Williamson, J., Pajewski, N., Auchus, A., Bryan, R., Chelune, G., Cheung, A., Cleveland, M., Coker, L., & Crowe, M. (2019). Effect of intensive vs. standard blood pressure control on probable dementia: A randomized clinical trial. *JAMA*, 321(6), 553–561.
- Stack, S. (2021). Contributing factors to suicide: Political, social, cultural and economic. *Prevent Med*, 152, 106498. <https://doi.org/https://doi.org/10.1016/j.ypmed.2021.106498>
- Stafford, M. R., Mayo-Wilson, E., Loucas, C. E., et al. (2015). Efficacy and safety of pharmacological and psychological interventions for the treatment of psychosis and schizophrenia in children, adolescents and young adults: A systematic review and meta-analysis. *PLoS ONE*, 10(2), e0117166. <https://doi.org/10.1371/journal.pone.0117166>
- Stalker, K., & McArthur, K. (2012). Child abuse, child protection and disabled children: A review of recent research. *Child Abuse Rev*, 21(1), 24–40.
- Stallard, P. J. E., Ukraintseva, S. V., Doraiswamy, P. M. (2025). Changing story of the dementia epidemic. *J Am Med Assoc*, 333, 1579–1580.
- Stamelou, M., Respondek, G., Giagkou, N., et al. (2021). Evolving concepts in progressive supranuclear palsy and other 4-repeat tauopathies. *Nat Rev Neurol*, 17(10), 601–620. doi: 10.1038/s41582-021-00541-5
- Stanek, G., Wormser, G. P., Gray, J., & Strle, F. (2012). Lyme borreliosis. *Lancet*, 379(9814), 461–473.
- Stanghellini, G., Broome, M., Raballo, A., et al. (Eds.). (2019). *The Oxford Handbook of Phenomenological Psychopathology*. Oxford University Press.
- Stanghellini, G., & Broome, M. R. (2014). Psychopathology as the basic science of psychiatry. *BJPsych*, 205(3), 169–170.
- Staniloiu, A., & Markowitsch, H. J. (2014). Dissociative amnesia. *Lancet Psychiatry*, 1(3), 226–241.
- Statista. (2022). *Crime in the UK: Statistics and facts*. <https://www.statista.com/topics/3793/crime-in-the-uk/#topicOverview>
- Stedal, K., Broomfield, C., Hay, P., Touyz, S., & Scherer, R. (2021). Neuropsychological functioning in adult anorexia nervosa: A meta-analysis. *Neurosci Biobehav Rev*, 130, 214–226. <https://doi.org/10.1016/j.neubiorev.2021.08.021>
- Steadman, H. J., Gounis, K., Dennis, D., et al. (2001). Assessing the New York City involuntary outpatient commitment pilot program. *Psychiatr Serv*, 52, 330–336. <http://dx.doi.org/10.1176/appi.ps.52.3.330>
- Stein, A., Pearson, R. M., Goodman, S. H., et al. (2014). Effects of perinatal mental disorders on the fetus and child. *Lancet*, 384(9956), 1800–1819.
- Stein, A. T., Carl, E., Cuijpers, P., et al. (2021). Looking beyond depression: A meta-analysis of the effect of behavioral activation on depression, anxiety, and activation. *Psychol Med*, 51(9), 1491–1504.
- Stein, D. J., Shoptaw, S. J., Vigo, D. V., et al. (2022). Psychiatric diagnosis and treatment in the 21st century: Paradigm shifts versus incremental integration. *World Psychiatry*, 21(3), 393–414. <https://doi.org/10.1002/wps.20998>
- Stein, D. J., Szatmari, P., Gaebel, W., et al. (2020). Mental, behavioral and neurodevelopmental disorders in the ICD-11: An international perspective on key changes and controversies. *BMC Med*, 18(1), 1–24.
- Stein, L. I., & Test, M. A. (1980). Alternative to mental hospital treatment. I. Conceptual model, treatment program, and clinical evaluation. *Arch Gen Psychiatry*, 37(4), 392–397. <https://doi.org/10.1001/archpsyc.1980.01780170034003>
- Steinbrecher, N., Koerber, S., Frieser, D., & Hiller, W. (2011). The prevalence of medically unexplained symptoms in primary care. *Psychosomatics*, 52(3), 263–271.
- Steingard, R., Taskiran, S., Connor, D. F., et al. (2019). New formulations of stimulants: An update for clinicians. *J Child Adolesc Psychopharmacol*, 29(5), 324–339.
- Stengel, E. (1959). Classification of mental disorders. *Bull World Health Organ*, 21, 601–603.
- Stephan, K. E., Friston, K. J., & Frith, C. D. (2009). Dysconnection in schizophrenia: From abnormal synaptic plasticity to failures of self-monitoring. *Schizophr Bull*, 35(3), 509–527.
- Stiffel, M., Bergeron, D., Mourabit Amari, K., et al. (2022). Use of Alzheimer's disease cerebrospinal fluid biomarkers in a tertiary care memory clinic. *Can J Neurol Sci*, 49(2), 203–209. <https://doi.org/10.1017/cjn.2021.67>
- Stinton, C., McKeith, I., Taylor, J.-P., et al. (2015). Pharmacological management of Lewy body dementia: A systematic review and meta-analysis. *Am J Psychiatry*, 172(8), 731–742.
- Stockings, E., Hall, W. D., Lynskey, M., et al. (2016). Prevention, early intervention, harm reduction, and treatment of substance use in young people. *Lancet Psychiatry*, 3(3), 280–296.
- Stöckl, H., Dekel, B., Morris-Gehring, A., et al. (2017). Child homicide perpetrators worldwide: A systematic review. *BMJ Paediatr Open*, 1(1), e000112. <https://doi.org/10.1136/bmjpo-2017-000112>
- Stockmeier, C. A., & Rajkowska, G. (2004). Cellular abnormalities in depression: Evidence from postmortem brain tissue. *Dialogues Clin Neurosci*, 6(2), 185–197.
- Stolte, M., Hodgetts, S., & Smith, V. (2016). A critical review of outcome measures used to evaluate the effectiveness of comprehensive, community-based treatment for young children with ASD. *Res Autism Spectr Disord*, 23, 221–234. <https://doi.org/10.1016/j.rasd.2015.12.009>
- Stone, M. H., Hurt, S. W., & Stone, D. K. (1987). The PI 500: Long-term follow-up of borderline inpatients meeting DSM-III Criteria I. Global outcome. *J Personal Disord*, 1, 291–298.
- Strang, J., Hall, W., Hickman, M., & Bird, S. M. (2010). Impact of supervision of methadone consumption on deaths related to methadone overdose (1993–2008): Analyses using OD4 index in England and Scotland. *BMJ*, 341, 640.

- Strathdee, G. (1987). Primary care–psychiatry interaction: A British perspective. *Gen Hosp Psychiatry*, 9(2), 102–110. [https://doi.org/10.1016/0163-8343\(87\)90021-1](https://doi.org/10.1016/0163-8343(87)90021-1)
- Strauss, B., Gawlytta, R., Schleu, A., & Frenzl, D. (2021). Negative effects of psychotherapy: Estimating the prevalence in a random national sample. *BJPsych Open*, 7(6), e186, Article e186. <https://doi.org/10.1192/bjo.2021.1025>
- Strömberg, E. (1985). World-wide issues in psychiatric diagnosis and classification and the Scandinavian point of view. In *Mental Disorders, Alcohol and Drug Related Problems: International Perspectives on Their Diagnosis and Classification* (pp. 3–12). Elsevier Science.
- Stroup, T. S., & Gray, N. (2018). Management of common adverse effects of antipsychotic medications. *World Psychiatry*, 17(3), 341–356. <https://doi.org/10.1002/wps.20567>
- Stroup, T. S., Olsson, M., Huang, C., et al. (2021). Age-specific prevalence and incidence of dementia diagnoses among older US adults with schizophrenia. *JAMA Psychiatry*, 78(6), 632–641. <https://doi.org/10.1001/jamapsychiatry.2021.0042>
- Strydom, A., Hassiotis, A., King, M., & Livingston, G. (2009). The relationship of dementia prevalence in older adults with intellectual disability (ID) to age and severity of ID. *Psychol Med*, 39(01), 13–21.
- Stulz, N., Wyder, L., Maeck, L., et al. (2020). Home treatment for acute mental healthcare: Randomised controlled trial. *BJPsych*, 216(6), 323–330.
- Styron, W. (2001). *Darkness Visible: A Memoir of Madness*. Vintage.
- Su, T., Gong, J., Tang, G., et al. (2021). Structural and functional brain alterations in anorexia nervosa: A multimodal meta-analysis of neuroimaging studies. *Hum Brain Mapp*, 42, 5154–5169. <https://doi.org/10.1002/hbm.25602>
- Substance Abuse and Mental Health Services Administration. (2020). *Key Substance Use and Mental Health Indicators in the United States: Results from the 2019 National Survey on Drug Use and Health (HHS Publication No. PEP 20–07-01-001, NSDUH Series H-55)*. Center for Behavioral Health Statistics and Quality, Substance Abuse and Mental Health Services Administration. <https://www.samhsa.gov/data/>
- Sullivan, E. V., & Pfefferbaum, A. (2019). Brain-behavior relations and effects of aging and common comorbidities in alcohol use disorder: A review. *Neuropsychol*, 33(6), 760.
- Sullivan, P. F., Kendler, K. S., & Neale, M. C. (2003). Schizophrenia as a complex trait: Evidence from a meta-analysis of twin studies. *Arch Gen Psychiatry*, 60(12), 1187–1192.
- Sumnall, H. (2022). Recreational use of nitrous oxide. *BMJ*, 378, o2297.
- Sun, S., Qian, J., Wang, F., et al. (2023). Impact of contact with the baby following stillbirth on parental mental health and well-being: A systematic review and meta-analysis. *Int J Nurs Pract*, 29(6), e13146. <https://doi.org/10.1111/ijn.13146>
- Surtees, A. D. R., Oliver, C., Jones, C. A., et al. (2018). Sleep duration and sleep quality in people with and without intellectual disability: A meta-analysis. *Sleep Med Rev*, 40, 135–150. <https://doi.org/10.1016/j.smrv.2017.11.003>
- Surtees, P. G., Wainwright, N. W., Luben, R. N., et al. (2008). Depression and ischemic heart disease mortality: Evidence from the EPIC-Norfolk United Kingdom prospective cohort study. *Am J Psychiatry*, 165(4), 515–523.
- Sutton, T. E. (2019). Review of attachment theory: Familial predictors, continuity and change, and intrapersonal and relational outcomes. *Marriage Fam Rev*, 55(1), 1–22. <https://doi.org/10.1080/01494929.2018.1458001>
- Svaldi, J., Schmitz, F., Baur, J., et al. (2019). Efficacy of psychotherapies and pharmacotherapies for Bulimia nervosa. *Psychol Med*, 49, 898–910. <https://doi.org/10.1017/S0033291718003525>
- Swain, R. S., Taylor, L. G., Braver, E. R., et al. (2019). A systematic review of validated suicide outcome classification in observational studies. *Int J Epidemiol*, 48(5), 1636–1649. <https://doi.org/10.1093/ije/dyz038>
- Swanson, J. M. (2019). Debate: Are stimulant medications for attention-deficit/hyperactivity disorder effective in the long term? (Against). *J Am Acad Child Adolesc Psychiatry*, 58(10), 936–938.
- Swartz, M. S., Swanson, J. W., Wagner, H. R., et al. (1999). Can involuntary outpatient commitment reduce hospital recidivism? Findings from a randomized trial with severely mentally ill individuals. *Am J Psychiatry*, 156, 1968–1975.
- Swendsen, J., Burstein, M., Case, B., et al. (2012). Use and abuse of alcohol and illicit drugs in US adolescents: Results of the National Comorbidity Survey–Adolescent Supplement. *Arch Gen Psychiatry*, 69(4), 390–398.
- Szasz, T. (1960). The myth of mental illness. *Am Psychologist*, 15, 113–118.
- Szmukler, G., & Appelbaum, P. S. (2008). Treatment pressures, leverage, coercion, and compulsion in mental health care. *J Ment Health*, 17(3), 233–244. <https://doi.org/10.1080/09638230802052203>
- Szuhany, K. L., Malgaroli, M., Miron, C. D., & Simon, N. M. (2021). Prolonged grief disorder: Course, diagnosis, assessment, and treatment. *Focus*, 19(2), 161–172. <https://doi.org/10.1176/appi.focus.20200052>
- Taddei, R. N., & Duff, K. E. (2025). Synapse vulnerability and resilience across the clinical spectrum of dementias. *Nat Rev Neurol*, 21, 353–369.
- Taipale, H., Lähteenhuo, M., Tanskanen, A., et al. (2021). Comparative effectiveness of antipsychotics for risk of attempted or completed suicide among persons with schizophrenia. *Schizophr Bull*, 47(1), 23–30. <https://doi.org/10.1093/schbul/sbaa111>
- Taipale, H., Schneider-Thoma, J., Pinzón-Espinosa, J., et al. (2022). Representation and outcomes of individuals with schizophrenia seen in everyday practice who are ineligible for randomized clinical trials. *JAMA Psychiatry*, 79(3), 210–218. <https://doi.org/10.1001/jamapsychiatry.2021.3990>
- Talbot, P., Astbury, G., & Mason, T. (2010). *Key Concepts in Learning Disabilities*. Sage.
- Talbott, J. A., Jefferies, M., & Arana, J. D. (1987). CMHC's: Relationships with academia and the state. *Community Ment Health J*, 23(4), 271–281. <https://doi.org/10.1007/BF00769839>
- Tam, V., Patel, N., Turcotte, M., et al. (2019). Benefits and limitations of genome-wide association studies. *Nat Rev Genet*, 20(8), 467–484. <https://doi.org/10.1038/s41576-019-0127-1>
- Tampi, R. R., & Jeste, D. V. (2022). Dementia is more than memory loss: Neuropsychiatric symptoms of dementia and their

- nonpharmacological and pharmacological management. *Am J Psychiatry*, 179(8), 528–543.
- Tansella, M. (2002). The scientific evaluation of mental health treatments: An historical perspective*. *Evid Based Ment Health*, 5(1), 4–6.
- Tao, Y., Li, H., Li, L., et al. (2022). Comparing the efficacy of pharmacological and psychological treatment, alone and in combination, in children and adolescents with obsessive-compulsive disorder: A network meta-analysis. *Am J Res*, 148, 95–102.
- Taquet, M., Dercon, Q., Todd, J. A., & Harrison, P. J. (2024). The recombinant shingles vaccine is associated with lower risk of dementia. *Nature Med*, 30, 2777–2781.
- Taquet, M., Geddes, J. R., Luciano, S., & Harrison, P. J. (2021). Incidence and outcomes of eating disorders during the COVID-19 pandemic. *Br J Psychiatry*, 220, 262–264. <https://doi.org/10.1192/bjp.2021.105>
- Taquet, M., Sillett, R., Zhu, L., et al. (2022). Neurological and psychiatric risk trajectories after SARS-CoV-2 infection: An analysis of 2-year retrospective cohort studies including 1 284 437 patients. *Lancet Psychiatry*, 9(10), 815–827. [https://doi.org/10.1016/s2215-0366\(22\)00260-7](https://doi.org/10.1016/s2215-0366(22)00260-7)
- Tarrier, N., Yusupoff, L., Kinney, C., et al. (1998). Randomised controlled trial of intensive cognitive behaviour therapy for patients with chronic schizophrenia. *BMJ*, 317(7154), 303–307.
- Taylor, D. (2008). Psychoanalytic and psychodynamic therapies for depression: The evidence base. *Adv Psychiatr Treat*, 14(6), 401–413.
- Taylor, D., Meader, N., Bird, V., et al. (2011). Pharmacological interventions for people with depression and chronic physical health problems: Systematic review and meta-analyses of safety and efficacy. *BJPsych*, 198(3), 179–188.
- Taylor, D. B., Barnes, T. R. E., & Young, A. H. (2021). *The Maudsley Prescribing Guidelines in Psychiatry*, 14th edn. John Wiley and Sons.
- Taylor, F. (1981). On pseudo-hallucinations. *Psychol Med*, 11, 265–272.
- Taylor, J. E., & Harvey, S. T. (2010). A meta-analysis of the effects of psychotherapy with adults sexually abused in childhood. *Clin Psychol Rev*, 30(6), 749–767.
- Taylor, J. L., Thorne, I., Robertson, A., & Avery, G. (2002). Evaluation of a group intervention for convicted arsonists with mild and borderline intellectual disabilities. *Crim Behav Ment Health*, 12(4), 282–293. <https://doi.org/10.1002/cbm.506>
- Taylor, M., & Perera, U. (2015). NICE CG178—Psychosis and schizophrenia in adults: Treatment and management—an evidence-based guideline? *BJPsych*, 206(5), 357–359.
- Taylor, P., & Dunn, E. (2009). Management of offenders with mental disorder in specialist forensic mental health services. In M. Gelder, N. Andreasen, J. Lopez-Ibor, Jr., & J. R. Geddes (Eds.), *New Oxford Textbook of Psychiatry*, 2nd edn. Oxford University Press.
- Taylor, P., Mahendra, B., & Gunn, J. (1983). Erotomania in males. *Psychol Med*, 13(3), 645–650.
- Teunissen, C. E., Verberk, I. M., Thijssen, E. H., et al. (2022). Blood-based biomarkers for Alzheimer's disease: Towards clinical implementation. *Lancet Neurol*, 21(1), 66–77.
- Tevis, C., & Matson, J. L. (2022). Challenging behaviour in children with developmental disabilities: An overview of behavioural assessment and treatment methods. *Br J Psych Advances*, 28(6), 401–409. <https://doi.org/10.1192/bja.2022.59>
- Thapar, A., Eyre, O., Patel, V., & Brent, D. (2022). Depression in young people. *Lancet*, 400(10352), 617–631.
- Thapar, A., Pine, D. S., Leckman, J. F., et al. (Eds.). (2015). *Rutter's Child and Adolescent Psychiatry*, 6th edn. John Wiley & Sons.
- Thibaut, F., Cosyns, P., Fedoroff, J. P., et al. (2020). The World Federation of Societies of Biological Psychiatry (WFSBP) 2020 guidelines for the pharmacological treatment of paraphilic disorders. *World J Biol Psychiatry*, 21, 412–490. <https://doi.org/10.1080/15622975.2020.1744723>
- Thiebaut de Schotten, M., Foulon, C., & Nachev, P. (2020). Brain disconnections link structural connectivity with function and behaviour. *Nature Comm*, 11(1), 5094.
- Thijs, R. D., Surges, R., O'Brien, T. J., & Sander, J. W. (2019). Epilepsy in adults. *Lancet*, 393(10172), 689–701.
- Thom, R. P., Levy-Carrick, N. C., Bui, M., & Silbersweig, D. (2019). Delirium. *Am J Psychiatry*, 176(10), 785–793. <https://doi.org/10.1176/appi.ajp.2018.18070893>
- Thomas, A. (2020). Depression in older people. In T. Denning, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 595–622). Oxford University Press.
- Thornicroft, G., Farrelly, S., Szumukler, G., et al. (2013). Clinical outcomes of Joint Crisis Plans to reduce compulsory treatment for people with psychosis: A randomised controlled trial. *Lancet*, 381(9878), 1634–1641. [https://doi.org/10.1016/S0140-6736\(13\)60105-1](https://doi.org/10.1016/S0140-6736(13)60105-1)
- Thornicroft, G., Sunkel, C., Alikhon Aliev, A., et al. (2022). The Lancet Commission on ending stigma and discrimination in mental health. *Lancet*, 400(10361), 1438–1480. [https://doi.org/10.1016/s0140-6736\(22\)01470-2](https://doi.org/10.1016/s0140-6736(22)01470-2)
- Thornicroft, G., & Tansella, M. (2013). The balanced care model: The case for both hospital-and community-based mental healthcare. *BJPsych*, 202(4), 246–248.
- Thurber, K. A., Brinckley, M.-M., Jones, R., et al. (2022). Population-level contribution of interpersonal discrimination to psychological distress among Australian Aboriginal and Torres Strait Islander adults, and to Indigenous–non-Indigenous inequities: Cross-sectional analysis of a community-controlled First Nations cohort study. *Lancet*, 400(10368), 2084–2094.
- Tienari, P., Wynne, L. C., Läksy, K., et al. (2003). Genetic boundaries of the schizophrenia spectrum: Evidence from the Finnish Adoptive Family Study of Schizophrenia. *Am J Psychiatry*, 160(9), 1587–1594.
- Tiihonen, J., Mittendorfer-Rutz, E., Majak, M., et al. (2017). Real-world effectiveness of antipsychotic treatments in a nationwide cohort of 29 823 patients with schizophrenia. *JAMA Psychiatry*, 74(7), 686–693. <https://doi.org/10.1001/jamapsychiatry.2017.1322>
- Tiley, C., & Kyriakopoulos, M. (2021). The prevalence of autism spectrum traits and autism spectrum disorders in children and adolescents with obsessive compulsive disorder: A systematic review. *BJPsych Open*, 7(S1), S56–S56.
- Tjak, A., Davis, M. L., Morina, N., et al. (2015). A meta-analysis of the efficacy of acceptance and commitment therapy for clinically relevant mental and physical health problems. *Psychother Psychosom*, 84(1), 30–36. <https://doi.org/10.1159/000365764>

- Tol, W. A., Barbui, C., Galappatti, A., et al. (2011). Mental health and psychosocial support in humanitarian settings: Linking practice and research. *Lancet*, 378(9802), 1581–1591.
- Tondo, L., Vázquez, G., & Baldessarini, R. (2020). Melancholic versus nonmelancholic major depression compared. *J Affect Disord*, 266, 760–765.
- Tonge, B. J., Bull, K., Brereton, A., & Wilson, R. (2014). A review of evidence-based early intervention for behavioural problems in children with autism spectrum disorder: The core components of effective programs, child-focused interventions and comprehensive treatment models. *Curr Opin Psychiatry*, 27(2), 158–165. <https://doi.org/10.1097/ycp.0000000000000043>
- Torgersen, S., Myers, J., Reichborn-Kjennerud, T., et al. (2012). The heritability of Cluster B personality disorders assessed both by personal interview and questionnaire. *J Personal Disord*, 26(6), 848.
- Torok, M., Calear, A. L., Smart, A., et al. (2019). Preventing adolescent suicide: A systematic review of the effectiveness and change mechanisms of suicide prevention gatekeeping training programs for teachers and parents. *J Adolesc*, 73, 100–112. <https://doi.org/10.1016/j.adolescence.2019.04.005>
- Torous, J., Bucci, S., Bell, I. H., et al. (2021). The growing field of digital psychiatry: current evidence and the future of apps, social media, chatbots, and virtual reality. *World Psychiatry*, 20(3), 318–335. <https://doi.org/10.1002/wps.20883>
- Torrey, E. F., & Yolken, R. H. (2010). Psychiatric genocide: Nazi attempts to eradicate schizophrenia. *Schizophr Bull*, 36(1), 26–32.
- Tørring, N., Sanghani, S., Petrides, G., et al. (2017). The mortality rate of electroconvulsive therapy: A systematic review and pooled analysis. *Acta Psychiatr Scand*, 135(5), 388–397.
- Townsend, L., Pillinger, T., Selvaggi, P., et al. (2023). Brain glucose metabolism in schizophrenia: A systematic review and meta-analysis of (18)FDG-PET studies in schizophrenia. *Psychol Med*, 53(11), 4880–4897. <https://doi.org/10.1017/s00332917200174x>
- Treasure, J. (2004). Motivational interviewing. *Adv Psychiatr Treat*, 10(5), 331–337.
- Treasure, J., Duarte, T. A., & Schmidt, U. (2020). Eating disorders. *Lancet*, 395, 899–911. [https://doi.org/10.1016/S0140-6736\(20\)30059-3](https://doi.org/10.1016/S0140-6736(20)30059-3)
- Trim, R. S., Schuckit, M. A., & Smith, T. L. (2013). Predictors of initial and sustained remission from alcohol use disorders: Findings from the 30-year follow-up of the San Diego Prospective Study. *Alcohol Clin Exp Res*, 37(8), 1424–1431.
- Trubetskoy, V., Pardiñas, A. F., Qi, T., et al. (2022). Mapping genomic loci implicates genes and synaptic biology in schizophrenia. *Nature*, 604(7906), 502–508. <https://doi.org/10.1038/s41586-022-04434-5>
- True, W. R., Rice, J., Eisen, S. A., et al. (1993). A twin study of genetic and environmental contributions to liability for posttraumatic stress symptoms. *Arch Gen Psychiatry*, 50(4), 257–264.
- Tschorn, M., Lorenz, R. C., O'Reilly, P. F., et al. (2021). Differential predictors for alcohol use in adolescents as a function of familial risk. *Transl Psychiatry*, 11(1), 157.
- Tseng, W.-S. (1999). Culture and psychotherapy: Review and practical guidelines. *Transcultur Psychiatry*, 36(2), 131–179. <https://doi.org/10.1177/136346159903600201>
- Tseng, W.-S. (2006). From peculiar psychiatric disorders through culture-bound syndromes to culture-related specific syndromes. *Transcultur Psychiatry*, 43(4), 554–576.
- Tundo, A., Musetti, L., Del Grande, C., et al. (2021). The relationship between depression with anxious distress DSM-5 specifier and mixed depression: A network analysis. *CNS Spectr*, 26(3), 251–257.
- Turecki, G., Brent, D. A., Gunnell, D., et al. (2019). Suicide and suicide risk. *Nat Rev Dis Primers*, 5(1), 74. <https://doi.org/10.1038/s41572-019-0121-0>
- Turkington, D., Kingdon, D., Rathod, S., et al. (2006). Outcomes of an effectiveness trial of cognitive-behavioural intervention by mental health nurses in schizophrenia. *BJPsych*, 189, 36–40. <https://doi.org/10.1192/bjp.bp.105.010884>
- Turkington, D., Spencer, H., Jassal, I., & Cummings, A. (2015). Cognitive behavioural therapy for the treatment of delusional systems. *Psychosis*, 7(1), 48–59. <https://doi.org/10.1080/17522439.2014.899385>
- Turner, D., Petermann, J., Harrison, K., Krueger, R., & Briken, P. (2019). Pharmacological treatment of patients with paraphilic disorders and risk of sexual offending: An international perspective. *World J Biol Psychiatry*, 20, 616–625. <https://doi.org/10.1080/15622975.2017.1395069>
- Turner, D. T., van der Gaag, M., Karyotaki, E., & Cuijpers, P. (2014). Psychological interventions for psychosis: a meta-analysis of comparative outcome studies. *Am J Psychiatry*, 171(5), 523–538.
- Tyler, K. L. (2018). Acute viral encephalitis. *N Engl J Med*, 379(6), 557–566.
- Tyler, N., & Barnoux, M.-F. (2022). Adult perpetrated firesetting. In C. Garofalo & J. J. Sijtsma (Eds.), *Clinical Forensic Psychology: Introductory Perspectives on Offending* (pp. 397–413). Springer International Publishing. https://doi.org/10.1007/978-3-030-80882-2_21
- Tyrer, P. (2013). *Models for Mental Disorder* (5th ed.). Wiley-Blackwell.
- Tyrer, P., & Baldwin, D. (2006). Generalised anxiety disorder. *Lancet*, 368(9553), 2156–2166.
- Tyrer, P., & Bateman, A. W. (2004). Drug treatment for personality disorders. *Adv Psychiatr Treat*, 10(5), 389–398. <https://doi.org/10.1192/apt.10.5.389>
- Tyrer, P., & Methuen, C. (2007). *Rating Scales in Psychiatry*. Royal College of Psychiatrists Publications.
- Tyrer, P., Mulder, R., Kim, Y. R., & Crawford, M. J. (2019). The development of the ICD-11 Classification of Personality Disorders: An amalgam of science, pragmatism, and politics. *Annu Rev Clin Psychol*, 15, 481–502. <https://doi.org/10.1146/annurev-clinpsy-050718-095736>
- Tyrer, P., Reed, G. M., & Crawford, M. J. (2015). Classification, assessment, prevalence, and effect of personality disorder. *Lancet*, 385(9969), 717–726.
- Tyrer, P., Seivewright, H., & Johnson, T. (2004). The Nottingham Study of Neurotic Disorder: Predictors of 12-year outcome of dysthymic, panic and generalized anxiety disorder. *Psychol Med*, 34(8), 1385–1394.
- Tyrer, P., Tyrer, H., Yang, M., & Guo, B. (2016). Long-term impact of temporary and persistent personality disorder on anxiety and depressive disorders. *Personal Ment Health*, 10(2), 76–83. <https://doi.org/10.1002/pmh.1324>

- Tyrer, R. A., & Fazel, M. (2014). School and community-based interventions for refugee and asylum-seeking children: A systematic review. *PLoS ONE*, 9(2), e89359. <https://doi.org/10.1371/journal.pone.0089359> [pii]
- Uhlhaas, P. J., McGorry, P. D., & Wood, S. J. (2021). Toward a paradigm for youth mental health. *JAMA Psychiatry*, 78(5), 473–474. <https://doi.org/10.1001/jamapsychiatry.2020.3905>
- UKATT Research Team. (2005). Effectiveness of treatment for alcohol problems: Findings of the randomised UK alcohol treatment trial (UKATT). *BMJ*, 331(7516), 541.
- UK ECT Review Group. (2003). Efficacy and safety of electroconvulsive therapy in depressive disorders: A systematic review and meta-analysis. *Lancet*, 361(9360), 799–808.
- United Nations. (2020). Selected results of the 2019 UN World Population projections. *Popul Dev Rev*, 45, 687–694.
- Unterholzner, J., Millischer, V., Wotawa, C., et al. (2021). Making sense of patient-derived iPSCs, transdifferentiated neurons, olfactory neuronal cells, and cerebral organoids as models for psychiatric disorders. *Int J Neuropsychopharmacol*, 24(10), 759–775. <https://doi.org/10.1093/ijnp/pyab037>
- Upthegrove, R., Chard, C., Jones, L., et al. (2015). Adverse childhood events and psychosis in bipolar affective disorder. *BJPsych*, 206(3), 191–197.
- Ursini, G., Punzi, G., Chen, Q., et al. (2018). Convergence of placenta biology and genetic risk for schizophrenia. *Nat Med*, 24(6), 792–801. <https://doi.org/10.1038/s41591-018-0021-y>
- Vaillant, G. E. (2003). A 60-year follow-up of alcoholic men. *Addiction*, 98(8), 1043–1051.
- Valkanova, V., & Ebmeier, K. P. (2013). Vascular risk factors and depression in later life: A systematic review and meta-analysis. *Biol Psychiatry*, 73(5), 406–413.
- Van Alphen, S., Derksen, J., Sadavoy, J., & Rosowsky, E. (2012). Features and challenges of personality disorders in late life. *Ageing Ment Health*, 16(7), 805–810.
- van de Wouw, E., Evenhuis, H., & Echteld, M. (2012). Prevalence, associated factors and treatment of sleep problems in adults with intellectual disability: A systematic review. *Res Develop Disabil*, 33(4), 1310–1332.
- van den Brink, W., & Kiefer, F. (2020). Alcohol use disorder. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 498–506). Oxford University Press.
- van den Eynde, V., Abdelmoemin, W. R., Abraham, M. M., et al. (2023). The prescriber's guide to classic MAO inhibitors (phenelzine, tranylcypromine, isocarboxazid) for treatment-resistant depression. *CNS Spect*, 28(4), 427–440.
- van der Naalt, J., Timmerman, M. E., de Koning, M. E., et al. (2017). Early predictors of outcome after mild traumatic brain injury (UPFRONT): An observational cohort study. *Lancet Neurol*, 16(7), 532–540.
- van der Zweerde, T., Bisdounis, L., Kyle, S. D., et al. (2019). Cognitive behavioral therapy for insomnia: A meta-analysis of long-term effects in controlled studies. *Sleep Med Rev*, 48, 101208.
- van Dis, E. A. M., van Veen, S. C., Hagenars, M. A., et al. (2020). Long-term outcomes of cognitive behavioral therapy for anxiety-related disorders: A systematic review and meta-analysis. *JAMA Psychiatry*, 77(3), 265–273. <https://doi.org/10.1001/jamapsychiatry.2019.3986>
- van Duijn, E. (2017). Medical treatment of behavioral manifestations of Huntington disease. *Clin Neurol*, 144, 129–139.
- van Kessel, K., Moss-Morris, R., Willoughby, E., et al. (2008). A randomized controlled trial of cognitive behavior therapy for multiple sclerosis fatigue. *Psychosomat Med*, 70(2), 205–213.
- Van Os, J., Hanssen, M., Bak, M., et al. (2003). Do urbanicity and familial liability coparticipate in causing psychosis? *Am J Psychiatry*, 160(3), 477–482.
- Vancampfort, D., Firth, J., Correll, C. U., et al. (2019). The impact of pharmacological and non-pharmacological interventions to improve physical health outcomes in people with schizophrenia: A meta-review of meta-analyses of randomized controlled trials. *World Psychiatry*, 18(1), 53–66. <https://doi.org/10.1002/wps.20614>
- Vancampfort, D., Stubbs, B., Mitchell, A. J., et al. (2015). Risk of metabolic syndrome and its components in people with schizophrenia and related psychotic disorders, bipolar disorder and major depressive disorder: A systematic review and meta-analysis. *World Psychiatry*, 14(3), 339–347.
- Vassos, E., Pedersen, C. B., Murray, R. M., et al. (2012). Meta-analysis of the association of urbanicity with schizophrenia. *Schizophr Bull*, 38(6), 1118–1123. <https://doi.org/10.1093/schbul/sbs096>
- Vaughn, M. G., Salas-Wright, C. P., DeLisi, M., & Larson, M. (2015). Deliberate self-harm and the nexus of violence, victimization, and mental health problems in the United States. *Psychiatry Res*, 225(3), 588–595.
- Veale, D. (2007). Cognitive-behavioural therapy for obsessive-compulsive disorder. *Adv Psychiatr Treat*, 13(6), 438–446.
- Veale, D., Miles, S., Smallcombe, N., et al. (2014). Atypical antipsychotic augmentation in SSRI treatment refractory obsessive-compulsive disorder: A systematic review and meta-analysis. *BMC Psychiatry*, 14(1), 1–13.
- Velayudhan, L., Ryu, S.-H., Raczek, M., et al. (2014). Review of brief cognitive tests for patients with suspected dementia. *Int Psychogeriatr*, 26(8), 1247–1262.
- Veltman, D. (2021). Structural imaging in OCD. In N. A. Fineberg & T. W. Robbins (Eds.), *The Neurobiology and Treatment of OCD: Accelerating Progress* (pp. 201–229). Springer.
- Velupillai, S., Hadlaczky, G., Baca-García, E., et al. (2019). Risk assessment tools and data-driven approaches for predicting and preventing suicidal behavior. *Front Psychiatry*, 10, 36.
- Vergani, F., Landi, A., Pirillo, D., et al. (2010). Surgical, medical, and hardware adverse events in a series of 141 patients undergoing subthalamic deep brain stimulation for Parkinson disease. *World Neurosurg*, 73(4), 338–344.
- Verkuijl, N., Perkins, M., & Fazel, M. (2015). Childhood attention-deficit/hyperactivity disorder. *BMJ*, 350, h2168. <https://doi.org/10.1136/bmj.h2168>
- Vicens, V., Radua, J., Salvador, R., et al. (2016). Structural and functional brain changes in delusional disorder. *BJPsych*, 208(2), 153–159.
- Victor, M., Adams, R. D., & Collins, G. H. (1971). The Wernicke-Korsakoff syndrome. A clinical and pathological study of 245 patients, 82 with post-mortem examinations. *Contemp Neurol Ser*, 7: 1–206.
- Vigod, S. N., Wilson, C. A., & Howard, L. M. (2016). Depression in pregnancy. *BMJ*, 352, i1547.

- Villa, R. F., Ferrari, F., & Moretti, A. (2018). Post-stroke depression: mechanisms and pharmacological treatment. *Pharmacol Therapeut*, 184, 131–144.
- Visscher, P. M., Hill, W. G., & Wray, N. R. (2008). Heritability in the genomics era—concepts and misconceptions. *Nat Rev Genet*, 9(4), 255–266.
- Vissers, L. E., Gilissen, C., & Veltman, J. A. (2016). Genetic studies in intellectual disability and related disorders. *Nat Rev Genet*, 17, 9–18.
- Vita, A., Barlati, S., Ceraso, A., et al. (2021). Effectiveness, core elements, and moderators of response of cognitive remediation for schizophrenia: A systematic review and meta-analysis of randomized clinical trials. *JAMA Psychiatry*, 78(8), 848–858. <https://doi.org/10.1001/jamapsychiatry.2021.0620>
- Vizard, T., Sadler, K., Ford, T., et al. (2020). *Mental Health of Children and Young People in England, 2020: Wave 1 Follow Up to the 2017 Survey*. National Health Service.
- Vöhringer, P. A., & Perlis, R. H. (2016). Discriminating between bipolar disorder and major depressive disorder. *Psychiatr Clin North Am*, 39(1), 1–10. <https://doi.org/10.1016/j.psc.2015.10.001>
- von Soest, T., Kvalem, I. L., Roald, H. E., & Skolteberg, K. C. (2009). The effects of cosmetic surgery on body image, self-esteem, and psychological problems. *J Plast Reconstr Aesthet Surg*, 62(10), 1238–1244. <https://doi.org/10.1016/j.bjps.2007.12.093>
- Wagenaar, A. C., Tobler, A. L., & Komro, K. A. (2010). Effects of alcohol tax and price policies on morbidity and mortality: a systematic review. *Am J Public Health*, 100(11), 2270–2278.
- Wahbeh, M. H., & Avramopoulos, D. (2021). Gene–environment interactions in schizophrenia: A literature review. *Genes*, 12(12), 1850. <https://doi.org/10.3390/genes12121850>
- Waite, F. (2019). Patients', carers' and the public's perspective on ECT and related treatments. In I. Ferrier & J. Waite (Eds.), *The ECT Handbook* (pp. 267–273). Cambridge University Press.
- Wakefield, J. C. (1992). The concept of mental disorder: on the boundary between biological facts and social values. *Am Psychol*, 47(3), 373.
- Walker, J., Holm Hansen, C., Martin, P., et al. (2013). Prevalence of depression in adults with cancer: A systematic review. *Ann Oncol*, 24(4), 895–900. <https://doi.org/10.1093/annonc/mds575>
- Walker, J., Sawhney, A., Hansen, C. H., et al. (2014). Treatment of depression in adults with cancer: A systematic review of randomized controlled trials. *Psychol Med*, 44(5), 897–907.
- Walker, W. H., 2nd, Walton, J. C., DeVries, A. C., & Nelson, R. J. (2020). Circadian rhythm disruption and mental health. *Transl Psychiatry*, 10, 28. <https://doi.org/10.1038/s41398-020-0694-0>
- Walker, Z., Possin, K. L., Boeve, B. F., & Aarsland, D. (2015). Lewy body dementias. *Lancet*, 386(10004), 1683–1697.
- Walther, S., Stegmayer, K., Wilson, J. E., & Heckers, S. (2019). Structure and neural mechanisms of catatonia. *Lancet Psychiatry*, 6(7), 610–619. [https://doi.org/10.1016/s2215-0366\(18\)30474-7](https://doi.org/10.1016/s2215-0366(18)30474-7)
- Wang, H.-F., Yu, J.-T., Tang, S.-W., et al. (2015). Efficacy and safety of cholinesterase inhibitors and memantine in cognitive impairment in Parkinson's disease, Parkinson's disease dementia, and dementia with Lewy bodies: Systematic review with meta-analysis and trial sequential analysis. *J Neurol Neurosurg Psychiatry*, 86, 135–143.
- Wang, L., Zhou, C., Zhu, D., et al. (2016). Serotonin-1A receptor alterations in depression: A meta-analysis of molecular imaging studies. *BMC Psychiatry*, 16(1), 319. <https://doi.org/10.1186/s12888-016-1025-0>
- Wang, W., Corominas, R., & Lin, G. N. (2019). De novo mutations from whole exome sequencing in neurodevelopmental and psychiatric disorders: From discovery to Application. *Front Genet*, 10, 258. <https://doi.org/10.3389/fgene.2019.00258>
- Watson, C., & Bhugra, D. (2020). Transcultural psychiatric assessment. *Medicine*, 48(11), 3.
- Watson, H. J., Yilmaz, Z., Thornton, L. M., et al. (2019). Genome-wide association study identifies eight risk loci and implicates metabo-psychiatric origins for anorexia nervosa. *Nat Genet*, 51, 1207–1214. <https://doi.org/10.1038/s41588-019-0439-2>
- Watson, J. B., & Rayner, R. (1920). Conditioned emotional reactions. *J Exp Psychol*, 3, 1–14. <https://doi.org/10.1037/h0069608>
- Watts, A. L., Wood, P. K., Jackson, K. M., et al. (2021). Incipient alcohol use in childhood: Early alcohol sipping and its relations with psychopathology and personality. *Dev Psychopathol*, 33(4), 1338–1350.
- Webb, R. T., Kontopantelis, E., Doran, T., et al. (2012). Suicide risk in primary care patients with major physical diseases: A case-control study. *Arch Gen Psychiatry*, 69(3), 256–264.
- Webber, C. (2017). Epistasis in neuropsychiatric disorders. *Trends Genet*, 33(4), 256–265. <https://doi.org/10.1016/j.tig.2017.01.009>
- Webster, M. (2011). Introduction to acceptance and commitment therapy. *Adv Psychiatr Treat*, 17(4), 309–316. <https://doi.org/10.1192/apt.bp.107.005256>
- Wei, J., Hou, R., Zhang, X., et al. (2019). The association of late-life depression with all-cause and cardiovascular mortality among community-dwelling older adults: Systematic review and meta-analysis. *BJPsych*, 215(2), 449–455.
- Weinberger, D. R. (1987). Implications of normal brain development for the pathogenesis of schizophrenia. *Arch Gen Psychiatry*, 44(7), 660–669.
- Weinberger, D. R. (2022). It's dopamine and schizophrenia all over again. *Biol Psychiatry*, 92(10), 757–759. <https://doi.org/10.1016/j.biopsych.2022.08.001>
- Weinberger, D. R., & Radulescu, E. (2016). Finding the elusive psychiatric 'lesion' with 21st-century neuroanatomy: A note of caution. *Am J Psychiatry*, 173(1), 27–33.
- Weinberger, J. M., Houman, J., Caron, A. T., & Anger, J. (2019). Female sexual dysfunction: A systematic review of outcomes across various treatment modalities. *Sex Med Rev*, 7, 223–250. <https://doi.org/10.1016/j.sxmr.2017.12.004>
- Weintraub, D., Aarsland, D., Chaudhuri, K. R., et al. (2022). The neuropsychiatry of Parkinson's disease: Advances and challenges. *Lancet Neurology*, 21(1), 89–102.
- Wells, A. (2013). Advances in metacognitive therapy. *Int J Cognitive Ther*, 6(2), 186–201.
- Werbeloff, N., Chang, C. K., Broadbent, M., et al. (2017). Admission to acute mental health services after contact with crisis resolution and home treatment teams: An investigation in two large mental health-care providers. *Lancet Psychiatry*, 4(1), 49–56. [https://doi.org/10.1016/s2215-0366\(16\)30416-3](https://doi.org/10.1016/s2215-0366(16)30416-3)

- Wertheimer, J. (1997). Psychiatry of the elderly: A consensus statement. *Int J Geriatr Psychiatry*, 12(4), 432–435.
- Wesseloo, R., Kamperman, A. M., Munk-Olsen, T., et al. (2016). Risk of postpartum relapse in bipolar disorder and postpartum psychosis: A systematic review and meta-analysis. *Am J Psychiatry*, 173(2), 117–127.
- Westby, C. L., Erlandsen, A. R., Nilsen, S. A., et al. (2021). Depression, anxiety, PTSD, and OCD after stillbirth: A systematic review. *BMC Preg Childbirth*, 21(1), 782. <https://doi.org/10.1186/s12884-021-04254-x>
- Wheat, K. (2020). The law relating to mental capacity and mental health. In T. Denning, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 907–916). Oxford University Press.
- Whipp, A. M., Korhonen, T., Raevuori, A., et al. (2019). Early adolescent aggression predicts antisocial personality disorder in young adults: A population-based study. *Eur Child Adolesc Psychiatry*, 28(3), 341–350. <https://doi.org/10.1007/s00787-018-1198-9>
- Whitaker, S. (2015). Guidance on the assessment and diagnosis of intellectual disabilities in adulthood: Getting there but still some way to go. *FPID Bull*, 13(3), 16–21.
- White, P., Goldsmith, K., Johnson, A., et al. (2011). Comparison of adaptive pacing therapy, cognitive behaviour therapy, graded exercise therapy, and specialist medical care for chronic fatigue syndrome (PACE): A randomised trial. *Lancet*, 377(9768), 823–836.
- White, P., Rickards, H., & Zeman, A. (2012). Time to end the distinction between mental and neurological illnesses. *BMJ*, 344, e3454.
- White, P. D., Goldsmith, K. A., Johnson, A. L., et al. (2011). Comparison of adaptive pacing therapy, cognitive behaviour therapy, graded exercise therapy, and specialist medical care for chronic fatigue syndrome (PACE): A randomised trial. *Lancet*, 377(9768), 823–836.
- Whiteford, H. A., Degenhardt, L., Rehm, J., et al. (2013). Global burden of disease attributable to mental and substance use disorders: Findings from the Global Burden of Disease Study 2010. *Lancet*, 382(9904), 1575–1586. [https://doi.org/10.1016/s0140-6736\(13\)61611-6](https://doi.org/10.1016/s0140-6736(13)61611-6)
- Whitehurst, T. S., Osugo, M., Townsend, L., et al. (2020). Proton magnetic resonance spectroscopy of N-acetyl aspartate in chronic schizophrenia, first episode of psychosis and high-risk of psychosis: A systematic review and meta-analysis. *Neurosci Biobehav Rev*, 119, 255–267. <https://doi.org/10.1016/j.neubio.2020.10.001>
- Whiting, D., & Cowen, P. J. (2013). Drug information update: Agomelatine. *Psychiatrist*, 37(11), 356–358.
- Whiting, D., Gulati, G., Geddes, J. R., & Fazel, S. (2022). Association of schizophrenia spectrum disorders and violence perpetration in adults and adolescents from 15 countries: A systematic review and meta-analysis. *JAMA Psychiatry*, 79(2), 120–132. <https://doi.org/10.1001/jamapsychiatry.2021.3721>
- Whiting, D., Lichtenstein, P., & Fazel, S. (2021). Violence and mental disorders: A structured review of associations by individual diagnoses, risk factors, and risk assessment. *Lancet Psychiatry*, 8(2), 150–161. [https://doi.org/10.1016/s2215-0366\(20\)30262-5](https://doi.org/10.1016/s2215-0366(20)30262-5)
- Wichstrøm, L., Berg-Nielsen, T. S., Angold, A., et al. (2012). Prevalence of psychiatric disorders in preschoolers. *J Child Psychol Psychiatry*, 53(6), 695–705. <https://doi.org/10.1111/j.1469-7610.2011.02514.x>
- Wickersham, A., Sugg, H. V., Epstein, S., et al. (2021). Systematic review and meta-analysis: The association between child and adolescent depression and later educational attainment. *J Am Acad Child Adolesc Psychiatry*, 60(1), 105–118.
- Wilbur, C., Bitnun, A., Kronenberg, S., et al. (2019). PANDAS/PANS in childhood: Controversies and evidence. *Paediatr Child Health*, 24(2), 85–91.
- Wiles, N., Thomas, L., Abel, A., et al. (2013). Cognitive behavioural therapy as an adjunct to pharmacotherapy for primary care-based patients with treatment resistant depression: results of the CoBaIT randomised controlled trial. *Lancet*, 381(9864), 375–384.
- Wilhelm, S., Phillips, K. A., Didie, E., et al. (2014). Modular cognitive-behavioral therapy for body dysmorphic disorder: A randomized controlled trial. *Behav Ther*, 45(3), 314–327. <https://doi.org/10.1016/j.beth.2013.12.007>
- Wilkinson, P., & Laidla, K. (2020). Psychological treatments In T. Denning, A. Thomas, R. Stewart, & J-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 249–252). Oxford University Press.
- Wilkinson, S. T., Sanacora, G., & Bloch, M. H. (2017). Hippocampal volume changes following electroconvulsive therapy: A systematic review and meta-analysis. *Biol Psychiatry Cogn Neurosci Neuroimaging*, 2(4), 327–335.
- Williams, J. A., Burgess, S., Suckling, J., et al. (2022). Inflammation and brain structure in schizophrenia and other neuropsychiatric disorders: A Mendelian randomization study. *JAMA Psychiatry*, 79(5), 498–507. <https://doi.org/10.1001/jamapsychiatry.2022.0407>
- Williams, M. J., McManus, F., Muse, K., & Williams, J. M. (2011). Mindfulness-based cognitive therapy for severe health anxiety (hypochondriasis): An interpretative phenomenological analysis of patients' experiences. *Br J Clin Psychol*, 50(4), 379–397. <https://doi.org/10.1111/j.2044-8260.2010.02000.x>
- Williams, N. R., Heifets, B. D., Blasey, C., et al. (2018). Attenuation of antidepressant effects of ketamine by opioid receptor antagonism. *Am J Psychiatry*, 175(12), 1205–1215.
- Williamson, A. A., Mindell, J. A., Hiscock, H., & Quach, J. (2019, Nov). Child sleep behaviors and sleep problems from infancy to school-age. *Sleep Med*, 63, 5–8. <https://doi.org/10.1016/j.sleep.2019.05.003>
- Willner, P., Rose, J., Jahoda, A., et al. (2013). Group-based cognitive-behavioural anger management for people with mild to moderate intellectual disabilities: Cluster randomised controlled trial. *BJPsych*, 203(3), 288–296.
- Wilson, J. E., Mart, M. F., Cunningham, C., et al. (2020). Delirium. *Nat Rev Dis Primers*, 6(1), 90.
- Wilson, R. P., & Bhattacharyya, S. (2016). Antipsychotic efficacy in psychosis with co-morbid cannabis misuse: A systematic review. *J Psychopharmacol*, 30(2), 99–111.
- Wilson, S., Anderson, K., Baldwin, D., et al. (2019). British Association for Psychopharmacology consensus statement on evidence-based treatment of insomnia, parasomnias and circadian rhythm disorders: An update. *J Psychopharmacol*, 33(8), 923–947. <https://doi.org/10.1177/0269881119855343>

- Wilson, S., & Pinner, G. (2013). Driving and dementia: A clinician's guide. *Adv Psychiatr Treat*, 19(2), 89–96.
- Wilson, S., Roberts, L., Roalfe, A., et al. (2004). Prevalence of irritable bowel syndrome: A community survey. *Br J Gen Pract*, 54(504), 495–502.
- Wincze, J. P., & Weisberg, R. B. (2015). *Sexual Dysfunction: A Guide for Assessment and Treatment*. Guildford Publications.
- Wing, J., Beevor, A., Curtis, R., et al. (1998). Health of the Nation Outcome Scales (HoNOS): Research and development. *BJPsych*, 172(1), 11–18.
- Wing, J., & Brown, G. (1970). *Institutionalism and Schizophrenia*. Cambridge University Press.
- Wing, J., Cooper, J., & Sartorius, N. (1974). *Measurement and Classification of Psychiatric Symptoms: An Instruction Manual for the PSE and the CATEGO Programme*. Cambridge University Press.
- Wing, J. K., & Furlong, R. (1986). A haven for the severely disabled within the context of a comprehensive psychiatric community service. *BJPsych*, 149(4), 449–457. <https://doi.org/10.1192/bjp.149.4.449>
- Wisper, C. (2018). The aetiology of borderline personality disorder (BPD): Contemporary theories and putative mechanisms. *Curr Opin Psychol*, 21, 105–110. <https://doi.org/10.1016/j.copsyc.2017.10.005>
- Wisper, C., Bilgin, A., Thompson, A., et al. (2020). The prevalence of personality disorders in the community: A global systematic review and meta-analysis. *BJPsych*, 216(2), 69–78. <https://doi.org/10.1192/bjp.2019.166>
- Winstock, A., & Flechais, R. (2020). Stimulants, ecstasy and other 'party drugs'. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 525–538). Oxford University Press.
- Winstock, A., & Rucker, J. (2020). Psychedelics and dissociative substances. In J. Geddes, N. Andreasen, & G. Goodwin (Eds.), *New Oxford Textbook of Psychiatry*, 3rd edn. (pp. 536–549). Oxford University Press.
- Winter, S., Diamond M, Green, J., Karasic, D., Reed, T., Whittle, S., Wylie, K. (2016). Transgender people: Health at the margins of society. *Lancet* 388, 390–400. doi: 10.1016/S0140-6736(16)00683-8.
- Wisniewski, S. R., Rush, A. J., Nierenberg, A. A., et al. (2009). Can phase III trial results of antidepressant medications be generalized to clinical practice? A STAR* D report. *Am J Psychiatry*, 166(5), 599–607.
- Witlox, J., Eurelings, L. S., de Jonghe, J. F., et al. (2010). Delirium in elderly patients and the risk of postdischarge mortality, institutionalization, and dementia: A meta-analysis. *JAMA*, 304(4), 443–451.
- Witt, K. G., Hetrick, S. E., Rajaram, G., et al. (2021). Interventions for self-harm in children and adolescents. *Cochrane Database Syst Rev*, 3(3), CD013667. <https://doi.org/10.1002/14651858.CD013667.pub2>
- Witt, S. H., Streit, F., Jungkunz, M., et al. (2017). Genome-wide association study of borderline personality disorder reveals genetic overlap with bipolar disorder, major depression and schizophrenia. *Transl Psychiatry*, 7(6), e1155. <https://doi.org/10.1038/tp.2017.115>
- Wittchen, H. U., Gloster, A. T., Beesdo-Baum, K., et al. (2010). Agoraphobia: A review of the diagnostic classificatory position and criteria. *Depress Anxiety*, 27(2), 113–133.
- Witteveen, A., Young, S., Cuijpers, P., et al. (2022). Remote mental health care interventions during the COVID-19 pandemic: An umbrella review. *Behav Res Therapy*, 159, 104226.
- Wolf, A., Leucht, S., & Pajonk, F.-G. (2017). Do antipsychotics lead to cognitive impairment in dementia? A meta-analysis of randomised placebo-controlled trials. *Eur Arch Psychiatry Clin Neurosci*, 267, 187–198.
- Wolitzky-Taylor, K. B., Horowitz, J. D., Powers, M. B., & Telch, M. J. (2008). Psychological approaches in the treatment of specific phobias: A meta-analysis. *Clin Psychol Rev*, 28(6), 1021–1037.
- Wolk, D. A., Nelson, P. T., Apostolova, L., et al. (2025). Clinical criteria for limbic-predominant age-related TDP-43 encephalopathy. *Alzheimers Dement*, 21(1), e14202. doi: 10.1002/alz.14202
- Wolpe, J. (1968). Psychotherapy by reciprocal inhibition. *Cond Reflex*, 3(4), 234–240. <https://doi.org/10.1007/BF03000093>
- Wolpert, L. (1999). *Malignant Sadness: The Anatomy of Depression*, 2nd edn. Faber.
- Wong, E. K. O., Krishnadas, R., & Cavanagh, J. (2013). The interface between neurology and psychiatry: The case of multiple sclerosis. *Adv Psychiatr Treat*, 19(5), 370–377.
- Wooden, M. D., Air, T. M., Schrader, G. D., et al. (2009). Frequent attenders with mental disorders at a general hospital emergency department. *Emerg Med Australas*, 21(3), 191–195.
- Woods, B., & Windle, G. (2020). The effects of ageing on personality. In T. Denning, A. Thomas, R. Stewart, & J.-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 771–784). Oxford University Press.
- Wooff, K., & Goldberg, D. (1988). Further observations on the practice of community care in Salford: Differences between community psychiatric nurses and mental health social workers. *BJPsych*, 153(1), 30–37.
- Woolfenden, S., Sarkozy, V., Ridley, G., & Williams, K. (2012). A systematic review of the diagnostic stability of autism spectrum disorder. *Res Autism Spect Disord*, 6(1), 345–354.
- World Health Organization (WHO). (2014). *Global Status Report on Alcohol and Health*. World Health Organization.
- World Health Organization (WHO). (2024). *International Classification of Diseases 11th Revision (ICD-11)*. World Health Organization.
- World Health Organization (WHO). (1984). *Mental Health Care in Developing Countries: A Critical Appraisal of Research Findings, Report of a WHO Study Group*. World Health Organization.
- World Health Organization (WHO). (1973). *Report of the International Pilot Study of Schizophrenia*. World Health Organization.
- World Health Organization (WHO). (2022a). Social determinants of health. <https://www.who.int/health-topics/social-determinants-of-health>
- World Health Organization (WHO). (2022b). *World Mental Health Report: Transforming Mental Health for All*. World Health Organization.
- Woulfe, J., Gray, D., & Proctor, C. (2020). The ageing brain. In T. Denning, A. Thomas, R. Stewart, & J.-P. Taylor (Eds.), *Oxford Textbook of Old Age Psychiatry*, 3rd edn. (pp. 15–26). Oxford University Press.
- Wozniak, J. R., Riley, E. P., & Charness, M. E. (2019). Clinical presentation, diagnosis, and management of fetal alcohol spectrum disorder. *Lancet Neurol*, 18(8), 760–770.

- Wright, A. J., & Jackson, J. J. (2022). Childhood temperament and adulthood personality differentially predict life outcomes. *Sci Rep*, 12(1), 1–9.
- Wright, C., Catty, J., Watt, H., & Burns, T. (2004). A systematic review of home treatment services—classification and sustainability. *Soc Psychiatry Psychiatr Epidemiol*, 39(10), 789–796. <https://doi.org/10.1007/s00127-004-0818-5>
- Wright, C. F., Fitzgerald, T. W., Jones, W. D., et al. (2015). Genetic diagnosis of developmental disorders in the DDD study: A scalable analysis of genome-wide research data. *Lancet*, 385(9975), 1305–1314.
- Wu, Y., Zhong, Y., Ma, Z., et al. (2018). Gray matter changes in panic disorder: A voxel-based meta-analysis and meta-analytic connectivity modeling. *Psychiatry Res Neuroimaging*, 282, 82–89.
- Wu, Y.-T., Beiser, A. S., Breteler, M. M., et al. (2017). The changing prevalence and incidence of dementia over time—current evidence. *Nat Rev Neurol*, 13(6), 327–339.
- Wylie, K., Knudson, G., Khan, S.I., Bonierbale, M., Watanyusakul, S., Baral, S. (2016). Serving transgender people: Clinical care considerations and service delivery models in transgender health. *Lancet* 388, 401–411. doi: 10.1016/S0140-6736(16)00682-6
- Xu, G., Strathearn, L., Liu, B., et al. (2019). Prevalence and treatment patterns of autism spectrum disorder in the United States, 2016. *JAMA Pediatr*, 173(2), 153–159. <https://doi.org/10.1001/jamapediatrics.2018.4208>
- Yalom, I., & Leszcz, M. (2005). *The Theory and Practice of Group Psychotherapy*, 5th edn. Basic Books.
- Yamazaki, Y., Zhao, N., Caulfield, T. R., et al. (2019). Apolipoprotein E and Alzheimer disease: Pathobiology and targeting strategies. *Nat Rev Neurol*, 15(9), 501–518.
- Yap, M. B., Cheong, T. W., Zaravinos-Tsakos, F., et al. (2017). Modifiable parenting factors associated with adolescent alcohol misuse: A systematic review and meta-analysis of longitudinal studies. *Addiction*, 112(7), 1142–1162.
- Yee, C. S., Bahji, A., Lolich, M., et al. (2022). Comparative efficacy and tolerability of antipsychotics for juvenile psychotic disorders: A systematic review and network meta-analysis. *J Clin Psychopharmacol*, 42(2), 198–208. <https://doi.org/10.1097/jcp.0000000000001506>
- Yilmaz, A. Y., & Jankovic, J. (2025). Epidemiology of Tourette syndrome. *Brain Sciences*, 15, e426.
- Yokoi, F., Gründer, G., Biziere, K., et al. (2002). Dopamine D2 and D3 receptor occupancy in normal humans treated with the antipsychotic drug aripiprazole (OPC 14597): A study using positron emission tomography and [¹¹C] raclopride. *Neuropsychopharmacol*, 27(2), 248–259.
- Yonkers, K. A., Gotman, N., Smith, M. V., et al. (2011). Does antidepressant use attenuate the risk of a major depressive episode in pregnancy? *Epidemiol*, 22(6), 848.
- Yonkers, K. A., Warshaw, M. G., Massion, A. O., & Keller, M. B. (1996). Phenomenology and course of generalised anxiety disorder. *BJPsych*, 168(3), 308–313.
- Young, A. H., & Juruena, M. F. (Eds.). (2021). *Bipolar Disorder: From Neuroscience to Treatment*. Springer Nature.
- Young, R. C., Biggs, J. T., Ziegler, V. E., & Meyer, D. A. (1978). A rating scale for mania: Reliability, validity and sensitivity. *BJPsych*, 133(5), 429–435.
- Young, S., & Church, R. (2015). Forensic psychiatry. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 636–647). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch49>
- Yule, W., & Smith, P. (2015). Post traumatic stress disorder. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 806–821). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118381953.ch59>
- Yunusa, I., Alsumali, A., Garba, A. E., et al. (2019). Assessment of reported comparative effectiveness and safety of atypical antipsychotics in the treatment of behavioral and psychological symptoms of dementia: A network meta-analysis. *JAMA Netw Open*, 2(3), e190828–e190828.
- Zachar, P., & Kendler, K. S. (2007). Psychiatric disorders: A conceptual taxonomy. *Am J Psychiatry*, 164(4), 557–565.
- Zalsman, G., Hawton, K., Wasserman, D., et al. (2016). Suicide prevention strategies revisited: 10-year systematic review. *Lancet Psychiatry*, 3(7), 646–659.
- Zanarini, M. C., Frankenburg, F. R., Hennen, J., et al. (2006). Prediction of the 10-year course of borderline personality disorder. *Am J Psychiatry*, 163(5), 827–832. <https://doi.org/10.1176/ajp.2006.163.5.827>
- Zeanah, C. H., & Smyke, A. T. (2015). Disorders of attachment and social engagement related to deprivation. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 793–805). John Wiley & Sons, Ltd.
- Zerr, I. (2022). Laboratory diagnosis of Creutzfeldt–Jakob disease. *N Engl J Med*, 386(14), 1345–1350.
- Zetterqvist, M. (2015). The DSM-5 diagnosis of nonsuicidal self-injury disorder: A review of the empirical literature. *Child Adolesc Psychiatry Ment Health*, 9, 31. <https://doi.org/10.1186/s13034-015-0062-7>
- Zhao, G., Yang, Y., Zhang, S., et al. (2008). A cross-sectional study on the mental status of 780 survivors after Wenchuan earthquake. *Chinese J Evid Based Med*, 8(10), 815–819.
- Zheng, W., Cao, X.-L., Ungvari, G. S., et al. (2016). Electroconvulsive therapy added to non-clozapine antipsychotic medication for treatment resistant schizophrenia: Meta-analysis of randomized controlled trials. *PLoS ONE*, 11(6), e0156510.
- Zhong, S., Senior, M., Yu, R., et al. (2021). Risk factors for suicide in prisons: A systematic review and meta-analysis. *Lancet Public Health*, 6(3), e164–e174.
- Zhou, X., Teng, T., Zhang, Y., et al. (2020). Comparative efficacy and acceptability of antidepressants, psychotherapies, and their combination for acute treatment of children and adolescents with depressive disorder: A systematic review and network meta-analysis. *Lancet Psychiatry*, 7(7), 581–601.
- Zigman, W. B. (2013). Atypical aging in Down syndrome. *Dev Disabil Res Rev*, 18(1), 51–67. <https://doi.org/10.1002/ddrr.1128>
- Zigmond, A. S., & Snaith, R. P. (1983). The hospital anxiety and depression scale. *Acta Psychiatr Scand*, 67(6), 361–370.
- Zimbron, J., Stahl, D., Hutchinson, G., et al. (2014). Pre-morbid fertility in psychosis: Findings from the AESOP first episode study. *Schizophr Res*, 156(2–3), 168–173.
- Zimmerman, M., Balling, C., Chelminski, I., & Dalrymple, K. (2021). Patients with borderline personality disorder and bipolar disorder: A descriptive and comparative study. *Psychol*

- Med*, 51(9), 1479–1490. <https://doi.org/10.1017/s0033291720000215>
- Zimmerman, M., Chelminski, I., & Young, D. (2008). The frequency of personality disorders in psychiatric patients. *Psychiatr Clin North Am*, 31(3), 405–420, vi. <https://doi.org/10.1016/j.psc.2008.03.015>
- Zinkstok, J. R., Boot, E., Bassett, A. S., et al. (2019). Neurobiological perspective of 22q11.2 deletion syndrome. *Lancet Psychiatry*, 6(11), 951–960. [https://doi.org/10.1016/s2215-0366\(19\)30076-8](https://doi.org/10.1016/s2215-0366(19)30076-8)
- Zoghbi, A. W., Lieberman, J. A., & Girgis, R. R. (2023). The neurobiology of duration of untreated psychosis: A comprehensive review. *Mol Psychiatry*, 28(1), 168–190. <https://doi.org/10.1038/s41380-022-01718-0>
- Zucker, K. J., & Seto, M. C. (2015). Gender dysphoria and paraphilic sexual disorders. In A. Thapar, D. S. Pine, J. F. Leckman, et al. (Eds.), *Rutter's Child and Adolescent Psychiatry*, 6th edn. (pp. 983–998). John Wiley & Sons, Ltd.
- Zuckerman, M. (2005). *Psychobiology of Personality*. Cambridge University Press.

Index

For the benefit of digital users, indexed terms that span two pages (e.g., 52–53) may, on occasion, appear on only one of those pages.

Tables, figures, and boxes are indicated by an italic *t*, *f*, and *b* following the page number.

A

- abdominal pain, recurrent 447
- ABI *see* absolute benefit increase (ABI)
- abnormal behaviour 94
- abnormal grief 157
- abnormal personalities
 - classification of 386–87
 - comorbidity 386
 - general issues 386
 - ICD-11 classification 386–87
- absolute benefit increase (ABI) 122–23
- acceptance and commitment therapy (ACT) 420, 673
- accidents 637–38
- acculturation 101–2
- N-acetylaspartate (NAA) 275
- N-acetylcysteine 191
- acquired brain injury (ABI) 468
- ACT *see* acceptance and commitment therapy (ACT)
- ACTH *see* adrenocorticotrophic hormone (ACTH)
- acute dystonia 705
- acute psychiatric disorder 772
 - generic versus specialized services 772, 773*t*
 - patients referred to specialist care 772
- acute stress disorder/
 - reaction 140, 141–43
- aetiology 142
- clinical picture 141
- debriefing 142–43
- diagnostic conventions/
 - groupings 141–42
- epidemiology 142
- management 143
- planning for disaster 142
- prolonged exposure therapy 143
- PTSD 143
- SSRIs 143
- trauma-focused CBT 143
- treatment 142–43
- variations in the clinical picture 141
- ADHD (attention deficit/hyperactivity disorder) 124–25, 433–36
 - in adults 436
 - aetiology 434
 - clinical features 433
 - diagnostic criteria 433–34
 - dietary advice 435
 - epidemiology 434
 - genetic studies 434
 - medications 435–36
 - neuroimaging studies 434
 - neurological findings 434
 - prognosis 434–35
 - psychosocial interventions 435
 - treatment 435–36
- adjustment disorders 152–60
 - aetiology 151
 - clinical features 150–51
 - diagnostic conventions 151
 - epidemiology 151
 - grief and bereavement *see* bereavement; grief
 - onset of physical illness 152–53
 - physical illness and disability 151–52
 - priorities for care 155*b*
 - prognosis 151
 - sexual abuse in childhood 159–60
 - terminal illness 153–55
 - treatment 151
 - types of adjustment 152–60
- Adler, Alfred 383
- admission notes 62–63
- adolescents
 - adolescence as a unique period in development 403
 - alcohol and substance use disorders 456
 - antipsychotic medication 281
 - assessment 456
 - bulimia nervosa 314
 - circadian rhythm sleep disorder, 320
 - clinical features of psychiatric disorder 455–56
 - development process 405–6
 - eating disorders 455–56
 - epidemiology of psychiatric disorder 455
 - exhibitionism 330
 - gender dysphoria 331–32
 - irritability 196
 - National Comorbidity Study 409
 - peer groups 406
 - puberty, effects on 320
 - review of disorders 455–56
 - services for 456
 - sleep-onset insomnia 316
 - social media, influence on 256, 308
 - see also* child and adolescent psychiatry
- adoption 422–23
 - embryo 422
 - foster care 422
 - international 422
 - private domestic 422
 - in schizophrenia 265–66
 - studies 105
- adrenocorticotrophic hormone (ACTH) 211, 212
- ADRT *see* Advance Decision to Refuse Treatment (ADRT)
- Adult ADHD Self-Report Rating Scale (ASRS) 436
- Adult Attachment Interview 383–84
- Adult Psychiatric Morbidity Survey 165–66
- Advance Decision to Refuse Treatment (ADRT) 78–79
- aetiological models 92–96
 - behavioural 94
 - biopsychosocial 94
 - developmental 94
 - medical 93–94
 - neuroscience 93
 - political 94–96
 - reductionist and non-reductionist 92–93, 94
- aetiology *see* aetiological models; psychiatric aetiology
- affect
 - mixed mood (affective) states, in mania 233
 - and mood 4–5
- affective personality disorders 392
- agitated depression 196
- agomelatine 217, 723–24
- agoraphobia 175–78
 - aetiology 176–77
 - age of onset 175
 - anticipatory anxiety 175
 - anxiety 175
 - and behavioural experiments 665*b*
 - behavioural management 178
 - biological theory 177
 - CBT 177
 - circumstances of onset 175
 - cognitive hypothesis 176
 - and depressive disorders/symptoms 175, 176
 - diagnostic conventions 175–76
 - diagnostic criteria 176*b*
 - differential diagnosis 176
 - effect on family 175
 - epidemiology 176
 - exposure treatment 177
 - and GAD 176
 - management 178
 - medications 177–78
 - onset and course 175
 - and panic disorder 176
 - and paranoid disorders 176
 - prognosis 177

- agoraphobia (*cont.*)
 psychological treatment 177
 required patient knowledge 178
 situations 175
 and social phobia 172, 176
 theories of onset 176–77
 theories of spread and maintenance 177
 treatment 177–78
- akathisia 705
- alcohol use disorders
 age 549
 alcohol withdrawal syndrome 550
 associated psychiatric disorder 553
 cardiovascular system 551
 causes 554
 childhood factors 555
 consumption in society 555–56
 course of 550
 damage to foetus 551
 dependence and withdrawal 549–50
 detection 556–58
 epidemiology 548–49
 excessive consumption 547–50, 554
 family factors 555
 family problems 553–54
 gastrointestinal disorders 551
 gender 549
 genetic factors 554
 harmful use in women 551
 individual factors 554
 intoxication 552
 moral and medical models 545–46
 mortality 551–52
 nervous system 551
 neurobiological factors 555
 neurocognitive syndromes 552–53
 occupation 549
 personality factors 555
 physical damage 551–53
 population characteristics 549
 prevalence 548–49
 prevention 564–65
 psychiatric disorder 555
 psychiatric disorders 552–53
 recognition of harmful alcohol use and dependence 556–58
 religion, culture and ethnicity 549
 social damage 553–54
 terminology 545
 treatment 559–64
 withdrawal phenomena 552
- alprazolam 177, 181
- Altman Self-Rating Mania Scale 68
- Alzheimer's disease 110*b*
 ADAS-cog test 350
 aetiology and pathogenesis 355–57
 amyloid cascade hypothesis 356
 amyloid precursor protein (APP) 356*b*, 356
 amyloid (senile) plaques 354–55
 apolipoprotein E gene, apoE4 allele 354
 behaviour 354
 cholinergic hypothesis 357
 cholinesterase inhibitors 539
 clinical features 353–54, 354*b*
 Consortium to Establish a Registry for Alzheimer's Disease (CERAD) 354–55
 course 354
 depression 354
 drug treatment in the elderly 539–40
 early onset 354
 environmental factors 356
 genes 355–56
 granulovascular accumulation 355
 Hirano bodies 355
 infectious agents 357
 inflammatory mechanisms 356–57
 investigations 354
 memantine 539–40
 neurofibrillary tangles 354–55
 neuropathology 354–55
 novel treatment approaches 540
 preclinical, biomarkers of 352, 352*t*
 progression of neuropathology 355
 psychotic symptoms 354
 suspected 350
 vascular amyloid or congophilic angiopathy 355
 with vascular dementia 363
 wandering 354
- ambitendence 17
- American Psychiatric Association, DSM classification scheme 32
- AMHP *see* approved mental health professional (AMHP)
- amisulpride 704
- amnesia/amnesic disorders 18, 344–46
 aetiology and pathology 345
 causes 345*b*
 clinical features 345
 course/prognosis 346
 differential diagnosis 346*b*
 dissociative amnesia 614
 forensic psychiatry 511
 investigation and management 345–46
 Korsakoff syndrome 345
 transient 346*b*, 346
- amnesic syndrome 335
- amphetamines 580–81
- AMPT *see* α -methyl-para-tyrosine
- amygdala 102, 103
 antisocial behaviour 394
 bilateral lesions 102
 and bipolar disorder 239
 brain imaging 102, 145*b*
 fear conditioning 102
 and GAD 167
 and phobias 171, 173
 and PTSD 145*b*, 145–46
- amyloid precursor protein (APP) 107
- anankastic (obsessive-compulsive) personality disorder 391–92, 392*b*
- anger 153
- angina 633
- Anglicus, Bartholomeus 96
- anhedonia 5–6, 194
- animal experiment models 102
- anisotropy 117–18
- anorexia nervosa 306–12
 aetiology 308–9
 amenorrhoea 306
 assessment 309*b*–10*b*, 309
 CBT 310
 differential diagnosis 311*b*
 epidemiology 307–8
 familial causes 309
 family therapy 310
 features 306, 307*b*
 genetic factors 308
 individual psychological causes 308
 management 311–12
 medications 310–11
 neurobiology 308
 NICE guidelines 311*b*
 onset, course and prognosis 308
 patterns 306–7
 physical consequences 307*b*, 307
 psychotherapy 310
 pursuit of thinness 306
 restoring weight 312
 setting of treatment 312
 sociocultural factors 308
 starting treatment 311–12
 treatment 310–11
 urgent follow-up required 310*b*
- anosognosia 17
- anoxia 378
- anterior cingulate cortex 167
- anterograde amnesia 18
- anticholinergic drugs 711
- anticipatory anxiety 169–70, 173
- anticonvulsants 218, 692
- antidepressants 711–29
 agomelatine 723–24
 for agoraphobia 177–78
 for anorexia nervosa 311
 for anxiety 697
 augmentation of 227–29
 bipolar depression 242, 248
 for bulimia nervosa 314
 bupropion 727–28
 change in treatment 227
 choice/use of 224–25
 clinical characteristics 217, 217*t*
 combination treatment 227
 for depression 217–18
 esketamine 728–29
 ketamine 728–29
 mechanism of action 712–13
 mirtazapine 724
 new 217
 for panic disorder 181, 698
 pregnant women 692
 for PTSD 147
 for schizophrenia 282
 St John's wort 727
 trazodone 724–25
 tricyclic 114, 147

- venlafaxine 725–26
 - see also selective serotonin reuptake inhibitors (SSRIs)
- antihistamines 699
- antipsychiatry 94
- antipsychotics 700–10
 - acute behavioural disturbance 710
 - antiadrenergic effects 706
 - anticholinergic effects 706
 - antihistaminic effects 706
 - anxiety 697
 - atypical drugs 218, 244, 702–3
 - augmentation 228
 - bipolar disorder 244
 - cardiac conduction defects 706–7
 - in children and adolescents 453
 - contraindications 709
 - depot (long-acting injectable) 282, 287b, 703
 - depression 707
 - diabetes 707
 - discontinuation rate 228
 - dosage 281, 709–10
 - drug interactions 704
 - effectiveness and side effects in schizophrenia 280–81
 - endocrine changes 707–8
 - 5-HT₂ receptor antagonists . 703
 - for mania 245
 - metabolic syndrome 707
 - in older patients 538
 - onset of action 281
 - PET imaging 709–10
 - pharmacokinetics 702, 704
 - pharmacology 700–1
 - phenothiazines 702
 - pregnant women 692
 - and risk of sudden death 710
 - selective D2-receptor antagonists 702–3
 - side effects 228
 - tardive dyskinesia 705–6
 - tardive dystonia 706
 - typical 244, 702
 - and atypical 701–3
 - unwanted effects 704–6, 708
- antisocial personality disorders (ASPD) 385, 388–89
 - aetiology 394–95
 - cerebral pathology and maturation 394
 - and concept of psychopathy 389
 - course of 396
 - developmental theories 394–95
 - features 389b
 - 5-hydroxytryptamine and aggression 394
 - genetic causes 394
 - management 400
 - pharmacotherapy 400
 - probands 394
- anxiety 57, 161–91
 - agoraphobia 175
 - anticipatory 169–70, 173
 - antidepressants 697
 - antipsychotics 697
 - beta-adrenoceptor antagonists 697
 - clinical associations 5
 - components of response 5
 - emergencies 629
 - GAD 163–69
 - Hamilton Anxiety Scale 68
 - hoarding 175–78
 - Hospital Anxiety and Depression Scale 68
 - mental state examination 48
 - mixed depressive and anxiety disorder 164, 198–99
 - obsessive-compulsive disorder 183–88
 - OCD-related disorders 188–91
 - panic disorder 178–81
 - phobias 169–78
 - prevalence rates of anxiety/OCD 163, 163t
 - psychiatric interviews 56
 - State-Trait Anxiety Inventory 68
 - symptoms 162b
 - terminal illness, adjustment to 153
 - transcultural variations 182
 - trichotillomania 190
 - types 161
- anxiety disorders 161–63
 - in children and adolescents 442–46
 - aetiology 443
 - at different ages 443
 - management 443–44
 - phobias 444
 - prevalence 443
 - prognosis 443
 - separation and anxiety disorder 444
 - classification of 163, 163t
 - and depressive disorders 203, 204
 - development of ideas about 162
 - and individuals with DID 472
 - minor 204
 - older patients 525, 534
 - prevalence rates 163, 163t
 - transcultural variations 182
 - see also anxiety; depressive disorders
- anxiolytics 694–98
 - advice on management 698
 - agoraphobia 177
 - antidepressants for anxiety 697
 - antipsychotics for anxiety 697
 - benzodiazepines see benzodiazepines
 - beta-adrenoceptor antagonists for anxiety 697
 - GAD 168
 - grief 158
 - management advice 698
 - misuse of 577–78
 - OCD 187
 - pregabalin 174, 698
 - pregnant women 692
- anxious (avoidant) personality disorder 391, 399
- apathy 5
- apolipoprotein E (*apoE*) gene, dementia, polymorphisms 110b, 355
- appearance
 - emotional expression 46
 - facial 46
 - general 46
 - in mania 232
 - posture/movement 46
 - severe depressive episode 194
- applied relaxation 168
- approved mental health professional (AMHP) 83
- arbitrary interference 208
- ARFID see avoidant/restrictive food intake disorder (ARFID)
- aripiprazole 703
- ARR see absolute risk reduction (ARR)
- arson 507
- arterial spin labelling (ASL) 115
- ASD see autism spectrum disorders (ASD)
- ASL see arterial spin labelling (ASL)
- assertive outreach teams 765–67
 - functional assertive community treatment teams 765–66
 - purpose 765
- assessment, psychiatric see psychiatric assessment
- atomoxetine 435
- attachment theory 657
 - Adult Attachment Interview 383–84
 - adverse caregiving environments 413
 - effect of separation 413
 - insecure attachment 412
 - maternal deprivation 412–13
 - personality 383–84
 - reactive attachment disorder 425
 - secure attachment 383–84, 412
 - Strange Situations Procedure 383–84, 412
- attention and concentration, disturbances of 19–20, 50
- attention deficit and hyperactivity disorder
 - and individuals with DID 473
 - see also ADHD (attention deficit/hyperactivity disorder)
- atypical antipsychotic drugs 218, 242
- atypical depression
 - bright light treatment 220
 - and SSRIs
- auditory hallucinations 7–8, 50, 252
- augmentation
 - of antidepressants 227–29
 - of antipsychotics 228
 - of lithium 218, 228
- Autism Diagnostic Observation Schedule (ADOS) 431–32
- autism spectrum disorders (ASD) 427–33
 - abnormal behaviour, management 432
 - abnormalities of communication 428
 - abnormalities of social development 428
 - assessment 431–32

- autism spectrum disorders (ASD) (*cont.*)
 clinical characteristics 427–29
 clinical clues 428*b*
 cognitive and emotional processes 430–31
 co-occurring mental health problems, management 432
 differential diagnosis 431
 educational issues 432
 epidemiology 429
 family support 432–33
 genetic factors 430*b*, 433
 and individuals with DID 473
 intelligence level 429
 and learning disability 84
 management 432–33
 non-verbal communication and play 428
 and OCD 184
 prognosis 431
 restriction of interests and behaviour 428–29
 risk factors/possible aetiology 429–30
 autonomy, respect for 72
 autoscopic hallucinations 7
 autosomes 95*b*
 avoidance behaviour 5, 102, 143, 144, 148, 170
 avoidant personality disorder *see* anxious (avoidant) personality disorder
 avoidant/restrictive food intake disorder (ARFID) 315
- B**
 bariatric surgery 631
 BDI *see* Beck Depression Inventory (BDI)
 BDNF *see* brain-derived neurotrophic factor (BDNF)
 Beck Depression Inventory (BDI) 68
 behaviour 47–48
 abnormal 94
 aggressive 138
 Alzheimer's disease 354
 avoidance 5
 challenging, in DID 473–74, 474*b*
 illness behaviour 152
 in mania 232
 mental state examination 47–48
 mood 47–48
 neologisms 47
 social 47, 100, 100*t*
 and speech
 difficulty in speaking 47
 flow of 47
 rate/quantity of 47
 behavioural model 94
 behavioural therapy 658
 defining 664
 desensitisation 666
 enuresis alarms 667
 exposure 666
 exposure with response prevention 666–67
- EMDR 668
 flooding 666
 in routine practice 666
 relaxation training 666
 self-control techniques 667
 social skills training 667
 anger management 667
 assertiveness training 667
 techniques 666–68
- beliefs
 body weight and shape 15
 challenging 668*b*, 670
 dysfunctional 208, 664*b*, 664
 false 9*b*
- beneficence 72
- benzodiazepines
 agoraphobia 177
 bipolar disorder 241, 246
 compounds available 695
 dependence and withdrawal 696
 drug interactions 696
 GAD 168, 169
 misuse of 577–78
 pharmacokinetics 695
 pharmacology 695
 schizophrenia 282
 short-acting 699
 social phobia 174
 toxic effects 696
 unwanted effects 695–96
- bereavement
 and child development 414
 defining 156
 morbidity following 157
 persistent complex bereavement disorder 141
see also grief
- beta-adrenergic antagonists 168, 697
 beta-adrenergic blockers 174
- big data 114, 133
- binge eating disorder 315
 avoidant/restrictive food intake disorder 315
 other specified feeding or eating disorder 315
- PICA 315
- rumination–regurgitation disorder 315
- biochemical studies 111–14
 biochemical abnormality 111
 brain biochemistry and brain imaging 112–13
 CSF and blood 113
 indirect methods 111–12
 magnetic resonance imaging (MRI) 112–13
 neurotransmitter release *in vivo* 112
 peripheral measures 113–14
 post-mortem studies 112
 receptor binding with PET and SPET 112
- biological psychiatry 98
 biomedical versus social concepts 23
 Bion, W. 675
 biopsychosocial model 94
- bipolar depression 241–43
 antidepressants 242, 248
 assessment 247
 medications 247–48
 antidepressants 242
 atypical antipsychotic drugs 242
 lithium 242
 placebo-controlled trials 242, 242*t*
 psychotherapy 248
- bipolar disorder
 aetiology 237–39
 assessment 245
 bipolar I 235, 236
 bipolar II 235, 236
 bipolar spectrum 235
 brain imaging 238–39
 CBT 244
 classification of 200, 234–35
 clinical descriptive studies 98
 clinical features 231–34
 cortisol 238
 course/prognosis 239–40
 differential diagnosis 235–36
 dopamine metabolism 238
 ECT 242–43
 mania 241
 endocrine abnormalities 233–38
 and the environment 237–38, 239
 epidemiology 236
 of bipolar I and unipolar depressive disorder 231, 232*t*
 family-focused approaches 244
 functional imaging 239
 genetic factors 237
 longer-term treatment 241–43
 and major depression 239
 mania 231–33
 acute 245–47
 treatment 240–41
 medications 242
 antipsychotics 244
 bipolar depression 247–48
 carbamazepine 243–44
 lamotrigine 243
 lithium 242, 243, 249, 686–87
 longer-term 243–44
 mania 240–41
 prevention of relapse/recurrence 248–49
 valproate 243
 mixed mood (affective) states 233
 molecular genetics 237
 mortality of 240
 neurobiological approaches to 238–39
 neurochemistry 238
 neuropsychological changes 239
 NICE guidance 247*b*, 248*b*
 oestrogen 238
 older patients 534
 as an organic brain disorder 236
 practical management 245–50
 predisposing factors 239
 prevention of relapse/recurrence 248–50

- progesterone 238
 psychoeducation 244
 psychological approaches to 250*b*, 250
 psychosocial prevention 249–50
 psychotherapy 244
 bipolar depression 248
 rapid cycling disorders 234
 and recurrent depression 236
 and schizophrenia 235
 severity of 245
 sleep and circadian rhythm 238, 245
 and suicide risk 593
 transcultural factors 234
 treatment 241–43
 longer-term 243–44
 mania 240–41
 blindness 634
 blood oxygenation-level-dependent (BOLD) imaging 115, 116, 275
 bodily distress disorders 611–12
 body dysmorphic disorder 188–89, 627–28
 body image, disturbances of 17
 BOLD *see* blood oxygenation-level-dependent (BOLD) imaging
 borderline personality disorder (BPD) 32, 148, 262, 385, 389–90
 abbreviated criteria 390, 390*t*
 aetiology 395
 and bipolar disorder 236
 classification of 386
 course of 396
 DBT for 672–73
 dialectical behaviour therapy 399
 evidence-based treatments 400*b*
 generalist approach to treating 400*b*
 general population rates 393
 management 399
 mentalization 399
 psychotherapy 400
 and sexual abuse in childhood 159
 boundaries of the self, disturbances of 18
 Bowlby, John 383–84, 412, 657
 see also attachment theory
 BPRS *see* Brief Psychiatric Rating Scale (BPRS)
 Bradford Hill, Sir Austin 119–20
 brain
 biopsy 341
 changes in volume 213
 disorders in children and adolescents 410–11
 imaging *see* brain imaging
 regional pathology, associated symptoms 338–40
 basal ganglia 339–40
 cerebellum 339–40
 corpus callosum 339
 cortico-subcortical circuits 339
 frontal lobe 338
 memory systems and neuroanatomy 340
 occipital lobe 339
 parietal lobe 338
 rostral brainstem 340
 subcortical structures 339
 temporal lobe 338–39
 thalamus 339–40
 structural changes, in schizophrenia 273–74
 see also amygdala; brain biochemistry and brain imaging; hippocampus; traumatic brain injury (TBI); white matter hyperintensities
 brain biochemistry and brain imaging 112–13
 see also brain imaging
 brain-derived neurotrophic factor (BDNF) 113
 brain imaging
 ADHD 434
 bipolar disorder 238–39
 and brain biochemistry 112–13
 CT *see* computed tomography (CT)
 functional imaging/
 neuroimaging 173, 180
 bipolar disorder 239
 neuropsychiatric patient 341
 schizophrenia 274–75
 functional MRI *see* fMRI (functional magnetic resonance imaging)
 in mood disorder 213–14
 MRI *see* magnetic resonance imaging (MRI)
 MRS *see* magnetic resonance spectroscopy (MRS)
 neurochemical 210
 and OCD 185
 PET *see* positron emission tomography (PET)
 SPET *see* single-photon emission tomography (SPET)
 structural 213, 238–39, 273–74, 274*b*
 white matter hyperintensities 213
 brain lesions 102
 brain tumours 375–76
 cognitive impairments in cancer 375–76, 376*b*
 brief dynamic therapy 220, 664*b*
 Brief Psychiatric Rating Scale (BPRS) 68
 bright light treatment 220–21, 751–52
 adverse effects 752
 clinical use of *phototherapy* 752
 equipment 751
 mechanism of action 751
 therapeutic efficacy 752
 British Medical Association 76
 bulimia nervosa 312–15
 aetiology 314
 depressive syndromes 313
 epidemiology 313
 management 314–15
 medications 314
 onset, course and prognosis 313
 physical consequences 313*b*, 313
 psychotherapy 314
 treatment 314
 bulimia nervosa, CBT for 671*b*
 bupropion 727–28
 Burton, Robert, *The Anatomy of Melancholy* 96
 buspirone 168, 696–97
 drug interactions 697
 indications and pharmacology 696–97
 pharmacokinetics 697
 butyrophenones 702
- C**
 CACNA1C (voltage-gated calcium channel) 237
 caesarian section 640
 cancer 619–37
 cognitive impairments in 375–76, 376*b*
 candidate gene studies 108–9, 205
 CANDID (Camberwell Assessment of Need for Adults with Developmental and Intellectual Disabilities) 485
 cannabinoid signalling 277
 cannabis 578–80
 Capgras syndrome 301
 carbamazepine 218, 240–41, 736–38
 clinical use 737
 dosage and plasma concentrations 736
 drug interactions 737
 pharmacokinetics 736
 unwanted effects 736–37
 carbon monoxide poisoning 378
 cardiac disorders 632–33
 care in the community 75
 Care Programme Approach, schizophrenia 286
 cariprazine 242
 case management 763, 767–68
 case notes 62
 case summaries 63, 64*b*
 catalepsy 16–17
 catatonia 16
 schizophrenia 260–61, 261*b*, 284
 catecholamine depletion 211
 CATEGO 29–30, 46
 CBT (cognitive behavioural therapy) *see* cognitive therapy/cognitive behavioural therapy (CBT)
 Centre for Reviews and Dissemination (UK) 120
 cerebral blood flow and metabolism 116
 depression 213–14
 schizophrenia 275
 cerebral metabolism 116
 cerebrospinal fluid (CSF)
 and blood 113
 dopamine function 211
 neuropsychiatric patient 341
 studies 210
 subarachnoid haemorrhage 368
 subdural haematoma 368
 cerebrovascular disorders 349–68
 stroke 349–68
 CGI *see* Clinical Global Impression (CGI)

- CGT *see* complicated grief treatment (CGT)
- Charcot, Pierre 97–98
- Charles Bonnet syndrome 6
- child abduction 507
- child and adolescent psychiatry
 acceptance and commitment therapy 420
 aetiology 410–15
 assessment 58, 465–66
 behavioural methods 420
 behaviour and mental state 416
 behaviour problems and antisocial personality 394–95
 bullying 414–15
 CBT 420
 child development 270–71
 childhood psychiatric disorders 101
 child protection 76
 classification of psychiatric disorders 407–9
 comprehension impairment, reading 426
 confidentiality in the care of 75
 court reports 418*b*, 418
 death of parent 413–14
 delayed effects 411
 developmental assessments 416
 developmental disorders 425
 difference in psychiatric practice from adult care 403–4
 disorders of communication, speech and language 426–27
 dyslexia 426
 eating disorders 305
 ending of assessment 417
 environmental factors 412–15
 epidemiology 409
 epilepsy 411–12
 ethical and legal problems 464
 family and parenting interventions 420–21
 family risk factors 413
 family therapy 310
 formulation 417–18
 frequency of psychiatric disorders 409, 410*t*
 genetic factors 410, 411*b*
 head injury, consequences 411
 and individuals with DID 473
 intensive treatments 423
 interviewing parents 415
 learning disorders 425–26
 maltreatment *see* child maltreatment
 management, psychological aspects 419–20
 maternal deprivation, attachment 412–13
 medications 419
 mental capacity 77
 neighbourhood, effects of 414
 normal development 404–7
 occupational therapy 421
 with OCD 185
- overall rates of disorder 409
- parental mental disorder, effects 413
- parental separation, effects 413
- personal history 44
- physical examination 417–18
- prescribing advice 691
- primary care team, role of 419
- protective factors 413
- psychiatric aspects of physical illness 454–55
 effect on parents 454–55
 hospital treatment 455
 impact 454–55
- psychiatric assessment of children and families 415–17
- psychiatric team 419
- psychodynamic therapies 421
- psychological assessment 416, 417*t*
- psychological treatment 419–21
- psychotherapy for children 681
- reading disorder 425–26
- refugee families 150
- remote and digital interventions 421
- review of disorders 423–63
 ADHD 433–36
 anorexia nervosa 309, 310, 311*b*, 312
 anxiety 442–46
 ASPD 394
 autism spectrum 427–33
 avoidant/restrictive food intake disorder (ARFID) 315
 bipolar disorder 449–50
 brain disorders 410–11
 chronic fatigue syndrome 446–47
 communication, speech and language 426–27
 faecal soiling 451–52
 forensic child psychiatry and juvenile delinquency 440–42
 functional enuresis 451
 gender dysphoria 453
 learning 425–26
 mood disorders 447–51
 oppositional and conduct 436
 preschool children and families 423–25
 psychiatric aspects of physical illness 454–55
 psychosis and early-onset schizophrenia 452–53
 recurrent abdominal pain 447
 Rett syndrome 412
 schizophrenia 263, 270–71, 280, 289
 school refusal 450–51
 selective mutism 452
 somatoform and related disorders 446–47
 specific developmental disorders 425
 speech and language disorder 426–27
 stammering 452
- suicidal behaviour/self-harm 453
 Tourette syndrome 367
- school, effects of 414–15
- self-harm 603
- social and cultural factors 414
- social learning in childhood 394
- social work 421
- special education 421
- substitute care 421–23
- temperament 410
- treatment for children and families 419–23
 variations with gender and age 409, 410*t*
 see also adolescents; attachment theory
- Child Autism Rating Scale (CARS) 431–32
- childhood cancer 637
- child maltreatment 457–63
 aetiology 458–59
 emotional abuse 460
 environmental factors 458
 foetal abuse 458
 growth faltering 460–61
 and individuals with DID 475
 neglect 460
 physical abuse 459–60
 recognition of 457, 457*t*
 risk factors 458*b*, 458
 sexual abuse 461–63
- chlorpromazine 240, 702
- cholinesterase inhibitors and migraine 538
- chronic fatigue syndrome (CFS) 446–47, 623–25
 aetiology 623–24
 clinical features 623
 course/prognosis 624
 epidemiology 623
 management 624–25
 treatment 624
- CIDI *see* Composite International Diagnostic Interview (CIDI)
- circadian rhythm 238
- sleep disorders 320
- CISD *see* critical incident stress debriefing (CISD)
- citalopram 181
- civil law 85–87
 family law 86
 fitness to drive 87
 personal injury 87
 power of attorney 86
 receivership 86
 testamentary capacity 86
 torts and contracts 86
- classification, psychiatric *see* psychiatric classification
- Cleckley, Hervey, *The Mask of Sanity* 388
- Clinical Global Impression (CGI) 69
- clinical investigations 98
- clinical records 62
- clinical utility of interventions 126
- clomethiazole edisylate 699

- clomipramine 177–78, 181, 187, 717
- clonazepam 241, 246
- clouding of consciousness 19, 342
 - see also* delirium
- clozapine 703, 706
 - adverse effects 708
- clubhouse model 774
- CNV *see* copy number variation (CNV)
- cocaine 581–82
- Cochrane, Archibald 120
 - Effectiveness and Efficiency: Random Reflections on Health Services* 120
- coenestopathic states 17
- cognitive analytic therapy 664b
- cognitive distortions 208
- cognitive errors in depressive disorders 658b
- cognitive functions 50–51
 - attention and concentration, disturbances of 19–20, 50
 - memory, disturbances of 50–51
 - orientation 50
- cognitive impairment
 - mild *see* mild cognitive impairment/ neurocognitive disorder
 - ratings used in 48
 - traumatic brain injury 369
- cognitive remediation, schizophrenia 284
- cognitive restructuring 669
- cognitive therapy/cognitive behavioural therapy (CBT) 122
 - ABC approach 665
 - agoraphobia 177
 - anorexia nervosa 310
 - anxiety disorders 669
 - anxiety management 669
 - automatic thoughts 664
 - avoidance behaviour 664
 - behavioural therapy *see* behavioural therapy
 - bipolar disorder 244
 - challenging beliefs 668b
 - child and adolescent psychiatry 420
 - depression 213, 219–20, 222
 - in children and adolescents 449
 - depressive disorders 669–70
 - activities in severe depression 670
 - beliefs, challenging 670
 - mindfulness 670
 - errors of logic 670
 - monitoring 670
 - development 658–59
 - behavioural therapy 658
 - CBT 658
 - cognitive therapy
 - distraction 668
 - dysfunctional beliefs and attitudes 664b, 664
 - general features of CBT 664–65
 - GAD 167–68
 - hypochondriasis 671–72
 - illogical thinking 664
 - information sources 665–66
 - formulation 666
 - self-monitoring 665
 - special interviewing 665–66
 - see also* behavioural techniques 666–68,
 - neutralizing 668
 - OCD 187
 - panic disorder neutralizing 181, 668
 - personality disorders 672
 - PTSD 146–47, 150, 669
 - reassessing patient's responsibility 668
 - safety seeking 669
 - schizophrenia 283, 672
 - Socratic questioning 664
 - treatments 668–73
 - trauma-focused CBT 146
 - topics to consider during assessment 665b, 665
 - worry 669
- coma 19
- combatants, PTSD in 149
- communicating with patients and relatives 41b
- communication and liaison 764
- communication skills
 - child and adolescent psychiatry 403
 - communicating with patients and relatives 41b
 - lacking social skills, in autism 173
- pain management 612
- training 244, 561b
- community care, problems with 777
 - burden on relatives 777
 - coordination of services 777
 - resource distribution 777
- community mental health teams (CMHTs) 392
- Community Treatment Orders (CTOs) 83
- comorbidity 28–29
 - disorders 29
- compensation neurosis 638
- complex PTSD (CPTSD) 147–48
- complicated grief treatment (CGT) 158–59
- Composite International Diagnostic Interview (CIDI) 67, 100
- Comprehensive Assessment of At-Risk Mental State (CAARMS) 256
- compulsions
 - checking rituals 14
 - cleaning rituals 14
 - counting rituals 14
 - defining 14
 - dressing rituals 14
 - mental state examination 49
 - problems caused 14
 - thinking processes, disturbances of 14
 - see also* obsessions; OCD (obsessive-compulsive disorder)
- compulsivity 186
- conduct disorders *see* oppositional and conduct disorders
- confabulation 18
- confidentiality 74–76
 - assessment on behalf of a third party 75
 - care in the community 75
 - in the care of children 75
 - child and adolescent psychiatry 464
 - consent to disclosure of information 74–75
 - disclosing information to others 75
 - ethical principles relating to 74–75
 - exceptions to rule 76
 - group therapy 75
 - potential problems 75
 - safeguarding information 74
 - seeking information from others 75
 - therapy with couples 75
- confused patients 57
- confusion 19
- consciousness, disturbances of 19
- consent by proxy 79
- consent to treatment 76–77
 - child and adolescent psychiatry 464
 - competent patients, refusal by 77
 - emergency 77
 - implied 77
 - incompetent patients, refusal by 77
 - and individuals with DID 490–91
 - maintaining 76
 - mentally disordered patients, refusal by 77
 - necessity 77
 - obtaining 76
 - by proxy 79
 - when explicit consent is not required 76–77
 - when refused 77
- Consortium to Establish a Registry for Alzheimer's Disease (CERAD) 354–55
- contamination thoughts, in OCD 14
- continuation (progress) notes 63
- control, delusions of 11, 49
- coping strategies 138
 - aggressive behaviour 138
 - and defence mechanisms 138–40, 141
 - emotion-reducing strategies 138
 - maladaptive 138
 - self-harm 138
 - styles 138
 - worry 167
- copy number variation (CNV) 108, 261, 267
- cortico-striatal-thalamic loops 186
- corticotropin-releasing hormone 167
- cortisol 211, 212, 238
- Cotard's syndrome 12, 196
- counselling
 - adjustment disorders 151
 - approaches to 661–62
 - interpersonal counselling 661
 - problem-solving 661
 - psychodynamic counselling 661
 - bereavement 158, 475, 662

- counselling (*cont.*)
 client-centred approach 661
 and crisis intervention 661–62
 in depression 219
 doctor–patient relationship 73–74
 in GAD 168
 genetic 366, 487, 637
 grief 663–64
 humanitarian emergencies 654
 interpersonal 661, 662
 non-directive 168, 219
 post-natal depression 641–42
 in primary care 662
 problem-solving 151, 399, 661
 psychodynamic 661
 sexual abuse in childhood 159
 for specific purposes 661–62
 bereavement 662
 debriefing 661
 in primary care 662
 for relationship problems 661–62
 countertransference 660
 couples therapy 75, 220, 679
 Covid-19 618–36
 and Crime Survey for England and Wales (2022) 471–72, 495
 and individuals with DID 476
 CPTSD *see* complex PTSD (CPTSD)
 Creutzfeldt–Jakob disease (CJD) 362
 crime, association with mental disorder 497–501
 epilepsy 500
 intellectual disability 499
 intermittent explosive disorder 500
 kleptomania 501
 mood disorders 499
 morbid jealousy 483
 nature of association 497–98
 organic mental disorders 500
 pathological gambling 500
 post-traumatic stress disorder 499–500
 psychiatric disorder and offending 498*b*
 pyromania 501
 relative risk estimates 497, 498*t*
 schizophrenia and related disorders 499
 specific psychiatric disorders 498–501
 crime, causes of 496–97
 genetic and physiological factors 496
 psychiatric 497
 psychosocial factors 496–97, 497*b*
 criminal justice system 84
 and individuals with DID 476
 criminology, general 494–96
 crime time trends 495
 national differences 495–96
 ethnicity 495
 gender 495
 victims 495–96
 prevalence 494–95
see also criminal justice system;
 forensic child psychiatry and juvenile delinquency; forensic psychiatry
- crisis intervention 662*b*, 662
 critical incident stress debriefing (CISD) 142
 critical psychiatry 24, 95–96
 cross-fostering study 105
 CTOs *see* Community Treatment Orders (CTOs)
 Cullen, William 26, 96
 culture/cultural variations
 anxiety 182
 bipolar disorder 234
 depression 199
 descriptive psychopathology 4
 DSM-5 classification scheme 34
 misconceptions 9*b*
 psychiatric classification 34
 cyclothymia 234, 386
 cyclothymic personality 392
- D**
 daycare 761
 D-cycloserine 171–72
 deafness 634
 deep brain stimulation 188, 748–49
 adverse effects 749
 ethical aspects 749
 default mode network (DMN) 116, 214
 defence mechanisms 139*b*
 and coping strategies 138–40, 141
 degeneration, theory of 97
 déjà vu 18
 delirium 261, 335
 aetiology 343
 clinical features 342–43, 343*b*
 and dementia 344
 epidemiology 342
 and individuals with DID 473
 management 343–44
 outcome 344
 predisposing and precipitating factors 342*b*
 prevention 343
 treatment 343–44
 delusional disorders 296–97
 aetiology 297
 Capgras syndrome 301
 classification of 296
 epidemiology 296–97
 erotomania/erotic delusions 299–300
 family studies 297
 intermetamorphosis 301
 neurobiological approaches 297
 pathological jealousy 297–300
 prognosis 303
 querulant 300
 reformist 300
 and schizophrenia 262
 shared/induced 301
 somatic 300
 stalking 299–300
 subjective doubles 301
 treatment 302–3
 general principles 302
 medications 302
 psychological 303
- delusional misidentification syndrome 300–1
 delusions 8–11, 9*b*
 autochthonous 10
 bizarre 12
 concerning possession of thought 11–12
 of control 11, 49
 cultural misconceptions 9*b*
 definitional issues 9*b*
 delusional memory 11
 delusional misidentification 10–11
 delusional mood 10
 delusional themes 11–13
 depressive 223
 descriptions 10*b*
 false beliefs, *seen as* 9*b*
 Fregoli syndrome 301
 grandiose 12, 232, 292*b*
 of guilt 12
 hypochondriacal 12
 of jealousy 12–13
 mental state examination 49
 mood-congruent/incongruent 12
 nihilistic 12
 paranoid 176
 of persecution 232, 292*b*
 persecutory 11, 232
 primary and secondary 10
 of reference 11, 235, 292*b*
 in schizophrenia 252–53
 sexual/amorous 13
 shared 11
 of thought broadcasting 12, 49
 of thought insertion 11, 49
 thought withdrawal 11–12, 49
 dementia 347–63
 assessment 349–52
 of cause 350–52, 351*b*
 of risk 352
 screening tests 350, 351*b*
 of severity and clinical profile 350
 behavioural and psychiatric symptoms 536–39
 drug treatment 537–39
 non-pharmacological treatment 537
 causes 347*b*, 347–48
 clinical features 348–49
 clinical differentiation 348, 349*t*
 and delirium 344
 and depressive disorders 203
 early detection 352
 emerging concepts 363
 epidemiology in the elderly 523–25
 frontotemporal 359–61
 and GAD 165
 and individuals with DID 473
 with Lewy bodies 358–59
 limbic-predominant age-related TDP-43 encephalopathy (LATE) 363
 mixed 363
 neurodegenerative diseases, classification as
 proteinopathies 363
 Parkinson's disease 364

- presenile and senile 348–49
- prion diseases 361–63
- ratings used in 48
- risk factors 352–53, 353*b*
- subcortical and cortical 348, 350*t*
- treatment 536–41
- vascular 357*b*, 357–58
- see also* Alzheimer's disease; amnesia/
amnesic disorders; memory,
disturbances of
- denial 139*b*
- deoxyglucose 116
- dependent personality disorder 391*b*,
391, 399
- depersonalization 16, 48
- and OCD 184
- depression 193–230
 - activity 226
 - acute treatment 216–21
 - antidepressants 217–18
 - aetiology 204–15
 - early environment 206
 - genetic causes 205–6
 - neurobiological approaches 209–15
 - personality 206
 - precipitating factors 207–8
 - psychological approaches to 208
 - agitated 196
 - altered cerebral perfusion 214*b*
 - Alzheimer's disease 354
 - amino acid neurotransmitters 211–12
 - and anxiety 203
 - atypical 197
 - Beck Depression Inventory 68
 - behavioural activation 220
 - bipolar *see* bipolar disorder
 - brief episodes, in mania 231
 - in children and adolescents 447–51
 - aetiology 448, 448*t*
 - clinical picture 447
 - epidemiology 447
 - management 448–49
 - medications 449
 - prognosis 449
 - psychological treatment 449
 - classification of 199–203
 - by course 200–1
 - in DSM and ICD 200*b*, 201–2
 - in everyday practice 202
 - mixed depressive and anxiety
disorder 198
 - by presumed aetiology 199
 - by symptomatic picture 199–200
 - clinical associations 5–16
 - clinical features 193–99
 - cognitions 195
 - cognitive theories 208, 209*f*
 - cognitive therapy/cognitive
behavioural therapy 213, 219–
20, 222
 - course/prognosis 215–16
 - major depression 215–16
 - and dementia 203
 - depressive stupor 196–97
 - depressive syndromes 193–99
 - differential diagnosis 203
 - dopamine function 211
 - dynamic psychotherapy 220
 - dysthymia 204, 216, 217
 - ECT
 - long-term treatment 225–26
 - resistant depression 229
 - endocrine abnormalities 212
 - endogenous 199
 - epidemiology 203–4
 - 5-HT function 210–11
 - genetic causes 205–6
 - and grief 157
 - guilt 195
 - Hamilton Rating Scale for
Depression 68
 - Hospital Anxiety and Depression
Scale 68
 - and hypomania 234
 - in ICD-11 194*b*
 - and immune system 212*b*, 212–13
 - inter-ictal disturbances 374*b*, 374
 - interpersonal psychotherapy 220, 222
 - ketamine 228–29
 - long-term treatment 221–30
 - augmentation 227–29
 - changes in 227
 - combination 227
 - continuation 221
 - ECT 225–26, 229
 - failure to respond to initial
treatment 226–29
 - maintenance treatment 221
 - management 223–30
 - medications 221, 222*f*
 - prevention of relapse/
recurrence 221
 - psychotherapy 222, 226
 - transcranial magnetic
stimulation 229
 - major 203–4, 215–16
 - management 223–30
 - guidelines 225*b*
 - less severe depression 225*b*
 - and mania 234
 - medications
 - anticonvulsants 218
 - antidepressants 217–18, 224–25
 - atypical antipsychotic drugs 218
 - continuation treatment 221
 - ECT 225–26
 - electroconvulsive therapy 218–19
 - lithium 218
 - maintenance treatment 221, 222*f*
 - unipolar depression 221
 - melancholic 194, 199–200, 200*b*
 - mental state examination 47
 - mild depressive states 197–98
 - mindfulness-based cognitive
therapy 222
 - minor affective disorders 198–99
 - minor depressive-anxiety disorder 204
 - mixed 197
 - mode of inheritance 205
 - monamine hypothesis 114, 210*b*
 - role of monamines in
depression 211
 - Montgomery-Åsberg Depression
Rating Scale inventory 68
 - neurobiological approaches to 209–14
 - neurotrophic hypothesis 213
 - neurovegetative symptoms 195–96
 - non-melancholic 200
 - noradrenaline function 211
 - and OCD 184
 - parental deprivation 206
 - Parkinson's disease 364
 - pathophysiological mechanism 215
 - and personality 206
 - and physical illness 207–8
 - postnatal 641–42
 - post-schizophrenia depression 255
 - post-stroke depression 367–68
 - precipitating factors 207–8
 - predisposing factors 214
 - prevention of relapse/
recurrence 221, 229–30
 - psychiatric aetiology 204–15
 - cognitive theories 208, 209*f*
 - early environment 206
 - immune changes 212*b*
 - multifactorial origin 205*b*
 - neurological approaches 209–14
 - personality 206
 - precipitating factors 207–8
 - psychoanalytical theory 208*b*
 - psychological approaches 208
 - psychiatric interviews 56
 - psychological approaches to
aetiology 208
 - psychological treatment 219–20
 - psychomotor retardation 195
 - psychotic 196, 200, 228
 - reactive 199
 - recent life events 207
 - recurrent 236
 - recurrent brief 201
 - reduced energy 195
 - rejection sensitivity 197
 - relationships with parents 206
 - residual symptoms after treatment 222
 - resistant 229
 - change in antidepressant
treatment 227
 - combination treatment with
antidepressants 227
 - ECT 229
 - failure to respond to initial
treatment 226–29
 - TMS 52
 - retarded 196
 - role of monamines in 211
 - and schizophrenia 203
 - severe depressive episode 194–96
 - sleep disturbances 195, 213
 - supportive psychotherapy 219
 - symptoms 56
 - terminal illness, adjustment
to 153, 155
 - transcultural factors 199

- depression (*cont.*)
 traumatic brain injury 369
 unipolar 200
 vulnerability factors/life
 difficulties 207
 weight loss 195
see also depressive disorders;
 melancholia
- depressive disorders 193
 agitated depression 196
 and agoraphobia 176
 and anxiety disorders 203
 assessment 222–23
 classification of 201–2, 201*t*
 cognitive errors in 658*b*
 and dementia 203
 differential diagnosis 203
 dialectical behavioural therapy
 for BPD 672–73
 mindfulness 672
 skills training 672
 dysthymia 204
 epidemiology 203–4
 and GAT 165
 incidence of 101
 management 223–30
 and mania 247
 milder depressive states 197–98
 minor 216
 mixed depressive and anxiety
 disorder 164, 198–99
 minor 204
 mortality of 216
 older patients 532–34
 persistent 204
 and phobias 171
 predisposed individuals 101
 prevalence 100
 prognostic factors 216
 required patient knowledge 224*b*
 retarded depression 196
 and schizophrenia 203
 severity of 223–24
 social resources of patient 223–24
 and suicide risk 593
 systematic scheme for clinical
 description 202*t*, 202
see also anxiety disorders; depression
- depressive personality disorder 392
- deprivation of liberty
 Deprivation of Liberty Safeguards 79
 Liberty Protection Safeguards 79
- derailment 16
- derealization 16, 48
- descriptive psychopathology
 aim 2
 cultural variations 4
 defining 2
 form and content 3
 patient's experience 4
 primary and secondary symptoms and
 signs 3
 significance of individual
 symptoms 3–4
- subjective and objective symptoms
 and signs 2–3
 symptoms and signs 2
 terms and concepts 2–4
 understanding and explanation 3
- developmental models 94
- developmental psychopathology 406–7
- Development, Dimensional and
 Diagnostic Interview (3di) 431–32
- development process 404–7
- developmental psychopathology 406–7
 first year of life 404–5
 middle childhood 405
 preschool years 405
 specific developmental disorders 425
 year two 405
see also adolescents; child and
 adolescent psychiatry
- dexamethasone 128
- diabetes 631–32
- diagnosis-specific wards 761
- Diagnostic and Statistical Manual (DSM)
 and CIDI 67
 history/revisions 32
 version 5 *see* DSM-5
 classification scheme
- Diagnostic Interview for Social and
 Communication Disorders
 (DISCO) 431–32
- diazepam 168
- DID *see* disorders of intellectual
 development (DID)
- diffusion tensor imaging (DTI) 117–18
- diminished responsibility 510
- Diogenes syndrome 189
- disability 23
- discharge notes 63
- disorders of intellectual development
 (DID) 57, 467–91
 aetiology 476–83, 478*t*
 ASD and ADHD 473
 assessment and classification 483–85
 autonomy 490
 behaviour modification 490
 care of people with 486–90
 genetic screening and
 counselling 487
 historical perspective 486–87
 prenatal care 487–88
 preventive services 487–88
 specific services 487–90
 challenging behaviour 473–74, 474*b*
 clinical features 469–70
 conflict of interests 490
 consent to treatment 490–91
 and COVID 476
 criminal justice system 476
 delirium 473
 dementia 473
 developmental assessments 485, 486*t*
 development of ideas about 468
 Down syndrome 477
 education, training and
 occupation 488
- environmental factors 477
- epidemiology 469, 471–74
- epilepsy 471
- ethical and legal problems 490–91
- follow-up assessments 485
- functional behavioural assessment 485
- genetic and laboratory
 investigations 484
- genetic factors 477, 479*b*, 480*t*
 and growing old 475
 help for families 488
 history-taking 484
 maltreatment and abuse 475
 medical services 489
 medications 489
 mental state examination 484
 mild ID (IQ 50–70) 469
 moderate ID (IQ 35–49) 469–70
 needs assessment 485
 non-genetic causes 482*t*
 normalization policy 490
 physical disorders 470–71
 physical examination 484
 postnatal prevention 488
 prenatal environment 477
 profound ID (IQ below 20) 470
 psychiatric disorders 471–74
 anxiety disorders 472
 eating disorders 473
 effects on family 475–76
 first diagnosed in childhood and
 adolescence 473
 fragile X syndrome 479–83, 483*b*
 mood disorders 472
 movement disorders 473
 neurodevelopmental disorders 468
 schizophrenia 472
 sleep disorders 474
 treatment 489–90
 psychiatric services 489
 psychological treatment 489–90
 psychotic disorders 472
 residential care 488–89
 sensory and motor disabilities 471
 severe ID (IQ 20–34) 470
 sex, relationships and
 parenthood 474–75
 sexuality 475
 social interaction and adjustment,
 assessment 485
 special educational needs 468
 syndromic and non-syndromic
 forms 468
 terminology 467–69
 WHO on 468
- displacement 139*b*
- dissociation 141
- dissociative disorders 613–16
 amnesia 614
 depersonalization disorder 615
 dissociative fugue 616
 dissociative identity disorder 614–15
 Ganser syndrome 616
 neurological disorders 614

- stupor 616
 - trance and possession disorder 614
 - types 613
 - dissociative identity disorder 614–15
 - DNA 108–9
 - doctor–patient relationship and abuse of 73–74
 - domestic violence 503–4
 - ethical and legal issues 503*b*, 503
 - violence towards partners 503–4
 - dopamine
 - dopamine function 211
 - dopamine hypothesis, schizophrenia 114
 - excessive activity 238
 - and glutamate 277
 - and schizophrenia 276
 - double orientation 8–9, 9*b*
 - Down syndrome 477–79
 - aetiology 477–79
 - features 477, 483*b*
 - dream analysis 95*b*
 - dream anxiety disorder 321
 - drug interactions 689–90
 - benzodiazepines 696
 - bupropion 728
 - buspirone 697
 - carbamazepine 737
 - duloxetine 726
 - lithium 733
 - monoamine oxidase inhibitors (MAOIs) 723
 - pharmacodynamic 689–90
 - pharmacokinetic 689
 - reboxetine 727
 - St John's wort 727
 - trazodone 725
 - venlafaxine 725–26
 - vortioxetine 727
 - DSM-5 classification scheme 21–22, 24, 27
 - acute stress disorder/reaction 140, 142
 - adjustment disorders 141
 - child psychiatric disorders 407, 408*b*
 - classification of reactions to stressful events 140
 - compared with ICD-11 32–34, 32*t*, 33*t*
 - cultural explanation/perceived cause 34
 - cultural idiom of distress 34
 - cultural syndrome 34
 - definitions of mental disorder 23
 - features of version 5 32
 - grief reactions, coding 141, 156
 - mental disorders 23
 - specific conditions
 - ADHD 433
 - adjustment disorders 150–51, 156
 - agoraphobia 175–76
 - amnesia/amnestic disorders 344
 - anxiety disorders 161
 - autism spectrum disorders 427
 - bipolar and related disorder 235
 - BPD 386
 - delusional disorders 296
 - depression 199–200, 201*b*, 201–2
 - disorders of intellectual development (DID) 467–68
 - excoriation (skin picking) disorder 191
 - feeding and eating disorders 305–6
 - GAD 164–65
 - gender dysphoria 330–31
 - hoarding 189
 - hypomania 233
 - insomnia 316
 - mania/manic episodes 233, 234–35
 - neurocognitive disorders 336
 - OCD 161, 184
 - oppositional and conduct disorders 436–37
 - panic disorder 178, 179
 - paranoid personality disorder 294
 - phobias 170
 - psychosis 260
 - schizoaffective disorder 260
 - schizophrenia 257, 259, 259*t*, 275–59
 - schizophreniform disorder 260
 - schizotypal disorder 260
 - seasonal affective disorder (SAD) 201
 - sexual function disorders 322
 - social phobia 172
 - substance use/misuse 544
 - trichotillomania 190
 - DTI *see* diffusion tensor imaging (DTI)
 - dualism
 - limitations of 610
 - medicine and psychiatry 610
 - practical consequences of 610–11
 - duloxetine 217, 726
 - duty-based approach 71–72
 - dynamic psychotherapy 174, 220, 284
 - dysfunctional beliefs 208
 - dysthymia 204, 216, 217
 - dystonias 366
- E**
- early intervention services 767–68
 - eating disorders 305–15
 - anorexia nervosa *see* anorexia nervosa
 - binge eating *see* binge eating disorder
 - bulimia nervosa *see* bulimia nervosa
 - classification of 305–6, 306*f*
 - diagnostic classification 305–6, 306*t*
 - and individuals with DID 473
 - temporal movement between 305–6, 306*f*
 - echopraxia 17
 - ECT (electroconvulsive therapy)
 - administration technique 742–45
 - bipolar disorder 242–43
 - cognitive problems after 741–42
 - comparison with other treatments 219
 - comparison with simulated ECT 218–19
 - consent to 745–46
 - contraindications 742
 - depression 218–19
 - long-term treatment 225–26
 - resistant 229
 - ethical aspects 715
 - frequency and number of treatments 745
 - history of physical treatments 686, 738
 - indications for 219, 738–39
 - mania 241, 247
 - mixed affective states 241, 247
 - mode of action 710–18
 - mortality of 742
 - neurobiological effects 740–41
 - older patients 530
 - patient attitude 746
 - physiological changes 741
 - relapse prevention 745
 - schizophrenia 282–83
 - and seizures 740
 - unilateral or bilateral 741
 - unwanted effects 741
 - use of 225–26
 - EEG *see* electroencephalogram (EEG)
 - EER *see* experimental event rate (EER)
 - elation, clinical associations 6
 - elderly patients *see* older patients
 - electroconvulsive therapy (ECT) *see* ECT (electroconvulsive therapy)
 - electroencephalogram (EEG) 116–17
 - coma 19
 - evoked potentials 117, 275
 - magnetoencephalography 117
 - methods 116
 - neuropsychiatric patient 341
 - P300 response 117, 275
 - quantified 117
 - schizophrenia 275
 - supporting of findings 259*b*, 277–78
 - sleep 117
 - standard EEG 116–17
 - electromyographic (EMG) activity 115
 - electronic records 66–67
 - EMDR *see* eye movement desensitization and reprocessing (EMDR)
 - emergency interviews 55
 - EMG *see* electromyographic (EMG)
 - emotional incontinence 5
 - emotionally unstable personality disorder 389–90
 - see also* borderline personality disorder (BPD)
 - emotion-reducing strategies 138
 - emotions and mood disturbances, symptoms and signs 4–6
 - encephalitis 377–78
 - endocrine abnormalities
 - in depression 212
 - endocrine pathology 212
 - HPA axis 9199–9202
 - endocrinology 115
 - endophenotypes 109–10
 - Engel, George 94

- environmental factors
 - adverse caregiving environments 425
 - Alzheimer's disease 356
 - anorexia nervosa 308
 - ASPD 394
 - bipolar disorder 237–38, 239, 245
 - child and adolescent
 - psychiatry 412–15
 - child maltreatment 458
 - dementia 352–54, 356
 - depression 206
 - epidemiology 99–100
 - family influences 412–14
 - gene–environment interactions
 - and correlations 111, 265, 268, 269, 394
 - hospital admission 285
 - and individuals with DID 477
 - interaction between person and environment 102
 - maternal deprivation,
 - attachment 412–13
 - narcolepsy 319
 - neighbourhood, effects of 414
 - Parkinson's disease 365
 - peer groups 403
 - prenatal environment 266
 - psychosocial environment 239
 - schizophrenia 263, 264, 265–66, 268, 269*b*, 269–71
 - school, effects of 414–15
 - sexual dysfunction 324
 - social environment 405
 - twin studies 265
- epidemiology
 - aims of enquiries 99
 - case, defining 99
 - case finding 99
 - causes in the environment 99–100
 - clinical and social variables 99
 - cohort studies 99
 - concepts and methods 99
 - declared cases 99
 - depression 203–4
 - lifetime expectation or risk 99
 - period prevalence 99
 - point prevalence 99
 - population at risk, defining 99
 - precipitating factors 99–100
 - predisposing factors 99–100
 - prevalence rates 99
 - scientific disciplines, contribution to
 - psychiatric aetiology 99–100
 - see also specific conditions*
- epigenetics/genetic factors 111
- epilepsy 6, 370–75
 - absences 371
 - aetiology 371–72
 - in children 411–12
 - crime, association with mental disorder 500
 - epidemiology 371
 - ictal disturbances 373
 - and individuals with DID 471
 - inter-ictal disturbances 374
 - cognitive impairment 374
 - depression 374*b*, 374
 - personality 374
 - psychosis 374
 - suicide risk 374
 - making the diagnosis 372
 - post-ictal disturbances 373
 - pre-ictal disturbances 373
 - psychiatric aspects 373–74
 - and psychological disturbance 373*b*, 373
 - social aspects 374–75
 - sudden unexplained death in
 - epilepsy 471
 - treatment 375
 - see also seizures*
- epistasis 268
- Erikson, Erik 382
- erotomania/erotic delusions 299–300
- escitalopram 181, 217, 715
- Esquirol, Jean-Étienne 97
- ESRS *see* Extrapyramidal Symptoms Rating Scale (ESRS)
- ethical problems 72*b*
 - in child and adolescent psychiatry 464
 - confidentiality *see* confidentiality
 - consent to treatment 76–77
 - doctor–patient relationship 73–74
 - mental capacity 77–79
 - personality disorder 400–1
 - prescribing advice 761–62
 - in psychiatric practice 73–80
 - research ethics 79–80
- ethics
 - codes of practice 72–73
 - duty-based approach 71–72
 - ethical approaches 71–72
 - ethical principles 72
 - ethical problems 73–80
 - general issues 71–73
 - mental capacity 77–79
 - Mental Health Act 81–85
 - of research 79–80
- ethnic stratification 109
- ethology and evolutionary psychology 104
- Eugenics Movement 97
- EuroQol EQ-SD 69
- evidence-based medicine (EBM) 119–35
 - absolute risk reduction 122, 126
 - applications 131
 - big data 133
 - clinical utility of interventions 122–23
 - absolute risk reduction 122
 - clinical relevance of effect size 123
 - computing the NNT from ORs 123
 - indices for translating
 - research results into clinical practice 122*b*, 122
 - risk reduction and number needed to treat 122*t*, 122
 - concealment of randomization list 121
 - defining evidence-based medicine 119
 - diagnosis 131–32
 - effect sizes 121
 - clinical relevance 123
 - ethical aspects of therapeutic trials 123–24
 - informed consent 124
 - placebo, use of 123–24
 - randomization 123
 - evaluation of 131
 - hierarchy of quality of research 95*b*, 120
 - history 119–20
 - implementation of 129–31
 - clinical practice guidelines 130–31
 - effectiveness 130
 - efficacy 130
 - efficiency 130
 - for individual patients 129–30
 - at a service level 130
 - indices 122*b*
 - individual treatment studies 121–24
 - number to treat 122, 122*t*
 - presentation of results 121
 - odds ratios 121
 - relative risk 121
 - prognosis 131–32
 - qualitative research methods 133–35
 - rationale for 120
 - systematic reviews *see* systematic reviews
 - types of evidence 120–21
 - validity 121
- evidence-based practice 73
- evolutionary psychology 104
- exceptions to confidentiality rule
 - for legal representatives 76
 - in the patient's interest 76
 - in the public interest 76
- excessive daytime sleepiness 318*b*, 318
- excoriation (skin picking)
 - disorder 190–91
 - clinical features 190–91
 - diagnosis 191
 - epidemiology 191
 - management 191
- exercise 284
- exhibitionism 330
- exome and genome sequencing 109
- expansive ideas 232
- experimental and clinical psychology 102–4
 - continuity between normal and abnormal 102
 - factors maintaining abnormal behaviour 102
 - information processing 102–4
 - interaction between person and environment 102
 - neuropsychology 102
- experimental event rate (EER) 122
- explicit (conscious) declarative memory 102
- exploitation of patients by doctors 73–74

- exposure and response prevention (ERP) 187
 Extrapyramidal Symptoms Rating Scale (ESRS) 69
 eye movement desensitization and reprocessing (EMDR) 146, 147, 668
 Eysenck Personality Questionnaire 206
- F**
- fabricated/induced illness 627
 facial expression 46
 factitious disorder 625–26
 faecal soiling 451–52
 false accusations 511
 false confessions 511
 false memories 18
 family homicide 503
 family interviews 56
 family law 86
 family studies 104–5
 delusional disorders 297
 depression 205
 index cases 104
 probands 104, 265*t*
 schizophrenia 264–65
 family therapy
 anorexia nervosa 310
 eclectic 680–81
 assessment 680
 intervention 680–81
 schizophrenia 283*b*, 283
 systematic 679
 fatal familial insomnia 362
 female orgasmic disorder 327
 female sexual arousal disorder 326
 fetishistic disorder 329–30
 fibromyalgia 625
 fitness to drive 87
 5-HT *see* serotonin (5-HT)
 5-hydroxyindoleacetic acid (5-HIAA) 113
 5-hydroxytryptamine and aggression 394
 flashbacks, acute stress disorder 141
 flight of ideas 15
 18-fluorodeoxyglucose (FDG) 275
 fluoxetine 242, 449, 715
 fluphenazine 702
 fluvoxamine 715
 fMRI (functional magnetic resonance imaging) 115, 116
 BOLD signal 116, 275
 depression 213, 214
 schizophrenia 275
 task-based 214
 forensic child psychiatry and juvenile delinquency 440–42
 assessment 440–41, 441*b*
 causes 440
 interventions for juvenile offenders 441–42, 442*b*
 multidimensional treatment foster care 442
 multisystemic therapy 442
 structured programs 442
 forensic psychiatry 493–519
 advice on medical treatment 519
 arson 507
 assessment 518
 causes of crime 496–97
 child abduction 507
 crimes of violence 502–4
 dangerousness, associated factors 516*b*
 dangerous severe personality disorder units 513
 ethical issues 494*b*
 ethnic minorities 501–2
 female offenders 501
 headings for a court report 517*b*
 involvement in psychiatrists in stages of legal process 517, 518*t*
 management 508
 of violence in health care settings 514–15
 preparing report 518
 psychiatric aspects of being a victim 508–9
 burglary and robbery 508
 murder 508
 psychological impact 508
 rape 508
 terrorist crimes 508
 types of crime 508
 psychiatric aspects of specific crimes 502–7
 psychiatric report 517–19
 psychiatrist appearing in court 519
 risk assessment 515–17
 role of psychiatrist in criminal courts 509–11
 absence of intention 510–11
 amnesia 511
 assessment 510
 children 509
 competence to stand trial 509–11
 diminished responsibility 510
 false accusations 511
 false confessions 511
 fitness to be punished 511
 legal insanity 509–10
 mental state, intention and responsibility 509
 role of psychiatrist in relation to court 517–18
 routine psychiatric care 508
 sexual offences 504–7
 shoplifting 507
 specialist services 508
 specific offender groups 501–2
 treatment in the community 513
 treatment of offenders with mental disorders 511–13
 general issues 511–12
 mentally abnormal in prison 512
 offenders in hospital 512–13
 settings of treatment 512
 special hospitals and secure units in the UK 513
 young people 501
 see also criminal justice system; criminology, general; forensic child psychiatry and juvenile delinquency
 formal thought disorder 15, 48
 formation 8
 formulations 63–65, 65*b*
 example 65*b*
 fostering 422–23
 foster care adoption 422
 multidimensional treatment foster care 442
 Foucault, Michel 94
 fragile X syndrome 482–83, 483*b*
 free association 95*b*
 Regoli syndrome 301
 Freud, Anna 138–40, 421, 681
 Freud, Sigmund 94
 on anxiety 162
 case studies 133
 contribution of writings 26
 on defence mechanisms 138–40
 on depression 208
 and development of psychological treatments 657
 on dissociation 613*b*
 on dream work and free association 95*b*
 and experimental psychopathology 2
 on fetishism 328
 on hypnosis 656–57
 on the id, ego and superego 657
 on mourning 208
 on nervous diseases 98
 on personality development 383
 and psychoanalysis 95*b*, 208, 328
 on PTSD 149
 on schizoid personality disorder 395
 on sexual impulses 95*b*
 on the unconscious 95*b*, 298, 657
 frontotemporal dementias 359–61
 clinical features and subtypes 359–61, 360*b*
 diagnostic categories subsumed within 359
 features characteristic of main subtypes 360*b*, 360
 drug treatment 540–41
 Pick's disease 361
 frotteuristic disorder 330
 functional brain imaging 102
 functional disorders 26
 functional enuresis 451
 funnel plots 128
- G**
- GABA *see* gamma-aminobutyric acid (GABA)
 gabapentin 174, 698
 misuse of 578

- GAD *see* generalized anxiety disorder (GAD)
- GAF *see* Global Assessment of Functioning (GAF)
- Galen 96
- Gall, Franz 96
- gamma-aminobutyric acid (GABA) 112–13, 167
- and depression/depressive disorders 205, 211–12
 - and schizophrenia 277
- Ganser syndrome 616
- garrulous patients 57
- gender dysphoria 322, 330–32
- aetiology 331
 - associated psychopathology 331
 - in children and adolescents 453
 - counselling 332
 - detransitioning 332
 - epidemiology 331
 - gender-affirming sex reassignment surgery 332
 - hormone therapy 332
 - psychotherapy 332
 - real-life experience and social transition 332
 - terminology 331*b*
 - treatment 331–32
- gender identity disorder 322
- gender incongruence 322, 323*t*
- General Health Questionnaire (GHQ) 69, 99
- Generalized Anxiety Disorder Assessment (GAD-7) 68
- generalized anxiety disorder (GAD) 163–69
- anxiety and depression 164
 - autonomic overactivity 163–64
 - clinical picture 163–64
 - cognitive behavioural theories 166
 - cognitive behavioural therapy 168
 - cognitive theory 166
 - comorbidity of anxiety symptoms 164–65
 - conditioning theories 166
 - continuous anxiety 161
 - and dementia 165
 - and depressive disorders 165
 - diagnostic conventions 164
 - differential diagnosis 165
 - early experiences 166
 - genetic causes 166
 - hyperventilation 164
 - long-term drugs 168
 - management 168–69
 - medications 168
 - mistaken for physical illness 165–67
 - aetiology 166–67
 - epidemiology 165–66
 - psychiatric aetiology 166–67 - muscle tension 164
 - neurobiological mechanisms 167
 - and other anxiety disorders 164–65
 - and panic disorder 179
 - personality traits 166–67
 - and physical illness 165
 - pregabalin 698
 - prognosis 167
 - psychological arousal 163
 - required patient knowledge 168–69
 - and schizophrenia 165
 - self-help and psychoeducation 167–68
 - short-term drugs 168
 - sleep disturbances 164
 - and social phobia 176
 - stepped-care approach 169*b*
 - stressful events 166
 - and substance misuse 165
 - symptoms 164*b*
 - treatment 167–68
 - and worry 163, 166, 167
- generalized social phobia 172
- General Medical Council (GMC) 46, 72–73, 79–80
- on confidentiality 74
 - on consent to treatment 76
- general paresis 377
- general practice, interviews in 55–56
- general practice liaison 764
- genetic association studies 108–9
- candidate gene studies 108–9, 205
 - exome and genome sequencing 109
 - genome-wide association studies 107, 109
- genetic counselling 637
- genetic polymorphisms 107, 116, 185
- genetics/genetic factors 104–11
- ADHD 434
 - adoption studies 105
 - allelic heterogeneity 95*b*
 - Alzheimer's disease 355–56
 - amyloid precursor protein 107
 - aneuploidy 107
 - anorexia nervosa 308
 - apolipoprotein E gene, apoE4 allele in Alzheimer's 110*b*, 354
 - ASD 430*b*, 433
 - candidate genes 108–9, 205
 - child and adolescent psychiatry 410, 411*b*
 - chromosomal (cytogenetic) abnormalities 107
 - chromosome 21 107
 - clinical and ethical implications 111
 - common disease–variant model 107
 - copy number variation 108
 - and crime 496
 - depression 205–6
 - difficulting inf finding genes for psychiatric disorders 95*b*
 - distinction between Mendelian and non-Mendelian disorders 107
 - Down syndrome 107
 - epigenetics/genetic factors 111
 - gene–environment interactions and correlations 111, 265, 268, 269, 394
 - genetic contribution to psychiatric disorders 104–5
 - family studies 104–5
 - genetic heterogeneity 95*b*
 - genetic linkage 107
 - genetic testing 341
 - and individuals with DID 477
 - karyotyping 107
 - Klinefelter syndrome 107
 - linkage and association 107
 - mode of inheritance 106–7, 205
 - molecular genetics 205–6, 237, 383
 - mutations 107–8
 - non-Mendelian genetic disorders 106, 107
 - non-protein-coding genes 110
 - and OCD 185
 - oppositional and conduct disorders 437
 - phenotypes 104
 - polygenic inheritance 205
 - polymorphisms *see* polymorphisms
 - relationships between genotype and phenotype 109–10
 - schizophrenia 264–69
 - terminology of molecular genetic variation 95*b*
 - Turner syndrome 107
 - twin studies 95*b*
 - types of genetic variation 107–8
 - velocardiofacial syndrome 107
 - see also* family studies; genetic association studies; heritability; *specific conditions*
- genome-wide association studies (GWAS) 107, 109, 114
- depression 205
 - panic disorder 180
- genomics 114
- Gerstmann–Straussler–Scheinker syndrome 362
- GHQ *see* General Health Questionnaire (GHQ)
- Global Assessment of Functioning (GAF) 69
- global burden of mental illness 651–54
- children and adolescents 653–54
 - HIV/AIDS pandemic 653
 - humanitarian emergencies 654
 - integration into primary health care 652
 - lay mental health workers 651
 - Lancet Commission on adolescent health and wellbeing 654
 - Mental Health Gap Action Programme (mhGAP) 651
 - perinatal mental illness 653
 - proportions of age-standardized estimated disability adjusted life years 651, 652*t*
- global mental health, concepts and missions 650–51
- global rating scales 69
- glutamate 211–12, 238

- brain levels 238
- and dopamine 277
- schizophrenia 276–77
- Grades of Recommendation, Assessment, Development, and Evaluation (GRADE) system 125
- grandiose delusions 12, 232, 292*b*
- grief
 - abnormal 157
 - complicated 157
 - complicated grief treatment 158–59
 - counselling 158
 - defining 156
 - and depression 157
 - first stage 156
 - management 157–59
 - medication 158
 - normal reactions 156*b*
 - persistent complex bereavement disorder 157
 - prolonged grief disorder 141
 - psychotherapy 158–59
 - second stage 156
 - for stillborn child 158
 - support groups 158
 - third stage 156–57
 - see also* bereavement
- Griesinger, Wilhelm 97
 - Pathology and Therapy of Mental Disorders* 97
- group psychoeducation 167–68
- group psychotherapy
 - encounter groups/psychodrama action techniques 678
 - group analysis 678
 - interpersonal 677–78
 - problems in 677–57
 - setting up the group 677
 - origins 675
 - small-group 675–76
 - classification 675–76
 - self-help groups 676–77
 - supportive groups 676
 - terminology 676
 - therapeutic factors 676*b*, 676
 - word groups 678
- group therapy 75
- growth faltering 460–61
- guanfacine 436
- guilt 153, 195
- GWAS *see* genome-wide association studies (GWAS)
- H**
- habit learning 186
- HADS *see* Hospital Anxiety and Depression Scale
- hallucinations 6–8, 7*b*
 - auditory 7–8, 50, 252
 - autoscopic 7
 - clinical associations 7–8
 - complex 7
 - of deep sensation 7
 - elementary 7
 - extracampine 7
 - gustatory 7
 - hypnopompic/hypnagogic 6, 7
 - in mania 232
 - olfactory 7, 8
 - pseudohallucinations 8
 - reflex 7
 - second-person/third-person 7
 - tactile 7, 8
 - visual 7, 8, 50
- hallucinogens 583
- haloperidol 240, 245
- Hamilton Anxiety Scale (HAS) 68
- Hamilton Rating Scale for Depression (HAM-D) 68, 121
- handicap 23
- Hare Psychopathy Checklist-Revised (PCL-R) 389
- head injury
 - and boxing 375
 - in children 411
 - treatment 369–70
 - see also* traumatic brain injury (TBI)
- health anxiety
 - (hypochondriasis) 671–72
- Health of the Nation Outcome Scales (HoNOS) 69
- Health Regulatory Authority, UK 79–80
- Helsinki Declaration 79–80
- heritability 105–6
 - estimates 106, 106*t*
 - penetrance 106
 - phenocopies 106
- hippocampus 102, 145–46, 167, 173
 - and anxiety 167
 - brain imaging studies 145*b*, 185
 - decreased volume 185
 - LATE 363
 - neurofibrillary tangles 355
 - neurogenesis 741
 - neurone loss 355
 - Pick bodies 361
 - and social phobia 173
- history, psychiatric *see* psychiatric history
- histrionic personality disorder 390*b*–95, 390–91, 395
- HIV infection 635
- hoarding 175–78, 188–89
 - aetiology 189
 - clinical features 189
 - diagnosis 189
 - epidemiology 189
 - management 189–90
- home interviews 56
- homicide 502–3
 - family 503
 - followed by suicide 502–3
 - infanticide 503
 - multiple 502
 - parents killing children 503
- homovanillic acid (HVA) 211
- HoNOS *see* Health of the Nation Outcome Scales (HoNOS)
- Hospital Anxiety and Depression Scale (HADS) 68
- hospital interviews 56
- humours, body 96
- Huntington's disease 365–66
 - aetiology 366
 - clinical features 365*b*
 - neuropathology 366
- HVA *see* homovanillic acid (HVA)
- hydrocephalus, normal pressure 378
- hyperarousal 144, 147
- hypersomnolence 318*b*, 318
- hyperventilation
 - GAD 164
 - panic disorder 179*b*, 179, 180
- hypnotics 699–700
 - compounds available 699
 - interactions 700
 - management advice 700
 - pharmacology 699
 - unwanted effects 699–700
- hypoactive sexual desire disorder 326
- hypochondriasis 12
- hypoglycaemia 165, 378
- hypomania 231
 - and depression 234
 - distinguished from mania 234–35
 - ICD-11/DSM-5 criteria 233
- hypothalamic–pituitary–adrenal (HPA) axis 145*b*
 - bipolar disorder 238
 - depression 193–212, 205*b*
- hysterectomy 643
- I**
- ICD-10 classification scheme
 - ADHD 433
 - affective personality disorders 392
 - anankastic (obsessive-compulsive) personality disorder 391–92
 - anxious (avoidant) personality disorder 391*b*, 391
 - comparison with ICD-11 30, 31*t*
 - dependent personality disorder 391
 - emotionally unstable personality disorder 390
 - histrionic personality disorder 390
 - impulsive personality disorder 390
 - personality disorder 385, 386, 387, 387*t*
- ICD-11 classification scheme 21–22, 24, 27
 - acute reactions to stress/acute stress disorder 140, 141–42
 - adjustment disorders 141, 156
 - child psychiatric disorders 407, 408*b*
 - classification of reactions to stressful events 140
 - clinical features of a manic episode 233*b*
 - compared with DSM-5 32–34, 32*t*, 33*t*
 - comparison with ICD-10 30, 31*t*
 - definitions of mental disorder 23
 - depression 194*b*

- ICD-11 classification scheme (*cont.*)
 diagnostic conventions/groups 30,
 31*t*, 33*t*, 141–42
 entry into force (2022) 30
 grief reactions, coding 141, 156
 mental disorders 23
 organization 30
 personality disorder 387, 387*t*
 specific conditions
 abnormal personalities 386–87
 acute and transient psychotic
 disorder 260
 ADHD 433
 adjustment disorders 150–51
 agoraphobia 175–76, 176*b*
 amnesia/amnesic disorders 344
 anxiety disorders 161
 autism spectrum disorders 427
 BPD 386
 bulimia nervosa 312
 catatonia 260–61
 complex PTSD 147, 148*b*–39*b*
 delusional disorders 296
 depression 199–200, 201*b*, 201–2
 disorders of intellectual
 development (DID) 467–68
 dissociative disorders 616
 DSM-5 143
 excoriation (skin picking)
 disorder 191
 feeding and eating disorders 305–6
 GAD 164–65
 gender dysphoria 330–31
 hoarding 189
 hypomania 233
 impulsive personality disorder 390
 insomnia 316
 mania/manic episodes 233, 234–35
 neurocognitive disorders 336
 OCD 161, 184
 oppositional and conduct
 disorders 436–37
 panic disorder 178, 179
 pathological jealousy 500
 personality disorder 385, 386–87
 phobias 170
 psychosis 260
 PTSD 143, 148*b*
 schizoaffective disorder 260
 schizophrenia 257, 258, 259, 259*t*
 schizotypal disorder 260
 seasonal affective disorder
 (SAD) 201
 sexual function disorders 322
 social phobia 172*b*, 172
 trichotillomania 190
see also International Classification of
 Diseases (ICD)
- ideas
 common-sense 89
 development of 162
 flight of 15
 overvalued 14–15
- identification 139*b*
 illusions 6
- imagery
 acute stress disorder 141
 and perception 6
- imipramine 177–78, 181
- immune system, and
 depression 212*b*, 212–13
- impairment 23
- implicit emotional memory system 102
- impulsive personality disorder 390
- incapacity 23
- indecent assault 505
- indecent exposure 505
- index cases, family studies 104
- Index of Definition 67
- indoleamine 2, 3-dioxygenase. 212–13
- induced pluripotent stem cells
 (iPSCs) 114
- infanticide 503
- infections 634–35
- information
 collection of 37
 consent to disclosure of 74–75
 disclosing from others 75
 disclosure 75
 evaluation of 37–38
 integrating/evaluating
 information 58–61
 drawing conclusions/
 decision-making 59–60
 issues affecting how information is
 evaluated 60–61
 open and explicit sharing 124
 recording of *see* recording of
 information
 safeguarding 74
 seeking from others 75
 use of 38
- information processing, experimental
 and clinical psychology 102–4
- informed consent 80
 evidence-based medicine 124
 research involving people who cannot
 give 80
- inpatient notes 62–63
- inpatient wards 759–62
- insight 19–20
 impaired in mania 232
 impaired in schizophrenia 253
- insomnia 316–18
 assessment 317
 stimulus control 317
 transient 316
 treatment 317–18, 318*b*
- intellectual development, patients
 with disorders of *see* disorders of
 intellectual development (DID)
- intellectual disability 467–68
 crime, association with mental
 disorder 499
- intellectual impairment starting in early
 childhood (ID) 468
- inter-ictal disturbances 374
 cognitive impairment 374
 depression 374*b*, 374
 personality 374
- psychosis 374
 suicide risk 374
- intermetamorphosis 301
- intermittent explosive disorder 500
- International Classification of
 Diseases (ICD)
 editions/revisions 30
 ICD-10 on personality
 disorder 385, 386
 version 11 *see* ICD-11
 classification scheme
- International Classification of Sleep
 Disorders (ICSD 3) 316
- International Personality Disorder
 Examination 386
- International Pilot Study of
 Schizophrenia (IPSS) 29
- interpersonal psychotherapy 220,
 222, 663–64
- interviews, psychiatric *see* psychiatric
 interviews
- intoxication 544
- irritability and anger, clinical
 associations 6
- irritable bowel syndrome 625
- J**
- jamais vu 18
- Jung, Carl 383
- justice 72
- K**
- ketamine 728–29
 in depression 228–29
- Kleine–Levin syndrome 319–20
- Klein, Melanie 95*b*, 395, 421
- kleptomania 501
- knight's move, thought disorder 16, 252
- Korsakoff syndrome 345, 552
- L**
- laboratory investigations 52
- Laing, R. D. 94
- lamotrigine 218, 242, 248, 735–36
- Lancashire Quality of Life Scale 69
- language disorder, in children 427
- learning disorders 425–26
- Lennox–Gastaut syndrome 471
- letters to GP 66
- Lewis, Aubrey 98
- Lewy bodies, dementia with 358–59
 aetiology 359
 clinical criteria 359*b*
 clinical features 358
 drug treatment 540
 neuropathology 358–59
- Liberty Protection Safeguards (LPS) 79
see also deprivation of liberty
- libido 95*b*
- life events, study of 100–1
- lilliputian 7
- limb amputation 631
- limbic-predominant age-related TDP-43
 encephalopathy (LATE) 363
- linkage disequilibrium 95*b*

- lisdexamfetamine 315
 lithium 218, 729–34
 for bipolar depression 242
 for bipolar disorder 240, 243, 249, 686–87
 contraindications 733
 for depression
 augmentation 218, 228
 in combination with antidepressants 218
 long-term treatment 230
 as a sole treatment 218
 dosage and plasma concentrations 730
 drug interactions 733
 management 733–34
 mechanism of action 729
 pharmacokinetics 730
 toxic effects 732
 unwanted effects 730–32
 withdrawal from 733
 lofepramine 181, 717–18
 lorazepam 241, 246
 lormetazepam 699
 LPS *see* Liberty Protection Safeguards (LPS)
 lurasidone 242, 703
- M**
 magnetic resonance imaging (MRI) 102
 bipolar disorder 238
 cerebral pathology and maturation 394
 depression 213
 hyperintense 213
 neuronal metabolites and transmitters measured by 112–13, 113t
 magnetic resonance spectroscopy (MRS) 112, 113t
 bipolar disorder 238
 depression 211–12
 neuropsychiatric patient 341
 magnetoencephalography (MEG) 117
 major depression 203–4
 and bipolar disorder 239
 course/prognosis 215–16
 see also depression
 male erectile dysfunction 326–27
 male orgasmic disorder 327
 malingering 627
 mania 231–33
 acute 245–47
 Altman Self-Rating Mania Scale 68
 appearance and behaviour 232
 clinical features of a manic episode in ICD-11 233b
 continuation treatment 241
 cyclothymia 234
 delusions of persecution 232
 and depression 234
 and depressive disorders 247
 distinguished from hypomania 234–35
 ECT 241
 elevation of mood 231
 expansive ideas 232
 flight of ideas 232
 general care 240
 general clinical management 245
 grandiose delusions 232
 hospital admission 245
 ICD-11/DSM-5 criteria 233
 increased activity 231
 insight impairment 232
 management 245–47
 manic stupor 308
 medications 240–41, 245–47
 antipsychotics 245, 246
 atypical antipsychotics 240
 benzodiazepines 241
 carbamazepine 240–41
 continuation treatment 241, 247
 ECT 241
 lithium 240
 mood stabilizers 245, 246
 typical antipsychotics 240
 valproate 241, 246
 mental state examination 47
 mixed mood (affective) states 233 and mood 231–32
 NICE guidance 247b, 247
 perceptual disturbances 232
 psychotherapy 245
 rapid cycling disorders 234
 ratings of 68
 self-important ideas 231
 speech 232
 treatment 240–41
 see also bipolar disorder
 mannerisms 16
 Marmot, M 648
 maternal deprivation, attachment theory 383–84
 maternal mental health, effect on child 642
 Maudsley, Henry 232
 The Pathology of Mind 97
 MBCT *see* mindfulness-based cognitive therapy (MBCT)
 MDMA 582
 MDQ *see* Mood Disorder Questionnaire (MDQ)
 medical models 93–94
 medications 685–752
 ADHD 435–36
 agoraphobia 177–78
 antidepressants 177–78
 anxiolytics 177
 anorexia nervosa 310–11
 anticonvulsants 218
 antidepressants *see* antidepressants
 antipsychotics *see* antipsychotics
 anxiolytics 158, 177
 behavioural problems in people with DID 489
 benzodiazepines 181
 bipolar disorder 242, 243
 antipsychotics 244
 bipolar depression 247–48
 carbamazepine 243–44
 lamotrigine 243
 longer-term treatment 243–44
 mania 240–41
 prevention of relapse/recurrence 248–49
 valproate 243
 bulimia nervosa 314
 catatonia induced by 261
 child and adolescent psychiatry 419
 choice of 398
 classification of 693–94, 694t
 depression
 anticonvulsants 218
 antidepressants 217–18
 atypical antipsychotic drugs 218
 continuation treatment 221
 lithium 218
 maintenance treatment 221, 222f
 unipolar depression 221
 drug interactions 689–90
 drug withdrawal 690
 GAD 168
 grief reactions 158
 history of physical treatments 685–87
 hypnotics 158
 mania 240–41, 245–47
 atypical antipsychotics 240
 benzodiazepines 241
 carbamazepine 240–41
 continuation treatment 241
 ECT 241
 lithium 240
 typical antipsychotics 240
 valproate 241
 panic disorder 181
 paranoid symptoms and delusional disorder 302
 pharmacogenetics 689
 pharmacokinetics 688–89
 prescribing advice 690–91
 PTSD 147
 schizophrenia 280, 281b
 acute episode 286
 antidepressants 282
 antipsychotics *see above*
 side effects 228, 280–81, 435
 social phobia 174
 see also specific drugs
 MEG *see* magnetoencephalography (MEG)
 melancholia 96
 melancholic depression 194, 199–200, 200b
 melatonin 699
 memory, disturbances of 18
 in depression 195
 immediate, recent and long-term 18, 51
 mental state examination 50–51
 mild cognitive impairment/neurocognitive disorder 347, 349–50, 352
 subjective cognitive decline 349–50
 see also amnesia/amnesic disorders; delirium; dementia; neuropsychiatric patient
 meningovascular syphilis 377

- menopause 643
- mental capacity 77–79
 - Advance Decision to Refuse Treatment 78–79
 - advanced statements 78–79
 - Bournewood case 79
 - consent by proxy 79
 - Deprivation of Liberty Safeguards 79
 - lack of capacity 77
 - legal concept 77
- Mental Capacity Act (2005) 78–79
- mental deficiency 467–68
- mental disorders
 - autism and learning disability 84
 - biologically determined features 100
 - compulsory treatment for disorders 81
 - concepts 22–23
 - absence of health 22
 - biological disadvantage 22
 - biomedical versus social 23
 - definitions 22
 - diagnoses 23
 - disability 23
 - disease 22
 - disorders 23
 - handicap 23
 - impairment 23
 - incapacity 23
 - pathological process 22
 - presence of suffering 22
 - and crime 497–501
 - criminal justice system 84
 - DSM-5 classification 23
 - greater rights of appeal and better choice of care experience 83–84
 - hierarchy of pressures and leverage 84–85
 - ICD-11 classification 23
 - informal coercion 84
 - political and social perceptions 94
 - race equality 84
 - treatment of offenders with 511–13
 - see also* mental capacity; Mental Capacity Act (2005); Mental Health Act
- mental handicap 467–68
- Mental Health Act 71, 81–85
 - assessment and treatment sections 82
 - community care 758
 - Community Treatment Orders 83
 - compulsory treatment for mental disorders 81
 - detention criteria 81–82
 - informal coercion 84
 - key clinical procedures 72*b*
 - new legislation 83–84
 - police powers/power of entry 83
 - pressures and leverage, hierarchy of 84–85
 - restriction orders and transfers 82–83
 - sections used by clinicians 82, 82*t*
- Mental Health Gap Action Programme (mhGAP) 651
- Mental Health National Service Framework 1999 764–65
- mental retardation 467–68
- mental state examination 46–51
 - appearance 46
 - behaviour 47–48
 - cognitive functions 50–51
 - headings 46*b*
 - and individuals with DID 484
 - mood 47–48
 - neuropsychiatry 341
 - perceptions 49–50
 - social behaviour 47
 - speech *see* speech
 - suicide risk 596
 - thoughts 48–49
- mental subnormality 467–68
- mentalization 658
- Mesmer, A. C 656
- meta-analysis
 - accuracy, determining 128
 - biased conclusions 127–28
 - duplication of publication 127
 - patient level 127
 - problems with 127–28
 - publication bias 127
- metabolic and immune disorders 632
- metabolic syndrome 286
- metabolism
 - and altered cerebral perfusion 214*b*
 - and cerebral blood flow 213–14, 275
 - drug interactions 689
 - noradrenaline 211
 - pharmacokinetics of psychotropic drugs 688
- metabolomics 114
- methadone 574, 576
- N-methyl-D-aspartate (NMDA) 171–72, 228
- α-methyl-para-tyrosine (AMPT) 211
- methylphenidate 435
- Meyer, Adolf 32, 97, 211
- MHPG (3-methoxy-4-hydroxy-phenylethylene glycol) 211
- migration 101–2, 776
 - and ethnicity 271–72
 - selective 101
- mild cognitive impairment/
 - neurocognitive disorder 347, 349–50, 352
- mindfulness-based cognitive therapy (MBCT) 222
- Mini International Neuropsychiatric Interview (MINI) 67
- Mini-Mental State Examination (MMSE) 68, 350, 357, 529, 539, 559
- minor depressive-anxiety disorder 204
- mirtazapine 174, 217, 225, 724
- Mitgehen 17
- mixed depressive and anxiety disorders 164, 198–99
 - classification of 198
 - clinical picture 198–99
- MMSE *see* Mini-Mental State Examination (MMSE)
- MoCA *see* Montreal Cognitive Assessment scale (MoCA)
- moclobemide 174
- molecular genetics 205–6
- monamine hypothesis, depression 114
- monamine neurotransmitters 145–46
- monoamine oxidase A (MAOA) 394
- monoamine oxidase inhibitors (MAOIs) 147, 218, 719–29
 - clinical use in depression 722
 - compounds available 719–20
 - contraindications 722
 - drug interactions 723
 - efficacy in depression 720
 - interactions with other drugs 720–22
- moclobemide 721, 723
- pharmacokinetics 720
- pharmacological properties 719
- unwanted effects 720
- withdrawal from 722–23
- Montgomery-Åsberg Depression Rating Scale inventory 68
- Montreal Cognitive Assessment scale (MoCA) 50, 68, 350, 357, 529
- Mood Disorder Questionnaire (MDQ) 68
- mood disorders
 - and affect 4–5
 - anxious mood *see* anxiety
 - blunting/flattening 5
 - brain imaging 213–14
 - in children and adolescents 447–51
 - depression 447–51
 - crime, association with mental disorder 499
 - cyclothemia 234
 - delusional mood 10
 - depression and mania 47
 - disturbances 4–6
 - fluctuating and incongruous 48
 - and individuals with DID 472
 - lability of 5
 - mania 231–32
 - mixed mood (affective) states 233
 - mental state examination 47–48
 - multifactorial origin of 205*b*
 - neurophysiological changes 214
 - older patients 524–25
 - and paranoid symptoms 295
 - persistent 392
 - and personality 45
 - previous mood disturbance 223
 - psychiatric aetiology of 202*t*
 - psychotic features, with 262
 - rapid cycling 234
 - schizophrenia 253
 - severe depressive episode 194
 - suicidal ideation 48
- mood stabilizers 729–34
 - lamotrigine 735–36
 - mania 245, 246
 - pregnant women 763
 - schizophrenia 282
 - sodium valproate 734–35
 - see also* carbamazepine
- morbid jealousy 483

- Morel, Benedict Augustin 97
 Morgagni, Giovanni Battista 96
 mosaicism 477–79
 motor symptoms and signs 16–17
 ratings 68–69
 mourning 156
 movement disorders
 assessment 46
 dystonias 366
 and individuals with DID 473
 neuropsychiatric patient 364–65
 Parkinson's disease 363–65
 sleep-related 320
 tics 366–67
 Tourette syndrome 367
 MRS *see* magnetic resonance spectroscopy (MRS)
 multiaxial approach 28
 multiple sclerosis 376
 psychiatric features 377*b*
 Munchausen's syndrome 626–27
 murder 508
 muscarinic agonists 282
 mutations 95*b*, 107–8
 mutism 452
 myocardial infarction 633
- N**
 narcissistic personality disorder 391
 narcolepsy 318–19
 aetiology and pathology 319
 treatment 319
 narrative exposure therapy (NET) 146,
 147, 150
 National Comorbidity Survey
 Replication 173
 National Institute for Mental Health
 (NIMH), US 449
 National Institute of Clinical
 Excellence *see* NICE (National
 Institute of Clinical Excellence),
 guidance by
 needs assessment 54
 and risk assessment 54
 negativism 17
 neighbourhood, effects of 414
 neologisms 16, 47
 neonatal toxicity 692–93
 network meta-analysis (NMA) 126–27,
 127*f*, 181
 neurasthenia 97–98
 neurobiological approaches
 to anorexia nervosa 308
 to bipolar disorder 238–39
 to delusional disorders 297
 to depression 209–14
 to schizophrenia 273–77
 neurochemistry 238
 neurocognitive disorders 336, 337*t*
 neuroendocrine tests 210
 neuroimaging studies 434
 neuroleptic malignant syndrome 708–9
 neurological examination 51–52, 255–
 56, 346, 529, 613
 neuropathology 117–18
 neuropsychiatric patient 335–79
 amnesia/amnesic disorders 344–46
 anoxia 378
 assessment 340–41
 computed tomography 341
 history and mental state
 examination 341
 investigations 341
 physical examination 341
 structural brain imaging 341
 brain tumours 375–76
 carbon monoxide poisoning 378
 cerebrovascular disorders 349–68
 classification of 336, 337*t*
 delirium *see* delirium
 dementia *see* Alzheimer's disease;
 dementia; frontotemporal
 dementias; Lewy bodies, dementia
 with; vascular dementia
 disorders 336*b*
 EEG studies 341
 encephalitis 377–78
 epilepsy *see* epilepsy; seizures
 hypoglycaemia 378
 movement disorders 363–50
 multiple sclerosis 376
 neurophysiological testing 341
 neurosyphilis 376–77
 normal pressure hydrocephalus 378
 prion diseases *see* prion diseases
 secondary or symptomatic
 syndromes 379, 379*t*
 systemic lupus erythematosus 378
 neuropsychological assessment 52–53
 neuropsychology 102
 neurosis, psychiatric classification 27
 neurosurgery 188, 749–51
 adverse effects 750
 clinical use 750–51
 current approaches 749–50
 effectiveness 750
 history 749
 types of operation 750
 neurosyphilis 376–77
 neuroticism 206, 382
 neurotransmitters 114
 amino acid 211–12
 NICE (National Institute of Clinical
 Excellence), guidance by
 agoraphobia 177
 anorexia nervosa 311*b*
 ASPD 400
 and evidence-based medicine 130–31
 GAD 169
 mania 247*b*, 247
 PTSD 147*b*, 147
 social phobia 174
 night terror disorder 321
 NMDS (neurological, mental health,
 developmental and substance use
 disorders) 651
 NNT *see* number needed to treat (NNT)
 non-malevolence 72
 noradrenaline 114
 catecholamine depletion 211
 function 211
 metabolism 211
 neuroendocrine tests 211
 noradrenergic neurons 167
 normal pressure hydrocephalus 378
 nosology 96
 notes
 admission 62–63
 case notes 62
 continuation (progress) 63
 discharge 63
 on history taking 43–45
 inpatient 62–63
 number needed to treat (NNT) 122,
 122*t*, 123, 126
- O**
 obsessional doubts 13
 obsessional impulses 13, 183
 obsessional personality 184
 obsessional personality
 disorder 395, 399
 obsessional phobias 13, 183–84
 obsessional rituals 183
 obsessional ruminations 13, 183
 obsessional slowness 13, 183
 obsessional thoughts 13, 14, 183
 obsessions 13–14
 and compulsions 13
 forms 7*b*
 mental state examination 48
 see also compulsions; OCD (obsessive-
 compulsive disorder)
 obsessive-compulsive disorder *see* OCD
 (obsessive-compulsive disorder)
 obstetrics 638–43
 antenatal death 639
 caesarian section 640
 Couvade syndrome 639
 hyperemesis gravidarum 639
 pregnancy 638–39
 pseudocyesis 639
 spontaneous abortion 639
 termination of pregnancy 639
 in vitro fertilization 639
 obstructive sleep apnoea 320
 occupation 774–75
 occupational dystonia 366
 OCD (obsessive-compulsive
 disorder) 183–88
 abnormal serotonergic
 function 185–86
 aetiology 185–86
 and anxiety/anxiety
 disorders 162, 184
 anxiolytics 187
 and brain disorders 185
 in children and adolescents 445–46
 aetiology 445
 associated disorders 446
 childhood-onset disorders 184
 clinical features 445
 epidemiology 446
 management 446
 prognosis 446

- OCD (obsessive-compulsive disorder)
 (cont.)
 clinical picture 183–84
 clomipramine 187
 cognitive theory 186
 deep brain stimulation 188
 and depersonalization 184
 and depression/depressive disorders 184
 diagnostic criteria 184
 differential diagnosis 184
 DSM-5/ICD-11 classification 161
 dynamic psychotherapy 187
 epidemiology 184–85
 exposure and response prevention 187
 genetics/genetic factors 185
 key cognitive processes 186*b*
 lifetime prevalence 184–85
 management 188
 medications 187
 neuropsychological function 186
 neurosurgery 188
 OCD-related disorders 162, 188–91
 and organic disorders 184
 prevalence rates 163, 163*t*
 principal features 183*b*
 prognosis 186–87
 psychiatric aetiology 185–86
 and schizophrenia 184
 SSRIs 187
 treatment 187–88
see also compulsions; obsessions
- ODD/CD *see* oppositional and conduct disorders (ODD/CD)
- odds ratios (ORs) 121
 combining 125–26
 and effect sizes 126
 fixed effects model 125–26
 heterogeneity, ORs 125–26
 pooled estimates 125
 pooled random effects model 125–26
- Oedipus complex 95*b*
- olanzapine 228, 240, 245, 246, 310–11, 703
- older patients 521–41
 abuse and neglect of the elderly 532
 Alzheimer's disease, drug treatment 539–40
 anxiety disorders 525, 534
 assessment 58
 bipolar disorder 534
 carers 528
 day hospitals and centres 527
 dementia treatment 536–41
 demographics 521–22
 depressive disorders 532–34
 domiciliary psychiatric care 526–27
 domiciliary services 527
 driving 532
 ECT 530
 epidemiology of psychiatric disorders 523–25
 financial affairs 531–32
 geriatric medicine 527
- growing old and DID 475
 health services 526–27
 inpatient notes 527
 late-onset schizophrenia-like disorders 525
 long-term care 527
 mood disorders 524–25
 neuropsychology of ageing 522–23
 normal aging 521–23
 organization of services 525–28
 personality disorder 525
 physical changes, brain 522
 physical health 523
 physical treatments 530
 prescribing advice 691
 primary care 526
 principles and practice of old age psychiatry 525–32
 psychiatric assessment 528–29
 psychiatric services 526
 psychiatric treatment 530–31
 psychological treatment 530–31
 psychosocial approaches 531
 psychotherapy for 681
 residential care 528
 schizophrenia-like disorder 534–35
 social circumstances 523
 voluntary and third-sector services 528
- oneiroid states 19
- opioids 573–77
 clinical effects 574
 community treatment 577
 crises 574–75
 epidemiology 573–74
 maintenance treatment 576–77
 methadone 574, 576
 natural course of dependence 574
 planned withdrawal 575–76
 and pregnancy 576
 prevention 574
 use and misuse 574
 withdrawal from 574
- oppositional and conduct disorders (ODD/CD) 436
 aetiology 437–38
 brain functioning 437
 child mistreatment 438
 classification of 436–37
 clinical features 436
 family-level influences 437–38
 genetic factors 437
 individual-level influences 437
 language, IQ and educational attainment 437
 management 438–40
 parent training programmes 438
 poor outcome, predicting 438, 439*b*
 pregnancy and perinatal complications 437
 prevalence 437
 prognosis 438
 temperament 437
- orexin antagonists 699
- organic disorders 26
 organic mental disorders 500
 organic psychiatry 335
 organ transplantation 631
 orientation 50
 ORs *see* odds ratios (ORs)
 overgeneralization 208
 overvalued ideas 14–15
- P**
- P450 enzymes 689
- paedophilic disorder 329
- pain management 612
- paliperidone 703
- panic attacks 178, 179*b*
 hyperventilation symptoms 179*b*
- panic disorder 178–81
 aetiology 180
 and agoraphobia 176
 antidepressants 698
 benzodiazepines 181
 biochemistry 180
 choice of treatment 181
 clinical features 179
 cognitive theories 180
 cognitive therapy 181
 course/prognosis 180
 diagnostic criteria 179
 differential diagnosis 179
 epidemiology 179
 and GAD 179
 genetics/genetic factors 180
 hyperventilation 179*b*, 179, 180
 management 181
 neural mechanisms 180
 psychiatric aetiology 180
 required patient knowledge 181
 and social phobia 172
 symptoms 179*b*
 treatment 180–81
- PANSS *see* Positive and Negative Syndrome Scale (PANSS)
- Paracelsus 96
- paramnesia 18
- paranoid disorders/paranoid personality disorder 294, 297–301, 387
 aetiology 395
 and agoraphobia 176
 assessment of symptoms 301–2
 causes 292
 delusional, types 295*b*
 features 388*b*
 history 293*b*–94*b*
 ideas of reference 292*b*
 mood disorders 295
 organic disorders 295
 paranoid personality disorder 294
 paranoid schizophrenia 296
 paranoid symptoms in mood disorders 295
 paranoid symptoms in psychiatric disorders 295–97
 and paraphrenia 293*b*–94*b*
 pathological jealousy 297–99

- premorbid personality 292*b*, 292
- prognosis 303
- psychological treatment 303
- schizophrenia 254
- social isolation and deafness 294
- and substance misuse 295
- symptoms 292*b*, 292–94
 - assessment 301–2
 - causes 292
 - drug treatment 302
 - and paranoid schizophrenia 296
 - premorbid personality 292
 - principles of treatment 447
 - in psychiatric practice 295–97
 - in schizophrenia-like syndrome 296
 - treatment 302–3
- syndromes 291–303
- terminology 12*b*
- treatment 302–3
 - general principles 302
 - medications 302
 - psychological 303
- paranoid schizophrenia 296
- paraphilic disorders 322, 327–30
 - abnormalities in preference for sexual act 330
 - abnormalities of preference for sexual object 329–30
 - abnormalities of sexual preference 328–29
- aetiology 328
- behavioural models 328
- biological models 328
- disease models 328
- fetishistic disorder 329–30
- management 329
- paedophilic disorder 329
- psychoanalytical models 328
- transvestic disorder 330
- parasomnias 320–21
- pareidolia 6
- Parkinson's disease 363–65
 - aetiology 364–65
 - clinical features 363–64
 - dementia 364
 - depression 364
 - idiopathic 364–65
 - neuropsychiatric manifestations 364
 - Parkinsonian syndrome 705
 - psychiatric manifestations 364*b*
 - psychotic symptoms 364
 - risk factors 365*b*, 365
- pathological gambling 500
- pathological jealousy 297–99
 - aetiology 298
 - assessment 298–99
 - clinical features 297–98
 - delusional misidentification syndrome 300–1
 - erotomania/erotic delusions 299–300
 - prognosis 298
 - querulant delusions 300
 - reformist delusions 300
 - somatic delusional disorder 300
- treatment 299
- violence, risk of 298
- Patient Health Questionnaire 9 (PHQ-9) 68
- patient–interviewer language
 - differences 58
- Patient-Reported Outcome Measures (PROMS) 69
- patients
 - anxious *see* anxiety
 - characteristics 56–58, 60
 - children and adolescents *see* child and adolescent psychiatry
 - communicating with 41*b*
 - competence, assessing 78*b*
 - competent, refusal to treatment 77
 - confused 57
 - depressed *see* bipolar disorder; depression; major depression
 - descriptive psychopathology 4
 - discussing diagnosis and management plan with 61–62
 - effects of problems on lives of 59
 - elderly *see* older patients
 - evidence-based medicine, implementing for individual patients 129–30
 - exceptions to confidentiality rule in patient's interest 76
 - exploitation by doctors 73–74
 - garrulous 57
 - inpatient notes 62–63
 - intellectual development disorders *see* disorders of intellectual development (DID)
 - interviews in homes of 56
 - knowledge required of 60
 - lack of capacity 77
 - manic *see* mania
 - mentally disordered, refusal by 77
 - neuropsychiatric *see* neuropsychiatric patient
 - patient–interviewer language
 - difficulties 8
 - patient level meta-analysis 127
- Patient-Reported Outcome Measures (PROMS) 69
 - right to respect 95–96
 - social resources 223–24
 - taciturn 57
 - uncommunicative 155
 - uncooperative 57, 155
 - unresponsive 57
 - see also* doctor–patient relationship; specific conditions
- perception, disturbances of 6–8
 - defining perception 6
 - delusions/delusional perception 8, 10
 - hallucinations 6–8
 - illusions 6
 - imagery 6
 - mental state examination 49–50
 - see also* delusions
- persecution, delusions of 232, 292*b*
- persecutory delusions 11, 232
- perseveration 15
- personal history 42, 43–45
 - child development 43–44
 - children 44
 - education 44
 - marital 44
 - occupation 44
 - pregnancy/date of birth 43
 - sexual 44
 - social circumstances 44
 - substance use/misuse 45
- personal injury 87
- personality 381–82
 - assessment 42–43, 45, 53, 384
 - attachment theory 383–84
 - change in 385
 - character 45
 - childhood experience and personality development 383
 - childhood temperament and adult personality 383
 - cyclothymic 392
 - depression risk 206
 - extraversion-introversion 382
 - factor schemes 382
 - Freudian theory of development 383
 - and GAD 166–67
 - genetic influences 383
 - harm avoidance 382
 - importance 381–82
 - mood/emotional tone 45
 - neuroticism 166, 382
 - novelty-seeking 382
 - origins 383–84
 - as pathoplastic factor 381
 - personality types, history 382–83
 - and predisposing factors 91
 - as predisposition 381
 - relationships 45
 - in relation to treatment 381–82
 - reward dependence 382
 - temperaments 382
- personality disorder 167, 262, 384–85
 - aetiology 393–95
 - affective personality disorders 392
 - anankastic (obsessive-compulsive) personality disorder 391–92, 392*b*
 - antisocial personality disorders 385, 388–89
 - anxious (avoidant) personality disorder 391, 395, 399
 - assessment 397
 - classification of abnormal personalities 381–87
 - clinical population rates 392–99
 - concept of abnormal personality 384–85
 - course of 396
 - dependent personality disorder 391*b*, 391, 395, 399
 - depressive personality disorder 392
 - descriptions/diagnostic criteria 387–92
 - ethical problems 400–1

- personality disorder (*cont.*)
 general issues 393–94
 general population rates 393
 histrionic personality disorder 390*b*
 impulsive personality disorder 390
 management 397–98
 assessment 397
 general aims 397–98
 organization of services 398
 progress 398
 specific disorders 398–400
 supportive therapy 397
 medication choice 398
 narcissistic personality disorder 391
 obsessional personality disorder 395, 399
 older patients 525
 paranoid personality disorder 294, 387, 395
 passive-aggressive 395
 persistent 396
 psychopathic disorder 388–89, 389*b*
 psychotherapy 398
 rates of in clinic 392–93
 related conditions, classified elsewhere 386
 schizoid personality disorder 387–88, 388*b*, 395
 supportive therapy 397
 treatment 396–97
 see also borderline personality disorder (BPD)
- personalization 208
 PET *see* positron emission tomography (PET)
- phaeochromocytoma 165
 phantom limb 17
 pharmacodynamic measures 689
 pharmacogenetics 689
 pharmacokinetics of psychotropic drugs 688–89
 absorption 688
 distribution 688
 excretion 688
 measurement of circulating drug concentrations 689
 metabolism 688
 plasma half-life 688
 pharmacology 114
 phenelzine 174
 phenomenology/phenomenological psychopathology 2
 phenothiazines 702
 phobias 5, 162, 169–78
 aetiology 171
 blood and injury 170
 in children and adolescents 444
 choking 170
 classification of 170
 clinical picture 170
 conditioning and cognitive theories 171
 dental treatment 170
 differential diagnosis 171
 epidemiology 170
 exposure treatment 171
 flying 170
 genetic factors 171
 intermittent anxiety 161
 neural mechanisms 171
 prepared learning 171
 prognosis 171
 psychiatric aetiology 171
 specific 170–72
 agoraphobia 175–78
 clinical picture 170
 social phobia 172–74
 treatment 171–72
- PHQ-9 *see* Patient Health Questionnaire 9 (PHQ-9)
- phrenitis 96
 physical abuse of children 459–60
 physical examination 51–52, 52*b*, 146
 of children 417
 and individuals with DID 484
 of men presenting with sexual function disorders 325*b*
 neurological 51–52, 52*b*, 255–56, 346, 529, 613
 neuropsychiatric patient 341
- physical illness
 adjustment to 151–52
 appraisal of 152
 counselling 153
 and depression/depressive disorders 207–8
 direct cause of psychiatric symptoms 152
 effects 207–8
 GAD mistaken for 165–67
 help for people adjusting to 153
 illness behaviour 152
 and individuals with DID 17–33-6
 onset of 152–53
 psychiatric symptoms due to treatments for 152
 sick role 152
 terminal, adjustment to 153–55
- physiology 115–17
 default mode network 116
 dynamic susceptibility contrast 115
 functional connectivity 116
 functional MRI 115
 measurement of cerebral blood flow, metabolism and neuronal function 115–16
 PET imaging 116
 psychophysiological measures 115
- Pica 315
 in preschool years 425
- Pick's disease 361
 Picture Vocabulary Test 485
 plasma half-life 688
 plasma tryptophan 210
 plastic surgery 631
 pleiotropy 109–10
 political models 94–96
 polygenic risk scores (PRS) 109, 206
- polymerase chain reaction (PCR) 108–9
 polymorphisms
 apolipoprotein E (*apoE*)
 gene 110*b*, 355
 common disease–variant model 107
 examples in psychiatry 110*b*
 genetic 107, 116, 185
 and mutations 107–8
 nucleotide 106
 single nucleotide polymorphisms 95*b*, 106, 108, 109, 116, 266, 267
- polysomnogram 117
- Positive and Negative Syndrome Scale (PANSS) 68
- positron emission tomography (PET) 112, 116
 antipsychotics 709–10
 depression 213
 neuropsychiatric patient 341
 panic disorder 180
 schizophrenia 275
- post-concussional syndrome 368
 post-mortem studies 112
 postnatal depression 641–42
 post-operative period 630–31
 postpartum mental disorders 640–42
 postpartum psychosis 640–41
 post-schizophrenia depression 255
 post-traumatic stress disorder (PTSD) 143–48
 aetiology 145–46, 444–45
 assessment 146
 CBT 146–47
 in children and adolescents 444–45
 clinical picture 144
 complex 147–48
 cognitive theories 146
 combatants 149
 diagnosis 144
 differential diagnosis 144
 fear conditioning 146
 ICD-11 criteria 148*b*
 maintaining factors 146
 management 445
 neurological abnormalities 145*b*
 NICE guidance 147*b*
 prevalence 145
 principal symptoms 144*b*
 prognosis 445
 psychiatric aetiology 145–46
 psychological factors 146
 rape/physical assault, response to 148–49
 refugees/victims of torture 149–50
 treatment 146–48, 149, 150
- posture, assessing 46
 posturing 17
 power of attorney 86
- precipitating factors
 epidemiology 99–100
 psychiatric aetiology 91, 99–100
- predisposing factors
 aetiology 91
 bipolar disorder 239

- delirium 342*b*
- depression/depressive disorders 101, 214
- epidemiology 99–100
- and personality 91
- Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines 125
- pregabalin 174, 698
- misuse of 578
- pregnant women, prescribing advice 692–93
- premature ejaculation 327
- premenstrual syndrome/premenstrual dysphoric disorder 642–43
- premorbid personality 292*b*, 292, 348
- preschool years
 - adverse caregiving environments 425
 - assessment and treatment options 425
 - common problems of preschool children 424–25
 - development process 405
 - feeding problems 424–25
 - PICA 425
 - prevalence of disorders 424
 - review of disorders in 423–25
 - sleep problems 424
 - temper tantrums/disruptive behaviours 405, 424
- prescribing advice 690–91
 - adequate dose 690
 - breastfeeding 693
 - children and the elderly 691
 - compliance, concordance and collaboration 691
 - dosing and treatment duration 686–90
 - drug combinations, cautious use 690
 - ethical aspects 761–62
 - no therapeutic response 693
 - pregnant women 692–93
 - required patient knowledge 690–91
 - special groups 691–94
- Present State Examination (PSE) 29–30, 67
- primary care
 - acute disorders 770–71
 - advising and training GPs (Balint groups) 771–72
 - chronic disorders 771
 - classification of psychiatric disorders 769
 - counselling 662
 - disorders referred to psychiatric services 770
 - identification 769–70
 - improving access to psychological therapies 770–71
 - older patients 526
 - psychiatric disorders treated in 770
 - psychiatric team 771–72
 - role of primary care team 419
 - services for psychiatric disorders in 769–72
- prion diseases 361–63
 - aetiology 362–63
 - Creutzfeldt–Jakob disease (CJD) 362
 - fatal familial insomnia 362
 - Gerstmann–Straussler–Scheinker syndrome 362
 - suspected, investigation of 362
 - variant Creutzfeldt–Jakob disease (vCJD) 362
- PRISMA *see* Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines
- probands, family studies 104
- problem lists 66, 66*t*
- problem-solving strategies 138
- and supportive psychotherapy 219
- projection 139*b*
- prolonged exposure therapy 143
- PROMS *see* Patient-Reported Outcome Measures (PROMS)
- proteomics 114
- PSE *see* Present State Examination (PSE)
- pseudohallucinations 8
- psychiatric aetiology
 - approaches 89
 - cause and effect 90
 - causes remote in time from effects produced 90
 - classification of causes 91
 - complexity of causes 90–91
 - general issues 89–92
 - historical development of ideas of 96–98
 - indirect mechanisms 90–91
 - and intuitive understanding 89–90
 - maintaining factors 91
 - phobias 171
 - precipitating factors 91
 - predisposing factors 91
 - psychoanalysis 2, 95*b*
 - psychological approaches to 208
 - psychological reactions, concept 92
 - scientific disciplines, contribution of 98–118
 - stress, concept of 91–92
 - understanding and explanation 92
- see also specific conditions*
- psychiatric assessment 37–69
 - admission notes 62–63
 - on behalf of a third party 75
 - case notes and clinical records, importance 62
 - of children and families 415–17
 - continuing and completing 39–40
 - current circumstances 59–60
 - depressive disorders 222–23
 - diagnosis 59, 61–62
 - discussing diagnosis and management plan with patient 61–62
 - disease and illness, affecting evaluation 60–61
 - effective, techniques for 40*b*
 - effects of problems on patient's life 59
 - ending of 417
- evaluation of information 37–38
- evaluations by experienced psychiatrists 61
- goals 37
- history *see* psychiatric history
- information collection 37
- integrating/evaluating information 58–61
- interviews *see* psychiatric interviews
- laboratory investigations 52
- letters to GP 66
- mental state examination 41, 46–51
- needs assessment 54
- older patients 528–29
- pathological jealousy 298–99
- of personality 384
- physical/neurological examination 51–52, 52*b*
- preparation 37, 38
- previous mood disturbance 223
- problem lists 66*t*, 66
- process 37–38
- psychiatric services 763
- psychological 52–53
- recording/communicating information 38, 61–67
- required further information 60
- risk assessment 53–54, 54*b*, 59
- standardised assessment methods 67–69
- starting 38–39
- suicide risk 596–97
- therapeutic relationship, establishing 37
- types 54–56
- use of information 38
- what treatment is indicated 60
- why disorder has occurred 60
- see also* mental state examination
- psychiatric classification 21–35
 - categorical 28
 - comorbidity 28–29
 - concepts of mental disorders 22–23
 - contemporary, organizational principles 26–29
 - multiaxial approach 28
 - organic and functional 26–27
- criticisms 24
- cultural issues 34
- current classifications 30
- dimensional 28
- history 26
- multiaxial approach 28
- national systems 34*b*
- need for in psychiatry 21
- psychiatric aetiology 91
- purposes 21
- reactions to stressful experiences 140–41
- Research Domain Criteria 35
- substance use/misuse 543–44
- syndromal categories 21–22
- see also* DSM-5 classification scheme; ICD-11 classification scheme

- psychiatric diagnosis
 - concept 23
 - construct validity 30
 - context 37
 - dual diagnosis 28–29
 - evidence-based medicine 131–32
 - face validity 30
 - form and content 3
 - hierarchies of 28
 - international studies of diagnostic criteria 29
 - making 37
 - reliability 29–30
 - validity 30
 - see also individual conditions*
- psychiatric history 41–45
 - family history 42, 43
 - informant(s) 41
 - notes on history taking 43–45
 - outline 41b
 - past psychiatric and medical 42
 - personal 42, 43–45
 - personality 42–43, 45
 - present condition 41–42, 43
 - reason for referral 43
 - scheme for history taking 41–43
- psychiatric interviews
 - assessment 38
 - continuing and completing 39–40
 - Diagnostic Interview Schedule 67
 - emergency 55
 - family interviews 56
 - in general practice 55–56
 - in hospital 56
 - with informants 40–41
 - interviewing parents 415
 - and observing child 415–16
 - and observing family 416
 - patient characteristics affecting 56–58
 - in patient's home 56
 - preparation 38
 - screening of questions 55, 55t
 - standardised interview
 - schedules 29–30
 - starting 38–39
 - techniques 416
 - see also child and adolescent psychiatry; mental state examination; psychiatric assessment*
- psychiatric services 755–78
 - accommodation 774
 - assertive outreach teams 765–67
 - assessment 762–63
 - asylum movement 756
 - carers 774
 - case management 763, 767–68
 - communication and liaison 764
 - community care, problems with 777
 - components of a mental health service 759–69
 - continuity of care 775
 - continuity of responsibility 760–61
 - crisis houses and respite care 767
 - crisis resolution/home treatment teams 766
 - crisis teams 766
 - daycare 761
 - diagnosis-specific teams 768
 - diagnosis-specific wards 761
 - early intervention services 767–68
 - early treatment, arrangements for 756–57
 - ethnic minorities 776
 - forensic and rehabilitation teams 768
 - generic sector CMHT 762–64
 - history 755–59
 - homeless mentally ill people 776–77
 - hospital closure 757–58
 - inpatient care 776
 - inpatient wards 759–62
 - international service principles 778
 - longer inpatient care 761
 - long-term care 772–76
 - characteristics of patients requiring 772–73
 - diagnosis 772
 - problems 772–73
 - requirements of a community service providing 773–75
 - mental health services research 764
 - migrants 776
 - moral management 756
 - multidisciplinary community mental health teams 762
 - occupation 774–75
 - older patients 526
 - outpatient clinics 762
 - for people with particular needs 776–77
 - primary care 769–72
 - reassessment 775
 - under-recognition/over-admission of psychiatric disorder 776
 - recovery movement 775–76
 - refugees 756–57
 - rehabilitation and recovery 775
 - responses to crises 775
 - rise of community care 758–59
 - single-sex accommodation 760
 - social psychiatry and beginning of community care 757
 - staffing and management 762
 - team meetings 763–64
 - variations in practice and sustainability 766–67
 - working with families and volunteers 775
- psychiatry and medicine 609–43
 - bodily distress disorders 611–12
 - bodily symptoms not explained by physical pathology 620–23
 - body dysmorphic disorder 627–28
 - chronic fatigue syndrome 623–25
 - comorbidity 617–19
 - and dualism 610
 - limitations of 610
 - practical consequences of 610–11
 - effectiveness of psychiatric treatments in the medically ill 619–20
 - epidemiology 616–17
 - fabricated/induced illness 627
 - factitious disorder 625–26
 - fibromyalgia 625
 - integrated approach 611
 - irritable bowel syndrome 625
 - malingerer 627
 - management of psychiatric disorders in 620
 - medical diagnosis 609
 - medical settings 616–28
 - Munchausen's syndrome 626–27
 - presentation in medical settings 617
 - presenting as apparent worsening of a medical condition 617
 - prevention of psychiatric disorder in the medically ill 619
 - psychiatric diagnosis 610
 - psychological treatment
 - angry or aggressive patient 629
 - consent to treatment 630
 - psychoanalysis 2, 95b
 - and depression 208b
 - formal 95b
 - influence of 98
 - models of treatment 95b
 - and the unconscious 95b
 - psychoanalytical theory 208b
 - psychobiology 97
 - psychodynamic psychopathology 2
 - psychoeducation 245
 - psychological activation paradigms 116
 - psychological assessment 52–53
 - cognitive and behavioural 53
 - neuropsychological 52–53
 - personality assessment 53
 - psychological medicine services 628–38
 - anxious patient 629
 - composition 628–29
 - emergencies 629–30
 - psychological reactions, concept 92
 - psychological treatments 655–83
 - classification of 659
 - cognitive errors in depressive disorders 658b
 - behaviour therapy 658
 - CBT 658–59
 - cognitive therapy 658
 - development 656
 - early departures from Freud's original group 657–58
- psychomotor retardation 5–6
- psychopathic disorder 388–89, 389b
- psychopathology 2
- psychopathy, concept 389
- psychosis/psychotic symptoms
 - Alzheimer's disease 354
 - in children and adolescents 452–53
 - family history 100
 - and individuals with DID 472
 - insulin coma treatment 686
 - Parkinson's disease 364

pharmacology 114
 postpartum psychosis 640–41
 psychiatric classification 27
 traumatic brain injury 369
see also psychotic depression;
 schizophrenia
 psychostimulants 737–38
 psychotherapy
 anorexia nervosa 310
 bipolar depression 248
 bipolar disorder 244
 bulimia nervosa 314
 for children 681
 choice of 398
 cognitive therapy 222
 common factors in 659–60, 660*b*
 cross-cultural 682
 depression 222, 226
 dynamic 174, 284
 gender dysphoria 332
 grief 158–59
 group treatment *see* group
 psychotherapy
 individual dynamic
 psychotherapies 673–74
 interpersonal 220, 222
 long-term individual dynamic
 psychotherapy 674–75
 mania 245
 for older people 681
 personality disorder 398
 in public services 659
 small-group psychotherapy 675–76
 supportive, for depression 219
 treatments of mainly historical and
 cultural interest 681–82
 abreaction 682
 autogenic training 682
 diviners 682
 herbalists 682
 meditation and yoga 682
 shamans 682
 traditional healing 682
 psychotic depression 196, 200, 228
 in bipolar disorder 234
 public mental health
 cost effectiveness 649–50
 defining 645, 646
 life course 646
 populations 645, 646
 attributable fractions for public
 health interventions 648–49
 prevention of mental illness 646
 rationale for 647
 role of psychiatrists in managing 649
 pyromania 501

Q

QIDS *see* Quick Inventory of Depressive
 Symptomatology (QIDS)
 qualitative research methods 133–35
 quetiapine 228, 240, 245, 246, 248
 Quick Inventory of Depressive
 Symptomatology (QIDS) 47

R

race equality 84
 racial discrimination 101–2
 RAD *see* reactive attachment
 disorder (RAD)
 randomized trials *see* RCTs (randomized
 trials)
 rape 506–7
 causes 506
 epidemiology 506
 prognosis 506–7
 psychological impact 508
 response to 148–49
 rapid cycling disorders, mania 234
 rationalization 139*b*
 RCTs (randomized trials) 120–21, 123
 clear endpoint 128
 large-scale 128
 mega-trials 128
 reaction formation 139*b*
 reactions to stressful experiences 137–60
 acute stress disorder 141–43
 adjustment disorders 152–60
 anxiety responses 138
 classification of 140–41
 concept of a psychological reaction 92
 coping strategies 138
 and defence mechanisms 137
 coping styles 138
 defence mechanisms 137
 depressive responses 138
 emotional and somatic responses 138
 endocrine 91–92
 maladaptive coping strategies 138
 post-traumatic stress disorder 143–48
 psychological first aid 142*b*
 psychological responses 91–92
 response to stressful events 137–40
 severe stress, response to 148–50
 stress reaction 91–92
 reactive attachment disorder (RAD) 425
 reboxetine 181, 217, 224–25, 727
 receivership 86
 recording of information 61–67
 admission notes 62–63
 case notes and clinical records,
 importance 62
 case summaries 63, 64*b*
 communicating with patients and
 relatives 41*b*
 continuation (progress) notes 63
 discharge notes 63
 discussing diagnosis and management
 plan with patient 61–62
 electronic records 66–67
 forensic psychiatry 518
 formulations 63–65, 65*b*
 inpatient notes 62–63
 problem lists 66, 66*t*
 recovery movement 775–76
 reductionist and non-reductionist
 models 92–93, 94
 reduplication phenomenon 17
 reference, delusions of 292*b*

refugees 149–50, 756–57
 regression 139*b*
 rehabilitation and recovery 775
 relatives, communicating with 41*b*
 relaxation techniques 168
 reliability of psychiatric diagnoses 29–30
 categorical classification 28
 international studies of diagnostic
 criteria 30
 standardised interview
 schedules 29–30
 remote memory 18, 51
 REM-sleep behaviour disorder 320–21
 arousal disorders 321
 repression 139*b*
 Research Domain Criteria (RDoC) 35
 research ethics 79–80, 80*b*
 informed consent to research 80
 residential care
 for children 421–22
 older patients 528
 people with DID 488–89
 and supported accommodation 762
 response to stressful events,
 autonomic 91–92
 restless legs syndrome 320
 retarded depression 196
 retrograde amnesia 18
 Rett syndrome 412
 rhesus incompatibility 487–88
 risk assessment 53–54
 and needs assessment 54
 risk factors and harm to others 54*b*
 risperidone 228, 240, 245, 246, 703
 robaxetine 181
 Rogers, C. 661
 Royal College of Psychiatrists 76,
 111, 269
 rumination–regurgitation disorder 315

S

saccadic eye movements 668
 SCAN *see* Schedules for
 Clinical Assessment in
 Neuropsychiatry (SCAN)
 Schedules for Clinical Assessment in
 Neuropsychiatry (SCAN) 67
 schizoaffective disorder 284
 schizoid personality disorder 387–88
 aetiology 395
 features 388*b*
 schizophrenia 251–90
 aberrant connectivity 275
 acute 252*b*, 252–53, 261
 drug treatment 286
 management 285
 management after 287–88
 acute and transient psychotic
 disorder 284
 adherence therapy 284
 adherence to treatment 282
 adoption studies 265–66
 aetiology 264–73
 age at onset 263

- schizophrenia (*cont.*)
- antipsychotics 280–81
 - depot antipsychotics 282
 - arts therapies 284
 - assessment 288*b*
 - biochemical findings 276–77
 - biology 268–69
 - and bipolar disorder 235
 - brief psychotic disorders 260
 - Care Programme Approach 286
 - catatonia 254, 260–61, 261*b*, 284
 - CBT 283
 - child development 270–71
 - childhood adversity 272
 - in children and adolescents 263, 270–71, 289, 452–53
 - chronic 252, 253–54, 254*b*, 255–56
 - classification of 258–59
 - clinical features 251–56
 - factors modifying 256
 - modifying factors 256
 - clinical guidelines for treatment 280
 - clinical sub-syndromes 254
 - cognitive features 254–55
 - cognitive remediation 284
 - comorbidity 261
 - concepts and terminologies, development 257*b*–58*b*
 - course/prognosis 278–80
 - crime, association with mental disorder 499
 - delusions 252–53
 - and depressive disorders 203, 255
 - diagnosis 257–62
 - differential diagnosis 261–62, 262*b*
 - discussing with patients and carers 290
 - disorganized/disorganized thinking 251, 254
 - disrupted in schizophrenia-1 (*DISC-1*) 267
 - dopamine hypothesis 114, 276
 - double orientation 9*b*
 - drug-induced 262
 - dynamic psychotherapy 284
 - early intervention services 288–89
 - early-onset 452–53
 - EEG 275
 - effects on family 279–80
 - environmental risk factors 263, 264, 265–66, 269*b*, 269–71
 - epidemiology 262–73, 499
 - exercise 284
 - expressed emotion and family life 279
 - family dynamics 273*b*–53*b*
 - family life and expressed emotion 279
 - family studies 264–65
 - family therapy 283*b*, 283
 - features and theories 264, 264*t*
 - and fertility 263
 - first-rank symptoms 10, 251, 253*b*
 - functional brain imaging 274–75
 - and GAD 165
 - gender factors 263
 - genetic factors 264–69
 - biology of schizophrenia genes 268–69
 - clinical and therapeutic implications 269
 - gene–environment interactions 268
 - gene–gene interactions 268
 - ‘missing heritability’ 268
 - susceptibility genes 266–67
 - twin studies 265
 - glutamate 276–77
 - hallucinations 8
 - hebephrenic 254
 - historical development of ideas 257–58
 - incidence of 262–63
 - and individuals with DID 472
 - infective and inflammatory factors 270
 - inflammatory markers 277
 - and intelligence 312
 - International Pilot Study of Schizophrenia 29
 - lifetime risk 264, 265*t*
 - live events and difficulties 272
 - maintenance treatment and relapse prevention 281
 - management 285–89
 - maternal malnutrition 270
 - medications 280, 281*b*
 - acute episode 286
 - antidepressants 282
 - antipsychotics *see above*
 - migration and ethnicity 271–72
 - mode of inheritance 266
 - negative symptoms 251, 253
 - neurobiology 273–77
 - neurodevelopmental model 277–78
 - neurological signs 255–56
 - neuropathology 118, 274
 - neurophysiological findings 275
 - neuropsychological findings 276–77
 - obstetric complications 269–70
 - occupation, social class and income 271
 - and OCD 184
 - olfactory dysfunction 256
 - organic syndromes 261
 - pain insensitivity 256
 - paranoid 254
 - paternal age 270
 - physical health, mortality and suicide risk 278–80
 - place of birth/residence 271
 - positive symptoms 251
 - post-schizophrenia depression 255
 - predictors of outcome 278–79, 279*b*
 - prevalence of 263
 - prodrome of 256
 - psychological factors 252–73
 - psychological theories 272–73
 - psychosocial approaches 283*b*, 283–84
 - rare variants 267
 - ratings of symptoms 68
 - residual 254
 - schizoaffective disorder 235, 237, 260, 284
 - schizophrenia-like disorder 259–61, 534–35
 - Schneider’s symptoms of the first rank 232, 253*b*
 - secondary 261–62
 - social and psychosocial factors 271–72
 - social skills training 284
 - social stimulation 279
 - structural brain changes 273–74
 - substance use 271, 289
 - subtypes 254
 - suicide risk 289
 - and suicide risk 593
 - supported employment 284
 - thought disorder 252
 - treatment 280–84
 - treatment-resistant 288
 - undifferentiated 254
 - velocardiofacial syndrome (*VCFS*) 267*b*
 - violence 289
 - virtual reality 284
 - winter birth 270
 - see also* psychosis/psychotic symptoms
- schizophreniform disorder 258–59, 260
- schizotypal disorder/personality disorder 260, 284
- aetiology 395
 - conditions related to personality disorder and classified elsewhere 386
 - descriptions and diagnostic criteria 388
- school, effects of 414–15
- bullying 414–15
- school refusal
- aetiology 450–51
 - clinical picture 450
 - epidemiology 450
 - prognosis 451
 - treatment 451
- SCID *see* Structured Clinical Interview for Diagnosis (SCID)
- scientific disciplines, contribution to psychiatric aetiology 98*b*, 98–118
- biochemical studies 111–14
 - clinical descriptive studies 98–99
 - endocrinology 115
 - epidemiology 99–100
 - ethology and evolutionary psychology 104
 - experimental and clinical psychology 104
 - genetics/genetic factors 104–11
 - neuropathology 117–18
 - pharmacology 114
 - physiology 115–17
 - social sciences 100–2, 100*t*
- screening questions for psychiatric disorders 55, 55*t*
- seasonal affective disorder (SAD) 200–1

- secondary catatonia syndrome 261
secure attachment 383–84
seizures
 atonic 371
 classification of types 370*b*, 370–71
 complex partial 370–73, 371*t*
 in ECT 740
 generalized tonic–clonic 371
 myoclonic 371
 psychogenic non-epileptic 372
 simple partial 370
 see also epilepsy
selective abstraction 208
selective migration 101
selective mutism 452
selective serotonin reuptake inhibitors
 (SSRIs) 125, 143, 713–16
 agoraphobia 178
 atypical depression 197
 augmenting with antipsychotics 228
 bipolar depression 242
 choice and use of drug 224–25
 clinical use 715–16
 depression 217, 228
 efficacy in depression 713
 GAD 168
 interactions with other drugs 714–15
 OCD 187
 panic disorder 181
 pharmacokinetics 713
 pharmacological properties 713
 phobias 177–78
 PTSD 147
 social phobia 174
 and suicidal behaviour 714
 unwanted effects 713–14
self, disturbances of 17–18
self-harm 389, 453
 act of 600–1
 assessment of patients 604–6
 children and adolescents 603
 epidemiology 601
 frequent repeaters 607
 management 603–8
 methods 601
 mothers of young children 603
 non-fatal 599–603
 organization of services 603–4
 outcome 601–2
 patients refusing assessment 607
 repetition 601–2
 self-cutting 608
 self-poisoning methods 600–1
 treatment results 602–3
senile squalor syndrome 189
sensory disorders 634
separation and anxiety disorder 444
serotonin (5-HT) 113
 function 210–11
 low brain function 210
 role in pathology of
 depression 211
 schizophrenia, role in 277
 transporter gene 110*b*
serotonin and noradrenaline reuptake
 inhibitors (SNRIs) 147, 168,
 217, 725
sertindole 703
sertraline 181
severe stress, response to 148–50
sexual abuse in childhood 461–
 63, 504–5
 adulthood, effects in 462
 aetiology 160, 462
 assessment 462, 463*b*, 505
 childhood adversity 160
 clinical features 461–62
 effects 462
 epidemiology 159, 461
 long-term adjustment to 159–60
 management 462–63, 463*b*
 parental maladjustment 160
 recovered/false memory 159
 treatment 160, 505
 see also sexual offences
sexual dysfunction 322
 assessment 324–26, 325*b*
 biological/medical causes 323–24
 causes 323–24
 dyspareunia 327
 female orgasmic disorder 327
 female sexual arousal disorder 326
 lack/loss of female desire 326
 male erectile dysfunction 326–27
 male orgasmic disorder 327
 medical and surgical procedures
 associated with 324*b*
 premature ejaculation 327
 prevalence 322–23
 psychological causes 324
 sex therapy 326
 sociocultural factors 324
 treatment 325–26, 326*b*
 types 326–27
 see also sexual function disorders
sexual function disorders 321–32
 assessment 325*b*
 classification of 322, 323*t*
 drugs impairing sexual function 326
 dysfunction 322
 frequency 324*t*
 medical and surgical procedures
 associated with 324*b*
 physical examination of men
 presenting with 325*b*
 treatment 325–26
 types 326–27
 see also sexual dysfunction
sexual offences 504–7
 indecent assault 505
 indecent exposure 505
 rape 506–7
 stalking 505–6
 see also sexual abuse in childhood
sheltered work 774
shoplifting 507
single nucleotide polymorphisms
 (SNPs) 95*b*, 106, 108, 109, 116
 schizophrenia 266, 267
single-photon emission tomography
 (SPET) 112, 213
skin picking *see* excoriation
Skinner, B. F. 658
sleep disturbances/sleep-wake
 disorders 317*b*
 arousal from REM sleep,
 disorders of 321
 assessment 316, 317*b*
 in bipolar disorder 238
 breathing-related 320
 circadian rhythm 320
 classification of 316
 sleep-wake disorders 317*t*
 in depression 195, 213
 deprivation 245
 dyssomnia 316
 epidemiology 316
 in GAD 164
 hypersomnolence/excessive daytime
 sleepiness 318*b*, 318
 and individuals with DID 474
 insomnia 316–18
 Kleine–Levin syndrome 319–20
 movement disorders 320
 narcolepsy 318–19
 nightmares 321
 parasomnias 316, 320–21
 in preschool years 424
 REM-sleep behaviour disorder 320–21
 restless legs syndrome 320
 sleep hygiene 318*b*
 treatments 318*b*
sleep hygiene 317, 318*b*
sleep paralysis 321
sleepwalking 321
social behaviour 47
 applications of social theory to
 psychiatry 100*t*, 100
social exclusion 101–2
social phobia 172–74
 aetiology 173
 and agoraphobia 172, 176
 and avoidant personality disorder 173
 and body dysmorphic disorder 173
 choice of treatment 174
 clinical picture 172
 cognitive factors 173
 conditioning 173
 course and prognosis 174
 and depressive disorders 172
 diagnostic conventions 172
 differential diagnosis 172–73
 epidemiology 173
 and GAD 172
 generalized 172
 genetic factors 173
 inadequate social skills 173
 management 174
 neural mechanisms 173
 and normal shyness 173

- social phobia (*cont.*)
 - and panic disorder 172
 - psychological treatment 174
 - and schizophrenia 172–73
 - transcultural variations 182
 - treatment 174
- social sciences and mental disorders 100–2
 - childhood psychiatric disorders 101
 - family causes 101
 - family history 101
 - life events, study of 100–1
 - transcultural studies 100
 - vulnerability and protective factors 101
- social versus biomedical concepts 23
- sociology 100
- sodium valproate 734–35
- somatiform and related disorders 446–47
- speech
 - difficulty in speaking 47
 - flow of 47
 - in mania 232
 - neologisms 47
 - pressure of 47
 - rate/quantity of 47
 - recording in MSE 47
 - speech and language disorder in children, causes 426–27
 - assessment 427
 - classification of 426–27
 - language 427
 - talking past the point 16
 - verberation 16
- SPET *see* single-photon emission tomography (SPET)
- Spurzheim, Johann 96
- SSRIs *see* selective serotonin reuptake inhibitors (SSRIs)
- STAI *see* State–Trait Anxiety Inventory (STAI)
- stalking 299–300, 505–6
- stammering 452
- standardized assessment methods 67–69
 - Composite International Diagnostic Interview 67
 - global rating scales 69
 - instruments for measuring symptoms 68–69
 - Mini International Neuropsychiatric Interview 67
 - Present State Examination 29–30, 67
 - quality-of-life scales 69
 - ratings of affective symptoms 68
 - ratings of anxiety symptoms 68
 - ratings of broad groups of symptoms 48–46
 - ratings of motor systems 68–69
 - ratings of other symptoms 48
 - Schedules for Clinical Assessment in Neuropsychiatry 67
 - Structured Clinical Interview for Diagnosis 67
- State–Trait Anxiety Inventory (STAI) 68
- stereotypies 16
- sterilization 643
- stigma 25*b*
 - patient involvement 400–1
- stimulants 580–82
 - for ADHD 435
- St John's wort 727
- Strange Situations Procedure 383–84, 412
- stress, concept of 91–92
- stressors 91
- stroke 349–68
 - cognitive deficits 367
 - personality change 367
 - post-stroke depression 367–68
- structural brain imaging 213
- Structured Clinical Interview for Diagnosis (SCID) 67
- stupor 19
- subarachnoid haemorrhage 368
- subdural haematoma 368
- subjective doubles, delusion of 301
- sublimation 139*b*
- substance use/misuse 93–94
 - in adolescents 456
 - adverse effects 568–69
 - amphetamines 580–81
 - availability of drugs 566
 - and bipolar disorder 236
 - cannabis 578–80
 - causes 566–68
 - classification of 543–44
 - cocaine 581–82
 - definitions in DSM-5 and ICD-10 544
 - diagnosis of 569–70
 - drug maintenance 572
 - dual diagnosis 573
 - epidemiology 565–66
 - and GAD 165
 - hallucinogens 583
 - harmful use 544
 - harm reduction 572
 - illicit drug use 565
 - intoxication 544
 - ketamine 583–84
 - MDMA 582
 - misuse and dependence 565–66
 - nitrous oxide 584–85
 - opiods 573–77
 - and paranoid disorders 295
 - personal factors 566–67
 - pharmacological factors 567–68
 - phencyclidine 583–84
 - physical health 568
 - prevention 571
 - psychiatric comorbidity 569
 - psychosocial approaches 572–73
 - rehabilitation 573
 - schizophrenia 271, 289
 - social consequences 569
 - social environment 567
 - stimulants 580–82
 - and suicide risk 593–94
 - tolerance 544
 - treatment 571–73
- use disorder 544
- volatile substances 585–86
- withdrawal state 544
- in women 568–69
 - see also* alcohol use disorders
- substitute care, children 421–23
 - fostering and adoption 422–23
 - residential 421–22
 - see also* adoption
- sudden unexplained death in epilepsy (SUDEP) 471
- SUDEP *see* sudden unexplained death in epilepsy (SUDEP)
- suicide risk 589–608
 - act of suicide 590–91
 - among those in contact with psychiatric services 594–95
 - assessment 596–97
 - assisted/rational 595
 - biological factors 604
 - changes in rates of 591
 - in children and adolescents 453
 - community management 597
 - demographic characteristics 592–93
 - epidemiology 591–92
 - epilepsy 374
 - following homicide 502–3
 - gender and sexuality 593
 - geographic variations 591–92
 - high-risk occupational groups 595
 - history-taking 596
 - hospital management 597–98
 - inpatients 597
 - interpersonal risk factors 594
 - management of suicidal patients 597–99
 - mental state examination 596
 - pacts 595
 - psychiatric disorders 593–94
 - psychological risk factors 594
 - reducing rates 598–99
 - schizophrenia 278–80, 289
 - societal risk factors 594
 - SSRIs 714
 - statistics, accuracy of 591
 - suicidal ideation 48, 449
- sulpiride 704
- supported employment 774
- surgical treatment 630–31
- susceptibility genes 106
- symptoms and signs of psychiatric disorders 1–20
 - attention and concentration, disturbances of 19–20, 50
 - body image 17
 - changes in the nature of 5
 - changes in the way that they vary 5
 - clinical associations 5–6
 - anxiety 5
 - components 4–5
 - depression 5–6
 - elation 6
 - hallucinations 7–8
 - irritability and anger 6
 - consciousness, disturbances of 19

- descriptive psychopathology 2–4
- emotions and mood disturbances 4–6
- experimental psychopathology 2
- memory, disturbances of 18
- motor 16–17
- self, disturbances of 17–18
- thoughts, disturbances of 8–15
- systematic reviews 124–29
 - accuracy of meta-analysis, determining 128
 - applicability 129
 - biased conclusions 127–28
 - clinical utility 126
 - funnel plots 128
 - large-scale randomized trials 128
 - level of supervision 129
 - network meta-analysis 126–27, 127f
 - patient level meta-analysis 127
 - patient population 129
 - pooled estimates 125
 - pragmatic trials 129
 - presentation of results 125–26
 - problems with meta-analysis 127–28
 - questions 125
 - randomized trials, large-scale 128
 - statistical power 124
 - therapist variables 129
 - umbrella reviews 128
 - validity 124–25
- systemic lupus erythematosus 378
- Szasz, Thomas 94
- T**
- tabes dorsalis 377
- taciturn patients 57
- TBI *see* traumatic brain injury (TBI)
- team meetings 763–64
 - allocation of referrals 763
 - managing waiting lists and caseloads 763–64
 - patient reviews 763
- temazepam 699
- temperament
 - child and adolescent psychiatry 410, 437
 - oppositional and conduct disorders 437
 - and personality 382
 - see also* personality; personality disorder
- temporal lobe epilepsy 8
- terminal illness, adjustment to
 - aims of treatment 154
 - anxiety 153
 - depression 153
 - guilt and anger 153
 - helping patient to adjust 154
 - informing staff 154
 - informing/supporting relatives 155
 - liaison with medical/nursing staff 155
 - priorities for care 155b
 - reducing symptoms 154
 - referral to a psychiatrist 155
 - specialist services 155
 - symptoms induced directly by 153–54
 - treatment 154–55
- terrorist crimes 508
- testamentary capacity 86
- testimonial therapy 150
- therapeutic alliance 245
- therapeutic groups 677–79
- therapeutic communities 678b, 678
- therapeutic day hospitals 679
- thinking processes, disturbances of 8
 - depersonalization 16
 - derealization 16
 - flight of ideas 15
 - form of thought 15–16
 - loosening of associations 15–16
 - neologisms 16
 - overinclusion 16
 - perseveration 15
 - stream of thought 15
 - thought block 15
- thioxanthenes 702
- thought broadcasting 12, 49
- thought insertion 11, 49
- thoughts, disturbances of 8–15
 - compulsions 14
 - delusions 8–12, 9b
 - delusional themes 11–13
 - dysfunctional beliefs 208
 - formal thought disorder 15, 48
 - mental state examination 48–49
 - negative automatic thoughts 208
 - obsessional 13–14
 - overvalued ideas 14–15
 - pessimistic thoughts 195
 - in schizophrenia 252
 - thinking processes 15–16
- thought withdrawal 11–12, 49
- thyrotoxicosis 165
- tics 16, 366–67
- tinnitus 634
- topiramate 314
- torpor 19
- torts and contracts 86
- Tourette syndrome 184, 367
- trance and possession disorder 614
- transcranial direct current stimulation 747
- transcranial magnetic stimulation (TMS) 229, 746–49
 - adverse effects 747
 - uses 746–47
- transcriptomics 114
- transcultural studies
 - psychiatric aetiology 100
 - see also* culture/cultural variations
- transgender individuals 330–31
- transference 660
- tranlycypromine 242
- trauma-focused CBT 143, 146
- traumatic brain injury (TBI) 368–70
 - acute psychological effects 368–69
 - chronic psychological effects 369
 - cognitive impairment 369
 - depression 369
 - personality change 369
 - post-concussional syndrome 368
 - see also* head injury
- trazodone 217, 724–25
- Treatment for Adolescents with Depression (TADS) 449
- trichotillomania 188–89, 190
 - clinical features 190
 - diagnosis 190
 - epidemiology 190
 - management 190
- tricyclic antidepressants (TCAs) 114, 147, 217–18, 225, 717–19
 - clinical use 718–19
 - compounds available 717–18
 - contraindications 718
 - ineffective 228
 - interactions with other drugs 718
 - maintenance and prophylaxis 719
 - pharmacokinetics 717
 - pharmacology 717
 - toxic effects 718
- trifluoperazine 702
- tri-iodothyronine augmentation 228
- trisomy 21 477–79
- tryptophan 113
- tryptophan depletion 210–11
- twilight states 19
- twin studies 95b, 105, 145
 - depression 205
 - GAD 166
 - personality 383
 - schizophrenia 265
- U**
- umbrella reviews 128
- unconscious, the
 - Freud on 95b, 298, 657
 - and psychoanalysis 95b
- uncooperative patients 57
- unity of the self, disturbances of 17–18
- unresponsive patients 57
- US National Survey of Children's Health 429
- US–UK Diagnostic Project 257
- utilitarian approach 72
- V**
- vagal nerve stimulation 747–48
- validity
 - evidence-based approaches 121
 - systematic reviews 124–25
- validity of psychiatric diagnoses 30
- valproate 218, 241, 242, 246
- value-based practice 73
- variant Creutzfeldt–Jakob disease (vCJD) 362
- vascular dementia 357b, 357–58
 - aetiology 358
 - with Alzheimer's disease 363
 - clinical features 357–58
 - drug treatment 540
 - risk factors 358b
- velocardiofacial syndrome (VCFS) 107, 267b
- venlafaxine 177–78, 181, 217, 725–26
 - bipolar depression 242
- verbigeneration 16

violence
 crimes of 502–4
 de-escalation techniques 514–15, 515*b*
 domestic 503–4
 homicide 502–3
 management in health care
 settings 514*f*, 514–15
 pathological jealousy 298
 schizophrenia 289
 towards partners 503–4
 visual hallucinations 7, 8, 50
 vocational rehabilitation 774
 volatile substances 585–86
 voluntary activities 774
 vortioxetine 726–27
 voyeurism 330

W

Wechsler Abbreviated Scales of
 Intelligence (WASI) 485
 Wechsler Adult Intelligence Scale
 (WAIS) 485
 Wernicke's encephalopathy 552
 West syndrome 471

Weyer, Johan, *De Praestigiis
 Daemonum* 96
 WHO Disability Assessment Schedule 2.0
 (WHODAS 2.0) 69
 whole genome sequencing 109
 WHO (World Health Organization)
 on disorders of intellectual
 development (DID) 468
 production of ICD 30
 on psychiatric disorders 23
 see also ICD-11 classification scheme
 Willis, Thomas 96
 Wolp J. 658
 word salad 16, 252
 World Health Survey 100
 worry
 about worry 166
 adjustment disorders 141, 150
 affective personality disorders 392
 and anxiety 5, 56
 anxious personality disorder 391
 and CBT 669
 diagnostic conventions 164
 as emotional coping strategy 167

excessive but normal 166
 future potential threats 161
 and GAD 163, 164–65, 166, 167, 168
 health anxiety 621
 HIV infection 635
 loss of income 523
 and overvalued ideas 15
 and panic disorder 179
 and personality 384
 sleep disturbances/sleep-wake
 disorders 164
 stressful life events 151
see also anxiety; anxiety disorders;
 reactions to stressful experiences
 worthlessness, feelings of 160, 195, 196

Y

Young Mania Rating Scale (YMRS) 68

Z

Z drugs
 misuse of 577–78
 zolpidem 699
 zopiclone 699

